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Distributions, Partial Differential Equations, and Harmonic Analysis

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Dorina Mitrea

Distributions, Partial Differential Equations, and Harmonic Analysis

Springer

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Dedicată cu drag lui Diana şi Adrian,

Love, Mom

Preface

This book has been written from the personal perspective of a mathematician working at the interface between partial differential equations and harmonic analysis. Its aim is to offer, in a concise, rigorous, and largely self-contained form, a rapid introduction to the theory of distributions and its applications to partial differential equations and harmonic analysis. This is done in a format suitable for a graduate course spanning either a one-semester period, when the focus is primarily on the foundational aspects, or over a two-semester period, which allows for the proper amount of time to cover all intended applications as well.

Throughout, a special effort has been made to develop the theory of distributions not as an abstract edifice but rather to give the reader a chance to see the rationale behind various seemingly technical definitions, as well as the opportunity to apply the newly developed tools (in the natural build-up of the theory) to concrete problems in partial differential equations and harmonic analysis, at the earliest opportunity.

In addition to being suitable as a textbook for a graduate course, this book has been designed so that it may also be used for independent study since the presentation is reader-friendly, mostly self-sufficient (e.g., all auxiliary results originating outside the scope of this book have been carefully collected and presented in the appendix), and a large number of the suggested exercises have complete solutions.

Columbia, MO, USA — Dorina Irena Rita Mitrea

Contents

Preface vii

Introduction xiii

Common Notational Conventions xix

1 Weak Derivatives 1
- 1.1 The Cauchy Problem for a Vibrating Infinite String . . . 1
- 1.2 Weak Derivatives . . . 3
- 1.3 The Spaces $\mathcal{E}(\Omega)$ and $\mathcal{D}(\Omega)$ . . . 8
- 1.4 Additional Exercises for Chap. 1 . . . 13

2 The Space $\mathcal{D}'(\Omega)$ of Distributions 17
- 2.1 The Definition of Distributions . . . 17
- 2.2 The Topological Vector Space $\mathcal{D}'(\Omega)$ . . . 25
- 2.3 Multiplication of a Distribution with a C^∞ Function . . . 27
- 2.4 Distributional Derivatives . . . 29
- 2.5 The Support of a Distribution . . . 34
- 2.6 Compactly Supported Distributions and the Space $\mathcal{E}'(\Omega)$ . . . 37
- 2.7 Tensor Product of Distributions . . . 48
- 2.8 The Convolution of Distributions in $\mathbb{R}^n$ . . . 59
- 2.9 Distributions with Higher-Order Gradients Continuous or Bounded . . . 72
- 2.10 Additional Exercises for Chap. 2 . . . 80

3 The Schwartz Space and the Fourier Transform 89
- 3.1 The Schwartz Space of Rapidly Decreasing Functions . . . 89
- 3.2 The Action of the Fourier Transform on the Schwartz Class . . . 99
- 3.3 Additional Exercises for Chap. 3 . . . 106

4 The Space of Tempered Distributions 109
- 4.1 Definition and Properties of Tempered Distributions . . . 109
- 4.2 The Fourier Transform Acting on Tempered Distributions . . . 119
- 4.3 Homogeneous Distributions . . . 129

4.4 Principal Value Tempered Distributions . . . 135
4.5 The Fourier Transform of Principal Value Distributions . . . 141
4.6 Tempered Distributions Associated with $|x|^{-n}$ . . . 146
4.7 A General Jump-Formula in the Class of Tempered Distributions . . . 151
4.8 The Harmonic Poisson Kernel . . . 162
4.9 Singular Integral Operators . . . 167
4.10 Derivatives of Volume Potentials . . . 175
4.11 Additional Exercises for Chap. 4 . . . 184

5 The Concept of a Fundamental Solution 189
5.1 Constant Coefficient Linear Differential Operators . . . 189
5.2 A First Look at Fundamental Solutions . . . 191
5.3 The Malgrange–Ehrenpreis Theorem . . . 195
5.4 Additional Exercises for Chap. 5 . . . 200

6 Hypoelliptic Operators 201
6.1 Definition and Properties . . . 201
6.2 Hypoelliptic Operators with Constant Coefficients . . . 203
6.3 Integral Representation Formulas and Interior Estimates . . . 210
6.4 Additional Exercises for Chap. 6 . . . 216

7 The Laplacian and Related Operators 217
7.1 Fundamental Solutions for the Laplace Operator . . . 217
7.2 The Poisson Equation and Layer Potential Representation Formulas . . . 223
7.3 Fundamental Solutions for the Bi-Laplacian . . . 233
7.4 The Poisson Equation for the Bi-Laplacian . . . 237
7.5 Fundamental Solutions for the Poly-Harmonic Operator . . . 240
7.6 Fundamental Solutions for the Cauchy–Riemann Operator . . . 248
7.7 Fundamental Solutions for the Dirac Operator . . . 253
7.8 Fundamental Solutions for General Second-Order Operators . . . 260
7.9 Layer Potential Representation Formulas Revisited . . . 271
7.10 Additional Exercises for Chap. 7 . . . 274

8 The Heat Operator and Related Versions 277
8.1 Fundamental Solutions for the Heat Operator . . . 277
8.2 The Generalized Cauchy Problem for the Heat Operator . . . 280
8.3 Fundamental Solutions for General Second Order Parabolic Operators . . . 282
8.4 Fundamental Solution for the Schrödinger Operator . . . 287

9 The Wave Operator 289
9.1 Fundamental Solution for the Wave Operator . . . 289
9.2 The Generalized Cauchy Problem for the Wave Operator . . . 307
9.3 Additional Exercises for Chap. 9 . . . 308

10 The Lamé and Stokes Operators **309**
10.1 General Remarks About Vector and Matrix Distributions 309
10.2 Fundamental Solutions and Regularity for General Systems . . . 314
10.3 Fundamental Solutions for the Lamé Operator 319
10.4 Mean Value Formulas and Interior Estimates for the Lamé Operator 326
10.5 The Poisson Equation for the Lamé Operator 332
10.6 Fundamental Solutions for the Stokes Operator 334
10.7 Additional Exercises for Chap. 10 338

11 More on Fundamental Solutions for Systems **341**
11.1 Computing a Fundamental Solution for the Lamé Operator . . . 341
11.2 Computing a Fundamental Solution for the Stokes Operator . . . 343
11.3 Fundamental Solutions for Higher-Order Systems 344
11.4 Interior Estimates and Real-Analyticity for Null-Solutions of Systems 356
11.5 Reverse Hölder Estimates for Null-Solutions of Systems 361
11.6 Layer Potentials and Jump Relations for Systems 365
11.7 Additional Exercises for Chap. 11 373

12 Solutions to Selected Exercises **375**
12.1 Solutions to Exercises from Sect. 1.4 375
12.2 Solutions to Exercises from Sect. 2.10 381
12.3 Solutions to Exercises from Sect. 3.3 397
12.4 Solutions to Exercises from Sect. 4.11 400
12.5 Solutions to Exercises from Sect. 5.4 407
12.6 Solutions to Exercises from Sect. 6.4 408
12.7 Solutions to Exercises from Sect. 7.10 409
12.8 Solutions to Exercises from Sect. 9.3 413
12.9 Solutions to Exercises from Sect. 10.7 413
12.10 Solutions to Exercises from Sect. 11.7 414

13 Appendix **415**
13.1 Summary of Topological and Functional Analytic 415
13.2 Summary of Basic Results from Calculus, Measure Theory, and Topology 424
13.3 Custom-Designing Smooth Cut-Off Functions 428
13.4 Partition of Unity 429
13.5 The Gamma and Beta Functions 434
13.6 Surfaces in $\mathbb{R}^n$ and Surface Integrals 435
13.7 Integration by Parts and Green's Formula 437
13.8 Polar Coordinates and Integrals on Spheres 437
13.9 Tables of Fourier Transforms 445

Bibliography **449**

Subject Index **455**

Symbol Index **459**

Introduction

It has long been recognized that there is a large overlap and intricate interplay among distribution theory (DT), partial differential equations (PDE), and harmonic analysis (HA). The purpose of this book is to guide a reader with a background in basic real analysis through the journey taking her/him to the stage when such connections become self-evident.

Another goal of the present book is to convince the reader that traditional distinctions made among these branches of mathematics are largely artificial and are often simply a matter of choice in focus. Indeed, given the manner in which they complement, motivate, and draw inspiration from one another, it is not necessarily a stretch to attempt to pursue their development virtually simultaneously.

Concerning the triumvirate DT, PDE, HA, while there exist a number of good reference texts available on the market, they are by and large conceived in such a way that they either emphasize more one of these topics, typically at the detriment of the others, or are simply not particularly well-suited for a nonspecialist. By way of contrast, not only is the present text written in a way that brings together and blends the aforementioned topics in a unified, coherent body of results, but the resulting exposition is also sufficiently detailed and reader-friendly so that it may be read independently, outside the formal classroom setting. Indeed, the book is essentially self-contained, presents a balanced treatment of the topics involved, and contains a large number of exercises (upwards of 200, more than half of which are accompanied by solutions), which have been carefully chosen to amplify the effect, and substantiate the power and scope, of the theory discussed here.

While the topics treated are classical, the material is not entirely standard since a number of results are new even for a seasoned practitioner, and the overall architectural design of the monograph (including the way in which certain topics are covered) is original.

Regarding its inception, the present monograph is an expanded version of the notes prepared for a course on distribution theory that I taught in the Spring of 2007 and the Spring of 2011 at the University of Missouri. My intention was to present the theory of distributions not as an abstract edifice but rather to give the student a chance to instantaneously see the justification and practical benefits of the multitude of seemingly technical definitions and results, as well

as give her/him the opportunity to immediately see how the newly introduced concepts (in the natural build-up of the theory) apply to concrete problems in PDE and harmonic analysis.

Special care has been paid to the pedagogical aspect of the presentation of the material in the book. For example, a notable feature of the present monograph is the fact that fundamental solutions for some of the most basic differential operators in mathematical physics and engineering, including Laplace, heat, wave, poly-harmonic, Dirac, Lamé, Stokes, and Schrödinger, are systematically deduced starting from first principles. This stands in contrast with the more common practice in the literature in which one starts with a certain distribution (the origins of which are fairly obscure) and simply checks that the distribution in question is a fundamental solution for a given differential operator. Another feature is the emphasis placed on the interrelations between topics. For example, a clear picture is presented as to how DT vastly facilitates the computation of fundamental solutions, and the development of singular integral operators, tools which, in turn, are used to solve PDE as well as represent and estimate solutions of PDE.

The presentation is also conceived in such a way as to avoid having to confront heavy-duty topology/functional analysis up front, in the main narrative. For example, the jargon associated with the multitude of topologies on various spaces of test functions and distributions is minimized by deferring to an appendix the technical details while retaining in the main body of the discussion only those consequences that are most directly relevant to the fluency of the exposition.

While the core material I had in mind deals primarily with the theory of distributions, this book is ultimately devised in such a way as to make the present material a solid launching pad for a number of subsequent courses dealing with allied topics, including:

- Harmonic analysis
- Partial differential equations
- Boundary integral methods
- Sobolev spaces
- Pseudodifferential operators

For example, the theory of singular integral operators of convolution type in $L^2(\mathbb{R}^n)$ is essentially developed here, in full detail, up to the point where more specialized tools from harmonic analysis (such as the Hardy–Littlewood maximal operator and the Calderón–Zygmund lemma) are typically involved in order to further extend this theory (via a weak-$(1,1)$ estimate, interpolation with L^2, and then duality) to L^p spaces with $p \in (1,\infty)$, as in [6, 9, 16, 23, 47, 62–64], among others.

Regarding connections with PDE, the Poisson problem in the whole space,

$$Lu = f \quad in \quad \mathcal{D}'(\mathbb{R}^n), \tag{0.0.1}$$

is systematically treated here for a variety of differential operators L, including the Laplacian, the bi-harmonic operator, and the Lamé system. While Sobolev spaces are not explicitly considered (the interested reader may consult in this regard [1, 2, 13, 40, 43, 44, 76], to cite just a small fraction of a large body of literature on this topic), they lurk just beneath the surface as their presence is implicit in estimates of the form

$$\sum_{|\alpha|=m} \|\partial^\alpha u\|_{L^p(\mathbb{R}^n)} \leq C(L,p)\|f\|_{L^p(\mathbb{R}^n)}, \qquad 1 < p < \infty, \tag{0.0.2}$$

where m denotes the order of the elliptic differential operator L.

Such estimates are deduced from an integral representation formula for u, involving a fundamental solution of L, and estimates for singular integral operators of convolution type in $\mathbb{R}^n$. In particular, this justifies two features of the present monograph: (1) the emphasis placed on finding explicit formulas for fundamental solutions for a large number of operators of basic importance in mathematics, physics, and engineering; (2) the focus on the theory of singular integral operators of convolution type, developed along side the distribution theory. In addition, the analysis of the Cauchy problem formulated and studied for the heat and wave operators once more underscores the significance of the fundamental solutions for the named operators. As a whole, this material is designed to initiate the reader into the field of PDE. At the same time, it complements, and works well in tandem with, the treatment of this subject in [3, 12, 21, 30–32, 46, 68, 69, 75].

Whenever circumstances permit it, other types of problems are brought into play, such as the Dirichlet and Neumann problem in the upper-half space for the Laplacian, as well as more general second-order systems. In turn, the latter genre of boundary value problems motivates introducing and developing boundary integral methods, and serves as an opportunity to highlight the basic role that layer potential operators play in this context. References dealing with the latter topic include [34, 36, 46, 48, 74].

Analysis of the structure of the boundary layer potential operators naturally intervening in this context also points to the possibility of considering larger classes of operators where the latter may be composed, inverted, etc., in a stable fashion. This serves as an excellent motivation for the introduction of such algebras of operators as pseudodifferential and Fourier integral operators, a direction that the interested reader may then pursue in, for example, [24, 67, 69, 75], to name a few sources.

A brief description of this book's contents is as follows.

Chapters 1–2 are devoted to the development of the most basic aspects of the theory of distributions. Starting from the discussion of the Cauchy problem for a vibrating infinite string as a motivational example, the notion of a weak derivative is introduced as a mean of extending the notion of solution to a more general setting, where the functions involved may lack standard pointwise differentiability properties. After touching upon classes of test functions, the space of distributions is then introduced and studied from the perspective of a topological vector space with various other additional features (such as the concept of support, and a partially defined convolution product). Chapter 3 contains material pertaining to the Schwartz space of functions rapidly decaying at infinity and the Fourier transform in such a setting.

In Chap. 4 the action of the Fourier transform is further extended to the setting of tempered distributions, and several distinguished subclasses of tempered distributions are introduced and studied (including homogeneous and principal value distributions). The foundational material developed up to this point already has significant applications to harmonic analysis and PDE. For example, a general, higher-dimensional jump-formula is deduced in this chapter for a certain class of tempered distributions (that includes the classical harmonic Poisson kernel) which is latter used as the main tool in deriving information about the boundary behavior of layer potential operators associated with various partial differential operators and systems. Also, one witnesses here how singular integral operators of central importance to harmonic analysis (such as the Riesz transforms) naturally arise as an extension to L^2 of the convolution product of tempered distributions of principal value type with Schwartz functions.

The first explicit encounter with the notion of fundamental solution takes place in Chap. 5, where the classical Malgrange–Ehrenpreis theorem is presented. Subsequently, in Chap. 6, the concept of hypoelliptic operator is introduced and studied. In particular, here a classical result, due to L. Schwartz, is proved to the effect that a necessary and sufficient condition for a linear, constant coefficient differential operator to be hypoelliptic in the entire ambient space is that the named operator possesses a fundamental solution with singular support consisting of the origin alone. In Chap. 6 we also prove an integral representation formula and interior estimates for a subclass of hypoelliptic operators, which are subsequently used to show that null-solutions of these operators are real-analytic.

One of the main goals in Chap. 7 is identifying (starting from first principles) all fundamental solutions that are tempered distributions for scalar elliptic operators. While the natural starting point is the Laplacian, this study encompasses a variety of related operators, such as the bi-Laplacian, the polyharmonic operator, the Cauchy–Riemann operator, the Dirac operator, as well as general second-order constant coefficient strongly elliptic operators. Having

accomplished this task then makes it possible to prove the well-posedness of the Poisson problem (0.0.1) (equipped with a boundary condition at infinity), and derive qualitative/quantitative properties for the solution such as (0.0.2). Along the way, Cauchy-like integral operators are also introduced and their connections with Hardy spaces is brought to light in the setting of both complex and Clifford analysis.

Chapter 8 has a twofold aim: determine all fundamental solutions that are tempered distributions for the heat operator and related versions (including the Schrödinger operator), then use this as a toll in the solution of the generalized Cauchy problem for the heat operator. The same type of program is then carried out in Chap. 9, this time in connection with the wave operator.

While the analysis up to this point has been largely confined to scalar operators, the final two chapters in the book are devoted to studying systems of differential operators. The material in Chap. 10 is centered around two such basic systems: the Lamé operator arising in the theory of elasticity, and the Stokes operator arising in hydrodynamics. Among other things, all their fundamental solutions that are tempered distributions are identified, and the well-posedness of the Poisson problem for the Lamé system is established. The former issue is then revisited in the first part of Chap. 11 from a different perspective, and subsequently generalized to the case of (homogeneous) constant coefficient systems of arbitrary order. In Chap. 11 we also show that integral representation formulas and interior estimates hold for null-solutions of homogeneous systems with non-vanishing full symbol. As a consequence, we prove that such null-solutions are real-analytic and satisfy reverse Hölder estimates. The final topic addressed in Chap. 11 pertains to layer potentials associated with arbitrary constant coefficient second order systems in the upper-half space, and the relevance of these operators vis-a-vis to the solvability of boundary value problems for such systems in this setting.

For completeness, a summary of topological and functional analysis results in reference to the description of the topology and equivalent characterizations of convergence in spaces of test functions and in spaces of distributions is included in the appendix (which also contains a variety of foundational results from calculus, measure theory, and special functions originating outside the scope of this book). One aspect worth noting in this regard is that the exposition in the main body of the book may be followed even without being fully familiar with all these details by alternatively taking, as the starting point, the characterization of convergence in the various topologies considered here (summarized in the main text under the heading **Fact**) as definitions. Such an approach makes the topics covered in the present monograph accessible to a larger audience while, at the same time, provides a full treatment of the topological and functional analysis background accompanying the theory of distributions for the reader interested in a more in-depth treatment.

Finally, each book chapter ends with bibliographical references tailored to its respective contents under the heading **Further Notes**, as well as with a number of additional exercises, selectively solved in Chap. 12.

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Common Notational Conventions

Throughout this book the set of natural numbers will be denoted by $\mathbb{N}$, that is $\mathbb{N} := \{1, 2, \dots\}$, while $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$. For each $k \in \mathbb{N}$ set $k! := 1 \cdot 2 \cdots (k-1) \cdot k$, and make the convention that $0! := 1$. The letter $\mathbb{C}$ will denote the set of complex numbers, and $\overline{z}$ denotes the complex conjugate of $z \in \mathbb{C}$. Also the real and imaginary parts of a complex number z are denoted by $\operatorname{Re} z$ and $\operatorname{Im} z$, respectively. The symbol i is reserved for the complex imaginary unit $\sqrt{-1} \in \mathbb{C}$. The letter $\mathbb{R}$ will denote the set of real numbers and its n-fold Cartesian product of $\mathbb{R}$ with itself (where $n \in \mathbb{N}$) is denoted by $\mathbb{R}^n$. That is,

$$\mathbb{R}^n := \big\{x = (x_1, \dots, x_n) : x_1, \dots, x_n \in \mathbb{R}\big\} \tag{0.0.3}$$

considered with the usual vector space and inner product structure, that is,

$$\begin{gathered} x + y := (x_1 + y_1, \dots, x_n + y_n), \quad cx := (cx_1, \dots, cx_n), \quad x \cdot y := \sum_{j=1}^{n} x_j y_j, \\ \forall\, x = (x_1, \dots, x_n) \in \mathbb{R}^n, \ \ \forall\, y = (y_1, \dots, y_n) \in \mathbb{R}^n, \ \ \forall\, c \in \mathbb{R}. \end{gathered} \tag{0.0.4}$$

The standard orthonormal basis of vectors in $\mathbb{R}^n$ is denoted by $\{\mathbf{e}_j\}_{1 \le j \le n}$, where we have set $\mathbf{e}_j := (0, \dots, 0, 1, 0, \dots, 0) \in \mathbb{R}^n$ with the only nonzero component on the j-th slot. We shall also consider the two canonical (open) half-spaces of $\mathbb{R}^n$, denoted by $\mathbb{R}^n_\pm := \{x = (x_1, \dots, x_n) \in \mathbb{R}^n : \pm x_n > 0\}$. Hence, $\mathbb{R}^n_+ = \mathbb{R}^{n-1} \times (0, \infty)$ and $\mathbb{R}^n_- = \mathbb{R}^{n-1} \times (-\infty, 0)$.

Given a multi-index $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$, we set

$$\alpha! := \alpha_1! \alpha_2! \cdots \alpha_n! \quad \text{and} \quad |\alpha| := \sum_{j=1}^{n} \alpha_j, \tag{0.0.5}$$

$$\partial^\alpha := \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n}, \quad \text{where} \quad \partial_j := \frac{\partial}{\partial x_j} \ \text{ for } j = 1, \dots, n, \tag{0.0.6}$$

$$x^\alpha := \prod_{j=1}^{n} x_j^{\alpha_j} \quad \text{for every } \ x = (x_1, \dots, x_n) \in \mathbb{C}^n. \tag{0.0.7}$$

Also if $\beta = (\beta_1, \dots, \beta_n) \in \mathbb{N}_0^n$ is another multi-index we shall write $\beta \le \alpha$ provided $\beta_j \le \alpha_j$ for each $j \in \{1, \dots, n\}$, in which case we set $\alpha - \beta := (\alpha_1 - \beta_1, \dots, \alpha_n - \beta_n)$. We shall also say that $\beta < \alpha$ if $\beta \le \alpha$ and $\beta \ne \alpha$. Recall that the Kronecker symbol is defined by $\delta_{jk} := 1$ if $j = k$ and $\delta_{jk} := 0$ if $j \ne k$.

All functions in this monograph are assumed to be complex valued unless otherwise indicated. Derivatives of a function f defined on the real line are going to be denoted using f', f'', etc., or $f^{(k)}$, or $\frac{\mathrm{d}^k f}{\mathrm{d}x^k}$.

Throughout the book, Ω denotes an arbitrary open subset of $\mathbb{R}^n$. If A is an arbitrary subset of $\mathbb{R}^n$, then $\mathring{A}$, $\overline{A}$, and ∂A denote its interior, its closure, and its boundary, respectively. In addition, if B is another arbitrary subset of $\mathbb{R}^n$, then their set theoretic difference is denoted by $A \setminus B := \{x \in A : x \notin B\}$. In particular, the complement of A is $A^c := \mathbb{R}^n \setminus A$. For any $E \subset \mathbb{R}^n$ we let χ_E stand for the characteristic function of the set E (i.e., $\chi_E(x) = 1$ if $x \in E$ and $\chi_E(x) = 0$ if $x \in \mathbb{R}^n \setminus E$).

For $k \in \mathbb{N}_0 \cup \{\infty\}$, we will work with the following classes of functions that are vector spaces over $\mathbb{C}$:

$$C^k(\Omega) := \big\{\varphi : \Omega \to \mathbb{C} : \partial^\alpha \varphi \text{ continuous } \forall\, \alpha \in \mathbb{N}_0^n,\ |\alpha| \leq k\big\}, \tag{0.0.8}$$

$$C^k(\overline{\Omega}) := \big\{\varphi|_{\overline{\Omega}} : \varphi \in C^k(U),\ U \subseteq \mathbb{R}^n \text{ open set containing } \overline{\Omega}\,\big\}, \tag{0.0.9}$$

$$C_0^k(\Omega) := \big\{\varphi \in C^k(\Omega) : \operatorname{supp}\varphi \text{ compact subset of } \Omega\big\}. \tag{0.0.10}$$

As usual, for any Lebesgue-measurable (complex-valued) function f defined on a Lebesgue measurable set $E \subseteq \mathbb{R}^n$ and any $p \in [1, \infty]$ we write

$$\|f\|_{L^p(E)} := \begin{cases} \big(\int_E |f|^p \,\mathrm{d}x\big)^{1/p} & \text{if } 1 \leq p < \infty, \\ \text{ess-sup}_E |f| & \text{if } p = \infty, \end{cases} \tag{0.0.11}$$

and denote by $L^p(E)$ the Banach space of (equivalence classes of) Lebesgue-measurable functions f on E satisfying $\|f\|_{L^p(E)} < \infty$. Also, we will work with locally integrable functions and with compactly supported integrable functions. For $p \in [1, \infty]$ these are defined as

$$L^p_{\text{loc}}(\Omega) := \big\{f : \Omega \to \mathbb{C} : f \text{ Lebesgue measurable such that } \|f\|_{L^p(K)} < \infty, \quad \forall\, K \subset \Omega \text{ compact set}\big\}, \tag{0.0.12}$$

and, respectively, as

$$L^p_{\text{comp}}(\Omega) := \big\{f \in L^p(\Omega) : \operatorname{supp} f \text{ compact subset of } \Omega\big\}, \tag{0.0.13}$$

where $\operatorname{supp} f$ is defined in (2.5.10).

Given a measure space (X, μ), a measurable set $A \subseteq X$ with $0 < \mu(A) < \infty$, and a function $f \in L^1(A, \mu)$, we define the integral average of f over A by

$$\fint_A f \,\mathrm{d}\mu := \frac{1}{\mu(A)} \int_A f \,\mathrm{d}\mu. \tag{0.0.14}$$

If E is a Lebesgue measurable subset of $\mathbb{R}^n$, the Lebesgue measure of E is denoted by $|E|$. If $x \in \mathbb{R}^n$ and radius $r > 0$, set $B(x, r) := \{y \in \mathbb{R}^n : |y - x| < r\}$ for the ball of center x and radius R and its boundary is denoted by $\partial B(0, r)$. The unit sphere in $\mathbb{R}^n$ centered at zero is $S^{n-1} := \{x \in \mathbb{R}^n : |x| = 1\} = \partial B(0, 1)$ and its surface measure is denoted by ω_{n-1}.

For $n, m \in \mathbb{N}$ and $\mathfrak{R}$, an arbitrary commutative ring with multiplicative units, we denote by $\mathcal{M}_{n\times m}(\mathfrak{R})$ the collection of all $n \times m$ matrices with entries from $\mathfrak{R}$. If $B \in \mathcal{M}_{n\times m}(\mathfrak{R})$, then $B^{\top}$ denotes its transpose and if $n = m$, then $\det B$ denotes the determinant of the matrix B, while $I_{n\times n}$ denotes the identity matrix.

Regarding semi-orthodox notational conventions, $A := B$ stands for "A is defined as being B." while $A =: B$ stands for "B is defined as being A." Also, the letter C when used as a multiplicative constant in various inequalities, is allowed to vary from line to line. Whenever necessary, its dependence on the other parameters $a, b, \ldots$ implicit in the estimate in question is stressed by writing $C(a, b, \ldots)$ or $C_{a,b,\ldots}$ in place of just C.

Chapter 1

Weak Derivatives

1.1 The Cauchy Problem for a Vibrating Infinite String

The partial differential equation

$$\partial_1^2 u - \partial_2^2 u = 0 \quad \text{in} \quad \mathbb{R}^2 \tag{1.1.1}$$

was derived by Jean d'Alembert in 1747 to describe the displacement $u(x_1, x_2)$ of a violin string as a function of time and distance along the string. Assuming that the string is infinite and that at time $x_2 = 0$ the displacement is given by some function $\varphi_0 \in C^2(\mathbb{R})$ leads to the following global Cauchy problem

$$\begin{cases} u \in C^2(\mathbb{R}^2), \\ \partial_1^2 u - \partial_2^2 u = 0 & \text{in } \mathbb{R}^2, \\ u(\cdot, 0) = \varphi_0 & \text{in } \mathbb{R}, \\ (\partial_2 u)(\cdot, 0) = 0 & \text{in } \mathbb{R}. \end{cases} \tag{1.1.2}$$

Thanks to the regularity assumption on φ_0, it may be checked without difficulty that the function

$$u(x_1, x_2) := \tfrac{1}{2}[\varphi_0(x_1 + x_2) + \varphi_0(x_1 - x_2)], \qquad \text{for every } (x_1, x_2) \in \mathbb{R}^2, \tag{1.1.3}$$

is a solution of (1.1.2). This being said, the expression of u in (1.1.3) continues to be meaningful under much less restrictive assumptions on φ_0. For example, u is a well-defined continuous function in $\mathbb{R}^2$ whenever $\varphi_0 \in C^0(\mathbb{R})$. While in this case expression (1.1.3) is no longer a classical solution of (1.1.2), it is natural to ask whether it is possible to identify a new (and possibly weaker) sense in which (1.1.3) would continue to satisfy $\partial_1^2 u - \partial_2^2 u = 0$.

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To answer this question, fix a function $u \in C^2(\mathbb{R}^2)$ satisfying $\partial_1^2 u - \partial_2^2 u = 0$ pointwise in $\mathbb{R}^2$. If $\varphi \in C_0^\infty(\mathbb{R}^2)$ is an arbitrary function and $R \in (0,\infty)$ is a number such that $\operatorname{supp}\varphi \subseteq (-R,R)\times(-R,R)$, then integration by parts gives

$$\begin{aligned} 0 &= \iint_{\mathbb{R}^2} (\partial_1^2 u - \partial_2^2 u)\varphi \,\mathrm{d}x \\ &= \int_{-R}^{R}\int_{-R}^{R} \partial_1^2 u(x_1,x_2)\varphi(x_1,x_2)\,\mathrm{d}x_1\mathrm{d}x_2 \\ &\quad - \int_{-R}^{R}\int_{-R}^{R} \partial_2^2 u(x_1,x_2)\varphi(x_1,x_2)\,\mathrm{d}x_1\mathrm{d}x_2 \\ &= \iint_{\mathbb{R}^2} (\partial_1^2\varphi - \partial_2^2\varphi)u \,\mathrm{d}x. \end{aligned} \tag{1.1.4}$$

Note that the condition $\iint_{\mathbb{R}^2}(\partial_1^2\varphi - \partial_2^2\varphi)u\,\mathrm{d}x = 0$ for all $\varphi \in C_0^\infty(\mathbb{R}^2)$ is meaningful even if $u \in C^0(\mathbb{R}^2)$, which suggests the following definition.

Definition 1.1. *A function $u \in C^0(\mathbb{R}^2)$ is called a* `weak` *(*`generalized`*) solution of the equation $\partial_1^2 u - \partial_2^2 u = 0$ in $\mathbb{R}^2$ if*

$$\iint_{\mathbb{R}^2} (\partial_1^2\varphi - \partial_2^2\varphi)u\,\mathrm{d}x = 0 \quad \textit{for all } \varphi \in C_0^\infty(\mathbb{R}^2). \tag{1.1.5}$$

Returning to (1.1.3), let us now check that, under the assumption $\varphi_0 \in C^0(\mathbb{R})$, the function u defined in (1.1.3) is a generalized solution of $\partial_1^2 u - \partial_2^2 u = 0$ in $\mathbb{R}^2$. Concretely, fix $\varphi \in C_0^\infty(\mathbb{R}^2)$ and write

$$\begin{aligned} &\iint_{\mathbb{R}^2} (\partial_1^2\varphi - \partial_2^2\varphi)u\,\mathrm{d}x \\ &= \frac{1}{2}\int_{\mathbb{R}}\int_{\mathbb{R}} \big[\partial_1^2\varphi(x_1,x_2) - \partial_2^2\varphi(x_1,x_2)\big]\big(\varphi_0(x_1+x_2)+\varphi_0(x_1-x_2)\big)\,\mathrm{d}x_1\mathrm{d}x_2 \\ &= \frac{1}{4}\int_{\mathbb{R}}\int_{\mathbb{R}} (\partial_1^2\varphi)\left(\frac{y_1+y_2}{2},\frac{y_1-y_2}{2}\right)\big(\varphi_0(y_1)+\varphi_0(y_2)\big)\,\mathrm{d}y_1\mathrm{d}y_2 \\ &\quad - \frac{1}{4}\int_{\mathbb{R}}\int_{\mathbb{R}} (\partial_2^2\varphi)\left(\frac{y_1+y_2}{2},\frac{y_1-y_2}{2}\right)\big(\varphi_0(y_1)+\varphi_0(y_2)\big)\,\mathrm{d}y_1\mathrm{d}y_2, \end{aligned} \tag{1.1.6}$$

where for the last equality in (1.1.6) we have made the change of variables $x_1 + x_2 = y_1$, $x_1 - x_2 = y_2$. If we now let $\psi(y_1,y_2) := \varphi\big(\frac{y_1+y_2}{2},\frac{y_1-y_2}{2}\big)$ for $(y_1,y_2)\in\mathbb{R}^2$, then

$$\partial_1\psi(y_1,y_2) = \frac{1}{2}(\partial_1\varphi)\left(\frac{y_1+y_2}{2},\frac{y_1-y_2}{2}\right) + \frac{1}{2}(\partial_2\varphi)\left(\frac{y_1+y_2}{2},\frac{y_1-y_2}{2}\right), \tag{1.1.7}$$

and

$$\partial_2\partial_1\psi(y_1,y_2)=\frac{1}{4}(\partial_1^2\varphi)\left(\frac{y_1+y_2}{2},\frac{y_1-y_2}{2}\right)-\frac{1}{4}(\partial_2\partial_1\varphi)\left(\frac{y_1+y_2}{2},\frac{y_1-y_2}{2}\right)$$
$$+\frac{1}{4}(\partial_1\partial_2\varphi)\left(\frac{y_1+y_2}{2},\frac{y_1-y_2}{2}\right)-\frac{1}{4}(\partial_2^2\varphi)\left(\frac{y_1+y_2}{2},\frac{y_1-y_2}{2}\right) \tag{1.1.8}$$

which, when used in (1.1.6), give

$$\iint_{\mathbb{R}^2}(\partial_1^2\varphi-\partial_2^2\varphi)u\,\mathrm{d}x=\int_{\mathbb{R}}\int_{\mathbb{R}}\partial_1\partial_2\psi(y_1,y_2)[\varphi_0(y_1)+\varphi_0(y_2)]\,\mathrm{d}y_1\,\mathrm{d}y_2. \tag{1.1.9}$$

Let $R\in(0,\infty)$ be such that $\operatorname{supp}\varphi\subset(-R,R)\times(-R,R)$. Then the support of ψ is contained in the set of points $(y_1,y_2)\in\mathbb{R}^2$ satisfying $-2R\le y_1+y_2\le 2R$ and $-2R\le y_1-y_2\le 2R$. Hence, if $R'>2R$ we have $\operatorname{supp}\psi\subset(-R',R')\times(-R',R')$ and integration by parts yields

$$\begin{aligned}
&\iint_{\mathbb{R}^2}(\partial_1^2\varphi-\partial_2^2\varphi)u\,\mathrm{d}x\\
&\quad=\int_{-R'}^{R'}\varphi_0(y_1)\int_{-R'}^{R'}\partial_2\partial_1\psi(y_1,y_2)\,\mathrm{d}y_2\,\mathrm{d}y_1\\
&\qquad+\int_{-R'}^{R'}\varphi_0(y_2)\int_{-R'}^{R'}\partial_1\partial_2\psi(y_1,y_2)\,\mathrm{d}y_1\,\mathrm{d}y_2\\
&\quad=\int_{-R'}^{R'}\varphi_0(y_1)\big[\partial_1\psi(y_1,R')-\partial_1\psi(y_1,-R')\big]\,\mathrm{d}y_1\\
&\qquad+\int_{-R'}^{R'}\varphi_0(y_2)\big[\partial_2\psi(R',y_2)-\partial_2\psi(-R',y_2)\big]\,\mathrm{d}y_2=0.
\end{aligned} \tag{1.1.10}$$

In summary, this proves that

$$\begin{aligned}&\text{for each } \varphi_0\in C^0(\mathbb{R}) \text{ the function } u \text{ defined as in (1.1.3)}\\&\text{is a weak solution of the equation } \partial_1^2u-\partial_2^2u=0 \text{ in } \mathbb{R}^2.\end{aligned} \tag{1.1.11}$$

We emphasize that in general, there is no reason to expect that u has any pointwise differentiability properties if φ_0 is merely continuous.

1.2 Weak Derivatives

Convention. Unless otherwise specified, Ω denotes an arbitrary open subset of $\mathbb{R}^n$.

Before proceeding with the definition of weak (Sobolev) derivatives we discuss a phenomenon that serves as motivation for the definition. Let $f\in C^m(\Omega)$,

where $m \in \mathbb{N}_0$. Given an arbitrary $\varphi \in C_0^\infty(\Omega)$, consider a function $F \in C_0^m(\mathbb{R}^n)$ that agrees with f in a neighborhood of $\operatorname{supp}\varphi$. For example, we may take $F = \psi\widetilde{f}$, where tilde denotes the extension by zero outside Ω and $\psi \in C_0^\infty(\mathbb{R}^n)$ is identically one in a neighborhood of $\operatorname{supp}\varphi$ (see Proposition 13.26 for the construction of such a function). Also, pick $R > 0$ large enough so that $\operatorname{supp}\varphi \subset B(0,R)$. Then for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq m$, integration by parts (cf. Theorem 13.41 for a precise formulation) and support considerations yield

$$\begin{aligned}\int_\Omega (\partial^\alpha f)\varphi\,\mathrm{d}x &= \int_{B(0,R)} (\partial^\alpha F)\varphi\,\mathrm{d}x \\ &= (-1)^{|\alpha|}\int_{B(0,R)} F\,\partial^\alpha\varphi\,\mathrm{d}x = (-1)^{|\alpha|}\int_\Omega f\,\partial^\alpha\varphi\,\mathrm{d}x. \end{aligned} \tag{1.2.1}$$

This computation suggests the following definition.

Definition 1.2. *If $f \in L^1_{\mathrm{loc}}(\Omega)$ and $\alpha \in \mathbb{N}_0^n$, we say that $\partial^\alpha f$ belongs to $L^1_{\mathrm{loc}}(\Omega)$ in a* `weak` *(*`Sobolev`*)* `sense` *provided there exists some $g \in L^1_{\mathrm{loc}}(\Omega)$ with the property that*

$$\int_\Omega g\varphi\,\mathrm{d}x = (-1)^{|\alpha|}\int_\Omega f\,\partial^\alpha\varphi\,\mathrm{d}x \qquad \text{for every} \quad \varphi \in C_0^\infty(\Omega). \tag{1.2.2}$$

Whenever this happens, we shall write $\partial^\alpha f = g$ and call g the `weak derivative` *of order α of f.*

The fact that the concept of weak derivative is unambiguously defined is then ensured by the next theorem.

Theorem 1.3. *If $g \in L^1_{\mathrm{loc}}(\Omega)$ and $\int_\Omega g\varphi\,\mathrm{d}x = 0$ for each $\varphi \in C_0^\infty(\Omega)$ then $g = 0$ almost everywhere on Ω.*

Proof. From the start, we note that the real and imaginary parts of g enjoy the same type of property as g itself (as seen by integrating g against real-valued test functions φ). Thus, without any loss of generality we may assume that g is real-valued.

We make the claim $\int_\Omega gu\,\mathrm{d}x = 0$ for all $u \in L^\infty_{\mathrm{comp}}(\Omega)$. To see that this is the case, pick an arbitrary $u \in L^\infty_{\mathrm{comp}}(\Omega)$. By working with the extension of u by zero outside Ω, we may assume that $u \in L^\infty_{\mathrm{comp}}(\mathbb{R}^n)$ and $\operatorname{supp} u \subset \Omega$, to begin with. To proceed, consider a function ϕ satisfying (see (13.3.3) for a concrete example)

$$\phi \in C_0^\infty(\mathbb{R}^n), \quad \phi \geq 0, \quad \operatorname{supp}\phi \subseteq \overline{B(0,1)} \ \text{ and } \ \int_{\mathbb{R}^n}\phi(x)\mathrm{d}x = 1, \tag{1.2.3}$$

and for each $\varepsilon > 0$ define

$$\phi_\varepsilon(x) := \frac{1}{\varepsilon^n}\phi\left(\frac{x}{\varepsilon}\right), \qquad \text{for each } x \in \mathbb{R}^n. \tag{1.2.4}$$

Then

$$\phi_\varepsilon \in C_0^\infty(\mathbb{R}^n), \quad \phi_\varepsilon \geq 0, \quad \operatorname{supp}\phi_\varepsilon \subseteq \overline{B(0,\varepsilon)}, \quad \text{and} \quad \int_{\mathbb{R}^n} \phi_\varepsilon(x)\,\mathrm{d}x = 1. \tag{1.2.5}$$

Let us now introduce

$$u_\varepsilon(x) := \int_{\mathbb{R}^n} u(y)\phi_\varepsilon(x-y)\,\mathrm{d}y, \qquad \forall\, x \in \mathbb{R}^n, \tag{1.2.6}$$

and set $K_\varepsilon := \big\{x \in \mathbb{R}^n : \operatorname{dist}(x, \operatorname{supp} u) \leq \varepsilon\big\}$. Since u has compact support contained in Ω it follows that there exists $\varepsilon_0 > 0$ such that

$$u_\varepsilon \in C^\infty(\mathbb{R}^n) \ \text{ and } \ \operatorname{supp} u_\varepsilon \subseteq K_{\varepsilon_0} \subset \Omega \ \text{ if } \ 0 < \varepsilon \leq \varepsilon_0. \tag{1.2.7}$$

In particular, the restriction of u_ε to Ω satisfies $u_\varepsilon\big|_\Omega \in C_0^\infty(\Omega)$ provided we select $0 < \varepsilon \leq \varepsilon_0$. Granted the current assumptions on g, it follows that

$$\int_\Omega g u_\varepsilon\,\mathrm{d}x = 0 \quad \text{whenever} \ \ 0 < \varepsilon \leq \varepsilon_0. \tag{1.2.8}$$

Next, using the properties of ϕ_ε and the definition of u_ε, we may write

$$\begin{aligned} |u_\varepsilon(x) - u(x)| &= \left| \int_{B(0,\varepsilon)} u(x-y)\phi_\varepsilon(y)\,\mathrm{d}y - u(x) \right| \\ &= \left| \int_{B(0,\varepsilon)} [u(x-y) - u(x)]\phi_\varepsilon(y)\,\mathrm{d}y \right| \\ &\leq \frac{1}{\varepsilon^n} \int_{B(0,\varepsilon)} |u(x-y) - u(x)| \phi\left(\frac{y}{\varepsilon}\right)\,\mathrm{d}y \\ &\leq \frac{c}{|B(0,\varepsilon)|} \int_{B(0,\varepsilon)} |u(x-y) - u(x)|\mathrm{d}y \underset{\varepsilon\to 0^+}{\longrightarrow} 0 \quad \text{for a.e. } x \in \mathbb{R}^n, \end{aligned} \tag{1.2.9}$$

where $c := \frac{\omega_{n-1}}{n}\|\phi\|_{L^\infty(\mathbb{R}^n)}$, and for the convergence in (1.2.9) we used Lebesgue's differentiation theorem (cf. Theorem 13.11). Hence,

$$u_\varepsilon \to u \ \text{ as } \ \varepsilon \to 0^+ \ \text{ pointwise for almost every } x \in \mathbb{R}^n. \tag{1.2.10}$$

In addition, since $\int_{\mathbb{R}^n} \phi_\varepsilon(x-y)\,\mathrm{d}y = 1$ for every $x \in \mathbb{R}^n$, it follows that $|u_\varepsilon(x)| \leq \|u\|_{L^\infty(\mathbb{R}^n)}$ for every $x \in \mathbb{R}^n$. Together with (1.2.7), this implies

$$|g u_\varepsilon| = |u_\varepsilon||g| \leq \|u\|_{L^\infty(\mathbb{R}^n)} |g| \chi_{K_{\varepsilon_0}} \in L^1(\Omega). \tag{1.2.11}$$

Making use of (1.2.8) and Lebesgue's dominated convergence theorem (cf. Theorem 13.12 that applies thanks to (1.2.10) and (1.2.11)), we therefore obtain

$$0 = \lim_{\varepsilon\to 0^+} \int_\Omega g u_\varepsilon\,\mathrm{d}x = \int_\Omega g u\,\mathrm{d}x, \tag{1.2.12}$$

finishing the proof of the claim made earlier.

To proceed, pick a sequence of compact sets $\{K_j\}_{j\in\mathbb{N}}$ such that $\Omega = \bigcup\limits_{j\in\mathbb{N}} K_j$. For example,

$$K_j := \left\{x \in \Omega : \operatorname{dist}(x, \partial\Omega) \geq \tfrac{1}{j}\right\} \cap \overline{B(0,j)}, \qquad j \in \mathbb{N}, \tag{1.2.13}$$

will do. Also, for each $j \in \mathbb{N}$, define

$$u_j := \chi_{\{x\in K_j:\, g(x)>0\}} - \chi_{\{x\in K_j:\, g(x)<0\}}. \tag{1.2.14}$$

Then for each $j \in \mathbb{N}$ we have $u_j \in L^\infty_{\rm comp}(\Omega)$; hence, based on what we proved so far, $0 = \int_\Omega g u_j \, \mathrm{d}x = \int_{K_j} |g| \, \mathrm{d}x$. The latter condition implies that $g = 0$ almost everywhere in each K_j hence, $g = 0$ almost everywhere in $\bigcup\limits_{j\in\mathbb{N}} K_j = \Omega$. □

Next we present a few examples related to the notion of weak (Sobolev) derivative.

Example 1.4. Consider the function

$$f : \mathbb{R} \longrightarrow \mathbb{R}, \qquad f(x) := \begin{cases} x, & x > 0, \\ 0, & x \leq 0, \end{cases} \qquad \forall\, x \in \mathbb{R}. \tag{1.2.15}$$

Note that f is continuous on $\mathbb{R}$ but not differentiable at 0. Nonetheless, f has a weak derivative of order one that is equal to the Heaviside function

$$H : \mathbb{R} \longrightarrow \mathbb{R}, \qquad H(x) := \begin{cases} 1, & x > 0, \\ 0, & x \leq 0, \end{cases} \qquad \forall\, x \in \mathbb{R}. \tag{1.2.16}$$

Indeed, if $\varphi \in C^\infty_0(\mathbb{R})$, then integration by parts yields

$$\begin{aligned} -\int_{-\infty}^{\infty} f(x)\varphi'(x)\,\mathrm{d}x &= -\int_0^\infty x\varphi'(x)\,\mathrm{d}x = -x\varphi(x)\Big|_0^\infty + \int_0^\infty \varphi(x)\,\mathrm{d}x \\ &= \int_{-\infty}^{\infty} H(x)\varphi(x)\,\mathrm{d}x, \end{aligned} \tag{1.2.17}$$

which shows that

$$\int_{\mathbb{R}} H\varphi \,\mathrm{d}x = -\int_{\mathbb{R}} f\varphi' \mathrm{d}x, \qquad \forall\, \varphi \in C^\infty_0(\mathbb{R}). \tag{1.2.18}$$

Note that $H \in L^1_{\rm loc}(\mathbb{R})$ hence H is the weak (or Sobolev) derivative of order one of the function f in $\mathbb{R}$.

Example 1.5. Does there exist a function $g \in L^1_{\rm loc}(\mathbb{R})$ such that g is a weak derivative (of order one) of the Heaviside function? To answer this question, first observe that for each $\varphi \in C^\infty_0(\mathbb{R})$ we have

$$-\int_{-\infty}^{\infty} H(x)\varphi'(x)\,\mathrm{d}x = -\int_0^\infty \varphi'(x)\,\mathrm{d}x = \varphi(0). \tag{1.2.19}$$

Suppose that H has a weak derivative of order one, and call this $g \in L^1_{\rm loc}(\mathbb{R})$. Then, by Definition 1.2 and (1.2.19), we have $\int_{\mathbb{R}} g\,\varphi\,dx = \varphi(0)$ for all $\varphi \in C_0^\infty(\mathbb{R})$. This forces $\int_0^\infty g\varphi\,dx = 0$ for all $\varphi \in C_0^\infty((0,\infty))$. In concert with Theorem 1.3, the latter yields $g = 0$ almost everywhere on $(0,\infty)$. Similarly, we obtain that $g = 0$ almost everywhere on $(-\infty, 0)$, hence $g = 0$ almost everywhere on $\mathbb{R}$. When combined with (1.2.19), this gives that $0 = \int_{\mathbb{R}} g\varphi\,dx = \varphi(0)$ for all $\varphi \in C_0^\infty(\mathbb{R})$, leading to a contradiction (as there are functions $\varphi \in C_0^\infty(\mathbb{R})$ with $\varphi(0) \neq 0$). Thus, a weak (Sobolev) derivative of order one of H does not exist.

Having defined the notion of weak (Sobolev) derivatives for locally integrable functions, we return to the notion of weak solution considered in Definition 1.1 in a particular case, and extend this to more general partial differential equations. To set the stage, let $P(x,\partial)$ be a linear partial differential operator of order $m \in \mathbb{N}$ of the form

$$P(x,\partial) := \sum_{|\alpha|\leq m} a_\alpha(x)\partial^\alpha, \quad a_\alpha \in C^{|\alpha|}(\Omega), \quad \alpha \in \mathbb{N}_0^n, \quad |\alpha| \leq m. \tag{1.2.20}$$

Also, suppose $f \in C^0(\Omega)$ is a given function and that u is a classical solution of the partial differential equation $P(x,\partial)u = f$ in Ω. That is, assume $u \in C^m(\Omega)$ and the equation holds pointwise in Ω. Then, for each $\varphi \in C_0^\infty(\Omega)$, integration by parts gives

$$\int_\Omega f\varphi\,dx = \sum_{|\alpha|\leq m} \int_\Omega \varphi\, a_\alpha(\partial^\alpha u)\,dx = \int_\Omega \Big[\sum_{|\alpha|\leq m} (-1)^{|\alpha|}\partial^\alpha(a_\alpha\varphi)\Big] u\,dx. \tag{1.2.21}$$

Hence, if we define

$$P^\top(x,\partial)\varphi := \sum_{|\alpha|\leq m} (-1)^{|\alpha|}\partial^\alpha(a_\alpha\varphi), \tag{1.2.22}$$

and call it the `transpose` of the operator $P(x,\partial)$, the resulting equation from (1.2.21) becomes

$$\int_\Omega f\varphi\,dx = \int_\Omega [P^\top(x,\partial)\varphi]u\,dx, \qquad \forall\,\varphi \in C_0^\infty(\mathbb{R}^n). \tag{1.2.23}$$

Thus, any classical solution u of $P(x,\partial)u = f$ in Ω satisfies (1.2.23). On the other hand, there might exist functions $u \in L^1_{\rm loc}(\Omega)$ that satisfy (1.2.23) but are not classical solutions of the given equation. Such a scenario has been already encountered in (1.1.11) (cf. also the subsequent comment). This motivates the following general definition (compare with Definition 1.1 in the case when $P(x,\partial) = \partial_1^2 - \partial_2^2$).

Definition 1.6. *Let $u, f \in L^1_{\rm loc}(\Omega)$ be given and assume that $P(x,\partial)$ is as in (1.2.20). Then $P(x,\partial)u = f$ is said to hold in the weak (or Sobolev) sense if*

$$\int_\Omega f\varphi\,dx = \int_\Omega [P^\top(x,\partial)\varphi]u\,dx, \qquad \forall\,\varphi \in C_0^\infty(\Omega). \tag{1.2.24}$$

From the comments in the preamble to Definition 1.6 we know that if u is a classical solution of $P(x,\partial)u = f$ in Ω for some $f \in C^0(\Omega)$, then u is also a weak (Sobolev) solution of the same equation. Conversely, if $u \in C^m(\Omega)$ is a weak solution of the partial differential equation $P(x,\partial)u = f$ for some given function $f \in C^0(\Omega)$, then by Definition 1.6 and integration by parts we obtain

$$\int_\Omega f\varphi \,\mathrm{d}x = \int_\Omega [P^\top(x,\partial)\varphi]u \,\mathrm{d}x = \int_\Omega \varphi\, P(x,\partial)u \,\mathrm{d}x. \tag{1.2.25}$$

Since $\varphi \in C_0^\infty(\Omega)$ is arbitrary, Theorem 1.3 then forces $f = P(x,\partial)u$ almost everywhere in Ω, hence ultimately everywhere in Ω, since the functions in question are continuous.

In summary, the above discussion shows that the notion of weak solution of a partial differential equation is a natural, unambiguous, and genuine generalization of the concept of classical solution, in the following precise sense:

- any classical solution is a weak solution,
- any sufficiently regular weak solution is classical,
- weak solutions may exist even in the absence of classical ones.

1.3 The Spaces $\mathcal{E}(\Omega)$ and $\mathcal{D}(\Omega)$

A major drawback of Definition 1.2 is that while the right hand-side of (1.2.2) is always meaningful, it cannot always be written in the form given by the left-hand side of (1.2.2). In addition, it might be the case that some locally integrable function in Ω may admit weak (Sobolev) derivatives of a certain order and not of some intermediate lower order (see the example in Exercise 1.26). The remedy is to focus on the portion of (1.2.2) that always makes sense. Specifically, given $f \in L^1_{\mathrm{loc}}(\Omega)$ and $\alpha \in \mathbb{N}_0^n$, define the mapping

$$g_\alpha : C_0^\infty(\Omega) \to \mathbb{C}, \quad g_\alpha(\varphi) := (-1)^{|\alpha|} \int_\Omega f(\partial^\alpha \varphi)\,\mathrm{d}x, \quad \forall\, \varphi \in C_0^\infty(\Omega). \tag{1.3.1}$$

The functional in (1.3.1) has the following properties.

1. g_α is linear, i.e., $g_\alpha(\lambda_1\varphi_1 + \lambda_2\varphi_2) = \lambda_1 g_\alpha(\varphi_1) + \lambda_2 g_\alpha(\varphi_2)$ for every λ_1, $\lambda_2 \in \mathbb{C}$, and every φ_1, $\varphi_2 \in C_0^\infty(\Omega)$.
2. For each $\varphi \in C_0^\infty(\Omega)$ we may estimate

$$|g_\alpha(\varphi)| \le \int_\Omega |f||\partial^\alpha\varphi|\,\mathrm{d}x \le \Big(\int_{\operatorname{supp}\varphi} |f|\,\mathrm{d}x\Big) \sup_{x\in\operatorname{supp}\varphi} |\partial^\alpha\varphi(x)|. \tag{1.3.2}$$

The fact that the term $\int_{\operatorname{supp}\varphi} |f|\mathrm{d}x$ in (1.3.2) depends on φ is inconvenient if we want to considering the continuity of g_α in some sense. Nonetheless, if a priori

a compact set $K \subset \Omega$ is fixed and the requirement $\operatorname{supp}\varphi \subseteq K$ is imposed, then (1.3.2) becomes

$$|g_\alpha(\varphi)| \leq \Big(\int_K |f|\mathrm{d}x\Big) \sup_{x\in K} |\partial^\alpha \varphi(x)| \tag{1.3.3}$$

and, this time, $\int_K |f|\mathrm{d}x$ is a constant independent of φ. This observation motivates considering an appropriate topology τ on $C^\infty(\Omega)$. For the exact definition of this topology see Sect. 13.1. We will not elaborate here more on this subject other than highlighting those key features of τ that are particularly important for our future investigations. To record the precise statements of these features, introduce

$$\mathcal{E}(\Omega) := \big(C^\infty(\Omega), \tau\big), \tag{1.3.4}$$

a notation that emphasizes that $\mathcal{E}(\Omega)$ is the vector space $C^\infty(\Omega)$ equipped with the topology τ. We then have:

Fact 1.7. *A sequence $\{\varphi_j\}_{j\in\mathbb{N}} \subset C^\infty(\Omega)$ converges in $\mathcal{E}(\Omega)$ to some $\varphi \in C^\infty(\Omega)$ as $j \to \infty$ if and only if*

$$\forall\, K \subset \Omega \text{ compact},\ \forall\, \alpha \in \mathbb{N}_0^n, \quad \text{we have} \quad \lim_{j\to\infty} \sup_{x\in K} |\partial^\alpha(\varphi_j - \varphi)(x)| = 0, \tag{1.3.5}$$

in which case we use the notation $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\Omega)} \varphi$.

Fact 1.8. *$\mathcal{E}(\Omega)$ is a locally convex, metrizable, and complete topological vector space over $\mathbb{C}$.*

It is easy to see that as a consequence of Fact 1.7 we have the following result.

Remark 1.9. *A sequence $\{\varphi_j\}_{j\in\mathbb{N}} \subset C^\infty(\Omega)$ converges in $\mathcal{E}(\Omega)$ to $\varphi \in C^\infty(\Omega)$ as $j \to \infty$ if and only if for any compact set $K \subset \Omega$ and any $m \in \mathbb{N}_0$ one has*

$$\lim_{j\to\infty} \sup_{\alpha\in\mathbb{N}_0^n,\, |\alpha|\leq m} \sup_{x\in K} |\partial^\alpha(\varphi_j - \varphi)(x)| = 0. \tag{1.3.6}$$

Exercise 1.10. *Prove that if $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\Omega)} \varphi$ then the following also hold:*

(1) $\partial^\alpha\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\Omega)} \partial^\alpha\varphi$ for each $\alpha \in \mathbb{N}_0^n$;

(2) $a\,\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\Omega)} a\,\varphi$ for each $a \in C_0^\infty(\Omega)$.

A standard way of constructing a sequence of smooth functions in $\mathbb{R}^n$ that converges in $\mathcal{E}(\mathbb{R}^n)$ to a given $f \in C^\infty(\mathbb{R}^n)$ is by taking the convolution of f with dilations of a function as in (1.2.3). This construction is discussed in detail next.

Example 1.11. Let $f \in C^\infty(\mathbb{R}^n)$ be given. Then a sequence of functions from $C^\infty(\mathbb{R}^n)$ that converges to f in $\mathcal{E}(\mathbb{R}^n)$ may be constructed as follows. Recall ϕ from (1.2.3) and define

$$\phi_j(x) := j^n \phi(jx) \qquad \text{for } x \in \mathbb{R}^n \text{ and each } j \in \mathbb{N}. \tag{1.3.7}$$

Clearly, for each $j \in \mathbb{N}$ we have

$$\phi_j \in C_0^\infty(\mathbb{R}^n), \quad \operatorname{supp}\phi_j \subseteq \overline{B(0,1/j)}, \quad \text{and} \quad \int_{\mathbb{R}^n} \phi_j \, dx = 1. \tag{1.3.8}$$

Now if we further set for each $j \in \mathbb{N}$

$$f_j(x) := \int_{\mathbb{R}^n} f(x-y)\phi_j(y)\,dy = \int_{B(0,1)} f(x - z/j)\phi(z)\,dz, \qquad \text{for } x \in \mathbb{R}^n, \tag{1.3.9}$$

then $f_j \in C^\infty(\mathbb{R}^n)$. Also, if K is an arbitrary compact set in $\mathbb{R}^n$ and $\alpha \in \mathbb{N}_0^n$, then

$$\begin{aligned} |\partial^\alpha f_j(x) - \partial^\alpha f(x)| &\leq \int_{B(0,1)} |\partial^\alpha f(x - z/j) - \partial^\alpha f(x)|\,\phi(z)\,dz \\ &\leq \frac{1}{j} \max_{|\beta|=|\alpha|+1} \|\partial^\beta f\|_{L^\infty(\widetilde{K})} \qquad \forall\, x \in K, \end{aligned} \tag{1.3.10}$$

where $\widetilde{K} := \{x \in \mathbb{R}^n : \operatorname{dist}(x, K) \leq 1\}$. Hence $f_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} f$, as desired.

The previous approximation result may be further strengthened as indicated in the next two exercises.

Exercise 1.12. *Prove that $C_0^\infty(\mathbb{R}^n)$ is sequentially dense in $\mathcal{E}(\mathbb{R}^n)$. That is, show that for each $f \in C^\infty(\mathbb{R}^n)$ there exists a sequence $\{f_j\}_{j\in\mathbb{N}} \subset C_0^\infty(\mathbb{R}^n)$ with the property that $f_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} f$.*

Hint: Let $\psi \in C_0^\infty(\mathbb{R}^n)$ be such that $\psi(x) = 1$ whenever $|x| < 1$. Then if $f \in C^\infty(\mathbb{R}^n)$ define $f_j(x) := \psi(x/j)f(x)$, for each $x \in \mathbb{R}^n$ and each $j \in \mathbb{N}$.

Exercise 1.13. *Prove that $C_0^\infty(\Omega)$ is sequentially dense in $\mathcal{E}(\Omega)$.*

Hint: Recall the sequence of compacts from (1.2.13). Then $\bigcup_{j\in\mathbb{N}} K_j = \Omega$ and $K_j \subset \mathring{K}_{j+1}$ for each $j \in \mathbb{N}$. For each $j \in \mathbb{N}$, let $\psi_j \in C_0^\infty(\Omega)$ be such that $\psi_j = 1$ on a neighborhood of K_j and $\operatorname{supp}\psi_j \subseteq K_{j+1}$ (cf. Proposition 13.26). If $f \in C^\infty(\Omega)$, define $f_j := \psi_j f$ for every $j \in \mathbb{N}$.

Moving on, we focus on defining a topology on $C_0^\infty(\Omega)$ that suits the purposes we have in mind. Since $C_0^\infty(\Omega) \subset \mathcal{E}(\Omega)$, one option would be to consider the topology induced by this larger ambient on $C_0^\infty(\Omega)$. However, this topology has the distinct drawback of not preserving the property of being compactly supported under convergence. Here is an example to that effect.

Example 1.14. Consider the function

$$\varphi(x) := \begin{cases} \mathrm{e}^{\frac{1}{\left|x-\frac{1}{2}\right|^2-\frac{1}{4}}} & \text{if } \ 0<x<1, \\ 0 & \text{if } \ x\leq 0 \ \text{ or } \ x\geq 1, \end{cases} \qquad \text{for each } x\in\mathbb{R}, \tag{1.3.11}$$

that satisfies $\varphi\in C^\infty(\mathbb{R})$, $\operatorname{supp}\varphi=[0,1]$, and $\varphi>0$ in $(0,1)$. For each $j\in\mathbb{N}$ define

$$\varphi_j(x) := \varphi(x-1)+\tfrac{1}{2}\varphi(x-2)+\cdots+\tfrac{1}{j}\varphi(x-j), \qquad x\in\mathbb{R}. \tag{1.3.12}$$

Then $\varphi_j\in C^\infty(\mathbb{R})$, $\operatorname{supp}\varphi_j=[1,j+1]$, and $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} \varphi$ where

$$\varphi(x) := \sum_{k=1}^{\infty}\tfrac{1}{k}\varphi\big(x-j\big) \quad \text{for} \quad x\in\mathbb{R}. \tag{1.3.13}$$

Clearly this limit function does not have compact support.

The flaw just highlighted is remedied by introducing a different topology on $C_0^\infty(\Omega)$ that is finer than the one inherited from $\mathcal{E}(\Omega)$. First, for each $K\subset\Omega$ compact, denote by $\mathcal{D}_K(\Omega)$ the vector space consisting of functions from $C^\infty(\Omega)$ supported in K endowed with the topology induced by $\mathcal{E}(\Omega)$. Second, consider on $C_0^\infty(\Omega)$ the inductive limit topology of the spaces $\Big\{\mathcal{D}_K(\Omega)\Big\}_{\substack{K\subset\Omega\\ \text{compact}}}$ and denote the resulting topological vector space by $\mathcal{D}(\Omega)$. For precise definitions see Sect. 13.1. Two features that are going to be particularly important for our analysis are singled out below (see Sect. 13.1 in this regard).

Fact 1.15. *$\mathcal{D}(\Omega)$ is a locally convex and complete topological vector space over $\mathbb{C}$.*

Fact 1.16. *A sequence $\{\varphi_j\}_{j\in\mathbb{N}}\subset C_0^\infty(\Omega)$ converges in $\mathcal{D}(\Omega)$ to some function $\varphi\in C_0^\infty(\Omega)$ as $j\to\infty$ if and only if the following two conditions are satisfied:*

(1) there exists a compact set $K\subset\Omega$ such that $\operatorname{supp}\varphi_j\subseteq K$ for all $j\in\mathbb{N}$ and $\operatorname{supp}\varphi\subseteq K$;

(2) for any $\alpha\in\mathbb{N}_0^n$ we have $\lim_{j\to\infty}\sup_{x\in K}|\partial^\alpha(\varphi_j-\varphi)(x)|=0$.

We abbreviate (1)–(2) by simply writing $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} \varphi$.

In view of Fact 1.7 one obtains the following consequence of Fact 1.16.

Remark 1.17. *$\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} \varphi$ if and only if*

(1) there exists a compact set $K\subset\Omega$ such that $\operatorname{supp}\varphi_j\subseteq K$ for all $j\in\mathbb{N}$, and

(2) $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\Omega)} \varphi$.

If one now considers the identity map from $\mathcal{D}(\Omega)$ into $\mathcal{E}(\Omega)$, a combination of Remark 1.17, and Theorem 13.5 yields that this map is continuous. Hence, if we also take into account Exercise 1.12, it follows that

$$\mathcal{D}(\Omega) \text{ is continuously and densely embedded into } \mathcal{E}(\Omega). \tag{1.3.14}$$

Exercise 1.18. *Suppose ω is an open subset of Ω and consider the map*

$$\iota : C_0^\infty(\omega) \to C_0^\infty(\Omega), \qquad \iota(\varphi) := \begin{cases} \varphi & \text{on } \omega, \\ 0 & \text{on } \Omega \setminus \omega, \end{cases} \qquad \forall\, \varphi \in C_0^\infty(\omega). \tag{1.3.15}$$

Prove that if $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\omega)} \varphi$ then $\iota(\varphi_j) \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} \iota(\varphi)$. Use Theorem 13.5 to conclude that $\iota : \mathcal{D}(\omega) \to \mathcal{D}(\Omega)$ is continuous.

Exercise 1.19. *Let $x_0 \in \mathbb{R}^n$ and consider the translation by x_0 map defined as*

$$\begin{aligned} t_{x_0} &: \mathcal{D}(\mathbb{R}^n) \longrightarrow \mathcal{D}(\mathbb{R}^n) \\ t_{x_0}(\varphi) &:= \varphi(\cdot - x_0), \quad \forall\, \varphi \in C_0^\infty(\mathbb{R}^n). \end{aligned} \tag{1.3.16}$$

Prove that t_{x_0} is linear and continuous.

Hint: Use Theorem 13.5.

Exercise 1.20. *Prove that if $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} \varphi$ then the following also hold:*

(1) $\partial^\alpha \varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} \partial^\alpha \varphi$ for each $\alpha \in \mathbb{N}_0^n$;

(2) $a\,\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} a\,\varphi$ for each $a \in C_0^\infty(\Omega)$.

Exercise 1.21. *Prove that if $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\Omega)} \varphi$ then $a\,\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} a\,\varphi$ for each function $a \in C_0^\infty(\Omega)$. Also show that if ω is an open subset of Ω, $a \in C_0^\infty(\omega)$, and we have $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\Omega)} \varphi$, then $a\,\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\omega)} a\,\varphi$.*

We have already noted that the topology $\mathcal{D}(\Omega)$ is finer than the topology $C_0^\infty(\Omega)$ inherits from $\mathcal{E}(\Omega)$, and an example of a sequence of smooth, compactly supported functions in Ω convergent in $\mathcal{E}(\Omega)$ to a limit which does not belong to $\mathcal{D}(\Omega)$ has been given in Example 1.14. The example below shows that even if the limit function is in $\mathcal{D}(\Omega)$, one should still not expect that convergence in $\mathcal{E}(\Omega)$ of a sequence of smooth, compactly supported functions in Ω implies convergence in $\mathcal{D}(\Omega)$.

Example 1.22. Let φ be as in (1.3.11) and for each $j \in \mathbb{N}$ set $\varphi_j(x) := \varphi(x-j)$ for $x \in \mathbb{R}$. Clearly, $\varphi_j \in C^\infty(\mathbb{R})$ and $\operatorname{supp}\varphi_j = [j, j+1]$ for all $j \in \mathbb{N}$. If K is some compact subset of $\mathbb{R}$, then there exists $j_0 \in \mathbb{N}$ such that $K \subseteq [-j_0, j_0]$.

Consequently, $\operatorname{supp}\varphi_j \cap K = \emptyset$ for $j \geq j_0$. Thus, trivially, $\sup_{x\in K} \left|\varphi_j^{(k)}(x)\right| = 0$ if $j \geq j_0$ which shows that $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R})} 0$. Consider next the issue whether $\{\varphi_j\}_{j\in\mathbb{N}}$ converge in $\mathcal{D}(\mathbb{R})$. If this were to be the case, there would exist $r \in (0,\infty)$ such that $\operatorname{supp}\varphi_j \subseteq [-r,r]$ for every j. However, $\bigcup_{j=1}^{\infty} \operatorname{supp}\varphi_j = [1,\infty)$ which leads to a contradiction. Thus, $\{\varphi_j\}_{j\in\mathbb{N}}$ does not converge in $\mathcal{D}(\mathbb{R})$.

Convention. In what follows, we will often identify a function $f \in C_0^\infty(\Omega)$ with its extension by zero outside its support, which makes such an extension belong to $C_0^\infty(\mathbb{R}^n)$.

Further Notes for Chap. 1. The concept of weak derivative goes back to the pioneering work of the Soviet mathematician Sergei Lvovich Sobolev (1908–1989). Although we shall later extend the scope of taking derivatives in a generalized sense to the larger class of distributions, a significant portion of partial differential equations may be developed solely based on the notion of a weak derivative. For example, this is the approach adopted in [12], where distributions are avoided all together. A good reference for the topological aspects that are most pertinent to the spaces of test functions considered here is [70], though there are many other monographs dealing with these issues. The interested reader may consult [10, 59, 71], and the references therein.

1.4 Additional Exercises for Chap. 1

Exercise 1.23. *Given $\varphi_0 \in C^2(\mathbb{R})$, $\varphi_1 \in C^1(\mathbb{R})$, and $F \in C^1(\mathbb{R}^2)$, show that the function $u : \mathbb{R}^2 \to \mathbb{R}$ defined by*

$$u(x_1,x_2) := \tfrac{1}{2}\big[(\varphi_0(x_1+x_2)+\varphi_0(x_1-x_2)\big] + \tfrac{1}{2}\int_{x_1-x_2}^{x_1+x_2} \varphi_1(t)\,\mathrm{d}t$$
$$- \tfrac{1}{2}\int_0^{x_2}\left(\int_{x_1-(x_2-t)}^{x_1+(x_2-t)} F(\xi,t)\,\mathrm{d}\xi\right)\mathrm{d}t, \qquad \forall\,(x_1,x_2)\in\mathbb{R}^2, \quad (1.4.1)$$

is a classical solution of the problem

$$\begin{cases} u \in C^2(\mathbb{R}^2), \\ \partial_1^2 u - \partial_2^2 u = F \quad \text{in} \quad \mathbb{R}^2, \\ u(\cdot,0) = \varphi_0 \quad \text{in} \quad \mathbb{R}, \\ (\partial_2 u)(\cdot,0) = \varphi_1 \quad \text{in} \quad \mathbb{R}. \end{cases} \qquad (1.4.2)$$

Exercise 1.24. *Determine the values of $a \in \mathbb{R}$ for which the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) := \begin{cases} x, & \text{if } x \geq a, \\ 0, & \text{if } x < a, \end{cases}$ for each $x \in \mathbb{R}$, has a weak derivative.*

Exercise 1.25. *Does the function* $f : \mathbb{R} \to \mathbb{R}$ *defined by* $f(x) := \begin{cases} 1, & \text{if } x \geq 2, \\ 0, & \text{if } x < 2, \end{cases}$ *for each* $x \in \mathbb{R}$*, have a weak derivative?*

Exercise 1.26. *Let* $f : \mathbb{R}^2 \to \mathbb{R}^2$ *be defined by* $f(x, y) := H(x) + H(y)$ *for each* $(x, y) \in \mathbb{R}^2$*, where* H *is the Heaviside function from* (1.2.16)*. Prove that for* $\alpha = (1, 1)$ *and* $\beta = (1, 0)$ *the weak derivatives* $\partial^\alpha f$ *and* $\partial^{\alpha+\beta} f$ *exist, while the weak derivative* $\partial^\beta f$ *does not exist.*

Exercise 1.27. *Compute the weak derivative of order one of* $f : (-1, 1) \to \mathbb{R}$ *defined by* $f(x) := \operatorname{sgn}(x)\sqrt{|x|}$ *for every* $x \in (-1, 1)$*, where*

$$\operatorname{sgn}(x) := \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x = 0, \\ -1 & \text{if } x < 0. \end{cases} \tag{1.4.3}$$

Exercise 1.28. *Let* $f : \mathbb{R}^2 \to \mathbb{R}^2$ *be the function defined by* $f(x, y) := x|y|$ *for each* $(x, y) \in \mathbb{R}^2$*. Prove that the weak derivative* $\partial_1^2 \partial_2 f$ *exists, while the weak derivative* $\partial_1 \partial_2^2 f$ *does not.*

Exercise 1.29. *Let* $f : \mathbb{R}^2 \to \mathbb{R}^2$ *be defined by* $f(x, y) := H(x) - \operatorname{sgn}(y)$ *for each* $(x, y) \in \mathbb{R}^2$*, and let* $\alpha = (\alpha_1, \alpha_2) \in \mathbb{N}_0^2$*. Prove that* $\partial^\alpha f$ *exists in the weak sense if and only if* $\alpha_1 \geq 1$ *and* $\alpha_2 \geq 1$*.*

Exercise 1.30. *Let* $f : \mathbb{R} \to \mathbb{R}$ *be defined by* $f(x) := \sin|x|$ *for every* $x \in \mathbb{R}$*. Does* f' *exist in the weak sense? How about* f''*?*

Exercise 1.31. *Let* ω, Ω *be open subsets of* $\mathbb{R}^n$ *with* $\omega \subseteq \Omega$*. Suppose* $f \in L^1_{\mathrm{loc}}(\Omega)$ *is such that* $\partial^\alpha f$ *exists in the weak sense in* Ω *for some* $\alpha \in \mathbb{N}_0^n$*. Show that the weak derivative* $\partial^\alpha(f|_\omega)$ *exists and equals* $(\partial^\alpha f)|_\omega$ *almost everywhere in* ω*.*

Exercise 1.32. *Let* $\varepsilon \in (0, 1)$ *and consider the function* $f : \mathbb{R}^n \to \mathbb{R}$ *defined by*

$$f(x) := \begin{cases} |x|^{-\varepsilon} & \text{if } x \in \mathbb{R}^n \setminus \{0\}, \\ 1 & \text{if } x = 0, \end{cases} \qquad \forall\, x \in \mathbb{R}^n.$$

Prove that $\partial_j f$ *exists in the weak sense for each* $j \in \{1, \ldots, n\}$ *if and only if* $n \geq 2$*. Also compute the weak derivatives* $\partial_j f$*,* $j \in \{1, \ldots, n\}$*, in the case when* $n \geq 2$*.*

Exercise 1.33. *Assume that* $a, b \in \mathbb{R}$ *are such that* $a < b$*.*

(a) Prove that if $f \in L^1_{\mathrm{loc}}\big((a, b)\big)$ *is such that the weak derivative* f' *exists and is equal to zero almost everywhere on* (a, b)*, then there exists some complex number* c *such that* $f = c$ *almost everywhere on* (a, b)*.*

Hint: Fix $\varphi_0 \in C_0^\infty\big((a, b)\big)$ with $\int_a^b \varphi_0(t)\,\mathrm{d}t = 1$. Then each $\varphi \in C_0^\infty\big((a, b)\big)$ is of the form $\varphi = \varphi_0 \int_a^b \varphi(t)\,\mathrm{d}t + \psi'$, for some $\psi \in C_0^\infty\big((a, b)\big)$.

(b) Assume that $g \in L^1_{\mathrm{loc}}\big((a,b)\big)$ *and* $x_0 \in (a,b)$. *Prove that the function defined by* $f(x) := \int_{x_0}^{x} g(t)\,\mathrm{d}t$, *for* $x \in (a,b)$, *belongs to* $L^1_{\mathrm{loc}}((a,b))$ *and has a weak derivative that is equal to* g *almost everywhere on* (a,b).

(c) Let $f \in L^1_{\mathrm{loc}}\big((a,b)\big)$ *be such that the weak derivative* $f^{(k)}$ *exists for some* $k \in \mathbb{N}$. *Prove that all the weak derivatives* $f^{(j)}$ *exist for each* $j \in \mathbb{N}$ *with* $j < k$.

Hint: Prove that if $g(x) := \int_{x_0}^{x} h(t)\,\mathrm{d}t$ where $x_0 \in (a,b)$ is a fixed point and $h := f^{(k)}$, and if φ_0 is as in the hint to *(a)*, then

$$f^{(k-1)} = g - \int_a^b g(t)\varphi_0(t)\,\mathrm{d}t + (-1)^{k-1}\int_a^b f(t)\varphi_0^{(k-1)}(t)\mathrm{d}t.$$

(d) Let $f \in L^1_{\mathrm{loc}}\big((a,b)\big)$ *such that* $f^{(k)} = 0$ *for some* $k \in \mathbb{N}$. *Prove that there exist* $a_0, a_1, \ldots, a_{k-1} \in \mathbb{C}$ *such that* $f(x) = \sum_{j=0}^{k-1} a_j x^j$ *for almost every* $x \in (a,b)$.

(e) If Ω *is an open set in* $\mathbb{R}^n$ *for* $n \geq 2$ *and* $f \in L^1_{\mathrm{loc}}(\Omega)$ *is such that the weak derivative* $\partial^\alpha f$ *exists for some* $\alpha \in \mathbb{N}_0^n$, *does it follow that* $\partial^\beta f$ *exists in a weak sense for all* $\beta \leq \alpha$*?*

Exercise 1.34. *Let* $\theta \in C_0^\infty(\mathbb{R}^n)$ *and* $m \in \mathbb{N}$. *Prove that the sequence*

$$\varphi_j(x) := \mathrm{e}^{-j} j^m \theta(jx), \qquad \forall\, x \in \mathbb{R}^n, \quad j \in \mathbb{N},$$

converges in $\mathcal{D}(\mathbb{R}^n)$.

Exercise 1.35. *Let* $\theta \in C_0^\infty(\mathbb{R}^n)$, $h \in \mathbb{R}^n \setminus \{0\}$, *and set*

$$\varphi_j(x) := \theta(x + jh), \qquad \forall\, x \in \mathbb{R}^n, \quad j \in \mathbb{N}.$$

Prove that $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} 0$. *Is it true that* $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\mathbb{R}^n)} 0$*?*

Exercise 1.36. *Let* $\theta \in C_0^\infty(\mathbb{R}^n)$ *be not identically zero, and for each* $j \in \mathbb{N}$ *define*

$$\varphi_j(x) := \frac{1}{j}\,\theta(jx), \qquad \forall\, x \in \mathbb{R}^n.$$

Prove that the sequence $\{\varphi_j\}_{j\in\mathbb{N}}$ *does not converge in* $\mathcal{D}(\mathbb{R}^n)$.

Exercise 1.37. *Let* $\theta \in C_0^\infty(\mathbb{R}^n)$, $h \in \mathbb{R}^n \setminus \{0\}$. *Prove that the sequence*

$$\varphi_j(x) := \mathrm{e}^{-j}\theta(j(x - jh)), \quad \forall\, x \in \mathbb{R}^n, \quad j \in \mathbb{N},$$

converges in $\mathcal{D}(\mathbb{R}^n)$ *if and only if* $\theta(x) = 0$ *for all* $x \in \mathbb{R}^n$.

Exercise 1.38. *Consider $\theta \in C_0^\infty(\mathbb{R}^n)$ not identically zero, and for each $j \in \mathbb{N}$ define*

$$\varphi_j(x) := \frac{1}{j}\,\theta\Big(\frac{x}{j}\Big), \qquad \forall\, x \in \mathbb{R}^n.$$

Does $\{\varphi_j\}_{j\in\mathbb{N}}$ converge in $\mathcal{D}(\mathbb{R}^n)$? How about in $\mathcal{E}(\mathbb{R}^n)$?

Exercise 1.39. *Suppose that $\{\varphi_j\}_{j\in\mathbb{N}}$ is a sequence in $C_0^\infty(\Omega)$ with the property that $\lim_{j\to\infty} \int_{\mathbb{R}^n} f(x)\varphi_j(x)\,\mathrm{d}x = 0$ for every $f \in L^1_{\rm loc}(\Omega)$. Is it true that $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\mathbb{R}^n)} 0$?*

Chapter 2

The Space $\mathcal{D}'(\Omega)$ of Distributions

2.1 The Definition of Distributions

Building on the idea emerging in (1.3.1), we now make the following definition that is central to all subsequent considerations.

Definition 2.1. *$u : \mathcal{D}(\Omega) \to \mathbb{C}$ is called a* `distribution on` Ω *if u is linear and continuous.*

By design, distributions are simply elements of the dual space of the topological vector space $\mathcal{D}(\Omega)$. Given a functional $u : \mathcal{D}(\Omega) \to \mathbb{C}$ and a function $\varphi \in C_0^\infty(\Omega)$, we use the traditional notation $\langle u, \varphi\rangle$ in place of $u(\varphi)$ (in particular, $\langle u, \varphi\rangle$ is a complex number).

While any linear and continuous functional is sequentially continuous, the converse is not always true. Nonetheless, for linear functionals on $\mathcal{D}(\Omega)$, continuity is equivalent with sequential continuity. This remarkable property, itself a consequence of Theorem 13.5, is formally recorded below.

Fact 2.2. *Let $u : \mathcal{D}(\Omega) \to \mathbb{C}$ be a linear map. Then u is a distribution on Ω if and only if for every sequence $\{\varphi_j\}_{j\in\mathbb{N}} \subset C_0^\infty(\Omega)$ with $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} \varphi$ for some $\varphi \in C_0^\infty(\Omega)$, we have $\lim_{j\to\infty} \langle u, \varphi_j\rangle = \langle u, \varphi\rangle$ (where the latter limit is considered in $\mathbb{C}$).*

Remark 2.3. *In general, if X, Y are topological vector spaces and $\Lambda : X \to Y$ is a linear map, then Λ is sequentially continuous on X if and only if Λ is sequentially continuous at the zero vector $0 \in X$. When combined with Fact 2.2,*

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this shows that a linear map $u : \mathcal{D}(\Omega) \to \mathbb{C}$ is a distribution on Ω if and only if $\lim_{j\to\infty} \langle u, \varphi_j\rangle = 0$ for every sequence $\{\varphi_j\}_{j\in\mathbb{N}} \subset C_0^\infty(\Omega)$ with $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} 0$.

Another important characterization of continuity of complex-valued linear functionals u defined on $\mathcal{D}(\Omega)$ is given by the following proposition.

Proposition 2.4. *Let $u : \mathcal{D}(\Omega) \to \mathbb{C}$ be a linear map. Then u is a distribution if and only if for each compact set $K \subset \Omega$ there exist $k \in \mathbb{N}_0$ and $C \in (0, \infty)$ such that*

$$|\langle u, \varphi\rangle| \le C \sup_{\substack{x\in K\\ |\alpha|\le k}} |\partial^\alpha \varphi(x)| \quad \text{for all } \varphi \in C_0^\infty(\Omega) \quad \text{with} \quad \operatorname{supp}\varphi \subseteq K. \tag{2.1.1}$$

Proof. Fix $u : \mathcal{D}(\Omega) \to \mathbb{C}$ linear and suppose that for each compact set $K \subset \Omega$ there exist $k \in \mathbb{N}_0$ and $C \in (0, \infty)$ satisfying (2.1.1). To show that u is a distribution, let $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} 0$. Then, there exists a compact set $K \subseteq \Omega$ such that $\operatorname{supp}\varphi_j \subseteq K$ for all $j \in \mathbb{N}$, and $\partial^\alpha \varphi_j \xrightarrow[j\to\infty]{} 0$ uniformly on K for any $\alpha \in \mathbb{N}_0^n$. For this compact set K, by our hypotheses, there exist $C > 0$ and $k \in \mathbb{N}_0$ such that (2.1.1) holds, and hence,

$$|\langle u, \varphi_j\rangle| \le C \sup_{\substack{x\in K\\ |\alpha|\le k}} |\partial^\alpha \varphi_j(x)| \xrightarrow[j\to\infty]{} 0, \tag{2.1.2}$$

which implies that $\langle u, \varphi_j\rangle \xrightarrow[j\to\infty]{} 0$. From this and Remark 2.3 it follows that u is a distribution in Ω.

To prove the converse implication we reason by contradiction. Suppose that there exists a compact set $K \subseteq \Omega$ such that for every $j \in \mathbb{N}$, there exists a function $\varphi_j \in C_0^\infty(\Omega)$ with $\operatorname{supp}\varphi_j \subseteq K$ and

$$|\langle u, \varphi_j\rangle| > j \sup_{\substack{x\in K\\ |\alpha|\le j}} |\partial^\alpha \varphi_j(x)|. \tag{2.1.3}$$

Define $\psi_j := \frac{1}{\langle u, \varphi_j\rangle}\varphi_j$. Then for each $j \in \mathbb{N}$ we have $\psi_j \in C^\infty(\Omega)$, $\operatorname{supp}\psi_j \subseteq K$, and $\langle u, \psi_j\rangle = 1$. On the other hand, from (2.1.3) we see that

$$\sup_{\substack{x\in K\\ |\alpha|\le j}} |\partial^\alpha \psi_j(x)| < \frac{1}{j} \qquad \forall\, j \in \mathbb{N}. \tag{2.1.4}$$

Now let $\alpha \in \mathbb{N}_0^n$ be arbitrary. Then (2.1.4) implies that

$$\sup_{\substack{x\in K\\ |\alpha|\le j}} |\partial^\alpha \psi_j| < \frac{1}{j} \quad \text{whenever} \quad j \ge |\alpha|, \tag{2.1.5}$$

thus $\psi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} 0$. Since u is a distribution in Ω, the latter implies $\lim_{j\to\infty} \langle u, \psi_j\rangle = 0$, contradicting the fact that $\langle u, \psi_j\rangle = 1$ for each $j \in \mathbb{N}$. This completes the proof of the proposition. □

Remark 2.5. *Recall that for each compact set $K \subset \Omega$ we denote by $\mathcal{D}_K(\Omega)$ the vector space of functions in $C^\infty(\Omega)$ with support contained in K endowed with the topology inherited from $\mathcal{E}(\Omega)$.*

A closer look at the topology in $\mathcal{D}_K(\Omega)$ reveals that Proposition 2.4 may be rephrased as saying that a linear map $u : \mathcal{D}(\Omega) \to \mathbb{C}$ is a distribution in Ω if and only if $u\big|_{\mathcal{D}_K(\Omega)}$ is continuous for each compact set $K \subset \Omega$. In fact, the topology on $\mathcal{D}(\Omega)$ is the smallest topology on $C_0^\infty(\Omega)$ with this property.

Definition 2.6. *Let u be a distribution in Ω. If the nonnegative integer k intervening in (2.1.1) may be taken to be independent of K, then u is called a* `distribution of finite order`*. If u is a distribution of finite order, then the* `order of` *u is by definition the smallest $k \in \mathbb{N}_0$ satisfying condition (2.1.1) for every compact set $K \subset \Omega$.*

Here are a few important examples of distributions.

Example 2.7. For each $f \in L^1_{\rm loc}(\Omega)$ define the functional $u_f : \mathcal{D}(\Omega) \to \mathbb{C}$ by

$$u_f(\varphi) := \int_\Omega f(x)\varphi(x)\,\mathrm{d}x, \qquad \forall\, \varphi \in C_0^\infty(\Omega). \tag{2.1.6}$$

Then clearly u_f is linear and, if K is an arbitrary compact set contained in Ω, then

$$\begin{gathered} |u_f(\varphi)| \leq \sup_{x\in K} |\varphi(x)| \int_K |f|\,\mathrm{d}x = C \sup_{x\in K} |\varphi(x)|, \\ \text{for all } \varphi \in C_0^\infty(\Omega) \text{ with } \operatorname{supp}\varphi \subseteq K. \end{gathered} \tag{2.1.7}$$

Hence, by Proposition 2.4, u_f is a distribution in Ω. Moreover, (2.1.7) also shows that u_f is a distribution of order 0.

Remark 2.8. *A distribution with an action defined as in (2.1.6) will be referred to as a* `distribution of function type`*.*

To simplify notation, if $f \in L^1_{\rm loc}(\Omega)$ we will often simply use f (in place of u_f) for the distribution of function type defined as in (2.1.6). This is justified by the fact that the injection

$$\begin{gathered} \iota : L^1_{\rm loc}(\Omega) \to \{u : \mathcal{D}(\Omega) \to \mathbb{C} : u \text{ linear and continuous}\} \\ \iota(f) := u_f \quad \text{for each } f \in L^1_{\rm loc}(\Omega), \end{gathered} \tag{2.1.8}$$

is one-to-one. Indeed, if $\iota(f) = 0$ for some $f \in L^1_{\rm loc}(\Omega)$, then $\int_\Omega f\varphi\,\mathrm{d}x = 0$ for all functions $\varphi \in C_0^\infty(\Omega)$, which in turn, based on Theorem 1.3, implies that $f = 0$ almost everywhere in Ω. Since ι is also linear, the desired conclusion follows.

Example 2.9. We have that $\ln|x| \in L^1_{\rm loc}(\mathbb{R}^n)$, thus $\ln|x|$ is a distribution in $\mathbb{R}^n$.

To see that indeed $\ln|x|$ is locally integrable in $\mathbb{R}^n$, observe first that

$$\big|\ln t\big| \leq \frac{1}{\varepsilon} \max\left\{t^\varepsilon, t^{-\varepsilon}\right\} \quad \text{for all } t > 0 \text{ and } \varepsilon > 0. \tag{2.1.9}$$

This is justified by starting with the elementary inequality $\ln t \leq t$ for all $t > 0$, then replacing t by $t^{\pm\varepsilon}$. In turn, for each $R > 0$ and $\varepsilon \in (0,1)$, estimate (2.1.9) gives

$$\int_{B(0,R)} \big|\ln|x|\big|\,\mathrm{d}x \leq \varepsilon^{-1} \int_{B(0,R)} \max\left\{|x|^{\varepsilon}, |x|^{-\varepsilon}\right\}\mathrm{d}x < \infty. \tag{2.1.10}$$

Let us now revisit the functional from (1.3.1).

Example 2.10. For a given $f \in L^1_{\mathrm{loc}}(\Omega)$ and multi-index $\alpha \in \mathbb{N}_0^n$, consider the functional $g_\alpha : \mathcal{D}(\Omega) \to \mathbb{C}$ defined by

$$g_\alpha(\varphi) := (-1)^{|\alpha|} \int_\Omega f\,\partial^\alpha\varphi\,\mathrm{d}x \quad \text{for each} \quad \varphi \in C_0^\infty(\Omega). \tag{2.1.11}$$

Clearly this is a linear mapping. Moreover, if $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} 0$, then there exists K compact subset of Ω such that $\operatorname{supp}\varphi_j \subseteq K$ for each $j \in \mathbb{N}$, hence

$$\begin{aligned} |g_\alpha(\varphi_j)| &\leq \int_K |f||\partial^\alpha\varphi_j|\,\mathrm{d}x \\ &\leq \|\partial^\alpha\varphi_j\|_{L^\infty(K)}\|f\|_{L^1(K)} \to 0 \quad \text{as} \quad j \to \infty. \end{aligned} \tag{2.1.12}$$

By invoking Remark 2.3, it follows that g_α is a distribution in Ω.

Next we consider a set of examples of distributions that are not of function type. As a preamble, observe that $f(x) := \frac{1}{x}$ for $x \neq 0$, is not locally integrable on $\mathbb{R}$. Nonetheless, it is possible to associate to this function a certain distribution, not as in (2.1.6), but in the specific manner described below.

Example 2.11. Consider the mapping $\mathrm{P.V.}\,\frac{1}{x} : \mathcal{D}(\mathbb{R}) \to \mathbb{C}$ defined by

$$\Big(\mathrm{P.V.}\,\frac{1}{x}\Big)(\varphi) := \lim_{\varepsilon\to 0^+} \int_{|x|\geq\varepsilon} \frac{\varphi(x)}{x}\,\mathrm{d}x, \qquad \forall\,\varphi \in C_0^\infty(\mathbb{R}). \tag{2.1.13}$$

We claim that $\mathrm{P.V.}\,\frac{1}{x}$ is a distribution of order one in $\mathbb{R}$.

First, we prove that the mapping (2.1.13) is well-defined. Let $\varphi \in C_0^\infty(\mathbb{R})$ and suppose $R > 0$ is such that $\operatorname{supp}\varphi \subset (-R, R)$. Fix $\varepsilon \in (0, R)$ and observe that since $\frac{1}{x}$ is odd on $\mathbb{R}\setminus\{0\}$, we have

$$\int_{|x|\geq\varepsilon} \frac{\varphi(x)}{x}\,\mathrm{d}x = \int_{\varepsilon\leq|x|\leq R} \frac{\varphi(x)}{x}\,\mathrm{d}x = \int_{\varepsilon\leq|x|\leq R} \frac{\varphi(x)-\varphi(0)}{x}\,\mathrm{d}x. \tag{2.1.14}$$

In addition, $\big|\frac{\varphi(x)-\varphi(0)}{x}\big| \leq \sup_{|y|\leq R}|\varphi'(y)|$ for each $x \in \mathbb{R}\setminus\{0\}$. Thus, Lebesgue's dominated convergence theorem implies that $\lim_{\varepsilon\to 0^+}\int_{|x|\geq\varepsilon}\frac{\varphi(x)}{x}$ exists and is equal to $\int_{|x|\leq R}\frac{\varphi(x)-\varphi(0)}{x}\,\mathrm{d}x$. This shows that the mapping $\mathrm{P.V.}\,\frac{1}{x}$ is well-defined, and

from definition it is clear that P.V. $\frac{1}{x}$ is linear. Furthermore, it is implicit in the argument above that

$$\Big|\Big(\text{P.V.}\,\frac{1}{x}\Big)(\varphi)\Big| \leq 2R \sup_{|x|\leq R} |\varphi'(x)|, \qquad \forall\,\varphi \in C_0^\infty\big((-R,R)\big). \tag{2.1.15}$$

In concert with Proposition 2.4 this shows that P.V. $\frac{1}{x}$ is a distribution in $\mathbb{R}$ of order at most one. We are left with showing that P.V. $\frac{1}{x}$ does not have order 0. Consider the compact $K = [0,1]$ and for each $j \in \mathbb{N}$ let $\varphi_j \in C_0^\infty\big((0,1)\big)$ be such that $0 \leq \varphi_j \leq 1$ and $\varphi_j = 1$ on $\big[\frac{1}{j+2}, 1 - \frac{1}{j+2}\big]$. Then from the very definition of P.V. $\frac{1}{x}$ and the fact that φ_j vanishes near zero,

$$\Big|\Big\langle \text{P.V.}\,\frac{1}{x}, \varphi_j \Big\rangle\Big| = \int_0^1 \frac{\varphi_j(x)}{x}\,dx \geq \int_{\frac{1}{j+2}}^{1-\frac{1}{j+2}} \frac{1}{x}\,dx = \ln(j+1), \qquad \forall\, j \in \mathbb{N}. \tag{2.1.16}$$

Since $\sup_{x\in K} |\varphi_j(x)| \leq 1$ and $\lim_{j\to\infty} \ln(j+1) = \infty$, the inequality in (2.1.16) shows that there is no constant $C \in (0,\infty)$ such that $|\langle \text{P.V.}\,\frac{1}{x}, \varphi\rangle| \leq C \sup_{x\in K} |\varphi(x)|$ for all $\varphi \in C_0^\infty(\mathbb{R})$ with $\operatorname{supp}\varphi \subseteq K$. This proves that the distribution P.V. $\frac{1}{x}$ does not have order 0.

Remark 2.12. *An inspection of the proof in Example 2.11 shows that we have*

$$\Big\langle \text{P.V.}\,\frac{1}{x}, \varphi \Big\rangle = \int_{|x|\leq 1} \frac{\varphi(x) - \varphi(0)}{x}\,dx + \int_{|x|>1} \frac{\varphi(x)}{x}\,dx, \qquad \forall\,\varphi \in C_0^\infty(\mathbb{R}). \tag{2.1.17}$$

Example 2.13. An important distribution is the `Dirac distribution` δ defined by

$$\delta(\varphi) := \varphi(0), \qquad \forall\,\varphi \in C_0^\infty(\mathbb{R}^n). \tag{2.1.18}$$

It is not difficult to check that δ is a distribution in $\mathbb{R}^n$ of order 0. A natural question to ask is whether δ is a distribution of function type. To answer this question, suppose there exists $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ such that $\varphi(0) = \langle \delta, \varphi\rangle = \int_{\mathbb{R}^n} f\varphi\,dx$ for all $\varphi \in C_0^\infty(\mathbb{R}^n)$. The latter implies that $\int_{\mathbb{R}^n} f\varphi\,dx = 0$ for every $\varphi \in C_0^\infty(\mathbb{R}^n)$ satisfying $\operatorname{supp}\varphi \cap \{0\} = \emptyset$. Hence, by Theorem 1.3 we have $f = 0$ almost everywhere on $\mathbb{R}^n\backslash\{0\}$, thus $f = 0$ almost everywhere in $\mathbb{R}^n$. Consequently,

$$\varphi(0) = \langle \delta, \varphi\rangle = \int_{\mathbb{R}^n} f\varphi\,dx = 0, \qquad \forall\,\varphi \in C_0^\infty(\mathbb{R}^n), \tag{2.1.19}$$

which is false. This proves that the Dirac distribution is not of function type.

Example 2.14. The Dirac distribution δ is sometimes referred to as having "mass at zero" since, for each $x_0 \in \mathbb{R}^n$, one may similarly define the map $\delta_{x_0} : C_0^\infty(\mathbb{R}^n) \to \mathbb{C}$ by setting $\delta_{x_0}(\varphi) := \varphi(x_0)$. Then it is easy to see that δ_{x_0} is a distribution in $\mathbb{R}^n$, called the Dirac distribution with mass at x_0, and the convention we make is to drop the subscript x_0 if $x_0 = 0 \in \mathbb{R}^n$.

Example 2.15. Let μ be either a complex Borel measure on Ω, or a Borel positive measure on Ω that is locally finite (i.e., satisfies $\mu(K) < \infty$ for every compact $K \subset \Omega$). Consider

$$\mu : \mathcal{D}(\Omega) \to \mathbb{C}, \quad \mu(\varphi) := \int_\Omega \varphi \, d\mu, \quad \forall\, \varphi \in C_0^\infty(\Omega). \tag{2.1.20}$$

The mapping in (2.1.20) is well-defined, linear, and if K is an arbitrary compact set in Ω, then

$$|\mu(\varphi)| \leq |\mu|(K) \sup_{x\in K} |\varphi(x)|, \qquad \forall\, \varphi \in C^\infty(\Omega), \ \operatorname{supp}\varphi \subset K. \tag{2.1.21}$$

By Proposition 2.4 we have that μ induces a distribution in Ω. The estimate in (2.1.21) also shows that μ is a distribution of order zero.

In the last part of this section we discuss the validity of a converse implication to the implication in Example 2.15.

Proposition 2.16. *Let u be a distribution in Ω of order zero. Then the distribution u extends uniquely to a linear map $\Lambda_u : C_0^0(\Omega) \to \mathbb{C}$ that is locally bounded, in the following sense: for each compact set $K \subset \Omega$ there exists $C_K \in (0, \infty)$ with the property that*

$$|\Lambda_u(\varphi)| \leq C_K \sup_{x\in K} |\varphi(x)|, \qquad \forall\, \varphi \in C_0^0(\Omega) \ \text{ with } \ \operatorname{supp}\varphi \subseteq K. \tag{2.1.22}$$

In addition, the functional Λ_u satisfies the following properties.

(i) Let $\{K_j\}_{j\in\mathbb{N}}$ be a sequence of compact subsets of Ω satisfying $K_j \subset \mathring{K}_{j+1}$ for $j \in \mathbb{N}$ and $\Omega = \bigcup_{j=1}^{\infty} K_j$. Then there exists a sequence of complex regular Borel measures μ_j on K_j, $j \in \mathbb{N}$, with the following properties:

(a) $\mu_j(E) = \mu_\ell(E)$ for each $\ell \in \mathbb{N}$, each Borel set $E \subset \mathring{K}_\ell$, and each $j \geq \ell$;

(b) for each $j \in \mathbb{N}$ one has

$$\Lambda_u(\varphi) = \int_{K_j} \varphi \, d\mu_j, \quad \forall\, \varphi \in C_0^0(\Omega) \ \text{ with } \ \operatorname{supp}\varphi \subset K_j. \tag{2.1.23}$$

(ii) Then there exist two Radon measures μ_1, μ_2, taking Borel sets from Ω into $[0, \infty]$ (i.e., measures satisfying the regularity properties (ii)–(iv) in Theorem 13.17), such that

$$\operatorname{Re}\big[\Lambda_u(\varphi)\big] = \int_\Omega \varphi \, d\mu_1 - \int_\Omega \varphi \, d\mu_2, \quad \forall\, \varphi \in C_0^0(\Omega) \ \text{real-valued.} \tag{2.1.24}$$

Furthermore, a similar conclusion is valid for $\operatorname{Im}\Lambda_u$.

Proof. Let u be a distribution in Ω of order zero and let K be a compact set contained in Ω. Fix $\varphi \in C_0^0(\Omega)$ such that $\operatorname{supp}\varphi \subseteq K$. We claim that there exists a sequence of functions $\phi_j \in C_0^\infty(\Omega)$, $j \in \mathbb{N}$, supported in a fixed compact neighborhood K_0 of K such that

$$\lim_{j\to\infty} \sup_{x\in K_0} |\phi_j(x) - \varphi(x)| = 0. \tag{2.1.25}$$

To justify this claim consider ϕ_ε as in (1.2.3)–(1.2.5) then define

$$\varphi_\varepsilon(x) := \int_{\mathbb{R}^n} \varphi(y)\phi_\varepsilon(x-y)\,dy, \qquad \forall\, x \in \mathbb{R}^n, \tag{2.1.26}$$

Then the same type of reasoning as in (1.2.9) (using that a continuous function on a compact set is uniformly continuous as a replacement for the use of Lebesgue's differential theorem), then shows that $\phi_j := \varphi_{1/j}$ do the job in (2.1.25).

Because u is a distribution of order zero, there exists $C = C(K_0) \in (0,\infty)$ such that (2.1.1) holds with $k = 0$. The latter combined with (2.1.25) implies

$$|\langle u, \phi_j\rangle - \langle u, \phi_k\rangle| \leq C \sup_{x\in K_0} |\phi_j(x) - \phi_k(x)| \xrightarrow[j,k\to\infty]{} 0. \tag{2.1.27}$$

Hence, the sequence of complex numbers $\{\langle u, \phi_j\rangle\}_{j\in\mathbb{N}}$ is Cauchy, thus convergent in $\mathbb{C}$, which allows us to define

$$\Lambda_u(\varphi) := \lim_{j\to\infty} \langle u, \phi_j\rangle. \tag{2.1.28}$$

Proving that this definition is independent of the selection of $\{\phi_j\}_{j\in\mathbb{N}}$ is done by interlacing sequences. Specifically, if $\{\phi_j'\}_{j\in\mathbb{N}}$ is another sequence of functions in $C_0^\infty(\Omega)$, supported in K_0 and satisfying (2.1.25), then by considering the sequence $\{\psi_j\}_{j\geq 2}$ defined by $\psi_{2j+1} := \phi_j$ and $\psi_{2j} := \phi_j'$ for every $j \geq 2$, we obtain that $\{\langle u, \psi_j\rangle\}_{j\geq 2}$ is convergent, thus its subsequences $\{\langle u, \phi_j\rangle\}_{j\in\mathbb{N}}$ and $\{\langle u, \phi_j'\rangle\}_{j\in\mathbb{N}}$ are also convergent and have the same limit. In turn, the independence of the definition of $\Lambda_u(\varphi)$ on the approximating sequence $\{\phi_j\}_{j\in\mathbb{N}}$ readily implies that $\Lambda_u : C_0^0(\Omega) \to \mathbb{C}$ is a linear mapping.

In addition, the fact that u is a distribution of order zero implies the existence of a finite constant $C > 0$ such that

$$|\langle u, \phi_j\rangle| \leq C \sup_{x\in K_0} |\phi_j(x)|, \qquad \forall\, j \in \mathbb{N}. \tag{2.1.29}$$

Taking the limit as $j \to \infty$ in (2.1.29) gives $|\Lambda_u(\varphi)| \leq C \sup_{x\in K} |\varphi(x)|$ on account of (2.1.28). Let us now show that the linear extension Λ_u of u to $C_0^0(\Omega)$ is unique in the class of locally bounded mappings. By linearity, this comes down to proving that if $\Lambda : C_0^0(\Omega) \to \mathbb{C}$ is a linear locally bounded mapping that vanishes on $C_0^\infty(\Omega)$ then Λ is identically zero on $C_0^0(\Omega)$. To this end, pick an arbitrary $\varphi \in C_0^\infty(\Omega)$, set $K := \operatorname{supp}\varphi$ and, as before, let $\phi_j \in C_0^\infty(\Omega)$, $j \in \mathbb{N}$,

be a sequence of functions supported in a fixed compact neighborhood K_0 of K such that (2.1.25) holds. Then

$$|\Lambda(\varphi)| = |\Lambda(\varphi - \phi_j)| \leq C \sup_{x\in K_0} |\varphi(x) - \phi_j(x)| \xrightarrow[j\to\infty]{} 0, \tag{2.1.30}$$

proving that $\Lambda(\varphi) = 0$, as wanted.

Moving on, fix a sequence of compacts as in the hypothesis of (i) (e.g., the sequence from (1.2.13) will do). Based on what we have proved so far and Riesz's representation theorem for complex measures (see Theorem 13.18), it follows that there exists a sequence of complex regular Borel measures μ_j on K_j, $j \in \mathbb{N}$, such that (2.1.23) holds.

Now fix $\ell \in \mathbb{N}$ and a Borel set $E \subset \mathring{K}_\ell$. If $j \geq \ell$, since the measures μ_j, μ_ℓ are regular, in order to prove that $\mu_j(E) = \mu_\ell(E)$ it suffices to show that $\mu_j(F) = \mu_\ell(F)$ for every compact set $F \subseteq E$. Fix such a compact set and choose a sequence of compact sets $\{F_k\}_{k\in\mathbb{N}}$ contained in $\mathring{K}_\ell$, with the property that $F_{k+1} \subset \mathring{F}_k$ for $k \in \mathbb{N}$ and $\bigcap_{k=1}^{\infty} F_k = F$. Applying Uryshon's lemma (see Proposition 13.20) we obtain a sequence $\varphi_k \in C_0^0(\Omega)$, $0 \leq \varphi_k \leq 1$, $\operatorname{supp}\varphi_k \subseteq F_k$ and $\varphi_k = 1$ on F, for each $k \in \mathbb{N}$. Then $\lim_{k\to\infty} \varphi_k(x) = \chi_F(x)$ for $x \in \Omega$ and we may write

$$\begin{aligned}\mu_j(F) &= \int_{K_j} \chi_F \,\mathrm{d}\mu_j = \lim_{k\to\infty} \int_{K_j} \varphi_k \,\mathrm{d}\mu_j = \lim_{k\to\infty} \langle u, \varphi_k\rangle = \lim_{k\to\infty} \int_{K_\ell} \varphi_k \,\mathrm{d}\mu_\ell \\ &= \int_{K_\ell} \chi_F \,\mathrm{d}\mu_\ell = \mu_\ell(F).\end{aligned} \tag{2.1.31}$$

This completes the proof of *(i)*. Finally, the claim in *(ii)* follows by invoking Riesz's representation theorem for locally bounded functionals (cf. Theorem 13.19). □

Among other things, Proposition 2.16 is a useful ingredient in the following representation theorem for positive distributions.

Theorem 2.17. *Let u be a distribution in Ω such that $\langle u, \varphi\rangle \geq 0$ for every nonnegative function $\varphi \in C_0^\infty(\Omega)$. Then there exists a unique positive Borel regular measure μ on Ω such that*

$$\langle u, \varphi\rangle = \int_\Omega \varphi \,\mathrm{d}\mu, \quad \forall\, \varphi \in C_0^\infty(\Omega). \tag{2.1.32}$$

Proof. First we prove that u has order zero. To do so, let K be a compact set in Ω and fix $\psi \in C_0^\infty(\Omega)$, $\psi \geq 0$ and satisfying $\psi \equiv 1$ on K. Then, if $\varphi \in C_0^\infty(\Omega)$ has $\operatorname{supp}\varphi \subseteq K$ and is real valued, it follows that

$$(\sup_{x\in K} |\varphi(x)|)\psi \pm \varphi \geq 0, \tag{2.1.33}$$

hence by the positivity of u, we have

$$(\sup_{x\in K}|\varphi(x)|)\langle u,\psi\rangle \pm \langle u,\varphi\rangle \geq 0.$$

Thus,

$$\begin{gathered}|\langle u,\varphi\rangle| \leq \langle u,\psi\rangle \sup_{x\in K}|\varphi(x)|,\\ \text{for all real valued } \varphi\in C_0^0(\Omega) \text{ with } \operatorname{supp}\varphi\subseteq K.\end{gathered} \tag{2.1.34}$$

If now $\varphi\in C_0^\infty(\Omega)$ with $\operatorname{supp}\varphi\subseteq K$ is complex valued, say $\varphi=\varphi_1+i\varphi_2$ and φ_1, φ_2, are real valued, then by using (2.1.34) we may write

$$\begin{aligned}|\langle u,\varphi\rangle| &= \sqrt{|\langle u,\varphi_1\rangle|^2+|\langle u,\varphi_2\rangle|^2} \leq \langle u,\psi\rangle\big[\sup_{x\in K}|\varphi_1(x)|+\sup_{x\in K}|\varphi_2(x)|\big]\\ &\leq 2\,\langle u,\psi\rangle \sup_{x\in K}|\varphi(x)|.\end{aligned} \tag{2.1.35}$$

The fact that the distribution u has order zero now follows from (2.1.35). Having established this, we may apply Proposition 2.16 to conclude that u may be uniquely extended to a linear map $\Lambda_u : C_0^0(\Omega)\to\mathbb{C}$ that is locally bounded.

We next propose to show that this linear map is positive. In this respect, we note that if $\varphi\geq 0$ then the functions $\{\phi_j\}_{j\in\mathbb{N}}$ used in the proof of Proposition 2.16 (which are constricted by convolving φ with a nonnegative mollifier) may also be taken to be nonnegative. When combined with (2.1.28) and the fact that u is positive, this shows that the extension Λ_u of u to $C_0^0(\Omega)$ as defined in (2.1.28) is a positive functional. Consequently, Riesz's representation theorem for positive functionals (see Theorem 13.17) may be may invoked to conclude that there exists a unique positive Borel regular measure μ on Ω such that

$$\Lambda_u(\varphi)=\int_\Omega \varphi\,\mathrm{d}\mu, \quad \forall\,\varphi\in C_0^0(\Omega). \tag{2.1.36}$$

Now (2.1.32) follows by specializing (2.1.36) to $\varphi\in C_0^\infty(\Omega)$. This finishes the proof of the theorem. □

2.2 The Topological Vector Space $\mathcal{D}'(\Omega)$

The space of distributions in Ω endowed with the natural addition and scalar multiplication of linear mappings becomes a vector space over $\mathbb{C}$. Indeed, if u_1, u_2, u are distributions in Ω and λ_1, λ_2, $\lambda\in\mathbb{C}$, we define $u_1+u_2 : \mathcal{D}(\Omega)\to\mathbb{C}$ and $\lambda u : \mathcal{D}(\Omega)\to\mathbb{C}$ by setting

$$(u_1+u_2)(\varphi) := \langle u_1,\varphi\rangle+\langle u_2,\varphi\rangle, \quad (\lambda u)(\varphi) := \lambda\langle u,\varphi\rangle, \qquad \forall\,\varphi\in C_0^\infty(\Omega). \tag{2.2.1}$$

It is not difficult to check that u_1+u_2 and λu are distributions in Ω.

The topology we consider on the vector space of distributions in Ω is the weak$*$-topology induced by $\mathcal{D}(\Omega)$ (for details see Sect. 13.1), which makes it a

topological vector space over $\mathbb{C}$ and we denote this topological vector space by $\mathcal{D}'(\Omega)$. In particular, $\mathcal{D}'(\Omega)$ is a locally convex topological vector space over $\mathbb{C}$. We record next an important equivalent characterization of convergence in $\mathcal{D}'(\Omega)$.

Fact 2.18. *A sequence $\{u_j\}_{j\in\mathbb{N}}$ in $\mathcal{D}'(\Omega)$ converges to some $u \in \mathcal{D}'(\Omega)$ as $j \to \infty$ in $\mathcal{D}'(\Omega)$ if and only if $\langle u_j, \varphi\rangle \xrightarrow[j\to\infty]{} \langle u, \varphi\rangle$ for every $\varphi \in C_0^\infty(\Omega)$, in which case we use the notation $u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\Omega)} u$.*

Moreover, the topological space $\mathcal{D}'(\Omega)$ is complete, in the following sense. If the sequence $\{u_j\}_{j\in\mathbb{N}} \subset \mathcal{D}'(\Omega)$ is such that $\lim_{j\to\infty}\langle u_j, \varphi\rangle$ exists (in $\mathbb{C}$) for every $\varphi \in C_0^\infty(\Omega)$ then the functional $u : \mathcal{D}(\Omega) \to \mathbb{C}$ defined by $u(\varphi) := \lim_{j\to\infty}\langle u_j, \varphi\rangle$ for every $\varphi \in C_0^\infty(\Omega)$ is a distribution in Ω.

Note that from Fact 2.18 it is easy to see that if a sequence $\{u_j\}_{j\in\mathbb{N}}$ in $\mathcal{D}'(\Omega)$ is convergent, then its limit is unique. Indeed, if such a sequence would have two limits, say $u, v \in \mathcal{D}'(\mathbb{R}^n)$, then it would follow that for each $\varphi \in C_0^\infty(\mathbb{R}^n)$, the sequence of numbers $\{\langle u_j, \varphi\rangle\}_{j\in\mathbb{N}}$ would converge to both $\langle u, \varphi\rangle$ and $\langle v, \varphi\rangle$, thus $\langle u, \varphi\rangle = \langle v, \varphi\rangle$. Hence, $u = v$.

Remark 2.19. *Assume that we are given $u \in \mathcal{D}'(\Omega)$ and $u_\varepsilon \in \mathcal{D}'(\Omega)$ for each $\varepsilon \in (0, \infty)$. We make the convention that $u_\varepsilon \xrightarrow[\varepsilon\to 0^+]{\mathcal{D}'(\Omega)} u$ is understood in the sense that for every sequence of positive numbers $\{\varepsilon_j\}_{j\in\mathbb{N}}$ satisfying $\lim_{j\to\infty}\varepsilon_j = 0$ we have $u_{\varepsilon_j} \xrightarrow[j\to\infty]{\mathcal{D}'(\Omega)} u$.*

Example 2.20. Let ϕ be as in (1.2.3) and recall the sequence of functions $\{\phi_j\}_{j\in\mathbb{N}}$ from (1.3.7). Interpreting each $\phi_j \in L^1_{\rm loc}(\mathbb{R}^n)$ as distribution in $\mathbb{R}^n$, for each function $\varphi \in C_0^\infty(\mathbb{R}^n)$ we have

$$\langle \phi_j, \varphi\rangle = \int_{\mathbb{R}^n} \phi_j(x)\varphi(x)\,dx = \int_{\mathbb{R}^n} \phi(y)\varphi(y/j)\,dy, \qquad \forall\, j \in \mathbb{N}. \tag{2.2.2}$$

Thus, by Lebesgue dominated convergence theorem (cf. Theorem 13.12),

$$\langle \phi_j, \varphi\rangle = \int_{\mathbb{R}^n} \phi(y)\varphi(y/j)\,dy \xrightarrow[j\to\infty]{} \varphi(0)\int_{\mathbb{R}^n}\phi(y)dy = \varphi(0) = \langle \delta, \varphi\rangle. \tag{2.2.3}$$

This proves that $\phi_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} \delta$.

Exercise 2.21. *Prove that if $p \in [1, \infty]$ and a sequence of functions $\{f_j\}_{j\in\mathbb{N}}$ from $L^p(\Omega)$ converges in $L^p(\Omega)$ to some $f \in L^p(\Omega)$ then $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\Omega)} f$.*

Hint: Use Hölder's inequality.

Exercise 2.22.

(a) Assume that $f \in L^1(\mathbb{R}^n)$ and for each $\varepsilon > 0$ define $f_\varepsilon(x) := \varepsilon^{-n} f(x/\varepsilon)$ for $x \in \mathbb{R}^n$. Prove that for each $g \in L^\infty(\mathbb{R}^n)$ that is continuous at $0 \in \mathbb{R}^n$ we have

$$\int_{\mathbb{R}^n} f_\varepsilon(x) g(x)\,\mathrm{d}x \xrightarrow[\varepsilon\to 0^+]{} \Big(\int_{\mathbb{R}^n} f(x)\,\mathrm{d}x\Big) g(0). \tag{2.2.4}$$

(b) Use part (a) to prove that if $f \in L^1(\mathbb{R}^n)$ is given and for each $j \in \mathbb{N}$ we define $f_j(x) := j^n f(jx)$ for each $x \in \mathbb{R}^n$, then each f_j belongs to $L^1(\mathbb{R}^n)$ (hence $f_j \in \mathcal{D}'(\mathbb{R}^n)$) and

$$f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} c\,\delta \quad \text{where} \quad c := \int_{\mathbb{R}^n} f(x)\,\mathrm{d}x. \tag{2.2.5}$$

Hint: To justify the claim in part *(a)*, make a change of variables to write the integral $\int_{\mathbb{R}^n} f_\varepsilon(x) g(x)\,\mathrm{d}x$ as $\int_{\mathbb{R}^n} f(y) g(\varepsilon y)\,\mathrm{d}y$, then use Lebesgue's dominated convergence theorem.

2.3 Multiplication of a Distribution with a C^∞ Function

The issue we discuss in this section is the definition of the multiplication of a distribution $u \in \mathcal{D}'(\Omega)$ with a smooth function $a \in C^\infty(\Omega)$. First we consider the case when u is of function type, i.e., $u = u_f$ for some $f \in L^1_{\mathrm{loc}}(\Omega)$. In this particular case, $af \in L^1_{\mathrm{loc}}(\Omega)$; thus, it defines a distribution u_{af} on Ω and

$$\langle u_{af}, \varphi\rangle = \int_\Omega (af)\varphi\,\mathrm{d}x = \int_\Omega f(a\,\varphi)\,\mathrm{d}x = \langle f, a\,\varphi\rangle, \qquad \forall\,\varphi \in C_0^\infty(\Omega). \tag{2.3.1}$$

In the general case, this suggests defining au as in the following proposition.

Proposition 2.23. *Let $u \in \mathcal{D}'(\Omega)$ and $a \in C^\infty(\Omega)$. Then the mapping*

$$au : \mathcal{D}(\Omega) \to \mathbb{C} \quad \text{defined by} \quad (au)(\varphi) := \langle u, a\varphi\rangle, \quad \forall\,\varphi \in C_0^\infty(\Omega), \tag{2.3.2}$$

is linear and continuous, hence a distribution on Ω.

Proof. The fact that au is linear is immediate from its definition. To show that au is also continuous we make use of Remark 2.3. To this end, consider a sequence $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} 0$. By *(2)* in Exercise 1.20 we have $a\,\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} 0$. Since u is a distribution on Ω, the latter implies $\lim_{j\to\infty} \langle u, a\varphi_j\rangle = 0$. Moreover, from (2.3.2) we have $(au)(\varphi_j) = \langle u, a\varphi_j\rangle$ for each $j \in \mathbb{N}$. Hence, $\lim_{j\to\infty} (au)(\varphi_j) = 0$ proving that au is continuous. □

Exercise 2.24. *Let $f \in L^1_{\mathrm{loc}}(\Omega)$ and $a \in C^\infty(\Omega)$. With the notation from (2.1.6), prove that $au_f = u_{af}$ in $\mathcal{D}'(\Omega)$.*

Remark 2.25.

(1) When more information about $u \in \mathcal{D}'(\Omega)$ is available, (2.3.2) may continue to yield a distribution under weaker regularity demands on a than the current assumption that $a \in C^\infty(\Omega)$. In general, however, the condition $a \in C^\infty(\Omega)$ may not be weakened if (2.3.2) is to yield a distribution for arbitrary $u \in \mathcal{D}'(\Omega)$.

(2) As observed later (see Remark 2.30), one may not define the product of two arbitrary distributions in a way that ensures associativity.

(3) Based on (2.3.2) and (2.2.1), it follows that if u, u_1, $u_2 \in \mathcal{D}'(\Omega)$, and if a, a_1, $a_2 \in C^\infty(\Omega)$, then $a(u_1+u_2) = au_1 + au_2$, $(a_1+a_2)u = a_1u + a_2u$, and $a_1(a_2u) = (a_1a_2)u$, where the equalities are considered in $\mathcal{D}'(\Omega)$.

Exercise 2.26. *The following properties hold.*

1. *If $u \in \mathcal{D}'(\Omega)$ and $a_j \xrightarrow[j\to\infty]{\mathcal{E}(\Omega)} a$, then $a_j u \xrightarrow[j\to\infty]{\mathcal{D}'(\Omega)} au$.*
2. *If $a \in C^\infty(\Omega)$ and $u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\Omega)} u$, then $au_j \xrightarrow[j\to\infty]{\mathcal{D}'(\Omega)} au$.*

Example 2.27. Recall the Dirac distribution defined in (2.1.18) and assume that some function $a \in C^\infty(\Omega)$ has been given. Then, for each $\varphi \in C_0^\infty(\mathbb{R}^n)$ we may write

$$\langle a\delta, \varphi\rangle = \langle \delta, a\varphi\rangle = (a\varphi)(0) = a(0)\varphi(0) = \big\langle a(0)\delta, \varphi\big\rangle.$$

This shows that

$$a\delta = a(0)\delta \quad \text{in} \quad \mathcal{D}'(\mathbb{R}^n) \quad \text{for each } a \in C^\infty(\Omega). \tag{2.3.3}$$

As a consequence,

$$x^m\delta = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}), \qquad \text{for every } m \in \mathbb{N}. \tag{2.3.4}$$

Example 2.28. The goal is to solve the equation

$$xu = 1 \quad \text{in } \mathcal{D}'(\mathbb{R}). \tag{2.3.5}$$

Clearly, this equation does not have a solution u of function type. Recall the distribution defined in Example 2.11. Then for every $\varphi \in C_0^\infty(\mathbb{R})$ we may write

$$\begin{aligned}\Big\langle x\Big(\mathrm{P.V.}\,\frac{1}{x}\Big), \varphi\Big\rangle &= \Big\langle \mathrm{P.V.}\,\frac{1}{x}\,, x\varphi(x)\Big\rangle = \lim_{\varepsilon\to 0^+}\int_{|x|\ge\varepsilon} \frac{x\,\varphi(x)}{x}\,\mathrm{d}x \\ &= \int_{\mathbb{R}} \varphi(x)\,\mathrm{d}x = \langle 1, \varphi\rangle. \end{aligned} \tag{2.3.6}$$

Thus,

$$x\Big(\text{P.V.}\,\frac{1}{x}\Big) = 1 \quad \text{in} \quad \mathcal{D}'(\mathbb{R}). \tag{2.3.7}$$

Given (2.3.4), it follows that $u := \text{P.V.}\,\frac{1}{x} + c\,\delta$ will also be a solution of (2.3.5) for any $c \in \mathbb{C}$. We will see later (c.f. Remark 2.70) that in fact any solution of (2.3.5) is of the form $\text{P.V.}\,\frac{1}{x} + c\,\delta$, where $c \in \mathbb{C}$.

Exercise 2.29. *Let $\psi \in C^\infty(\mathbb{R})$. Determine a solution $u \in \mathcal{D}'(\mathbb{R})$ of the equation $xu = \psi$ in $\mathcal{D}'(\mathbb{R})$.*

Remark 2.30. *Suppose one could define the product of distributions as an associative operation, in a manner compatible with the multiplication by a smooth function. Considering then δ, x, and $\text{P.V.}\,\frac{1}{x} \in \mathcal{D}'(\mathbb{R})$, one would then necessarily have*

$$0 = (\delta \cdot x)\Big(\text{P.V.}\,\frac{1}{x}\Big) = \delta\Big[x \cdot \Big(\text{P.V.}\,\frac{1}{x}\Big)\Big] = \delta \cdot 1 = \delta \quad \text{in } \mathcal{D}'(\mathbb{R}), \tag{2.3.8}$$

thanks to (2.3.4) and (2.3.7), leading to the false conclusion that $\delta = 0$ in $\mathcal{D}'(\mathbb{R})$.

2.4 Distributional Derivatives

We are now ready to define derivatives of distributions. One of the most basic attributes of the class of distributions, compared with other classes of locally integrable functions, is that distributions may be differentiated unrestrictedly within this environment (with the resulting objects being still distributions), and that the operation of distributional differentiation retains some of the most basic properties as in the case of ordinary differentiable functions (such as, a suitable product formula, symmetry of mixed derivatives, etc.). In addition, the differentiation of distributions turns out to be compatible with the pointwise differentiation in the case when the distribution in question is of function type, given by a sufficiently regular function.

To develop some sort of intuition, we shall start our investigation by looking first at a distribution of function type, and try to generalize the notion of weak (Sobolev) derivative from Definition 1.2. As noted earlier, if $f \in L^1_{\text{loc}}(\Omega)$, the mapping defined in (1.3.1) is a distribution on Ω. This suggests making the following definition.

Definition 2.31. *For each $\alpha \in \mathbb{N}_0^n$ and $u \in \mathcal{D}'(\Omega)$, the* `distributional derivative` *(or the derivative in the sense of distributions) of order α of the distribution u is the mapping $\partial^\alpha u : \mathcal{D}(\Omega) \to \mathbb{C}$ defined by*

$$\partial^\alpha u(\varphi) := (-1)^{|\alpha|}\langle u, \partial^\alpha \varphi\rangle, \qquad \forall\, \varphi \in C_0^\infty(\Omega). \tag{2.4.1}$$

Note that if $u \in \mathcal{D}'(\Omega)$ is of function type, say $u = u_f$ for some $f \in L^1_{\text{loc}}(\Omega)$, and if the weak derivative $\partial^\alpha f$ exists, that is, one can find $g \in L^1_{\text{loc}}(\Omega)$ such that (1.2.2) holds, then according to Definition 2.31 we have that the distributional derivative $\partial^\alpha u$ is equal to the distribution u_g in $\mathcal{D}'(\Omega)$. In short,

$\partial^\alpha u_f = u_{\partial^\alpha f}$ in this case. Thus, Definition 2.31 generalizes Definition 1.2. That the class of distributions in Ω is stable under taking distributional derivatives is proved in the next proposition.

Proposition 2.32. *For each $\alpha \in \mathbb{N}_0^n$ and each $u \in \mathcal{D}'(\Omega)$ we have $\partial^\alpha u \in \mathcal{D}'(\Omega)$.*

Proof. Fix $\alpha \in \mathbb{N}_0^n$ and a distribution $u \in \mathcal{D}'(\Omega)$. The linearity of the mapping $\partial^\alpha u : \mathcal{D}(\Omega) \to \mathbb{C}$ is immediate. Regarding its continuity, let $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} 0$. Then, since $\partial^\alpha \varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\Omega)} 0$ (see *(1)* in Exercise 1.20) and $u \in \mathcal{D}'(\Omega)$, we may write

$$\partial^\alpha u(\varphi_j) = (-1)^{|\alpha|} \langle u, \partial^\alpha \varphi_j \rangle \xrightarrow[j\to\infty]{} 0. \tag{2.4.2}$$

Remark 2.3 then shows that $\partial^\alpha u \in \mathcal{D}'(\Omega)$. □

Exercise 2.33. *Suppose that $m \in \mathbb{N}$ and $f \in C^m(\Omega)$. Prove that for any $\alpha \in \mathbb{N}_0^n$ satisfying $|\alpha| \leq m$, the distributional derivative of order α of u_f is the distribution of function type given by the derivative, in the classical sense, of order α of f, that is, $\partial^\alpha(u_f) = u_{\partial^\alpha f}$ in $\mathcal{D}'(\Omega)$.*

Proposition 2.34. *Let $m \in \mathbb{N}_0$ and assume that*

$$P(x,\partial) := \sum_{|\alpha|\leq m} a_\alpha(x)\partial^\alpha, \quad a_\alpha \in C^\infty(\Omega), \quad \alpha \in \mathbb{N}_0^n, \quad |\alpha| \leq m. \tag{2.4.3}$$

Also, suppose that $u \in C^m(\Omega)$. Then $P(x,\partial)u$, computed in $\mathcal{D}'(\Omega)$, coincides as a distribution with the distribution induced by $P(x,\partial)u$, computed pointwise in Ω.

Proof. This follows from (2.4.3), Exercises 2.33, and 2.24. □

Remark 2.35. *Recall that a function $f : \mathbb{R}^n \to \mathbb{C}$ is called* `Lipschitz` *provided there exists $M \in [0,\infty)$ such that*

$$|f(x) - f(y)| \leq M|x-y| \qquad \forall\, x, y \in \mathbb{R}^n. \tag{2.4.4}$$

If f is Lipschitz, then the constant $C := \inf\{M$ for which (2.4.4) holds$\}$ is called the `Lipschitz constant` *of f. We will prove (see Theorem 2.104) that if f is Lipschitz then the distributional derivatives $\partial_k f$, $k = 1,\ldots,n$, belong to $L^\infty(\mathbb{R}^n)$. Consequently,*

$$f : \mathbb{R}^n \to \mathbb{R} \text{ Lipschitz} \implies \partial_k(u_f) = u_{\partial_k f} \text{ in } \mathcal{D}'(\mathbb{R}^n) \text{ for } k = 1,\ldots,n. \tag{2.4.5}$$

Some basic properties of differentiation in the distributional sense are summarized below.

Proposition 2.36. *The following properties of distributional differentiation hold.*

(1) Any distribution is infinitely differentiable (i.e. $\mathcal{D}'(\Omega)$ is stable under the action of ∂^α for any $\alpha \in \mathbb{N}_0$).

(2) If $u \in \mathcal{D}'(\Omega)$ and $k, \ell \in \{1, \dots, n\}$ then $\partial_k \partial_\ell u = \partial_\ell \partial_k u$ in $\mathcal{D}'(\Omega)$.

(3) If $u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\Omega)} u$ and $\alpha \in \mathbb{N}_0^n$, then $\partial^\alpha u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\Omega)} \partial^\alpha u$.

(4) For any $u \in \mathcal{D}'(\Omega)$ and any $a \in C^\infty(\Omega)$ we have $\partial_j(au) = (\partial_j a)u + a(\partial_j u)$ in $\mathcal{D}'(\Omega)$.

Proof. The first property follows immediately from the definition of distributional derivatives. To prove the remaining properties, fix an arbitrary function $\varphi \in C_0^\infty(\Omega)$. Then, using (2.4.1) repeatedly and the symmetry of mixed partial derivatives for smooth functions (Schwarz's theorem), we have

$$\begin{aligned}\langle \partial_k \partial_\ell u, \varphi\rangle &= -\langle \partial_\ell u, \partial_k \varphi\rangle = \langle u, \partial_\ell \partial_k \varphi\rangle = \langle u, \partial_k \partial_\ell \varphi\rangle \\ &= -\langle \partial_k u, \partial_\ell \varphi\rangle = \langle \partial_\ell \partial_k u, \varphi\rangle, \quad \forall\, k, \ell \in \{1, \dots, n\},\end{aligned}$$

which implies *(2)*. Let now $\{u_j\}_{j\in\mathbb{N}}$, u, and α satisfy the hypotheses in *(3)*. Based on (2.4.1) and Fact 2.18 we may write

$$\langle \partial^\alpha u_j, \varphi\rangle = (-1)^{|\alpha|}\langle u_j, \partial^\alpha \varphi\rangle \xrightarrow[j\to\infty]{} (-1)^{|\alpha|}\langle u, \partial^\alpha \varphi\rangle = \langle \partial^\alpha u, \varphi\rangle,$$

hence $\partial^\alpha u_j$ converges to $\partial^\alpha u$ in $\mathcal{D}'(\Omega)$ as $j \to \infty$ and the proof of *(3)* is complete. Finally, for $u \in \mathcal{D}'(\Omega)$ and $a \in C^\infty(\Omega)$, using (2.4.1) and Leibniz's product formula for derivatives of smooth functions we can write

$$\begin{aligned}\langle \partial_j(au), \varphi\rangle &= -\langle au, \partial_j \varphi\rangle = -\langle u, a(\partial_j \varphi)\rangle = -\langle u, \partial_j(a\varphi)\rangle + \langle u, (\partial_j a)\varphi\rangle \\ &= \langle \partial_j u, a\varphi\rangle + \langle (\partial_j a)u, \varphi\rangle = \big\langle a(\partial_j u) + (\partial_j a)u, \varphi\big\rangle, \qquad (2.4.6)\end{aligned}$$

from which *(4)* follows. □

Example 2.37. Recall the Heaviside function H from (1.2.16). This is a locally integrable function thus it defines a distribution on $\mathbb{R}$ that we denote also by H. Then, the computation in (1.2.19) implies

$$\langle H', \varphi\rangle = -\langle H, \varphi'\rangle = \langle \delta, \varphi\rangle, \qquad \forall\, \varphi \in C_0^\infty(\mathbb{R}). \qquad (2.4.7)$$

Hence,

$$H' = \delta \quad \text{in} \quad \mathcal{D}'(\mathbb{R}). \qquad (2.4.8)$$

Exercise 2.38. *Prove that for every function $a \in C^\infty(\Omega)$ and every $\alpha \in \mathbb{N}_0^n$ we have*

$$a(\partial^\alpha \delta) = \sum_{\beta \le \alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!}(-1)^{|\beta|}(\partial^\beta a)(0)\partial^{\alpha-\beta}\delta \quad \text{in } \mathcal{D}'(\Omega). \qquad (2.4.9)$$

Hint: Use formula (13.2.4) when computing $\partial^\alpha(a\varphi)$ for $\varphi \in C_0^\infty(\Omega)$.

Exercise 2.39. *Prove that for every $c \in \mathbb{R}$ one has*

$$\left(\mathrm{e}^{-c|x|}\right)' = -c\,\mathrm{e}^{-cx}H(x) + c\,\mathrm{e}^{cx}H(-x) \quad \text{in} \quad \mathcal{D}'(\mathbb{R}). \tag{2.4.10}$$

Hint: Show $\mathrm{e}^{-c|x|} = \mathrm{e}^{-cx}H(x) + \mathrm{e}^{cx}H(-x)$ in $\mathcal{D}'(\mathbb{R})$ and then use part *(4)* in Proposition 2.36 and (2.4.8).

Next, we look at the issue of existence of antiderivatives for distributions on open intervals.

Proposition 2.40. *Let I be an open interval in $\mathbb{R}$ and suppose $u_0 \in \mathcal{D}'(I)$.*

(1) The equation $u' = u_0$ in $\mathcal{D}'(I)$ admits at least one solution.

(2) If $u_1, u_2 \in \mathcal{D}'(I)$ are such that $(u_1)' = u_0$ in $\mathcal{D}'(I)$ and $(u_1)' = u_0$ in $\mathcal{D}'(I)$, then there exists $c \in \mathbb{C}$ such that $u_1 - u_2 = c$ in $\mathcal{D}'(I)$.

Proof. Suppose $I = (a, b)$, where $a \in \mathbb{R} \cup \{-\infty\}$ and $b \in \mathbb{R} \cup \{+\infty\}$, and define the set $\mathcal{A}(I) := \{\varphi' : \varphi \in C_0^\infty(I)\}$. We claim that

$$\text{if} \quad \varphi \in C_0^\infty(I), \quad \text{then} \quad \varphi \in \mathcal{A}(I) \iff \int_I \varphi(x)\,\mathrm{d}x = 0. \tag{2.4.11}$$

The left-to-right implication in (2.4.11) is immediate from the fundamental theorem of calculus. To prove the converse implication, suppose $\varphi \in C_0^\infty(I)$ is such that $\int_I \varphi(x)\,\mathrm{d}x = 0$. Then the function $\psi(x) := \int_a^x \varphi(t)\,\mathrm{d}t$, $x \in I$, satisfies $\psi \in C^\infty(I)$, $\operatorname{supp}\psi \subseteq \operatorname{supp}\varphi$, and $\psi' = \varphi$ on I, thus $\varphi \in \mathcal{A}(I)$. This finishes the justification of (2.4.11).

Next, fix $\varphi_0 \in C_0^\infty(I)$ with the property that $\int_I \varphi_0(x)\,\mathrm{d}x = 1$ and consider the map $\Theta : \mathcal{D}(I) \to \mathcal{D}(I)$ defined for each $\varphi \in C_0^\infty(I)$ by

$$\Theta(\varphi) := \theta_\varphi \quad \text{where} \quad \theta_\varphi(x) := \int_a^x \left[\varphi(t) - \Big(\int_I \varphi(x)\,\mathrm{d}x\Big)\varphi_0(t)\right] \mathrm{d}t, \quad \forall\, x \in I. \tag{2.4.12}$$

Since the integral of $\varphi - \Big(\int_I \varphi(x)\,\mathrm{d}x\Big)\varphi_0$ over I is zero, our earlier discussion shows that Θ is well-defined. Since Θ is linear, Theorem 13.5 and Fact 1.15 imply that Θ is continuous if and only if it is sequentially continuous. The latter property may be verified from definitions. Finally, from (2.4.12) and the fundamental theorem of calculus we have

$$\Theta(\varphi') = \varphi \quad \text{for every} \quad \varphi \in C_0^\infty(I). \tag{2.4.13}$$

Suppose now that an arbitrary distribution u_0 on I has been fixed. Define $u := -u_0 \circ \Theta$, which is a distribution on I thanks to the properties of Θ. In concert with (2.4.13), this definition implies that $\langle u, \varphi' \rangle = -\langle u_0, \varphi \rangle$ for every function $\varphi \in C_0^\infty(I)$, proving that $u' = u_0$ in $\mathcal{D}'(I)$. This finishes the proof of the statement in *(1)*.

Moving on, suppose $u \in \mathcal{D}'(I)$ is such that $u' = 0$ in $\mathcal{D}'(I)$. Then if φ_0 is as earlier in the proof, for any $\varphi \in C_0^\infty(I)$ we may write

$$\begin{aligned}\langle u, \varphi\rangle &= \Big\langle u, \varphi - \Big(\int_I \varphi(x)\,\mathrm{d}x\Big)\varphi_0\Big\rangle + \Big(\int_I \varphi(x)\,\mathrm{d}x\Big)\langle u, \varphi_0\rangle \\ &= \langle u, (\theta_\varphi)'\rangle + \big\langle \langle u, \varphi_0\rangle, \varphi\big\rangle = \big\langle c, \varphi\big\rangle, \end{aligned} \tag{2.4.14}$$

where $c := \langle u, \varphi_0\rangle \in \mathbb{C}$. Hence, $u = c$ in $\mathcal{D}(I)$. By linearity, this readily implies the statement in *(2)*. The proof of the proposition is now complete. □

Proposition 2.41. *Let I be an open interval in $\mathbb{R}$, $g \in C^\infty(I)$, and $f \in C^k(I)$ for some $k \in \mathbb{N}_0$. If $u \in \mathcal{D}'(I)$ satisfies $u' + gu = f$ in $\mathcal{D}'(I)$, then $u \in C^{k+1}(I)$.*

Proof. Fix $a \in I$ and define $F(x) := \mathrm{e}^{\int_a^x g_0(t)\,\mathrm{d}t}$ for $x \in I$. Then $F \in C^\infty(I)$ and we may use *(4)* in Proposition 2.36 and the equation satisfied by u to write (keeping in mind that f is continuous)

$$\Big(Fu - \int_a^x F(t)f(t)\,\mathrm{d}t\Big)' = F'u + Fu' - Ff = 0 \quad \text{in } \mathcal{D}'(I). \tag{2.4.15}$$

By Proposition 2.40, it follows that there exists some constant $c \in \mathbb{C}$ with the property that $Fu = \int_a^x F(t)f(t)\,\mathrm{d}t + c$ in $\mathcal{D}'(I)$. Since $\int_a^x F(t)f(t)\,\mathrm{d}t \in C^{k+1}(I)$ and $\frac{1}{F} \in C^\infty(I)$, we conclude that $u = \frac{1}{F(x)}\int_a^x F(t)f(t)\,\mathrm{d}t + \frac{c}{F(x)} \in C^{k+1}(I)$, as desired. □

We close this section by presenting a higher-degree version of the product formula for differentiation from part *(4)* in Proposition 2.36.

Proposition 2.42 (Generalized Leibniz Formula)**.** *Suppose $f \in C^\infty(\Omega)$ and $u \in \mathcal{D}'(\Omega)$. Then for each $\alpha \in \mathbb{N}_0^n$ one has*

$$\partial^\alpha(fu) = \sum_{\beta \le \alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!}\,(\partial^\beta f)(\partial^{\alpha-\beta}u) \quad \text{in} \quad \mathcal{D}'(\Omega). \tag{2.4.16}$$

Proof. The first step is to observe that for each $j \in \{1, \ldots, n\}$ and each $k \in \mathbb{N}_0$ we have

$$\partial_j^k(fu) = \sum_{0 \le \ell \le k} \frac{k!}{\ell!(k-\ell)!}\,(\partial_j^\ell f)(\partial_j^{k-\ell}u) \quad \text{in} \quad \mathcal{D}'(\Omega), \tag{2.4.17}$$

which is proved by induction on k making use of part *(4)* in Proposition 2.36. Hence, given any $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n) \in \mathbb{N}_0^n$, via repeated applications of (2.4.17) we obtain

$$\partial_1^{\alpha_1}(fu) = \sum_{0 \le \beta_1 \le \alpha_1} \frac{\alpha_1!}{\beta_1!(\alpha_1-\beta_1)!}\,(\partial_1^{\beta_1} f)(\partial_1^{\alpha_1-\beta_1}u), \tag{2.4.18}$$

$$\partial_1^{\alpha_1}\partial_2^{\alpha_2}(fu)$$
$$= \sum_{0\leq\beta_1\leq\alpha_1}\sum_{0\leq\beta_2\leq\alpha_2}\frac{\alpha_1!}{\beta_1!(\alpha_1-\beta_1)!}\,\frac{\alpha_2!}{\beta_2!(\alpha_2-\beta_2)!}\,(\partial_1^{\beta_1}\partial_2^{\beta_2}f)(\partial_1^{\alpha_1-\beta_1}\partial_2^{\alpha_2-\beta_2}u), \tag{2.4.19}$$

and by induction

$$\partial^\alpha(fu) = \sum_{0\leq\beta_1\leq\alpha_1}\cdots\sum_{0\leq\beta_n\leq\alpha_n}\frac{\alpha_1!}{\beta_1!(\alpha_1-\beta_1)!}\cdots\frac{\alpha_n!}{\beta_n!(\alpha_n-\beta_n)!}\times$$
$$\times(\partial_1^{\beta_1}\cdots\partial_n^{\beta_n}f)(\partial_1^{\alpha_1-\beta_1}\cdots\partial_n^{\alpha_n-\beta_n}u)$$
$$= \sum_{0\leq\beta\leq\alpha}\frac{\alpha!}{\beta!(\alpha-\beta)!}\,(\partial^\beta f)(\partial^{\alpha-\beta}u), \tag{2.4.20}$$

as claimed. □

2.5 The Support of a Distribution

In preparation to discussing the notion of support of a distribution, we first define the restriction of a distribution to an open subset of the Euclidean domain on which the distribution is considered. Necessarily, such a definition should generalize restrictions at the level of locally integrable functions. We start from the observation that if $f\in L^1_{\rm loc}(\Omega)$ and ω is a non-empty set open subset of Ω, then $f\big|_\omega\in L^1_{\rm loc}(\omega)$. Thus,

$$\langle f\big|_\omega,\varphi\rangle = \int_\omega f\varphi\,{\rm d}x = \int_\Omega f\,\iota(\varphi)\,{\rm d}x, \qquad \forall\,\varphi\in C_0^\infty(\omega), \tag{2.5.1}$$

where ι is the map from (1.3.15).

Proposition 2.43. *Let Ω be a nonempty open subset of $\mathbb{R}^n$ and suppose ω is a nonempty open subset of Ω. Also, recall the map ι from (1.3.15). Then for each $u\in\mathcal{D}'(\Omega)$, the mapping arising as the* `restriction` *of the distribution u to ω, i.e.,*

$$u\big|_\omega\colon\mathcal{D}(\omega)\to\mathbb{C} \quad \text{defined by} \quad \big(u\big|_\omega\big)(\varphi) := \langle u,\iota(\varphi)\rangle, \qquad \forall\,\varphi\in C_0^\infty(\omega), \tag{2.5.2}$$

is linear and continuous. Hence, $u\big|_\omega\in\mathcal{D}'(\omega)$.

Proof. It is immediate that the map in (2.5.2) is well-defined and linear. To see that it is also continuous we use Proposition 2.4. Let K be a compact set contained in ω. Then $K\subset\Omega$ and, since $u\in\mathcal{D}'(\Omega)$, Proposition 2.4 applies and gives $k\in\mathbb{N}_0$ and $C\in(0,\infty)$ such that (2.1.1) holds. In particular, for each $\varphi\in C_0^\infty(\omega)$ with $\operatorname{supp}\varphi\subseteq K$,

$$\Big|\big(u\big|_\omega\big)(\varphi)\Big| = \big|\langle u,\iota(\varphi)\rangle\big| \leq C\sup_{\substack{x\in K\\|\alpha|\leq k}}|\partial^\alpha\varphi(x)|. \tag{2.5.3}$$

The conclusion that $u\big|_\omega \in \mathcal{D}'(\omega)$ now follows. For an alternative proof of the continuity of $u\big|_\omega$ one may use Fact 2.2 and Exercise 1.18. □

Exercise 2.44.

(1) Prove that the definition of the restriction of a distribution from (2.5.2) generalizes the usual restriction of functions. More specifically, using the notation introduced in (2.1.6), show that if ω is an open subset of Ω and $f \in L^1_{\rm loc}(\Omega)$, then $(u_f)\big|_\omega = u_{f|_\omega}$ in $\mathcal{D}'(\omega)$.

(2) Prove that the operation of differentiation of a distribution commutes with the operation of restriction of a distribution to open sets, that is, if ω is an open subset of Ω, then

$$\partial^\alpha \big(u\big|_\omega\big) = (\partial^\alpha u)\big|_\omega, \qquad \forall\, u \in \mathcal{D}'(\Omega), \quad \forall\, \alpha \in \mathbb{N}_0^n. \tag{2.5.4}$$

The next proposition shows that a distribution is uniquely determined by its local behavior.

Proposition 2.45. *If $u_1, u_2 \in \mathcal{D}'(\Omega)$ are such that for each $x_0 \in \Omega$ there exists an open subset ω of Ω with $x_0 \in \omega$ and satisfying $u_1\big|_\omega = u_2\big|_\omega$ in $\mathcal{D}'(\omega)$, then $u_1 = u_2$ in $\mathcal{D}'(\Omega)$.*

Proof. Observe that this proposition may be viewed as a reconstruction problem; thus, it is meaningful to try to use a partition of unity. Let $\varphi \in C_0^\infty(\Omega)$ be arbitrary, fixed and set $K := \operatorname{supp}\varphi$. The goal is to prove that $\langle u_1, \varphi\rangle = \langle u_2, \varphi\rangle$. From hypotheses it follows that for each $x \in K$ there exists an open neighborhood $\omega_x \subset \Omega$ of x with the property that $u_1\big|_{\omega_x} = u_2\big|_{\omega_x}$. Based on the fact that K is compact, the cover $\{\omega_x\}_{x\in K}$ of K may be refined to a finite one, consisting of, say $\omega_1, \ldots, \omega_N$. These are open subsets of Ω and satisfy

$$K \subset \bigcup_{j=1}^N \omega_j \quad \text{and} \quad u_1\big|_{\omega_j} = u_2\big|_{\omega_j} \quad \text{for } j = 1, \ldots, N. \tag{2.5.5}$$

Consider a partition of unity $\{\psi_j : j = 1, \ldots, N\}$ subordinate to the cover $\{\omega_j\}_{j=1}^N$ of K, as given by Theorem 13.29. Hence, we have $\psi_j \in C_0^\infty(\Omega)$ with $\operatorname{supp}\psi_j \subset \omega_j$ for each $j = 1, \ldots, N$, and $\sum_{j=1}^N \psi_j = 1$ on K. Consequently, using the linearity of distributions and (2.5.5), we obtain

$$\begin{aligned}\langle u_1, \varphi\rangle &= \Big\langle u_1, \varphi \sum_{j=1}^N \psi_j \Big\rangle = \sum_{j=1}^N \langle u_1, \varphi\psi_j\rangle = \sum_{j=1}^N \langle u_2, \varphi\psi_j\rangle \\ &= \Big\langle u_2, \sum_{j=1}^N \varphi\psi_j \Big\rangle = \langle u_2, \varphi\rangle. \end{aligned} \tag{2.5.6}$$

Thus, $\langle u_1, \varphi\rangle = \langle u_2, \varphi\rangle$ and the proof of the proposition is complete. □

Exercise 2.46. *Let $k \in \mathbb{N}_0 \cup \{\infty\}$ and suppose $u \in \mathcal{D}'(\Omega)$ is such that for each $x \in \Omega$ there exists a number $r_x > 0$ and a function $f_x \in C^k(B(x, r_x))$ such that $B(x, r_x) \subset \Omega$ and $u\big|_{B(x,r_x)} = f_x$ in $\mathcal{D}'(B(x, r_x))$. Prove that $u \in C^k(\Omega)$.*

Hint: Use Theorem 13.34 to obtain a partition of unity $\{\psi_j\}_{j\in J}$ subordinate to the cover $\{B(x, r_x)\}_{x\in\Omega}$ of Ω, then show that $f := \sum_{j\in J} \psi_j f_j$ is a function in $C^k(\Omega)$ satisfying $u = f$ in $\mathcal{D}'(\Omega)$.

Exercise 2.47. *Let $u, v \in \mathcal{D}'(\Omega)$ be such that $\operatorname{supp} u \cap \operatorname{supp} v = \emptyset$ and $u + v = 0$ in $\mathcal{D}'(\Omega)$. Prove that $u = 0$ and $v = 0$ in $\mathcal{D}'(\Omega)$.*

Hint: Note that $v\big|_{\Omega\setminus\operatorname{supp} u} = (u + v)\big|_{\Omega\setminus\operatorname{supp} u} = 0$ and $v\big|_{\Omega\setminus\operatorname{supp} v} = 0$. Combine this with the fact that $(\Omega \setminus \operatorname{supp} u) \cup (\Omega \setminus \operatorname{supp} v) = \Omega$ and Proposition 2.45 to deduce that $v = 0$.

Now we are ready to define the notion of support of a distribution. Recall that if $f \in C^0(\Omega)$ then its support is defined to be the closure relative to Ω of the set $\{x \in \Omega : f(x) \neq 0\}$. However, the value of an arbitrary distribution at a point is not meaningful. The fact that $\Omega\setminus\operatorname{supp} f$ is the largest open set contained in Ω on which $f = 0$ suggests the introduction of the following definition.

Definition 2.48. *The* `support of a distribution` *$u \in \mathcal{D}'(\Omega)$ is defined as*

$$\operatorname{supp} u := \big\{x \in \Omega : \text{there is no } \omega \text{ open such that } x \in \omega \subseteq \Omega \text{ and } u\big|_{\omega} = 0\big\}. \tag{2.5.7}$$

Based on (2.5.7), it follows that

$$\Omega \setminus \operatorname{supp} u = \big\{x \in \Omega : \exists\, \omega \text{ open set such that } x \in \omega \subseteq \Omega \text{ and } u\big|_{\omega} = 0\big\}, \tag{2.5.8}$$

which is an open set. Hence, $\operatorname{supp} u$ is relatively closed in Ω. Moreover, if we apply Proposition 2.45 to the distributions u and $0 \in \mathcal{D}'(\Omega \setminus \operatorname{supp} u)$ we obtain that

$$u\big|_{\Omega\setminus\operatorname{supp} u} = 0, \qquad \forall\, u \in \mathcal{D}'(\Omega). \tag{2.5.9}$$

In other words, $\Omega\setminus\operatorname{supp} u$ is the largest open subset of Ω on which the restriction of u is zero.

Example 2.49. Recall the Dirac distribution from (2.1.18). We claim that $\operatorname{supp}\delta = \{0\}$. Indeed, if $\varphi \in C_0^\infty(\mathbb{R}^n \setminus \{0\})$ it follows that $\langle\delta, \varphi\rangle = \varphi(0) = 0$. By Proposition 2.45, $\delta\big|_{\mathbb{R}^n\setminus\{0\}} = 0$, thus $\operatorname{supp}\delta \subseteq \{0\}$. To prove the opposite inclusion, consider an arbitrary open subset ω of Ω such that $0 \in \omega$. Then there exists $\varphi \in C_0^\infty(\omega)$ such that $\varphi(0) = 1$, and hence, $\langle\delta, \varphi\rangle = 1 \neq 0$, which in turn implies that $\delta\big|_{\omega} \neq 0$. Consequently, $0 \in \operatorname{supp}\delta$ as desired. Similarly, if $x_0 \in \mathbb{R}^n$, then $\operatorname{supp}\delta_{x_0} = \{x_0\}$, where δ_{x_0} is as in Example 2.14.

Example 2.50. If $f \in C^0(\Omega)$ then $\operatorname{supp} u_f = \operatorname{supp} f$, where u_f is the distribution from (2.1.6). Indeed, since $f = 0$ in $\Omega \setminus \operatorname{supp} f$, we have $\int_\Omega f(x)\varphi(x)\,dx = 0$ for every $\varphi \in C_0^\infty(\Omega \setminus \operatorname{supp} f)$, hence $\operatorname{supp} u_f \subseteq \operatorname{supp} f$. Also, if $x \in \Omega \setminus \operatorname{supp} u_f$ then there exists an open neighborhood ω of x with $\omega \subseteq \Omega$ and such that $u_f\big|_{\omega} = 0$. Thus, for every $\varphi \in C_0^\infty(\Omega)$ one has $0 = \langle u_f, \varphi\rangle = \int_\Omega f(x)\varphi(x)\,dx$.

Invoking Theorem 1.3 we arrive at the conclusion that $f = 0$ almost everywhere in ω hence, ultimately, $f = 0$ in ω (since f is continuous in ω). Consequently, $x \notin \operatorname{supp} f$ and this proves that $\operatorname{supp} f \subseteq \operatorname{supp} u_f$.

We propose to extend the scope of the discussion in Example 2.50 as to make it applicable to functions that are merely locally integrable (instead of continuous). This requires defining a suitable notion of support for functions that lack continuity, and we briefly address this issue first.

Given an arbitrary set $E \subseteq \mathbb{R}^n$ and an arbitrary function $f : E \to \mathbb{C}$, we define the support of f as

$$\operatorname{supp} f := \big\{x \in E : \text{ there is no } r > 0 \text{ so that } \ f = 0 \ \text{ a.e. in } \ B(x,r) \cap E\big\}. \tag{2.5.10}$$

From this definition one may check without difficulty that

$$E \setminus \operatorname{supp} f = E \cap \bigcup_{x \in E \setminus \operatorname{supp} f} B(x, r_x) \tag{2.5.11}$$

where for each $x \in E \setminus \operatorname{supp} f$ the number $r_x > 0$ is such that $f = 0$ a.e. in $B(x, r_x) \cap E$. Moreover, since $\mathbb{R}^n$ has the Lindelöf property, the above union can be refined to a countable one. Based on these observations, the following basic properties of the support may be deduced:

$$\operatorname{supp} f \ \text{ is a relatively closed subset of } E, \tag{2.5.12}$$

$$f = 0 \ \text{ a.e. in } \ E \setminus \operatorname{supp} f, \tag{2.5.13}$$

$$\operatorname{supp} f \subseteq F \text{ if } F \subseteq E \text{ is relatively closed and } f = 0 \text{ a.e. on } E \setminus F, \tag{2.5.14}$$

$$\operatorname{supp} f = \operatorname{supp} g \ \text{ if } g : E \to \mathbb{C} \text{ is such that } f = g \text{ a.e. on } E. \tag{2.5.15}$$

In addition, if the set $E \subseteq \mathbb{R}^n$ is open and the function $f : E \to \mathbb{C}$ is continuous, then $\operatorname{supp} f$ may be described as the closure in E of the set $\{x \in E : f(x) \neq 0\}$, which is precisely our earlier notion of support in this context.

Exercise 2.51. *If $f \in L^1_{\rm loc}(\Omega)$, then $\operatorname{supp} u_f = \operatorname{supp} f$, where u_f is the distribution from (2.1.6).*

Hint: Use (2.5.10), (2.5.8), part *(1)* in Exercise 2.44, and the fact that the injection in (2.1.8) is one-to-one.

2.6 Compactly Supported Distributions and the Space $\mathcal{E}'(\Omega)$

Next we discuss the issue of extending the action of a distribution $u \in \mathcal{D}'(\Omega)$ to a subclass of $C^\infty(\Omega)$ that is possibly larger than $C^\infty_0(\Omega)$. Observe that if $f \in L^1_{\rm loc}(\Omega)$, the expression $\int_\Omega f\varphi\,dx$ is meaningful for functions $\varphi \in C^\infty(\Omega)$

such that $\operatorname{supp} f \cap \operatorname{supp} \varphi$ is a compact subset of Ω. A particular case is when we have $\operatorname{supp} \varphi \cap \operatorname{supp} f = \emptyset$ in which scenario $\int_\Omega f\varphi \, dx = 0$. This observation is the motivation behind the following theorem.

Theorem 2.52. *Let $u \in \mathcal{D}'(\Omega)$ and consider a relatively closed subset F of Ω satisfying $\operatorname{supp} u \subseteq F$. Set*

$$\mathcal{M}_F := \{\varphi \in C^\infty(\Omega) : \operatorname{supp}\varphi \cap F \text{ is a compact set in } \mathbb{R}^n\}. \tag{2.6.1}$$

Then there exists a unique linear map $\widetilde{u} : \mathcal{M}_F \to \mathbb{C}$ satisfying the following conditions:

(i) $\langle \widetilde{u}, \varphi\rangle = \langle u, \varphi\rangle$ for every $\varphi \in C_0^\infty(\Omega)$, and

(ii) $\langle \widetilde{u}, \varphi\rangle = 0$ for every $\varphi \in C^\infty(\Omega)$ with $\operatorname{supp}\varphi \cap F = \emptyset$,

where, if $\psi \in \mathcal{M}_F$, then $\langle \widetilde{u}, \psi\rangle$ denotes $\widetilde{u}(\psi)$.

Moreover, extensions of u constructed with respect to different choices of F act in a compatible fashion. More precisely, if $F_1, F_2 \subseteq \Omega$ are two relatively closed sets in Ω with the property that $\operatorname{supp} u \subseteq F_j$, $j = 1, 2$, then $\langle \widetilde{u}_1, \varphi\rangle = \langle \widetilde{u}_2, \varphi\rangle$ for every $\varphi \in \mathcal{M}_{F_1} \cap \mathcal{M}_{F_2}$, where $\widetilde{u}_1$, $\widetilde{u}_2$, are the extensions of u constructed as above relative to the sets F_1 and F_2, respectively.

Before presenting the proof of this theorem, a few comments are in order.

Remark 2.53. *Retain the context of Theorem 2.52.*

(a) One has $C_0^\infty(\Omega) \subseteq \mathcal{M}_F$ and $\{\varphi \in C^\infty(\Omega) : \operatorname{supp}\varphi \cap F = \emptyset\} \subseteq \mathcal{M}_F$.

(b) $\mathcal{M}_F$ is a vector subspace of $C^\infty(\Omega)$, albeit not a topological subspace of $\mathcal{E}(\Omega)$.

(c) If $F = \operatorname{supp} u$ we are in the setting discussed prior to the statement of Theorem 2.52. Also, if $F_1 \subseteq F_2$ then $\mathcal{M}_{F_2} \subseteq \mathcal{M}_{F_1}$. In particular, the largest $\mathcal{M}_F$ corresponds to the case when $F = \operatorname{supp} u$.

(d) If $\operatorname{supp} u$ is compact and we take $F = \operatorname{supp} u$ then $\mathcal{M}_F = C^\infty(\Omega)$. In such a scenario, Theorem 2.52 gives an extension of u, originally defined as linear functional on $C_0^\infty(\Omega)$, to a linear functional defined on the larger space $C^\infty(\Omega)$. From a topological point of view, this extension turns out to be a continuous mapping of $\mathcal{E}(\Omega)$ into $\mathbb{C}$ (as we will see later, in Theorem 2.59).

Proof of Theorem 2.52. Fix a relatively closed subset F of Ω satisfying the condition $\operatorname{supp} u \subseteq F$. First we prove the uniqueness statement in the first part of the theorem. Suppose $\widetilde{u}_1$, $\widetilde{u}_2 : \mathcal{M}_F \to \mathbb{C}$ satisfy (i) and (ii). Fix $\varphi \in \mathcal{M}_F$ and consider $\psi \in C_0^\infty(\Omega)$ such that $\psi \equiv 1$ in an open neighborhood W of the set $F \cap \operatorname{supp}\varphi$. That such a function ψ exists is guaranteed by Proposition 13.26. Decompose $\varphi = \varphi_0 + \varphi_1$ where $\varphi_0 := \psi\varphi \in C_0^\infty(\Omega)$ and $\varphi_1 := (1-\psi)\varphi \in C^\infty(\Omega)$.

In general, if $A \subseteq \mathbb{R}^n$ and $f \in C^0(\mathbb{R}^n)$, it may be readily verified that $f = 0$ on A if and only if $\operatorname{supp} f \subseteq \overline{(A^c)}$. Making use of this observation we obtain

that $\operatorname{supp}(1-\psi) \subseteq \overline{W^c} = \big(\mathring{W}\big)^c = W^c$. It follows that $\operatorname{supp}\varphi_1 \subseteq W^c \cap \operatorname{supp}\varphi$, hence $\operatorname{supp}\varphi_1 \cap F = \emptyset$. Thus, by (i) and (ii) written for $\widetilde{u}_1$ and $\widetilde{u}_2$, we have

$$\begin{aligned}\langle \widetilde{u}_1, \varphi\rangle &= \langle \widetilde{u}_1, \varphi_0\rangle + \langle \widetilde{u}_1, \varphi_1\rangle = \langle u, \varphi_0\rangle + 0 = \langle \widetilde{u}_2, \varphi_0\rangle \\ &= \langle \widetilde{u}_2, \varphi_0\rangle + \langle \widetilde{u}_2, \varphi_1\rangle = \langle \widetilde{u}_2, \varphi\rangle, \end{aligned} \tag{2.6.2}$$

which implies that $\widetilde{u}_1 = \widetilde{u}_2$.

To prove the existence of an extension satisfying properties (i) and (ii), we make use of the decomposition of φ already employed in the proof of uniqueness. The apparent problem is that such a decomposition is not unique. Let $\varphi \in \mathcal{M}_F$ and suppose that $\varphi = \varphi_0' + \varphi_1' = \varphi_0'' + \varphi_1''$, with $\varphi_0', \varphi_0'' \in C_0^\infty(\Omega)$, $\varphi_1', \varphi_1'' \in C^\infty(\Omega)$ and

$$\operatorname{supp}\varphi_1' \cap F = \emptyset = \operatorname{supp}\varphi_1'' \cap F.$$

Thus, $\varphi_0' - \varphi_0'' = \varphi_1'' - \varphi_1'$, and since $\operatorname{supp}(\varphi_1'' - \varphi_1') \cap F = \emptyset$, we also have

$$\operatorname{supp}(\varphi_0' - \varphi_0'') \cap F = \emptyset,$$

which in turn implies $\operatorname{supp}(\varphi_0' - \varphi_0'') \subseteq \Omega \setminus \operatorname{supp} u$. The latter condition entails

$$0 = \langle u, \varphi_0' - \varphi_0''\rangle = \langle u, \varphi_0'\rangle - \langle u, \varphi_0''\rangle.$$

This suggests defining the extension

$$\begin{aligned}&\widetilde{u} : \mathcal{M}_F \longrightarrow \mathbb{C}, \quad \langle \widetilde{u}, \varphi\rangle := \langle u, \psi\varphi\rangle \quad \text{for each} \quad \varphi \in \mathcal{M}_F \quad \text{and} \\ &\text{each } \psi \in C_0^\infty(\Omega) \text{ with } \psi \equiv 1 \text{ in a neighborhood of } \operatorname{supp}\varphi \cap F. \end{aligned} \tag{2.6.3}$$

Clearly $\widetilde{u}$ as in (2.6.3) is linear and, based on the previous reasoning, independent of the choice of ψ, thus well-defined. We claim that this extension also satisfies (i) and (ii). Indeed, if $\varphi \in C_0^\infty(\Omega)$, we choose $\psi \equiv 1$ on $\operatorname{supp}\varphi$. Then $\langle \widetilde{u}, \varphi\rangle = \langle u, \varphi\rangle$, so the extension in (2.6.3) satisfies (i). Also, if $\varphi \in C^\infty(\Omega)$ is such that $\operatorname{supp}\varphi \cap F = \emptyset$, we may choose $\psi \in C_0^\infty(\Omega)$ such that $\operatorname{supp}\psi \cap F = \emptyset$ which forces $\langle \widetilde{u}, \varphi\rangle = \langle u, \psi\varphi\rangle = 0$, hence our extension satisfies (ii) as well. This proves the claim.

We are left with proving the compatibility of extensions. Let $F_1, F_2 \subseteq \Omega$ be relatively closed sets in Ω each containing $\operatorname{supp} u$. Denote by $\widetilde{u}_1$ and $\widetilde{u}_2$ the linear extensions of u to $\mathcal{M}_{F_1}$ and $\mathcal{M}_{F_2}$, respectively, constructed as above relative to the sets F_1 and F_2. For $\varphi \in \mathcal{M}_{F_1} \cap \mathcal{M}_{F_2}$ let $\psi \in C_0^\infty(\Omega)$ be such that $\psi = 1$ on an open neighborhood of $\operatorname{supp}\varphi \cap F_1$ and on an open neighborhood of $\operatorname{supp}\varphi \cap F_2$. Then by (2.6.3), $\langle \widetilde{u}_1, \varphi\rangle = \langle u, \psi\varphi\rangle = \langle \widetilde{u}_2, \varphi\rangle$. The proof of the theorem is now complete. □

Remark 2.54. *In the context of Theorem 2.52 consider $u \in \mathcal{D}'(\Omega)$, $a \in C^\infty(\Omega)$, and $\alpha \in \mathbb{N}_0^n$. Then the extension given in Theorem 2.52 satisfies the following properties:*

(1) $\langle \widetilde{au}, \varphi \rangle = \langle \widetilde{u}, a\varphi \rangle$ *for every* $\varphi \in \mathcal{M}_F$;

(2) $\langle \widetilde{\partial^\alpha u}, \varphi \rangle = (-1)^{|\alpha|} \langle \widetilde{u}, \partial^\alpha \varphi \rangle$ *for every* $\varphi \in \mathcal{M}_F$.

Indeed, since by Theorem 2.52 an extension with properties (i) *and* (ii) *is unique, the statement in (1) above will follow if one proves that the actions of the linear functionals considered in the left- and right-hand sides of the equality in (1) coincide on* $C_0^\infty(\Omega)$ *and on* $C^\infty(\Omega)$ *functions with supports outside* F*, which are immediate from* (2.6.3) *and properties of distributions. A similar approach works for the proof of (2).*

We introduce the following notation

$$\mathcal{D}'_c(\Omega) := \{u \in \mathcal{D}'(\Omega) : \operatorname{supp} u \text{ is a compact subset of } \Omega\}. \tag{2.6.4}$$

By applying Theorem 2.52 to a distribution $u \in \mathcal{D}'_c(\Omega)$ and the set $F := \operatorname{supp} u$, in which case we have $\mathcal{M}_F = C^\infty(\Omega)$, it follows that there exists a linear map $\widetilde{u} : C^\infty(\Omega) \to \mathbb{C}$ satisfying (i) and (ii) in the statement of this theorem. In fact, this extension turns out to be continuous with respect to the topology $\mathcal{E}(\Omega)$, an issue that we will address shortly.

The dual of $\mathcal{E}(\Omega)$ is the space

$$\{v : \mathcal{E}(\Omega) \to \mathbb{C} : \ v \text{ linear and continuous}\}. \tag{2.6.5}$$

Whenever $v : \mathcal{E}(\Omega) \to \mathbb{C}$ is linear and continuous, and whenever $\varphi \in C^\infty(\Omega)$, we use the notation $\langle v, \varphi \rangle$ in place of $v(\varphi)$. The following is an equivalent characterization of continuity for linear functionals on $\mathcal{E}(\Omega)$ (see Sect. 13.1 for more details).

Fact 2.55. *A linear functional* $v : \mathcal{E}(\Omega) \to \mathbb{C}$ *is continuous (for details see* (13.1.22)*) if and only if there exist a compact* $K \subset \Omega$*, a number* $m \in \mathbb{N}_0$*, and a constant* $C \in (0, \infty)$*, such that*

$$|v(\varphi)| \leq C \sup_{\alpha \in \mathbb{N}_0^n, \, |\alpha| \leq m} \sup_{x \in K} |\partial^\alpha \varphi(x)|, \qquad \forall\, \varphi \in C^\infty(\Omega). \tag{2.6.6}$$

In the current setting, functionals on $\mathcal{E}(\Omega)$ are continuous if and only if they are sequentially continuous. This can be seen by combining the general result presented in Theorem 13.14 with Fact 1.8. A direct proof, applicable to the specific case of linear functionals on $\mathcal{E}(\Omega)$, is given in the next proposition.

Proposition 2.56. *Let* $v : \mathcal{E}(\Omega) \to \mathbb{C}$ *be a linear map. Then* v *is continuous if and only if* v *is sequentially continuous.*

Proof. The general fact that any linear and continuous functional on topological vector spaces is sequentially continuous gives the left-to-right implication. To prove the converse implication, it suffices to check continuity at zero. This is done reasoning by contradiction. Assume that

$$v(\varphi_j) \xrightarrow[j \to \infty]{} 0 \quad \text{whenever} \quad \varphi_j \xrightarrow[j \to \infty]{\mathcal{E}(\Omega)} 0, \tag{2.6.7}$$

but that v is not continuous at $0 \in \mathcal{E}(\Omega)$. Then for each compact subset K of Ω and every $j \in \mathbb{N}$, there exists $\varphi_j \in \mathcal{E}(\Omega)$ such that

$$|v(\varphi_j)| > j \sup_{\substack{x\in K\\ |\alpha|\leq j}} |\partial^\alpha \varphi_j(x)|. \tag{2.6.8}$$

Consider now a nested sequence of compact sets $\{K_j\}_{j\in\mathbb{N}}$ such that $\bigcup_{j=1}^{\infty} K_j = \Omega$. For each $j \in \mathbb{N}$, let φ_j be as given by (2.6.8) corresponding to $K := K_j$ and define the function $\psi_j := \frac{\varphi_j}{v(\varphi_j)}$ which belongs to $\mathcal{E}(\Omega)$. Then

$$v(\psi_j) = 1 \quad \text{and} \quad \sup_{x\in K_j,\ |\alpha|\leq j} |\partial^\alpha \psi_j(x)| \leq \frac{1}{j} \quad \text{for every} \quad j \in \mathbb{N}. \tag{2.6.9}$$

Thus, for each fixed $\alpha \in \mathbb{N}_0^n$ and every compact subset K of Ω there exists some $j_0 \geq |\alpha|$ with the property that $K \subset K_{j_0}$ and $\sup_{x\in K} |\partial^\alpha \psi_j(x)| < \frac{1}{j}$ for all $j \geq j_0$. The latter implies $\psi_j \xrightarrow[j\to\infty]{\mathcal{E}(\Omega)} 0$ which, in light of (2.6.7), further implies $v(\psi_j) \xrightarrow[j\to\infty]{} 0$. Since this contradicts the fact that $v(\psi_j) = 1$ for every $j \in \mathbb{N}$, the proof is finished. □

The topology we consider on the dual of $\mathcal{E}(\Omega)$ is the weak$*$-topology, and we denote the resulting topological vector space by $\mathcal{E}'(\Omega)$ (see Sect. 13.1 for more details). A significant byproduct of this set up is singled out next.

Fact 2.57. *$\mathcal{E}'(\Omega)$ is a locally convex topological vector space over $\mathbb{C}$, which is not metrizable, but is complete.*

In addition, we have the following important characterization of continuity in $\mathcal{E}'(\Omega)$.

Fact 2.58. *A sequence $\{u_j\}_{j\in\mathbb{N}} \subset \mathcal{E}'(\Omega)$ converges to $u \in \mathcal{E}'(\Omega)$ as $j \to \infty$ in $\mathcal{E}'(\Omega)$, something we will indicate by writing $u_j \xrightarrow[j\to\infty]{\mathcal{E}'(\Omega)} u$, if and only if $\langle u_j, \varphi\rangle \xrightarrow[j\to\infty]{} \langle u, \varphi\rangle$ for every $\varphi \in \mathcal{E}(\Omega)$.*

We are now ready to state and prove a result that gives a complete characterization of the class of functionals that are extensions as in Theorem 2.52 of distributions $u \in \mathcal{D}'(\Omega)$ with compact support.

Theorem 2.59. *The spaces $\mathcal{D}_c'(\Omega)$ and $\mathcal{E}'(\Omega)$ are algebraically isomorphic.*

Proof. Consider the mapping $\iota : \mathcal{D}_c'(\Omega) \to \mathcal{E}'(\Omega)$, $\iota(u) := \widetilde{u}$, where $\widetilde{u}$ is the extension of u given by Theorem 2.52 corresponding to $F := \operatorname{supp} u$. Then $\mathcal{M}_{\operatorname{supp} u} = C^\infty(\Omega)$ and, to conclude that ι is well defined, there remains to show that the functional $\widetilde{u}$ is continuous on $\mathcal{E}(\Omega)$. With this goal in mind, note that while in general the function $\psi \in C_0^\infty(\Omega)$ used in the construction of $\widetilde{u}$ as in (2.6.3) depends on φ, given that we are currently assuming that $\operatorname{supp} u$

is compact, we may take $\psi = 1$ on a neighborhood of $\operatorname{supp} u$ (originally we only needed $\psi = 1$ on a neighborhood of $\operatorname{supp} u \cap \operatorname{supp} \varphi \subseteq \operatorname{supp} u$). Then $\langle \widetilde{u}, \varphi\rangle = \langle u, \psi\varphi\rangle$ for all $\varphi \in C^\infty(\Omega)$. Now let $K_0 := \operatorname{supp} \psi$. It follows that $\varphi\psi \in C_0^\infty(\Omega)$ and $\operatorname{supp}(\varphi\psi) \subseteq K_0$, for all $\varphi \in C^\infty(\Omega)$. Fix $\varphi \in C^\infty(\Omega)$. Since $u \in \mathcal{D}'(\Omega)$, corresponding to the compact set K_0 there exist $k_0 \in \mathbb{N}_0$ and a finite constant $C \geq 0$ such that

$$|\langle \widetilde{u}, \varphi\rangle| = |\langle u, \psi\varphi\rangle| \leq C \sup_{\substack{x\in K_0\\ |\alpha|\leq k_0}} |\partial^\alpha(\psi\varphi)|. \tag{2.6.10}$$

Starting with Leibniz's formula (13.2.4) applied to $\psi\varphi$, we estimate

$$|\partial^\alpha(\psi\varphi)| = \left|\sum_{\beta\leq\alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!}\partial^{\alpha-\beta}\psi\partial^\beta\varphi\right| \leq C' \sup_{\substack{x\in K_0\\ |\beta|\leq k_0}} |\partial^\beta\varphi(x)|, \tag{2.6.11}$$

for some finite constant $C' = C'(\alpha, \psi) > 0$. Combining (2.6.10) and (2.6.11), we obtain

$$|\langle \widetilde{u}, \varphi\rangle| \leq C \cdot C' \sup_{\substack{x\in K_0\\ |\beta|\leq k_0}} |(\partial^\beta\varphi)(x)|, \tag{2.6.12}$$

hence $\widetilde{u} \in \mathcal{E}'(\Omega)$, proving that ι is well-defined.

Moving on, it is clear that ι is linear, hence to conclude that it is injective, it suffices to show that if $\iota(u) = 0$ for some $u \in \mathcal{D}'_c(\Omega)$, then $u = 0$. Consider $u \in \mathcal{D}'_c(\Omega)$ such that $\iota(u) = 0$. Then by *(i)* in Theorem 2.52, $u = \iota(u)\big|_{C_0^\infty(\Omega)} = 0$, as desired.

Consider now the task of proving that ι is surjective. To get started, let $v \in \mathcal{E}'(\Omega)$ be arbitrary, and set $u := v\big|_{C_0^\infty(\Omega)}$. Clearly $u : \mathcal{D}(\Omega) \to \mathbb{C}$ is linear. Since $v \in \mathcal{E}'(\Omega)$, Fact 2.55 ensures the existence of a compact set $K \subset \Omega$, nonnegative integer k, and finite constant $C > 0$, such that

$$|\langle v, \varphi\rangle| \leq C \sup_{\substack{x\in K\\ |\alpha|\leq k}} |\partial^\alpha\varphi(x)|, \qquad \forall\, \varphi \in C^\infty(\Omega). \tag{2.6.13}$$

Then, for each compact subset A of Ω and $\varphi \in C_0^\infty(\Omega)$ with $\operatorname{supp}\varphi \subseteq A$, by regarding φ as being in $\mathcal{E}(\Omega)$ we may use (2.6.13) to write

$$|\langle u, \varphi\rangle| \leq C \sup_{\substack{x\in K\\ |\alpha|\leq k}} |\partial^\alpha\varphi(x)| = C \sup_{\substack{x\in K\cap A\\ |\alpha|\leq k}} |\partial^\alpha\varphi(x)| \leq C \sup_{\substack{x\in A\\ |\alpha|\leq k}} |\partial^\alpha\varphi(x)|. \tag{2.6.14}$$

From (2.6.14) we may now conclude (invoking Proposition 2.4) that $u \in \mathcal{D}'(\Omega)$.

Next, we claim that $\operatorname{supp} u \subseteq K$. Indeed, if the function $\varphi \in C_0^\infty(\Omega)$ is such that $\operatorname{supp}\varphi \cap K = \emptyset$ then from (2.6.13) we obtain $|\langle u, \varphi\rangle| = 0$, thus $u = 0$ on the set $\Omega \setminus K$. Hence, the claim is proved which, in turn, shows that $u \in \mathcal{D}'_c(\Omega)$.

To finish the proof of the surjectivity of ι, it suffices to show that $\iota(u) = v$. Denote by $\widetilde{u}_K$ the extension of u given by Theorem 2.52 with $F := K$. Then

reasoning as in the proof of the fact that ι is well-defined, we obtain that $\widetilde{u}_K \in \mathcal{E}'(\Omega)$. By part (i) in Theorem 2.52 it follows that $\widetilde{u}_K\big|_{C_0^\infty(\Omega)} = u$. Also, if $\varphi \in C^\infty(\Omega)$ satisfies the condition $\operatorname{supp}\varphi \cap K = \emptyset$, then by (ii) in Theorem 2.52 we have $\langle \widetilde{u}_K, \varphi\rangle = 0$, while (2.6.13) implies $\langle v, \varphi\rangle = 0$. Hence, the uniqueness result in Theorem 2.52 yields $\widetilde{u}_K = v$. On the other hand, since K and $\operatorname{supp} u$ are compact, we have $\mathcal{M}_K = C^\infty(\Omega) = \mathcal{M}_{\operatorname{supp} u}$. Now the last conclusion in Theorem 2.52 gives $\widetilde{u}_K = \iota(u)$. Consequently, $\iota(u) = v$, and the surjectivity of ι is proved. This finishes the proof of the theorem. □

In light of the significance of $\mathcal{D}'_c(\Omega)$, Theorem 2.59 provides a natural algebraic identification

$$\mathcal{E}'(\Omega) = \big\{u \in \mathcal{D}'(\Omega) : \operatorname{supp} u \ \text{ is a compact subset of } \ \Omega\big\}. \tag{2.6.15}$$

Remark 2.60. *The spaces $\mathcal{D}'_c(\Omega)$ and $\mathcal{E}'(\Omega)$ are not topologically isomorphic since there exist sequences of distributions with compact support that converge in $\mathcal{D}'(\Omega)$ but not in $\mathcal{E}'(\Omega)$. For example, take the sequence $\{\delta_j\}_{j\in\mathbb{N}} \subset \mathcal{D}'(\mathbb{R})$ of Dirac distributions with mass at $j \in \mathbb{N}$, that have been defined in Example 2.14. Then it is easy to check that the sequence $\{\delta_j\}_j$ converges to 0 in $\mathcal{D}'(\mathbb{R})$ but not in $\mathcal{E}'(\mathbb{R})$.*

Theorem 2.59 nonetheless proves that the identity mapping is well-defined from $\mathcal{E}'(\Omega)$ into $\mathcal{D}'(\Omega)$. Keeping this in mind and relying on (1.3.14) and Proposition 13.3, we see that

$$\mathcal{E}'(\Omega) \text{ is continuously embedded into } \mathcal{D}'(\Omega). \tag{2.6.16}$$

This corresponds to the dual version of (1.3.14). In particular, the operation of restriction to an open subset ω of Ω is a well-defined linear mapping

$$\mathcal{E}'(\Omega) \ni u \mapsto u\big|_\omega \in \mathcal{D}'(\omega). \tag{2.6.17}$$

Moreover, $u\big|_\omega \in \mathcal{E}'(\omega)$ whenever the support of $u \in \mathcal{E}'(\Omega)$ is contained in ω.

Remark 2.61.

(1) In the sequel, we will often drop $\widetilde{\cdot}$ from the notation of the extension (as defined in the proof of Theorem 2.59 or Proposition 2.63) of a compactly supported distribution. More precisely, if $u \in \mathcal{D}'_c(\Omega)$ we will simply use u for the extension of u to a functional in $\mathcal{E}'(\Omega)$, as well as for its extension to a functional in $\mathcal{E}'(\mathcal{O})$, where $\mathcal{O}$ is an open subset of $\mathbb{R}^n$ containing Ω.

(2) Whenever necessary, if $u \in \mathcal{E}'(\Omega)$, $\varphi \in C_0^\infty(\Omega)$, and $\psi \in C^\infty(\Omega)$, we will use the notation ${}_{\mathcal{D}'}\langle u, \varphi\rangle_{\mathcal{D}}$ for the action of u on φ as a functional in $\mathcal{D}'(\Omega)$, and the notation ${}_{\mathcal{E}'}\langle u, \psi\rangle_{\mathcal{E}}$ for the action of u on ψ as a functional in $\mathcal{E}'(\Omega)$.

Proposition 2.62. *Let $u \in \mathcal{D}'(\Omega)$ and $\psi \in C_0^\infty(\Omega)$. Then $\psi u \in \mathcal{E}'(\Omega)$ and*

$${}_{\mathcal{E}'}\langle \psi u, \varphi\rangle_{\mathcal{E}} = {}_{\mathcal{D}'}\langle u, \psi\varphi\rangle_{\mathcal{D}}, \quad \forall\, \varphi \in \mathcal{E}(\Omega). \tag{2.6.18}$$

Proof. Since $\psi u \in \mathcal{D}'(\Omega)$ and $\operatorname{supp} u \subseteq \operatorname{supp} \psi$, we have $\psi u \in \mathcal{D}'_c(\Omega)$, thus $\psi u \in \mathcal{E}'(\Omega)$ (i.e., ψu extends as an element in $\mathcal{E}'(\Omega)$). Let $\phi \in C_0^\infty(\Omega)$ be such that $\phi = 1$ in a neighborhood of $\operatorname{supp} \psi$. Then for every $\varphi \in \mathcal{E}(\Omega)$,

$$\begin{aligned} {}_{\mathcal{E}'}\langle \psi u, \varphi\rangle_{\mathcal{E}} &= {}_{\mathcal{E}'}\langle \phi\psi u, \varphi\rangle_{\mathcal{E}} = {}_{\mathcal{E}'}\langle \psi u, \phi\varphi\rangle_{\mathcal{E}} = {}_{\mathcal{D}'}\langle \psi u, \phi\varphi\rangle_{\mathcal{D}} \\ &= {}_{\mathcal{D}'}\langle u, \psi\phi\varphi\rangle_{\mathcal{D}} = {}_{\mathcal{D}'}\langle u, \psi\varphi\rangle_{\mathcal{D}} \end{aligned} \tag{2.6.19}$$

proving (2.6.18). □

Proposition 2.63. *Let ω and Ω be open subsets of $\mathbb{R}^n$ such that $\omega \subseteq \Omega$. Then every $u \in \mathcal{E}'(\omega)$ extends to a functional $\widetilde{u} \in \mathcal{E}'(\Omega)$ by setting*

$$\widetilde{u} : \mathcal{E}(\Omega) \to \mathbb{C}, \quad \langle \widetilde{u}, \varphi\rangle := \langle u, \psi\varphi\rangle, \quad \forall\, \varphi \in C^\infty(\Omega), \tag{2.6.20}$$

where $\psi \in C_0^\infty(\omega)$ is such that $\psi = 1$ in a neighborhood of $\operatorname{supp} u$.

Proof. We first claim that the mapping in (2.6.20) is well-defined. Indeed, if $\psi_j \in C_0^\infty(\omega)$, $\psi_j = 1$ in a neighborhood of $\operatorname{supp} u$, for $j = 1, 2$, then for each function $\varphi \in C^\infty(\Omega)$ we have $(\psi_1 - \psi_2)\varphi \in C_0^\infty(\omega)$ and $\operatorname{supp} u \cap \operatorname{supp}[(\psi_1 - \psi_2)\varphi] = \emptyset$. Consequently, $\langle u, (\psi_1 - \psi_2)\varphi\rangle = 0$, so that the definition of $\widetilde{u}$ is independent of the choice of ψ with the given properties. The functional defined in (2.6.20) is also linear, its continuity is a consequence of Proposition 2.56 and Exercise 1.21. In addition, if $\varphi \in C^\infty(\omega)$ is given and if $\psi \in C_0^\infty(\omega)$ is such that $\psi = 1$ in a neighborhood of $\operatorname{supp} u$, then $\operatorname{supp}[(1 - \psi)\varphi] \cap \operatorname{supp} u = \emptyset$. Hence

$$\langle \widetilde{u}, \varphi\rangle = \langle u, \psi\varphi\rangle = \langle u, (1 - \psi)\varphi\rangle + \langle u, \varphi\rangle = \langle u, \varphi\rangle, \tag{2.6.21}$$

proving that $\widetilde{u}$ is an extension of u. □

Exercise 2.64. *In the context of Proposition 2.63 prove that*

$$\partial^\alpha \widetilde{u} = \widetilde{\partial^\alpha u} \quad \text{in } \mathcal{D}'(\Omega) \tag{2.6.22}$$

for every $u \in \mathcal{E}'(\omega)$ and every $\alpha \in \mathbb{N}_0^n$.

Exercise 2.65. *Let $u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\Omega)} u$ be such that there exists a compact K in $\mathbb{R}^n$ that is contained in Ω and with the property that $\operatorname{supp} u_j \subseteq K$ for every $j \in \mathbb{N}$. Prove that $\operatorname{supp} u \subseteq K$ and $u_j \xrightarrow[j\to\infty]{\mathcal{E}'(\mathbb{R}^n)} u$. Consequently, we also have $u_j \xrightarrow[j\to\infty]{\mathcal{E}'(\Omega)} u$.*

Exercise 2.66. *Let $k \in \mathbb{N}_0$ and assume that $c_\alpha \in \mathbb{C}$ for $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \le k$. Prove that*

$$\sum_{|\alpha|\le k} c_\alpha \partial^\alpha \delta = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^n) \iff \text{each } c_\alpha = 0. \tag{2.6.23}$$

Hint: Use (13.2.5).

Exercise 2.67. *Prove that if* $u \in \mathcal{D}'(\mathbb{R}^n)$ *and* $\operatorname{supp} u \subseteq \{a\}$ *for some* $a \in \mathbb{R}^n$, *then* u *has a unique representation of the form*

$$u = \sum_{|\alpha| \leq k} c_\alpha \partial^\alpha \delta_a, \tag{2.6.24}$$

for some $k \in \mathbb{N}_0$ *and coefficients* $c_\alpha \in \mathbb{C}$.

Sketch of proof:

(I) Via a translation, reduce matters to the case $a = 0$.

(II) Use Fact 2.55 to determine $k \in \mathbb{N}_0$.

(III) Fix $\psi \in C_0^\infty(B(0,1))$ such that $\psi = 1$ on $B(0, \frac{1}{2})$ and for $\varepsilon > 0$ define the function $\psi_\varepsilon(x) := \psi(\frac{x}{\varepsilon})$ for every $x \in \mathbb{R}^n$. Prove that $u = \psi_\varepsilon u$ in $\mathcal{D}'(\mathbb{R}^n)$.

(IV) For $\varphi \in C_0^\infty(\mathbb{R}^n)$ consider the kth order Taylor polynomial for φ at 0, i.e.,

$$\varphi_k(x) := \sum_{|\beta| \leq k} \frac{1}{\beta!} \partial^\beta \varphi(0)\, x^\beta, \qquad \forall\, x \in \mathbb{R}^n. \tag{2.6.25}$$

Prove that for $\alpha \in \mathbb{N}_0^n$ satisfying $|\alpha| \leq k$ one has

$$\partial^\alpha(\varphi - \varphi_k) = \partial^\alpha \varphi - (\partial^\alpha \varphi)_{k-|\alpha|}.$$

(V) Show that for each $\varphi \in C_0^\infty(\mathbb{R}^n)$ there exists a constant $c \in (0, \infty)$ such that $|\langle u, (\varphi - \varphi_k)\psi_\varepsilon \rangle| \leq c\,\varepsilon$.

(VI) Combine all the above to obtain that

$$\langle u, \varphi \rangle = \Big\langle \sum_{|\alpha| \leq k} \Big[\frac{(-1)^{|\alpha|}}{\alpha!} \langle u, x^\alpha \rangle \Big] \partial^\alpha \delta\,,\, \varphi \Big\rangle \qquad \forall\, \varphi \in C_0^\infty(\mathbb{R}^n). \tag{2.6.26}$$

(VII) Prove that the representation in (VI) is unique.

Example 2.68. Let $m \in \mathbb{N}$. We are interested in solving the equation

$$x^m u = 0 \quad \text{in} \quad \mathcal{D}'(\mathbb{R}). \tag{2.6.27}$$

In this regard, assume that $u \in \mathcal{D}'(\mathbb{R})$ solves (2.6.27) and note that if we have $\varphi \in C_0^\infty(\mathbb{R} \setminus \{0\})$, then $\frac{1}{x^m}\varphi \in C_0^\infty(\mathbb{R} \setminus \{0\})$. This observation permits us to write

$$\langle u, \varphi \rangle = \Big\langle x^m u, \frac{1}{x^m}\varphi \Big\rangle = 0, \qquad \forall\, \varphi \in C_0^\infty(\mathbb{R} \setminus \{0\}), \tag{2.6.28}$$

which proves that $\operatorname{supp} u \subseteq \{0\}$. In particular, $u \in \mathcal{E}'(\mathbb{R})$. Applying Exercise 2.67 we conclude that there exists $N \in \mathbb{N}_0$ such that $u = \sum_{k=0}^{N} c_k \delta^{(k)}$ in $\mathcal{D}'(\mathbb{R})$, for some $c_k \in \mathbb{C}$, $k = 0, 1, 2, \ldots, N$. We claim that

$$c_\ell = 0 \quad \text{whenever} \quad m \leq \ell \leq N. \tag{2.6.29}$$

To see why this is true, observe that since $u \in \mathcal{E}'(\mathbb{R})$ it makes sense to apply u to any function in $C^\infty(\mathbb{R})$. In particular, it is meaningful to apply u to any polynomial. Concerning (2.6.29), if $N \leq m-1$ there is nothing to prove, while in the case when $N \geq m$ for each $\ell \in \{m, \dots, N\}$ we may write

$$0 = \langle x^m u, x^{\ell-m}\rangle = \langle u, x^\ell\rangle = \sum_{k=0}^{N} c_k \langle \delta^{(k)}, x^\ell\rangle$$

$$= \sum_{k=0}^{N} (-1)^k c_k \Big[\frac{\mathrm{d}^k}{\mathrm{d}x^k}(x^\ell)\Big]\Big|_{x=0} = (-1)^\ell \ell!\, c_\ell. \tag{2.6.30}$$

This proves (2.6.29) which, in turn, forces u to have the form

$$u = \sum_{k=0}^{m-1} c_k \delta^{(k)} \quad \text{for some} \quad c_k \in \mathbb{C},\ k = 0, 1, \dots, m-1. \tag{2.6.31}$$

Conversely, one may readily verify that any distribution u as in (2.6.31) solves (2.6.27). In conclusion, any solution u of (2.6.27) is as in (2.6.31).

Exercise 2.69. *Let $m \in \mathbb{N}$, $a \in \mathbb{R}$, and $u \in \mathcal{D}'(\mathbb{R})$. Prove u is a solution of the equation $(x-a)^m u = 0$ in $\mathcal{D}'(\mathbb{R})$ if and only if u is of the form $u = \sum_{k=0}^{m-1} c_k \delta_a^{(k)}$ for some $c_k \in \mathbb{C}$, $k = 0, 1, \dots, m-1$.*

Remark 2.70. *You have seen in Example 2.28 that $\mathrm{P.V.}\,\frac{1}{x}$ is a solution of the equation $xu = 1$ in $\mathcal{D}'(\mathbb{R})$. Hence, if $v \in \mathcal{D}'(\mathbb{R})$ is another solution of this equation, then the distribution $v - \mathrm{P.V.}\,\frac{1}{x}$ is a solution of the equation $xu = 0$ in $\mathcal{D}'(\mathbb{R})$. By Example 2.68, it follows that $v - \mathrm{P.V.}\,\frac{1}{x} = c\,\delta$, where $c \in \mathbb{C}$. Thus, the general solution of the equation $xu = 1$ in $\mathcal{D}'(\mathbb{R})$ is $u = \mathrm{P.V.}\,\frac{1}{x} + c\,\delta$, for $c \in \mathbb{C}$.*

Example 2.71. Let $N \in \mathbb{N}$ and $a_j \in \mathbb{R}^n$, $j \in \{1, \dots, N\}$, be distinct points. If $u \in \mathcal{D}'(\mathbb{R}^n)$ is such that $\operatorname{supp} u \subseteq \{a_1, a_2, \dots, a_N\}$, then u has a unique representation of the form

$$u = \sum_{j=1}^{N} \sum_{|\alpha| \leq k_j} c_{\alpha,j} \partial^\alpha \delta_{a_j}, \qquad k_j \in \mathbb{N}_0, \quad c_{\alpha,j} \in \mathbb{C}. \tag{2.6.32}$$

To justify formula (2.6.32), fix a family of pairwise disjoint balls $B_j := B(a_j, r_j)$, $j \in \{1, \dots, N\}$, and for each j select a function $\psi_j \in C_0^\infty(B_j)$ satisfying $\psi_j = 1$ in a neighborhood of $\overline{B(a_j, r_j/2)}$. Then for each $j \in \{1, \dots, N\}$ we have that $\psi_j u \in \mathcal{E}'(\mathbb{R}^n)$ and $\operatorname{supp}(\psi_j u) \subseteq \{a_j\}$. Exercise 2.67 then gives that there exist $k_j \in \mathbb{N}_0$ and $c_{\alpha,j} \in \mathbb{C}$ such that $\psi_j u = \sum_{|\alpha| \leq k_j} c_{\alpha,j} \partial^\alpha \delta_{a_j}$ in $\mathcal{D}'(\mathbb{R}^n)$. In addition, since $\sum_{j=1}^{N} \psi_j = 1$ in a neighborhood of $\operatorname{supp} u$, we have

$$\Big\langle \sum_{j=1}^{N} u\psi_j, \varphi \Big\rangle = \sum_{j=1}^{N} \langle \psi_j u, \varphi\rangle = \sum_{j=1}^{N} \langle u, \psi_j \varphi\rangle = \Big\langle u, \varphi \sum_{j=1}^{N} \psi_j \Big\rangle = \langle u, \varphi\rangle, \tag{2.6.33}$$

for each $\varphi \in C_0^\infty(\mathbb{R}^n)$. Hence, $u = \sum_{j=1}^{N} \psi_j u$ in $\mathcal{D}'(\mathbb{R}^n)$ which, given (2.6.33), proves (2.6.32).

Example 2.72. Let $a, b \in \mathbb{R}$ be such that $a \neq b$. We are interested in solving the equation

$$(x-a)(x-b)u = 0 \quad \text{in} \quad \mathcal{D}'(\mathbb{R}). \tag{2.6.34}$$

The first observation is that any solution u of this equation satisfies the condition $\operatorname{supp} u \subseteq \{a, b\}$. Indeed, if we take an arbitrary $\varphi \in C_0^\infty(\mathbb{R} \setminus \{a, b\})$, then the function $\frac{1}{(x-a)(x-b)}\varphi$ belongs to the space $C_0^\infty(\mathbb{R} \setminus \{a, b\})$ and

$$\langle u, \varphi \rangle = \Big\langle (x-a)(x-b)u\,,\, \tfrac{1}{(x-a)(x-b)}\,\varphi \Big\rangle = 0. \tag{2.6.35}$$

Hence, we may apply Example 2.71 to conclude that

$$u = \sum_{j=0}^{N_1} c_j \delta_a^{(j)} + \sum_{j=0}^{N_2} d_j \delta_b^{(j)} \quad \text{in} \quad \mathcal{D}'(\mathbb{R}), \tag{2.6.36}$$

where $N_1, N_2 \in \mathbb{N}$, $\{c_j\}_{0 \leq j \leq N_1} \subset \mathbb{C}$ and $\{d_j\}_{0 \leq j \leq N_2} \subset \mathbb{C}$. Moreover, by dropping terms with zero coefficients, there is no loss of generality in assuming that

$$c_{N_1} \neq 0 \quad \text{and} \quad d_{N_2} \neq 0. \tag{2.6.37}$$

In this scenario, we make the claim that $N_1 = N_2 = 0$. To prove this claim, suppose first that $N_1 \geq 1$. Then, using (2.6.36) and the hypotheses on u, we obtain

$$\begin{aligned}
0 &= \big\langle (x-a)(x-b)u, (x-a)^{N_1-1}(x-b)^{N_2} \big\rangle = \big\langle u, (x-a)^{N_1}(x-b)^{N_2+1} \big\rangle \\
&= \sum_{j=0}^{N_1} c_j \big\langle \delta_a^{(j)}, (x-a)^{N_1}(x-b)^{N_2+1} \big\rangle + \sum_{j=0}^{N_2} d_j \big\langle \delta_b^{(j)}, (x-a)^{N_1}(x-b)^{N_2+1} \big\rangle \\
&= \sum_{j=0}^{N_1} (-1)^j c_j \Big[\frac{\mathrm{d}^j}{\mathrm{d}x^j} \Big((x-a)^{N_1}(x-b)^{N_2+1} \Big) \Big] \Big|_{x=a} \\
&\quad + \sum_{j=0}^{N_2} (-1)^j d_j \Big[\frac{\mathrm{d}^j}{\mathrm{d}x^j} \Big((x-a)^{N_1}(x-b)^{N_2+1} \Big) \Big] \Big|_{x=b} \\
&= (-1)^{N_1} c_{N_1} N_1! \, (a-b)^{N_2+1}.
\end{aligned} \tag{2.6.38}$$

Since by assumption $a \neq b$, from (2.6.38) we obtain $c_{N_1} = 0$. This contradicts (2.6.37) and shows that necessarily $N_1 = 0$. Similarly, we obtain that $N_2 = 0$, hence any solution of (2.6.34) has the form

$$u = c\,\delta_a + d\,\delta_b \quad \text{in} \quad \mathcal{D}'(\mathbb{R}), \quad c, d \in \mathbb{C}. \tag{2.6.39}$$

Conversely, it is clear that any distribution as in (2.6.39) solves (2.6.34). To sum up, (2.6.39) describes all solutions of (2.6.34).

2.7 Tensor Product of Distributions

Let $m, n \in \mathbb{N}$, U be an open subset of $\mathbb{R}^m$, V be an open subset of $\mathbb{R}^n$, and consider two complex-valued functions $f \in L^1_{\rm loc}(U)$ and $g \in L^1_{\rm loc}(V)$. Then the `tensor product` of the functions f and g is defined as

$$f\otimes g : U\times V \to \mathbb{C}, \quad (f\otimes g)(x,y) := f(x)g(y) \quad \text{for each} \quad (x,y)\in U\times V. \tag{2.7.1}$$

In particular, it follows from (2.7.1) that $f \otimes g \in L^1_{\rm loc}(U \times V)$. When f, g, and $f \otimes g$ are regarded as distributions, for each $\varphi \in C^\infty_0(U \times V)$ we obtain that

$$\begin{aligned}\langle f \otimes g, \varphi\rangle &= \iint_{U\times V} f(x)g(y)\varphi(x,y)\,{\rm d}x\,{\rm d}y = \int_U f(x)\left(\int_V g(y)\varphi(x,y)\,{\rm d}y\right){\rm d}x \\ &= \int_V g(y)\left(\int_U f(x)\varphi(x,y)\,{\rm d}x\right){\rm d}y \end{aligned} \tag{2.7.2}$$

or, concisely,

$$\langle f \otimes g, \varphi\rangle = \big\langle f(x), \langle g(y), \varphi(x,y)\rangle\big\rangle = \big\langle g(y), \langle f(x), \varphi(x,y)\rangle\big\rangle. \tag{2.7.3}$$

If, in addition, the test function φ has the form $\varphi_1 \otimes \varphi_2$, for some $\varphi_1 \in C^\infty_0(U)$ and $\varphi_2 \in C^\infty_0(V)$, then (2.7.3) becomes

$$\langle f \otimes g, \varphi_1 \otimes \varphi_2\rangle = \langle f, \varphi_1\rangle\langle g, \varphi_2\rangle. \tag{2.7.4}$$

This suggests a natural way to define tensor products of general distributions granted the availability of the following density result for $\mathcal{D}(U \times V)$.

Proposition 2.73. *Let $m, n \in \mathbb{N}$, U be an open subset of $\mathbb{R}^m$, and V be an open subset of $\mathbb{R}^n$. Then the set*

$$C^\infty_0(U)\otimes C^\infty_0(V) := \left\{\sum_{j=1}^N \varphi_j\otimes\psi_j : \varphi_j\in C^\infty_0(U),\ \psi_j\in C^\infty_0(V),\ N\in\mathbb{N}\right\} \tag{2.7.5}$$

is sequentially dense in $\mathcal{D}(U \times V)$.

Before proceeding with the proof of Proposition 2.73 we state and prove two lemmas.

Lemma 2.74. *Suppose that the sequence $\{f_j\}_{j\in\mathbb{N}} \subset \mathcal{E}(\mathbb{R}^n)$ and $f \in \mathcal{E}(\mathbb{R}^n)$ are such that*

$$\|\partial^\alpha f_j - \partial^\alpha f\|_{L^\infty(B(0,j))} < \frac{1}{j} \quad \text{for all } \alpha\in\mathbb{N}_0^n \text{ satisfying } |\alpha|\le j. \tag{2.7.6}$$

Then $f_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} f$.

Proof. Suppose $\{f_j\}_{j\in\mathbb{N}}$ and f satisfy the current hypotheses, and let $\varepsilon \in (0,\infty)$, $\alpha \in \mathbb{N}_0^n$, and a compact subset K of $\mathbb{R}^n$ be fixed. Then there exists $j_0 \in \mathbb{N}$ such that $K \subset B(0,j_0)$. If we now fix some $j^* \in \mathbb{N}$ with the property that $j^* > \max\{\frac{1}{\varepsilon}, |\alpha|, j_0\}$, it follows that for each $j \geq j^*$ we have $|\alpha| \leq j^* \leq j$ and

$$\|\partial^\alpha f_j - \partial^\alpha f\|_{L^\infty(K)} \leq \|\partial^\alpha f_j - \partial^\alpha f\|_{L^\infty(B(0,j))} < \frac{1}{j} < \frac{1}{j^*} < \varepsilon. \tag{2.7.7}$$

Hence, $\partial^\alpha f_j$ converges uniformly on K to $\partial^\alpha f$. Since α and K are arbitrary, we conclude that $f_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} f$. □

Lemma 2.75. *For every $f \in C_0^\infty(\mathbb{R}^n)$ there exists a sequence $\{P_j\}_{j\in\mathbb{N}}$ of polynomials in $\mathbb{R}^n$ such that $P_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} f$.*

Proof. For each $t > 0$ define the function

$$f_t(x) := (4\pi t)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \mathrm{e}^{-\frac{|x-y|^2}{4t}} f(y)\,\mathrm{d}y, \qquad \forall\, x \in \mathbb{R}^n. \tag{2.7.8}$$

The first goal is to prove that

$$f_t \xrightarrow[t\to 0^+]{\mathcal{E}(\mathbb{R}^n)} f. \tag{2.7.9}$$

To this end, consider the function u defined by

$$u(x,t) := \begin{cases} f_t(x), & x \in \mathbb{R}^n,\ t > 0, \\ f(x), & x \in \mathbb{R}^n,\ t = 0, \end{cases} \qquad \forall\,(x,t) \in \mathbb{R}^n \times [0,\infty). \tag{2.7.10}$$

From definition it is clear that u is continuous on $\mathbb{R}^n \times (0,\infty)$. We claim that, in fact, u is continuous on $\mathbb{R}^n \times [0,\infty)$. Indeed, by making use of the change of variables $x - y = 2\sqrt{t}z$, we may write

$$u(x,t) = \pi^{-\frac{n}{2}} \int_{\mathbb{R}^n} \mathrm{e}^{-|z|^2} f(x - 2\sqrt{t}z)\,\mathrm{d}z, \qquad \forall\, x \in \mathbb{R}^n,\ \forall\, t > 0. \tag{2.7.11}$$

Hence, for each $x_* \in \mathbb{R}^n$, Lebesgue dominated convergence theorem (cf. Theorem 13.12) gives

$$\lim_{\substack{x\to x_* \\ t\to 0^+}} u(x,t) = f(x_*)\pi^{-\frac{n}{2}} \int_{\mathbb{R}^n} \mathrm{e}^{-|z|^2}\mathrm{d}z = f(x_*) = u(x_*,0), \tag{2.7.12}$$

proving that u is continuous at points of the form $(x_*,0)$.

Being continuous on $\mathbb{R}^n \times [0,\infty)$, u is uniformly continuous on every compact subset of $\mathbb{R}^n \times [0,\infty)$, thus uniformly continuous on sets of the form $K \times [0,1]$, where $K \subset \mathbb{R}^n$ is compact. Fix such a compact K and fix $\varepsilon \in (0,1)$ arbitrary. Then, there exists $\delta > 0$ such that if (x_1,t_1), $(x_2,t_2) \in K \times [0,1]$ satisfy

$\|(x_1,t_1)-(x_2,t_2)\|\leq\delta$ then $\|u(x_1,t_1)-u(x_2,t_2)\|\leq\varepsilon$. In particular, if $x\in K$ and $t\in(0,\delta)$, then $\|u(x,t)-u(x,0)\|\leq\varepsilon$, that is $\|f_t-f\|_{L^\infty(K)}\leq\varepsilon$ for all $0<t<\delta$. This proves that $\lim_{t\to0^+}f_t(x)=f(x)$ uniformly on compact sets in $\mathbb{R}^n$.

The derivatives of f_t enjoy the same type of properties as f_t. More precisely, by a direct computation (involving also the change of variables $x-y=2\sqrt{t}z$) we see that for each $t>0$ and each $\alpha\in\mathbb{N}_0^n$ we have

$$\partial_x^\alpha f_t(x)=(4\pi t)^{-\frac{n}{2}}\int_{\mathbb{R}^n}\mathrm{e}^{-\frac{|x-y|^2}{4t}}(\partial^\alpha f)(y)\,\mathrm{d}y,\qquad\forall\,x\in\mathbb{R}^n. \tag{2.7.13}$$

In addition, as before, we obtain that $\lim_{t\to0^+}(\partial_x^\alpha f_t)=\partial^\alpha f$ uniformly on compact sets in $\mathbb{R}^n$. This completes the proof of (2.7.9).

Next, recall that the Taylor expansion of the function e^x, $x\in\mathbb{R}$, about the origin is $\mathrm{e}^x=\sum_{j=0}^{\infty}\frac{x^j}{j!}$ with the series converging uniformly on compact subsets of $\mathbb{R}$. In addition, for each $N\in\mathbb{N}$, the remainder $R_N(x):=\mathrm{e}^x-\sum_{j=0}^{N}\frac{x^j}{j!}$ satisfies $|R_N(x)|\leq\mathrm{e}^C\frac{C^{N+1}}{(N+1)!}$ whenever $|x|\leq C$. Fix $t>0$ and a compact subset K of $\mathbb{R}^n$. Then there exists $C>0$ such that $\frac{|x-y|^2}{4t}\leq C$ for every $x\in K$ and every $y\in\operatorname{supp}f$, so

$$\lim_{N\to\infty}\Big|R_N\Big(-\frac{|x-y|^2}{4t}\Big)\Big|\leq\lim_{N\to\infty}\mathrm{e}^C\frac{C^{N+1}}{(N+1)!}=0. \tag{2.7.14}$$

Consequently,

$$\partial_x^\alpha f_t(x)=(4\pi t)^{-\frac{n}{2}}\sum_{j=0}^{\infty}\frac{1}{j!}\int_{\mathbb{R}^n}\left(-\frac{|x-y|^2}{4t}\right)^j\partial^\alpha f(y)\,\mathrm{d}y, \tag{2.7.15}$$

for each $\alpha\in\mathbb{N}_0^n$ and each $t>0$, and the series in (2.7.15) converges uniformly for x in a compact set in $\mathbb{R}^n$. In addition, integrating by parts, we may write

$$\partial_x^\alpha f_t(x)=(4\pi t)^{-\frac{n}{2}}\sum_{j=0}^{\infty}\frac{(-1)^{|\alpha|}}{j}\int_{\mathbb{R}^n}\partial_y^\alpha\Big[\Big(-\frac{|x-y|^2}{4t}\Big)^j\Big]f(y)\,\mathrm{d}y, \tag{2.7.16}$$

where, for each $t>0$ fixed, the series in (2.7.16) converges uniformly on compact sets in $\mathbb{R}^n$. Hence, if for each $t>0$ we define the sequence of polynomials

$$P_{t,k}(x):=(4\pi t)^{-\frac{n}{2}}\sum_{j=0}^{k}\frac{1}{j!}\int_{\mathbb{R}^n}\left(-\frac{|x-y|^2}{4t}\right)^j f(y)\,\mathrm{d}y,\qquad\forall\,x\in\mathbb{R}^n,\quad\forall\,k\in\mathbb{N}, \tag{2.7.17}$$

then the above proof implies that for each $t>0$ we have $P_{t,k}\xrightarrow[k\to\infty]{\mathcal{E}(\mathbb{R}^n)}f_t$.

Next, we claim that there exists a sequence of positive numbers $\{t_j\}_{j\in\mathbb{N}}$ with the property that for each $j\in\mathbb{N}$ we have

$$\|\partial^\alpha f_{t_j}-\partial^\alpha f\|_{L^\infty(B(0,j))}<\tfrac{1}{2j}\qquad\text{for every }\alpha\in\mathbb{N}_0^n\text{ with }|\alpha|\leq j. \tag{2.7.18}$$

To construct a sequence $\{t_j\}_{j\in\mathbb{N}}$ satisfying (2.7.18) we proceed by induction. First, consider the compact set $\overline{B(0,1)}$. For each $\alpha\in\mathbb{N}_0^n$ satisfying $|\alpha|\leq 1$, based on (2.7.9), there exists $\ell_\alpha^1\in\mathbb{N}$ with the property that

$$\left\|\partial^\alpha f_t-\partial^\alpha f\right\|_{L^\infty(B(0,1))}<\tfrac{1}{2}\quad\text{for all }\ t\in\big(0,1/\ell_\alpha^1\big]. \tag{2.7.19}$$

Define $t_1:=\min\left\{\dfrac{1}{\ell_\alpha^1}:\alpha\in\mathbb{N}_0^n,\ |\alpha|\leq 1\right\}$.

Suppose that, for some $j\geq 2$, we have already selected $t_1,\dots,t_{j-1}$ satisfying (2.7.18). Let $\alpha\in\mathbb{N}_0^n$ be such that $|\alpha|\leq j$. Based on (2.7.9), there exists $\ell_\alpha^j\in\mathbb{N}$ with the property that $\ell_\alpha^j\geq\ell_\alpha^{j-1}$ whenever $|\alpha|\leq j-1$, and such that

$$\left\|\partial^\alpha f_t-\partial^\alpha f\right\|_{L^\infty(B(0,j))}<\tfrac{1}{2j}\quad\text{for all }\ t\in\big(0,1/\ell_\alpha^j\big]. \tag{2.7.20}$$

Now define $t_j:=\min\left\{\dfrac{1}{\ell_\alpha^j}:\alpha\in\mathbb{N}_0^n,\ |\alpha|\leq j\right\}$. In particular, this choice ensures that $t_j\leq t_{j-1}$. Proceeding by induction it follows that the sequence $\{t_j\}_{j\in\mathbb{N}}$ constructed in this manner satisfies (2.7.18).

Our next claim is that for each $t>0$ and each $j\in\mathbb{N}$ there exists $k_{t,j}\in\mathbb{N}$ such that

$$\|\partial^\alpha P_{t,k_{t,j}}-\partial^\alpha f_t\|_{L^\infty(B(0,j))}<\tfrac{1}{2j}\quad\text{for every }\alpha\in\mathbb{N}_0^n\text{ with }|\alpha|\leq j. \tag{2.7.21}$$

To prove this, fix $t>0$ and $j\in\mathbb{N}$. Since $P_{t,k}\xrightarrow[k\to\infty]{\mathcal{E}(\mathbb{R}^n)}f_t$ and $\overline{B(0,j)}$ is a compact subset of $\mathbb{R}^n$, it follows that for each $\alpha\in\mathbb{N}_0^n$ satisfying $|\alpha|\leq j$ there exists $k_\alpha^*\in\mathbb{N}$ such that

$$\|\partial^\alpha P_{t,k}-\partial^\alpha f_t\|_{L^\alpha(B(0,j))}<\tfrac{1}{2j},\qquad\text{for }k\geq k_\alpha^*. \tag{2.7.22}$$

If we now define $k_{t,j}:=\max\left\{k_\alpha^*:\alpha\in\mathbb{N}_0^n,\ |\alpha|\leq j\right\}$, then estimate (2.7.21) holds for this $k_{t,j}$. This completes the proof the claim.

Here is the end-game in the proof of the lemma. For each $j\in\mathbb{N}$, let $t_j>0$ be as constructed above so that (2.7.18) holds, for this t_j let $k_{t_j,j}$ be as defined above so that (2.7.21) holds, and set $P_j:=P_{t_j,k_{t_j,j}}$. Hence, for each $j\in\mathbb{N}$ and each $\alpha\in\mathbb{N}_0^n$ satisfying $|\alpha|\leq j$ we have

$$\|\partial^\alpha P_j-\partial^\alpha f\|_{L^\infty(B(0,j))}=\|\partial^\alpha P_{t_j,k_{t_j,j}}-\partial^\alpha f\|_{L^\infty(B(0,j))}\leq\tfrac{1}{2j}+\tfrac{1}{2j}=\tfrac{1}{j}. \tag{2.7.23}$$

The fact that $P_j\xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)}f$ now follows from (2.7.23) by invoking Lemma 2.74. □

Before turning to the proof of Proposition 2.73 we introduce some notation. For $m,n\in\mathbb{N}$, if U is an open subset of $\mathbb{R}^m$, V is an open subset of $\mathbb{R}^n$, and $A\subseteq U\times V$, the projections of A on U and V, respectively, are

$$\begin{aligned}\pi_U(A)&:=\{x\in U:\exists\,y\in V\,\text{such that }(x,y)\in A\},\\ \pi_V(A)&:=\{y\in V:\exists\,x\in U\,\text{such that }(x,y)\in A\}.\end{aligned} \tag{2.7.24}$$

We are ready to present the proof of the density result stated at the beginning of this section.

Proof of Proposition 2.73. Let $\varphi \in C_0^\infty(U \times V)$. By Lemma 2.75, there exists a sequence of polynomials $\{P_j\}_{j\in\mathbb{N}}$ in $\mathbb{R}^{n+m}$ with the property that $P_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^{n+m})} \varphi$. Set $K := \operatorname{supp}\varphi$, $K_1 := \pi_U(K)$, and $K_2 := \pi_V(K)$. Then K_1 and K_2 are compact sets in $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively. Fix a compact set $L_1 \subset U$ such that $K_1 \subset \mathring{L}_1$ and a compact set $L_2 \subset U$ such that $K_2 \subset \mathring{L}_2$. Then there exists a function $\varphi_1 \in C_0^\infty(U)$ with $\operatorname{supp}\varphi_1 \subseteq L_1$, $\varphi_1 = 1$ in a neighborhood of K_1, and a function $\varphi_2 \in C_0^\infty(V)$ satisfying $\operatorname{supp}\varphi_2 \subseteq L_2$ and $\varphi_2 = 1$ in a neighborhood of K_2. Consequently,

$$\varphi_1 \otimes \varphi_2 \in C_0^\infty(\mathbb{R}^{n+m}) \quad \text{and} \quad \operatorname{supp}(\varphi_1 \otimes \varphi_2) \subseteq L_1 \times L_2. \tag{2.7.25}$$

By *(2)* in Exercise 1.10 it follows that

$$(\varphi_1 \otimes \varphi_2)P_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^{n+m})} (\varphi_1 \otimes \varphi_2)\varphi. \tag{2.7.26}$$

Hence, since

$$\operatorname{supp}[(\varphi_1 \otimes \varphi_2)P_j] \subseteq L_1 \times L_2 \quad \text{for every} \quad j \in \mathbb{N}$$

and since $(\varphi_1 \otimes \varphi_2)\varphi = \varphi$, we obtain

$$(\varphi_1 \otimes \varphi_2)P_j \xrightarrow[j\to\infty]{\mathcal{D}(U\times V)} \varphi. \tag{2.7.27}$$

Upon observing that $(\varphi_1 \otimes \varphi_2)P_j \in C_0^\infty(U) \otimes C_0^\infty(V)$ for every $j \in \mathbb{N}$, the desired conclusion follows. □

The next proposition is another important ingredient used to define the tensor product of two distributions.

Proposition 2.76. *Let $m, n \in \mathbb{N}$, U be an open subset of $\mathbb{R}^m$, and V be an open subset of $\mathbb{R}^n$. For each $u \in \mathcal{D}'(U)$ the following properties hold.*

(a) If for each $\varphi \in C_0^\infty(U \times V)$ we define the mapping

$$\psi : V \to \mathbb{C}, \qquad \psi(y) := \big\langle u(x), \varphi(x, y)\big\rangle \quad \forall\, y \in V, \tag{2.7.28}$$

then $\psi \in C_0^\infty(V)$.

(b) The mapping $\mathcal{D}(U \times V) \ni \varphi \mapsto \psi \in \mathcal{D}(V)$, with ψ as defined in (a), is linear and continuous.

Remark 2.77. *In the definition of ψ in part (a) of Proposition 2.76, the use of the notation $u(x)$ does NOT mean that the distributions u is evaluated at x since the latter is not meaningful. The notation $\big\langle u(x), \varphi(x, y)\big\rangle$ should be understood in the following sense: for each $y \in V$ fixed, the distribution u acts on the function $\varphi(\cdot, y)$.*

Proof of Proposition 2.76. Fix $\varphi \in C_0^\infty(U \times V)$ and let $K := \operatorname{supp}\varphi$ that is a compact subset of $U \times V$. Also, consider ψ as in (2.7.28) and recall the projections π_U, π_V from (2.7.24). Then clearly $\operatorname{supp}\psi \subseteq \pi_U(K)$, thus ψ has compact support. Next we prove that ψ is continuous on V. Let $\{y_j\}_{j\in\mathbb{N}}$ be a sequence in V such that $\lim_{j\to\infty} y_j = y_0$ for some $y_0 \in V$. Since $u \in \mathcal{D}'(U)$, based on the definition of ψ and Fact 2.2, in order to conclude that $\lim_{j\to\infty} \psi(y_j) = \psi(y_0)$ it suffices to show that $\varphi(\cdot, y_j) \xrightarrow[j\to\infty]{\mathcal{D}(U)} \varphi(\cdot, y_0)$. It is clear that for every $j \in \mathbb{N}$ we have $\varphi(\cdot, y_j) \in C_0^\infty(U)$ and $\operatorname{supp}\varphi(\cdot, y_j) \subseteq \pi_U(K)$. Moreover, since $\varphi \in C^\infty(U \times V)$ it follows that $\partial_x^\alpha\varphi$ is continuous on K for every $\alpha \in \mathbb{N}_0^m$, thus uniformly continuous on K. Consequently,

$$(\partial_x^\alpha\varphi)(\cdot, y^j) \xrightarrow[j\to\infty]{} (\partial_x^\alpha\varphi)(\cdot, y^0) \quad \text{uniformly on } \pi_U(K).$$

This completes the proof of the fact that ψ is continuous on V.

To continue, we claim that ψ is of class C^1 on V. Fix $y \in V$ and some $j \in \{1, \ldots, n\}$. Recall that $\mathbf{e}_j$ is the unit vector in $\mathbb{R}^n$ with the jth component equal to 1, and let $h \in \mathbb{R} \setminus \{0\}$. Since V is open, there exists $\varepsilon_0 > 0$ such that if $|h| < \varepsilon_0$ then $y + h\mathbf{e}_j \in V$. Make the standing assumption that $|h| < \varepsilon_0$ and set

$$R_h(x, y) := \frac{\varphi(x, y + h\mathbf{e}_j) - \varphi(x, y)}{h} - \frac{\partial\varphi}{\partial y_j}(x, y), \qquad \forall\, x \in U. \tag{2.7.29}$$

Then

$$\frac{\psi(y + h\mathbf{e}_j) - \psi(y)}{h} - \Big\langle u(x), \frac{\partial\varphi}{\partial y_j}(x, y)\Big\rangle = \langle u(x), R_h(x, y)\rangle, \qquad \forall\, x \in U. \tag{2.7.30}$$

Suppose

$$\lim_{h\to 0} R_h(\cdot, y) = 0 \quad \text{in} \quad \mathcal{D}(U). \tag{2.7.31}$$

Then $\lim_{h\to 0} \big\langle u, R_h(\cdot, y)\big\rangle = 0$, which in view of (2.7.30) implies

$$\partial_j\psi(y) = \Big\langle u, \frac{\partial\varphi}{\partial y_j}(\cdot, y)\Big\rangle. \tag{2.7.32}$$

Moreover, since $\frac{\partial\varphi}{\partial y_j} \in C_0^\infty(U \times V)$ by reasoning as in the proof of the continuity of ψ on V, we also obtain that $\partial_j\psi$ is continuous on V. Hence, since j in $\{1, \ldots, n\}$ is arbitrary, to complete the proof of the claim, we are left with showing (2.7.31).

Clearly $\operatorname{supp}[R_h(\cdot, y)] \subseteq \pi_U(K)$. Applying Taylor's formula to φ in the variable y for each fixed $x \in U$ we obtain

$$\varphi(x, y + h\mathbf{e}_j) = \varphi(x, y) + h\frac{\partial\varphi}{\partial y_j}(x, y) + h^2\int_0^1 (1-t)\frac{\partial^2\varphi}{\partial y_j^2}(x, y + th\mathbf{e}_j)\,\mathrm{d}t. \tag{2.7.33}$$

Hence, (2.7.29) and (2.7.33) imply

$$R_h(x,y) = h\int_0^1 (1-t)\frac{\partial^2\varphi}{\partial y_j^2}(x, y+th\mathbf{e}_j)\,\mathrm{d}t. \tag{2.7.34}$$

Consequently, for every $\beta \in \mathbb{N}_0^m$, we have

$$\partial_x^\beta R_h(x,y) = h\int_0^1 (1-t)\partial_x^\beta\frac{\partial^2\varphi}{\partial y_j^2}(x, y+th\mathbf{e}_j)\,\mathrm{d}t, \qquad \forall\, x \in U. \tag{2.7.35}$$

Since the integral in the right-hand side of (2.7.35) is bounded by a constant independent of h, x and y, it follows that $\lim_{h\to 0}\partial_x^\beta R_h(\cdot,y) = 0$ uniformly on $\pi_U(K)$. Combined with the support information on $R_h(\cdot,y)$, this implies (2.7.31) and completes the proof of the claim that $\psi \in C^1(V)$. By induction, we obtain $\psi \in C^\infty(V)$, completing the proof of the statement in part *(a)* of the proposition.

The linearity of the mapping in part *(b)* is immediate since u is a linear mapping. To show that the mapping in *(b)* is also continuous, since $\mathcal{D}(V)$ is locally convex, by Theorem 13.5 it suffices to prove that it is sequentially continuous. To this end, let $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(U\times V)} \varphi$. In particular, there exists a compact subset K of $U\times V$ such that $\operatorname{supp}\varphi_j \subseteq K$ for all $j\in\mathbb{N}$ and

$$\partial^\alpha\varphi_j \xrightarrow[j\to\infty]{} \partial^\alpha\varphi \quad \text{uniformly on } K, \text{ for each } \alpha \in \mathbb{N}_0^{m+n}. \tag{2.7.36}$$

To proceed, for each $y\in V$ set $\psi(y) := \langle u, \varphi(\cdot,y)\rangle$ and $\psi_j(y) := \langle u, \varphi_j(\cdot,y)\rangle$ for each $j\in\mathbb{N}$. The goal is to prove that $\psi_j \xrightarrow[j\to\infty]{\mathcal{D}(V)} \psi$. Applying Proposition 2.4 to the distribution u and compact $\pi_U(K)$ yields $k\in\mathbb{N}_0$ and $C>0$ for which (2.1.1) holds with K replaced by $\pi_U(K)$. Then, for each $\beta\in\mathbb{N}_0^n$, we have

$$\begin{aligned}
\sup_{y\in\pi_V(K)}\left|\partial^\beta\psi_j(y) - \partial^\beta\psi(y)\right| &= \sup_{y\in\pi_V(K)}\left|\left\langle u(x), \partial_y^\beta\varphi_j(\cdot,y) - \partial_y^\beta\varphi(\cdot,y)\right\rangle\right| \\
&\le \sup_{y\in\pi_V(K)}\sup_{\substack{x\in\pi_U(K)\\ |\gamma|\le k}}\left|\partial_x^\gamma\partial_y^\beta\varphi_j(x,y) - \partial_x^\gamma\partial_y^\beta\varphi(x,y)\right| \\
&= C\sup_{\substack{(x,y)\in K\\ |\gamma|\le k}}\left|\partial_x^\gamma\partial_y^\beta\varphi_j(x,y) - \partial_x^\gamma\partial_y^\beta\varphi(x,y)\right| \xrightarrow[j\to\infty]{} 0,
\end{aligned} \tag{2.7.37}$$

where $\gamma\in\mathbb{N}_0^m$ and the convergence to zero in (2.7.37) is due to (2.7.36) applied for $\alpha := (\gamma,\beta)$. Thus, $\psi_j \xrightarrow[j\to\infty]{\mathcal{D}(V)} \psi$ and the proof of the statement in part *(b)* is complete. □

We are now ready to define the tensor product of distributions.

Theorem 2.78. *Let $m,n\in\mathbb{N}$, U be an open subset of $\mathbb{R}^m$, and V be an open subset of $\mathbb{R}^n$. Consider $u\in\mathcal{D}'(U)$ and $v\in\mathcal{D}'(V)$. Then the following statements are true.*

(i) There exists a unique distribution $u \otimes v \in \mathcal{D}'(U \times V)$*, called the* `tensor product` *of* u *and* v*, with the property that*

$$\big\langle u \otimes v, \varphi_1 \otimes \varphi_2 \big\rangle = \langle u, \varphi_1 \rangle \langle v, \varphi_2 \rangle, \quad \forall\, \varphi_1 \in C_0^\infty(U),\ \forall\, \varphi_2 \in C_0^\infty(V). \tag{2.7.38}$$

(ii) The tensor product just defined satisfies $u \otimes v = v \otimes u$ *in* $\mathcal{D}'(U \times V)$.

Proof. For each $\varphi \in C_0^\infty(U \times V)$ consider the function

$$\psi(y) := \langle u(x), \varphi(x, y) \rangle \quad \text{for} \quad y \in V.$$

By Proposition 2.76, we have $\psi \in C_0^\infty(V)$ and the mapping

$$\mathcal{D}(U \times V) \ni \varphi \mapsto \psi \in \mathcal{D}(V) \quad \text{ is linear and continuous.} \tag{2.7.39}$$

Hence, $\langle v, \psi \rangle$ is meaningful and we may define

$$\begin{aligned} &u \otimes v : \mathcal{D}(U \times V) \longrightarrow \mathbb{C} \\ &\langle u \otimes v, \varphi \rangle := \Big\langle v(y), \langle u(x), \varphi(x, y) \rangle \Big\rangle \quad \text{for every} \quad \varphi \in C_0^\infty(U \times V). \end{aligned} \tag{2.7.40}$$

As defined, the mapping $u \otimes v$ is the composition of two linear and continuous mappings, linear and continuous. In addition, if $\varphi_1 \in C_0^\infty(U)$ and $\varphi_2 \in C_0^\infty(V)$, then

$$\begin{aligned} \langle u \otimes v, \varphi_1 \otimes \varphi_2 \rangle &= \Big\langle v(y)\,,\, \langle u(x), \varphi_1(x)\varphi_2(y) \rangle \Big\rangle \\ &= \Big\langle v(y)\,,\, \varphi_2(y) \langle u(x), \varphi_1(x) \rangle \Big\rangle \\ &= \big\langle v(y), \varphi_2(y) \big\rangle \big\langle u(x), \varphi_1(x) \big\rangle, \end{aligned} \tag{2.7.41}$$

thus the mapping $u \otimes v$ defined in (2.7.40) satisfies (2.7.38).

To prove the uniqueness statement in part *(i)*, suppose $w_1, w_2 \in \mathcal{D}'(U \times V)$ are such that

$$\begin{aligned} &\langle w_j, \varphi_1 \otimes \varphi_2 \rangle = \langle u, \varphi_1 \rangle \langle v, \varphi_2 \rangle, \quad j = 1, 2, \\ &\text{for every} \quad \varphi_1 \in C_0^\infty(U), \quad \varphi_2 \in C_0^\infty(V). \end{aligned} \tag{2.7.42}$$

Then it follows that $\langle w_1, \varphi \rangle = \langle w_2, \varphi \rangle$ for every $\varphi \in C_0^\infty(U) \otimes C_0^\infty(V)$, which in concert with Proposition 2.73 and the continuity of w_1 and w_2 implies $w_1 = w_2$ in $\mathcal{D}'(U \times V)$. This completes the proof of the statement in *(i)*.

As for the statement in *(ii)*, observe that based on (2.7.40) we have

$$\begin{aligned} &v \otimes u : \mathcal{D}(U \times V) \to \mathbb{C} \quad \text{and} \\ &\langle v \otimes u, \varphi \rangle = \big\langle u(x), \langle v(y), \varphi(x, y) \rangle \big\rangle \quad \text{for every} \quad \varphi \in C_0^\infty(U \times V). \end{aligned} \tag{2.7.43}$$

In particular, for every $\varphi_1 \in C_0^\infty(U)$ and $\varphi_2 \in C_0^\infty(U)$ we have

$$\langle v \otimes u, \varphi_1 \otimes \varphi_2 \rangle = \langle u, \varphi_1 \rangle \langle v, \varphi_2 \rangle = \langle u \otimes v, \varphi_1 \otimes \varphi_2 \rangle. \tag{2.7.44}$$

The uniqueness result from part *(i)* now implies $u \otimes v = v \otimes u$ in $\mathcal{D}'(U \times V)$. □

Remark 2.79. *If $u \in \mathcal{D}'(U)$ and $v \in L^1_{\rm loc}(V)$, then the statement in part (ii) in Theorem 2.78 becomes*

$$\Big\langle u(x), \int_V v(y)\varphi(x,y)\,{\rm d}y \Big\rangle = \int_V v(y)\big\langle u(x), \varphi(x,y)\big\rangle\,{\rm d}y, \quad \forall\,\varphi \in C_0^\infty(U\times V). \tag{2.7.45}$$

The interpretation of (2.7.45) is that the distribution u commutes with the integral.

We next establish a number of basic properties for the tensor products of distributions.

Theorem 2.80. *Let $m,n\in\mathbb{N}$, U be an open subset of $\mathbb{R}^m$, and V be an open subset of $\mathbb{R}^n$. Assume that $u\in\mathcal{D}'(U)$ and $v\in\mathcal{D}'(V)$. Then the following properties hold.*

(a) $\operatorname{supp} u\otimes v = \operatorname{supp} u\times\operatorname{supp} v$.

(b) $\partial_x^\alpha\partial_y^\beta(u\otimes v) = (\partial_x^\alpha u)\otimes(\partial_y^\beta v)$ *for every* $\alpha\in\mathbb{N}_0^m$ *and every* $\beta\in\mathbb{N}_0^n$.

(c) $(f\otimes g)\cdot(u\otimes v) = (fu)\otimes(gv)$ *for every* $f\in C^\infty(U)$ *and every* $g\in C^\infty(V)$.

(d) The mapping $\mathcal{D}'(U)\times\mathcal{D}'(V)\ni(u,v)\mapsto u\otimes v\in\mathcal{D}'(U\times V)$ *is bilinear and separately continuous.*

(e) The tensor product of distributions is associative.

Proof. We start by proving the set theoretic equality from *(a)*. For the left-to-right inclusion, fix $(x_0,y_0)\in\operatorname{supp} u\times\operatorname{supp} v$. If $C\subseteq U\times V$ is an open neighborhood of (x_0,y_0), then there exist an open set $A\subseteq U$ containing x_0 and an open set $B\subseteq V$ containing y_0 such that $A\times B\subset C$. In particular, since $x_0\in\operatorname{supp} u$ and $y_0\in\operatorname{supp} v$, there exist $\varphi_1\in C_0^\infty(A)$ and $\varphi_2\in C_0^\infty(B)$ with the property that $\langle u,\varphi_1\rangle\neq 0$ and $\langle v,\varphi_2\rangle\neq 0$. If we now set $\varphi:=\varphi_1\otimes\varphi_2$, then $\varphi\in C_0^\infty(C)$ and $\langle u\otimes v,\varphi\rangle = \langle u,\varphi_1\rangle\langle v,\varphi_2\rangle\neq 0$. Hence $(x_0,y_0)\in\operatorname{supp}(u\otimes v)$, finishing the proof of the left-to-right inclusion in *(a)*.

To prove the opposite inclusion, observe that $\operatorname{supp}(u\otimes v)\subseteq\operatorname{supp} u\times\operatorname{supp} v$ is equivalent to

$$(U\times V)\setminus(\operatorname{supp} u\times\operatorname{supp} v)\subseteq(U\times V)\setminus\operatorname{supp}(u\otimes v). \tag{2.7.46}$$

Write the left hand-side of (2.7.46) as $D_1\cup D_2$, where $D_1:=(U\setminus\operatorname{supp} u)\times V$ and $D_2:=U\times(V\setminus\operatorname{supp} v)$. Note that D_1 and D_2 are open sets in $\mathbb{R}^m\times\mathbb{R}^n$. Since the support of a distribution is the smallest relatively closed set outside of which the distribution vanishes, for (2.7.46) to hold it suffices to show that $\langle u\otimes v,\varphi\rangle = 0$ for every $\varphi\in C_0^\infty(D_1\cup D_2)$. Fix such a function φ, set $K:=\operatorname{supp}\varphi$, and consider a partition of unity subordinate to the covering $\{D_1,D_2\}$ of K, say

$$\psi_j\in C_0^\infty(D_j),\ j\in\{1,2\},\quad \psi_1+\psi_2=1 \text{ in a neighborhood of } K. \tag{2.7.47}$$

Then $\varphi\psi_1 \in C_0^\infty(D_1)$, $\varphi\psi_2 \in C_0^\infty(D_2)$ (with the understanding that ψ_1 and ψ_2 have been extended by zero outside their supports), and $\varphi = \varphi\psi_1 + \varphi\psi_2$ on $U \times V$. Since $\pi_U(D_1) \cap \operatorname{supp} u = \emptyset$ and $\pi_V(D_2) \cap \operatorname{supp} v = \emptyset$, we may write

$$\begin{aligned}\langle u \otimes v, \varphi\rangle &= \big\langle v(y), \langle u(x), \varphi(x,y)\psi_1(x,y)\rangle\big\rangle \\ &\quad + \big\langle u(x), \langle v(y), \varphi(x,y)\psi_2(x,y)\rangle\big\rangle = 0. \end{aligned} \tag{2.7.48}$$

This completes the proof of the equality of sets from part *(a)*.

To prove the identity in *(b)*, fix $\alpha \in \mathbb{N}_0^m$, $\beta \in \mathbb{N}_0^n$, and let $\varphi_1 \in C_0^\infty(U)$, $\varphi_2 \in C_0^\infty(V)$. Then starting with the definition of distributional derivatives and then using (2.7.38) we may write

$$\begin{aligned}\big\langle \partial_x^\alpha \partial_y^\beta (u \otimes v), \varphi_1 \otimes \varphi_2\big\rangle &= (-1)^{|\alpha|+|\beta|}\big\langle u \otimes v, \partial_x^\alpha \partial_y^\beta(\varphi_1 \otimes \varphi_2)\big\rangle \\ &= (-1)^{|\alpha|+|\beta|}\langle u \otimes v, \partial_x^\alpha \varphi_1 \otimes \partial_y^\beta \varphi_2\rangle \\ &= (-1)^{|\alpha|+|\beta|}\langle u, \partial_x^\alpha \varphi_1\rangle\langle v, \partial_y^\beta \varphi_2\rangle \\ &= \langle \partial_x^\alpha u, \varphi_1\rangle\langle \partial_y^\beta u, \varphi_2\rangle \\ &= \big\langle (\partial_x^\alpha u) \otimes (\varphi_y^\beta v), \varphi_1 \otimes \varphi_2\big\rangle. \end{aligned} \tag{2.7.49}$$

By the uniqueness statement in part *(i)* of Theorem 2.78 we deduce from (2.7.49) that $(\partial_x^\alpha u) \otimes (\varphi_y^\beta v) = \partial_x^\alpha \partial_y^\beta (u \otimes v)$, completing the proof of the identity in *(b)*.

Moving on to the proof of the statement in *(c)*, note that if $f \in C^\infty(U)$ and $g \in C^\infty(V)$, then $f \otimes g \in C^\infty(U \times V)$. The latter, combined with the definition of multiplication of a distribution with a smooth function and (2.7.38), permits us to write

$$\begin{aligned}\big\langle (f \otimes g) \cdot (u \otimes v), \varphi_1 \otimes \varphi_2\big\rangle &= \langle u \otimes v, (f \otimes g) \cdot (\varphi_1 \otimes \varphi_2)\rangle \\ &= \big\langle u \otimes v, (f\varphi_1) \otimes (g\varphi_2)\big\rangle = \langle u, f\varphi_1\rangle\langle v, g\varphi_2\rangle \\ &= \langle fu, \varphi_1\rangle\langle gv, \varphi_2\rangle = \big\langle (fu) \otimes (gv), \varphi_1 \otimes \varphi_2\big\rangle, \end{aligned} \tag{2.7.50}$$

for every $\varphi_1 \in C_0^\infty(U)$, $\varphi_2 \in C_0^\infty(V)$. The identity in *(c)* now follows from (2.7.50) by once again invoking the uniqueness result from part *(i)* in Theorem 2.78.

The bilinearity of the mapping $(u, v) \mapsto u \otimes v$ is a consequence of the definition of $u \otimes v$ and (2.2.1). To prove that this mapping is also separately continuous, let $u_j \xrightarrow[j\to\infty]{\mathcal{D}'(U)} u$ and fix $v \in \mathcal{D}'(V)$. If $\varphi \in C_0^\infty(U \times V)$, then $\langle v(y), \varphi(x,y)\rangle \in C_0^\infty(U)$ by Proposition 2.76, and we may use Fact 2.18 to write

$$\langle u_j \otimes v, \varphi\rangle = \langle u_j(x), \langle v(y), \varphi(x,y)\rangle \xrightarrow[j\to\infty]{} \langle u(x), \langle v(y), \varphi(x,y)\rangle = \langle u \otimes v, \varphi\rangle. \tag{2.7.51}$$

Similarly, if $u \in \mathcal{D}'(U)$ is fixed and $v_j \xrightarrow[j\to\infty]{\mathcal{D}'(v)} v$, then $u \otimes v_j \xrightarrow[j\to\infty]{\mathcal{D}'(U\times V)} u \otimes v$. Upon recalling that the topological vector space $\mathcal{D}(U \times V)$ is locally convex, Theorem 13.5 may now be invoked to finish the proof of *(d)*.

Finally, we are left with proving the associativity of the tensor product of distributions. To this end, let $k \in \mathbb{N}$, W be an open subset of $\mathbb{R}^k$, and $w \in \mathcal{D}'(W)$ be arbitrary. By Theorem 2.78, $u \otimes v \in \mathcal{D}'(U \times V)$, $v \otimes w \in \mathcal{D}'(V \times W)$ and, furthermore,

$$(u \otimes v) \otimes w \in \mathcal{D}'(U \times V \times W), \qquad u \otimes (v \otimes w) \in \mathcal{D}'(U \times V \times W). \tag{2.7.52}$$

The goal is to prove that

$$(u \otimes v) \otimes w = u \otimes (v \otimes w) \quad \text{in} \quad \mathcal{D}'(U \times V \times W). \tag{2.7.53}$$

In this regard we first note that, for each $\varphi \in C_0^\infty(U)$, $\psi \in C_0^\infty(V)$, and $\eta \in C_0^\infty(W)$, we may write

$$\begin{aligned} \big\langle (u \otimes v) \otimes w, (\varphi \otimes \psi) \otimes \eta \big\rangle &= \langle u, \varphi \rangle \langle v, \psi \rangle \langle w, \eta \rangle \\ &= \big\langle u \otimes (v \otimes w), \varphi \otimes (\varphi \otimes \eta) \big\rangle. \end{aligned} \tag{2.7.54}$$

Define $C_0^\infty(U) \otimes C_0^\infty(V) \otimes C_0^\infty(W)$ as

$$\left\{ \sum_{j=1}^{N} \varphi_j \otimes \psi_j \otimes \eta_j : \varphi_j \in C_0^\infty(U),\, \psi_j \in C_0^\infty(V),\, \eta_j \in C_0^\infty(W),\, N \in \mathbb{N} \right\}, \tag{2.7.55}$$

and note that this set is sequentially dense in $\mathcal{D}(U \times V \times W)$ (which can be proved by reasoning as in the proof of Proposition 2.73). Granted this, (2.7.53) is implied by (2.7.54), completing the proof of the theorem. □

Exercise 2.81. *Let $n, m \in \mathbb{N}$ and $x_0 \in \mathbb{R}^n$, $y_0 \in \mathbb{R}^m$. Prove that $\delta_{x_0} \otimes \delta_{y_0} = \delta_{(x_0, y_0)}$ in $\mathcal{D}'(\mathbb{R}^{n+m})$.*

We close this section by revisiting the result proved in Proposition 2.76 and establishing a related version that is going to be useful later on.

Proposition 2.82. *Let $m, n \in \mathbb{N}$, U be an open subset of $\mathbb{R}^m$, and V be an open subset of $\mathbb{R}^n$. Assume that $u \in \mathcal{E}'(U)$, $\varphi \in C^\infty(U \times V)$, and define the function*

$$\psi : V \to \mathbb{C}, \qquad \psi(y) := \langle u(x), \varphi(x, y) \rangle, \qquad \forall\, y \in V. \tag{2.7.56}$$

Then $\psi \in C^\infty(V)$ and for every $\alpha \in \mathbb{N}_0^n$ we have

$$\partial^\alpha \psi(y) = \big\langle u(x)\,,\, \partial_y^\alpha \varphi(x, y) \big\rangle, \quad \forall\, y \in V. \tag{2.7.57}$$

Proof. Fix some $\eta \in C_0^\infty(U)$ that satisfies $\eta = 1$ in a neighborhood of $\operatorname{supp} u$. Then for each $\theta \in C_0^\infty(V)$ we may write

$$(\theta\psi)(y) = \big\langle (\eta u)(x), \theta(y)\varphi(x,y)\big\rangle = \big\langle u(x), (\eta\otimes\theta)(x,y)\varphi(x,y)\big\rangle$$
$$= \big\langle u(x), \big[(\eta\otimes\theta)\varphi\big](x,y)\big\rangle, \qquad \forall\, y\in V. \tag{2.7.58}$$

Given that $(\eta\otimes\theta)\varphi\in C^\infty_0(U\times V)$, Proposition 2.76 applies and gives that the right-most side of (2.7.58) depends in a C^∞ manner on the variable $y\in V$. Hence, $\theta\psi\in C^\infty(V)$ and since $\theta\in C^\infty_0(V)$ has been arbitrarily chosen we deduce that $\psi\in C^\infty(V)$. This takes care of the first claim in the statement of the proposition.

As regards (2.7.57), observe that it suffices to prove this formula in the case $|\alpha|=1$, since the general case then follows by iteration. With this in mind, fix $j\in\{1,\dots,n\}$ and pick an arbitrary point $y^*\in V$. Also, select some function $\theta\in C^\infty_0(V)$ such that $\theta=1$ near y^*. These properties of θ permit us to compute

$$\partial_{y_j}\big[(\eta\otimes\theta)\varphi\big](x,y^*) = \eta(x)\Big\{(\partial_j\theta)(y^*)\varphi(x,y^*) + \theta(y^*)(\partial_{y_j}\varphi)(x,y^*)\Big\}$$
$$= \eta(x)(\partial_{y_j}\varphi)(x,y^*), \qquad \forall\, x\in U. \tag{2.7.59}$$

Making use of (2.7.32), (2.7.58), and (2.7.59), we may then write

$$\partial_j\psi(y^*) = \partial_j(\theta\psi)(y^*) = \big\langle u(x), \partial_{y_j}\big[(\eta\otimes\theta)\varphi\big](x,y^*)\big\rangle$$
$$= \big\langle u(x), \eta(x)(\partial_{y_j}\varphi)(x,y^*)\big\rangle$$
$$= \big\langle u(x), (\partial_{y_j}\varphi)(x,y^*)\big\rangle. \tag{2.7.60}$$

This corresponds precisely to formula (2.7.57) written at the point $y=y^*$ and for the multi-index $\alpha=(0,\dots,0,1,0,\dots,0)\in\mathbb{N}^n_0$ with the nonzero component on the jth slot. As remarked earlier, this suffices to finish the proof. □

2.8 The Convolution of Distributions in $\mathbb{R}^n$

Recall that if $f,g\in L^1(\mathbb{R}^n)$ then, as a consequence of Fubini's theorem, the function $h:\mathbb{R}^n\times\mathbb{R}^n\to\mathbb{C}$, defined by $h(x,y):=f(x-y)g(y)$ for every $(x,y)\in\mathbb{R}^n\times\mathbb{R}^n$, is absolutely integrable on $\mathbb{R}^n\times\mathbb{R}^n$ and

$$\iint_{\mathbb{R}^n\times\mathbb{R}^n}|h(x,y)|\,dx\,dy = \int_{\mathbb{R}^n}\int_{\mathbb{R}^n}|f(x-y)g(y)|\,dx\,dy$$
$$= \int_{\mathbb{R}^n}|g(y)|\Big(\int_{\mathbb{R}^n}|f(x-y)|\,dx\Big)\,dy$$
$$= \|f\|_{L^1(\mathbb{R}^n)}\|g\|_{L^1(\mathbb{R}^n)}. \tag{2.8.1}$$

Hence, the `convolution` of f and g defined as

$$f*g:\mathbb{R}^n\to\mathbb{C}, \quad (f*g)(x) := \int_{\mathbb{R}^n} f(x-y)g(y)\,dy \quad \text{for each} \quad x\in\mathbb{R}^n, \tag{2.8.2}$$

satisfies $f*g\in L^1(\mathbb{R}^n)$ (and a natural estimate). We would like to extend this definition to functions that are not necessarily in $L^1(\mathbb{R}^n)$. Specifically, assume that $f,g\in L^1_{\rm loc}(\mathbb{R}^n)$ have the property that

$$\begin{aligned} &M_{B(0,r)}:=\Big\{(x,y)\in\operatorname{supp}f\times\operatorname{supp}g:\ x+y\in\overline{B(0,r)}\Big\}\\ &\text{is a compact set in }\mathbb{R}^n\times\mathbb{R}^n\text{ for every }r\in(0,\infty). \end{aligned} \tag{2.8.3}$$

In this scenario, consider the function $G:\mathbb{R}^n\to[0,\infty]$ defined by

$$G(x):=\int_{\mathbb{R}^n}|f(x-y)|\,|g(y)|\,{\rm d}y\quad\text{for each}\quad x\in\mathbb{R}^n. \tag{2.8.4}$$

Note that for every $r\in(0,\infty)$, by making a natural change of variables and using the fact that $f\otimes g\in L^1_{\rm loc}(\mathbb{R}^n\times\mathbb{R}^n)$, we obtain

$$\int_{|x|\leq r}G(x)\,{\rm d}x=\iint_{M_{B(0,r)}}|f(z)||g(y)|\,{\rm d}y\,{\rm d}z<\infty. \tag{2.8.5}$$

Thus, the function G is locally integrable, hence finite almost everywhere in $\mathbb{R}^n$.

In conclusion, whenever $f,g\in L^1_{\rm loc}(\mathbb{R}^n)$ satisfy (2.8.3), for almost every $x\in\mathbb{R}^n$ the integral $(f*g)(x):=\int_{\mathbb{R}^n}f(x-y)g(y)\,{\rm d}y$ is absolutely convergent and $f*g\in L^1_{\rm loc}(\mathbb{R}^n)$. Furthermore, having fixed an arbitrary $\varphi\in C^\infty_0(\mathbb{R}^n)$ we may write

$$\begin{aligned} \langle f*g,\varphi\rangle&=\int_{\mathbb{R}^n}\int_{\mathbb{R}^n}f(x-y)g(y)\varphi(x)\,{\rm d}y\,{\rm d}x\\ &=\int_{\mathbb{R}^n}\int_{\mathbb{R}^n}f(z)g(y)\varphi(z+y)\,{\rm d}y\,{\rm d}z. \end{aligned} \tag{2.8.6}$$

To proceed, observe that the function φ^Δ defined by

$$\varphi^\Delta:\mathbb{R}^n\times\mathbb{R}^n\to\mathbb{C},\qquad\varphi^\Delta(x,y):=\varphi(x+y)\quad\text{for every}\quad x,y\in\mathbb{R}^n, \tag{2.8.7}$$

satisfies $\varphi^\Delta\in C^\infty(\mathbb{R}^n\times\mathbb{R}^n)$ though, in general, the support of φ^Δ is not compact. Formally, the last double integral in (2.8.6) has the same expression as $\langle f\otimes g,\varphi^\Delta\rangle$. However, under the current assumptions on φ, f, and g, it is not clear that this pairing may be interpreted in the standard distributional sense. Indeed, even though $f\otimes g$ is a well-defined distribution in $\mathcal{D}'(\mathbb{R}^n\times\mathbb{R}^n)$ (cf. Theorem 2.78), the function φ^Δ does not belong to $C^\infty_0(\mathbb{R}^n\times\mathbb{R}^n)$, as it lacks the compact support property. Nonetheless, (2.8.3) implies

$$\operatorname{supp}\varphi^\Delta\cap\operatorname{supp}(f\otimes g)\quad\text{is a compact set in }\mathbb{R}^n\times\mathbb{R}^n. \tag{2.8.8}$$

Theorem 2.52 applies with $F:=\operatorname{supp}(f\otimes g)$ and allows us to uniquely extend the action of the distribution $f\otimes g$ to the set of functions $\psi\in C^\infty(\mathbb{R}^n\times\mathbb{R}^n)$ satisfying the property that $\operatorname{supp}\psi\cap\operatorname{supp}(f\otimes g)$ is a compact set in $\mathbb{R}^n\times\mathbb{R}^n$. Denote this unique extension by $\widetilde{f\otimes g}$. Then $\langle\widetilde{f\otimes g},\varphi^\Delta\rangle$ is well-defined, and it is meaningful to set $\langle f*g,\varphi\rangle:=\langle\widetilde{f\otimes g},\varphi^\Delta\rangle$. This discussion justifies making the following definition.

Definition 2.83. *Suppose u, $v \in \mathcal{D}'(\mathbb{R}^n)$ are such that*

$$\text{for each compact subset } K \text{ of } \mathbb{R}^n \text{ the set}$$
$$M_K := \{(x,y) \in \operatorname{supp} u \times \operatorname{supp} v : x+y \in K\} \text{ is compact in } \mathbb{R}^n \times \mathbb{R}^n. \tag{2.8.9}$$

Granted this, define the convolution *of the distributions u and v as the functional $u * v : \mathcal{D}(\mathbb{R}^n) \to \mathbb{C}$ whose action on each $\varphi \in C_0^\infty(\mathbb{R}^n)$ is given by*

$$\langle u * v, \varphi \rangle := \langle \widetilde{u \otimes v}, \varphi^\Delta \rangle \tag{2.8.10}$$

where $\varphi^\Delta(x,y) := \varphi(x+y)$ for every $x, y \in \mathbb{R}^n$, and $\widetilde{u \otimes v}$ is the unique extension of $u \otimes v$ obtained by applying Theorem 2.52 with $F := \operatorname{supp}(u \otimes v)$.

Remark 2.84. *Retain the context of Definition 2.83.*

(1) If $\varphi \in C_0^\infty(\mathbb{R}^n)$ and $\psi \in C_0^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ is such that $\psi = 1$ on a neighborhood of $M_{\operatorname{supp}\varphi}$ then

$$\langle u * v, \varphi \rangle = \big\langle u \otimes v, \psi \varphi^\Delta \big\rangle. \tag{2.8.11}$$

(2) If (2.8.9) holds for the compacts $K_j = \overline{B(0,j)}$, $j \in \mathbb{N}$, then (2.8.9) holds for arbitrary compact sets $K \subset \mathbb{R}^n$.

(3) Condition (2.8.9) is always satisfied if either u or v is compactly supported.

The issue of continuity of the convolution map introduced in Definition 2.83 is discussed next.

Theorem 2.85. *If u, $v \in \mathcal{D}'(\mathbb{R}^n)$ are such that (2.8.9) holds, then one has $u * v \in \mathcal{D}'(\mathbb{R}^n)$. In particular, the convolution between two distributions in $\mathbb{R}^n$, one of which is compactly supported, is always well-defined and is a distribution in $\mathbb{R}^n$.*

Proof. Let u, $v \in \mathcal{D}'(\mathbb{R}^n)$ be such that (2.8.9) is satisfied. From Theorem 2.52, we have that $\widetilde{u \otimes v}$ is linear, hence $u * v$ is linear as well. Let $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\mathbb{R}^n)} 0$. Then there exists a compact subset K of $\mathbb{R}^n$ such that $\operatorname{supp}\varphi_j \subseteq K$ for each $j \in \mathbb{N}$, and $\lim_{j\to\infty} \partial^\alpha \varphi_j = 0$ uniformly on K, for every $\alpha \in \mathbb{N}_0^n$. In particular, $M_{\operatorname{supp}\varphi_j} \subseteq M_K$ for every $j \in \mathbb{N}$. Hence, if we fix $\psi \in C_0^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ such that $\psi = 1$ in a neighborhood of M_K, then part *(1)* in Remark 2.84 gives

$$\langle u * v, \varphi_j \rangle = \big\langle u \otimes v, \psi \varphi_j^\Delta \big\rangle, \qquad \forall\, j \in \mathbb{N}. \tag{2.8.12}$$

Moreover, we claim that

$$\psi\, \varphi_j^\Delta \xrightarrow[j\to\infty]{\mathcal{D}(\mathbb{R}^n \times \mathbb{R}^n)} 0. \tag{2.8.13}$$

To prove this claim, note that $\operatorname{supp}(\psi\, \varphi_j^\Delta) \subseteq \operatorname{supp}\psi$ for each $j \in \mathbb{N}$ and for each α_1, α_2, β_1, $\beta_2 \in \mathbb{N}_0^n$ we have

$$\sup_{(x,y)\in\operatorname{supp}\psi} \Big| (\partial_x^{\alpha_1} \partial_y^{\beta_1} \psi)(x,y) (\partial_x^{\alpha_2} \partial_y^{\beta_2} \varphi_j^\Delta)(x,y) \Big|$$

$$\leq \sup_{(x,y)\in \operatorname{supp}\psi} \left|\partial_x^{\alpha_1}\partial_y^{\beta_1}\psi(x,y)\right| \|\partial^{\alpha_2+\beta_2}\varphi_j\|_{L^\infty(\mathbb{R}^n)}$$

$$\leq \|\partial_x^{\alpha_1}\partial_y^{\beta_1}\psi\|_{L^\infty(\operatorname{supp}\psi)} \|\partial^{\alpha_2+\beta_2}\varphi_j\|_{L^\infty(K)} \xrightarrow[j\to\infty]{} 0. \tag{2.8.14}$$

Hence, (2.8.13) follows by combining (2.8.14) with Leibniz's formula (13.2.4). Since $u \otimes v \in \mathcal{D}'(\mathbb{R}^n \times \mathbb{R}^n)$, from (2.8.12) and (2.8.13) we deduce

$$\langle u * v, \varphi_j\rangle = \left\langle u \otimes v, \psi\varphi_j^{\Delta}\right\rangle \xrightarrow[j\to\infty]{} 0. \tag{2.8.15}$$

On account of Remark 2.3, this proves that $u * v \in \mathcal{D}'(\mathbb{R}^n)$. Finally, the last statement of the theorem is a consequence of what we proved so far and part *(3)* in Remark 2.84. □

Remark 2.86. *A combination of Theorem 2.85 and the discussion regarding* (2.8.6) *yields the following result. If $f, g \in L^1_{\rm loc}(\mathbb{R}^n)$ are such that for each compact $K \subset \mathbb{R}^n$ the set $\{(x,y) : x \in \operatorname{supp} f,\ y \in \operatorname{supp} g,\ x+y \in K\}$ is compact, then $u_f * u_g = u_{f*g}$ in $\mathcal{D}'(\mathbb{R}^n)$. That is, $f * g \in L^1_{\rm loc}(\mathbb{R}^n)$, the convolution between the distributions u_f and u_g [recall* (2.1.6)*] is well-defined, and $u_f * u_g$ is a distribution of function type that is equal to the distribution u_{f*g}. In particular,*

$$\left.\begin{array}{c} f \in L^1_{\rm loc}(\mathbb{R}^n),\ \ g \in L^1_{\rm comp}(\mathbb{R}^n) \\ \text{or} \\ f \in L^1_{\rm comp}(\mathbb{R}^n),\ \ g \in L^1_{\rm loc}(\mathbb{R}^n) \end{array}\right\} \Longrightarrow u_f * u_g = u_{f*g}. \tag{2.8.16}$$

The main properties of the convolution of distributions, whenever meaningfully defined, are stated and proved in the next theorem. Recall that for any $A, B \subseteq \mathbb{R}^n$ the set $A \pm B$ is defined as $\{x \pm y : x \in A,\ y \in B\}$.

Theorem 2.87. *The following statements are true.*

(a) If $u, v \in \mathcal{D}'(\mathbb{R}^n)$ are such that (2.8.9) *is satisfied, then*

$$\operatorname{supp}(u * v) \subseteq \operatorname{supp} u + \operatorname{supp} v. \tag{2.8.17}$$

(b) If $u, v \in \mathcal{D}'(\mathbb{R}^n)$ are such that (2.8.9) *is satisfied, then $u * v = v * u$.*

(c) If $u, v, w \in \mathcal{D}'(\mathbb{R}^n)$ are such that

$$\left\{\begin{array}{l} \text{for each compact subset } K \text{ of } \mathbb{R}^n \text{ the set} \\ M'_K := \{(x,y,z) \in \operatorname{supp} u \times \operatorname{supp} v \times \operatorname{supp} w : x+y+z \in K\} \\ \text{is compact in } \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n, \end{array}\right. \tag{2.8.18}$$

then $(u * v) * w$ and $u * (v * w)$ are well-defined, belong to $\mathcal{D}'(\mathbb{R}^n)$, and are equal.

(d) Let $u \in \mathcal{D}'(\mathbb{R}^n)$, $\alpha \in \mathbb{N}_0^n$. Then $(\partial^\alpha u) * \delta = \partial^\alpha u = u * \partial^\alpha \delta$. In particular, $u * \delta = u$.

(e) If the distributions $u, v \in \mathcal{D}'(\mathbb{R}^n)$ are such that (2.8.9) is satisfied and $\alpha \in \mathbb{N}_0^n$, then $\partial^\alpha(u * v) = (\partial^\alpha u) * v = u * (\partial^\alpha v)$.

Proof. Let $u, v \in \mathcal{D}'(\mathbb{R}^n)$ be such that (2.8.9) holds. Since $\operatorname{supp} u + \operatorname{supp} v$ is closed, the inclusion in *(a)* will follow as soon as we show that

$$u * v\big|_{\mathbb{R}^n \setminus (\operatorname{supp} u + \operatorname{supp} v)} = 0.$$

Pick some arbitrary function $\varphi \in \mathcal{D}(\mathbb{R}^n \setminus (\operatorname{supp} u + \operatorname{supp} v))$. Then necessarily $\operatorname{supp} \varphi^\Delta \cap \operatorname{supp} (u \otimes v) = \emptyset$, hence $\langle \widetilde{u \otimes v}, \varphi^\Delta \rangle = 0$ which, in turn, implies $\langle u * v, \varphi \rangle = 0$, as wanted. The statement in *(b)* is an immediate consequence of Definition 2.83 and the fact that the tensor product is commutative (see *(ii)* in Theorem 2.78).

To prove the statement in *(c)*, suppose $u, v, w \in \mathcal{D}'(\mathbb{R}^n)$ satisfy (2.8.18). Define the functional $u * v * w : \mathcal{D}(\mathbb{R}^n) \to \mathbb{C}$ by setting

$$\langle u * v * w, \varphi \rangle := \big\langle \widetilde{u \otimes v \otimes w}, \varphi^{\widetilde{\Delta}} \big\rangle, \qquad \text{for each } \varphi \in C_0^\infty(\mathbb{R}^n), \tag{2.8.19}$$

where $\varphi^{\widetilde{\Delta}}(x, y, z) := \varphi(x + y + z)$ for each $x, y, x \in \mathbb{R}^n$, and where $\widetilde{u \otimes v \otimes w}$ is the unique extension of $u \otimes v \otimes w$ obtained by applying Theorem 2.52 for the set $F := \operatorname{supp} (u \otimes v \otimes w)$. The mapping in (2.8.19) is well-defined since if $\varphi \in C_0^\infty(\mathbb{R}^n)$ then $\varphi^{\widetilde{\Delta}} \in C^\infty(\mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n)$ and, based on (2.8.18), the set

$$M'_{\operatorname{supp} \varphi} = \operatorname{supp} (u \otimes v \otimes w) \cap \operatorname{supp} \varphi^{\widetilde{\Delta}} \quad \text{is compact in } \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n. \tag{2.8.20}$$

Reasoning as in the proof of Theorem 2.85, it follows that $u * v * w \in \mathcal{D}'(\mathbb{R}^n)$ and

$$\begin{aligned} &\langle u * v * w, \varphi \rangle = \big\langle u \otimes v \otimes w, \psi \varphi^{\widetilde{\Delta}} \big\rangle \quad \text{for } \varphi \in C_0^\infty(\mathbb{R}^n) \text{ and for each} \\ &\psi \in C_0^\infty(\mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n) \text{ with } \psi = 1 \text{ in a neighborhood of } M'_{\operatorname{supp} \varphi}. \end{aligned} \tag{2.8.21}$$

Given the freedom in selecting ψ as in (2.8.21), we choose to take ψ as follows. Let $\pi_j : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$, $j = 1, 2, 3$, be the projections defined by $\pi_1\big((x, y, z)\big) := x$, $\pi_2\big((x, y, z)\big) := y$, $\pi_3\big((x, y, z)\big) := z$, for all $x, y, z \in \mathbb{R}^n$. Given $\varphi \in C_0^\infty(\mathbb{R}^n)$, fix

$$\psi_j \in C_0^\infty(\mathbb{R}^n) \text{ with } \psi_j = 1 \text{ near } \pi_j(M'_{\operatorname{supp} \varphi}), \quad j = 1, 2, 3, \tag{2.8.22}$$

then choose

$$\psi := \psi_1 \otimes \psi_2 \otimes \psi_3 \in C_0^\infty(\mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n). \tag{2.8.23}$$

The next two claims are designed to proving that $u * (v * w)$ exists.

Claim 1. For every compact K in $\mathbb{R}^n$ the set

$$N_K := \{(y, z) \in \operatorname{supp} v \times \operatorname{supp} w : \ y + z \in K\}$$

is compact in $\mathbb{R}^n \times \mathbb{R}^n$.

To see why this is true, start by observing that, for every $x_0 \in \operatorname{supp} u$, the set $K + x_0$ is compact in $\mathbb{R}^n$ and

$$B := \{(x, y, z) \in \{x_0\} \times \operatorname{supp} v \times \operatorname{supp} w : \ x + y + z \in K + x_0\} \tag{2.8.24}$$

is closed and contained in M'_{K+x_0}. Since (2.8.18) ensures that M'_{K+x_0} is compact in $\mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n$, it follows that B is compact in $\mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n$. In addition, the mapping $\theta : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}^n$, defined by $\theta(x, y, z) := (y, z)$ for $x, y, z \in \mathbb{R}^n$, is continuous and $\theta(B) = N_K$. Therefore, N_K must be compact, and Claim 1 is proved. The latter ensures that $v * w$ exists.

Claim 2. For every $K \subset \mathbb{R}^n$ compact, the set

$$P_K := \{(x, z) \in \operatorname{supp} u \times \operatorname{supp} (v * w) : \ x + z \in K\}$$

is compact in $\mathbb{R}^n \times \mathbb{R}^n$.

By part *(a)* in the theorem, we have $\operatorname{supp} (v * w) \subseteq \operatorname{supp} v + \operatorname{sup} w$. Thus

$$\begin{aligned} P_K &\subseteq \{(x, z) \in \operatorname{supp} u \times (\operatorname{supp} v + \operatorname{supp} w) : x + z \in K\} \\ &= \big\{(x, y + t) : \ x \in \operatorname{supp} u, \ y \in \operatorname{supp} v, \ t \in \operatorname{supp} w, \ x + y + t \in K\big\} \end{aligned} \tag{2.8.25}$$

and the last set in (2.8.25) is closed in $\mathbb{R}^n \times \mathbb{R}^n$. If we now set

$$\begin{aligned} &\sigma : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}^n, \\ &\sigma(x, y, t) := (x, y + t) \quad \text{for every } x, y, t \in \mathbb{R}^n, \end{aligned} \tag{2.8.26}$$

then σ is continuous and $P_K \subseteq \sigma(M'_K)$. By (2.8.18), the set M'_K is compact in $\mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n$, hence $\sigma(M'_K)$ is compact in $\mathbb{R}^n \times \mathbb{R}^n$. The set P_K being closed in $\mathbb{R}^n \times \mathbb{R}^n$, we may conclude that P_K is compact in $\mathbb{R}^n \times \mathbb{R}^n$. This proves Claim 2 and, as a consequence, the fact that $u * (v * w)$ exists.

With an eye toward proving $u * (v * w) = u * v * w$, we dispense with two more claims.

Claim 3. For each $K \subset \mathbb{R}^n$ compact, $A := (\operatorname{supp} v + \operatorname{supp} w) \cap (K - \operatorname{supp} u)$ is a compact set in $\mathbb{R}^n$.

Rewrite A as

$$A = \{t \in \operatorname{supp} v + \operatorname{supp} w : \ t = \omega - x, \ \text{ for some } \omega \in K, \ x \in \operatorname{supp} u\}.$$

Then if $t \in A$, it follows that there exist $y \in \operatorname{supp} v$, $z \in \operatorname{supp} w$, $\omega \in K$ and $x \in \operatorname{supp} u$ such that $t = y + z = \omega - x$. Hence, $(x, y, z) \in \operatorname{supp} u \times \operatorname{supp} v \times \operatorname{supp} w$ and $x + y + z = \omega$, which implies that $(x, y, z) \in M'_K$ and $A \subseteq \nu(M'_K)$, where

$$\begin{aligned} &\nu : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n, \\ &\nu(x_1, x_2, x_3) := x_2 + x_3 \quad \text{for every } x_1, x_2, x_3 \in \mathbb{R}^n. \end{aligned} \tag{2.8.27}$$

We may now conclude that A is compact since the map ν is continuous, M'_K is compact and A is closed. This proves Claim 3.

Claim 4. Fix $\varphi \in C_0^\infty(\mathbb{R}^n)$ and set $K := \operatorname{supp}\varphi$. Also, let A be as in Claim 3 corresponding to this K, and suppose $\eta \in C_0^\infty(\mathbb{R}^n)$ is such that $\eta = 1$ in a neighborhood of A. Then, with ψ_1 as in (2.8.22), we have

$$\psi_1 \otimes \eta = 1 \quad \text{on} \quad (\operatorname{supp} u \times \operatorname{supp}(v * w)) \cap \operatorname{supp} \varphi^\Delta \tag{2.8.28}$$

where, as before, $\varphi^\Delta(x, y) = \varphi(x + y)$ for $x, y \in \mathbb{R}^n$.

To prove this claim, since $\psi_1 = 1$ on $\pi_1(M'_K)$ and $\eta = 1$ on A, it suffices to show that

$$\big(\operatorname{supp} u \times \operatorname{supp}(v * w)\big) \cap \operatorname{supp} \varphi^\Delta \subseteq \pi_1(M'_K) \times A. \tag{2.8.29}$$

To justify the latter inclusion, let $x \in \operatorname{supp} u$, $y \in \operatorname{supp} v$, and $z \in \operatorname{supp} w$, be such that $(x, y + z) \in \operatorname{supp} \varphi^\Delta$. Then $x + y + z \in K$ which forces $x \in \pi_1(M'_K)$ as well as $y + z \in K - \operatorname{supp} u \subseteq A$. Thus, $(x, y + z) \in \pi_1(M'_K) \times A$ and this completes the proof of Claim 4.

Consider now an arbitrary function $\varphi \in C_0^\infty(\mathbb{R}^n)$ and set $K := \operatorname{supp}\varphi$. Also, assume that ψ_1, ψ_2, ψ_3 are as in (2.8.22), and let η be as in Claim 4. Making use of the definition of the convolution and tensor products, and keeping in mind (2.8.28), we may write

$$\begin{aligned}
\big\langle u * (v * w), \varphi \big\rangle &= \Big\langle u(x) \otimes (v * w)(t), \psi_1(x)\eta(t)\varphi(x + t) \Big\rangle \\
&= \Big\langle u(x), \big\langle (v * w)(t), \psi_1(x)\eta(t)\varphi(x + t) \big\rangle \Big\rangle \\
&= \Big\langle u(x), \big\langle v(y) \otimes w(z), \psi_1(x)\eta(y + z)\varphi(x + y + z)\psi_2(y)\psi_3(z) \big\rangle \Big\rangle.
\end{aligned} \tag{2.8.30}$$

A few words explaining the origin of the last equality are in order. According to the definition of the convolution, passing from $v * w$ to $v \otimes w$ requires that we consider the set

$$C := (\operatorname{supp} v \times \operatorname{supp} w) \cap \operatorname{supp} \eta^\Delta \cap \operatorname{supp}[\varphi(x + \cdot)]^\Delta. \tag{2.8.31}$$

Since C is closed and satisfies $C \subset \pi_2(M'_K) \times \pi_3(M'_K)$, it follows that C is compact. Now, the fact that $\psi_2 \otimes \psi_3 = 1$ in a neighborhood of C justifies the presence of $\psi_2 \otimes \psi_3$ in the last term in (2.8.30).

Going further, since

$$\psi_1(x)\psi_2(y)\psi_3(z) = \psi_1(x)\psi_2(y)\psi_3(z)\eta(y + z) \quad \text{for } (x, y, z) \text{ near } M'_K, \tag{2.8.32}$$

referring to (2.8.21) and (2.8.23) allows us to rewrite (2.8.30) in the form

$$\begin{aligned}
\langle u * (v * w), \varphi \rangle &= \Big\langle u(x), \big\langle v(y) \otimes w(z), \psi_1(x)\psi_2(y)\psi_3(z)\varphi(x + y + z) \big\rangle \Big\rangle \\
&= \Big\langle u \otimes v \otimes w, (\psi_1 \otimes \psi_2 \otimes \psi_3)\varphi^{\widetilde{\Delta}} \Big\rangle = \langle u * v * w, \varphi \rangle.
\end{aligned} \tag{2.8.33}$$

Since $\varphi \in C_0^\infty(\mathbb{R}^n)$ was arbitrary, it follows that $u*(v*w) = u*v*w$. Similarly, it can be seen that $(u*v)*w = u*v*w$ and this completes the proof of the statement in *(c)*.

Moving on to *(d)*, fix $u \in \mathcal{D}'(\mathbb{R}^n)$ and $\alpha \in \mathbb{N}_0^n$. Since the Dirac distribution δ has compact support, by Theorem 2.85 it follows that both $\delta * u$ and $(\partial^\alpha \delta) * u$ are well-defined and belong to $\mathcal{D}'(\mathbb{R}^n)$. Let $\varphi \in C_0^\infty(\mathbb{R}^n)$ be arbitrary and consider $\psi \in C_0^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ such that $\psi = 1$ on a neighborhood of the set $(\{0\} \times \operatorname{supp} u) \cap \operatorname{supp} \varphi^\Delta$. Starting with Definition 2.83, then using (2.4.1) combined with Proposition 2.32, then Leibniz's formula (13.2.4), then (2.1.18), the support condition for ψ and then (2.4.1) again, we may write

$$\begin{aligned}
\big\langle \partial^\alpha \delta * u, \varphi \big\rangle &= \big\langle (\partial_x^\alpha \delta) \otimes u(y), \psi(x,y)\varphi(x+y) \big\rangle \\
&= \Big\langle u(y), \big\langle \partial^\alpha \delta(x), \psi(x,y)\varphi(x+y) \big\rangle \Big\rangle \\
&= (-1)^{|\alpha|} \Big\langle u(y), \big\langle \delta(x), \partial_x^\alpha (\psi(x,y)\varphi(x+y)) \big\rangle \Big\rangle \\
&= (-1)^{|\alpha|} \Big\langle u(y), \Big\langle \delta(x), \sum_{\beta \le \alpha} \tfrac{\alpha!}{\beta!(\alpha-\beta)!} \partial_x^\beta \psi(x,y) (\partial_x^{\alpha-\beta}\varphi)(x+y) \Big\rangle \Big\rangle \\
&= (-1)^{|\alpha|} \Big\langle u(y)\,, \sum_{\beta \le \alpha} \tfrac{\alpha!}{\beta!(\alpha-\beta)!} (\partial_x^\beta \psi)(0,y) (\partial_x^{\alpha-\beta}\varphi)(y) \Big\rangle \\
&= (-1)^{|\alpha|} \big\langle u(y), \psi(0,y) \partial^\alpha \varphi(y) \big\rangle \\
&= (-1)^{|\alpha|} \big\langle u, \partial^\alpha \varphi \big\rangle = \big\langle \partial^\alpha u, \varphi \big\rangle. \qquad (2.8.34)
\end{aligned}$$

In particular, if $|\alpha| = 0$, the above implies $\delta * u = u$. When combined with *(b)*, this finishes the proof of the statement in *(d)*.

Finally, by making use of the results from *(d)* and *(c)* we have

$$\partial^\alpha (u*v) = \partial^\alpha(\delta * (u*v)) = \partial^\alpha \delta * (u*v) = (\partial^\alpha \delta * u) * v = (\partial^\alpha u) * v. \qquad (2.8.35)$$

A similar argument also shows that $\partial^\alpha(u*v) = u * (\partial^\alpha v)$. The proof of the theorem is now complete. □

Exercise 2.88. *Prove that for a distribution $u \in \mathcal{D}'(\mathbb{R}^n)$,*

$$u = \delta \iff u * f = f \quad \text{for each} \quad f \in C_0^\infty(\mathbb{R}^n). \qquad (2.8.36)$$

Hint: For the right-to-left implication use $f = \phi_j$, where ϕ_j is as in Example 2.20, and let $j \to \infty$.

Next, we extend the translation map (1.3.16) to distributions.

Proposition 2.89. *For each $x_0 \in \mathbb{R}^n$ and each $u \in \mathcal{D}'(\mathbb{R}^n)$ fixed, the translation mapping $\mathcal{D}(\mathbb{R}^n) \ni \varphi \mapsto \langle u, t_{-x_0}(\varphi)\rangle \in \mathbb{C}$ is linear and continuous.*

Denoting this map by $t_{x_0}u$ thus yields a distribution in $\mathbb{R}^n$ that satisfies

$$\langle t_{x_0}u, \varphi\rangle = \langle u, t_{-x_0}(\varphi)\rangle, \qquad \forall\, \varphi \in C_0^\infty(\mathbb{R}^n). \qquad (2.8.37)$$

Proof. This follows by observing that the mapping in question is the composition $u \circ t_{-x_0}$ where the latter translation operator is consider in the sense of Exercise 1.19. □

Exercise 2.90. *Fix $x_0 \in \mathbb{R}^n$ and recall the distribution δ_{x_0} from Example 2.14. Prove that $\delta_{x_0} = t_{x_0}\delta$ in $\mathcal{D}'(\mathbb{R}^n)$. Also show that $\delta_{x_0} * u = t_{x_0}u$ for every $u \in \mathcal{D}'(\mathbb{R}^n)$. In particular, if $x_1 \in \mathbb{R}^n$ is arbitrary, then $\delta_{x_0} * \delta_{x_1} = \delta_{x_0+x_1}$ in $\mathcal{D}'(\mathbb{R}^n)$.*

Remark 2.91.

(1) If $u, v \in \mathcal{E}'(\mathbb{R}^n)$ then $u * v \in \mathcal{E}'(\mathbb{R}^n)$. This is an immediate consequence of (2.8.17).

(2) There exists a sequence $\{u_j\}_{j\in\mathbb{N}}$ of compactly supported distributions in $\mathbb{R}$ that converges to some $u \in \mathcal{D}'(\mathbb{R})$ and such that $u_j * v$ does not necessarily converge to $u * v$ in $\mathcal{D}'(\mathbb{R})$ for each $v \in \mathcal{D}'(\mathbb{R})$.

To see an example in this regard, consider the sequence of compactly supported distributions $\{\delta_j\}_{j\in\mathbb{N}}$ that satisfies $\delta_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} 0$. Then if 1 denotes the distribution on $\mathbb{R}$ given by the constant function 1, Exercise 2.90 gives $\delta_j * 1 = 1$ for each j, and the constant sequence $\{1\}_{j\in\mathbb{N}} \subset \mathcal{D}'(\mathbb{R})$ does not converge in $\mathcal{D}'(\mathbb{R})$ to $0 * 1 = 0$. This shows that sequential continuity for convolution of distributions cannot be expected in general.

(3) Condition (2.8.18) is necessary for the operation of convolution of distributions to be associative. To see this, consider the distributions 1, δ', and H on $\mathbb{R}$. Then we have $\operatorname{supp}\delta' = \{0\}$, $\operatorname{supp} 1 = \mathbb{R}$, $\operatorname{supp} H = [0,\infty)$. If K is a compact set in $\mathbb{R}^n$, the set

$$M'_K = \big\{(x,0,z) : x \in \mathbb{R},\, z \in [0,\infty),\, x+z \in K\big\} \tag{2.8.38}$$

is not compact in $\mathbb{R}\times\mathbb{R}\times\mathbb{R}$, thus (2.8.18) does not hold. Furthermore, $1 * \delta' = 1' * \delta = 0 * \delta = 0$ so $(1 * \delta') * H = 0$, while $1 * (\delta' * H) = 1 * (\delta * H') = 1 * (\delta * \delta) = 1 * \delta = 1$ and clearly $(1 * \delta') * H \neq 1 * (\delta' * H)$ in $\mathcal{D}'(\mathbb{R})$.

Proposition 2.92. *The following statements are true.*

(1) If $u \in \mathcal{E}'(\mathbb{R}^n)$ and $v_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} v$, then $u * v_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} u * v$.

(2) If $u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} u$ and there exists $K \subset \mathbb{R}^n$ compact with $\operatorname{supp} u_j \subseteq K$ for every $j \in \mathbb{N}$, then $u_j * v \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} u * v$ for every $v \in \mathcal{D}'(\mathbb{R}^n)$.

Proof. To see why *(1)* is true, fix $\varphi \in C_0^\infty(\mathbb{R}^n)$. Then by definition, for each $j \in \mathbb{N}$ we have $\langle u * v_j, \varphi\rangle = \langle u \otimes v_j, \psi_j\varphi^\Delta\rangle$ for any smooth compactly supported

function ψ_j with $\psi_j = 1$ on a neighborhood of $(\operatorname{supp} u \times \operatorname{supp} v_j) \cap \operatorname{supp} \varphi^\Delta$. Note that

$$(\operatorname{supp} u \times \operatorname{supp} v_j) \cap \operatorname{supp} \varphi^\Delta \subseteq \operatorname{supp} u \times (\operatorname{supp} \varphi - \operatorname{supp} u) \tag{2.8.39}$$

and $\operatorname{supp} u \times (\operatorname{supp} \varphi - \operatorname{supp} u)$ is compact since both $\operatorname{supp} u$ and $\operatorname{supp} \varphi$ are compact. Hence, if we fix $\psi \in C_0^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ such that $\psi = 1$ in a neighborhood of the set $\operatorname{supp} u \times (\operatorname{supp} \varphi - \operatorname{supp} u)$, then $\langle u * v_j, \varphi\rangle = \langle u \otimes v_j, \psi\varphi^\Delta\rangle$ for every $j \in \mathbb{N}$. Based on *(d)* in Theorem 2.80, we may write

$$\langle u * v_j, \varphi\rangle = \langle u \otimes v_j, \psi\varphi^\Delta\rangle \xrightarrow[j\to\infty]{} \langle u \otimes v, \psi\varphi^\Delta\rangle = \langle u * v, \varphi\rangle, \tag{2.8.40}$$

and the desired conclusion follows.

Assume next the hypotheses in part *(2)* of the proposition. In particular, these entail $\operatorname{supp} u \subseteq K$. Let $\varphi \in C_0^\infty(\mathbb{R}^n)$ and note that

$$(\operatorname{supp} u_j \times \operatorname{supp} v) \cap \operatorname{supp} \varphi^\Delta \subseteq K \times (\operatorname{supp} \varphi - K), \quad \forall\, j \in \mathbb{N}. \tag{2.8.41}$$

Then if $\psi \in C_0^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ is a function with the property that $\psi = 1$ in a neighborhood of $K \times (\operatorname{supp} \varphi - K)$, we have $\langle u_j * v, \varphi\rangle = \langle u_j \otimes v, \psi\varphi^\Delta\rangle$ for every $j \in \mathbb{N}$. Hence,

$$\langle u_j * v, \varphi\rangle = \langle u_j \otimes v, \psi\varphi^\Delta\rangle \xrightarrow[j\to\infty]{} \langle u \otimes v, \psi\varphi^\Delta\rangle = \langle u * v, \varphi\rangle, \tag{2.8.42}$$

where for the convergence in (2.8.42) we used part *(d)* in Theorem 2.80. □

When convolving an arbitrary distribution with a distribution of function type given by a compactly supported smooth function, the resulting distribution is of function type. This fact is particularly useful in applications and we prove it next.

Proposition 2.93. *If $u \in \mathcal{D}'(\mathbb{R}^n)$ and $g \in C_0^\infty(\mathbb{R}^n)$, then the distribution $u * g$ is of function type given by the function*

$$f : \mathbb{R}^n \to \mathbb{C}, \qquad f(x) := \langle u(y), g(x-y)\rangle, \qquad \forall\, x \in \mathbb{R}^n, \tag{2.8.43}$$

that satisfies $f \in C^\infty(\mathbb{R}^n)$. Moreover, if u is compactly supported then so is f. In short,

$$\mathcal{D}'(\mathbb{R}^n) * C_0^\infty(\mathbb{R}^n) \subseteq C^\infty(\mathbb{R}^n) \quad \text{and} \quad \mathcal{E}'(\mathbb{R}^n) * C_0^\infty(\mathbb{R}^n) \subseteq C_0^\infty(\mathbb{R}^n). \tag{2.8.44}$$

Proof. Let $\phi : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{C}$ be defined by $\phi(x,y) := g(x-y)$ for each $(x,y) \in \mathbb{R}^n \times \mathbb{R}^n$. Then $\phi \in C^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ and $\phi(x,\cdot) \in C_0^\infty(\mathbb{R}^n)$ for each $x \in \mathbb{R}^n$. This shows that the function f in (2.8.43) is well-defined.

To prove that f is of class C^∞ in $\mathbb{R}^n$, pick an arbitrary point $x^* \in \mathbb{R}^n$ and pick a function $\psi \in C_0^\infty(\mathbb{R}^n)$ with the property that $\psi = 1$ in a neighborhood of $\overline{B(x^*,1)}$. In addition, select $\eta \in C_0^\infty(\mathbb{R}^n)$ such that $\eta = 1$ near $\overline{B(x^*,1)} - \operatorname{supp} g$. Then

$$\begin{aligned}\Big(f\big|_{B(x^*,1)}\Big)(x) &= \big\langle u(y), g(x-y)\big\rangle = \big\langle u(y), \psi(x)g(x-y)\big\rangle \\ &= \big\langle (\eta u)(y), \psi(x)g(x-y)\big\rangle, \qquad \forall\, x \in B(x^*,1).\end{aligned} \tag{2.8.45}$$

Since $\eta u \in \mathcal{E}'(\mathbb{R}^n)$ and the function $\mathbb{R}^n \times \mathbb{R}^n \ni (x,y) \mapsto \psi(x)g(x-y) \in \mathbb{C}$ is of class C^∞, we may further invoke Proposition 2.82 to conclude that $f\big|_{B(x^*,1)} \in C^\infty\big(B(x^*,1)\big)$. Given that $x^* \in \mathbb{R}^n$ has been arbitrarily chosen, it follows that $f \in C^\infty(\mathbb{R}^n)$.

We now turn to the task of showing that the distribution $u * g$ is of function type and is given by f. To this end, fix $\varphi \in C_0^\infty(\mathbb{R}^n)$ and consider a function

$$\begin{aligned} &\psi \in C_0^\infty(\mathbb{R}^n \times \mathbb{R}^n) \ \text{ such that } \ \psi = 1 \text{ in a neighborhood of} \\ &\text{the set } \ \big\{(x,y) : \, x \in \operatorname{supp} u, \, y \in \operatorname{supp} g, \, x+y \in \operatorname{supp} \varphi\big\}. \end{aligned} \tag{2.8.46}$$

Then, starting with Definition 2.83 we write

$$\begin{aligned} \langle u * g, \varphi\rangle &= \big\langle u \otimes g, \psi\varphi^\Delta\big\rangle \\ &= \Big\langle u(x), \big\langle g(y), \psi(x,y)\varphi(x+y)\big\rangle\Big\rangle \\ &= \Big\langle u(x)\,, \int_{\mathbb{R}^n} g(y)\psi(x,y)\varphi(x+y)\,\mathrm{d}y\Big\rangle \\ &= \Big\langle u(x)\,, \int_{\mathbb{R}^n} g(y)\varphi(x+y)\,\mathrm{d}y\Big\rangle \\ &= \Big\langle u(x)\,, \int_{\mathbb{R}^n} g(z-x)\varphi(z)\,\mathrm{d}z\Big\rangle \\ &= \int_{\mathbb{R}^n} \langle u(x), g(z-x)\rangle\varphi(z)\,\mathrm{d}z \\ &= \langle f, \varphi\rangle. \end{aligned} \tag{2.8.47}$$

For the third equality in (2.8.47) we have used the fact that g is a distribution of function type, for the forth equality we used condition (2.8.46), the fifth equality is based on a change of variables, the sixth equality follows from (2.7.45), while the last equality is a consequence of the definition of f.

To complete the proof of the proposition there remains to notice that, by part (2.8.17), we have $\operatorname{supp} f \subseteq \operatorname{supp} u + \operatorname{supp} g$. In particular, if u is compactly supported, then so is f. □

Exercise 2.94. *Prove that if $u \in \mathcal{E}'(\mathbb{R}^n)$ and $g \in C^\infty(\mathbb{R}^n)$ then the distribution $u * g$ is of function type given by the function*

$$f : \mathbb{R}^n \to \mathbb{C}, \qquad f(x) := \langle u(y), g(x-y)\rangle, \qquad \forall\, x \in \mathbb{R}^n, \tag{2.8.48}$$

that satisfies $f \in C^\infty(\mathbb{R}^n)$. In short, $\mathcal{E}'(\mathbb{R}^n) * C^\infty(\mathbb{R}^n) \subseteq C^\infty(\mathbb{R}^n)$.

Hint: Use Proposition 2.82 to show that $f \in C^\infty(\mathbb{R}^n)$, then reason as in Proposition 2.93 to take care of the remaining claims.

Exercise 2.95. *Let Ω be a bounded open set in $\mathbb{R}^n$. Suppose $u \in L^1(\Omega)$ and define $\widetilde{u}$ to be the extension by zero outside Ω of u. Also assume that a function $g \in L^1_{\rm loc}(\mathbb{R}^n)$ is given. Prove that $\widetilde{u} \in \mathcal{D}'(\mathbb{R}^n)$, it satisfies $\operatorname{supp} u \subseteq \overline{\Omega}$, hence $\widetilde{u} * g$ is well-defined in $\mathcal{D}'(\mathbb{R}^n)$, and that the distribution $\widetilde{u} * g$ is of function type given by the function*

$$(\widetilde{u} * g)(x) = \int_{\Omega} g(x-y)u(y)\,{\rm d}y, \qquad \forall\, x \in \mathbb{R}^n. \tag{2.8.49}$$

Hint: Apply (2.8.16) with $f = \widetilde{u}$.

Exercise 2.96. *Prove that if $u \in \mathcal{D}'(\mathbb{R}^n)$ and $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} \varphi$, then one has $u * \varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} u * \varphi$.*

Next, we prove that distributions of function type given by smooth, compactly supported functions are dense in the class of distributions. First we treat the simpler case when the distributions are considered in $\mathbb{R}^n$. The case when the distributions are considered on arbitrary open subsets of $\mathbb{R}^n$ needs to be handled with a bit more care.

Theorem 2.97. *The set $C^\infty_0(\mathbb{R}^n)$ is sequentially dense in $\mathcal{D}'(\mathbb{R}^n)$.*

Proof. First we will show that $C^\infty(\mathbb{R}^n)$ is sequentially dense in $\mathcal{D}'(\mathbb{R}^n)$. Let ϕ be as in (1.2.3) and recall the sequence of functions $\{\phi_j\}_{j\in\mathbb{N}}$ from (1.3.7). In particular, we have that $\operatorname{supp}\phi_j \subseteq \overline{B(0,1)}$ for every $j \in \mathbb{N}$. Recall from Example 2.20 that $\phi_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} \delta$.

Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be arbitrary and define

$$u_j := u * \phi_j \quad \text{in} \quad \mathcal{D}'(\mathbb{R}^n), \qquad \forall\, j \in \mathbb{N}. \tag{2.8.50}$$

By Proposition 2.93 we have $u_j \in C^\infty(\mathbb{R}^n)$ for all $j \in \mathbb{N}$. Also, by part *(2)* in Proposition 2.92 and part *(d)* in Theorem 2.87 we obtain

$$u_j = u * \phi_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} u * \delta = u. \tag{2.8.51}$$

This completes the proof of the fact that $C^\infty(\mathbb{R}^n)$ is sequentially dense in $\mathcal{D}'(\mathbb{R}^n)$.

Moving on, and keeping the notation introduced so far, fix $\psi \in C^\infty_0(\mathbb{R}^n)$ satisfying $\psi(x) = 1$ if $|x| < 1$ and set

$$w_j(x) := \psi(x/j)(u * \phi_j)(x) \qquad \forall\, x \in \mathbb{R}^n, \quad \forall\, j \in \mathbb{N}. \tag{2.8.52}$$

It is immediate that $w_j \in C^\infty_0(\mathbb{R}^n)$. Moreover, if $\varphi \in C^\infty_0(\mathbb{R}^n)$ is given, then there exists $j_0 \in \mathbb{N}$ with the property that $\operatorname{supp}\varphi \subset B(0, j_0)$. Therefore, for all $j \geq j_0$ we may write

$$\langle w_j,\varphi\rangle=\int_{\mathbb{R}^n}\psi\Big(\frac{x}{j}\Big)(u*\phi_j)(x)\varphi(x)\,dx=\int_{\mathbb{R}^n}(u*\phi_j)(x)\varphi(x)\,dx$$
$$=\langle u*\phi_j,\varphi\rangle\xrightarrow[j\to\infty]{}\langle u,\varphi\rangle. \tag{2.8.53}$$

This shows that $w_j\xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)}u$. Hence, ultimately, $C^\infty(\mathbb{R}^n)$ is sequentially dense in $\mathcal{D}'(\mathbb{R}^n)$, finishing the proof of the theorem. □

The same type of result as in Theorem 2.97 actually holds in arbitrary open subsets of the Euclidean ambient.

Theorem 2.98. *The set $C_0^\infty(\Omega)$ is sequentially dense in $\mathcal{D}'(\Omega)$.*

Proof. Fix $u\in\mathcal{D}'(\Omega)$ and recall the sequence of compact sets introduced in (1.2.13). Then $\bigcup_{j\in\mathbb{N}}K_j=\Omega$ and $K_j\subset\mathring{K}_{j+1}$ for every $j\in\mathbb{N}$. For each $j\geq 2$ consider a function

$$\psi_j\in C_0^\infty(\Omega),\quad \psi_j=1\text{ on a neighborhood of }K_{j-1},\quad \operatorname{supp}\psi_j\subseteq K_j, \tag{2.8.54}$$

and define $u_j:=\psi_j u\in\mathcal{D}'(\Omega)$. Since $\operatorname{supp}u_j\subseteq K_j$, Proposition 2.63 states that each u_j may be extended to a distribution in $\mathcal{E}'(\mathbb{R}^n)$, which we continue to denote by u_j. If we now set

$$\varepsilon_j:=\frac{1}{4}\operatorname{dist}(K_j,\partial K_{j+1})>0,\qquad\forall\,j\in\mathbb{N}, \tag{2.8.55}$$

then the definition of the compacts in (1.2.13) forces $\varepsilon_j\searrow 0$ as $j\to\infty$. Having fixed some $\phi\in C_0^\infty(\mathbb{R}^n)$ satisfying $\operatorname{supp}\phi\subseteq\overline{B(0,1)}$ and $\int_{\mathbb{R}^n}\phi(x)\,dx=1$, define

$$\phi_j(x):=\varepsilon_j^{-n}\phi(x/\varepsilon_j),\qquad\forall\,x\in\mathbb{R}^n,\quad\forall\,j\in\mathbb{N}. \tag{2.8.56}$$

Thus,

$$\phi_j\in C_0^\infty(\mathbb{R}^n),\quad\operatorname{supp}\phi_j\subseteq\overline{B(0,\varepsilon_j)}\subseteq\overline{B(0,1)},\qquad\forall\,j\in\mathbb{N}, \tag{2.8.57}$$

and reasoning as in Example 2.20 we see that

$$\phi_j\xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)}\delta. \tag{2.8.58}$$

Finally, for each $j\geq 2$ introduce $w_j:=u_j*\phi_j$. By combining part *(1)* in Remark 2.91 with Proposition 2.93 we obtain that $w_j\in C_0^\infty(\mathbb{R}^n)$ for every $j\geq 2$. In addition, (2.8.17) and (2.8.55) imply that $\operatorname{supp}w_j\subseteq K_j+\overline{B(0,\varepsilon_j)}\subset K_{j+1}$, so in fact we have $w_j\in C_0^\infty(\Omega)$.

To complete the proof of the theorem it suffices to show that $w_j\xrightarrow[j\to\infty]{\mathcal{D}'(\Omega)}u$. To this end, fix $\varphi\in C_0^\infty(\Omega)$ and observe that based on (1.2.13) there exists $j_0\in\mathbb{N}$ such that $\operatorname{supp}\varphi\subseteq K_{j_0}$. Then for $j>j_0+1$ we may write

$$\langle w_j,\varphi\rangle=\langle u_j*\phi_j,\varphi\rangle=\int_\Omega(u_j*\phi_j)(x)\varphi(x)\,dx$$

$$= \int_\Omega \langle u_j(y), \phi_j(x-y)\rangle \varphi(x)\,\mathrm{d}x = \int_\Omega \langle u(y), \psi_j(y)\phi_j(x-y)\rangle \varphi(x)\,\mathrm{d}x$$

$$= \int_\Omega \langle u(y), \psi_{j_0+2}(y)\phi_j(x-y)\rangle \varphi(x)\,\mathrm{d}x = \langle u_{j_0+2} * \phi_j, \varphi\rangle. \tag{2.8.59}$$

For the second equality in (2.8.59) we used the fact that $u_j * \phi_j \in C_0^\infty(\mathbb{R}^n)$ for $j \geq 2$, for the third equality we used Proposition 2.93, while for the forth the fact that $u_j = \psi_j u$ for every $j \in \mathbb{N}$. These observations also give the last equality in (2.8.59). As for the penultimate equality in (2.8.59), observe that if $j > j_0 + 1$, $x \in \operatorname{supp}\varphi \subseteq K_{j_0}$ and $x - y \in \operatorname{supp}\phi_j \subseteq \overline{B(0,\varepsilon_j)}$, then $y \in K_{j_0} - \overline{B(0,\varepsilon_j)} \subseteq K_{j_0} - \overline{B(0,\varepsilon_{j_0})} \subset K_{j_0+1}$, thus $\psi_j(y) = 1 = \psi_{j_0+2}(y)$.

If we now combine (2.8.58) with (2.8.57) and part *(2)* in Proposition 2.92, it follows that $u_{j_0+2} * \phi_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} u_{j_0+2} * \delta = u_{j_0+2}$. Hence, if we return with this in (2.8.59), we may write

$$\lim_{j\to\infty} \langle w_j, \varphi\rangle = \lim_{j\to\infty} \langle u_{j_0+2} * \phi_j, \varphi\rangle = \langle u_{j_0+2}, \varphi\rangle = \langle \psi_{j_0+2} u, \varphi\rangle = \langle u, \varphi\rangle, \tag{2.8.60}$$

since $\psi_{j_0+2} = 1$ in a neighborhood of the support of φ. This finishes the proof of the theorem. □

2.9 Distributions with Higher-Order Gradients Continuous or Bounded

We have seen that if $u \in C^m(\mathbb{R}^n)$ for some $m \in \mathbb{N}$, then its distributional derivatives up to order m are distributions of function type, each given by the corresponding pointwise derivative of u. A more subtle question pertains to the possibility of deducing regularity results for distributions with distributional derivatives of a certain order that are of function type, and these functions exhibit a certain amount of smoothness. In this section we prove two main results in this regard. In the first result (see Theorem 2.102), we show that if a distribution $u \in \mathcal{D}'(\Omega)$ has all distributional derivatives of order $m \in \mathbb{N}$ continuous, then in fact $u \in C^m(\Omega)$. In the second main result (see Theorem 2.104) we prove that a distribution in $\mathcal{D}'(\mathbb{R}^n)$ is of function type given by a Lipschitz function if and only if all its first-order distributional derivatives are bounded functions in $\mathbb{R}^n$.

We start by proving a weaker version of Theorem 2.102.

Proposition 2.99. *If $u \in \mathcal{D}'(\mathbb{R}^n)$ and there exists $m \in \mathbb{N}_0$ such that for each multi-index $\alpha \in \mathbb{N}_0^n$ satisfying $|\alpha| \leq m$ the distributional derivative $\partial^\alpha u$ belongs to $C^0(\mathbb{R}^n)$, then $u \in C^m(\mathbb{R}^n)$.*

Proof. Recall the sequence of distributions $\{\phi_j\}_{j\in\mathbb{N}}$ from Example 2.20 and for u satisfying the hypothesis of the proposition let $\{u_j\}_{j\in\mathbb{N}}$ be as in (2.8.50). In particular, (2.8.51) holds, thus the distributional and classical derivatives of each u_j coincide.

Next we make the following claim:

$$\begin{aligned}&\lim_{j\to\infty}\partial^\alpha u_j=\partial^\alpha u\quad\text{uniformly on compact sets in }\mathbb{R}^n\\&\text{for every multi-index }\alpha\in\mathbb{N}_0^n\text{ satisfying }|\alpha|\leq m.\end{aligned}\tag{2.9.1}$$

To prove this claim, observe that by part *(e)* in Theorem 2.87, for each $j\in\mathbb{N}$ we have $\partial^\alpha u_j=(\partial^\alpha u)*\phi_j$ for every $\alpha\in\mathbb{N}_0^n$. Fix $\alpha\in\mathbb{N}_0^n$ satisfying $|\alpha|\leq m$. Since by the current hypotheses the distributional derivative $\partial^\alpha u$ is continuous, by invoking Proposition 2.93 we may write

$$\begin{aligned}\partial^\alpha u_j(x)&=((\partial^\alpha u)*\phi_j)(x)=\langle\partial^\alpha u(y),\phi_j(x-y)\rangle\\&=\int_{\mathbb{R}^n}(\partial^\alpha u)(y)\phi_j(x-y)\,\mathrm{d}y=\int_{\mathbb{R}^n}(\partial^\alpha u)(x-z)\phi_j(z)\,\mathrm{d}z\\&=\int_{\mathbb{R}^n}(\partial^\alpha u)\big(x-y/j\big)\phi(y)\,\mathrm{d}y,\qquad\forall\,x\in\mathbb{R}^n.\end{aligned}\tag{2.9.2}$$

Fix a compact set K in $\mathbb{R}^n$. Making use of (2.9.2) and the properties of ϕ (recall (1.2.3)) we estimate

$$\begin{aligned}\sup_{x\in K}|\partial^\alpha u_j(x)-\partial^\alpha u(x)|&=\sup_{x\in K}\left|\int_{\mathbb{R}^n}[(\partial^\alpha u)\big(x-y/j\big)-(\partial^\alpha u)(x)]\phi(y)\,\mathrm{d}y\right|\\&\leq\sup_{\substack{x\in K\\ y\in\overline{B(0,1)}}}\Big|(\partial^\alpha u)\big(x-y/j\big)-(\partial^\alpha u)(x)\Big|.\end{aligned}\tag{2.9.3}$$

Since $\partial^\alpha u$ is continuous in $\mathbb{R}^n$ it follows that $\partial^\alpha u$ is uniformly continuous on compact subsets of $\mathbb{R}^n$, thus

$$\lim_{j\to\infty}\sup_{x\in K}|\partial^\alpha u_j(x)-\partial^\alpha u(x)|\leq\lim_{j\to\infty}\sup_{\substack{x\in K\\ y\in\overline{B(0,1)}}}\Big|(\partial^\alpha u)\big(x-z/j\big)-(\partial^\alpha u)(x)\Big|=0,\tag{2.9.4}$$

completing the proof of the claim.

With (2.9.1) in hand, we may invoke Lemma 2.100 below and proceed by induction on $|\alpha|$ to conclude that, as desired, $u\in C^m(\mathbb{R}^n)$. □

Lemma 2.100. *Suppose the functions $\{u_j\}_{j\in\mathbb{N}}$ and u are such that:*

(i) $u_j\in C^1(\Omega)$ *for every* $j\in\mathbb{N}$,

(ii) $\lim_{j\to\infty}u_j=u$ *uniformly on compact subsets of* $\mathbb{R}^n$ *contained in* Ω, *and*

(iii) for each $k\in\{1,\dots,n\}$ *there exists a function* $v_k\in C^0(\Omega)$ *with the property that* $\lim_{j\to\infty}\partial_k u_j=v_k$ *uniformly on compact subsets of* $\mathbb{R}^n$ *contained in* Ω.

Then $u\in C^1(\Omega)$ *and* $\partial_k u=v_k$ *for each* $k\in\{1,\dots,n\}$.

Proof. From the start, since uniform convergence on compact sets preserves continuity, we have that $u\in C^0(\Omega)$. Fix $x\in\Omega$ and $k\in\{1,2,\dots,n\}$ and let $t_0>0$ be such that $x+t\mathbf{e}_k\in\Omega$ whenever $t\in[-t_0,t_0]$ where, as before, $\mathbf{e}_k$ is

the unit vector in $\mathbb{R}^n$ where the kth component is equal to 1. Then, for each $t \in [-t_0, t_0]$, we may write

$$u(x+te_k)-u(x) = \lim_{j\to\infty}[u_j(x+t\mathbf{e}_k)-u_j(x)] = \lim_{j\to\infty}\int_0^t \frac{\mathrm{d}}{\mathrm{d}s}\big[u_j(x+s\,\mathbf{e}_k)\big]\,\mathrm{d}s$$
$$= \lim_{j\to\infty}\int_0^t (\partial_k u_j)(x+s\,\mathbf{e}_k)\,\mathrm{d}s = \int_0^t v_k(x+s\,\mathbf{e}_k)\,\mathrm{d}s. \quad (2.9.5)$$

The first equality in (2.9.5) is based on *(ii)* while the last one is a consequence of *(iii)*. Next, since v_k is continuous on Ω, by the mean value theorem for integrals it follows that there exists some s_t, belonging to the interval with endpoints 0 and t, such that $\int_0^t v_k(x+s\,\mathbf{e}_k)\,\mathrm{d}s = tv_k(x+s_t\mathbf{e}_k)$. Hence,

$$\lim_{t\to 0}\frac{u(x+t\mathbf{e}_k)-u(x)}{t} = \lim_{t\to 0} v_k(x+s_t\mathbf{e}_k) = v_k(x), \quad (2.9.6)$$

which proves that $(\partial_k u)(x)$ exists and is equal to $v_k(x)$. Thus, $\partial_k u = v_k \in C^0(\Omega)$ for every k, which shows that $u \in C^1(\Omega)$. □

Lemma 2.101. *Let $u \in \mathcal{D}'(\Omega)$ be such that for each $j \in \{1,\dots,n\}$, the distributional derivatives $\partial_j u$ are of function type and belong to $C^0(\Omega)$. Then $u \in C^0(\Omega)$.*

Proof. Since $\nabla u \in [C^0(\Omega)]^n$ the function $v(x) := \int_0^1 (\nabla u)(tx)\cdot x\,\mathrm{d}t$ for $x \in \Omega$ (where "$\cdot$" denotes the dot product of vectors) is well-defined and belongs to $C^0(\Omega)$. Given the current goals, by Exercise 2.46 it suffices to show that for each $x_0 \in \Omega$ there exists an open set $\omega \subset \Omega$ such that $x_0 \in \omega$ and $u\big|_\omega = v\big|_\omega$ in $\mathcal{D}'(\omega)$.

To this end, fix $x_0 \in \Omega$ and let $r \in (0, \mathrm{dist}(x_0, \partial\Omega))$. Then $B(x_0, r) \subset \Omega$. Without loss of generality in what follows we may assume that $x_0 = 0$ (since translations interact favorably, in a reversible manner, with both hypotheses and conclusion). Let $\varphi \in C_0^\infty(\Omega)$ be such that $\mathrm{supp}\,\varphi \subset B(0,r)$ and fix some $j \in \{1,\dots,n\}$. Then we have

$$\langle \partial_j v, \varphi\rangle = -\langle v, \partial_j\varphi\rangle = -\int_\Omega \Big(\int_0^1 (\nabla u)(tx)\cdot x\,\partial_j\varphi(x)\,\mathrm{d}t\Big)\mathrm{d}x$$
$$= -\int_0^1 \Big(\int_\Omega (\nabla u)(tx)\cdot x\,\partial_j\varphi(x)\,\mathrm{d}x\Big)\mathrm{d}t$$
$$= -\lim_{\varepsilon\to 0^+}\int_\varepsilon^1 \Big(\int_\Omega (\nabla u)(tx)\cdot x\,\partial_j\varphi(x)\,\mathrm{d}x\Big)\mathrm{d}t$$
$$= -\lim_{\varepsilon\to 0^+}\int_\varepsilon^1 \Big(\int_\Omega \nabla u(y)\cdot \frac{y}{t^{n+1}}\,(\partial_j\varphi)(y/t)\,\mathrm{d}y\Big)\mathrm{d}t$$
$$= -\lim_{\varepsilon\to 0^+}\int_\Omega \sum_{k=1}^n \partial_k u(y)\Big(\int_\varepsilon^1 \frac{y_k}{t^{n+1}}\,(\partial_j\varphi)(y/t)\,\mathrm{d}t\Big)\mathrm{d}y$$

$$= \lim_{\varepsilon\to 0^+} \Big\langle u(y),\ \sum_{k=1}^{n} \int_{\varepsilon}^{1} \partial_{y_k}\Big[\frac{y_k}{t^{n+1}}\,(\partial_j\varphi)(y/t)\Big]\,\mathrm{d}t\Big\rangle. \tag{2.9.7}$$

For the fourth equality in (2.9.7) we have used Lebesgue's dominated convergence theorem (cf. Theorem 13.12) and for the fifth one the change of variables $tx = y$ (note that since $\operatorname{supp}\varphi \in B(0,r)$ and $t \in (0,1]$, we have $\operatorname{supp}\varphi(\cdot/t) \subset B(0,r) \subset \Omega$). Furthermore, for each $\varepsilon > 0$, differentiating with respect to y and then integrating by parts in t, gives

$$\begin{aligned}
&\sum_{k=1}^{n} \int_{\varepsilon}^{1} \partial_{y_k}\Big[\frac{y_k}{t^{n+1}}\,(\partial_j\varphi)(y/t)\Big]\,\mathrm{d}t \\
&\qquad = \int_{\varepsilon}^{1} \sum_{k=1}^{n} \Big[\frac{1}{t^{n+1}}\,(\partial_j\varphi)(y/t) + \frac{y_k}{t^{n+2}}\,(\partial_k\partial_j\varphi)(y/t)\Big]\,\mathrm{d}t \\
&\qquad = \int_{\varepsilon}^{1} \Big[\frac{n}{t^{n+1}}\,(\partial_j\varphi)(y/t) - \frac{1}{t^n}\,\frac{d}{\mathrm{d}t}\big[(\partial_j\varphi)(y/t)\big]\Big]\,\mathrm{d}t \\
&\qquad = -\frac{1}{t^n}\,(\partial_j\varphi)(y/t)\Big|_{t=\varepsilon}^{t=1} = -\partial_j\varphi(y) + \frac{1}{\varepsilon^n}(\partial_j\varphi)(y/\varepsilon).
\end{aligned} \tag{2.9.8}$$

By combining (2.9.7) and (2.9.8) we obtain

$$\begin{aligned}
\langle \partial_j v, \varphi\rangle &= -\langle u, \partial_j\varphi\rangle + \lim_{\varepsilon\to 0^+} \Big\langle u(y), \frac{1}{\varepsilon^{n-1}}\partial_{y_j}\big[\varphi(y/\varepsilon)\big]\Big\rangle \\
&= \langle \partial_j u, \varphi\rangle + \lim_{\varepsilon\to 0^+} \Big\langle \partial_j u(y), \frac{1}{\varepsilon^{n-1}}\varphi(y/\varepsilon)\Big\rangle.
\end{aligned} \tag{2.9.9}$$

Since $\partial_j u \in C^0(\Omega)$, the pairing under the limit in the right-most term in (2.9.9) is given by an integral in which we make the change of variables $x = y/\varepsilon$ to further compute

$$\begin{aligned}
\lim_{\varepsilon\to 0^+} \Big\langle \partial_j u(y), \frac{1}{\varepsilon^{n-1}}\varphi(y/\varepsilon)\Big\rangle &= \lim_{\varepsilon\to 0^+} \frac{1}{\varepsilon^{n-1}} \int_{\Omega} (\partial_j u)(y)\varphi(y/\varepsilon)\,\mathrm{d}y \\
&= \lim_{\varepsilon\to 0^+} \varepsilon \int_{\Omega} (\partial_j u)(\varepsilon x)\varphi(x)\,\mathrm{d}x = 0
\end{aligned} \tag{2.9.10}$$

given that, by the continuity of $\partial_j u$ at $0 \in \Omega$,

$$\lim_{\varepsilon\to 0^+} \int_{\Omega} (\partial_j u)(\varepsilon x)\varphi(x)\,\mathrm{d}x = \partial_j u(0) \int_{\Omega} \varphi(x)\,\mathrm{d}x. \tag{2.9.11}$$

In summary, from (2.9.9)–(2.9.11), the fact that φ is an arbitrary element in $\mathcal{D}(B(0,r))$, and that j is arbitrary in $\{1,\dots,n\}$, we conclude that necessarily $\nabla v\big|_{B(0,r)} = \nabla u\big|_{B(0,r)}$ in $\mathcal{D}'(B(0,r))$. By Exercise 2.140 there exists $c \in \mathbb{C}$ such that $u\big|_{B(0,r)} - v\big|_{B(0,r)} = c$ in $\mathcal{D}'(B(0,r))$. Since $v \in C^0(\Omega)$, the latter implies that $u\big|_{B(0,r)}$ belongs to $C^0(B(0,r))$ as desired. This completes the proof of the lemma. □

After these preparations, we are ready to state and prove our first main result.

Theorem 2.102. *Let $u \in \mathcal{D}'(\Omega)$ and suppose that there exists $m \in \mathbb{N}_0$ such that for each $\alpha \in \mathbb{N}_0^n$ satisfying $|\alpha| = m$ the distributional derivative $\partial^\alpha u$ is continuous on Ω. Then $u \in C^m(\Omega)$.*

Proof. We prove the theorem by induction on m. For $m = 0$ there is nothing to prove. Suppose $m = 1$. Applying Lemma 2.101, we obtain that $u \in C^0(\Omega)$. To prove that $u \in C^1(\Omega)$, it suffices to show that u is of class C^1 in a neighborhood of any point $x_0 \in \Omega$. Fix $x_0 \in \Omega$, take $r > 0$ such that $B(x_0, r) \subset \Omega$ and a function $\psi \in C_0^\infty(\mathbb{R}^n)$ such that $\operatorname{supp}\psi \subset B(x_0, r)$ and $\psi \equiv 1$ on $B(x_0, r/2)$. Then by Proposition 2.63 the distribution ψu extends to $v := \widetilde{\psi u} \in \mathcal{E}'(\mathbb{R}^n)$. Also, $v \in C^0(\mathbb{R}^n)$ since $u \in C^0(\Omega)$ and ψ is compactly supported in Ω. Using (2.5.4) and part *(4)* in Proposition 2.36 we obtain

$$\partial_j v = \widetilde{\partial_j(\psi u)} = \widetilde{(\partial_j \psi)u} + \psi\widetilde{\partial_j u} \in C^0(\mathbb{R}^n). \tag{2.9.12}$$

Applying Proposition 2.99 with $m = 1$ then gives $v \in C^1(\mathbb{R}^n)$. In addition, since by design $v\big|_{B(x_0,r/2)} = u\big|_{B(x_0,r/2)}$, we conclude that $u\big|_{B(x_0,r/2)} \in C^1(B(x_0, r/2))$, as wanted.

Assume now that the theorem is true for all nonnegative integers up to, and including, some $m \in \mathbb{N}$. Take $u \in \mathcal{D}'(\Omega)$ with the property that $\partial^\alpha u \in C^0(\Omega)$ for all $\alpha \in \mathbb{N}_0^n$ satisfying $|\alpha| = m + 1$ and fix $j \in \{1, \ldots, n\}$. Then $\partial_j u \in \mathcal{D}'(\Omega)$ satisfies $\partial^\alpha(\partial_j u) \in C^0(\Omega)$ for all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = m$. By the induction hypothesis, it follows that $\partial_j u \in C^m(\Omega)$. In particular, $\partial_j u \in C^0(\Omega)$. This being true for all $j \in \{1, \ldots, n\}$, what we already proved for $m = 1$ implies $u \in C^1(\Omega)$. Thus, u is a C^1 function in Ω with first-order partial derivatives that are of class C^m in Ω. Then necessarily $u \in C^{m+1}(\Omega)$ as wanted. The proof of the theorem is finished. □

Exercise 2.103. *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be such that for some $N \in \mathbb{N}_0$ we have $\partial^\alpha u = 0$ in $\mathcal{D}'(\mathbb{R}^n)$ for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| > N$. Prove that u is a polynomial of degree less than or equal to N.*

Hint: Use Theorem 2.102 to conclude that $u \in C^\infty(\mathbb{R}^n)$ then invoke Taylor's formula (13.2.6).

Moving on to the second issue discussed at the beginning of this section we recall (cf. Remark 2.35) that a function $f : \Omega \to \mathbb{C}$ is called Lipschitz in Ω provided there exists some constant $M \in [0, \infty)$ such that

$$|f(x) - f(y)| \leq M|x - y|, \qquad \forall\, x, y \in \Omega, \tag{2.9.13}$$

and that the Lipschitz constant of f is the smallest M for which (2.9.13) holds. Our next task is to prove the following theorem.

Theorem 2.104. *For $f \in \mathcal{D}'(\mathbb{R}^n)$ and a number $M \in [0, \infty)$ the following two statements are equivalent:*

(i) f *is given by a Lipschitz function in* $\mathbb{R}^n$ *with Lipschitz constant less than or equal to* M*;*

(ii) for each $k \in \{1,\dots,n\}$*, the distributional derivative* $\partial_k f$ *belongs to* $L^\infty(\mathbb{R}^n)$ *and* $\|\partial_k f\|_{L^\infty(\mathbb{R}^n)} \leq M$*.*

Proof. Fix a distribution $f \in \mathcal{D}'(\mathbb{R}^n)$ and let ϕ be as in (1.2.3). Consider the sequence of functions $\{\phi_j\}_{j\in\mathbb{N}}$ from (1.3.7) that satisfies the properties listed in (1.3.8). Furthermore, set

$$f_j := f * \phi_j \quad \text{in} \quad \mathcal{D}'(\mathbb{R}^n), \qquad \forall\, j \in \mathbb{N}. \tag{2.9.14}$$

By Proposition 2.93 we have

$$f_j \in C^\infty(\mathbb{R}^n) \quad \text{and} \quad f_j(x) = \langle f, \phi_j(x-\cdot)\rangle, \quad \forall\, j \in \mathbb{N}. \tag{2.9.15}$$

Also, since (as proved in Example 2.20) one has $\phi_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} \delta$, by part *(2)* in Proposition 2.92 and part *(d)* in Theorem 2.87 one obtains

$$f_j = f * \phi_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} f * \delta = f. \tag{2.9.16}$$

Next we proceed with the proof of *(i)* $\Longrightarrow$ *(ii)*. Suppose f is Lipschitz with Lipschitz constant $\leq M$. In particular, the formula in (2.9.15) becomes

$$f_j(x) = \int_{\mathbb{R}^n} f(y)\phi_j(x-y)\,dy = \int_{\mathbb{R}^n} f(x-y)\phi_j(y)\,dy, \quad \forall\, x \in \mathbb{R}^n, \ \forall\, j \in \mathbb{N}. \tag{2.9.17}$$

We claim that for each $j \in \mathbb{N}$ one has

$$|f_j(x) - f_j(y)| \leq M|x-y|, \quad \forall\, x, y \in \mathbb{R}^n \quad \text{and} \quad \|\nabla f_j\|_{L^\infty(\mathbb{R}^n)} \leq M. \tag{2.9.18}$$

Indeed, if $j \in \mathbb{N}$ is fixed, then from (2.9.17) we obtain

$$|f_j(x) - f_j(y)| \leq \int_{\mathbb{R}^n} |f(x-z) - f(y-z)|\phi_j(z)\,dz \leq M|x-y|, \tag{2.9.19}$$

for every $x, y \in \mathbb{R}^n$. In turn, since f_j is smooth, we have

$$(\partial_k f_j)(x) = \lim_{h\to 0} \frac{f_j(x+h\mathbf{e}_k) - f_j(x)}{h}, \quad \forall\, x \in \mathbb{R}^n, \ \forall\, k \in \{1,\dots,n\}. \tag{2.9.20}$$

In combination with (2.9.19) this implies $\|\nabla f_j\|_{L^\infty(\mathbb{R}^n)} \leq M$, completing the proof of the claims made in (2.9.18).

Next, fix $k \in \{1,\dots,n\}$ and $\varphi \in C_0^\infty(\mathbb{R}^n)$. Then based on (2.9.16) we may write

$$\langle \partial_k f, \varphi\rangle = -\langle f, \partial_k\varphi\rangle = -\lim_{j\to\infty} \langle f_j, \partial_k\varphi\rangle = \lim_{j\to\infty} \langle \partial_k f_j, \varphi\rangle$$

$$= \lim_{j\to\infty} \int_{\mathbb{R}^n} (\partial_k f_j)(x)\varphi(x)\,dx. \tag{2.9.21}$$

Using the second estimate in (2.9.18) we obtain

$$\left| \int_{\mathbb{R}^n} (\partial_k f_j)(x)\varphi(x)\,dx \right| \leq \|\nabla f_j\|_{L^\infty(\mathbb{R}^n)} \|\varphi\|_{L^1(\mathbb{R}^n)} \leq M\|\varphi\|_{L^1(\mathbb{R}^n)}. \tag{2.9.22}$$

Hence, from (2.9.22) and (2.9.21) it follows that

$$\left| \langle \partial_k f, \varphi\rangle \right| \leq \limsup_{j\to\infty} \left| \int_{\mathbb{R}^n} (\partial_k f_j)(x)\varphi(x)\,dx \right| \leq M\|\varphi\|_{L^1(\mathbb{R}^n)}. \tag{2.9.23}$$

Consequently, the linear assignment

$$C_0^\infty(\mathbb{R}^n) \ni \varphi \mapsto \langle \partial_k f, \varphi\rangle \in \mathbb{R} \tag{2.9.24}$$

is continuous in the L^1-norm and has norm less than or equal to M. Since $C_0^\infty(\mathbb{R}^n)$ is dense in $L^1(\mathbb{R}^n)$, the linear functional in (2.9.24) extends to a linear, bounded functional $\Lambda_k : L^1(\mathbb{R}^n) \to \mathbb{C}$ with norm less or equal to M. Thus, $\Lambda_k \in \big(L^1(\mathbb{R}^n)\big)^* = L^\infty(\mathbb{R}^n)$ has norm less than or equal to M, which implies that there exists a unique

$$g_k \in L^\infty(\mathbb{R}^n) \quad \text{with} \quad \|g_k\|_{L^\infty(\mathbb{R}^n)} \leq M \tag{2.9.25}$$

and such that

$$\Lambda_k(h) = \int_{\mathbb{R}^n} g_k(x)h(x)\,dx, \qquad \forall\, h \in L^1(\mathbb{R}^n). \tag{2.9.26}$$

Granted (2.9.26) and keeping in mind that Λ_k is an extension of the linear assignment in (2.9.24), we arrive at the conclusion that

$$\langle \partial_k f, \varphi\rangle = \int_{\mathbb{R}^n} g_k(x)\varphi(x)\,dx, \qquad \forall\, \varphi \in C_0^\infty(\mathbb{R}^n). \tag{2.9.27}$$

The identity in (2.9.27) yields $\partial_k f = g_k$ in $\mathcal{D}'(\mathbb{R}^n)$, which proves *(ii)* in view of (2.9.25).

Conversely, suppose *(ii)* is true. Fix $j \in \mathbb{N}$ and note that from (2.9.14) and part *(e)* in Theorem 2.87, for every $k \in \{1,\dots,n\}$ one has $\partial_k f_j = (\partial_k f) * \phi_j$ in $\mathcal{D}'(\mathbb{R}^n)$. Thus, using Proposition 2.93, the current assumptions on f, and the properties of ϕ_j, we have

$$\begin{aligned} |(\partial_k f_j)(x)| = |\langle \partial_k f, \phi_j(x-\cdot)\rangle| &= \left| \int_{\mathbb{R}^n} (\partial_k f)(y)\phi_j(x-y)\,dy \right| \\ &\leq \|\partial_k f\|_{L^\infty(\mathbb{R}^n)} \int_{\mathbb{R}^n} \phi_j(x-y)\,dy \leq M, \qquad \forall\, x \in \mathbb{R}^n, \end{aligned} \tag{2.9.28}$$

for each $k \in \{1, \dots, n\}$. Now fix $x_0 \in \mathbb{R}^n$ and consider the sequence of functions $\{g_j\}_{j\in\mathbb{N}}$ given by

$$g_j(x) := f_j(x) - f_j(x_0), \quad \text{for} \quad x \in \mathbb{R}^n. \tag{2.9.29}$$

Then $g_j \in C^\infty(\mathbb{R}^n)$ and based on the mean value theorem, (2.9.29), and (2.9.28) we also obtain

$$|g_j(x) - g_j(y)| \leq M|x-y|, \quad \forall\, x, y \in \mathbb{R}^n, \quad \text{and} \tag{2.9.30}$$

$$|g_j(x)| \leq M|x - x_0|, \quad \forall\, x \in \mathbb{R}^n. \tag{2.9.31}$$

By a corollary of the classical Arzelá–Ascoli theorem (see Theorem 13.16), there exists a subsequence $\{g_{j_\ell}\}_{\ell\in\mathbb{N}}$ that converges uniformly on any compact subset of $\mathbb{R}^n$ to some function $g \in C^0(\mathbb{R}^n)$. As such, for each $\varphi \in C_0^\infty(\mathbb{R}^n)$ we may write

$$\lim_{\ell\to\infty} \langle g_{j_\ell}, \varphi\rangle = \lim_{\ell\to\infty} \int_{\operatorname{supp}\varphi} g_{j_\ell}(x)\varphi(x)\,dx = \int_{\operatorname{supp}\varphi} g(x)\varphi(x)\,dx = \langle g, \varphi\rangle \tag{2.9.32}$$

which goes to show that $g_{j_\ell} \xrightarrow[\ell\to\infty]{\mathcal{D}'(\mathbb{R}^n)} g$. The latter, (2.9.16), and (2.9.29), state that whenever $\psi \in C_0^\infty(\mathbb{R}^n)$ is such that $\int_{\mathbb{R}^n} \psi(x)\,dx = 0$ we have

$$\begin{aligned} \langle g, \psi\rangle &= \lim_{\ell\to\infty} \langle g_{j_\ell}, \psi\rangle = \lim_{\ell\to\infty} \int_{\mathbb{R}^n} g_{j_\ell}(x)\psi(x)\,dx \\ &= \lim_{\ell\to\infty} \int_{\mathbb{R}^n} f_{j_\ell}(x)\psi(x)\,dx = \lim_{\ell\to\infty} \langle f_{j_\ell}, \psi\rangle = \langle f, \psi\rangle. \end{aligned} \tag{2.9.33}$$

Thus, we may apply Exercise 2.139 to conclude that there exists some constant $c \in \mathbb{C}$ with the property that $f = g + c$ in $\mathcal{D}'(\mathbb{R}^n)$. This proves that the distribution f is of function type and is given by the function $g + c$. Moreover, writing the estimate in (2.9.30) with j replaced by j_ℓ and then taking the limit as $\ell \to \infty$ (recall that $\{g_{j_\ell}\}_\ell$ converges pointwise to g) leads to

$$|g(x) - g(y)| \leq M|x-y|, \quad \forall\, x, y \in \mathbb{R}^n, \tag{2.9.34}$$

which in concert with the fact that $f(x) - f(y) = g(x) - g(y)$ proves that f is a Lipschitz function with Lipschitz constant $\leq M$. Now the proof of *(ii)* $\Longrightarrow$ *(i)* is finished. □

Further Notes for Chap. 2. The material in this chapter is at the very core of the theory of distributions since it provides a versatile calculus for distributions that naturally extends the scope of the standard calculus for ordinary functions. The definition of distributions used here is essentially that of the French mathematician Laurent Schwartz (1915–2002), cf. [60], though nowadays there are many books dealing

at length with the classical topics discussed here. These include [8, 15, 17–20, 24, 25, 30, 31, 59, 65, 68, 70–72], and the reader is referred to these sources for other angles of exposition. In particular, in [24, 31, 68], distributions are defined on smooth manifolds, while in [53] the notion of distributions is adapted to rough settings.

2.10 Additional Exercises for Chap. 2

Exercise 2.105. *Consider the mapping $u : \mathcal{D}(\mathbb{R}) \to \mathbb{C}$ by setting*

$$u(\varphi) := \sum_{j=1}^{\infty} \Big[\varphi\Big(\frac{1}{j^2}\Big) - \varphi(0)\Big], \qquad \forall\, \varphi \in C_0^\infty(\mathbb{R}). \tag{2.10.1}$$

Prove that u is well-defined. Is u a distribution? If yes, what is the order of u?

Exercise 2.106. *Prove that there exists $u \in \mathcal{D}'(\Omega)$ for which the following statement is false:*

For each compact K contained in Ω there exist $C > 0$ and $k \in \mathbb{N}_0$ such that

$$|\langle u, \varphi\rangle| \leq C \sup_{x\in K,\, |\alpha|\leq k} |\partial^\alpha \varphi(x)|, \qquad \forall\, \varphi \in C_0^\infty(\Omega). \tag{2.10.2}$$

Exercise 2.107. *Prove that $|x|^N \ln|x| \in L^1_{\rm loc}(\mathbb{R}^n)$ whenever N is a real number satisfying $N > -n$. Thus, in particular, $|x|^N \ln|x| \in \mathcal{D}'(\mathbb{R}^n)$ when $N > -n$.*

Exercise 2.108. *Suppose $n \geq 2$ and given $\xi \in S^{n-1}$ define $f(x) := \ln|x \cdot \xi|$ for each $x \in \mathbb{R}^n \setminus \{0\}$. Prove that $f \in L^1_{\rm loc}(\mathbb{R}^n)$. In particular, $\ln|x \cdot \xi| \in \mathcal{D}'(\mathbb{R}^n)$.*

Exercise 2.109. *Prove that $(\ln|x|)' = \text{P.V.}\, \frac{1}{x}$ in $\mathcal{D}'(\mathbb{R})$, where $\text{P.V.}\, \frac{1}{x}$ is the distribution defined in (2.1.13).*

Exercise 2.110. *Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x \ln|x| - x$ if $x \in \mathbb{R} \setminus \{0\}$ and $f(0) = 0$. Prove that $f \in C^0(\mathbb{R})$ and the distributional derivative of f in $\mathcal{D}'(\mathbb{R})$ equals $\ln|x|$.*

Exercise 2.111. *Suppose $n \geq 2$. Prove that $\partial_j(\ln|x|) = \frac{x_j}{|x|^2}$ in $\mathcal{D}'(\mathbb{R}^n)$ for every $j \in \{1, \ldots, n\}$.*

Exercise 2.112. *Let $\theta \in C_0^\infty(\mathbb{R})$ be supported in the interval $(-1, 1)$ and such that $\int_{\mathbb{R}} \theta(t)\, dt = 1$. For each $j \in \mathbb{N}$, define*

$$\psi_j(x) := \int_{1/j}^{j} j\, \theta(jx - jt)\, dt, \qquad \forall\, x \in \mathbb{R}.$$

Prove that $\psi_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} H$, where H is the Heaviside function.

Exercise 2.113. *Let $u : \mathcal{D}(\mathbb{R}) \to \mathbb{C}$ defined by $u(\varphi) := \sum_{j=1}^{\infty} \varphi^{(j)}(j)$ for each $\varphi \in \mathcal{D}(\mathbb{R})$. Prove that $u \in \mathcal{D}'(\mathbb{R})$ and that this distribution does not have finite order.*

Exercise 2.114. *For each $j \in \mathbb{N}$ define*

$$f_j(x) := \frac{1}{j|x|^{n-\frac{1}{j}}} \qquad \text{for } x \in \mathbb{R}^n \setminus \{0\}.$$

Prove that $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} \omega_{n-1}\delta$, where ω_{n-1} is the area of the unit sphere in $\mathbb{R}^n$.

Exercise 2.115. *For each $\varepsilon > 0$ define*

$$f_\varepsilon(x) := \frac{1}{\pi}\frac{\varepsilon}{x^2+\varepsilon^2} \qquad \text{for } x \in \mathbb{R}.$$

Prove that $f_\varepsilon \xrightarrow[\varepsilon\to 0^+]{\mathcal{D}'(\mathbb{R})} \delta$.

Exercise 2.116. *For each $\varepsilon > 0$ define*

$$f_\varepsilon(x) := (4\pi\varepsilon)^{-\frac{n}{2}} \mathrm{e}^{-\frac{|x|^2}{4\varepsilon}} \qquad \text{for } x \in \mathbb{R}^n.$$

Prove that $f_\varepsilon \xrightarrow[\varepsilon\to 0^+]{\mathcal{D}'(\mathbb{R}^n)} \delta$.

Exercise 2.117. *Recall that $i = \sqrt{-1} \in \mathbb{C}$ and, for each $\varepsilon \in (0,\infty)$, define*

$$f_\varepsilon^{\pm}(x) := \frac{1}{x \pm i\,\varepsilon} \qquad \text{for } x \in \mathbb{R}.$$

Also, recall the distribution from (2.1.13). Prove that

$$\frac{1}{x \pm i\,\varepsilon} \longrightarrow \mp i\pi\delta + \mathrm{P.V.}\,\frac{1}{x} \quad \text{as} \quad \varepsilon \to 0^+ \ \text{ in } \ \mathcal{D}'(\mathbb{R}). \tag{2.10.3}$$

This is the so-called Sokhotsky's formula.

Exercise 2.118. *Prove that the sequence $\left\{\frac{\sin(jx)}{\pi x}\right\}_{j\in\mathbb{N}}$ converges to δ in $\mathcal{D}'(\mathbb{R})$ as $j \to \infty$.*

Exercise 2.119. *In each case determine if the given sequence of distributions in $\mathcal{D}'(\mathbb{R})$ indexed over $j \in \mathbb{N}$ converges and determine its limit whenever convergent. Below $m \in \mathbb{N}$ is fixed and $i = \sqrt{-1}$.*

(a) $f_j(x) := \dfrac{\sqrt{j}}{\sqrt{\pi}}\mathrm{e}^{-jx^2}$ *for $x \in \mathbb{R}$;*

(b) $f_j(x) ::= j^m \cos(jx)$ *for $x \in \mathbb{R}$;*

(c) $f_j(x) = \dfrac{2j^3x^2}{\pi(1+j^2x^2)^2}$ *for* $x \in \mathbb{R}$;

(d) $u_j := (-1)^j \delta_{\frac{1}{j}}$;

(e) $u_j := \dfrac{j}{2}\Big[\delta_{\frac{1}{j}} - \delta_{-\frac{1}{j}}\Big]$;

(f) $f_j(x) := \dfrac{1}{x}\chi_{|x|\geq\frac{1}{j}}$ *for* $x \in \mathbb{R}\setminus\{0\}$;

(g) $f_j(x) := \dfrac{1}{j\pi}\dfrac{\sin^2(jx)}{x^2}$ *for* $x \in \mathbb{R}\setminus\{0\}$;

(h) $f_j(x) := j^m \mathrm{e}^{ijx}$ *for* $x \in \mathbb{R}$;

(j) $f_j(x) := \begin{cases} j\mathrm{e}^{ijx} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0, \end{cases}$ *for* $x \in \mathbb{R}$.

Exercise 2.120. *Let* $a \in \mathbb{R}$. *Compute* $(H(\cdot - a))'$ *in* $\mathcal{D}'(\mathbb{R})$.

Exercise 2.121. *Consider the function*

$$f : \mathbb{R} \longrightarrow \mathbb{R} \quad \text{defined by} \quad f(x) := \begin{cases} x & \text{if } x > a, \\ 0 & \text{if } x \leq a, \end{cases} \qquad \forall\, x \in \mathbb{R}, \tag{2.10.4}$$

where $a \in \mathbb{R}$ *is fixed. Compute* $(u_f)'$ *in* $\mathcal{D}'(\mathbb{R})$, *where* u_f *is defined as in* (2.1.6).

Exercise 2.122. *Let* $f : \mathbb{R} \to \mathbb{R}$ *be defined by* $f(x) := \sin|x|$ *for every* $x \in \mathbb{R}$. *Compute* $(u_f)'$ *and* $(u_f)''$ *in* $\mathcal{D}'(\mathbb{R})$.

Exercise 2.123. *Let* $I \subseteq \mathbb{R}$ *be an open interval,* $x_0 \in I$, *and* $f \in C^1(I\setminus\{x_0\})$ *be such that* $f' \in L^1_{\mathrm{loc}}(I)$ *(here* f' *is the pointwise derivative of* f *in* $I\setminus\{x_0\}$*). Prove that the one-sided limits* $\lim\limits_{x\to x_0^+} f(x)$, $\lim\limits_{x\to x_0^-} f(x)$ *exist and are finite, that* $f \in L^1_{\mathrm{loc}}(I)$, *and that*

$$(u_f)' = u_{f'} + \Big[\lim_{x\to x_0^+} f(x) - \lim_{x\to x_0^-} f(x)\Big]\delta_{x_0} \qquad \text{in} \quad \mathcal{D}'(I).$$

Remark 2.124. *Prove that there exist pointwise differentiable functions on* $\mathbb{R}$ *for which the distributional derivative does not coincide with the classical derivative. You may consider the function* f *defined by* $f(x) := x^2\cos\big(\frac{1}{x^2}\big)$ *for* $x \neq 0$ *and* $f(0) := 0$, *and show that* $f \in C^1(\mathbb{R}\setminus\{0\})$ *and also* f *is differentiable at the origin, while* $f' \notin L^1_{\mathrm{loc}}(\mathbb{R})$.

Exercise 2.125. *Let* $I \subseteq \mathbb{R}$ *be an open interval,* $x_0 \in I$, *and let* $m \in \mathbb{N}$. *Suppose that* $f \in C^\infty(I\setminus\{x_0\})$ *is such that the pointwise derivatives* $f', f'', \ldots, f^{(m)}$, *belong to* $L^1_{\mathrm{loc}}(I)$. *Prove that for each* $k \in \{0,, 1, \ldots, m-1\}$ *the limits* $\lim\limits_{x\to x_0^+} f^{(k)}(x)$

and $\lim_{x\to x_0^-} f^{(k)}(x)$ *exist, are finite, and that*

$$\begin{aligned} u_f^{(m)} =& u_{f^{(m)}} + \Big[\lim_{x\to x_0^+} f(x) - \lim_{x\to x_0^-} f(x)\Big]\delta_{x_0}^{(m-1)} \\ &+ \Big[\lim_{x\to x_0^+} f'(x) - \lim_{x\to x_0^-} f'(x)\Big]\delta_{x_0}^{(m-2)} + \cdots \\ &+ \Big[\lim_{x\to x_0^+} f^{(m-1)}(x) - \lim_{x\to x_0^-} f^{(m-1)}(x)\Big]\delta_{x_0} \qquad \text{in } \mathcal{D}'(I). \end{aligned}$$

Exercise 2.126. *Let $I \subseteq \mathbb{R}$ be an open interval and consider a sequence $\{x_k\}_{k\in\mathbb{N}} \subset I$ with no accumulation point in I. Suppose $f \in C^1(I\setminus\{x_k : k\in\mathbb{N}\})$ is such that the pointwise derivative f' belongs to $L^1_{\rm loc}(I)$. Prove that for each $k\in\mathbb{N}$ the limits $\lim_{x\to x_k^\pm} f(x)$ exist and are finite, that $f\in L^1_{\rm loc}(I)$, and that*

$$(u_f)' = u_{f'} + \sum_{k=1}^{\infty}\Big[\lim_{x\to x_k^+} f(x) - \lim_{x\to x_k^-} f(x)\Big]\delta_{x_k} \qquad \text{in} \quad \mathcal{D}'(I).$$

Exercise 2.127. *Let $f:\mathbb{R}\to\mathbb{R}$ be the function defined by $f(x) := \lfloor x\rfloor$ for each $x\in\mathbb{R}$, where $\lfloor x\rfloor$ is the integer part of x. Determine $(u_f)'$.*

Exercise 2.128. *Let $\Sigma\subset\mathbb{R}^n$ be a surface of class C^1 as in Definition 13.36, and denote by σ the surface measure on Σ. Define the mapping $\delta_\Sigma : C_0^\infty(\mathbb{R}^n)\to\mathbb{C}$ by $\delta_\Sigma(\varphi) := \int_\Sigma \varphi(x)\,d\sigma(x)$ for each $\varphi\in C_0^\infty(\mathbb{R}^n)$. Prove that $\delta_\Sigma\in\mathcal{D}'(\mathbb{R}^n)$, it has order zero, and $\operatorname{supp}\delta_\Sigma = \Sigma$. Also show that if $g\in L^\infty(K\cap\Sigma)$ for each compact set K in $\mathbb{R}^n$, and if we define*

$$\big(g\delta_\Sigma\big)(\varphi) := \int_\Sigma g(x)\varphi(x)\,d\sigma(x), \qquad \forall\,\varphi\in C_0^\infty(\mathbb{R}^n), \tag{2.10.5}$$

then $g\delta_\Sigma\in\mathcal{D}'(\mathbb{R}^n)$.

Exercise 2.129. *Let $\Omega\subset\mathbb{R}^n$ be a domain of class C^1 (recall Definition 13.40) and denote by $\nu = (\nu_1,\ldots,\nu_n)$ its outward unit normal. Denote by $\delta_{\partial\Omega}$ the distribution defined as in Exercise 2.128 corresponding to $\Sigma := \partial\Omega$.*

Set $\Omega_+ := \Omega$ and $\Omega_- := \mathbb{R}^n\setminus\overline{\Omega}$. Suppose that $f\in L^1_{\rm loc}(\mathbb{R}^n)$ has the property that the distributional derivative $\partial_k f$ belongs to $L^1_{\rm loc}(\mathbb{R}^n)$ for each number $k\in\{1,2\ldots,n\}$. In addition, assume that the restrictions $f_\pm := f|_{\Omega_\pm}$ satisfy $f_\pm\in C^1(\Omega_\pm)$ and that they may be extended so that $f_\pm\in C^0(\overline{\Omega_\pm})$. Prove that for each $k\in\{1,2\ldots,n\}$ the following equality holds:

$$\partial_k u_f = u_{\partial_k f} + s_{\partial\Omega}(f)\nu_k\delta_{\partial\Omega} \qquad \text{in } \mathcal{D}'(\mathbb{R}^n),$$

where $s_{\partial\Omega}(f):\partial\Omega\to\mathbb{C}$ is defined by

$$\begin{aligned} s_{\partial\Omega}(f)(x) &:= f_-(x) - f_+(x) \\ &= \lim_{\mathbb{R}^n\setminus\overline{\Omega}\ni y\to x} f(y) - \lim_{\Omega\ni y\to x} f(y) \quad \text{for every } x\in\partial\Omega. \end{aligned} \tag{2.10.6}$$

Exercise 2.130. *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain of class C^1 with outward unit normal $\nu = (\nu_1, \dots, \nu_n)$. Prove that for each $k \in \{1, 2 \dots, n\}$ one has $\partial_k \chi_\Omega = -\nu_k \delta_{\partial\Omega}$ in $\mathcal{D}'(\mathbb{R}^n)$.*

Exercise 2.131. *Suppose $R \in (0, \infty)$ and let $u \in \mathcal{D}'(\mathbb{R}^n)$ be such that*

$$(|x|^2 - R^2)u = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^n). \tag{2.10.7}$$

Prove that u has compact support. Give an example of a distribution u satisfying condition (2.10.7).

Exercise 2.132. *Let $f \in C^0(\Omega)$ be such that $u_f \in \mathcal{E}'(\Omega)$. Prove that f has compact support and $\operatorname{supp} u_f = \operatorname{supp} f$.*

Exercise 2.133. *Compute the derivatives of order $m \in \mathbb{N}$ of each distribution on $\mathbb{R}$ given below.*

(a) $|x|$ *(b)* $\operatorname{sgn} x$ *(c)* $\cos x\, H$ *(d)* $\sin x\, H$ *(e)* $x^2 \chi_{[-1,1]}$

Exercise 2.134. *Consider the set $A := \{(x, y) \in \mathbb{R}^2 : |x-2|+|y-1| < 1\} \subset \mathbb{R}^2$. Compute $(\partial_1^2 - \partial_2^2)\chi_A$ in $\mathcal{D}'(\mathbb{R}^2)$.*

Exercise 2.135. *Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x, y) := \chi_{[0,1]}(x - y)$ for $x, y \in \mathbb{R}$. Compute $\partial_1(u_f)$, $\partial_2(u_f)$ in $\mathcal{D}'(\mathbb{R}^2)$. Prove that $\partial_1^2(u_f) - \partial_2^2(u_f) = 0$ in $\mathcal{D}'(\mathbb{R}^2)$.*

Exercise 2.136. *Let $\psi \in C^\infty(\Omega)$ be such that $\psi(x) \neq 0$ for every $x \in \Omega$. Prove that for each $v \in \mathcal{D}'(\Omega)$ there exists a unique solution $u \in \mathcal{D}'(\Omega)$ of the equation $\psi u = v$ in $\mathcal{D}'(\Omega)$.*

Exercise 2.137. *Let $\psi \in C^\infty(\Omega)$ and suppose $u_1, u_2 \in \mathcal{D}'(\Omega)$ are such that $u_1 \neq u_2$ and $\psi u_1 = \psi u_2$ in $\mathcal{D}'(\Omega)$. Prove that the set $\{x \in \Omega : \psi(x) = 0\}$ is not empty.*

Exercise 2.138. *Suppose $\{\Omega_j\}_{j \in I}$ is an open cover of the open set $\Omega \subseteq \mathbb{R}^n$ and there exists a family of distributions $\{u_j\}_{j \in I}$ such that $u_j \in \mathcal{D}'(\Omega_j)$ for each $j \in I$ and $u_j\big|_{\Omega_j \cap \Omega_k} = u_k\big|_{\Omega_j \cap \Omega_k}$ in $\mathcal{D}'(\Omega_j \cap \Omega_k)$ for every $j, k \in I$ such that $\Omega_j \cap \Omega_k \neq \emptyset$. Prove that there exists a unique distribution $u \in \mathcal{D}'(\Omega)$ with the property that $u\big|_{\Omega_j} = u_j$ in $\mathcal{D}'(\Omega_j)$ for every $j \in I$.*

Exercise 2.139. *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be such that $\langle u, \varphi \rangle = 0$ for every $\varphi \in C_0^\infty(\mathbb{R}^n)$ satisfying $\int_{\mathbb{R}^n} \varphi(x)\, \mathrm{d}x = 0$. Prove that there exists $c \in \mathbb{C}$ such that $u = c$ in $\mathcal{D}'(\mathbb{R}^n)$.*

Exercise 2.140. *Let $\Omega \subseteq \mathbb{R}^n$ be open and connected and let $u \in \mathcal{D}'(\Omega)$ be such that $\partial_1 u = \partial_2 u = \cdots = \partial_n u = 0$ in $\mathcal{D}'(\Omega)$. Prove that there exists $c \in \mathbb{C}$ such that $u = c$ in $\mathcal{D}'(\Omega)$.*

Exercise 2.141. *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be such that $x_n u = 0$ in $\mathcal{D}'(\mathbb{R}^n)$. Prove that there exists $v \in \mathcal{D}'(\mathbb{R}^{n-1})$ such that $u(x', x_n) = v(x') \otimes \delta(x_n)$ in $\mathcal{D}'(\mathbb{R}^n)$.*

Exercise 2.142. *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be such that $x_1 u = \cdots = x_n u = 0$ in $\mathcal{D}'(\mathbb{R}^n)$. Determine the expression for u.*

Exercise 2.143. *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be such that $\partial_n u = 0$ in $\mathcal{D}'(\mathbb{R}^n)$. Prove that there exists $v \in \mathcal{D}'(\mathbb{R}^{n-1})$ such that $u(x', x_n) = v(x') \otimes 1$ in $\mathcal{D}'(\mathbb{R}^n)$, where 1 denotes the constant function (equal to 1) in $\mathbb{R}$.*

Exercise 2.144. *Let $v, w \in \mathcal{D}'(\mathbb{R})$ and define the distribution*

$$u(x_1, x_2) := 1 \otimes v(x_2) + w(x_1) \otimes 1 \quad \text{in} \quad \mathcal{D}'(\mathbb{R}^2),$$

where 1 denotes the constant function (equal to 1) in $\mathbb{R}$. Prove that $\partial_1 \partial_2 u = 0$ in $\mathcal{D}'(\mathbb{R}^2)$.

Exercise 2.145. *Let $u(x_1, x_2, x_3) := H(x_1) \otimes \delta(x_2) \otimes \delta(x_3)$ in $\mathcal{D}'(\mathbb{R}^3)$, where H is the Heaviside function on the real line. Compute $\partial_1 u$, $\partial_2 u$, $\partial_3 u$ in $\mathcal{D}'(\mathbb{R}^3)$.*

Exercise 2.146. *Consider the sequence*

$$f_j(x) := (2\pi)^{-n} \int_{[-j,j]\times\cdots\times[-j,j]} \mathrm{e}^{ix\cdot\xi} \mathrm{d}\xi, \qquad \forall\, x \in \mathbb{R}^n, \ \forall\, j \in \mathbb{N}. \tag{2.10.8}$$

Prove that $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} \delta$.

Exercise 2.147. *Solve each equation in $\mathcal{D}'(\mathbb{R})$ for u.*

(1) $(x-1)u = \delta$;

(2) $xu = a$, *where* $a \in C^\infty(\mathbb{R})$;

(3) $xu = v$, *where* $v \in \mathcal{D}'(\mathbb{R})$.

Exercise 2.148. *Prove that the given convolutions exist and then compute them.*

(a) $H * H$

(b) $H(-x) * H(-x)$

(c) $x^2 H * (\sin x\, H)$

(d) $\chi_{[0,1]} * (xH)$

(e) $|x|^2 * \delta_{\partial B(0,r)}$ *where* $r > 0$ *and* $\delta_{\partial B(0,r)}$ *is as defined in Exercise 2.128 corresponding to the surface* $\Sigma := \partial B(0, r)$.

Exercise 2.149. *Let $a \in \mathbb{R}^n \setminus \{0\}$, $u_j := \delta_{ja}$, $v_j := \delta_{-ja}$, for each $j \in \mathbb{N}$. Determine $\lim_{j\to\infty} u_j$, $\lim_{j\to\infty} v_j$, $\lim_{j\to\infty} (u_j * v_j)$, in $\mathcal{D}'(\mathbb{R}^n)$.*

Exercise 2.150. *For each $j \in \mathbb{N}$, consider the functions defined by $f_j(x) := \frac{(-1)^j}{2}\chi_{\left[-\frac{1}{j},\frac{1}{j}\right]}(x)$ and $g_j(x) := (-1)^j$, for every $x \in \mathbb{R}$. Determine if the given limits exist in $\mathcal{D}'(\mathbb{R})$.*

(a) $\lim_{j\to\infty} f_j$

(b) $\lim_{j\to\infty} g_j$

(c) $\lim_{j\to\infty} (f_j * g_j)$

Exercise 2.151. *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ and consider the map $\Lambda : \mathcal{D}(\mathbb{R}^n) \to \mathcal{E}(\mathbb{R}^n)$ given by $\Lambda(\varphi) := u * \varphi$, for every $\varphi \in C_0^\infty(\mathbb{R}^n)$. Prove that Λ is a well-defined, linear and continuous map. Also prove that Λ commutes with translations, that is, if $x_0 \in \mathbb{R}^n$ and $\varphi \in C_0^\infty(\mathbb{R}^n)$, then $t_{x_0}\big(\Lambda(\varphi)\big) = \Lambda\big(t_{x_0}\varphi\big)$, where t_{x_0} is the map from (1.3.16).*

Exercise 2.152. *Suppose $\Lambda : \mathcal{D}(\mathbb{R}^n) \to \mathcal{E}(\mathbb{R}^n)$ is a linear, continuous map that commutes with translations (in the sense explained in Exercise 2.151). Prove that there exists a unique $u \in \mathcal{D}'(\mathbb{R}^n)$ such that $\Lambda(\varphi) = u * \varphi$ for every function $\varphi \in C_0^\infty(\mathbb{R}^n)$.*

Exercise 2.153. *Let $u \in \mathcal{E}'(\mathbb{R}^n)$ be such that $\langle u, x^\alpha\rangle = 0$ for every $\alpha \in \mathbb{N}^n$. Prove that $u = 0$ in $\mathcal{E}'(\mathbb{R}^n)$.*

Exercise 2.154. *Let $u : \mathcal{E}(\mathbb{R}) \to \mathbb{C}$ be the functional defined by*

$$u(\varphi) := \lim_{k\to\infty} \left[\Big(\sum_{j=1}^{k} \varphi\big(\tfrac{1}{j}\big) \Big) - k\varphi(0) - \varphi'(0)\ln k \right], \qquad \forall\, \varphi \in C^\infty(\mathbb{R}).$$

Prove that $u \in \mathcal{E}'(\mathbb{R})$ and determine $\operatorname{supp} u$.

Exercise 2.155. *For each $j \in \mathbb{N}$ consider the function $f_j : \mathbb{R} \to \mathbb{R}$ defined by $f_j(x) := \frac{j}{2}$ if $|x| \le \frac{1}{j}$ and $f_j(x) := 0$ if $|x| > \frac{1}{j}$. Prove that $f_j \xrightarrow[j\to\infty]{\mathcal{E}'(\mathbb{R})} \delta$.*

Exercise 2.156. *For each $j \in \mathbb{N}$ consider the function $f_j : \mathbb{R} \to \mathbb{R}$ defined by $f_j(x) := \frac{1}{j}$ if $|x| \le j$ and $f_j(x) := 0$ if $|x| > j$. Prove that the sequence $\{f_j\}_{j\in\mathbb{N}}$ converges in $\mathcal{D}'(\mathbb{R})$ but not in $\mathcal{E}'(\mathbb{R})$.*

Exercise 2.157. *Let $\psi \in C_0^\infty(\mathbb{R}^n)$ be such that $\int_{\mathbb{R}^n} \psi(x)\,\mathrm{d}x = 1$ and for each $j \in \mathbb{N}$ define $f_j : \mathbb{R}^n \to \mathbb{C}$ by $f_j(x) := j^n\psi(jx)$ for each $x \in \mathbb{R}^n$. Prove that $f_j \xrightarrow[j\to\infty]{\mathcal{E}'(\mathbb{R}^n)} \delta$.*

Exercise 2.158. *Let $\{x_j\}_{j\in\mathbb{N}}$ be a sequence of points in $\mathbb{R}^n$. Prove that $\{x_j\}_{j\in\mathbb{N}}$ is convergent in $\mathbb{R}^n$ if and only if $\{\delta_{x_j}\}_{j\in\mathbb{N}}$ is convergent in $\mathcal{E}'(\mathbb{R}^n)$.*

Exercise 2.159. *Let $a \in \mathbb{R}$ and $k \in \mathbb{N}_0$ be given. Prove that*

$$(x+a)\delta_a^{(k)} = 2a\,\delta_a^{(k)} + k\delta_a^{(k-1)} \quad \text{in} \quad \mathcal{D}'(\mathbb{R}), \tag{2.10.9}$$

$$(x^2-a^2)\delta_a^{(k)} = -2k\,a\,\delta_a^{(k-1)} + k(k-1)\delta_a^{(k-2)} \quad \text{in} \quad \mathcal{D}'(\mathbb{R}), \tag{2.10.10}$$

with the convention that $\delta_a^{(-m)} := 0 \in \mathcal{D}'(\mathbb{R})$ for each $m \in \mathbb{N}$.

Chapter 3

The Schwartz Space and the Fourier Transform

3.1 The Schwartz Space of Rapidly Decreasing Functions

Recall that if $f \in L^1(\mathbb{R}^n)$ then the `Fourier transform` of f is the mapping $\widehat{f} : \mathbb{R}^n \to \mathbb{C}$ defined by

$$\widehat{f}(\xi) := \int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi} f(x)\,\mathrm{d}x \quad \text{for each} \quad \xi \in \mathbb{R}^n, \tag{3.1.1}$$

where $i := \sqrt{-1} \in \mathbb{C}$. Note that under the current assumptions the integral in (3.1.1) is absolutely convergent (which means that $\widehat{f}$ is well-defined pointwise in $\mathbb{R}^n$) and one has

$$\sup_{\xi\in\mathbb{R}^n} |\widehat{f}(\xi)| \leq \|f\|_{L^1(\mathbb{R}^n)} \quad \text{and} \quad \widehat{f} \in C^0(\mathbb{R}^n). \tag{3.1.2}$$

where the second condition is seen by applying Lebesgue's dominated convergence theorem. Hence, the mapping

$$\mathcal{F} : L^1(\mathbb{R}^n) \to \{g \in C^0(\mathbb{R}^n) : g \text{ is bounded}\}, \quad \mathcal{F}f := \widehat{f}, \quad \forall\, f \in L^1(\mathbb{R}^n), \tag{3.1.3}$$

called the Fourier transform, is well-defined. Besides being continuous, functions belonging to the image of $\mathcal{F}$ also vanish at infinity. This property is proved next.

Proposition 3.1. *If $f \in L^1(\mathbb{R}^n)$, then* $\lim_{|\xi|\to\infty} \widehat{f}(\xi) = 0$.

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Proof. First, consider the case when $f \in C_0^\infty(\mathbb{R}^n)$. In this scenario, integrating by parts gives $|\xi|^2 \widehat{f}(\xi) = -\widehat{\Delta f}(\xi)$ for each $\xi \in \mathbb{R}^n \setminus \{0\}$, where $\Delta f := \sum_{j=1}^{n} \partial_j^2 f$. Hence,

$$|\widehat{f}(\xi)| \leq \frac{|\widehat{\Delta f}(\xi)|}{|\xi|^2} \leq \frac{\|\Delta f\|_{L^1(\mathbb{R}^n)}}{|\xi|^2}, \qquad \forall\, \xi \in \mathbb{R}^n \setminus \{0\}, \tag{3.1.4}$$

from which it is clear that $\lim_{|\xi|\to\infty} \widehat{f}(\xi) = 0$ in this case.

Consider now the case when f is an arbitrary function in $L^1(\mathbb{R}^n)$. Since $C_0^\infty(\mathbb{R}^n)$ is dense in the latter space, for each $\varepsilon > 0$ fixed there exists $g \in C_0^\infty(\mathbb{R}^n)$ such that $\|f - g\|_{L^1(\mathbb{R}^n)} \leq \frac{\varepsilon}{2}$. Also, from what we proved so far, there exists $R > 0$ with the property that $|\widehat{g}(\xi)| \leq \frac{\varepsilon}{2}$ whenever $|\xi| > R$. Keeping this in mind and using the linearity of the Fourier transform as well as the estimate in (3.1.2) we may write

$$|\widehat{f}(\xi)| \leq |\widehat{(f-g)}(\xi)| + |\widehat{g}(\xi)| \leq \|f - g\|_{L^1(\mathbb{R}^n)} + \frac{\varepsilon}{2} \leq \varepsilon, \quad \text{if } |\xi| > R. \tag{3.1.5}$$

From this, the desired conclusion follows. The proof of the proposition is therefore complete. □

We are very much interested in the possibility of extending the action of the Fourier transform from functions to distributions, though this is going to be accomplished later. For now, we note the following consequence of Fubini's theorem:

$$\int_{\mathbb{R}^n} \widehat{f}(\xi) g(\xi)\, \mathrm{d}\xi = \int_{\mathbb{R}^n} f(x) \widehat{g}(x)\, \mathrm{d}x, \qquad \forall\, f, g \in L^1(\mathbb{R}^n). \tag{3.1.6}$$

Identity (3.1.6) might suggest defining the Fourier transform of a distribution based on duality. However, there is a serious impediment in doing so. Specifically, while for every $\varphi \in C_0^\infty(\mathbb{R}^n)$ we have $\widehat{\varphi} \in C^\infty(\mathbb{R}^n)$ (as may be seen directly from (3.1.1)) and $\widehat{\varphi}$ decays at infinity (as proved in Proposition 3.1), we nonetheless have

$$\mathcal{F}(C_0^\infty(\mathbb{R}^n)) \not\subset C_0^\infty(\mathbb{R}^n). \tag{3.1.7}$$

In fact, we claim that

$$\varphi \in C_0^\infty(\mathbb{R}^n) \ \text{ and } \ \widehat{\varphi} \in C_0^\infty(\mathbb{R}^n) \implies \varphi = 0. \tag{3.1.8}$$

To see that this is the case, suppose $\varphi \in C_0^\infty(\mathbb{R}^n)$ is such that $\widehat{\varphi}$ has compact support in $\mathbb{R}^n$, and pick an arbitrary point $x^* = (x_1^*, \ldots, x_n^*) \in \mathbb{R}^n$. Define the function $\Phi : \mathbb{C} \to \mathbb{C}$ by setting

$$\Phi(z) := \int_{\mathbb{R}^n} \mathrm{e}^{-iz x_1 + \sum_{j=2}^n x_j^* x_j}\, \varphi(x_1, \ldots, x_n)\, \mathrm{d}x_1 \cdots \mathrm{d}x_n, \quad \text{for} \quad z \in \mathbb{C}. \tag{3.1.9}$$

Then Φ is analytic in $\mathbb{C}$ and $\Phi(t) = \widehat{\varphi}(t, x_2^*, \ldots, x_n^*)$ for each $t \in \mathbb{R}$. Given that $\widehat{\varphi}$ has compact support, this implies that Φ vanishes on $\mathbb{R} \setminus [-R, R]$ if

$R > 0$ is suitably large. The identity theorem for ordinary analytic functions of one complex variable then forces $\Phi = 0$ everywhere in $\mathbb{C}$. In particular, $\widehat{\varphi}(x^*) = \Phi(x_1^*) = 0$. Since $x^* \in \mathbb{R}^n$ has been chosen arbitrarily, we conclude that $\widehat{\varphi} = 0$ in $\mathbb{R}^n$. However, as we will see in the sequel, the Fourier transform is injective on $C_0^\infty(\mathbb{R}^n)$, so (3.1.8) follows.

To overcome the difficulty highlighted in (3.1.7), we introduce a new (topological vector) space of functions, that contains $C_0^\infty(\mathbb{R}^n)$, is invariant under $\mathcal{F}$, and that has a dual that is a subspace of $\mathcal{D}'(\mathbb{R}^n)$. This is the space of Schwartz functions, named after the French mathematician Laurent–Moïse Schwartz (1915–2002) who pioneered the theory of distributions and first considered this space in connection with the Fourier transform.

Before presenting the definition of Schwartz functions, we introduce some notation, motivated by the observation that each time a partial derivative of $\widehat{\varphi}$ is taken, the exponential introduces i as a multiplicative factor. To adjust for this factor, it is therefore natural to re-normalize the ordinary partial differentiation operators as follows:

$$\begin{aligned} &D_j := \tfrac{1}{i}\partial_j, \quad j = 1, \dots, n, \quad D := (D_1, \dots, D_n), \\ &D^\alpha := D_1^{\alpha_1} \cdots D_n^{\alpha_n}, \quad \forall\, \alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n. \end{aligned} \tag{3.1.10}$$

At times, we will also use subscripts to specify the variable with respect to which the differentiation is taken. For example, D_x^α stands for D^α with the additional specification that the differentiation is taken with respect to the variable $x \in \mathbb{R}^n$.

Fix now $\alpha, \beta \in \mathbb{N}_0^n$ and observe that for each $\varphi \in C_0^\infty(\mathbb{R}^n)$ integration by parts implies

$$\begin{aligned} \xi^\beta D_\xi^\alpha \widehat{\varphi}(\xi) &= \int_{\mathbb{R}^n} \xi^\beta \mathrm{e}^{-ix\cdot\xi}(-x)^\alpha \varphi(x)\,dx = \int_{\mathbb{R}^n} \Big[(-D_x)^\beta (\mathrm{e}^{-ix\cdot\xi})\Big](-x)^\alpha \varphi(x)\,dx \\ &= \int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi} D_x^\beta \Big[(-x)^\alpha \varphi(x)\Big]\,dx. \end{aligned} \tag{3.1.11}$$

Hence,

$$\sup_{\xi\in\mathbb{R}^n} \big|\xi^\beta D_\xi^\alpha \widehat{\varphi}(\xi)\big| \leq \int_{\mathbb{R}^n} |D_x^\beta (x^\alpha \varphi(x))|\,dx < \infty. \tag{3.1.12}$$

The conclusion from (3.1.12) is that derivatives of any order of $\widehat{\varphi}$ decrease at ∞ faster than any polynomial. This suggests making the following definition.

Definition 3.2. *The* `Schwartz class` *of rapidly decreasing functions is defined as*

$$\mathcal{S}(\mathbb{R}^n) := \big\{\varphi \in C^\infty(\mathbb{R}^n) : \sup_{x\in\mathbb{R}^n} |x^\beta \partial^\alpha \varphi(x)| < \infty,\ \forall\, \alpha, \beta \in \mathbb{N}_0^n\big\}. \tag{3.1.13}$$

We shall simply say that φ is a `Schwartz function` if $\varphi \in \mathcal{S}(\mathbb{R}^n)$. Obviously,

$$C_0^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \subset C^\infty(\mathbb{R}^n) \tag{3.1.14}$$

though both inclusions are strict. An example of a Schwartz function that is not compactly supported is provided below.

Exercise 3.3. *Prove that for each fixed number $a \in (0,\infty)$, the function f, defined by $f(x) := \mathrm{e}^{-a|x|^2}$ for each $x \in \mathbb{R}^n$, belongs to $\mathcal{S}(\mathbb{R}^n)$ and has the property that* $\operatorname{supp} f = \mathbb{R}^n$.

Other elementary observations pertaining to the Schwartz class from Definition 3.2 are recorded below.

Remark 3.4. *One has*

$$\mathcal{S}(\mathbb{R}^n) = \big\{\varphi \in C^\infty(\mathbb{R}^n) : \sup_{x\in\mathbb{R}^n} |x^\beta D^\alpha \varphi(x)| < \infty,\ \forall\, \alpha,\beta \in \mathbb{N}_0^n\big\}, \tag{3.1.15}$$

and if $\varphi \in C^\infty(\mathbb{R}^n)$, then $\varphi \in \mathcal{S}(\mathbb{R}^n)$ if and only if

$$\sup_{x\in\mathbb{R}^n} \big[(1+|x|)^m |\partial^\alpha \varphi(x)|\big] < \infty, \qquad \forall\, m \in \mathbb{N}_0,\ \forall\, \alpha \in \mathbb{N}_0^n,\ |\alpha| \le m. \tag{3.1.16}$$

Indeed, (3.1.15) is immediate from Definition 3.2. Also, the second claim in the remark readily follows from the observation that for each $m \in \mathbb{N}_0$ there exists $C \in [1,\infty)$ with the property that

$$C^{-1}|x|^m \le \sum_{|\gamma|=m} |x^\gamma| \le C|x|^m, \qquad \forall\, x \in \mathbb{R}^n. \tag{3.1.17}$$

In turn, the second inequality in (3.1.17) is seen by noting that for each $\alpha \in \mathbb{N}_0^n$ and $x = (x_1,\dots,x_n) \in \mathbb{R}^n$, we have

$$|x^\alpha| = |x_1|^{\alpha_1}\cdots|x_n|^{\alpha_n} \le |x|^{\alpha_1}\cdots|x|^{\alpha_n} = |x|^{|\alpha|}. \tag{3.1.18}$$

To justify the first inequality in (3.1.17), consider the function $g(x) := \sum_{|\gamma|=m} |x^\gamma|$ for $x \in \mathbb{R}^n$. Then its restriction to S^{n-1} attains a nonzero minimum, and the desired inequality follows by rescaling.

Exercise 3.5. *Prove that if $f \in \mathcal{S}(\mathbb{R}^n)$ then for each $\alpha,\beta \in \mathbb{N}_0^n$ and $N \in \mathbb{N}$ there exists $C = C_{f,N,\alpha,\beta} \in (0,\infty)$ such that*

$$\big|x^\alpha \partial^\beta f(x)\big| \le \frac{C}{(1+|x|)^N} \quad \text{for each } x \in \mathbb{R}^n. \tag{3.1.19}$$

Use this to deduce that $\mathcal{S}(\mathbb{R}^n) \subset L^p(\mathbb{R}^n)$ for every $p \in [1,\infty]$.

In particular, $\mathcal{S}(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$ which, in concert with (3.1.3), allows us to consider the Fourier transform on $\mathcal{S}(\mathbb{R}^n)$. Also, $\mathcal{F}\big(C_0^\infty(\mathbb{R}^n)\big) \subseteq \mathcal{S}(\mathbb{R}^n)$ as seen from the computation in (3.1.12).

Clearly, $\mathcal{S}(\mathbb{R}^n)$ is a vector space when endowed with the canonical operations of addition of functions and multiplication by complex numbers. For a detailed discussion regarding the topology we consider on $\mathcal{S}(\mathbb{R}^n)$ see Sect. 13.1. We continue to denote by $\mathcal{S}(\mathbb{R}^n)$ the respective topological vector space and we single out here a few important facts that are useful for our analysis.

Fact 3.6. *$\mathcal{S}(\mathbb{R}^n)$ is a Frechét space, that is, $\mathcal{S}(\mathbb{R}^n)$ is a locally convex, metrizable, complete, topological vector space over $\mathbb{C}$.*

Fact 3.7. *A sequence $\{\varphi_j\}_{j\in\mathbb{N}} \subset \mathcal{S}(\mathbb{R}^n)$ converges in $\mathcal{S}(\mathbb{R}^n)$ to some $\varphi \in \mathcal{S}(\mathbb{R}^n)$ provided*

$$\sup_{x\in\mathbb{R}^n} \left|x^\beta \partial^\alpha \big[\varphi_j(x) - \varphi(x)\big]\right| \xrightarrow[j\to\infty]{} 0, \qquad \forall\, \alpha, \beta \in \mathbb{N}_0^n, \tag{3.1.20}$$

in which case we use the notation $\varphi_j \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} \varphi$.

Exercise 3.8. *Use (3.1.17) to prove that $\varphi_j \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} \varphi$ if and only if*

$$\sup_{\substack{x\in\mathbb{R}^n \\ \alpha\in\mathbb{N}_0^n,\, |\alpha|\leq m}} \Big[(1+|x|)^m \big|\partial^\alpha[\varphi_j(x) - \varphi(x)]\big|\Big] \xrightarrow[j\to\infty]{} 0, \qquad \forall\, m \in \mathbb{N}_0. \tag{3.1.21}$$

It is useful to observe that the Schwartz class embeds continuously into Lebesgue spaces.

Exercise 3.9. *Prove that if $p \in [1,\infty]$ and a sequence of functions $\{\varphi_j\}_{j\in\mathbb{N}}$ in $\mathcal{S}(\mathbb{R}^n)$ converges in $\mathcal{S}(\mathbb{R}^n)$ to some $\varphi \in \mathcal{S}(\mathbb{R}^n)$ then $\{\varphi_j\}_{j\in\mathbb{N}}$ also converges in $L^p(\mathbb{R}^n)$ to φ.*

Hint: Use Exercise 3.8.

Definition 3.10. *The space of* `slowly increasing functions` *in $\mathbb{R}^n$ is defined as*

$$\mathcal{L}(\mathbb{R}^n) := \big\{a \in C^\infty(\mathbb{R}^n) : \forall\, \alpha \in \mathbb{N}_0^n \ \ \exists\, k \in \mathbb{N}_0 \quad \textit{such that}$$
$$\sup_{x\in\mathbb{R}^n} \big[(1+|x|)^{-k} |\partial^\alpha a(x)|\big] < \infty\big\}. \tag{3.1.22}$$

Note that an immediate consequence of Definition 3.10 is that $\mathcal{L}(\mathbb{R}^n)$ is stable under differentiation (i.e., if $a \in \mathcal{L}(\mathbb{R}^n)$ then $\partial^\alpha a \in \mathcal{L}(\mathbb{R}^n)$ for every $\alpha \in \mathbb{N}_0^n$). Also,

$$\mathcal{S}(\mathbb{R}^n) \subset \mathcal{L}(\mathbb{R}^n), \tag{3.1.23}$$

though $\mathcal{L}(\mathbb{R}^n)$ contains many additional functions that lack decay as, for example, the class of polynomials (other examples are contained in the two exercises below).

Exercise 3.11. *Prove that the function $f(x) := \mathrm{e}^{i|x|^2}$, $x \in \mathbb{R}^n$, belongs to $\mathcal{L}(\mathbb{R}^n)$.*

Exercise 3.12. *Prove that for each $k \in \mathbb{R}$ the function $f(x) := (1+|x|^2)^k$, $x \in \mathbb{R}^n$, belongs to $\mathcal{L}(\mathbb{R}^n)$.*

Some other basic properties of the Schwartz class are collected in the next theorem.

Theorem 3.13. *The following statements are true.*

(a) For each $a \in \mathcal{L}(\mathbb{R}^n)$, the mapping $\mathcal{S}(\mathbb{R}^n) \ni \varphi \mapsto a\varphi \in \mathcal{S}(\mathbb{R}^n)$ is well-defined, linear, and continuous.

(b) For each $\alpha \in \mathbb{N}_0^n$, the mapping $\mathcal{S}(\mathbb{R}^n) \ni \varphi \mapsto \partial^\alpha \varphi \in \mathcal{S}(\mathbb{R}^n)$ is well-defined, linear, and continuous.

(c) $\mathcal{D}(\mathbb{R}^n) \hookrightarrow \mathcal{S}(\mathbb{R}^n) \hookrightarrow \mathcal{E}(\mathbb{R}^n)$ and the embeddings are continuous.

(d) $C_0^\infty(\mathbb{R}^n)$ is sequentially dense in $\mathcal{S}(\mathbb{R}^n)$. Also, the Schwartz class $\mathcal{S}(\mathbb{R}^n)$ is sequentially dense in $\mathcal{E}(\mathbb{R}^n)$.

(e) If $m, n \in \mathbb{N}$ and $f \in \mathcal{S}(\mathbb{R}^m)$, $g \in \mathcal{S}(\mathbb{R}^n)$, then $f \otimes g \in \mathcal{S}(\mathbb{R}^m \times \mathbb{R}^n)$ and the mapping

$$\mathcal{S}(\mathbb{R}^m) \times \mathcal{S}(\mathbb{R}^n) \ni (f, g) \mapsto f \otimes g \in \mathcal{S}(\mathbb{R}^m \times \mathbb{R}^n) \tag{3.1.24}$$

is bilinear and continuous.

(f) If $f, g \in \mathcal{S}(\mathbb{R}^n)$ then $f * g \in \mathcal{S}(\mathbb{R}^n)$ and the mapping

$$\mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n) \ni (f, g) \mapsto f * g \in \mathcal{S}(\mathbb{R}^n) \tag{3.1.25}$$

is bilinear and continuous.

Proof. Clearly, the mappings in *(a)* and *(b)* are linear. By Fact 3.6 and Theorem 13.14, their continuity is equivalent with sequential continuity at 0, something that can be easily checked using Fact 3.7. Moving on to the statement in *(c)*, we first prove that $\mathcal{D}(\mathbb{R}^n)$ embeds continuously into $\mathcal{S}(\mathbb{R}^n)$. Let $\iota : \mathcal{D}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ be defined by $\iota(\varphi) := \varphi$ for each $\varphi \in C_0^\infty(\mathbb{R}^n)$. From (3.1.14) this is a well-defined and linear mapping. To see that ι is sequentially continuous at $0 \in \mathcal{D}(\mathbb{R}^n)$, consider $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\mathbb{R}^n)} 0$. Then there exists a compact set $K \subset \mathbb{R}^n$ with the property that $\operatorname{supp}\varphi_j \subseteq K$ for every $j \in \mathbb{N}$, and $\lim_{j\to\infty} \sup_{x\in K} |\partial^\alpha \varphi_j| = 0$ for every $\alpha \in \mathbb{N}_0^n$. Hence, for any $\alpha, \beta \in \mathbb{N}_0^n$,

$$\sup_{x\in\mathbb{R}^n} \left|x^\beta \partial^\alpha \varphi_j(x)\right| = \sup_{x\in K} \left|x^\beta \partial^\alpha \varphi_j(x)\right| \leq C \sup_{x\in K} \left|\partial^\alpha \varphi_j(x)\right| \xrightarrow[j\to\infty]{} 0, \tag{3.1.26}$$

proving that ι is sequentially continuous at the origin. Recalling now Fact 3.6 and Theorem 13.5, we conclude that ι is continuous.

Our next goal is to show that $\mathcal{S}(\mathbb{R}^n)$ embeds continuously in $\mathcal{E}(\mathbb{R}^n)$. From (3.1.14) we have that the identity $\iota : \mathcal{S}(\mathbb{R}^n) \to \mathcal{E}(\mathbb{R}^n)$ given by $\iota(f) := f$, for each $f \in \mathcal{S}(\mathbb{R}^n)$, is a well-defined linear map. By Fact 3.6, Fact 1.8, and Theorem 13.14, ι is continuous if and only if it is sequentially continuous at zero. However, if $f_j \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} 0$, then for any compact set $K \subset \mathbb{R}^n$ and any $\alpha \in \mathbb{N}_0^n$,

$$\lim_{j\to\infty} \sup_{x\in K} |\partial^\alpha f_j(x)| \leq \lim_{j\to\infty} \sup_{x\in\mathbb{R}^n} |\partial^\alpha f_j(x)| = 0. \tag{3.1.27}$$

This shows that ι is sequentially continuous at zero, finishing the proof *(c)*.

Next, we prove the statement in *(d)*. Let $f\in\mathcal{S}(\mathbb{R}^n)$ be arbitrary and, for some fixed $\psi\in C_0^\infty(\mathbb{R}^n)$ satisfying $\psi=1$ in a neighborhood of $\overline{B(0,1)}$, define the sequence of functions $f_j:\mathbb{R}^n\to\mathbb{C}$ by setting $f_j(x):=\psi\big(\frac{x}{j}\big)f(x)$ for every $x\in\mathbb{R}^n$ and every $j\in\mathbb{N}$. Then $f_j\in C_0^\infty(\mathbb{R}^n)$ and $f_j=f$ on $\overline{B(0,j)}$ for each $j\in\mathbb{N}$. We claim that

$$f_j\xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)}f. \tag{3.1.28}$$

To see that this is the case, if $\alpha,\beta\in\mathbb{N}_0^n$ are arbitrary, by making use of Leibniz's formula (13.2.4) and the fact that $\psi\big(\frac{x}{j}\big)=1$ for each $x\in\overline{B(0,j)}$, we may write

$$\begin{aligned}\sup_{x\in\mathbb{R}^n}\big|x^\beta\partial^\alpha\big(f_j(x)-f(x)\big)\big| &= \sup_{x\in\mathbb{R}^n}\left|x^\beta\sum_{\gamma\leq\alpha}\frac{\alpha!}{\gamma!(\alpha-\gamma)!}\partial^\gamma f(x)\partial^{\alpha-\gamma}\Big[\psi\Big(\frac{x}{j}\Big)-1\Big]\right|\\ &\leq \sup_{|x|\geq j}\left|x^\beta\sum_{\gamma<\alpha}\frac{\alpha!}{\gamma!(\alpha-\gamma)!}\partial^\gamma f(x)\partial^{\alpha-\gamma}\Big[\psi\Big(\frac{x}{j}\Big)\Big]\right|\\ &\quad+\sup_{|x|\geq j}\left|x^\beta\partial^\alpha f(x)\Big[\psi\Big(\frac{x}{j}\Big)-1\Big]\right|.\end{aligned} \tag{3.1.29}$$

Since $\psi\in C_0^\infty(\mathbb{R}^n)$, it follows that there exists a finite constant $C>0$, depending only on ψ and α, such that

$$\sup_{|x|\geq j}\left|\partial^{\alpha-\gamma}\Big[\psi\Big(\frac{x}{j}\Big)\Big]\right|\leq\frac{C}{j},\qquad\forall\,\gamma\in\mathbb{N}_0^n,\ \gamma<\alpha,\ \forall\,j\in\mathbb{N}. \tag{3.1.30}$$

Also, since $f\in\mathcal{S}(\mathbb{R}^n)$, we may invoke (3.1.19) to conclude that there exists some $C=C_{f,\alpha,\beta}\in(0,\infty)$ such that

$$\sup_{|x|\geq j}\left|x^\beta\partial^\alpha f(x)\Big[\psi\Big(\frac{x}{j}\Big)-1\Big]\right|\leq\frac{C}{j}\big[1+\|\psi\|_{L^\infty(\mathbb{R}^n)}\big]. \tag{3.1.31}$$

Combining (3.1.29)–(3.1.31), and keeping in mind that $f\in\mathcal{S}(\mathbb{R}^n)$, we obtain

$$\begin{aligned}&\sup_{x\in\mathbb{R}^n}|x^\beta\partial^\alpha(f_j(x)-f(x))|\\ &\qquad\leq\frac{C}{j}\sup_{x\in\mathbb{R}^n}\left|x^\beta\sum_{\gamma\leq\alpha}\frac{\alpha!}{\gamma!(\alpha-\gamma)!}\partial^\gamma f(x)\right|+\frac{C}{j}\xrightarrow[j\to\infty]{}0.\end{aligned} \tag{3.1.32}$$

This shows that $f_j\xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)}f$ and completes the proof of the fact that $C_0^\infty(\mathbb{R}^n)$ is sequentially dense in $\mathcal{S}(\mathbb{R}^n)$. The sequential continuity of $\mathcal{S}(\mathbb{R}^n)$ in $\mathcal{E}(\mathbb{R}^n)$ is a consequence of Exercise 1.13 and (3.1.14). This completes the proof of *(d)*.

The claims in part *(e)* follow using the observation that

$$\big|(x,y)^{(\alpha,\beta)}\partial_x^\gamma\partial_y^\mu(f\otimes g)(x,y)\big|=\big|x^\alpha\partial^\gamma f(x)\big|\big|y^\beta\partial^\mu g(y)\big|, \tag{3.1.33}$$

for every $(x,y)\in\mathbb{R}^m\times\mathbb{R}^n$, for every $f\in\mathcal{S}(\mathbb{R}^m)$, $g\in\mathcal{S}(\mathbb{R}^n)$, and every $\alpha,\gamma\in\mathbb{N}_0^m$, $\beta,\mu\in\mathbb{N}_0^n$.

Consider now the statement in *(g)*. From Exercise 3.5 we know that $\mathcal{S}(\mathbb{R}^n)\subset L^1(\mathbb{R}^n)$, thus $\mathcal{S}(\mathbb{R}^n)*\mathcal{S}(\mathbb{R}^n)$ is meaningful. To see that $\mathcal{S}(\mathbb{R}^n)*\mathcal{S}(\mathbb{R}^n)\subset\mathcal{S}(\mathbb{R}^n)$, fix some arbitrary $f,g\in\mathcal{S}(\mathbb{R}^n)$ and $\alpha,\beta\in\mathbb{N}_0^n$. Then, making use of the binomial theorem (cf. Theorem 13.7) as well as Exercise 3.5, we may estimate

$$\begin{aligned}
\sup_{x\in\mathbb{R}^n}\left|x^\beta\partial_x^\alpha(f*g)(x)\right| &= \sup_{x\in\mathbb{R}^n}\left|\int_{\mathbb{R}^n}((x-y)+y)^\beta\partial_x^\alpha f(x-y)g(y)\mathrm{d}y\right| \\
&\leq \sup_{x\in\mathbb{R}^n}\sum_{\gamma\leq\beta}\frac{\beta!}{\gamma!(\beta-\gamma)!}\int_{\mathbb{R}^n}|(x-y)^{\beta-\gamma}(\partial^\alpha f)(x-y)||y^\gamma g(y)|\,\mathrm{d}y \qquad (3.1.34)\\
&\leq C_{\alpha,\beta}\sup_{z\in\mathbb{R}^n}\left[(1+|z|)^{|\beta|}\left|\partial^\alpha f(z)\right|\right]\int_{\mathbb{R}^n}(1+|y|)^{|\beta|}|g(y)|\,\mathrm{d}y \\
&\leq C_{\alpha,\beta}\Big(\sup_{z\in\mathbb{R}^n}\left[(1+|z|)^{|\beta|}\left|\partial^\alpha f(z)\right|\right]\Big)\Big(\sup_{y\in\mathbb{R}^n}\left[(1+|y|)^{|\beta|+n+1}\left|g(y)\right|\right]\Big)<\infty.
\end{aligned}$$

This implies $f*g\in\mathcal{S}(\mathbb{R}^n)$. The fact that the mapping in *(e)* is bilinear is immediate from definitions. As regards its continuity, we may invoke again Theorem 13.14 and Fact 3.6 to reduce matters to proving sequential continuity instead. However, the latter is apparent from the estimate in (3.1.34). This finishes the proof of the theorem. □

Exercise 3.14. *Assume that $\psi\in\mathcal{S}(\mathbb{R}^n)$ is given and, for each $j\in\mathbb{N}$, define the function $\psi_j(x):=\psi\big(\frac{x}{j}\big)$ for every $x\in\mathbb{R}^n$. Prove that*

$$\psi_j f\xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)}\psi(0)f\qquad\text{for every}\quad f\in\mathcal{S}(\mathbb{R}^n). \tag{3.1.35}$$

Hint: Adapt estimates (3.1.29) and (3.1.30) to the current setting and, in place of (3.1.31), this time use the mean value theorem for the term $\psi(x/j)-\psi(0)$ to get a decay factor of the order $1/j$.

Proposition 3.15. *Let $m,n\in\mathbb{N}$. Then $C_0^\infty(\mathbb{R}^m)\otimes C_0^\infty(\mathbb{R}^n)$ is sequentially dense in $\mathcal{S}(\mathbb{R}^m\times\mathbb{R}^n)$.*

Proof. Since the topology on $\mathcal{S}(\mathbb{R}^m\times\mathbb{R}^n)$ is metrizable (recall Fact 3.6), there exists a distance function $d:\mathcal{S}(\mathbb{R}^m\times\mathbb{R}^n)\times\mathcal{S}(\mathbb{R}^m\times\mathbb{R}^n)\to[0,\infty)$ that induces its topology. Hence,

$$f_j\xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^m\times\mathbb{R}^n)}f\quad\text{if and only if}\quad\lim_{j\to\infty}d(f_j,f)=0. \tag{3.1.36}$$

Now let $f\in\mathcal{S}(\mathbb{R}^m\times\mathbb{R}^n)$. Then by part *(d)* in Theorem 3.13, there exists a sequence $\{f_j\}_{j\in\mathbb{N}}\subset C_0^\infty(\mathbb{R}^m\times\mathbb{R}^n)$ with the property that $d(f_j,f)<\frac{1}{j}$ for every $j\in\mathbb{N}$. Furthermore, by Proposition 2.73, for each fixed number $j\in\mathbb{N}$, there

exists a sequence $\{g_{j_k}\}_{k\in\mathbb{N}} \subset C_0^\infty(\mathbb{R}^m)\otimes C_0^\infty(\mathbb{R}^n)$ such that $g_{j_k} \xrightarrow[k\to\infty]{\mathcal{D}(\mathbb{R}^m\times\mathbb{R}^n)} f_j$. In particular, by *(c)* in Theorem 3.13,

$$g_{j_k} \xrightarrow[k\to\infty]{\mathcal{S}(\mathbb{R}^m\times\mathbb{R}^n)} f_j \quad \text{for each } j\in\mathbb{N}, \tag{3.1.37}$$

thus

$$\lim_{k\to\infty} d(g_{j_k}, f_j) = 0 \quad \text{for each } j\in\mathbb{N}. \tag{3.1.38}$$

Condition (3.1.38) implies that for each $j\in\mathbb{N}$ there exists $k_j\in\mathbb{N}$ with the property that $d(g_{j_{k_j}}, f_j) < \frac{1}{j}$. Now the sequence $\{g_{j_{k_j}}\}_{j\in\mathbb{N}} \subset C_0^\infty(\mathbb{R}^m)\otimes C_0^\infty(\mathbb{R}^n)$ satisfies

$$d(g_{j_{k_j}}, f) \le d(g_{j_{k_j}}, f_j) + d(f_j, f) < \frac{2}{j} \quad \text{for every} \quad j\in\mathbb{N}. \tag{3.1.39}$$

In turn, this forces $g_{j_{k_j}} \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^m\times\mathbb{R}^n)} f$, from which the desired conclusion follows. □

For $n, m\in\mathbb{N}$, denote by $\mathcal{M}_{n\times m}(\mathbb{R})$ the collection of all $n\times m$ matrices with entries in $\mathbb{R}$. Recall that a map $L:\mathbb{R}^m\to\mathbb{R}^n$ is linear if and only if there exists a matrix $A\in\mathcal{M}_{n\times m}(\mathbb{R})$ such that $L(x) = Ax$ for every $x\in\mathbb{R}^m$, where Ax denotes the multiplication of the matrix A with the vector x viewed as an element in $\mathcal{M}_{m\times 1}(\mathbb{R})$. Moreover, such a matrix is unique. In the sequel, we follow the standard practice of denoting by A the linear map associated with a matrix A.

Exercise 3.16. *Suppose $A\in\mathcal{M}_{n\times n}(\mathbb{R})$ is such that $\det A\neq 0$. Prove that the composition mapping*

$$\mathcal{S}(\mathbb{R}^n)\ni\varphi\mapsto\varphi\circ A\in\mathcal{S}(\mathbb{R}^n) \quad \textit{is well-defined, linear, and continuous.} \tag{3.1.40}$$

Hint: To prove continuity you may use the linearity of the map in (3.1.40), Fact 3.6, and Theorem 13.14, to reduce matters to proving sequential continuity at 0.

We conclude this section by proving that $\mathcal{L}(\mathbb{R}^n) * \mathcal{S}(\mathbb{R}^n)\subseteq C^\infty(\mathbb{R}^n)$.

Proposition 3.17. *For every function $f\in\mathcal{L}(\mathbb{R}^n)$ and every function $g\in\mathcal{S}(\mathbb{R}^n)$ one has $\int_{\mathbb{R}^n}|f(x-y)||g(y)|\,dy<\infty$ for each $x\in\mathbb{R}^n$, and the convolution $f*g$ defined by*

$$(f*g)(x) := \int_{\mathbb{R}^n} f(x-y)g(y)\,dy \quad \textit{for each} \quad x\in\mathbb{R}^n, \tag{3.1.41}$$

has the property that $f*g\in C^\infty(\mathbb{R}^n)$.

Proof. If f, g are as in the statement, then from (3.1.22) and Exercise 3.5 it follows that there exists $M \in \mathbb{N}$ such that for every $N \in \mathbb{N}$ there exists a constant $C \in (0, \infty)$

$$\int_{\mathbb{R}^n} |f(x-y)||g(y)|\,\mathrm{d}y \leq C \int_{\mathbb{R}^n} (1+|x-y|)^M (1+|y|)^{-N}\,\mathrm{d}y, \qquad x \in \mathbb{R}^n. \tag{3.1.42}$$

For each fixed point $x \in \mathbb{R}^n$ choose now $N \in \mathbb{N}$ such that $N > M + n$ and note that this ensures $\int_{\mathbb{R}^n} (1+|x-y|)^M (1+|y|)^{-N}\,\mathrm{d}y < \infty$, proving the first claim in the statement. The fact that $f * g \in C^\infty(\mathbb{R}^n)$ is now seen in a similar fashion given that $\partial^\alpha f$ continues to be in $\mathcal{L}(\mathbb{R}^n)$ for every $\alpha \in \mathbb{N}_0^n$. □

Exercise 3.18. *Prove that $L^p(\mathbb{R}^n) * \mathcal{S}(\mathbb{R}^n) \subseteq C^\infty(\mathbb{R}^n)$ for every $p \in [1, \infty]$.*

Hint: Use the blueprint as in the proof of Proposition 3.17, using Hölder's inequality in place of estimates for slowly growing functions, and arrange matters so that all derivatives fall on the Schwartz function.

We conclude this section with an integration by parts formula that will be useful shortly.

Lemma 3.19. *If $f \in \mathcal{L}(\mathbb{R}^n)$ and $g \in \mathcal{S}(\mathbb{R}^n)$, then for every $\alpha \in \mathbb{N}_0^n$ the following integration by parts formula holds:*

$$\int_{\mathbb{R}^n} (\partial^\alpha g)(x) f(x)\,\mathrm{d}x = (-1)^{|\alpha|} \int_{\mathbb{R}^n} g(x)(\partial^\alpha f)(x)\,\mathrm{d}x. \tag{3.1.43}$$

Proof. Fix $f \in \mathcal{L}(\mathbb{R}^n)$ and $g \in \mathcal{S}(\mathbb{R}^n)$. Since the classes $\mathcal{L}(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^n)$ are stable under differentiation, it suffices to show that for each $j \in \{1, \ldots, n\}$ we have

$$\int_{\mathbb{R}^n} (\partial_j g)(x) f(x)\,\mathrm{d}x = -\int_{\mathbb{R}^n} g(x)(\partial_j f)(x)\,\mathrm{d}x, \tag{3.1.44}$$

since (3.1.43) then follows by iterating (3.1.44). To this end, fix some integer $j \in \{1, \ldots, n\}$ along with some arbitrary $R \in (0, \infty)$. The classical integration by parts formula in the bounded, smooth, domain $B(0,R) \subset \mathbb{R}^n$ then reads (cf. (13.7.4))

$$\begin{aligned} \int_{B(0,R)} (\partial_j g)(x) f(x)\,\mathrm{d}x &= -\int_{B(0,R)} g(x)(\partial_j f)(x)\,\mathrm{d}x \\ &\quad + \int_{\partial B(0,R)} g(x) f(x)(x_j/R)\,\mathrm{d}\sigma(x). \end{aligned} \tag{3.1.45}$$

From part *(a)* in Theorem 3.13 we know that $fg \in \mathcal{S}(\mathbb{R}^n)$. Based on this and Exercise 3.5, it follows that

$$\begin{aligned} \Big| \int_{\partial B(0,R)} g(x) f(x)(x_j/R)\,\mathrm{d}\sigma(x) \Big| &\leq \omega_{n-1} R^{n-1} \sup_{|x|=R} |(fg)(x)| \\ &\leq C R^{-1} \xrightarrow[R\to\infty]{} 0. \end{aligned} \tag{3.1.46}$$

On the other hand, $(\partial_j g)f$, $g\partial_j f \in \mathcal{S}(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$. As such, passing to limit $R \to \infty$ in (3.1.45) yields (3.1.44) on account of Lebesgue's dominated convergence theorem and (3.1.46). □

3.2 The Action of the Fourier Transform on the Schwartz Class

Originally, we defined the Fourier transform in (3.1.3) as a mapping acting on functions from $L^1(\mathbb{R}^n)$. Since $\mathcal{S}(\mathbb{R}^n)$ is contained in $L^1(\mathbb{R}^n)$, it makes sense to consider the Fourier transform acting on the Schwartz class. In this section we study the main properties of the Fourier transform in this setting. The reader is advised that we use the symbols $\mathcal{F}$ and $\widehat{\cdot}$ interchangeably to denote this Fourier transform.

To state our first major result pertaining to the Fourier transform in this setting, recall the notation introduced in (3.1.10).

Theorem 3.20. *The following statements are true.*

(a) If $f \in \mathcal{S}(\mathbb{R}^n)$ and $\alpha \in \mathbb{N}_0^n$ are arbitrary, then $\widehat{D^\alpha f}(\xi) = \xi^\alpha \widehat{f}(\xi)$ for every $\xi \in \mathbb{R}^n$.

(b) If $f \in \mathcal{S}(\mathbb{R}^n)$ and $\alpha \in \mathbb{N}_0^n$ are arbitrary, then $\widehat{x^\alpha f}(\xi) = (-D)^\alpha \widehat{f}(\xi)$ for every $\xi \in \mathbb{R}^n$.

(c) The Fourier transform, originally introduced in the context of (3.1.3), induces a mapping $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ that is linear and continuous.

(d) If $m, n \in \mathbb{N}$, $f \in \mathcal{S}(\mathbb{R}^m)$ and $g \in \mathcal{S}(\mathbb{R}^n)$, then $\widehat{f \otimes g} = \widehat{f} \otimes \widehat{g}$.

Proof. Fix $f \in \mathcal{S}(\mathbb{R}^n)$ and $\alpha \in \mathbb{N}_0^n$. Then the decay of f (cf. (3.1.19)) ensures that we may differentiate under the integral sign in (3.1.1) and obtain

$$\begin{aligned} D^\alpha \widehat{f}(\xi) &= \int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi}(-x)^\alpha f(x)\,\mathrm{d}x \\ &= \widehat{(-x)^\alpha f}\,(\xi), \qquad \forall\,\xi \in \mathbb{R}^n. \end{aligned} \tag{3.2.1}$$

From this, the statement in *(b)* readily follows. Also, if $\beta \in \mathbb{N}_0^n$ is arbitrary, then using the first identity in (3.2.1), the fact that $\xi^\beta \mathrm{e}^{-ix\cdot\xi} = (-D_x)^\beta(\mathrm{e}^{-ix\cdot\xi})$, and the integration by parts formula from Lemma 3.19, we obtain

$$\begin{aligned} \xi^\beta D^\alpha \widehat{f}(\xi) &= \int_{\mathbb{R}^n} (-D_x)^\beta(\mathrm{e}^{-ix\cdot\xi})(-x)^\alpha f(x)\,\mathrm{d}x \\ &= \int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi} D_x^\beta\big[(-x)^\alpha f(x)\big]\,\mathrm{d}x, \qquad \forall\,\xi \in \mathbb{R}^n. \end{aligned} \tag{3.2.2}$$

The formula in *(a)* follows by specializing (3.2.2) to the case $\alpha = (0, \dots, 0) \in \mathbb{N}_0^n$. In addition, starting with (3.2.2) we may estimate

$$\sup_{\xi\in\mathbb{R}^n} |\xi^\beta D^\alpha \widehat{f}(\xi)| \leq \left(\int_{\mathbb{R}^n} (1+|x|^2)^{-n} \mathrm{d}x \right) \times \\ \times \sup_{x\in\mathbb{R}^n} \left\{ (1+|x|^2)^n \left| D_x^\beta \left[x^\alpha f(x) \right] \right| \right\} < \infty, \tag{3.2.3}$$

where the finiteness condition is a consequence of the membership of f to $\mathcal{S}(\mathbb{R}^n)$. Clearly, (3.2.1) also implies that $\widehat{f} \in C^\infty(\mathbb{R}^n)$ which, in combination with (3.2.3), shows that $\widehat{f} \in \mathcal{S}(\mathbb{R}^n)$. Hence, the mapping in *(c)* is well-defined. The fact that this mapping is linear is immediate from definition. In addition, if $f_j \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} 0$, then based on the first inequality in (3.2.3) we have that, for each $m, k \in \mathbb{N}_0$,

$$\sup_{\substack{\xi\in\mathbb{R}^n \\ |\alpha|\leq m,\, |\beta|\leq k}} |\xi^\beta \partial^\alpha \widehat{f_j}(\xi)| \\ \leq C \sup_{\substack{x\in\mathbb{R}^n \\ |\alpha|\leq m,\, |\beta|\leq k}} \left[(1+|x|^2)^n \left| \partial^\beta \left(x^\alpha f_j(x) \right) \right| \right] \to 0 \ \text{ as } \ j \to \infty. \tag{3.2.4}$$

In view of Exercise 3.8, this proves $\mathcal{F} f_j \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} 0$. The latter combined with Fact 3.6 and Theorem 13.14 then implies that $\mathcal{F}$ is continuous from $\mathcal{S}(\mathbb{R}^n)$ into $\mathcal{S}(\mathbb{R}^n)$.

At this stage we are left with proving the statement in *(d)*. To this end, fix $f \in \mathcal{S}(\mathbb{R}^m)$ and $g \in \mathcal{S}(\mathbb{R}^n)$. Then by part *(e)* in Theorem 3.13, we obtain that $f \otimes g \in \mathcal{S}(\mathbb{R}^m \times \mathbb{R}^n)$, so $\mathcal{F}(f \otimes g)$ is well-defined. Furthermore, by applying Fubini's theorem, we may write

$$\begin{aligned} \widehat{f \otimes g}(\xi, \eta) &= \int_{\mathbb{R}^m} \int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi - iy\cdot\eta} (f \otimes g)(x, y) \, \mathrm{d}y \, \mathrm{d}x \\ &= \int_{\mathbb{R}^m} \mathrm{e}^{-ix\cdot\xi} f(x) \, \mathrm{d}x \int_{\mathbb{R}^n} \mathrm{e}^{-iy\cdot\eta} g(y) \, \mathrm{d}y \\ &= \widehat{f}(\xi)\widehat{g}(\eta) = (\widehat{f} \otimes \widehat{g})(\xi, \eta), \qquad \forall\, \xi \in \mathbb{R}^m, \ \forall\, \eta \in \mathbb{R}^n. \end{aligned} \tag{3.2.5}$$

This finishes the proof of the theorem. □

Example 3.21. Suppose $\lambda \in \mathbb{C}$ satisfies $\mathrm{Re}(\lambda) > 0$ and if $\lambda = re^{i\theta}$ for $r > 0$ and $-\pi/2 < \theta < \pi/2$, set $\lambda^{\frac{1}{2}} := \sqrt{r}\mathrm{e}^{i\theta/2}$. Consider the function $f(x) := \mathrm{e}^{-\lambda|x|^2}$ for $x \in \mathbb{R}^n$. Then $f \in \mathcal{S}(\mathbb{R}^n)$ and

$$\widehat{f}(\xi) = \left(\tfrac{\pi}{\lambda}\right)^{\frac{n}{2}} \mathrm{e}^{-\frac{|\xi|^2}{4\lambda}} \quad \text{ for each } \ \xi \in \mathbb{R}^n. \tag{3.2.6}$$

Proof. Fix $\lambda \in \mathbb{C}$ satisfying the given hypotheses. Then Exercise 3.3 ensures that f is a Schwartz function. Also, $f(x) = \mathrm{e}^{-\lambda x_1^2} \otimes \cdots \otimes \mathrm{e}^{-\lambda x_n^2}$ for each $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$. As such, part *(d)* in Theorem 3.20 shows that it suffices to prove (3.2.6) when $n = 1$, in which case $f(x) = \mathrm{e}^{-\lambda x^2}$ for every $x \in \mathbb{R}$. Suppose that this is the case and observe that f satisfies $f' + 2\lambda x f = 0$ in $\mathbb{R}$. By taking the Fourier transform of both sides of this differential equation, and using *(a)* and *(b)* in Theorem 3.20, we arrive at $\xi\widehat{f} + 2\lambda\big(\widehat{f}\big)' = 0$. The solution to this latter ordinary differential equation is $\widehat{f}(\xi) = \widehat{f}(0)\mathrm{e}^{-\frac{\xi^2}{4\lambda}}$ for $\xi \in \mathbb{R}$. There remains to show that $\widehat{f}(0) = \big(\frac{\pi}{\lambda}\big)^{\frac{1}{2}}$. Since by definition, $\widehat{f}(0) = \int_{\mathbb{R}} f(x)\,\mathrm{d}x = \int_{\mathbb{R}} \mathrm{e}^{-\lambda x^2}\,\mathrm{d}x$, we are left with showing that

$$\int_{\mathbb{R}} \mathrm{e}^{-\lambda x^2}\,\mathrm{d}x = \Big(\frac{\pi}{\lambda}\Big)^{\frac{1}{2}} \quad \text{whenever } \lambda \in \mathbb{C} \text{ has } \mathrm{Re}(\lambda) > 0. \tag{3.2.7}$$

Corresponding to the case when $\lambda \in \mathbb{R}_+$, the identity $\int_{\mathbb{R}} \mathrm{e}^{-\lambda x^2}\,\mathrm{d}x = \big(\frac{\pi}{\lambda}\big)^{\frac{1}{2}}$ is a standard exercise in basic calculus. To extend this to complex λ's, observe that the function

$$h(z) := \int_{\mathbb{R}} \mathrm{e}^{-zx^2}\,\mathrm{d}x - \Big(\frac{\pi}{z}\Big)^{\frac{1}{2}} \quad \text{for } z \in \mathbb{C} \text{ with } \mathrm{Re}(z) > 0,$$

is analytic and equal to zero for every $z \in \mathbb{R}_+$. This forces $h(z) = 0$ for all $z \in \mathbb{C}$ with $\mathrm{Re}(z) > 0$. Thus, $\widehat{f}(0) = \big(\frac{\pi}{\lambda}\big)^{\frac{1}{2}}$, as desired. □

Exercise 3.22. *Let $a \in (0, \infty)$ and $b \in \mathbb{R}$ be fixed. Show that if $x \in \mathbb{R}$ then*

$$\mathcal{F}(\mathrm{e}^{-ax^2+ibx})(\xi) = \big(\tfrac{\pi}{a}\big)^{\frac{1}{2}}\mathrm{e}^{-\frac{(\xi-b)^2}{4a}} \qquad \text{for every } \xi \in \mathbb{R}. \tag{3.2.8}$$

Hint: First prove that $\mathcal{F}(\mathrm{e}^{-ax^2+ibx})(\xi) = (\mathcal{F}(\mathrm{e}^{-ax^2}))(\xi - b)$ for every $\xi \in \mathbb{R}$, then use Example 3.21.

Exercise 3.23. *Prove that if $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is such that $\det A \neq 0$, then*

$$\widehat{\varphi \circ A^{-1}} = |\det A|\,(\widehat{\varphi} \circ A^{\top}), \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n), \tag{3.2.9}$$

where A^{-1} and $A^{\top}$ denote, respectively, the inverse and the transpose of the matrix A.

Next we note a consequence of Theorem 3.20 of basic importance. As motivation, suppose $P(D) = \sum_{|\alpha|\leq m} a_\alpha D^\alpha$ is a differential operator of order $m \in \mathbb{N}$, with constant coefficients $a_\alpha \in \mathbb{C}$, for every $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq m$. Furthermore, assume that $f \in \mathcal{S}(\mathbb{R}^n)$ has been given. Then any solution $u \in \mathcal{S}(\mathbb{R}^n)$ of the differential equation $P(D)u = f$ in $\mathbb{R}^n$ also satisfies $P(\xi)\widehat{u}(\xi) = \widehat{f}(\xi)$ for each $\xi \in \mathbb{R}^n$, where we have set $P(\xi) := \sum_{|\alpha|\leq m} a_\alpha \xi^\alpha$. In particular, if $P(\xi)$ has no zeros, then $\widehat{u}(\xi) = \frac{\widehat{f}(\xi)}{P(\xi)}$, for every $\xi \in \mathbb{R}^n$. This gives a formula for the Fourier

transform of u. In order to find a formula for u itself, the natural question that arises is whether we can reconstruct u from $\widehat{u}$. The next theorem provides a positive answer to this question in the class of Schwartz functions.

Theorem 3.24. *The mapping $\mathcal{F}:\mathcal{S}(\mathbb{R}^n)\to\mathcal{S}(\mathbb{R}^n)$ is an algebraic and topologic isomorphism, that is, it is bijective, continuous, and its inverse is also continuous. In addition, its inverse is the operator $\mathcal{F}^{-1}:\mathcal{S}(\mathbb{R}^n)\to\mathcal{S}(\mathbb{R}^n)$ given by the formula*

$$(\mathcal{F}^{-1}g)(x)=(2\pi)^{-n}\int_{\mathbb{R}^n}\mathrm{e}^{ix\cdot\xi}g(\xi)\,\mathrm{d}\xi,\qquad\forall\,x\in\mathbb{R}^n,\quad\forall\,g\in\mathcal{S}(\mathbb{R}^n).\tag{3.2.10}$$

Proof. The proof of the fact that the mapping $\mathcal{F}^{-1}:\mathcal{S}(\mathbb{R}^n)\to\mathcal{S}(\mathbb{R}^n)$ defined as in (3.2.10) is well-defined, linear, and continuous is similar to the proof of part *(c)* in Theorem 3.20. There remains to show that $\mathcal{F}^{-1}\circ\mathcal{F}=I=\mathcal{F}\circ\mathcal{F}^{-1}$ on $\mathcal{S}(\mathbb{R}^n)$, where I is the identity operator on $\mathcal{S}(\mathbb{R}^n)$. To proceed, observe that the identity $\mathcal{F}^{-1}\circ\mathcal{F}=I$ is equivalent to

$$(2\pi)^{-n}\int_{\mathbb{R}^n}\mathrm{e}^{ix\cdot\xi}\widehat{f}(\xi)\,\mathrm{d}\xi=f(x),\qquad\forall\,x\in\mathbb{R}^n,\quad\forall\,f\in\mathcal{S}(\mathbb{R}^n).\tag{3.2.11}$$

As regards (3.2.11), fix some function $f\in\mathcal{S}(\mathbb{R}^n)$ along with some point $x\in\mathbb{R}^n$. Recall (cf. (3.1.1)) that

$$\widehat{f}(\xi)=\int_{\mathbb{R}^n}\mathrm{e}^{-iy\cdot\xi}f(y)\,\mathrm{d}y,\quad\forall\,\xi\in\mathbb{R}^n.\tag{3.2.12}$$

As such, one is tempted to directly replace $\widehat{f}(\xi)$ in (3.2.11) by the right-hand side of (3.2.12) and then use Fubini's theorem to reverse the order of integration in the variables ξ and y. There is, however, a problem in carrying out this approach, since the function $\mathrm{e}^{i(x-y)\cdot\xi}f(y)$, considered jointly in the variable $(\xi,y)\in\mathbb{R}^n\times\mathbb{R}^n$, does not belong to $L^1(\mathbb{R}^n\times\mathbb{R}^n)$, hence Fubini's theorem is not necessarily applicable. To remedy this problem, we introduce a "convergence factor" in the form of a suitable family of Schwartz functions ψ^ε, indexed by $\varepsilon>0$ (to be specified shortly), designed to provide control in the variable ξ, thus ensuring the applicability of Fubini's theorem.

Hence, in place of $\int_{\mathbb{R}^n}\mathrm{e}^{ix\cdot\xi}\widehat{f}(\xi)\,\mathrm{d}\xi$ the idea is to consider $\int_{\mathbb{R}^n}\mathrm{e}^{ix\cdot\xi}\psi^\varepsilon(\xi)\widehat{f}(\xi)\,\mathrm{d}\xi$, for which we may write (granted that $\psi^\varepsilon\in\mathcal{S}(\mathbb{R}^n)$)

$$\begin{aligned}\int_{\mathbb{R}^n}\mathrm{e}^{ix\cdot\xi}\psi^\varepsilon(\xi)\widehat{f}(\xi)\,\mathrm{d}\xi&=\int_{\mathbb{R}^n}\mathrm{e}^{ix\cdot\xi}\psi^\varepsilon(\xi)\int_{\mathbb{R}^n}\mathrm{e}^{-iy\cdot\xi}f(y)\,\mathrm{d}y\,\mathrm{d}\xi\\&=\int_{\mathbb{R}^n\times\mathbb{R}^n}\mathrm{e}^{-i(y-x)\cdot\xi}\psi^\varepsilon(\xi)f(y)\,\mathrm{d}y\,\mathrm{d}\xi\\&=\int_{\mathbb{R}^n}\left(\int_{\mathbb{R}^n}\mathrm{e}^{-i(y-x)\cdot\xi}\psi^\varepsilon(\xi)\,\mathrm{d}\xi\right)f(y)\,\mathrm{d}y\\&=\int_{\mathbb{R}^n}\widehat{\psi^\varepsilon}(y-x)f(y)\,\mathrm{d}y=\int_{\mathbb{R}^n}\widehat{\psi^\varepsilon}(z)f(x+z)\,\mathrm{d}z.\end{aligned}\tag{3.2.13}$$

Given the goal we have in mind (cf. (3.2.11)), as well as the format of the current identity, we find it convenient to define ψ^{ε} by setting

$$\psi^{\varepsilon}(\xi) := \varphi(\varepsilon\,\xi) \quad \text{for each } \xi \in \mathbb{R}^n \text{ and } \varepsilon > 0, \tag{3.2.14}$$

where $\varphi \in \mathcal{S}(\mathbb{R}^n)$ is to be specified momentarily. The rationale behind this choice is that both $\lim_{\varepsilon\to 0^+} \psi^{\varepsilon}(\xi)$ and $\lim_{\varepsilon\to 0^+} \widehat{\psi^{\varepsilon}}(x)$ are reasonably easy to compute. This will eventually allow us to deduce (3.2.11) from (3.2.13) by letting $\varepsilon \to 0^+$. Concretely, from (3.2.14) we have $\lim_{\varepsilon\to 0^+} \psi^{\varepsilon}(\xi) = \varphi(0)$, while from the definition of the Fourier transform it is immediate that

$$\widehat{\psi^{\varepsilon}}(z) = \varepsilon^{-n}\widehat{\varphi}\left(\frac{z}{\varepsilon}\right) = \big(\widehat{\varphi}\big)_{\varepsilon}(z) \quad \text{for every} \quad z \in \mathbb{R}^n. \tag{3.2.15}$$

Keeping this in mind and employing part *(a)* in Exercise 2.22 we obtain

$$\begin{aligned}\lim_{\varepsilon\to 0^+} \int_{\mathbb{R}^n} \widehat{\psi^{\varepsilon}}(z) f(x+z)\,\mathrm{d}z &= \lim_{\varepsilon\to 0^+} \int_{\mathbb{R}^n} \big(\widehat{\varphi}\big)_{\varepsilon}(z) f(x+z)\,\mathrm{d}z \\ &= \Big(\int_{\mathbb{R}^n} \widehat{\varphi}(z)\,\mathrm{d}z\Big) f(x).\end{aligned} \tag{3.2.16}$$

Also, on account of (3.2.14) and the fact that $\widehat{f} \in \mathcal{S}(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$, Lebesgue's dominated convergence theorem gives

$$\begin{aligned}\lim_{\varepsilon\to 0^+} \int_{\mathbb{R}^n} \mathrm{e}^{ix\cdot\xi}\psi^{\varepsilon}(\xi)\widehat{f}(\xi)\,\mathrm{d}\xi &= \lim_{\varepsilon\to 0^+} \int_{\mathbb{R}^n} \mathrm{e}^{ix\cdot\xi}\varphi(\varepsilon\,\xi)\widehat{f}(\xi)\,\mathrm{d}\xi \\ &= \varphi(0)\int_{\mathbb{R}^n} \mathrm{e}^{ix\cdot\xi}\widehat{f}(\xi)\,\mathrm{d}\xi.\end{aligned} \tag{3.2.17}$$

In summary, (3.2.13), (3.2.16), and (3.2.17), show that whenever $\varphi \in \mathcal{S}(\mathbb{R}^n)$ is such that $\int_{\mathbb{R}^n} \widehat{\varphi}(z)\,\mathrm{d}z \neq 0$ we have

$$C\int_{\mathbb{R}^n} \mathrm{e}^{ix\cdot\xi}\widehat{f}(\xi)\,\mathrm{d}\xi = f(x), \qquad \forall\, x \in \mathbb{R}^n, \quad \forall\, f \in \mathcal{S}(\mathbb{R}^n), \tag{3.2.18}$$

where the normalization constant C is given by

$$C := \frac{\varphi(0)}{\int_{\mathbb{R}^n} \widehat{\varphi}(z)\,\mathrm{d}z}. \tag{3.2.19}$$

As such, (3.2.11) will follow as soon as we prove that $C = (2\pi)^{-n}$. For this task, we have the freedom of choosing the function $\varphi \in \mathcal{S}(\mathbb{R}^n)$ and a candidate that springs to mind is the Schwartz function from Example 3.21 (say, in the particular case when $\lambda = 1$). Hence, if $\varphi(x) := \mathrm{e}^{-|x|^2}$ for each $x \in \mathbb{R}^n$, formula (3.2.6) gives

$$\widehat{\varphi}(\xi) = \pi^{\frac{n}{2}}\mathrm{e}^{-\frac{|\xi|^2}{4}} \quad \text{for each} \quad \xi \in \mathbb{R}^n. \tag{3.2.20}$$

Consequently,

$$\int_{\mathbb{R}^n} \widehat{\varphi}(\xi)\,\mathrm{d}\xi = \pi^{\frac{n}{2}} \int_{\mathbb{R}^n} \mathrm{e}^{-\frac{|\xi|^2}{4}}\,\mathrm{d}\xi = \pi^{\frac{n}{2}} \Big(\int_{\mathbb{R}} \mathrm{e}^{-\frac{|t|^2}{4}}\,\mathrm{d}t \Big)^n = \pi^{\frac{n}{2}} (4\pi)^{\frac{n}{2}} = (2\pi)^n. \tag{3.2.21}$$

where the second equality is simply Fubini's theorem, while the third equality is provided by (3.2.7) with $\lambda := 1/4$. Since in this case we also have $\varphi(0) = 1$, it follows that $C = (2\pi)^{-n}$, as wanted. This finishes the justification of the identity $\mathcal{F}^{-1} \circ \mathcal{F} = I$ on $\mathcal{S}(\mathbb{R}^n)$. The same approach also works for the proof of $\mathcal{F} \circ \mathcal{F}^{-1} = I$, completing the proof of the theorem. □

In what follows, for an arbitrary function $f : \mathbb{R}^n \to \mathbb{C}$ we define

$$f^{\vee}(x) := f(-x), \qquad \forall\, x \in \mathbb{R}^n. \tag{3.2.22}$$

Exercise 3.25. *Prove that the mapping*

$$\mathcal{S}(\mathbb{R}^n) \ni f \mapsto f^{\vee} \in \mathcal{S}(\mathbb{R}^n) \tag{3.2.23}$$

is well-defined, linear, and continuous.

Hint: Use Fact 3.6 and Theorem 13.14.

Recall that $\overline{z}$ denotes the complex conjugate of $z \in \mathbb{C}$.

Exercise 3.26. *Let $f \in \mathcal{S}(\mathbb{R}^n)$. Then the following formulas hold:*

(1) $\widehat{f^{\vee}} = (\widehat{f}\,)^{\vee}$.

(2) $\overline{\widehat{f}} = \widehat{\overline{f}^{\vee}}$.

(3) $\widehat{\widehat{f}} = (2\pi)^n f^{\vee}$.

(4) $\int_{\mathbb{R}^n} f(x)\,\mathrm{d}x = \widehat{f}(0)$ *and* $\int_{\mathbb{R}^n} \widehat{f}(\xi)\,\mathrm{d}\xi = (2\pi)^n f(0)$.

Proposition 3.27. *Let f, $g \in \mathcal{S}(\mathbb{R}^n)$. Then the following identities hold:*

(a) $\int_{\mathbb{R}^n} f(x)\widehat{g}(x)\,\mathrm{d}x = \int_{\mathbb{R}^n} \widehat{f}(\xi) g(\xi)\,\mathrm{d}\xi$.

(b) $\int_{\mathbb{R}^n} f(x)\overline{g(x)}\,\mathrm{d}x = (2\pi)^{-n} \int_{\mathbb{R}^n} \widehat{f}(\xi)\,\overline{\widehat{g}(\xi)}\,\mathrm{d}\xi$ *an identity referred to in the literature as* `Parseval's identity`.

(c) $\widehat{f * g} = \widehat{f} \cdot \widehat{g}$.

(d) $\widehat{f \cdot g} = (2\pi)^{-n} \widehat{f} * \widehat{g}$.

Proof. The identity in *(a)* follows via a direct computation using Fubini's theorem. Also, based on Exercise 3.26, we have $\widehat{\widehat{g}} = \widehat{\widehat{g}^{\vee}} = (2\pi)^n \overline{g}$ which, when

combined with *(a)* gives *(b)*. The identity in *(c)* follows using Fubini's theorem. Specifically, for each $\xi \in \mathbb{R}^n$ we may write

$$\begin{aligned}\widehat{f * g}(\xi) &= \int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi}(f*g)(x)\,\mathrm{d}x = \int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi}\int_{\mathbb{R}^n} f(x-y)g(y)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_{\mathbb{R}^u} g(y)\int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi} f(x-y)\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_{\mathbb{R}} \mathrm{e}^{-iy\cdot\xi} g(y)\int_{\mathbb{R}^n} \mathrm{e}^{-iz\cdot\xi} f(z)\,\mathrm{d}z\,\mathrm{d}y = \widehat{f}(\xi)\widehat{g}(\xi), \end{aligned} \tag{3.2.24}$$

as wanted. Next, the identity from *(c)* combines with Exercise 3.26 to yield

$$\widehat{\widehat{f} * \widehat{g}} = \widehat{\widehat{f}}\cdot\widehat{\widehat{g}} = (2\pi)^{2n} f^{\vee}\cdot g^{\vee} = (2\pi)^{2n}(f\cdot g)^{\vee}. \tag{3.2.25}$$

Applying now the Fourier transform to the most extreme sides of (3.2.25) and once again invoking Exercise 3.26, we obtain

$$(2\pi)^{-n}\,\widehat{f} * \widehat{g} = (2\pi)^{-2n}\,\widehat{\widehat{\widehat{f} * \widehat{g}}}^{\vee} = \big(\widehat{f\cdot g}\big)^{\vee\vee} = \widehat{f\cdot g}. \tag{3.2.26}$$

This completes the proof of the proposition. □

Remark 3.28.

(1) It is not difficult to see via a direct computation that we also have

$$\int_{\mathbb{R}^n} f(x)\widehat{g}(x)\,\mathrm{d}x = \int_{\mathbb{R}^n} \widehat{f}(\xi)g(\xi)\,\mathrm{d}\xi, \qquad \forall\, f\in L^1(\mathbb{R}^n),\ \forall\, g\in \mathcal{S}(\mathbb{R}^n). \tag{3.2.27}$$

(2) Parseval's identity written for $f = g \in \mathcal{S}(\mathbb{R}^n)$ becomes

$$\|\widehat{f}\|_{L^2(\mathbb{R}^n)} = (2\pi)^{\frac{n}{2}}\|f\|_{L^2(\mathbb{R}^n)}. \tag{3.2.28}$$

As a consequence, since $C_0^\infty(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$, the Fourier transform $\mathcal{F}$ may be extended to a linear operator from $L^2(\mathbb{R}^n)$ into itself, and the latter identity continues to hold for every $f \in L^2(\mathbb{R}^n)$. In summary, this extension of $\mathcal{F}$, originally considered as in part (c) of Theorem 3.20, satisfies

$$\begin{aligned}&\mathcal{F} : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n) \text{ is linear and continuous and} \\ &\|\mathcal{F}f\|_{L^2(\mathbb{R}^n)} = (2\pi)^{\frac{n}{2}}\|f\|_{L^2(\mathbb{R}^n)}, \quad \forall\, f\in L^2(\mathbb{R}^n).\end{aligned} \tag{3.2.29}$$

Based on this, part (3) in Exercise 3.26, the continuity of the linear mapping $L^2(\mathbb{R}^n) \ni f \mapsto f^{\vee} \in L^2(\mathbb{R}^n)$, and the density of Schwartz functions in $L^2(\mathbb{R}^n)$, we further deduce that

$$\mathcal{F}\big(\mathcal{F}f\big) = (2\pi)^n f^{\vee}, \quad \forall\, f\in L^2(\mathbb{R}^n). \tag{3.2.30}$$

Combined with (3.2.29), *this proves that*

$$\begin{aligned}&\mathcal{F}: L^2(\mathbb{R}^n)\to L^2(\mathbb{R}^n) \text{ is a linear, continuous, isomorphism,}\\ &\text{and}\quad \mathcal{F}^{-1}f=(2\pi)^{-n}\big(\mathcal{F}f^{\vee}\big)=(2\pi)^{-n}\big(\mathcal{F}f\big)^{\vee},\quad \forall\, f\in L^2(\mathbb{R}^n).\end{aligned} \tag{3.2.31}$$

We will continue to use the notation $\widehat{f}$ *for* $\mathcal{F}f$ *whenever* $f\in L^2(\mathbb{R}^n)$. *The identity in* (3.2.29) *is called* `Planchérel's identity`. *The same type of density argument shows that formula from part (b) of Proposition* 3.27 *extends to*

$$\int_{\mathbb{R}^n} f(x)\overline{g(x)}\,\mathrm{d}x=(2\pi)^{-n}\int_{\mathbb{R}^n}\widehat{f}(\xi)\,\overline{\widehat{g}(\xi)}\,\mathrm{d}\xi,\qquad \forall\, f,g\in L^2(\mathbb{R}^n), \tag{3.2.32}$$

to which we continue to refer as Parseval's identity.

(3) An inspection of the computation in (3.2.24) *reveals that the identity* $\widehat{f*g}=\widehat{f}\cdot\widehat{g}$ *remains valid if* $f,g\in L^1(\mathbb{R}^n)$.

Exercise 3.29. *Prove that* $\int_{\mathbb{R}^n} f(x)\widehat{g}(x)\,\mathrm{d}x=\int_{\mathbb{R}^n}\widehat{f}(\xi)g(\xi)\,\mathrm{d}\xi$ *for every* $f,g\in L^2(\mathbb{R}^n)$.

Hint: Use part *(a)* in Proposition 3.27, (3.2.29), and the fact that $C_0^\infty(\mathbb{R}^n)$ is sequentially dense in $L^2(\mathbb{R}^n)$ to first prove the identity for $f\in L^2(\mathbb{R}^n)$ and $g\in\mathcal{S}(\mathbb{R}^n)$.

Further Notes for Chap. 3. The basic formalism associated with the Fourier transform goes back to the French mathematician and physicist Joseph Fourier (1768–1830) in a more or less precise form. A distinguished attribute of this tool, of fundamental importance in the context of partial differential equations, is the ability to render the action of a constant coefficient differential operator simply as ordinary multiplication by its symbol on the Fourier transform side. As the name suggest, the Schwartz space of rapidly decreasing functions has been formally introduced by Laurent Schwartz who was the first to recognize its significance in the context of the Fourier transform. Much of the elegant theory presented here is due to him.

3.3 Additional Exercises for Chap. 3

Exercise 3.30. *Prove that if* $f\in L^1_{\rm comp}(\mathbb{R}^n)$ *then* $\widehat{f}\in C^\infty(\mathbb{R}^n)$.

Exercise 3.31. *Prove that if* $f\in L^1(\mathbb{R}^n)$ *is real-valued and odd, then so is* $\widehat{f}$.

Exercise 3.32. *Let* $\varphi\in C_0^\infty(\mathbb{R}^n)$ *be such that* $\varphi\neq 0$ *and for each* $j\in\mathbb{N}$ *set*

$$\varphi_j(x):=\mathrm{e}^{-j}\varphi(x/j),\qquad \forall\, x\in\mathbb{R}^n.$$

Prove that $\varphi_j\xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)}0$ *but the sequence* $\{\varphi_j\}_{j\in\mathbb{N}}$ *does not converge in* $\mathcal{D}(\mathbb{R}^n)$.

Exercise 3.33. *Let $\varphi \in C_0^\infty(\mathbb{R}^n)$ be such that $\varphi \neq 0$ and for each $j \in \mathbb{N}$ set*

$$\varphi_j(x) := \frac{1}{j}\varphi(x/j), \qquad \forall\, x \in \mathbb{R}^n.$$

Prove that $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} 0$ but the sequence $\{\varphi_j\}_{j\in\mathbb{N}}$ does not converge in $\mathcal{S}(\mathbb{R}^n)$.

Exercise 3.34. *Let $\theta \in C_0^\infty(\mathbb{R})$ be such that $\theta(x) = 1$ for $|x| \leq 1$, and let $\psi \in C^\infty(\mathbb{R})$ be such that $\psi(x) = 0$ for $x \leq -1$ and $\psi(x) = \mathrm{e}^{-x}$ for $x \geq 0$. For each $j \in \mathbb{N}$ then set*

$$\varphi_j(x) := \frac{1}{j}\psi(x)\theta(x/j), \qquad \forall\, x \in \mathbb{R}.$$

Prove that the sequence $\{\varphi_j\}_{j\in\mathbb{N}}$ converges in $\mathcal{S}(\mathbb{R})$.

Exercise 3.35. *Determine which of the following functions belongs to $\mathcal{S}(\mathbb{R}^n)$.*

(a) $\mathrm{e}^{-(x_1+x_2^2+\cdots+x_n^2)}$

(b) $(x_1^2 + x_2^2 + \cdots + x_n^2)^{n!}\mathrm{e}^{-|x|^2}$

(c) $(1 + |x|^2)^{-2^n}$

(d) $\frac{\sin(\mathrm{e}^{-|x|^2})}{1+|x|^2}$

(e) $\frac{\cos(\mathrm{e}^{-|x|^2})}{(1+|x|^2)^n}$

(f) $\mathrm{e}^{-|x|^2}\sin(\mathrm{e}^{x_1^2})$

(g) $\mathrm{e}^{-(Ax)\cdot x}$, *where $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is symmetric and satisfies $(Ax)\cdot x > 0$ for all $x \in \mathbb{R}^n \setminus \{0\}$ (as before, "$\cdot$" denotes the dot product of vectors in $\mathbb{R}^n$).*

Exercise 3.36. *Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ be symmetric and such that $(Ax)\cdot x > 0$ for every $x \in \mathbb{R}^n \setminus \{0\}$. Prove that if $f(x) := \mathrm{e}^{-(Ax)\cdot x}$ for $x \in \mathbb{R}^n$, then $\widehat{f}(\xi) = \frac{\pi^{\frac{n}{2}}}{\sqrt{\det A}}\mathrm{e}^{-\frac{(A^{-1}\xi)\cdot\xi}{4}}$ for every $\xi \in \mathbb{R}^n$.*

Exercise 3.37. *Prove that $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) := \mathrm{e}^{-(x_1^2+x_1x_2+x_2^2)}$ for $(x_1, x_2) \in \mathbb{R}^2$ belongs to $\mathcal{S}(\mathbb{R}^2)$, then compute its Fourier transform.*

Exercise 3.38. *If $P(x)$ is a polynomial in $\mathbb{R}^n$, compute the Fourier transform of the function defined by $f(x) := P(x)\mathrm{e}^{-|x|^2}$ for each $x \in \mathbb{R}^n$.*

Exercise 3.39. *If $a \in (0, \infty)$ and $x_0 \in \mathbb{R}^n$ are fixed, compute the Fourier transform of the function defined by $f(x) := \mathrm{e}^{-a|x|^2}\sin(x \cdot x_0)$ for each $x \in \mathbb{R}^n$.*

Exercise 3.40. *Let $\varphi \in \mathcal{S}(\mathbb{R})$. Prove that the equation $\psi' = \varphi$ has a solution $\psi \in \mathcal{S}(\mathbb{R})$ if and only if $\int_{\mathbb{R}} \varphi(x)\,\mathrm{d}x = 0$.*

Exercise 3.41. *Does the equation* $\psi' = \mathrm{e}^{-x^2}$ *have a solution in* $\mathcal{S}(\mathbb{R})$*?*

Exercise 3.42. *Fix* $x_0 \in \mathbb{R}^n$*. Prove that the translation map* t_{x_0} *from* (1.3.16) *extends linearly and continuously as a map from* $\mathcal{S}(\mathbb{R}^n)$ *into itself. More precisely, show that the translation map* $t_{x_0} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ *defined by* $t_{x_0}(\varphi) := \varphi(\cdot - x_0)$ *for every* $\varphi \in \mathcal{S}(\mathbb{R}^n)$*, is linear and continuous.*

Also, prove that for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$ *the following identities hold in* $\mathcal{S}(\mathbb{R}^n)$

$$\mathcal{F}\big(t_{x_0}(\varphi)\big)(\xi) = \mathrm{e}^{-ix_0\cdot\xi}\,\widehat{\varphi}(\xi) \quad \text{and} \quad t_{x_0}\big(\widehat{\varphi}\big) = \mathcal{F}\big(\mathrm{e}^{ix_0\cdot x}\varphi\big). \tag{3.3.1}$$

Chapter 4

The Space of Tempered Distributions

4.1 Definition and Properties of Tempered Distributions

The algebraic dual of $\mathcal{S}(\mathbb{R}^n)$ is the vector space

$$\big\{u : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C} : \ u \text{ is linear and continuous}\big\}. \tag{4.1.1}$$

Functionals u belonging to this space are called `tempered distributions` (a piece of terminology justified a little later). An important equivalent condition for a linear functional on $\mathcal{S}(\mathbb{R}^n)$ to be a tempered distribution is stated next (see (13.1.32)).

Fact 4.1. *A linear functional $u : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$ is continuous if and only if there exist $m, k \in \mathbb{N}_0$, and a finite constant $C > 0$, such that*

$$|u(\varphi)| \leq C \sup_{\alpha,\beta\in\mathbb{N}_0^n,\, |\alpha|\leq m,\, |\beta|\leq k} \ \sup_{x\in\mathbb{R}^n} \left|x^\beta \partial^\alpha \varphi(x)\right|, \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.1.2}$$

From Fact 3.6 and Theorem 13.14 we also know that any $f : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$ (linear or not) is continuous if and only if it is sequentially continuous. As a consequence, we have the following characterization of tempered distributions.

Proposition 4.2. *Let $u : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$ be linear. Then u is a tempered distribution if and only $u(\varphi_j) \xrightarrow[j\to\infty]{} 0$ whenever $\varphi_j \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} 0$.*

As discussed in Example 2.7, to any locally integrable function f in $\mathbb{R}^n$ one can associate an "ordinary" distribution $u_f \in \mathcal{D}'(\mathbb{R}^n)$. This being said, it is not always the case that u_f is actually a tempered distribution (for more on this, see Remark 4.15). This is, however, true if the locally integrable function f becomes integrable at infinity after being tempered by a polynomial. We elaborate on this issue in the next example.

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Example 4.3. Let $f \in L^1_{\rm loc}(\mathbb{R}^n)$ be such that there exist $m \in [0,\infty)$ and $R \in (0,\infty)$ with the property that

$$\int_{|x|\geq R} |x|^{-m}|f(x)|\,{\rm d}x < \infty. \tag{4.1.3}$$

We claim that the distribution of function type defined by f is a tempered distribution, that is, the mapping

$$u_f : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}, \qquad u_f(\varphi) := \int_{\mathbb{R}^n} f\varphi\,{\rm d}x, \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n), \tag{4.1.4}$$

is a well-defined tempered distribution. To see that this is the case, pick $N \in \mathbb{N}_0$ such that $N \geq m$ and, for an arbitrary $\varphi \in \mathcal{S}(\mathbb{R}^n)$, estimate

$$\begin{aligned}\int_{\mathbb{R}^n} |f\varphi|\,{\rm d}x &\leq \int_{|x|<R} |f(x)\varphi(x)|\,{\rm d}x + \int_{|x|\geq R} |x|^{-m}|f(x)||x|^m|\varphi(x)|\,{\rm d}x \\ &\leq C \sup_{x\in\mathbb{R}^n,\,0\leq k\leq N} \big[|x|^k|\varphi(x)|\big] \left[\int_{|x|<R} |f(x)|\,{\rm d}x\right. \\ &\quad \left. + \int_{|x|\geq R} |x|^{-m}|f(x)|\,{\rm d}x\right] \leq C \sup_{x\in\mathbb{R}^n,\,0\leq k\leq N} \big[|x|^k|\varphi(x)|\big], \end{aligned} \tag{4.1.5}$$

where we have used the easily checked fact that, since $N \geq m$, we have

$$\sup_{|x|\geq R} \big[|x|^m|\varphi(x)|\big] \leq R^{m-N} \sup_{x\in\mathbb{R}^n} \big[|x|^N|\varphi(x)|\big]. \tag{4.1.6}$$

Estimate (4.1.5) shows that the functional u_f is well-defined on $\mathcal{S}(\mathbb{R}^n)$. Clearly, u_f is linear, and (4.1.5) also implies (based on Fact 4.1) that u_f is continuous on $\mathcal{S}(\mathbb{R}^n)$. Hence, u_f is a tempered distribution.

In the sequel, we will often use the notation f instead of u_f, whenever f is such that the operator u_f as in (4.1.4) is linear and continuous.

Exercise 4.4. *Prove that if $a \in (-n,\infty)$ then $|x|^a$ is a tempered distribution in $\mathbb{R}^n$.*

Exercise 4.5. *Let $f : \mathbb{R}^n \to \mathbb{C}$ be a Lebesgue measurable function with the property that there exists $m \in \mathbb{R}$ such that*

$$\int_{\mathbb{R}^n} (1+|x|)^m|f(x)|\,{\rm d}x < \infty. \tag{4.1.7}$$

Then the mapping u_f as in (4.1.4) is well-defined and is a tempered distribution.

Hint: Use Example 4.3.

Exercise 4.6. *Prove that $\mathcal{L}(\mathbb{R}^n)$, the space of slowly increasing functions defined in (3.1.22), is contained in the space of tempered distributions.*

Example 4.7. Let $p \in [1,\infty]$ and $f \in L^p(\mathbb{R}^n)$ be arbitrary. We claim that u_f defined in (4.1.4) is a tempered distribution. To prove this claim, since f is measurable, by Exercise 4.5, it suffices to show that f satisfies (4.1.7). If $p = 1$, then (4.1.7) holds for $m = 0$, while if $p = \infty$, (4.1.7) holds for any $m < -n$. If $p \in (1,\infty)$, by applying Hölder's inequality, we have

$$\int_{\mathbb{R}^n} (1+|x|)^m |f(x)| \mathrm{d}x \leq \|f\|_{L^p(\mathbb{R}^n)} \left(\int_{\mathbb{R}^n} (1+|x|)^{\frac{mp}{p-1}} \, \mathrm{d}x \right)^{\frac{p-1}{p}} < \infty$$

provided we choose $m < -\frac{n(p-1)}{p}$. In summary, we proved that

$$\begin{array}{c} \text{for each } p \in [1,\infty] \text{ the space } L^p(\mathbb{R}^n) \text{ is} \\ \text{a subspace of the space of tempered distributions.} \end{array} \tag{4.1.8}$$

As an immediate consequence of Exercise 3.5 and (4.1.8) we therefore obtain

$$\mathcal{S}(\mathbb{R}^n) \text{ is a subspace of the space of tempered distributions.} \tag{4.1.9}$$

Example 4.8. From Example 2.9 we know that $\ln|x| \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ and, hence, defines a distribution in $\mathbb{R}^n$. We claim that the distribution $\ln|x|$ is, in fact, a tempered distribution. Indeed, this follows from Exercise 4.5, since (4.1.7) holds for any $m < -n$ (something seen by using estimate (2.1.9) with $0 < \varepsilon < \min\{n, -m-n\}$).

The topology we consider on the space of tempered distributions is the weak$*$-topology and we denote this topological vector space by $\mathcal{S}'(\mathbb{R}^n)$. In particular, from the general discussion in Sect. 13.1 we have:

Fact 4.9. *$\mathcal{S}'(\mathbb{R}^n)$ is a locally convex topological space.*

Fact 4.10. *A sequence $\{u_j\}_{j\in\mathbb{N}} \subset \mathcal{S}'(\mathbb{R}^n)$ converges to some $u \in \mathcal{S}'(\mathbb{R}^n)$ in $\mathcal{S}'(\mathbb{R}^n)$ if and only if $\langle u_j, \varphi\rangle \xrightarrow[j\to\infty]{} \langle u, \varphi\rangle$ for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$, in which case we write $u_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} u$.*

It is useful to note that, for each $p \in [1,\infty]$, the space of pth-power integrable functions in $\mathbb{R}^n$ embeds continuously in the space of tempered distributions.

Exercise 4.11. *Prove that if $p \in [1,\infty]$ and a sequence of functions $\{f_j\}_{j\in\mathbb{N}}$ in $L^p(\mathbb{R}^n)$ converges in $L^p(\mathbb{R}^n)$ to some $f \in L^p(\mathbb{R}^n)$ then $f_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} f$.*

Hint: Use Hölder's inequality and (4.1.8).

Exercise 4.12. *Assume that ϕ is as in (1.2.3) and recall the sequence of functions $\{\phi_j\}_{j\in\mathbb{N}}$ from (1.3.7). Prove that $\phi_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} \delta$.*

Hint: Reason as in Example 2.20.

Theorem 4.13. *For each $n, m \in \mathbb{N}$ the following statements are true.*

(a) $\mathcal{E}'(\mathbb{R}^n) \hookrightarrow \mathcal{S}'(\mathbb{R}^n) \hookrightarrow \mathcal{D}'(\mathbb{R}^n)$, *where all the embeddings are injective and continuous.*

(b) For each $a \in \mathcal{L}(\mathbb{R}^n)$ and each $u \in \mathcal{S}'(\mathbb{R}^n)$, the distribution $au \in \mathcal{D}'(\mathbb{R}^n)$ extends uniquely to a tempered distribution, which will be denoted by au and its action is given by $\langle au, \varphi\rangle = \langle u, a\varphi\rangle$ for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$. Moreover, the mapping

$$\mathcal{S}'(\mathbb{R}^n) \ni u \mapsto au \in \mathcal{S}'(\mathbb{R}^n) \tag{4.1.10}$$

is linear, and continuous.

(c) For every $\alpha \in \mathbb{N}_0^n$ and each $u \in \mathcal{S}'(\mathbb{R}^n)$, the distribution $\partial^\alpha u \in \mathcal{D}'(\mathbb{R}^n)$ extends uniquely to a tempered distribution, that will be denoted by $\partial^\alpha u$ and its action is given by

$$\langle \partial^\alpha u, \varphi\rangle = (-1)^{|\alpha|}\langle u, \partial^\alpha \varphi\rangle \quad \text{for every} \quad \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.1.11}$$

Moreover, the mapping

$$\mathcal{S}'(\mathbb{R}^n) \ni u \mapsto \partial^\alpha u \in \mathcal{S}'(\mathbb{R}^n) \tag{4.1.12}$$

is linear and continuous.

(d) If $u \in \mathcal{S}'(\mathbb{R}^m)$ and $v \in \mathcal{S}'(\mathbb{R}^n)$ then the distribution $u \otimes v \in \mathcal{D}'(\mathbb{R}^m \times \mathbb{R}^n)$ extends uniquely to a tempered distribution, that will be denote by $u \otimes v$. In addition,

$$\langle u\otimes v, \varphi_1 \otimes \varphi_2\rangle = \langle u, \varphi_1\rangle\langle v, \varphi_2\rangle, \quad \forall\, \varphi_1 \in \mathcal{S}(\mathbb{R}^m),\ \forall\, \varphi_2 \in \mathcal{S}(\mathbb{R}^n), \tag{4.1.13}$$

and $u \otimes v = v \otimes u$ for every $u \in \mathcal{S}'(\mathbb{R}^m)$ and $v \in \mathcal{S}'(\mathbb{R}^n)$. Moreover, the mapping

$$\mathcal{S}'(\mathbb{R}^m) \times \mathcal{S}'(\mathbb{R}^n) \ni (u, v) \mapsto u \otimes v \in \mathcal{S}'(\mathbb{R}^m \times \mathbb{R}^n) \tag{4.1.14}$$

is bilinear and separately continuous.

Proof. The statement in *(a)* is a consequence of parts *(c)* and *(d)* in Theorem 3.13 and duality (cf. Proposition 13.3). Next, fix a function $a \in \mathcal{L}(\mathbb{R}^n)$ and let $u \in \mathcal{S}'(\mathbb{R}^n)$ be arbitrary. Then based on part *(a)* we have $u \in \mathcal{D}'(\mathbb{R}^n)$. Hence, by Proposition 2.23, au exists and belongs to $\mathcal{D}'(\mathbb{R}^n)$. We will show that au may be extended uniquely to a tempered distribution. Define

$$\widetilde{au} : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}, \qquad \widetilde{au}(\varphi) := \langle u, a\varphi\rangle \quad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.1.15}$$

Since $\widetilde{au}$ is the composition of $u \in \mathcal{S}'(\mathbb{R}^n)$ with the map in part *(a)* of Theorem 3.13, both of which are linear and continuous, it follows that $\widetilde{au} \in \mathcal{S}'(\mathbb{R}^n)$. In addition, $\widetilde{au}\Big|_{C_0^\infty(\mathbb{R}^n)} = au$, so if we invoke *(d)* in Theorem 3.13, we obtain that $\widetilde{au}$ is the unique continuous extension of au to $\mathcal{S}(\mathbb{R}^n)$.

The map in (4.1.10) is also continuous as seen from Theorem 3.13 and the general fact that the transpose of any linear and continuous operator between two topological vector spaces is continuous at the level of dual spaces equipped with weak∗-topologies (cf. Proposition 13.1). Re-denoting $\widetilde{au}$ simply as au now finishes the proof of the statement in *(b)*. The proof of *(c)* is similar to the proof of *(b)*.

Turning our attention to *(d)*, let $u \in \mathcal{S}'(\mathbb{R}^m)$ and $v \in \mathcal{S}'(\mathbb{R}^n)$. By part *(a)* we have $u \in \mathcal{D}'(\mathbb{R}^m)$ and $v \in \mathcal{D}'(\mathbb{R}^n)$, hence (by Theorem 2.78) $u \otimes v \in \mathcal{D}'(\mathbb{R}^m \times \mathbb{R}^n)$. We construct an extension $\widetilde{u \otimes v} : \mathcal{S}(\mathbb{R}^{n+m}) \to \mathbb{C}$ by setting

$$\big\langle \widetilde{u \otimes v}, \varphi \big\rangle := \big\langle u(x), \langle v(y), \varphi(x,y)\rangle \big\rangle, \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^m \times \mathbb{R}^n). \tag{4.1.16}$$

To see that this mapping is in $\mathcal{S}'(\mathbb{R}^m \times \mathbb{R}^n)$, fix $\varphi \in \mathcal{S}(\mathbb{R}^n)$. Clearly, we have $\varphi(x,\cdot) \in \mathcal{S}(\mathbb{R}^n)$ for each $x \in \mathbb{R}^m$. We claim that

$$\psi(x) := \langle v(y), \varphi(x,y)\rangle \ \text{ for each } \ x \in \mathbb{R}^m \implies \psi \in \mathcal{S}(\mathbb{R}^m). \tag{4.1.17}$$

Indeed, by reasoning as in the proof of Proposition 2.76, we obtain $\psi \in C^\infty(\mathbb{R}^m)$. Also, for any $\alpha, \beta \in \mathbb{N}_0^m$ we have $x^\beta \partial^\alpha \psi(x) = \big\langle v(y), x^\beta \partial_x^\alpha \varphi(x,y)\big\rangle$ and, since $v \in \mathcal{S}'(\mathbb{R}^n)$, there exist $C > 0$, and $\ell,\, k \in \mathbb{N}_0$, such that v satisfies (4.1.2). Thus, for every $x \in \mathbb{R}^m$, we have

$$\big|x^\beta \partial^\alpha \psi(x)\big| = \big|\big\langle v(y), x^\beta \partial_x^\alpha \varphi(x,y)\big\rangle\big| \leq C \sup_{\substack{y\in\mathbb{R}^n \\ |\gamma|\leq \ell,\, |\delta|\leq k}} \big|y^\gamma x^\beta \partial_x^\alpha \partial_y^\delta \varphi(x,y)\big|. \tag{4.1.18}$$

Therefore, since $\varphi \in \mathcal{S}(\mathbb{R}^m \times \mathbb{R}^n)$, estimate (4.1.18) further yields

$$\sup_{x\in\mathbb{R}^m} \big|x^\beta \partial^\alpha \psi(x)\big| \leq C \sup_{\substack{x\in\mathbb{R}^m,\, y\in\mathbb{R}^n \\ |\gamma|\leq \ell,\, |\delta|\leq k}} \big|y^\gamma x^\alpha \partial_x^\beta \partial_y^\delta \varphi(x,y)\big| < \infty. \tag{4.1.19}$$

This shows that $\psi \in \mathcal{S}(\mathbb{R}^m)$, finishing the proof of the claim. Consider next the mapping

$$\mathcal{S}(\mathbb{R}^m \times \mathbb{R}^n) \ni \varphi \mapsto \psi \in \mathcal{S}(\mathbb{R}^m). \tag{4.1.20}$$

This is linear by design, as well as continuous (as seen from (4.1.19), Fact 3.6, and Theorem 13.14). Thus, the composition between u and the map in (4.1.20) gives rise to a linear and continuous map, which proves that $\widetilde{u \otimes v}$ belongs to $\mathcal{S}'(\mathbb{R}^m \times \mathbb{R}^n)$. In addition $\widetilde{u \otimes v}\big|_{C_0^\infty(\mathbb{R}^m\times\mathbb{R}^n)} = u \otimes v$, which in combination with *(d)* in Theorem 3.13, implies that $\widetilde{u \otimes v}$ is the unique continuous extension of $u \otimes v$ to $\mathcal{S}(\mathbb{R}^m \times \mathbb{R}^n)$. Moreover, from (4.1.16) we see that

$$\big\langle \widetilde{u \otimes v}, \varphi_1 \otimes \varphi_2 \big\rangle = \langle u, \varphi_1\rangle \langle v, \varphi_2\rangle, \quad \forall\, \varphi_1 \in \mathcal{S}(\mathbb{R}^m),\ \forall\, \varphi_2 \in \mathcal{S}(\mathbb{R}^n). \tag{4.1.21}$$

Similarly, we define $\widetilde{v \otimes u} \in \mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^m)$. Recalling now part *(ii)* in Theorem 2.78 we may write

$$\widetilde{u \otimes v}\big|_{C_0^\infty(\mathbb{R}^m\times\mathbb{R}^n)} = u \otimes v = v \otimes u = \widetilde{v \otimes u}\big|_{C_0^\infty(\mathbb{R}^n\times\mathbb{R}^m)}. \tag{4.1.22}$$

Thus, using *(d)* in Theorem 3.13, we conclude $\widetilde{u\otimes v}=\widetilde{v\otimes u}$.

Clearly, $\mathcal{S}'(\mathbb{R}^m)\times\mathcal{S}'(\mathbb{R}^n)\ni(u,v)\mapsto\widetilde{u\otimes v}\in\mathcal{S}'(\mathbb{R}^m\times\mathbb{R}^n)$ is bilinear, and our goal is to show that this is also separately continuous. Since $\widetilde{u\otimes v}=\widetilde{v\otimes u}$ for every $u\in\mathcal{S}'(\mathbb{R}^m)$ and every $v\in\mathcal{S}'(\mathbb{R}^n)$, it suffices to prove that this map is continuous in the first variable. For this, we shall rely on the abstract description of open sets in the weak$*$-topology from (13.1.9). Specifically, having fixed $v\in\mathcal{S}'(\mathbb{R}^n)$, pick an arbitrary finite set $A\subset\mathcal{S}(\mathbb{R}^m\times\mathbb{R}^n)$ along with some number $\varepsilon\in(0,\infty)$, and introduce

$$\mathcal{O}_{A,\varepsilon}:=\big\{w\in\mathcal{S}'(\mathbb{R}^m\times\mathbb{R}^n):\,|\langle w,\psi\rangle|<\varepsilon,\ \forall\,\psi\in A\big\}. \tag{4.1.23}$$

If we now define $\widetilde{A}:=\big\{\langle v(y),\psi(\cdot,y)\rangle\,:\,\psi\in A\big\}$, then from what we proved earlier (cf. (4.1.17)) it follows that $\widetilde{A}$ is a subset of $\mathcal{S}'(\mathbb{R}^m)$. Also, $\widetilde{A}$ is finite since A is finite. Hence, if we now set

$$\widetilde{\mathcal{O}}_{\widetilde{A},\varepsilon}:=\big\{u\in\mathcal{S}'(\mathbb{R}^m):\,|\langle u,\varphi\rangle|<\varepsilon,\ \forall\,\varphi\in\widetilde{A}\big\}, \tag{4.1.24}$$

using (4.1.16) we have $\widetilde{u\otimes v}\in\mathcal{O}_{A,\varepsilon}$ for every $u\in\widetilde{\mathcal{O}}_{\widetilde{A},\varepsilon}$. In light of (13.1.9), this shows that $u\mapsto\widetilde{u\otimes v}$ is continuous.

Finally, abbreviating matters by simply writing $u\otimes v$ in place of $\widetilde{u\otimes v}$, all claims in part *(d)* of the statement of the theorem now follow. □

Remark 4.14.

(i) In view of (a) in Theorem 4.13 and (d) in Theorem 3.13 we have:

$$\text{if } u,v\in\mathcal{S}'(\mathbb{R}^n) \text{ and } u=v \text{ in } \mathcal{D}'(\mathbb{R}^n), \text{ then } u=v \text{ in } \mathcal{S}'(\mathbb{R}^n). \tag{4.1.25}$$

(ii) Whenever $u\in\mathcal{S}'(\mathbb{R}^n)$ its support is understood as defined in (2.5.7).

Remark 4.15. *The inclusion $\mathcal{S}'(\mathbb{R}^n)\hookrightarrow\mathcal{D}'(\mathbb{R}^n)$ (which goes to show that any tempered distribution is indeed an ordinary distribution) is actually strict. This follows from the observation that, in contrast to the case of ordinary distributions,*

$$L^1_{\rm loc}(\mathbb{R}^n)\not\subset\mathcal{S}'(\mathbb{R}^n). \tag{4.1.26}$$

To justify (4.1.26), take $\lambda>0$ arbitrary, fixed, and consider the function

$$f:\mathbb{R}^n\to\mathbb{C},\qquad f(x):=\mathrm{e}^{|x|^\lambda}\quad\text{for every}\quad x\in\mathbb{R}^n. \tag{4.1.27}$$

Clearly $f\in L^1_{\rm loc}(\mathbb{R}^n)$, hence $f\in\mathcal{D}'(\mathbb{R}^n)$. However, this distribution cannot be extended to a tempered distribution. To see why this is true, suppose there exists $u\in\mathcal{S}'(\mathbb{R}^n)$ such that $u\big|_{C^\infty_0(\mathbb{R}^n)}=f$ in $\mathcal{D}'(\mathbb{R}^n)$. Since u is a tempered distribution there exist some finite constant $C\geq0$ and numbers $k,m\in\mathbb{N}_0$ such that

$$\big|\langle u,\varphi\rangle\big|\leq C\sup_{x\in\mathbb{R}^n,\,|\alpha|\leq k,\,|\beta|\leq m}\big|x^\beta\partial^\alpha\varphi(x)\big|,\qquad\forall\,\varphi\in\mathcal{S}(\mathbb{R}^n). \tag{4.1.28}$$

Now fix a function

$$\psi\in C_0^\infty(\mathbb{R}^n),\quad \psi\geq 0,\quad \operatorname{supp}\psi\subseteq\overline{B(0,1)},\quad \int_{1/2\leq|x|\leq 1}\psi(x)\,dx=1, \tag{4.1.29}$$

and for each $j\in\mathbb{N}$ define $\psi_j(x):=\psi(x/j)$, for $x\in\mathbb{R}^n$. Then by (4.1.28), for each $j\in\mathbb{N}$, we have

$$\big|\langle u,\psi_j\rangle\big|\leq C\sup_{x\in\overline{B(0,j)},\,|\alpha|\leq k,\,|\beta|\leq m}\big|x^\beta\partial^\alpha\psi_j(x)\big|\leq C'j^{m-k}, \tag{4.1.30}$$

for some $C'\in[0,\infty)$ independent of j. On the other hand, (4.1.29) forces the lower bound

$$\begin{aligned}\big|\langle u,\psi_j\rangle\big| &= \int_{|x|\leq j}\mathrm{e}^{|x|^\lambda}\psi_j(x)\,dx\geq\int_{j/2\leq|x|\leq j}\mathrm{e}^{|x|^\lambda}\psi_j(x)\,dx\\ &\geq \mathrm{e}^{(j/2)^\lambda}j^{-n}\int_{1/2\leq|x|\leq 1}\psi(x)\,dx=\mathrm{e}^{(j/2)^\lambda}j^{-n},\qquad\forall\,j\in\mathbb{N}.\end{aligned} \tag{4.1.31}$$

Comparing (4.1.30) and (4.1.31) yields a contradiction choosing j large enough. Hence, f cannot be extended to a tempered distribution.

Remark 4.16. *What prevents locally integrable functions of the form (4.1.27) from belonging to $\mathcal{S}'(\mathbb{R}^n)$ is the fact that their growth at infinity is not tempered enough and, in fact, this observation justifies the very name "tempered distribution."*

Exercise 4.17. *Let $\lambda\in\mathbb{R}$ be such that $\lambda<n-1$, and fix some $j\in\{1,\dots,n\}$. For these choices, consider the functions f,g defined, respectively, by $f(x):=|x|^{-\lambda}$ and $g(x):=-\lambda x_j|x|^{-\lambda-2}$ for each $x\in\mathbb{R}^n\setminus\{0\}$. Prove that $f,g\in\mathcal{S}'(\mathbb{R}^n)$ and that $\partial_j f=g$ in $\mathcal{S}'(\mathbb{R}^n)$. In short,*

$$\begin{aligned}&\partial_j\big[|x|^{-\lambda}\big]=-\lambda x_j|x|^{-\lambda-2}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n)\\ &\text{whenever }\lambda<n-1\text{ and }j\in\{1,\dots,n\}.\end{aligned} \tag{4.1.32}$$

Hint: To prove (4.1.32) use (4.1.11), and integration by parts coupled with a limiting argument to extricate the singularity at the origin.

Given a tempered distribution, we may consider the convolution between its restriction to $C_0^\infty(\mathbb{R}^n)$ and any compactly supported distribution. A natural question, addressed in the next theorem, is whether the distribution obtained via such a convolution may be extended to a tempered distribution.

Theorem 4.18. *The following statements are true.*

(a) If $u\in\mathcal{E}'(\mathbb{R}^n)$ and $v\in\mathcal{S}'(\mathbb{R}^n)$, then the distribution $u*v\in\mathcal{D}'(\mathbb{R}^n)$ extends uniquely to a tempered distribution, that will be denoted by $u*v$. Also, $v*u\in\mathcal{D}'(\mathbb{R}^n)$ extends uniquely to a tempered distribution and $u*v=v*u$ in $\mathcal{S}'(\mathbb{R}^n)$. Moreover, the mapping

$$\mathcal{E}'(\mathbb{R}^n)\times\mathcal{S}'(\mathbb{R}^n)\ni(u,v)\mapsto u*v\in\mathcal{S}'(\mathbb{R}^n)$$

is bilinear, and for every $u\in\mathcal{E}'(\mathbb{R}^n)$, $v\in\mathcal{S}'(\mathbb{R}^n)$, we have

$$\partial^\alpha(u*v) = (\partial^\alpha u)*v = u*(\partial^\alpha v) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \quad \forall\, \alpha \in \mathbb{N}_0^n. \tag{4.1.33}$$

(b) If $u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} u$ *and there exists some compact set* $K \subset \mathbb{R}^n$ *such that* $\operatorname{supp} u_j \subset K$ *for every* $j \in \mathbb{N}$, *then* $u_j * v \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} u*v$ *for each* $v \in \mathcal{S}'(\mathbb{R}^n)$.

(c) If $v_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} v$, *then* $u * v_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} u*v$ *for each* $u \in \mathcal{E}'(\mathbb{R}^n)$.

(d) If $u \in \mathcal{S}'(\mathbb{R}^n)$ *then* $\delta * u = u = u * \delta$ *for each* $u \in \mathcal{S}'(\mathbb{R}^n)$.

(e) Let $a \in \mathcal{S}(\mathbb{R}^n)$ *and* $u \in \mathcal{S}'(\mathbb{R}^n)$. *Then the mapping*

$$a * u : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}, \quad (a*u)(\varphi) := \langle u,\, a^\vee * \varphi\rangle \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n), \tag{4.1.34}$$

is well-defined, linear, and continuous, hence $a*u$ *belongs to* $\mathcal{S}'(\mathbb{R}^n)$. *We also define* $u*a := a*u$. *If* $a \in C_0^\infty(\mathbb{R}^n)$, *then the map in* (4.1.34) *is the unique continuous extension of* $a*u \in \mathcal{D}'(\mathbb{R}^n)$ *to a tempered distribution. In addition, the mapping*

$$\mathcal{S}(\mathbb{R}^n) \times \mathcal{S}'(\mathbb{R}^n) \ni (a,u) \mapsto a*u \in \mathcal{S}'(\mathbb{R}^n) \tag{4.1.35}$$

is bilinear and separately sequentially continuous. Also, for every function $a \in \mathcal{S}(\mathbb{R}^n)$ *and every distribution* $u \in \mathcal{S}'(\mathbb{R}^n)$ *we have*

$$\partial^\alpha(a*u) = (\partial^\alpha a)*u = a*(\partial^\alpha u) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \quad \forall\, \alpha \in \mathbb{N}_0^n. \tag{4.1.36}$$

Proof. Fix $u \in \mathcal{E}'(\mathbb{R}^n)$ and $v \in \mathcal{S}'(\mathbb{R}^n)$. Because of *(a)* in Theorem 4.13 and Theorem 2.85, it follows that $u*v$ exists as an element in $\mathcal{D}'(\mathbb{R}^n)$. We will prove that $u*v$ extends uniquely to an element in $\mathcal{S}'(\mathbb{R}^n)$. Fix $\psi \in C_0^\infty(\mathbb{R}^n)$ such that $\psi = 1$ in a neighborhood of $\operatorname{supp} u$. Recall the notation introduces in (2.8.7) and let 1 denote the constant function equal to 1 in $\mathbb{R}^n$. We then claim that the map

$$\mathcal{S}(\mathbb{R}^n) \ni \varphi \mapsto (\psi\otimes 1)\varphi^\Delta \in \mathcal{S}(\mathbb{R}^n\times\mathbb{R}^n) \quad \text{is linear and continuous.} \tag{4.1.37}$$

Indeed, if $\varphi \in \mathcal{S}(\mathbb{R}^n)$, then $(\psi\otimes 1)\varphi^\Delta \in C^\infty(\mathbb{R}^n\times\mathbb{R}^n)$ and if $\alpha,\beta,\gamma,\delta \in \mathbb{N}_0^n$, then by Leibniz's formula (cf. Proposition 13.8), the binomial theorem (cf. Theorem 13.7), and the compactness of the support of ψ, we may write

$$\begin{aligned}
&\sup_{x\in\mathbb{R}^n,\, y\in\mathbb{R}^n} \left|x^\alpha y^\beta \partial_x^\gamma\big[\psi(x)\partial_y^\delta\varphi(x+y)\big]\right| \\
&\quad= \sup_{x\in\operatorname{supp}\psi,\, y\in\mathbb{R}^n} \left|x^\alpha y^\beta \partial_x^\gamma\big[\psi(x)(\partial^\delta\varphi)(x+y)\big]\right| \\
&\quad\le C\sum_{\mu\le\gamma} \sup_{x\in\operatorname{supp}\psi,\, y\in\mathbb{R}^n} \left|(y+x-x)^\beta(\partial^\mu\psi)(x)(\partial^{\gamma-\mu+\delta}\varphi)(x+y)\right|
\end{aligned}$$

$$\leq C\sum_{\mu\leq\gamma}\sum_{\eta\leq\beta}\sup_{x\in\operatorname{supp}\psi,\,y\in\mathbb{R}^n}\left|(y+x)^{\eta}(\partial^{\gamma-\mu+\delta}\varphi)(x+y)\right|$$

$$\leq C\sum_{\mu\leq\gamma}\sum_{\eta\leq\beta}\sup_{z\in\mathbb{R}^n}\left|z^{\eta}\partial^{\gamma-\mu+\delta}\varphi(z)\right|<\infty, \tag{4.1.38}$$

where all constants are independent of φ. Hence, $(\psi\otimes 1)\varphi^{\Delta}\in\mathcal{S}(\mathbb{R}^n\times\mathbb{R}^n)$ which proves that the map in (4.1.37) is well-defined. Since this map is also linear, it suffices to check its continuity at zero. This, in turn, follows from Fact 3.6, Theorem 13.14, and the fact that the final estimate in (4.1.38) implies that the map in (4.1.37) is sequentially continuous. If we now define

$$\begin{aligned}&\widetilde{u*v}:\mathcal{S}(\mathbb{R}^n)\longrightarrow\mathbb{C},\\ &\big(\widetilde{u*v}\big)(\varphi):=\big\langle u(x)\otimes v(y),\psi(x)\varphi(x+y)\big\rangle,\quad\forall\,\varphi\in\mathcal{S}(\mathbb{R}^n),\end{aligned} \tag{4.1.39}$$

then *(d)* in Theorem 4.13 combined with (4.1.37) gives that $\widetilde{u*v}$ belongs to the space $\mathcal{S}'(\mathbb{R}^n\times\mathbb{R}^n)$. In addition, this definition is independent of the choice of ψ selected as above. Indeed, if $\psi_1,\psi_2\in C_0^\infty(\mathbb{R}^n)$ are such that each equals 1 in some neighborhood of $\operatorname{supp}u$, then

$$\Big\langle u(x)\otimes v(y)\,,\,\big(\psi_1(x)-\psi_2(x)\big)\varphi(x+y)\Big\rangle=0\quad\text{for every}\quad\varphi\in\mathcal{S}(\mathbb{R}^n). \tag{4.1.40}$$

To see that the map in (4.1.39) is an extension of $u*v\in\mathcal{D}'(\mathbb{R}^n)$, take an arbitrary function $\varphi\in C_0^\infty(\mathbb{R}^n)$. If $\eta\in C_0^\infty(\mathbb{R}^n)$ is such that $\eta=1$ in a neighborhood of the set $\operatorname{supp}\varphi-\operatorname{supp}u$, then the function $\Psi(x,y):=\psi(x)\eta(y)$, for each $x,y\in\mathbb{R}^n$, satisfies $\Psi\in C_0^\infty(\mathbb{R}^n\times\mathbb{R}^n)$ and $\Psi=1$ in a neighborhood of the set $(\operatorname{supp}u\times\operatorname{supp}v)\cap\operatorname{supp}\varphi^{\Delta}$. Upon recalling (2.8.11), this permits us to write

$$\begin{aligned}\big\langle u(x)\otimes v(y),\psi(x)\varphi(x+y)\big\rangle&=\big\langle u(x)\otimes v(y),\Psi(x,y)\varphi(x+y)\big\rangle\\ &=\langle u*v,\varphi\rangle.\end{aligned} \tag{4.1.41}$$

This shows that the mapping in (4.1.39) is an extension of $u*v\in\mathcal{D}'(\mathbb{R}^n)$ which, together with *(d)* in Theorem 3.13, implies that $\widetilde{u*v}$ is the unique extension of $u*v\in\mathcal{D}'(\mathbb{R}^n)$ to a tempered distribution.

A similar construction realizes $\widetilde{v*u}$ as a tempered distribution. Moreover, since $u*v=v*u$ in $\mathcal{D}'(\mathbb{R}^n)$, from (4.1.25) we deduce that $\widetilde{u*v}=\widetilde{v*u}$ in $\mathcal{S}'(\mathbb{R}^n)$. It is also clear from (4.1.39) that the mapping

$$\mathcal{E}'(\mathbb{R}^n)\times\mathcal{S}'(\mathbb{R}^n)\ni(u,v)\mapsto\widetilde{u*v}\in\mathcal{S}'(\mathbb{R}^n)$$

is bilinear. After dropping the tilde, all claims in statement *(a)* now follow, with the exception of (4.1.33). Regarding the latter, we first note that that the equalities in (4.1.33) hold when interpreted $\mathcal{D}'(\mathbb{R}^n)$, thanks to part *(e)* in Theorem 2.87. Given that all distributions involved are tempered, we may invoke (4.1.25) to conclude that the named equalities also hold in $\mathcal{S}'(\mathbb{R}^n)$.

Moving on, let $u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} u$ be such that there exists a compact set $K \subset \mathbb{R}^n$ with the property that $\operatorname{supp} u_j \subseteq K$ for each $j \in \mathbb{N}$. Then, by Exercise 2.65, we have $\operatorname{supp} u \subseteq K$ and $u_j \xrightarrow[j\to\infty]{\mathcal{E}'(\mathbb{R}^n)} u$. The latter combined with *(a)* in Theorem 4.13 implies $u_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} u$. Fix now an arbitrary $v \in \mathcal{S}'(\mathbb{R}^n)$. From the current part *(a)* it follows that $u * v \in \mathcal{S}'(\mathbb{R}^n)$ and $u_j * v \in \mathcal{S}'(\mathbb{R}^n)$, for every $j \in \mathbb{N}$. If $\varphi \in \mathcal{S}(\mathbb{R}^n)$ then definition (4.1.39) implies that

$$\langle u_j * v, \varphi\rangle = \big\langle u_j(x) \otimes v(y), \psi(x)\varphi(x+y)\big\rangle, \qquad \forall\, j \in \mathbb{N}, \tag{4.1.42}$$

where $\psi \in C_0^\infty(\mathbb{R}^n)$ is a fixed function chosen so that $\psi = 1$ in a neighborhood of K. The proof of (4.1.37) shows that $(\psi\otimes 1)\varphi^\Delta \in \mathcal{S}(\mathbb{R}^n \times \mathbb{R}^n)$. Since the map in (4.1.14) is separately continuous, we may write

$$\langle u_j * v, \varphi\rangle = \big\langle u_j \otimes v, (\psi\otimes 1)\varphi^\Delta\big\rangle \xrightarrow[j\to\infty]{} \big\langle u\otimes v, (\psi\otimes 1)\varphi^\Delta\big\rangle = \langle u * v, \varphi\rangle. \tag{4.1.43}$$

This proves the statement in *(b)*. The proof of the statement in *(c)* is similar. Also the statement in *(d)* is an immediate consequence of part *(d)* in Theorem 2.87 and (4.1.25).

Turning our attention to the statement in *(e)*, we first note that, as noted in Exercise 3.25, the mapping $\mathcal{S}(\mathbb{R}^n) \ni a \mapsto a^\vee \in \mathcal{S}(\mathbb{R}^n)$ is well-defined, linear, and continuous. Based on this and *(f)* in Theorem 3.13 we may then conclude that the map in (4.1.34) is well-defined, linear and continuous, as a composition of linear and continuous maps. Assume next that $a \in C_0^\infty(\mathbb{R}^n)$. Then by Proposition 2.93, for each $u \in \mathcal{D}'(\mathbb{R}^n)$ we have $a * u \in C^\infty(\mathbb{R}^n)$ and $(a*u)(x) = \langle u(y), a(x-y)\rangle$ for every $x \in \mathbb{R}^n$, hence

$$\begin{aligned}\langle a * u, \varphi\rangle &= \int_{\mathbb{R}^n} \langle u(y), a(x-y)\rangle\varphi(x)\,\mathrm{d}x \\ &= \Big\langle u(y), \int_{\mathbb{R}^n} a(x-y)\varphi(x)\mathrm{d}x\Big\rangle = \langle u, a^\vee * \varphi\rangle, \quad \forall\,\varphi \in C_0^\infty(\mathbb{R}^n).\end{aligned} \tag{4.1.44}$$

This shows that, in this case, the map in (4.1.34) is an extension of $a*u \in \mathcal{D}'(\mathbb{R}^n)$ to a tempered distribution. The uniqueness of such an extension is then seen from *(d)* in Theorem 3.13.

Regarding the map in (4.1.35), its bilinearity is immediate. If $a_j \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} a$ then $a_j^\vee \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} a^\vee$, so by *(f)* in Theorem 3.13, $a_j^\vee * \varphi \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} a^\vee * \varphi$ for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$, hence

$$\langle u, a_j^\vee * \varphi\rangle \xrightarrow[j\to\infty]{} \langle u, a^\vee * \varphi\rangle \quad \text{for each } u \in \mathcal{S}'(\mathbb{R}^n),$$

proving that the map in (4.1.35) is sequentially continuous in the first variable. Moreover, if $u_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} u$ and $a \in \mathcal{S}(\mathbb{R}^n)$, then

$$\langle a*u_j, \varphi\rangle = \langle u_j, a^\vee * \varphi\rangle \xrightarrow[j\to\infty]{} \langle u, a^\vee * \varphi\rangle = \langle a*u, \varphi\rangle \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.1.45}$$

Thus, $a * u_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} a * u$. This finishes the proof of the fact that the mapping (4.1.35) is bilinear and separately sequentially continuous. Finally, (4.1.36) is a consequence of (4.1.33), part *(d)* in Theorem 3.13, the separate sequential continuity of (4.1.35), and part *(c)* in Theorem 4.13. This completes the proof of the theorem. □

In the last part of this section we present a density result.

Proposition 4.19. *The space $C_0^\infty(\mathbb{R}^n)$ is sequentially dense in $\mathcal{S}'(\mathbb{R}^n)$. In particular, the Schwartz class $\mathcal{S}(\mathbb{R}^n)$ is sequentially dense in $\mathcal{S}'(\mathbb{R}^n)$.*

Proof. Pick an arbitrary $u \in \mathcal{S}'(\mathbb{R}^n)$. Fix a function $\psi \in C_0^\infty(\mathbb{R}^n)$ such that $\psi = 1$ on $B(0,1)$ and, for each $j \in \mathbb{N}$, define $\psi_j(x) := \psi(x/j)$ for every $x \in \mathbb{R}^n$. We claim that

$$\psi_j u \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} u. \tag{4.1.46}$$

Indeed, if $\varphi \in \mathcal{S}(\mathbb{R}^n)$ is arbitrary, then by reasoning as in the proof of (3.1.28) we deduce that $\psi_j\varphi \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} \varphi$. Hence,

$$\langle \psi_j u, \varphi\rangle = \langle u, \psi_j\varphi\rangle \xrightarrow[j\to\infty]{} \langle u, \varphi\rangle, \tag{4.1.47}$$

from which (4.1.46) follows. In light of (4.1.46) it suffices to show that any tempered distribution with compact support is the limit in $\mathcal{S}'(\mathbb{R}^n)$ of a sequence from $C_0^\infty(\mathbb{R}^n)$. To this end, suppose that $u \in \mathcal{S}'(\mathbb{R}^n)$ is compactly supported. Also, choose a function ϕ as in (1.2.3) and recall the sequence $\{\phi_j\}_{j\in\mathbb{N}} \subset C_0^\infty(\mathbb{R}^n)$ from (1.3.7). Exercise 4.12 then gives $\phi_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} \delta$ which, in concert with parts *(d)–(e)* in Theorem 4.18, implies that $\phi_j * u \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)} u$. At this stage there remains to observe that $\phi_j * u \in C_0^\infty(\mathbb{R}^n)$ for every $j \in \mathbb{N}$, thanks to (2.8.44). □

4.2 The Fourier Transform Acting on Tempered Distributions

The goal here is to extend the action of the Fourier transform, considered earlier on the Schwartz class of rapidly decreasing functions, to the space of tempered distributions. To get some understanding of how this can be done in a natural fashion, we begin by noting that if $f \in L^1(\mathbb{R}^n)$ is arbitrary then (3.2.27) gives

$$\int_{\mathbb{R}^n} \widehat{f}(\xi)\varphi(\xi)\,d\xi = \int_{\mathbb{R}^n} f(x)\widehat{\varphi}(x)\,dx, \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.2.1}$$

Since $f \in L^1(\mathbb{R}^n)$ and $\widehat{f} \in L^\infty(\mathbb{R}^n)$, based on (4.1.8) we have $f, \widehat{f} \in \mathcal{S}'(\mathbb{R}^n)$. Hence, (4.2.1) may be rewritten as $\langle \widehat{f}, \varphi\rangle = \langle f, \widehat{\varphi}\,\rangle$, for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$. This

suggests the definition made below for the Fourier transform of a tempered distribution.

Proposition 4.20. *Let $u \in \mathcal{S}'(\mathbb{R}^n)$. Then the mapping*

$$\widehat{u} : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}, \quad \widehat{u}(\varphi) := \langle u, \widehat{\varphi} \rangle, \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n), \tag{4.2.2}$$

is well-defined, linear and continuous. Hence, $\widehat{u} \in \mathcal{S}'(\mathbb{R}^n)$.

Proof. This is an immediate consequence of the fact that $u \in \mathcal{S}'(\mathbb{R}^n)$, part *(c)* in Theorem 3.20, and the identity $\widehat{u} = u \circ \mathcal{F}$, where $\mathcal{F}$ is the Fourier transform on $\mathcal{S}(\mathbb{R}^n)$. □

Example 4.21. Since $\delta \in \mathcal{S}'(\mathbb{R}^n)$ we may write

$$\langle \widehat{\delta}, \varphi \rangle = \langle \delta, \widehat{\varphi} \rangle = \widehat{\varphi}(0) = \int_{\mathbb{R}^n} \varphi(x)\, dx = \langle 1, \varphi \rangle, \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n), \tag{4.2.3}$$

thus

$$\widehat{\delta} = 1 \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.2.4}$$

Also, if $x_0 \in \mathbb{R}^n$, then for each $\varphi \in \mathcal{S}(\mathbb{R}^n)$ we may write

$$\langle \widehat{\delta_{x_0}}, \varphi \rangle = \langle \delta_{x_0}, \widehat{\varphi} \rangle = \widehat{\varphi}(x_0) = \int_{\mathbb{R}^n} \mathrm{e}^{-ix_0 \cdot x} \varphi(x)\, dx = \langle \mathrm{e}^{-ix_0 \cdot x}, \varphi \rangle, \tag{4.2.5}$$

proving

$$\widehat{\delta_{x_0}} = \mathrm{e}^{-ix_0 \cdot x} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.2.6}$$

Remark 4.22. *The map $\mathcal{S}'(\mathbb{R}^n) \ni u \mapsto \widehat{u} \in \mathcal{S}'(\mathbb{R}^n)$ (which is further analyzed in part (a) of Theorem 4.25) is an extension of the map in (3.1.3). Indeed, if $f \in L^1(\mathbb{R}^n)$, then the map u_f from (4.1.4) is a tempered distribution and we may write*

$$\begin{aligned} \langle \widehat{u_f}, \varphi \rangle = \langle u_f, \widehat{\varphi} \rangle &= \int_{\mathbb{R}^n} f(x) \widehat{\varphi}(x)\, dx = \int_{\mathbb{R}^n} \widehat{f}(\xi) \varphi(\xi)\, d\xi \\ &= \langle \widehat{f}, \varphi \rangle \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n), \end{aligned} \tag{4.2.7}$$

where $\widehat{f}$ is as in (3.1.1) and for the third equality we used (3.2.27). This shows that

$$\widehat{u_f} = u_{\widehat{f}} \qquad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \quad \forall\, f \in L^1(\mathbb{R}^n). \tag{4.2.8}$$

In particular, (4.2.8) also holds if $f \in \mathcal{S}(\mathbb{R}^n)$ (since $\mathcal{S}(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$ by Exercise 3.5).

Example 4.23. Suppose $a \in (0, \infty)$ and consider the function $f(x) := \frac{1}{x^2 + a^2}$ for every $x \in \mathbb{R}$. Since $f \in L^1(\mathbb{R})$, by (4.1.8) we may regard f as a tempered distribution. The goal is to compute $\widehat{f}$ in $\mathcal{S}'(\mathbb{R})$. From (4.2.8) we know that this is the same as the Fourier transform of f as a function in $L^1(\mathbb{R})$, that is,

$$\widehat{f}(\xi) = \int_{\mathbb{R}} \frac{\mathrm{e}^{-ix\xi}}{x^2 + a^2}\, dx, \qquad \forall\, \xi \in \mathbb{R}. \tag{4.2.9}$$

To compute the integral in (4.2.9) we use the calculus of residues applied to the function

$$g(z) := \frac{\mathrm{e}^{-iz\xi}}{z^2+a^2} \quad \text{for} \quad z \in \mathbb{C} \setminus \{ia, -ia\}. \tag{4.2.10}$$

In this regard, denote the residue of g at z by $\mathrm{Res}_g(z)$. We separate the computation in two cases.

Case 1: Assume $\xi \in (-\infty, 0)$. For each $R \in (a, \infty)$ consider the contour $\Gamma := \Gamma_1 \cup \Gamma_2$, where

$$\Gamma_1 := \big\{t : -R \le t \le R\big\}, \tag{4.2.11}$$

$$\Gamma_2 := \big\{R\mathrm{e}^{i\theta} : 0 \le \theta \le \pi\big\}. \tag{4.2.12}$$

The function g has one pole $z = ia$ enclosed by Γ and the residue theorem gives

$$\int_{\Gamma_1} g(z)\,\mathrm{d}z + \int_{\Gamma_2} g(z)\,\mathrm{d}z = 2\pi i\,\mathrm{Res}_g(ia). \tag{4.2.13}$$

Let us analyze each term in (4.2.13). First,

$$\mathrm{Res}_g(ia) = \frac{\mathrm{e}^{-iz\xi}}{z+ia}\Big|_{z=ia} = \frac{\mathrm{e}^{a\xi}}{2ia} = \frac{\mathrm{e}^{-a|\xi|}}{2ia}. \tag{4.2.14}$$

Second, given that for each fixed $\xi \in \mathbb{R}$ the function $\mathbb{R} \ni t \mapsto \frac{\mathrm{e}^{-it\xi}}{t^2+a^2} \in \mathbb{C}$ is absolutely integrable, Lebesgue's dominated convergence theorem gives

$$\int_{\Gamma_1} g(z)\,\mathrm{d}z = \int_{-R}^{R} \frac{\mathrm{e}^{-it\xi}}{t^2+a^2}\,\mathrm{d}t \xrightarrow[R\to\infty]{} \int_{\mathbb{R}} \frac{\mathrm{e}^{-it\xi}}{t^2+a^2}\,\mathrm{d}t = \widehat{f}(\xi). \tag{4.2.15}$$

Third,

$$\left|\int_{\Gamma_2} g(z)\,\mathrm{d}z\right| = \left|\int_0^{\pi} \frac{\mathrm{e}^{\xi R\sin\theta - i\xi R\cos\theta}}{R^2\mathrm{e}^{i2\theta}+a^2}\, iR\mathrm{e}^{i\theta}\,d\theta\right| \le \frac{C}{R} \xrightarrow[R\to\infty]{} 0, \tag{4.2.16}$$

since $\mathrm{e}^{\xi R\sin\theta} \le 1$ for $\theta \in [0,\pi]$ (recall that we are assuming $\xi < 0$). Hence, letting $R \to \infty$ in (4.2.13) and making use of (4.2.14)–(4.2.16), we arrive at

$$\widehat{f}(\xi) = 2\pi i\,\frac{\mathrm{e}^{-a|\xi|}}{2ia} = \frac{\pi}{a}\mathrm{e}^{-a|\xi|} \qquad \text{whenever } \xi \in (0,\infty). \tag{4.2.17}$$

Case 2: Assume $\xi \in (0, \infty)$. This time we consider a contour that encloses the pole $z = -ia$. Specifically, we keep Γ_1 as before with $R \in (a, \infty)$ and set

$$\Gamma_3 := \big\{R\mathrm{e}^{i\theta} : \pi \le \theta \le 2\pi\big\}. \tag{4.2.18}$$

Applying the residue theorem for the contour $\Gamma_1 \cup \Gamma_3$ we then obtain

$$-\int_{\Gamma_1} g(z)\,\mathrm{d}z + \int_{\Gamma_3} g(z)\,\mathrm{d}z = 2\pi i\,\mathrm{Res}_g(-ia) = \frac{\mathrm{e}^{-iz\xi}}{z-ia}\Big|_{z=-ia} = -\frac{\mathrm{e}^{-a|\xi|}}{2ia}. \tag{4.2.19}$$

A computation similar to that in (4.2.16) (note that $\mathrm{e}^{\xi R\sin\theta}\leq 1$ for $\theta\in[\pi,2\pi]$ and $\xi>0$) gives $\lim_{R\to\infty}\big|\int_{\Gamma_2}g(z)\,\mathrm{d}z\big|=0$, which when used in (4.2.19) yields $\widehat{f}(\xi)=\frac{\pi}{a}\mathrm{e}^{-a|\xi|}$ for $\xi\in(0,\infty)$.

Case 3: Assume $\xi=0$. It is immediate from (4.2.9) that

$$\widehat{f}(0)=\int_{\mathbb{R}}\frac{1}{x^2+a^2}\mathrm{d}x=\frac{\pi}{a}.$$

In summary, we have proved

$$\mathcal{F}\Big(\frac{1}{x^2+a^2}\Big)(\xi)=\frac{\pi}{a}\mathrm{e}^{-a|\xi|}\quad\text{for every }\xi\in\mathbb{R},\text{ as well as in }\mathcal{S}'(\mathbb{R}).\tag{4.2.20}$$

In the next example we compute the Fourier transform of a tempered distribution induced by a slowly growing function that is not absolutely integrable in $\mathbb{R}^n$.

Example 4.24. Let $a\in(0,\infty)$ and consider the function $f(x):=\mathrm{e}^{-ia|x|^2}$, $x\in\mathbb{R}^n$. It is not difficult to check that $f\in\mathcal{L}(\mathbb{R}^n)$; thus, by Exercise 4.6 the function f may be regarded as a tempered distribution (identifying it with $u_f\in\mathcal{S}'(\mathbb{R}^n)$ associated with f as in (4.1.4)). We claim that

$$\widehat{\mathrm{e}^{-ia|x|^2}}=\Big(\frac{\pi}{ia}\Big)^{\frac{n}{2}}\mathrm{e}^{i\frac{|\xi|^2}{4a}}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n).\tag{4.2.21}$$

To prove (4.2.21), fix $\varphi\in\mathcal{S}(\mathbb{R}^n)$ and starting with the definition of the Fourier transform on $\mathcal{S}'(\mathbb{R}^n)$ write

$$\begin{aligned}
\Big\langle\widehat{\mathrm{e}^{-ia|x|^2}},\varphi\Big\rangle&=\big\langle\mathrm{e}^{-ia|x|^2},\widehat{\varphi}\,\big\rangle=\int_{\mathbb{R}^n}\mathrm{e}^{-ia|\xi|^2}\widehat{\varphi}(\xi)\,\mathrm{d}\xi\\
&=\lim_{\varepsilon\to0^+}\int_{\mathbb{R}^n}\mathrm{e}^{-(ia+\varepsilon)|\xi|^2}\widehat{\varphi}(\xi)\,\mathrm{d}\xi=\lim_{\varepsilon\to0^+}\big\langle\,\widehat{\varphi},\,\mathrm{e}^{-(ia+\varepsilon)|\xi|^2}\,\big\rangle\\
&=\lim_{\varepsilon\to0^+}\big\langle\varphi,\,\mathcal{F}\big(\mathrm{e}^{-(ia+\varepsilon)|x|^2}\big)\big\rangle=\lim_{\varepsilon\to0^+}\Big\langle\varphi,\,\big(\tfrac{\pi}{ia+\varepsilon}\big)^{\frac{n}{2}}\mathrm{e}^{-\frac{|\xi|^2}{4(ia+\varepsilon)}}\Big\rangle\\
&=\lim_{\varepsilon\to0^+}\Big(\frac{\pi}{ia+\varepsilon}\Big)^{\frac{n}{2}}\int_{\mathbb{R}^n}\mathrm{e}^{-\frac{|\xi|^2}{4(ia+\varepsilon)}}\varphi(\xi)\,\mathrm{d}\xi\\
&=\Big(\frac{\pi}{ia}\Big)^{\frac{n}{2}}\int_{\mathbb{R}^n}\mathrm{e}^{-\frac{|\xi|^2}{4ia}}\varphi(\xi)\,\mathrm{d}\xi\\
&=\Big\langle\Big(\frac{\pi}{ia}\Big)^{\frac{n}{2}}\mathrm{e}^{i\frac{|\xi|^2}{4a}},\,\varphi\Big\rangle.
\end{aligned}\tag{4.2.22}$$

Above, the first equality is based on (4.2.2), while the second equality is a consequence of the way in which the slowly growing function $\mathrm{e}^{-ia|x|^2}$ is regarded as a tempered distribution. The third and second-to-last equalities are based on Lebesgue's dominated convergence theorem. In the fourth equality we interpret the Schwartz function $\widehat{\varphi}$ as being in $\mathcal{S}'(\mathbb{R}^n)$ and rely on the fact that

$e^{-(ia+\varepsilon)|x|^2} \in \mathcal{S}(\mathbb{R}^n)$ for each $\varepsilon > 0$ (as noted in Example 3.21). The fifth equality once uses Remark 4.22 and once again (4.2.2). The sixth equality is a consequence of formula (3.2.6), while the rest is routine.

Theorem 4.25. *The following statements are true.*

(a) The mapping $\mathcal{F} : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ defined by $\mathcal{F}(u) := \widehat{u}$, for every $u \in \mathcal{S}'(\mathbb{R}^n)$, is bijective and continuous, and its inverse is also continuous.

(b) $\widehat{D^\alpha u} = \xi^\alpha \widehat{u}$ for all $\alpha \in \mathbb{N}_0^n$ and all $u \in \mathcal{S}'(\mathbb{R}^n)$.

(c) $\widehat{x^\alpha u} = (-D)^\alpha \widehat{u}$ for all $\alpha \in \mathbb{N}_0^n$ and all $u \in \mathcal{S}'(\mathbb{R}^n)$.

(d) If $u \in \mathcal{S}'(\mathbb{R}^m)$ and $v \in \mathcal{S}'(\mathbb{R}^n)$, then $\widehat{u \otimes v} = \widehat{u} \otimes \widehat{v}$.

Proof. Recall from Theorem 3.24 that the map $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is linear, continuous, and bijective, and its inverse is also continuous. Since the transpose of this map [in the sense of (13.1.10)] is precisely the Fourier transform in the context of part *(a)* of the current statement, from Propositions 13.1 and 13.2 it follows that $\mathcal{F} : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is also well-defined, linear, continuous, and bijective, and its inverse is also continuous.

Consider next $u \in \mathcal{S}'(\mathbb{R}^n)$ and $\alpha \in \mathbb{N}_0^n$. Then for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$, using part *(c)* in Theorem 4.13, part *(a)* in Theorem 3.20, (4.2.2), and part *(b)* in Theorem 4.13, we may write

$$\begin{aligned}\langle \widehat{D^\alpha u}, \varphi\rangle &= \langle D^\alpha u, \widehat{\varphi}\,\rangle = (-1)^{|\alpha|}\langle u, D^\alpha\widehat{\varphi}\,\rangle = \langle u, \widehat{\xi^\alpha\varphi}\,\rangle \\ &= \langle \widehat{u}, \xi^\alpha\varphi\rangle = \langle \xi^\alpha\widehat{u}, \varphi\rangle,\end{aligned} \tag{4.2.23}$$

and

$$\begin{aligned}\langle \widehat{x^\alpha u}, \varphi\rangle &= \langle x^\alpha u, \widehat{\varphi}\,\rangle = \langle u, x^\alpha\widehat{\varphi}\,\rangle = \langle u, \widehat{D^\alpha\varphi}\,\rangle \\ &= \langle \widehat{u}, D^\alpha\varphi\rangle = \langle (-D)^\alpha\widehat{u}, \varphi\rangle.\end{aligned} \tag{4.2.24}$$

This proves the claims in *(b)* and *(c)*.

We are left with the proof of the statement in *(d)*. Fix $u \in \mathcal{S}'(\mathbb{R}^m)$ and $v \in \mathcal{S}'(\mathbb{R}^n)$. Then based on *(d)* in Theorem 4.13 we have $u \otimes v \in \mathcal{S}'(\mathbb{R}^m \times \mathbb{R}^n)$. Hence, starting with (4.2.2), then using *(d)* in Theorem 3.20 and (4.1.13), we may write

$$\begin{aligned}\langle \widehat{u\otimes v}, \varphi\otimes\psi\rangle &= \langle u\otimes v, \widehat{\varphi\otimes\psi}\,\rangle = \langle u\otimes v, \widehat{\varphi}\otimes\widehat{\psi}\,\rangle \\ &= \langle u, \widehat{\varphi}\,\rangle\langle v, \widehat{\psi}\,\rangle = \langle \widehat{u}, \varphi\rangle\langle \widehat{v}, \psi\rangle \\ &= \langle \widehat{u}\otimes\widehat{v}, \varphi\otimes\psi\rangle, \quad \forall\,\varphi\in\mathcal{S}(\mathbb{R}^m),\ \forall\,\psi\in\mathcal{S}(\mathbb{R}^n).\end{aligned} \tag{4.2.25}$$

Consequently, $\widehat{u\otimes v}\big|_{C_0^\infty(\mathbb{R}^m)\otimes C_0^\infty(\mathbb{R}^n)} = \widehat{u}\otimes\widehat{v}\big|_{C_0^\infty(\mathbb{R}^m)\otimes C_0^\infty(\mathbb{R}^n)}$, which in combination with Proposition 3.15 proves the statement in *(d)*. □

Exercise 4.26. *Recall* (4.1.4) *and* (4.1.8). *Prove that* $\widehat{u_f} = u_{\widehat{f}}$ *in* $\mathcal{S}'(\mathbb{R}^n)$ *for each* $f \in L^2(\mathbb{R}^n)$ *and each* $f \in L^1(\mathbb{R}^n)$.

Hint: Use (3.2.27) and Exercise 3.29.

Lemma 4.27 (Riemann–Lebesgue Lemma)**.** *If* $f \in L^1(\mathbb{R}^n)$, *then the tempered distribution* u_f *satisfies* $\widehat{u_f} \in C^0(\mathbb{R}^n)$.

Proof. This is a consequence of Exercise 4.26 and (3.1.3). □

Exercise 4.28. *Prove that for every* $f, g \in L^2(\mathbb{R}^n)$ *one has* $\widehat{f * g} = \widehat{f} \cdot \widehat{g}$ *in* $\mathcal{S}'(\mathbb{R}^n)$.

Hint: Use a density argument based on the formula from part *(c)* of Proposition 3.27, Exercise 4.11, Young's inequality (cf. Theorem 13.13) in the particular case when $p = q = 2$, and the fact that the Fourier transform is continuous both on $L^2(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$.

Example 4.29. Let $a \in (0, \infty)$. We are interested in computing the Fourier transform of the bounded function $\frac{x}{x^2+a^2}$, viewed as a distribution in $\mathcal{S}'(\mathbb{R})$. In this vein, consider the auxiliary function $f(x) := \frac{1}{x^2+a^2}$ for $x \in \mathbb{R}$, and recall from (4.2.20) that $\widehat{f}(\xi) = \frac{\pi}{a}\mathrm{e}^{-a|\xi|}$ in $\mathcal{S}'(\mathbb{R})$. This formula, part *(c)* in Theorem 4.25, and (2.4.10), allows us to write

$$\begin{aligned}\mathcal{F}\Big(\frac{x}{x^2+a^2}\Big)(\xi) &= \mathcal{F}\big(xf\big)(\xi) = i\frac{\mathrm{d}}{\mathrm{d}\xi}\big[\widehat{f}(\xi)\big] = i\frac{\mathrm{d}}{\mathrm{d}\xi}\Big(\frac{\pi}{a}\mathrm{e}^{-a|\xi|}\Big)\\ &= \frac{\pi i}{a}\Big(-a\,\mathrm{e}^{-a\xi}H(\xi) + a\,\mathrm{e}^{a\xi}H(-\xi)\Big)\\ &= -\pi i(\operatorname{sgn}\xi)\mathrm{e}^{-a|\xi|} \qquad \text{in} \quad \mathcal{S}'(\mathbb{R}). \end{aligned} \tag{4.2.26}$$

Proposition 4.30. *For each* $u \in \mathcal{S}'(\mathbb{R}^n)$ *define the mapping*

$$u^\vee : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}, \qquad u^\vee(\varphi) := \langle u, \varphi^\vee\rangle, \quad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.2.27}$$

Then $u^\vee \in \mathcal{S}'(\mathbb{R}^n)$ *and*

$$\widehat{\widehat{u}} = (2\pi)^n u^\vee. \tag{4.2.28}$$

Proof. Fix $u \in \mathcal{S}'(\mathbb{R}^n)$. Then (4.2.27) is simply the composition of u with the mapping from Exercise 3.25. Since both are linear and continuous, it follows that $u^\vee \in \mathcal{S}'(\mathbb{R}^n)$. Formula (4.2.28) then follows by combining (4.2.2) with the identity in part *(3)* of Exercise 3.26. □

Exercise 4.31. *Show that*

$$\widehat{1} = (2\pi)^n\delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.2.29}$$

Exercise 4.32. *Prove that for any tempered distribution* u *the following equivalence holds:*

$$u^\vee = -u \quad \text{in } \mathcal{S}'(\mathbb{R}^n) \iff (\widehat{u}\,)^\vee = -\widehat{u} \quad \text{in } \mathcal{S}'(\mathbb{R}^n). \tag{4.2.30}$$

One suggestive way of summarizing (4.2.30) is to say that a tempered distribution is odd if and only if its Fourier transform is odd.

Theorem 4.33. *The following statements are true.*

(a) If $a \in \mathcal{S}(\mathbb{R}^n)$ and $u \in \mathcal{S}'(\mathbb{R}^n)$, then $\widehat{u * a} = \widehat{a}\,\widehat{u}$ in $\mathcal{S}'(\mathbb{R}^n)$, where $\widehat{a}$ is viewed as an element from $\mathcal{S}(\mathbb{R}^n)$.

(b) If $u \in \mathcal{E}'(\mathbb{R}^n)$ then the tempered distribution $\widehat{u}$ is of function type given by the formula $\widehat{u}(\xi) = \big\langle u(x), \mathrm{e}^{-ix\cdot\xi}\big\rangle$ for every $\xi \in \mathbb{R}^n$, and $\widehat{u} \in \mathcal{L}(\mathbb{R}^n)$.

(c) If $u \in \mathcal{E}'(\mathbb{R}^n)$, $v \in \mathcal{S}'(\mathbb{R}^n)$ then $\widehat{u * v} = \widehat{u}\,\widehat{v}$, where $\widehat{u}$ is viewed as an element in $\mathcal{L}(\mathbb{R}^n)$.

Proof. By *(d)* in Theorem 4.18 we have $u * a \in \mathcal{S}'(\mathbb{R}^n)$, hence $\widehat{u * a}$ exists and belongs to $\mathcal{S}'(\mathbb{R}^n)$. Also, since $\widehat{a} \in \mathcal{S}(\mathbb{R}^n)$, by (3.1.23) and *(b)* in Theorem 4.13 it follows that $\widehat{a}\,\widehat{u} \in \mathcal{S}'(\mathbb{R}^n)$. Then, we may write

$$\begin{aligned}\big\langle \widehat{u * a}, \varphi\big\rangle &= \big\langle u * a, \widehat{\varphi}\,\big\rangle = \big\langle u, a^{\vee} * \widehat{\varphi}\,\big\rangle = (2\pi)^{-n}\big\langle u, \widehat{\widehat{a}} * \widehat{\varphi}\,\big\rangle \\ &= \big\langle u, \widehat{\widehat{a}\,\varphi}\big\rangle = \big\langle \widehat{u}, \widehat{a}\,\varphi\big\rangle = \big\langle \widehat{a}\,\widehat{u}, \varphi\big\rangle, \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n). \end{aligned} \tag{4.2.31}$$

For the second equality in (4.2.31) we used (4.1.34), for the third we used part *(3)* in Exercise 3.26, while for the forth we used *(d)* in Proposition 3.27. This proves the statement in *(a)*.

Moving on to the proof of *(b)*, fix some $u \in \mathcal{E}'(\mathbb{R}^n)$ and introduce the function $f(\xi) := \big\langle u(x), \mathrm{e}^{-ix\cdot\xi}\big\rangle$ for every $\xi \in \mathbb{R}^n$. From Proposition 2.82 it follows that $f \in C^\infty(\mathbb{R}^n)$ and

$$\partial^\alpha f(\xi) = \Big\langle u(x)\,,\, \partial_\xi^\alpha[\mathrm{e}^{-ix\cdot\xi}]\Big\rangle, \quad \forall\,\xi \in \mathbb{R}^n, \quad \text{for every} \quad \alpha \in \mathbb{N}_0^n. \tag{4.2.32}$$

In addition, since $u \in \mathcal{E}'(\mathbb{R}^n)$, there exist a compact subset K of $\mathbb{R}^n$, along with a constant $C \in (0,\infty)$, and a number $k \in \mathbb{N}_0$, such that u satisfies (2.6.6). Combining all these facts, for each $\alpha \in \mathbb{N}_0^n$, we may estimate

$$\begin{aligned}\big|\partial^\alpha f(\xi)\big| &= \big|\langle u(x), \partial_\xi^\alpha \mathrm{e}^{-ix\cdot\xi}\rangle\big| \leq C \sup_{x\in K,\,|\beta|\leq k}\big|\partial_x^\beta\partial_\xi^\alpha \mathrm{e}^{-ix\cdot\xi}\big| \\ &\leq C \sup_{x\in K,\,|\beta|\leq k}\big|x^\alpha\xi^\beta\big| \leq C(1+|\xi|)^k. \end{aligned} \tag{4.2.33}$$

From (4.2.33) and the fact that f is smooth we deduce that $f \in \mathcal{L}(\mathbb{R}^n)$. Hence, if we now recall Exercise 4.6, it follows that $f \in \mathcal{S}'(\mathbb{R}^n)$.

We are left with proving that $\widehat{u} = f$ as tempered distributions. To this end, fix $\theta \in C_0^\infty(\mathbb{R}^n)$ such that $\theta = 1$ in a neighborhood of $\operatorname{supp} u$. Then, for every $\varphi \in C_0^\infty(\mathbb{R}^n)$ one has

$$\langle \widehat{u}, \varphi\rangle = \langle u, \widehat{\varphi}\,\rangle = \big\langle u(\xi)\,,\, \theta(\xi)\widehat{\varphi}(\xi)\,\big\rangle = \Big\langle u(\xi), \int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi}\theta(\xi)\varphi(x)\,\mathrm{d}x\Big\rangle. \tag{4.2.34}$$

At this point recall Remark 2.79 (note that the function $\mathrm{e}^{-ix\cdot\xi}\theta(\xi)\varphi(x)$ belongs to $C_0^\infty(\mathbb{R}^n\times\mathbb{R}^n)$ and one may take $v=1$ in (2.7.45)) which allows one to rewrite the last term in (4.2.34) and conclude that

$$\langle\widehat{u},\varphi\rangle=\int_{\mathbb{R}^n}\big\langle u(\xi),\mathrm{e}^{-ix\cdot\xi}\big\rangle\varphi(x)\,dx=\langle f,\varphi\rangle,\qquad\forall\,\varphi\in C_0^\infty(\mathbb{R}^n).\tag{4.2.35}$$

Hence the tempered distributions $\widehat{u}$ and f coincide on $C_0^\infty(\mathbb{R}^n)$. By (4.1.25), we therefore have $\widehat{u}=f$ in $\mathcal{S}'(\mathbb{R}^n)$, completing the proof of the statement in *(b)*.

Regarding the formula in part *(c)*, while informally this is similar to the formula proved in part *(a)*, the computation in (4.2.31) through which the latter has been deduced no longer works in the current case, as various ingredients used to justify it break down (given that $\widehat{u}$ is now only known to belong to $\mathcal{L}(\mathbb{R}^n)$ and not necessarily to $\mathcal{S}(\mathbb{R}^n)$). This being said, we may employ what has been established in part *(a)* together with a limiting argument to get the desired result.

Specifically, assume that $u\in\mathcal{E}'(\mathbb{R}^n)$ and $v\in\mathcal{S}'(\mathbb{R}^n)$. Then $u*v\in\mathcal{S}'(\mathbb{R}^n)$ (recall statement *(a)* in Theorem 4.18) hence, $\widehat{u*v}$ is well-defined in $\mathcal{S}'(\mathbb{R}^n)$. Also, from *(b)* we have $\widehat{u}\in\mathcal{L}(\mathbb{R}^n)$, thus $\widehat{u}\,\widehat{v}$ is meaningful and $\widehat{u}\,\widehat{v}\in\mathcal{S}'(\mathbb{R}^n)$. To proceed, recall the sequence $\{\phi_j\}_{j\in\mathbb{N}}$ from Example 1.11. In particular, (1.3.8) holds and $\phi_j\xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)}\delta$ (see Example 2.20). Thus, the statement from part *(b)* in Theorem 4.18 applies and gives that $u*\phi_j\xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)}u*\delta=u$. Moreover, since $u\in\mathcal{E}'(\mathbb{R}^n)$, by (2.8.17) it follows that $u*\phi_j\in\mathcal{E}'(\mathbb{R}^n)$ and $\operatorname{supp}(u*\phi_j)\subseteq\operatorname{supp}u+B(0,1)$ for every $j\in\mathbb{N}$ (by part *(1)* in Remark 2.91). Hence, one may apply *(b)* in Theorem 4.18 to further conclude that $(u*\phi_j)*v\xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R}^n)}u*v$. Given *(a)* in Theorem 4.25, the latter implies (recall Fact 4.10)

$$\lim_{j\to\infty}\big\langle\mathcal{F}\big((u*\phi_j)*v\big),\varphi\big\rangle=\langle\widehat{u*v},\varphi\rangle,\qquad\forall\,\varphi\in\mathcal{S}(\mathbb{R}^n).\tag{4.2.36}$$

Note that (2.8.44) gives $u*\phi_j\in C_0^\infty(\mathbb{R}^n)$ for every $j\in\mathbb{N}$. Hence, based on what we have proved already in part *(a)*, we obtain (keeping in mind that $\widehat{u}\in\mathcal{L}(\mathbb{R}^n)$)

$$\mathcal{F}\big((u*\phi_j)*v\big)=\widehat{u*\phi_j}\,\widehat{v}=\widehat{\phi_j}\,\widehat{u}\,\widehat{v}=\widehat{\phi}(\cdot/j)\,\widehat{u}\,\widehat{v}\quad\text{ in }\quad\mathcal{S}'(\mathbb{R}^n),\qquad\forall j\in\mathbb{N}.\tag{4.2.37}$$

In concert, (4.2.36) and (4.2.37) yield that for each $\varphi\in\mathcal{S}(\mathbb{R}^n)$

$$\big\langle\widehat{u*v},\varphi\big\rangle=\lim_{j\to\infty}\big\langle\widehat{u}\,\widehat{v},\widehat{\phi}(\cdot/j)\,\varphi\big\rangle=\big\langle\widehat{u}\,\widehat{v},\widehat{\phi}(0)\,\varphi\big\rangle=\big\langle\widehat{u}\,\widehat{v},\varphi\big\rangle\tag{4.2.38}$$

since $\widehat{\phi}(\cdot/j)\,\varphi\xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)}\widehat{\phi}(0)\,\varphi$ by Exercise 3.14, and $\widehat{\phi}(0)=1$. This proves the statement in *(c)* and finishes the proof of the theorem. □

Example 4.34. If $a\in(0,\infty)$ then $\chi_{[-a,a]}$, the characteristic function of the interval $[-a,a]$, belongs to $\mathcal{E}'(\mathbb{R})$ and by statement *(b)* in Theorem 4.33 we have

$$\widehat{\chi_{[-a,a]}}(\xi)=\int_{-a}^{a}\mathrm{e}^{-ix\xi}\,dx=\begin{cases}2\frac{\sin(a\xi)}{\xi} & \text{for }\ \xi\in\mathbb{R}\setminus\{0\},\\ 2a & \text{for }\ \xi=0.\end{cases}\tag{4.2.39}$$

Exercise 4.35. *Use Exercise 2.65 and statement (b) in Theorem 4.33 to prove that if $u \in \mathcal{S}'(\mathbb{R}^n)$ is such that $\operatorname{supp}\widehat{u} \subseteq \{a\}$ for some $a \in \mathbb{R}^n$, then there exist $k \in \mathbb{N}_0$ and constants $c_\alpha \in \mathbb{C}$, for $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq k$, such that*

$$u = \sum_{|\alpha|\leq k} c_\alpha x^\alpha \mathrm{e}^{ix\cdot a} \qquad \text{in } \mathcal{S}'(\mathbb{R}^n). \tag{4.2.40}$$

In particular, if $a = 0$ then u is a polynomial in $\mathbb{R}^n$.

Example 4.36. Let $a \in \mathbb{R}$ and consider the function $f(x) := \sin(ax)$ for $x \in \mathbb{R}$. Then $f \in C^\infty(\mathbb{R}) \cap L^\infty(\mathbb{R})$, hence $f \in \mathcal{S}'(\mathbb{R})$. Suppose $a \neq 0$. We shall compute the Fourier transform of f in $\mathcal{S}'(\mathbb{R})$ by making use of a technique relying on the ordinary differential equation that f satisfies. More precisely, since $f'' + a^2 f = 0$ in $\mathbb{R}$, the same equation holds in $\mathcal{S}'(\mathbb{R})$, thus by *(a)* and *(b)* in Theorem 4.25 we have $(\xi^2 - a^2)\widehat{f} = 0$ in $\mathcal{S}'(\mathbb{R})$. By Example 2.72 this implies $\widehat{f} = C_1\delta_a + C_2\delta_{-a}$ in $\mathcal{S}'(\mathbb{R})$ for some $C_1, C_2 \in \mathbb{C}$. Applying again the Fourier transform to the last equation, using (4.2.28) and part *(b)* from Theorem 4.33, we have

$$\begin{aligned} 2\pi\sin(-ax) &= \widehat{\widehat{\sin(ax)}} = C_1\widehat{\delta_a} + C_2\widehat{\delta_{-a}} = C_1\langle\delta_a(\xi), \mathrm{e}^{-ix\xi}\rangle + C_2\langle\delta_{-a}(\xi), \mathrm{e}^{-ix\xi}\rangle \\ &= C_1\mathrm{e}^{-iax} + C_2\mathrm{e}^{iax} \\ &= (C_1 + C_2)\cos(ax) + i(C_1 - C_2)\sin(-ax). \end{aligned} \tag{4.2.41}$$

The resulting identity in (4.2.41) forces $C_1 = -i\pi$ and $C_2 = i\pi$. Plugging these constants back in the expression for $\widehat{f}$ yields

$$\widehat{\sin(ax)} = -i\pi\delta_a + i\pi\delta_{-a} \qquad \text{in } \mathcal{S}'(\mathbb{R}). \tag{4.2.42}$$

It is immediate that (4.2.42) also holds if $a = 0$.

Example 4.37. Let $a, b \in \mathbb{R}$. Then the function $g(x) := \sin(ax)\sin(bx)$ for $x \in \mathbb{R}$ satisfies $g \in L^\infty(\mathbb{R})$, thus $g \in \mathcal{S}'(\mathbb{R})$ (cf. (4.1.8)). Applying the Fourier transform to the identity in (4.2.42) and using (4.2.28) we obtain

$$\sin(ax) = \frac{i}{2}\widehat{\delta_a} - \frac{i}{2}\widehat{\delta_{-a}} \qquad \text{in } \mathcal{S}'(\mathbb{R}). \tag{4.2.43}$$

Also, making use of (4.2.43), part *(c)* in Theorem 4.33, and then Exercise 2.90, we may write

$$\begin{aligned} \sin(ax)\sin(bx) &= \Big(\frac{i}{2}\widehat{\delta_a} - \frac{i}{2}\widehat{\delta_{-a}}\Big)\Big(\frac{i}{2}\widehat{\delta_b} - \frac{i}{2}\widehat{\delta_{-b}}\Big) \\ &= -\frac{1}{4}\mathcal{F}\big(\delta_{a+b} - \delta_{a-b} - \delta_{b-a} + \delta_{-a-b}\big) \quad \text{in } \mathcal{S}'(\mathbb{R}). \end{aligned} \tag{4.2.44}$$

Hence, another application of the Fourier transform gives (relying on (4.2.28))

$$\mathcal{F}\big(\sin(ax)\sin(bx)\big) = -\frac{\pi}{2}\big(\delta_{a+b} - \delta_{a-b} - \delta_{b-a} + \delta_{-a-b}\big) \quad \text{in } \mathcal{S}'(\mathbb{R}). \tag{4.2.45}$$

Example 4.38. Let $a \in (0,\infty)$ and consider the function $f(x) := \mathrm{e}^{ia|x|^2}$ for $x \in \mathbb{R}^n$. Then $f \in L^\infty(\mathbb{R}^n)$; thus, $f \in \mathcal{S}'(\mathbb{R}^n)$ by (4.1.8). The goal is to compute the Fourier transform of f in $\mathcal{S}'(\mathbb{R}^n)$. The starting point is the observation that

$$f(x) = \mathrm{e}^{iax_1^2} \otimes \mathrm{e}^{iax_2^2} \otimes \cdots \otimes \mathrm{e}^{iax_n^2} \qquad \forall\, x = (x_1,\ldots,x_n) \in \mathbb{R}. \tag{4.2.46}$$

Invoking part *(d)* in Theorem 4.25 then reduces matters to computing the Fourier transform of f in the case $n = 1$.

Assume that $n = 1$, in which case $f(x) = \mathrm{e}^{iax^2}$ for $x \in \mathbb{R}$. Then f satisfies the differential equation: $f' - 2ia\,xf = 0$ in $\mathcal{S}'(\mathbb{R})$. Taking the Fourier transform in $\mathcal{S}'(\mathbb{R})$ and using the formulas from *(b)–(c)* in Theorem 4.25 we obtain

$$(\widehat{f}\,)' + \frac{i}{2a}\xi\widehat{f} = 0 \quad \text{in} \quad \mathcal{S}'(\mathbb{R}). \tag{4.2.47}$$

The format of (4.2.47) suggests that we consider the ordinary differential equation $y' + \frac{i}{2a}\xi y = 0$ in $\mathbb{R}$, in the unknown $y = y(\xi)$. One particular solution of this o.d.e. is $y(\xi) = \mathrm{e}^{-i\frac{\xi^2}{4a}}$. Note that both y and $1/y$ belong to $\mathcal{L}(\mathbb{R}^n)$. In particular, it makes sense to consider the tempered distribution $u := (1/y)\widehat{f}$, with the derivative

$$\begin{aligned} u' &= (1/y)'\widehat{f} + (1/y)(\widehat{f}\,)' = -(y'/y^2)\widehat{f} + (1/y)(\widehat{f}\,)' \\ &= \left(\left(\frac{i}{2a}\xi\right)/y\right)\widehat{f} - \frac{i}{2a}\xi(1/y)\widehat{f} = 0 \quad \text{in} \quad \mathcal{S}'(\mathbb{R}). \end{aligned} \tag{4.2.48}$$

From this and part *(2)* in Proposition 2.40 we then deduce that there exists $c \in \mathbb{C}$ such that $u = c$ in $\mathcal{S}'(\mathbb{R})$ which, given the significance of u and y, forces $\widehat{f} = c\,\mathrm{e}^{-i\frac{\xi^2}{4a}}$ in $\mathcal{S}'(\mathbb{R})$ for some $c \in \mathbb{C}$. We are therefore left with determining the constant c. This may be done by choosing a suitable Schwartz function and then computing the action of $\widehat{f}$ on it. Take $\varphi(\xi) := \mathrm{e}^{-\frac{\xi^2}{4a}}$ for $\xi \in \mathbb{R}$. Since $\widehat{\varphi}(x) = \sqrt{4a\pi}\mathrm{e}^{-ax^2}$ for $x \in \mathbb{R}$ (recall Example 3.21), we may write

$$c\int_{\mathbb{R}} \mathrm{e}^{-i\frac{\xi^2}{4a}-\frac{\xi^2}{4a}}\,\mathrm{d}\xi = \langle \widehat{f}, \varphi\rangle = \langle f, \widehat{\varphi}\,\rangle = \sqrt{4a\pi}\int_{\mathbb{R}} \mathrm{e}^{iax^2-ax^2}\,\mathrm{d}x. \tag{4.2.49}$$

The two integrals in (4.2.49) may be computed using formula (3.2.7) with $\lambda := \frac{1+i}{4a}$ and $\lambda := a(1-i)$, respectively. After some routine algebra (i.e., computing these integrals, replacing their values in (4.2.49), then solving for c), we find $c = \sqrt{\frac{\pi}{a}}\mathrm{e}^{i\frac{\pi}{4}}$. In summary, this analysis proves that

$$\widehat{\mathrm{e}^{ia|x|^2}}(\xi) = \left(\tfrac{\pi}{a}\right)^{\frac{n}{2}}\mathrm{e}^{i\frac{n\pi}{4}}\mathrm{e}^{-i\frac{|\xi|^2}{4a}} \qquad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.2.50}$$

Partial Fourier Transforms

In the last part of this section we define partial Fourier transforms. To set the stage, fix $m, n \in \mathbb{N}$. We shall denote by x, ξ generic variables in $\mathbb{R}^n$, and by

y, η generic variables in $\mathbb{R}^m$. The partial Fourier transform with respect to the variable x of a function $\varphi \in \mathcal{S}(\mathbb{R}^{n+m})$, denoted by $\widehat{\varphi}_x$ or $\mathcal{F}_x\varphi$, is defined by

$$\widehat{\varphi}_x(\xi, y) := \int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi}\varphi(x, y)\,\mathrm{d}x, \qquad \forall\,\xi \in \mathbb{R}^n,\ \forall\, y \in \mathbb{R}^m. \tag{4.2.51}$$

Reasoning in a similar manner as in the proof of Theorem 3.24, it follows that

$$\begin{gathered}\mathcal{F}_x : \mathcal{S}(\mathbb{R}^{n+m}) \to \mathcal{S}(\mathbb{R}^{n+m}) \ \text{ is bijective,} \\ \text{continuous, with continuous inverse}\end{gathered} \tag{4.2.52}$$

and its inverse is given by

$$\begin{gathered}(\mathcal{F}_x^{-1}\psi)(x, \eta) := (2\pi)^{-n} \int_{\mathbb{R}^n} \mathrm{e}^{ix\cdot\xi}\psi(\xi, \eta)\,\mathrm{d}\xi, \\ \text{for all } \ (x, \eta) \in \mathbb{R}^{n+m} \ \text{ and all } \ \psi \in \mathcal{S}(\mathbb{R}^{n+m}).\end{gathered} \tag{4.2.53}$$

Furthermore, analogously to Proposition 4.20, the partial Fourier transform $\mathcal{F}_x$ extends to $\mathcal{S}'(\mathbb{R}^{n+m})$ as a continuous map by setting

$$\langle \mathcal{F}_x u, \varphi\rangle := \langle u, \mathcal{F}_x\varphi\rangle, \qquad \forall\, u \in \mathcal{S}'(\mathbb{R}^{n+m}), \quad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^{n+m}), \tag{4.2.54}$$

and this extension is an isomorphism from $\mathcal{S}'(\mathbb{R}^{n+m})$ into itself, with continuous inverse denoted by $\mathcal{F}_x^{-1}$. Moreover, the action of $\mathcal{F}_x$ enjoys properties analogous to those established for the "full" Fourier transform in Theorem 3.20, Exercise 3.26, Theorem 4.25, and Proposition 4.30.

Exercise 4.39. *Let $\widehat{\ }$ denote the full Fourier transform in $\mathbb{R}^{n+m}$. Prove that for each function $\varphi \in \mathcal{S}(\mathbb{R}^n \times \mathbb{R}^m)$ we have*

$$\big(\mathcal{F}_x\mathcal{F}_y\varphi(x, y)\big)(\xi, \eta) = \big(\mathcal{F}_y\mathcal{F}_x\varphi(x, y)\big)(\xi, \eta) = \widehat{\varphi}(\xi, \eta) \quad \forall\,(\xi, \eta) \in \mathbb{R}^n \times \mathbb{R}^m. \tag{4.2.55}$$

Also, show that

$$\mathcal{F}_x\mathcal{F}_y u = \mathcal{F}_y\mathcal{F}_x u = \widehat{u} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n+m}), \quad \forall\, u \in \mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^m). \tag{4.2.56}$$

Exercise 4.40. *Prove that*

$$\mathcal{F}_x\big(\delta(x) \otimes \delta(y)\big) = 1(\xi) \otimes \delta(y) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n+m}). \tag{4.2.57}$$

4.3 Homogeneous Distributions

Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ be such that $\det A \neq 0$. Then for every $f \in L^1(\mathbb{R}^n)$ one has $f \circ A \in L^1(\mathbb{R}^n)$; thus, $f, f \circ A \in \mathcal{S}'(\mathbb{R}^n)$ by (4.1.8). Moreover,

$$\begin{aligned}\langle f \circ A, \varphi\rangle &= \int_{\mathbb{R}^n} f(Ax)\varphi(x)\,\mathrm{d}x = |\det A|^{-1} \int_{\mathbb{R}^n} f(y)\varphi(A^{-1}y)\,\mathrm{d}y \\ &= |\det A|^{-1}\langle f, \varphi \circ A^{-1}\rangle, \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n).\end{aligned} \tag{4.3.1}$$

This and Exercise 3.16 justifies extending the operator of composition with linear maps to $\mathcal{S}'(\mathbb{R}^n)$ as follows.

Proposition 4.41. *Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ be such that $\det A \neq 0$. For each $u \in \mathcal{S}'(\mathbb{R}^n)$, define the mapping $u \circ A : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$ by setting*

$$\big(u \circ A\big)(\varphi) := |\det A|^{-1}\big\langle u, \varphi \circ A^{-1}\big\rangle, \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.3.2}$$

Then $u \circ A \in \mathcal{S}'(\mathbb{R}^n)$.

Proof. This is an immediate consequence of (3.1.40). □

Exercise 4.42. *Let $A, B \in \mathcal{M}_{n\times n}(\mathbb{R})$ be such that $\det A \neq 0$ and $\det B \neq 0$. Then the following identities hold in $\mathcal{S}'(\mathbb{R}^n)$:*

(1) $(u \circ A) \circ B = u \circ (AB)$ for every $u \in \mathcal{S}'(\mathbb{R}^n)$.

(2) $u \circ (\lambda A) = \lambda u \circ A$ for every $u \in \mathcal{S}'(\mathbb{R}^n)$ and every $\lambda \in \mathbb{R}$.

(3) $(u + v) \circ A = u \circ A + v \circ A$ for every $u, v \in \mathcal{S}'(\mathbb{R}^n)$.

In the next proposition we study how the Fourier transform interacts with the operator of composition by an invertible matrix. Recall that the transpose of a matrix A is denoted by $A^\top$.

Proposition 4.43. *Assume that $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is such that $\det A \neq 0$. Then for each $u \in \mathcal{S}'(\mathbb{R}^n)$,*

$$\widehat{u \circ A} = |\det A|^{-1}\, \widehat{u} \circ \big(A^\top\big)^{-1}. \tag{4.3.3}$$

Proof. For each $\varphi \in \mathcal{S}(\mathbb{R}^n)$, based on (4.2.2), (4.3.2), and (3.2.9), we may write

$$\begin{aligned}
\langle \widehat{u \circ A}, \varphi\rangle &= \langle u \circ A, \widehat{\varphi}\,\rangle = |\det A|^{-1}\big\langle u, \widehat{\varphi} \circ A^{-1}\big\rangle \\
&= \big\langle u, \widehat{\varphi \circ A^\top}\big\rangle = \big\langle \widehat{u}, \varphi \circ A^\top\big\rangle \\
&= |\det A|^{-1}\big\langle \widehat{u} \circ \big(A^\top\big)^{-1}, \varphi\big\rangle.
\end{aligned} \tag{4.3.4}$$

This proves (4.3.3). □

The mappings in (3.1.40) and (4.3.2) corresponding to $A := tI_{n\times n}$, for some number $t \in (0,\infty)$, are called `dilations` and will be denoted by τ_t. More precisely, for each $t \in (0,\infty)$ we have

$$\tau_t : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n), \quad (\tau_t\varphi)(x) := \varphi(tx), \quad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n),\ \forall\, x \in \mathbb{R}^n, \tag{4.3.5}$$

and

$$\begin{aligned}
&\tau_t : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n), \\
&\langle \tau_t u, \varphi\rangle := t^{-n}\langle u, \tau_{\frac{1}{t}}\varphi\rangle, \quad \forall\, u \in \mathcal{S}'(\mathbb{R}^n),\ \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n).
\end{aligned} \tag{4.3.6}$$

Exercise 4.44. *Prove that for each $t \in (0,\infty)$ the following are true:*

$$\mathcal{F}(\tau_t\varphi) = t^{-n}\tau_{\frac{1}{t}}\mathcal{F}(\varphi) \quad \text{in} \quad \mathcal{S}(\mathbb{R}^n), \quad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n), \tag{4.3.7}$$

$$\mathcal{F}(\tau_t u) = t^{-n}\tau_{\frac{1}{t}}\mathcal{F}(u) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \quad \forall\, u \in \mathcal{S}'(\mathbb{R}^n). \tag{4.3.8}$$

Hint: Use (3.2.9) with $A = \frac{1}{t} I_{n\times n}$ to prove (4.3.7), then use (4.3.7) and (4.3.6) to prove (4.3.8).

To proceed, we make a couple of definitions.

Definition 4.45. *A linear transformation $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is called* `orthogonal` *provided A is invertible and $A^{-1} = A^{\top}$.*

Some of the most basic attributes of an orthogonal matrix A are

$$(A^{\top})^{-1} = A, \quad |\det A| = 1, \quad |Ax| = |x| \ \text{ for every } \ x \in \mathbb{R}^n. \tag{4.3.9}$$

Definition 4.46. *A distribution $u \in \mathcal{S}'(\mathbb{R}^n)$ is called* `invariant under orthogonal transformations` *provided $u = u \circ A$ in $\mathcal{S}'(\mathbb{R}^n)$ for every orthogonal matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$.*

Proposition 4.47. *Let $u \in \mathcal{S}'(\mathbb{R}^n)$. Then u is invariant under orthogonal transformations if and only if $\widehat{u}$ is invariant under orthogonal transformations.*

Proof. This is a direct consequence of (4.3.3) and the fact that any orthogonal matrix A satisfies (4.3.9). □

Next we take a look at homogeneous functions to gain some insight into how this notion may be defined in the setting of distributions.

Definition 4.48.

(1) A nonempty open set $\mathcal{O}$ in $\mathbb{R}^n$ is called a `cone-like region` *if $tx \in \mathcal{O}$ whenever $x \in \mathcal{O}$ and $t \in (0, \infty)$.*

(2) Given a cone-like region $\mathcal{O} \subseteq \mathbb{R}^n$, call a function $f : \mathcal{O} \to \mathbb{C}$ `positive homogeneous of degree` *$k \in \mathbb{R}$ if $f(tx) = t^k f(x)$ for every $t > 0$ and every $x \in \mathcal{O}$.*

Exercise 4.49. *Prove that if $\mathcal{O} \subseteq \mathbb{R}^n$ is a cone-like region, $N \in \mathbb{N}$, and $f \in C^N(\mathcal{O})$ is positive homogeneous of degree $k \in \mathbb{R}$ on $\mathcal{O}$, then $\partial^\alpha f$ is positive homogeneous of degree $k - N$ on $\mathcal{O}$ for every $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq N$.*

Exercise 4.50. *Prove that if $f \in C^0(\mathbb{R}^n \setminus \{0\})$ is positive homogeneous of degree $1 - n$ on $\mathbb{R}^n \setminus \{0\}$, and $g \in C^0(S^{n-1})$, then*

$$\int_{\partial B(0,R)} g(x/R) f(x)\, d\sigma(x) = \int_{S^{n-1}} g(x) f(x)\, d\sigma(x), \qquad \forall\, R \in (0, \infty). \tag{4.3.10}$$

Exercise 4.51. *Prove that if $f \in C^0(\mathbb{R}^n \setminus \{0\})$ is positive homogeneous of degree $k \in \mathbb{R}$, then $|f(x)| \leq \|f\|_{L^\infty(S^{n-1})} |x|^k$ for every $x \in \mathbb{R}^n \setminus \{0\}$.*

Hint: Write $f(x) = f\big(\frac{x}{|x|}\big)|x|^k$ for every $x \in \mathbb{R}^n \setminus \{0\}$.

Exercise 4.52. *Show that if $f \in C^0(\mathbb{R}^n \setminus \{0\})$ is positive homogeneous of degree $k \in \mathbb{R}$ with $k > -n$, then $f \in \mathcal{S}'(\mathbb{R}^n)$.*

Hint: Make use of Exercise 4.51 and the result discussed in Example 4.3.

After this preamble, we are ready to extend the notion of positive homogeneity to tempered distributions.

Definition 4.53. *A distribution $u \in \mathcal{S}'(\mathbb{R}^n)$ is called* `positive homogeneous of degree` *$k \in \mathbb{R}$ provided $\tau_t u = t^k u$ in $\mathcal{S}'(\mathbb{R}^n)$ for every $t > 0$.*

Exercise 4.54. *Prove that $\delta \in \mathcal{S}'(\mathbb{R}^n)$ is positive homogeneous of degree $-n$.*

Exercise 4.55. *Let $f \in L^1_{\rm loc}(\mathbb{R}^n)$ be such that (4.1.3) is satisfied for some $m \in \mathbb{N}$ and some $R \in (0,\infty)$ and let $k \in \mathbb{R}$. Show that the tempered distribution u_f is positive homogeneous of degree k, if and only if f is positive homogeneous of degree k.*

Exercise 4.56. *Prove that if $u \in \mathcal{S}'(\mathbb{R}^n)$ is positive homogeneous of degree $k \in \mathbb{R}$ then for every $\alpha \in \mathbb{N}_0^n$ the tempered distribution $\partial^\alpha u$ is positive homogeneous of degree $k - |\alpha|$. Deduce from this that for every $\alpha \in \mathbb{N}_0^n$ the tempered distribution $\partial^\alpha \delta$ is positive homogeneous of degree $-n - |\alpha|$.*

Proposition 4.57. *Let $k \in \mathbb{R}$. If $u \in \mathcal{S}'(\mathbb{R}^n)$ is positive homogeneous of degree k, then $\widehat{u}$ is positive homogeneous of degree $-n-k$.*

Proof. Let $u \in \mathcal{S}'(\mathbb{R}^n)$ be positive homogeneous of degree k, and fix $t > 0$. Then (4.3.8) and the assumption on u give

$$\tau_t \widehat{u} = t^{-n}\mathcal{F}\big(\tau_{\frac{1}{t}} u\big) = t^{-n}\mathcal{F}\big(t^{-k}u\big) = t^{-n-k}\widehat{u} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{4.3.11}$$

hence $\widehat{u}$ is positive homogeneous of degree $-n-k$. □

Proposition 4.58. *If $u \in \mathcal{S}'(\mathbb{R}^n)$, $u\big|_{\mathbb{R}^n\setminus\{0\}} \in C^\infty(\mathbb{R}^n \setminus \{0\})$, and $u\big|_{\mathbb{R}^n\setminus\{0\}}$ is positive homogeneous of degree k, for some $k \in \mathbb{R}$, then $\widehat{u}\big|_{\mathbb{R}^n\setminus\{0\}} \in C^\infty(\mathbb{R}^n\setminus\{0\})$.*

Proof. Fix u satisfying the hypotheses of the proposition. By *(c)* in Proposition 4.25, for each $\alpha \in \mathbb{N}_0^n$ one has $D_\xi^\alpha \widehat{u} = \widehat{(-x)^\alpha u}$ in $\mathcal{S}'(\mathbb{R}^n)$. Also, it is not difficult to check that $(-x)^\alpha u \in \mathcal{S}'(\mathbb{R}^n)$ continues to satisfy the hypotheses of the proposition with k replaced by $k+|\alpha|$. Hence, the desired conclusion follows once we prove that $\widehat{u}\big|_{\mathbb{R}^n\setminus\{0\}}$ is continuous on $\mathbb{R}^n \setminus \{0\}$.

To this end, assume first that $k < -n$ and fix $\psi \in C_0^\infty(\mathbb{R}^n)$ such that $\psi = 1$ on $B(0,1)$. Use this to decompose $u = \psi u + (1-\psi)u$. Since $\psi u \in \mathcal{E}'(\mathbb{R}^n)$ part *(b)* in Theorem 4.33 gives $\widehat{\psi u} \in C^\infty(\mathbb{R}^n)$. Furthermore, $(1-\psi)u$ vanishes near the origin while outside $\operatorname{supp}\psi$ becomes $u(x) = u\big(|x|\frac{x}{|x|}\big) = |x|^k u\big(\frac{x}{|x|}\big)$. Given the current assumption on k, this behavior implies $(1-\psi)u \in L^1(\mathbb{R}^n)$, hence $\widehat{(1-\psi)u} \in C^0(\mathbb{R}^n)$ by Lemma 4.27. To summarize, this analysis shows that

$$\widehat{u} \in C^0(\mathbb{R}^n) \quad \text{whenever} \quad k+n < 0. \tag{4.3.12}$$

To treat the case $k+n\geq 0$, let $\alpha\in\mathbb{N}_0^n$ be arbitrary and set $v_\alpha:=D^\alpha u\in\mathcal{S}'(\mathbb{R}^n)$. Since $u\big|_{\mathbb{R}^n\setminus\{0\}}\in C^\infty(\mathbb{R}^n\setminus\{0\})$ differentiating $u(tx)=t^k u(x)$ yields

$$t^{|\alpha|}(D^\alpha u)(tx)=t^k(D^\alpha u)(x)\quad\text{for}\quad x\in\mathbb{R}^n\setminus\{0\}\quad\text{and}\quad t>0. \tag{4.3.13}$$

Given that $v_\alpha\big|_{\mathbb{R}^n\setminus\{0\}}\in C^\infty(\mathbb{R}\setminus\{0\})$, the latter translates into

$$v_\alpha(tx)=t^{k-|\alpha|}v_\alpha(x)\quad\text{for}\quad x\in\mathbb{R}^n\setminus\{0\}\quad\text{and}\quad t>0. \tag{4.3.14}$$

Hence, $v_\alpha\big|_{\mathbb{R}^n\setminus\{0\}}$ is homogeneous of degree $k-|\alpha|$. Based on what we proved earlier (cf. (4.3.12)), it follows that $\widehat{v_\alpha}\in C^0(\mathbb{R}^n)$ whenever $k-|\alpha|<-n$. In terms of the original distribution u this amounts to saying that

$$\xi^\alpha\,\widehat{u}\in C^0(\mathbb{R}^n),\qquad\forall\,\alpha\in\mathbb{N}_0^n\quad\text{with}\quad|\alpha|>k+n. \tag{4.3.15}$$

The end-game in the proof is then as follows. Given an arbitrary $k\in\mathbb{R}$, pick a natural number N with the property that $2N>k+n$. Writing (cf. (13.2.1)) for each $\xi\in\mathbb{R}^n$

$$|\xi|^{2N}=\sum_{|\alpha|=N}\frac{N!}{\alpha!}\,\xi^{2\alpha}, \tag{4.3.16}$$

we obtain

$$|\xi|^{2N}\widehat{u}=\sum_{|\alpha|=N}\frac{N!}{\alpha!}\,\xi^{2\alpha}\,\widehat{u}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n). \tag{4.3.17}$$

Collectively, (4.3.17), (4.3.15), and the assumption on N imply

$$|\xi|^{2N}\widehat{u}\in C^0(\mathbb{R}^n). \tag{4.3.18}$$

Since $\frac{1}{|\xi|^{2N}}\big|_{\mathbb{R}^n\setminus\{0\}}$ belongs to $C^\infty(\mathbb{R}^n\setminus\{0\})$, condition (4.3.18) further implies that $\widehat{u}\big|_{\mathbb{R}^n\setminus\{0\}}\in C^0(\mathbb{R}^n\setminus\{0\})$. This completes the proof of the proposition. □

An inspection of the proof of Proposition 4.58 shows that several other useful versions could be derived, two of which are recorded below.

Exercise 4.59. *If $u\in\mathcal{S}'(\mathbb{R}^n)$, $u\big|_{\mathbb{R}^n\setminus\{0\}}\in C^\infty(\mathbb{R}^n\setminus\{0\})$ and $u\big|_{\mathbb{R}^n\setminus\{0\}}$ is positive homogeneous of degree k, for some $k\in\mathbb{R}$ satisfying $k<-n$, then $\widehat{u}\in C^{k_0}(\mathbb{R}^n)$, where $k_0:=\max\{j\in\mathbb{N}_0,\,j+k<-n\}$.*

Exercise 4.60. *Assume that $u\in\mathcal{S}'(\mathbb{R}^n)$, $u\big|_{\mathbb{R}^n\setminus\{0\}}\in C^N(\mathbb{R}^n\setminus\{0\})$ where $N\in\mathbb{N}$ is even, and $u\big|_{\mathbb{R}^n\setminus\{0\}}$ is positive homogeneous of degree k, for some $k\in\mathbb{R}$ satisfying $k<N-n$. Then $\widehat{u}\in C^m(\mathbb{R}^n\setminus\{0\})$ for every $m\in\mathbb{N}_0$ satisfying $m<N-n-k$.*

Next we take on the task of computing the Fourier transform of certain homogeneous tempered distributions that will be particularly important later in applications. Recall the gamma function Γ from (13.5.1).

Proposition 4.61. *Let $\lambda \in (0,n)$ and set $f_\lambda(x) := |x|^{-\lambda}$, for each $x \in \mathbb{R}^n \setminus \{0\}$. Then $f_\lambda \in \mathcal{S}'(\mathbb{R}^n)$, $\widehat{f_\lambda}\big|_{\mathbb{R}^n\setminus\{0\}} \in C^\infty(\mathbb{R}^n \setminus \{0\})$, and*

$$\widehat{f_\lambda}(\xi) = 2^{n-\lambda}\pi^{\frac{n}{2}}\frac{\Gamma\left(\frac{n-\lambda}{2}\right)}{\Gamma\left(\frac{\lambda}{2}\right)}|\xi|^{\lambda-n} \quad \text{for every} \quad \xi \in \mathbb{R}^n \setminus \{0\}. \tag{4.3.19}$$

Proof. Fix $\lambda \in (0,n)$. Exercise 4.4 then shows that $f_\lambda \in \mathcal{S}'(\mathbb{R}^n)$. Clearly, $|x|^{-\lambda}$ is invariant under orthogonal transformations and is positive homogeneous of degree $-\lambda$. Hence, by Proposition 4.47 and Proposition 4.57, it follows that $\widehat{f_\lambda}$ is invariant under orthogonal transformations and positive homogeneous of degree $-n+\lambda$. In addition, $\widehat{f_\lambda}\big|_{\mathbb{R}^n\setminus\{0\}} \in C^\infty(\mathbb{R}^n \setminus \{0\})$ by Proposition 4.58.

Fix $\xi \in \mathbb{R}^n \setminus \{0\}$ and choose an orthogonal matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ with the property that $A\xi = (0,0,\ldots,0,|\xi|)$ (such a matrix may be obtained by completing the vector $v_n := \frac{\xi}{|\xi|}$ to an orthonormal basis $\{v_1,\ldots,v_n\}$ in $\mathbb{R}^n$ and then taking A to be the matrix mapping each v_j into $\mathbf{e}_j$ for $j = 1,\ldots,n$). Then

$$\widehat{f_\lambda}(\xi) = \widehat{f_\lambda}(A\xi) = \widehat{f_\lambda}(0,\ldots,0,|\xi|\,) = c_\lambda|\xi|^{\lambda-n}, \tag{4.3.20}$$

where $c_\lambda := \widehat{f_\lambda}(0,\ldots,0,1) \in \mathbb{C}$. As such, we are left with determining the value of c_λ. We do so by apply $\widehat{f_\lambda}$ to the particular Schwartz function $\varphi(x) := \mathrm{e}^{-\frac{|x|^2}{2}}$ for $x \in \mathbb{R}^n$. From Example 3.21 we know that $\widehat{\varphi}(\xi) = (2\pi)^{\frac{n}{2}}\mathrm{e}^{-\frac{|\xi|^2}{2}}$ for every $\xi \in \mathbb{R}^n$. Based on this formula and (4.3.20), the identity $\langle\widehat{f_\lambda},\varphi\rangle = \langle f_\lambda,\widehat{\varphi}\,\rangle$ may be rewritten as

$$c_\lambda\int_{\mathbb{R}^n}|\xi|^{\lambda-n}\mathrm{e}^{-\frac{|\xi|^2}{2}}\,\mathrm{d}\xi = (2\pi)^{\frac{n}{2}}\int_{\mathbb{R}^n}|x|^{-\lambda}\mathrm{e}^{-\frac{|x|^2}{2}}\,\mathrm{d}x. \tag{4.3.21}$$

The two integrals in (4.3.21) may be computed simultaneously, by adopting a slightly more general point of view, as follows. For any $p > -n$ use polar coordinates (cf. (13.8.9)) and a natural change of variables to write (for the definition and properties of the gamma function Γ see (13.5.1) and the subsequent comments)

$$\begin{aligned}\int_{\mathbb{R}^n}|\xi|^p\mathrm{e}^{-\frac{|\xi|^2}{2}}\,\mathrm{d}\xi &= \omega_{n-1}\int_0^\infty \rho^{p+n-1}\mathrm{e}^{-\frac{\rho^2}{2}}\,\mathrm{d}\rho \\ &= \omega_{n-1}2^{\frac{p+n-2}{2}}\int_0^\infty t^{\frac{p+n-2}{2}}\mathrm{e}^{-t}\,\mathrm{d}t \\ &= 2^{\frac{p+n-2}{2}}\omega_{n-1}\Gamma\Big(\frac{p+n}{2}\Big).\end{aligned} \tag{4.3.22}$$

When used with $p := \lambda - n$ and $p := -\lambda$, formula (4.3.22) allows us to rewrite (4.3.21) as

$$c_\lambda 2^{\frac{\lambda-n+n-2}{2}}\omega_{n-1}\Gamma\Big(\frac{\lambda}{2}\Big) = (2\pi)^{\frac{n}{2}}2^{\frac{-\lambda+n-2}{2}}\omega_{n-1}\Gamma\Big(\frac{-\lambda+n}{2}\Big). \tag{4.3.23}$$

This gives $c_\lambda = 2^{n-\lambda}\pi^{\frac{n}{2}}\frac{\Gamma\left(\frac{n-\lambda}{2}\right)}{\Gamma\left(\frac{\lambda}{2}\right)}$, finishing the proof of (4.3.19). □

A remarkable consequence of Proposition 4.61 is singled out below.

Corollary 4.62. *Assume that $n \in \mathbb{N}$, $n \geq 2$, and fix $\lambda \in [0, n-1)$. Then for each $j \in \{1, \ldots, n\}$, we have*

$$\mathcal{F}\left(\frac{x_j}{|x|^{\lambda+2}}\right) = -i\, 2^{n-\lambda-1}\pi^{\frac{n}{2}} \frac{\Gamma\left(\frac{n-\lambda}{2}\right)}{\Gamma\left(\frac{\lambda}{2}+1\right)} \frac{\xi_j}{|\xi|^{n-\lambda}} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.3.24}$$

In particular, corresponding to the case when $\lambda = n-2$, formula (4.3.24) becomes

$$\mathcal{F}\left(\frac{x_j}{|x|^{n}}\right) = -i\, \omega_{n-1} \frac{\xi_j}{|\xi|^2} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.3.25}$$

Proof. Fix an integer $n \geq 2$, and suppose first that $\lambda \in (0, n-1)$. In this regime, both (4.3.19) and (4.1.32) hold. In concert with part *(b)* in Theorem 4.25, these give

$$\begin{aligned} -\lambda\, \mathcal{F}\left(\frac{x_j}{|x|^{\lambda+2}}\right) &= \mathcal{F}\left(\partial_j f_\lambda\right) = i\xi_j \widehat{f_\lambda}(\xi) \\ &= i\, 2^{n-\lambda}\pi^{\frac{n}{2}} \frac{\Gamma\left(\frac{n-\lambda}{2}\right)}{\Gamma\left(\frac{\lambda}{2}\right)} \xi_j |\xi|^{\lambda-n} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \end{aligned} \tag{4.3.26}$$

Hence, whenever $\lambda \in (0, n-1)$,

$$\mathcal{F}\left(\frac{x_j}{|x|^{\lambda+2}}\right) = C_\lambda \xi_j |\xi|^{\lambda-n} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{4.3.27}$$

where we have set

$$C_\lambda := -i\, 2^{n-\lambda-1}\pi^{\frac{n}{2}} \frac{\Gamma\left(\frac{n-\lambda}{2}\right)}{\frac{\lambda}{2}\Gamma\left(\frac{\lambda}{2}\right)} = -i\, 2^{n-\lambda-1}\pi^{\frac{n}{2}} \frac{\Gamma\left(\frac{n-\lambda}{2}\right)}{\Gamma\left(\frac{\lambda}{2}+1\right)} \tag{4.3.28}$$

and the last equality follows from (13.5.2). This proves formula (4.3.24) in the case when $\lambda \in (0, n-1)$. The case when $\lambda = 0$ then follows from what we have just proved, by passing to limit $\lambda \to 0^+$ in (4.3.24) and observing that all quantities involved depend continuously on λ, in an appropriate sense. Finally, (4.3.25) is a direct consequence of (4.3.24) and (13.5.6). □

4.4 Principal Value Tempered Distributions

Recall the distribution P.V. $\frac{1}{x} \in \mathcal{D}'(\mathbb{R})$ from Example 2.11. As seen from Exercise 4.103, we have P.V. $\frac{1}{x} \in \mathcal{S}'(\mathbb{R})$. The issue we address in this section is the generalization of this distribution to higher dimensions. The key features of the function $\Theta(x) := \frac{1}{x}$, $x \in \mathbb{R} \setminus \{0\}$, that allowed us to define P.V. $\frac{1}{x}$ as a tempered distribution on the real line are as follows: first, $\Theta \in C^0(\mathbb{R} \setminus \{0\})$, second, Θ is positive homogeneous of degree -1, and third, $\Theta(1) + \Theta(-1) = 0$.

Moving from one dimension to $\mathbb{R}^n$, this suggests considering the class of functions satisfying

$$\Theta\in C^0(\mathbb{R}^n\setminus\{0\}),\quad \text{positive homogeneous of degree } -n,\quad \int_{S^{n-1}}\Theta\,d\sigma=0. \tag{4.4.1}$$

It is worth noting that a function Θ as above typically fails to be in $L^1_{\rm loc}(\mathbb{R}^n)$ [though obviously $\Theta\in L^1_{\rm loc}(\mathbb{R}^n\setminus\{0\})$]. As such, associating a distribution with Θ necessarily has to be a more elaborate process than the one identifying functions from $L^1_{\rm loc}(\mathbb{R}^n)$ with distributions in $\mathbb{R}^n$. This has already been the case when defining P.V. $\frac{1}{x}$ and here we model the same type of definition in higher dimensions.

Specifically, given Θ as in (4.4.1) consider the linear mapping

$$\begin{aligned}&\text{P.V.}\,\Theta:\mathcal{S}(\mathbb{R}^n)\to\mathbb{C},\\ &\big(\text{P.V.}\,\Theta\big)(\varphi):=\lim_{\varepsilon\to0^+}\int_{|x|\geq\varepsilon}\Theta(x)\varphi(x)\,dx,\quad\forall\,\varphi\in\mathcal{S}(\mathbb{R}^n).\end{aligned} \tag{4.4.2}$$

That this definition does the job is proved next.

Proposition 4.63. *Let Θ be a function satisfying (4.4.1). Then the map* P.V. Θ *considered in (4.4.2) is well-defined and is a tempered distribution in $\mathbb{R}^n$. In addition, $\big(\text{P.V.}\,\Theta\big)\big|_{\mathbb{R}^n\setminus\{0\}}=\Theta\big|_{\mathbb{R}^n\setminus\{0\}}$ in $\mathcal{D}'(\mathbb{R}^n\setminus\{0\})$.*

Before proceeding with the proof of Proposition 4.63 we recall a definition and introduce a class of functions that will be used in the proof.

Definition 4.64. *A function $\psi:\mathbb{R}^n\to\mathbb{C}$ is called* `radial` *if $\psi(x)$ depends only on $|x|$ for every $x\in\mathbb{R}^n$, that is, $\psi(x)=f(|x|)$ for all $x\in\mathbb{R}^n$, where $f:\mathbb{R}\to\mathbb{C}$.*

Next, consider the class of functions

$$\begin{cases}\psi:\mathbb{R}^n\to\mathbb{C},\quad \psi\in C^1(\mathbb{R}^n),\\ \psi\ \text{ is radial and } \psi(0)=1,\\ \exists\,\varepsilon_o\in(0,\infty)\text{ such that }\psi\text{ decays like }|x|^{-\varepsilon_o}\text{ at }\infty,\end{cases} \tag{4.4.3}$$

(for example $\psi(x)=e^{-\frac{|x|^2}{2}}$, $x\in\mathbb{R}^n$, satisfies (4.4.3)) and set

$$Q:=\big\{\psi:\ \psi\text{ satisfies (4.4.3)}\big\}. \tag{4.4.4}$$

Now we are ready to return to Proposition 4.63.

Proof of Proposition 4.63. Fix an arbitrary ψ satisfying (4.4.3). Then, making use of formula (13.8.9) and the properties of Θ and ψ, for each $\varepsilon \in (0,\infty)$ we have

$$\begin{aligned}\int_{|x|\geq\varepsilon} \Theta(x)\psi(x)\,\mathrm{d}x &= \int_{|x|\geq\varepsilon} \frac{\Theta(\frac{x}{|x|})}{|x|^n}\psi(x)\,\mathrm{d}x \\ &= \int_{\varepsilon}^{\infty} \frac{\psi(\rho)}{\rho}\int_{S^{n-1}} \Theta(\omega)\,\mathrm{d}\sigma(\omega)\,\mathrm{d}\rho = 0. \end{aligned} \tag{4.4.5}$$

Hence, $\Theta[\varphi-\varphi(0)\psi] \in L^1(\mathbb{R}^n)$ for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$, which when combined with (4.4.5) and Lebesgue dominated convergence theorem 13.12 yields

$$\big(\mathrm{P.V.}\,\Theta\big)(\varphi) = \int_{\mathbb{R}^n} \Theta(x)[\varphi(x)-\varphi(0)\psi(x)]\,\mathrm{d}x, \qquad \forall\,\varphi\in\mathcal{S}(\mathbb{R}^n). \tag{4.4.6}$$

Note that because of (4.4.5), the right-hand side in (4.4.6) is independent of the choice of ψ. Estimating the right hand-side of (4.4.6) (using Exercise 4.51, the decay at infinity of functions from Q and the Schwartz class, and the mean value theorem near the origin) shows that there exists a constant $C \in (0,\infty)$ independent of φ with the property that

$$\big|\big(\mathrm{P.V.}\,\Theta\big)(\varphi)\big| \leq C \sup_{x\in\mathbb{R}^n,\,|\alpha|\leq 1,\,|\beta|\leq 1} \big|x^\alpha \partial^\beta\varphi(x)\big|, \qquad \forall\,\varphi\in\mathcal{S}(\mathbb{R}^n). \tag{4.4.7}$$

Since from (4.4.6) we see that $\mathrm{P.V.}\,\Theta$ is linear, in light of Fact 4.1 estimate (4.4.7) implies $\mathrm{P.V.}\,\Theta \in \mathcal{S}'(\mathbb{R}^n)$ as wanted. The fact that the restriction in the distributional sense of $\mathrm{P.V.}\,\Theta$ to $\mathbb{R}^n\setminus\{0\}$ is equal to the restriction of the function Θ to $\mathbb{R}^n\setminus\{0\}$ is immediate from definitions. □

Remark 4.65.

(1) As already alluded to, if $n=1$ and $\Theta(x) := \frac{1}{x}$, $x\in\mathbb{R}\setminus\{0\}$, then we have $\mathrm{P.V.}\Theta = \mathrm{P.V.}\,\frac{1}{x}$.

(2) Suppose Θ is as in (4.4.1). Since identity (4.4.6) holds for any $\psi\in Q$, we may select $\psi\in Q$ that also satisfies $\psi = 1$ on S^{n-1} and observe that for this choice of ψ we have

$$\begin{aligned}&\int_{\mathbb{R}^n} \Theta(x)[\varphi(x)-\varphi(0)\psi(x)]\,\mathrm{d}x \\ &\qquad = \int_{|x|\leq 1} \Theta(x)\big[\varphi(x)-\varphi(0)\big]\,\mathrm{d}x + \int_{|x|>1} \Theta(x)\varphi(x)\,\mathrm{d}x, \forall\,\varphi\in\mathcal{S}(\mathbb{R}^n).\end{aligned} \tag{4.4.8}$$

Hence,

$$\begin{aligned}\langle \mathrm{P.V.}\,\Theta,\varphi\rangle &= \int_{|x|\leq 1} \Theta(x)\big[\varphi(x)-\varphi(0)\big]\,\mathrm{d}x \\ &\quad + \int_{|x|>1} \Theta(x)\varphi(x)\,\mathrm{d}x, \quad \forall\,\varphi\in\mathcal{S}(\mathbb{R}^n).\end{aligned} \tag{4.4.9}$$

Example 4.66. If $j \in \{1, \dots, n\}$, the function Θ defined by $\Theta(x) := \frac{x_j}{|x|^{n+1}}$ for each $x \in \mathbb{R}^n \setminus \{0\}$ satisfies (4.4.1). By Proposition 4.63 we have $\text{P.V.} \frac{x_j}{|x|^{n+1}} \in \mathcal{S}'(\mathbb{R}^n)$ and part *(2)* in Remark 4.65 gives that for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$

$$\begin{aligned} \Big\langle \text{P.V.} \frac{x_j}{|x|^{n+1}}, \varphi \Big\rangle &= \lim_{\varepsilon \to 0^+} \int_{|x| \geq \varepsilon} \frac{x_j \varphi(x)}{|x|^{n+1}} \, \mathrm{d}x \qquad (4.4.10) \\ &= \int_{|x| \leq 1} \frac{x_j(\varphi(x) - \varphi(0))}{|x|^{n+1}} \, \mathrm{d}x + \int_{|x|>1} \frac{x_j \varphi(x)}{|x|^{n+1}} \, \mathrm{d}x. \end{aligned}$$

The next proposition elaborates on the manner in which principal value tempered distributions convolve with Schwartz functions.

Proposition 4.67. *Let Θ be a function satisfying the conditions in (4.4.1). Then for each $\varphi \in \mathcal{S}(\mathbb{R}^n)$ one has that $\big(\text{P.V.}\, \Theta\big) * \varphi \in \mathcal{S}'(\mathbb{R}^n) \cap C^\infty(\mathbb{R}^n)$ and*

$$\Big(\big(\text{P.V.}\, \Theta\big) * \varphi\Big)(x) = \lim_{\varepsilon \to 0^+} \int_{|x-y| \geq \varepsilon} \Theta(x-y)\varphi(y) \, \mathrm{d}y, \qquad \forall\, x \in \mathbb{R}^n. \quad (4.4.11)$$

Proof. Fix an arbitrary $\varphi \in \mathcal{S}(\mathbb{R}^n)$ and note that since $\text{P.V.}\, \Theta \in \mathcal{S}'(\mathbb{R}^n)$ (c.f. Proposition 4.63), part *(e)* in Theorem 4.18 gives $\big(\text{P.V.}\, \Theta\big) * \varphi \in \mathcal{S}'(\mathbb{R}^n)$. Let $\psi \in C_0^\infty(\mathbb{R}^n)$ be such that $\psi = 1$ near the origin. Then $1 - \psi \in \mathcal{L}(\mathbb{R}^n)$ and it makes sense to consider $(1-\psi)\big(\text{P.V.}\, \Theta\big)$ in $\mathcal{S}'(\mathbb{R}^n)$ (cf. part *(b)* in Theorem 4.13). Hence, we may decompose $\text{P.V.}\, \Theta = u + v$ where

$$u := \psi\, \text{P.V.}\, \Theta \in \mathcal{E}'(\mathbb{R}^n) \quad \text{and} \quad v := (1-\psi)\big(\text{P.V.}\, \Theta\big) \in \mathcal{S}'(\mathbb{R}^n). \qquad (4.4.12)$$

The last part in Proposition 4.63 also permits us to identify $v = (1-\psi)\Theta$ in $L^1_{\text{loc}}(\mathbb{R}^n)$. By Exercise 4.51 we have $v \in L^p(\mathbb{R}^n)$ for every $p \in (1, \infty)$ which, in combination with Exercise 3.18, allows us to conclude that $v * \varphi \in C^\infty(\mathbb{R}^n)$ and

$$(v * \varphi)(x) = \int_{\mathbb{R}^n} (1 - \psi(y))\Theta(y)\varphi(x-y) \, \mathrm{d}y, \qquad \forall\, x \in \mathbb{R}^n. \qquad (4.4.13)$$

Since the above integral is absolutely convergent, by Lebesgue's dominated convergence theorem we further express this as

$$(v * \varphi)(x) = \lim_{\varepsilon \to 0^+} \int_{|y| \geq \varepsilon} (1 - \psi(y))\Theta(y)\varphi(x-y) \, \mathrm{d}y, \qquad \forall\, x \in \mathbb{R}^n. \quad (4.4.14)$$

Thanks to Exercise 2.94 we also have $u * \varphi \in C^\infty(\mathbb{R}^n)$ and

$$(u * \varphi)(x) = \langle \psi\, \text{P.V.}\, \Theta, \varphi(x - \cdot) \rangle \quad \text{for each } x \in \mathbb{R}^n. \qquad (4.4.15)$$

On the other hand, the definition of the principal value gives that for each $x \in \mathbb{R}^n$

$$\begin{aligned}\big\langle \psi\,\text{P.V.}\,\Theta, \varphi(x-\cdot)\big\rangle &= \big\langle \text{P.V.}\,\Theta, \psi(\cdot)\varphi(x-\cdot)\big\rangle \\ &= \lim_{\varepsilon\to 0^+}\int_{|y|\geq\varepsilon}\Theta(y)\psi(y)\varphi(x-y)\,dy. \end{aligned} \tag{4.4.16}$$

Collectively, these arguments show that, for each $x\in\mathbb{R}^n$,

$$\begin{aligned}\Big(\big(\text{P.V.}\,\Theta\big)*\varphi\Big)(x) &= (u*\varphi)(x)+(v*\varphi)(x) \\ &= \lim_{\varepsilon\to 0^+}\int_{|y|\geq\varepsilon}\Theta(y)\psi(y)\varphi(x-y)\,dy \\ &\quad+\lim_{\varepsilon\to 0^+}\int_{|y|\geq\varepsilon}(1-\psi(y))\Theta(y)\varphi(x-y)\,dy \\ &= \lim_{\varepsilon\to 0^+}\int_{|x-y|\geq\varepsilon}\Theta(x-y)\varphi(y)\,dy, \end{aligned} \tag{4.4.17}$$

proving (4.4.11). □

The next example discusses a basic class of principal value tempered distributions arising naturally in applications.

Example 4.68. Let $\Phi\in C^1(\mathbb{R}^n\setminus\{0\})$ be positive homogeneous of degree $1-n$. Then for each $j\in\{1,\ldots,n\}$ it follows that $\partial_j\Phi$ satisfies the conditions in (4.4.1). Consequently, P.V.$(\partial_j\Phi)$ is a well-defined tempered distribution.

To see why this is true fix $j\in\{1,\ldots,n\}$ and note that $\partial_j\Phi\in C^0(\mathbb{R}^n\setminus\{0\})$ and $\partial_j\Phi$ is positive homogeneous of degree $-n$ (cf. Exercise 4.49). Moreover, using Exercise 4.50, then integrating by parts based on (13.7.4), and then using (13.8.5), we obtain

$$\begin{aligned} 0 &= \int_{|x|=2}\Phi(x)\frac{x_j}{2}\,d\sigma(x)-\int_{|x|=1}\Phi(x)x_j\,d\sigma(x)=\int_{1<|x|<2}\partial_j\Phi(x)\,dx \\ &= \int_1^2\int_{S^{n-1}}(\partial_j\Phi)(\rho\omega)\,\rho^{n-1}d\sigma(\omega)\,d\rho=\Big(\int_1^2\frac{d\rho}{\rho}\Big)\int_{S^{n-1}}(\partial_j\Phi)(\omega)\,d\sigma(\omega) \\ &= (\ln 2)\int_{S^{n-1}}(\partial_j\Phi)(\omega)\,d\sigma(\omega). \end{aligned} \tag{4.4.18}$$

This shows that $\int_{S^{n-1}}\partial_j\Phi\,d\sigma=0$, hence $\partial_j\Phi$ satisfies all conditions in (4.4.1).

Principal value tempered distributions often arise when differentiating certain types of functions exhibiting a point singularity. Specifically, we have the following theorem.

Theorem 4.69. *Let $\Phi\in C^1(\mathbb{R}^n\setminus\{0\})$ be a function that is positive homogeneous of degree $1-n$. Then for each $j\in\{1,\ldots,n\}$, the distributional derivative $\partial_j\Phi$ satisfies*

$$\partial_j\Phi=\Big(\int_{S^{n-1}}\Phi(\omega)\omega_j\,d\sigma(\omega)\Big)\delta+\text{P.V.}(\partial_j\Phi)\qquad \text{in}\ \ \mathcal{S}'(\mathbb{R}^n). \tag{4.4.19}$$

Proof. From the properties of Φ, Exercises 4.51 and 4.5 it follows $\Phi \in \mathcal{S}'(\mathbb{R}^n)$. Fix $j \in \{1, \dots, n\}$. By Example 4.68 we have that $\text{P.V.}(\partial_j \Phi) \in \mathcal{S}'(\mathbb{R}^n)$. Hence, invoking (4.1.25), to conclude (4.4.19) it suffices to prove that the equality in (4.4.19) holds in $\mathcal{D}'(\mathbb{R}^n)$. To this end, fix $\varphi \in C_0^\infty(\mathbb{R}^n)$ and using the Lebesgue dominated convergence theorem 13.12 and integration by parts (based on (13.7.4)) write

$$\begin{aligned}
\langle \partial_j \Phi, \varphi \rangle &= -\langle \Phi, \partial_j \varphi \rangle = -\int_{\mathbb{R}^n} \Phi(x) \partial_j \varphi(x)\, dx \\
&= -\lim_{\varepsilon \to 0^+} \int_{|x| \geq \varepsilon} \Phi(x) \partial_j \varphi(x)\, dx \\
&= \lim_{\varepsilon \to 0^+} \int_{|x| \geq \varepsilon} \partial_j \Phi(x) \varphi(x)\, dx + \lim_{\varepsilon \to 0^+} \int_{|x| = \varepsilon} \frac{x_j}{\varepsilon} \Phi(x) \varphi(x)\, d\sigma(x) \\
&= \Big\langle \text{P.V.}(\partial_j \Phi), \varphi \Big\rangle + \lim_{\varepsilon \to 0^+} \int_{|x| = \varepsilon} \frac{x_j}{\varepsilon} \Phi(x) \varphi(x)\, d\sigma(x). \qquad (4.4.20)
\end{aligned}$$

Also, for each $\varepsilon \in (0, \infty)$ we have

$$\begin{aligned}
&\int_{|x| = \varepsilon} \frac{x_j}{\varepsilon} \Phi(x) \varphi(x)\, d\sigma(x) \\
&= \int_{|x| = \varepsilon} \frac{x_j}{\varepsilon} \Phi(x) \big[\varphi(x) - \varphi(0)\big]\, d\sigma(x) + \varphi(0) \int_{|x| = \varepsilon} \frac{x_j}{\varepsilon} \Phi(x)\, d\sigma(x) \\
&= \int_{|x| = \varepsilon} \frac{x_j}{\varepsilon} \Phi(x) \big[\varphi(x) - \varphi(0)\big]\, d\sigma(x) + \varphi(0) \int_{S^{n-1}} \omega_j \Phi(\omega)\, d\sigma(\omega), \qquad (4.4.21)
\end{aligned}$$

where for the last equality in (4.4.21) we used Exercise 4.50. In addition, using the fact that $\varphi \in C_0^\infty(\mathbb{R}^n)$ and Exercise 4.51 we may estimate

$$\begin{aligned}
&\left| \int_{|x| = \varepsilon} \frac{x_j}{\varepsilon} \Phi(x) \big[\varphi(x) - \varphi(0)\big]\, d\sigma(x) \right| \\
&\leq \varepsilon \|\nabla \varphi\|_{L^\infty(\mathbb{R}^n)} \|\Phi\|_{L^\infty(S^{n-1})} \int_{|x| = \varepsilon} \frac{1}{|x|^{n-1}}\, d\sigma(x) \\
&= \|\nabla \varphi\|_{L^\infty(\mathbb{R}^n)} \|\Phi\|_{L^\infty(S^{n-1})}\, \omega_{n-1}\, \varepsilon \xrightarrow[\varepsilon \to 0^+]{} 0. \qquad (4.4.22)
\end{aligned}$$

Combining (4.4.20), (4.4.21), and (4.4.22), we conclude

$$\langle \partial_j \Phi, \varphi \rangle = \Big\langle \text{P.V.}(\partial_j \Phi), \varphi \Big\rangle + \Big(\int_{S^{n-1}} \omega_j \Phi(\omega)\, d\sigma(\omega) \Big) \langle \delta, \varphi \rangle. \qquad (4.4.23)$$

This yields that $\partial_j\Phi = \text{P.V.}(\partial_j\Phi) + \Big(\int_{S^{n-1}}\omega_j\Phi(\omega)\,d\sigma(\omega)\Big)\delta$ in $\mathcal{D}'(\mathbb{R}^n)$ and completes the proof of the theorem. □

4.5 The Fourier Transform of Principal Value Distributions

From Proposition 4.63 we know that whenever Θ is a function satisfying the conditions in (4.4.1), the principal value distribution $\text{P.V.}\,\Theta$ belongs to $\mathcal{S}'(\mathbb{R}^n)$. As such, its Fourier transform makes sense as a tempered distribution. This being said, in many applications (cf. the discussion in Remark 4.92), it is of basic importance to actually identify this distribution. The general aim of this section is to do just that, though as a warm-up, we deal with the following particular (yet relevant) case.

Proposition 4.70. *For each $k\in\{1,\dots,n\}$ consider the function*

$$\Phi_k(x) := \frac{x_k}{|x|^n}, \qquad \forall\, x\in\mathbb{R}^n\setminus\{0\}. \tag{4.5.1}$$

Then for each $j\in\{1,\dots,n\}$ the function $\partial_j\Phi_k$ satisfies the conditions in (4.4.1) and

$$\mathcal{F}\big(\text{P.V.}\,(\partial_j\Phi_k)\big) = \omega_{n-1}\frac{\xi_j\xi_k}{|\xi|^2} - \frac{\omega_{n-1}}{n}\delta_{jk} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.5.2}$$

Proof. Fix $j,k\in\{1,\dots,n\}$. From Example 4.68 it is clear that the function $\partial_j\Phi_k$ is as in (4.4.1). Moreover, Theorem 4.69 gives that

$$\partial_j\Phi_k = \Big(\int_{S^{n-1}}\omega_k\omega_j\,d\sigma(\omega)\Big)\delta + \text{P.V.}(\partial_j\Phi_k) \qquad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.5.3}$$

Taking into account (13.8.44), and applying the Fourier transform to both sides, this yields

$$\mathcal{F}\big(\text{P.V.}\,(\partial_j\Phi_k)\big) = \mathcal{F}(\partial_j\Phi_k) - \frac{\omega_{n-1}}{n}\delta_{jk} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.5.4}$$

On the other hand, since by part *(b)* in Theorem 4.25 and Corollary 4.62 we have

$$\mathcal{F}(\partial_j\Phi_k) = i\,\xi_j\mathcal{F}(\Phi_k) = \omega_{n-1}\frac{\xi_j\xi_k}{|\xi|^2} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{4.5.5}$$

formula (4.5.2) follows from (4.5.4) and (4.5.5). □

The next theorem shows that the Fourier transform of principal value distributions $\text{P.V.}\,\Theta$ is given by bounded functions. As we shall see in Sect. 4.9, such a result plays a key role in establishing the L^2 boundedness of singular integral operators (SIO).

Theorem 4.71. *Let Θ be a function satisfying the conditions in (4.4.1). Then the function given by the formula*

$$m_\Theta(\xi) := -\int_{S^{n-1}} \Theta(\omega)\log(i(\xi\cdot\omega))\,d\sigma(\omega) \qquad \text{for } \xi\in\mathbb{R}^n\setminus\{0\}, \tag{4.5.6}$$

is well-defined, positive homogeneous of degree zero, satisfies

$$\big\|m_\Theta\big\|_{L^\infty(\mathbb{R}^n)} \le C_n\|\Theta\|_{L^\infty(S^{n-1})}, \tag{4.5.7}$$

where $C_n\in(0,\infty)$ is defined as

$$C_n := \frac{\pi\omega_{n-1}}{2} + \int_{S^{n-1}} \Big|\ln\Big|\tfrac{\xi}{|\xi|}\cdot\omega\Big|\Big|\,d\sigma(\omega), \tag{4.5.8}$$

and

$$\big(m_\Theta\big)^\vee = m_{\Theta^\vee}. \tag{4.5.9}$$

Moreover, the Fourier transform of the tempered distribution P.V. Θ *is of function type and*

$$\mathcal{F}\big(\text{P.V.}\,\Theta\big) = m_\Theta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{4.5.10}$$

and

$$\overline{m_\Theta} = m_{\overline{\Theta}} \quad \text{if } \Theta \text{ is an even function.} \tag{4.5.11}$$

Proof. First, we show that the integral in (4.5.6) is absolutely convergent for each vector $\xi\in\mathbb{R}^n\setminus\{0\}$. To see this, fix $\xi\in\mathbb{R}^n\setminus\{0\}$ and observe that for each $\omega\in S^{n-1}$ we have

$$\begin{aligned}\log(i(\xi\cdot\omega)) &= \ln|\xi\cdot\omega| + i\frac{\pi}{2}\operatorname{sgn}(\xi\cdot\omega)\\ &= \ln|\xi| + \ln\Big|\tfrac{\xi}{|\xi|}\cdot\omega\Big| + i\frac{\pi}{2}\operatorname{sgn}(\xi\cdot\omega).\end{aligned} \tag{4.5.12}$$

Applying Proposition 13.46 with $f(t) := \ln|t|$, $t\in\mathbb{R}$, and $v := \frac{\xi}{|\xi|}$, yields

$$\int_{S^{n-1}} \Big|\ln\Big|\tfrac{\xi}{|\xi|}\cdot\omega\Big|\Big|\,d\sigma(\omega) = 2\omega_{n-2}\int_0^1 \big|\ln s\big|(\sqrt{1-s^2})^{n-3}\,ds < \infty. \tag{4.5.13}$$

As a by-product, we note that this implies that the constant C_n from (4.5.8) is finite. Next, from (4.5.13) and (4.5.8) we obtain

$$\begin{aligned}&\int_{S^{n-1}} \Big|\Theta(\omega)\Big(\ln\Big|\tfrac{\xi}{|\xi|}\cdot\omega\Big| + i\frac{\pi}{2}\operatorname{sgn}(\xi\cdot\omega)\Big)\Big|\,d\sigma(\omega)\\ &\qquad\le \|\Theta\|_{L^\infty(S^{n-1})}\left(\frac{\pi\omega_{n-1}}{2} + \int_{S^{n-1}}\Big|\ln\Big|\tfrac{\xi}{|\xi|}\cdot\omega\Big|\Big|\,d\sigma(\omega)\right)\\ &\qquad= C_n\|\Theta\|_{L^\infty(S^{n-1})} < \infty.\end{aligned} \tag{4.5.14}$$

From (4.5.12) and (4.5.14) it is now clear that the integral in (4.5.6) is absolutely convergent for each fixed $\xi \in \mathbb{R}^n \setminus \{0\}$. This proves that m_Θ is well-defined in $\mathbb{R}^n \setminus \{0\}$.

Going further, since $\int_{S^{n-1}} \Theta \, d\sigma = 0$, from (4.5.6) and (4.5.12) we see that, for each $\xi \in \mathbb{R}^n \setminus \{0\}$,

$$m_\Theta(\xi) = -\int_{S^{n-1}} \Theta(\omega)\Big(\ln\Big|\tfrac{\xi}{|\xi|}\cdot\omega\Big| + i\frac{\pi}{2}\operatorname{sgn}(\xi\cdot\omega)\Big)\, d\sigma(\omega). \tag{4.5.15}$$

Having justified this, we see that m_Θ is positive homogeneous of degree zero and that (4.5.7) follows based on (4.5.14). Also, (4.5.9) is obtained directly from (4.5.6) by changing variables $\omega \mapsto -\omega$.

To prove (4.5.10), fix an arbitrary function $\varphi \in \mathcal{S}(\mathbb{R}^n)$. Using (4.4.2), Lebesgue's dominated convergence theorem, Fubini's theorem, and (13.8.7), write

$$\begin{aligned}
\big\langle \mathcal{F}\big(\text{P.V.}\,\Theta\big), \varphi\big\rangle &= \big\langle \text{P.V.}\,\Theta, \widehat{\varphi}\,\big\rangle = \lim_{\varepsilon\to 0^+} \int_{|x|\geq\varepsilon} \Theta(x)\widehat{\varphi}(x)\,dx \qquad (4.5.16)\\
&= \lim_{\substack{\varepsilon\to 0^+\\ R\to\infty}} \int_{\varepsilon\leq|x|\leq R} \Theta(x)\widehat{\varphi}(x)\,dx\\
&= \lim_{\substack{\varepsilon\to 0^+\\ R\to\infty}} \int_{\mathbb{R}^n} \varphi(\xi) \int_{\varepsilon\leq|x|\leq R} \Theta(x)\, \mathrm{e}^{-ix\cdot\xi}\,dx\,d\xi\\
&= \lim_{\substack{\varepsilon\to 0^+\\ R\to\infty}} \int_{\mathbb{R}^n} \varphi(\xi) \int_{S^{n-1}} \Theta(\omega) \int_\varepsilon^R \mathrm{e}^{-i(\omega\cdot\xi)\rho}\,\frac{d\rho}{\rho}\,d\sigma(\omega)\,d\xi\\
&= \lim_{\substack{\varepsilon\to 0^+\\ R\to\infty}} \int_{\mathbb{R}^n} \varphi(\xi) \int_{S^{n-1}} \Theta(\omega) \int_\varepsilon^R \big[\mathrm{e}^{-i(\omega\cdot\xi)\rho} - \cos\rho\big]\,\frac{d\rho}{\rho}\,d\sigma(\omega)\,d\xi,
\end{aligned}$$

where the last equality uses $\int_{S^{n-1}} \Theta(\omega)\Big(\int_\varepsilon^R \frac{\cos\rho}{\rho}d\rho\Big)\,d\sigma(\omega) = 0$ for each $\varepsilon, R > 0$ (itself a consequence of the fact that Θ has mean value zero over S^{n-1}). At this stage, we wish to invoke Lebesgue's dominated convergence theorem in order to absorb the limit inside the integral. To see that this theorem is applicable in the current context, we first note that for each $\xi \in \mathbb{R}^n \setminus \{0\}$ and each $\omega \in S^{n-1}$ such that $\xi\cdot\omega \neq 0$, formulas (4.11.5) and (4.11.6) give

$$\begin{aligned}
&\lim_{\substack{\varepsilon\to 0^+\\ R\to\infty}} \int_\varepsilon^R \big[\mathrm{e}^{-i(\omega\cdot\xi)\rho)} - \cos\rho\big]\,\frac{d\rho}{\rho}\\
&\quad= \lim_{\substack{\varepsilon\to 0^+\\ R\to\infty}} \left[\int_\varepsilon^R \frac{\cos\big((\omega\cdot\xi)\rho\big) - \cos\rho}{\rho}\,d\rho - i\int_\varepsilon^R \frac{\sin\big((\omega\cdot\xi)\rho\big)}{\rho}\,dr\right]\\
&\quad= -\ln|(\omega\cdot\xi)| - i\frac{\pi}{2}\operatorname{sgn}(\omega\cdot\xi) = -\log\big(i(\omega\cdot\xi)\big).
\end{aligned} \tag{4.5.17}$$

This takes care of the pointwise convergence aspect of Lebesgue's theorem. To verify the uniform domination aspect, based on (4.11.7) and (4.11.8) we first estimate

$$\begin{aligned}
&\int_{\mathbb{R}^n} |\varphi(\xi)| \int_{S^{n-1}} |\Theta(\omega)| \Big(\sup_{0<\varepsilon<R} \Big| \int_\varepsilon^R \big[\mathrm{e}^{-i(\omega\cdot\xi)\rho} - \cos\rho \big] \frac{\mathrm{d}\rho}{\rho} \Big| \Big) \mathrm{d}\sigma(\omega)\, \mathrm{d}\xi \\
&\qquad \leq \int_{\mathbb{R}^n} |\varphi(\xi)| \int_{S^{n-1}} |\Theta(\omega)| \big(2 \big| \ln|\omega\cdot\xi| \big| + 4 \big) \mathrm{d}\sigma(\omega)\, \mathrm{d}\xi \qquad (4.5.18) \\
&\qquad \leq \|\Theta\|_{L^\infty(S^{n-1})} \Big[4\omega_{n-1} \|\varphi\|_{L^1(\mathbb{R}^n)} + 2 \int_{\mathbb{R}^n} |\varphi(\xi)| \int_{S^{n-1}} \big| \ln|\omega\cdot\xi| \big| \, \mathrm{d}\sigma(\omega)\, \mathrm{d}\xi \Big],
\end{aligned}$$

and note that, further,

$$\begin{aligned}
\int_{\mathbb{R}^n} |\varphi(\xi)| \int_{S^{n-1}} \big| \ln|\omega\cdot\xi| \big| \, \mathrm{d}\sigma(\omega)\, \mathrm{d}\xi &\leq \int_{\mathbb{R}^n} |\varphi(\xi)| \int_{S^{n-1}} \Big| \ln \Big| \tfrac{\xi}{|\xi|} \cdot \omega \Big| \Big| \, \mathrm{d}\sigma(\omega)\, \mathrm{d}\xi \\
&\quad + \omega_{n-1} \int_{\mathbb{R}^n} |\varphi(\xi)| \big| \ln|\xi| \big| \, \mathrm{d}\xi. \qquad (4.5.19)
\end{aligned}$$

From (4.5.18) and (4.5.19), the fact that $\varphi \in \mathcal{S}(\mathbb{R}^n)$, and (4.5.13), we may therefore conclude that

$$\int_{\mathbb{R}^n} |\varphi(\xi)| \int_{S^{n-1}} |\Theta(\omega)| \Big(\sup_{0<\varepsilon<R} \Big| \int_\varepsilon^R \big[\mathrm{e}^{-i(\omega\cdot\xi)\rho} - \cos\rho \big] \frac{\mathrm{d}\rho}{\rho} \Big| \Big) \mathrm{d}\sigma(\omega)\, \mathrm{d}\xi < \infty. \qquad (4.5.20)$$

Having established (4.5.17) and (4.5.20) we may now use Lebesgue's dominated convergence theorem in the context of (4.5.16) to obtain

$$\big\langle \mathcal{F}\big(\mathrm{P.V.}\,\Theta\big), \varphi \big\rangle = \int_{\mathbb{R}^n} \varphi(\xi)\, m_\Theta(\xi)\, \mathrm{d}\xi. \qquad (4.5.21)$$

From this, the fact that $m_\Theta \in L^\infty(\mathbb{R}^n)$, and keeping in mind that $\varphi \in \mathcal{S}(\mathbb{R}^n)$ was arbitrary, (4.5.10) follows.

Finally, to show (4.5.11), note that if Θ is assumed to be even, then

$$\begin{aligned}
\int_{S^{n-1}} \overline{\Theta(\omega)}\, \mathrm{sgn}\,(\xi\cdot\omega)\, \mathrm{d}\sigma(\omega) &= \int\limits_{\substack{\omega\in S^{n-1} \\ \omega\cdot\xi>0}} \overline{\Theta(\omega)}\, \mathrm{d}\sigma(\omega) - \int\limits_{\substack{\omega\in S^{n-1} \\ \omega\cdot\xi<0}} \overline{\Theta(\omega)}\, \mathrm{d}\sigma(\omega) \\
&= \int\limits_{\substack{\omega\in S^{n-1} \\ \omega\cdot\xi>0}} \overline{\Theta(\omega)}\, \mathrm{d}\sigma(\omega) - \int\limits_{\substack{\omega\in S^{n-1} \\ (-\omega)\cdot\xi<0}} \overline{\Theta(-\omega)}\, \mathrm{d}\sigma(\omega) \\
&= \int\limits_{\substack{\omega\in S^{n-1} \\ \omega\cdot\xi>0}} \big[\overline{\Theta(\omega)} - \overline{\Theta(-\omega)} \big] \mathrm{d}\sigma(\omega) = 0. \qquad (4.5.22)
\end{aligned}$$

Consequently,

$$\begin{aligned}\overline{m_{\Theta}(\xi)} &= -\int_{S^{n-1}} \overline{\Theta(\omega)}\Big(\ln\Big|\tfrac{\xi}{|\xi|}\cdot\omega\Big| - i\frac{\pi}{2}\operatorname{sgn}(\xi\cdot\omega)\Big)\,\mathrm{d}\sigma(\omega)\\ &= -\int_{S^{n-1}} \overline{\Theta(\omega)}\Big(\ln\Big|\tfrac{\xi}{|\xi|}\cdot\omega\Big| + i\frac{\pi}{2}\operatorname{sgn}(\xi\cdot\omega)\Big)\,\mathrm{d}\sigma(\omega)\\ &\quad + i\pi\int_{S^{n-1}} \overline{\Theta(\omega)}\operatorname{sgn}(\xi\cdot\omega)\,\mathrm{d}\sigma(\omega)\\ &= m_{\overline{\Theta}}(\xi),\end{aligned} \tag{4.5.23}$$

proving (4.5.11). The proof of the theorem is complete. □

Exercise 4.72. *Complete the following outline aimed at extending the convolution product to the class of principal value distributions in $\mathbb{R}^n$. Assume that Θ_1, Θ_2 are two given functions as in (4.4.1).*

Step 1. Pick an arbitrary function $\psi \in C_0^\infty(\mathbb{R}^n)$ with the property that $\psi = 1$ near the origin, and show that the following convolutions are meaningfully defined in $\mathcal{S}'(\mathbb{R}^n)$:

$$\begin{aligned}u_{00} &:= \Big(\psi\,\text{P.V.}\,\Theta_1\Big) * \Big(\psi\,\text{P.V.}\,\Theta_2\Big)\\ u_{01} &:= \Big(\psi\,\text{P.V.}\,\Theta_1\Big) * \Big((1-\psi)\,\text{P.V.}\,\Theta_2\Big)\\ u_{10} &:= \Big((1-\psi)\,\text{P.V.}\,\Theta_1\Big) * \Big(\psi\,\text{P.V.}\,\Theta_2\Big)\\ u_{11} &:= \Big((1-\psi)\,\text{P.V.}\,\Theta_1\Big) * \Big((1-\psi)\,\text{P.V.}\,\Theta_2\Big).\end{aligned} \tag{4.5.24}$$

For u_{00}, u_{01}, and u_{10}, use Proposition 4.63 and part (a) of Theorem 4.18. Show that $u_{11} = f_1 * f_2$ where $f_j := (1-\psi)\Theta_j$, $j = 1, 2$, are functions belonging to $L^2(\mathbb{R}^n)$ (here the behavior of Θ_1, Θ_2 at infinity is relevant). Use Young's inequality to conclude that u_{11} is meaningfully defined in $L^\infty(\mathbb{R}^n)$.

Step 2. With the same cutoff function ψ as in Step 1, define

$$\Big(\text{P.V.}\,\Theta_1\Big) * \Big(\text{P.V.}\,\Theta_2\Big) := u_{00} + u_{01} + u_{10} + u_{11} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{4.5.25}$$

and use this definition to show that

$$\mathcal{F}\Big[\Big(\text{P.V.}\,\Theta_1\Big) * \Big(\text{P.V.}\,\Theta_2\Big)\Big] = m_{\Theta_1} m_{\Theta_2} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{4.5.26}$$

where $m_{\Theta_1}, m_{\Theta_2}$ are associated with Θ_1, Θ_2 as in (4.5.6). To do so, compute first $\widehat{u_{jk}}$ for $j, k \in \{0, 1\}$, using part (c) in Theorem 4.33, Exercise 4.28, and Theorem 4.71.

Step 3. Use (4.5.26) to show that the definition in (4.5.25) is independent of the cutoff function ψ chosen at the beginning.

As an application, show that

$$\Big(\text{P.V.}\,\frac{1}{x}\Big) * \Big(\text{P.V.}\,\frac{1}{x}\Big) = -\pi^2\delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}). \tag{4.5.27}$$

4.6 Tempered Distributions Associated with $|x|^{-n}$

Let us consider the effect of dropping the cancellation condition in (4.4.1). Concretely, given a function $\Psi \in C^0(\mathbb{R}^n \setminus \{0\})$ that is positive homogeneous of degree $-n$, if one sets $C := \int_{S^{n-1}} \Psi \, d\sigma$, then $\Psi(x) = \Psi_0(x) + C|x|^{-n}$ for every $x \in \mathbb{R}^n \setminus \{0\}$, where Ψ_0 satisfies (4.4.1).

From Sect. 4.4 we know that one may associate to Ψ_0 the tempered distribution P.V. Ψ_0. As such, associating a tempered distribution to the original function Ψ hinges on how to meaningfully associat a tempered distribution to $\frac{1}{|x|^n}$. In fact, we shall associate to $\frac{1}{|x|^n}$ a family of tempered distributions in the manner described below.

Recall the class of functions Q from (4.4.4), fix $\psi \in Q$, and define

$$w_\psi(\varphi) := \int_{\mathbb{R}^n} \frac{1}{|x|^n}[\varphi(x) - \varphi(0)\psi(x)]\,dx, \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.6.1}$$

This gives rise to a well-defined, linear mapping $w_\psi : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$. In addition, much as in the proof of (4.4.7), there exists a constant $C \in [0,\infty)$ with the property that

$$\big|w_\psi(\varphi)\big| \leq C \sup_{x\in\mathbb{R}^n,\,|\alpha|\leq 1,\,|\beta|\leq 1} \big|x^\alpha \partial^\beta \varphi(x)\big|, \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.6.2}$$

In light of Fact 4.1 this shows that w_ψ is a tempered distribution. An inspection of (4.6.1) also reveals that $w_\psi\big|_{\mathbb{R}^n\setminus\{0\}} = |x|^{-n}$ in $\mathcal{D}'(\mathbb{R}^n \setminus \{0\})$. This is the sense in which we shall say that we have associated to $|x|^{-n}$ the family of tempered distributions $\{w_\psi\}_{\psi\in Q}$.

Our next goal is to determine a formula for the Fourier transform of the tempered distribution in (4.6.1). As a preamble, note that the class Q is invariant under dilations (recall (4.3.5)); that is, for each $t \in (0,\infty)$ we have

$$\tau_t(\psi) \in Q, \qquad \forall\, t \in (0,\infty), \quad \forall\,\psi \in Q. \tag{4.6.3}$$

Theorem 4.73. *For each $\psi \in Q$, the Fourier transform of the tempered distribution w_ψ defined in (4.6.1) is of function type and*

$$\widehat{w_\psi}(\xi) = -\omega_{n-1}\ln|\xi| + C_\psi, \quad \forall\,\xi \in \mathbb{R}^n \setminus \{0\}, \tag{4.6.4}$$

for some constant $C_\psi \in \mathbb{C}$.

Proof. Fix $\psi \in Q$. As noted earlier, $w_\psi\big|_{\mathbb{R}^n\setminus\{0\}} = \frac{1}{|x|^n}$, hence Proposition 4.58 applies and yields

$$\widehat{w_\psi}\big|_{\mathbb{R}^n\setminus\{0\}} \in C^\infty(\mathbb{R}^n \setminus \{0\}). \tag{4.6.5}$$

To establish (4.6.4) we proceed by proving a series of claims.
Claim 1: $\widehat{w_\psi}$ is invariant under orthogonal transformations. Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ be an orthogonal matrix. Then

$$\begin{aligned}\langle w_\psi \circ A, \varphi\rangle &= \langle w_\psi, \varphi \circ A^{-1}\rangle = \int_{\mathbb{R}^n} \frac{1}{|x|^n}\left[\varphi(A^{-1}x) - \varphi(0)\psi(A^{-1}x)\right] \mathrm{d}x \\ &= \langle w_\psi, \varphi\rangle, \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n).\end{aligned} \tag{4.6.6}$$

This shows that w_ψ is invariant under orthogonal transformations which, together with Proposition 4.47, implies that $\widehat{w_\psi}$ is also invariant under orthogonal transformations.

Claim 2: $\tau_t(w_\psi) = t^{-n} w_{\tau_t\psi}$ in $\mathcal{S}'(\mathbb{R}^n)$. To see why this is true, for any $t \in (0,\infty)$ and any $\varphi \in \mathcal{S}(\mathbb{R}^n)$ write

$$\begin{aligned}\langle \tau_t w_\psi, \varphi\rangle &= t^{-n}\langle w_\psi, \tau_{\frac{1}{t}}\varphi\rangle = t^{-n}\int_{\mathbb{R}^n} \frac{1}{|x|^n}\left[\varphi\left(\frac{x}{t}\right) - \varphi(0)\psi(x)\right] \mathrm{d}x \\ &= \int_{\mathbb{R}^n} \frac{1}{t^n|y|^n}\left[\varphi(y) - \varphi(0)\psi(ty)\right] \mathrm{d}y = t^{-n}\langle w_{\tau_t\psi}, \varphi\rangle,\end{aligned} \tag{4.6.7}$$

from which the desired identity follows.

Claim 3: For each $t \in (0,\infty)$ we have

$$\tau_t w_\psi - t^{-n} w_\psi = \left(t^{-n}\int_{\mathbb{R}^n} \frac{\psi(x) - \psi(tx)}{|x|^n}\,\mathrm{d}x\right)\delta \quad \text{in } \mathcal{S}'(\mathbb{R}^n). \tag{4.6.8}$$

To prove the current claim, consider $\psi_1, \psi_2 \in Q$ and write

$$\begin{aligned}\langle w_{\psi_1} - w_{\psi_2}, \varphi\rangle &= \int_{\mathbb{R}^n} \frac{\varphi(0)}{|x|^n}[\psi_2(x) - \psi_1(x)]\,\mathrm{d}x \\ &= \left\langle \left(\int_{\mathbb{R}^n} \frac{\psi_2(x) - \psi_1(x)}{|x|^n}\mathrm{d}x\right)\delta\,,\,\varphi\right\rangle \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n).\end{aligned} \tag{4.6.9}$$

As a result,

$$w_{\psi_1} - w_{\psi_2} = \left(\int_{\mathbb{R}^n} \frac{1}{|x|^n}[\psi_2(x) - \psi_1(x)]\,\mathrm{d}x\right)\delta \quad \text{in } \mathcal{S}'(\mathbb{R}^n). \tag{4.6.10}$$

Combining now the identity from Claim 2 and (4.6.10) yields

$$\tau_t w_\psi - t^{-n} w_\psi = t^{-n}[w_{\tau_t\psi} - w_\psi]$$
$$= \left(t^{-n} \int_{\mathbb{R}^n} \frac{\psi(x) - \psi(tx)}{|x|^n}\, \mathrm{d}x \right) \delta \quad in\, \mathcal{S}'(\mathbb{R}^n) \tag{4.6.11}$$

for every $t \in (0, \infty)$. This proves Claim 3.

Claim 4: The following identity is true:

$$\int_{\mathbb{R}^n} \frac{\psi(x) - \psi(tx)}{|x|^n}\, \mathrm{d}x = \omega_{n-1} \ln t \qquad \forall\, t \in (0, \infty). \tag{4.6.12}$$

To set the stage for the proof of Claim 4 observe that there exists some C^1 function $\eta : [0, \infty) \to \mathbb{R}$ decaying at ∞, satisfying $\eta(0) = 1$, and $\psi(x) = \eta(|x|)$ for every $x \in \mathbb{R}^n$. In particular, $(\nabla\psi)(x) = \eta'(|x|)\frac{x}{|x|}$ for every $x \in \mathbb{R}^n \setminus \{0\}$. Focusing attention on (4.6.12), note that the intervening integral is absolutely convergent. In turn, this allows us to write for each fixed $t \in (0, \infty)$

$$\begin{aligned}
\int_{\mathbb{R}^n} \frac{\psi(x) - \psi(tx)}{|x|^n}\, \mathrm{d}x &= \lim_{R\to\infty} \int_{|x|<R} \frac{1}{|x|^n} \Big(\int_t^1 \frac{\mathrm{d}}{\mathrm{d}s} \big[\psi(sx)\big]\, \mathrm{d}s \Big)\, \mathrm{d}x \\
&= \lim_{R\to\infty} \int_{|x|<R} \frac{1}{|x|^{n-1}} \Big(\int_t^1 \eta'(s|x|)\, \mathrm{d}s \Big)\, \mathrm{d}x \\
&= \omega_{n-1} \lim_{R\to\infty} \int_0^R \Big(\int_t^1 \eta'(s\rho)\, \mathrm{d}s \Big)\, \mathrm{d}\rho \\
&= \omega_{n-1} \lim_{R\to\infty} \int_t^1 \Big(\int_0^R \frac{1}{s} \frac{\mathrm{d}}{\mathrm{d}\rho} \big[\eta(s\rho)\big]\, \mathrm{d}\rho \Big)\, \mathrm{d}s \\
&= \omega_{n-1} \lim_{R\to\infty} \int_t^1 \frac{1}{s} \big[\eta(sR) - \eta(0)\big]\, \mathrm{d}s = \omega_{n-1} \ln t,
\end{aligned} \tag{4.6.13}$$

as wanted.

Claim 5: The following identity holds:

$$\tau_t \widehat{w_\psi} = \widehat{w_\psi} - \omega_{n-1} \ln t \quad in\ \mathbb{R}^n \setminus \{0\}, \quad \forall\, t \in (0, \infty). \tag{4.6.14}$$

To prove (4.6.14), we first combine (4.6.8) and (4.6.12) and obtain

$$\tau_t w_\psi = t^{-n} w_\psi + (\omega_{n-1} t^{-n} \ln t)\delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \quad \forall\, t \in (0, \infty). \tag{4.6.15}$$

Hence, for each $t \in (0, \infty)$ and each $\varphi \in \mathcal{S}(\mathbb{R}^n)$, we have

$$\begin{aligned}
\langle \tau_t \widehat{w_\psi}, \varphi \rangle &= t^{-n} \Big\langle \widehat{w}_\psi, \tau_{\frac{1}{t}} \varphi \Big\rangle = \langle w_\psi, \tau_t \widehat{\varphi}\, \rangle = t^{-n} \big\langle \tau_{\frac{1}{t}} w_\psi, \widehat{\varphi}\, \big\rangle \\
&= t^{-n} \big\langle t^n w_\psi - (\omega_{n-1} t^n \ln t)\delta\, , \widehat{\varphi}\, \big\rangle = \big\langle w_\psi - (\omega_{n-1} \ln t)\delta\, , \widehat{\varphi}\, \big\rangle \\
&= \big\langle \widehat{w_\psi} - \omega_{n-1} \ln t\, ,\, \varphi \big\rangle.
\end{aligned} \tag{4.6.16}$$

The first and third equality in (4.6.16) is based on the definition of τ_t acting on $\mathcal{S}'(\mathbb{R}^n)$, the second uses (4.3.7), while the forth makes use of (4.6.15). Now (4.6.14) follows from (4.6.16) and the fact that $\widehat{w_\psi}\big|_{\mathbb{R}^n\setminus\{0\}}\in C^\infty(\mathbb{R}^n\setminus\{0\})$.

We are ready to complete the proof of Theorem 4.73. First, Claim 1 ensures that $\widehat{w_\psi}\big|_{\mathbb{R}^n\setminus\{0\}}$ is constant on S^{n-1}, thus

$$C_\psi := \widehat{w_\psi}\big|_{S^{n-1}} \tag{4.6.17}$$

is a well-defined complex number. Hence, given any $\xi\in\mathbb{R}^n\setminus\{0\}$, taking $t:=\frac{1}{|\xi|}$ in (4.6.14) yields

$$C_\psi = \widehat{w_\psi}\left(\frac{\xi}{|\xi|}\right) = \tau_{\frac{1}{|\xi|}}\widehat{w_\psi}(\xi) = \widehat{w_\psi}(\xi)+\omega_{n-1}\ln|\xi|, \tag{4.6.18}$$

from which (4.6.4) follows. □

Remark 4.74. *It is possible to enlarge the scope of our earlier considerations by relaxing the hypotheses on the function ψ. Specifically, in place of (4.4.3) assume now that $\psi:\mathbb{R}^n\to\mathbb{C}$ is such that*

$$\begin{aligned}&\psi \text{ is radial, of class } C^1 \text{ near origin, satisfies } \psi(0)=1,\\ &\text{and } \exists\,\varepsilon_o, C\in(0,\infty) \text{ such that } |\psi(x)|\leq C(1+|x|)^{-\varepsilon_o}.\end{aligned} \tag{4.6.19}$$

Such a function ψ still induces a tempered distribution as in (4.6.1), and we claim that formula (4.6.4) continues to be valid in this more general case as well.

To justify this claim, observe that the proof of Theorem 4.73 goes through for ψ as in (4.6.19), albeit (4.6.12) requires more care, since we are now dropping the global C^1 requirement on η. However, this may be remedied via mollification, i.e., working with η_ε in place of η (constructed much as in (1.2.3)–(1.2.6)), then observing that if $\psi_\varepsilon(x):=\eta_\varepsilon(|x|)$ then $\psi_\varepsilon/\psi_\varepsilon(0)\in\mathcal{Q}$, hence

$$\int_{\mathbb{R}^n}\frac{\psi_\varepsilon(x)-\psi_\varepsilon(tx)}{|x|^n}\,dx=\omega_{n-1}\psi_\varepsilon(0)\ln t\qquad\forall\,t\in(0,\infty), \tag{4.6.20}$$

and finally passing to limit $\varepsilon\to 0^+$ in (4.6.20).

Remark 4.74 applies, in particular, to the function $\psi:=\chi_{B(0,1)}$. This gives that

$$\Big\langle w_{\chi_{B(0,1)}},\varphi\Big\rangle=\int_{|x|>1}\frac{\varphi(x)}{|x|^n}\,dx+\int_{|x|\leq 1}\frac{\varphi(x)-\varphi(0)}{|x|^n}\,dx,\quad\forall\,\varphi\in\mathcal{S}(\mathbb{R}^n), \tag{4.6.21}$$

is a tempered distribution, $\mathcal{F}\big(w_{\chi_{B(0,1)}}\big)\in\mathcal{S}'(\mathbb{R}^n)$ is of function type, and there exists a constant $c\in\mathbb{C}$ such that

$$\mathcal{F}\big(w_{\chi_{B(0,1)}}\big)(\xi)=-\omega_{n-1}\ln|\xi|+c,\quad\forall\,\xi\in\mathbb{R}^n\setminus\{0\}. \tag{4.6.22}$$

In the last part of this section we compute the constant c from (4.6.22) in the case $n = 1$. Before doing so we recall that `Euler's constant` γ is defined by

$$\gamma := \lim_{k\to\infty}\left[\sum_{j=1}^{k}\frac{1}{j} - \ln k\right]. \tag{4.6.23}$$

The number γ plays an important role in analysis, as it appears prominently in a number of basic formulas. Two such identities that are relevant for us here are as follows:

$$\int_0^\infty \mathrm{e}^{-x}\ln x\,\mathrm{d}x = -\gamma, \qquad \int_0^\infty \mathrm{e}^{-x^2}\ln x\,\mathrm{d}x = -\frac{\sqrt{\pi}}{4}(\gamma + 2\ln 2). \tag{4.6.24}$$

Example 4.75. We claim that

$$\mathcal{F}\big(w_{\chi_{(-1,1)}}\big)(\xi) = -2\ln|\xi| - 2\gamma \quad \text{ in } \ \mathcal{S}'(\mathbb{R}), \tag{4.6.25}$$

where $w_{\chi_{(-1,1)}}$ is defined in (4.6.21). To see why this is true, note that (4.6.22) written for $n = 1$ (in which case $\omega_{n-1} = 2$) becomes $\mathcal{F}\big(w_{\chi_{(-1,1)}}\big)(x) = -2\ln|x| + c$ in $\mathcal{S}'(\mathbb{R})$. Consequently, to obtain (4.6.25) it remains to prove that $c = -2\gamma$ when $n = 1$.

The idea for determining the value of c is to apply $\mathcal{F}\big(w_{\chi_{(-1,1)}}\big)$ to the Schwartz function e^{-x^2}. First, from (4.6.22) we have

$$\Big\langle \mathcal{F}\big(w_{\chi_{(-1,1)}}\big), \mathrm{e}^{-x^2}\Big\rangle = \langle -2\ln|x| + c, \mathrm{e}^{-x^2}\rangle. \tag{4.6.26}$$

Second, we compute the term in the left hand-side of (4.6.26). Specifically, based on the definition of the Fourier transform of a tempered distribution, (3.2.6), and (4.6.21) we may write

$$\begin{aligned}
\Big\langle \mathcal{F}\big(w_{\chi_{(-1,1)}}\big), \mathrm{e}^{-x^2}\Big\rangle &= \big\langle w_{\chi_{(-1,1)}}, \mathcal{F}(\mathrm{e}^{-x^2})\big\rangle = \big\langle w_{\chi_{(-1,1)}}, \sqrt{\pi}\mathrm{e}^{-x^2/4}\big\rangle \\
&= \sqrt{\pi}\int_{|x|>1}\frac{\mathrm{e}^{-x^2/4}}{|x|}\,\mathrm{d}x + \sqrt{\pi}\int_{|x|\le 1}\frac{\mathrm{e}^{-x^2/4}-1}{|x|}\,\mathrm{d}x \\
&= 2\sqrt{\pi}\int_{1/2}^{\infty}\frac{\mathrm{e}^{-x^2}}{x}\,\mathrm{d}x + 2\sqrt{\pi}\int_0^{1/2}\frac{\mathrm{e}^{-x^2}-1}{x}\,\mathrm{d}x =: I + II.
\end{aligned} \tag{4.6.27}$$

Going further, integrating by parts then changing variables gives

$$\begin{aligned}
I &= 2\sqrt{\pi}\mathrm{e}^{-1/4}\ln 2 + 4\sqrt{\pi}\int_{1/2}^{\infty}(\ln x)\mathrm{e}^{-x^2}x\,\mathrm{d}x \\
&= 2\sqrt{\pi}\mathrm{e}^{-1/4}\ln 2 + \sqrt{\pi}\int_{1/\sqrt{2}}^{\infty}(\ln x)\mathrm{e}^{-x}\,\mathrm{d}x.
\end{aligned} \tag{4.6.28}$$

Regarding II, for each $\varepsilon > 0$ an integration by parts and then a change of variables yield

$$\begin{aligned}\int_{\varepsilon}^{1/2} \frac{\mathrm{e}^{-x^2}-1}{x}\,\mathrm{d}x &= -(\mathrm{e}^{-1/4}-1)\ln 2 - (\mathrm{e}^{-\varepsilon^2}-1)\ln\varepsilon + 2\int_{\varepsilon}^{1/2}(\ln x)\mathrm{e}^{-x^2}x\,\mathrm{d}x \\ &= -(\mathrm{e}^{-1/4}-1)\ln 2 - (\mathrm{e}^{-\varepsilon^2}-1)\ln\varepsilon + \frac{1}{2}\int_{\sqrt{\varepsilon}}^{1/\sqrt{2}}(\ln x)\mathrm{e}^{-x}\,\mathrm{d}x.\end{aligned} \tag{4.6.29}$$

Observe that since

$$\lim_{\varepsilon\to 0^+}(\mathrm{e}^{-\varepsilon^2}-1)\ln\varepsilon = \frac{1}{2}\lim_{y\to\infty}\frac{\ln\big(\ln y - \ln(y-1)\big)}{y} = 0, \tag{4.6.30}$$

by taking the limit as $\varepsilon \to 0^+$ in (4.6.29) we may conclude that

$$II = -2\sqrt{\pi}(\mathrm{e}^{-1/4}-1)\ln 2 + \sqrt{\pi}\int_0^{1/\sqrt{2}}(\ln x)\mathrm{e}^{-x}\,\mathrm{d}x. \tag{4.6.31}$$

Hence, from (4.6.28), (4.6.31), and the first identity in (4.6.24), it follows that

$$I + II = 2\sqrt{\pi}\ln 2 + \sqrt{\pi}\int_0^{\infty}(\ln x)\mathrm{e}^{-x}\,\mathrm{d}x = 2\sqrt{\pi}\ln 2 - \sqrt{\pi}\gamma. \tag{4.6.32}$$

This takes care of the term in the left-hand side of (4.6.25). To compute the term in the right-hand side of (4.6.25), write

$$\begin{aligned}\langle -2\ln|\xi| + c, \mathrm{e}^{-x^2}\rangle &= -2\int_{\mathbb{R}}(\ln|x|)\mathrm{e}^{-x^2}\,\mathrm{d}x + c\int_{\mathbb{R}}\mathrm{e}^{-x^2}\,\mathrm{d}x \\ &= -4\int_0^{\infty}(\ln x)\mathrm{e}^{-x^2}\,\mathrm{d}x + c\sqrt{\pi} \\ &= \sqrt{\pi}(\gamma + 2\ln 2) + c\sqrt{\pi}.\end{aligned} \tag{4.6.33}$$

For the last equality in (4.6.33) we used the second identity in (4.6.24). A combination of (4.6.26), (4.6.27), (4.6.32), and (4.6.33), then implies $c = -2\gamma$, as desired.

4.7 A General Jump-Formula in the Class of Tempered Distributions

The aim in this section is to prove a very useful formula expressing the limits of certain sequences of tempered distributions $\{\Phi_\varepsilon\}_{\varepsilon\neq 0}$ as the index parameter $\varepsilon \in \mathbb{R}\setminus\{0\}$ approaches the origin from either side. The trademark feature (which also justifies the name) of this formula is the presence of a jump-term of

the form $\pm C\delta$ (where the sign is correlated to $\operatorname{sgn}\varepsilon$) in addition to a suitable principal value tempered distribution. A conceptually simple example of this phenomenon has been presented in Exercise 2.117, which our theorem contains as a simple special case (see Remark 4.79).

With the notational convention that for points $x \in \mathbb{R}^n$ we write $x = (x', t)$, where $x' = (x_1, \ldots, x_{n-1}) \in \mathbb{R}^{n-1}$ and $t \in \mathbb{R}$, our main result in this regard reads as follows.

Theorem 4.76. *If $\Phi \in C^4(\mathbb{R}^n \setminus \{0\})$ is odd and positive homogeneous of degree $1-n$, then*

$$\lim_{\varepsilon\to 0^{\pm}} \Phi(x',\varepsilon) = \pm\frac{i}{2}\,\widehat{\Phi}(0',1)\,\delta(x') + \text{P.V.}\,\Phi(x',0) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n-1}). \tag{4.7.1}$$

A few comments before presenting the proof of this result are in order. First, above we have employed the earlier convention of writing $u(x')$ for a distribution u in $\mathbb{R}^{n-1}$ simply to stress that the test functions to which u is applied are considered in the variable $x' \in \mathbb{R}^{n-1}$.

Second, for each fixed $\varepsilon \in \mathbb{R} \setminus \{0\}$, applying Exercise 4.51 yields

$$|\Phi(x',\varepsilon)| \leq \|\Phi\|_{L^\infty(S^{n-1})}\big(|x'|^2+\varepsilon^2\big)^{-(n-1)/2} \quad \text{for each} \quad x' \in \mathbb{R}^{n-1}. \tag{4.7.2}$$

Having observed this, the discussion in Example 4.3 then shows $\Phi(\cdot,\varepsilon) \in \mathcal{S}'(\mathbb{R}^{n-1})$.

Third, it is worth recalling an earlier convention to the effect that given a family of tempered distributions u_ε, indexed by $\varepsilon \in I$, $I = (a,b) \subseteq \mathbb{R}$ open interval, we say that $u_\varepsilon \to u \in \mathcal{D}'(\mathbb{R}^n)$ in $\mathcal{S}'(\mathbb{R}^n)$ as $I \ni \varepsilon \to a^+$ provided $u_{\varepsilon_j} \to u$ in $\mathcal{S}'(\mathbb{R}^n)$ for every sequence $\{\varepsilon_j\}_{j\in\mathbb{N}} \subseteq I$ such that $\varepsilon_j \to a^+$ as $j \to \infty$. In particular, this interpretation is in effect for (4.7.1).

Fourth, from Exercise 4.52 we know that $\Phi \in \mathcal{S}'(\mathbb{R}^n)$. In addition, Exercise 4.60 applied to Φ shows that $\widehat{\Phi}\big|_{\mathbb{R}^n\setminus\{0\}} \in C^1(\mathbb{R}^n\setminus\{0\})$. In particular, $\widehat{\Phi}(0',1)$ is meaningfully defined.

Fifth, $\Phi(\cdot,0)$ (viewed as a function in $\mathbb{R}^{n-1}\setminus\{0'\}$) is continuous and positive homogeneous of degree $-(n-1)$ in $\mathbb{R}^{n-1}\setminus\{0'\}$ and, being odd, satisfies $\int_{S^{n-2}} \Phi(\cdot,0)\,d\sigma = 0$. As such, the conditions in (4.4.1) are satisfied (with $n-1$ in place of n), hence Proposition 4.63 ensures that $\text{P.V.}\,\Phi(\cdot,0)$ is a well-defined tempered distribution in $\mathbb{R}^{n-1}$.

Proof of Theorem 4.76. Assume $\Phi \in C^4(\mathbb{R}^n\setminus\{0\})$ is odd and positive homogeneous of degree $1-n$, and let $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$. Then, for any fixed $\varepsilon > 0$, write

$$\begin{aligned}\lim_{t\to 0^+}\int_{\mathbb{R}^{n-1}} \Phi(x',t)\varphi(x')\,dx' &= \lim_{t\to 0^+}\int\limits_{|x'|>\varepsilon} \Phi(x',t)\varphi(x')\,dx' \\ &\quad + \lim_{t\to 0^+}\int\limits_{|x'|<\varepsilon} \Phi(x',t)[\varphi(x')-\varphi(0')]\,dx'\end{aligned}$$

$$+\,\varphi(0')\lim_{t\to 0^+}\int\limits_{|x'|<\varepsilon}\Phi(x',t)\,\mathrm{d}x'$$

$$=:I_\varepsilon+II_\varepsilon+III_\varepsilon. \tag{4.7.3}$$

Note that, making the change of variable $y':=x'/t$, we obtain

$$III_\varepsilon=\varphi(0')\lim_{t\to 0^+}\int\limits_{|y'|<\varepsilon/t}\Phi(y',1)\,\mathrm{d}y'=\varphi(0')\lim_{r\to\infty}\int\limits_{|y'|<r}\Phi(y',1)\,\mathrm{d}y', \tag{4.7.4}$$

that is independent of ε. Also, since for every $x'\in\mathbb{R}^{n-1}$ and $t\in\mathbb{R}\setminus\{0\}$,

$$|\Phi(x',t)|\leq\frac{\|\Phi\|_{L^\infty(S^{n-1})}}{|(x',t)|^{n-1}}\leq\frac{\|\Phi\|_{L^\infty(S^{n-1})}}{|x'|^{n-1}}, \tag{4.7.5}$$

$$|\varphi(x')-\varphi(0')|\leq\|\nabla\varphi\|_{L^\infty(\mathbb{R}^{n-1})}|x'|, \tag{4.7.6}$$

it follows that $|II_\varepsilon|\leq C\varepsilon$, hence

$$\lim_{\varepsilon\to 0^+}II_\varepsilon=0. \tag{4.7.7}$$

Finally, it is clear from Lebesgue's dominated convergence theorem that

$$I_\varepsilon=\int_{|x'|>\varepsilon}\Phi(x',0)\varphi(x')\,\mathrm{d}x'. \tag{4.7.8}$$

By passing to the limit $\varepsilon\to 0^+$ we therefore conclude that for any $\varphi\in\mathcal{S}(\mathbb{R}^{n-1})$

$$\lim_{t\to 0^+}\int_{\mathbb{R}^{n-1}}\Phi(x',t)\varphi(x')\,\mathrm{d}x'=\lim_{\varepsilon\to 0^+}\int\limits_{|x'|>\varepsilon}\Phi(x',0)\varphi(x')\,\mathrm{d}x'$$

$$+\,\varphi(0')\lim_{r\to\infty}\int\limits_{|x'|<r}\Phi(x',1)\,\mathrm{d}x'. \tag{4.7.9}$$

To proceed, make the change of variable $x'\mapsto -x'$ in each integral of (4.7.9) and use the fact that Φ is odd. After re-denoting t by $-t$ and $\varphi^\vee$ by φ this yields the identity

$$\lim_{t\to 0^-}\int_{\mathbb{R}^{n-1}}\Phi(x',t)\varphi(x')\,\mathrm{d}x'=\lim_{\varepsilon\to 0^+}\int\limits_{|x'|>\varepsilon}\Phi(x',0)\varphi(x')\,\mathrm{d}x'$$

$$+\,\varphi(0')\lim_{r\to\infty}\int\limits_{|x'|<r}\Phi(x',-1)\,\mathrm{d}x', \tag{4.7.10}$$

for any $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$. Collectively, (4.7.9) and (4.7.10) may then be written as

$$\lim_{t\to 0^{\pm}} \int_{\mathbb{R}^{n-1}} \Phi(x',t)\varphi(x')\,\mathrm{d}x' \tag{4.7.11}$$
$$= a_{\pm}\varphi(0') + \lim_{\varepsilon\to 0^{+}} \int\limits_{|x'|>\varepsilon} \Phi(x',0)\varphi(x')\,\mathrm{d}x', \quad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^{n-1}),$$

where

$$a_{\pm} := \lim_{r\to\infty} \int\limits_{|x'|<r} \Phi(x',\pm 1)\,\mathrm{d}x'. \tag{4.7.12}$$

Regarding (4.7.12), we note that the limits

$$\lim_{r\to\infty} \int_{|x'|<r} \Phi(x',\pm 1)\,\mathrm{d}x' \quad \text{exist in } \ \mathbb{C}. \tag{4.7.13}$$

To see that this is the case, first observe that by Exercise 4.51,

$$\int_{|x'|<r} |\Phi(x',t)|\,\mathrm{d}x' < \infty, \quad \forall\, t \neq 0,\ \forall\, r > 0. \tag{4.7.14}$$

Consider the case of the choice of the sign "plus" in (4.7.13) (the case of the sing "minus" is treated analogously). For this choice of sign we write for each fixed $r > 0$

$$\begin{aligned}\int_{|x'|<r} \Phi(x',1)\,\mathrm{d}x' &= \frac{1}{2}\int_{|x'|<r} \Phi(x',1)\,\mathrm{d}x' + \frac{1}{2}\int_{|x'|<r} \Phi(x',1)\,\mathrm{d}x' \\ &= \frac{1}{2}\int_{|x'|<r} \Phi(x',1)\,\mathrm{d}x' - \frac{1}{2}\int_{|x'|<r} \Phi(x',-1)\,\mathrm{d}x' \\ &= \int_{|x'|<r} \tfrac{1}{2}\big[\Phi(x',1) - \Phi(x',-1)\big]\,\mathrm{d}x', \end{aligned} \tag{4.7.15}$$

where we have used the fact that Φ is odd. Next, the mean value theorem and the fact that $\nabla\Phi$ is positive homogeneous of degree $-n$ allow us to estimate

$$|\Phi(x',1) - \Phi(x',-1)| \leq \frac{2\|\nabla\Phi\|_{L^{\infty}(S^{n-1})}}{|x'|^{n}} \quad \text{for } |x'| \text{ large}, \tag{4.7.16}$$

which implies that

$$\int_{\mathbb{R}^{n-1}} |\Phi(x',1)\,\mathrm{d}x' - \Phi(x',-1)|\,\mathrm{d}x' < \infty. \tag{4.7.17}$$

Consequently, by Lebesgue's dominated convergence theorem

$$\lim_{r\to\infty}\int_{|x'|<r}\Phi(x',1)\,dx'=\int_{\mathbb{R}^{n-1}}\tfrac{1}{2}\big[\Phi(x',1)\,dx'-\Phi(x',-1)\big]\,dx'\in\mathbb{C},\quad(4.7.18)$$

proving (4.7.13).

There remains to identify the actual values of $a_\pm$ and we organize the remainder of the proof as a series of claims, starting with:

Claim 1. If $u_0\in\mathcal{E}'(\mathbb{R}^n)$ and $u_1\in L^1(\mathbb{R}^n)$, then

$$u:=u_0+u_1\quad\text{has the property that }\widehat{u}\in C^0(\mathbb{R}^n).\qquad(4.7.19)$$

Proof of Claim 1. First note that $\widehat{u}=\widehat{u}_0+\widehat{u}_1$. Then, since $\widehat{u}_0\in C^\infty(\mathbb{R}^n)$ by part *(b)* in Theorem 4.33, and $\widehat{u}_1\in C^0(\mathbb{R}^n)$ by Lemma 4.27, it follows that $\widehat{u}\in C^0(\mathbb{R}^n)$.

Claim 2. Assume that $\xi_n\neq 0$ and that $\xi'\in\mathbb{R}^{n-1}$ is such that $|\xi'|\leq C'$ for some $C'\in(0,\infty)$. Also, suppose that $\Theta\in C^1(\mathbb{R}^n\setminus\{0\})$ satisfies

$$\Theta(\lambda x)=\lambda^{-1}\Theta(x),\qquad\forall\,x\in\mathbb{R}^n\setminus\{0\},\quad\forall\,\lambda\in\mathbb{R}\setminus\{0\}.\qquad(4.7.20)$$

Then there exists $C\in(0,\infty)$ independent of ξ_n such that

$$|\Theta(\xi',-\xi_n)+\Theta(0',\xi_n)|\leq\frac{C}{|\xi_n|^2}.\qquad(4.7.21)$$

Proof of Claim 2. From (4.7.20) it follows that $(\nabla\Theta)(\lambda x)=\lambda^{-2}(\nabla\Theta)(x)$ for every $x\in\mathbb{R}^n\setminus\{0\}$ and every $\lambda\in\mathbb{R}\setminus\{0\}$. Based on (4.7.20) and this observation, we may estimate

$$\begin{aligned}|\Theta(\xi',-\xi_n)+\Theta(0',\xi_n)|&=\frac{1}{|\xi_n|}|\Theta(-\xi'/\xi_n,1)-\Theta(0',1)|\\&\leq\frac{1}{|\xi_n|}\cdot\left|\frac{\xi'}{\xi_n}\right|\cdot\sup_{t\in[0,1]}|(\nabla\Theta)(-t\xi'/\xi_n,1)|\\&\leq\frac{1}{|\xi_n|}\cdot\left|\frac{\xi'}{\xi_n}\right|\cdot\sup_{t\in[0,1]}\frac{\|\nabla\Theta\|_{L^\infty(S^{n-1})}}{|(-t\xi'/\xi_n,1)|^2}\\&\leq\frac{C'}{|\xi_n|^2}\cdot\|\nabla\Theta\|_{L^\infty(S^{n-1})},\end{aligned}\qquad(4.7.22)$$

where the last line uses the assumption that $|\xi'|\leq C'$. Since $\nabla\Theta$ is continuous, hence bounded, on S^{n-1}, the desired result follows by taking $C:=C'\|\nabla\Theta\|_{L^\infty(S^{n-1})}<\infty$.

Claim 3. If $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$ is such that $\widehat{\varphi}$ has compact support, then

$$F(\xi_n) := -(2\pi)^{1-n} \int_{\mathbb{R}^{n-1}} \big[\widehat{\Phi}(\xi', -\xi_n) + \widehat{\Phi}(0, \xi_n)\big] \widehat{\varphi}(\xi')\, \mathrm{d}\xi'. \tag{4.7.23}$$

is well-defined and integrable on $\mathbb{R} \setminus [-1, 1]$.

Proof of Claim 3. From Proposition 4.57 it follows that $\widehat{\Phi}$ is positive homogeneous tempered distribution of degree $-n-(1-n) = -1$. This readily implies that $\widehat{\Phi}\big|_{\mathbb{R}^n\setminus\{0\}}$, viewed as a continuous function in $\mathbb{R}^n \setminus \{0\}$, is also positive homogeneous of degree -1. Moreover, from (4.2.30) we deduce that $\widehat{\Phi}$ is an odd function in $\mathbb{R}^n \setminus \{0\}$. As a consequence, the function $\Theta := \widehat{\Phi}\big|_{\mathbb{R}^n\setminus\{0\}} \in C^1(\mathbb{R}^n \setminus \{0\})$ satisfies (4.7.20).

Hence, Claim 2 applies to this Θ and, granted the compact support condition on $\widehat{\varphi}$, it follows that there exists $C \in (0, \infty)$ such that

$$|F(\xi_n)| \leq \frac{C}{|\xi_n|^2}, \quad \forall\, \xi_n \neq 0, \tag{4.7.24}$$

from which the desired conclusion follows.

Claim 4. Assume $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$ is such that $\widehat{\varphi}$ has compact support. Then the function

$$f(t) := \int_{\mathbb{R}^{n-1}} \Phi(x', t)\varphi(x')\, \mathrm{d}x' + \frac{1}{2i}(\operatorname{sgn} t)\widehat{\Phi}(0', 1)\varphi(0'), \tag{4.7.25}$$

originally defined for $t \in \mathbb{R} \setminus \{0\}$, has a continuous extension to all of $\mathbb{R}$.

Proof of Claim 4. Decompose $f = f_1 + f_2$ where $f_1(t) := \int_{\mathbb{R}^{n-1}} \Phi(x', t)\varphi(x')\, \mathrm{d}x'$ and $f_2(t) := \frac{1}{2i}(\operatorname{sgn} t)\widehat{\Phi}(0', 1)\varphi(0')$ for each $t \in \mathbb{R} \setminus \{0\}$. From (12.4.18) and (3.2.10) we know that

$$\widehat{\operatorname{sgn}}(\xi_n) = -2i\, \text{P.V.} \left(\frac{1}{\xi_n}\right), \quad \text{and} \quad \varphi(0') = (2\pi)^{1-n} \int_{\mathbb{R}^{n-1}} \widehat{\varphi}(\xi')\, \mathrm{d}\xi'. \tag{4.7.26}$$

Also, $\widehat{\Phi}$ is odd and positive homogeneous of degree -1 in $\mathbb{R}^n \setminus \{0\}$, and since P.V. $\left(\frac{1}{\xi_n}\right)$ restricted to $\mathbb{R} \setminus \{0\}$ is simply $\frac{1}{\xi_n}$, it follows that

$$\text{P.V.} \left(\frac{1}{\xi_n}\right) \widehat{\Phi}(0', 1) = \widehat{\Phi}(0', \xi_n) \quad \text{on } \mathbb{R} \setminus \{0\}.$$

Keeping these in mind we conclude that

$$\widehat{f_2}(\xi_n) = -(2\pi)^{1-n} \int_{\mathbb{R}^{n-1}} \widehat{\Phi}(0', \xi_n)\widehat{\varphi}(\xi')\, \mathrm{d}\xi' \quad \text{for} \quad \xi_n \in \mathbb{R} \setminus \{0\}. \tag{4.7.27}$$

As far as f_1 is concerned, observe first that $f_1 \in \mathcal{S}'(\mathbb{R})$. Indeed, for every $\psi \in \mathcal{S}(\mathbb{R})$ we have $\langle f_1, \psi\rangle = \langle \Phi, \varphi \otimes \psi\rangle$. Since $\varphi \otimes \psi \in \mathcal{S}(\mathbb{R}^n)$ and, as already

noted, $\Phi\in\mathcal{S}'(\mathbb{R}^n)$, the desired conclusion follows. The next order of business is to compute the Fourier transform of f_1. With this goal in mind, pick an arbitrary $\psi\in\mathcal{S}(\mathbb{R})$ and write (keeping in mind Exercise 3.26)

$$\begin{aligned}\langle\widehat{f_1},\psi\rangle&=\langle f_1,\widehat{\psi}\,\rangle=\langle\Phi,\varphi\otimes\widehat{\psi}\,\rangle\\&=(2\pi)^{1-n}\big\langle\Phi,\mathcal{F}(\widehat{\varphi}^{\vee}\otimes\psi)\big\rangle=(2\pi)^{1-n}\big\langle\widehat{\Phi},\widehat{\varphi}^{\vee}\otimes\psi\big\rangle\\&=(2\pi)^{1-n}\Big\langle\big\langle\widehat{\Phi}(\xi',\xi_n),\widehat{\varphi}(-\xi')\big\rangle\,,\,\psi(\xi_n)\Big\rangle\end{aligned}\tag{4.7.28}$$

which proves that

$$\widehat{f_1}(\xi_n)=(2\pi)^{1-n}\int_{\mathbb{R}^{n-1}}\widehat{\Phi}(\xi',\xi_n)\widehat{\varphi}(-\xi')\,d\xi'\quad\text{for}\quad\xi_n\in\mathbb{R}\setminus\{0\}.\tag{4.7.29}$$

By combining (4.7.27) with (4.7.29) we arrive at the conclusion that, for $\xi_n\in\mathbb{R}\setminus\{0\}$,

$$\begin{aligned}\widehat{f}(\xi_n)&=(2\pi)^{1-n}\int_{\mathbb{R}^{n-1}}\widehat{\Phi}(\xi',\xi_n)\widehat{\varphi}(-\xi')\,d\xi'-(2\pi)^{1-n}\int_{\mathbb{R}^{n-1}}\widehat{\Phi}(0',\xi_n)\widehat{\varphi}(\xi')\,d\xi'\\&=-(2\pi)^{1-n}\int_{\mathbb{R}^{n-1}}\widehat{\Phi}(\xi',-\xi_n)\widehat{\varphi}(\xi')\,d\xi'-(2\pi)^{1-n}\int_{\mathbb{R}^{n-1}}\widehat{\Phi}(0',\xi_n)\widehat{\varphi}(\xi')\,d\xi'\\&=F(\xi_n),\end{aligned}\tag{4.7.30}$$

where the second equality uses the fact that $\widehat{\Phi}$ is odd in $\mathbb{R}^n\setminus\{0\}$. Hence, $\widehat{f}=F$ on $\mathbb{R}\setminus\{0\}$, where F is defined in Claim 3. Now select $\theta\in C^\infty_0(\mathbb{R})$ with $\theta\equiv1$ on $[-1,1]$ and write

$$\widehat{f}=(1-\theta)\widehat{f}+\theta\widehat{f}.\tag{4.7.31}$$

Since $(1-\theta)\widehat{f}=(1-\theta)F\in L^1(\mathbb{R})$ by Claim 3, and $\theta\widehat{f}\in\mathcal{E}'(\mathbb{R})$, we may conclude from Claim 1 that the Fourier transform of $\widehat{f}$ belongs to $C^0(\mathbb{R})$ hence, ultimately, that f itself belongs to $C^0(\mathbb{R})$. This completes the proof of Claim 4.

Claim 5. Assume that $\varphi\in\mathcal{S}(\mathbb{R}^{n-1})$ is such that $\widehat{\varphi}$ has compact support. Then

$$\lim_{t\to0^+}\int_{\mathbb{R}^{n-1}}\Phi(x',t)\varphi(x')\,dx'-\lim_{t\to0^-}\int_{\mathbb{R}^{n-1}}\Phi(x',t)\varphi(x')\,dx'=i\,\widehat{\Phi}(0',1)\varphi(0').\tag{4.7.32}$$

Proof of Claim 5. This follows from Claim 4 by writing for each $t\in\mathbb{R}\setminus\{0\}$

$$\int_{\mathbb{R}^{n-1}}\Phi(x',t)\varphi(x')\,dx'=-\frac{1}{2i}(\operatorname{sgn}t)\widehat{\Phi}(0',1)\varphi(0')+f(t)\tag{4.7.33}$$

with f continuous on $\mathbb{R}$.

Claim 6. For $a_\pm$ originally defined in (4.7.12) one has

$$a_\pm=\pm\frac{i}{2}\,\widehat{\Phi}(0',1).\tag{4.7.34}$$

Proof of Claim 6. Let $\psi \in C_0^\infty(\mathbb{R}^{n-1})$ be such that

$$\int_{\mathbb{R}^{n-1}} \psi(x')\,dx' = 1, \tag{4.7.35}$$

and set $\varphi := \widehat{\psi}$. Then $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$, $\widehat{\varphi} = (2\pi)^{n-1}\psi^\vee$ has compact support, and

$$\varphi(0') = \widehat{\psi}(0') = \int_{\mathbb{R}^{n-1}} \psi(x')\,dx' = 1. \tag{4.7.36}$$

From (4.7.11) and Claim 5, one obtains

$$i\,\widehat{\Phi}(0',1) = a_+ - a_-. \tag{4.7.37}$$

However, since Φ is odd, from the definition of $a_\pm$ in (4.7.12) we see that $a_- = -a_+$. This forces the equalities in (4.7.34), and finishes the proof of Claim 6.

At this stage, we note that (4.7.1) is a consequence of (4.7.11) and (4.7.34). The proof of Theorem 4.76 is therefore complete. □

The proof just completed offers a bit more and, below, we bring to the forefront one such by-product.

Proposition 4.77. *Assume that $\Phi \in C^4(\mathbb{R}^n \setminus \{0\})$ is odd and positive homogeneous of degree $1-n$. Also, let $\xi \in S^{n-1}$ be arbitrary and set $H_\xi := \{x \in \mathbb{R}^n : x \cdot \xi = 0\}$. Then*

$$\widehat{\Phi}(\xi) = -2i \lim_{r\to\infty} \int_{x\in H_\xi,\,|x|<r} \Phi(x+\xi)\,d\sigma(x), \tag{4.7.38}$$

where σ denotes the surface measure on H_ξ (viewed as surface in $\mathbb{R}^n$).

Proof. We start by observing that (4.7.12) and (4.7.34) imply that

$$\widehat{\Psi}(\mathbf{e}_n) = -2i \lim_{r\to\infty} \int_{x'\in\mathbb{R}^{n-1},\,|x'|<r} \Psi\big((x',0)+\mathbf{e}_n\big)\,dx' \quad \text{for each function } \Psi$$
$$\text{of class } C^4, \text{ odd, positive homogeneous of degree } 1-n, \text{ in } \mathbb{R}^n\setminus\{0\}. \tag{4.7.39}$$

To proceed, fix a vector $\xi \in S^{n-1}$ and denote by H_ξ the hyperplane in $\mathbb{R}^n$ orthogonal to ξ. Consider then an orthogonal matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ satisfying

$$AH_\xi = \mathbb{R}^{n-1}\times\{0\} \quad \text{and} \quad A\xi = \mathbf{e}_n. \tag{4.7.40}$$

For example, if $v_1, \ldots, v_{n-1}$ is an orthonormal basis in H_ξ and we set $v_n := \xi$, then the conditions $Av_j = \mathbf{e}_j$ for $j \in \{1,\ldots,n\}$ define an orthogonal matrix A such that the conditions in (4.7.40) hold. If we now introduce $\Psi : \mathbb{R}^n\setminus\{0\} \to \mathbb{C}$ by setting $\Psi := \Phi\circ A^{-1}$ then $\Psi \in C^4(\mathbb{R}^n\setminus\{0\})$ is odd, positive homogeneous of

degree $1-n$, and $\widehat{\Psi}=\widehat{\Phi}\circ A^{-1}$ [as seen from (4.3.3)]. Using these observations and (4.7.39), we may write

$$\widehat{\Phi}(\xi)=\widehat{\Psi}(A\xi)=\widehat{\Psi}(\mathbf{e}_n)=-2i\lim_{r\to\infty}\int\limits_{x'\in\mathbb{R}^{n-1},\,|x'|<r}\Psi\big((x',0)+\mathbf{e}_n\big)\,dx'. \tag{4.7.41}$$

Given a number $r\in(0,\infty)$, consider now the surface $\Sigma:=\{x\in H_\xi:\,|x|<r\}$ and note that if $\mathcal{O}:=\{x'\in\mathbb{R}^{n-1}:\,|x'|<r\}$ then

$$P:\mathcal{O}\to\Sigma,\quad P(x'):=A^{-1}(x',0)\ \text{ for each }\ x'\in\mathcal{O}, \tag{4.7.42}$$

is a global C^∞ parametrization of Σ, satisfying $|\partial_1P\times\cdots\times\partial_{n-1}P|=1$ at each point in the set $\mathcal{O}$. Consequently, (13.6.6) gives

$$\begin{aligned}\int\limits_{x\in H_\xi,\,|x|<r}\Phi(x+\xi)\,d\sigma(x)&=\int\limits_{\mathcal{O}}\Phi(P(x')+\xi)\,dx'\\&=\int\limits_{\mathcal{O}}\Phi\big(A^{-1}(x',0)+A^{-1}\mathbf{e}_n\big)\,dx'\\&=\int\limits_{x'\in\mathbb{R}^{n-1},\,|x'|<r}\Psi\big((x',0)+\mathbf{e}_n\big)\,dx'.\end{aligned} \tag{4.7.43}$$

At this stage, (4.7.38) is clear from (4.7.41) and (4.7.43). □

The power of Theorem 4.76 is most apparent by considering the consequence discussed in Corollary 4.78 below, which sheds light on the boundary behavior of the form $\lim_{t\to0^+}F^\pm(x',t)$ of functions of the following type

$$F^\pm(x',t):=\int_{\mathbb{R}^{n-1}}\Phi(x'-y',\pm t)\varphi(y')\,dy',\qquad(x',t)\in\mathbb{R}^n_+, \tag{4.7.44}$$

where Φ is as in the statement of Theorem 4.76 and φ is a Schwartz function on $\mathbb{R}^{n-1}$ (which is canonically identified with $\partial\mathbb{R}^n_+$). Functions of the form (4.7.44) play a prominent role in partial differential equations and harmonic analysis as they arise naturally in the treatment of boundary value problems via boundary integral methods. We shall return to this topic in Sect. 11.6.

Corollary 4.78. *Let the function $\Phi\in C^4(\mathbb{R}^n\setminus\{0\})$ be odd and positive homogeneous of degree $1-n$, and assume that $\varphi\in\mathcal{S}(\mathbb{R}^{n-1})$. Then for every $x'\in\mathbb{R}^{n-1}$ one has*

$$\begin{aligned}\lim_{t\to0^\pm}\int_{\mathbb{R}^{n-1}}\Phi(x'-y',t)\varphi(y')\,dy'&=\pm\frac{i}{2}\,\widehat{\Phi}(0',1)\varphi(x')\\&\quad+\lim_{\varepsilon\to0^+}\int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\|x'-y'|>\varepsilon}}\Phi(x'-y',0)\varphi(y')\,dy'.\end{aligned} \tag{4.7.45}$$

Proof. Given any $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$ and any $x' \in \mathbb{R}^{n-1}$, write

$$\begin{aligned}\lim_{t\to 0^{\pm}} \int_{\mathbb{R}^{n-1}} \Phi(x'-y',t)\varphi(y')\,\mathrm{d}y' &= \lim_{t\to 0^{\pm}} \int_{\mathbb{R}^{n-1}} \Phi(z',t)\varphi(x'-z')\,\mathrm{d}z' \\ &= \lim_{t\to 0^{\pm}} \big\langle \Phi(\cdot,t), \varphi(x'-\cdot)\big\rangle \qquad (4.7.46)\\ &= \pm\frac{i}{2}\,\widehat{\Phi}(0',1)\varphi(x') + \Big\langle \mathrm{P.V.}\,\Phi(\cdot,0), \varphi(x'-\cdot)\Big\rangle\end{aligned}$$

then notice that

$$\begin{aligned}\Big\langle \mathrm{P.V.}\,\Phi(\cdot,0), \varphi(x'-\cdot)\Big\rangle &= \lim_{\varepsilon\to 0^{+}} \int\limits_{\substack{z'\in\mathbb{R}^{n-1}\\ |z'|>\varepsilon}} \Phi(z',0)\varphi(x'-z')\,\mathrm{d}z' \\ &= \lim_{\varepsilon\to 0^{+}} \int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\ |x'-y'|>\varepsilon}} \Phi(x'-y',0)\varphi(y')\,\mathrm{d}y'. \qquad (4.7.47)\end{aligned}$$

Now (4.7.45) follows from (4.7.46) and (4.7.47). □

Remark 4.79. *Consider the function* $\Phi : \mathbb{R}^2 \setminus \{(0,0)\} \to \mathbb{C}$ *given by* $\Phi(x,y) := \frac{1}{x+iy}$ *for all* $(x,y) \in \mathbb{R}^2 \setminus \{(0,0)\}$. *Then* Φ *is odd and homogeneous of degree* -1. *Moreover, since under the canonical identification* $\mathbb{R}^2 \equiv \mathbb{C}$ *the function* Φ *becomes* $\Phi(z) = \frac{1}{z}$ *for* $z \in \mathbb{C}\setminus\{0\}$, *it follows that* $\widehat{\Phi}(\xi) = \frac{2\pi}{i}\cdot\frac{1}{\xi}$ *for all* $\xi \in \mathbb{C}\setminus\{0\}$ *(for details, see Proposition 7.33). In particular, this yields* $\widehat{\Phi}(0,1) = \frac{2\pi}{i}\cdot\frac{1}{i} = -2\pi$. *Consequently,* (4.7.1) *becomes in this case*

$$\lim_{\varepsilon\to 0^{+}} \frac{1}{x \pm i\varepsilon} = \mp i\pi\,\delta + \mathrm{P.V.}\,\frac{1}{x} \quad \text{in } \mathcal{S}'(\mathbb{R}), \qquad (4.7.48)$$

which is in agreement with Sokhotsky's formula from (2.10.3).

Theorem 4.76 also suggests a natural procedure for computing the Fourier transform of certain principal value distributions. While the latter topic has been treated in Sect. 4.5, where the general formula (4.5.10) has been established, such a procedure remains of interest since the integral in (4.5.6) may not always be readily computed from scratch. Specifically, we have the following result.

Corollary 4.80. *Assume that the function* $\Phi \in C^4(\mathbb{R}^n\setminus\{0\})$ *is odd and positive homogeneous of degree* $1-n$. *Then*

$$\mathcal{F}'\Big(\mathrm{P.V.}\,\Phi(\,\cdot\,,0)\Big) = \mp\frac{i}{2}\,\widehat{\Phi}(0',1) + \lim_{\varepsilon\to 0^{\pm}} \mathcal{F}'\big(\Phi(\,\cdot\,,\varepsilon)\big) \quad \text{in } \mathcal{S}'(\mathbb{R}^{n-1}), \qquad (4.7.49)$$

where $\mathcal{F}'$ *denotes the Fourier transform in* $\mathbb{R}^{n-1}$.

Proof. Formula (4.7.49) is a direct consequence of Theorem 4.76, the continuity of $\mathcal{F}'$ on $\mathcal{S}'(\mathbb{R}^{n-1})$, and the fact that $\mathcal{F}'\big(\delta(x')\big)=1$. □

Here is an example implementing this procedure in a case that is going to be useful later on, when discussing the Riesz transforms in $\mathbb{R}^n$.

Proposition 4.81. *For each $j\in\{1,\dots,n\}$ we have (with $\mathcal{F}$ denoting the Fourier transform in $\mathbb{R}^n$)*

$$\mathcal{F}\Big(\text{P.V.}\,\frac{x_j}{|x|^{n+1}}\Big)=-\frac{i\omega_n}{2}\,\frac{\xi_j}{|\xi|}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n). \tag{4.7.50}$$

Proof. Fix some $j\in\{1,\dots,n\}$ and consider the function defined by

$$\Phi(x,t):=\frac{x_j}{\big(|x|^2+t^2\big)^{\frac{n+1}{2}}},\qquad\forall\,(x,t)\in\mathbb{R}^{n+1}\setminus\{0\}. \tag{4.7.51}$$

Note that Φ is C^∞, odd, and positive homogeneous of degree $-n$ in $\mathbb{R}^{n+1}\setminus\{0\}$. Moreover, if hat denotes the Fourier transform in $\mathbb{R}^{n+1}$, from Corollary 4.62 we have

$$\widehat{\Phi}(\xi,\eta)=-i\omega_n\frac{\xi_j}{|\xi|^2+\eta^2}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^{n+1}), \tag{4.7.52}$$

from which we see that $\widehat{\Phi}(0,1)=0$. Keeping this in mind, formula (4.7.49) (used with $n+1$ in place on n) then yields

$$\mathcal{F}\Big(\text{P.V.}\,\frac{x_j}{|x|^{n+1}}\Big)=\lim_{t\to0^+}\mathcal{F}_x\Phi(x,t)\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n), \tag{4.7.53}$$

where $\mathcal{F}_x$ denotes the Fourier transform in the variable x in $\mathbb{R}^n$. Next, recall the discussion at the end of Sect. 4.2 regarding partial Fourier transforms and compute

$$\begin{aligned}\big(\mathcal{F}_x\Phi(x,t)\big)(\xi)&=\mathcal{F}_\eta^{-1}\big[\,\widehat{\Phi}(\xi,\eta)\big](t)=-i\omega_n\xi_j\mathcal{F}_\eta^{-1}\Big[\frac{1}{|\xi|^2+\eta^2}\Big](t)\\&=-i\omega_n\,(2\pi)^{-1}\xi_j\mathcal{F}_\eta\Big[\frac{1}{|\xi|^2+\eta^2}\Big](t)\\&=-\frac{i\omega_n}{2}\,\frac{\xi_j}{|\xi|}\mathrm{e}^{-|\xi||t|}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n),\end{aligned} \tag{4.7.54}$$

where the last equality makes uses of (4.2.20). Since by Lebesgue's dominated convergence theorem

$$\lim_{t\to0}\Big[\frac{\xi_j}{|\xi|}\mathrm{e}^{-|\xi||t|}\Big]=\frac{\xi_j}{|\xi|}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n), \tag{4.7.55}$$

formula (4.7.50) now follows from (4.7.53)–(4.7.55). □

Remark 4.82. *Alternatively, one may prove* (4.7.50) *by combining* (4.3.19) *and* (4.4.19). *Indeed, if* $j \in \{1,\dots,n\}$ *is fixed, identity* (4.4.19) *written for* $\Phi(x) := \frac{1}{1-n}|x|^{-(n-1)}$, $x \in \mathbb{R}^n \setminus \{0\}$, *becomes*

$$\frac{1}{1-n}\partial_j\big[|x|^{-(n-1)}\big] = \text{P.V.}\Big(\frac{x_j}{|x|^{n+1}}\Big) \qquad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.7.56}$$

Hence, taking the Fourier transform of (4.7.56), *then using (b) in Theorem 4.25, then* (4.3.19) *with* $\lambda := n-1$, *and formulas* (13.5.2) *and* (13.5.6), *we obtain*

$$\begin{aligned}\mathcal{F}\Big(\text{P.V.}\,\frac{x_j}{|x|^{n+1}}\Big) &= \frac{1}{1-n}\mathcal{F}\Big(\partial_j\big[|x|^{-(n-1)}\big]\Big) = \frac{i}{1-n}\,\xi_j\,\mathcal{F}\big(|x|^{-(n-1)}\big)\\ &= -\frac{i}{n-1}\,\frac{2\pi^{\frac{n}{2}}\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right)}\,\frac{\xi_j}{|\xi|} = -\frac{i\omega_n}{2}\,\frac{\xi_j}{|\xi|} \qquad \text{in} \quad \mathcal{S}'(\mathbb{R}^n).\end{aligned} \tag{4.7.57}$$

Corollary 4.80 may also be combined with Theorem 4.71 to obtain the following result pertaining to certain limits of Fourier transforms of tempered distributions.

Corollary 4.83. *Suppose that* $\Phi \in C^4(\mathbb{R}^n \setminus \{0\})$ *is odd and positive homogeneous of degree* $1-n$. *Then in* $\mathcal{S}'(\mathbb{R}^{n-1})$,

$$\lim_{\varepsilon\to 0^{\pm}} \mathcal{F}'\big[\Phi(\,\cdot\,,\varepsilon)\big](\xi') = \pm\frac{i}{2}\,\widehat{\Phi}(0',1) - \int_{S^{n-2}} \Phi(\theta,0)\log(i(\xi'\cdot\theta))\,d\sigma(\theta) \tag{4.7.58}$$

where $\mathcal{F}'$ *denotes the Fourier transform in* $\mathbb{R}^{n-1}$ *and* S^{n-2} *denotes the unit sphere centered at the origin in* $\mathbb{R}^{n-1}$.

Proof. This follows from Corollary 4.80, (4.5.10), and (4.5.6). □

4.8 The Harmonic Poisson Kernel

The goal of this section is to introduce and discuss the harmonic Poisson kernel.

Definition 4.84. *Define the* `harmonic Poisson kernel` $P : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}$ *by setting*

$$P(x',x_n) := \frac{2}{\omega_{n-1}}\,\frac{x_n}{|x|^n}, \qquad \forall\, x = (x',x_n) \in \mathbb{R}^n \setminus \{0\}. \tag{4.8.1}$$

Furthermore, for each $x' \in \mathbb{R}^{n-1}$ *set* $p(x') := P(x',1)$, *that is, consider*

$$p(x') := \frac{2}{\omega_{n-1}}\,\frac{1}{\big(1+|x'|^2\big)^{\frac{n}{2}}}, \qquad \forall\, x' \in \mathbb{R}^{n-1}. \tag{4.8.2}$$

In our next result we discuss the boundary behavior of a mapping $\mathscr{P}$ defined below, taking Schwartz functions from $\mathbb{R}^{n-1}$ into functions defined in $\mathbb{R}^n_{\pm}$, which in partial differential equations is referred to as the harmonic double layer potential operator.

Proposition 4.85. *Given any $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$ and any $x = (x', x_n) \in \mathbb{R}^n$ with $x_n \neq 0$, define (where P denotes the harmonic Poisson kernel)*

$$(\mathscr{P}\varphi)(x) := \int_{\mathbb{R}^{n-1}} P(x' - y', x_n)\varphi(y')\,\mathrm{d}y'$$
$$= \frac{2}{\omega_{n-1}} \int_{\mathbb{R}^{n-1}} \frac{x_n}{\left(|x' - y'|^2 + x_n^2\right)^{\frac{n}{2}}} \varphi(y')\,\mathrm{d}y'. \tag{4.8.3}$$

Then for each $x' \in \mathbb{R}^{n-1}$ one has

$$\lim_{x_n \to 0^{\pm}} (\mathscr{P}\varphi)(x', x_n) = \pm\,\varphi(x'). \tag{4.8.4}$$

Proof. The function $P : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}$ from (4.8.1) is C^∞, odd, and positive homogeneous of degree $1 - n$. Moreover, Corollary 4.62 gives

$$\widehat{P}(\xi) = -2i\frac{\xi_n}{|\xi|^2} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.8.5}$$

Consequently, $\widehat{P}(0', 1) = -2i$. Also, since $P(x', 0)$=0 we have P.V. $P(x', 0) = 0$. At this point, Corollary 4.78 applies (with P playing the role of Φ) and yields (4.8.4). □

Remark 4.86. *When specializing Theorem 4.76 to the case $\Phi := P$, with P defined as in (4.8.1), we obtain (making use of (4.8.5))*

$$\lim_{x_n \to 0^{\pm}} P(x', x_n) = \pm\,\delta(x') \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n-1}). \tag{4.8.6}$$

This may be regarded as a higher dimensional generalization of the result presented in Exercise 2.115, to which (4.8.6) reduces in the case when $n = 2$ and $x_n = \varepsilon > 0$.

Among other things, the following proposition sheds light on the normalization of the harmonic Poisson kernel introduced in (4.8.1).

Proposition 4.87. *The function p defined in (4.8.2) satisfies the following properties.*

(1) One has $p \in \bigcap_{1 \leq q \leq \infty} L^q(\mathbb{R}^{n-1})$ and

$$\int_{\mathbb{R}^{n-1}} p(x')\,\mathrm{d}x' = 1. \tag{4.8.7}$$

(2) For each $t > 0$ set

$$p_t(x') := t^{1-n} p(x'/t) \quad \text{for} \quad x' \in \mathbb{R}^{n-1}. \tag{4.8.8}$$

Then for each $t \in (0, \infty)$ we have $p_t \in \bigcap_{1 \leq q \leq \infty} L^q(\mathbb{R}^{n-1})$ and its Fourier transform is

$$\widehat{p_t}(\xi') = \mathrm{e}^{-t|\xi'|} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n-1}). \tag{4.8.9}$$

(3) The family $\{p_t\}_{t>0}$ has the semigroup property, that is,

$$p_{t_1} * p_{t_2} = p_{t_1+t_2} \qquad \forall\, t_1, t_2 \in (0,\infty). \tag{4.8.10}$$

Proof. That $p \in L^q(\mathbb{R}^{n-1})$ for any $q \in [1,\infty]$ is immediate from its expression. Fix $t \in (0,\infty)$ and let p_t be as in part *(2)*. Applying Exercise 2.22 (with $n-1$ in place of n, p in place of f, and t in place of $1/j$), we obtain

$$p_t(x') \xrightarrow[t\to 0^+]{\mathcal{D}'(\mathbb{R}^{n-1})} c\,\delta(x') \quad \text{where} \quad c := \int_{\mathbb{R}^{n-1}} p(x')\,\mathrm{d}x'. \tag{4.8.11}$$

On the other hand, as seen from (4.8.2) and (4.8.1),

$$p_t(x') = P(x',t) \qquad \text{for each } t>0 \text{ and } x' \in \mathbb{R}^{n-1}, \tag{4.8.12}$$

so (4.8.6) gives

$$p_t(x') \to \delta(x') \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n) \quad \text{as} \quad t \to 0^+. \tag{4.8.13}$$

Now (4.8.7) follows by combining (4.8.13) with (4.8.11).

Moving on to the proof of part *(2)*, observe first that $p_t \in L^q(\mathbb{R}^{n-1})$ for each $t>0$, if $q \in [1,\infty]$, thus $p_t \in \mathcal{S}'(\mathbb{R}^{n-1})$ (as a consequence of (4.1.8)). In particular, its Fourier transform in $\mathcal{S}'(\mathbb{R}^{n-1})$ is meaningfully defined and equal to its Fourier transform as a function in $L^1(\mathbb{R}^{n-1})$ (by Remark 4.22) and belongs to $C^0(\mathbb{R}^{n-1})$ (by (3.1.3)).

In what follows we will be using partial Fourier transforms (cf. the discussion at the end of Sect. 4.2) and the notation $\xi' \in \mathbb{R}^{n-1}$ and $\eta \in \mathbb{R}$. The idea we will pursue is to compute $\mathcal{F}_{x'}\big[p_t(x')\big](\xi')$ by making use of (4.8.12) and the fact that we know $\widehat{P}(\xi',\eta)$, the Fourier transform of P in $\mathcal{S}'(\mathbb{R}^n)$ (cf. (4.8.5)). With this in mind, for each $\xi' \in \mathbb{R}^{n-1}\setminus\{0\}$ and $t \in \mathbb{R}$ we write

$$\begin{aligned}
\mathcal{F}_{x'}\big[P(x',t)\big](\xi') &= \mathcal{F}_{\eta}^{-1}\big[\widehat{P}(\xi',\eta)\big](t) = \mathcal{F}_{\eta}^{-1}\Big[-2i\frac{\eta}{|\xi'|^2+\eta^2}\Big](t) \\
&= 2i(2\pi)^{-1}\mathcal{F}_{\eta}\Big[\frac{\eta}{|\xi'|^2+\eta^2}\Big](t) = \frac{i}{\pi}(-\pi i)(\operatorname{sgn} t)\mathrm{e}^{-|t|\,|\xi'|} \\
&= (\operatorname{sgn} t)\mathrm{e}^{-|t|\,|\xi'|}.
\end{aligned} \tag{4.8.14}$$

For the second equality in (4.8.14) we have used (4.8.5), for the third the fact that $\mathcal{F}_{\eta}^{-1}g = (2\pi)^{-1}\mathcal{F}_{\eta}g^{\vee}$ for every $g \in \mathcal{S}'(\mathbb{R}^n)$, while for the fourth we have used (4.2.26) (applied with $a := |\xi'|$). In particular, (4.8.14) implies that

$$\mathcal{F}_{x'}\big[p_t(x')\big](\xi') = \mathrm{e}^{-t|\xi'|} \qquad \text{for } \xi' \in \mathbb{R}^{n-1}\setminus\{0\} \text{ and } t>0. \tag{4.8.15}$$

Now (4.8.9) follows from (4.8.15) and the fact that $p_t \in C^0(\mathbb{R}^{n-1})$ for every $t>0$. Regarding (4.8.10), fix $t_1,t_2 \in (0,\infty)$ and by using the property of the Fourier transform singled out in part *(3)* of Remark 3.28 and (4.8.10) we obtain

$$\mathcal{F}_{x'}\big(p_{t_1} * p_{t_2}\big)(\xi') = \mathcal{F}_{x'}\big(p_{t_1}\big)(\xi')\mathcal{F}_{x'}\big(p_{t_2}\big)(\xi')$$

$$= \mathrm{e}^{-t_1|\xi'|}\mathrm{e}^{-t_2|\xi'|} = \mathrm{e}^{-(t_1+t_2)|\xi'|}$$

$$= \mathcal{F}_{x'}\big(p_{t_1+t_2}\big) \qquad \forall\,\xi' \in \mathbb{R}^{n-1},\ \forall\, t>0. \tag{4.8.16}$$

Now (4.8.10) follows from (4.8.16) by recalling that $\mathcal{F}_{x'}$ is an isomorphism on $\mathcal{S}'(\mathbb{R}^{n-1})$. □

We wish to note that the harmonic Poisson kernel presented above plays a basic role in the treatment of boundary value problems for the Laplacian $\Delta := \partial_1^2 + \cdots + \partial_n^2$ in the upper-half space. More specifically, for any given function $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$ (called the `boundary datum`), the `Dirichlet problem`

$$\begin{cases} u \in C^\infty(\mathbb{R}^n_+), \\ \Delta u = 0 \qquad\qquad \text{in} \quad \mathbb{R}^n_+, \\ u\Big|^{ver}_{\partial\mathbb{R}^n_+} = \varphi \qquad \text{on} \quad \mathbb{R}^{n-1} \equiv \partial\mathbb{R}^n_+, \end{cases} \tag{4.8.17}$$

has as a solution the function

$$u := \mathscr{P}\varphi \quad \text{in} \quad \mathbb{R}^n_+, \tag{4.8.18}$$

i.e., (cf. (4.8.3))

$$u(x) = \frac{2}{\omega_{n-1}} \int_{\mathbb{R}^{n-1}} \frac{x_n}{\big(|x'-y'|^2 + x_n^2\big)^{\frac{n}{2}}}\, \varphi(y')\, \mathrm{d}y', \qquad x = (x', x_n) \in \mathbb{R}^n_+. \tag{4.8.19}$$

In the last line of (4.8.17), the symbol $u\Big|^{ver}_{\partial\mathbb{R}^n_+}$ stands for the "vertical limit" of u to the boundary of the upper-half space, understood at each point $x' \in \mathbb{R}^{n-1}$ as

$$\Big(u\Big|^{ver}_{\partial\mathbb{R}^n_+}\Big)(x') := \lim_{x_n \to 0^+} u(x', x_n). \tag{4.8.20}$$

To see why u as in (4.8.19) is a solution of (4.8.17) note that (4.8.1) implies $u \in C^\infty(\mathbb{R}^n_+)$, that u has the limit to the boundary equal to φ based on (4.8.4), while the fact that $\Delta u = 0$ in $\mathbb{R}^n_+$ follows from (4.8.19) by checking directly that, for each $y' \in \mathbb{R}^{n-1}$ fixed, $\Delta\Big[x_n\big(|x'-y'|^2 + x_n^2\big)^{-\frac{n}{2}}\Big] = 0$ for all $(x', x_n) \in \mathbb{R}^n_+$.

Definition 4.88. *The mappings $Q_j : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}$, $j \in \{1, \ldots, n-1\}$, defined by*

$$Q_j(x', x_n) := \frac{2}{\omega_{n-1}} \frac{x_j}{|x|^n}, \qquad \forall\, x = (x', x_n) \in \mathbb{R}^n \setminus \{0\}, \quad j \in \{1, \ldots, n-1\}, \tag{4.8.21}$$

are called the `conjugate harmonic Poisson kernels`.

Furthermore, for each $x' \in \mathbb{R}^{n-1}$ we set $q_j(x') := Q_j(x', 1)$, that is,

$$q_j(x') := \frac{2}{\omega_{n-1}} \frac{x_j}{\big(1+|x'|^2\big)^{\frac{n}{2}}}, \qquad \forall\, x' \in \mathbb{R}^{n-1}, \quad j \in \{1, \ldots, n-1\}. \tag{4.8.22}$$

Proposition 4.89. *The functions defined in* (4.8.22) *satisfy the following properties.*

(1) For each $j \in \{1,\dots,n-1\}$ *and each* $p \in (1,\infty)$ *one has* $q_j \in L^p(\mathbb{R}^{n-1})$.

(2) For each $t > 0$ *set*

$$(q_j)_t(x') := t^{1-n} q_j(x'/t) \quad \text{for} \quad x' \in \mathbb{R}^{n-1}, \quad j \in \{1,\dots,n-1\}. \tag{4.8.23}$$

Then for each $j \in \{1,\dots,n-1\}$, *each* $t \in (0,\infty)$, *and each* $p \in (1,\infty)$, *we have* $(q_j)_t \in L^p(\mathbb{R}^{n-1})$ *and its Fourier transform is*

$$\widehat{(q_j)_t}(\xi') = -i\frac{\xi_j}{|\xi'|}\mathrm{e}^{-t|\xi'|} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n-1}). \tag{4.8.24}$$

(3) The following identity holds

$$\frac{\mathrm{d}}{\mathrm{d}t}\big[p_t(x')\big] = -\sum_{j=1}^{n-1}\partial_j\Big[(q_j)_t(x')\Big] \qquad \forall\, t \in (0,\infty), \ \forall\, x' \in \mathbb{R}^{n-1}, \tag{4.8.25}$$

where p_t *is as in* (4.8.23).

Proof. The claim in *(1)* is an immediate consequence of (4.8.22). To prove *(2)*, fix $j \in \{1,\dots,n-1\}$. Using (4.8.23) we obtain

$$(q_j)_t(x') = \frac{2}{\omega_{n-1}}\frac{x_j}{(t^2+|x'|^2)^{\frac{n}{2}}} = Q_j(x',t) \qquad \forall\, x' \in \mathbb{R}^{n-1}, \quad \forall\, t \in (0,\infty). \tag{4.8.26}$$

As in the proof of Proposition 4.87, we will be using partial Fourier transforms (cf. the discussion at the end of Sect. 4.2) and the notation $\xi' \in \mathbb{R}^{n-1}$ and $\eta \in \mathbb{R}$. The idea we will pursue is to compute $\mathcal{F}_{x'}\big[(q_j)_t(x')\big](\xi')$ by making use of (4.8.26) and the fact that, by (4.3.25), the Fourier transform of Q_j is

$$\widehat{Q_j}(\xi',\eta) = -2i\frac{\xi_j}{|\xi'|^2+\eta^2} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.8.27}$$

Hence, for each $t \in (0,\infty)$ we may write

$$\begin{aligned}\mathcal{F}_{x'}\big[(q_j)_t(x')\big](\xi') &= \mathcal{F}_\eta^{-1}\big[\widehat{Q_j}(\xi',\eta)\big](t) = \mathcal{F}_\eta^{-1}\Big[-2i\frac{\xi_j}{|\xi'|^2+\eta^2}\Big](t)\\ &= -2i(2\pi)^{-1}\xi_j\mathcal{F}_\eta\Big[\frac{1}{|\xi'|^2+\eta^2}\Big](t) = -\frac{i\xi_j}{\pi}\,\frac{\pi}{|\xi'|}\,\mathrm{e}^{-t\,|\xi'|}\\ &= -\frac{i\xi_j}{|\xi'|}\,\mathrm{e}^{-t\,|\xi'|} \qquad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n-1}).\end{aligned} \tag{4.8.28}$$

For the third equality in (4.8.28) we used the fact that $\mathcal{F}_\eta^{-1}g = (2\pi)^{-1}\mathcal{F}_\eta g^\vee$ for every $g \in \mathcal{S}'(\mathbb{R}^n)$, while for the fourth we used (4.2.20) (applied with $a := |\xi'|$). This completes the proof of *(2)*. Finally, (4.8.25) follows by a direct computation based on the chain rule. □

4.9 Singular Integral Operators

In Example 4.66 we have already encountered the principal value tempered distributions P.V. $\frac{x_j}{|x|^{n+1}}$, for $j \in \{1,\dots,n\}$. The operators R_j, $j \in \{1,\dots,n\}$, defined by convolving with these distributions, that is,

$$R_j\varphi := \Big(\text{P.V.}\,\frac{x_j}{|x|^{n+1}}\Big) * \varphi, \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n), \tag{4.9.1}$$

are called the `Riesz transforms` in $\mathbb{R}^n$. In the particular case when $n = 1$ the corresponding operator, that is,

$$H\varphi := \Big(\text{P.V.}\,\frac{1}{x}\Big) * \varphi, \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}), \tag{4.9.2}$$

is called the `Hilbert transform`. These operators play a fundamental role in harmonic analysis and here the goal is to study a larger class of operators containing the aforementioned examples. We begin by introducing this class.

The format of Proposition 4.67 suggests the following definition.

Definition 4.90. *For each function $\Theta \in C^0(\mathbb{R}^n \setminus \{0\})$ that is positive homogeneous of degree $-n$ and such that $\int_{S^{n-1}} \Theta(\omega)\,d\sigma(\omega) = 0$, define the* `SIO`

$$(T_\Theta\varphi)(x) := \lim_{\varepsilon\to 0^+} \int_{|x-y|\geq\varepsilon} \Theta(x-y)\varphi(y)\,dy, \qquad \forall\,x \in \mathbb{R}^n,\ \ \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.9.3}$$

Proposition 4.67 ensures that the above definition is meaningful. Moreover, for the class of SIO just defined, the following result holds (further properties are deduced in Theorem 4.96).

Proposition 4.91. *Let Θ be a just completed" the conditions in (4.4.1) and consider the SIO T_Θ associated with Θ as in (4.9.3). Then*

$$T_\Theta : \mathcal{S}(\mathbb{R}^n) \longrightarrow \mathcal{S}'(\mathbb{R}^n) \quad \textit{is linear and sequentially continuous.} \tag{4.9.4}$$

Moreover, for each $\varphi \in \mathcal{S}(\mathbb{R}^n)$ we have

$$T_\Theta\varphi \in C^\infty(\mathbb{R}^n) \quad \textit{and} \quad T_\Theta\varphi = \big(\text{P.V.}\,\Theta\big) * \varphi \quad \textit{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{4.9.5}$$

as well as

$$\widehat{T_\Theta\varphi} = \widehat{\varphi}\,\mathcal{F}\big(\text{P.V.}\,\Theta\big) \quad \textit{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.9.6}$$

Proof. All claims are consequences of Definition 4.90, Proposition 4.67, part *(e)* in Theorem 4.18, and part *(a)* in Theorem 4.33. □

Remark 4.92. *In the harmonic analysis parlance, a mapping*

$$T : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$$

with the property that there exists $m \in \mathcal{S}'(\mathbb{R}^n)$ such that $\widehat{T\varphi} = \widehat{\varphi}\,m$ in $\mathcal{S}'(\mathbb{R}^n)$ for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$ is called a `multiplier` *(and the tempered distribution m*

is referred to as the `symbol of the multiplier`*). Note that any multiplier T is necessarily a linear and sequentially bounded mapping from $\mathcal{S}(\mathbb{R}^n)$ into $\mathcal{S}'(\mathbb{R}^n)$. Also, with this piece of terminology, for any function Θ as in (4.4.1),*

$$\text{the operator } T_\Theta \text{ is a multiplier with symbol } \mathcal{F}\big(\text{P.V.}\,\Theta\big). \tag{4.9.7}$$

Regarding the nature of the symbol $\mathcal{F}\big(\text{P.V.}\,\Theta\big)$, we have already seen that this Fourier transform may be computed explicitly in the case $\Theta(x) := \frac{x_j}{|x|^{n+1}}$, $j \in \{1,\dots,n\}$ (see (4.7.50)). The fact that such functions appear in the definition of the Riesz transforms in $\mathbb{R}^n$ (cf. (4.9.1)), warrants revisiting these operators. To set the stage for the subsequent discussion, we recall that given a linear and bounded map (also referred to as a bounded operator) $T : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$, its adjoint is the unique map $T^* : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ with the property that

$$\int_{\mathbb{R}^n} (Tf)(x)\overline{g(x)}\,dx = \int_{\mathbb{R}^n} f(x)\overline{(T^*g)(x)}\,dx, \qquad \forall\, f, g \in L^2(\mathbb{R}^n). \tag{4.9.8}$$

All ingredients are now in place for proving the following important theorem.

Theorem 4.93. *Consider the Riesz transforms, originally introduced as in (4.9.1). Then for each $j \in \{1,\dots,n\}$ the following properties hold.*

(a) The operator

$$R_j : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n) \quad \text{is well-defined, linear, and sequentially continuous.} \tag{4.9.9}$$

(b) For each $\varphi \in \mathcal{S}(\mathbb{R}^n)$ we have $R_j\varphi \in C^\infty(\mathbb{R}^n)$ and R_j may be expressed as the SIO

$$(R_j\varphi)(x) = \lim_{\varepsilon\to 0^+} \int_{|x-y|\geq\varepsilon} \frac{x_j - y_j}{|x-y|^{n+1}}\varphi(y)\,dy, \qquad \forall\, x \in \mathbb{R}^n. \tag{4.9.10}$$

(c) The jth Riesz transform is a multiplier with symbol $m_j(\xi) := -\frac{i\omega_n}{2}\frac{\xi_j}{|\xi|}$ which belongs to $L^\infty(\mathbb{R}^n)$. That is, for each $\varphi \in \mathcal{S}(\mathbb{R}^n)$,

$$\widehat{R_j\varphi}(\xi) = -\frac{i\omega_n}{2}\,\frac{\xi_j}{|\xi|}\,\widehat{\varphi}(\xi) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.9.11}$$

(d) For each $\varphi \in \mathcal{S}(\mathbb{R}^n)$, we have that $R_j\varphi$ originally viewed as a tempered distribution belongs to the subspace $L^2(\mathbb{R}^n)$ of $\mathcal{S}'(\mathbb{R}^n)$, and

$$\|R_j\varphi\|_{L^2(\mathbb{R}^n)} \leq (\omega_n/2)\|\varphi\|_{L^2(\mathbb{R}^n)}. \tag{4.9.12}$$

(e) The jth Riesz transform R_j, originally considered as in (4.9.9) extends, by density to a linear and bounded operator

$$R_j : L^2(\mathbb{R}^n) \longrightarrow L^2(\mathbb{R}^n) \tag{4.9.13}$$

and the operator R_j is skew-symmetric (i.e., $R_j^* = -R_j$) in this context. In addition,

$$\widehat{R_j f}(\xi) = -\frac{i\omega_n}{2}\,\frac{\xi_j}{|\xi|}\,\widehat{f}(\xi) \quad \text{a.e. in} \quad \mathbb{R}^n, \qquad \forall\, f \in L^2(\mathbb{R}^n). \tag{4.9.14}$$

(f) For every $k \in \{1,\dots,n\}$, we have

$$R_j R_k = R_k R_j \quad \text{as operators on} \quad L^2(\mathbb{R}^n), \tag{4.9.15}$$

and

$$\sum_{k=1}^{n} R_k^2 = -\left(\tfrac{\omega_n}{2}\right)^2 I \quad \text{as operators on} \quad L^2(\mathbb{R}^n), \tag{4.9.16}$$

where I denotes the identity operator on $L^2(\mathbb{R}^n)$.

Proof. The claims in *(a)* and *(b)* are immediate consequence of (4.9.1) and (4.9.2) and Proposition 4.67, upon recalling the discussion in Example 4.66. Also, formula (4.9.11) follows from (4.9.1), (4.9.6), and Proposition 4.81.

Turning our attention to *(d)*, fix an index $j \in \{1,\dots,n\}$ along with two arbitrary functions $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$. Based on Exercise 3.26, (4.2.2), (4.9.11), Cauchy–Schwarz's inequality, and (3.2.28), we may then write

$$\begin{aligned}
\big|\langle R_j\varphi, \psi\rangle\big| &= (2\pi)^{-n}\big|\big\langle \widehat{R_j\varphi}, \widehat{\psi}^{\vee}\big\rangle\big| = (2\pi)^{-n}\frac{\omega_n}{2}\left|\left\langle \frac{\xi_j}{|\xi|}\,\widehat{\varphi}\,,\, \widehat{\psi}^{\vee}\right\rangle\right| \\
&= (2\pi)^{-n}\frac{\omega_n}{2}\left|\int_{\mathbb{R}^n} \frac{\xi_j}{|\xi|}\,\widehat{\varphi}(\xi)\widehat{\psi}(-\xi)\,d\xi\right| \\
&\leq (2\pi)^{-n}\frac{\omega_n}{2}\int_{\mathbb{R}^n} \big|\widehat{\varphi}(\xi)\big|\big|\widehat{\psi}(-\xi)\big|\,d\xi \\
&\leq (2\pi)^{-n}\frac{\omega_n}{2}\|\widehat{\varphi}\|_{L^2(\mathbb{R}^n)}\|\widehat{\psi}\|_{L^2(\mathbb{R}^n)} \\
&= \tfrac{\omega_n}{2}\|\varphi\|_{L^2(\mathbb{R}^n)}\|\psi\|_{L^2(\mathbb{R}^n)}.
\end{aligned} \tag{4.9.17}$$

This computation may be summarized by saying that, for each $\varphi \in \mathcal{S}(\mathbb{R}^n)$ fixed, the linear functional

$$\Lambda_\varphi : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}, \quad \Lambda_\varphi(\psi) := \langle R_j\varphi, \psi\rangle, \quad \forall\, \psi \in \mathcal{S}(\mathbb{R}^n), \tag{4.9.18}$$

satisfies $\big|\Lambda_\varphi(\psi)\big| \leq \frac{\omega_n}{2}\|\varphi\|_{L^2(\mathbb{R}^n)}\|\psi\|_{L^2(\mathbb{R}^n)}$ for every $\psi \in \mathcal{S}(\mathbb{R}^n)$. Given that $\mathcal{S}(\mathbb{R}^n)$ is a dense subspace of $L^2(\mathbb{R}^n)$, it follows that the mapping Λ_φ from (4.9.18) has a unique extension $\widetilde{\Lambda}_\varphi : L^2(\mathbb{R}^n) \to \mathbb{C}$ satisfying

$$\big|\widetilde{\Lambda}_\varphi(g)\big| \leq \tfrac{\omega_n}{2}\|\varphi\|_{L^2(\mathbb{R}^n)}\|g\|_{L^2(\mathbb{R}^n)} \quad \text{for every} \quad g \in L^2(\mathbb{R}^n). \tag{4.9.19}$$

According to Riesz's representation theorem for such functionals, there exists a unique $f_\varphi \in L^2(\mathbb{R}^n)$ satisfying the following two properties:

$$\widetilde{\Lambda}_\varphi(g) = \int_{\mathbb{R}^n} f_\varphi(x)g(x)\,dx \quad \text{for every} \quad g \in L^2(\mathbb{R}^n), \tag{4.9.20}$$

and

$$\|f_\varphi\|_{L^2(\mathbb{R}^n)} \leq \tfrac{\omega_n}{2}\|\varphi\|_{L^2(\mathbb{R}^n)}. \tag{4.9.21}$$

Since $L^2(\mathbb{R}^n) \subseteq \mathcal{S}'(\mathbb{R}^n)$ (cf. (4.1.8)), it follows that f_φ may be regarded as a tempered distribution. At this stage we make the claim that $R_j\varphi = f_\varphi$ as tempered distributions. Indeed, for every $\psi \in \mathcal{S}(\mathbb{R}^n)$ we have

$$\langle R_j\varphi, \psi\rangle = \Lambda_\varphi(\psi) = \widetilde{\Lambda}_\varphi(\psi) = \int_{\mathbb{R}^n} f_\varphi(x)\psi(x)\,dx = \langle f_\varphi, \psi\rangle, \tag{4.9.22}$$

from which the claim follows. With this in hand, (4.9.12) now follows from (4.9.21), finishing the proof of part *(d)*. In turn, estimate (4.9.12) and a standard density argument give that R_j extends to a linear and bounded operator in the context of (4.9.13).

The next item in part *(e)* is showing that R_j in (4.9.13) is skew-symmetric. To this end, first note that, given any $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$, by reasoning much as in (4.9.17) we obtain

$$\begin{aligned}
\langle R_j\varphi, \overline{\psi}\,\rangle &= (2\pi)^{-n}\big\langle \widehat{R_j\varphi}, \widehat{\overline{\psi}}^{\vee}\,\big\rangle = -i(2\pi)^{-n}\frac{\omega_n}{2}\Big\langle \frac{\xi_j}{|\xi|}\,\widehat{\varphi}\,, \widehat{\overline{\psi}}^{\vee}\Big\rangle \\
&= -i(2\pi)^{-n}\frac{\omega_n}{2}\int_{\mathbb{R}^n}\frac{\xi_j}{|\xi|}\,\widehat{\varphi}(\xi)\widehat{\overline{\psi(-\xi)}}\,d\xi \\
&= i(2\pi)^{-n}\frac{\omega_n}{2}\int_{\mathbb{R}^n}\frac{\xi_j}{|\xi|}\,\widehat{\overline{\psi(\xi)}}\widehat{\varphi}(-\xi)\,d\xi \\
&= i(2\pi)^{-n}\frac{\omega_n}{2}\Big\langle \frac{\xi_j}{|\xi|}\,\widehat{\overline{\psi}}\,, \widehat{\varphi}^{\vee}\Big\rangle = -(2\pi)^{-n}\big\langle \widehat{R_j\overline{\psi}}, \widehat{\varphi}^{\vee}\,\big\rangle \\
&= -\langle R_j\overline{\psi}, \varphi\rangle = -\langle \overline{R_j\psi}, \varphi\rangle,
\end{aligned} \tag{4.9.23}$$

where the last step uses the fact that, as seen from (4.9.10), $R_j\overline{\psi} = \overline{R_j\psi}$. Since all functions involved are (by part *(d)*) in $L^2(\mathbb{R}^n)$, the distributional pairings may be interpreted as pairings in $L^2(\mathbb{R}^n)$. With this in mind, the equality of the most extreme sides of (4.9.23) reads

$$\int_{\mathbb{R}^n}(R_j\varphi)(x)\overline{\psi(x)}\,dx = -\int_{\mathbb{R}^n}\varphi(x)\overline{(R_j\psi)(x)}\,dx. \tag{4.9.24}$$

By density and part *(d)*, we deduce from (4.9.24) that

$$\int_{\mathbb{R}^n}(R_jf)(x)\overline{g(x)}\,dx = -\int_{\mathbb{R}^n}f(x)\overline{(R_jg)(x)}\,dx, \qquad \forall\, f, g \in L^2(\mathbb{R}^n). \tag{4.9.25}$$

In light of (4.9.8), this shows that $R_j^* = -R_j$ as mappings on $L^2(\mathbb{R}^n)$, as wanted. To complete the treatment of part *(e)*, there remains to observe that, collectively, the boundedness of R_j and of the Fourier transform on $L^2(\mathbb{R}^n)$, (4.9.11), and the density of the Schwartz class in $L^2(\mathbb{R}^n)$, readily yield (4.9.14) via a limiting argument.

In turn, given $k \in \{1,\dots,n\}$, for each $f \in L^2(\mathbb{R}^n)$ formula (4.9.14) allows us to compute

$$\mathcal{F}\big(R_j(R_k f)\big) = -\frac{i\omega_n}{2}\,\frac{\xi_j}{|\xi|}\,\widehat{R_k f} = -\Big(\frac{\omega_n}{2}\Big)^2\frac{\xi_j\xi_k}{|\xi|^2}\,\widehat{f} \quad \text{in} \quad L^2(\mathbb{R}^n). \tag{4.9.26}$$

A similar computation also gives that $\mathcal{F}\big(R_k(R_j f)\big) = -\big(\frac{\omega_n}{2}\big)^2\frac{\xi_j\xi_k}{|\xi|^2}\,\widehat{f}$ in $L^2(\mathbb{R}^n)$. From these, (4.9.15) follows upon recalling (cf. (3.2.31)) that the Fourier transform is an isomorphism of $L^2(\mathbb{R}^n)$.

Finally, as regards (4.9.16), for any function $f \in L^2(\mathbb{R}^n)$, formula (4.9.26) (with $j = k$) permits us to write

$$\begin{aligned}\mathcal{F}\Big(\sum_{k=1}^n R_k^2 f\Big) &= \sum_{k=1}^n \mathcal{F}\big(R_k(R_k f)\big) = -\Big(\frac{\omega_n}{2}\Big)^2\Big(\sum_{k=1}^n \frac{\xi_k^2}{|\xi|^2}\Big)\widehat{f}\\ &= -\Big(\frac{\omega_n}{2}\Big)^2\widehat{f} \quad \text{in} \quad L^2(\mathbb{R}^n),\end{aligned} \tag{4.9.27}$$

where the last equality uses the fact that $\sum_{k=1}^n \frac{\xi_k^2}{|\xi|^2} = 1$ a.e. in $\mathbb{R}^n$. Now identity (4.9.16) follows from (4.9.27) and (3.2.31). □

Corollary 4.94. *Let $j \in \{1,\dots,n-1\}$ and $t \in (0,\infty)$, and recall the functions p_t and $(q_j)_t$ from (4.8.8) and (4.8.23), respectively. Then*

$$(q_j)_t = \frac{2}{\omega_{n-1}} R_j p_t \quad \text{in} \quad L^2(\mathbb{R}^{n-1}), \tag{4.9.28}$$

where R_j in (4.9.28) is the jth Riesz transform from Theorem 4.93 corresponding to n replaced by $n-1$.

Proof. By (3.2.31), it suffices to check identity (4.9.28) on the Fourier transform side. That the latter holds is an immediate consequence of part *(2)* in Proposition 4.89, (4.9.14), and part *(2)* in Proposition 4.87. □

In the one-dimensional setting, Theorem 4.93 yields the following type of information about the Hilbert transform.

Corollary 4.95. *The Hilbert transform defined in (4.9.2) satisfies the following properties.*

(a) The operator

$$\begin{aligned}&H : \mathcal{S}(\mathbb{R}) \to \mathcal{S}'(\mathbb{R}) \quad \textit{is well-defined,}\\ &\textit{linear, and sequentially continuous.}\end{aligned} \tag{4.9.29}$$

(b) For each $\varphi \in \mathcal{S}(\mathbb{R})$ we have $H\varphi \in C^\infty(\mathbb{R})$ and H may be expressed as the SIO

$$(H\varphi)(x) = \lim_{\varepsilon\to 0^+}\int_{|x-y|\geq\varepsilon}\frac{1}{x-y}\varphi(y)\,dy, \qquad \forall\, x \in \mathbb{R}. \tag{4.9.30}$$

(c) *The Hilbert transform is a multiplier with symbol* $m(\xi) := -i\pi\,(\operatorname{sgn}\xi)$ *belonging to* $L^\infty(\mathbb{R})$. *In other words, for each* $\varphi \in \mathcal{S}(\mathbb{R})$,

$$\widehat{H\varphi}(\xi) = -i\pi\,(\operatorname{sgn}\xi)\widehat{\varphi}(\xi) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}). \tag{4.9.31}$$

(d) *For each* $\varphi \in \mathcal{S}(\mathbb{R})$, *we have that* $H\varphi$ *originally viewed as a tempered distribution belongs to the subspace* $L^2(\mathbb{R})$ *of* $\mathcal{S}'(\mathbb{R})$, *and*

$$\|H\varphi\|_{L^2(\mathbb{R}^n)} \leq \pi\|\varphi\|_{L^2(\mathbb{R}^n)}. \tag{4.9.32}$$

(e) *The Hilbert transform* H, *originally considered as in* (4.9.29) *extends, by density to a linear and bounded operator*

$$H : L^2(\mathbb{R}) \longrightarrow L^2(\mathbb{R}) \tag{4.9.33}$$

and the operator H *is skew-symmetric (i.e.,* $H^* = -H$*) in this context. In addition,*

$$\widehat{Hf}(\xi) = -i\pi\,(\operatorname{sgn}\xi)\,\widehat{f}(\xi) \quad \text{a.e. in} \quad \mathbb{R}, \qquad \forall\, f \in L^2(\mathbb{R}). \tag{4.9.34}$$

(f) *In the context of* (4.9.33),

$$H^2 = -\pi^2\, I \quad \text{in} \quad L^2(\mathbb{R}), \tag{4.9.35}$$

where I *denotes the identity operator on* $L^2(\mathbb{R})$.

Proof. This corresponds to Theorem 4.93 in the case when $n = 1$. □

The Riesz transforms are prototypes for the more general class of operators introduced in Definition 4.90 and most of the properties deduced for the Riesz transforms in Theorem 4.93 have natural counterparts in this more general setting.

Theorem 4.96. *Assume that* Θ *is a function satisfying the conditions in* (4.4.1) *and let* T_Θ *be the SIO associated with* Θ *as in* (4.9.3). *Then the following statements are true.*

(a) *The operator*

$$\begin{gathered} T_\Theta : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n) \quad \textit{is well-defined,} \\ \textit{linear, and sequentially continuous.} \end{gathered} \tag{4.9.36}$$

(b) *For each* $\varphi \in \mathcal{S}(\mathbb{R}^n)$ *we have* $T_\Theta\varphi \in C^\infty(\mathbb{R}^n)$ *and the SIO* T_Θ *may be expressed as the convolution*

$$T_\Theta\varphi = \big(\text{P.V.}\,\Theta\big) * \varphi \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.9.37}$$

(c) The function m_Θ, associated with Θ as in (4.5.6), is the symbol of the multiplier T_Θ. This means that for each $\varphi \in \mathcal{S}(\mathbb{R}^n)$,

$$\widehat{T_\Theta \varphi}(\xi) = m_\Theta(\xi)\,\widehat{\varphi}(\xi) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{4.9.38}$$

(d) For each $\varphi \in \mathcal{S}(\mathbb{R}^n)$, we have that $T_\Theta\varphi$ originally viewed as a tempered distribution belongs to the subspace $L^2(\mathbb{R}^n)$ of $\mathcal{S}'(\mathbb{R}^n)$, and

$$\|T_\Theta\varphi\|_{L^2(\mathbb{R}^n)} \leq C_n \|\Theta\|_{L^\infty(S^{n-1})} \|\varphi\|_{L^2(\mathbb{R}^n)}, \tag{4.9.39}$$

where the constant $C_n \in (0,\infty)$ is as in (4.5.8).

(e) The operator T_Θ, originally considered as in (4.9.36) extends, by density to a linear and bounded operator

$$T_\Theta : L^2(\mathbb{R}^n) \longrightarrow L^2(\mathbb{R}^n). \tag{4.9.40}$$

Moreover, in this context,

$$\widehat{T_\Theta f}(\xi) = m_\Theta(\xi)\,\widehat{f}(\xi) \quad \text{a.e. in} \quad \mathbb{R}^n, \qquad \forall\, f \in L^2(\mathbb{R}^n), \tag{4.9.41}$$

and the operator T_Θ satisfies

$$\big(T_\Theta\big)^* = T_{\overline{\Theta}^\vee} \quad \text{in} \quad L^2(\mathbb{R}^n). \tag{4.9.42}$$

In particular, if Θ is odd and real-valued, then the operator T_Θ is skew-symmetric (i.e., $\big(T_\Theta\big)^* = -T_\Theta$), while if Θ is even and real-valued then the operator T_Θ is self-adjoint (i.e., $\big(T_\Theta\big)^* = T_\Theta$).

Proof. Parts *(a)* and *(b)* are contained in Proposition 4.91, while part *(c)* follows from (4.9.6) and Theorem 4.71. Part *(d)* is justified by reasoning as in the proof of part *(d)* in Theorem 4.93, keeping in mind (4.5.7). As for part *(e)*, in a first stage we note that, for any $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$, by reasoning much as in (4.9.23) while relying on (4.9.38) and (4.5.9), we obtain

$$\begin{aligned}
\langle T_\Theta\varphi, \overline{\psi}\,\rangle &= (2\pi)^{-n}\big\langle\, \widehat{T_\Theta\varphi}, \widehat{\overline{\psi}}^{\,\vee}\,\big\rangle = (2\pi)^{-n}\Big\langle\, m_\Theta\,\widehat{\varphi}\,,\, \widehat{\overline{\psi}}^{\,\vee}\,\Big\rangle \\
&= (2\pi)^{-n}\int_{\mathbb{R}^n} m_\Theta(\xi)\,\widehat{\varphi}(\xi)\widehat{\overline{\psi(-\xi)}}\,d\xi \\
&= (2\pi)^{-n}\int_{\mathbb{R}^n} m_\Theta(-\xi)\,\widehat{\overline{\psi(\xi)}}\widehat{\varphi}(-\xi)\,d\xi \\
&= (2\pi)^{-n}\Big\langle\, m_{\Theta^\vee}\,\widehat{\overline{\psi}}\,,\, \widehat{\varphi}^{\,\vee}\,\Big\rangle = (2\pi)^{-n}\big\langle\, \widehat{T_{\Theta^\vee}\overline{\psi}}, \widehat{\varphi}^{\,\vee}\,\big\rangle \\
&= \langle T_{\Theta^\vee}\overline{\psi}, \varphi\rangle = \langle\, \overline{T_{\overline{\Theta}^\vee}\,\psi}, \varphi\rangle,
\end{aligned} \tag{4.9.43}$$

where the last step uses the fact that, as seen from (4.9.3), $T_{\Theta^\vee}\overline{\psi} = \overline{T_{\overline{\Theta}^\vee}\,\psi}$. Passing from (4.9.43) to (4.9.42) may now be done by arguing as in the proof of part *(e)* of Theorem 4.93 (a pattern of reasoning which also gives all remaining claims in the current case). □

We conclude this section with a discussion of the following result of basic importance from Calderón–Zygmund theory (originally proved in [5]; for a more timely presentation see, e.g., [23, 62, 63]).

Theorem 4.97. *Let Θ be a function satisfying (4.4.1) and, in analogy with (4.9.3), given some $p \in (1,\infty)$ set*

$$(T_\Theta f)(x) := \lim_{\varepsilon\to 0^+} \int_{|x-y|\geq\varepsilon} \Theta(x-y) f(y)\,dy, \qquad x \in \mathbb{R}^n, \quad f \in L^p(\mathbb{R}^n). \tag{4.9.44}$$

Then for every $f \in L^p(\mathbb{R}^n)$ we have that $(T_\Theta f)(x)$ exists for almost every $x \in \mathbb{R}^n$ and

$$\|T_\Theta f\|_{L^p(\mathbb{R}^n)} \leq C_\Theta \|f\|_{L^p(\mathbb{R}^n)}, \tag{4.9.45}$$

where C_Θ is a finite positive constant independent of f.

Compared with Theorem 4.96, a basic achievement of Calderón and Zygmund is allowing functions from $L^p(\mathbb{R}^n)$ in lieu of $\mathcal{S}(\mathbb{R}^n)$. While the estimate in (4.9.45) for $p = 2$ is essentially contained in part *(d)* of Theorem 4.96, the fact that for each $f \in L^2(\mathbb{R}^n)$ the limit defining $(T_\Theta f)(x)$ as in (4.9.44) exists for almost every $x \in \mathbb{R}^n$ is a bit more subtle. To elaborate on this issue, recall that we have already proved that the limit in question exists at every point when f is a Schwartz function (cf. Proposition 4.67). Starting from this, one can pass to arbitrary functions in $L^2(\mathbb{R}^n)$ granted the availability of additional tools from harmonic analysis (more specifically, the boundedness in $L^2(\mathbb{R}^n)$ of the so-called maximal operator $T_* f(x) := \Big|\sup_{\varepsilon>0} \int_{|x-y|\geq\varepsilon} \Theta(x-y) f(y)\,dy\Big|$, for $x \in \mathbb{R}^n$). Having dealt with (4.9.45) in the case $p = 2$, its proof for $p \in (1,\infty)$ proceeds as follows. In a first step, a suitable version of (4.9.45) is established for $p = 1$ which, when combined with the case $p = 2$ already treated yields (via a technique called interpolation) (4.9.45) for $p \in (1, 2)$. In a second step, one uses duality to handle the case $p \in (2,\infty)$. There are two important factors at play here: that the dual of $L^p(\mathbb{R}^n)$ with $p \in (1, 2)$ is $L^{p'}(\mathbb{R}^n)$ with $p' = \frac{p}{p-1} \in (2,\infty)$, and the formula for the adjoint of T_Θ we deduced in (4.9.42).

In particular, Theorem 4.97 gives that, given $p \in (1,\infty)$, the jth Riesz transform R_j, $j \in \{1,\dots,n\}$, originally considered as in (4.9.9) extends by density so that

$$R_j : L^p(\mathbb{R}^n) \longrightarrow L^p(\mathbb{R}^n) \quad \text{is linear and bounded,} \tag{4.9.46}$$

and for each $f \in L^p(\mathbb{R}^n)$ we have

$$(R_j f)(x) = \lim_{\varepsilon\to 0^+} \int_{|x-y|\geq\varepsilon} \frac{x_j - y_j}{|x-y|^{n+1}} f(y)\,dy \quad \text{for a.e. } x \in \mathbb{R}^n. \tag{4.9.47}$$

Of course, the same type of result is true for the Hilbert transform on the real line.

4.10 Derivatives of Volume Potentials

One basic integral operator in analysis is the so-called `Newtonian potential`, given by

$$(\mathbf{N}_\Omega f)(x) := \frac{-1}{(n-2)\omega_{n-1}} \int_\Omega \frac{1}{|x-y|^{n-2}} f(y)\,\mathrm{d}y, \qquad x \in \mathbb{R}^n, \tag{4.10.1}$$

where Ω is an open set in $\mathbb{R}^n$, $n \geq 3$, and $f \in L^\infty(\Omega)$ with bounded support. In this regard, an important issue is that of computing $\partial_j \partial_k \mathbf{N}_\Omega f$ in the sense of distributions in $\mathbb{R}^n$, where $j,k \in \{1,\dots,n\}$. First, we will show (cf. Theorem 4.101) that $\mathbf{N}_\Omega f$ belongs to $C^1(\mathbb{R}^n)$ and we have

$$\partial_k(\mathbf{N}_\Omega f)(x) = \frac{1}{\omega_{n-1}} \int_\Omega \frac{x_k - y_k}{|x-y|^n} f(y)\,\mathrm{d}y, \qquad x \in \mathbb{R}^n. \tag{4.10.2}$$

Hence, $\partial_k(\mathbf{N}_\Omega f)(x) = \int_\Omega \Phi(x-y) f(y)\,\mathrm{d}y$ where $\Phi(z) := \frac{1}{\omega_{n-1}} \frac{z_k}{|z|^n}$ for $z \in \mathbb{R}^n \setminus \{0\}$. Note that for x fixed, $\Phi(x-\cdot)$ is locally integrable, though this is no longer the case for $(\partial_j \Phi)(x-\cdot)$. This makes the job of computing $\partial_j\big(\partial_k(\mathbf{N}_\Omega f)\big)$ considerably more subtle. There is a good reason for this since, as it turns out, the latter distributional derivative involves (as we shall see momentarily) SIO. We note that the function Φ considered above is C^∞ and positive homogeneous of degree $1-n$ in $\mathbb{R}^n \setminus \{0\}$. These are going to be key features in our subsequent analysis.

Our most general results encompassing the discussion about the Newtonian potential introduced above are contained in Theorems 4.100 and 4.101. We begin by first considering the case when $\Omega = \mathbb{R}^n$ and f is a Schwartz function.

Proposition 4.98. *Let $\Phi \in C^1(\mathbb{R}^n \setminus \{0\})$ be a function that is positive homogeneous of degree $1-n$ and let $f \in \mathcal{S}(\mathbb{R}^n)$. Then $\int_{\mathbb{R}^n} \Phi(x-y) f(y)\,\mathrm{d}y$ is differentiable in $\mathbb{R}^n$ and for each $j \in \{1,\dots,n\}$ we have*

$$\begin{aligned} \partial_{x_j}\Big[\int_{\mathbb{R}^n} \Phi(x-y) f(y)\,\mathrm{d}y\Big] &= \Big(\int_{S^{n-1}} \Phi(\omega)\omega_j\,\mathrm{d}\sigma(\omega)\Big) f(x) \\ &\quad + \lim_{\varepsilon \to 0^+} \int_{|y-x|\geq\varepsilon} (\partial_j \Phi)(x-y) f(y)\,\mathrm{d}y \quad \forall\, x \in \mathbb{R}^n. \end{aligned} \tag{4.10.3}$$

Proof. The key ingredient in the proof of (4.10.3) is formula (4.4.19). To see how (4.4.19) applies, first note that by using Exercise 4.51 we have for each $R > 0$,

$$\begin{aligned} &\Big|\int_{\mathbb{R}^n} \Phi(x-y) f(y)\,\mathrm{d}y\Big| \\ &\qquad \leq \|\Phi\|_{L^\infty(S^{n-1})} \int_{\mathbb{R}^n} \frac{|f(y)|}{|x-y|^{n-1}}\,\mathrm{d}y \end{aligned}$$

$$\leq \|\Phi\|_{L^\infty(S^{n-1})}\|f\|_{L^\infty(\mathbb{R}^n)}\int_{|y-x|\leq R}\frac{\mathrm{d}y}{|x-y|^{n-1}}$$

$$+\|\Phi\|_{L^\infty(S^{n-1})}\|(1+|y|)^2 f\|_{L^\infty(\mathbb{R}^n)}\int_{|y-x|>R}\frac{\mathrm{d}y}{(1+|y|^2)|x-y|^{n-1}}<\infty, \tag{4.10.4}$$

thus $\int_{\mathbb{R}^n}\Phi(x-y)f(y)\,\mathrm{d}y$ is well-defined. Second, since

$$\int_{\mathbb{R}^n}\Phi(x-y)f(y)\,\mathrm{d}y=\int_{\mathbb{R}^n}\Phi(y)f(x-y)\,\mathrm{d}y\qquad\forall\,x\in\mathbb{R}^n, \tag{4.10.5}$$

we have that $\int_{\mathbb{R}^n}\Phi(x-y)f(y)\,\mathrm{d}y$ is differentiable and

$$\partial_{x_j}\int_{\mathbb{R}^n}\Phi(x-y)f(y)\,\mathrm{d}y=\int_{\mathbb{R}^n}\Phi(y)(\partial_j f)(x-y)\,\mathrm{d}y\qquad\forall\,x\in\mathbb{R}^n. \tag{4.10.6}$$

Third, for each $x\in\mathbb{R}^n$, with t_x as in (2.8.37) (and recalling (3.2.22)), we may write

$$\int_{\mathbb{R}^n}\Phi(x-y)\varphi(y)\,\mathrm{d}y=\langle t_x(\Phi^\vee),\varphi\rangle,\qquad\forall\,\varphi\in\mathcal{S}(\mathbb{R}^n). \tag{4.10.7}$$

Now fix $j\in\{1,\dots,n\}$. Then for $x\in\mathbb{R}^n$ we have

$$\begin{aligned}
&\partial_{x_j}\Big[\int_{\mathbb{R}^n}\Phi(x-y)f(y)\,\mathrm{d}y\Big]\\
&\quad=\langle t_x(\Phi^\vee),\partial_j f\rangle=\langle\Phi^\vee,t_{-x}(\partial_j f)\rangle=\langle\Phi^\vee,\partial_j[t_{-x}f]\rangle=-\langle\partial_j[\Phi^\vee],t_{-x}f\rangle\\
&\quad=-\Big(\int_{S^{n-1}}\omega_j\Phi^\vee(\omega)\,\mathrm{d}\sigma(\omega)\Big)\langle\delta,t_{-x}f\rangle-\Big\langle\mathrm{P.V.}\big(\partial_j(\Phi^\vee)\big),t_{-x}f\Big\rangle\\
&\quad=f(x)\Big(\int_{S^{n-1}}\omega_j\Phi(\omega)\,\mathrm{d}\sigma(\omega)\Big)+\lim_{\varepsilon\to0^+}\int_{|y|\geq\varepsilon}(\partial_j\Phi)(-y)f(x+y)\,\mathrm{d}y\\
&\quad=f(x)\Big(\int_{S^{n-1}}\omega_j\Phi(\omega)\,\mathrm{d}\sigma(\omega)\Big)+\lim_{\varepsilon\to0^+}\int_{|y|\geq\varepsilon}(\partial_j\Phi)(x-y)f(y)\,\mathrm{d}y.
\end{aligned} \tag{4.10.8}$$

The first equality in (4.10.8) uses (4.10.6) and (4.10.7), the fifth uses (4.4.19), the sixth the fact that $\partial_j(\Phi^\vee)=-(\partial_j\Phi)^\vee$, and the last one a suitable change of variables. This completes the proof of the corollary. □

Next, we present a version of Proposition 4.98 when the function f is lacking any type of differentiability properties. Here we make use of the basic Calderón–Zygmund result recorded in Theorem 4.97.

Theorem 4.99. *Let $\Phi \in C^1(\mathbb{R}^n\setminus\{0\})$ be a function that is positive homogeneous of degree $1-n$ and let $f \in L^p(\mathbb{R}^n)$ for some $p \in (1,n)$. Then $\Phi * f \in L^1_{\rm loc}(\mathbb{R}^n)$ and for each $j \in \{1,\dots,n\}$ we have $T_{\partial_j\Phi} f \in L^p(\mathbb{R}^n)$ and*

$$\partial_j(\Phi * f) = \Big(\int_{S^{n-1}} \Phi(\omega)\omega_j \,{\rm d}\sigma(\omega)\Big) f + T_{\partial_j\Phi} f \qquad \text{in } \ \mathcal{D}'(\mathbb{R}^n), \tag{4.10.9}$$

where $T_{\partial_j\Phi}$ is the operator from (4.9.45) with ψ replaced by $\partial_j\Phi$.

Proof. Fix $p \in (1,n)$, $f \in L^p(\mathbb{R}^n)$, and $R \in (0,\infty)$. Then we write

$$\begin{aligned}\int_{B(0,R)}\int_{\mathbb{R}^n} \frac{|f(y)|}{|x-y|^{n-1}}\,{\rm d}y\,{\rm d}x &= \int_{|y|\leq 2R} |f(y)| \int_{B(0,R)} \frac{1}{|x-y|^{n-1}}\,{\rm d}x\,{\rm d}y \\ &\quad + \int_{|y|>2R} |f(y)| \int_{B(0,R)} \frac{1}{|x-y|^{n-1}}\,{\rm d}x\,{\rm d}y =: I + II.\end{aligned} \tag{4.10.10}$$

If $y \in B(0,2R)$ and $x \in B(0,R)$, then $|x-y| \leq 3R$, thus

$$\begin{aligned} I &\leq \int_{|y|\leq 2R} |f(y)| \int_{|x-y|\leq 3R} \frac{1}{|x-y|^{n-1}}\,{\rm d}x\,{\rm d}y \\ &\leq \Big(\int_{|z|\leq 3R} \frac{1}{|z|^{n-1}}\,{\rm d}z\Big) |B(0,2R)|^{1-\frac{1}{p}} \|f\|_{L^p(B(0,2R))} < \infty, \end{aligned} \tag{4.10.11}$$

where for the second inequality in (4.10.11) Hölder's inequality has been used. Also, if $y \in \mathbb{R}^n \setminus B(0,2R)$ and $x \in B(0,R)$, then

$$|y| \leq |y-x| + |x| \leq |y-x| + R \leq |y-x| + \frac{|y|}{2},$$

which implies $|y-x| \geq |y|/2$. Using this, II is estimated as

$$\begin{aligned} II &\leq 2^{n-1} \int_{|y|>2R} |f(y)| \int_{|x|\leq R} \frac{1}{|y|^{n-1}}\,{\rm d}x\,{\rm d}y \\ &\leq 2^{n-1}|B(0,R)| \int_{|y|>2R} \frac{|f(y)|}{|y|^{n-1}}\,{\rm d}y \\ &\leq 2^{n-1}|B(0,R)| \|f\|_{L^p(\mathbb{R}^n)} \Big(\int_{|y|>2R} \frac{{\rm d}y}{|y|^{(n-1)p'}}\Big)^{1/p'} < \infty, \end{aligned} \tag{4.10.12}$$

where $p' := \frac{p}{p-1}$ is the Hölder conjugate exponent for p. In the third inequality in (4.10.12) the assumption $p \in (1,n)$ has been used to ensure that $(n-1)p' > n$.

A combination of (4.10.10), (4.10.11), and (4.10.12) yields the following conclusion: for every $R \in (0,\infty)$ there exists some finite positive constant C, depending on R, n, and p, such that

$$\int_{B(0,R)}\int_{\mathbb{R}^n} \frac{|f(y)|}{|x-y|^{n-1}}\,{\rm d}y\,{\rm d}x \leq C\|f\|_{L^p(\mathbb{R}^n)}, \quad \forall\, f \in L^p(\mathbb{R}^n). \tag{4.10.13}$$

In turn, (4.10.13) entails several useful conclusions. Recall that the assumptions on Φ imply $|\Phi(x)| \leq \|\Phi\|_{L^\infty(S^{n-1})}|x|^{1-n}$ for each $x \in \mathbb{R}^n \setminus \{0\}$ (cf. Exercise 4.51). The first conclusion is that for each $f \in L^p(\mathbb{R}^n)$ and each $R \in (0,\infty)$,

$$\int_{\mathbb{R}^n} |\Phi(x-y)||f(y)|\,dy < \infty \quad \text{for a.e.} \quad x \in B(0,R). \tag{4.10.14}$$

This shows that $(\Phi * f)(x)$ is well-defined for a.e. $x \in \mathbb{R}^n$. Second, from what we have just proved and (4.10.13) we may conclude that $\Phi * f \in L^1_{\rm loc}(\mathbb{R}^n)$.

Next, recall that $C_0^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$ for each $p \in (1,\infty)$. As such there exists a sequence $\{f_k\}_{k\in\mathbb{N}}$ of function in $C_0^\infty(\mathbb{R}^n)$ with the property that $\lim_{k\to\infty} f_k = f$ in $L^p(\mathbb{R}^n)$. Based on (4.10.13) we may conclude that the sequence $\{\Phi * f_k\}_{k\in\mathbb{N}}$ converges in $L^p(\mathbb{R}^n)$ to $\Phi * f$. Therefore, by Exercise 2.21,

$$\Phi * f_k \xrightarrow[k\to\infty]{\mathcal{D}'(\mathbb{R}^n)} \Phi * f. \tag{4.10.15}$$

Moving on, fix $j \in \{1,\dots,n\}$ and note that, by Exercise 4.49, $\partial_j\Phi$ is positive homogeneous of degree $-n$. The function $\Phi * f_k$ belongs to $C^\infty(\mathbb{R}^n)$ and Proposition 4.98 applies and implies that

$$\partial_j\big(\Phi * f_k\big) = \Big(\int_{S^{n-1}} \Phi(\omega)\omega_j\,d\sigma(\omega)\Big) f_k + T_{\partial_j\Phi} f_k, \quad \forall\, k \in \mathbb{N}, \tag{4.10.16}$$

pointwise in $\mathbb{R}^n$. In particular, from (4.10.16) we infer that $T_{\partial_j\Phi} f_k \in C^\infty(\mathbb{R}^n)$ for all $k \in \mathbb{N}$, and that the equality in (4.10.16) also holds in $\mathcal{D}'(\mathbb{R}^n)$. Since Theorem 4.97 gives that $T_{\partial_j\Phi} f_k \xrightarrow[k\to\infty]{L^p(\mathbb{R}^n)} T_{\partial_j\Phi} f$, based on Exercise 2.21 it follows that

$$T_{\partial_j\Phi} f_k \xrightarrow[k\to\infty]{\mathcal{D}'(\mathbb{R}^n)} T_{\partial_j\Phi} f \qquad \text{and} \qquad f_k \xrightarrow[k\to\infty]{\mathcal{D}'(\mathbb{R}^n)} f. \tag{4.10.17}$$

Since an immediate consequence of (4.10.15) is that $\partial_j(\Phi * f_k) \xrightarrow[k\to\infty]{\mathcal{D}'(\mathbb{R}^n)} \partial_j(\Phi * f)$, combining this, (4.10.17), and the fact that (4.10.16) holds in $\mathcal{D}'(\mathbb{R}^n)$, we obtain (4.10.9). □

We are prepared to state and prove our most general results regarding distributional derivatives of volume potentials. In particular, the next two theorems contain solutions to the questions formulated at the beginning of this section.

Theorem 4.100. *Let $\Phi \in C^1(\mathbb{R}^n \setminus \{0\})$ be a function that is positive homogeneous of degree $1-n$. Consider a measurable set $\Omega \subseteq \mathbb{R}^n$ and assume that $f \in L^p(\Omega)$ for some $p \in (1,n)$. Then $\int_\Omega \Phi(x-y)f(y)\,dy$ is absolutely convergent for a.e. $x \in \mathbb{R}^n$ and belongs to $L^1_{\rm loc}(\mathbb{R}^n)$ as a function of the variable x. Moreover, if $\widetilde{f}$ denotes the extension of f by zero to $\mathbb{R}^n$, then for each $j \in \{1,\dots,n\}$ we have*

$$\partial_{x_j}\Big[\int_\Omega \Phi(x-y)f(y)\,dy\Big] = \Big(\int_{S^{n-1}} \Phi(\omega)\omega_j\,d\sigma(\omega)\Big)\widetilde{f}(x) + \lim_{\varepsilon\to 0^+}\int_{y\in\Omega\setminus B(x,\varepsilon)} (\partial_j\Phi)(x-y)f(y)\,dy \quad (4.10.18)$$

where the derivative in the left-hand side is taken in $\mathcal{D}'(\mathbb{R}^n)$ and the equality is also understood in $\mathcal{D}'(\mathbb{R}^n)$.

In particular, formula (4.10.18) holds for any function f belonging to some $L^q(\Omega)$, with $q\in(1,\infty)$, that vanishes outside of a measurable subset of Ω of finite measure.

Proof. The main claim in the statement follows by applying Theorem 4.99 to the function

$$\widetilde{f} := \begin{cases} f & \text{in } \Omega, \\ 0 & \text{in } \mathbb{R}^n\setminus\Omega, \end{cases} \quad (4.10.19)$$

upon observing that $\widetilde{f}\in L^p(\mathbb{R}^n)$. Moreover, if f is as in the last claim in the statement then Hölder's inequality may be invoked to show that f belongs to $L^p(\Omega)$ for some $p\in(1,n)$. □

Recall that if $a\in\mathbb{R}$, the integer part of a is denoted by $\lfloor a\rfloor$ and is by definition the largest integer that is less than or equal to a. To state the next theorem we introduce

$$\langle a\rangle := \begin{cases} \lfloor a\rfloor & \text{if } a\notin\mathbb{Z}, \\ a-1 & \text{if } a\in\mathbb{Z}. \end{cases} \quad (4.10.20)$$

Hence, $\langle a\rangle$ is the largest integer strictly less than a (thus, in particular, $\langle a\rangle < a$ for every $a\in\mathbb{R}$).

Theorem 4.101. *Let $\Phi\in C^\infty(\mathbb{R}^n\setminus\{0\})$ be a function that is positive homogeneous of degree $m\in\mathbb{R}$ where $m>-n$. Define the* `generalized volume potential` *associated with Φ by setting for each $f\in L^\infty_{\rm comp}(\mathbb{R}^n)$*

$$(\Pi_\Phi f)(x) := \int_{\mathbb{R}^n}\Phi(x-y)f(y)\,dy, \qquad \forall\, x\in\mathbb{R}^n. \quad (4.10.21)$$

Then for each $f\in L^\infty_{\rm comp}(\mathbb{R}^n)$ one has $\Pi_\Phi f\in C^{\langle m+n\rangle}(\mathbb{R}^n)$ and

$$\partial^\alpha(\Pi_\Phi f) = \Pi_{\partial^\alpha\Phi} f \quad \text{pointwise in } \mathbb{R}^n, \quad \forall\,\alpha\in\mathbb{N}_0^n \ \text{ with } \ |\alpha|\le\langle m+n\rangle. \quad (4.10.22)$$

Moreover, if $\alpha\in\mathbb{N}_0^n$ is such that $|\alpha|=m+n$, then for each $f\in L^\infty_{\rm comp}(\mathbb{R}^n)$, the distributional derivative $\partial^\alpha\big[\Pi_\Phi f\big]$ is of function type and satisfies

$$\partial^\alpha\big[\Pi_\Phi f\big](x) = \Big(\int_{S^{n-1}} (\partial^\beta\Phi)(\omega)\omega_j\,{\rm d}\sigma(\omega)\Big) f(x) + \lim_{\varepsilon\to 0^+}\int_{\mathbb{R}^n\setminus B(x,\varepsilon)} (\partial^\alpha\Phi)(x-y)\,f(y)\,{\rm d}y \quad \text{in } \mathcal{D}'(\mathbb{R}^n), \tag{4.10.23}$$

for any $\beta\in\mathbb{N}_0^n$ *and* $j\in\{1,\dots,n\}$ *such that* $\beta+\mathbf{e}_j=\alpha$.

Proof. Fix $f\in L^\infty_{\rm comp}(\mathbb{R}^n)$ and let $K:=\operatorname{supp} f$ which is a compact set in $\mathbb{R}^n$. By Exercises 4.49 and 4.51,

$$\begin{aligned} &\text{for every } \alpha\in\mathbb{N}_0^n \quad \text{we have that} \quad \partial^\alpha\Phi\in C^\infty(\mathbb{R}^n\setminus\{0\}),\\ &\partial^\alpha\Phi \quad \text{is positive homogeneous of degree} \quad m-|\alpha|, \quad \text{and}\\ &\big|\partial^\alpha\Phi(x-y)\big|\leq\|\partial^\alpha\Phi\|_{L^\infty(S^{n-1})}|x-y|^{m-|\alpha|} \quad \forall\,x,\,y\in\mathbb{R}^n,\,x\neq y. \end{aligned} \tag{4.10.24}$$

Fix now $\alpha\in\mathbb{N}_0^n$ such that $|\alpha|\leq\langle m+n\rangle$. Then $m-|\alpha|\geq\langle m+n\rangle>-n$, hence (4.10.24) further yields

$$\int_{\mathbb{R}^n}|(\partial^\alpha\Phi)(x-y)f(y)|\,{\rm d}y\leq C\|f\|_{L^\infty(\mathbb{R}^n)}\int_K|x-y|^{m-|\alpha|}\,{\rm d}y<\infty. \tag{4.10.25}$$

This proves that $\Pi_{\partial^\alpha\Phi}f$ is well-defined.

Next, we focus on proving

$$\Pi_{\partial^\alpha\Phi}f\in C^0(\mathbb{R}^n). \tag{4.10.26}$$

To see this, fix $x_0\in\mathbb{R}^n$ and pick an arbitrary sequence $\{x_k\}_{k\in\mathbb{N}}$ of points in $\mathbb{R}^n$ satisfying $\lim_{k\to\infty}x_k=x_0$. Consider the following functions defined a.e. in $\mathbb{R}^n$:

$$v_k:=(\partial^\alpha\Phi)(x_k-\cdot)f,\quad\forall\,k\in\mathbb{N},\quad\text{and}\quad v:=(\partial^\alpha\Phi)(x_0-\cdot)f. \tag{4.10.27}$$

To conclude that $\Pi_{\partial^\alpha\Phi}f$ is continuous at x_0, it suffices to show that

$$\lim_{k\to\infty}\int_K v_k(y)\,{\rm d}y=\int_K v(y)\,{\rm d}y. \tag{4.10.28}$$

The strategy for proving (4.10.28) is to apply Vitali's theorem (cf. Theorem 13.21) with $X:=K$ and μ being the restriction to K of the Lebesgue measure in $\mathbb{R}^n$. Since K is compact we have $\mu(X)<\infty$ and from (4.10.25) we know that $v_k\in L^1(X,\mu)$ for all $k\in\mathbb{N}$. Clearly, $|v(x)|<\infty$ for μ-a.e. $x\in K$. Also, $\lim_{k\to\infty}v_k(y)=v(y)$ for μ-almost every $y\in K$. Hence, in order to obtain (4.10.28), the only hypothesis left to verify in Vitali's theorem is that the sequence $\{v_k\}_{k\in\mathbb{N}}$ is uniformly integrable in (X,μ). With this goal in mind, let $\varepsilon>0$ be fixed and consider a μ-measurable set $A\subset K$ such that $\mu(A)$ is sufficiently small, to a degree to be specified later. Then for every $k\in\mathbb{N}$, based on (4.10.24), we have

$$\Big|\int_A v_k(x)\,d\mu(x)\Big| \leq \|\partial^\alpha\Phi\|_{L^\infty(S^{n-1})}\|f\|_{L^\infty(\mathbb{R}^n)}\int_A |x_k-y|^{m-|\alpha|}\,dy$$
$$= C\int_{A-x_k}|x|^{m-|\alpha|}\,dx, \tag{4.10.29}$$

where $C:=\|\partial^\alpha\Phi\|_{L^\infty(S^{n-1})}\|f\|_{L^\infty(\mathbb{R}^n)}$. Note that $\mu(A-x_k)=\mu(A)$ for $k\in\mathbb{N}$. Also, since the sequence $\{x_k\}_{k\in\mathbb{N}}$ and the set K are bounded,

$$\exists\, R\in(0,\infty) \ \text{ such that } \ A-x_k\subset K-x_k\subset B(0,R) \quad \forall\, k\in\mathbb{N}. \tag{4.10.30}$$

Given that $m-|\alpha|>-n$, we have $|x|^{m-|\alpha|}\in L^1(B(0,R))$ and we may invoke Proposition 13.22 to conclude that there exists $\delta>0$ such that for every Lebesgue measurable set $E\subset B(0,R)$ with Lebesgue measure less than δ we have $\int_E|x|^{m-|\alpha|}\,dx<\varepsilon/C$. At this point return with the latter estimate in (4.10.29) to conclude that

$$\text{if } \ \mu(A)<\delta \ \text{ then } \ \Big|\int_A v_k(x)\,d\mu(x)\Big|<\varepsilon \quad \forall\, k\in\mathbb{N}. \tag{4.10.31}$$

This proves that the sequence $\{v_k\}_{k\in\mathbb{N}}$ is uniformly integrable in (X,μ) and finishes the proof of the fact that $\Pi_{\partial^\alpha\Phi}f$ is continuous at x_0. Since x_0 was arbitrary in $\mathbb{R}^n$ the membership in (4.10.26) follows.

Our next goal is to show (4.10.22). First note that based on what we proved so far we have $\Pi_\Phi f,\Pi_{\partial^\alpha\Phi}f\in L^1_{\rm loc}(\mathbb{R}^n)$, thus they define distributions in $\mathbb{R}^n$. We claim that the distribution $\Pi_{\partial^\alpha\Phi}f$ is equal to the distributional derivative $\partial^\alpha[\Pi_\Phi f]$. To see this, fix $\varphi\in C^\infty_0(\mathbb{R}^n)$ and using the definition of distributional derivatives and the definition of $\Pi_\Phi f$ write

$$\begin{aligned}\big\langle\partial^\alpha\Pi_\Phi f,\varphi\big\rangle &= (-1)^{|\alpha|}\big\langle\Pi_\Phi f,\partial^\alpha\varphi\big\rangle\\ &= (-1)^{|\alpha|}\int_{\mathbb{R}^n}\Big(\int_{\mathbb{R}^n}\Phi(x-y)f(y)\,dy\Big)\partial^\alpha\varphi(x)\,dx\\ &= (-1)^{|\alpha|}\int_{\mathbb{R}^n}f(y)\Big(\int_{\mathbb{R}^n}\Phi(x-y)\partial^\alpha\varphi(x)\,dx\Big)\,dy.\end{aligned} \tag{4.10.32}$$

Based on (4.10.24), the assumptions on φ, and the fact that $m-|\alpha|>-n$, we may use Lebesgue's dominated convergence theorem and formula (13.7.4) repeatedly to further write

$$\begin{aligned}\int_{\mathbb{R}^n}\Phi(x-y)\partial^\alpha\varphi(x)\,dx &= \lim_{\varepsilon\to0^+}\int_{\mathbb{R}^n\setminus\overline{B(y,\varepsilon)}}\Phi(x-y)\partial^\alpha\varphi(x)\,dx\\ &= \lim_{\varepsilon\to0^+}\Bigg\{(-1)^{|\alpha|}\int_{\mathbb{R}^n\setminus\overline{B(y,\varepsilon)}}(\partial^\alpha\Phi)(x-y)\varphi(x)\,dx\\ &\quad+\sum_{\alpha=\beta+\gamma+\mathbf{e}_j}c^j_{\beta\gamma}\int_{\partial B(y,\varepsilon)}(\partial^\beta\Phi)(x-y)\frac{x_j-y_j}{\varepsilon}\,\partial^\gamma\varphi(x)\,d\sigma(x)\Bigg\},\end{aligned} \tag{4.10.33}$$

where $c^{j}_{\beta\gamma}$ are suitable constants independent of ε. Note that, for each β, γ, and j such that $\alpha=\beta+\gamma+\mathbf{e}_j$, in light of (4.10.24) we have

$$\begin{aligned}
&\left|\int_{\partial B(y,\varepsilon)}(\partial^{\beta}\Phi)(x-y)\frac{x_j-y_j}{\varepsilon}\,\partial^{\gamma}\varphi(x)\,d\sigma(x)\right| \\
&\qquad\leq\int_{\partial B(y,\varepsilon)}|(\partial^{\beta}\Phi)(x-y)|\,|\partial^{\gamma}\varphi(x)|\,d\sigma(x) \\
&\qquad\leq\|\partial^{\gamma}\varphi\|_{L^{\infty}(\mathbb{R}^n)}\|\partial^{\beta}\Phi\|_{L^{\infty}(S^{n-1})}\int_{\partial B(y,\varepsilon)}|x-y|^{m-|\beta|}\,d\sigma(x) \\
&\qquad=C\varepsilon^{m-|\beta|+n-1}\xrightarrow[\varepsilon\to 0^{+}]{}0,
\end{aligned}\tag{4.10.34}$$

since $m-|\beta|+n-1\geq m+n-|\alpha|>0$. Returning with (4.10.34) to (4.10.33), and applying one more time Lebesgue's dominated convergence theorem, it follows that

$$\int_{\mathbb{R}^n}\Phi(x-y)\partial^{\alpha}\varphi(x)\,dx=(-1)^{|\alpha|}\int_{\mathbb{R}^n}(\partial^{\alpha}\Phi)(x-y)\varphi(x)\,dx.\tag{4.10.35}$$

This, when used in (4.10.32) further yields

$$\begin{aligned}
\big\langle\partial^{\alpha}\Pi_{\Phi}f,\varphi\big\rangle&=\int_{\mathbb{R}^n}f(y)\Big(\int_{\mathbb{R}^n}(\partial^{\alpha}\Phi)(x-y)\varphi(x)\,dx\Big)\,dy \\
&=\int_{\mathbb{R}^n}\Big(\int_{\mathbb{R}^n}(\partial^{\alpha}\Phi)(x-y)f(y)\,dy\Big)\varphi(x)\,dx \\
&=\big\langle\Pi_{\partial^{\alpha}\Phi}f,\varphi\big\rangle.
\end{aligned}\tag{4.10.36}$$

Since $\varphi\in C^{\infty}_{0}(\mathbb{R}^n)$ is arbitrary, from (4.10.36) we conclude

$$\partial^{\alpha}\Pi_{\Phi}f=\Pi_{\partial^{\alpha}\Phi}f\quad\text{in}\quad\mathcal{D}'(\mathbb{R}^n),\quad\forall\,\alpha\in\mathbb{N}^n_0\ \text{ with }\ |\alpha|\leq\langle m+n\rangle.\tag{4.10.37}$$

Upon observing that (4.10.26) and (4.10.37) hold for every $\alpha\in\mathbb{N}^n_0$ with the property that $|\alpha|=\langle m+n\rangle$, we may invoke Theorem 2.102 to infer that $\Pi_{\Phi}f\in C^{\langle m+n\rangle}(\mathbb{R}^n)$ and that (4.10.22) is valid.

We are left with proving the very last statement in the theorem. To this end, suppose there exists $\alpha\in\mathbb{N}^n_0$ satisfying $|\alpha|=m+n$. In particular, we have $m\in\mathbb{Z}$ and $|\alpha|\geq 1$. Hence, there exists $j\in\{1,\dots,n\}$ with the property that $\alpha_j\geq 1$. Set $\beta:=(\alpha_1,\dots,\alpha_{j-1},\alpha_j-1,\alpha_{j+1},\dots,\alpha_n)$, so that

$$\alpha=\beta+\mathbf{e}_j\quad\text{ and }\quad|\beta|=m+n-1=\langle m+n\rangle.\tag{4.10.38}$$

Based on what we proved earlier, we have $\partial^{\beta}[\Pi_{\Phi}f]=\Pi_{\partial^{\beta}\Phi}f$ pointwise in $\mathbb{R}^n$. Also, $\partial^{\beta}\Phi$ is of class C^{∞} and positive homogeneous of degree $1-n$ in $\mathbb{R}^n\setminus\{0\}$. Thus, we may apply Theorem 4.100 with Φ replaced by $\partial^{\beta}\Phi$ and Ω replaced by K and obtain

$$\begin{aligned}\partial^{\alpha}\big[\Pi_{\Phi}f\big](x) &= \partial_j\big[\partial^{\beta}[\Pi_{\Phi}f]\big](x) = \partial_j\big[\Pi_{\partial^{\beta}\Phi}f\big](x)\\ &= \partial_{x_j}\Big[\int_K(\partial^{\beta}\Phi)(x-y)f(y)\,dy\Big]\\ &= \Big(\int_{S^{n-1}}(\partial^{\beta}\Phi)(\omega)\omega_j\,d\sigma(\omega)\Big)f(x)\\ &\quad+\lim_{\varepsilon\to 0^+}\int_{\mathbb{R}^n\setminus B(x,\varepsilon)}(\partial^{\alpha}\Phi)(x-y)\,f(y)\,dy,\end{aligned}\tag{4.10.39}$$

where the derivative in the left hand-side of (4.10.39) is taken in $\mathcal{D}'(\mathbb{R}^n)$ and the equality is understood in $\mathcal{D}'(\mathbb{R}^n)$. This proves (4.10.23) and finishes the proof of the theorem. □

Exercise 4.102. *In the context of Theorem 4.101 prove directly, without relying on Theorem 2.102, that whenever $f\in L^{\infty}_{\rm comp}(\mathbb{R}^n)$ one has $\Pi_{\Phi}f\in C^{\langle m+n\rangle}(\mathbb{R}^n)$.*

Hint: Show that for each $x\in\mathbb{R}^n$ and each $j\in\{1,\dots,n\}$ fixed, one has

$$\lim_{h\to 0}\Big[(\Pi_{\Phi}f)(x+h\mathbf{e}_j)-(\Pi_{\Phi}f)(x)\Big]=(\Pi_{\partial_j\Phi}f)(x)\tag{4.10.40}$$

The proof of (4.10.40) may be done by using Vitali's theorem (cf. Theorem 13.21) in a manner analogous to the proof of (4.10.26). Once (4.10.40) is established, iterate to allow higher-order partial derivatives.

Further Notes for Chap. 4. The significance of the class of tempered distributions stems from the fact that this class is stable under the action of the Fourier transform. The topics discussed in Sects. 4.1–4.6 are classical and a variety of expositions is present in the literature, though they differ in terms of length and depth, and the current presentation is no exception. For example, while the convolution product of distributions is often confined to the case in which one of the distributions in question is compactly supported, that is, $\mathcal{E}'(\mathbb{R}^n)*\mathcal{D}'(\mathbb{R}^n)$, in Theorem 4.18 we have seen that $\mathcal{S}(\mathbb{R}^n)*\mathcal{S}'(\mathbb{R}^n)$ continues to be meaningfully defined in $\mathcal{S}'(\mathbb{R}^n)$. For us, extending the action of the convolution product in this manner is motivated by the discussion in Sect. 4.9, indicating how SIO may be interpreted as multipliers.

The main result in Sect. 4.7, Theorem 4.76, appears to be new at least in the formulation and the degree of generality in which it has been presented. This result may be regarded as a far-reaching generalization of Sokhotsky's formula (2.10.3); cf. Remark 4.79 for details. Theorem 4.76 has a number of remarkable consequences, and we use it to offer a new perspective on the treatment of the classical harmonic Poisson kernel in Sect. 4.8. Later on, in Sect. 11.6, Theorem 4.76 resurfaces as the key ingredient in the study of boundary behavior of layer potential operators in the upper-half space.

The treatment of the SIO from in Sect. 4.9 highlights the interplay between distribution theory, harmonic analysis, and partial differential equations. Specifically, first the action of a SIO of the form T_{Θ} on a Schwartz function φ is interpreted as $\big(\text{P.V.}\,\Theta\big)*\varphi$ (which, as pointed out earlier, is a well-defined object in $\mathcal{S}(\mathbb{R}^n)*\mathcal{S}'(\mathbb{R}^n)\subset\mathcal{S}'(\mathbb{R}^n)$). Second, the Fourier analysis of tempered distributions is invoked to conclude

that $\widehat{T_\Theta\varphi} = \widehat{\varphi}\,\mathcal{F}\big(\text{P.V.}\,\Theta\big)$ which shifts the focus of understanding the properties of the SIO T_Θ to clarifying the nature of the tempered distribution $m_\Theta := \mathcal{F}\big(\text{P.V.}\,\Theta\big)$. Third, it turns out that the distribution m_Θ is of function type and, in fact, a suitable pointwise formula may be deduced for it. In particular, it is apparent from this formula that $m_\Theta \in L^\infty(\mathbb{R}^n)$. Having established this, in a fourth step we may return to the original focus of our investigation, namely the SIO T_Θ, and use the fact that $\widehat{T_\Theta\varphi} = m_\Theta\widehat{\varphi}$, along with Planchérel's formula and the boundedness of m_Θ to eventually conclude (via a density argument) that T_Θ extends to a linear and bounded operator on $L^2(\mathbb{R}^n)$. In turn, SIO naturally intervene in the derivatives of volume potentials discussed in Sect. 4.10, and the boundedness result just derived ultimately becomes the key tool in obtaining estimates for the solution of the Poisson problem, treated later.

The discussion in the previous paragraph also sheds light on some of the main aims of the theory of SIO, as a subbranch of harmonic analysis. For example, one would like to extend the action of SIO to $L^p(\mathbb{R}^n)$ for any $p \in (1,\infty)$, rather than just $L^2(\mathbb{R}^n)$. Also, it is desirable to consider SIO which are not necessarily of convolution type, and/or in a setting in which $\mathbb{R}^n$ is replaced by a more general type of ambient (e.g., some type of surface in $\mathbb{R}^{n+1}$). The execution of this program, essentially originating in the work of A. P. Calderón, A. Zygmund, and S. G. Mikhlin in the 1950s, stretches all the way to the present day. The reader is referred to the exposition in the monographs [6, 7, 16, 23, 47, 62, 63], where references to specific research articles may be found. Here, we only wish to mention that, in turn, such progress in harmonic analysis has led to significant advances in those areas of partial differential equations where SIO play an important role. In this regard, the reader is referred to the discussion in [28, 36, 47, 50].

4.11 Additional Exercises for Chap. 4

Exercise 4.103. *Prove that* $\text{P.V.}\,\frac{1}{x} \in \mathcal{S}'(\mathbb{R})$.

Exercise 4.104. *Prove that* $|x|^N \ln|x| \in \mathcal{S}'(\mathbb{R}^n)$ *if* N *is a real number satisfying* $N > -n$.

Exercise 4.105. *Prove that* $(\ln|x|)' = \text{P.V.}\,\frac{1}{x}$ *in* $\mathcal{S}'(\mathbb{R})$.

Exercise 4.106. *Let* $a \in (0,\infty)$ *and recall the Heaviside function* H *from* (1.2.16). *Prove that the function* $\mathrm{e}^{ax}H(x)$, $x \in \mathbb{R}$, *does not belong to* $\mathcal{S}'(\mathbb{R})$. *Do any of the functions* $\mathrm{e}^{-ax}H(-x)$, $\mathrm{e}^{ax}H(-x)$, *or* $\mathrm{e}^{-ax}H(x)$, *defined for* $x \in \mathbb{R}$, *belong to* $\mathcal{S}'(\mathbb{R})$?

Exercise 4.107. *In each case below determine if the given sequence of tempered distributions indexed over* $j \in \mathbb{N}$ *converges in* $\mathcal{S}'(\mathbb{R})$. *Whenever convergent, determine its limit.*

(a) $f_j(x) = \frac{x}{x^2+j^{-2}}$, $x \in \mathbb{R}$

(b) $f_j(x) = \frac{1}{j}\cdot\frac{1}{x^2+j^{-2}}$, $x \in \mathbb{R}$

(c) $f_j(x) = \frac{1}{\pi}\frac{\sin jx}{x}$, $x \in \mathbb{R} \setminus \{0\}$

(d) $f_j(x) = \mathrm{e}^{j^2}\delta_j$

Exercise 4.108. *For each $j \in \mathbb{N}$ let $f_j(x) := j^n\theta(jx)$, $x \in \mathbb{R}^n$, where $\theta \in \mathcal{S}(\mathbb{R}^n)$ is fixed and satisfies $\int_{\mathbb{R}^n}\theta(x)\,\mathrm{d}x = 1$. Does the sequence $\{f_j\}_{j\in\mathbb{N}}$ converge in $\mathcal{S}'(\mathbb{R}^n)$? If yes, determine its limit.*

Exercise 4.109. *For each $j \in \mathbb{N}$ consider the function $f_j(x) := \mathrm{e}^{x^2}\chi_{[-j,j]}(x)$ defined for $x \in \mathbb{R}$. Prove that the sequence $\{f_j\}_{j\in\mathbb{N}}$ converges in $\mathcal{D}'(\mathbb{R})$ but not in $\mathcal{S}'(\mathbb{R})$.*

Exercise 4.110. *For each $j \in \mathbb{N}$ let $f_j(x) := \chi_{[j-1,j]}(x)$ for $x \in \mathbb{R}$. Prove that the sequence $\{f_j\}_{j\in\mathbb{N}}$ does not converge in $L^p(\mathbb{R})$ for any $p \geq 1$ but it converges to zero in $\mathcal{S}'(\mathbb{R})$.*

Exercise 4.111. *For each $j \in \mathbb{N}$ consider the tempered distribution $u_j \in \mathcal{S}'(\mathbb{R})$ defined by $u_j := \chi_{[-j,j]}\mathrm{e}^x\sin(\mathrm{e}^x)$. Prove that $u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} \mathrm{e}^x\sin(\mathrm{e}^x)$. Does the sequence $\{u_j\}_{j\in\mathbb{N}}$ converge in $\mathcal{S}'(\mathbb{R})$? If yes, what is the limit?*

Exercise 4.112. *Let $m \in \mathbb{N}$. Prove that any solution u of the equation $x^m u = 0$ in $\mathcal{D}'(\mathbb{R})$ satisfies $u \in \mathcal{S}'(\mathbb{R})$. Use the Fourier transform to show that the general solution to this equation is $u = \sum_{k=0}^{m-1} c_k\delta^{(k)}$ in $\mathcal{S}'(\mathbb{R})$, where $c_k \in \mathbb{C}$, for every $k = 0, 1, \ldots, m-1$.*

Exercise 4.113. *Prove that for any $f \in \mathcal{S}'(\mathbb{R})$ there exists $u \in \mathcal{S}'(\mathbb{R})$ such that $xu = f$ in $\mathcal{S}'(\mathbb{R})$.*

Exercise 4.114. *Does the equation $\mathrm{e}^{-|x|^2}u = 1$ in $\mathcal{S}'(\mathbb{R}^n)$ have a solution?*

Exercise 4.115. *Prove that the Heaviside function H belongs to $\mathcal{S}'(\mathbb{R})$ and compute its Fourier transform in $\mathcal{S}'(\mathbb{R})$.*

Exercise 4.116. *Compute the Fourier transform of* $\mathrm{P.V.}\,\frac{1}{x}$ *in $\mathcal{S}'(\mathbb{R})$.*

Exercise 4.117. *Compute the Fourier transform in $\mathcal{S}'(\mathbb{R})$ of each of the following tempered distribution (all are considered in $\mathcal{S}'(\mathbb{R})$ and recall the definition of the* sgn *function from (1.4.3)):*

(a) $\operatorname{sgn} x$.

(b) $|x|^k$ *for* $k \in \mathbb{N}$.

(c) $\frac{\sin(ax)}{x}$ *for* $a \in \mathbb{R} \setminus \{0\}$.

(d) $\frac{\sin(ax)\sin(bx)}{x}$ *for* $a, b \in \mathbb{R} \setminus \{0\}$.

(e) $\sin(x^2)$.

(f) $\ln|x|$.

Exercise 4.118. *Let $n = 3$ and $R \in (0, \infty)$. Compute the Fourier transform of the tempered distribution $\delta_{\partial B(0,R)}$ in $\mathcal{S}'(\mathbb{R}^3)$, where $\delta_{\partial B(0,R)}$ the distribution defined as in Exercise 2.128 corresponding to $\Sigma := \partial B(0, R)$.*

Exercise 4.119. *Let $k \in \mathbb{N}_0$ and suppose $f \in L^1(\mathbb{R}^n)$ has the property that $x^\alpha f \in L^1(\mathbb{R}^n)$ for every $\alpha \in \mathbb{N}_0^n$ satisfying $|\alpha| \leq k$. Prove that $\widehat{f}$, the Fourier transform of f in $\mathcal{S}'(\mathbb{R}^n)$ satisfies $\widehat{f} \in C^k(\mathbb{R}^n)$.*

Exercise 4.120. *Show that $\chi_{[-1,1]} \in \mathcal{S}'(\mathbb{R})$ and compute $\widehat{\chi_{[-1,1]}}$ in $\mathcal{S}'(\mathbb{R})$. Do we have $\widehat{\chi_{[-1,1]}} \in L^1(\mathbb{R})$?*

Exercise 4.121. *Fix $x_0 \in \mathbb{R}^n$ and set*

$$\begin{aligned} t_{x_0} &: \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n) \\ \langle t_{x_0} u, \varphi \rangle &:= \langle u, t_{-x_0}(\varphi) \rangle, \quad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n), \end{aligned} \tag{4.11.1}$$

where $t_{-x_0}(\varphi)$ is understood as in Exercise 3.42. Prove that the map in (4.11.1) is well-defined, linear, and continuous, and is an extension of the map from Exercise 3.42 in the sense that, if $f \in \mathcal{S}(\mathbb{R}^n)$, then $t_{x_0} u_f = u_{t_{x_0}(f)}$ in $\mathcal{S}'(\mathbb{R}^n)$. Also show that

$$\mathcal{F}\big(t_{x_0} u\big) = \mathrm{e}^{-ix_0 \cdot (\cdot)} \widehat{u} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n) \quad \text{for every } u \in \mathcal{S}'(\mathbb{R}^n). \tag{4.11.2}$$

Exercise 4.122. *Let $a \in \mathbb{R}$ and consider the function $g(x) := \cos(ax)$ for every $x \in \mathbb{R}$. Prove that $g \in \mathcal{S}'(\mathbb{R})$ and compute $\widehat{g}$ in $\mathcal{S}'(\mathbb{R})$.*

Exercise 4.123. *Let $m \in \mathbb{N}$ and suppose P is a polynomial in $\mathbb{R}^n$ of degree $2m$ that has no real roots. Prove that $\frac{1}{P} \in \mathcal{S}'(\mathbb{R}^n)$, and that if $2m > n$ then $\mathcal{F}\big(\frac{1}{P}\big) \in C^{2m-n-1}(\mathbb{R}^n)$.*

Exercise 4.124. *Let P be a polynomial in $\mathbb{R}^n$ and suppose $k \in \mathbb{N}$. Prove that if P is homogeneous of degree $-k$ then $P \equiv 0$.*

Exercise 4.125. *Let $\zeta, \eta \in \mathbb{R}^n$, $n \geq 2$, be two unit vectors such that $\zeta \cdot \eta \neq 1$, and consider the linear mapping $R : \mathbb{R}^n \to \mathbb{R}^n$ define by*

$$R\xi := \xi - \Big[\frac{\xi \cdot (\eta + \zeta)}{1 + \eta \cdot \zeta}\Big]\zeta + \Big[\frac{\xi \cdot [(1 + 2\eta \cdot \zeta)\zeta - \eta]}{1 + \eta \cdot \zeta}\Big]\eta, \qquad \forall\, \xi \in \mathbb{R}^n. \tag{4.11.3}$$

Show that this is an orthogonal transformation in $\mathbb{R}^n$ that satisfies

$$R\zeta = \eta \quad \text{and} \quad R^\top \eta = \zeta. \tag{4.11.4}$$

Exercise 4.126. *Prove that for every $c \in \mathbb{R} \setminus \{0\}$ the following formulas hold*

$$\lim_{\substack{\varepsilon \to 0^+ \\ R \to \infty}} \int_\varepsilon^R \frac{\cos(c\rho) - \cos\rho}{\rho} \, \mathrm{d}\rho = -\ln|c|, \tag{4.11.5}$$

$$\lim_{\substack{\varepsilon \to 0^+ \\ R \to \infty}} \int_\varepsilon^R \frac{\sin(c\rho)}{\rho} \, \mathrm{d}\rho = \frac{\pi}{2} \operatorname{sgn} c, \tag{4.11.6}$$

$$\sup_{0<\varepsilon<R} \left| \int_\varepsilon^R \frac{\cos(c\rho) - \cos\rho}{\rho} \, \mathrm{d}\rho \right| \leq 2\left|\ln|c|\right|, \tag{4.11.7}$$

$$\sup_{0<\varepsilon<R} \left| \int_\varepsilon^R \frac{\sin(c\rho)}{\rho} \, \mathrm{d}\rho \right| \leq 4. \tag{4.11.8}$$

Chapter 5

The Concept of a Fundamental Solution

In this chapter we consider constant coefficient linear differential operators and discuss the existence of a fundamental solution for such operators.

5.1 Constant Coefficient Linear Differential Operators

Recall (3.1.10). To fix notation, for $m \in \mathbb{N}_0$, let

$$P(D) := \sum_{|\alpha|\leq m} a_\alpha D^\alpha, \quad a_\alpha \in \mathbb{C}, \quad \forall\, \alpha \in \mathbb{N}_0^n, \quad |\alpha| \leq m. \tag{5.1.1}$$

Also, whenever $P(D)$ is as above, define $P(-D) := \sum_{|\alpha|\leq m} (-1)^{|\alpha|} a_\alpha D^\alpha$, and set

$$P(\xi) := \sum_{|\alpha|\leq m} a_\alpha \xi^\alpha, \quad \forall\, \xi \in \mathbb{R}^n. \tag{5.1.2}$$

Typically, an object $P(D)$ as in (5.1.1) is called a constant coefficient linear differential operator, while $P(\xi)$ from (5.1.2) is referred to as its total (or full) symbol. Moreover,

$$\begin{aligned} &P(D) \text{ is said to have order } m \text{ provided} \\ &\text{there exists } \alpha_* \in \mathbb{N}_0^n \text{ with } |\alpha_*| = m \text{ and } a_{\alpha_*} \neq 0. \end{aligned} \tag{5.1.3}$$

Whenever $P(D)$ is a constant coefficient linear differential operator of order m we define its `principal symbol` to be

$$P_m(\xi) := \sum_{|\alpha|=m} a_\alpha \xi^\alpha, \qquad \text{for} \quad \xi \in \mathbb{R}^n. \tag{5.1.4}$$

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Obviously, any operator of the form

$$Q(\partial) := \sum_{|\alpha|\leq m} b_\alpha \partial^\alpha, \tag{5.1.5}$$

may be expressed as $P(D)$ for the choice of coefficients $a_\alpha := i^{|\alpha|} b_\alpha$ for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq m$. Hence, we may move freely back and forth between the writings of a differential operator in (5.1.1) and (5.1.5). While we shall naturally define in this case $Q(\xi) := \sum_{|\alpha|\leq m} b_\alpha \xi^\alpha$, the reader is advised that the full symbol of $Q(\partial)$ from (5.1.5) is the expression $\sum_{|\alpha|\leq m} i^{|\alpha|} b_\alpha \xi^\alpha$, for $\xi \in \mathbb{R}^n$, while its principal symbol is $\sum_{|\alpha|=m} i^m b_\alpha \xi^\alpha$, for $\xi \in \mathbb{R}^n$.

The terminology and formalism described above will play a significant role in our subsequent work. In the last part of this section we prove a regularity result involving constant coefficient linear differential operators with nonvanishing total symbol outside the origin, that is further used to establish a rather general Liouville-type theorem for this class of operators.

Proposition 5.1. *Suppose $P(D)$ is a constant coefficient linear differential operator in $\mathbb{R}^n$ with the property that*

$$P(\xi) \neq 0 \quad \text{for every} \quad \xi \in \mathbb{R}^n \setminus \{0\}. \tag{5.1.6}$$

Then if $u \in \mathcal{S}'(\mathbb{R}^n)$ is such that $P(D)u = 0$ in $\mathcal{S}'(\mathbb{R}^n)$ it follows that u is a polynomial in $\mathbb{R}^n$ and $P(D)u = 0$ pointwise in $\mathbb{R}^n$.

Proof. If $u \in \mathcal{S}'(\mathbb{R}^n)$ is such that $P(D)u = 0$ in $\mathcal{S}'(\mathbb{R}^n)$, taking the Fourier transform, we obtain $P(\xi)\widehat{u} = 0$ in $\mathcal{S}'(\mathbb{R}^n)$. In view of (5.1.6), the latter identity implies $\operatorname{supp} \widehat{u} \subseteq \{0\}$. By Exercise 4.35, it follows that u is a polynomial in $\mathbb{R}^n$. Since $u \in C^\infty(\mathbb{R}^n)$, the condition $P(D)u = 0$ in $\mathcal{S}'(\mathbb{R}^n)$ further yields $P(D)u = 0$ pointwise in $\mathbb{R}^n$. □

In preparation for the general Liouville-type theorem advertised earlier, we prove the following result relating the growth of a polynomial to its degree.

Lemma 5.2. *Let $P(x)$ be a polynomial in $\mathbb{R}^n$ with the property that there exist $N \in \mathbb{N}_0$ and $C, R \in [0, \infty)$ such that*

$$|P(x)| \leq C|x|^N \quad \text{whenever} \quad |x| \geq R. \tag{5.1.7}$$

Then $P(x)$ has degree at most N. In particular, the only bounded polynomials in $\mathbb{R}^n$ are constants.

Proof. We begin by noting that the case $n = 1$ is easily handled. In the multidimensional setting, assume that $P(x) = \sum_{|\alpha|\leq m} a_\alpha x^\alpha$ is a polynomial in $\mathbb{R}^n$ satisfying (5.1.7) for some $N \in \mathbb{N}_0$ and $C, R \in [0, \infty)$. Suppose $m \geq N + 1$,

since otherwise there is nothing to prove. Fix an arbitrary $\omega \in S^{n-1}$ and consider the one-dimensional polynomial $p_\omega(t) := P(t\omega)$ for $t \in \mathbb{R}$, that is,

$$p_\omega(t) = \sum_{j=0}^{m} \Big(\sum_{|\alpha|=j} a_\alpha \omega^\alpha \Big) t^j, \qquad t \in \mathbb{R}. \tag{5.1.8}$$

Given that $|p_\omega(t)| = |P(t\omega)| \leq C|t\omega|^N = Ct^N$ whenever $|t| \geq R$, the one-dimensional result applies and gives that the coefficient of t^j in (5.1.8) vanishes if $j \geq N+1$. Thus,

$$P(t\omega) = p_\omega(t) = \sum_{j=0}^{N} \Big(\sum_{|\alpha|=j} a_\alpha \omega^\alpha \Big) t^j = \sum_{|\alpha|\leq N} a_\alpha (t\omega)^\alpha, \qquad \forall\, t \in \mathbb{R}. \tag{5.1.9}$$

Given that $\omega \in S^{n-1}$ was arbitrary, it follows that $P(x) = \sum_{|\alpha|\leq N} a_\alpha x^\alpha$ for every $x \in \mathbb{R}^n$. □

All ingredients are now in place for dealing with the following result.

Theorem 5.3 (A General Liouville-Type Theorem)**.** *Assume $P(D)$ is a constant coefficient linear differential operator in $\mathbb{R}^n$ such that (5.1.6) holds. Also, suppose $u \in L^1_{loc}(\mathbb{R}^n)$ satisfies $P(D)u = 0$ in $\mathcal{S}'(\mathbb{R}^n)$ and has the property that there exist $N \in \mathbb{N}_0$ and $C, R \in [0, \infty)$ such that*

$$|u(x)| \leq C|x|^N \quad \text{whenever} \quad |x| \geq R. \tag{5.1.10}$$

Then u is a polynomial in $\mathbb{R}^n$ of degree at most N.

In particular, if $u \in L^\infty(\mathbb{R}^n)$ satisfies $P(D)u = 0$ in $\mathcal{S}'(\mathbb{R}^n)$ then u is necessarily a constant.

Proof. Since (5.1.10) implies that the locally integrable function u belongs to $\mathcal{S}'(\mathbb{R}^n)$ (cf. Example 4.3), Proposition 5.1 implies that u is a polynomial in $\mathbb{R}^n$. Moreover, Lemma 5.2 gives that the degree of this polynomial is at most N. □

5.2 A First Look at Fundamental Solutions

Definition 5.4. *Let $P(D)$ be a constant coefficient linear differential operator in $\mathbb{R}^n$. Then $E \in \mathcal{D}'(\mathbb{R}^n)$ is called a* `fundamental solution` *for $P(D)$ provided*

$$P(D)E = \delta \quad \text{in} \quad \mathcal{D}'(\mathbb{R}^n). \tag{5.2.1}$$

A simple, useful observation is that $E \in \mathcal{D}'(\mathbb{R}^n)$ is a fundamental solution for $P(D)$ if and only if

$$\langle E, P(-D)\varphi \rangle = \varphi(0), \qquad \forall\, \varphi \in C_0^\infty(\mathbb{R}^n). \tag{5.2.2}$$

Indeed, this is immediate from the fact that $\langle P(D)E, \varphi \rangle = \langle E, P(-D)\varphi \rangle$ for each function $\varphi \in C_0^\infty(\mathbb{R}^n)$.

Remark 5.5. *Suppose $P(D)$ is a constant coefficient linear differential operator in $\mathbb{R}^n$. Then the equation*

$$P(D)u = f \tag{5.2.3}$$

is called the `Poisson equation for the operator` $P(D)$. *If the operator $P(D)$ has a fundamental solution $E \in \mathcal{D}'(\mathbb{R}^n)$, then for any $f \in \mathcal{D}'(\mathbb{R}^n)$ for which $E*f$ exists as an element of $\mathcal{D}'(\mathbb{R}^n)$ (for example, this is the case if $f \in \mathcal{E}'(\mathbb{R}^n)$) the Poisson equation (5.2.3) is solvable in $\mathcal{D}'(\mathbb{R}^n)$. Indeed, if we set $u := E * f$, then*

$$P(D)u = (P(D)E) * f = \delta * f = f \quad \text{in} \quad \mathcal{D}'(\mathbb{R}^n). \tag{5.2.4}$$

This is one of the many reasons for which fundamental solutions are important when solving partial differential equations.

Remark 5.6. *Let $P(D)$ and $Q(D)$ be two constant coefficient linear differential operators in $\mathbb{R}^n$ and let $E \in \mathcal{D}'(\mathbb{R}^n)$ be a fundamental solution for the operator $Q(D)P(D)$ in $\mathcal{D}'(\mathbb{R}^n)$. Then $P(D)E$ is a fundamental solution for $Q(D)$ in $\mathcal{D}'(\mathbb{R}^n)$. Moreover, $P(D)E \in \mathcal{S}'(\mathbb{R}^n)$ if $E \in \mathcal{S}'(\mathbb{R}^n)$.*

In general, there are two types of questions we are going to be concerned with in this regard, namely whether fundamental solutions exist, and if so whether they can all be cataloged. The proposition below shows that for any differential operator with a nonvanishing total symbol on $\mathbb{R}^n \setminus \{0\}$ a fundamental solution which is a tempered distribution is unique up to polynomials.

Proposition 5.7. *Suppose $P(D)$ is a constant coefficient linear differential operator in $\mathbb{R}^n$ with the property that $P(\xi) \neq 0$ for every $\xi \in \mathbb{R}^n \setminus \{0\}$.*

If $P(D)$ has a fundamental solution E that is a tempered distribution, then any other fundamental solution of $P(D)$ belonging to $\mathcal{S}'(\mathbb{R}^n)$ differs from E by a polynomial Q satisfying $P(D)Q = 0$ pointwise in $\mathbb{R}^n$.

Proof. Suppose $u \in \mathcal{S}'(\mathbb{R}^n)$ is such that $P(D)u = \delta$ in $\mathcal{S}'(\mathbb{R}^n)$. Then $P(D)$(u-E)=0 in $\mathcal{S}'(\mathbb{R}^n)$ and the desired conclusion is provided by Proposition 5.1. □

Consider now the task of finding a formula for a fundamental solution. In a first stage, suppose $P(D)$ is as in (5.1.1) and make the additional assumption that

$$\text{there exists } c \in (0,\infty) \text{ such that } |P(\xi)| \geq c \text{ for every } \xi \in \mathbb{R}^n. \tag{5.2.5}$$

In such a scenario, $\frac{1}{P(\xi)} \in \mathcal{L}(\mathbb{R}^n)$, hence $\frac{1}{P(\xi)} \in \mathcal{S}'(\mathbb{R}^n)$ by Exercise 4.6. Thus, if some $E \in \mathcal{S}'(\mathbb{R}^n)$ happens to be a fundamental solution of $P(D)$ (we will see later in Theorem 5.13 that such an E always exists), then $P(D)E = \delta$ in $\mathcal{S}'(\mathbb{R}^n)$. After applying the Fourier transform, this implies $P(\xi)\widehat{E} = 1$ in $\mathcal{S}'(\mathbb{R}^n)$ (recall part *(b)* in Theorem 4.25 and Example 4.21). In particular, since $P(\xi)\,\frac{1}{P(\xi)} = 1$ in $\mathcal{S}'(\mathbb{R}^n)$, we obtain

$$P(\xi)\Big(\widehat{E}(\xi) - \frac{1}{P(\xi)}\Big) = 0 \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{5.2.6}$$

Multiplying by $\frac{1}{P(\xi)} \in \mathcal{L}(\mathbb{R}^n)$ then gives $\widehat{E}(\xi) = \frac{1}{P(\xi)}$ hence, further,

$$\begin{aligned}\langle E, \varphi\rangle &= (2\pi)^{-n}\Big\langle \widehat{\widehat{E}}^{\vee}, \varphi\Big\rangle = (2\pi)^{-n}\big\langle \widehat{E}, \widehat{\varphi}^{\vee}\big\rangle \\ &= (2\pi)^{-n}\int_{\mathbb{R}^n} \frac{\widehat{\varphi}(-\xi)}{P(\xi)}\,\mathrm{d}\xi, \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^n). \end{aligned} \tag{5.2.7}$$

Formula (5.2.7), derived under the assumption that (5.2.5) holds, is the departure point for the construction of a fundamental solution for a constant coefficient linear differential operator $P(D)$ in the more general case when (5.2.5) is no longer assumed. Of course, we would have to assume that $P(D) \neq 0$ (since otherwise (5.2.1) clearly does not have any solution).

The following comments are designed to shed some light into how the issues created by the zeros of $P(\xi)$ [in the context of (5.2.7)] may be circumvented when condition (5.2.5) is dropped. The first observation is that if $\varphi \in C_0^\infty(\mathbb{R}^n)$ we may extend its Fourier transform $\widehat{\varphi}$ to $\mathbb{C}^n$ by setting

$$\widehat{\varphi}(\zeta) := \int_{\mathbb{R}^n} \mathrm{e}^{-i\sum_{j=1}^n x_j\zeta_j}\varphi(x)\,\mathrm{d}x \quad \text{for every} \quad \zeta = (\zeta_1, \ldots, \zeta_n) \in \mathbb{C}^n. \tag{5.2.8}$$

Note that the compact support condition on φ is needed since for each fixed $\zeta \in \mathbb{C}^n \setminus \mathbb{R}^n$ the function $|\mathrm{e}^{-i\sum_{j=1}^n x_j\zeta_j}|$ grows rapidly as $|x| \to \infty$.

Second, if (5.2.5) holds then $P(\xi) \neq 0$ for each $\xi \in \mathbb{R}^n$. Consequently, for each $\xi \in \mathbb{R}^n$ there exists a small number $r(\xi) > 0$ such that $P(z) \neq 0$ whenever $z \in \mathbb{C}^n$ satisfies $|z - \xi| < 2r(\xi)$. Hence, for every $\eta \in S^{n-1}$ the mapping

$$\big\{\tau \in \mathbb{C} : |\tau| < 2r(\xi)\big\} \ni \tau \mapsto \frac{\widehat{\varphi}(-\xi - \tau\eta)}{P(\xi + \tau\eta)} \in \mathbb{C} \tag{5.2.9}$$

is well-defined and analytic. Using Cauchy's formula [and keeping in mind the convention (5.2.8)] we may therefore re-write (5.2.7) in the form

$$\begin{aligned}\langle E, \varphi\rangle &= (2\pi)^{-n}\int_{\mathbb{R}^n} \frac{\widehat{\varphi}(-\xi)}{P(\xi)}\mathrm{d}\xi \\ &= (2\pi)^{-n}\int_{\mathbb{R}^n}\Big[\frac{1}{2\pi i}\int_{|\tau|=r(\xi)} \frac{\widehat{\varphi}(-\xi-\tau\eta)}{P(\xi+\tau\eta)}\cdot\frac{\mathrm{d}\tau}{\tau}\Big]\,\mathrm{d}\xi, \qquad \forall\,\varphi \in C_0^\infty(\mathbb{R}^n).\end{aligned} \tag{5.2.10}$$

The advantage of the expression in the last line of (5.2.10), compared to the expression in the last line of (5.2.7), is the added flexibility in the choice of the set over which $\frac{1}{P}$ is integrated. This discussion suggests that we try to construct a fundamental solution for $P(D)$ in a manner similar to the expression in the last line of (5.2.10), with the integration taking place on a set avoiding the zeros of P, even in the case when (5.2.5) is dropped.

We shall implement this strategy in Sect. 5.3. For the time being, we discuss a preliminary result that will play a basic role in this endeavor. To state it recall the notion of principal symbol from (5.1.4).

Lemma 5.8. *Let $m \in \mathbb{N}$ and let $P(D)$ be a differential operator of order m as in (5.1.1). In addition, suppose that $P_m(\eta) \neq 0$ for some $\eta \in S^{n-1}$. Then there exist $\varepsilon > 0$, r_0, $r_1, \ldots, r_m > 0$ and closed sets F_0, $F_1, \ldots, F_m$ in $\mathbb{R}^n$ such that*

(a) $\mathbb{R}^n = \bigcup_{j=0}^{m} F_j$ *and*

(b) If $\xi \in F_j$, $\tau \in \mathbb{C}$, $|\tau| = r_j$, then $|P(\xi + \tau\eta)| \geq \varepsilon$, for each $j = 0, 1, \ldots, m$.

Proof. Choose $r_j := \frac{j+1}{m}$, for $j \in \{0, 1, \ldots, m\}$ and set $\varepsilon := (2m)^{-m}|P_m(\eta)| > 0$. Also, for each $j \in \{1, \ldots, m\}$ set

$$F_j := \big\{\xi \in \mathbb{R}^n : |P(\xi + \tau\eta)| \geq \varepsilon, \ \forall\, \tau \in \mathbb{C}, \ |\tau| = r_j\big\}. \tag{5.2.11}$$

These sets are closed in $\mathbb{R}^n$ and clearly satisfy *(b)*. There remains to show that *(a)* holds.

Fix $\xi \in \mathbb{R}^n$, and consider the mapping $\mathbb{C} \ni \tau \mapsto P(\xi + \tau\eta) \in \mathbb{C}$. This is a polynomial in τ of order m with the coefficient of τ^m equal to $P_m(\eta)$. Let $\tau_j(\xi)$, $j = 1, \ldots, m$, be the zeros of this polynomial. Then

$$P(\xi + \tau\eta) = P_m(\eta) \prod_{j=1}^{m} \big(\tau - \tau_j(\xi)\big), \qquad \forall\, \tau \in \mathbb{C}. \tag{5.2.12}$$

Consider

$$M_j := \Big\{\tau \in \mathbb{C} : \operatorname{dist}\big(\tau, \partial B(0, r_j)\big) < \frac{1}{2m}\Big\}, \quad \text{for } \ j \in \{0, 1, \ldots, m\}. \tag{5.2.13}$$

Given that $r_j - r_{j-1} = 1/m$ for $j = 1, \ldots, m$, the sets in the collection $\{M_j\}_{j=0}^{m}$ are pairwise disjoint. Consequently, among these $m+1$ sets there is at least one that does not contain any of the numbers $\tau_j(\xi)$, $j = 1, \ldots, m$ (a simple form of the pigeon hole principle). Hence, there exists $k \in \{0, 1, \ldots, m\}$ such that

$$\operatorname{dist}\big(\tau_j(\xi), \partial B(0, r_k)\big) \geq \frac{1}{2m}, \qquad \forall\, j \in \{1, \ldots, m\}. \tag{5.2.14}$$

Then, $|\tau_j(\xi) - \tau| \geq \frac{1}{2m}$, for every $|\tau| = r_k$ and every $j = 1, \ldots, m$, which when used in (5.2.12) implies

$$|P(\xi + \tau\eta)| = \Big|P_m(\eta) \prod_{j=1}^{m} \big(\tau - \tau_j(\xi)\big)\Big| \geq \frac{|P_m(\eta)|}{(2m)^m} = \varepsilon, \qquad \text{if} \quad |\tau| = r_k. \tag{5.2.15}$$

This shows that $\xi \in F_k$, thus *(a)* is satisfied for the sets defined in (5.2.11). □

5.3 The Malgrange–Ehrenpreis Theorem

The Malgrange–Ehrenpreis theorem reads as follows:

Theorem 5.9 (Malgrange–Ehrenpreis)**.** *Let $P(D)$ be a linear constant coefficient differential operator that is not identically zero. Then there exists $E \in \mathcal{D}'(\mathbb{R}^n)$ which is a fundamental solution for $P(D)$.*

Proof. Let $m \in \mathbb{N}_0$ and assume that $P(D)$ is as in (5.1.1) of order m. If $m = 0$ then $P(D) = a \in \mathbb{C}\setminus\{0\}$ and the distribution $E := \frac{1}{a}\delta$ is a fundamental solution for $P(D)$. Suppose now that $m \geq 1$. Since the principal symbol P_m does not vanish identically in $\mathbb{R}^n \setminus \{0\}$ and since

$$P_m(\xi) = |\xi|^m P_m\Big(\frac{\xi}{|\xi|}\Big) \quad \text{for all} \quad \xi \in \mathbb{R}^n \setminus \{0\}, \tag{5.3.1}$$

we conclude that there exists $\eta \in S^{n-1}$ such that $P_m(\eta) \neq 0$. Fix such an η and retain the notation introduced in the proof of Lemma 5.8. Then the functions

$$\chi_j : \mathbb{R}^n \to \mathbb{R}, \quad \chi_j(\xi) := \begin{cases} 1, & \xi \in F_j, \\ 0, & \xi \in \mathbb{R}^n \setminus F_j, \end{cases} \qquad j \in \{0, 1, \ldots, m\}, \tag{5.3.2}$$

and

$$f_j : \mathbb{R}^n \to \mathbb{R}, \quad f_j := \frac{\chi_j}{\sum\limits_{k=0}^{m} \chi_k}, \qquad j \in \{0, 1, \ldots, m\}, \tag{5.3.3}$$

are well-defined, measurable (recall that the F_j's are closed), and $\sum\limits_{j=0}^{m} f_j = 1$ on $\mathbb{R}^n$. In addition, $0 \leq f_j \leq 1$, and $f_j = 0$ on $\mathbb{R}^n \setminus F_j$ for each $j \in \{0, 1, \ldots, m\}$.

Define now the linear functional $E : \mathcal{D}(\mathbb{R}^n) \to \mathbb{C}$ by setting, for each function $\varphi \in C_0^\infty(\mathbb{R}^n)$,

$$E(\varphi) := (2\pi)^{-n} \sum_{j=0}^{m} \int_{\mathbb{R}^n} f_j(\xi) \Big[\frac{1}{2\pi i} \int_{|\tau|=r_j} \frac{\widehat{\varphi}(-\xi - \tau\eta)}{P(\xi + \tau\eta)} \cdot \frac{\mathrm{d}\tau}{\tau}\Big] \mathrm{d}\xi. \tag{5.3.4}$$

In order to prove that E is well-defined, fix an arbitrary compact set $K \subset \mathbb{R}^n$ and suppose $\varphi \in C_0^\infty(\mathbb{R}^n)$ is such that $\operatorname{supp}\varphi \subseteq K$. First, observe that for each $j = 0, 1, \ldots, m$, if $\xi \in F_j = \operatorname{supp} f_j$, then $|P(\xi + \tau\eta)| \geq \varepsilon$ for every $\tau \in \mathbb{C}$, $|\tau| = r_j$, which makes the integral in (5.3.4) over $|\tau| = r_j$ finite. Second, we claim that the integrals over $\mathbb{R}^n$ in (5.3.4) are convergent. Indeed, since φ is compactly supported and

$$\widehat{\varphi}(-\xi - \tau\eta) = \int_{\mathbb{R}^n} \mathrm{e}^{ix\cdot\xi + i\tau x\cdot\eta} \varphi(x)\, \mathrm{d}x, \tag{5.3.5}$$

it follows that $\widehat{\varphi}(-\xi - \tau\eta)$ is analytic in τ. Then, if we fix $j \in \{0, 1, \ldots, m\}$, and take $N \in \mathbb{N}_0$, using (5.3.5) and integration by parts, for each $|\tau| = r_j$ we may write

$$(1+|\xi|^2)^N\widehat{\varphi}(-\xi-\tau\eta)=\int_{\mathbb{R}^n}\Big[\Big(1-\sum_{\ell=1}^n\partial_{x_\ell}^2\Big)^N\mathrm{e}^{ix\cdot\xi}\Big]\mathrm{e}^{i\tau x\cdot\eta}\varphi(x)\,\mathrm{d}x$$

$$=\int_{\mathbb{R}^n}\mathrm{e}^{ix\cdot\xi}\Big(-1-\sum_{\ell=1}^n\partial_{x_\ell}^2\Big)^N\big[\mathrm{e}^{i\tau x\cdot\eta}\varphi(x)\big]\,\mathrm{d}x, \tag{5.3.6}$$

for each $\xi\in\mathbb{R}^n$. An inspection of the last integral in (5.3.6) then yields

$$|\widehat{\varphi}(-\xi-\tau\eta)|\leq C_j(1+|\xi|^2)^{-N}\sup_{\substack{x\in K\\|\alpha|\leq 2N}}\big|\partial_x^\alpha\varphi(x)\big|\qquad\text{if }\ |\tau|=r_j, \tag{5.3.7}$$

for some constant $C_j=C_j(K,r_j,\eta)\in(0,\infty)$. In turn, we may use (5.3.7) in (5.3.4) to obtain

$$|E(\varphi)|\leq(2\pi)^{-n}\sum_{j=0}^m C_j\int_{\mathbb{R}^n}f_j(\xi)\frac{1}{2\pi}\cdot\frac{2\pi r_j}{\varepsilon r_j}(1+|\xi|^2)^{-N}\sup_{\substack{x\in K\\|\alpha|\leq 2N}}\big|\partial^\alpha\varphi(x)\big|\,\mathrm{d}\xi$$

$$\leq C\sup_{\substack{x\in K\\|\alpha|\leq 2N}}\big|\partial^\alpha\varphi(x)\big|\int_{\mathbb{R}^n}(1+|\xi|^2)^{-N}\,\mathrm{d}\xi$$

$$=C\sup_{\substack{x\in K\\|\alpha|\leq 2N}}\big|\partial^\alpha\varphi(x)\big| \tag{5.3.8}$$

for some $C\in(0,\infty)$ independent of φ, by choosing $N>\frac{n}{2}$. This proves that E is well-defined.

Clearly, E is linear which, together with (5.3.8) and Proposition 2.4, implies that $E\in\mathcal{D}'(\mathbb{R}^n)$. To finish the proof of the theorem we are left with showing that $P(D)E=\delta$ in $\mathcal{D}'(\mathbb{R}^n)$, that is, that (5.2.2) holds. Since $\mathcal{F}\big(P(-D)\varphi\big)=P^\vee\widehat{\varphi}$, recalling (5.3.4) we may write

$$\langle E,P(-D)\varphi\rangle=(2\pi)^{-n}\sum_{j=0}^m\int_{\mathbb{R}^n}f_j(\xi)\left[\frac{1}{2\pi i}\int_{|\tau|=r_j}\frac{\widehat{\varphi}(-\xi-\tau\eta)}{\tau}\,\mathrm{d}\tau\right]\mathrm{d}\xi$$

$$=(2\pi)^{-n}\sum_{j=0}^m\int_{\mathbb{R}^n}f_j(\xi)\widehat{\varphi}(-\xi)\,\mathrm{d}\xi$$

$$=(2\pi)^{-n}\int_{\mathbb{R}^n}\widehat{\varphi}(\xi)\,\mathrm{d}\xi$$

$$=\varphi(0). \tag{5.3.9}$$

For the second equality in (5.3.9) we used the fact that $\widehat{\varphi}(-\xi-\tau\eta)$ is analytic in τ, hence by Cauchy's formula we have $\frac{1}{2\pi i}\int_{|\tau|=r_j}\frac{\widehat{\varphi}(-\xi-\tau\eta)}{\tau}\mathrm{d}\tau=\widehat{\varphi}(-\xi)$. The third equality relies on $\sum_{j=0}^m f_j=1$ on $\mathbb{R}^n$, while the last one follows from part *(4)* in Exercise 3.26. On account of (5.3.9) and (5.2.2) we may therefore conclude that E is a fundamental solution for $P(D)$. □

One of the consequences of Theorem 5.9 is the fact that the Poisson problem for nonzero constant coefficient linear differential operators is locally solvable. Specifically, we have the following result.

Corollary 5.10. *Let $P(D)$ be a constant coefficient linear differential operator that is not identically zero. Suppose Ω is a nonempty, open subset of $\mathbb{R}^n$ and that some $f \in \mathcal{D}'(\Omega)$ is given. Then for each nonempty, open subset ω of Ω with the property that $\overline{\omega}$ is a compact subset of Ω, there exists $u \in \mathcal{D}'(\Omega)$ such that $P(D)u = f$ in $\mathcal{D}'(\omega)$.*

Proof. Let $E \in \mathcal{D}'(\mathbb{R}^n)$ be a fundamental solution of $P(D)$, which is known to exist by Theorem 5.9. Also, fix $\psi \in C_0^\infty(\Omega)$ such that $\psi = 1$ on ω. Then $\psi f \in \mathcal{E}'(\Omega)$ hence, by Proposition 2.63, it has a unique extension $\widetilde{\psi f} \in \mathcal{E}'(\mathbb{R}^n)$ and $\operatorname{supp}(\widetilde{\psi f}) \subseteq \operatorname{supp}\psi$. Now let $v := E * (\widetilde{\psi f}) \in \mathcal{D}'(\mathbb{R}^n)$. By Proposition 2.43 we have $v\big|_\Omega \in \mathcal{D}'(\Omega)$ and we claim that $u := v\big|_\Omega$ is a solution of $P(D)u = f$ in $\mathcal{D}'(\omega)$ (in the sense that $P(D)\big(u\big|_\omega\big) = f\big|_\omega$ in $\mathcal{D}'(\omega)$). Indeed,

$$\begin{aligned} P(D)\big(u\big|_\omega\big) &= \big[P(D)u\big]\big|_\omega = \big[P(D)(v\big|_\Omega)\big]\big|_\omega = \big[P(D)(E * \widetilde{\psi f})\big]\big|_\omega \\ &= \Big[\big(P(D)E\big) * \widetilde{\psi f}\Big]\Big|_\omega = \big[\delta * \widetilde{\psi f}\big]\big|_\omega \\ &= [\widetilde{\psi f}]\big|_\omega = [\psi f]\big|_\omega = f\big|_\omega \qquad \text{in } \mathcal{D}'(\omega). \end{aligned} \tag{5.3.10}$$

For the first and third equality in (5.3.10) we used (2.5.4), for the fourth equality we used *(e)* in Theorem 2.87, while for the sixth equality we used *(d)* in Theorem 2.87. □

The next example gives an approach for computing fundamental solutions for linear constant coefficient operators on the real line.

Example 5.11. Let $m \in \mathbb{N}$ and let $P := \big(\frac{\mathrm{d}}{\mathrm{d}x}\big)^m + \sum_{j=0}^{m-1} a_j \big(\frac{\mathrm{d}}{\mathrm{d}x}\big)^j$ be a differential operator of order m in $\mathbb{R}$ with constant coefficients. Suppose that v is the solution to

$$\begin{cases} v \in C^\infty(\mathbb{R}), \\ Pv = 0 \quad \text{in } \mathbb{R}, \\ v^{(j)}(0) = 0, \qquad 0 \leq j \leq m-2, \\ v^{(m-1)}(0) = 1, \end{cases} \tag{5.3.11}$$

with the understanding that the third condition is void if $m = 1$. Recall the Heaviside function H from (1.2.16) and define the distributions

$$u_+ := vH \quad \text{and} \quad u_- := -vH^\vee \quad \text{in} \quad \mathcal{D}'(\mathbb{R}). \tag{5.3.12}$$

We claim that u_+ and u_- are fundamental solutions for P in $\mathcal{D}'(\mathbb{R})$. To see that this is the case, express P as $P(D)=i^m\big(\frac{1}{i}\frac{\mathrm{d}}{\mathrm{d}x}\big)^m+\sum\limits_{j=0}^{m-1}i^ja_j\big(\frac{1}{i}\frac{\mathrm{d}}{\mathrm{d}x}\big)^j$, in line with (5.1.1), and note that this forces

$$\begin{aligned}P(-D)&=(-i)^m\left(\frac{1}{i}\frac{\mathrm{d}}{\mathrm{d}x}\right)^m+\sum_{j=0}^{m-1}(-i)^ja_j\left(\frac{1}{i}\frac{\mathrm{d}}{\mathrm{d}x}\right)^j\\&=(-1)^m\left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^m+\sum_{j=0}^{m-1}(-1)^ja_j\left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^j.\end{aligned}\tag{5.3.13}$$

As such, for every $\varphi\in C_0^\infty(\mathbb{R})$ we have

$$\begin{aligned}\big\langle P(D)u_+,\varphi\big\rangle&=\big\langle u_+,P(-D)\varphi\big\rangle=\int_0^\infty v(x)\big(P(-D)\varphi\big)(x)\,\mathrm{d}x\\&=v^{(m-1)}(0)\varphi(0)+\int_0^\infty(Pv)(x)\varphi(x)\,\mathrm{d}x=\langle\delta,\varphi\rangle,\end{aligned}\tag{5.3.14}$$

where the third equality uses (5.3.11) and repeated integrations by parts. This proves that $Pu_+=\delta$ in $\mathcal{D}'(\mathbb{R})$. Similarly, one obtains that $Pu_-=\delta$ in $\mathcal{D}'(\mathbb{R})$. It is worth pointing out that the two fundamental solutions just described satisfy the support conditions $\operatorname{supp}u_+\subseteq[0,\infty)$ and $\operatorname{supp}u_-\subseteq(-\infty,0]$.

Example 5.12. Let $c\in\mathbb{C}$ and consider the differential operator $P:=\frac{\mathrm{d}}{\mathrm{d}x}+c$ in $\mathbb{R}$. It is easy to see that $v(x)=\mathrm{e}^{-cx}$, $x\in\mathbb{R}$ is the solution (in the classical sense) of the initial value problem $Pv=0$ in $\mathbb{R}$, $v(0)=1$. Hence, by Example 5.11, the distributions $u_+:=vH$ and $u_-:=-vH^\vee$ in $\mathcal{D}'(\mathbb{R})$ are fundamental solutions for P. Note that instead of using Example 5.11, one may show that $Pu_+=\delta$ (and similarly that $Pu_-=\delta$) in $\mathcal{D}'(\mathbb{R})$ by applying *(4)* in Proposition 2.36 and the fact that $H'=\delta$ in $\mathcal{D}'(\mathbb{R})$ to write

$$P(vH)=(vH)'+cvH=v'H+vH'+cvH=v\delta=\delta\quad\text{in }\mathcal{D}'(\mathbb{R}).\tag{5.3.15}$$

We close this section by proving a strengthened version of Theorem 5.9 that addresses the issue of existence of fundamental solutions that are tempered distributions. Such a result is important in the study of partial differential equations as it opens the door for using the Fourier transform.

Theorem 5.13. *Let $P(D)$ be a nonzero constant coefficient linear differential operator in $\mathbb{R}^n$. Then there exists $E\in\mathcal{S}'(\mathbb{R}^n)$, which is a fundamental solution for $P(D)$. In particular, the equality $P(D)E=\delta$ holds in $\mathcal{S}'(\mathbb{R}^n)$.*

We prove this theorem by relying on a deep result due to L. Hörmander and S. Lojasiewicz. For a proof of Theorem 5.14 the interested reader is referred to [29, 41].

Theorem 5.14 (Hörmander–Lojasiewicz). *Let P be a polynomial in $\mathbb{R}^n$ that does not vanish identically and consider the pointwise multiplication mapping*

$$M_P : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n), \qquad M_P(\varphi) := P\varphi, \quad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{5.3.16}$$

Then M_P is linear, continuous, injective, with closed range $\mathcal{S}_P(\mathbb{R}^n)$ in $\mathcal{S}(\mathbb{R}^n)$. Moreover, the mapping $M_P : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}_P(\mathbb{R}^n)$ has a linear and continuous inverse $T : \mathcal{S}_P(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$.

Granted this, we can now turn to the proof of the theorem stated earlier.

Proof of Theorem 5.13. Let $P(D)$ be as in the statement and consider the polynomial $P(\xi)$ associated to it as in (5.1.2). Denote by $\mathcal{S}_P(\mathbb{R}^n)$ the range of the mapping M_P from (5.3.16) corresponding to this polynomial and by T the inverse of $M_P : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}_P(\mathbb{R}^n)$. Then Theorem 5.14 combined with Proposition 13.1 give that the transpose of T is linear and continuous as a mapping

$$T^t : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}_P(\mathbb{R}^n)' \tag{5.3.17}$$

where $\mathcal{S}_P(\mathbb{R}^n)'$ is the dual of $\mathcal{S}_P(\mathbb{R}^n)$ endowed with the weak-$*$ topology.

Next, assume that some $f \in \mathcal{S}'(\mathbb{R}^n)$ has been fixed and define $u := T^t f$. Then $u \in \mathcal{S}_P(\mathbb{R}^n)'$ and, making use of (13.1.3), (13.1.7), and Theorem 13.4, it is possible to extend u to an element in $\mathcal{S}'(\mathbb{R}^n)$, which we continue to denote by u. We claim that $Pu = f$ in $\mathcal{S}'(\mathbb{R}^n)$, where Pu is the tempered distribution obtained by multiplying u with the Schwartz function $P(\xi)$. To see this, fix $\varphi \in \mathcal{S}(\mathbb{R}^n)$ and write

$$\langle Pu, \varphi\rangle = \langle u, P\varphi\rangle = \big\langle T^t f, M_P\varphi\big\rangle = \langle f, TM_P\varphi\rangle = \langle f, \varphi\rangle, \tag{5.3.18}$$

proving the claim. In particular, the above procedure may be implemented for the distribution $f := 1 \in \mathcal{S}'(\mathbb{R}^n)$. Corresponding to this choice, we obtain $u \in \mathcal{S}'(\mathbb{R}^n)$ with the property that $Pu = 1$ in $\mathcal{S}'(\mathbb{R}^n)$. Define $E := \mathcal{F}^{-1}u$. Then $E \in \mathcal{S}'(\mathbb{R}^n)$ and

$$\widehat{P(D)E} = P(\xi)\widehat{E} = Pu = 1 = \widehat{\delta} \qquad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{5.3.19}$$

Taking the Fourier transform we see that, as desired, E is a tempered distribution fundamental solution for $P(D)$. □

Further Notes for Chap. 5. The existence of fundamental solutions for arbitrary nonzero, constant coefficient, linear, partial differential operators was first established in this degree of generality by L. Ehrenpreis and B. Malgrange (cf. [11, 42]), via proofs relying on the Hahn–Banach theorem. This basic result served as the first compelling piece of evidence of the impact of the theory of distributions in the field of partial differential equations. The general discussion initiated here is going to be further augmented by considering in later chapters concrete examples of fundamental solutions associated with basic operators arising in mathematics, physics, and engineering.

5.4 Additional Exercises for Chap. 5

Exercise 5.15. *Let $m \in \mathbb{N}$. Prove that*

$$\Big\{u \in \mathcal{S}'(\mathbb{R}) : u^{(m)} = \delta \ \ in \ \ \mathcal{S}'(\mathbb{R})\Big\} \tag{5.4.1}$$

$$= \Big\{\tfrac{1}{(m-1)!}\, x^{m-1} H + \sum_{k=0}^{m-1} c_k x^k : c_k \in \mathbb{C},\ k = 1, \ldots, m-1\Big\}.$$

Exercise 5.16. *Let $n, m \in \mathbb{N}$ and $k_1, k_2 \in \mathbb{N}_0$. Consider two scalar operators*

$$P_1(D_x) := \sum_{|\alpha| \le k_1} a_\alpha D_x^\alpha, \quad a_\alpha \in \mathbb{C},\ \forall\, \alpha \in \mathbb{N}_0^n,\ |\alpha| \le k_1, \tag{5.4.2}$$

$$P_2(D_y) := \sum_{|\beta| \le k_2} b_\alpha D_y^\alpha, \quad b_\beta \in \mathbb{C},\ \forall\, \beta \in \mathbb{N}_0^m,\ |\beta| \le k_2, \tag{5.4.3}$$

and assume that $E_1 \in \mathcal{D}'(\mathbb{R}^n)$ and $E_2 \in \mathcal{D}'(\mathbb{R}^m)$ are fundamental solutions for P_1 in $\mathbb{R}^n$ and P_2 in $\mathbb{R}^m$, respectively. Prove that $E_1 \otimes E_2$ is a fundamental solution for $P_1(D_x) \otimes P_2(D_y) := \sum\limits_{|\alpha| \le k_1} \sum\limits_{|\beta| \le k_2} a_\alpha b_\alpha D_x^\alpha D_y^\alpha$ in $\mathbb{R}^n \times \mathbb{R}^m$.

Chapter 6

Hypoelliptic Operators

As usual, assume that $\Omega \subseteq \mathbb{R}^n$ is an open set, and fix $m \in \mathbb{N}$. From Theorem 2.102 we know that if $u \in \mathcal{D}'(\Omega)$ is such that for each $\alpha \in \mathbb{N}_0^n$ satisfying $|\alpha| = m$ the distributional derivative $\partial^\alpha u$ is continuous on Ω, then $u \in C^m(\Omega)$. To rephrase this result in a manner that lends itself more naturally to generalizations, define $\mathcal{P}^m u := \big(\partial^\alpha u\big)_{|\alpha|=m}$ for each $u \in \mathcal{D}'(\Omega)$. In this notation, the earlier result reads

$$u \in \mathcal{D}'(\Omega) \quad \text{and} \quad \mathcal{P}^m u \in C^0(\Omega) \Longrightarrow u \in C^m(\Omega). \tag{6.0.1}$$

We are interested in proving results in the same spirit with the role of mapping $\mathcal{P}^m$ played by (certain) linear differential operators.

Recall the notation from (3.1.10) and consider the linear differential operator

$$P(x,D) = \sum_{|\alpha|\leq m} a_\alpha(x) D^\alpha, \ a_\alpha \in C^\infty(\Omega), \text{ for all } \alpha \in \mathbb{N}_0^n \text{ with } |\alpha| \leq m. \tag{6.0.2}$$

By Proposition 2.34, if $f \in C^0(\Omega)$ and $u \in C^m(\Omega)$ is a solution of $P(x,D)u = f$ in $\mathcal{D}'(\Omega)$, then u is a solution of $P(x,D)u = f$ in Ω in the classical sense. However, this reasoning requires knowing in advance that $u \in C^m(\Omega)$. It is desirable to have a more general statement to the effect that if $u \in \mathcal{D}'(\Omega)$ is a solution of $P(x,D)u = f$ in $\mathcal{D}'(\Omega)$ and f has certain regularity properties, then the distribution u also belongs to an appropriate smoothness class (e.g., if $f \in C^\infty(\Omega)$ then $u \in C^\infty(\Omega)$). While such a phenomenon is not to be expected in general, there exist classes of operators $P(x,D)$ for which there are a priori regularity results for the solution u based on the regularity of the datum f. Such a class of operators is singled out in this chapter.

6.1 Definition and Properties

Definition 6.1. *An operator $P(x,D)$ as in (6.0.2) is called* `hypoelliptic` *in Ω if for all open subsets ω of Ω the following property holds: whenever $u \in \mathcal{D}'(\omega)$ is such that $P(x,D)u \in C^\infty(\omega)$ then $u \in C^\infty(\omega)$.*

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Remark 6.2. *It is easy to see from Definition 6.1 that if an operator is hypoelliptic in Ω, then it is also hypoelliptic in any other open subset of Ω.*

Definition 6.3. *Let $u \in \mathcal{D}'(\Omega)$ be arbitrary. Then the* `singular support` *of u, denoted by* $\operatorname{sing\,supp} u$*, is defined by*

$$\operatorname{sing\,supp} u := \big\{x \in \Omega : \text{there is no open set } \omega \text{ such that } x \in \omega \subseteq \Omega \text{ and } u\big|_{\omega} \in C^{\infty}(\omega)\big\}. \tag{6.1.1}$$

Example 6.4. Not surprisingly,

$$\operatorname{sing\,supp} \delta = \{0\}. \tag{6.1.2}$$

To see why this is true, note that clearly $\operatorname{sing\,supp} \delta \subset \{0\}$ and $\operatorname{sing\,supp} \delta \neq \emptyset$ since, by Example 2.13, the distribution δ is not of function type.

Remark 6.5. *(1) Since $\Omega \setminus \operatorname{sing\,supp} u$ is an open set, it follows that $\operatorname{sing\,supp} u$ is relatively closed in Ω.*

(2) By Exercise 2.46, we have that $u\big|_{\Omega\setminus \operatorname{sing\,supp} u} \in C^{\infty}(\Omega \setminus \operatorname{sing\,supp} u)$.

(3) The singular support of a distribution $u \in \mathcal{D}'(\Omega)$ is the smallest relatively closed set in Ω with the property that u restricted to its complement in Ω is of class C^{∞}.

(4) If the operator $P(x, D)$ is as in (6.0.2), then

$$\operatorname{sing\,supp}\,[P(x, D)u] \subseteq \operatorname{sing\,supp} u, \qquad \forall\, u \in \mathcal{D}'(\Omega). \tag{6.1.3}$$

Indeed, if $x_0 \in \Omega \setminus \operatorname{sing\,supp} u$, then there exists an open neighborhood ω of x_0 contained in Ω and such that $u\big|_{\omega} \in C^{\infty}(\omega)$. Hence,

$$[P(x, D)u]\big|_{\omega} = P(x, D)[u\big|_{\omega}] \in C^{\infty}(\omega), \tag{6.1.4}$$

which implies that $x_0 \in \Omega \setminus \operatorname{sing\,supp}\,[P(x, D)u]$.

The quality of being a hypoelliptic operator turns out to be equivalent to (6.1.3) being valid with equality.

Proposition 6.6. *Let $P(x, D)$ be as in (6.0.2). Then $P(x, D)$ is hypoelliptic in Ω if and only if*

$$\operatorname{sing\,supp}\,[P(x, D)u] = \operatorname{sing\,supp} u, \qquad \forall\, u \in \mathcal{D}'(\Omega). \tag{6.1.5}$$

Proof. Suppose $P(x, D)$ is hypoelliptic in Ω. The left-to-right inclusion in (6.1.5) has been proved in part *(4)* of Remark 6.5. To prove the opposite inclusion, fix a point $x_0 \in \Omega \setminus \operatorname{sing\,supp}\,[P(x, D)u]$. Then there exists an open subset ω of Ω with $x_0 \in \omega$ and $[P(x, D)u]\big|_{\omega} \in C^{\infty}(\omega)$. Since $P(x, D)$ is hypoelliptic in Ω it is also hypoelliptic in ω and, as such, $u\big|_{\omega} \in C^{\infty}(\omega)$. This proves that $x_0 \in \Omega \setminus \operatorname{sing\,supp} u$, as wanted.

Now suppose (6.1.5) holds and let ω be an open subset of Ω. Fix some $u \in \mathcal{D}'(\omega)$ such that $P(x,D)u \in C^\infty(\omega)$. The desired conclusion will follow as soon as we show that $u \in C^\infty(\omega)$. By Exercise 2.46, it suffices to prove that for each $x_0 \in \omega$ there exists an open neighborhood $\omega_0 \subseteq \omega$ of x_0 such that $u\big|_{\omega_0} \in C^\infty(\omega_0)$. To this end, fix $x_0 \in \omega$ along with some open neighborhood ω_0 of x_0 with a closure that is a compact subset of ω. Furthermore, pick a function $\psi \in C_0^\infty(\omega)$ with the property that $\psi = 1$ in a neighborhood of ω_0, and note that $\psi u \in \mathcal{E}'(\omega)$. By Proposition 2.63, every $v \in \mathcal{E}'(\omega)$ has a unique extension to $\mathcal{E}'(\Omega)$ that we will denote by $\text{Ext}(v)$. Since the operation of taking the extension Ext commutes with applying $P(x,D)$ (recall Exercise 2.64) we may write

$$\begin{aligned} P(x,D)[\text{Ext}(\psi u)] &= \text{Ext}\big(P(x,D)(\psi u)\big) = \text{Ext}\big(\psi P(x,D)u + g\big) \\ &= \text{Ext}\big(\psi P(x,D)u\big) + \text{Ext}(g) \quad \text{in } \mathcal{D}'(\Omega), \end{aligned} \tag{6.1.6}$$

where $g := P(x,D)(\psi u) - \psi P(x,D)u \in \mathcal{E}'(\omega)$. On the other hand, from part *(4)* in Proposition 2.36 it follows that $g\big|_{\omega_0} = 0$, thus

$$(\text{sing supp}\, g) \cap \omega_0 = \emptyset. \tag{6.1.7}$$

In concert, (6.1.7), the fact that $\psi(P(x,D)u) \in C_0^\infty(\omega)$ [recall that, by assumption, $P(x,D)u \in C^\infty(\omega)$], and (6.1.6), imply that

$$\text{sing supp}\,\big(P(x,D)[\text{Ext}(\psi u)]\big) \cap \omega_0 = \emptyset. \tag{6.1.8}$$

Invoking the hypothesis (6.1.5) then yields $\text{sing supp}\,[\text{Ext}(\psi u)] \cap \omega_0 = \emptyset$ which, in turn, forces $[\text{Ext}(\psi u)]\big|_{\omega_0} \in C^\infty(\omega_0)$. The desired conclusion now follows upon observing that $[\text{Ext}(\psi u)]\big|_{\omega_0} = (\psi u)\big|_{\omega_0} = u\big|_{\omega_0}$. □

Remark 6.7. *A noteworthy consequence of Proposition 6.6 is as follows: any two fundamental solutions of a hypoelliptic operator in $\mathbb{R}^n$ differ by a C^∞ function. More specifically, if an operator $P(x,D)$ as in (6.0.2) is hypoelliptic in $\mathbb{R}^n$, and $E_1, E_2 \in \mathcal{D}'(\mathbb{R}^n)$ are such that $P(x,D)E_1 = \delta$ and $P(x,D)E_2 = \delta$ in $\mathcal{D}'(\mathbb{R}^n)$, then we have $E_1 - E_2 \in C^\infty(\mathbb{R}^n)$.*

In particular, if an operator $P(x,D)$ as in (6.0.2) is hypoelliptic in $\mathbb{R}^n$ and has a fundamental solution $E \in \mathcal{D}'(\mathbb{R}^n)$ with $\text{sing supp}\, E = \{0\}$, then the singular support of any other fundamental solution of $P(x,D)$ is also $\{0\}$.

6.2 Hypoelliptic Operators with Constant Coefficients

In this subsection we further analyze hypoelliptic operators with constant coefficients. First we present a result due to L. Schwartz regarding a necessary and sufficient condition for a linear, constant coefficient operator to be hypoelliptic in the entire space.

Theorem 6.8. *Let $P(D)$ be a constant coefficient linear differential operator in $\mathbb{R}^n$. Then $P(D)$ is hypoelliptic in $\mathbb{R}^n$ if and only if there exists $E \in \mathcal{D}'(\mathbb{R}^n)$ fundamental solution of $P(D)$ in $\mathcal{D}'(\mathbb{R}^n)$ satisfying* $\operatorname{sing\,supp} E = \{0\}$.

Proof. Suppose $P(D)$ is hypoelliptic. Then necessarily $P(D)$ is not identically zero (since the zero operator sends the Dirac distribution to the C^∞ function zero). Granted this, Theorem 5.9 applies and gives that the operator $P(D)$ has a fundamental solution $E \in \mathcal{D}'(\mathbb{R}^n)$. In concert with the hypoellipticity of $P(D)$, Proposition 6.6, and (6.1.2), this allows us to write

$$\operatorname{sing\,supp} E = \operatorname{sing\,supp}[P(D)E] = \operatorname{sing\,supp}\delta = \{0\}, \tag{6.2.1}$$

as wanted.

Consider next the converse implication. Suppose that there exists some $E \in \mathcal{D}'(\mathbb{R}^n)$ with $P(D)E = \delta$ and $\operatorname{sing\,supp} E = \{0\}$. Fix $u \in \mathcal{D}'(\mathbb{R}^n)$ arbitrary. Combining *(4)* in Remark 6.5 with Proposition 6.6, in order to conclude that $P(D)$ is hypoelliptic, it suffices to prove that

$$\operatorname{sing\,supp} u \subseteq \operatorname{sing\,supp}[P(D)u]. \tag{6.2.2}$$

To this end, fix a point $x_0 \in \mathbb{R}^n \setminus \operatorname{sing\,supp}[P(D)u]$. Then, there exists an open neighborhood ω of the point x_0 such that $[P(D)u]\big|_\omega \in C^\infty(\omega)$. The fact that $x_0 \notin \operatorname{sing\,supp} u$ will follow if we are able to show that there exists some number $r \in (0,\infty)$ with the property that $u\big|_{B(x_0,r)} \in C^\infty(B(x_0,r))$.

Pick a number $r_1 \in (0,\infty)$ with the property that $\overline{B(x_0,r_1)} \subset \omega$. Also, fix $r_0 \in (0,r_1)$ and select a function $\psi \in C_0^\infty(\omega)$ such that $\psi = 1$ on $B(x_0,r_1)$. Set $v := \psi u$. Then $v\big|_{B(x_0,r_1)} = u\big|_{B(x_0,r_1)}$ and $v \in \mathcal{E}'(\omega)$. By virtue of Proposition 2.63 we may extend v to a distribution in $\mathcal{E}'(\mathbb{R}^n)$ (and we continue to denote this extension by v).

Define $g := P(D)v - \psi[P(D)u]$. Since $[P(D)u]\big|_\omega \in C^\infty(\omega)$ and $\operatorname{supp}\psi \subset \omega$, we deduce that $\psi[P(D)u] \in C_0^\infty(\omega)$ and $\operatorname{supp}\big(\psi[P(D)u]\big) \subset \omega$. In particular, $\psi[P(D)u] \in C_0^\infty(\mathbb{R}^n)$. Consequently, $g \in \mathcal{E}'(\mathbb{R}^n)$. In addition, combining *(4)* in Proposition 2.36 with the fact that $\psi = 1$ on $B(x_0,r_1)$ shows that $g\big|_{B(x_0,r_1)} = 0$. Thus,

$$\operatorname{sing\,supp} g \cap B(x_0,r_1) = \emptyset. \tag{6.2.3}$$

On the other hand, invoking *(d)* and *(e)* in Theorem 2.87 we may write

$$\begin{aligned} v &= \delta * v = (P(D)E) * v = E * (P(D)v) \\ &= E * g + E * (\psi P(D)u) \quad \text{in } \mathcal{D}'(\mathbb{R}^n). \end{aligned} \tag{6.2.4}$$

We claim that

$$\operatorname{sing\,supp}(E * g) \cap B(x_0,r_0) = \emptyset. \tag{6.2.5}$$

To see why (6.2.5) is true, fix $\varepsilon \in (0, r_1 - r_0)$ and $\phi \in C_0^\infty(B(0,\varepsilon))$ with the property that $\phi = 1$ on $B\left(0, \frac{\varepsilon}{2}\right)$. Then $E * g = [(1-\phi)E] * g + (\phi E) * g$. Hence, by (2.8.17), we may write

$$\begin{aligned} \operatorname{supp}[(\phi E) * g] &\subseteq \operatorname{supp}(\phi E) + \operatorname{supp} g \subseteq B(0,\varepsilon) + [\mathbb{R}^n \setminus B(x_0, r_1)] \\ &\subseteq \mathbb{R}^n \setminus B(x_0, r_0). \end{aligned} \tag{6.2.6}$$

This shows that

$$\operatorname{sing\,supp}[(\phi E) * g] \cap B(x_0, r_0) = \emptyset. \tag{6.2.7}$$

Since by assumption $\operatorname{sing\,supp} E = \{0\}$, it follows that $(1-\phi)E \in C^\infty(\mathbb{R}^n)$ and invoking Exercise 2.94, we obtain that $[(1-\phi)E] * g \in C^\infty(\mathbb{R}^n)$. The latter combined with (6.2.7) now yields (6.2.5).

After using (6.2.5) back in (6.2.4) and observing that $E * (\psi P(D)u) \in C^\infty(\mathbb{R}^n)$ (itself a consequence of Proposition 2.93) we conclude that $v\big|_{B(x_0,r_0)} \in C^\infty(B(x_0,r_0))$. Upon recalling that $\psi = 1$ on $B(x_0,r_0)$, we necessarily have $u\big|_{B(x_0,r_0)} \in C^\infty(B(x_0,r_0))$. The proof of the theorem is now complete. □

Exercise 6.9. *Show that if $P(D)$ is a constant coefficient linear differential operator which is hypoelliptic in $\mathbb{R}^n$, then any fundamental solution $E \in \mathcal{D}'(\mathbb{R}^n)$ of $P(D)$ has the property that* $\operatorname{sing\,supp} E = \{0\}$.

Definition 6.10. *Let $P(D)$ be a constant coefficient linear differential operator in $\mathbb{R}^n$. Then $F \in \mathcal{D}'(\mathbb{R}^n)$ is called a* `parametrix` *of $P(D)$ if there exists some $w \in C^\infty(\mathbb{R}^n)$ with the property that $P(D)F = \delta + w$ in $\mathcal{D}'(\mathbb{R}^n)$.*

It is natural to think of the concept of a parametrix as a relaxation of the notion of a fundamental solution since, clearly, any fundamental solution is a parametrix. We have seen that Theorem 6.8 is an important tool in determining if a given operator $P(D)$ is hypoelliptic. One apparent drawback of this particular result is the fact that constructing a fundamental solution for a given operator $P(D)$ may be a delicate task. It is therefore desirable to have a more flexible criterion for deciding whether a certain operator is hypoelliptic and, remarkably, Theorem 6.11 shows that one can use a parametrix in place of a fundamental solution to the same effect.

Theorem 6.11. *Let $P(D)$ be a constant coefficient linear differential operator in $\mathbb{R}^n$. Then $P(D)$ is hypoelliptic in $\mathbb{R}^n$ if and only if there exists $F \in \mathcal{D}'(\mathbb{R}^n)$ which is a parametrix of $P(D)$ and satisfies* $\operatorname{sing\,supp} F = \{0\}$.

Proof. In one direction, if $P(D)$ is hypoelliptic in $\mathbb{R}^n$, then by Theorem 6.8 there exists a fundamental solution E of $P(D)$ with $\operatorname{sing\,supp} E = \{0\}$. The desired conclusion follows by taking $F := E$. Conversely, suppose there exists a parametrix $F \in \mathcal{D}'(\mathbb{R}^n)$ of $P(D)$ satisfying $\operatorname{sing\,supp} F = \{0\}$. Let $w \in C^\infty(\mathbb{R}^n)$ be such that $P(D)F - \delta = w$ in $\mathcal{D}'(\mathbb{R}^n)$. To show that $P(D)$ is hypoelliptic in $\mathbb{R}^n$, we adjust the argument used in the proof of Theorem 6.8. Specifically, starting with $u \in \mathcal{D}'(\mathbb{R}^n)$ arbitrary, as before, matters are reduced to proving

that if $x_0 \in \mathbb{R}^n \setminus \operatorname{sing\,supp}[P(D)u]$, then there exists $r \in (0,\infty)$ such that $u\big|_{B(x_0,r)} \in C^\infty(B(x_0,r))$. Retaining the notation and context from the proof of Theorem 6.8, the reasoning there applies and allows us to write (in place of (6.2.4))

$$\begin{aligned} v &= \delta * v = (P(D)F - w) * v = F * (P(D)v) - w * v \\ &= F * g + F * (\psi P(D)u) - w * v \quad \text{in } \mathcal{D}'(\mathbb{R}^n). \end{aligned} \tag{6.2.8}$$

Since $\operatorname{sing\,supp} F = \{0\}$, the same argument as in the proof of (6.2.5) shows that

$$\operatorname{sing\,supp}(F * g) \cap B(x_0, r_0) = \emptyset, \tag{6.2.9}$$

while since $\psi[P(D)u] \in C_0^\infty(\mathbb{R}^n)$, Proposition 2.93 gives that

$$F * (\psi P(D)u) \in C^\infty(\mathbb{R}^n). \tag{6.2.10}$$

The new observation is that from $v \in \mathcal{E}'(\mathbb{R}^n)$, $w \in C^\infty(\mathbb{R}^n)$, and Exercise 2.94, we have

$$w * v \in C^\infty(\mathbb{R}^n). \tag{6.2.11}$$

Combining (6.2.8)-(6.2.11) it follows that $v\big|_{B(x_0,r_0)} \in C^\infty(B(x_0,r_0))$. The latter forces $u\big|_{B(x_0,r_0)} \in C^\infty(B(x_0,r_0))$ (recall that $\psi = 1$ on $B(x_0,r_0)$). The proof of the theorem is now complete. □

In the class of constant coefficient linear differential operators, Theorem 6.8 and Theorem 6.11 provide useful characterizations of hypoellipticity. In order to offer a wider perspective on this matter, we state, without proof, another such characterization[1] due to L. Hörmander.

Theorem 6.12. *Let $P(D)$ be a nonzero constant coefficient linear differential operator in $\mathbb{R}^n$. Then $P(D)$ is hypoelliptic if and only if there exist some constants $C, R, c \in (0,\infty)$ with the property that*

$$\left|\frac{\partial^\beta P(\xi)}{P(\xi)}\right| \leq C|\xi|^{-c|\beta|}, \qquad \forall\, \xi \in \mathbb{R}^n \quad \text{with} \quad |\xi| \geq R, \tag{6.2.12}$$

for every $\beta \in \mathbb{N}_0^n$.

An important subclass of linear, constant coefficient operators is that of `elliptic operators`. Assume that an operator $P(D)$ of order $m \in \mathbb{N}_0$ as in (5.1.1) has been given and recall that its principal symbol is $P_m(\xi) = \sum_{|\alpha|=m} a_\alpha \xi^\alpha$, $\xi \in \mathbb{R}^n$.

Definition 6.13. *Let $P(D)$ be a constant coefficient linear differential operator of order $m \in \mathbb{N}_0$ in $\mathbb{R}^n$. Then $P(D)$ is called* `elliptic` *if its principal symbol vanishes only at zero, that is, if $P_m(\xi) \neq 0$ for every $\xi \in \mathbb{R}^n \setminus \{0\}$.*

[1] This characterization is not going to play a significant role for us here. For a proof, the interested reader is referred to [32, Theorem 11.1.3, p. 62].

Remark 6.14. *An example of a constant coefficient elliptic operator playing a significant role in partial differential equations is the* `poly-harmonic` *operator of order* $2m$*, where* $m \in \mathbb{N}$*, defined as*

$$\Delta^m := \sum_{j_1,\dots,j_m=1}^{n} \partial_{j_1}^2 \cdots \partial_{j_m}^2 \quad \text{in} \quad \mathbb{R}^n. \tag{6.2.13}$$

Note that Δ^m *is simply* $\big(\sum_{j=1}^n \partial_j^2\big)^m$*, hence, the principal symbol of* Δ^m *is* $(-1)^m|\xi|^{2m}$ *that does not vanish for any* $\xi \in \mathbb{R}^n\setminus\{0\}$*. The operators obtained for* $m = 1$ *and* $m = 2$ *are called, respectively, the* `Laplacian` *and the* `bi-Laplacian` *operator. By the above discussion they are also elliptic.*

The relevance of the class of elliptic operators is apparent from the following important consequence of Theorem 6.11.

Theorem 6.15. *Linear, constant coefficient elliptic operators are hypoelliptic in* $\mathbb{R}^n$*.*

Proof. Let $P(D)$ be a constant coefficient linear elliptic operator of order $m \in \mathbb{N}_0$. By Theorem 6.11, to conclude that $P(D)$ is hypoelliptic, matters reduce to showing that $P(D)$ has a parametrix F with $\operatorname{sing\,supp} F = \{0\}$.

Starting from the ellipticity condition $P_m(\xi) \neq 0$ for every $\xi \in \mathbb{R}^n \setminus \{0\}$, we note that if $C_0 := \inf_{|\xi|=1} |P_m(\xi)|$, then $C_0 > 0$. Keeping in mind that $P_m(\xi)$ is a homogeneous polynomial of degree m we may then estimate

$$|P_m(\xi)| = \left|P_m\left(|\xi|\frac{\xi}{|\xi|}\right)\right| = |\xi|^m \left|P_m\left(\frac{\xi}{|\xi|}\right)\right| \geq C_0|\xi|^m, \qquad \forall\,\xi \in \mathbb{R}^n \setminus \{0\}. \tag{6.2.14}$$

In addition, there exists $C_1 \in (0,\infty)$ such that

$$|P(\xi) - P_m(\xi)| \leq C_1(1+|\xi|)^{m-1} \quad \forall\,\xi \in \mathbb{R}^n. \tag{6.2.15}$$

Hence, from (6.2.14) and (6.2.15) it follows that for every $\xi \in \mathbb{R}^n \setminus \{0\}$ we have

$$|P(\xi)| \geq |P_m(\xi)| - |P(\xi) - P_m(\xi)| \geq C_0|\xi|^m - C_1(1+|\xi|)^{m-1}. \tag{6.2.16}$$

If $|\xi|$ is sufficiently large, then $C_1(1+|\xi|)^{m-1} \leq \frac{C_0}{2}|\xi|^m$, which implies that there exists $R \in (0,\infty)$ such that

$$|P(\xi)| \geq \frac{C_0}{2}|\xi|^m \quad \text{whenever} \quad |\xi| \geq R. \tag{6.2.17}$$

Hence, if we take $\psi \in C_0^\infty(\mathbb{R}^n)$ satisfying $\psi = 1$ on $B(0,R)$, then

$$\frac{1-\psi(\xi)}{P(\xi)} \in C^\infty(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n). \tag{6.2.18}$$

Recalling (4.1.8) and *(a)* in Theorem 4.25, it follows that there exists $F \in \mathcal{S}'(\mathbb{R}^n)$ with the property that

$$\widehat{F} = \frac{1-\psi(\xi)}{P(\xi)} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{6.2.19}$$

Since $P(\xi)\in\mathcal{L}(\mathbb{R}^n)$, we further have $P(\xi)\widehat{F}=1-\psi(\xi)$ in $\mathcal{S}'(\mathbb{R}^n)$. Making use of *(b)* in Theorem 4.25 and (4.2.4), then taking the Fourier transform of the latter identity and invoking (4.2.28), we arrive at

$$P(D)F-\delta=-(2\pi)^{-n}\widehat{\psi}^{\vee}\quad\text{ in }\quad\mathcal{S}'(\mathbb{R}^n). \tag{6.2.20}$$

Since $\widehat{\psi}\in\mathcal{S}(\mathbb{R}^n)$, identity (6.2.20) shows that F is a parametrix for $P(D)$.

There remains to prove $\operatorname{sing\,supp}F=\{0\}$. The inclusion $\{0\}\subseteq\operatorname{sing\,supp}F$ is immediate from (6.2.20) given (6.1.2) and (6.1.3). The opposite inclusion will follow once we show

$$F\big|_{\mathbb{R}^n\setminus\{0\}}\in C^\infty(\mathbb{R}^n\setminus\{0\}). \tag{6.2.21}$$

To this end, we claim that if $\beta,\alpha\in\mathbb{N}_0^n$ then

$$\mathcal{F}\big(D^\beta x^\alpha F\big)\in L^1(\mathbb{R}^n)\qquad\text{whenever}\quad|\beta|-|\alpha|-m<-n. \tag{6.2.22}$$

Assume (6.2.22) for now. Then, recalling (3.1.3), it follows that

$$D^\beta(x^\alpha F)\in C^0(\mathbb{R}^n)\qquad\text{for}\quad\beta,\alpha\in\mathbb{N}_0^n,\ \ |\beta|-|\alpha|-m<-n. \tag{6.2.23}$$

Next, fix $k\in\mathbb{N}_0$ and choose $N\in\mathbb{N}_0$ with the property that $2N>n+k-m$. Since $|x|^{2N}=\sum\limits_{|\alpha|=N}\frac{N!}{\alpha!}\,x^{2\alpha}$ for every $x\in\mathbb{R}^n$ [cf. (13.2.1)] from (6.2.23) we may conclude that $D^\beta(|x|^{2N}F)\in C^0(\mathbb{R}^n)$ for every $\beta\in\mathbb{N}_0^n$ such that $|\beta|=k$. Hence, $|x|^{2N}F\in C^k(\mathbb{R}^n)$. Because $\frac{1}{|x|^{2N}}\big|_{\mathbb{R}^n\setminus\{0\}}\in C^\infty(\mathbb{R}^n\setminus\{0\})$, the latter further gives $F\big|_{\mathbb{R}^n\setminus\{0\}}\in C^k(\mathbb{R}^n\setminus\{0\})$. The membership in (6.2.21) now follows since $k\in\mathbb{N}_0$ is arbitrary.

Returning to the proof of (6.2.22), fix $\beta,\alpha\in\mathbb{N}_0^n$ with $|\beta|-|\alpha|-m<-n$. In light of *(b)*–*(c)* in Theorem 4.25, it suffices to show $\xi^\beta D^\alpha\widehat{F}\in L^1(\mathbb{R}^n)$. Note that based on (6.2.19) and (2.4.16) we may write

$$D^\alpha\widehat{F}=\sum_{\gamma\leq\alpha}\frac{\alpha!}{\gamma!(\alpha-\gamma)!}\,D^\gamma\Big[\frac{1}{P(\xi)}\Big]D^{\alpha-\gamma}\big[1-\psi(\xi)\big]\qquad\text{in}\quad\mathcal{S}'(\mathbb{R}^n). \tag{6.2.24}$$

We now claim that for each $\mu,\nu\in\mathbb{N}_0^n$ and with R as in (6.2.17), there exists a finite constant $C>0$ independent of ξ such that

$$\Big|\xi^\mu D^\nu\Big[\frac{1}{P(\xi)}\Big]\Big|\leq C|\xi|^{|\mu|-|\nu|-m}\qquad\text{for}\quad|\xi|>R. \tag{6.2.25}$$

Indeed, since by induction one can see that $D^\nu\Big[\frac{1}{P(\xi)}\Big]=\frac{Q(\xi)}{P(\xi)^{|\nu|+1}}$ for some polynomial Q of degree at most $(m-1)|\nu|$, by making also use of (6.2.17), for $|\xi|\geq R$ we may write

$$\Big|\xi^\mu D^\nu\Big[\frac{1}{P(\xi)}\Big]\Big|=|\xi|^{|\mu|}\frac{|Q(\xi)|}{|P(\xi)|^{|\nu|+1}}\leq C|\xi|^{|\mu|}\frac{|\xi|^{(m-1)|\nu|}}{|\xi|^{(|\nu|+1)m}}=C|\xi|^{|\mu|-|\nu|-m}, \tag{6.2.26}$$

where $C\in(0,\infty)$ is independent of ξ. This proves (6.2.25).

At this point, we combine (6.2.24), (6.2.25), (6.2.17), and the properties of ψ to estimate

$$\begin{aligned}\left|\xi^{\beta}D^{\alpha}\widehat{F}\right| &\leq \sum_{\gamma<\alpha}\frac{\alpha!}{\gamma!(\alpha-\gamma)!}\left|\xi^{\beta}D^{\gamma}\Big[\frac{1}{P(\xi)}\Big]D^{\alpha-\gamma}\big[1-\psi(\xi)\big]\right| \\ &\quad+\left|\xi^{\beta}D^{\alpha}\Big[\frac{1}{P(\xi)}\Big]\big[1-\psi(\xi)\big]\right| \\ &\leq \sum_{\gamma<\alpha}C_{\alpha,\gamma}\,|\xi|^{|\beta|-|\gamma|-m}\chi_{\operatorname{supp}\psi\setminus B(0,R)} \\ &\quad+C|\xi|^{|\beta|-|\alpha|-m}\chi_{\mathbb{R}^n\setminus B(0,R)}, \qquad \forall\,\xi\in\mathbb{R}^n. \end{aligned} \tag{6.2.27}$$

Recalling that $|\beta|-|\alpha|-m<-n$, from (6.2.27) we obtain that $\xi^{\beta}D^{\alpha}\widehat{F}\in L^1(\mathbb{R}^n)$, as desired. This completes the proof of (6.2.22), and with it the proof of the theorem. □

Exercise 6.16. *Use Theorem 6.12 to give an alternative proof of Theorem 6.15, that is, that linear, constant coefficient elliptic operators are hypoelliptic in* $\mathbb{R}^n$.

Hint: Show that C_1 may be chosen sufficiently large so that, in addition to (6.2.15), one also has $|\partial^{\beta}P(\xi)|\leq C_1(1+|\xi|)^{m-|\beta|}$. Then use the latter and (6.2.17) to show that (6.2.12) is verified by taking $C:=\frac{2C_1}{C_0}(1+\frac{1}{R})^m$ and $c:=1$.

Proposition 6.17. *The poly-harmonic operator is hypoelliptic. In particular, the operators* Δ *and* Δ^2 *are hypoelliptic.*

Proof. This is a consequence of Theorem 6.15 and Remark 6.14. □

Corollary 6.18. *Let* $P(D)$ *be a linear, constant coefficient elliptic operator in* $\mathbb{R}^n$ *and let* Ω *be an open subset of* $\mathbb{R}^n$. *If* $u\in\mathcal{D}'(\Omega)$ *is such that for some open subset* ω *of* Ω *we have* $P(D)u\big|_{\omega}\in C^{\infty}(\omega)$, *then* $u\big|_{\omega}\in C^{\infty}(\omega)$.

Proof. From Theorem 6.15 and Remark 6.2 we know that the restriction of $P(D)$ to ω is a hypoelliptic operator. Since $P(D)u\big|_{\omega}\in\mathcal{D}'(\omega)$ has an empty singular support, Proposition 6.6 gives that the singular support of $u\big|_{\omega}$ is also empty. Consequently, $u\big|_{\omega}\in C^{\infty}(\omega)$. □

An example of a linear, constant coefficient, differential operator that is not hypoelliptic is the operator $P=\partial_1^2-\partial_2^2$ used in (1.1.1) to describe the equation governing the displacement of a vibrating string in $\mathbb{R}^2$. Indeed, as we shall see later (cf. (9.1.28)), the distribution u_f associated with the locally integrable function

$$f(x_1,x_2):=H(x_2-|x_1|), \qquad \forall\,(x_1,x_2)\in\mathbb{R}^2, \tag{6.2.28}$$

(where H stands for the Heaviside function) satisfies $(\partial_1^2-\partial_2^2)u_f=2\delta$ in $\mathcal{D}'(\mathbb{R}^2)$ and $\operatorname{sing\,supp}u_f=\{(x_1,x_2)\in\mathbb{R}^2:\,x_2=|x_1|\}$. Hence,

$$\operatorname{sing\,supp}u_f\neq\operatorname{sing\,supp}Pu_f \tag{6.2.29}$$

which, in light of Proposition 6.6, shows that P is not hypoelliptic.

6.3 Integral Representation Formulas and Interior Estimates

Consider the constant coefficient, linear, differential operator

$$P(\partial)=\sum_{|\alpha|\leq m}a_\alpha\partial^\alpha,\quad a_\alpha\in\mathbb{C},\quad \text{for all }\alpha\in\mathbb{N}_0^n\text{ with }|\alpha|\leq m. \tag{6.3.1}$$

Theorem 6.19 (A general integral representation formula)**.** *Assume that the constant coefficient, linear, differential operator $P(\partial)$ is as in (6.3.1) and is hypoelliptic in $\mathbb{R}^n$. Let $E\in\mathcal{D}'(\mathbb{R}^n)$ be a fundamental solution for $P(\partial)$ as provided by Theorem 6.8 (hence, in particular, $E\in C^\infty(\mathbb{R}^n\setminus\{0\})$).*

Let Ω be an open subset of $\mathbb{R}^n$ and suppose $u\in\mathcal{D}'(\Omega)$ satisfies $P(\partial)u=0$ in $\mathcal{D}'(\Omega)$. Then $u\in C^\infty(\Omega)$ and for each $x_0\in\Omega$, each $r\in\big(0,\operatorname{dist}(x_0,\partial\Omega)\big)$, and each function $\psi\in C_0^\infty\big(B(x_0,r)\big)$ such that $\psi=1$ near $\overline{B(x_0,r/2)}$, we have

$$u(x)=-\sum_{|\alpha|\leq m}\sum_{\gamma<\alpha}(-1)^{|\alpha|+|\gamma|}a_\alpha\,\frac{\alpha!}{(\alpha-\gamma)!\gamma!}\times \tag{6.3.2}$$

$$\times\int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}}(\partial^\gamma E)(x-y)(\partial^{\alpha-\gamma}\psi)(y)u(y)\,dy,$$

for each $x\in B(x_0,r/2)$. In particular, for every $\mu\in\mathbb{N}_0^n$,

$$(\partial^\mu u)(x)=-\sum_{|\alpha|\leq m}\sum_{\gamma<\alpha}(-1)^{|\alpha|+|\gamma|}\,a_\alpha\,\frac{\alpha!}{(\alpha-\gamma)!\gamma!}\times \tag{6.3.3}$$

$$\times\int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}}(\partial^{\gamma+\mu}E)(x-y)(\partial^{\alpha-\gamma}\psi)(y)u(y)\,dy,$$

for each $x\in B(x_0,r/2)$.

Proof. The fact that $u\in C^\infty(\Omega)$ is a consequence of the hypoellipticity of the operator $P(\partial)$ (cf. Definition 6.1 and Remark 6.2). As regards (6.3.2), pick an arbitrary $x_0\in\Omega$, fix $r\in\big(0,\operatorname{dist}(x_0,\partial\Omega)\big)$, and let $\psi\in C_0^\infty\big(B(x_0,r)\big)$ be such that $\psi=1$ near $\overline{B(x_0,r/2)}$. Then $\psi u\in C_0^\infty(\mathbb{R}^n)$ and since $P(\partial)E=\delta$ in $\mathcal{D}'(\mathbb{R}^n)$ we may write, for each point $x\in B(x_0,r/2)$,

$$\begin{aligned}u(x)&=(\psi u)(x)=\big(\delta*(\psi u)\big)(x)\\&=\big((P(\partial)E)*(\psi u)\big)(x)=\big(E*(P(\partial)(\psi u))\big)(x)\\&=\sum_{|\alpha|\leq m}\sum_{0<\beta\leq\alpha}a_\alpha\frac{\alpha!}{\beta!(\alpha-\beta)!}\big(E*(\partial^\beta\psi\partial^{\alpha-\beta}u)\big)(x),\end{aligned} \tag{6.3.4}$$

where we have used (6.3.1), part *(e)* of Theorem 2.87, and that $P(\partial)u=0$ in Ω. Note that for each $\beta>0$ the function $\partial^\beta\psi$ is compactly supported in

$B(x_0,r)\setminus\overline{B(x_0,r/2)}$. Keeping this in mind, we may then integrate by parts in order to further write the last expression in (6.3.4) as

$$\sum_{|\alpha|\leq m}\sum_{0<\beta\leq\alpha} a_\alpha\frac{\alpha!}{\beta!(\alpha-\beta)!}\big(E*(\partial^\beta\psi\partial^{\alpha-\beta}u)\big)(x) \tag{6.3.5}$$

$$=\sum_{|\alpha|\leq m}\sum_{0<\beta\leq\alpha} a_\alpha\frac{\alpha!}{\beta!(\alpha-\beta)!}\int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}}E(x-y)\partial^\beta\psi(y)\partial^{\alpha-\beta}u(y)\,dy$$

$$=\sum_{|\alpha|\leq m}\sum_{0<\beta\leq\alpha} a_\alpha\frac{(-1)^{|\alpha|+|\beta|}\alpha!}{\beta!(\alpha-\beta)!}\int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}}\partial_y^{\alpha-\beta}\big[E(x-y)\partial^\beta\psi(y)\big]u(y)\,dy$$

$$=\sum_{|\alpha|\leq m}\sum_{0<\beta\leq\alpha}\sum_{\gamma\leq\alpha-\beta} a_\alpha\frac{(-1)^{|\alpha|+|\beta|+|\gamma|}\alpha!}{\beta!(\alpha-\beta)!}\,\frac{(\alpha-\beta)!}{\gamma!(\alpha-\beta-\gamma)!}\times$$

$$\times\int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}}(\partial^\gamma E)(x-y)\partial^{\alpha-\gamma}\psi(y)\,u(y)\,dy.$$

To proceed, observe that whenever $\gamma<\alpha$ formula (13.2.3) gives

$$\sum_{0<\beta\leq\alpha-\gamma}(-1)^{|\beta|}\frac{(\alpha-\gamma)!}{\beta!(\alpha-\gamma-\beta)!}=\Big(\sum_{\beta\leq\alpha-\gamma}(-1)^{|\beta|}\frac{(\alpha-\gamma)!}{\beta!(\alpha-\gamma-\beta)!}\Big)-1$$

$$=0-1=-1. \tag{6.3.6}$$

Making use of (6.3.6) back in (6.3.5) yields

$$\sum_{|\alpha|\leq m}\sum_{0<\beta\leq\alpha} a_\alpha\frac{\alpha!}{\beta!(\alpha-\beta)!}\big(E*(\partial^\beta\psi\partial^{\alpha-\beta}u)\big)(x)$$

$$=-\sum_{|\alpha|\leq m}\sum_{\gamma<\alpha} a_\alpha\frac{(-1)^{|\alpha|+|\gamma|}\alpha!}{\gamma!(\alpha-\gamma)!}\times \tag{6.3.7}$$

$$\times\int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}}(\partial^\gamma E)(x-y)\partial^{\alpha-\gamma}\psi(y)\,u(y)\,dy,$$

and (6.3.2) follows from (6.3.4) and (6.3.7). Finally, (6.3.3) is obtained by differentiating (6.3.2). □

To state our next result, recall that the operator $P(\partial)$ from (6.3.1) is said to be homogeneous (of degree m) whenever $a_\alpha=0$ for all multi-indices α with $|\alpha|<m$.

Theorem 6.20 (Interior estimates)**.** *Let $P(\partial)$ be a constant coefficient, linear, differential operator, of order $m\in\mathbb{N}_0$, which is hypoelliptic in $\mathbb{R}^n$. Suppose $E\in\mathcal{D}'(\mathbb{R}^n)$ is a fundamental solution for $P(\partial)$ as provided by Theorem 6.8 (thus, in particular, $E\in C^\infty(\mathbb{R}^n\setminus\{0\})$).*

Then for each $\mu\in\mathbb{N}_0^n$ there exists a constant $C_\mu\in(0,\infty)$ (which also depends on the coefficients of $P(\partial)$ and n) with the following significance.

If Ω be an open subset of $\mathbb{R}^n$ and $u \in \mathcal{D}'(\Omega)$ satisfies $P(\partial)u = 0$ in $\mathcal{D}'(\Omega)$, then $u \in C^\infty(\Omega)$ and for each $x_0 \in \Omega$ and each $r \in \big(0, \operatorname{dist}(x_0, \partial\Omega)\big)$ we have

$$\sup_{x\in B(x_0,r/2)} |(\partial^\mu u)(x)| \leq C_\mu \max_{\substack{|\alpha|\leq m\\ \gamma\leq\alpha}} \Big\{ r^{|\gamma|-|\alpha|} \cdot \sup_{r/4<|z|<r} |(\partial^{\gamma+\mu}E)(z)| \Big\} \times$$

$$\times \int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}} |u(y)|\,dy. \tag{6.3.8}$$

In the particular case when $P(\partial)$ is homogeneous (of degree m) and one also assumes that there exists $k \in \mathbb{N}$ such that the fundamental solution E satisfies

$$|\partial^\beta E(x)| \leq C_\beta |x|^{-n+m-|\beta|}, \ \forall\, x \in \mathbb{R}^n \setminus \{0\}, \ \forall\, \beta \in \mathbb{N}_0^n \ \text{ with } \ |\beta| \geq k, \tag{6.3.9}$$

then whenever $|\mu| \geq k$ we have

$$|(\partial^\mu u)(x_0)| \leq \frac{C_\mu}{r^{|\mu|}} \fint_{B(x_0,r)} |u(y)|\,dy. \tag{6.3.10}$$

Proof. Pick a function $\phi \in C_0^\infty\big(B(0,1)\big)$ such that $\phi = 1$ near $\overline{B(0,3/4)}$, and note that if we set

$$\psi(x) := \phi\big((x-x_0)/r\big), \qquad \forall\, x \in \mathbb{R}^n, \tag{6.3.11}$$

then $\psi \in C_0^\infty\big(B(x_0,r)\big)$ and $\psi = 1$ near $\overline{B(x_0,3r/4)}$. In particular, for each $\gamma \in \mathbb{N}_0^n$ with $|\gamma| > 0$ there exists $c_\gamma \in (0,\infty)$ independent of r such that

$$\operatorname{supp}(\partial^\gamma\psi) \subseteq B(x_0,r) \setminus \overline{B(x_0,3r/4)} \ \text{ and } \ \|\partial^\gamma\psi\|_{L^\infty(\mathbb{R}^n)} \leq c_\gamma\, r^{-|\gamma|}. \tag{6.3.12}$$

For such a choice of ψ, estimate (6.3.8) then follows in a straightforward manner from the integral representation formula (6.3.3). In turn, (6.3.8) readily implies (6.3.10) in the case when $P(\partial)$ is homogeneous (of degree m), E satisfies (6.3.9) for some $k \in \mathbb{N}$, and $|\mu| \geq k$. □

The procedure described abstractly in the following lemma is useful for deriving higher-order interior estimates, with precise control of the constants involved, for various classes of functions.

Lemma 6.21. *Let $\mathcal{A} \subset C^\infty(\Omega)$ be a family of functions satisfying the following two conditions:*

(1) $\partial^\alpha u \in \mathcal{A}$ for every $u \in \mathcal{A}$ and every $\alpha \in \mathbb{N}_0^n$;

(2) There exists $C \in (0,\infty)$ such that for every $u \in \mathcal{A}$ we have

$$|\partial_j u(x)| \leq \frac{C}{r} \max_{y\in\overline{B(x,r)}} |u(y)|, \qquad \forall\, j \in \{1,\dots,n\}, \tag{6.3.13}$$

whenever $x \in \Omega$ and $r \in \big(0, \operatorname{dist}(x,\partial\Omega)\big)$.

Then given any $u \in \mathcal{A}$, for every $x \in \Omega$, every $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, every $k \in \mathbb{N}$, and every $\lambda \in (0,1)$, we have (with C as in (6.3.13))

$$\max_{y\in\overline{B(x,\lambda r)}} |\partial^\alpha u(y)| \leq \frac{C^k(1-\lambda)^{-k}\mathrm{e}^{k-1}k!}{r^k} \max_{y\in\overline{B(x,r)}} |u(y)|, \tag{6.3.14}$$

$$\text{for every multi-index } \alpha \in \mathbb{N}_0^n \text{ with } |\alpha| = k.$$

Proof. In a first stage, we propose to prove by induction over k that, given any $u \in \mathcal{A}$, for every $x \in \Omega$, every $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, and every $k \in \mathbb{N}$ we have (with C as in (6.3.13))

$$|\partial^\alpha u(x)| \leq \frac{C^k\mathrm{e}^{k-1}k!}{r^k} \max_{y\in\overline{B(x,r)}} |u(y)|, \qquad \forall\, \alpha \in \mathbb{N}_0^n \text{ with } |\alpha| = k. \tag{6.3.15}$$

Note that if $k = 1$, this is contained in (6.3.13). Suppose now that (6.3.15) holds for some $k \in \mathbb{N}$ and every $u \in \mathcal{A}$, every $x \in \Omega$, and every $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, and pick an arbitrary $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = k+1$. Then there exist $j \in \{1, \ldots, n\}$ and $\beta \in \mathbb{N}_0^n$ for which $\partial^\alpha = \partial_j\partial^\beta$. Fix such j and β and note that, in particular, $|\beta| = k$. Next, take $x \in \Omega$ and $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$ arbitrary, and pick some $\varepsilon \in (0,1)$ to be specified shortly. Since $\partial^\beta u \in \mathcal{A}$ we may use (6.3.13) with u replaced by $\partial^\beta u$ and r replaced by $(1-\varepsilon)r$ to obtain

$$|\partial^\alpha u(x)| \leq \frac{C}{(1-\varepsilon)r} \max_{y\in\overline{B(x,(1-\varepsilon)r)}} |\partial^\beta u(y)|. \tag{6.3.16}$$

Note that if $y \in B\big(x, (1-\varepsilon)r\big)$ then $\overline{B(y,\varepsilon r)} \subset B(x,r) \subset \Omega$. Thus, for each point $y \in B\big(x, (1-\varepsilon)r\big)$, we may use the induction hypothesis to estimate

$$|\partial^\beta u(y)| \leq \frac{C^k\mathrm{e}^{k-1}k!}{(\varepsilon r)^k} \max_{z\in\overline{B(y,\varepsilon r)}} |u(z)| \leq \frac{C^k\mathrm{e}^{k-1}k!}{\varepsilon^k r^k} \max_{z\in\overline{B(x,r)}} |u(z)|. \tag{6.3.17}$$

Combined, (6.3.16) and (6.3.17) yield

$$|\partial^\alpha u(x)| \leq \frac{C^{k+1}\mathrm{e}^{k-1}k!}{(1-\varepsilon)\varepsilon^k r^{k+1}} \max_{z\in\overline{B(x,r)}} |u(z)|. \tag{6.3.18}$$

Set now $\varepsilon := \frac{k}{k+1} \in (0,1)$ in (6.3.18) which, given that $\varepsilon^{-k} < e$, further implies

$$|\partial^\alpha u(x)| \leq \frac{C^{k+1}\mathrm{e}^k(k+1)!}{r^{k+1}} \max_{z\in\overline{B(x,r)}} |u(z)|. \tag{6.3.19}$$

Hence, (6.3.15) holds with $k+1$ in place of k, as desired.

Finally, as far as (6.3.14) is concerned, pick $x_0 \in \Omega$, $r \in \big(0, \operatorname{dist}(x_0, \partial\Omega)\big)$, $k \in \mathbb{N}$, as well as $\lambda \in (0,1)$. Next, select an arbitrary point $x \in B(x_0, \lambda r)$ and note that this forces $(1-\lambda)r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$. In concert with the fact that $B\big(x, (1-\lambda)r\big) \subseteq B(x_0, r)$, this permits us to invoke (6.3.15) in order to estimate

$$|\partial^\alpha u(x)| \leq \frac{C^k\mathrm{e}^{k-1}k!}{\big((1-\lambda)r\big)^k} \max_{y\in\overline{B(x,(1-\lambda)r)}} |u(y)|$$

$$\leq \frac{C^k(1-\lambda)^{-k}\mathrm{e}^{k-1}k!}{r^k}\max_{y\in\overline{B(x_0,r)}}|u(y)|, \tag{6.3.20}$$

whenever $\alpha\in\mathbb{N}_0^n$ satisfies $|\alpha|=k$. Taking the supremum over $x\in B(x_0,\lambda r)$ then yields (6.3.14) (written with x_0 in place of x). □

Definition 6.22. *A function $u\in C^\infty(\Omega)$ is called* `real-analytic` *in Ω provided for every $x\in\Omega$ there exists $r_x\in(0,\infty)$ with the property that $B(x,r_x)\subseteq\Omega$ and the Taylor series for u at x converges uniformly to u on $B(x,r_x)$, that is,*

$$u(y)=\sum_{\alpha\in\mathbb{N}_0^n}\frac{1}{\alpha!}(y-x)^\alpha(\partial^\alpha u)(x)\quad \text{uniformly for } y\in B(x,r_x). \tag{6.3.21}$$

The following observation justifies the term "real-analytic" used for the class of functions introduced above.

Remark 6.23. *In the context of Definition 6.22, it is clear that, for each $x\in\Omega$,*

$$\widetilde{u}(z):=\sum_{\alpha\in\mathbb{N}_0^n}\frac{1}{\alpha!}(z-x)^\alpha(\partial^\alpha u)(x)\quad \text{for } z\in\mathbb{C}^n \text{ with } |z-x|<r_x, \tag{6.3.22}$$

is a holomorphic function in $\{z\in\mathbb{C}^n:|z-x|<r_x\}$ which locally extends the original real-analytic function u. Such local extensions may be constructed near each point $x\in\Omega$ and Lemma 7.53 ensures that any two local extensions coincide on their common domain. The conclusion is that the real-analytic function u has a well-defined extension to a holomorphic function defined in a neighborhood of Ω in $\mathbb{C}^n$.

Our next lemma gives a sufficient condition (which is in the nature of best possible) ensuring real-analyticity.

Lemma 6.24. *Suppose $u\in C^\infty(\Omega)$ is a function with the property that for each $x\in\Omega$ there exist $r=r(x)\in\big(0,\operatorname{dist}(x,\partial\Omega)\big)$, $M=M(x)\in(0,\infty)$, and $C=C(x)\in(0,\infty)$, such that for every $k\in\mathbb{N}$ the following estimate holds:*

$$\max_{y\in\overline{B(x,r)}}|\partial^\alpha u(y)|\leq MC^k k!\qquad\forall\,\alpha\in\mathbb{N}_0^n \text{ with } |\alpha|=k. \tag{6.3.23}$$

Then u is real-analytic in Ω.

Proof. Fix $x\in\Omega$ and let $r\in\big(0,\operatorname{dist}(x,\partial\Omega)\big)$ be such that (6.3.23) holds for every $k\in\mathbb{N}$. Write Taylor's formula (13.2.7) for u at x to obtain that for each $N\in\mathbb{N}$ and each $y\in B(x,r/2)$ there exists $\theta\in(0,1)$ such that

$$u(y)=\sum_{|\alpha|\leq N-1}\frac{1}{\alpha!}(y-x)^\alpha(\partial^\alpha u)(x)+R_{N,u}(y), \tag{6.3.24}$$

where

$$R_{N,u}(y):=\sum_{|\alpha|=N}\frac{1}{\alpha!}(y-x)^\alpha(\partial^\alpha u)(x+\theta(y-x)). \tag{6.3.25}$$

Using (6.3.23), for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = N$ and $y \in B(x, r/2)$, we may estimate

$$\big|(\partial^\alpha u)(x+\theta(y-x))\big| \leq \max_{z\in\overline{B(x,r)}} |(\partial^\alpha u)(z)| \leq MC^N N! \tag{6.3.26}$$

since $B\big(x+\theta(y-x), r\big) \subset B(x,r)$. Together, (6.3.25) and (6.3.26) imply that, for each $y \in B(x, r/2)$,

$$|R_{N,u}(y)| \leq MC^N |y-x|^N \sum_{|\alpha|=N} \frac{N!}{\alpha!} = M\left(nC|y-x|\right)^N. \tag{6.3.27}$$

For the equality in (6.3.27) we used formula (13.2.1) for $x_1 = \cdots = x_n = 1$. Hence, if say, $|y-x| < \min\big\{\frac{1}{2nC}, \frac{r}{2}\big\}$, then $\lim_{N\to\infty} R_{N,u}(y) = 0$ uniformly with respect to y. Consequently, the Taylor series for u converges uniformly to u in a neighborhood of x. Since x is arbitrary in Ω it follows that u is real-analytic in Ω. □

Theorem 6.25 (Unique continuation)**.** *Suppose $\Omega \subseteq \mathbb{R}^n$ is an open and connected set. Then any real-analytic function u in Ω with the property that there exists $x_0 \in \Omega$ such that $\partial^\alpha u(x_0) = 0$ for all $\alpha \in \mathbb{N}_0^n$ (which is the case if, e.g., u is zero in a neighborhood of+ x_0) vanishes identically in Ω.*

Proof. Suppose u satisfies the hypotheses of the theorem and define the set

$$U := \big\{x \in \Omega : \partial^\alpha u(x) = 0 \text{ for all } \alpha \in \mathbb{N}_0^n\big\}. \tag{6.3.28}$$

Since $x_0 \in U$ we have $U \neq \emptyset$. Also, U is relatively closed in Ω given that it is the intersection of the relatively closed sets $(\partial^\alpha u)^{-1}\big(\{0\}\big)$, $\alpha \in \mathbb{N}_0^n$. In addition, if $x \in U$ then, on the one hand, the Taylor series of u at x is identically zero, while on the other hand, this series converges to u in an open neighborhood of x. Thus, u is identically zero in that neighborhood of x, which ultimately proves that U is also open. Recalling that Ω is connected, it follows that $U = \Omega$ as desired. □

Theorem 6.25 highlights the much more restrictive nature of real-analyticity compared to indefinite differentiability, since obviously there are plenty of C^∞ functions that vanish in a neighborhood of a point without being identically zero.

Theorem 6.26. *Let $P(\partial)$ be a constant coefficient, linear, differential operator in $\mathbb{R}^n$, which is homogeneous of degree $m \in \mathbb{N}_0$. Assume that $P(\partial)$ has a fundamental solution $E \in \mathcal{D}'(\mathbb{R}^n)$ with the property that $E \in C^\infty(\mathbb{R}^n \setminus \{0\})$ and*

$$|\partial^\beta E(x)| \leq C_\beta |x|^{-n+m-|\beta|}, \qquad \forall\, x \in \mathbb{R}^n \setminus \{0\}, \quad \forall\, \beta \in \mathbb{N}_0^n \ \text{ with } \ |\beta| > 0. \tag{6.3.29}$$

Finally, suppose that $\Omega \subseteq \mathbb{R}^n$ is open and $u \in \mathcal{D}'(\Omega)$ satisfies $P(\partial)u = 0$ in $\mathcal{D}'(\Omega)$.

Then $u \in C^\infty(\Omega)$ and there exists a constant $C \in (0,\infty)$ with the property that if $\lambda \in (0,1)$ we have

$$\max_{y\in\overline{B(x,\lambda r)}} |\partial^\alpha u(y)| \leq \frac{C^{|\alpha|}(1-\lambda)^{-|\alpha|}|\alpha|!}{r^{|\alpha|}} \max_{y\in\overline{B(x,r)}} |u(y)|, \qquad \forall\,\alpha\in\mathbb{N}_0^n, \quad (6.3.30)$$

whenever $x \in \Omega$ and $r \in \big(0, \operatorname{dist}(x,\partial\Omega)\big)$. In particular, u is real-analytic in Ω.

Proof. The fact that $P(\partial)$ has a fundamental solution $E \in \mathcal{D}'(\mathbb{R}^n)$ with the property that $E \in C^\infty(\mathbb{R}^n \setminus \{0\})$ implies $\operatorname{sing\,supp} E = \{0\}$, hence $P(\partial)$ is hypoelliptic in $\mathbb{R}^n$ by Theorem 6.8. Granted the properties enjoyed by $P(\partial)$ and condition (6.3.29), estimate (6.3.30) follows from (6.3.9) (used with $|\mu| = 1$) and Lemma 6.21 (in which we take $\mathcal{A} := \{u \in C^\infty(\Omega) : P(\partial)u = 0\}$). Finally, the last claim in the statement of the theorem is a consequence of (6.3.30) and Lemma 6.24. □

Further Notes for Chap. 6. For more extensive discussion of the of notion of hypoellipticity the interested reader is referred to the excellent presentation in [32]. Further information about real analytic functions may be found in, e.g., [37].

6.4 Additional Exercises for Chap. 6

Exercise 6.27. *Let $a > 0$ be fixed.*

(a) Compute a fundamental solution for the operator $\frac{d^2}{dx^2} - a^2$ in $\mathbb{R}$.

(b) Use the result from part (a) to compute the Fourier transform in $\mathcal{S}'(\mathbb{R})$ of the tempered distribution associated with the function $f(x) := \frac{1}{x^2+a^2}$, $x \in \mathbb{R}$ (compare with Example 4.23).

Exercise 6.28. *Prove that* $\operatorname{sing\,supp} u \subseteq \operatorname{supp} u$ *for every $u \in \mathcal{D}'(\Omega)$.*

Exercise 6.29. *Give an example of a distribution u for which the inclusion from Exercise 6.28 is strict. Give an example of a distribution u for which* $\operatorname{sing\,supp} u = \operatorname{supp} u$.

Exercise 6.30. *Let $x_0 \in \mathbb{R}^n$. Prove that* $\operatorname{sing\,supp} \delta^\alpha_{x_0} = \{x_0\}$ *for every $\alpha \in \mathbb{N}_0^n$.*

Exercise 6.31. *Let $a \in C^\infty_0(\Omega)$ and $u \in \mathcal{D}'(\Omega)$. Prove that*

$$\operatorname{sing\,supp}(au) \subseteq \operatorname{sing\,supp} u.$$

Exercise 6.32. *Recall the distribution from (2.1.13). Determine* $\operatorname{sing\,supp}$ $(\text{P.V.}\,\frac{1}{x})$.

Exercise 6.33. *Let $m \in \mathbb{N}$ and let P be a polynomial in $\mathbb{R}^n$ of degree $2m$ with no real roots. Recall from Exercise 4.123 that $\frac{1}{P} \in \mathcal{S}'(\mathbb{R}^n)$. Prove that* $\operatorname{sing\,supp}\Big(\mathcal{F}\big(\frac{1}{P}\big)\Big) \neq \emptyset$.

Exercise 6.34. *Let $u \in \mathcal{D}'(\mathbb{R}^2)$ be the distribution defined by $u := \text{P.V.}\,\frac{1}{x} \otimes \delta(y)$, where $\text{P.V.}\,\frac{1}{x} \in \mathcal{D}'(\mathbb{R})$ is as in (2.1.13). Determine* $\operatorname{sing\,supp} u$.

Chapter 7

The Laplacian and Related Operators

One of the most important operators in partial differential equations is the Laplace operator[1] Δ (also called the Laplacian) in $\mathbb{R}^n$ which is defined as $\Delta := \sum_{j=1}^{n} \partial_j^2$. Functions f satisfying $\Delta f = 0$ pointwise in $\mathbb{R}^n$ are called harmonic in $\mathbb{R}^n$. The Laplace operator arises in many applications such as in the modeling of heat conduction, electrical conduction, and chemical concentration, to name a few. The focus for us will be on finding all fundamental solutions for Δ that are tempered distributions. Part of the motivation is that, as explained in Remark 5.5, this greatly facilitates the study of the Poisson equation $\Delta u = f$ in $\mathcal{D}'(\mathbb{R}^n)$, a task we take up in Sect. 7.2.

In addition to treating the Laplace operator, in this chapter we also consider a number of other related operators, such as the bi-Laplacian, the poly-harmonic operator, the Cauchy–Riemann operator, the Dirac operator, as well as general constant coefficient (homogeneous) second-order strongly elliptic operators.

7.1 Fundamental Solutions for the Laplace Operator

The goal in this section is to determine all fundamental solutions of Δ that are also in $\mathcal{S}'(\mathbb{R}^n)$. This is done by employing the properties of the Fourier transform we have proved so far. To get started, fix some $E \in \mathcal{S}'(\mathbb{R}^n)$ that is a fundamental solution for Δ. Note that the existence of such a tempered distribution is guaranteed by Theorem 5.13. Since $\Delta E = \delta$ in $\mathcal{D}'(\mathbb{R}^n)$ and $\delta \in \mathcal{S}'(\mathbb{R}^n)$, by (4.1.25) it follows that

$$\Delta E = \delta \quad \text{in } \mathcal{S}'(\mathbb{R}^n). \tag{7.1.1}$$

[1] Named after the French mathematician and astronomer Pierre Simon de Laplace (1749–1827.)

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Applying $\mathcal{F}$ to the equation in (7.1.1) (and using that $-\Delta = \sum_{j=1}^{n} D_{x_j}^2$, part *(b)* in Theorem 4.25, and (4.2.3)) we obtain

$$-|\xi|^2 \widehat{E} = 1 \quad \text{in } \mathcal{S}'(\mathbb{R}^n). \tag{7.1.2}$$

To proceed with determining E, we discuss separately the cases $n \geq 3$, $n = 2$ and $n = 1$.

Case $n \geq 3$. From Exercise 4.4 we have that $\frac{1}{|\xi|^2} \in \mathcal{S}'(\mathbb{R}^n)$. In addition, $|\xi|^2 \in \mathcal{L}(\mathbb{R}^n)$, thus $|\xi|^2 \cdot \frac{1}{|\xi|^2} \in \mathcal{S}'(\mathbb{R}^n)$ [recall *(b)* in Theorem 4.13], and it is not difficult to check that $|\xi|^2 \cdot \frac{1}{|\xi|^2} = 1$ in $\mathcal{S}'(\mathbb{R}^n)$. Hence, $|\xi|^2\big(\widehat{E} + \frac{1}{|\xi|^2}\big) = 0$ in $\mathcal{S}'(\mathbb{R}^n)$. Thus, $\operatorname{supp}\big(\widehat{E} + \frac{1}{|\xi|^2}\big) \subseteq \{0\}$ and by Exercise 4.35, it follows that

$$\widehat{E} + \frac{1}{|\xi|^2} = \widehat{P}(\xi) \quad \text{in } \mathcal{S}'(\mathbb{R}^n), \tag{7.1.3}$$

where P is a polynomial in $\mathbb{R}^n$ satisfying $|\xi|^2 \widehat{P} = 0$ in $\mathcal{S}'(\mathbb{R}^n)$. The latter implies that $\Delta P = 0$ in $\mathcal{S}'(\mathbb{R}^n)$. Since $P \in C^\infty(\mathbb{R}^n)$, we may conclude that $\Delta P = 0$ pointwise in $\mathbb{R}^n$, hence P is a harmonic polynomial in $\mathbb{R}^n$. To compute E (which is equal to $\mathcal{F}^{-1}(\widehat{E})$) apply $\mathcal{F}^{-1}$ to the identity in (7.1.3) and then recall Proposition 4.61 with $\lambda = 2$ to write

$$\begin{aligned} E(x) &= -\mathcal{F}^{-1}\Big(\frac{1}{|\xi|^2}\Big)(x) + P(x) \\ &= -2^{-2}\pi^{-\frac{n}{2}} \frac{\Gamma\left(\frac{n-2}{2}\right)}{\Gamma(1)} |x|^{2-n} + P(x) \\ &= -\frac{1}{(n-2)\omega_{n-1}} \cdot \frac{1}{|x|^{n-2}} + P(x), \qquad \forall\, x \in \mathbb{R}^n \setminus \{0\}, \end{aligned} \tag{7.1.4}$$

where the last equality in (7.1.4) is also based on the fact that $\Gamma(1) = 1$ and $\Gamma\left(\frac{n}{2}\right) = \left(\frac{n}{2} - 1\right)\Gamma\left(\frac{n}{2} - 1\right)$, and on (13.5.6). Moreover,

$$\Delta\left(-\frac{1}{(n-2)\omega_{n-1}} \cdot \frac{1}{|x|^{n-2}}\right) = \delta \quad \text{in } \mathcal{S}'(\mathbb{R}^n), \tag{7.1.5}$$

since (7.1.5) is equivalent [via the Fourier transform, as a consequence of *(a)* and *(b)* in Theorem 4.25, as well as (4.2.3)] with $|\xi|^2 \mathcal{F}\Big(\frac{1}{(n-2)\omega_{n-1}} \cdot \frac{1}{|x|^{n-2}}\Big) = 1$ in $\mathcal{S}'(\mathbb{R}^n)$, or equivalently, with $|\xi|^2 \cdot \frac{1}{|\xi|^2} = 1$ in $\mathcal{S}'(\mathbb{R}^n)$, which we know to be true.

Case $n = 2$. Since $\frac{1}{|\xi|^2} \notin L^1_{loc}(\mathbb{R}^2)$, we cannot proceed as in the case $n \geq 3$. Instead, we rely on Theorem 4.73. Retaining the notation from Theorem 4.73, fix $\psi \in Q$ and observe that

$$|\xi|^2 w_\psi = 1 \quad \text{in } \mathcal{S}'(\mathbb{R}^2). \tag{7.1.6}$$

To see why this is true, first note that since $|\xi|^2 \in \mathcal{L}(\mathbb{R}^2)$, by *(b)* in Theorem 4.13 it follows that $|\xi|^2 w_\psi \in \mathcal{S}'(\mathbb{R}^2)$ while by part *(a)* in Theorem 3.13 we have $|\xi|^2\varphi \in \mathcal{S}(\mathbb{R}^2)$ for every $\varphi \in \mathcal{S}(\mathbb{R}^2)$. Hence, we may write

$$\begin{aligned}\langle |\xi|^2 w_\psi, \varphi\rangle = \langle w_\psi, |\xi|^2\varphi\rangle &= \int_{\mathbb{R}^n} \frac{|\xi|^2\varphi(\xi) - (|\xi|^2\varphi(\xi))\big|_{\xi=0}\psi(\xi)}{|\xi|^2}\,\mathrm{d}\xi \\ &= \int_{\mathbb{R}^2} \varphi(\xi)\,\mathrm{d}\xi = \langle 1, \varphi\rangle, \qquad \forall\,\varphi \in \mathcal{S}(\mathbb{R}^2),\end{aligned} \tag{7.1.7}$$

finishing the proof of (7.1.6).

A combination of (7.1.2) and (7.1.6) yields $|\xi|^2(\widehat{E} + w_\psi) = 0$ in $\mathcal{S}'(\mathbb{R}^2)$. Hence, by reasoning as in the case $n \geq 3$, the latter implies

$$\widehat{E} = -w_\psi + \widehat{P} \quad \text{ in } \mathcal{S}'(\mathbb{R}^n), \tag{7.1.8}$$

for some harmonic polynomial P. After applying the Fourier transform to the equality in (7.1.8) and recalling (4.6.4), we arrive at

$$E(x) = \frac{1}{2\pi}\ln|x| + P(x), \qquad \forall\, x \in \mathbb{R}^2 \setminus \{0\}. \tag{7.1.9}$$

In addition, by making use of the Fourier transform (recall *(a)* and *(c)* in Theorem 4.25, (4.2.3), and Proposition 4.30), Eq. (7.1.6) is equivalent with

$$-\Delta(\widehat{w_\psi}) = (2\pi)^2\delta \quad \text{ in } \mathcal{S}'(\mathbb{R}^2). \tag{7.1.10}$$

Combining (7.1.10) with (4.6.4), we arrive at

$$\Delta\Big(\tfrac{1}{2\pi}\ln|x|\Big) = \delta \quad \text{ in } \mathcal{S}'(\mathbb{R}^2). \tag{7.1.11}$$

Case $n = 1$. In this case we use Exercise 5.15 to conclude that $E = xH + c_1 x + c_0$ in $\mathcal{S}'(\mathbb{R})$, for some $c_0, c_1 \in \mathbb{C}$. Hence, E is of function type and we also have $E(x) = xH + c_1 x + c_0$, for every $x \in \mathbb{R}$.

Remark 7.1. *From Example 5.11 we know that* $-xH^\vee$ *is another fundamental solution for* Δ *in* $\mathcal{D}'(\mathbb{R})$. *This is in agreement with what we proved above since* $H + H^\vee = 1$ *in* $\mathcal{S}'(\mathbb{R})$; *thus* $-xH^\vee = xH + x$ *in* $\mathcal{S}'(\mathbb{R})$.

In summary, we have proved the following result.

Theorem 7.2. *The function* $E \in L^1_{loc}(\mathbb{R}^n)$ *defined as*

$$E(x) := \begin{cases} \dfrac{-1}{(n-2)\omega_{n-1}}\dfrac{1}{|x|^{n-2}} & \text{if } \ x \in \mathbb{R}^n \setminus \{0\},\ n \geq 3, \\[2ex] \frac{1}{2\pi}\ln|x| & \text{if } \ x \in \mathbb{R}^2 \setminus \{0\},\ n = 2, \\[2ex] xH(x) & \text{if } \ x \in \mathbb{R},\ n = 1, \end{cases} \tag{7.1.12}$$

belongs to $\mathcal{S}'(\mathbb{R}^n)$ *and is a fundamental solution*[2] *for the Laplace operator* Δ *in* $\mathbb{R}^n$. *Moreover,*

$$\begin{aligned}\big\{u\in\mathcal{S}'(\mathbb{R}^n):\,\Delta u=\delta\ \ in\ \ \mathcal{S}'(\mathbb{R}^n\big\}\\ =\big\{E+P:\,P\ \ harmonic\ polynomial\ in\ \ \mathbb{R}^n\big\}.\end{aligned}\tag{7.1.13}$$

Remark 7.3. *Note that, as a consequence of Proposition 5.1 and Example 4.3, for a harmonic function in* $\mathbb{R}^n$, *the respective function is a polynomial if and only if it is a tempered distribution.*

The collection of harmonic functions is strictly larger that the collection of harmonic polynomials. For example, the function $u(x,y):=e^x\cos y$ *for* $(x,y)\in\mathbb{R}^2$, *is harmonic in* $\mathbb{R}^2$ *without being a polynomial. In particular, we obtain that* $u\notin\mathcal{S}'(\mathbb{R}^2)$.

Moving on, we recall that if $\Omega\subset\mathbb{R}^n$ is a domain of class C^1 (see Definition 13.40) with outward unit normal $\nu=(\nu_1,\ldots,\nu_n)$ and u is a function of class C^1 in an open neighborhood of $\partial\Omega$, then the normal derivative of u on $\partial\Omega$, denoted by $\frac{\partial u}{\partial\nu}$, is the directional derivative of the function u along the unit vector ν, that is,

$$\frac{\partial u}{\partial\nu}(y):=\sum_{j=1}^{n}\nu_j(y)\partial_j u(y),\qquad\forall\,y\in\partial\Omega.\tag{7.1.14}$$

Remark 7.4. *Via a direct computation, one may check that* E *as in* (7.1.12) *defines a fundamental solution for the Laplacian. We outline the computation for* $n\geq 3$ *and leave the cases* $n=2$ *and* $n=1$ *as an exercise.*

First, observe that $E\in C^\infty(\mathbb{R}^n\setminus\{0\})$ *and* $\Delta E=0$ *in* $\mathbb{R}^n\setminus\{0\}$. *Next, let* $\varphi\in C_0^\infty(\mathbb{R}^n)$ *be such that* $\operatorname{supp}\varphi\subset B(0,R)$, *for some* $R\in(0,\infty)$. *Then, starting with the definition of distributional derivatives and then using Lebesgue's dominated convergence theorem, we write*

$$\begin{aligned}\langle\Delta E,\varphi\rangle&=\langle E,\Delta\varphi\rangle=\lim_{\varepsilon\to 0^+}\int_{\varepsilon\leq|x|\leq R}E(x)\Delta\varphi(x)\,dx\\ &=\lim_{\varepsilon\to 0^+}\bigg[\int_{\varepsilon\leq|x|\leq R}\Delta E(x)\varphi(x)\,dx+\int_{\partial B(0,R)}E\frac{\partial\varphi}{\partial\nu}\,d\sigma\\ &\quad-\int_{\partial B(0,\varepsilon)}E\frac{\partial\varphi}{\partial\nu}\,d\sigma-\int_{\partial B(0,R)}\frac{\partial E}{\partial\nu}\varphi\,d\sigma+\int_{\partial B(0,\varepsilon)}\frac{\partial E}{\partial\nu}\varphi\,d\sigma\bigg]\\ &=\lim_{\varepsilon\to 0^+}\bigg[\int_{\partial B(0,\varepsilon)}\bigg(\frac{\partial E}{\partial\nu}\varphi-E\frac{\partial\varphi}{\partial\nu}\bigg)\,d\sigma\bigg]=\varphi(0),\end{aligned}\tag{7.1.15}$$

[2] In the case $n=3$, the expression for E was used in 1789 by Pierre Simon de Laplace to show that for f smooth, compactly supported, $\Delta(E*f)=0$ outside the support of f (cf. [38]).

where for each integral considered over $\partial B(0,r)$, for some $r\in(0,\infty)$, ν denotes the outward unit normal to $B(0,r)$. For the third equality in (7.1.15) we used (13.7.5) while for the fourth equality in (7.1.15) we used the fact that the support of φ is contained inside $B(0,R)$. In addition, to see why the last equality in (7.1.15) holds, use the fact that $\nabla E(x)=\frac{1}{\omega_{n-1}}\cdot\frac{x}{|x|^n}$ for every $x\in\mathbb{R}^n\setminus\{0\}$ to write

$$\begin{aligned}\int_{\partial B(0,\varepsilon)}\frac{\partial E}{\partial\nu}(x)\,\varphi(x)\,d\sigma(x) &= \frac{1}{\omega_{n-1}\varepsilon^{n-1}}\int_{\partial B(0,\varepsilon)}\varphi(x)d\sigma(x) \\ &= \frac{1}{\omega_{n-1}\varepsilon^{n-1}}\int_{\partial B(0,\varepsilon)}[\varphi(x)-\varphi(0)]d\sigma(x)+\varphi(0) \\ &= \frac{1}{\omega_{n-1}}\int_{S^{n-1}}[\varphi(\varepsilon y)-\varphi(0)]d\sigma(y)+\varphi(0)\xrightarrow[\varepsilon\to0^+]{}\varphi(0)\end{aligned} \tag{7.1.16}$$

where the convergence follows by invoking Lebesgue's dominated convergence theorem. Also,

$$\begin{aligned}\left|\int_{\partial B(0,\varepsilon)}E(x)\frac{\partial\varphi}{\partial\nu}(x)\,d\sigma(x)\right| &\leq \frac{\|\nabla\varphi\|_{L^\infty}}{(n-2)\omega_{n-1}}\int_{\partial B(0,\varepsilon)}\frac{1}{|x|^{n-2}}\,d\sigma(x) \\ &= \varepsilon\,C(n,\varphi)\xrightarrow[\varepsilon\to0^+]{}0.\end{aligned} \tag{7.1.17}$$

This computation shows that if $n\geq3$, the $L^1_{loc}(\mathbb{R}^n)$ function E from (7.1.12) satisfies $\langle\Delta E,\varphi\rangle=\langle\delta,\varphi\rangle$ for every $\varphi\in C^\infty_0(\mathbb{R}^n)$; thus, E is a fundamental solution for the Laplacian.

We conclude this section by presenting a result relating the composition of Riesz transforms to singular integral operators which have kernels that are two derivatives on the fundamental solution for the Laplacian.

Proposition 7.5. *Let E be the fundamental solution for the Laplacian from (7.1.12) and recall the Riesz transforms in $\mathbb{R}^n$ (cf. Theorem 4.93). Then for each $j,k\in\{1,\dots,n\}$ and each $\varphi\in\mathcal{S}(\mathbb{R}^n)$, we have*

$$\begin{aligned}R_j\big(R_k\varphi\big)(x) &= -\Big(\frac{\omega_n}{2}\Big)^2\lim_{\varepsilon\to0^+}\int_{|x-y|>\varepsilon}(\partial_j\partial_kE)(x-y)\varphi(y)\,dy \\ &\quad-\Big(\frac{\omega_n}{2}\Big)^2\frac{\delta_{jk}}{n}\varphi(x)\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n),\end{aligned} \tag{7.1.18}$$

with the left-hand side initially understood in $L^2(\mathbb{R}^n)$ (cf. Theorem 4.93) and the right-hand side considered as a tempered distribution (cf. Proposition 4.67).

Moreover, if $T_{\partial_j\partial_kE}$ is the operator as in part (e) of Theorem 4.96 (for the choice $\Theta:=\partial_j\partial_kE$), then for every $f\in L^2(\mathbb{R}^n)$ we have

$$R_j\big(R_kf\big)=-\Big(\frac{\omega_n}{2}\Big)^2T_{\partial_j\partial_kE}f-\Big(\frac{\omega_n}{2}\Big)^2\frac{\delta_{jk}}{n}f\quad\text{in}\quad L^2(\mathbb{R}^n). \tag{7.1.19}$$

Proof. Fix $j,k\in\{1,\dots,n\}$ along with some $\varphi\in\mathcal{S}(\mathbb{R}^n)$. Since the Fourier transform is an isomorphism of $\mathcal{S}'(\mathbb{R}^n)$, it suffices to show that (7.1.18) holds on the Fourier transform side. With this in mind, note that on the one hand,

$$\mathcal{F}\big(R_j\big(R_k\varphi\big)\big)=-i\frac{\omega_n}{2}\frac{\xi_j}{|\xi|}\mathcal{F}\big(R_k\varphi\big)=-\Big(\frac{\omega_n}{2}\Big)^2\frac{\xi_j\xi_k}{|\xi|^2}\,\widehat{\varphi}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n),\tag{7.1.20}$$

by a twofold application of (4.9.14). On the other hand, since the function $\partial_j\partial_k E$ is as in (4.4.1) (here Example 4.68 is also used), we may invoke Proposition 4.67, (4.9.6), and (4.5.2), to conclude that the Fourier transform of the right-hand side of (7.1.18) is

$$\begin{aligned}
&-\Big(\frac{\omega_n}{2}\Big)^2\Big[\widehat{\varphi}\,\mathcal{F}\big(\text{P.V.}\,(\partial_j\partial_k E)\big)+\frac{\delta_{jk}}{n}\widehat{\varphi}\Big]\\
&\qquad=-\Big(\frac{\omega_n}{2}\Big)^2\Big[\frac{1}{\omega_{n-1}}\widehat{\varphi}\,\mathcal{F}\big(\text{P.V.}\,(\partial_j\Phi_k)\big)+\frac{\delta_{jk}}{n}\widehat{\varphi}\Big]\\
&\qquad=-\Big(\frac{\omega_n}{2}\Big)^2\Big[\Big(\frac{\xi_j\xi_k}{|\xi|^2}-\frac{\delta_{jk}}{n}\Big)\widehat{\varphi}+\frac{\delta_{jk}}{n}\widehat{\varphi}\Big]\\
&\qquad=-\Big(\frac{\omega_n}{2}\Big)^2\frac{\xi_j\xi_k}{|\xi|^2}\widehat{\varphi}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n).
\end{aligned}\tag{7.1.21}$$

That (7.1.18) holds now follows from (7.1.20) and (7.1.21).

Finally, (7.1.19) is a consequence of what we have proved so far, part *(e)* in Theorem 4.93, part *(e)* in Theorem 4.96, and the fact that $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$. □

Making use of the full force of Theorem 4.97 we obtain the following L^p version, $1<p<\infty$, of Proposition 7.5.

Proposition 7.6. *Let E be the fundamental solution for the Laplacian from (7.1.12) and recall the Riesz transforms in $\mathbb{R}^n$ (cf. Theorem 4.93). Also, fix $p\in(1,\infty)$. Then for each $j,k\in\{1,\dots,n\}$ and $f\in L^p(\mathbb{R}^n)$ we have*

$$\begin{aligned}
R_j\big(R_kf\big)(x)=&-\Big(\frac{\omega_n}{2}\Big)^2\lim_{\varepsilon\to0^+}\int_{|x-y|>\varepsilon}(\partial_j\partial_kE)(x-y)f(y)\,dy\\
&-\Big(\frac{\omega_n}{2}\Big)^2\frac{\delta_{jk}}{n}f(x)\quad\text{for a.e. }\ x\in\mathbb{R}^n,
\end{aligned}\tag{7.1.22}$$

where the Riesz transforms are understood to be bounded operators on $L^p(\mathbb{R}^n)$ (cf. (4.9.46)–(4.9.47)) while the singular integral operator on the right-hand side is also considered as a bounded mapping on $L^p(\mathbb{R}^n)$ (cf. Theorem 4.97).

In particular, if $T_{\partial_j\partial_kE}$ is interpreted in the sense of Theorem 4.97 (for the choice $\Theta:=\partial_j\partial_kE$), then for every $f\in L^p(\mathbb{R}^n)$ we have

$$R_j\big(R_kf\big)=-\Big(\frac{\omega_n}{2}\Big)^2T_{\partial_j\partial_kE}f-\Big(\frac{\omega_n}{2}\Big)^2\frac{\delta_{jk}}{n}f\quad\text{in}\quad L^p(\mathbb{R}^n).\tag{7.1.23}$$

Proof. All claims are consequences of Proposition 7.5, Theorem 4.97, and the density of $\mathcal{S}(\mathbb{R}^n)$ in $L^p(\mathbb{R}^n)$. □

7.2 The Poisson Equation and Layer Potential Representation Formulas

In this section we use the fundamental solutions for the Laplacian determined in the previous section in a number of applications. The first application concerns the Poisson equation[3] for the Laplacian in $\mathbb{R}^n$:

$$\Delta u = f \qquad \text{in } \mathbb{R}^n. \tag{7.2.1}$$

In (7.2.1) the unknown is u and f is a given function. As a consequence of Remark 5.5, this equation is always solvable in $\mathcal{D}'(\mathbb{R}^n)$ if $f \in C_0^\infty(\mathbb{R}^n)$. We will see that in fact the solution obtained via the convolution of such an f with the fundamental solution for Δ from (7.1.12) is smooth and that we can solve the Poisson equation in $\mathcal{D}'(\mathbb{R}^n)$ for a less restrictive class of functions f. The second application is an integral representation formula, involving layer potentials, for functions of class C^2 on a neighborhood of the closure of a domain of class C^1.

Proposition 7.7. *Let E be the fundamental solution for the Laplacian from (7.1.12). Suppose $f \in C_0^\infty(\mathbb{R}^n)$. Then the function*

$$u(x) := \int_{\mathbb{R}^n} E(x-y) f(y)\, dy, \qquad \forall\, x \in \mathbb{R}^n, \tag{7.2.2}$$

satisfies $u \in C^\infty(\mathbb{R}^n)$ and is a classical solution of the Poisson equation (7.2.1) for the Laplacian in $\mathbb{R}^n$.

Proof. Fix $f \in C_0^\infty(\mathbb{R}^n)$ and let E be as in (7.1.12). Define $u := E * f$ in $\mathcal{D}'(\mathbb{R}^n)$. Then Remark 5.5 implies that u is a solution of the equation $\Delta u = f$ in $\mathcal{D}'(\mathbb{R}^n)$. In addition, by Proposition 2.93 and the fact that $E \in L^1_{loc}(\mathbb{R}^n)$, we have $u \in C^\infty(\mathbb{R}^n)$ and

$$u(x) = \langle E(x-y), f(y)\rangle = \int_{\mathbb{R}^n} E(x-y) f(y)\, dy, \qquad \forall\, x \in \mathbb{R}^n, \tag{7.2.3}$$

or even more explicitly (based on the expressions in (7.1.12)), that

$$u(x) = \begin{cases} \dfrac{-1}{(n-2)\omega_{n-1}} \displaystyle\int_{\mathbb{R}^n} \frac{f(y)}{|x-y|^{n-2}}\, dy, & \text{if } n \geq 3, \\[2ex] \dfrac{1}{2\pi} \displaystyle\int_{\mathbb{R}^2} \ln|x-y| f(y)\, dy, & \text{if } n = 2, \\[2ex] \displaystyle\int_{\mathbb{R}} (x-y) H(x-y) f(y)\, dy, & \text{if } n = 1, \end{cases} \qquad \forall\, x \in \mathbb{R}^n. \tag{7.2.4}$$

Furthermore, based on Proposition 2.34, we may conclude that $\Delta u = f$ pointwise in $\mathbb{R}^n$. This proves that u is a solution of (7.7) as desired. □

[3] In 1813, the French mathematician, geometer, and physicist Siméon Denis Poisson (1781–1840) proved that $\Delta(E * f) = f$ in dimension $n = 3$ (cf. [55]), where $E(x) = -\frac{1}{4\pi|x|}$, $x \in \mathbb{R}^3 \setminus \{0\}$, and $f \in C_0^\infty(\mathbb{R}^3)$.

An inspection of the right-hand side of (7.2.2) reveals that this expression continues to be meaningful under weaker assumptions on f. This observation may be used to prove that (7.7) is solvable in the class of distributions for f belonging to a larger class of functions than $C_0^\infty(\mathbb{R}^n)$.

Proposition 7.8. *Let E be the fundamental solution for the Laplacian from (7.1.12). Suppose Ω is an open set in $\mathbb{R}^n$, $n \geq 2$, and $f \in L^\infty(\Omega)$ vanishes outside a bounded measurable subset of Ω. Then the function*

$$u(x) := \int_\Omega E(x-y)f(y)\,\mathrm{d}y, \qquad \forall\, x \in \Omega, \tag{7.2.5}$$

is a distributional solution of the Poisson equation for the Laplacian in Ω, i.e., $\Delta u = f$ in $\mathcal{D}'(\Omega)$. In addition, $u \in C^1(\Omega)$ and for each $j \in \{1,\dots,n\}$,

$$\partial_j u(x) = \int_\Omega (\partial_j E)(x-y)f(y)\,\mathrm{d}y, \qquad \forall\, x \in \Omega. \tag{7.2.6}$$

Proof. Suppose first that $n \geq 3$. Note that $E \in C^\infty(\mathbb{R}^n)$ and is positive homogeneous of degree $2-n$. Also the extension of f by zero outside Ω, which we continue to denote by f, satisfies $f \in L^\infty_{comp}(\mathbb{R}^n)$. From Theorem 4.101 it follows that $u \in C^1(\Omega)$ and (7.2.6) holds for each $j \in \{1,\dots,n\}$.

If $n = 2$, a reasoning based on Vitali's theorem, similar to the one used in the proof of Theorem 4.101, yields $(\partial_j E) * f \in C^0(\Omega)$ for each $j \in \{1,\dots,n\}$. Also, the computation in (4.10.33) may be adapted to give that the distributional derivative $\partial_j u$ is equal to $(\partial_j E)*f$ for each $j \in \{1,\dots,n\}$. Hence, Theorem 2.102 applies and yields $u \in C^1(\Omega)$, as desired.

To finish the proof of the proposition, there remains to show that $\Delta u = f$ in $\mathcal{D}'(\Omega)$. Since $f \in L^1_{loc}(\Omega)$ we have $f \in \mathcal{D}'(\Omega)$. Thus, for each $\varphi \in C_0^\infty(\Omega)$ we may write

$$\begin{aligned}
\langle \Delta u, \varphi\rangle = \langle u, \Delta\varphi\rangle &= \int_\Omega u(x)\Delta\varphi(x)\,\mathrm{d}x = \int_\Omega \Big(\int_\Omega E(x-y)f(y)\,\mathrm{d}y\Big)\Delta\varphi(x)\,\mathrm{d}x \\
&= \int_\Omega f(y)\Big(\int_\Omega E(x-y)\Delta\varphi(x)\,\mathrm{d}x\Big)\mathrm{d}y \\
&= \int_\Omega f(y)\Big(\int_{\Omega-y} E(x)\Delta\varphi(x+y)\,\mathrm{d}x\Big)\mathrm{d}y \\
&= \int_\Omega f(y)\big\langle E, \Delta\varphi(\cdot+y)\big\rangle\,\mathrm{d}y = \int_\Omega f(y)\big\langle \Delta E, \varphi(\cdot+y)\big\rangle\,\mathrm{d}y \\
&= \int_\Omega f(y)\langle \delta, \varphi(\cdot+y)\rangle\,\mathrm{d}y = \int_\Omega f(y)\varphi(y)\,\mathrm{d}y = \langle f, \varphi\rangle,
\end{aligned} \tag{7.2.7}$$

where for each $y \in \Omega$ we set $\Omega - y := \{x - y : x \in \Omega\}$. Note that for each $y \in \Omega$ fixed one has $\varphi(\cdot+y) \in C_0^\infty(\Omega-y)$, $0 \in \Omega - y$, and $E \in \mathcal{D}'(\Omega-y)$, thus the sixth equality in (7.2.7) is justified. The eight equality in (7.2.7) is a consequence of the fact that E is a fundamental solution for the Laplacian and that $0 \in \Omega - y$. This completes the proof of the proposition. □

Exercise 7.9. *Assume the hypothesis of Proposition 7.8 and recall the fundamental solution E for the Laplacian from (7.1.12). By using Vitali's Theorem 13.21, show that $u(x) = \int_\Omega E(\cdot - y)f(y)dy$ is differentiable in Ω (without making use of Theorem 2.102). As a by-product, show that, for each $j \in \{1, \ldots, n\}$,*

$$\partial_j u(x) = \int_\Omega \partial_j E(x-y) f(y)\, \mathrm{d}y, \qquad \forall\, x \in \Omega. \tag{7.2.8}$$

We next establish `mean value formulas` for `harmonic` functions. A clarification of the terminology employed is in order. Traditionally, the name harmonic function in an open set $\Omega \subseteq \mathbb{R}^n$ has been used for functions $u \in C^2(\Omega)$ with the property that $\sum_{j=1}^{n} \partial_j^2 u = 0$ in a pointwise sense, everywhere in Ω. Such a function is called a classical solution of the equation $\Delta u = 0$ in Ω.

Theorem 7.10. *For an open set $\Omega \subseteq \mathbb{R}^n$ the following are equivalent:*

(i) u is harmonic in Ω;

(ii) $u \in \mathcal{D}'(\Omega)$ and $\Delta u = 0$ in $\mathcal{D}'(\Omega)$;

(iii) $u \in C^\infty(\Omega)$ and $\Delta u = 0$ in a pointwise sense in Ω.

Proof. This is a direct consequence of Theorem 6.8 and Theorem 7.2 (or, alternatively, of Remark 6.14 and Corollary 6.18). □

Recall the symbol for integral averages from (0.0.14).

Theorem 7.11. *Let Ω be an open set in $\mathbb{R}^n$ and u be a harmonic function in Ω. Then for every $x \in \Omega$ and every $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$ we have*

$$u(x) = \fint_{B(x,r)} u(y)\, \mathrm{d}y \quad \text{and} \quad u(x) = \fint_{\partial B(x,r)} u(y)\, \mathrm{d}\sigma(y). \tag{7.2.9}$$

Proof. From Theorem 7.10 we know that $u \in C^\infty(\Omega)$ and $\Delta u = 0$ in a pointwise sense in Ω. Fix $x \in \Omega$ and define the function

$$\begin{aligned} &\phi : \big(0, \operatorname{dist}(x, \partial\Omega)\big) \to \mathbb{R}, \\ &\phi(r) := \fint_{\partial B(x,r)} u(y)\, \mathrm{d}\sigma(y), \quad \forall\, r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big). \end{aligned} \tag{7.2.10}$$

A change of variables gives that $\phi(r) = \frac{1}{\omega_{n-1}} \int_{S^{n-1}} u(x + r\omega)\, \mathrm{d}\sigma(\omega)$. Taking the derivative of ϕ and then using the integration by parts formula from Theorem 13.41 gives that

$$\begin{aligned} \phi'(r) &= \frac{1}{\omega_{n-1}} \int_{S^{n-1}} (\nabla u)(x + r\omega) \cdot \omega\, \mathrm{d}\sigma(\omega) \\ &= \frac{r}{\omega_{n-1}} \int_{B(0,1)} (\Delta u)(x + ry)\, \mathrm{d}y = \frac{r}{n} \fint_{B(x,r)} \Delta u(z)\, \mathrm{d}z. \end{aligned} \tag{7.2.11}$$

Since $\Delta u = 0$ in Ω we have that $\phi'(r) = 0$, thus $\phi(r) = C$ for some constant C. We claim that $\lim_{r\to 0^+} \phi(r) = u(x)$. Then this claim implies $C = u(x)$ and the first formula in (7.2.9) follows. To see the claim, fix $\varepsilon > 0$ and by the continuity of u at x find $\delta \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$ such that $|u(y) - u(x)| < \varepsilon$ if $|y - x| < \delta$. Consequently,

$$r \in (0, \delta) \implies \big|\phi(r) - u(x)\big| = \Big| \fint_{B(x,r)} [u(y) - u(x)]\, dy \Big| \leq \varepsilon, \tag{7.2.12}$$

as desired.

With the help of (13.5.6) and (13.8.5), the first formula in (7.2.9) may be rewritten as

$$\frac{\omega_{n-1} r^n}{n} u(x) = \int_{B(x,r)} u(y)\, d\sigma(y) = \int_0^r \int_{\partial B(x,\,\rho)} u(\omega)\, d\sigma(\omega)\, d\rho \tag{7.2.13}$$

for $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$. Differentiating the left- and rightmost terms in (7.2.13) with respect to r and then dividing by $\omega_{n-1} r^{n-1}$ gives the second formula in (7.2.9). □

Interior estimates for harmonic functions, as in (6.3.30), may be obtained as a particular case of Theorem 6.26 by observing that the fundamental solution for the Laplacian from (7.1.12) satisfies (6.3.29) (with $m = 2$). Below we take a slightly more direct route which also yields explicit constants.

Theorem 7.12 (Interior estimates for the Laplacian)**.** *Suppose u is harmonic in Ω. Then for each $x \in \Omega$, each $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, and each $k \in \mathbb{N}$, we have*

$$\max_{y \in \overline{B(x,r/2)}} |\partial^\alpha u(y)| \leq \frac{(2n)^k \mathrm{e}^{k-1} k!}{r^k} \max_{y \in \overline{B(x,r)}} |u(y)|, \qquad \forall\, \alpha \in \mathbb{N}_0^n \ \text{ with } \ |\alpha| = k. \tag{7.2.14}$$

Proof. Let $j \in \{1, \ldots, n\}$ be arbitrary and note that, by Theorem 7.10, we have $u \in C^\infty(\Omega)$ and hence, $\partial_j u$ is harmonic in Ω. Thus, by (7.2.9) and the integration by parts formula (13.7.4), for each $x \in \Omega$ and each $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, we may write

$$\begin{aligned} |\partial_j u(x)| &= \Big| \frac{n}{\omega_{n-1} r^n} \int_{B(x,r)} \partial_j u(y)\, dy \Big| \\ &= \Big| \frac{n}{\omega_{n-1} r^n} \int_{\partial B(x,r)} \frac{y_j - x_j}{r}\, u(y)\, d\sigma(y) \Big| \\ &\leq \frac{n}{r} \max_{y \in \overline{B(x,r)}} |u(y)|. \end{aligned} \tag{7.2.15}$$

With this in hand, Lemma 6.21 applies (with $\mathcal{A}$ the class of harmonic functions in Ω and $C := n$) and yields (7.2.14). □

Theorem 7.13. *Any harmonic function in Ω is real-analytic in Ω.*

Proof. This is an immediate consequence of Theorem 7.12 and Lemma 6.24 (or, alternatively, Theorem 6.26). □

Corollary 7.14. *Suppose u is harmonic in an open, connected set $\Omega \subseteq \mathbb{R}^n$ with the property that there exists $x_0 \in \Omega$ such that $\partial^\alpha u(x_0) = 0$ for all $\alpha \in \mathbb{N}_0^n$. Then u vanishes identically in Ω.*

As a consequence, if a harmonic function defined in an open connected set $\Omega \subseteq \mathbb{R}^n$ vanishes on an open subset of Ω, then the respective function vanishes identically in Ω.

Proof. This is an immediate consequence of Theorem 6.25 and Theorem 7.13. □

Theorem 7.15 (Liouville's theorem for the Laplacian)**.** *If u is a bounded harmonic function in $\mathbb{R}^n$ then there exists $c \in \mathbb{C}$ such that $u = c$ in $\mathbb{R}^n$.*

Proof. This may be justified in several ways. For example, it suffices to note that this is a particular case of Theorem 5.3. Another proof may be given based on interior estimates. Specifically, let $j \in \{1, \ldots, n\}$ and using (7.2.14), for each $x \in \mathbb{R}^n$, we may write

$$\lim_{r\to\infty} |\partial_j u(x)| \leq \lim_{r\to\infty} \frac{n}{r} \|u\|_{L^\infty(\mathbb{R}^n)} = 0. \tag{7.2.16}$$

Hence, $\nabla u = 0$ proving that u is locally constant in $\mathbb{R}^n$. Since $\mathbb{R}^n$ is connected, the desired conclusion follows. □

All ingredients are now in place for proving the following basic well-posedness result for the Poisson problem for the Laplacian in $\mathbb{R}^n$.

Theorem 7.16. *Assume $n \geq 3$. Then for each $f \in L^\infty_{comp}(\mathbb{R}^n)$ and each $c \in \mathbb{C}$ the Poisson problem for the Laplacian in $\mathbb{R}^n$,*

$$\begin{cases} u \in C^0(\mathbb{R}^n), \\ \Delta u = f \text{ in } \mathcal{D}'(\mathbb{R}^n), \\ \lim\limits_{|x|\to\infty} u(x) = c, \end{cases} \tag{7.2.17}$$

has a unique solution. Moreover, the solution u satisfies the following additional properties.

(1) The function u belongs to $C^1(\mathbb{R}^n)$ and has the integral representation formula

$$u(x) = c + \int_{\mathbb{R}^n} E(x-y) f(y)\, \mathrm{d}y, \qquad x \in \mathbb{R}^n, \tag{7.2.18}$$

where E is the fundamental solution for the Laplacian in $\mathbb{R}^n$ from (7.1.12).

(2) If, in fact, $f \in C^\infty_0(\mathbb{R}^n)$, then actually $u \in C^\infty(\mathbb{R}^n)$.

(3) For every $j,k\in\{1,\dots,n\}$, we have

$$\partial_j\partial_k u=-\Big(\frac{2}{\omega_n}\Big)^2 R_j\big(R_k f\big)\quad\text{in}\quad\mathcal{D}'(\mathbb{R}^n),\tag{7.2.19}$$

where R_j and R_k are (jth and kth) Riesz transforms in $\mathbb{R}^n$.

(4) For every $p\in(1,\infty)$, the solution u of (7.2.17) satisfies $\partial_j\partial_k u\in L^p(\mathbb{R}^n)$ for each $j,k\in\{1,\dots,n\}$, where the derivatives are taken in $\mathcal{D}'(\mathbb{R}^n)$, and there exists a constant $C=C(p,n)\in(0,\infty)$ with the property that

$$\sum_{j,k=1}^{n}\|\partial_j\partial_k u\|_{L^p(\mathbb{R}^n)}\leq C\|f\|_{L^p(\mathbb{R}^n)}.\tag{7.2.20}$$

Proof. The fact that the function u defined as in (7.2.18) is of class $C^1(\mathbb{R}^n)$ and solves $\Delta u=f$ in $\mathcal{D}'(\mathbb{R}^n)$ has been established in Proposition 7.8. To proceed, let $R\in(0,\infty)$ be such that $\operatorname{supp}f\subset B(0,R)$, and note that if $|x|\geq 2R$, then for every $y\in B(0,R)$ we have $|x-y|\geq|x|-|y|\geq R\geq|y|$. Hence, using (7.1.12) (recall that we are assuming $n\geq 3$) and Lebesgue's dominated convergence theorem we obtain

$$\Big|\int_{\mathbb{R}^n}E(x-y)f(y)\,\mathrm{d}y\Big|\leq C\|f\|_{L^\infty(\mathbb{R}^n)}\int_{B(0,R)}\frac{\mathrm{d}y}{|x-y|^{n-2}}\to 0\quad\text{as}\quad|x|\to\infty.\tag{7.2.21}$$

It is then clear from (7.2.21) that the function u from (7.2.18) also satisfies the limit condition in (7.2.17). That (7.2.18) is the unique solution of (7.2.17) follows from linearity and Theorem 7.15. Next, that $u\in C^\infty(\mathbb{R}^n)$ if $f\in C_0^\infty(\mathbb{R}^n)$, is a consequence of Proposition 7.7 (or, alternatively, Corollary 6.18 and the ellipticity of the Laplacian). This proves *(1)–(2)*. As regards *(3)*, start by fixing $j,k\in\{1,\dots,n\}$. Then from (7.2.6), Theorem 4.99, (13.8.44), and Proposition 7.6, we deduce that

$$\begin{aligned}(\partial_j\partial_k u)(x)&=\partial_j\big(\partial_k u(x)\big)=\partial_j\Big[\int_{\mathbb{R}^n}(\partial_k E)(x-y)f(y)\,\mathrm{d}y\Big]\\&=\frac{1}{\omega_{n-1}}\Big(\int_{S^{n-1}}\omega_k\omega_j\,\mathrm{d}\sigma(\omega)\Big)f(x)+\lim_{\varepsilon\to 0^+}\int_{|x-y|>\varepsilon}(\partial_j\partial_k E)(x-y)f(y)\,\mathrm{d}y\\&=\frac{\delta_{jk}}{n}f(x)+\lim_{\varepsilon\to 0^+}\int_{|x-y|>\varepsilon}(\partial_j\partial_k E)(x-y)f(y)\,\mathrm{d}y\\&=-\Big(\frac{2}{\omega_n}\Big)^2 R_j\big(R_k f\big)(x)\quad\text{in}\quad\mathcal{D}'(\mathbb{R}^n).\end{aligned}\tag{7.2.22}$$

Finally, the claim in *(4)* follows from *(3)* and the boundedness of the Riesz transforms on $L^p(\mathbb{R}^n)$ (cf. (4.9.46)). □

In the last part of this section, we prove an integral representation formula involving layer potentials (a piece of terminology we elaborate on a little later) for arbitrary (as opposed to harmonic) functions $u \in C^2(\overline{\Omega})$, where Ω is a bounded domain of class C^1 in $\mathbb{R}^n$ (as in Definition 13.40). To state this recall the definition of the normal derivative from (7.1.14).

Proposition 7.17. *Suppose $n \geq 2$ and let $\Omega \subset \mathbb{R}^n$ be a bounded domain of class C^1 and let ν denote its outward unit normal. Also let E be the fundamental solution for the Laplacian from (7.1.12). If $u \in C^2(\overline{\Omega})$, then*

$$\begin{aligned}
&\int_\Omega \Delta u(y) E(x-y)\,dy - \int_{\partial\Omega} E(x-y)\frac{\partial u}{\partial \nu}(y)\,d\sigma(y) \\
&\quad - \int_{\partial\Omega} [\nu(y)\cdot(\nabla E)(x-y)]u(y)\,d\sigma(y) \\
&\quad = \begin{cases} u(x), & x \in \Omega, \\ 0, & x \in \mathbb{R}^n \setminus \overline{\Omega}. \end{cases}
\end{aligned} \tag{7.2.23}$$

Several strategies may be employed to prove Proposition 7.17, and here we choose an approach that highlights the role of distribution theory. Another approach (based on isolating the singularity in the fundamental solution and a limiting argument), more akin to the proof of (7.1.15), is going to be used in the proof of Theorem 7.57 (presented later) where a result of similar flavor to (7.2.23) is established for more general differential operators than the Laplacian.

Proof of Proposition 7.17. In this proof, for a function v defined in Ω, we let $\widetilde{v}$ denote the extension of v with zero outside Ω. Fix some function $u \in C^2(\overline{\Omega})$ and observe that $\widetilde{u} \in L^1_{loc}(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$. In addition, for each $\varphi \in C^\infty_0(\mathbb{R}^n)$ one has

$$\langle \Delta\widetilde{u}, \varphi\rangle = \langle \widetilde{u}, \Delta\varphi\rangle = \int_\Omega u\Delta\varphi\,dx = \int_\Omega \varphi\Delta u\,dx + \int_{\partial\Omega}\left(u\frac{\partial\varphi}{\partial\nu} - \varphi\frac{\partial u}{\partial\nu}\right)d\sigma, \tag{7.2.24}$$

where the last equality in (7.2.24) is based on (13.7.5).

For $a \in C^0(\partial\Omega)$ define the mappings $a\delta_{\partial\Omega}, \frac{\partial}{\partial\nu}(a\delta_{\partial\Omega}) : C^\infty_0(\mathbb{R}^n) \to \mathbb{C}$ by setting

$$\langle a\delta_{\partial\Omega}, \varphi\rangle := \int_{\partial\Omega} a(x)\varphi(x)\,d\sigma(x), \qquad \forall\,\varphi \in C^\infty_0(\mathbb{R}^n), \tag{7.2.25}$$

and

$$\left\langle \frac{\partial}{\partial\nu}(a\delta_{\partial\Omega}), \varphi\right\rangle := -\int_{\partial\Omega} a(x)\frac{\partial\varphi}{\partial\nu}(x)\,d\sigma(x), \qquad \forall\,\varphi \in C^\infty_0(\mathbb{R}^n). \tag{7.2.26}$$

By Exercise 2.128 corresponding to $\Sigma := \partial\Omega$ we have $a\delta_{\partial\Omega} \in \mathcal{D}'(\mathbb{R}^n)$. By a similar reasoning, one also has $\frac{\partial}{\partial\nu}(a\delta_{\partial\Omega}) \in \mathcal{D}'(\mathbb{R}^n)$. In addition, it is easy to see that the supports of the distributions in (7.2.25) and (7.2.26) are contained

in $\partial\Omega$, hence $a\delta_{\partial\Omega}$, $\frac{\partial}{\partial\nu}(a\delta_{\partial\Omega}) \in \mathcal{E}'(\mathbb{R}^n)$. In light of these definitions, (7.2.24) is equivalent with

$$\Delta\widetilde{u} = \widetilde{\Delta u} - \frac{\partial u}{\partial\nu}\delta_{\partial\Omega} - \frac{\partial}{\partial\nu}(u\delta_{\partial\Omega}) \qquad \text{in } \mathcal{D}'(\mathbb{R}^n). \tag{7.2.27}$$

Note that the supports of the distributions $\Delta\widetilde{u}$ and $\widetilde{\Delta u}$ are contained in the compact set $\overline{\Omega}$. Hence, all distributions in (7.2.27) are compactly supported and their convolutions with E are well-defined. Furthermore, since

$$\Delta\widetilde{u} * E = \widetilde{u} * \Delta E = \widetilde{u} * \delta = \widetilde{u} \quad \text{in } \mathcal{D}'(\mathbb{R}^n),$$

after convolving the left- and right-hand sides of (7.2.27) with E we arrive at

$$\widetilde{u} = \widetilde{\Delta u} * E - \Big(\frac{\partial u}{\partial\nu}\delta_{\partial\Omega}\Big) * E - \Big(\frac{\partial}{\partial\nu}(u\delta_{\partial\Omega})\Big) * E \quad \text{in } \mathcal{D}'(\mathbb{R}^n). \tag{7.2.28}$$

Moreover, since $\mathbb{R}^n \setminus \partial\Omega$ is open, it follows that the equality in (7.2.28) also holds in $\mathcal{D}'(\mathbb{R}^n \setminus \partial\Omega)$. The goal is to show that all distributions in (7.2.28) when restricted to $\mathbb{R}^n \setminus \partial\Omega$ are of function type and to determine the respective functions that define them.

The fact that the distribution $\widetilde{u}\big|_{\mathbb{R}^n\setminus\partial\Omega}$ is of function type follows from the definition of $\widetilde{u}$. Also, it is immediate that $\widetilde{u}\big|_{\mathbb{R}^n\setminus\partial\Omega}$ is given by the function equal to u in Ω and equal to zero in $\mathbb{R}^n \setminus \overline{\Omega}$, which is precisely the function in the right-hand side of (7.2.23).

Since $\widetilde{\Delta u}$, $E \in L^1_{loc}(\mathbb{R}^n)$, with $\widetilde{\Delta u}$ being compactly supported in $\overline{\Omega}$, by Exercise 2.95 we have that $\widetilde{\Delta u} * E$ is of function type, determined by the function $\widetilde{\Delta u} * E \in L^1_{loc}(\mathbb{R}^n)$ defined as

$$(\widetilde{\Delta u} * E)(x) = \int_\Omega E(x-y)\Delta u(y)\mathrm{d}y, \qquad \forall\, x \in \mathbb{R}^n. \tag{7.2.29}$$

To proceed with identifying the restrictions to $\mathbb{R}^n \setminus \partial\Omega$ of the other two convolutions in the right-hand side of (7.2.28), let $\varphi \in C^\infty_0(\mathbb{R}^n \setminus \partial\Omega)$ and set $K := \operatorname{supp}\varphi$. Then, the set

$$M_K := \big\{(x,y) \in \mathbb{R}^n \times \partial\Omega : \, x + y \in K\big\}$$

is compact in $\mathbb{R}^{2n}$ and if we take $\psi \in C^\infty_0(\mathbb{R}^{2n})$ such that $\psi = 1$ in a neighborhood of M_K, we have

$$\Big\langle \Big[\Big(\frac{\partial u}{\partial\nu}\Big)\delta_{\partial\Omega}\Big] * E, \varphi\Big\rangle = \Big\langle \Big(\frac{\partial u}{\partial\nu}\delta_{\partial\Omega}\Big)(y), \langle E(x), \psi(x,y)\varphi(x+y)\rangle\Big\rangle \tag{7.2.30}$$

$$= \Big\langle \Big(\frac{\partial u}{\partial\nu}\delta_{\partial\Omega}\Big)(y), \int_{\mathbb{R}^n} E(x)\psi(x,y)\varphi(x+y)\,\mathrm{d}x\Big\rangle.$$

Note that via a change of variables we have

$$\int_{\mathbb{R}^n} E(x)\psi(x,y)\varphi(x+y)\,\mathrm{d}x = \int_{\mathbb{R}^n} E(z-y)\psi(z-y,y)\varphi(z)\,\mathrm{d}z, \quad \forall\, y \in \mathbb{R}^n. \tag{7.2.31}$$

If we now consider the function

$$h(y) := \int_{\mathbb{R}^n} E(z-y)\psi(z-y,y)\varphi(z)\,\mathrm{d}z, \quad \forall\, y\in\mathbb{R}^n, \tag{7.2.32}$$

then h is well-defined. Moreover, since derivatives of order 2 or higher of E are not integrable near the origin, it follows that h does not belong to $C^\infty(\mathbb{R}^n)$, hence (7.2.31) may not be used to rewrite the last term in (7.2.30). To fix this drawback, we impose an additional restriction on ψ. Specifically, since $\partial\Omega\cap K=\emptyset$, there exists $\varepsilon>0$ sufficiently small such that $(\partial\Omega+B(0,\varepsilon))\cap(K+B(0,\varepsilon))=\emptyset$. Then, if U, the neighborhood of M_K where $\psi=1$, is such that whenever $(x,y)\in U$ we have $y\in\partial\Omega+B(0,\varepsilon/2)$, we may further require that $\psi(\cdot,y)=0$ for $y\in K+B(0,\varepsilon/2)$. Under this requirement, if $z\in K=\operatorname{supp}\varphi$ and $y\in\mathbb{R}^n$ is such that $|z-y|<\varepsilon/2$, then $\psi(z-y,y)=0$. This ensures that derivatives in y of any order of the function $E(z-y)\psi(z-y,y)\varphi(z)$ are integrable in z over $\mathbb{R}^n$, thus $h\in C^\infty(\mathbb{R}^n)$. Now (7.2.31) may be used in the last term in (7.2.30) to obtain

$$\Big\langle\Big[\Big(\frac{\partial u}{\partial\nu}\Big)\delta_{\partial\Omega}\Big]*E,\varphi\Big\rangle=\Big\langle\Big(\frac{\partial u}{\partial\nu}\delta_{\partial\Omega}\Big)(y),\int_{\mathbb{R}^n}E(z-y)\psi(z-y,y)\varphi(z)\,\mathrm{d}z\Big\rangle. \tag{7.2.33}$$

Since $\frac{\partial u}{\partial\nu}\in C^0(\partial\Omega)$, by (7.2.25) one has

$$\begin{aligned}
&\Big\langle\Big(\frac{\partial u}{\partial\nu}\delta_{\partial\Omega}\Big)(y),\int_{\mathbb{R}^n}E(x-y)\psi(x-y,y)\varphi(x)\,\mathrm{d}x\Big\rangle\\
&\quad=\int_{\partial\Omega}\frac{\partial u}{\partial\nu}(y)\Big[\int_{\mathbb{R}^n}\varphi(x)E(x-y)\psi(x-y,y)\,\mathrm{d}x\Big]\,\mathrm{d}\sigma(y)\\
&\quad=\int_{\mathbb{R}^n}\Big[\int_{\partial\Omega}\frac{\partial u}{\partial\nu}(y)E(x-y)\,\mathrm{d}\sigma(y)\Big]\varphi(x)\,\mathrm{d}x.
\end{aligned}\tag{7.2.34}$$

Combining (7.2.33) and (7.2.34) it follows that $\big(\big[\big(\frac{\partial u}{\partial\nu}\big)\,\delta_{\partial\Omega}\big]*E\big)\,\big|_{\mathbb{R}^n\setminus\partial\Omega}$ is of function type and

$$\left(\left[\left(\frac{\partial u}{\partial\nu}\right)\delta_{\partial\Omega}\right]*E\right)(x)=\int_{\partial\Omega}\frac{\partial u}{\partial\nu}(y)E(x-y)\,\mathrm{d}\sigma(y),\qquad\forall\,x\in\mathbb{R}^n\setminus\partial\Omega. \tag{7.2.35}$$

Similarly, with φ and ψ as previously specified, since $u\in C^0(\partial\Omega)$, definition (7.2.26) applies and we obtain

$$\begin{aligned}
\Big\langle\Big(\frac{\partial}{\partial\nu}(u\delta_{\partial\Omega})\Big)*E,\varphi\Big\rangle&=\Big\langle\frac{\partial}{\partial\nu}(u\delta_{\partial\Omega})(y),\int_{\mathbb{R}^n}E(x)\psi(x,y)\varphi(x+y)\,\mathrm{d}x\Big\rangle\\
&=\Big\langle\frac{\partial}{\partial\nu}(u\delta_{\partial\Omega})(y),\int_{\mathbb{R}^n}E(x-y)\psi(x-y,y)\varphi(x)\,\mathrm{d}x\Big\rangle\\
&=-\int_{\partial\Omega}u(y)\int_{\mathbb{R}^n}\varphi(x)\left[\nu(y)\cdot\frac{\partial}{\partial y}(E(x-y))\right]\mathrm{d}x\,\mathrm{d}\sigma(y)
\end{aligned}$$

$$= \int_{\mathbb{R}^n} \varphi(x) \int_{\partial\Omega} u(y)[\nu(y)\cdot(\nabla E)(x-y)]\,d\sigma(y)\,dx. \tag{7.2.36}$$

Hence $\Big(\Big(\frac{\partial}{\partial\nu}(u\delta_{\partial\Omega})\Big) * E\Big)\Big|_{\mathbb{R}^n\setminus\partial\Omega}$ is of function type and

$$\Big(\Big(\frac{\partial}{\partial\nu}(u\delta_{\partial\Omega})\Big) * E\Big)(x) \tag{7.2.37}$$

$$= \int_{\partial\Omega} u(y)[\nu(y)\cdot(\nabla E)(x-y)]\,d\sigma(y), \qquad \forall\, x\in\mathbb{R}^n\setminus\partial\Omega.$$

Combining (7.2.28), earlier comments about $\widetilde{u}\big|_{\mathbb{R}^n\setminus\partial\Omega}$, (7.2.29), (7.2.35), and (7.2.37), we may conclude that

$$\widetilde{u}(x) = \int_\Omega E(x-y)\Delta u(y)\,dy - \int_{\partial\Omega} E(x-y)\frac{\partial u}{\partial\nu}(y)\,d\sigma(y)$$

$$- \int_{\partial\Omega}[\nu(y)\cdot(\nabla E)(x-y)]u(y)\,d\sigma(y), \qquad \forall\, x\in\mathbb{R}^n\setminus\partial\Omega, \tag{7.2.38}$$

proving (7.2.23). □

As the observant reader has perhaps noticed, the solid integral appearing in (7.2.23) is simply the volume (or Newtonian) potential defined in (4.10.1) acting on $f := \Delta u$. Starting from this observation, formula (7.2.23) also suggests the consideration of two other types of integral operators associated with a given bounded domain $\Omega\subset\mathbb{R}^n$. Specifically, for each given complex-valued function $\varphi\in C^0(\partial\Omega)$ set

$$\big(\mathscr{D}\varphi\big)(x) := -\int_{\partial\Omega}\big[\nu(y)\cdot(\nabla E)(x-y)\big]\varphi(y)\,d\sigma(y)$$

$$= \frac{1}{\omega_{n-1}}\int_{\partial\Omega}\frac{\nu(y)\cdot(y-x)}{|x-y|^n}\varphi(y)\,d\sigma(y), \qquad x\in\mathbb{R}^n\setminus\partial\Omega, \tag{7.2.39}$$

and $\big(\mathscr{S}\varphi\big)(x) := \int_{\partial\Omega} E(x-y)\varphi(y)\,d\sigma(y)$, for $x\in\mathbb{R}^n\setminus\partial\Omega$. Thus,

$$\big(\mathscr{S}\varphi\big)(x) = \begin{cases} \frac{-1}{(n-2)\omega_{n-1}}\int_{\partial\Omega}\frac{1}{|x-y|^{n-2}}\varphi(y)\,d\sigma(y) & \text{if } n\geq 3,\\[2ex] \frac{1}{2\pi}\int_{\partial\Omega}\ln|y-x|\varphi(y)\,d\sigma(y) & \text{if } n=2. \end{cases} \tag{7.2.40}$$

The operators $\mathscr{D}$ and $\mathscr{S}$ are called, respectively, the double- and single-layer potentials for the Laplacian. In this notation, (7.2.23) reads

$$\big(\mathbf{N}_\Omega(\Delta u)\big)(x) - \mathscr{S}\Big(\frac{\partial u}{\partial\nu}\Big)(x) + \mathscr{D}\big(u\big|_{\partial\Omega}\big)(x) = \begin{cases} u(x), & x\in\Omega,\\ 0, & x\in\mathbb{R}^n\setminus\overline{\Omega}, \end{cases} \tag{7.2.41}$$

for every $u\in C^2(\overline{\Omega}\,)$. In particular, this formula shows that we may recover any function $u\in C^2(\overline{\Omega}\,)$ knowing Δu in Ω as well as $\frac{\partial u}{\partial\nu}$ and u on $\partial\Omega$.

The above layer potential operators play a crucial role in the treatment of boundary value problems. In this vein, we invite the reader to check that the version of the double-layer operator (7.2.39) corresponding to the case when $\Omega=\mathbb{R}^n_+$ is precisely twice the operator $\mathscr{P}$ from (4.8.3), the relevance of which, in the context of boundary value problems for the Laplacian, has been highlighted in (4.8.17)–(4.8.18).

7.3 Fundamental Solutions for the Bi-Laplacian

The `bi-Laplacian` in $\mathbb{R}^n$ is the operator $\Delta^2=\Delta\Delta=\big(\sum_{j=1}^{n}\partial_j^2\big)^2$, sometimes also referred to as the `bi-harmonic operator`. Functions u of class C^4 in an open set Ω of $\mathbb{R}^n$ satisfying $\Delta^2 u=0$ pointwise in Ω are called bi-harmonic in Ω.

Theorem 5.13 guarantees the existence of $E\in\mathcal{S}'(\mathbb{R}^n)$ such that $\Delta^2 E=\delta$ in $\mathcal{S}'(\mathbb{R}^n)$. The goal in this section is to determine all such fundamental solution for Δ^2. In the case when $n\geq 4$, these may be computed by following the approach from Sect. 7.1 (corresponding to $n\geq 2$ there). Such a line of reasoning will then require treating the cases $n=1,2,3$ via another method. In what follows, we will proceed differently by employing the fact that we have a complete description of all fundamental solutions in $\mathcal{S}'(\mathbb{R}^n)$ for the Laplacian and that $\Delta^2=\Delta\Delta$. This latter approach, will take care of the cases $n\geq 2$, leaving $n=1$ to be treated separately.

Fix $E\in\mathcal{S}'(\mathbb{R}^n)$ satisfying $\Delta^2 E=\delta$ in $\mathcal{D}'(\mathbb{R}^n)$. By (4.1.25) we have $\Delta^2 E=\delta$ in $\mathcal{S}'(\mathbb{R}^n)$. Keeping in mind that $\Delta^2=\Delta\Delta$, it is natural to consider the following equation:

$$\Delta E=E_\Delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{7.3.1}$$

where E_Δ is the fundamental solution for the Laplacian from (7.1.12). Note that any E as in (7.3.1) is a fundamental solution for Δ^2. We proceed by analyzing three cases.

Case $n\geq 3$. Under the current assumptions, $E_\Delta(x)=-\frac{1}{(n-2)\omega_{n-1}}|x|^{2-n}$ for each $x\in\mathbb{R}^n\setminus\{0\}$. The key observation is that the following identity holds

$$\Delta\big[|x|^m\big]=m(m+n-2)|x|^{m-2},\qquad \text{pointwise in } \mathbb{R}^n\setminus\{0\},\quad \forall\, m\in\mathbb{R}. \tag{7.3.2}$$

This suggests investigating the validity of a similar identity when derivatives are taken in the distributional sense. As seen in the next lemma, the proof of which we postpone for later, a version of identity (7.3.2) holds in $\mathcal{S}'(\mathbb{R}^n)$ for a suitable range of exponents that depends on the dimension n.

Lemma 7.18. *Let N be a real number such that $N>2-n$. Then*

$$\Delta\big[|x|^N\big]=N(N+n-2)|x|^{N-2}\qquad \text{in}\qquad \mathcal{S}'(\mathbb{R}^n). \tag{7.3.3}$$

In view of Lemma 7.18 and the definition of E_Δ, it is justified to look for a solution to (7.3.1) of the form $E(x)=c_n|x|^{4-n}$, $x\in\mathbb{R}^n\setminus\{0\}$, where c_n is a constant

to be determined. This candidate is in $\mathcal{S}'(\mathbb{R}^n)$ (recall Exercise 4.4) and satisfies (7.3.1) if $2c_n(4-n) = -\frac{1}{(n-2)\omega_{n-1}}$. Hence, we obtain $c_n = \frac{1}{2(n-2)(n-4)\omega_{n-1}}$ if $n \neq 4$.

To handle the case $n = 4$, we use another result that we state next (for a proof see the last part of this section).

Lemma 7.19. *Let $n \geq 3$. Then $\Delta\big[\ln|x|\big] = (n-2)|x|^{-2}$ in $\mathcal{S}'(\mathbb{R}^n)$.*

Lemma 7.19 suggests to take when $n = 4$ the candidate $E(x) = c\ln|x|$, $x \in \mathbb{R}^n \setminus \{0\}$, where c is a constant to be determined. From Example 4.8 we know that $E \in \mathcal{S}'(\mathbb{R}^n)$. Also, Lemma 7.19 and the expression of E_Δ corresponding to $n = 4$ imply that $E(x) = c\ln|x|$ satisfies (7.3.1) if $2c = -\frac{1}{2\omega_3}$. Since $\omega_3 = 2\pi^2$ (recall (13.5.6) and (13.5.3)), the latter implies $c = -\frac{1}{8\pi^2}$.

In summary, we proved that

$$E(x) = \begin{cases} \dfrac{1}{2(n-2)(n-4)\omega_{n-1}}|x|^{4-n} & \text{for } n \geq 3,\ n \neq 4, \\ -\frac{1}{8\pi^2}\ln|x| & \text{for } n = 4, \end{cases} \qquad x \in \mathbb{R}^n \setminus \{0\},$$

$$\text{satisfies } E \in \mathcal{S}'(\mathbb{R}^n) \text{ and } \Delta^2 E = \delta \text{ in } \mathcal{S}'(\mathbb{R}^n),\ n \geq 3. \tag{7.3.4}$$

The treatment of the case $n = 2$ is different from the above considerations since, in such a setting, $E_\Delta(x) = \frac{1}{2\pi}\ln|x|$ for $x \in \mathbb{R}^2 \setminus \{0\}$. Given the format of E_Δ, for some insight on how to choose a candidate E we start with the readily verified identity

$$\Delta\big[|x|^m \ln|x|\big] = |x|^{m-2}\big[m(m+n-2)\ln|x| + 2m + n - 2\big] \quad \text{pointwise in } \mathbb{R}^n \setminus \{0\},\ \forall\, m \in \mathbb{R}. \tag{7.3.5}$$

The function $|x|^a$ belongs to $\mathcal{L}(\mathbb{R}^n)$ only when $a \in \mathbb{N}_0$ and is even, in which case $|x|^a \ln|x| \in \mathcal{S}'(\mathbb{R}^n)$ (by Example 4.8 and *(b)* in Theorem 4.13). Hence, these restrictions should be taken into account when considering the analogue of identity (7.3.5) in the distributional sense in $\mathcal{S}'(\mathbb{R}^n)$. The latter is stated next and its proof, which is very much in the spirit of the proofs for Lemmas 7.18 and 7.19 (here Exercise 4.104 is also relevant), is left as an exercise.

Lemma 7.20. *Let N be a real number such that $N > 2 - n$. Then*

$$\Delta\big[|x|^N \ln|x|\big] = |x|^{N-2}\big[N(N+n-2)\ln|x| + 2N + n - 2\big] \quad \text{in } \mathcal{S}'(\mathbb{R}^n). \tag{7.3.6}$$

We are ready to proceed in earnest with considering:

Case $n = 2$. In view of Lemma 7.20, it is natural to start with a candidate of the form $E(x) := c|x|^2\ln|x|$, $x \in \mathbb{R}^2 \setminus \{0\}$, with c a constant yet to be determined. Lemma 7.20 applied for $N = 2 = n$ implies $\Delta E = 4c\ln|x| + 4c$ in $\mathcal{S}'(\mathbb{R}^2)$, hence (7.3.1) is satisfied for this E provided $4c = \frac{1}{2\pi}$, or equivalently, if $c = \frac{1}{8\pi}$. The

bottom line is that, for this value of c, we have $E \in \mathcal{S}'(\mathbb{R}^2)$ and $\Delta E = E_\Delta + \frac{1}{2\pi}$ in $\mathcal{S}'(\mathbb{R}^2)$. Hence, since $\Delta c = 0$ in $\mathcal{S}'(\mathbb{R}^2)$, we conclude that

$$\begin{aligned} &E(x) = \tfrac{1}{8\pi}|x|^2 \ln|x| \quad \text{ for } \quad x \in \mathbb{R}^2 \setminus \{0\}, \\ &\text{satisfies } \ E \in \mathcal{S}'(\mathbb{R}^2) \ \text{ and } \ \Delta^2 E = \delta \ \text{ in } \ \mathcal{S}'(\mathbb{R}^2). \end{aligned} \tag{7.3.7}$$

Finally, we are left with:

Case $n = 1$. In this situation we use Exercise 5.15 to conclude that any fundamental solution for Δ^2 that is a tempered distribution has the form $\frac{1}{6}x^3 H + P$ for some polynomial P in $\mathbb{R}$ of degree less than or equal to 3.

The main result emerging from this analysis is summarized next.

Theorem 7.21. *The function $E \in L^1_{loc}(\mathbb{R}^n)$ defined as*

$$E(x) := \begin{cases} \dfrac{1}{2(n-2)(n-4)\omega_{n-1}}|x|^{4-n} & \text{if } \ x \in \mathbb{R}^n \setminus \{0\}, \quad n \geq 3, \ n \neq 4, \\ -\dfrac{1}{8\pi^2}\ln|x| & \text{if } \ x \in \mathbb{R}^4 \setminus \{0\}, \quad n = 4, \\ \dfrac{1}{8\pi}|x|^2 \ln|x| & \text{if } \ x \in \mathbb{R}^2 \setminus \{0\}, \quad n = 2, \\ \dfrac{1}{6}x^3 H & \text{if } \ x \in \mathbb{R}, \quad n = 1, \end{cases} \tag{7.3.8}$$

belongs to $\mathcal{S}'(\mathbb{R}^n)$ and is a fundamental solution for the operator Δ^2 in $\mathbb{R}^n$. Moreover,

$$\begin{aligned} &\big\{u \in \mathcal{S}'(\mathbb{R}^n) : \Delta^2 u = \delta \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n)\big\} \\ &\qquad = \big\{E + P : P \ \text{ bi-harmonic polynomial in } \ \mathbb{R}^n\big\}. \end{aligned} \tag{7.3.9}$$

Proof. It is clear from (7.3.8) that $E \in L^1_{loc}(\mathbb{R}^n) \cap \mathcal{S}'(\mathbb{R}^n)$. That $\Delta^2 E = \delta$ in $\mathcal{S}'(\mathbb{R}^n)$ has been checked in (7.3.4), (7.3.7), and Exercise 5.15. Finally, (7.3.9) is a consequence of Proposition 5.7. □

We now turn to the task of proving Lemmas 7.18 and 7.19.

Proof of Lemma 7.18. Fix $N > 2 - n$. Based on Exercise 4.4 we may conclude that $|x|^{N-2}, |x|^N \in \mathcal{S}'(\mathbb{R}^n)$. Thus, invoking (4.1.25), there remains to show that

$$\big\langle \Delta\big[|x|^N\big], \varphi\big\rangle = N(N+n-2)\langle |x|^{N-2}, \varphi\rangle, \qquad \forall\, \varphi \in C^\infty_0(\mathbb{R}^n). \tag{7.3.10}$$

To this end, fix $\varphi \in C^\infty_0(\mathbb{R}^n)$ arbitrary and let $R \in (0,\infty)$ be such that $\operatorname{supp}\varphi \subset B(0,R)$. Then, starting with the definition of distributional derivatives, then using the support condition for φ and Lebesgue's dominated convergence theorem, and then (13.7.5) [keeping in mind that $\operatorname{supp}\varphi \subset B(0,R)$], we have

$$\begin{aligned}\big\langle\Delta\big[|x|^N\big],\varphi\big\rangle &= \int_{\mathbb{R}^n}|x|^N\Delta\varphi(x)\,dx=\lim_{\varepsilon\to 0^+}\int_{\varepsilon<|x|<R}|x|^N\Delta\varphi(x)\,dx\\ &=\lim_{\varepsilon\to 0^+}\Big[\int_{\varepsilon<|x|<R}\varphi(x)\Delta|x|^N\,dx-\int_{\partial B(0,\varepsilon)}|x|^N\frac{\partial\varphi}{\partial\nu}(x)\,d\sigma(x)\\ &\qquad+\int_{\partial B(0,\varepsilon)}\varphi(x)\frac{\partial|x|^N}{\partial\nu}\,d\sigma(x)\Big],\end{aligned}\tag{7.3.11}$$

where $\nu(x)=\frac{x}{\varepsilon}$, for $x\in\partial B(0,\varepsilon)$. By (7.3.2) and the Lebesgue dominated convergence theorem 13.12 it follows that

$$\begin{aligned}\lim_{\varepsilon\to 0^+}\int_{\varepsilon<|x|<R}\varphi(x)\Delta|x|^N\,dx&=N(N+n-2)\int_{\mathbb{R}^n}|x|^{N-2}\varphi(x)\,dx\\ &=N(N+n-2)\big\langle|x|^{N-2},\varphi\big\rangle.\end{aligned}\tag{7.3.12}$$

Since $\nabla(|x|^N)=N|x|^{N-2}x$ for every $x\neq 0$, we have $\frac{\partial|x|^N}{\partial\nu}=N\varepsilon^{N-1}$ on $\partial B(0,\varepsilon)$. Hence,

$$\begin{aligned}&\Big|-\int_{\partial B(0,\varepsilon)}|x|^N\frac{\partial\varphi}{\partial\nu}(x)\,d\sigma(x)+\int_{\partial B(0,\varepsilon)}\varphi(x)\frac{\partial|x|^N}{\partial\nu}\,d\sigma(x)\Big|\\ &\quad\leq\|\nabla\varphi\|_{L^\infty(\mathbb{R}^n)}\varepsilon^{N+n-1}+N\|\varphi\|_{L^\infty(\mathbb{R}^n)}\varepsilon^{N+n-2}\xrightarrow[\varepsilon\to 0^+]{}0\end{aligned}\tag{7.3.13}$$

given that $N>2-n$. Now (7.3.10) follows from (7.3.11)–(7.3.13). The proof of the lemma is complete. □

Proof of Lemma 7.19. From Exercise 4.4 and the assumption $n\geq 3$ it follows that $|x|^{-2}\in\mathcal{S}'(\mathbb{R}^n)$, while from Example 4.8 we have $\ln|x|\in\mathcal{S}'(\mathbb{R}^n)$. Hence, by (4.1.25) matters reduce to showing

$$\big\langle\Delta\big[\ln|x|\big],\varphi\big\rangle=(n-2)\big\langle|x|^{-2},\varphi\big\rangle,\qquad\forall\,\varphi\in C_0^\infty(\mathbb{R}^n).\tag{7.3.14}$$

Fix $\varphi\in C_0^\infty(\mathbb{R}^n)$ and let $R\in(0,\infty)$ be such that $\operatorname{supp}\varphi\subset B(0,R)$. Using in the current setting a reasoning similar to that applied when deriving (7.3.11) one obtains

$$\begin{aligned}\big\langle\Delta\big[\ln|x|\big],\varphi\big\rangle&=\int_{\mathbb{R}^n}\ln|x|\Delta\varphi(x)\,dx=\lim_{\varepsilon\to 0^+}\int_{\varepsilon<|x|<R}\ln|x|\Delta\varphi(x)\,dx\\ &=\lim_{\varepsilon\to 0^+}\Big[\int_{\varepsilon<|x|<R}\varphi(x)\Delta(\ln|x|)\,dx-\int_{\partial B(0,\varepsilon)}\ln|x|\frac{\partial\varphi}{\partial\nu}(x)\,d\sigma(x)\\ &\qquad+\int_{\partial B(0,\varepsilon)}\varphi(x)\frac{\partial\ln|x|}{\partial\nu}\,d\sigma(x)\Big].\end{aligned}\tag{7.3.15}$$

It is easy to see that $\Delta(\ln|x|) = (n-2)|x|^{-2}$ pointwise in $\mathbb{R}^n \setminus \{0\}$, thus invoking the Lebesgue dominated convergence theorem 13.12 we obtain

$$\lim_{\varepsilon\to 0^+} \int_{\varepsilon<|x|<R} \varphi(x)\Delta(\ln|x|)\,dx = (n-2)\int_{\mathbb{R}^n} |x|^{-2}\varphi(x)\,dx$$
$$= (n-2)\langle |x|^{-2}, \varphi\rangle. \tag{7.3.16}$$

Also, since $\nabla(\ln|x|) = \frac{x}{|x|^2}$ pointwise in $\mathbb{R}^n \setminus \{0\}$, we have $\frac{\partial \ln|x|}{\partial \nu} = \varepsilon^{-1}$ on $\partial B(0,\varepsilon)$. Hence,

$$\left| -\int_{\partial B(0,\varepsilon)} \ln|x| \frac{\partial\varphi}{\partial\nu}(x)\,d\sigma(x) + \int_{\partial B(0,\varepsilon)} \varphi(x)\frac{\partial \ln|x|}{\partial\nu}\,d\sigma(x)\right|$$
$$\leq \|\nabla\varphi\|_{L^\infty(\mathbb{R}^n)}\varepsilon^{n-1}|\ln\varepsilon| + \|\varphi\|_{L^\infty(\mathbb{R}^n)}\varepsilon^{n-2} \xrightarrow[\varepsilon\to 0^+]{} 0, \tag{7.3.17}$$

where for the convergence in (7.3.17) we used the fact that $n \geq 3$. To finish the proof of the lemma we combine (7.3.15)–(7.3.17). □

Exercise 7.22. *Let E be the fundamental solution for the bi-Laplacian from (7.3.8). Prove that for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = 3$ the function $\partial^\alpha E$ is C^∞ and positive homogeneous of degree $1-n$ in $\mathbb{R}^n \setminus \{0\}$.*

7.4 The Poisson Equation for the Bi-Laplacian

Analogously to Theorem 7.10 we have the following regularity result for the bi-Laplacian.

Theorem 7.23. *For an open set $\Omega \subseteq \mathbb{R}^n$ the following are equivalent:*

(i) *u is* `bi-harmonic` *in Ω;*

(ii) *$u \in \mathcal{D}'(\Omega)$ and $\Delta^2 u = 0$ in $\mathcal{D}'(\Omega)$;*

(iii) *$u \in C^\infty(\Omega)$ and $\Delta^2 u = 0$ in a pointwise sense in Ω.*

Proof. This is a direct consequence of Theorem 6.8 and Theorem 7.21 (or, alternatively, of Remark 6.14 and Corollary 6.18). □

We begin by establishing mean value formulas for biharmonic functions. Recall the notation from (0.0.4).

Theorem 7.24. *Let Ω be an open set in $\mathbb{R}^n$ and let u be a biharmonic function in Ω. Then for each $x \in \Omega$ and each $r \in \big(0, \operatorname{dist}(x,\partial\Omega)\big)$ the following formulas hold:*

$$u(x) = \fint_{\partial B(x,r)} u(y)\,d\sigma(y) - \frac{r^2}{2n}(\Delta u)(x), \tag{7.4.1}$$

$$u(x) = \fint_{B(x,r)} u(y)\,dy - \frac{r^2}{2(n+2)}(\Delta u)(x). \tag{7.4.2}$$

Proof. Fix $x \in \Omega$ and recall the function ϕ defined in (7.2.10). Then (7.2.11) holds. Since $\Delta(\Delta u) = 0$ in Ω we have that Δu is harmonic in Ω and we can apply (7.2.9) to obtain that $\phi'(r) = \frac{r}{n}\Delta u(x)$. Consequently, $\phi(r) = \frac{r^2}{2n}\Delta u(x) + C$ for some constant C. To determine C we use the fact that $\lim_{r\to 0^+} \phi(r) = u(x)$ (for the latter see (7.2.15)). Hence formula (7.4.1) follows.

Next, write (7.4.1) as

$$\omega_{n-1} r^{n-1} u(x) = \int_{\partial B(x,r)} u(y)\, d\sigma(y) - \frac{\omega_{n-1} r^{n+1}}{n} \Delta u(x). \tag{7.4.3}$$

Integrating (7.4.3) with respect to r for $r \in (0, R)$ and $R \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, and then dividing by $\omega_{n-1} R^n/n$ gives (7.4.2) with r replaced by R. □

Theorem 7.25 (Liouville's theorem for Δ^2)**.** *Any bounded bi-harmonic function in $\mathbb{R}^n$ is constant.*

Proof. One way to justify this result is by observing that it is a particular case of Theorem 5.3. Another proof based on interior estimates goes as follows. Let u be a bounded bi-harmonic function in $\mathbb{R}^n$. Formula (7.4.2) in the current setting gives that

$$\Delta u(x) = \frac{2(n+2)}{r^2} \fint_{B(x,r)} u(y)\, dy - \frac{2(n+2)}{r^2} u(x) \tag{7.4.4}$$

for every $x \in \mathbb{R}^n$ and every $r \in (0, \infty)$. Since $\big|\fint_{B(x,r)} u(y)\, dy\big| \leq \|u\|_{L^\infty(\mathbb{R}^n)}$, if we let $r \to \infty$ in (7.4.4), we see that $\Delta u(x) = 0$, thus u is harmonic. By Liouville's Theorem 7.15 for harmonic functions it follows that there exists $c \in \mathbb{R}$ such that $u(x) = c$ for all $x \in \mathbb{R}^n$. □

We shall next show that bi-harmonic functions satisfy interior estimates and are real-analytic.

Theorem 7.26. *Suppose $u \in \mathcal{D}'(\Omega)$ satisfies $\Delta^2 u = 0$ in $\mathcal{D}'(\Omega)$. Then u is real-analytic in Ω and there exists a dimensional constant $C \in (0, \infty)$ such that*

$$\max_{y \in \overline{B(x,r/2)}} |\partial^\alpha u(y)| \leq \frac{C^{|\alpha|}|\alpha|!}{r^{|\alpha|}} \max_{y \in \overline{B(x,r)}} |u(y)|, \qquad \forall\, \alpha \in \mathbb{N}_0^n, \tag{7.4.5}$$

for each $x \in \Omega$ and each $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$.

Proof. In the case when $n = 1$ or $n \geq 3$, all claims are direct consequences of Theorem 6.26 and Theorem 7.21. To treat the case $n = 2$ we shall introduce a "dummy" variable. Specifically, in this setting consider the open set $\widetilde{\Omega} := \Omega \times \mathbb{R}$ in $\mathbb{R}^3$ and define the function $\widetilde{u}(x_1, x_2, x_3) := u(x_1, x_2)$ for $(x_1, x_2, x_3) \in \widetilde{\Omega}$. Observe that $\widetilde{u}$ is bi-harmonic in $\widetilde{\Omega}$. As such, the higher-dimensional theory applies and yields that $\widetilde{u}$ satisfies, for some universal constant $C \in (0, \infty)$,

$$\max_{y \in \overline{B(x,r/2)}} |\partial^\alpha \widetilde{u}(y)| \leq \frac{C^{|\alpha|}|\alpha|!}{r^{|\alpha|}} \max_{y \in \overline{B(x,r)}} |\widetilde{u}(y)|, \qquad \forall\, \alpha \in \mathbb{N}_0^3, \tag{7.4.6}$$

for each $x \in \widetilde{\Omega}$ and each $r \in \big(0, \operatorname{dist}(x, \partial\widetilde{\Omega})\big)$. Applying (7.4.6) for points of the form $x = (x_1, x_2, 0) \in \widetilde{\Omega}$ and for multi-indices $\alpha = (\alpha_1, \alpha_2, 0)$ then yields (7.4.5) for the original function u. In turn, this and Lemma 6.24 imply that u is real-analytic in Ω. □

We are now prepared to discuss the well-posedness of the Poisson problem for the bi-Laplacian in $\mathbb{R}^n$.

Theorem 7.27. *Assume $n \in \mathbb{N}$ satisfies $n \geq 3$ and $n \neq 4$. Then for each function $f \in L^\infty_{comp}(\mathbb{R}^n)$ and each $c \in \mathbb{C}$ the Poisson problem for the bi-Laplacian in $\mathbb{R}^n$,*

$$\begin{cases} u \in C^0(\mathbb{R}^n), \\ \Delta^2 u = f \ \text{ in } \ \mathcal{D}'(\mathbb{R}^n), \\ \lim\limits_{|x|\to\infty} u(x) = c, \end{cases} \tag{7.4.7}$$

has a unique solution. Moreover, the solution u satisfies the following additional properties.

(1) The function u has the integral representation formula

$$u(x) = c + \int_{\mathbb{R}^n} E(x-y) f(y)\,\mathrm{d}y, \qquad x \in \mathbb{R}^n, \tag{7.4.8}$$

where E is the fundamental solution for the bi-Laplacian in $\mathbb{R}^n$ from (7.3.8). Moreover, u belongs to $C^3(\mathbb{R}^n)$ and for each $\alpha \in \mathbb{N}_0^n$ with the property that $0 < |\alpha| \leq 3$ we have

$$\partial^\alpha u(x) = \int_{\mathbb{R}^n} (\partial^\alpha E)(x-y) f(y)\,\mathrm{d}y, \qquad x \in \mathbb{R}^n. \tag{7.4.9}$$

(2) If in fact $f \in C^\infty_0(\mathbb{R}^n)$ then $u \in C^\infty(\mathbb{R}^n)$.

(3) For every $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = 4$ there exists a constant $c_\alpha \in \mathbb{C}$ such that

$$\partial^\alpha u = c_\alpha\, f + T_{\partial^\alpha E} f \quad \text{in } \ \mathcal{D}'(\mathbb{R}^n), \tag{7.4.10}$$

where $T_{\partial^\alpha E}$ is the singular integral operator associated with $\Theta := \partial^\alpha E$ as in Definition 4.90.

(4) For each $p \in (1, \infty)$, the solution u of (7.4.7) satisfies $\partial^\alpha u \in L^p(\mathbb{R}^n)$ for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = 4$, where the derivatives are taken in $\mathcal{D}'(\mathbb{R}^n)$. Moreover, there exists a constant $C = C(p, n) \in (0, \infty)$ with the property that

$$\sum_{|\alpha|=4}^{n} \|\partial^\alpha u\|_{L^p(\mathbb{R}^n)} \leq C \|f\|_{L^p(\mathbb{R}^n)}. \tag{7.4.11}$$

Proof. That the function u defined as in (7.4.8) is of class $C^3(\mathbb{R}^n)$ and formula (7.4.9) holds for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq 3$ can be established much as in the proof of Proposition 7.8 keeping in mind that $\partial^\alpha E \in L^1_{loc}(\mathbb{R}^n$ for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq 3$. Also, this function solves $\Delta^2 u = f$ in $\mathcal{D}'(\mathbb{R}^n)$ by reasoning in a similar fashion to the computation in (7.2.7). Given the format of E from (7.3.8) under the current assumptions on n, and the conditions on f, it is clear that the function u from (7.4.8) satisfies the limit condition in (7.4.7) [see (7.2.21) in the case of the Laplace operator]. The fact that (7.4.8) is the unique solution of (7.4.7) is a consequence of Theorem 7.25. Moving on, that $u \in C^\infty(\mathbb{R}^n)$ if $f \in C_0^\infty(\mathbb{R}^n)$ follows from Corollary 6.18 and the ellipticity of Δ^2. The above arguments cover the claims in parts *(1)-(2)*.

Turning to part *(3)*, start by fixing $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = 4$. Then there exists $\beta \in \mathbb{N}_0^n$ with $|\beta| = 3$ and $j \in \{1, \ldots, n\}$ such that $\partial^\alpha = \partial_j \partial^\beta$. Then the function $\Phi := \partial^\beta E$ is C^∞ and positive homogeneous of degree $1-n$ in $\mathbb{R}^n \setminus \{0\}$ (cf. Exercise 7.22). Consequently, the function $\Theta := \partial^\alpha E$ is as in (4.4.1), thanks to the discussion in Example 4.68. Then from what we have proved in part *(1)* and Theorem 4.99 we deduce that the following formula holds in $\mathcal{D}'(\mathbb{R}^n)$:

$$\partial^\alpha u(x) = \partial_j\big[\partial^\beta u(x)\big] = \partial_j\Big[\int_{\mathbb{R}^n} (\partial^\beta E)(x-y) f(y)\,dy\Big] \tag{7.4.12}$$

$$= \Big(\int_{S^{n-1}} (\partial^\beta E)(\omega)\omega_j\,d\sigma(\omega)\Big) f(x) + \lim_{\varepsilon\to 0^+} \int_{|x-y|>\varepsilon} (\partial^\alpha E)(x-y) f(y)\,dy.$$

Choosing $c_\alpha := \int_{S^{n-1}} (\partial^\beta E)(\omega)\omega_j\,d\sigma(\omega)$ the above formula may be written as in (7.4.10), and this finishes the proof of part *(3)*. Finally, the claim in *(4)* follows from *(3)* and the boundedness of the singular integral operators $T_{\partial^\alpha E}$ on $L^p(\mathbb{R}^n)$ (cf. Theorem 4.97). □

7.5 Fundamental Solutions for the Poly-Harmonic Operator

Let $m \in \mathbb{N}$ and consider the `poly-harmonic` operator

$$\Delta^m := \left(\sum_{j=1}^n \partial_j^2\right)^m = \sum_{|\gamma|=m} \frac{m!}{\gamma!}\,\partial^{2\gamma}, \quad \text{in } \mathbb{R}^n. \tag{7.5.1}$$

The goal in this section is to identify all fundamental solutions for this operator in $\mathbb{R}^n$ that are tempered distributions. The case $m = 1$ has been treated in Theorem 7.2 and the case $m = 2$ is discussed in Theorem 7.21. We remark that the formulas from Lemma 7.18, Lemma 7.19, and Lemma 7.20, suggest that we should look for a fundamental solution for Δ^m that is a constant multiple of $|x|^{2m-n}$ or a constant multiple of $|x|^{2m-n}\ln|x|$. In precise terms, we will prove the following result (recall that H denotes the Heaviside function).

Theorem 7.28. *Let $m, n \in \mathbb{N}$ and consider the function*

$$F_{m,n}(x) := \begin{cases} \dfrac{(-1)^m \Gamma(n/2-m)}{\pi^{n/2} 4^m (m-1)!} \, |x|^{2m-n} & \text{if } n > 2m \text{ or } n \text{ is odd and } n \neq 1, \\ \dfrac{(-1)^{n/2+1}}{2\pi^{n/2} 4^{m-1} (m-n/2)!(m-1)!} \, |x|^{2m-n} \ln |x| & \text{if } n \text{ is even and } \leq 2m, \\ \dfrac{1}{(2m-1)!} x^{2m-1} H & \text{if } n = 1, \end{cases} \tag{7.5.2}$$

defined for $x \in \mathbb{R}^n \setminus \{0\}$ if $n \geq 2$ and for $x \in \mathbb{R}$ if $n = 1$. Then $F_{m,n}$ belongs to $L^1_{loc}(\mathbb{R}^n) \cap \mathcal{S}'(\mathbb{R}^n)$ and is a fundamental solution for Δ^m in $\mathbb{R}^n$. Moreover,

$$\begin{aligned} &\big\{ u \in \mathcal{S}'(\mathbb{R}^n) : \Delta^m u = \delta \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n) \big\} \\ &\qquad = \big\{ F_{m,n} + P : P \ \text{ poly-harmonic polynomial in } \ \mathbb{R}^n \big\}. \end{aligned} \tag{7.5.3}$$

Proof. The case $n = 1$ is immediate from Exercise 5.15. Assume in what follows that $n \geq 2$. Since $n - 2m < n$, we clearly have $|x|^{2m-n} \in L^1_{loc}(\mathbb{R}^n)$ and furthermore, by Exercise 4.4, that $|x|^{2m-n} \in \mathcal{S}'(\mathbb{R}^n)$. By Exercise 2.107 and Exercise 4.104 we also have $|x|^{2m-n} \ln |x| \in L^1_{loc}(\mathbb{R}^n) \cap \mathcal{S}'(\mathbb{R}^n)$. This proves that $F_{m,n} \in L^1_{loc}(\mathbb{R}^n) \cap \mathcal{S}'(\mathbb{R}^n)$. Once we show that $F_{m,n}$ is a fundamental solution for Δ^m in $\mathbb{R}^n$, Proposition 5.7 readily yields (7.5.3).

There remains to prove $\Delta^m F_{m,n} = \delta$ in $\mathcal{S}'(\mathbb{R}^n)$. If $m = 1$, then it is easy to see that $F_{1,n}$ is the same as the distribution in (7.1.12) (corresponding to $n \geq 2$). Thus, by Theorem 7.2 we have

$$\Delta F_{1,n} = \delta \quad \text{ in } \quad \mathcal{S}'(\mathbb{R}^n). \tag{7.5.4}$$

We proceed by breaking up the remaining part of our analysis in a few cases.

The case $n > 2m$ or n is odd.
Because of (7.5.4) we may assume $m \geq 2$. Applying Lemma 7.18 and (13.5.2) we obtain

$$\begin{aligned} \Delta F_{m,n} &= \frac{(-1)^m \Gamma(n/2-m)}{\pi^{n/2} 4^m (m-1)!} \, \Delta(|x|^{2m-n}) \\ &= \frac{(-1)^m \Gamma(n/2-m)}{\pi^{n/2} 4^m (m-1)!} \, (2m-n)(2m-2)|x|^{2m-n-2} \\ &= \frac{(-1)^{m-1} \Gamma(n/2-m)(n/2-m)(m-1)}{\pi^{n/2} 4^{m-1} (m-1)!} \, |x|^{2(m-1)-n} \\ &= F_{m-1,n} \quad \text{ in } \mathcal{S}'(\mathbb{R}^n) \text{ and pointwise in } \mathbb{R}^n \setminus \{0\}. \end{aligned} \tag{7.5.5}$$

Hence, inductively we have $\Delta^{m-1} F_{m,n} = F_{1,n}$ in $\mathcal{S}'(\mathbb{R}^n)$, which when combined with (7.5.4) yields the desired conclusion in the current case.

The case $n = 2m$.
In this scenario n is even, and the goal is to prove

$$\Delta^{\frac{n}{2}} F_{\frac{n}{2},n} = \delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{7.5.6}$$

Note that (7.5.6) holds for $n = 2$ as seen from (7.5.4). Consequently, we may assume that $n \geq 4$. Inductively, using Lemma 7.18, it is not difficult to show that

$$\Delta^k \big[|x|^{-2}\big] = \frac{(-1)^k 4^k k! \big(\frac{n}{2} - 2\big)!}{\big(\frac{n}{2} - k - 2\big)!} |x|^{-2-2k} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n) \tag{7.5.7}$$

if n is even, $n \geq 4$ and $k \in \mathbb{N}_0$ is such that $k < \frac{n}{2} - 1$.

In particular, (7.5.7) specialized to $k = \frac{n}{2} - 2$ yields

$$\Delta^{\frac{n}{2}-2} \big[|x|^{-2}\big] = (-1)^{\frac{n}{2}} 4^{\frac{n}{2}} \Big[\Big(\frac{n}{2} - 2\Big)!\Big]^2 |x|^{2-n} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{7.5.8}$$

Starting with the expression from (7.5.2) we the obtain

$$\begin{aligned}
\Delta^{\frac{n}{2}-1} F_{\frac{n}{2},n} &= \frac{(-1)^{\frac{n}{2}+1}}{2\pi^{\frac{n}{2}} 4^{\frac{n}{2}-1} \big(\frac{n}{2} - 1\big)!} \Delta^{\frac{n}{2}-1} \big[\ln |x|\big] \\
&= \frac{(-1)^{\frac{n}{2}+1}}{2\pi^{\frac{n}{2}} 4^{\frac{n}{2}-1} \big(\frac{n}{2} - 1\big)!} (n-2) \Delta^{\frac{n}{2}-2} \big[|x|^{-2}\big] \\
&= \frac{(-1)^{\frac{n}{2}+1}}{2\pi^{\frac{n}{2}} 4^{\frac{n}{2}-1} \big(\frac{n}{2} - 1\big)!} (n-2)(-1)^{\frac{n}{2}} 4^{\frac{n}{2}} \Big[\Big(\frac{n}{2} - 2\Big)!\Big]^2 |x|^{2-n} \\
&= \frac{-1}{(n-2)\omega_{n-1}} |x|^{2-n} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n).
\end{aligned} \tag{7.5.9}$$

The second equality in (7.5.9) is based on Lemma 7.19, the third equality uses (7.5.8), while the last equality follows from straightforward calculations and (13.5.6). In summary, we proved that if n is even and $n \geq 4$, then $\Delta^{\frac{n}{2}-1} F_{\frac{n}{2},n} = E_\Delta$ in $\mathcal{S}'(\mathbb{R}^n)$, where E_Δ is the fundamental solution for the Laplacian from (7.1.12). The latter combined with Theorem 7.2 now yields (7.5.6) for $n \geq 4$, and finishes the proof of (7.5.6).

The case $n \leq 2m$ and n is even.
Fix an even number $n \in \mathbb{N}$. We will prove that if $m \geq \frac{n}{2}$ then

$$\Delta^m F_{m,n} = \delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n) \tag{7.5.10}$$

by induction on m. The case $m = \frac{n}{2}$ has been treated in the previous case. Suppose (7.5.10) holds for some $m \geq \frac{n}{2}$. Then clearly $2(m+1) > n$ and invoking Lemma 7.20 we obtain

$$\Delta F_{m+1,n} = F_{m,n} + C_{m+1,n}(4m - n + 2)|x|^{2m-n} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{7.5.11}$$

where $C_{m+1,n}$ is the coefficient of $|x|^{2(m+1)-n}\ln|x|$ in the expression for $F_{m+1,n}$. Note that since n is even the expression $|x|^{2m-n}$ is a polynomial of degree $2m-n$. Hence, $\Delta^m[|x|^{2m-n}]=0$ given that Δ^m is a homogeneous differential operator of degree $2m>2m-n$. Combined with (7.5.11) and the induction hypothesis on $F_{m,n}$, this gives

$$\Delta^{m+1}F_{m+1,n}=\Delta^m F_{m,n}+C_{m+1,n}(4m-n+2)\Delta^m[|x|^{2m-n}]=\delta \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n). \tag{7.5.12}$$

By induction, it follows that (7.5.10) holds in the current setting. This finishes the proof of the theorem. □

We remark that Theorem 7.28 specialized to $m=2$ gives another proof of the fact that when $n\geq 2$ the expression from (7.3.8) is a fundamental solution for Δ^2 in $\mathbb{R}^n$.

Proposition 7.29. *Let $m,n\in\mathbb{N}$ be such that $n\geq 2$ and let $F_{m,n}$ be the fundamental solution for Δ^m in $\mathbb{R}^n$ as defined in (7.5.2). Then if $N\in\mathbb{N}_0$ is such that $0\leq N<m$ the following identity holds*

$$\Delta^N\Big[F_{m,n}\Big]=F_{m-N,n}+\mathcal{P}_{m,N} \quad \text{in } \ \mathcal{S}'(\mathbb{R}^n) \quad \text{and pointwise in } \mathbb{R}^n\setminus\{0\}, \tag{7.5.13}$$

where $\mathcal{P}_{m,N}$ is a homogeneous polynomial in $\mathbb{R}^n$ of degree less than or equal to $\max\{2m-n-2N,0\}$ which is identically zero if $n>2m$ or n is odd, or if n is even such that $n\leq 2m$ and $N>m-\frac{n}{2}$ (in short, $\mathcal{P}_{m,N}\equiv 0$ if either n is odd or $2N+n>2m$).

Proof. We start by observing that using an inductive reasoning and Lemma 7.20 we obtain that for each $N\in\mathbb{N}_0$

$$\Delta^N\big[|x|^M\ln|x|\big]=C^{(1)}_{M,N}|x|^{M-2N}\ln|x|+C^{(2)}_{M,N}|x|^{M-2N} \quad \text{in } \ \mathcal{S}'(\mathbb{R}^n)$$
$$\text{if } M \text{ is a real number such that } M-2N>-n, \tag{7.5.14}$$

for some real constants $C^{(1)}_{M,N}$, $C^{(2)}_{M,N}$, depending on n, M and N.

Now suppose n is even and $n\leq 2m$. Specializing (7.5.14) to the case when $M:=2m-n$ and taking $N\in\big\{0,1,\ldots,m-\frac{n}{2}\big\}$, we conclude

$$\Delta^N F_{m,n}=c^{(1)}_{m,N}F_{m-N,n}+c^{(2)}_{m,N}|x|^{2m-n-2N} \quad \text{in } \ \mathcal{S}'(\mathbb{R}^n), \tag{7.5.15}$$

for some real constants $c^{(1)}_{m,N}$, $c^{(2)}_{m,N}$, depending only on n, m and N. For each number $N\in\big\{0,1,\ldots,m-\frac{n}{2}\big\}$ set

$$\mathcal{P}_{m,N}(x):=c^{(2)}_{m,N}|x|^{2m-n-2N}, \quad x\in\mathbb{R}^n.$$

Note that since n is even and $n\leq 2m$, each $\mathcal{P}_{m,N}$ is a homogeneous polynomial of degree at most $2m-n-2N$. Also, we have that for each $N\in\big\{0,1,\ldots,m-\frac{n}{2}\big\}$

the operator Δ^{m-N} is a homogeneous differential operator of degree $2m-2N > 2m-n-2N$, thus $\Delta^{m-N}\mathcal{P}_{m,N} = 0$ both in $\mathcal{S}'(\mathbb{R}^n)$ and pointwise in $\mathbb{R}^n$. The latter, (7.5.15), and Theorem 7.28 imply

$$\delta = \Delta^{m-N}\big[\Delta^N F_{m,n}\big] = c^{(1)}_{m,N}\Delta^{m-N}\big[F_{m-N,n}\big] + \Delta^{m-N}\big[\mathcal{P}_{m,N}\big] = c^{(1)}_{m,N}\delta \tag{7.5.16}$$

in $\mathcal{S}'(\mathbb{R}^n)$ for each $N \in \big\{0, 1, \ldots, m - \frac{n}{2}\big\}$. This implies that $c^{(1)}_{m,N} = 0$ for each $N \in \big\{0, 1, \ldots, m - \frac{n}{2}\big\}$. Consequently, (7.5.15) becomes

$$\Delta^N F_{m,n} = F_{m-N,n} + \mathcal{P}_{m,N} \quad \text{in } \mathcal{S}'(\mathbb{R}^n), \ \forall\, N \in \Big\{0, 1, \ldots, m - \frac{n}{2}\Big\}. \tag{7.5.17}$$

That the above equality also holds pointwise in $\mathbb{R}^n$ is immediate, hence we proved (7.5.13) for n even such that $n \leq 2m$ and $0 \leq N \leq m - \frac{n}{2}$. Note that if $n = 2$ then $m - \frac{n}{2} = m - 1$ so we actually have proved (7.5.13) for all $N \in \big\{0, 1, \ldots, m-1\big\}$.

To finish with the proof of (7.5.13) when n is even and satisfies $n \leq 2m$, there remains the case $N \in \big\{m - \frac{n}{2} + 1, \ldots, m-1\big\}$ under the assumption that $n \geq 4$. For starters, observe that if we write (7.5.17) for $N = m - \frac{n}{2}$, then $\mathcal{P}_{m,N}$ is just a constant and using the expression for $F_{\frac{n}{2},n}$ we have

$$\Delta^{m-\frac{n}{2}} F_{m,n} = F_{\frac{n}{2},n} + C^{(2)}_{m,m-\frac{n}{2}} = c_n \ln|x| + c^{(2)}_{m,m-\frac{n}{2}} \quad \text{in } \mathcal{S}'(\mathbb{R}^n), \tag{7.5.18}$$

for some real constant c_n. Therefore, (7.5.18), Lemma 7.19, and Lemma 7.18 (which we can apply $N - m + \frac{n}{2} - 1$ times since $N < m$) allow us to conclude that for each $N \in \big\{m - \frac{n}{2} + 1, \ldots, m-1\big\}$ there exists some constant $c_{m,N}$ such that

$$\Delta^N F_{m,n} = \Delta^{N-m+\frac{n}{2}}\big[\Delta^{m-\frac{n}{2}} F_{m,n}\big] = c_{m,N}|x|^{-2N+2m-2n} \quad \text{in } \mathcal{S}'(\mathbb{R}^n). \tag{7.5.19}$$

Upon observing that the condition $N > m - \frac{n}{2}$ is equivalent with $n > 2(m-N)$, we may conclude that $c_{m,N}|x|^{-2N+2m-2n} = c^{(1)}_{m,N} F_{m-N,n}$ for some constant $c^{(1)}_{m,N}$. Hence, for each $N \in \big\{m - \frac{n}{2} + 1, \ldots, m-1\big\}$ we have

$$\Delta^N F_{m,n} = c^{(1)}_{m,N} F_{m-N,n} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{7.5.20}$$

Applying Δ^{m-N} to (7.5.20) and invoking Theorem 7.28 we obtain $c^{(1)}_{m,N} = 1$ for each $N \in \big\{m - \frac{n}{2} + 1, \ldots, m-1\big\}$, so that

$$\Delta^N F_{m,n} = F_{m-N,n} \quad \text{in } \mathcal{S}'(\mathbb{R}^n), \ \forall\, N \in \Big\{m - \frac{n}{2} + 1, \ldots, m-1\Big\}. \tag{7.5.21}$$

This proves (7.5.13) when n is even, satisfies $n \leq 2m$, and $m - \frac{n}{2} < N < m$ if we take $\mathcal{P}_{m,N} := 0$.

Suppose next that $n > 2m$ or n is odd. Then as seen in the proof of Theorem 7.28, identity (7.5.12) holds in $\mathcal{S}'(\mathbb{R}^n)$ and pointwise in $\mathbb{R}^n \setminus \{0\}$. Iterating this identity we arrive at

$$\Delta^N \big[F_{m,n}\big] = F_{m-N,n} \quad \text{in } \mathcal{S}'(\mathbb{R}^n) \text{ and pointwise in } \mathbb{R}^n \setminus \{0\}. \tag{7.5.22}$$

Consequently, formula (7.5.13) holds in the current setting if we take $\mathcal{P}_{m,N} := 0$ for all $N \in \{0, \dots, m-1\}$. This completes the proof of the proposition. □

We next address the issue of interior estimates and real-analyticity for poly-harmonic functions.

Theorem 7.30. *Whenever $u \in \mathcal{D}'(\Omega)$ satisfies $\Delta^m u = 0$ in $\mathcal{D}'(\Omega)$, it follows that u is real-analytic in Ω and there exists a constant $C = C(n,m) \in (0,\infty)$ with the property that*

$$\max_{y \in \overline{B(x,r/2)}} |\partial^\alpha u(y)| \leq \frac{C^{|\alpha|}|\alpha|!}{r^{|\alpha|}} \max_{y \in \overline{B(x,r)}} |u(y)|, \qquad \forall\, \alpha \in \mathbb{N}_0^n, \tag{7.5.23}$$

for each $x \in \Omega$ and each $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$.

Proof. In the case when either n is odd, or $n \geq 2m$, all claims are consequences of Theorem 6.26 and Theorem 7.28. The remaining cases may be reduced to the ones just treated by introducing "dummy" variables, as in the proof of Theorem 7.26. □

In the last part of this section we prove an integral representation formula for a fundamental solution for the poly-harmonic operator in $\mathbb{R}^n$ involving a sufficiently large power of the Laplacian (i.e., Δ^m with m positive integer bigger than or equal to $n/2$). The difference between this fundamental solution and that in (7.5.2) turns out to be a homogeneous polynomial that is a null solution of the respective poly-harmonic operator. In what follows log denotes the principal branch of the complex logarithm that is defined for points $z \in \mathbb{C} \setminus (-\infty, 0]$.

Lemma 7.31. *Let $n \in \mathbb{N}$, $n \geq 2$, and suppose $q \in \mathbb{N}_0$ is such that $n+q$ is an even number. Then the function*

$$E_q(x) := -\frac{1}{(2\pi i)^n q!} \int_{S^{n-1}} (x\cdot\xi)^q \log\Big(\frac{x\cdot\xi}{i}\Big)\, d\sigma(\xi), \qquad \forall\, x \in \mathbb{R}^n \setminus \{0\}, \tag{7.5.24}$$

is a fundamental solution for the poly-harmonic operator $\Delta^{\frac{n+q}{2}}$ in $\mathbb{R}^n$.

Moreover, $E_q \in \mathcal{S}'(\mathbb{R}^n)$ and

$$E_q(x) = F_{\frac{n+q}{2},n}(x) + C(n,q)|x|^q, \qquad \forall\, x \in \mathbb{R}^n \setminus \{0\}, \tag{7.5.25}$$

where $F_{\frac{n+q}{2},n}$ is the fundamental solution for $\Delta^{\frac{n+q}{2}}$ in $\mathbb{R}^n$ given in (7.5.2) and

$$C(n,q) := \begin{cases} -\dfrac{2\omega_{n-2}}{(2\pi i)^n q!} \displaystyle\int_0^1 s^q (\ln s)(\sqrt{1-s^2})^{n-3}\, ds & \text{if } n \text{ is even,} \\ 0 & \text{if } n \text{ is odd.} \end{cases} \tag{7.5.26}$$

Proof. Fix n and q as in the hypotheses of the lemma. Fix $x \in \mathbb{R}^n \setminus \{0\}$ and use the fact that $\log\left(\frac{x\cdot\xi}{i}\right) = \ln|x\cdot\xi| - i\frac{\pi}{2}\mathrm{sgn}\,(x\cdot\xi)$ for every $\xi \in S^{n-1}$ to write

$$\begin{aligned}\int_{S^{n-1}} (x\cdot\xi)^q \log\left(\frac{x\cdot\xi}{i}\right) d\sigma(\xi) &= \int_{\xi\in S^{n-1},\, x\cdot\xi>0} (x\cdot\xi)^q \Big[\ln|x\cdot\xi| - i\tfrac{\pi}{2}\Big]\, d\sigma(\xi)\\ &\quad + (-1)^q \int_{\xi\in S^{n-1},\, x\cdot\xi>0} (x\cdot\xi)^q \Big[\ln|x\cdot\xi| + i\frac{\pi}{2}\Big]\, d\sigma(\xi)\\ &= \frac{1}{2}\int_{S^{n-1}} |x\cdot\xi|^q \Big[\ln|x\cdot\xi| - i\frac{\pi}{2}\Big]\, d\sigma(\xi)\\ &\quad + \frac{(-1)^q}{2}\int_{S^{n-1}} |x\cdot\xi|^q \Big[\ln|x\cdot\xi| + i\tfrac{\pi}{2}\Big]\, d\sigma(\xi)\\ &= \frac{1+(-1)^q}{2}\int_{S^{n-1}} |x\cdot\xi|^q \ln|x\cdot\xi|\, d\sigma(\xi)\\ &\quad + \frac{((-1)^q-1)\pi i}{4}\int_{S^{n-1}} |x\cdot\xi|^q\, d\sigma(\xi) \end{aligned} \tag{7.5.27}$$

To compute the integrals in the rightmost side of (7.5.27) we use Proposition 13.46. First, applying (13.8.15) with $f(t) := |t|^q$ for $t \in \mathbb{R}\setminus\{0\}$ and $v := x$, we obtain

$$\begin{aligned}\int_{S^{n-1}} |x\cdot\xi|^q\, \mathrm{d}\sigma(\xi) &= 2\omega_{n-2}|x|^q \int_0^1 s^q(\sqrt{1-s^2})^{n-3}\, \mathrm{d}s\\ &= 2\omega_{n-2}|x|^q \int_0^{\frac{\pi}{2}} \big(\sin\theta\big)^q \big(\cos\theta\big)^{n-2}\, \mathrm{d}\theta\\ &= \omega_{n-2}\frac{\Gamma\big(\frac{q+1}{2}\big)\Gamma\big(\frac{n-1}{2}\big)}{\Gamma\big(\frac{q+n}{2}\big)}\,|x|^q = \frac{2\pi^{\frac{n-1}{2}}\Gamma\big(\frac{q+1}{2}\big)}{\Gamma\big(\frac{q+n}{2}\big)}\,|x|^q. \end{aligned} \tag{7.5.28}$$

For the third equality in (7.5.28) we used (13.5.10), while the last one is based on (13.5.6). Second, formula (13.8.15) with $f(t) := |t|^q \ln|t|$ for $t \in \mathbb{R}$ and $v := x$, in concert with (7.5.28) further yields

$$\begin{aligned}\int_{S^{n-1}} |x\cdot\xi|^q \ln|x\cdot\xi|\, \mathrm{d}\sigma(\xi) &= 2\omega_{n-2}|x|^q \int_0^1 s^q\big[\ln|x| + \ln s\big](\sqrt{1-s^2})^{n-3}\, \mathrm{d}s\\ &= \frac{2\pi^{\frac{n-1}{2}}\Gamma\big(\frac{q+1}{2}\big)}{\Gamma\big(\frac{q+n}{2}\big)}\,|x|^q \ln|x| + c(n,q)|x|^q, \end{aligned} \tag{7.5.29}$$

where

$$c(n,q) := 2\omega_{n-2}\int_0^1 s^q(\ln s)(\sqrt{1-s^2})^{n-3}\, \mathrm{d}s < \infty. \tag{7.5.30}$$

Hence, a combination of (7.5.27), (7.5.28), (7.5.29), and (7.5.24) implies

$$E_q(x)=\begin{cases}\dfrac{(-1)^{\frac{n}{2}+1}\Gamma\left(\frac{q+1}{2}\right)}{2^{n-1}\pi^{\frac{n+1}{2}}q!\,\Gamma\left(\frac{q+n}{2}\right)}\,|x|^q\ln|x|-\dfrac{c(n,q)}{(2\pi i)^n q!}\,|x|^q & \text{if } n \text{ is even},\\[2ex] \dfrac{(-1)^{\frac{n-1}{2}}\Gamma\left(\frac{q+1}{2}\right)}{2^n\pi^{\frac{n-1}{2}}q!\,\Gamma\left(\frac{q+n}{2}\right)}\,|x|^q & \text{if } n \text{ is odd},\end{cases} \tag{7.5.31}$$

for every $x\in\mathbb{R}^n\setminus\{0\}$. That this expression belongs to $\mathcal{S}'(\mathbb{R}^n)$ follows from Exercise 4.104.

Let $F_{\frac{n+q}{2},n}$ be as in (7.5.2). The goal is to prove that

$$E_q-F_{\frac{n+q}{2},n}=\frac{c(n,q)}{(2\pi i)^n q!}|x|^q \quad \text{pointwise in } \ \mathbb{R}^n\setminus\{0\}. \tag{7.5.32}$$

Once such an identity is established, Theorem 7.28 may be used to conclude that E_q is a fundamental solution for $\Delta^{\frac{n+q}{2}}$ in $\mathbb{R}^n$ (note that if n is even, then q is even and $|x|^q$ is a homogeneous polynomial of order q that is annihilated by $\Delta^{\frac{n+q}{2}}$). Also, it is easy to see that $C(n,q)=\frac{c(n,q)}{(2\pi i)^n q!}$.

With an eye toward proving (7.5.32), suppose first that n is even. Choosing $m:=\frac{n+q}{2}$ in (7.5.2), we have $n\leq 2m$, hence

$$F_{\frac{n+q}{2},n}(x)=\frac{(-1)^{\frac{n}{2}+1}}{\pi^{\frac{n}{2}}2^{n+q-1}\left(\frac{q}{2}\right)!\left(\frac{n+q}{2}-1\right)!}\,|x|^q\,\ln|x|,\qquad \forall\,x\in\mathbb{R}^n\setminus\{0\}. \tag{7.5.33}$$

Applying (13.5.4) with $q/2$ in place of n, and using (13.5.3) (recall that $n+q$ is even) we have

$$\Gamma\Big(\frac{q+1}{2}\Big)=\frac{q!\,\pi^{1/2}}{2^q(q/2)!}\quad\text{and}\quad\Gamma\Big(\frac{q+n}{2}\Big)=\Big(\frac{q+n}{2}-1\Big)!. \tag{7.5.34}$$

Using now (7.5.34) in (7.5.33) and the expression for E_q, it follows that (7.5.32) holds.

If next we assume that n is odd, then

$$F_{\frac{n+q}{2},n}(x)=\frac{(-1)^{\frac{n+q}{2}}\Gamma\left(-\frac{q}{2}\right)}{\pi^{\frac{n}{2}}2^{n+q}\left(\frac{n+q}{2}-1\right)!}\,|x|^q\,\ln|x|,\qquad \forall\,x\in\mathbb{R}^n\setminus\{0\}. \tag{7.5.35}$$

Applying (13.5.5) and (13.5.3) with $(q+1)/2$ in place of n we see that

$$\Gamma\Big(\frac{q+1}{2}\Big)=\frac{q!\,\pi^{1/2}}{2^q(q/2)!}\quad\text{and}\quad\Gamma\Big(-\frac{q}{2}\Big)=\frac{(-1)^{(q+1)/2}2^{q+1}\left(\frac{q+1}{2}\right)!\pi^{1/2}}{(q+1)!} \tag{7.5.36}$$

which in concert with the second equality in (7.5.34), the expression in (7.5.35), and the formula for E_q, implies $E_q(x)=F_{\frac{n+q}{2},n}(x)$ in $\mathbb{R}^n\setminus\{0\}$. The proof of the lemma is now complete. □

7.6 Fundamental Solutions for the Cauchy–Riemann Operator

In this section we determine all fundamental solutions for the Cauchy-Riemann operator $\frac{\partial}{\partial\overline{z}} := \frac{1}{2}(\partial_1 + i\partial_2)$ and its conjugate, $\frac{\partial}{\partial z} := \frac{1}{2}(\partial_1 - i\partial_2)$ that belong to $\mathcal{S}'(\mathbb{R}^2)$. It is immediate that $\frac{\partial}{\partial\overline{z}}\frac{\partial}{\partial z} = \frac{1}{4}\Delta$ when acting on distributions from $\mathcal{D}'(\mathbb{R}^2)$. Given this, Remark 5.6 is particularly useful. Specifically, since by Theorem 7.2 we have that $E(x) = \frac{1}{2\pi}\ln|x|$, $x \in \mathbb{R}^2 \setminus \{0\}$, is a fundamental solution for Δ in $\mathbb{R}^2$ and $E \in \mathcal{S}'(\mathbb{R}^2)$, Remark 5.6 implies that

$$\frac{\partial}{\partial\overline{z}}\Big(\frac{2}{\pi}\ln|x|\Big) \in \mathcal{S}'(\mathbb{R}^2) \quad \text{is a fundamental solution for} \quad \frac{\partial}{\partial z} \quad \text{in } \mathcal{S}'(\mathbb{R}^2), \tag{7.6.1}$$

$$\frac{\partial}{\partial z}\Big(\frac{2}{\pi}\ln|x|\Big) \in \mathcal{S}'(\mathbb{R}^2) \quad \text{is a fundamental solution for} \quad \frac{\partial}{\partial\overline{z}} \quad \text{in } \mathcal{S}'(\mathbb{R}^2). \tag{7.6.2}$$

We proceed with computing $\frac{\partial}{\partial\overline{z}}\big(\frac{2}{\pi}\ln|x|\big)$ in the distributional sense in $\mathcal{S}'(\mathbb{R}^2)$. By *(c)* in Theorem 4.13 it suffices to compute the distributional derivative $\frac{\partial}{\partial\overline{z}}\big(\frac{2}{\pi}\ln|x|\big)$ in $\mathcal{D}'(\mathbb{R}^2)$. We will show that this distributional derivative is equal to the distribution given by the function obtained by taking $\frac{\partial}{\partial\overline{z}}\big(\frac{2}{\pi}\ln|x|\big)$ in the classical sense for $x \neq 0$. To this end, fix a function $\varphi \in C_0^\infty(\mathbb{R}^2)$. Since $\ln|x| \in L^1_{loc}(\mathbb{R}^2)$ (recall Example 2.9) we may write

$$\Big\langle \frac{\partial}{\partial\overline{z}}\Big(\frac{2}{\pi}\ln|x|\Big), \varphi\Big\rangle = -\Big\langle \frac{2}{\pi}\ln|x|, \frac{\partial\varphi}{\partial\overline{z}}\Big\rangle = -\frac{1}{\pi}\int_{\mathbb{R}^2} \ln|x|[(\partial_1 + i\partial_2)\varphi(x)]\,\mathrm{d}x. \tag{7.6.3}$$

Let $R > 0$ be such that $\operatorname{supp}\varphi \subset B(0,R)$. By the Lebesgue dominated convergence theorem 13.12 we further have

$$\int_{\mathbb{R}^2} \ln|x|\,[(\partial_1 + i\partial_2)\varphi(x)]\,\mathrm{d}x = \lim_{\varepsilon\to 0^+}\int_{\varepsilon\le|x|\le R} \ln|x|\,[(\partial_1 + i\partial_2)\varphi(x)]\,\mathrm{d}x. \tag{7.6.4}$$

Fix $\varepsilon \in (0, R/2)$ and with $x = (x_1, x_2)$ use (13.7.4) to write

$$\begin{aligned}&\int_{\varepsilon\le|x|\le R} \ln|x|\,(\partial_1 + i\partial_2)\varphi(x)\,\mathrm{d}x \\ &= -\int_{|x|\ge\varepsilon} \varphi(x)(\partial_1 + i\partial_2)\ln|x|\,\mathrm{d}x - \frac{\ln\varepsilon}{\varepsilon}\int_{|x|=\varepsilon}(x_1 + ix_2)\varphi(x)\,\mathrm{d}\sigma(x),\end{aligned} \tag{7.6.5}$$

where we have used the fact that the outward unit normal to $B(0,\varepsilon)$ at a point x on $\partial B(0,\varepsilon)$ is $\frac{1}{\varepsilon}(x_1, x_2)$ and that φ vanishes on $\partial B(0,R)$. Using polar coordinates, we further write

$$\Big|\frac{\ln\varepsilon}{\varepsilon}\int_{|x|=\varepsilon}(x_1 + ix_2)\varphi(x)\,\mathrm{d}\sigma(x)\Big|$$

$$= \Big|\varepsilon \ln \varepsilon \int_0^{2\pi} (\cos\theta + i\sin\theta)\varphi(\varepsilon\cos\theta, \varepsilon\sin\theta)\,\mathrm{d}\theta\Big|$$

$$\leq C\varepsilon|\ln\varepsilon \to 0 \quad \text{as} \quad \varepsilon \to 0^+. \tag{7.6.6}$$

Since $(\partial_1 + i\partial_2)[\ln|x|] \in L^1_{loc}(\mathbb{R}^2)$ and φ is compactly supported, by Lebesgue's dominated convergence theorem we have

$$\lim_{\varepsilon\to 0^+} \int_{|x|\geq\varepsilon} \varphi(x)(\partial_1 + i\partial_2)\ln|x|\,\mathrm{d}x = \int_{\mathbb{R}^2} \varphi(x)(\partial_1 + i\partial_2)\ln|x|\,\mathrm{d}x. \tag{7.6.7}$$

By combining (7.6.3)–(7.6.7) we conclude that

$$\frac{\partial}{\partial\overline{z}}\left(\frac{2}{\pi}\ln|x|\right) = \frac{1}{\pi}(\partial_1 + i\partial_2)[\ln|x|] = \frac{1}{\pi}\cdot\frac{1}{x_1 - ix_2} \quad \text{in } \mathcal{S}'(\mathbb{R}^2). \tag{7.6.8}$$

Similarly,

$$\frac{\partial}{\partial z}\left(\frac{2}{\pi}\ln|x|\right) = \frac{1}{\pi}(\partial_1 - i\partial_2)[\ln|x|] = \frac{1}{\pi}\cdot\frac{1}{x_1 + ix_2} \quad \text{in } \mathcal{S}'(\mathbb{R}^2). \tag{7.6.9}$$

With this at hand, by invoking Proposition 5.7, we have a complete description of all fundamental solutions for $\frac{\partial}{\partial z}$ and $\frac{\partial}{\partial\overline{z}}$ that are also tempered distributions.

In summary, we proved the following theorem.

Theorem 7.32. *Consider the functions*

$$E(x_1,x_2) := \frac{1}{\pi}\cdot\frac{1}{x_1 - ix_2}, \qquad F(x_1,x_2) := \frac{1}{\pi}\cdot\frac{1}{x_1 + ix_2}, \tag{7.6.10}$$

defined for $x = (x_1, x_2) \in \mathbb{R}^2 \setminus \{0\}$. Then $E \in \mathcal{S}'(\mathbb{R}^2)$ and is a fundamental solution for the operator $\frac{\partial}{\partial z}$. Also $F \in \mathcal{S}'(\mathbb{R}^2)$ and is a fundamental solution for the operator $\frac{\partial}{\partial\overline{z}}$.

Moreover,

$$\big\{u \in \mathcal{S}'(\mathbb{R}^2) :\ \tfrac{\partial}{\partial z}u = \delta \text{ in } \mathcal{S}'(\mathbb{R}^2)\big\} \tag{7.6.11}$$
$$= \big\{E + P :\ P \text{ polynomial in } \mathbb{R}^2 \text{ satisfying } \tfrac{\partial}{\partial z}P = 0 \text{ in } \mathbb{R}^2\big\}$$

and

$$\big\{u \in \mathcal{S}'(\mathbb{R}^2) :\ \tfrac{\partial}{\partial\overline{z}}u = \delta \text{ in } \mathcal{S}'(\mathbb{R}^2)\big\} \tag{7.6.12}$$
$$= \big\{F + P :\ P \text{ polynomial in } \mathbb{R}^2 \text{ satisfying } \tfrac{\partial}{\partial\overline{z}}P = 0 \text{ in } \mathbb{R}^2\big\}.$$

Having identified the fundamental solutions in $\mathbb{R}^2$ for the Cauchy–Riemann operator $\overline{\partial} = \frac{1}{2}(\partial_x + i\partial_y)$, we shall now compute the Fourier transform of $1/z$. Throughout, we shall repeatedly identify $\mathbb{R}^2$ with $\mathbb{C}$.

Proposition 7.33. *Let $u(z) := \frac{1}{z}$ for all $z \in \mathbb{C}\setminus\{0\}$. Then $u \in \mathcal{S}'(\mathbb{R}^2)$ and*

$$\widehat{u}(\xi) = \frac{2\pi}{i\xi} \quad \text{in } \mathcal{S}'(\mathbb{R}^2). \tag{7.6.13}$$

Proof. That $u \in \mathcal{S}'(\mathbb{R}^2)$ follows by observing that u is locally integrable near the origin, while $|u(z)|$ decays like $|z|^{-1}$ for $|z| > 1$. To prove (7.6.13), first note that, since $\overline{\partial}\left(\frac{1}{\pi z}\right) = \delta$ in $\mathcal{S}'(\mathbb{R}^2)$ by Theorem 7.32, it follows that $\overline{\partial} u = \pi\,\delta$ in $\mathcal{S}'(\mathbb{R}^2)$. Hence

$$\pi = \widehat{\overline{\partial} u}(\xi) = \frac{i}{2}\,\xi\,\widehat{u}(\xi) \Longrightarrow \xi\,\widehat{u}(\xi) = \frac{2\pi}{i}. \tag{7.6.14}$$

Thus, $\xi\left(\frac{2\pi}{i\xi} - \widehat{u}(\xi)\right) = 0$ in $\mathcal{S}'(\mathbb{R}^2)$, which, when combined with Example 2.68, implies

$$\frac{2\pi}{i\xi} - \widehat{u}(\xi) = c\,\delta, \quad \text{for some } \ c \in \mathbb{C}. \tag{7.6.15}$$

Thus,

$$\frac{2\pi}{i} u - \widehat{u} = c\,\delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^2). \tag{7.6.16}$$

Taking another Fourier transform yields

$$\frac{2\pi}{i}\widehat{u} + (2\pi)^2 u = c \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^2), \tag{7.6.17}$$

given that $\widehat{\widehat{u}} = -(2\pi)^2 u$ (keeping in mind that u is odd). A linear combination of (7.6.16) and (7.6.17), which eliminates $\widehat{u}$, then leads us to the conclusion that $(-2i\pi\delta + 1)\,c = 0$ in $\mathcal{S}'(\mathbb{R}^2)$. In turn, this forces $c = 0$, concluding the proof of the proposition. □

Exercise 7.34. *Consider $u(z) := 1/\overline{z}$ for all $z \in \mathbb{C} \setminus \{0\}$. Show that $u \in \mathcal{S}'(\mathbb{R}^2)$ and*

$$\widehat{u}(\xi) = \frac{2\pi}{i\overline{\xi}} \quad \textit{in} \quad \mathcal{S}'(\mathbb{R}^2). \tag{7.6.18}$$

Hint: Use Proposition 7.33 and Exercise 3.26.

Proposition 7.35. *Let Ω be an open set in $\mathbb{C}$ that contains the origin. Suppose $u \in C^0(\Omega)$ is such that $\frac{\partial u}{\partial \overline{z}} \in L^1_{loc}(\Omega)$ and $\frac{1}{z}\frac{\partial u}{\partial \overline{z}} \in L^1_{loc}(\Omega)$. Then $\frac{u}{z} \in L^1_{loc}(\Omega)$ and*

$$\frac{\partial}{\partial \overline{z}}\left[\frac{u}{z}\right] = \pi u(0)\delta + \frac{1}{z}\,\frac{\partial u}{\partial \overline{z}} \qquad \textit{in} \quad \mathcal{D}'(\Omega). \tag{7.6.19}$$

Proof. Pick a function $\theta \in C^\infty(\mathbb{C})$ with the property that $\theta = 0$ on $B(0,1)$ and $\theta = 1$ on $\mathbb{C} \setminus B(0,2)$. For each $\varepsilon \in (0,1)$ define $\theta_\varepsilon : \mathbb{C} \to \mathbb{C}$ by setting $\theta_\varepsilon(z) := \theta(z/\varepsilon)$ for each $z \in \mathbb{C}$. Then

$$\theta_\varepsilon \in C^\infty(\mathbb{C}), \quad \operatorname{supp}(\nabla\theta_\varepsilon) \subseteq \overline{B(0,2\varepsilon)} \setminus B(0,\varepsilon) \quad \forall\, z \in \mathbb{C} \setminus \{0\}, \tag{7.6.20}$$

$$\lim_{\varepsilon \to 0^+} \theta_\varepsilon(z) = 1, \quad \text{and} \quad |\theta_\varepsilon(z)| \leq 1 \quad \forall\, z \in \mathbb{C}. \tag{7.6.21}$$

Next, fix $\varphi \in C_0^\infty(\Omega)$ and, with $\mathcal{L}^2$ denoting the Lebesgue measure in $\mathbb{R}^2$, write

$$\begin{aligned}\Big\langle \frac{\partial}{\partial \overline{z}}\Big[\frac{u}{z}\Big], \varphi\Big\rangle &= -\Big\langle \frac{u}{z}, \frac{\partial \varphi}{\partial \overline{z}}\Big\rangle = -\int_\Omega \frac{u}{z}\,\frac{\partial \varphi}{\partial \overline{z}}\,\mathrm{d}\mathcal{L}^2 \\ &= -\lim_{\varepsilon\to 0^+}\int_\Omega \frac{u}{z}\,\theta_\varepsilon\,\frac{\partial \varphi}{\partial \overline{z}}\,\mathrm{d}\mathcal{L}^2, \end{aligned} \tag{7.6.22}$$

where for the last equality in (7.6.22) we used (7.6.21) and Lebesgue's dominated convergence theorem. Note that

$$\frac{1}{z}\varphi\,\theta_\varepsilon \in C_0^\infty(\Omega) \quad \text{and} \quad \frac{\partial}{\partial \overline{z}}\Big[\frac{1}{z}\Big] = 0 \quad \text{in} \quad \mathbb{C}\setminus\{0\}. \tag{7.6.23}$$

Therefore,

$$\frac{\partial}{\partial \overline{z}}\Big[\frac{1}{z}\varphi\,\theta_\varepsilon\Big] = \frac{1}{z}\theta_\varepsilon\frac{\partial \varphi}{\partial \overline{z}} + \frac{1}{z}\varphi\frac{\partial \theta_\varepsilon}{\partial \overline{z}} \quad \text{in} \quad \Omega. \tag{7.6.24}$$

Combining (7.6.22) and (7.6.24) we obtain

$$\begin{aligned}\Big\langle \frac{\partial}{\partial \overline{z}}\Big[\frac{u}{z}\Big], \varphi\Big\rangle &= -\lim_{\varepsilon\to 0^+}\int_\Omega u\,\frac{\partial}{\partial \overline{z}}\Big[\frac{1}{z}\,\varphi\,\theta_\varepsilon\Big]\,\mathrm{d}\mathcal{L}^2 + \lim_{\varepsilon\to 0^+}\int_\Omega \frac{u}{z}\,\varphi\,\frac{\partial \theta_\varepsilon}{\partial \overline{z}}\,\mathrm{d}\mathcal{L}^2 \\ &= \lim_{\varepsilon\to 0^+}\Big\langle \frac{\partial u}{\partial \overline{z}}, \frac{1}{z}\,\varphi\,\theta_\varepsilon\Big\rangle + \lim_{\varepsilon\to 0^+}\int_\Omega \frac{u}{z}\,\varphi\,\frac{\partial \theta_\varepsilon}{\partial \overline{z}}\,\mathrm{d}\mathcal{L}^2 =: I + II. \end{aligned} \tag{7.6.25}$$

Using the hypotheses on u, (7.6.21), and Lebesgue's dominated convergence theorem, we obtain

$$I = \lim_{\varepsilon\to 0^+}\int_\Omega \frac{\partial u}{\partial \overline{z}}\,\frac{1}{z}\,\varphi\,\theta_\varepsilon\,\mathrm{d}\mathcal{L}^2 = \int_\Omega \frac{\partial u}{\partial \overline{z}}\,\frac{1}{z}\,\varphi\,\mathrm{d}\mathcal{L}^2 = \Big\langle \frac{1}{z}\,\frac{\partial u}{\partial \overline{z}}, \varphi\Big\rangle. \tag{7.6.26}$$

To compute II, first write

$$\begin{aligned}\int_\Omega \frac{u}{z}\,\varphi\,\frac{\partial \theta_\varepsilon}{\partial \overline{z}}\,\mathrm{d}\mathcal{L}^2 &= \int_\Omega \frac{u-u(0)}{z}\,\varphi\,\frac{1}{\varepsilon}\,\frac{\partial \theta}{\partial \overline{z}}(\cdot/\varepsilon)\,\mathrm{d}\mathcal{L}^2 + u(0)\int_\Omega \frac{1}{z}\,\varphi\,\frac{\partial \theta_\varepsilon}{\partial \overline{z}}\,\mathrm{d}\mathcal{L}^2 \\ &=: III + IV. \end{aligned} \tag{7.6.27}$$

Using the support condition from (7.6.20), the continuity of u and the properties of φ, term III may be estimated by

$$|III| \le C\|\varphi\|_{L^\infty(\Omega)}\|\nabla\theta\|_{L^\infty(\Omega)}\left(\sup_{B(0,2\varepsilon)}|u-u(0)|\right) \xrightarrow[\varepsilon\to 0^+]{} 0. \tag{7.6.28}$$

As for IV, we have

$$\begin{aligned}\lim_{\varepsilon\to 0^+} IV &= u(0)\lim_{\varepsilon\to 0^+}\int_\Omega \frac{1}{z}\,\varphi\,\frac{\partial\theta_\varepsilon}{\partial\overline{z}}\,d\mathcal{L}^2 = -u(0)\lim_{\varepsilon\to 0^+}\int_\Omega \frac{1}{z}\,\frac{\partial\varphi}{\partial\overline{z}}\,\theta_\varepsilon\,d\mathcal{L}^2\\ &= -u(0)\int_\Omega \frac{1}{z}\,\frac{\partial\varphi}{\partial\overline{z}}\,d\mathcal{L}^2 = -u(0)\left\langle\frac{1}{z},\,\frac{\partial\varphi}{\partial\overline{z}}\right\rangle = u(0)\left\langle\frac{\partial}{\partial\overline{z}}\left[\frac{1}{z}\right],\,\varphi\right\rangle\\ &= \pi u(0)\langle\delta,\varphi\rangle. \end{aligned}\tag{7.6.29}$$

For the second equality in (7.6.29), for each $\varepsilon\in(0,1)$ fixed, we used integration by parts (cf. (13.7.4)) on $D\setminus\overline{B(0,\varepsilon/2)}$, where D is a bounded open subset of Ω with the property that $\operatorname{supp}\varphi\subset D$. At this step is also useful to recall (7.6.23). For the third equality in (7.6.29), we applied Lebesgue's dominated convergence theorem, while the fifth is based on Theorem 7.32. Now (7.6.19) follows by combining (7.6.25)–(7.6.29), since φ is arbitrary in $C_0^\infty(\Omega)$. □

Exercise 7.36. *Let Ω be an open set in $\mathbb{C}$ and let $z_0\in\Omega$. Suppose $u\in C^0(\Omega)$ is such that $\frac{\partial u}{\partial\overline{z}}\in L^1_{loc}(\Omega)$ and $\frac{1}{z-z_0}\,\frac{\partial u}{\partial\overline{z}}\in L^1_{loc}(\Omega)$. Then $\frac{u}{z-z_0}\in L^1_{loc}(\Omega)$ and*

$$\frac{\partial}{\partial\overline{z}}\left[\frac{u}{z-z_0}\right] = \pi u(z_0)\delta_{z_0} + \frac{1}{z-z_0}\,\frac{\partial u}{\partial\overline{z}} \qquad \text{in}\quad \mathcal{D}'(\Omega).\tag{7.6.30}$$

Proposition 7.37. *Given any complex-valued function $\varphi\in\mathcal{S}(\mathbb{R})$, define the* `Cauchy operator`

$$(\mathscr{C}\varphi)(z) := \frac{1}{2\pi i}\int_{\mathbb{R}}\frac{\varphi(x)}{x-z}\,dx, \qquad \forall\, z\in\mathbb{C}\setminus\mathbb{R}.\tag{7.6.31}$$

Then the following Plemelj jump-formula holds at every $x\in\mathbb{R}$:

$$\lim_{y\to 0^\pm}(\mathscr{C}\varphi)(x+iy) = \pm\frac{1}{2}\,\varphi(x) + \lim_{\varepsilon\to 0^+}\frac{1}{2\pi i}\int\limits_{\substack{y\in\mathbb{R}\\ |x-y|>\varepsilon}}\frac{\varphi(y)}{y-x}\,dy.\tag{7.6.32}$$

Proof. Apply Corollary 4.78 to the function $\Phi:\mathbb{R}^2\setminus\{(0,0)\}\to\mathbb{C}$ given by $\Phi(x,y) := \frac{-1}{2\pi i(x+iy)}$ for all $(x,y)\in\mathbb{R}^2\setminus\{(0,0)\}$. Note that Φ is C^∞, odd, and homogeneous of degree -1 and, under the canonical identification $\mathbb{R}^2\equiv\mathbb{C}$, takes the form $\Phi(z)=\frac{-1}{2\pi iz}$ for $z\in\mathbb{C}\setminus\{0\}$. Proposition 7.33 then gives that $\widehat{\Phi}(\xi)=\frac{1}{\xi}$ for all $\xi\in\mathbb{C}\setminus\{0\}$ which, in particular, yields $\widehat{\Phi}(0,1)=-i$. Having established this, (7.6.32) follows directly from (4.7.45). □

Remark 7.38. *Upon recalling formula (4.9.30) for the Hilbert transform H on the real line, we may recast the version of Plemelj jump-formula (7.6.32) corresponding to considering the Cauchy operator in the upper half-plane in the form*

$$\left[\mathscr{C}\varphi\right]\Big|^{ver}_{\partial\mathbb{R}^2_+} = \frac{1}{2}\,\varphi + \frac{i}{2\pi}H\varphi \quad \text{in}\quad \mathbb{R}, \qquad \forall\,\varphi\in\mathcal{S}(\mathbb{R}),\tag{7.6.33}$$

where the "vertical limit" of $\mathscr{C}\varphi$ to the boundary of the upper half-plane is understood as in (4.8.20). In turn, formula (7.6.33) suggests the consideration of the operator (with I denoting the identity)

$$P := \frac{1}{2}\,I + \frac{i}{2\pi}H. \tag{7.6.34}$$

From Corollary 4.95 it follows that P is a well-defined, linear and bounded operator on $L^2(\mathbb{R})$. Using the fact that $H^2 = -\pi^2 I$ and $H^* = -H$ on $L^2(\mathbb{R})$ (again, see Corollary 4.95), we may then compute

$$\begin{aligned} P^2 &= \Big(\frac{1}{2}\,I + \frac{i}{2\pi}H\Big)^2 = \frac{1}{4}\,I + \frac{i}{2\pi}H + \Big(\frac{i}{2\pi}\Big)^2 H^2 \\ &= \frac{1}{4}\,I + \frac{i}{2\pi}H + \frac{1}{4}\,I = \frac{1}{2}\,I + \frac{i}{2\pi}H \\ &= P, \end{aligned} \tag{7.6.35}$$

and

$$P^* = \frac{1}{2}\,I - \frac{i}{2\pi}H^* = \frac{1}{2}\,I + \frac{i}{2\pi}H = P. \tag{7.6.36}$$

Any linear and bounded operator on $L^2(\mathbb{R})$ satisfying these two properties (i.e., $P^2 = P$ and $P^* = P$) is called a projection. Then one may readily verify that $I - P = \frac{1}{2}\,I - \frac{i}{2\pi}H$ is also a projection and if we introduce (what are commonly referred to as Hardy spaces on the real line)

$$\mathscr{H}^2_{\pm}(\mathbb{R}) := \Big\{\Big(\frac{1}{2}\,I \pm \frac{i}{2\pi}H\Big)f :\ f \in L^2(\mathbb{R})\Big\}, \tag{7.6.37}$$

then any complex-valued function $f \in L^2(\mathbb{R})$ may be uniquely decomposed as $f = f_+ + f_-$ with $f_\pm \in \mathscr{H}^2_{\pm}(\mathbb{R})$ and, moreover, any two functions $f_\pm \in \mathscr{H}^2_{\pm}(\mathbb{R})$ are orthogonal, in the sense that

$$\int_{\mathbb{R}} f_+(x)\overline{f_-(x)}\,dx = 0. \tag{7.6.38}$$

7.7 Fundamental Solutions for the Dirac Operator

In a nutshell, `Dirac operators` are first order differential operators factoring the Laplacian. When $n = 1$, $\Delta = d^2/dx^2$, hence if we set $D := i(d/dx)$, then $D^2 = -\Delta$. We seek a higher-dimensional generalization of the latter factorization formula. The natural context in which such a generalization may be carried out is the Clifford algebra setting.

The Clifford algebra with n generators $\mathcal{C}\ell_n$ is the associative algebra with unit $\big(\mathcal{C}\ell_n, \odot, +, 1\big)$ freely generated over $\mathbb{R}$ by the family $\{\mathbf{e}_j\}_{1\leq j\leq n}$, of the standard orthonormal base in $\mathbb{R}^n$, now called imaginary units, subject to the following axioms:

$$\mathbf{e}_j \odot \mathbf{e}_k + \mathbf{e}_k \odot \mathbf{e}_j = -2\delta_{jk}, \qquad \forall\, j,k \in \{1,\dots,n\}. \tag{7.7.1}$$

Hence,

$$\begin{aligned} &\mathbf{e}_j \odot \mathbf{e}_k = -\mathbf{e}_k \odot \mathbf{e}_j \quad \text{if} \quad 1 \leq j \neq k \leq n, \\ &\text{and} \quad \mathbf{e}_j \odot \mathbf{e}_j = -1 \quad \text{for} \quad j \in \{1,\dots,n\}. \end{aligned} \tag{7.7.2}$$

The first condition above indicates that $\mathcal{C}\ell_n$ is noncommutative if $n > 1$, while the second condition justifies calling $\{\mathbf{e}_j\}_{1\leq j\leq n}$ imaginary units. Elements in the Clifford algebra $\mathcal{C}\ell_n$ can be uniquely written in the form

$$a = \sum_{l=0}^{n} {\sum_{|I|=l}}' a_I\, \mathbf{e}_I \tag{7.7.3}$$

with $a_I \in \mathbb{C}$, where $\mathbf{e}_I$ stands for the product $\mathbf{e}_{i_1} \odot \mathbf{e}_{i_2} \odot \cdots \odot \mathbf{e}_{i_l}$ whenever $I = (i_1, i_2, \dots, i_l)$ with $1 \leq i_1 < i_2 < \cdots < i_l \leq n$, $\mathbf{e}_\emptyset := 1 \in \mathbb{R}$ (which plays the role of the multiplicative unit in $\mathcal{C}\ell_n$) and ${\sum_{|I|=l}}'$ indicates that the sum is performed over strictly increasingly ordered indexes I with l components (selected from the set $\{1,\dots,n\}$). In the writing (7.7.3) we shall refer to the numbers $a_I \in \mathbb{C}$ as the scalar components of a.

Exercise 7.39. *Given any $a, b \in \mathcal{C}\ell_n$ with scalar components $a_I, b_I \in \mathbb{C}$ define*

$$(a,b) := \sum_{l=0}^{n} {\sum_{|I|=l}}' a_I b_I, \qquad \overline{a} := \sum_{l=0}^{n} {\sum_{|I|=l}}' \overline{a_I}\, \mathbf{e}_I, \tag{7.7.4}$$

and abbreviate $M_a b := a \odot b$. Prove that

$$\overline{M_a b} = M_{\overline{a}}\, \overline{b} \quad \text{and} \quad (M_a b\,,\, c) = (b\,,\, M_a c) \quad \text{for every} \quad a,b,c \in \mathcal{C}\ell_n. \tag{7.7.5}$$

Clifford algebra-valued functions defined in an open set $\Omega \subseteq \mathbb{R}^n$ may be defined naturally. Specifically, any function $f : \Omega \to \mathcal{C}\ell_n$ is an object of the form

$$f = \sum_{l=0}^{n} {\sum_{|I|=l}}' f_I\, \mathbf{e}_I, \tag{7.7.6}$$

where each f_I is a real-valued function defined in Ω. Given any $k \in \mathbb{N}_0 \cup \{\infty\}$, we shall denote by $C^k(\Omega, \mathcal{C}\ell_n)$ the collection of all Clifford algebra-valued functions f whose scalar components f_I are of class C^k in Ω. In a similar manner we may define $C_0^\infty(\Omega, \mathcal{C}\ell_n)$, $L^1_{loc}(\Omega, \mathcal{C}\ell_n)$, $L^p(\Omega, \mathcal{C}\ell_n)$, etc.

In fact, we may also consider Clifford algebra-valued distributions in an open set $\Omega \subseteq \mathbb{R}^n$. Specifically, write $u \in \mathcal{D}'(\Omega, \mathcal{C}\ell_n)$ provided

$$u = \sum_{l=0}^{n} {\sum_{|I|=l}}' u_I\, \mathbf{e}_I \qquad \text{with} \quad u_I \in \mathcal{D}'(\Omega) \quad \text{for each} \quad I. \tag{7.7.7}$$

Much of the theory originally developed for scalar-valued distributions naturally extends to this setting. For example, if $u \in \mathcal{D}'(\Omega, \mathcal{C}\ell_n)$ is as in (7.7.7), we may define

$$\partial^\alpha u := \sum_{l=0}^{n} {\sum_{|I|=l}}' \partial^\alpha u_I \, \mathbf{e}_I, \qquad \forall\, \alpha \in \mathbb{N}_0^n, \tag{7.7.8}$$

and

$$fu := \sum_{\ell=0}^{n} {\sum_{|J|=\ell}}' \sum_{k=0}^{n} {\sum_{|I|=k}}' f_J u_I \, \mathbf{e}_J \odot \mathbf{e}_I, \tag{7.7.9}$$

for any $f = \sum_{k=0}^{n} {\sum_{|J|=k}}' f_J \, \mathbf{e}_J \in C^\infty(\Omega, \mathcal{C}\ell_n)$.

We are ready to introduce the Dirac operator D associated with $\mathcal{C}\ell_n$. Specifically, given a Clifford algebra-valued distribution $u \in \mathcal{D}'(\Omega, \mathcal{C}\ell_n)$ as in (7.7.7), we define $Du \in \mathcal{D}'(\Omega, \mathcal{C}\ell_n)$ by setting

$$Du := \sum_{j=1}^{n} \sum_{l=0}^{n} {\sum_{|I|=l}}' \partial_j u_I \, \mathbf{e}_j \odot \mathbf{e}_I. \tag{7.7.10}$$

In other words,

$$D := \sum_{j=1}^{n} M_{\mathbf{e}_j} \partial_j, \tag{7.7.11}$$

where $M_{\mathbf{e}_j}$ denotes the operator of Clifford algebra multiplication by $\mathbf{e}_j$ from the left.

Proposition 7.40. *Let $\Omega \subseteq \mathbb{R}^n$ be an open set. Then the Dirac operator D satisfies*

$$D^2 = -\Delta \quad \text{in} \quad \mathcal{D}'(\Omega, \mathcal{C}\ell_n), \tag{7.7.12}$$

where Δ is the Laplacian.

Proof. Pick an arbitrary $u \in \mathcal{D}'(\Omega, \mathcal{C}\ell_n)$, say $u = \sum_{l=0}^{n} {\sum_{|I|=l}}' u_I \, \mathbf{e}_I$ with $u_I \in \mathcal{D}'(\Omega)$ for each I. Then using (7.7.10) twice yields

$$D^2 u = D(Du) = \sum_{j,k=1}^{n} \sum_{l=0}^{n} {\sum_{|I|=l}}' \partial_k \partial_j u_I \, \mathbf{e}_k \odot \mathbf{e}_j \odot \mathbf{e}_I. \tag{7.7.13}$$

Observe that, on the one hand,

$$\sum_{1 \leq j \neq k \leq n} \sum_{l=0}^{n} {\sum_{|I|=l}}' \partial_k \partial_j u_I \, \mathbf{e}_k \odot \mathbf{e}_j \odot \mathbf{e}_I = 0, \tag{7.7.14}$$

since for each I, we have $\partial_k\partial_j u_I = \partial_j\partial_k u_I$ and $\mathbf{e}_k \odot \mathbf{e}_j \odot \mathbf{e}_I = -\mathbf{e}_j \odot \mathbf{e}_k \odot \mathbf{e}_I$ whenever $1 \leq j \neq k$ by the first formula in (7.7.2). On the other hand, corresponding to the case when $j = k$

$$\sum_{k=1}^{n}\sum_{l=0}^{n}{\sum_{|I|=l}}' \partial_k^2 u_I\, \mathbf{e}_k \odot \mathbf{e}_k \odot \mathbf{e}_I = -\sum_{k=1}^{n}\sum_{l=0}^{n}{\sum_{|I|=l}}' \partial_k^2 u_I\, \mathbf{e}_I = -\Delta u, \tag{7.7.15}$$

since $\mathbf{e}_k \odot \mathbf{e}_k = -1$ by the second formula in (7.7.2). □

Exercise 7.41. *Consider the embedding*

$$\mathbb{R}^n \hookrightarrow \mathcal{C}\ell_n, \qquad \mathbb{R}^n \ni x = (x_j)_{1\leq j\leq n} \equiv \sum_{j=1}^{n} x_j \mathbf{e}_j \in \mathcal{C}\ell_n, \tag{7.7.16}$$

which identifies vectors from $\mathbb{R}^n$ with elements in the Clifford algebra $\mathcal{C}\ell_n$. With this identification in mind, show that

$$x \odot x = -|x|^2 \quad \text{for any } \ x \in \mathbb{R}^n, \tag{7.7.17}$$

$$x \odot y + y \odot x = -2x \cdot y \quad \text{for any } \ x, y \in \mathbb{R}^n. \tag{7.7.18}$$

For example, in light of the embedding described in (7.7.16) we may regard the assignment $\mathbb{R}^n \setminus \{0\} \ni x \mapsto \frac{x}{|x|^n}$ as a Clifford algebra-valued function.

Proposition 7.40 combined with Remark 5.6 yields the following result.

Theorem 7.42. *The Clifford algebra-valued function*

$$E(x) := -\frac{1}{\omega_{n-1}}\frac{x}{|x|^n} \in L^1_{loc}(\mathbb{R}^n, \mathcal{C}\ell_n) \cap \mathcal{S}'(\mathbb{R}^n, \mathcal{C}\ell_n) \tag{7.7.19}$$

is a fundamental solution for the Dirac operator D in $\mathbb{R}^n$, i.e.,

$$DE = \delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n, \mathcal{C}\ell_n). \tag{7.7.20}$$

Moreover, any $u \in \mathcal{S}'(\mathbb{R}^n, \mathcal{C}\ell_n)$ satisfying $Du = \delta$ in $\mathcal{S}'(\mathbb{R}^n, \mathcal{C}\ell_n)$ is of the form $E + P$ where E as in (7.7.19) and P is a Clifford algebra-valued function whose components are polynomials and such that $DP = 0$ in $\mathbb{R}^n$.

Proof. Let E_Δ be the fundamental solution for Δ as described in (7.1.12) for $n \geq 2$. Then

$$-(DE_\Delta)(x) = -\frac{1}{\omega_{n-1}}\frac{x}{|x|^n}, \qquad x \in \mathbb{R}^n \setminus \{0\}, \tag{7.7.21}$$

becomes, thanks to (7.7.12) and Remark 5.6, a fundamental solution for the Dirac operator D in $\mathbb{R}^n$.

To justify the last claim in the statement of the theorem, let $F \in \mathcal{S}'(\mathbb{R}^n, \mathcal{C}\ell_n)$ be an arbitrary fundamental solution for the Dirac operator D in $\mathbb{R}^n$. Then the tempered distribution $P := F - E$ satisfies $DP = 0$ in $\mathcal{S}'(\mathbb{R}^n, \mathcal{C}\ell_n)$ and, on the

Fourier transform side, we have $\xi \odot \widehat{P} = 0$ in $\mathcal{S}'(\mathbb{R}^n, \mathcal{C}\ell_n)$. Multiplying (in the Clifford algebra sense) this equality with the Clifford algebra-valued function with polynomial growth ξ then yields $-|\xi|^2 \widehat{P} = 0$ in $\mathcal{S}'(\mathbb{R}^n, \mathcal{C}\ell_n)$ (cf. (7.7.17)). The latter implies $\operatorname{supp} \widehat{P} \subseteq \{0\}$, hence the components of P are polynomials in $\mathbb{R}^n$ (cf. Exercise 4.35). □

There are other versions of the Dirac operator D from (7.7.11) that are more in line with the classical Cauchy–Riemann operator $\frac{\partial}{\partial \overline{z}} := \frac{1}{2}(\partial_1 + i\partial_2)$. Specifically, in $\mathbb{R}^{n+1}$ set $x = (x_0, x_1, x_2, \ldots, x_n) \in \mathbb{R}^{n+1}$ and consider

$$D^{\pm} := \partial_0 \pm D = \partial_0 \pm \sum_{j=1}^{n} M_{\mathbf{e}_j} \partial_j, \tag{7.7.22}$$

Note that in the case when $n = 1$ the Dirac operator D^- corresponds to a constant multiple of the Cauchy–Riemann operator $\frac{\partial}{\partial \overline{z}}$. A reasoning similar to that used above for D also yields fundamental solutions for $D^{\pm}$. We leave this as an exercise for the interested reader.

Exercise 7.43. *Let $\Omega \subseteq \mathbb{R}^{n+1}$ be an open set and let $x = (x_0, x_1, x_2, \ldots, x_n) \in \mathbb{R}^{n+1}$. Then the Dirac operators $D^{\pm}$ satisfy*

$$D^+ D^- = D^- D^+ = \Delta_{n+1} \quad \text{in} \quad \mathcal{D}'(\Omega, \mathcal{C}\ell_n), \tag{7.7.23}$$

where Δ_{n+1} is the Laplacian in $\mathbb{R}^{n+1}$.

Moreover, use a reasoning similar to the one used in the proof of Theorem 7.42 to show that the functions

$$E^+(x) := \frac{1}{\omega_n} \frac{x_0 + \sum_{j=1}^{n} x_j \mathbf{e}_j}{|x|^{n+1}} \in L^1_{loc}(\mathbb{R}^{n+1}, \mathcal{C}\ell_n) \hookrightarrow \mathcal{D}'(\mathbb{R}^{n+1}, \mathcal{C}\ell_n) \tag{7.7.24}$$

$$E^-(x) := \frac{1}{\omega_n} \frac{x_0 - \sum_{j=1}^{n} x_j \mathbf{e}_j}{|x|^{n+1}} \in L^1_{loc}(\mathbb{R}^{n+1}, \mathcal{C}\ell_n) \hookrightarrow \mathcal{D}'(\mathbb{R}^{n+1}, \mathcal{C}\ell_n) \tag{7.7.25}$$

are fundamental solution for the Dirac operators D^- and D^+, respectively, in $\mathbb{R}^{n+1}$.

We next introduce the `Cauchy-Clifford operator` and discuss its jump formulas in the upper and lower half-spaces.

Theorem 7.44. *For each $\mathcal{C}\ell_n$-valued function $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$ define the Cauchy–Clifford operator*

$$(\mathscr{C}\varphi)(x) := -\frac{1}{\omega_{n-1}} \int_{\mathbb{R}^{n-1}} \frac{\sum_{j=1}^{n-1}(x_j - y_j)\mathbf{e}_j + x_n \mathbf{e}_n}{\big|x - (y', 0)\big|^n} \odot \mathbf{e}_n \odot \varphi(y')\, dy', \tag{7.7.26}$$

for each $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ with $x_n \neq 0$. Then for each $x' \in \mathbb{R}^{n-1}$ one has

$$\lim_{x_n\to 0^{\pm}}(\mathscr{C}\varphi)(x',x_n)$$

$$=\pm\frac{1}{2}\,\varphi(x')+\lim_{\varepsilon\to 0^+}\frac{1}{\omega_{n-1}}\int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\ |x'-y'|>\varepsilon}}\frac{\sum_{j=1}^{n-1}(y_j-x_j)\mathbf{e}_j}{|x'-y'|^n}\odot\mathbf{e}_n\odot\varphi(y')\,\mathrm{d}y'. \tag{7.7.27}$$

Proof. Consider the Clifford algebra-value function $\Phi:\mathbb{R}^n\setminus\{0\}\to\mathcal{C}\ell_n$ given by

$$\Phi(x):=-\frac{\sum_{j=1}^{n}x_j\mathbf{e}_j}{\omega_{n-1}|x|^n}\odot\mathbf{e}_n\quad\text{for each}\quad x=(x_1,\dots,x_n)\in\mathbb{R}^n\setminus\{0\}. \tag{7.7.28}$$

Then Φ is C^∞, odd, and positive homogeneous of degree $1-n$ in $\mathbb{R}^n\setminus\{0\}$. As such, Corollary 4.78 may be applied (to each component of Φ). In this regard note from Corollary 4.62 that

$$\widehat{\Phi}(\xi)=-\frac{1}{\omega_{n-1}}\left[\sum_{j=1}^{n}\mathcal{F}\left(\frac{x_j}{|x|^n}\right)(\xi)\,\mathbf{e}_j\right]\odot\mathbf{e}_n=i\left[\sum_{j=1}^{n}\frac{\xi_j}{|\xi|^2}\,\mathbf{e}_j\right]\odot\mathbf{e}_n \tag{7.7.29}$$

in $\mathcal{S}'(\mathbb{R}^n)$. In particular, since $\mathbf{e}_n\odot\mathbf{e}_n=-1$, we obtain

$$\widehat{\Phi}(0',1)=i\,\mathbf{e}_n\odot\mathbf{e}_n=-i. \tag{7.7.30}$$

Given that, as seen from (7.7.26) and (7.7.28), we have

$$(\mathscr{C}\varphi)(x)=\int_{\mathbb{R}^{n-1}}\Phi(x'-y',x_n)\odot\varphi(y')\,\mathrm{d}y', \tag{7.7.31}$$

for each $x=(x_1,\dots,x_n)\in\mathbb{R}^n$ with $x_n\neq 0$, the jump formulas for the Cauchy–Clifford operator in (7.7.27) follow from (4.7.45) and (7.7.30). □

In parallel with the discussion in Remark 7.38, in the higher-dimensional setting we have the following connection between the Cauchy–Clifford operator and Riesz transforms.

Remark 7.45. *Let R_j, $j\in\{1,\dots,n-1\}$, be the Riesz transforms in $\mathbb{R}^{n-1}$ (i.e., singular integral operators defined as in (4.9.10) with $n-1$ in place of n). Also, recall the definition of the "vertical limit" of a function defined in $\mathbb{R}^n_+$ to the boundary of the upper half-space from (4.8.20). Then we may express the version of the jump-formula (7.7.27) corresponding to considering the Cauchy–Clifford operator in the upper half-space as*

$$\big[\mathscr{C}\varphi\big]\Big|^{ver}_{\partial\mathbb{R}^n_+}=\frac{1}{2}\,\varphi-\frac{1}{\omega_{n-1}}\sum_{j=1}^{n-1}\mathbf{e}_j\odot\mathbf{e}_n\odot(R_j\varphi)\quad\text{in}\quad\mathbb{R}^{n-1} \tag{7.7.32}$$

for each $\mathcal{C}\ell_n$-valued function $\varphi\in\mathcal{S}(\mathbb{R}^{n-1})$, where the Riesz transforms act on φ componentwise.

The format of the jump formula displayed in (7.7.32) suggests considering the operator acting on $\mathcal{C}\ell_n$-valued functions according to

$$P := \frac{1}{2}\,I - \frac{1}{\omega_{n-1}}\sum_{j=1}^{n-1}\mathbf{e}_j\odot\mathbf{e}_n\odot R_j$$

$$= \frac{1}{2}\,I + \frac{1}{\omega_{n-1}}\,\mathbf{e}_n\odot\Big(\sum_{j=1}^{n-1}\mathbf{e}_j\odot R_j\Big), \tag{7.7.33}$$

where I stands for the identity operator. In the second line of (7.7.33) the change in sign is due to the formula (cf. (7.7.2))

$$\mathbf{e}_j\odot\mathbf{e}_n = -\mathbf{e}_n\odot\mathbf{e}_j \quad \text{for each} \quad j\in\{1,\dots,n-1\}. \tag{7.7.34}$$

Theorem 4.93 then gives that P is a well-defined, linear, and bounded operator on $L^2(\mathbb{R}^{n-1},\mathcal{C}\ell_n)$. Moreover, since each R_j has a real-valued kernel, its action commutes with multiplication by elements from $\mathcal{C}\ell_n$ (i.e., for each $a\in\mathcal{C}\ell_n$ we have $R_jM_a = M_aR_j$, in the notation from Exercise 7.39). Keeping this in mind and relying on (7.7.34) and the fact that $\mathbf{e}_n^2 = -1$ (cf. (7.7.2)), we may then write

$$P^2 = \Big(\frac{1}{2}\,I + \frac{1}{\omega_{n-1}}\,\mathbf{e}_n\odot\Big(\sum_{j=1}^{n-1}\mathbf{e}_j\odot R_j\Big)\Big)^2$$

$$= \frac{1}{4}\,I + \frac{1}{\omega_{n-1}}\,\mathbf{e}_n\odot\Big(\sum_{j=1}^{n-1}\mathbf{e}_j\odot R_j\Big) + \Big(\frac{1}{\omega_{n-1}}\Big)^2\Big(\sum_{j=1}^{n-1}\mathbf{e}_j\odot R_j\Big)^2. \tag{7.7.35}$$

Furthermore, using the fact that, as proved in Theorem 4.93, the Riesz transforms commute with one another and satisfy $\sum\limits_{j=1}^{n-1}R_j^2 = -\big(\frac{\omega_{n-1}}{2}\big)^2 I$ on $L^2(\mathbb{R}^{n-1})$, we may expand

$$\Big(\sum_{j=1}^{n-1}\mathbf{e}_j\odot R_j\Big)^2 = \sum_{j,k=1}^{n-1}\mathbf{e}_j\odot\mathbf{e}_k\odot R_jR_k$$

$$= \sum_{j=1}^{n-1}\mathbf{e}_j^2R_j^2 + \sum_{1\leq j\neq k\leq n-1}\mathbf{e}_j\odot\mathbf{e}_k\odot R_jR_k$$

$$= -\sum_{j=1}^{n-1}R_j^2 = \big(\tfrac{\omega_{n-1}}{2}\big)^2 I, \tag{7.7.36}$$

where the source of the cancellation taking place in the third equality above is the observation that $\mathbf{e}_j\odot\mathbf{e}_k\odot R_jR_k = -\mathbf{e}_k\odot\mathbf{e}_j\odot R_kR_j$ whenever we have that $1\leq j\neq k\leq n-1$. Combining (7.7.35)–(7.7.36) and recalling (7.7.33) then yields

$$P^2 = P \quad \text{on} \quad L^2(\mathbb{R}^{n-1},\mathcal{C}\ell_n). \tag{7.7.37}$$

Note that this is in agreement with the result obtained in (7.6.35) in the case of the two-dimensional setting. Let us also consider the higher-dimensional analogue of (7.6.36). In this regard, we first observe that based on Exercise 7.39 for any $f, g \in L^2(\mathbb{R}^{n-1}, \mathcal{C}\ell_n)$ we may write

$$\begin{aligned}
&\int_{\mathbb{R}^{n-1}} \Big(M_{\mathbf{e}_n} \sum_{j=1}^{n-1} M_{\mathbf{e}_j} (R_j f)(x') \,,\, \overline{g(x')} \Big) \, \mathrm{d}x' \\
&\qquad = \int_{\mathbb{R}^{n-1}} \sum_{j=1}^{n-1} \Big((R_j f)(x') \,,\, \overline{M_{\mathbf{e}_j} M_{\mathbf{e}_n} g(x')} \Big) \, \mathrm{d}x' \\
&\qquad = - \int_{\mathbb{R}^{n-1}} \sum_{j=1}^{n-1} \Big(f(x') \,,\, \overline{M_{\mathbf{e}_j} M_{\mathbf{e}_n} (R_j g)(x')} \Big) \, \mathrm{d}x' \\
&\qquad = \int_{\mathbb{R}^{n-1}} \Big(f(x') \,,\, \overline{M_{\mathbf{e}_n} \sum_{j=1}^{n-1} M_{\mathbf{e}_j} (R_j g)(x')} \Big) \, \mathrm{d}x'. \qquad (7.7.38)
\end{aligned}$$

From this and (7.7.33) it follows that for every $f, g \in L^2(\mathbb{R}^{n-1}, \mathcal{C}\ell_n)$ we have

$$\int_{\mathbb{R}^{n-1}} \big((Pf)(x'), \overline{g(x')} \big) \, \mathrm{d}x' = \int_{\mathbb{R}^{n-1}} \big(f(x'), \overline{(Pg)(x')} \big) \, \mathrm{d}x', \qquad (7.7.39)$$

a condition that we shall interpret simply as

$$P^* = P \quad \text{on} \quad L^2(\mathbb{R}^{n-1}, \mathcal{C}\ell_n). \qquad (7.7.40)$$

In summary, the above analysis shows that the operator P defined as in (7.7.33) is a projection on $L^2(\mathbb{R}^{n-1}, \mathcal{C}\ell_n)$. Starting from this result, a corresponding higher-dimensional Hardy space theory may be developed in the Clifford-algebra setting as well.

7.8 Fundamental Solutions for General Second-Order Operators

Consider a constant, complex coefficient, homogeneous, second-order differential operator $L = \sum_{j,k=1}^{n} a_{jk} \partial_j \partial_k$ in $\mathbb{R}^n$. In the first stage, our goal is to find necessary and sufficient conditions, that can be expressed without reference to the theory of distributions, guaranteeing that a function $E \in L^1_{loc}(\mathbb{R}^n)$ is a fundamental solution for L. Several necessary conditions readily present themselves. First, it is clear that LE is the zero distribution in $\mathbb{R}^n \setminus \{0\}$. In addition, if L is elliptic then this necessarily implies that $E \in C^\infty(\mathbb{R}^n \setminus \{0\})$. In the absence of ellipticity as a hypothesis for L, we may wish to assume that E is reasonably regular, say $E \in C^2(\mathbb{R}^n \setminus \{0\})$. Second, if the fundamental solution E is a priori

known to be a tempered distribution, then $-L(\xi)\widehat{E}(\xi) = 1$ in $\mathcal{S}'(\mathbb{R}^n)$. If L is elliptic and $n \geq 3$, this forces $E = E_0 + P$ where $E_0 := -\mathcal{F}^{-1}\big(L(\xi)^{-1}\big)$ is homogeneous of degree $2-n$ and P is a polynomial that is annihilated by L. Hence, working with E_0 in place of E, there is no loss of generality in assuming that E is homogeneous of degree $2-n$. The case $n = 2$ may also be included in this discussion by demanding that ∇E is homogeneous of degree $1-n$.

In summary, it is reasonable to restrict our search for a fundamental solution for L in the class of functions satisfying $E \in C^2(\mathbb{R}^n \setminus \{0\}) \cap L^1_{loc}(\mathbb{R}^n)$ with the property that ∇E is positive homogeneous of degree $1-n$ in $\mathbb{R}^n \setminus \{0\}$. However, these conditions do not rule out such trivial candidates as the zero distribution. In Theorem 7.46 we identify the key nondegeneracy property (7.8.2), guaranteeing that E is in fact a fundamental solution. Theorem 7.46 is then later used to find an explicit formula for such a fundamental solution, under a strong ellipticity assumption on L (cf. Theorem 7.54).

Theorem 7.46. *Assume that $n \geq 2$ and consider*

$$L = \sum_{j,k=1}^{n} a_{jk}\partial_j\partial_k, \qquad a_{jk} \in \mathbb{C}. \tag{7.8.1}$$

Then for a function $E \in C^2(\mathbb{R}^n \setminus \{0\}) \cap L^1_{loc}(\mathbb{R}^n)$ with the property that ∇E is positive homogeneous of degree $1-n$ in $\mathbb{R}^n \setminus \{0\}$, the following statements are equivalent:

(1) When viewed in $L^1_{loc}(\mathbb{R}^n)$, the function E is a fundamental solution for L in $\mathbb{R}^n$;

(2) One has $LE = 0$ pointwise in $\mathbb{R}^n \setminus \{0\}$ and

$$\sum_{j,k=1}^{n} \int_{S^{n-1}} a_{jk}\omega_j \partial_k E(\omega)\, d\sigma(\omega) = 1. \tag{7.8.2}$$

Remark 7.47. *(i) In the partial differential equation parlance, the integrand in (7.8.2), that is, the expression $\sum_{j,k=1}^{n} a_{jk}\omega_j\partial_k E(\omega)$, is referred to as the conormal derivative of E on S^{n-1}.*

(ii) One remarkable aspect of Theorem 7.46 is that the description of a fundamental solution from part (2) is purely in terms of ordinary calculus (i.e., without any reference to the theory of distributions).

Proof of Theorem 7.46. Let $E \in C^2(\mathbb{R}^n \setminus \{0\}) \cap L^1_{loc}(\mathbb{R}^n)$ with the property that ∇E is positive homogeneous of degree $1-n$ in $\mathbb{R}^n \setminus \{0\}$. Exercise 4.51 then implies that $E \in L^1_{loc}(\mathbb{R}^n)$. Fix an arbitrary $f \in C^\infty_0(\mathbb{R}^n)$. Then making use of (4.4.19) and Proposition 4.67 (applied to each $\partial_k E$) we obtain

$$
\begin{aligned}
(LE)*f &= \sum_{j,k=1}^{n} a_{jk}\big(\partial_j(\partial_k E)\big)*f \\
&\quad \sum_{j,k=1}^{n} a_{jk}\Big(\int_{S^{n-1}} \partial_k E(\omega)\omega_j \,\mathrm{d}\sigma(\omega)\Big)(\delta * f) + \sum_{j,k=1}^{n} a_{jk}\Big(\text{P.V.}(\partial_j\partial_k E)\Big) * f \\
&= \Big(\int_{S^{n-1}} \sum_{j,k=1}^{n} a_{jk}\partial_k E(\omega)\omega_j \,\mathrm{d}\sigma(\omega)\Big) f \qquad (7.8.3) \\
&\quad + \lim_{\varepsilon\to 0^+} \int_{|y-\cdot\,|>\varepsilon} \sum_{j,k=1}^{n} a_{jk}(\partial_j\partial_k E)(\cdot - y) f(y)\,\mathrm{d}y \qquad \text{in} \quad \mathcal{D}'(\mathbb{R}^n),
\end{aligned}
$$

where we have also used the fact that $\delta * f = f$. Having proved this, we now turn in earnest to the proof of the equivalence in the statement of the theorem.

First, assume that E is a fundamental solution for L in $\mathbb{R}^n$. Then $LE = \delta$ in $\mathcal{D}'(\mathbb{R}^n)$ implies that $L\big(E\big|_{\mathbb{R}^n\setminus\{0\}}\big) = 0$ in $\mathcal{D}'(\mathbb{R}^n)$. Since by assumption $E\big|_{\mathbb{R}^n\setminus\{0\}}$ belongs to $C^2(\mathbb{R}^n\setminus\{0\})$, we arrive at the conclusion that $LE = 0$ pointwise in $\mathbb{R}^n\setminus\{0\}$. Explicitly,

$$
\sum_{j,k=1}^{n} a_{jk}\partial_j\partial_k E(x) = 0, \qquad \forall\, x\in\mathbb{R}^n\setminus\{0\}. \tag{7.8.4}
$$

This proves the first claim in *(2)*. Next, for each function $f\in C_0^\infty(\mathbb{R}^n)$ we may write $f = \delta * f = (LE)*f$ in $\mathcal{D}'(\mathbb{R}^n)$ which, in light of (7.8.3) and (7.8.4), forces

$$
f = \Big(\int_{S^{n-1}} \sum_{j,k=1}^{n} a_{jk}\partial_k E(\omega)\omega_j \,\mathrm{d}\sigma(\omega)\Big) f. \tag{7.8.5}
$$

Since $f\in C_0^\infty(\mathbb{R}^n)$ was arbitrary, (7.8.2) follows. This finishes the proof of $(1)\Rightarrow(2)$.

Conversely, suppose that $E\in C^2(\mathbb{R}^n\setminus\{0\})\cap L^1_{loc}(\mathbb{R}^n)$ with the property that ∇E is positive homogeneous of degree $1-n$ in $\mathbb{R}^n\setminus\{0\}$, such that $LE = 0$ pointwise in $\mathbb{R}^n\setminus\{0\}$, and (7.8.2) holds. Then for each $f\in C_0^\infty(\mathbb{R}^n)$ formula (7.8.3) simply reduces to $(LE)*f = f$. Now Exercise 2.88 may be invoked to conclude that $LE = \delta$ in $\mathcal{D}'(\mathbb{R}^n)$, as wanted. □

Next, we turn to the task of finding all fundamental solutions that are tempered distributions for general homogeneous, second-order, constant coefficient operators that are strongly elliptic. We begin by defining this stronger (than originally introduced in Definition 6.13) notion of ellipticity.

Let $A = \big(a_{jk}\big)_{1\le j,k\le n}\in\mathcal{M}_{n\times n}(\mathbb{C})$ and associated to such a matrix A, consider the operator

$$
L_A := L_A(\partial) := \sum_{j,k=1}^{n} a_{jk}\partial_j\partial_k. \tag{7.8.6}
$$

This is a homogeneous, second-order, constant coefficient operator for which the ellipticity condition reads

$$L_A(\xi) := \sum_{j,k=1}^{n} a_{jk}\xi_j\xi_k \neq 0, \qquad \forall\, \xi = (\xi_1, \ldots, \xi_n) \in \mathbb{R}^n \setminus \{0\}. \tag{7.8.7}$$

As a trivial consequence of the Malgrange–Ehrenpreis theorem (cf. Theorem 5.9), any elliptic operator has a fundamental solution. The goal is to obtain explicit formulas for such fundamental solutions for a subclass of homogeneous, elliptic, second-order, constant coefficient operators satisfying a stronger condition than (7.8.7).

Definition 7.48. *Call an operator L_A as in (7.8.6)* `strongly elliptic`*, if there exists a constant $C \in (0, \infty)$ such that*

$$\mathrm{Re}\Big[\sum_{j,k=1}^{n} a_{jk}\xi_j\xi_k\Big] \geq C|\xi|^2, \quad \forall\, \xi = (\xi_1, \ldots, \xi_n) \in \mathbb{R}^n. \tag{7.8.8}$$

By extension, call a matrix $A = \big(a_{jk}\big)_{1\leq j,k\leq n} \in \mathcal{M}_{n\times n}(\mathbb{C})$ `strongly elliptic` *provided there exists some $C \in (0, \infty)$ with the property that (7.8.8) holds.*

Remark 7.49.

(1) It is obvious that any operator L_A as in (7.8.6) that is strongly elliptic is elliptic.

(2) Up to changing L to $-L$, any elliptic, homogeneous, second-order, constant coefficient differential operator L with real coefficients is strongly elliptic. To see why this is the case let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ and suppose that L_A is elliptic. Consider the function $f : S^{n-1} \to \mathbb{R}$ defined by $f(\xi) := \sum_{j,k=1}^{n} a_{jk}\xi_j\xi_k$ for $\xi \in S^{n-1}$. Then f is continuous and since L_A is elliptic the number 0 is not in the image of f. The unit sphere S^{n-1} being compact and connected, it is mapped by f in a compact, connected, subset of $\mathbb{R}$, not containing 0. This forces the image of f to be a compact interval that does not contain 0. Hence, there exists $c \in (0, \infty)$ with the property that either $f(\xi) \geq c$ for every $\xi \in S^{n-1}$ or $-f(\xi) \geq c$ for every $\xi \in S^{n-1}$. This implies that either $f(\xi/|\xi|) \geq c$ for every $\xi \in \mathbb{R}^n \setminus \{0\}$ or $-f(\xi/|\xi|) \geq c$ for every $\xi \in \mathbb{R}^n \setminus \{0\}$, or equivalently, that either L_A or $-L_A$ is strongly elliptic.

(3) Consider the operator $L = \partial_1^2 + i\partial_2^2$ in $\mathbb{R}^2$. Then L is a homogeneous, second-order, constant coefficient differential operator with $L(\xi) = \xi_1^2 + i\xi_2^2$, $\xi = (\xi_1, \xi_2) \in \mathbb{R}^2$. Clearly $L(\xi) \neq 0$ if $\xi \neq 0$, so L is elliptic. However, L is not strongly elliptic since $\mathrm{Re}\,[L(\xi)] = \xi_1^2$, which cannot be bounded from below by a constant multiple of $|\xi|^2$ since the latter blows up if $|\xi_2| \to \infty$.

Fix $A = \big(a_{jk}\big)_{1\leq j,k\leq n} \in \mathcal{M}_{n\times n}(\mathbb{C})$ and consider the operator L_A as in (7.8.6). Due to the symmetry of mixed partial derivatives in the sense of distributions, it is immediate that

$$L_A = L_{A_{sym}}, \quad \text{where} \quad A_{sym} := \frac{A+A^\top}{2}. \tag{7.8.9}$$

As such, any fundamental solution for $L_{A_{sym}}$ is also a fundamental solution for L_A. Also, since $(A_{sym}\xi)\cdot\xi = (A\xi)\cdot\xi$ for each $\xi\in\mathbb{R}^n$, we have that

$$L_A \quad \text{is strongly elliptic if and only if} \quad L_{A_{sym}} \quad \text{is strongly elliptic.} \tag{7.8.10}$$

Consequently, when computing the fundamental solution for L_A we may assume without loss of generality that A is symmetric, that is, $A = A^\top$.

For further reference we summarize a few basic properties of symmetric matrices (throughout, the symbol dot denotes the real inner product of vectors with complex components).

$$A\in\mathcal{M}_{n\times n}(\mathbb{R}),\ \ A=A^\top \implies (A\zeta)\cdot\overline{\zeta}\in\mathbb{R},\quad \forall\,\zeta\in\mathbb{C}^n, \tag{7.8.11}$$

$$A\in\mathcal{M}_{n\times n}(\mathbb{C}),\ \ A=A^\top \implies \operatorname{Re}A=(\operatorname{Re}A)^\top \ \text{ and } \ \operatorname{Im}A=(\operatorname{Im}A)^\top, \tag{7.8.12}$$

$$A\in\mathcal{M}_{n\times n}(\mathbb{C}),\ \ A=A^\top \implies \operatorname{Re}\big[(A\zeta)\cdot\overline{\zeta}\,\big]=\big((\operatorname{Re}A)\zeta\big)\cdot\overline{\zeta},\quad \forall\,\zeta\in\mathbb{C}^n. \tag{7.8.13}$$

It is easy to see that (7.8.11)–(7.8.12) hold, while (7.8.13) follows from (7.8.11)–(7.8.12) after writing $A=\operatorname{Re}A+i\operatorname{Im}A$.

Also, recall that a matrix $A\in\mathcal{M}_{n\times n}(\mathbb{C})$ is said to be `positive definite` provided

$$(A\zeta)\cdot\overline{\zeta} \quad \text{is real and strictly positive for each} \quad \zeta\in\mathbb{C}^n\setminus\{0\}. \tag{7.8.14}$$

It is easy to see that any positive definite matrix $A\in\mathcal{M}_{n\times n}(\mathbb{C})$ satisfies $\big(\overline{A}\,\big)^\top = A$, and there exists $c\in(0,\infty)$ such that

$$(A\zeta)\cdot\overline{\zeta}\geq c|\zeta|^2,\qquad \forall\,\zeta\in\mathbb{C}^n. \tag{7.8.15}$$

Remark 7.50. *Fix a matrix $A\in\mathcal{M}_{n\times n}(\mathbb{C})$ that is symmetric and satisfies (7.8.8). Then, for each $\zeta\in\mathbb{C}^n$ we have (with $C\in(0,\infty)$ as in (7.8.8))*

$$\begin{aligned}\operatorname{Re}\big[(A\zeta)\cdot\overline{\zeta}\,\big] &= \operatorname{Re}\big[\big(A(\operatorname{Re}\zeta+i\operatorname{Im}\zeta)\big)\cdot(\operatorname{Re}\zeta-i\operatorname{Im}\zeta)\big]\\ &= \operatorname{Re}\big[(A\operatorname{Re}\zeta)\cdot\operatorname{Re}\zeta+(A\operatorname{Im}\zeta)\cdot\operatorname{Im}\zeta\big]\\ &\geq C|\operatorname{Re}\zeta|^2+C|\operatorname{Im}\zeta|^2=C|\zeta|^2.\end{aligned} \tag{7.8.16}$$

The second equality in (7.8.16) uses the fact that A is symmetric, while (7.8.8) is used for the inequality in (7.8.16). Thus, combining (7.8.16) with the Cauchy–Schwarz inequality, we obtain

$$|A\zeta|\geq C|\zeta| \quad \textit{for every} \quad \zeta\in\mathbb{C}^n, \tag{7.8.17}$$

proving that the linear map $A : \mathbb{C}^n \to \mathbb{C}^n$ is injective, thus invertible. In particular, $\det A \neq 0$. *Also, thanks to* (7.8.13) *and* (7.8.16) *we have that* $\operatorname{Re} A$ *is a positive definite matrix. More precisely, with* C *as in* (7.8.8), *we have*

$$\big((\operatorname{Re} A)\zeta\big) \cdot \overline{\zeta} \geq C|\zeta|^2 \qquad \forall\, \zeta \in \mathbb{C}^n. \tag{7.8.18}$$

From Remark 7.50 and definitions we see that, given any $A \in \mathcal{M}_{n\times n}(\mathbb{C})$, the following implications hold:

$$\operatorname{Re} A \text{ positive definite} \implies A \text{ strongly elliptic}, \tag{7.8.19}$$

$$A \text{ strongly elliptic} \iff A_{sym} \text{ strongly elliptic}, \tag{7.8.20}$$

$$A \text{ strongly elliptic and symmetric} \implies \operatorname{Re} A \text{ positive definite}. \tag{7.8.21}$$

Remark 7.51. *Assume that* $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ *is symmetric and satisfies* (7.8.8). *From Remark 7.50 it follows that* A *is invertible. Moreover, if we define*

$$\|A\| := \sup\big\{|A\zeta| : \zeta \in \mathbb{C}^n,\ |\zeta| = 1\big\}, \tag{7.8.22}$$

then (7.8.17) *ensures that* $\|A\| > 0$. *We claim that*

$$\operatorname{Re}\big[(A^{-1}\zeta) \cdot \overline{\zeta}\,\big] \geq \frac{C}{\|A\|^2}|\zeta|^2 \qquad \forall\, \zeta \in \mathbb{C}^n, \tag{7.8.23}$$

where C *is as in* (7.8.8). *To justify this, first note that* $|\zeta|^2 = |AA^{-1}\zeta|^2 \leq \|A\|^2|A^{-1}\zeta|^2$ *for every* $\zeta \in \mathbb{C}^n$. *In turn, this and* (7.8.16) *permit us to estimate*

$$\begin{aligned}\operatorname{Re}\big[(A^{-1}\zeta) \cdot \overline{\zeta}\,\big] &= \operatorname{Re}\big[\,\overline{(A^{-1}\zeta)} \cdot \zeta\big] = \operatorname{Re}\big[\,\overline{(A^{-1}\zeta)} \cdot (AA^{-1}\zeta)\big] \geq C|A^{-1}\zeta|^2 \\ &\geq \frac{C}{\|A\|^2}|\zeta|^2 \qquad \forall\, \zeta \in \mathbb{C}^n. \end{aligned} \tag{7.8.24}$$

This proves (7.8.23). *In particular,* (7.8.23) *yields*

$$\|A^{-1}\||\xi|^2 \geq |(A^{-1}\xi) \cdot \xi| \geq \frac{C}{\|A\|^2}|\xi|^2 \qquad \forall\, \xi \in \mathbb{R}^n. \tag{7.8.25}$$

Remark 7.52. *Consider the set*

$$\mathfrak{M} := \big\{A \in \mathcal{M}_{n\times n}(\mathbb{C}) : A = A^{\top},\ \operatorname{Re} A \text{ is positive definite}\big\}. \tag{7.8.26}$$

Since the $n \times n$ *symmetric matrices* $A = (a_{jk})_{1\leq j,k\leq n}$ *with complex entries are uniquely determined by the elements* a_{jk} *with* $1 \leq j \leq k \leq n$, *we may naturally identify* $\mathfrak{M}$ *with an open convex subset of* $\mathbb{C}^{n(n+1)/2}$. *Throughout, this identification is implicitly assumed. Moreover, every* $A \in \mathfrak{M}$ *satisfies* $\det A \neq 0$, *since if* $A\zeta = 0$ *for some* $\zeta \in \mathbb{C}^n$, *then* (7.8.13) *and the fact that* $\operatorname{Re} A$ *is positive definite force* $\zeta = 0$. *The fact that* $\mathfrak{M}$ *is convex, implies that there is a unique analytic branch of the mapping* $\mathfrak{M} \ni A \mapsto (\det A)^{\frac{1}{2}} \in \mathbb{C}$ *such that* $(\det A)^{\frac{1}{2}} > 0$ *when* A *is real. Thus* $(\det A)^{\frac{1}{2}}$ *is unambiguously defined for* $A \in \mathfrak{M}$.

To proceed with the discussion regarding determining a fundamental solution for a strongly elliptic operator L_A we first analyze the case when A has real entries.

The case when A is real, symmetric and satisfies (7.8.8).
Since the matrix A is real, symmetric and positive definite, A is diagonalizable with $A = U^{-1}DU$ for some orthogonal matrix $U \in \mathcal{M}_{n\times n}(\mathbb{R})$ and some diagonal $n \times n$ matrix D where the entries on the main diagonal are strictly positive real numbers. Hence, $D^{\frac{1}{2}}$ is meaningfully defined as the $n \times n$ diagonal matrix with the entries on the diagonal being equal to the square roots of the entries on the main diagonal in D. In addition, $A^{\frac{1}{2}} := U^{-1}D^{\frac{1}{2}}U$ is well-defined, symmetric, invertible, $A^{\frac{1}{2}}A^{\frac{1}{2}} = A$ and $\det(A^{\frac{1}{2}}) = \sqrt{\det A}$. Next, fix a function

$$u \in L^1_{loc}(\mathbb{R}^n) \quad \text{such that} \quad \partial_k u \in L^1_{loc}(\mathbb{R}^n) \quad \text{for each} \quad k \in \{1,\dots,n\}, \tag{7.8.27}$$

where the derivatives are taken in the sense of distributions. Also, consider $\varphi \in C^\infty_0(\mathbb{R}^n)$. Then, using the fact that A is symmetric, we obtain

$$\begin{aligned}\langle L_A u,\varphi\rangle &= \sum_{j,k=1}^{n} a_{jk}\big\langle \partial_j\partial_k u,\varphi\big\rangle = -\sum_{j,k=1}^{n} a_{jk}\big\langle \partial_k u,\partial_j\varphi\big\rangle \\ &= -\int_{\mathbb{R}^n} \nabla u(x)\cdot\big(A\nabla\varphi(x)\big)\,dx. \end{aligned}\tag{7.8.28}$$

In the integral in (7.8.28) we make the change of variables $x = A^{\frac{1}{2}}y$. Since for every invertible matrix $B \in \mathcal{M}_{n\times n}(\mathbb{R})$ and any function f the chain rule gives

$$\begin{aligned}&(\nabla f)(By) = (B^\top)^{-1}\nabla[f(By)] \quad \text{for each} \quad y\in\mathbb{R}^n \\ &\text{with the property that } f \text{ is differentiable at } By,\end{aligned}\tag{7.8.29}$$

we obtain

$$\begin{aligned}\int_{\mathbb{R}^n}(\nabla u)(x)\cdot\big(A(\nabla\varphi)(x)\big)\,dx &= |\det A^{\frac{1}{2}}|\int_{\mathbb{R}^n}(\nabla u)(A^{\frac{1}{2}}y)\cdot\big(A(\nabla\varphi)(A^{\frac{1}{2}}y)\big)\,dy \\ &= \sqrt{\det A}\int_{\mathbb{R}^n}\nabla\big[u(A^{\frac{1}{2}}y)\big] \\ &\quad\cdot\big((A^{\frac{1}{2}})^{-1}A(A^{\frac{1}{2}})^{-1}\nabla\big[\varphi(A^{\frac{1}{2}}y)\big]\big)\,dy \\ &= \sqrt{\det A}\int_{\mathbb{R}^n}\nabla v(y)\cdot\nabla\psi(y)\,dy \\ &= -\sqrt{\det A}\int_{\mathbb{R}^n} v(y)\Delta\psi(y)\,dy,\end{aligned}\tag{7.8.30}$$

where we have set

$$v := u\circ A^{\frac{1}{2}} \quad \text{and} \quad \psi := \varphi\circ A^{\frac{1}{2}} \in C^\infty_0(\mathbb{R}^n), \tag{7.8.31}$$

and for the last equality in (7.8.30) we integrated by parts one more time. Hence, (7.8.28) and (7.8.30) imply

$$\langle L_A u, \varphi\rangle = \sqrt{\det A}\int_{\mathbb{R}^n} v(y)\Delta\psi(y)\,dy. \tag{7.8.32}$$

In particular, if we now choose

$$u(y) := \frac{1}{\sqrt{\det A}}\big(E_\Delta \circ A^{-\frac{1}{2}}\big)(y) \quad \text{for} \quad y\in\mathbb{R}^n\setminus\{0\}, \tag{7.8.33}$$

where E_Δ is the fundamental solution for the Laplacian from (7.1.12), then u satisfies the conditions listed in (7.8.27), and the function v from (7.8.31) becomes

$$v(y) = \frac{1}{\sqrt{\det A}}E_\Delta(y) \quad \text{for} \quad y\in\mathbb{R}^n\setminus\{0\}. \tag{7.8.34}$$

Since $A^{\frac{1}{2}}(0)=0$, we may write

$$\begin{aligned}
\varphi(0) &= \big(\varphi\circ A^{\frac{1}{2}}\big)(0) = \psi(0) = \langle\delta,\psi\rangle = \langle\Delta E_\Delta,\psi\rangle = \langle E_\Delta,\Delta\psi\rangle \\
&= \sqrt{\det A}\int_{\mathbb{R}^n}\frac{1}{\sqrt{\det A}}E_\Delta(y)\Delta\psi(y)\,dy = \sqrt{\det A}\int_{\mathbb{R}^n} v(y)\Delta\psi(y)\,dy \\
&= \langle L_A u,\varphi\rangle = \frac{1}{\sqrt{\det A}}\big\langle L_A\big(E_\Delta\circ A^{-\frac{1}{2}}\big),\varphi\big\rangle,
\end{aligned} \tag{7.8.35}$$

where for the second to the last equality in (7.8.35) we used (7.8.32). Since (7.8.35) holds for every $\varphi\in C_0^\infty(\mathbb{R}^n)$, we may conclude that

$$\frac{1}{\sqrt{\det A}}\,E_\Delta\circ A^{-\frac{1}{2}} \quad \text{is a fundamental solution for } L_A \text{ in } \mathbb{R}^n. \tag{7.8.36}$$

Denoting this fundamental solution by E_A and keeping in mind that $|A^{-\frac{1}{2}}x|^2 = \big(A^{-\frac{1}{2}}x\big)\cdot\big(A^{-\frac{1}{2}}x\big) = \big(A^{-1}x\big)\cdot x$ for every $x\in\mathbb{R}^n$, we obtain that the function defined by

$$E_A(x) := \begin{cases} \dfrac{-1}{(n-2)\omega_{n-1}\sqrt{\det A}}\cdot\dfrac{1}{[(A^{-1}x)\cdot x]^{\frac{n-2}{2}}} & \text{if } n\geq 3, \\[2ex] \dfrac{1}{4\pi\sqrt{\det A}}\ln\big[(A^{-1}x)\cdot x\big] & \text{if } n=2, \end{cases} \tag{7.8.37}$$

for every $x\in\mathbb{R}^n\setminus\{0\}$ is a fundamental solution for L_A in the current case. Moreover, as is apparent from (7.8.37), (7.8.25), and Exercise 4.5, the function E_A is locally integrable and a tempered distribution in $\mathbb{R}^n$.

In preparation for dealing with the case of matrices with complex entries, we state and prove the following useful complex analysis result.

Lemma 7.53. *Let $N \in \mathbb{N}$ and assume $\mathcal{O}$ is an open and convex subset of $\mathbb{C}^N$ with the property that $\mathcal{O} \cap \mathbb{R}^N \neq \emptyset$ (where $\mathbb{R}^N$ is canonically embedded into $\mathbb{C}^N$). Also, suppose $f, g : \mathcal{O} \to \mathbb{C}$ are two functions that are separately holomorphic (i.e., in each scalar complex component in $\mathbb{C}^N$) such that*

$$f\big|_{\mathcal{O}\cap\mathbb{R}^N} = g\big|_{\mathcal{O}\cap\mathbb{R}^N}. \tag{7.8.38}$$

Then $f = g$ in $\mathcal{O}$.

Proof. Fix an arbitrary point $(x_1, \ldots, x_N) \in \mathcal{O} \cap \mathbb{R}^N$ and consider

$$\mathcal{O}^1 := \big\{z_1 \in \mathbb{C} : (z_1, x_2, \ldots, x_N) \in \mathcal{O}\big\}. \tag{7.8.39}$$

Then $\mathcal{O}^1$ is an open convex subset of $\mathbb{C}$, which contains x_1, hence $\mathcal{O}^1 \cap \mathbb{R} \neq \emptyset$. Define the functions $f_1, g_1 : \mathcal{O}^1 \to \mathbb{C}$ by

$$f_1(z) := f(z, x_2, \ldots, x_N) \ \text{ and } \ g_1(z) := g(z, x_2, \ldots, x_N), \quad \forall\, z \in \mathcal{O}^1. \tag{7.8.40}$$

Then f_1 and g_1 are holomorphic functions in $\mathcal{O}^1$ that coincide on $\mathcal{O}^1 \cap \mathbb{R}$. Since the latter contains an accumulation point in the convex (hence connected) set $\mathcal{O}^1$, it follows that $f_1 = g_1$ on $\mathcal{O}^1$ by the coincidence theorem for holomorphic functions of one complex variable. Since $(x_1, \ldots, x_N) \in \mathcal{O} \cap \mathbb{R}^N$ was arbitrary, we may conclude that

$$f\big|_{\mathcal{O}\cap(\mathbb{C}\times\mathbb{R}^{N-1})} = g\big|_{\mathcal{O}\cap(\mathbb{C}\times\mathbb{R}^{N-1})}. \tag{7.8.41}$$

Next, fix $(z_1, x_2, \ldots, x_N) \in \mathcal{O} \cap (\mathbb{C} \times \mathbb{R}^{N-1})$ and define

$$\mathcal{O}^2 := \big\{z_2 \in \mathbb{C} : (z_1, z_2, x_3, \ldots, x_N) \in \mathcal{O}\big\}. \tag{7.8.42}$$

Once again, $\mathcal{O}^2$ is an open convex subset of $\mathbb{C}$, which contains x_2, hence $\mathcal{O}^2 \cap \mathbb{R} \neq \emptyset$. If we now define the functions $f_2, g_2 : \mathcal{O}^2 \to \mathbb{C}$ by

$$f_2(z) := f(z_1, z, x_3, \ldots, x_N) \ \text{ and } \ g_2(z) := g(z_1, z, x_3, \ldots, x_N), \quad \forall\, z \in \mathcal{O}^2, \tag{7.8.43}$$

it follows that f_2, g_2 are holomorphic in $\mathcal{O}^2$, which, by (7.8.41), coincide on $\mathcal{O}^2 \cap \mathbb{R}$. Given that the latter set contains an accumulation point in the convex set $\mathcal{O}^1$, we deduce that $f_2 = g_2$ on $\mathcal{O}^2$ by once again invoking the coincidence theorem for holomorphic functions of one complex variable. Upon recalling that $(z_1, x_2, \ldots, x_N) \in \mathcal{O} \cap (\mathbb{C} \times \mathbb{R}^{N-1})$ was arbitrary, we conclude that

$$f\big|_{\mathcal{O}\cap(\mathbb{C}^2\times\mathbb{R}^{N-2})} = g\big|_{\mathcal{O}\cap(\mathbb{C}^2\times\mathbb{R}^{N-2})}. \tag{7.8.44}$$

Continuing this process inductively, we arrive at the conclusion that $f = g$ in $\mathcal{O}$. □

After this preamble, we are ready to consider the general case.

The case when A has complex entries, is symmetric and satisfies (7.8.8). As observed in Remark 7.50, under the current assumptions, A continues to be

invertible. Also, (7.8.25) holds. In addition, under the current assumptions we have that $A \in \mathfrak{M}$ and $(\det A)^{\frac{1}{2}}$ is unambiguously defined (see in Remark 7.52).

These comments show that the function E_A from (7.8.37) continues to be well-defined under the current assumption on A if ln is replaced by the principal branch of the complex log (defined for points $z \in \mathbb{C}\backslash(-\infty, 0]$ so that $z^a = e^{a \log z}$ for each $a \in \mathbb{R}$). In addition, E_A continues to belong to $L^1_{loc}(\mathbb{R}^n)$ and we have $E_A \in C^\infty(\mathbb{R}^n \setminus \{0\})$. Furthermore, from (7.8.25) and Exercise 4.5 it follows that the function E_A continues to be a tempered distribution in $\mathbb{R}^n$. The goal is to prove that this expression is a fundamental solution for L_A in the current case.

First, observe that since A^{-1} is symmetric, for each $j, k \in \{1, \ldots, n\}$ we have

$$\partial_k\big[(A^{-1}x)\cdot x\big] = 2(A^{-1}x)_k \quad \text{and} \quad \partial_j\big[(A^{-1}x)_k\big] = (A^{-1})_{kj}. \tag{7.8.45}$$

Hence, for every $x \in \mathbb{R}^n \setminus \{0\}$, differentiating pointwise we obtain

$$\begin{aligned} L_A\big((A^{-1}x)\cdot x\big)^{\frac{2-n}{2}} &= \sum_{j,k=1}^n a_{jk}\partial_j\Big[(2-n)\big((A^{-1}x)\cdot x\big)^{\frac{-n}{2}}(A^{-1}x)_k\Big] \\ &= -n(2-n)\sum_{j,k=1}^n a_{jk}\big((A^{-1}x)\cdot x\big)^{\frac{-n-2}{2}}(A^{-1}x)_j(A^{-1}x)_k \\ &\quad + (2-n)\sum_{j,k=1}^n a_{jk}\big((A^{-1}x)\cdot x\big)^{\frac{-n}{2}}(A^{-1})_{kj} = 0. \end{aligned} \tag{7.8.46}$$

Similarly, $L_A \log\big[(A^{-1}x)\cdot x\big] = 0$ for $x \in \mathbb{R}^2 \setminus \{0\}$. Thus, we may conclude that

$$L_A E_A(x) = 0 \qquad \forall\, x \in \mathbb{R}^n \setminus \{0\}. \tag{7.8.47}$$

Second, by making use of (7.8.45) and the expression for E_A we obtain

$$\nabla E_A(x) = \frac{1}{\omega_{n-1}\sqrt{\det A}}\cdot\frac{A^{-1}x}{[(A^{-1}x)\cdot x]^{\frac{n}{2}}} \qquad \forall\, x \in \mathbb{R}^n \setminus \{0\} \tag{7.8.48}$$

which, in particular, shows that ∇E_A is positive homogeneous of degree $1-n$ in $\mathbb{R}^n \setminus \{0\}$. Furthermore, for each $x \in S^{n-1}$ we have

$$\begin{aligned} \sum_{j,k=1}^n a_{jk}x_j\partial_k E_A(x) &= \frac{1}{\omega_{n-1}\sqrt{\det A}}\cdot\frac{\sum_{j,k=1}^n a_{jk}x_j(A^{-1}x)_k}{[(A^{-1}x)\cdot x]^{\frac{n}{2}}} \\ &= \frac{1}{\omega_{n-1}\sqrt{\det A}}\cdot\frac{(A^{\top}x)\cdot(A^{-1}x)}{[(A^{-1}x)\cdot x]^{\frac{n}{2}}} \\ &= \frac{1}{\omega_{n-1}\sqrt{\det A}}\cdot\frac{1}{[(A^{-1}x)\cdot x]^{\frac{n}{2}}}, \end{aligned} \tag{7.8.49}$$

where the last equality uses the fact that $|x| = 1$.

Invoking Theorem 7.46 we conclude that E_A is a fundamental solution for L_A in $\mathbb{R}^n$ if and only if

$$\omega_{n-1}\sqrt{\det A} = \int_{S^{n-1}} \frac{\mathrm{d}\sigma(x)}{[(A^{-1}x)\cdot x]^{\frac{n}{2}}}. \tag{7.8.50}$$

The fact that we already know that E_A is a fundamental solution for L_A in $\mathbb{R}^n$ in the case when $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ satisfies $A = A^\top$ and condition (7.8.8), implies that formula (7.8.50) holds for this class of matrices.

We make the claim that in fact (7.8.50) actually holds for the larger class of matrices $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ satisfying $A = A^\top$ and condition (7.8.8). To see why this is true, recall the open subset $\mathfrak{M}$ of $\mathbb{C}^{n(n+1)/2}$ from (7.8.26) and consider the functions $f, g : \mathfrak{M} \to \mathbb{C}$ defined for every $A = \big(a_{jk}\big)_{1\leq j,k\leq n} \in \mathfrak{M}$ by

$$f\big((a_{jk})_{1\leq j\leq k\leq n}\big) := \omega_{n-1}\sqrt{\det A}, \tag{7.8.51}$$

$$g\big((a_{jk})_{1\leq j\leq k\leq n}\big) := \int_{S^{n-1}} \frac{\mathrm{d}\sigma(x)}{[(A^{-1}x)\cdot x]^{\frac{n}{2}}}. \tag{7.8.52}$$

Then f and g are analytic (as functions of several complex variables) on $\mathfrak{M}$, which is an open convex set in $\mathbb{C}^{n(n+1)/2}$. If $A \in \mathfrak{M}$ has real entries, then A satisfies (7.8.8), and (7.8.50) holds for such A, hence $f = g$ on $\mathfrak{M} \cap \mathcal{M}_{n\times n}(\mathbb{R})$. Invoking Lemma 7.53 we may therefore conclude that $f = g$ on $\mathfrak{M}$. Thus, (7.8.50) holds for every $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ satisfying $A = A^\top$ and condition (7.8.8).

Finally, we note that thanks to Proposition 5.7 and the current strong ellipticity assumption, any other fundamental solution of L_A belonging to $\mathcal{S}'(\mathbb{R}^n)$ differs from E_A by a polynomial that L_A annihilates.

In summary, the above analysis proves the following result.

Theorem 7.54. *Suppose $A = \big(a_{jk}\big)_{1\leq j,k\leq n} \in \mathcal{M}_{n\times n}(\mathbb{C})$ and consider the operator L_A associated to A as in (7.8.6). If L_A is strongly elliptic, then the function defined by*

$$E_A(x) := \begin{cases} -\dfrac{1}{(n-2)\omega_{n-1}\sqrt{\det A_{sym}}} \cdot \dfrac{1}{[((A_{sym})^{-1}x)\cdot x]^{\frac{n-2}{2}}} & \text{if } n \geq 3, \\[2ex] \dfrac{1}{4\pi\sqrt{\det A_{sym}}} \log\big[((A_{sym})^{-1}x)\cdot x\big] & \text{if } n = 2, \end{cases} \tag{7.8.53}$$

for $x \in \mathbb{R}^n\backslash\{0\}$ belongs to $L^1_{loc}(\mathbb{R}^n)\cap\mathcal{S}'(\mathbb{R}^n)\cap C^\infty(\mathbb{R}^n\backslash\{0\})$ and is a fundamental solution for L_A in $\mathbb{R}^n$. Above, $A_{sym} := \frac{1}{2}(A + A^\top)$, $\sqrt{\det A_{sym}}$ is defined as in Remark 7.52, and log *denotes the principal branch of the complex logarithm (defined for complex numbers $z \in \mathbb{C}\setminus(-\infty, 0]$ so that $z^a = e^{a\log z}$ for each $a \in \mathbb{R}$). Moreover,*

$$\begin{aligned} &\big\{u \in \mathcal{S}'(\mathbb{R}^n) : L_A u = \delta \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n)\big\} \\ &\qquad = \big\{E_A + P : P \ \text{ polynomial such that } L_A P = 0 \text{ in } \mathbb{R}^n\big\}. \end{aligned} \tag{7.8.54}$$

We conclude this section with a couple of related exercises about fundamental solutions for second-order, constant coefficient, differential operators.

Exercise 7.55. *Let $n \geq 2$ and consider a differential operator $L = \sum_{j,k=1}^{n} a_{jk}\partial_j\partial_k$ with complex coefficients. Assume that $E \in C^2(\mathbb{R}^n \setminus \{0\}) \cap L^1_{loc}(\mathbb{R}^n)$ is a function with the property that ∇E is positive homogeneous of degree $1-n$ in $\mathbb{R}^n \setminus \{0\}$. In addition, suppose that*

$$\lambda := \sum_{j,k=1}^{n} \int_{S^{n-1}} a_{jk}\omega_j \partial_k E(\omega)\, d\sigma(\omega) \neq 0. \tag{7.8.55}$$

Prove that $\lambda^{-1}E$ is a fundamental solution for L in $\mathbb{R}^n$. Use this result to find the proper normalization for the standard fundamental solution for the Laplacian in $\mathbb{R}^n$, starting with $E(x) := |x|^{2-n}$ when $n \geq 3$, and with $E(x) := \ln|x|$ for $n = 2$.

Exercise 7.56. *Let $n \geq 2$ and suppose that $E \in C^2(\mathbb{R}^n \setminus \{0\}) \cap L^1_{loc}(\mathbb{R}^n)$ has the property that ∇E is odd and positive homogeneous of degree $1-n$ in $\mathbb{R}^n \setminus \{0\}$. In addition, assume that the function E is a fundamental solution for the complex constant coefficient differential operator $L = \sum_{j,k=1}^{n} a_{jk}\partial_j\partial_k$ in $\mathbb{R}^n$. Prove that for every $\xi \in S^{n-1}$ one has*

$$\int\limits_{\substack{\omega\in S^{n-1}\\ \omega\cdot\xi>0}} \sum_{j,k=1}^{n} a_{jk}\omega_j\partial_k E(\omega)\, d\sigma(\omega) = \frac{1}{2}. \tag{7.8.56}$$

7.9 Layer Potential Representation Formulas Revisited

The goal here is to derive a layer potential representation formula generalizing the identity from Proposition 7.17 for the Laplacian. We begin by describing the setting in which we intend to work.

Given a matrix $A = \big(a_{jk}\big)_{1\leq j,k\leq n} \in \mathcal{M}_{n\times n}(\mathbb{C})$, we associate the homogeneous second-order differential operator

$$L_A = \sum_{j,k=1}^{n} a_{jk}\partial_j\partial_k \quad \text{in} \quad \mathbb{R}^n, \tag{7.9.1}$$

and for every unit vector $\nu = (\nu_1, \ldots, \nu_n)$ and any complex-valued function u of class C^1, define the conormal derivative of u associated with the matrix A (along ν) as

$$\partial^A_\nu u := \sum_{j,k=1}^{n} \nu_j a_{jk}\partial_k u. \tag{7.9.2}$$

Theorem 7.57. *Suppose $n \geq 2$ and let $\Omega \subset \mathbb{R}^n$ be a bounded domain of class C^1, with outward unit normal ν and surface measure σ. In addition, assume that the matrix $A = \big(a_{jk}\big)_{1\leq j,k\leq n} \in \mathcal{M}_{n\times n}(\mathbb{C})$ is such that the operator L_A associated with A as in (7.9.1) is strongly elliptic, and recall the fundamental solution E_A for L_A defined in (7.8.53).*

Then for every complex-valued function $u \in C^2(\overline{\Omega}\,)$ one has

$$\begin{aligned}
&\int_\Omega (L_A u)(y) E_A(x-y)\,{\rm d}y - \int_{\partial\Omega} E_A(x-y)(\partial_\nu^A u)(y)\,{\rm d}\sigma(y)\\
&\quad - \int_{\partial\Omega} u(y)(\partial_\nu^{A^\top} E)(x-y)\,{\rm d}\sigma(y)\\
&\qquad = \begin{cases} u(x), & x\in\Omega,\\ 0, & x\in\mathbb{R}^n\setminus\overline{\Omega}, \end{cases}
\end{aligned} \tag{7.9.3}$$

where is the conormal derivative associated with the matrix $A^\top$ (along ν).

Proof. When $x \in \mathbb{R}^n \setminus \overline{\Omega}$, it clear from (7.8.53) that $E_A(x-\cdot) \in C^\infty(\overline{\Omega}\,)$. Also, since E_A is a fundamental solution for L_A in $\mathbb{R}^n$ we have that $L_A[E_A(x-\cdot)] = 0$ in $\mathbb{R}^n \setminus \{x\}$, hence $(L_A E_A)(x-\cdot) = 0$ in Ω. Based on these and repeated use of (13.7.4) we may then write

$$\begin{aligned}
\int_\Omega (L_A u)(y) E_A(x-y)\,{\rm d}y &= \sum_{j,k=1}^n \int_\Omega a_{jk}(\partial_j\partial_k u)(y) E_A(x-y)\,{\rm d}y\\
&= \sum_{j,k=1}^n \int_\Omega a_{jk}(\partial_k u)(y)(\partial_j E_A)(x-y)\,{\rm d}y\\
&\quad + \sum_{j,k=1}^n \int_{\partial\Omega} a_{jk}\nu_j(y)(\partial_k u)(y) E_A(x-y)\,{\rm d}\sigma(y)\\
&= \int_\Omega u(y)(L_A E_A)(x-y)\,{\rm d}y + \int_{\partial\Omega} u(y)(\partial_\nu^{A^\top} E_A)(x-y)\,{\rm d}\sigma(y)\\
&\quad + \int_{\partial\Omega} E_A(x-y)(\partial_\nu^A u)(y)\,{\rm d}\sigma(y).
\end{aligned} \tag{7.9.4}$$

Upon recalling that $(L_A E_A)(x-\cdot) = 0$ in Ω, the last solid integral drops out and the resulting identity is in agreement with (7.9.3).

Consider now the case when $x \in \Omega$. Since Ω is open, there exists $r > 0$ such that $B(x,r) \subseteq \Omega$. For each $\varepsilon \in (0,r)$ define $\Omega_\varepsilon := \Omega \setminus \overline{B(x,\varepsilon)}$ which is a bounded domain of class C^1. Since $E_A(x-\cdot) \in C^\infty(\overline{\Omega_\varepsilon}\,)$ and $(L_A E_A)(x-\cdot) = 0$ in Ω_ε, the same type of reasoning as above gives (keeping in mind that $\partial\Omega_\varepsilon = \partial\Omega \cup \partial B(x,\varepsilon)$)

$$\begin{aligned}
\int_{\Omega_\varepsilon} (L_A u)(y) E_A(x-y)\,{\rm d}y &= \int_{\partial\Omega} u(y)(\partial_\nu^{A^\top} E_A)(x-y)\,{\rm d}\sigma(y)\\
&\quad - \int_{\partial B(x,\varepsilon)} u(y)(\partial_\nu^{A^\top} E_A)(x-y)\,{\rm d}\sigma(y)
\end{aligned}$$

$$+\int_{\partial\Omega} E_A(x-y)(\partial_\nu^A u)(y)\,d\sigma(y)$$
$$-\int_{\partial B(x,\varepsilon)} E_A(x-y)(\partial_\nu^A u)(y)\,d\sigma(y) =: I+II+III+IV. \tag{7.9.5}$$

As seen from (7.8.53), we have $|IV| \leq C_A\|\nabla u\|_{L^\infty(\Omega)}\,\varepsilon\max\{1,|\ln|\varepsilon||\}$, from which we deduce that $\lim_{\varepsilon\to 0^+} IV = 0$. Next, split $II = II' + u(x)II''$ where

$$II' := -\int_{\partial B(x,\varepsilon)} [u(y)-u(x)](\partial_\nu^{A^\top} E_A)(x-y)\,d\sigma(y), \tag{7.9.6}$$

and observe that

$$\begin{aligned} II'' &:= -\int_{\partial B(x,\varepsilon)} (\partial_\nu^{A^\top} E_A)(x-y)\,d\sigma(y) \\ &= -\int_{\partial B(x,\varepsilon)} \sum_{j,k=1}^n a_{jk}\frac{y_k-x_k}{\varepsilon}(\partial_j E_A)(x-y)\,d\sigma(y) \\ &= \int_{S^{n-1}} \sum_{j,k=1}^n a_{jk}\omega_k(\partial_j E_A)(\omega)\,d\sigma(\omega) \\ &= \int_{S^{n-1}} \sum_{j,k=1}^n (A^\top)_{jk}\,\omega_j(\partial_k E_{A^\top})(\omega)\,d\sigma(\omega) = 1. \end{aligned} \tag{7.9.7}$$

Above, the first equality defines II'', the second equality uses the definition of the conormal derivative and the outward unit normal to the ball, the third equality is based on the change of variables $\omega = (x-y)/\varepsilon$ and the fact that ∇E_A is positive homogeneous of degree $1-n$. Finally, in the fourth equality we have interchanged j and k in the summation and used the identities $E_A = E_{A^\top}$, $L_A = L_{A^\top}$, while the last equality is due to (7.8.2).

In addition, $|II'| \leq C_A\|\nabla u\|_{L^\infty(\Omega)}\,\varepsilon$, hence $\lim_{\varepsilon\to 0^+} II' = 0$, and

$$\lim_{\varepsilon\to 0^+}\int_{\Omega_\varepsilon} (L_A u)(y)E_A(x-y)\,dy = \int_\Omega (L_A u)(y)E_A(x-y)\,dy$$

by Lebesgue's dominated convergence theorem (recall that $E_A(x-\cdot)\in L^1(\Omega)$ since Ω is bounded). Collectively, the results deduced in the above analysis yield (7.9.3) in the case when $x\in\Omega$, finishing the proof of the theorem. □

Further Notes for Chap. 7. As evidenced by the treatment of the Poisson problem for the Laplacian and bi-Laplacian (from Sects. 7.2 and 7.4, respectively), fundamental solutions play a key role both for establishing integral representation formulas and for deriving estimates for the solution. This type of application to partial differential

equations amply substantiate the utility of the tools from distribution theory and harmonic analysis derived in Sect. 4.10 (dealing with derivatives of volume potentials) and Sect. 4.9 (dealing with singular integral operators). The aforementioned Poisson problems serve as a prototype for other types of boundary value problems formulated for other differential operators and with the entire space $\mathbb{R}^n$ replaced by an open set $\Omega \subset \mathbb{R}^n$. In the latter scenario one specifies boundary conditions on $\partial\Omega$ in place of ∞ as in the case of $\mathbb{R}^n$ (note that ∞ plays the role of the topological boundary of $\mathbb{R}^n$ regarded as an open subset of its compactification $\mathbb{R}^n \cup \{\infty\}$).

In Sect. 7.7 the Dirac operator has been considered in the natural setting of Clifford algebras. For more information pertaining to this topic, the interested reader is referred to the monographs [4, 22, 49]. The last two references also contain a discussion of Hardy spaces in the context of Clifford algebras (a topic touched upon in Sect. 7.7). A classical reference to Hardy spaces in the ordinary context of $\mathbb{C}$ (which appeared at the end of Sect. 7.6) is the book [27].

7.10 Additional Exercises for Chap. 7

Exercise 7.58. *Prove that there exists a unique $E \in \mathcal{S}'(\mathbb{R}^n)$ with the property that $\Delta E - E = \delta$ in $\mathcal{S}'(\mathbb{R}^n)$.*

Exercise 7.59. *Does there exist $E \in L^1(\mathbb{R}^n)$ such that $\Delta E = \delta$ in $\mathcal{D}'(\mathbb{R}^n)$?*

Exercise 7.60. *Prove that for every $u \in \mathcal{D}'(\Omega)$ we have*

$$\operatorname{div}(\nabla u) = \Delta u \quad \text{in} \quad \mathcal{D}'(\Omega). \tag{7.10.1}$$

Exercise 7.61. *Prove that if $f \in C^\infty(\Omega)$ and $u \in \mathcal{D}'(\Omega)$, then*

$$\Delta(fu) = (\Delta f)u + 2(\nabla f)\cdot(\nabla u) + f\Delta u \quad \text{in} \quad \mathcal{D}'(\Omega). \tag{7.10.2}$$

Exercise 7.62. *Suppose $n \geq 2$ and denote by E_Δ the fundamental solution for the Laplacian operator Δ given in (7.1.12). Without making use of Corollary 4.62, prove that for each $j \in \{1, \ldots, n\}$ one has*

$$\mathcal{F}(\partial_j E_\Delta) = -i\frac{\xi_j}{|\xi|^2} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{7.10.3}$$

In turn, use (7.10.3) to show that

$$\mathcal{F}\left(\frac{x_j}{|x|^n}\right) = -i\omega_{n-1}\frac{\xi_j}{|\xi|^2} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{7.10.4}$$

and

$$\mathcal{F}^{-1}\left(\frac{\xi_j}{|\xi|^2}\right) = \frac{i}{\omega_{n-1}}\cdot\frac{x_j}{|x|^n} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{7.10.5}$$

Exercise 7.63. *Suppose $n \geq 3$ and denote by E_{Δ^2} the fundamental solution for the bi-Laplacian operator Δ^2 given in (7.3.8). Prove that for each $j, k \in \{1, \ldots, n\}$ one has*

$$\mathcal{F}(\partial_j\partial_k E_{\Delta^2}) = -\frac{\xi_j\xi_k}{|\xi|^4} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{7.10.6}$$

Consequently,

$$\mathcal{F}^{-1}\left(\frac{\xi_j\xi_k}{|\xi|^4}\right)=\frac{1}{2(n-2)\omega_{n-1}}\cdot\frac{\delta_{jk}}{|x|^{n-2}}-\frac{1}{2\omega_{n-1}}\cdot\frac{x_jx_k}{|x|^n}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n),\tag{7.10.7}$$

and

$$\mathcal{F}\left(\frac{x_jx_k}{|x|^n}\right)=\omega_{n-1}\frac{\delta_{jk}}{|\xi|^2}-2\omega_{n-1}\frac{\xi_j\xi_k}{|\xi|^4}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n).\tag{7.10.8}$$

Exercise 7.64. *Let $P(D)$ be a nonzero linear constant coefficient operator of order $m\in\mathbb{N}_0$. Prove that $P(D)$ is elliptic if and only if there exist two constants $C,R\in(0,\infty)$ such that $|P(\xi)|\geq C|\xi|^m$ for every $\xi\in\mathbb{R}^n\setminus B(0,R)$.*

Exercise 7.65. *Reprove Theorem 7.54 without making any appeal to Theorem 7.46.*

Chapter 8

The Heat Operator and Related Versions

Throughout this chapter we use the notation $(x,t) := (x_1,\dots,x_n,t) \in \mathbb{R}^{n+1}$. The `heat operator`[1] is then defined as $L := \partial_t - \Delta_x = \partial_t - \sum_{j=1}^{n} \partial_{x_j}^2$ and will be at the center of our investigations in this chapter. The focus is on determining all fundamental solutions for the heat operator that are tempered distributions. As an application of the latter, we also discuss the generalized Cauchy problem for the heat operator. In addition, at the end of this section we compute a fundamental solution for the Schrödinger operator $\frac{1}{i}\partial_t - \Delta_x$.

8.1 Fundamental Solutions for the Heat Operator

The starting point in determining all fundamental solutions for the heat operator L that are tempered distributions is Theorem 5.13, which guarantees that such fundamental solutions do exist. As we have seen in the case of the Laplace operator, the Fourier transform is an important tool in determining explicit expressions for fundamental solutions that are tempered distributions. We will continue to make use of this tool in the case of the heat operator with the adjustment that, this time, we work with the partial Fourier transform $\mathcal{F}_x$ discussed at the end of Sect. 4.2.

Let $E \in \mathcal{S}'(\mathbb{R}^{n+1})$ be a fundamental solution for L. Thus, in view of Exercise 2.81 and (4.1.25), we have

$$\partial_t E - \Delta_x E = \delta(x)\otimes\delta(t) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n+1}). \tag{8.1.1}$$

[1] First considered in 1809 for $n = 1$ by Laplace (cf. [39]) and then for higher dimensions by Poisson (cf. [56] for $n = 2$.)

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Applying $\mathcal{F}_x$ to (8.1.1), denoting $\mathcal{F}_x(E)$ by $\widehat{E}_x$, and using Exercise 4.40, it follows that

$$\partial_t \widehat{E}_x + |\xi|^2 \widehat{E}_x = 1(\xi) \otimes \delta(t) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n+1}). \tag{8.1.2}$$

In particular, for each fixed $\xi \in \mathbb{R}^n$, we have

$$\left[\partial_t + |\xi|^2\right] \widehat{E}_x = \delta(t) \quad \text{in} \quad \mathcal{D}'(\mathbb{R}). \tag{8.1.3}$$

Applying Example 5.12, we obtain that $\widehat{E}_x(\xi, t) := H(t)\mathrm{e}^{-|\xi|^2 t}$ is a solution of (8.1.3), where as before, H denotes the Heaviside function from (1.2.16). Note that we have $\widehat{E}_x(\xi, t) \in \mathcal{S}'(\mathbb{R}^{n+1})$. Also, if $\varphi \in C_0^\infty(\mathbb{R}^{n+1})$ then, integration by parts yields

$$\begin{aligned}
\langle \partial_t \widehat{E}_x + |\xi|^2 \widehat{E}_x, \varphi \rangle &= -\langle H(t)\mathrm{e}^{-|\xi|^2 t}, \partial_t \varphi(\xi, t) \rangle + \langle H(t)\mathrm{e}^{-|\xi|^2 t}, |\xi|^2 \varphi(\xi, t) \rangle \\
&= -\int_{\mathbb{R}^n} \int_0^\infty \mathrm{e}^{-|\xi|^2 t} \partial_t \varphi(\xi, t)\, \mathrm{d}t\, \mathrm{d}\xi + \int_0^\infty \int_{\mathbb{R}^n} |\xi|^2 \mathrm{e}^{-|\xi|^2 t} \varphi(\xi, t)\, \mathrm{d}\xi\, \mathrm{d}t \\
&= \int_{\mathbb{R}^n} \varphi(\xi, 0)\, \mathrm{d}\xi = \langle 1(\xi) \otimes \delta(t), \varphi \rangle,
\end{aligned} \tag{8.1.4}$$

hence $\widehat{E}_x$ verifies $\partial_t \widehat{E}_x + |\xi|^2 \widehat{E}_x = 1(\xi) \otimes \delta(t)$ in $\mathcal{D}'(\mathbb{R}^{n+1})$. Invoking (4.1.25), it follows that $\widehat{E}_x(\xi, t) = H(t)\mathrm{e}^{-|\xi|^2 t}$ verifies (8.1.2). We remark here that while the distribution $-H(-t)\mathrm{e}^{-|\xi|^2 t}$ satisfies (8.1.3), based on our earlier discussion pertaining to the nature of the function in (4.1.27), it does not belong to $\mathcal{S}'(\mathbb{R}^{n+1})$, thus we cannot apply $\mathcal{F}_x^{-1}$ to it.

Starting from the identity $\mathcal{F}_x(\mathcal{F}_x E(x, t)) = (2\pi)^n E(-x, t)$ (which is easy to check), we may write

$$\begin{aligned}
E(-x, t) &= (2\pi)^{-n} \mathcal{F}_\xi(\widehat{E}_x(\xi))(x) = (2\pi)^{-n} H(t) \mathcal{F}_\xi(\mathrm{e}^{-t|\xi|^2})(x) \\
&= (2\pi)^{-n} H(t) \left(\frac{\pi}{t}\right)^{\frac{n}{2}} \mathrm{e}^{-\frac{|x|^2}{4t}} = H(t)(4\pi t)^{-\frac{n}{2}} \mathrm{e}^{-\frac{|x|^2}{4t}} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n+1}),
\end{aligned} \tag{8.1.5}$$

where for the third equality in (8.1.5) we used Remark 4.22 and (3.2.6). Hence, we may conclude that the tempered distribution from (8.1.5) is a fundamental solution for the heat operator.

Note that, with the notation from (3.1.10), we have $\partial_t - \Delta = iD_{n+1} + \sum_{j=1}^n D_j^2$, hence the heat operator satisfies the hypothesis of Proposition 5.7. Consequently, if $u \in \mathcal{S}'(\mathbb{R}^{n+1})$ is an arbitrary fundamental solution for the heat operator in $\mathbb{R}^{n+1}$, then $u - E = P(x, t)$ in $\mathcal{S}'(\mathbb{R}^{n+1})$, for some polynomial $P(x, t)$ in $\mathbb{R}^{n+1}$ satisfying $(\partial_t - \Delta_x)P(x, t) = 0$ in $\mathbb{R}^{n+1}$.

As a final remark, we claim that E as in (8.1.5) satisfies $E \in L^1_{loc}(\mathbb{R}^{n+1})$. Indeed, if K is a compact subset of $\mathbb{R}^n \times [-R, R]$ for some $R \in (0, \infty)$, then

$$0 \leq \int_K E(x,t)\,dx\,dt \leq \int_0^R \int_{\mathbb{R}^n} (4\pi t)^{-\frac{n}{2}} e^{-\frac{|x|^2}{4t}}\,dx\,dt$$
$$= \pi^{-\frac{n}{2}} \int_0^R \int_{\mathbb{R}^n} e^{-|y|^2}\,dy\,dt = R < \infty. \tag{8.1.6}$$

In summary, we proved the following theorem.

Theorem 8.1. *The function defined as*

$$E(x,t) := H(t)(4\pi t)^{-\frac{n}{2}} e^{-\frac{|x|^2}{4t}}, \qquad \forall\,(x,t) \in \mathbb{R}^{n+1}, \tag{8.1.7}$$

belongs to $\mathcal{S}'(\mathbb{R}^{n+1}) \cap L^1_{loc}(\mathbb{R}^{n+1}) \cap C^\infty(\mathbb{R}^{n+1}\setminus\{0\})$ *and is a fundamental solution for the heat operator* $\partial_t - \Delta_x$ *in* $\mathbb{R}^{n+1}$. *Moreover,*

$$\big\{u \in \mathcal{S}'(\mathbb{R}^{n+1}) : (\partial_t - \Delta_x)u = \delta(x)\otimes\delta(t) \ \text{ in } \ \mathcal{S}'(\mathbb{R}^{n+1})\big\} \tag{8.1.8}$$
$$= \big\{E + P : P \ \text{ polynomial in } \ \mathbb{R}^{n+1} \ \text{ satisfying } \ (\partial_t - \Delta_x)P(x,t) = 0\big\}.$$

Corollary 8.2. *The heat operator* $\partial_t - \Delta_x$ *is hypoelliptic in* $\mathbb{R}^{n+1}$.

Proof. This is a consequence of Theorem 6.8 and Theorem 8.1. □

Exercise 8.3. *Let* E *be the function defined in* (8.1.8) *and set* $F(x,t) := -E(x,-t)$, *for* $(x,t) \in \mathbb{R}^{n+1}$. *Prove that* F *is a fundamental solution for the operator* $\partial_t + \Delta$ *in* $\mathbb{R}^{n+1}$.

Remark 8.4. *Given the expression* $E(x,t) = H(t)(4\pi t)^{-\frac{n}{2}} e^{-\frac{|x|^2}{4t}}$ *defined for each* $(x,t) \in \mathbb{R}^{n+1}$, *then one may check via a direct computation that this is a fundamental solution for the heat operator* $L = \partial_t - \Delta_x$. *First, the computation in* (8.1.6) *gives that* $E \in L^1_{loc}(\mathbb{R}^{n+1})$, *which in turn implies* $E \in \mathcal{D}'(\mathbb{R}^{n+1})$.

Second, if $\varphi \in C_0^\infty(\mathbb{R}^{n+1})$ *is arbitrary, then using integration by parts we may write*

$$\langle LE, \varphi\rangle = -\langle E, \partial_t\varphi + \Delta_x\varphi\rangle$$
$$= -\lim_{\varepsilon\to 0^+}\left[\int_\varepsilon^\infty \int_{\mathbb{R}^n} E(x,t)(\partial_t\varphi(x,t) + \Delta_x\varphi(x,t))\,dx\,dt\right]$$
$$= \lim_{\varepsilon\to 0^+}\left[\int_{\mathbb{R}^n} E(x,\varepsilon)\varphi(x,\varepsilon)\,dx + \int_\varepsilon^\infty \int_{\mathbb{R}^n} LE(x,t)\varphi(x,t)\,dx\,dt\right]$$
$$= \lim_{\varepsilon\to 0^+} \int_{\mathbb{R}^n} (4\pi\varepsilon)^{-\frac{n}{2}} e^{-\frac{|x|^2}{4\varepsilon}}\varphi(x,\varepsilon)\,dx$$
$$= \lim_{\varepsilon\to 0^+} \int_{\mathbb{R}^n} \pi^{-\frac{n}{2}} e^{-|y|^2}\varphi(2\sqrt{\varepsilon}y,\varepsilon)\,dy$$
$$= \varphi(0) = \langle\delta,\varphi\rangle. \tag{8.1.9}$$

For the fourth equality in (8.1.9) *we have used the fact that* $LE = 0$ *pointwise in* $\mathbb{R}^n \times (0,\infty)$, *for the fifth a suitable change of variables, while for the sixth*

equality we applied Lebesgue's dominated convergence theorem. This proves that $LE = \delta$ *in* $\mathcal{D}'(\mathbb{R}^{n+1})$*, thus* E *is a fundamental solution for* L*.*

In closing we record a Liouville-type theorem for the operator $\partial_t - \Delta$, which is a particular case of Theorem 5.3.

Theorem 8.5 (Liouville's theorem for the heat operator)**.** *Any bounded function in* $\mathbb{R}^{n+1}$ *that is a null solution of the heat operator is constant.*

8.2 The Generalized Cauchy Problem for the Heat Operator

Let $F \in C^0\big(\mathbb{R}^n \times [0,\infty)\big)$, $f \in C^0(\mathbb{R}^n)$ and suppose u is a solution of the `Cauchy problem` for the heat operator:

$$\begin{cases} u \in C^0\big(\mathbb{R}^n \times [0,\infty)\big), \\ \partial_t u, \partial^2_{x_j} u \in C^0\big(\mathbb{R}^n \times (0,\infty)\big),\ j = 1,\dots,n, \\ \partial_t u - \Delta_x u = F \text{ in } \mathbb{R}^n \times (0,\infty), \\ u(\cdot,0) = f \text{ in } \mathbb{R}^n. \end{cases} \tag{8.2.1}$$

Denote by $\widetilde{u}$ and $\widetilde{F}$ the extensions by zero of u and F to the entire space $\mathbb{R}^{n+1}$. Then, if $\varphi \in C_0^\infty(\mathbb{R}^{n+1})$, integrating by parts and using Lebesgue's dominated convergence theorem we obtain

$$\begin{aligned} \big\langle (\partial_t - \Delta_x)\widetilde{u}, \varphi \big\rangle &= -\big\langle \widetilde{u}, \partial_t u + \Delta_x \varphi \big\rangle = -\lim_{\varepsilon \to 0^+} \int_\varepsilon^\infty \int_{\mathbb{R}^n} u(\partial_t \varphi + \Delta_x \varphi)\,\mathrm{d}x\,\mathrm{d}t \\ &= \lim_{\varepsilon \to 0^+} \left[\int_\varepsilon^\infty \int_{\mathbb{R}^n} (\partial_t u - \Delta_x u)\varphi\,\mathrm{d}x\,\mathrm{d}t + \int_{\mathbb{R}^n} u(x,\varepsilon)\varphi(x,\varepsilon)\,\mathrm{d}x \right] \\ &= \int_0^\infty \int_{\mathbb{R}^n} F(x,t)\varphi(x,t)\,\mathrm{d}x\,\mathrm{d}t + \int_{\mathbb{R}^n} f(x)\varphi(x,0)\,\mathrm{d}x \\ &= \big\langle \widetilde{F}, \varphi \big\rangle + \big\langle f(x) \otimes \delta(t), \varphi(x,t) \big\rangle. \end{aligned} \tag{8.2.2}$$

This proves that $(\partial_t - \Delta_x)\widetilde{u} = \widetilde{F} + f(x) \otimes \delta(t)$ in $\mathcal{D}'(\mathbb{R}^{n+1})$ and suggests the definition made below. As a preamble, we introduce the notation

$$\overline{\mathbb{R}^{n+1}_+} := \big\{(x,t) \in \mathbb{R}^{n+1} : t \geq 0\big\} \tag{8.2.3}$$

and the space

$$\mathcal{D}'_+(\mathbb{R}^{n+1}) := \{u \in \mathcal{D}'(\mathbb{R}^{n+1}) :\ \operatorname{supp} u \subseteq \overline{\mathbb{R}^{n+1}_+}\,\}. \tag{8.2.4}$$

Definition 8.6. *Given* $F_0 \in \mathcal{D}'_+(\mathbb{R}^{n+1})$ *and* $f \in \mathcal{D}'(\mathbb{R}^n)$*, a distribution* $u \in \mathcal{D}'_+(\mathbb{R}^{n+1})$ *is called a solution of the* `generalized Cauchy problem` *for the heat operator for the data* F_0 *and* f*, if* u *verifies*

$$(\partial_t - \Delta_x)u = F_0 + f(x) \otimes \delta(t) \quad \text{ in } \ \mathcal{D}'(\mathbb{R}^{n+1}). \tag{8.2.5}$$

The issue of solvability of Eq. (8.2.5) fits into the framework presented in Remark 5.5. More precisely, let E be the fundamental solution for the heat operator as given in (8.1.7). Then, if $F_0 \in \mathcal{D}'_+(\mathbb{R}^{n+1})$ and $f \in \mathcal{D}'(\mathbb{R}^n)$ are such that

$$u := E * [F_0 + f \otimes \delta] \quad \text{exists in} \quad \mathcal{D}'(\mathbb{R}^{n+1}) \tag{8.2.6}$$

the distribution u satisfies (8.2.5). For this u to be a solution of the generalized Cauchy problem for the heat operator, we would also need $\operatorname{supp} u \subseteq \overline{\mathbb{R}^{n+1}_+}$. Since it is not difficult to check that $\operatorname{supp} E \subseteq \overline{\mathbb{R}^{n+1}_+}$ and $\operatorname{supp}(f \otimes \delta) \subseteq \overline{\mathbb{R}^{n+1}_+}$, and since by assumption $\operatorname{supp} F_0 \subseteq \overline{\mathbb{R}^{n+1}_+}$, it follows that whenever condition (8.2.6) is verified, the distribution u defined in (8.2.6) also satisfies $\operatorname{supp} u \subseteq \overline{\mathbb{R}^{n+1}_+}$ (by (2.8.17)). While the convolution of two arbitrary distributions in $\mathcal{D}'_+(\mathbb{R}^n)$ does not always exist (you may want to check that an exception is the case $n = 1$), under the additional assumptions $f \in \mathcal{E}'(\mathbb{R}^n)$ and $F_0 \in \mathcal{E}'(\mathbb{R}^{n+1})$ condition (8.2.6) is verified.

The above discussion is the reason why we analyze in detail the following setting. Retain the notation introduced at the beginning of the section, and assume that

$$\widetilde{F} \in C^\infty_0(\mathbb{R}^{n+1}) \quad \text{and} \quad f \in C^\infty_0(\mathbb{R}^n). \tag{8.2.7}$$

Then

$$u := E * \widetilde{F} + E * \big(f(x) \otimes \delta(t)\big) \quad \text{exists, belongs to } \mathcal{D}'_+(\mathbb{R}^{n+1}), \tag{8.2.8}$$

and is a solution of (8.2.5). We proceed by rewriting the expression for u in a more explicit form. First, by Proposition 2.93 and the fact that $E \in L^1_{loc}(\mathbb{R}^{n+1})$, we have $E * \widetilde{F} \in C^\infty(\mathbb{R}^{n+1})$ and

$$\begin{aligned}
(E * \widetilde{F})(x,t) &= \big\langle E(y,\tau), \widetilde{F}(x-y,t-\tau)\big\rangle = \big\langle E(x-y,t-\tau), \widetilde{F}(y,\tau)\big\rangle \\
&= \int_0^\infty \int_{\mathbb{R}^n} H(t-\tau)\big[4\pi(t-\tau)\big]^{-\frac{n}{2}} \mathrm{e}^{-\frac{(x-y)^2}{4(t-\tau)}} F(y,\tau)\,\mathrm{d}y\,\mathrm{d}\tau \\
&= \int_0^t \int_{\mathbb{R}^n} E(x-y,t-\tau)F(y,\tau)\,\mathrm{d}y\,\mathrm{d}\tau.
\end{aligned} \tag{8.2.9}$$

To compute $E * (f \otimes \delta)$, fix an arbitrary compact set $K \subset \mathbb{R}^{n+1}$, consider a function $\varphi \in C^\infty_0(\mathbb{R}^{n+1})$ such that $\operatorname{supp}\varphi \subseteq K$, and pick some $\psi \in C^\infty_0(\mathbb{R}^{2n+2})$ with the property that $\psi = 1$ in a neighborhood of the set

$$\big\{(x,t,y,0) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n \times \mathbb{R} : \ y \in \operatorname{supp} f \ \text{ and } \ (x+y,t) \in K\big\}. \tag{8.2.10}$$

Relying on the definition of convolution we have

$$\begin{aligned}
\big\langle E * (f \otimes \delta), \varphi\big\rangle &= \Big\langle E(x,t), \big\langle f(y) \otimes \delta(\tau), \psi(x,t,y,\tau)\varphi(x+y,t+\tau)\big\rangle\Big\rangle \\
&= \left\langle E(x,t), \int_{\mathbb{R}^n} f(y)\varphi(x+y,t)\mathrm{d}y\right\rangle
\end{aligned}$$

$$= \int_{\mathbb{R}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} E(x,t) f(y) \varphi(x+y,t) \, dx \, dy \, dt$$

$$= \int_{\mathbb{R}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} E(z-y,t) f(y) \varphi(z,t) \, dz \, dy \, dt. \tag{8.2.11}$$

Hence, $E * \big(f(x) \otimes \delta(t)\big)$ is given by the function

$$\mathbb{R}^n \times \mathbb{R} \ni (x,t) \mapsto \int_{\mathbb{R}^n} E(x-y,t) f(y) \, dy \in \mathbb{C}, \tag{8.2.12}$$

whose restriction to $\mathbb{R}^n \times (0,\infty)$ is of class C^∞.

In summary, this analysis proves the following result.

Proposition 8.7. *Let $f \in C_0^\infty(\mathbb{R}^n)$ and assume that $F \in C^0\big(\mathbb{R}^n \times [0,\infty)\big)$ is such that its extension $\widetilde{F}$ by zero to $\mathbb{R}^{n+1}$ satisfies $\widetilde{F} \in C_0^\infty(\mathbb{R}^{n+1})$. Also let E be the fundamental solution for the heat operator $\partial_t - \Delta_x$ as given in (8.1.7).*

Then the generalized Cauchy problem for the heat operator for the data $\widetilde{F}$ and f has a solution $u \in \mathcal{D}'_+(\mathbb{R}^{n+1})$ that is of function type, whose restriction to $\mathbb{R}^n \times (0,\infty)$ is of class C^∞, and has the expression

$$u(x,t) = \int_{\mathbb{R}^n} E(x-y,t) f(y) \, dy + \int_0^t \int_{\mathbb{R}^n} E(x-y,t-\tau) \widetilde{F}(y,\tau) \, dy \, d\tau \tag{8.2.13}$$

for every $x \in \mathbb{R}^n$ and $t \in (0,\infty)$.

Note that the integrals in (8.2.13) are meaningfully defined under weaker assumptions on F and f. In fact, starting with u as in (8.2.13) one may prove that this is a solution to a version of (8.2.1) (corresponding to finite time, i.e., $t \in (0,T)$, for some $T > 0$) under suitable yet less stringent conditions on F and f.

8.3 Fundamental Solutions for General Second Order Parabolic Operators

Let $A = \big(a_{jk}\big)_{1 \le j,k \le n} \in \mathcal{M}_{n \times n}(\mathbb{C})$ and associated to such a matrix A, consider the `parabolic operator`

$$L_A := L_A(\partial) := \partial_t - \sum_{j,k=1}^n a_{jk} \partial_j \partial_k. \tag{8.3.1}$$

The goal is to obtain explicit formulas for all tempered distributions that are fundamental solutions for L_A under the additional assumption that there exists a constant $C \in (0,\infty)$ such that the matrix A satisfies the strict positiveness condition

$$\mathrm{Re}\Big[\sum_{j,k=1}^n a_{jk} \xi_j \xi_k \Big] \ge C|\xi|^2, \quad \forall \, \xi = (\xi_1, \ldots, \xi_n) \in \mathbb{R}^n. \tag{8.3.2}$$

The approach is an adaptation to the parabolic setting of the ideas used in Sect. 7.8 for the derivation of (7.8.53). In a first stage, we note that, via the same reasoning as in Sect. 7.8, when computing the fundamental solution for L_A we may assume without loss of generality that A is symmetric, that is, $A = A^\top$. Also, we treat first the case when A has real entries.

The case when A is real, symmetric,, and satisfies (8.3.2).
Since A is real, symmetric and positive definite, $A^{\frac12}$ is well-defined, real, symmetric, invertible, $A^{\frac12}A^{\frac12} = A$, and $\det(A^{\frac12}) = \sqrt{\det A}$.

Throughout this discussion, we agree to denote by $\nabla_{x,t}$ the global gradient in $\mathbb{R}^{n+1}$, and reserve ∇ for the gradient in the variable x only. As in the past, Δ refers to the Laplacian in the variable x. Fix

$$u \in L^1_{loc}(\mathbb{R}^{n+1}) \quad \text{such that} \quad \partial_k u \in L^1_{loc}(\mathbb{R}^{n+1}) \quad \text{for each} \quad k \in \{1,\dots,n\}, \tag{8.3.3}$$

Also, fix some $\varphi \in C^\infty_0(\mathbb{R}^{n+1})$. Then, using the fact that A is symmetric we obtain

$$\begin{aligned}
\langle L_A u, \varphi\rangle &= \langle \partial_t u, \varphi\rangle - \sum_{j,k=1}^n a_{jk}\big\langle \partial_j\partial_k u, \varphi\big\rangle = -\langle u, \partial_t\varphi\rangle + \sum_{j,k=1}^n a_{jk}\big\langle \partial_k u, \partial_j\varphi\big\rangle \\
&= -\int_{\mathbb{R}^{n+1}} u(x,t)\partial_t\varphi(x,t)\,dx\,dt + \sum_{j,k=1}^n \int_{\mathbb{R}^{n+1}} \partial_j u(x,t)a_{jk}\partial_k\varphi(x,t)\,dx\,dt \\
&= -\int_{\mathbb{R}}\int_{\mathbb{R}^n} \Big[u(x,t)\partial_t\varphi(x,t) - \nabla u(x)\cdot\big(A\nabla\varphi(x,t)\big)\Big]\,dx\,dt.
\end{aligned} \tag{8.3.4}$$

In the last integral over $\mathbb{R}^{n+1}$ in (8.3.4) we make the change of variables $x = A^{\frac12}y$. Since for every invertible matrix $B \in \mathcal{M}_{n\times n}(\mathbb{R})$ and any function f the chain rule gives

$$(\nabla f)(By) = (B^\top)^{-1}\nabla[f(By)] \tag{8.3.5}$$

for each $y \in \mathbb{R}^n$ such that f is differentiable at By, we obtain (the analogue of (7.8.30))

$$\begin{aligned}
&\int_{\mathbb{R}}\int_{\mathbb{R}^n} \Big[u(x,t)\partial_t\varphi(x,t) - \nabla u(x,t)\cdot\big(A\nabla\varphi(x,t)\big)\Big]\,dx\,dt \\
&\qquad = \sqrt{\det A}\int_{\mathbb{R}}\int_{\mathbb{R}^n} \Big[v(y,t)\partial_t\psi(y,t) - \nabla v(y,t)\cdot\nabla\psi(y,t)\Big]\,dy\,dt \\
&\qquad = \sqrt{\det A}\int_{\mathbb{R}}\int_{\mathbb{R}^n} v(y,t)\big[\partial_t\psi(y,t) + \Delta\psi(y,t)\big]\,dy\,dt,
\end{aligned} \tag{8.3.6}$$

where we have set

$$v(x,t) := u(A^{\frac12}x,t) \quad \text{and} \quad \psi(x,t) := \varphi(A^{\frac12}x,t) \quad \text{for} \quad (x,t) \in \mathbb{R}^{n+1}. \tag{8.3.7}$$

Note that $\psi \in C_0^\infty(\mathbb{R}^{n+1})$. Hence, (8.3.4) and (8.3.6) imply

$$\langle L_A u, \varphi\rangle = -\sqrt{\det A}\int_{\mathbb{R}^{n+1}} v(y,t)[\partial_t\psi(y,t)+\Delta\psi(y,t)]\,dy\,dt. \tag{8.3.8}$$

Choose u to be the function

$$u(y,t) := \frac{1}{\sqrt{\det A}}\,E_{\partial_t-\Delta}(A^{-\frac{1}{2}}y,t) \quad \text{for} \quad (y,t)\in\mathbb{R}^{n+1}, \tag{8.3.9}$$

where $E_{\partial_t-\Delta}$ is the fundamental solution for the heat operator from (8.1.7). In particular, u satisfies (8.3.3). For this choice of u, the function v from (8.3.7) becomes

$$v(y,t) = \frac{1}{\sqrt{\det A}}\,E_{\partial_t-\Delta}(y,t) \quad \text{for} \quad (y,t)\in\mathbb{R}^{n+1}. \tag{8.3.10}$$

Since $A^{\frac{1}{2}}(0)=0$, we may write

$$\begin{aligned}
\varphi(0) &= \varphi(A^{\frac{1}{2}}0,0) = \psi(0) = \big\langle \delta(x)\otimes\delta(t),\psi\big\rangle \\
&= \big\langle (\partial_t-\Delta)E_{\partial_t-\Delta},\psi\big\rangle = \big\langle E_{\partial_t-\Delta},-\partial_t\psi-\Delta\psi\big\rangle \\
&= -\sqrt{\det A}\int_{\mathbb{R}^{n+1}}\frac{1}{\sqrt{\det A}}\,E_{\partial_t-\Delta}(y,t)\big[\partial_t\psi(y,t)+\Delta\psi(y,t)\big]\,dy\,dt \\
&= -\sqrt{\det A}\int_{\mathbb{R}^{n+1}} v(y,t)\big[\partial_t\psi(y,t)+\Delta\psi(y,t)\big]\,dy\,dt \\
&= \langle L_A u,\varphi\rangle = \frac{1}{\sqrt{\det A}}\,\big\langle L_A\big(E_{\partial_t-\Delta}(A^{-\frac{1}{2}}x,t)\big)\,,\,\varphi(x,t)\big\rangle,
\end{aligned} \tag{8.3.11}$$

where the penultimate equality in (8.3.11) uses (8.3.8). From (8.3.11) we may conclude that the function

$$E_A(x,t) := \frac{1}{\sqrt{\det A}}E_{\partial_t-\Delta}(A^{-\frac{1}{2}}x,t) \quad \text{for} \quad (x,t)\in\mathbb{R}^{n+1}, \tag{8.3.12}$$

is a fundamental solution for L_A in $\mathbb{R}^{n+1}$. Keeping in mind that for each $x\in\mathbb{R}^n$ we have $|A^{-\frac{1}{2}}x|^2 = \big(A^{-\frac{1}{2}}x\big)\cdot\big(A^{-\frac{1}{2}}x\big) = \big(A^{-1}x\big)\cdot x$, the function E_A may be rewritten as

$$E_A(x) := \frac{1}{\sqrt{\det A}}\,H(t)(4\pi t)^{-\frac{n}{2}}\,\mathrm{e}^{-\frac{(A^{-1}x)\cdot x}{4t}} \quad \text{for} \quad (x,t)\in\mathbb{R}^{n+1}. \tag{8.3.13}$$

Moreover, from (8.3.12), Theorem 8.1, and Proposition 4.41, it follows that E_A belongs to $\mathcal{S}'(\mathbb{R}^{n+1})\cap L^1_{loc}(\mathbb{R}^{n+1})\cap C^\infty(\mathbb{R}^{n+1}\setminus\{0\})$.

The case when A has complex entries, is symmetric, and satisfies (8.3.2).
As observed in Remark 7.50, under the current assumptions, A continues to be invertible. Also, (7.8.25) holds. In addition, under the current assumptions $(\det A)^{\frac{1}{2}}$ is unambiguously defined (see in Remark 7.52).

These comments show that the function E_A from (8.3.13) continues to be well-defined under the current assumption on A if ln is replaced by the principal branch of the complex log (defined for points $z \in \mathbb{C}\backslash(-\infty, 0]$ so that $z^a = e^{a \log z}$ for each $a \in \mathbb{R}$). In addition, E_A continues to belong to $L^1_{loc}(\mathbb{R}^{n+1})$ (this can be seen by a computation similar to that in (8.1.6), keeping in mind (7.8.23)), and $E_A \in C^\infty(\mathbb{R}^{n+1} \setminus \{0\})$. Moreover, from (7.8.23) and Exercise 4.5 it follows that E_A belongs to $\mathcal{S}'(\mathbb{R}^{n+1})$.

The goal is to prove that this expression is a fundamental solution for L_A in the current case. Making use of (7.8.45) for every $(x,t) \in \mathbb{R}^{n+1}$ with $t \neq 0$ differentiating pointwise we obtain

$$\sum_{j,k=1}^{n} a_{jk}\partial_j\partial_k\Big[\mathrm{e}^{-\frac{(A^{-1}x)\cdot x}{4t}}\Big] = \mathrm{e}^{-\frac{(A^{-1}x)\cdot x}{4t}}\Big[\frac{(A^{-1}x)\cdot x}{4t^2} - \frac{n}{2t}\Big], \tag{8.3.14}$$

while

$$\partial_t\Big[t^{-n/2}\mathrm{e}^{-\frac{(A^{-1}x)\cdot x}{4t}}\Big] = t^{-n/2}\mathrm{e}^{-\frac{(A^{-1}x)\cdot x}{4t}}\Big[\frac{(A^{-1}x)\cdot x}{4t^2} - \frac{n}{2t}\Big]. \tag{8.3.15}$$

From (8.3.14), (8.3.15), and the expression for E_A we may conclude that

$$L_A E_A(x,t) = 0 \qquad \forall\, x \in \mathbb{R}^n,\ \forall\, t \in \mathbb{R}\setminus\{0\}. \tag{8.3.16}$$

In addition, for each $\varphi \in C^\infty_0(\mathbb{R}^{n+1})$ we may compute

$$\begin{aligned}
\langle L_A E_A, \varphi\rangle &= -\Big\langle E_A, \partial_t\varphi + \sum_{j,k=1}^{n} a_{jk}\partial_j\partial_k\varphi\Big\rangle \\
&= -\lim_{\varepsilon\to 0^+}\left[\int_\varepsilon^\infty\int_{\mathbb{R}^n} E_A(x,t)\Big(\partial_t\varphi(x,t) + \sum_{j,k=1}^{n} a_{jk}\partial_j\partial_k\varphi(x,t)\Big)\,dx\,dt\right] \\
&= \lim_{\varepsilon\to 0^+}\left[\int_{\mathbb{R}^n} E_A(x,\varepsilon)\varphi(x,\varepsilon)\,dx + \int_\varepsilon^\infty\int_{\mathbb{R}^n}(L_A E_A)(x,t)\varphi(x,t)\,dx\,dt\right] \\
&= \lim_{\varepsilon\to 0^+}\frac{1}{\sqrt{\det A}}\int_{\mathbb{R}^n}(4\pi\varepsilon)^{-\frac{n}{2}}\mathrm{e}^{-\frac{(A^{-1}x)\cdot x}{4\varepsilon}}\varphi(x,\varepsilon)\,dx \\
&= \lim_{\varepsilon\to 0^+}\frac{\pi^{-\frac{n}{2}}}{\sqrt{\det A}}\int_{\mathbb{R}^n}\mathrm{e}^{-(A^{-1}y)\cdot y}\,\varphi(2\sqrt{\varepsilon}y,\varepsilon)\,dy \\
&= \varphi(0)\frac{\pi^{-\frac{n}{2}}}{\sqrt{\det A}}\int_{\mathbb{R}^n}\mathrm{e}^{-(A^{-1}y)\cdot y}\,dy.
\end{aligned} \tag{8.3.17}$$

For the fourth equality in (8.3.17) we have used (8.3.16), for the fifth the change of variables $x = 2\sqrt{\varepsilon}y$, while for the sixth equality we applied Lebesgue's

dominated convergence theorem. From (8.3.17) we then conclude that E_A is a fundamental solution for L_A in $\mathbb{R}^{n+1}$ if and only if

$$\int_{\mathbb{R}^n} \mathrm{e}^{-(A^{-1}y)\cdot y}\,\mathrm{d}y = \pi^{\frac{n}{2}}\sqrt{\det A}. \tag{8.3.18}$$

The fact that we already know that E_A is a fundamental solution for L_A in $\mathbb{R}^{n+1}$ in the case when $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ satisfies $A = A^\top$ and condition (8.3.2), implies that formula (8.3.18) holds for this class of matrices. By using the same circle of ideas as the ones employed in proving (7.8.50) (based on Lemma 7.53), we conclude that (8.3.18) holds for the larger class of matrices $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ satisfying $A = A^\top$ and condition (8.3.2). Hence, E_A is indeed a fundamental solution for L_A under the current assumptions on A.

Next, we claim that the hypotheses of Proposition 5.7 hold in the case of the operator $P(D) := L_A$. To justify this, note that if $\xi = (\xi_1, \ldots, \xi_{n+1}) \in \mathbb{R}^{n+1}$ is such that $P(\xi) = 0$ then

$$i\xi_{n+1} + \sum_{j,k=1}^{n} a_{jk}\xi_j\xi_k = 0. \tag{8.3.19}$$

Taking reals parts, condition (8.3.2) implies $\xi_1 = \cdots = \xi_n = 0$ which, in combination with (8.3.19), also forces $\xi_{n+1} = 0$. Hence, $\xi = 0$ as wanted. Applying now Proposition 5.7 gives that if $u \in \mathcal{S}'(\mathbb{R}^{n+1})$ is an arbitrary fundamental solution for L_A in $\mathbb{R}^{n+1}$, then $u = E_A + P$ in $\mathcal{S}'(\mathbb{R}^{n+1})$, for some polynomial P in $\mathbb{R}^{n+1}$ satisfying $L_A P = 0$ pointwise in $\mathbb{R}^{n+1}$.

In summary, the above analysis proves the following result.

Theorem 8.8. *Suppose $A = \big(a_{jk}\big)_{1\leq j,k\leq n} \in \mathcal{M}_{n\times n}(\mathbb{C})$ satisfies (8.3.2) and consider the operator L_A associated to A as in (8.3.1). Then the function defined by*

$$E_A(x) := \frac{1}{\sqrt{\det A_{sym}}}\, H(t)(4\pi t)^{-\frac{n}{2}}\,\mathrm{e}^{-\frac{(A_{sym}^{-1}x)\cdot x}{4t}} \qquad \text{for all } (x,t) \in \mathbb{R}^{n+1}, \tag{8.3.20}$$

belongs to $\mathcal{S}'(\mathbb{R}^{n+1}) \cap L^1_{loc}(\mathbb{R}^{n+1}) \cap C^\infty(\mathbb{R}^{n+1}\setminus\{0\})$ and is a fundamental solution for L_A in $\mathbb{R}^{n+1}$. Above, $A_{sym} := \frac{1}{2}(A + A^\top)$, $\sqrt{\det A_{sym}}$ is defined as in Remark 7.50, and log *denotes the principal branch of the complex logarithm (defined for points $z \in \mathbb{C}\setminus(-\infty, 0]$ so that $z^a = e^{a\log z}$ for each $a \in \mathbb{R}$).*

Moreover,

$$\begin{aligned}&\big\{u \in \mathcal{S}'(\mathbb{R}^{n+1}) : L_A u = \delta \ \text{ in } \ \mathcal{S}'(\mathbb{R}^{n+1})\big\} \\ &\qquad = \big\{E_A + P : P \ \text{ polynomial in } \ \mathbb{R}^{n+1} \text{ satisfying } \ L_A P = 0\big\}.\end{aligned} \tag{8.3.21}$$

8.4 Fundamental Solution for the Schrödinger Operator

Let $x \in \mathbb{R}^n$ and $t \in \mathbb{R}$. The operator $\frac{1}{i}\partial_t - \Delta_x$ is called the (time-dependent) `Schrödinger operator` in $\mathbb{R}^{n+1}$ (with zero potential). In this section we determine a fundamental solution for this operator. By Theorem 5.13, there exists $E \in \mathcal{S}'(\mathbb{R}^{n+1})$ such that

$$\Big(\frac{1}{i}\partial_t - \Delta_x\Big)E = \delta(x) \otimes \delta(t) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n+1}). \tag{8.4.1}$$

Fix such a distribution E and take the partial Fourier transform $\mathcal{F}_x$ of (8.4.1) (recall the discussion of the partial Fourier transform at the end of Sect. 4.2) to obtain

$$\frac{1}{i}\partial_t \widehat{E}_x + |\xi|^2 \widehat{E}_x = 1(\xi) \otimes \delta(t) \quad \text{in } \mathcal{S}'(\mathbb{R}^{n+1}). \tag{8.4.2}$$

Fix $\xi \in \mathbb{R}^n$ and consider the equation

$$\frac{1}{i}\partial_t u + |\xi|^2 u = \delta \quad \text{in} \quad \mathcal{D}'(\mathbb{R}). \tag{8.4.3}$$

Using Example 5.11, we obtain that $iH(t)\mathrm{e}^{-i|\xi|^2 t}$ and $-iH(-t)\mathrm{e}^{-i|\xi|^2 t}$ are solution of this equation. This suggests considering $F := iH(t)\mathrm{e}^{-i|\xi|^2 t}$ that belongs to $\mathcal{S}'(\mathbb{R}^{n+1})$ and satisfies (based on a computation similar to that from (8.1.4))

$$\frac{1}{i}\partial_t F + |\xi|^2 F = 1(\xi) \otimes \delta(t) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n+1}). \tag{8.4.4}$$

Then $\mathcal{F}_x^{-1}(F) \in \mathcal{S}'(\mathbb{R}^{n+1})$ and

$$\Big(\frac{1}{i}\partial_t - \Delta_x\Big)\mathcal{F}_x^{-1}(F) = \delta(x) \otimes \delta(t) \quad \text{in } \mathcal{S}'(\mathbb{R}^{n+1}). \tag{8.4.5}$$

To compute $\mathcal{F}_x^{-1}(F)$, pick $\varphi \in \mathcal{S}(\mathbb{R}^{n+1})$ and, based on the properties of $\mathcal{F}_x$ as well as the expression for F, write

$$\begin{aligned}
\big\langle \mathcal{F}_x^{-1}(F), \varphi \big\rangle &= (2\pi)^{-n}\big\langle \mathcal{F}_x\big(F^\vee\big), \varphi\big\rangle = (2\pi)^{-n}\big\langle F^\vee, \mathcal{F}_x\varphi\big\rangle \\
&= (2\pi)^{-n} i \int_{\mathbb{R}^{n+1}} H(t)\mathrm{e}^{-i|\xi|^2 t}(\mathcal{F}_x\varphi)(\xi,t)\,\mathrm{d}\xi\,\mathrm{d}t \\
&= (2\pi)^{-n} i \int_0^\infty H(t)\left[\int_{\mathbb{R}^n} \mathrm{e}^{-i|\xi|^2 t}(\mathcal{F}_x\varphi)(\xi,t)\,\mathrm{d}\xi\right]\mathrm{d}t.
\end{aligned} \tag{8.4.6}$$

For each $t > 0$, the integral over $\mathbb{R}^n$ in (8.4.6) equals $\big\langle \mathcal{F}_x(\mathrm{e}^{-it|x|^2}), \varphi(\cdot,t)\big\rangle$, which by Example 4.24 (with $a = t$) is equal to $\big(\frac{\pi}{it}\big)^{\frac{n}{2}} \int_{\mathbb{R}^n} \mathrm{e}^{-\frac{|\xi|^2}{4it}}\varphi(\xi,t)\,\mathrm{d}\xi$. Hence, for each $\varphi \in \mathcal{S}(\mathbb{R}^{n+1})$ we have

$$\begin{aligned}\left\langle \mathcal{F}_x^{-1}(F), \varphi \right\rangle &= (2\pi)^{-n} i \int_0^\infty H(t) \left(\frac{\pi}{it}\right)^{\frac{n}{2}} \left[\int_{\mathbb{R}^n} \mathrm{e}^{\frac{-|\xi|^2}{4it}} \varphi(\xi, t)\, \mathrm{d}\xi \right] \mathrm{d}t \\ &= i^{1-\frac{n}{2}} \int_0^\infty H(t)(4\pi t)^{-\frac{n}{2}} \left[\int_{\mathbb{R}^n} \mathrm{e}^{-\frac{|\xi|^2}{4it}} \varphi(\xi, t)\, \mathrm{d}\xi \right] \mathrm{d}t. \end{aligned} \tag{8.4.7}$$

We remark here that $H(t)(4\pi t)^{-\frac{n}{2}} \mathrm{e}^{-\frac{|x|^2}{4it}} \in L^1_{loc}(\mathbb{R}^{n+1})$ only if $n = 1$. Hence $\mathcal{F}_x^{-1}(F)$ is of function type only if $n = 1$. In general, this distribution belongs to $\mathcal{S}'(\mathbb{R}^{n+1})$ and its action on $\varphi \in \mathcal{S}(\mathbb{R}^{n+1})$ is given as in (8.4.7).

In summary we proved the following result.

Theorem 8.9. *The distribution $E \in \mathcal{S}'(\mathbb{R}^{n+1})$ defined by*

$$\langle E, \varphi \rangle := i^{1-\frac{n}{2}} \int_0^\infty H(t)(4\pi t)^{-\frac{n}{2}} \left[\int_{\mathbb{R}^n} \mathrm{e}^{-\frac{|\xi|^2}{4it}} \varphi(\xi, t)\, \mathrm{d}\xi \right] \mathrm{d}t, \quad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^{n+1}), \tag{8.4.8}$$

is a fundamental solution for the Schrödinger operator $\frac{1}{i}\partial_t - \Delta_x$ in $\mathbb{R}^{n+1}$. In particular, if $n = 1$ then E is of function type and is given by the $L^1_{loc}(\mathbb{R}^2)$

$$E(x, t) = H(t)(4\pi t)^{-\frac{1}{2}} \mathrm{e}^{-\frac{|x|^2}{4it}}, \qquad x \in \mathbb{R},\ t \in \mathbb{R}. \tag{8.4.9}$$

Further Notes for Chap. 8. The heat equation is one example of what is commonly referred to as linear evolution equations. Originally derived in physics from Fourier's law and conservation of energy (see, e.g., [71] for details), the heat equation has come to play a role of fundamental importance in mathematics and applied sciences. In mathematics, the heat operator is the prototype for a larger class, called parabolic partial differential operators, that includes the operators studied in Sect. 8.3. The Schrödinger operator is named after the Austrian physicist Erwin Schrödinger who first introduced it in 1926. It plays a fundamental role in quantum mechanics, where it describes how the quantum state of certain physical systems changes over time.

Chapter 9

The Wave Operator

The operator $\Box := \partial_t^2 - \Delta_x$, $x \in \mathbb{R}^n$, $t \in \mathbb{R}$, is called the `wave operator` in $\mathbb{R}^{n+1}$. The wave operator arises from modeling vibrations in a string, membrane or elastic solid. The goal is to determine fundamental solutions for this operator that are tempered distributions. As an application we discuss the generalized Cauchy problem for the wave operator.

9.1 Fundamental Solution for the Wave Operator

By Theorem 5.13, we know that the wave operator $\Box$ admits a fundamental solution $E \in \mathcal{S}'(\mathbb{R}^{n+1})$. Hence, by (4.1.25) and Exercise 2.81, we have

$$\partial_t^2 E - \Delta_x E = \delta(x) \otimes \delta(t) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n+1}). \tag{9.1.1}$$

In this section we determine an explicit expression for two such fundamental solutions.

Fix $E \in \mathcal{S}'(\mathbb{R}^{n+1})$ that satisfies (9.1.1) and apply the partial Fourier transform $\mathcal{F}_x$ to this equation (recall the discussion about partial Fourier transforms from the last part of Sect. 4.2) to obtain

$$\partial_t^2 \widehat{E}_x + |\xi|^2 \widehat{E}_x = 1(\xi) \otimes \delta(t) \quad \text{in } \mathcal{S}'(\mathbb{R}^{n+1}), \tag{9.1.2}$$

where $\widehat{E}_x := \mathcal{F}_x(E) \in \mathcal{S}'(\mathbb{R}^{n+1})$. For $\xi \in \mathbb{R}^n \setminus \{0\}$ fixed consider the initial value problem (in the variable t)

$$\begin{cases} \frac{d^2}{dt^2} v + |\xi|^2 v = 0 \ \text{ in } \mathbb{R}, \\ (\partial_t v)(0) = 1, \ \ v(0) = 0, \end{cases} \tag{9.1.3}$$

which admits the solution $v(t) = \frac{\sin(t|\xi|)}{|\xi|}$ for $t \in \mathbb{R}$. By Example 5.11, it follows that the distributions vH and $-vH^{\vee}$ are fundamental solutions for the operator

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$\frac{d^2}{dt^2}+|\xi|^2$ in $\mathcal{D}'(\mathbb{R})$. In addition, vH and $-vH^\vee$ belong to $\mathcal{S}'(\mathbb{R})$ (based on (b) in Theorem 4.13 and Exercise 4.115), thus

$$\left(\frac{d^2}{dt^2}+|\xi|^2\right)(vH)=\delta \quad \text{and} \quad \left(\frac{d^2}{dt^2}+|\xi|^2\right)(-vH^\vee)=\delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}). \tag{9.1.4}$$

Moreover, there exists $c\in(0,\infty)$ such that

$$\left|H(\pm t)\frac{\sin(t|y|)}{|y|}\right|\le c|t| \quad \text{for} \quad (y,t)\in B(0,1)\setminus\{0\}\subset\mathbb{R}^{n+1}. \tag{9.1.5}$$

In particular, if we define the functions

$$f_\pm(y,t):=H(\pm t)\frac{\sin(t|y|)}{|y|} \qquad \text{for } (y,t)\in\mathbb{R}^{n+1} \text{ with } t\neq 0, \tag{9.1.6}$$

then (9.1.5) implies $f_\pm\in L^1_{\rm loc}(\mathbb{R}^{n+1})$. Furthermore,

$$(1+|y|^2+t^2)^{-n-2}f_\pm\in L^1(\mathbb{R}^{n+1}), \tag{9.1.7}$$

thus by Example 4.3, we obtain $H(\pm t)\frac{\sin(t|y|)}{|y|}\in\mathcal{S}'(\mathbb{R}^{n+1})$. Based on all these facts, if we introduce

$$F^+:=H(t)\frac{\sin(t|\xi|)}{|\xi|}, \qquad F^-:=-H(-t)\frac{\sin(t|\xi|)}{|\xi|}, \tag{9.1.8}$$

then

$$F^+,\,F^-\in\mathcal{S}'(\mathbb{R}^{n+1}) \quad \text{and} \tag{9.1.9}$$

$$\partial_t^2F^\pm+|\xi|^2F^\pm=1(\xi)\otimes\delta(t) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n+1}). \tag{9.1.10}$$

In particular, $\mathcal{F}_x^{-1}(F^+)$ and $\mathcal{F}_x^{-1}(F^-)$ are meaningfully defined in $\mathcal{S}'(\mathbb{R}^{n+1})$. Thus, if we set

$$E_+:=\mathcal{F}_x^{-1}\left(H(t)\frac{\sin(t|\xi|)}{|\xi|}\right), \tag{9.1.11}$$

$$E_-:=\mathcal{F}_x^{-1}\left(-H(-t)\frac{\sin(t|\xi|)}{|\xi|}\right), \tag{9.1.12}$$

we have

$$E_\pm\in\mathcal{S}'(\mathbb{R}^{n+1}), \quad (\partial_t^2-\Delta_x)E_\pm=\delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n+1}), \tag{9.1.13}$$

$$\operatorname{supp}E_+\subseteq\mathbb{R}^n\times[0,\infty), \quad \operatorname{supp}E_-\subseteq\mathbb{R}^n\times(-\infty,0]. \tag{9.1.14}$$

The next task is to find explicit expressions for $\mathcal{F}_x^{-1}(F^\pm)$. To this end, fix a function $\varphi\in\mathcal{S}(\mathbb{R}^{n+1})$ and, with the operation $\cdot^\vee$ considered only in the variable x, write

$$\langle E_+,\varphi\rangle=(2\pi)^{-n}\big\langle(\mathcal{F}_x\mathcal{F}_x(E_+))^\vee,\varphi\big\rangle=(2\pi)^{-n}\big\langle\mathcal{F}_x(E_+),(\mathcal{F}_x\varphi)^\vee\big\rangle$$

$$= (2\pi)^{-n} \int_{\mathbb{R}^{n+1}} H(t) \frac{\sin t|\xi|}{|\xi|} \, \widehat{\varphi}_x(-\xi, t) \, \mathrm{d}\xi \, \mathrm{d}t. \tag{9.1.15}$$

Note that if one replaces $\widehat{\varphi}_x(-\xi, t)$ above with $\int_{\mathbb{R}^n} e^{ix\cdot\xi} \varphi(x, t) \, \mathrm{d}x$ then the order of integration in the resulting iterated integral may not be switched since $\frac{1}{|\xi|}$ is not integrable at infinity, thus Fubini's theorem does not apply. This is why we should proceed with more care and, based on Lebesgue's dominated convergence theorem, we introduce a convergence factor that enables us to eventually make the use of Fubini's theorem. More precisely, we have

$$\begin{aligned}
\langle E_+, \varphi \rangle &= (2\pi)^{-n} \int_{\mathbb{R}^{n+1}} F^+(\xi, t) \, \widehat{\varphi}_x(-\xi, t) \, \mathrm{d}\xi \, \mathrm{d}t \\
&= \lim_{\varepsilon \to 0^+} (2\pi)^{-n} \int_{\mathbb{R}^{n+1}} F^+(\xi, t) e^{-\varepsilon|\xi|} \, \widehat{\varphi}_x(-\xi, t) \, \mathrm{d}\xi \, \mathrm{d}t \\
&= \lim_{\varepsilon \to 0^+} (2\pi)^{-n} \int_{\mathbb{R}^{n+1}} \varphi(x, t) \left(\int_{\mathbb{R}^n} e^{ix\cdot\xi - \varepsilon|\xi|} F^+(\xi, t) \, \mathrm{d}\xi \right) \mathrm{d}x \, \mathrm{d}t \\
&= \lim_{\varepsilon \to 0^+} \langle E_\varepsilon, \varphi \rangle,
\end{aligned} \tag{9.1.16}$$

where

$$E_\varepsilon(x, t) := (2\pi)^{-n} H(t) \int_{\mathbb{R}^n} e^{ix\cdot\xi - \varepsilon|\xi|} \, \frac{\sin(t|\xi|)}{|\xi|} \, \mathrm{d}\xi, \quad \forall \, x \in \mathbb{R}^n, \ \forall \, t \in \mathbb{R}. \tag{9.1.17}$$

To compute the last limit in (9.1.16), we separate our analysis into three cases: $n = 1$, $n = 2p + 1$ with $p \geq 1$, and $n = 2p$ with $p \geq 1$.

The Case $n = 1$

Fix $\varepsilon > 0$ and $x \in \mathbb{R}$ and define the function

$$f_\varepsilon(t) := \int_{-\infty}^{\infty} e^{ix\xi - \varepsilon|\xi|} \, \frac{\sin(t|\xi|)}{|\xi|} \, \mathrm{d}\xi \qquad \text{for} \quad t \in \mathbb{R}. \tag{9.1.18}$$

Then

$$\begin{aligned}
\partial_t f_\varepsilon(t) &= \int_{-\infty}^{\infty} e^{ix\xi - \varepsilon|\xi|} \cos(t|\xi|) \, \mathrm{d}\xi = \int_{-\infty}^{\infty} e^{ix\xi - \varepsilon|\xi|} \, \frac{e^{it|\xi|} + e^{-it|\xi|}}{2} \, \mathrm{d}\xi \\
&= \frac{1}{2} \left[\int_0^{\infty} e^{\xi(ix + it - \varepsilon)} \, \mathrm{d}\xi + \int_0^{\infty} e^{\xi(ix - it - \varepsilon)} \, \mathrm{d}\xi \right] \\
&\quad + \frac{1}{2} \left[\int_{-\infty}^{0} e^{\xi(ix - it + \varepsilon)} \, \mathrm{d}\xi + \int_{-\infty}^{0} e^{\xi(ix + it + \varepsilon)} \, \mathrm{d}\xi \right] \\
&= \frac{1}{2} \left[-\frac{1}{ix + it - \varepsilon} - \frac{1}{ix - it - \varepsilon} + \frac{1}{ix - it + \varepsilon} + \frac{1}{ix + it + \varepsilon} \right] \\
&= \frac{\varepsilon}{(x + t)^2 + \varepsilon^2} + \frac{\varepsilon}{(x - t)^2 + \varepsilon^2}, \qquad \forall \, t \in \mathbb{R}.
\end{aligned} \tag{9.1.19}$$

Consequently, (9.1.19) and the fact that $f_\varepsilon(0) = 0$ imply

$$f_\varepsilon(t) = \arctan\left(\frac{x+t}{\varepsilon}\right) - \arctan\left(\frac{x-t}{\varepsilon}\right) \qquad \text{for} \quad t \in \mathbb{R}. \tag{9.1.20}$$

Making use of (9.1.20) back in (9.1.17) (written for $n = 1$) then gives

$$E_\varepsilon(x,t) = \frac{1}{2\pi} H(t) \left(\arctan\left(\frac{x+t}{\varepsilon}\right) - \arctan\left(\frac{x-t}{\varepsilon}\right)\right), \qquad \forall\, x, t \in \mathbb{R}. \tag{9.1.21}$$

Hence, for $\varphi \in \mathcal{S}(\mathbb{R}^2)$ fixed, we may write

$$\begin{aligned} &\lim_{\varepsilon\to 0^+} \langle E_\varepsilon(x,t), \varphi(x,t)\rangle \qquad (9.1.22)\\ &\quad = \lim_{\varepsilon\to 0^+} \frac{1}{2\pi} \int_{\mathbb{R}} \int_0^\infty \left[\arctan\left(\frac{x+t}{\varepsilon}\right) - \arctan\left(\frac{x-t}{\varepsilon}\right)\right] \varphi(x,t)\, \mathrm{d}t\, \mathrm{d}x. \end{aligned}$$

To continue with our calculation, we further decompose

$$\begin{aligned} &\int_0^\infty \int_{\mathbb{R}} \arctan\left(\frac{x+t}{\varepsilon}\right) \varphi(x,t)\, \mathrm{d}x\, \mathrm{d}t\\ &= \int_0^\infty \int_{-\infty}^{-t} \arctan\left(\frac{x+t}{\varepsilon}\right) \varphi(x,t)\, \mathrm{d}x\, \mathrm{d}t + \int_0^\infty \int_{-t}^\infty \arctan\left(\frac{x+t}{\varepsilon}\right) \varphi(x,t)\, \mathrm{d}x\, \mathrm{d}t\\ &=: I_\varepsilon + II_\varepsilon. \end{aligned} \tag{9.1.23}$$

By Lebesgue's dominated convergence theorem,

$$\lim_{\varepsilon\to 0^+} I_\varepsilon = -\frac{\pi}{2} \int_0^\infty \int_{-\infty}^{-t} \varphi(x,t)\, \mathrm{d}x\, \mathrm{d}t \tag{9.1.24}$$

and

$$\lim_{\varepsilon\to 0^+} II_\varepsilon = \frac{\pi}{2} \int_0^\infty \int_{-t}^\infty \varphi(x,t)\, \mathrm{d}x\, \mathrm{d}t. \tag{9.1.25}$$

Similarly,

$$\begin{aligned} &\lim_{\varepsilon\to 0^+} \int_0^\infty \int_{\mathbb{R}} \arctan\left(\frac{x-t}{\varepsilon}\right) \varphi(x,t)\, \mathrm{d}x\, \mathrm{d}t\\ &\quad = -\frac{\pi}{2} \int_0^\infty \int_{-\infty}^{t} \varphi(x,t)\, \mathrm{d}x\, \mathrm{d}t + \frac{\pi}{2} \int_0^\infty \int_t^\infty \varphi(x,t)\, \mathrm{d}x\, \mathrm{d}t. \end{aligned} \tag{9.1.26}$$

Combining (9.1.22)–(9.1.26) permits us to write

$$\begin{aligned} &\lim_{\varepsilon\to 0^+} \langle E_\varepsilon(x,t), \varphi(x,t)\rangle \qquad (9.1.27)\\ &\quad = \frac{1}{4} \int_0^\infty \left[-\int_{-\infty}^{-t} \varphi(x,t)\, \mathrm{d}x + \int_{-t}^\infty \varphi(x,t)\, \mathrm{d}x\right] \mathrm{d}t \end{aligned}$$

$$+\frac{1}{4}\int_0^\infty\left[\int_{-\infty}^t\varphi(x,t)\,dx-\int_t^\infty\varphi(x,t)\,dx\right]dt$$

$$=\frac{1}{2}\int_0^\infty\int_{-t}^t\varphi(x,t)\,dx\,dt=\frac{1}{2}\int_{\mathbb{R}^2}H(t-|x|)\varphi(x,t)\,dx\,dt.$$

From (9.1.27) and (9.1.16) we then conclude that[1] $E_+(x,t)=\frac{H(t-|x|)}{2}$ for $(x,t)\in\mathbb{R}^2$. In summary, we proved that

$$\begin{cases} E_+(x,t)=\dfrac{H(t-|x|)}{2}, \quad \forall\,(x,t)\in\mathbb{R}^2, \quad \text{is a tempered distribution} \\ \text{and satisfies } \ (\partial_t^2-\partial_x^2)E_+=\delta \ \text{ in } \mathcal{S}'(\mathbb{R}^2). \end{cases} \tag{9.1.28}$$

Remark 9.1.

(1) The reasoning used to obtain E_+ also yields that

$$\begin{cases} E_-(x,t):=\dfrac{H(-t-|x|)}{2}, \quad \forall\,(x,t)\in\mathbb{R}^2, \quad \textit{is a tempered distribution} \\ \textit{and satisfies } \ (\partial_t^2-\partial_x^2)E_-=\delta \ \textit{ in } \mathcal{S}'(\mathbb{R}^2). \end{cases} \tag{9.1.29}$$

(2) An inspection of (9.1.28) and (9.1.29) reveals that

$$\begin{aligned} &\operatorname{supp}E_+=\{(x,t)\in\mathbb{R}^2:|x|\leq t\} \quad \textit{and} \\ &\operatorname{supp}E_-=\{(x,t)\in\mathbb{R}^2:|x|\leq -t\}. \end{aligned} \tag{9.1.30}$$

The Case $n=2p+1$, $p\geq 1$

It is immediate from (9.1.17) that $E_\varepsilon(x,t)$ is invariant under orthogonal transformations in the variable x. Hence, $E_\varepsilon(y,t)=E_\varepsilon(|y|,0,\dots,0,t)$ for every $y\in\mathbb{R}^n$ and $t\in\mathbb{R}$. As such, in what follows we may assume that $x=(|x|,0,\dots,0)$. Fix such an x and set $r:=|x|$. With $t\in\mathbb{R}$ fixed, using polar coordinates (see (13.8.1), (13.8.2), and (13.8.3)), rewrite the expression from (9.1.17) as

$$\begin{aligned} E_\varepsilon(x,t)&=(2\pi)^{-n}H(t)\int_0^\infty\int_{(0,\pi)^{n-2}}\int_0^{2\pi}e^{i\rho r\cos\theta_1-\varepsilon\rho}\,\frac{\sin t\rho}{\rho}\,\rho^{n-1} \\ &\quad\times(\sin\theta_1)^{n-2}\cdots\sin(\theta_{n-2})\,d\theta_{n-1}\cdots d\theta_1\,d\rho \\ &=(2\pi)^{-n}H(t)\int_0^\infty\rho^{n-2}\sin(t\rho)\int_0^\pi e^{i\rho r\cos\theta_1-\varepsilon\rho}(\sin\theta_1)^{n-2}\times \end{aligned}$$

[1] This expression for a fundamental solution for the wave operator when $n=1$ was first used by Jean d'Alembert in 1747 in connection with a vibrating string.

$$\times\left[\int_{(0,\pi)^{n-3}}\int_0^{2\pi}(\sin\theta_2)^{n-3}\cdots\sin(\theta_{n-2})\,d\theta_{n-1}\cdots d\theta_2\right]d\theta_1\,d\rho$$

$$=(2\pi)^{-n}\omega_{n-2}H(t)\int_0^\infty\int_0^\pi e^{i\rho r\cos\theta-\varepsilon\rho}\rho^{n-2}(\sin\theta)^{n-2}\sin(t\rho)\,d\theta\,d\rho \tag{9.1.31}$$

where ω_{n-2} denotes the surface area of the unit ball in $\mathbb{R}^{n-1}$ (see also (13.6.6) for more details on why the expression inside the right brackets in (9.1.31) is equal to ω_{n-2}).

To proceed with the computation of the integrals in the rightmost term in (9.1.31), recall that $n=2p+1$ and, for $\rho>0$ fixed, set

$$I_p:=\rho^{2p-1}\int_0^\pi e^{i\rho r\cos\theta}\,(\sin\theta)^{2p-1}\,d\theta,\qquad\forall\,p\in\mathbb{N}. \tag{9.1.32}$$

We claim that

$$I_{p+1}=-\frac{2p}{r}\partial_r I_p\quad\text{ for all }\ p\geq 1. \tag{9.1.33}$$

In order to prove (9.1.33) note that, for each $p\geq 1$, integration by parts yields

$$\begin{aligned}
I_{p+1}&=\rho^{2p+1}\int_0^\pi e^{i\rho r\cos\theta}(\sin\theta)^{2p+1}\,d\theta\\
&=\rho^{2p+1}\frac{-1}{i\rho r}\int_0^\pi\partial_\theta(e^{i\rho r\cos\theta})(\sin\theta)^{2p}\,d\theta\\
&=\frac{i}{r}\rho^{2p}\left[e^{i\rho r\cos\theta}(\sin\theta)^{2p}\Big|_{\theta=0}^{\theta=\pi}-2p\int_0^\pi e^{i\rho r\cos\theta}(\sin\theta)^{2p-1}\cos\theta\,d\theta\right]\\
&=-\frac{2pi}{r}\rho^{2p}\int_0^\pi e^{i\rho r\cos\theta}(\sin\theta)^{2p-1}\cos\theta\,d\theta\\
&=-\frac{2pi}{r}\rho^{2p}\frac{1}{i\rho}\partial_r\int_0^\pi e^{i\rho r\cos\theta}(\sin\theta)^{2p-1}\,d\theta=-\frac{2p}{r}(\partial_r I_p),
\end{aligned} \tag{9.1.34}$$

as wanted. By induction, from the recurrence relation in (9.1.33) it follows that

$$I_p=(-2)^{p-1}(p-1)!\left(\frac{1}{r}\partial_r\right)^{(p-1)}I_1,\qquad\forall\,p\in\mathbb{N}. \tag{9.1.35}$$

As for I_1, we have

$$I_1=\rho\int_0^\pi e^{i\rho r\cos\theta}\sin\theta\,d\theta=\frac{-1}{ir}e^{i\rho r\cos\theta}\Big|_{\theta=0}^{\theta=\pi}=\frac{2\sin(\rho r)}{r}. \tag{9.1.36}$$

Recalling (9.1.31) and the fact that $n=2p+1$ for some $p\in\mathbb{N}$, formulas (9.1.35) and (9.1.36) yield

$$E_\varepsilon(x,t)=(2\pi)^{-n}\omega_{n-2}H(t)\int_0^\infty I_p\,e^{-\varepsilon\rho}\sin(t\rho)\,d\rho \tag{9.1.37}$$

$$= (2\pi)^{-n}\omega_{n-2}H(t)(-2)^{p-1}(p-1)!\,2\times$$
$$\times\int_0^\infty e^{-\varepsilon\rho}\sin(t\rho)\left(\frac{1}{r}\partial_r\right)^{p-1}\left(\frac{\sin(\rho r)}{r}\right)\,\mathrm{d}\rho.$$

Furthermore, using (13.5.6) we have

$$\omega_{n-2}=\frac{2\pi^{\frac{n-1}{2}}}{\Gamma\left(\frac{n-1}{2}\right)}=\frac{2\pi^p}{\Gamma(p)}=\frac{2\pi^p}{(p-1)!} \tag{9.1.38}$$

which further simplifies the expression in (9.1.37) to

$$E_\varepsilon(x,t)=2(-2\pi)^{-p-1}H(t)\left(\frac{1}{r}\partial_r\right)^{p-1}\left[\frac{1}{r}\int_0^\infty e^{-\varepsilon\rho}\sin(t\rho)\sin(\rho r)\,\mathrm{d}\rho\right]. \tag{9.1.39}$$

Our next claim is that

$$\int_0^\infty e^{-\varepsilon\rho}\sin(t\rho)\sin(\rho r)\,\mathrm{d}\rho=\frac{1}{2}\left[\frac{\varepsilon}{\varepsilon^2+(t-r)^2}-\frac{\varepsilon}{\varepsilon^2+(t+r)^2}\right]. \tag{9.1.40}$$

Indeed,

$$\begin{aligned}\int_0^\infty e^{-\varepsilon\rho}\sin(t\rho)\sin(\rho r)\,\mathrm{d}\rho&=\frac{1}{2}\int_0^\infty e^{-\varepsilon\rho}[\cos(t-r)\rho-\cos(t+r)\rho]\,\mathrm{d}\rho\\&=\frac{1}{2}\mathrm{Re}\int_0^\infty[e^{-\varepsilon\rho+i(t-r)\rho}-e^{-\varepsilon\rho+i(t+r)\rho}]\,\mathrm{d}\rho\\&=\frac{1}{2}\mathrm{Re}\left[-\frac{1}{-\varepsilon+i(t-r)}+\frac{1}{-\varepsilon+i(t+r)}\right]\\&=\frac{1}{2}\mathrm{Re}\left[\frac{\varepsilon+i(t-r)}{\varepsilon^2+(t-r)^2}+\frac{-\varepsilon-i(t+r)}{\varepsilon^2+(t+r)^2}\right]\\&=\frac{1}{2}\left[\frac{\varepsilon}{\varepsilon^2+(t-r)^2}-\frac{\varepsilon}{\varepsilon^2+(t+r)^2}\right],\end{aligned} \tag{9.1.41}$$

proving (9.1.40). The identity resulting from (9.1.41) further simplifies the expression in (9.1.39) as

$$E_\varepsilon(x,t)=(-2\pi)^{-p-1}H(t)\left(\frac{1}{r}\partial_r\right)^{(p-1)}\left[\frac{1}{r}\left(\frac{\varepsilon}{\varepsilon^2+(t-r)^2}-\frac{\varepsilon}{\varepsilon^2+(t+r)^2}\right)\right] \tag{9.1.42}$$

for every $x\in\mathbb{R}^n$ and every $t\in\mathbb{R}$, where $r=|x|$.

Recall from (9.1.16) that in order to determine E_+ we further need to compute $\lim_{\varepsilon\to 0^+}E_\varepsilon(x,t)$ in $\mathcal{S}'(\mathbb{R}^{n+1})$. With this goal in mind, fix $\varphi\in\mathcal{S}(\mathbb{R}^{n+1})$ and use (9.1.42) in concert with (13.8.8) to write

$$\begin{aligned}\langle E_\varepsilon,\varphi\rangle &= (-2\pi)^{-p-1}\int_0^\infty\int_0^\infty\int_{\partial B(0,r)}\Big(\frac{1}{r}\partial_r\Big)^{p-1}\left[\frac{\varepsilon}{r}\left(\frac{1}{\varepsilon^2+(t-r)^2}-\frac{1}{\varepsilon^2+(t+r)^2}\right)\right]\times\\ &\qquad\times\varphi(\omega,t)\,d\sigma(\omega)\,dr\,dt\\ &= (-2\pi)^{-p-1}\varepsilon\int_0^\infty\int_0^\infty\Big(\frac{1}{r}\partial_r\Big)^{p-1}\left[\frac{1}{r}\left(\frac{1}{\varepsilon^2+(t-r)^2}-\frac{1}{\varepsilon^2+(t+r)^2}\right)\right]\times\\ &\qquad\times\int_{\partial B(0,r)}\varphi(\omega,t)\,d\sigma(\omega)\,dr\,dt.\end{aligned}\tag{9.1.43}$$

A natural question to ask is whether, if $p\geq 2$, we may integrate by parts $(p-1)$ times in the r variable in the rightmost expression in (9.1.43). Observe that, at least formally,

$$\int_0^\infty\left[\Big(\frac{1}{r}\frac{d}{dr}\Big)f(r)\right]g(r)\,dr=-\int_0^\infty f(r)\frac{d}{dr}\left[\frac{1}{r}\,g(r)\right]\,dr\tag{9.1.44}$$

if

$$\lim_{r\to\infty}\frac{f(r)g(r)}{r}=0=\lim_{r\to 0^+}\frac{f(r)g(r)}{r}.\tag{9.1.45}$$

In the setting of the last expression in (9.1.43), for each $\varepsilon,t>0$ fixed, and assuming $p\geq 2$, if we let

$$f(r):=\Big(\frac{1}{r}\partial_r\Big)^{p-2}\left[\frac{1}{r}\left(\frac{1}{\varepsilon^2+(t-r)^2}-\frac{1}{\varepsilon^2+(t+r)^2}\right)\right],\quad r>0,\tag{9.1.46}$$

$$g(r):=\int_{\partial B(0,r)}\varphi(\omega,t)\,d\sigma(\omega)=r^{n-1}\int_{S^{n-1}}\varphi(r\omega,t)\,d\sigma(\omega),\quad r>0,\tag{9.1.47}$$

then these functions satisfy (9.1.45) and (9.1.44). Proceeding by induction (with $p\geq 2$), we apply $(p-1)$ times formula (9.1.44), pick up $(-1)^{p-1}$ in the process (the last factor $\frac{1}{r}$ bundled up with the derivative) and write

$$\begin{aligned}\langle E_\varepsilon,\varphi\rangle &= (2\pi)^{-p-1}\int_0^\infty\int_0^\infty\left(\frac{\varepsilon}{\varepsilon^2+(t-r)^2}-\frac{\varepsilon}{\varepsilon^2+(t+r)^2}\right)\times\\ &\qquad\times\Big(\frac{1}{r}\partial_r\Big)^{p-1}\left[\frac{1}{r}\int_{\partial B(0,r)}\varphi(\omega,t)\,d\sigma(\omega)\right]dr\,dt.\end{aligned}\tag{9.1.48}$$

Note that (9.1.48) is also valid if $p=1$ without any need of integration by parts.

We are left with taking the limit as $\varepsilon \to 0^+$ in (9.1.48) a task we complete by using Lemma 9.3 (which is stated and proved at the end of this subsection). Specifically, we apply Lemma 9.3 with

$$h(r) := \left(\frac{1}{r}\partial_r\right)^{p-1}\left[\frac{1}{r}\int_{\partial B(0,r)}\varphi(\omega,t)\,d\sigma(\omega)\right]. \tag{9.1.49}$$

Note that the second equality in (9.1.47) and the fact that $\varphi \in \mathcal{S}(\mathbb{R}^{n+1})$ guarantee that h in (9.1.49) satisfies the hypothesis of Lemma 9.3. These facts combined with Lebesgue's dominated convergence theorem yield

$$\lim_{\varepsilon\to 0^+}\langle E_\varepsilon,\varphi\rangle = (2\pi)^{-p-1}\pi\int_0^\infty\left[\left(\frac{1}{r}\partial_r\right)^{p-1}\left(\frac{1}{r}\int_{\partial B(0,r)}\varphi(\omega,t)\,d\sigma(\omega)\right)\right]_{r=t}dt. \tag{9.1.50}$$

In summary, we proved the following result:

If $n = 2p+1$, for some $p \in \mathbb{N}$, then $E_+ \in \mathcal{S}'(\mathbb{R}^{n+1})$ defined by

$$\langle E_+,\varphi\rangle = (2\pi)^{-p-1}\pi\int_0^\infty\left[\left(\frac{1}{r}\partial_r\right)^{p-1}\left(\frac{1}{r}\int_{\partial B(0,r)}\varphi(\omega,t)\,d\sigma(\omega)\right)\right]_{r=t}dt \tag{9.1.51}$$

for $\varphi \in \mathcal{S}(\mathbb{R}^{n+1})$ is a fundamental solution for the wave operator in $\mathbb{R}^{n+1}$.

The reasoning used to obtain (9.1.51) also yields an expression for E_-. More precisely, similar to (9.1.16), we arrive at $\langle E_-,\varphi\rangle = \lim_{\varepsilon\to 0^+}\langle E_\varepsilon,\varphi\rangle$ for $\varphi \in \mathcal{S}(\mathbb{R}^{n+1})$ where, this time,

$$\begin{aligned}\langle E_\varepsilon,\varphi\rangle &= -(2\pi)^{-p-1}\int_{-\infty}^0\int_0^\infty\left(\frac{\varepsilon}{\varepsilon^2+(t-r)^2}-\frac{\varepsilon}{\varepsilon^2+(t+r)^2}\right)\times\\ &\qquad\times\left(\frac{1}{r}\partial_r\right)^{p-1}\left[\frac{1}{r}\int_{\partial B(0,r)}\varphi(\omega,t)\,d\sigma(\omega)\right]dr\,dt\\ &= (2\pi)^{-p-1}\int_0^\infty\int_0^\infty\left(\frac{\varepsilon}{\varepsilon^2+(t-r)^2}-\frac{\varepsilon}{\varepsilon^2+(t+r)^2}\right)\times\\ &\qquad\times\left(\frac{1}{r}\partial_r\right)^{p-1}\left[\frac{1}{r}\int_{\partial B(0,r)}\varphi(\omega,-t)\,d\sigma(\omega)\right]dr\,dt.\end{aligned} \tag{9.1.52}$$

Then applying Lemma 9.3, we obtain:

If $n=2p+1$, for some $p\in\mathbb{N}$, then $E_-\in\mathcal{S}'(\mathbb{R}^{n+1})$, defined by

$$\langle E_-,\varphi\rangle=(2\pi)^{-p-1}\pi\int_0^\infty\Big[\big(\tfrac{1}{r}\partial_r\big)^{p-1}\Big(\tfrac{1}{r}\int_{\partial B(0,r)}\varphi(\omega,-t)\,d\sigma(\omega)\Big)\Big]_{r=t}dt, \tag{9.1.53}$$

for $\varphi\in\mathcal{S}(\mathbb{R}^{n+1})$, is a fundamental solution for the wave operator in $\mathbb{R}^{n+1}$.

Remark 9.2.

(1) The distributions E_+ and E_- from (9.1.51) and (9.1.53), respectively, satisfy

$$\operatorname{supp}E_+=\{(x,t)\in\mathbb{R}^{2p+1}\times[0,\infty):\,|x|=t\}, \tag{9.1.54}$$

$$\operatorname{supp}E_+=\{(x,t)\in\mathbb{R}^{2p+1}\times(-\infty,0]:\,|x|=-t\}. \tag{9.1.55}$$

(2) In the case when $n=3$ (thus, for $p=1$), formulas (9.1.51) and (9.1.53) become

$$\begin{aligned}\langle E_+,\varphi\rangle&=\frac{1}{4\pi}\int_0^\infty\frac{1}{t}\int_{\partial B(0,t)}\varphi(\omega,t)\,d\sigma(\omega)\,dt\\&=\left\langle\frac{H(t)}{4\pi t}\delta_{\partial B(0,t)},\,\varphi\right\rangle,\end{aligned} \tag{9.1.56}$$

$$\begin{aligned}\langle E_-,\varphi\rangle&=\frac{1}{4\pi}\int_0^\infty\frac{1}{t}\int_{\partial B(0,t)}\varphi(\omega,-t)\,d\sigma(\omega)\,dt\\&=\left\langle-\frac{H(-t)}{4\pi t}\delta_{\partial B(0,-t)},\,\varphi\right\rangle,\end{aligned} \tag{9.1.57}$$

for every $\varphi\in\mathcal{S}(\mathbb{R}^4)$ where, for each $R\in(0,\infty)$, the symbol $\delta_{\partial B(0,R)}$ stands for the distribution defined as in Exercise 2.128 corresponding to $\Sigma:=\partial B(0,R)$.

(3) If $n=2p$, $p\in\mathbb{N}$, the approach used to obtain (9.1.51) works up to the point where the general formula for I_p was obtained. More precisely, with $\rho>0$ fixed, if we define

$$J_n:=\rho^n\int_0^\pi e^{i\rho r\cos\theta}(\sin\theta)^n\,d\theta,\quad\forall\,n\geq 2, \tag{9.1.58}$$

then the recurrence formula $J_n=-\frac{n-1}{r}\partial_r J_{n-2}$ is valid for all $n\geq 2$ (observe that $I_p=J_{2p-1}$). However, the integral $J_0=\int_0^\pi e^{i\rho r\cos\theta}\,d\theta$ cannot be computed explicitly (as opposed to the computation of I_1), hence the recurrence formula for J_n may not be used inductively to obtain an explicit expression for J_n. This is why, in order to obtain explicit expressions for $E_\pm$ when n is even we resort to a proof different than the one used when n is odd.

Lemma 9.3. *If $h : [0,\infty) \to \mathbb{R}$ is continuous and bounded, then for each $t \in (0,\infty)$ we have*

$$\lim_{\varepsilon\to 0^+} \int_0^\infty \left(\frac{\varepsilon}{\varepsilon^2+(t-r)^2} - \frac{\varepsilon}{\varepsilon^2+(t+r)^2} \right) h(r)\,\mathrm{d}r = \pi h(t). \tag{9.1.59}$$

Proof. If $t \geq 0$ is fixed, then via suitable changes of variables we obtain

$$\begin{aligned} &\lim_{\varepsilon\to 0^+} \int_0^\infty \left(\frac{\varepsilon}{\varepsilon^2+(t-r)^2} - \frac{\varepsilon}{\varepsilon^2+(t+r)^2} \right) h(r)\,\mathrm{d}r \\ &\qquad = \lim_{\varepsilon\to 0^+} \left[\int_{-\frac{t}{\varepsilon}}^\infty \frac{h(t+\varepsilon\lambda)}{1+\lambda^2}\,\mathrm{d}\lambda - \int_{\frac{t}{\varepsilon}}^\infty \frac{h(-t+\varepsilon\lambda)}{1+\lambda^2}\,\mathrm{d}\lambda \right] \qquad (9.1.60) \\ &\qquad = \int_{-\infty}^\infty \frac{h(t)}{1+\lambda^2}\,\mathrm{d}\lambda = \pi h(t), \end{aligned}$$

where for the second to the last equality in (9.1.60) we applied Lebesgue's dominated convergence theorem. □

The Method of Descent

In order to treat the case when n is even, we use a procedure called the method of descent. The ultimate goal is to use the method of descent to deduce from a fundamental solution for the wave operator in dimension $n+1$ with n even, a fundamental solution for the wave operator in dimension n. To set the stage we make the following definition.

Definition 9.4. *A sequence $\{\psi_j\}_{j\in\mathbb{N}} \subset C_0^\infty(\mathbb{R})$ is said to* converge in a dominated fashion to 1 *if the following two conditions are satisfied:*

(i) for each compact subset K of $\mathbb{R}$ there exists $j_0 = j_0(K) \in \mathbb{N}$ such that $\psi_j(x) = 1$ for all $x \in K$ if $j \geq j_0$;

(ii) for each $q \in \mathbb{N}_0$, the sequence $\{\psi_j^{(q)}\}_{j\in\mathbb{N}}^\infty$ is uniformly bounded on $\mathbb{R}$.

An example of a sequence $\{\psi_j\}_{j\in\mathbb{N}}$ converging in a dominated fashion to 1 is given by

$$\psi_j(x) := \psi\left(\frac{x}{j}\right), \quad \forall\, x \in \mathbb{R}, \qquad \forall\, j \in \mathbb{N}, \tag{9.1.61}$$

where

$$\psi \in C_0^\infty(\mathbb{R}) \quad \text{satisfied} \quad \psi(x) = 1 \text{ whenever } |x| \leq 1. \tag{9.1.62}$$

In what follows we will use the notation

$$x = (x', x_n) \in \mathbb{R}^n,\ x' \in \mathbb{R}^{n-1},\ x_n \in \mathbb{R}, \quad \partial = (\partial', \partial_n),\ \partial' = (\partial_1, \ldots, \partial_{n-1}). \tag{9.1.63}$$

Definition 9.5. *A distribution $u \in \mathcal{D}'(\mathbb{R}^n)$ is called* `integrable with respect to` x_n *if for any $\varphi \in C_0^\infty(\mathbb{R}^{n-1})$ and any sequence $\{\psi_j\}_{j\in\mathbb{N}} \subset C_0^\infty(\mathbb{R})$ converging in a dominated fashion to 1, the sequence $\big\{\langle u, \varphi \otimes \psi_j\rangle\big\}_{j\in\mathbb{N}}$ is convergent and its limit does not depend on the selection of the sequence $\{\psi_j\}_{j\in\mathbb{N}}$.*

Suppose $u \in \mathcal{D}'(\mathbb{R}^n)$ is integrable with respect to x_n and consider a sequence $\{\psi_j\}_{j\in\mathbb{N}} \subset C_0^\infty(\mathbb{R})$ that converges in a dominated fashion to 1. For each $j \in \mathbb{N}$ define the linear mapping

$$u_j : \mathcal{D}(\mathbb{R}^{n-1}) \to \mathbb{C}, \quad u_j(\varphi) := \langle u, \varphi \otimes \psi_j\rangle, \quad \forall\, \varphi \in C_0^\infty(\mathbb{R}^{n-1}). \tag{9.1.64}$$

Then $u_j \in \mathcal{D}'(\mathbb{R}^{n-1})$ for every $j \in \mathbb{N}$ and $\lim\limits_{j,k\to\infty} \langle u_j - u_k, \varphi\rangle = 0$ for every function $\varphi \in C_0^\infty(\mathbb{R}^{n-1})$, thus the sequence $\{u_j\}_{j\in\mathbb{N}}$ is Cauchy in $\mathcal{D}'(\mathbb{R}^{n-1})$ (see the definition after (13.1.27)). Since $\mathcal{D}'(\mathbb{R}^{n-1})$ is sequentially complete (recall Fact 2.18), it follows that

$$\begin{aligned} &\text{there exists some } u_0 \in \mathcal{D}'(\mathbb{R}^{n-1}) \text{ with the property that} \\ &u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^{n-1})} u_0 \quad \text{and} \quad \langle u_0, \varphi\rangle = \lim_{j\to\infty} \langle u_j, \varphi\rangle \quad \forall\, \varphi \in C_0^\infty(\mathbb{R}^{n-1}). \end{aligned} \tag{9.1.65}$$

Moreover, u_0 is independent of the choice of the sequence $\{\psi_j\}_{j\in\mathbb{N}}$ converging in a dominated fashion to 1 (prove this as an exercise). The distribution u_0 will be called `the integral of` u `with respect to` x_n and will be denoted by $\int_{-\infty}^{\infty} u\, \mathrm{d}x_n$. The reason for using this terminology and notation for u_0 is evident from the next proposition. We denote by $\mathcal{L}^{n-1}$ the Lebesgue measure in $\mathbb{R}^{n-1}$.

Proposition 9.6. *If $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ is a function with the property that*

$$\begin{aligned} &\int_{\mathbb{R}} |f(x', x_n)|\, \mathrm{d}x_n < \infty \quad \textit{for } \mathcal{L}^{n-1}\textit{-almost every } x' \in \mathbb{R}^{n-1} \\ &\textit{and} \quad f \in L^1(K \times \mathbb{R}) \quad \textit{for every compact set } K \subset \mathbb{R}^{n-1}, \end{aligned} \tag{9.1.66}$$

then the distribution $u_f \in \mathcal{D}'(\mathbb{R}^n)$ determined by f (recall (2.1.6)) is integrable with respect to x_n. In addition, the distribution $u_0 := \int_{-\infty}^{\infty} u_f\, \mathrm{d}x_n$ is of function type and is given by the function

$$g(x') := \int_{\mathbb{R}} f(x', x_n)\, \mathrm{d}x_n \quad \textit{defined for } \mathcal{L}^{n-1}\textit{-almost every } x' \in \mathbb{R}^{n-1}. \tag{9.1.67}$$

Proof. Since $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ we have that $u_f \in \mathcal{D}'(\mathbb{R}^n)$ as in (2.1.6) is well-defined. Fix $\varphi \in C_0^\infty(\mathbb{R}^{n-1})$. Since f is absolutely integrable on $\operatorname{supp}\varphi \times \mathbb{R}$, if $\{\psi_j\}_{j\in\mathbb{N}} \subset C_0^\infty(\mathbb{R})$ is a sequence converging in a dominated fashion to 1, we may apply Fubini's theorem and then Lebesgue's dominated convergence theorem to write

$$\lim_{j\to\infty}\big\langle u_f,\varphi\otimes\psi_j\big\rangle=\lim_{j\to\infty}\int_{\mathbb{R}^{n-1}}\varphi(x')\int_{\mathbb{R}}f(x',x_n)\psi_j(x_n)\,\mathrm{d}x_n\,\mathrm{d}x'$$
$$=\int_{\mathbb{R}^{n-1}}\varphi(x')\int_{\mathbb{R}}f(x',x_n)\,\mathrm{d}x_n\,\mathrm{d}x',\tag{9.1.68}$$

and the desired conclusion follows. □

The proposition that is the engine in the method of descent is proved next.

Proposition 9.7. *Let $m\in\mathbb{N}_0$ and let $P(\partial)=P(\partial',\partial_n)$ be a constant coefficient, linear operator of order m in $\mathbb{R}^n$. Define the differential operator $P_0(\partial'):=P(\partial',0)$ in $\mathbb{R}^{n-1}$. If $f\in\mathcal{D}'(\mathbb{R}^{n-1})$ and $u\in\mathcal{D}'(\mathbb{R}^n)$ are such that*

$$P(\partial)u=f(x')\otimes\delta(x_n)\quad\text{in}\quad\mathcal{D}'(\mathbb{R}^n)\tag{9.1.69}$$

and u is integrable with respect to x_n, then $u_0:=\int_{-\infty}^{\infty}u\,\mathrm{d}x_n$ is a solution of the equation $P_0(\partial')u_0=f$ in $\mathcal{D}'(\mathbb{R}^{n-1})$.

Proof. Fix a sequence $\{\psi_j\}_{j\in\mathbb{N}}\subset C_0^\infty(\mathbb{R})$ that converges in a dominated fashion to 1 and let $\varphi\in C_0^\infty(\mathbb{R}^{n-1})$. Using the definition of u_0 we may write

$$\big\langle P_0(\partial')u_0,\varphi\big\rangle=\big\langle u_0,P_0(-\partial')\varphi\big\rangle=\lim_{j\to\infty}\big\langle u,(P_0(-\partial')\varphi)\otimes\psi_j\big\rangle\tag{9.1.70}$$
$$=\lim_{j\to\infty}\Big[\Big\langle u,P(-\partial)(\varphi\otimes\psi_j)\Big\rangle+\Big\langle u,(P_0(-\partial')\varphi)\otimes\psi_j-P(-\partial)(\varphi\otimes\psi_j)\Big\rangle\Big].$$

We claim that

$$\lim_{j\to\infty}\Big\langle u\,,\,(P_0(-\partial')\varphi)\otimes\psi_j-P(-\partial)(\varphi\otimes\psi_j)\Big\rangle=0.\tag{9.1.71}$$

Assume the claim for now. Then returning to (9.1.70) we have

$$\big\langle P_0(\partial')u_0,\varphi\big\rangle=\lim_{j\to\infty}\big\langle u,\,P(-\partial)(\varphi\otimes\psi_j)\big\rangle=\lim_{j\to\infty}\big\langle P(\partial)u\,,\,\varphi\otimes\psi_j\big\rangle$$
$$=\lim_{j\to\infty}\big\langle f\otimes\delta,\varphi\otimes\psi_j\big\rangle=\lim_{j\to\infty}\big[\langle f,\varphi\rangle\psi_j(0)\big]=\langle f,\varphi\rangle,\tag{9.1.72}$$

where for the last equality in (9.1.72) we used property (i) in Definition 9.4 with $K=\{0\}$. Hence, the desired conclusion follows.

To prove (9.1.71), observe that

$$\big(P_0(-\partial')\varphi\big)\otimes\psi_j-P(-\partial)(\varphi\otimes\psi_j)=\sum_{q=1}^{m}\big(P_q(\partial')\varphi\big)\otimes\psi_j^{(q)}$$

where P_q is a differential operator of order $\leq m-q$. Then for each $q\in\{1,\dots m\}$, the sequence $\big\{\psi_j+\psi_j^{(q)}\big\}_{j\in\mathbb{N}}$ also converges in a dominated fashion to 1, which combined with the fact that u is integrable with respect to x_n, further yields

$$\begin{aligned}
&\lim_{j\to\infty}\big\langle u,\big(P_q(\partial')\varphi\big)\otimes\psi_j^{(q)}\big\rangle\\
&\quad=\lim_{j\to\infty}\big\langle u,\big(P_q(\partial')\varphi\big)\otimes(\psi_j+\psi_j^{(q)})\big\rangle-\lim_{j\to\infty}\big\langle u,\big(P_q(\partial')\varphi\big)\otimes\psi_j\big\rangle\\
&\quad=\big\langle u_0,P_q(\partial')\varphi\big\rangle-\big\langle u_0,P_q(\partial')\varphi\big\rangle=0,
\end{aligned}\tag{9.1.73}$$

proving (9.1.71). The proof of the proposition is now complete. □

The Case $n=2p$, $p\geq 1$

We are now ready to proceed with determining a fundamental solution for the wave operator in the case when $n=2p$, $p\geq 1$. The main idea is to use Proposition 9.7 with $P:=\partial_t^2-\sum_{j=1}^{2p+1}\partial_j^2$ being the wave operator in $\mathbb{R}^{n+2}$, $f:=\delta(x_1,\ldots,x_{2p})\otimes\delta(t)$, and u equal to E_{2p+1}, the fundamental solution from (9.1.51). Note that under these assumptions we have $P_0:=\partial_t^2-\sum_{j=1}^{2p}\partial_j^2$ which is the wave operator in $\mathbb{R}^{n+1}$. Thus, if E_{2p+1} is integrable with respect to x_{2p+1}, then by Proposition 9.7 it follows that $u:=\int_{-\infty}^{\infty}E_{2p+1}\,\mathrm{d}x_{2p+1}$ satisfies

$$\left(\partial_t^2-\sum_{j=1}^{2p}\partial_j^2\right)u=\delta(x_1,\ldots,x_{2p})\otimes\delta(t)\quad\text{in}\quad\mathcal{D}'(\mathbb{R}^{2p+1}).\tag{9.1.74}$$

Therefore, u is a fundamental solution for the wave operator corresponding to $n=2p$.

Let us first show that the distribution E_{2p+1} given by the formula in (9.1.51) is integrable with respect to x_{2p+1}. Fix an arbitrary function $\varphi\in C_0^\infty(\mathbb{R}^{2p+1})$ and let $\{\psi_j\}_{j\in\mathbb{N}}\subset C_0^\infty(\mathbb{R})$ be a sequence that converges in a dominated fashion to 1. Then, using the notation $x=(x',x_{2p+1})\in\mathbb{R}^{2p}\times\mathbb{R}$, we have

$$\begin{aligned}
&\lim_{j\to\infty}\big\langle E_{2p+1},\varphi\otimes\psi_j\big\rangle\\
&\quad=\lim_{j\to\infty}2^{-(p+1)}\pi^{-p}\int_0^\infty\left[\left(\frac{1}{r}\partial_r\right)^{p-1}\left(\frac{1}{r}\int\limits_{x\in\mathbb{R}^{2p+1},\,|x|=r}\varphi(x',t)\psi_j(x_{2p+1})\,\mathrm{d}\sigma(x)\right)\right]_{r=t}\mathrm{d}t\\
&\quad=2^{-(p+1)}\pi^{-p}\int_0^\infty\left[\left(\frac{1}{r}\partial_r\right)^{p-1}\left(\frac{1}{r}\int\limits_{x\in\mathbb{R}^{2p+1},\,|x|=r}\varphi(x',t)\,\mathrm{d}\sigma(x)\right)\right]_{r=t}\mathrm{d}t,
\end{aligned}\tag{9.1.75}$$

where for the last equality in (9.1.75) we used Lebesgue's dominated convergence theorem (here we note that the second equality in (9.1.47) and the properties satisfied by $\{\psi_j\}_{j\in\mathbb{N}}$ play an important role). With (9.1.75) in hand, we may conclude that E_{2p+1} is integrable with respect to x_{2p+1}. Consequently, if one

sets $E_{2p} := \int_{-\infty}^{\infty} E_{2p+1}\,dx_{2p+1}$, then $E_{2p} \in \mathcal{D}'(\mathbb{R}^{2p+1})$ and

$$\langle E_{2p},\varphi\rangle = 2^{-(p+1)}\pi^{-p}\int_0^\infty \left[\left(\frac{1}{r}\partial_r\right)^{p-1}\left(\frac{1}{r}\int\limits_{x\in\mathbb{R}^{2p+1},\,|(x',x_{2p+1})|=r}\varphi(x',t)\,d\sigma(x)\right)\right]_{r=t} dt \tag{9.1.76}$$

for every $\varphi \in C_0^\infty(\mathbb{R}^{2p+1})$. In addition, by Proposition 9.7, it follows that E_{2p} is a fundamental solution for the wave operator in $\mathbb{R}^{n+1}$.

Note that the function φ appearing under the second integral in (9.1.76) does not depend on the variable x_{2p+1}. Hence, it is natural to proceed further in order to rewrite (9.1.76) in a form that does not involve the variable x_{2p+1}. Fix $r > 0$ and denote by $B(0,r)$ the ball in $\mathbb{R}^n$ of radius r and centered at $0 \in \mathbb{R}^n$. Define the mappings $P_\pm : B(0,r) \to \mathbb{R}^{2p+1}$ by setting

$$P_\pm(x') := (x_1, x_2, \ldots, x_{2p}, \pm\sqrt{r^2-|x'|^2}), \forall\, x' = (x_1,\ldots,x_{2p}) \in B(0,r). \tag{9.1.77}$$

Then P_+ and P_- are parametrizations of the (open) upper and lower, respectively, hemispheres of the surface ball in $\mathbb{R}^{2p+1}$ of radius r and centered at 0. Hence (keeping in mind Definition 13.38 and Definition 13.39), we may write

$$\begin{aligned}\int\limits_{x\in\mathbb{R}^{2p+1},\,|(x',x_{2p+1})|=r}\varphi(x',t)\,d\sigma(x) &= \int_{B(0,r)}\varphi(x',t)\,|\partial_1 P_+\times\cdots\times\partial_{2p}P_+|\,dx'\\ &\quad+\int_{B(0,r)}\varphi(x',t)\,|\partial_1 P_-\times\cdots\times\partial_{2p}P_-|\,dx'.\end{aligned} \tag{9.1.78}$$

A direct computation based on (9.1.77) and (13.6.4) further yields

$$\partial_1 P_\pm\times\cdots\times\partial_{2p}P_\pm = \begin{vmatrix} 1 & 0 & \ldots & 0 & \mp\frac{x_1}{\sqrt{r^2-|x'|^2}}\\ 0 & 1 & \ldots & 0 & \mp\frac{x_2}{\sqrt{r^2-|x'|^2}}\\ \ldots & \ldots & \ldots & \ldots & \ldots\\ 0 & 0 & \ldots & 1 & \mp\frac{x_{2p}}{\sqrt{r^2-|x'|^2}}\\ \mathbf{e}_1 & \mathbf{e}_2 & \ldots & \mathbf{e}_{2p} & \mathbf{e}_{2p+1}\end{vmatrix} =: \det A^\pm, \tag{9.1.79}$$

where $\mathbf{e}_j$ is the unit vector in $\mathbb{R}^{2p+1}$ with 1 on the jth position, for each $j \in \{1,\ldots,2p+1\}$. Hence, the components of the vector $\partial_1 P_\pm\times\cdots\times\partial_{2p}P_\pm$ are

$$(-1)^{k+1}\det A_k^\pm,\quad k\in\{1,\ldots,2p,2p+1\}, \tag{9.1.80}$$

where $A_k^\pm$ is the $2p\times 2p$ matrix obtained from $A^\pm$ by deleting column k and row $2p+1$. It is easy to see from (9.1.79) that

$$\det A_{2p+1}^\pm = 1,\quad \det A_k^\pm = (-1)^k\frac{\mp x_k}{\sqrt{r^2-|x'|^2}},\quad \forall\, k\in\{1,\ldots,2p\}. \tag{9.1.81}$$

Consequently,

$$|\partial_1 P_\pm \times \cdots \times \partial_{2p} P_\pm| = \frac{r}{\sqrt{r^2 - |x'|^2}}. \tag{9.1.82}$$

Formula (9.1.82) combined with (9.1.78) and (9.1.76) gives

$$\begin{aligned}\langle E_{2p}, \varphi\rangle &= (2\pi)^{-p} \int_0^\infty \left[\left(\frac{1}{r}\partial r\right)^{p-1} \left(\frac{1}{r} \int_{x' \in B(0,r)} \varphi(x', r) \frac{r}{\sqrt{r^2 - |x'|^2}}\, \mathrm{d}x' \right) \right]_{r=t} \mathrm{d}t \\ &= (2\pi)^{-p} \int_0^\infty \left[\left(\frac{1}{r}\partial_r\right)^{p-1} \left(\int_{B(0,r)} \frac{\varphi(x', t)}{\sqrt{r^2 - |x'|^2}}\, \mathrm{d}x' \right) \right]_{r=t} \mathrm{d}t. \end{aligned} \tag{9.1.83}$$

In summary, we proved:

If $n = 2p$, for some $p \in \mathbb{N}$, then $E_+ \in \mathcal{S}'(\mathbb{R}^{n+1})$, defined by

$$\langle E_+, \varphi\rangle = (2\pi)^{-p} \int_0^\infty \left[\left(\frac{1}{r}\partial_r\right)^{p-1} \left(\int_{x \in \mathbb{R}^n,\, |x| < r} \frac{\varphi(x, t)}{\sqrt{r^2 - |x|^2}}\, \mathrm{d}x \right) \right]_{r=t} \mathrm{d}t, \tag{9.1.84}$$

for $\varphi \in \mathcal{S}(\mathbb{R}^{n+1})$, is a fundamental solution for the wave operator in $\mathbb{R}^{n+1}$.

The reasoning used to obtain E_+ also applies if one starts with E_{2p+1} being the distribution from (9.1.53). Under this scenario the conclusion is that

If $n = 2p$, for some $p \in \mathbb{N}$, then $E_- \in \mathcal{S}'(\mathbb{R}^{n+1})$, defined by

$$\langle E_-, \varphi\rangle = (2\pi)^{-p} \int_0^\infty \left[\left(\tfrac{1}{r}\partial_r\right)^{p-1} \left(\int_{x \in \mathbb{R}^n,\, |x| < r} \tfrac{\varphi(x, -t)}{\sqrt{r^2 - |x|^2}}\, \mathrm{d}x \right) \right]_{r=t} \mathrm{d}t, \tag{9.1.85}$$

for $\varphi \in \mathcal{S}(\mathbb{R}^{n+1})$, is a fundamental solution for the wave operator in $\mathbb{R}^{n+1}$.

Remark 9.8.

(1) If E_+ and E_- are as in (9.1.84) and (9.1.85), respectively, then

$$\operatorname{supp} E_+ = \{(x, t) \in \mathbb{R}^{2p} \times [0, \infty) : |x| \leq t\}, \quad \textit{and} \tag{9.1.86}$$

$$\operatorname{supp} E_- = \{(x, t) \in \mathbb{R}^{2p} \times (-\infty, 0] : |x| \leq -t\}. \tag{9.1.87}$$

(2) If $p = 1$, then for each $\varphi \in \mathcal{S}(\mathbb{R}^3)$ the expression[2] in (9.1.84) becomes

$$\langle E_+, \varphi\rangle = \frac{1}{2\pi} \int_0^\infty \int_{|x| < t} \frac{\varphi(x, t)}{\sqrt{t^2 - |x|^2}}\, \mathrm{d}x\, \mathrm{d}t = \left\langle \frac{H(t - |x|)}{2\pi\sqrt{t^2 - |x|^2}}, \varphi \right\rangle.$$

[2]This expression was first found by Vito Volterra (cf. [73]).

Hence, if $n = 2$ then E_+ is of function type and

$$E_+(x,t) = \frac{H(t-|x|)}{2\pi\sqrt{t^2-|x|^2}} \quad \text{for every} \quad x \in \mathbb{R}^2 \quad \text{and every} \quad t \in \mathbb{R}.$$

Similarly, when $n = 2$ it follows that E_- is of function type and

$$E_-(x,t) = \frac{H(-t-|x|)}{2\pi\sqrt{t^2-|x|^2}} \quad \text{for every} \quad x \in \mathbb{R}^2 \quad \text{and every} \quad t \in \mathbb{R}.$$

Summary for Arbitrary n

Here we combine the results obtained in this section regarding fundamental solutions for the heat operator. These results have been summarized in (9.1.28), Remark 9.1, (9.1.51), (9.1.53), Remark 9.2, (9.1.84), (9.1.85), and Remark 9.8.

Theorem 9.9. *Consider the wave operator $\Box = \partial_t^2 - \Delta_x$ in $\mathbb{R}^{n+1}$, where $x \in \mathbb{R}^n$ and $t \in \mathbb{R}$. Then the following are true.*

(1) Suppose $n = 1$. Then the $L^1_{\rm loc}(\mathbb{R}^2)$ functions

$$E_+(x,t) := \frac{H(t-|x|)}{2}, \qquad \forall\,(x,t) \in \mathbb{R}^2, \tag{9.1.88}$$

$$E_-(x,t) := \frac{H(-t-|x|)}{2}, \qquad \forall\,(x,t) \in \mathbb{R}^2, \tag{9.1.89}$$

satisfy $E_+, E_- \in \mathcal{S}'(\mathbb{R}^2)$ and are fundamental solutions for the wave operator in $\mathbb{R}^2$. Moreover,

$$\operatorname{supp} E_+ = \{(x,t) \in \mathbb{R}^2 : |x| \leq t\}, \tag{9.1.90}$$

$$\operatorname{supp} E_- = \{(x,t) \in \mathbb{R}^2 : |x| \leq -t\}. \tag{9.1.91}$$

(2) Suppose $n = 2p+1$, for some $p \in \mathbb{N}$. Then the distributions $E_+ \in \mathcal{S}'(\mathbb{R}^{n+1})$ and $E_- \in \mathcal{S}'(\mathbb{R}^{n+1})$ defined by

$$\langle E_+, \varphi\rangle := (2\pi)^{-p-1}\pi \int_0^\infty \left[\left(\tfrac{1}{r}\partial_r\right)^{p-1}\left(\tfrac{1}{r}\int_{\partial B(0,r)}\varphi(\omega,t)\,d\sigma(\omega)\right)\right]_{r=t} dt, \tag{9.1.92}$$

$$\langle E_-, \varphi\rangle := (2\pi)^{-p-1}\pi \int_0^\infty \left[\left(\tfrac{1}{r}\partial_r\right)^{p-1}\left(\tfrac{1}{r}\int_{\partial B(0,r)}\varphi(\omega,-t)\,d\sigma(\omega)\right)\right]_{r=t} dt, \tag{9.1.93}$$

for every $\varphi \in \mathcal{S}(\mathbb{R}^{n+1})$, are fundamental solutions for the wave operator in $\mathbb{R}^{n+1}$. Moreover,

$$\operatorname{supp} E_+ = \{(x,t) \in \mathbb{R}^{2p+1} \times [0,\infty) : |x| = t\}, \tag{9.1.94}$$

$$\operatorname{supp} E_+ = \{(x,t) \in \mathbb{R}^{2p+1} \times (-\infty, 0] : |x| = -t\}. \tag{9.1.95}$$

Corresponding to the case $n = 3$ (thus, for $p = 1$), formulas (9.1.92) and (9.1.93) become

$$\langle E_+, \varphi \rangle = \frac{1}{4\pi} \int_0^\infty \frac{1}{t} \int_{\partial B(0,t)} \varphi(\omega, t)\, \mathrm{d}\sigma(\omega)\, \mathrm{d}t = \left\langle \frac{H(t)}{4\pi t} \delta_{\partial B(0,t)},\, \varphi \right\rangle, \tag{9.1.96}$$

$$\langle E_-, \varphi \rangle = \frac{1}{4\pi} \int_0^\infty \frac{1}{t} \int_{\partial B(0,t)} \varphi(\omega, -t)\, \mathrm{d}\sigma(\omega)\, \mathrm{d}t = \left\langle -\frac{H(-t)}{4\pi t} \delta_{\partial B(0,-t)},\, \varphi \right\rangle, \tag{9.1.97}$$

for every $\varphi \in \mathcal{S}(\mathbb{R}^4)$.

(3) Suppose $n = 2p$, for some $p \in \mathbb{N}$. Then the distributions $E_+ \in \mathcal{S}'(\mathbb{R}^{n+1})$ and $E_- \in \mathcal{S}'(\mathbb{R}^{n+1})$ defined by

$$\langle E_+, \varphi \rangle := (2\pi)^{-p} \int_0^\infty \left[\left(\tfrac{1}{r}\partial_r\right)^{p-1} \left(\int_{x \in \mathbb{R}^n,\, |x| < r} \frac{\varphi(x,t)}{\sqrt{r^2 - |x|^2}}\, \mathrm{d}x \right) \right]_{r=t} \mathrm{d}t, \tag{9.1.98}$$

$$\langle E_-, \varphi \rangle := (2\pi)^{-p} \int_0^\infty \left[\left(\tfrac{1}{r}\partial_r\right)^{p-1} \left(\int_{x \in \mathbb{R}^n,\, |x| < r} \frac{\varphi(x,-t)}{\sqrt{r^2 - |x|^2}}\, \mathrm{d}x \right) \right]_{r=t} \mathrm{d}t, \tag{9.1.99}$$

for every $\varphi \in \mathcal{S}(\mathbb{R}^{n+1})$, are fundamental solutions for the wave operator in $\mathbb{R}^{n+1}$. Moreover,

$$\operatorname{supp} E_+ = \{(x,t) \in \mathbb{R}^{2p} \times [0, \infty) : |x| \le t\}, \quad \textit{and} \tag{9.1.100}$$

$$\operatorname{supp} E_- = \{(x,t) \in \mathbb{R}^{2p} \times (-\infty, 0] : |x| \le -t\}. \tag{9.1.101}$$

Corresponding to the case $n = 2$, thus $p = 1$, the distributions E_+ and E_- defined in (9.1.98) and (9.1.99) are of function type and given by the functions

$$E_+(x,t) = \tfrac{H(t-|x|)}{2\pi\sqrt{t^2-|x|^2}}, \qquad \forall\, x \in \mathbb{R}^2,\, \forall\, t \in \mathbb{R}, \tag{9.1.102}$$

$$E_-(x,t) = \tfrac{H(-t-|x|)}{2\pi\sqrt{t^2-|x|^2}}, \qquad \forall\, x \in \mathbb{R}^2,\, \forall\, t \in \mathbb{R}. \tag{9.1.103}$$

9.2 The Generalized Cauchy Problem for the Wave Operator

Let $F \in C^0(\mathbb{R}^{n+1})$, $f, g \in C^0(\mathbb{R}^n)$ and, as before, use the notation $x \in \mathbb{R}^n$, $t \in \mathbb{R}$. Suppose $u \in C^2(\mathbb{R}^{n+1})$ solves in the classical sense the Cauchy problem

$$\begin{cases} (\partial_t^2 - \Delta_x)u = F \quad \text{in } \mathbb{R}^{n+1}, \\ u\big|_{t=0} = f, \ \ \partial_t u\big|_{t=0} = g \quad \text{in } \mathbb{R}^n. \end{cases} \tag{9.2.1}$$

Define $\widetilde{u}(x,t) := H(t)u(x,t)$ and $\widetilde{F}(x,t) := H(t)F(x,t)$ for every $x \in \mathbb{R}^n$ and every $t \in \mathbb{R}$. Then $\widetilde{u}$, $\widetilde{F} \in L^1_{\rm loc}(\mathbb{R}^{n+1})$ and for each $\varphi \in C_0^\infty(\mathbb{R}^{n+1})$ we have

$$\begin{aligned} \big\langle (\partial_t^2 - \Delta_x)\widetilde{u}, \varphi \big\rangle &= \big\langle \widetilde{u}, (\partial_t^2 - \Delta_x)\varphi \big\rangle = \lim_{\varepsilon \to 0^+} \int_\varepsilon^\infty \int_{\mathbb{R}^n} u\,(\partial_t^2 - \Delta_x)\varphi \, \mathrm{d}x \, \mathrm{d}t \\ &= \lim_{\varepsilon \to 0^+} \int_\varepsilon^\infty \int_{\mathbb{R}^n} (\partial_t^2 - \Delta_x)u \, \varphi \, \mathrm{d}x \, \mathrm{d}t \\ &\quad + \lim_{\varepsilon \to 0^+} \int_{\mathbb{R}^n} \big(\partial_t u(x,\varepsilon)\varphi(x,\varepsilon) - u(x,\varepsilon)\partial_t\varphi(x,\varepsilon)\big) \, \mathrm{d}x \\ &= \big\langle \widetilde{F}, \varphi \big\rangle + \big\langle f(x) \otimes \delta'(t), \varphi(x,t) \big\rangle + \big\langle g(x) \otimes \delta(x), \varphi(x,t) \big\rangle. \end{aligned} \tag{9.2.2}$$

This suggests the following definition [recall (8.2.4)].

Definition 9.10. *Given $\widetilde{F} \in \mathcal{D}'_+(\mathbb{R}^{n+1})$ and $f, g \in C_0^\infty(\mathbb{R}^n)$ a distribution $\widetilde{u}$ in $\mathcal{D}'_+(\mathbb{R}^{n+1})$ is called a solution of the generalized Cauchy problem for the wave operator with data $\widetilde{F}$, f, and g, if*

$$(\partial_t^2 - \Delta_x)\widetilde{u} = \widetilde{F} + f(x) \otimes \delta'(t) + g(x) \otimes \delta(t) \quad \text{ in } \mathcal{D}'(\mathbb{R}^{n+1}). \tag{9.2.3}$$

Theorem 9.11. *Let $\widetilde{F} \in \mathcal{D}'_+(\mathbb{R}^{n+1})$ and $f, g \in C_0^\infty(\mathbb{R}^n)$ be given. Then the generalized Cauchy problem (9.2.3) has the unique solution*

$$\widetilde{u} = E * \widetilde{F} + E * (f \otimes \delta') + E * (g \otimes \delta), \tag{9.2.4}$$

where E is the fundamental solution for the wave operator in $\mathbb{R}^{n+1}$ as specified in (9.1.88) if $n = 1$, in (9.1.92) if $n \geq 3$ is odd, and in (9.1.98) if n is even.

Proof. Let E be as in the statement of the theorem. Then, by (9.1.90), by (9.1.94), and by (9.1.100), for any $G \in \mathcal{D}'_+(\mathbb{R}^{n+1})$ we have that, whenever K is a compact subset of $\mathbb{R}^{n+1}$, the set

$$M_K := \{((x,t),(y,s)) : (x,t) \in \operatorname{supp} E, \ (y,s) \in \operatorname{supp} G, \ (x+y, t+s) \in K\} \tag{9.2.5}$$

is compact in $\mathbb{R}^{n+1}$. Hence, by Theorem 2.85, the convolution $E * G$ exists. This proves that $\widetilde{u}$ as in (9.2.4) is a well-defined element of $\mathcal{D}'(\mathbb{R}^{n+1})$. Moreover, since

$\widetilde{F}$, $f \otimes \delta'$, $g \otimes \delta$, $E \in \mathcal{D}'_+(\mathbb{R}^{n+1})$, by (2.8.17), we may conclude that $\widetilde{u}$ belongs to $\mathcal{D}'_+(\mathbb{R}^{n+1})$. In addition, by Remark 5.5, we obtain that $\widetilde{u}$ as in (9.2.4) is a solution of the generalized Cauchy problem (9.2.3).

To prove uniqueness, observe that if $\widetilde{u} \in \mathcal{D}'_+(\mathbb{R}^{n+1})$ is such that $(\partial_t^2 - \Delta_x)\widetilde{u} = 0$ in $\mathcal{D}'(\mathbb{R}^{n+1})$, then

$$\widetilde{u} = \widetilde{u} * \delta = \widetilde{u} * ((\partial_t^2 - \Delta_x)E) = ((\partial_t^2 - \Delta_x)\widetilde{u}) * E = 0. \tag{9.2.6}$$

This completes the proof of the theorem. □

Further Notes for Chap. 9 The wave operator $\Box := \partial_t^2 - \Delta_x$, where $x \in \mathbb{R}^n$ and $t \in \mathbb{R}$, was originally discovered by the French mathematician and physicist Jean le Rond d'Alembert in 1747 in the case $n = 1$ in his studies of vibration of strings. For this reason, $\Box$ is also called the d'Alembert operator or, simply, the d'Alembertian. Like the heat operator discussed in Chap. 8, the wave operator is another basic example of a partial differential operator governing a linear evolution equation (though, unlike the heat operator, the wave operator belongs to a class of operators called hyperbolic operators).

9.3 Additional Exercises for Chap. 9

Exercise 9.12. *Use the method of descent to compute a fundamental solution for the Laplace operator* $\Delta = \sum_{j=1}^{n} \partial_j^2$ *in* $\mathbb{R}^n$, $n \geq 3$, *by starting from a fundamental solution for the heat operator* $L = \partial_t - \sum_{j=1}^{n} \partial_j^2$ *in* $\mathbb{R}^{n+1}$.

Chapter 10

The Lamé and Stokes Operators

With the exception of the Dirac operator, so far we have discussed only scalar differential operators. In this chapter we analyze two differential operators acting on vectors with components distributions in $\mathbb{R}^n$: the Lamé operator arising in the theory of elasticity and the Stokes operator arising in hydrodynamics.

Throughout this chapter, it is assumed that $n \in \mathbb{N}$ satisfies $n \geq 2$.

10.1 General Remarks About Vector and Matrix Distributions

The material developed up to this point may be regarded as a theory for scalar distributions. Nonetheless, practical considerations dictate the necessity of considering vectors/matrices with components/entries that are themselves distributions. It is therefore natural to refer to such objects as vector and matrix distributions. A significant portion of the theory of scalar distributions then readily extends to this more general setting. The philosophy in the vector/matrix case is that we perform the same type of analysis as in the scalar case, at the level of individual components, while at the same time obeying the natural algebraic rules that are now in effect (e.g., keeping in mind the algebraic mechanism according to which two matrices are multiplies, etc.).

To offer some examples, fix an open set $\Omega \subseteq \mathbb{R}^n$ and consider a matrix distribution

$$\mathbb{U} = \big(u_{k\ell}\big)_{\substack{1\leq k\leq N\\ 1\leq \ell\leq K}} \in \mathcal{M}_{N\times K}\big(\mathcal{D}'(\Omega)\big), \tag{10.1.1}$$

that is, $\mathbb{U}$ is an $N \times K$ matrix with entries from $\mathcal{D}'(\Omega)$. Naturally, equality of elements in $\mathcal{M}_{N\times K}\big(\mathcal{D}'(\Omega)\big)$ is understood entry by entry in $\mathcal{D}'(\Omega)$. We agree to

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define the support of $\mathbb{U}$ as

$$\operatorname{supp}\mathbb{U} := \bigcup_{k=1}^{K}\bigcup_{\ell=1}^{N}\operatorname{supp}u_{k\ell}. \tag{10.1.2}$$

Note that $\operatorname{supp}\mathbb{U}$ is the smallest relatively closed subset of Ω outside of which all entries of $\mathbb{U}$ vanish. Similarly, we define the singular support of $\mathbb{U}$ as

$$\operatorname{sing\,supp}\mathbb{U} := \bigcup_{k=1}^{K}\bigcup_{\ell=1}^{N}\operatorname{sing\,supp}u_{k\ell}. \tag{10.1.3}$$

Hence, $\operatorname{sing\,supp}\mathbb{U}$ is the smallest relatively closed subset of Ω outside of which all entries of $\mathbb{U}$ are C^∞.

Next, if $A = \big(a^{jk}\big)_{\substack{1\leq j\leq M\\ 1\leq k\leq N}} \in \mathcal{M}_{M\times N}\big(C^\infty(\Omega)\big)$, then we define (with $\mathbb{U}$ as in (10.1.1))

$$A\mathbb{U} := \left(\sum_{k=1}^{N} a^{jk}u_{k\ell}\right)_{\substack{1\leq j\leq M\\ 1\leq \ell\leq K}} \in \mathcal{M}_{M\times K}\big(\mathcal{D}'(\Omega)\big). \tag{10.1.4}$$

Some trivial, yet useful properties include

$$\begin{gathered}\text{if}\quad \mathbb{U},\mathbb{V}\in\mathcal{M}_{N\times K}\big(\mathcal{D}'(\Omega)\big),\quad A\in\mathcal{M}_{J\times M}\big(C^\infty(\Omega)\big),\quad B\in\mathcal{M}_{M\times N}\big(C^\infty(\Omega)\big),\\ \text{then}\quad A(B\mathbb{U}) = (AB)\mathbb{U}\quad\text{and}\\ \mathbb{U}=\mathbb{V}\implies A\mathbb{U}=A\mathbb{V}.\end{gathered} \tag{10.1.5}$$

Given an operator of the form

$$L = \sum_{|\alpha|\leq m} A_\alpha\partial^\alpha,\qquad A_\alpha = \big(a_\alpha^{jk}\big)_{\substack{1\leq j\leq M\\ 1\leq k\leq N}} \in \mathcal{M}_{M\times N}\big(C^\infty(\Omega)\big), \tag{10.1.6}$$

referred to as an $M\times N$ system (of differential operators) of order $m\in\mathbb{N}$, the natural way in which L acts on $\mathbb{U}$ as in (10.1.1), is to regard $L\mathbb{U}$ as the matrix distribution from $\mathcal{M}_{M\times K}\big(\mathcal{D}'(\Omega)\big)$ with entries given by

$$(L\mathbb{U})_{j\ell} := \sum_{|\alpha|\leq m}\sum_{k=1}^{N} a_\alpha^{jk}\partial^\alpha u_{k\ell}\quad\text{for all}\quad 1\leq j\leq M,\ 1\leq\ell\leq K. \tag{10.1.7}$$

A matrix distribution with a single column is referred to as a vector distribution and is simply denoted by $\mathbf{u} = (u_k)_{1\leq k\leq N}$ with components $u_k\in\mathcal{D}'(\Omega)$, $1\leq k\leq N$. The collection of all such vector distributions is denoted by $\big[\mathcal{D}'(\Omega)\big]^N$.

Regarding the convolution product for matrix distributions, if

$$\mathbb{U} = \big(u_{jk}\big)_{\substack{1\leq k\leq N\\ 1\leq \ell\leq K}} \in \mathcal{M}_{M\times N}\big(\mathcal{D}'(\mathbb{R}^n)\big)\quad\text{and}$$

$$\mathbb{V} = \big(v_{k\ell}\big)_{\substack{1\leq k\leq N\\ 1\leq \ell\leq K}} \in \mathcal{M}_{N\times K}\big(\mathcal{E}'(\mathbb{R}^n)\big)$$

(hence, $\mathbb{V}$ is an $N\times K$ matrix with entries from $\mathcal{E}'(\mathbb{R}^n)$), we define $\mathbb{U}*\mathbb{V}$ as being the $M\times K$ matrix with entries that are the distributions from $\mathcal{D}'(\mathbb{R}^n)$ given by

$$(\mathbb{U}*\mathbb{V})_{j\ell}:=\sum_{k=1}^{N}u_{jk}*v_{k\ell},\qquad 1\leq j\leq M,\ \ 1\leq\ell\leq K. \tag{10.1.8}$$

Formula (10.1.8) is also taken as definition for $\mathbb{U}*\mathbb{V}$ in the case when $\mathbb{U}=\big(u_{jk}\big)_{\substack{1\leq k\leq N\\ 1\leq\ell\leq K}}\in\mathcal{M}_{M\times N}\big(\mathcal{E}'(\mathbb{R}^n)\big)$ and $\mathbb{V}=\big(v_{k\ell}\big)_{\substack{1\leq k\leq N\\ 1\leq\ell\leq K}}\in\mathcal{M}_{N\times K}\big(\mathcal{D}'(\mathbb{R}^n)\big)$. The reader is advised that, as opposed to the scalar case, the commutativity property for the convolution product is lost in the setting of (say, square) matrix distributions.

If for each $M\in\mathbb{N}$ we denote by $\delta I_{M\times M}$ the matrix distribution belonging to $\mathcal{M}_{M\times M}\big(\mathcal{D}'(\mathbb{R}^n)\big)$ and with entries $\delta_{jk}\delta$, for all $j,k\in\{1,\dots,M\}$, where $\delta_{jk}=1$ if $j=k$ and 0 otherwise, is the Kronecker symbol, then one can check using (10.1.8) and part *(d)* in Theorem (2.87) that

$$(\delta I_{M\times M})*\mathbb{U}=\mathbb{U},\qquad\forall\,\mathbb{U}\in\mathcal{M}_{M\times N}\big(\mathcal{D}'(\mathbb{R}^n)\big), \tag{10.1.9}$$

$$\mathbb{V}*(\delta I_{M\times M})=\mathbb{V},\qquad\forall\,\mathbb{V}\in\mathcal{M}_{N\times M}\big(\mathcal{D}'(\mathbb{R}^n)\big). \tag{10.1.10}$$

Two important first-order systems of differential operators are the gradient ∇ and the divergence div. Specifically, the gradient $\nabla:=(\partial_1,\dots,\partial_n)$ acts on a scalar distribution according to $\nabla u:=(\partial_1 u,\dots,\partial_n u)\in\big[\mathcal{D}'(\Omega)\big]^n$ for every $u\in\mathcal{D}'(\Omega)$, while the divergence operator div acts on a vector distribution $\mathbf{v}:=(v_1,\dots,v_n)\in\big[\mathcal{D}'(\Omega)\big]^n$ according to $\mathrm{div}\mathbf{v}:=\sum\limits_{j=1}^{n}\partial_j v_j$.

To give an example of a specific result from the theory of scalar distributions and scalar differential operators that naturally carries over to the vector/matrix case, we note here that if L is an $J\times M$ system with constant coefficients, then for every matrix distributions $\mathbb{U}\in\mathcal{M}_{M\times N}\big(\mathcal{D}'(\mathbb{R}^n)\big)$ and $\mathbb{V}\in\mathcal{M}_{N\times K}\big(\mathcal{E}'(\mathbb{R}^n)\big)$ then

$$L(\mathbb{U}*\mathbb{V})=(L\mathbb{U})*\mathbb{V}=\mathbb{U}*(L\mathbb{V})\quad\text{in}\quad\mathcal{M}_{J\times K}\big(\mathcal{D}'(\mathbb{R}^n)\big). \tag{10.1.11}$$

Similar equalities hold if $\mathbb{U}\in\mathcal{M}_{M\times N}\big(\mathcal{E}'(\mathbb{R}^n)\big)$ and $\mathbb{V}\in\mathcal{M}_{N\times K}\big(\mathcal{D}'(\mathbb{R}^n)\big)$.

Definition 10.1. *Given an $M\times M$ system of order $m\in\mathbb{N}$,*

$$L=\sum_{|\alpha|\leq m}A_\alpha\partial^\alpha,\qquad A_\alpha=\big(a_\alpha^{jk}\big)_{1\leq j,k\leq M}\in\mathcal{M}_{M\times M}\big(C^\infty(\mathbb{R}^n)\big), \tag{10.1.12}$$

call a matrix distribution $\mathbb{E}\in\mathcal{M}_{M\times M}\big(\mathcal{D}'(\mathbb{R}^n)\big)$ a fundamental solution for L in $\mathbb{R}^n$ provided

$$L\mathbb{E}=\delta I_{M\times M}\quad\text{in}\quad\mathcal{M}_{M\times M}\big(\mathcal{D}'(\mathbb{R}^n)\big). \tag{10.1.13}$$

We record next the analogue of Proposition 5.1 and Proposition 5.7 in the current setting.

Proposition 10.2. *Suppose L is a constant coefficient $M\times M$ system of order $m\in\mathbb{N}$,*

$$L=L(\partial):=\sum_{|\alpha|\leq m}A_\alpha\partial^\alpha,\qquad A_\alpha=\big(a_\alpha^{jk}\big)_{1\leq j,k\leq M}\in\mathcal{M}_{M\times M}(\mathbb{C}),\tag{10.1.14}$$

with the property that

$$\det\big(L(i\xi)\big)\neq 0\quad\text{for every}\quad\xi\in\mathbb{R}^n\setminus\{0\}.\tag{10.1.15}$$

Then the following hold.

(1) If $\mathbb{U}\in\mathcal{M}_{M\times K}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ is such that $L\mathbb{U}=0$ in $\mathcal{M}_{M\times K}\big(\mathcal{S}'(\mathbb{R}^n)\big)$, then the entries of $\mathbb{U}$ are polynomials in $\mathbb{R}^n$ and $L\mathbb{U}=0\in\mathcal{M}_{M\times K}(\mathbb{C})$ pointwise in $\mathbb{R}^n$.

(2) If L has a fundamental solution $\mathbb{E}\in\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)$, then any other fundamental solution of L belonging to $\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ differs from $\mathbb{E}$ by an $M\times M$-matrix $\mathbb{P}$ which has entries that are polynomials in $\mathbb{R}^n$ satisfying $L\mathbb{P}=0\in\mathcal{M}_{M\times M}(\mathbb{C})$ pointwise in $\mathbb{R}^n$.

Proof. If $\mathbb{U}\in\mathcal{M}_{M\times K}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ is such that $L\mathbb{U}=0$ in $\mathcal{M}_{M\times K}\big(\mathcal{S}'(\mathbb{R}^n)\big)$, taking the Fourier transform, we obtain $L(i\xi)\widehat{\mathbb{U}}=0$ in $\mathcal{M}_{M\times K}\big(\mathcal{S}'(\mathbb{R}^n)\big)$. In particular,

$$L(i\xi)\widehat{\mathbb{U}}=0\quad\text{ in }\quad\mathcal{M}_{M\times K}\big(\mathcal{D}'(\mathbb{R}^n\setminus\{0\})\big).\tag{10.1.16}$$

Since (10.1.15) holds, the matrix $L(i\xi)$ is invertible for every $\xi\in\mathbb{R}^n\setminus\{0\}$, hence $\big(L(i\xi)\big)^{-1}\in\mathcal{M}_{M\times M}\big(C^\infty(\mathbb{R}^n\setminus\{0\})\big)$. Based on this, (10.1.16), and (10.1.5), we may write

$$\widehat{\mathbb{U}}=\big(L(i\xi)\big)^{-1}L(i\xi)\widehat{\mathbb{U}}=\big(L(i\xi)\big)^{-1}0=0\quad\text{ in }\quad\mathcal{M}_{M\times K}\big(\mathcal{D}'(\mathbb{R}^n\setminus\{0\})\big).\tag{10.1.17}$$

Hence, $\widehat{\mathbb{U}}=0$ in $\mathcal{M}_{M\times K}\big(\mathcal{D}'(\mathbb{R}^n\setminus\{0\})\big)$ which implies $\operatorname{supp}\widehat{\mathbb{U}}\subseteq\{0\}$. Exercise 4.35 (applied to each entry of $\mathbb{U}$) then gives that each component of $\mathbb{U}$ is a polynomial in $\mathbb{R}^n$. In addition, $L\mathbb{U}=0$ pointwise in $\mathbb{R}^n$. This proves *(1)*.

Suppose now $\mathbb{U}\in\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ is such that $L\mathbb{U}=\delta I_{M\times M}$ in $\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)$. Then $L(\mathbb{U}-\mathbb{E})=0$ in $\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ and the desired conclusion follows by applying part *(1)*. □

We also have the general Liouville-type theorem for a system with total symbol invertible outside the origin, which we prove next.

Theorem 10.3 (A general Liouville-type theorem for systems)**.** *Assume L is an $M\times M$ system of order $m\in\mathbb{N}$ of the form (10.1.14) which satisfies (10.1.15). Also, suppose $\mathbf{u}=(u_1,\ldots,u_M)\in\big[L^1_{\rm loc}(\mathbb{R}^n)\big]^M$ satisfies $L\mathbf{u}=0$ in $\big[\mathcal{S}'(\mathbb{R}^n)\big]^M$ and has the property that there exist $N\in\mathbb{N}_0$ and $C,R\in[0,\infty)$ such that*

$$|u_j(x)|\leq C|x|^N\quad\text{whenever}\quad|x|\geq R,\qquad\forall\,j\in\{1,\ldots,M\}.\tag{10.1.18}$$

Then u_j is a polynomial in $\mathbb{R}^n$ of degree at most N for each $j \in \{1,\dots,M\}$.

In particular, if $\mathbf{u} \in \big[L^\infty(\mathbb{R}^n)\big]^M$ satisfies $L\mathbf{u} = 0$ in $\big[\mathcal{S}'(\mathbb{R}^n)\big]^M$ then the components of $\mathbf{u}$ are necessarily constants.

Proof. Since (5.1.10) implies that the locally integrable function u_j belongs to $\mathcal{S}'(\mathbb{R}^n)$ (cf. Example 4.3) for each $j \in \{1,\dots,M\}$, Proposition 10.2 implies that all the entries of $\mathbf{u}$ are polynomials in $\mathbb{R}^n$. Moreover, Lemma 5.2 gives that the degree of these polynomials are at most N. □

We conclude this section by noting a couple of useful results.

Proposition 10.4. *Assume that*

$$L = \sum_{|\alpha|\leq m} A_\alpha \partial^\alpha, \qquad A_\alpha = \big(a_\alpha^{jk}\big)_{1\leq j,k\leq M} \in \mathcal{M}_{M\times M}\big(\mathbb{C}\big), \tag{10.1.19}$$

is an $M\times M$ system of order $m\in\mathbb{N}$ with constant complex coefficients and suppose that $\mathbb{E}\in\mathcal{M}_{M\times M}\big(\mathcal{D}'(\mathbb{R}^n)\big)$ is a fundamental solution for the system L in $\mathbb{R}^n$. Then for any $\mathbb{U}\in\mathcal{M}_{M\times N}\big(\mathcal{E}'(\mathbb{R}^n)\big)$ one has

$$L(\mathbb{E}*\mathbb{U}) = \mathbb{U} \quad \text{in} \quad \mathcal{M}_{M\times M}\big(\mathcal{D}'(\mathbb{R}^n)\big). \tag{10.1.20}$$

Proof. This follows from (10.1.11), (10.1.13) and (10.1.9). □

Our last result in this section is the following counterpart of Theorem 7.46 at the level of systems.

Theorem 10.5. *Assume that $n\geq 2$, $M\in\mathbb{N}$, and consider the $M\times M$ system*

$$L = \sum_{j,k=1}^{n} A_{jk}\partial_j\partial_k, \qquad A_{jk}\in\mathcal{M}_{M\times M}(\mathbb{C}). \tag{10.1.21}$$

Then for an $M\times M$ matrix-valued function $\mathbb{E}$ whose components are contained in $C^2(\mathbb{R}^n\setminus\{0\})\cap L^1_{\rm loc}(\mathbb{R}^n)$ and have the property that their gradients are positive homogeneous of degree $1-n$ in $\mathbb{R}^n\setminus\{0\}$, the following statements are equivalent:

(1) When viewed in $L^1_{\rm loc}(\mathbb{R}^n)$, the matrix-valued function $\mathbb{E}$ is a fundamental solution for the system L in $\mathbb{R}^n$;

(2) One has $L\mathbb{E}=0$ pointwise in $\mathbb{R}^n\setminus\{0\}$ and

$$\sum_{j,k=1}^{n}\int_{S^{n-1}} \omega_j A_{jk}\partial_k\mathbb{E}(\omega)\,d\sigma(\omega) = I_{M\times M}. \tag{10.1.22}$$

Proof. This is established much as in the scalar case, following the argument in the proof of Theorem 7.46, keeping in mind the matrix algebra formalism. □

10.2 Fundamental Solutions and Regularity for General Systems

Definition 10.6. *Let $\mathbb{K}$ be a field, and let $(\mathfrak{R},+,0)$ be a vector space over $\mathbb{K}$ equipped with an additional binary operation from $\mathfrak{R}\times\mathfrak{R}$ to $\mathfrak{R}$, called product (or multiplication). Then $\mathfrak{R}$ is said to be a* `commutative associative algebra` *over $\mathbb{K}$ if the following identities hold for any three elements $a,b,c\in\mathfrak{R}$, and any two scalars $x,y\in\mathbb{K}$:*

$$\text{commutativity}: \quad ab=ba, \tag{10.2.1}$$

$$\text{associativity}: \quad (ab)c=a(bc), \tag{10.2.2}$$

$$\text{distributivity}: \quad (a+b)c=ac+bc, \tag{10.2.3}$$

$$\text{compatibility with scalars}: \quad (xa)(yb)=(xy)(ab). \tag{10.2.4}$$

Finally, an element $e\in\mathfrak{R}$ is said to be `multiplicative unit` *provided one has $ea=ae=a$ for each $a\in\mathfrak{R}$.*

Let $\mathfrak{R}$ be a commutative associative algebra over a field $\mathbb{K}$, with multiplicative unit e and additive neutral element 0. We make the convention that $a-b:=a+(-b)$ if $a,b\in\mathfrak{R}$ and $-b$ is the inverse of b with respect to the addition operation in the algebra $\mathfrak{R}$. Also, for $j\in\mathbb{N}$, we define $(-1)^j a$ simply a when j is even, and as $-a$ when j is odd.

Given $M\in\mathbb{N}$, we shall let $\mathcal{M}_{M\times M}(\mathfrak{R})$ stand for the collection of $M\times M$ matrices with entries from $\mathfrak{R}$. Then, if $A=\big(a_{jk}\big)_{1\leq j,k\leq M}\in\mathcal{M}_{M\times M}(\mathfrak{R})$ and $c\in\mathfrak{R}$, one defines in the usual fashion the operations

$$A^{\top}:=\big(a_{kj}\big)_{1\leq j,k\leq M}\in\mathcal{M}_{M\times M}(\mathfrak{R}), \tag{10.2.5}$$

$$c\,A:=\big(c\,a_{jk}\big)_{1\leq j,k\leq M}\in\mathcal{M}_{M\times M}(\mathfrak{R}), \tag{10.2.6}$$

$$\det A:=\sum_{\sigma\in\mathcal{P}_M}(-1)^{\operatorname{sgn}\sigma}a_{1\sigma(1)}a_{2\sigma(2)}\cdots a_{M\sigma(M)}\in\mathfrak{R}, \tag{10.2.7}$$

where $\mathcal{P}_M$ is the collection of all permutations of the set $\{1,2,\ldots,M\}$, and the matrix of cofactors

$$\operatorname{adj}(A):=\big((-1)^{j+k}\det\big(A_{jk}\big)\big)_{1\leq j,k\leq M}, \tag{10.2.8}$$

where, for any given $j_0,k_0\in\{1,\ldots,M\}$, the minor $A_{j_0k_0}$ is defined by

$$A_{j_0k_0}:=\big(a_{jk}\big)_{1\leq j,k\leq M,\,j\neq j_0,\,k\neq k_0}\in\mathcal{M}_{(M-1)\times(M-1)}(\mathfrak{R}). \tag{10.2.9}$$

Furthermore, given another matrix $B=\big(b_{jk}\big)_{1\leq j,k\leq M}\in\mathcal{M}_{M\times M}(\mathfrak{R})$, we set

$$A\pm B:=\big(a_{jk}\pm b_{jk}\big)_{1\leq j,k\leq M}\quad\text{and}\quad A\cdot B:=\left(\sum_{r=1}^{M}a_{jr}b_{rk}\right)_{1\leq j,k\leq M}. \tag{10.2.10}$$

A number of basic properties satisfied by these operations are collected in the next proposition.

Proposition 10.7. *Let $\mathfrak{R}$ be a commutative associative algebra over a field $\mathbb{K}$ with multiplicative unit e and additive neutral element 0. Also, let $M \in \mathbb{N}$ be arbitrary. Then the following statements are true:*

(i) $(A \cdot B)^{\top} = B^{\top} \cdot A^{\top}$ *for all* $A, B \in \mathcal{M}_{M\times M}(\mathfrak{R})$;

(ii) *The identity matrix* $I_{M\times M} := \begin{pmatrix} e & 0 & \dots & 0 \\ 0 & e & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & e \end{pmatrix} \in \mathcal{M}_{M\times M}(\mathfrak{R})$ *has the property that* $I_{M\times M} \cdot A = A \cdot I_{M\times M} = A$ *for each* $A \in \mathcal{M}_{M\times M}(\mathfrak{R})$;

(iii) $\det(A^{\top}) = \det A$ *for each* $A \in \mathcal{M}_{M\times M}(\mathfrak{R})$;

(iv) $\det(c\,A) = c^{M} \det A$ *for each* $A \in \mathcal{M}_{M\times M}(\mathfrak{R})$ *and each* $c \in \mathbb{K}$;

(v) *Given any* $A \in \mathcal{M}_{M\times M}(\mathfrak{R})$, *for each* $j, k \in \{1, \dots, M\}$ *we have*

$$\det A = \sum_{\ell=1}^{M} a_{j\ell}(-1)^{j+\ell} \det(A_{j\ell}) = \sum_{\ell=1}^{M} a_{\ell k}(-1)^{\ell+k} \det(A_{\ell k}); \quad (10.2.11)$$

(vi) $\det A = 0$ *whenever* $A \in \mathcal{M}_{M\times M}(\mathfrak{R})$ *has either two identical columns, or two identical rows;*

(vii) $\operatorname{adj}(A^{\top}) \cdot A = A \cdot \operatorname{adj}(A^{\top}) = \big[\det A\big] I_{M\times M}$ *for each* $A \in \mathcal{M}_{M\times M}(\mathfrak{R})$;

(viii) $\det(A \cdot B) = (\det A)(\det B)$ *for all* $A, B \in \mathcal{M}_{M\times M}(\mathfrak{R})$;

(ix) $\operatorname{adj}(A \cdot B) = \operatorname{adj}(A) \cdot \operatorname{adj}(B)$ *for all* $A, B \in \mathcal{M}_{M\times M}(\mathfrak{R})$;

(x) $\mathcal{M}_{M\times M}(\mathfrak{R})$ *is a (in general, noncommutative) ring, with multiplicative unit* $I_{M\times M}$, *and* $A \in \mathcal{M}_{M\times M}(\mathfrak{R})$ *has a multiplicative inverse in* $\mathcal{M}_{M\times M}(\mathfrak{R})$ *if and only if* $\det(A)$ *has a multiplicative inverse in* $\mathfrak{R}$.

Proof. All properties are established much as in the standard case $\mathfrak{R} \equiv \mathbb{C}$. Here we only wish to mention that *(vii)* is a direct consequence of *(v)–(vi)* (complete proofs may be found in, e.g., [33]). □

An example that is relevant for our future considerations pertaining to the Lamé system is as follows. Suppose $\mathfrak{R}$ is a commutative associative algebra over a field $\mathbb{K}$ and that $n \in \mathbb{N}$, $\lambda, \mu \in \mathbb{K}$, with $\mu \neq 0$. Also, fix $a \in \mathfrak{R}^n$ (i.e., $a = (a_1, \dots, a_n)$ with $a_j \in \mathfrak{R}$ for each $j \in \{1, \dots, n\}$), and consider $A \in \mathcal{M}_{n\times n}(\mathfrak{R})$ given by

$$A := \mu\Big(\sum_{j=1}^{n} a_j a_j\Big) I_{n\times n} + (\lambda + \mu) a \otimes a, \quad (10.2.12)$$

where $a\otimes a\in\mathcal{M}_{n\times n}(\mathfrak{R})$ is defined as $a\otimes a:=\big(a_ja_k\big)_{1\leq j,k\leq n}$. Then a direct calculation (compare with Exercise 10.13) shows that

$$\det A=\mu^{n-1}(\lambda+2\mu)\bigg(\sum_{j=1}^{n}a_ja_j\bigg)^n, \tag{10.2.13}$$

and

$$\operatorname{adj}(A)=\mu^{n-2}\bigg(\sum_{j=1}^{n}a_ja_j\bigg)^{n-2}\bigg[(\lambda+2\mu)\bigg(\sum_{j=1}^{n}a_ja_j\bigg)I_{n\times n}-(\lambda+\mu)a\otimes a\bigg]. \tag{10.2.14}$$

Our main interest in the algebraic framework developed so far in this section lies with the particular case when, for some fixed $n\in\mathbb{N}$,

$$\mathfrak{R}:=\bigg\{\sum_{\alpha\in\mathbb{N}_0^n,\,|\alpha|\leq m}a_\alpha\partial^\alpha:\ a_\alpha\in\mathbb{C},\quad m\in\mathbb{N}_0\bigg\}, \tag{10.2.15}$$

with the natural operations of addition and multiplication, and with the convention that $\partial^{(0,\dots,0)}=1$. Then clearly $\mathfrak{R}$ is a commutative associative algebra over $\mathbb{C}$ with multiplicative unit 1 and, for each $M\in\mathbb{N}$, the set $\mathcal{M}_{M\times M}(\mathfrak{R})$ consists of all $M\times M$ systems of constant, complex coefficient differential operators. Thus, if $M\in\mathbb{N}$, $m\in\mathbb{N}_0$, are fixed an $M\times M$ system of constant, complex coefficient differential operators of degree m has the form $L(\partial)\in\mathcal{M}_{M\times M}(\mathfrak{R})$ withalign

$$L(\partial):=\bigg(\sum_{|\alpha|\leq m}a_\alpha^{jk}\partial^\alpha\bigg)_{1\leq j,k\leq M}. \tag{10.2.16}$$

We shall call $L(\partial)$ a homogeneous (linear) system (of order m) if $a_\alpha^{jk}=0$ whenever $|\alpha|<m$ and $j,k\in\{1,\dots,M\}$.

According to (10.2.5), the *algebraic* transpose of $L(\partial)$ from (10.2.16) in $\mathcal{M}_{M\times M}(\mathfrak{R})$ is given by

$$L(\partial)^\top=\bigg(\sum_{|\alpha|\leq m}a_\alpha^{kj}\partial^\alpha\bigg)_{1\leq j,k\leq M}. \tag{10.2.17}$$

Next, for each $L(\partial)$ as in (10.2.16) set

$$D_L(\partial):=\det(L(\partial)) \tag{10.2.18}$$

and notice that $D_L(\partial)$ is a scalar, constant (complex) coefficient, differential operator of order $\leq Mm$.

Consequently, statement *(vii)* in Proposition 10.7, (10.2.17), and (10.2.18) readily give that

$$L(\partial)\cdot \operatorname{adj}\left[L(\partial)^{\top}\right] = D_L(\partial) I_{M\times M}, \qquad \forall\, L(\partial)\in \mathcal{M}_{M\times M}(\mathfrak{R}). \tag{10.2.19}$$

Let us also observe that, at the level of symbols,

$$D_L(\xi) = \det(L(\xi)) \quad \text{for each} \quad \xi\in\mathbb{R}^n \tag{10.2.20}$$

and that

$$\begin{aligned} &D_L(\partial) \text{ is a homogeneous scalar operator} \\ &\text{whenever } L(\partial) \text{ is a homogeneous system.} \end{aligned} \tag{10.2.21}$$

In particular, from (10.2.20) and (10.2.21) we deduce that

$$\left.\begin{aligned} &\text{if } L(\partial) \text{ is a homogeneous system with} \\ &\det(L(\xi))\neq 0 \quad \text{for each} \quad \xi\in\mathbb{R}^n\setminus\{0\} \end{aligned}\right\} \Longrightarrow \; D_L(\partial) \text{ is elliptic.} \tag{10.2.22}$$

Our goal is to use (10.2.19) as a link between systems and scalar partial differential operators. We shall prove two results based on this scheme, the first of which is a procedure to reduce the task of finding a fundamental solution for a given system to the case of scalar operators. Specifically, we have the following proposition.

Theorem 10.8. *Let $L(\partial)$ be an $M\times M$ system with constant coefficients as in (10.2.16), and let $D_L(\partial)$ be the scalar differential operator associated with L as in (10.2.18). Then if $E\in\mathcal{D}'(\mathbb{R}^n)$ is a fundamental solution for $D_L(\partial)$ in $\mathbb{R}^n$, it follows that*

$$\mathbb{E} := \operatorname{adj}\left[L(\partial)^{\top}\right]\big(EI_{M\times M}\big) \in \mathcal{M}_{M\times M}(\mathcal{D}'(\mathbb{R}^n)) \tag{10.2.23}$$

is a fundamental solution for the system $L(\partial)$ in $\mathbb{R}^n$. Moreover,

$$E\in\mathcal{S}'(\mathbb{R}^n) \Longrightarrow \mathbb{E}\in\mathcal{M}_{M\times M}(\mathcal{S}'(\mathbb{R}^n)). \tag{10.2.24}$$

In particular, if $D_L(\partial)$ is not identically zero, then $L(\partial)$ has a fundamental solution that is a tempered distribution.

Proof. Let E be a fundamental solution for $D_L(\partial)$ and define $\mathbb{E}$ as in (10.2.23). That $\mathbb{E}$ is a fundamental solution for $L(\partial)$ follows from (10.2.19). Also it is clear that $E\in\mathcal{S}'(\mathbb{R}^n)$ forces $\mathbb{E}\in\mathcal{M}_{M\times M}(\mathcal{S}'(\mathbb{R}^n))$. As for the last claim in the statement of the theorem, note that if $D_L(\partial)$ is not identically zero, then Theorem 5.13 ensures the existence of a fundamental solution $E\in\mathcal{S}'(\mathbb{R}^n)$ for $D_L(\partial)$ which, in turn, yields a solution for $L(\partial)$ via the recipe (10.2.23). □

Next we record the following consequence of (10.2.19) and (10.2.22). As a preamble, recall the notion of singular support from (10.1.3).

Theorem 10.9. *Assume that $L = L(\partial)$ is an $M \times M$ homogeneous differential system in $\mathbb{R}^n$, with constant complex coefficients, such that*

$$\det\big[L(\xi)\big] \neq 0, \qquad \forall\, \xi \in \mathbb{R}^n \setminus \{0\}. \tag{10.2.25}$$

Then, for each open set $\Omega \subseteq \mathbb{R}^n$ and for each $\mathbf{u} \in \big[\mathcal{D}'(\Omega)\big]^M$ one has

$$\operatorname{sing\,supp} \mathbf{u} \subseteq \operatorname{sing\,supp}(L\mathbf{u}). \tag{10.2.26}$$

In particular, if $\mathbf{u} \in \big[\mathcal{D}'(\Omega)\big]^M$ is such that $L\mathbf{u} \in \big[C^\infty(\Omega)\big]^M$ then $\mathbf{u} \in \big[C^\infty(\Omega)\big]^M$.

Proof. Let L be a differential system as in the statement, and set $D_L(\partial) := \det[L(\partial)]$. Then (10.2.22) ensures that $D_L(\partial)$ is an elliptic, scalar differential operator, with constant coefficients. Next, fix $\mathbf{u} = (u_1, \ldots, u_M) \in \big[\mathcal{D}'(\Omega)\big]^M$ and let $\omega \subseteq \Omega$ be an open set such that $(L\mathbf{u})\big|_\omega \in C^\infty(\omega)$. Then, for each $j \in \{1, \ldots, M\}$ there holds

$$\big(D_L(\partial)u_j\big)\big|_\omega = \big(\operatorname{adj}\big[L(\partial)^\top\big]L(\partial)\mathbf{u}\big)_j\Big|_\omega = \big(\operatorname{adj}\big[L(\partial)^\top\big]L(\partial)\mathbf{u}\big|_\omega\big)_j \tag{10.2.27}$$

from which it follows that $D_L(\partial)u_j \in C^\infty(\omega)$ for each $j \in \{1, \ldots, M\}$. From this and Corollary 6.18 it follows that $u_j\big|_\omega \in C^\infty(\omega)$ for each $j \in \{1, \ldots, M\}$. Consequently $\omega \subseteq \Omega \setminus \operatorname{sing\,supp} \mathbf{u}$, from which (10.2.26) immediately follows. □

Corollary 10.10. *Let $L = L(\partial)$ be an $M \times M$ homogeneous differential system in $\mathbb{R}^n$, with constant complex coefficients, such that $\det[L(\xi)] \neq 0$ for all ξ in $\mathbb{R}^n \setminus \{0\}$. Then L has a fundamental solution $\mathbb{E} \in \mathcal{M}_{M\times M}(\mathcal{S}'(\mathbb{R}^n))$ with $\operatorname{sing\,supp} \mathbb{E} = \{0\}$.*

Proof. This is an immediate consequence of Theorem 10.8 and Theorem 10.9. □

Exercise 10.11. *Suppose $M \in \mathbb{N}$ and consider an $M \times M$ homogeneous second order system with constant complex coefficients $L(\partial) = \Big(\sum_{r,s=1}^{n} a_{rs}^{jk}\partial_r\partial_s\Big)_{1\leq j,k\leq M}$ in $\mathbb{R}^n$. Assume that there exists $c \in (0, \infty)$ such that this system satisfies the Legendre–Hadamard ellipticity condition*

$$\operatorname{Re}\big[\big(L(\xi)\eta\big) \cdot \overline{\eta}\big] \geq c|\xi|^2|\eta|^2, \qquad \forall\, \xi \in \mathbb{R}^n,\ \ \forall\, \eta \in \mathbb{C}^M. \tag{10.2.28}$$

Then, $|\det[L(\xi)]| \geq c^M|\xi|^{2M}$ for every $\xi \in \mathbb{R}^n$. In particular, $\det[L(\xi)] \neq 0$ for every $\xi \in \mathbb{R}^n \setminus \{0\}$.

Sketch of proof: Show that

$$A \in \mathcal{M}_{M\times M}(\mathbb{C}),\ \ |A\eta| \geq c|\eta|,\ \ \forall\, \eta \in \mathbb{C}^M \implies |\det A| \geq c^M. \tag{10.2.29}$$

This may be proved by considering the self-adjoint matrix $B := A^*A$ (where $A^* := (\overline{A}\,)^\top$ is the adjoint of A) that satisfies $(B\eta)\cdot\overline{\eta} = |A\eta|^2 \geq c^2|\eta|^2$ for every $\eta \in \mathbb{C}^M$. Use the latter and the fact that B is diagonalizable to conclude that $|\det A|^2 = \det B \in [c^{2M}, \infty)$. To finish the proof of the exercise, apply (10.2.29) with $A := L(\xi)$, $\xi \in \mathbb{R}^n$.

10.3 Fundamental Solutions for the Lamé Operator

The Lamé operator L in $\mathbb{R}^n$ is a differential operator that acts on vector distributions. Specifically, if $\mathbf{u} \in [\mathcal{D}'(\mathbb{R}^n)]^n$, i.e., $\mathbf{u} = (u_1, \ldots, u_n)$ where $u_k \in \mathcal{D}'(\mathbb{R}^n)$, $k = 1, \ldots, n$, then the action of the Lamé operator on $\mathbf{u}$ is defined by

$$\begin{aligned} L\mathbf{u} &:= \mu\Delta\mathbf{u} + (\lambda+\mu)\nabla\mathrm{div}\,\mathbf{u} \\ &= \left(\mu\Delta u_j + (\lambda+\mu)\sum_{\ell=1}^{n}\partial_j\partial_\ell u_\ell\right)_{1\leq j\leq n} \in \big[\mathcal{D}'(\mathbb{R}^n)\big]^n, \end{aligned} \tag{10.3.1}$$

where the constants $\lambda, \mu \in \mathbb{C}$ (typically called Lamé moduli) are assumed to satisfy

$$\mu \neq 0 \quad \text{and} \quad 2\mu + \lambda \neq 0. \tag{10.3.2}$$

To study in greater detail the structure of the Lamé operator, we need to discuss some useful algebraic formalism.

Definition 10.12. *Given, $a = (a_1, \ldots, a_n) \in \mathbb{C}^n$, $b = (b_1, \ldots, b_n) \in \mathbb{C}^n$, define $a \otimes b$ to be the matrix*

$$a \otimes b := \big(a_j b_k\big)_{1\leq j,k\leq n} \in \mathcal{M}_{n\times n}(\mathbb{C}). \tag{10.3.3}$$

Exercise 10.13. *Prove that:*

(1) $(a\otimes b)^\top = b\otimes a$ *for all* $a, b \in \mathbb{C}^n$;

(2) $\mathrm{Tr}\,(a\otimes b) = a\cdot b$ *for all* $a, b \in \mathbb{C}^n$;

(3) $(a\otimes b)c = (b\cdot c)a$ *for all* $a, b, c \in \mathbb{C}^n$;

(4) $\det\,(I_{n\times n} + a\otimes b) = 1 + a\cdot b$ *for all* $a, b \in \mathbb{C}^n$;

(5) $(a\otimes b)\,(c\otimes d) = (b\cdot c)\,a\otimes d$ *for all* $a, b, c, d \in \mathbb{C}^n$;

(6) for every $a \in \mathbb{R}^n$ *and every numbers* $\mu, \lambda \in \mathbb{C}$ *the matrix* $\mu I_{n\times n} + \lambda\, a\otimes a$ *is invertible if and only if* $\mu \neq 0$ *and* $\mu \neq -\lambda|a|^2$; *moreover, whenever* $\mu \neq 0$ *and* $\mu \neq -\lambda|a|^2$,

$$\big(\mu I_{n\times n} + \lambda\, a\otimes a\big)^{-1} = \frac{1}{\mu}\left[I_{n\times n} - \left(\frac{\lambda}{\mu+\lambda|a|^2}\right) a\otimes a\right]. \tag{10.3.4}$$

Hint for (4): Fix $a = (a_1, \dots, a_n) \in \mathbb{C}^n$ and $b = (b_1, \dots, b_n) \in \mathbb{C}^n$. By continuity, it suffices to prove the formula in *(4)* when $a_j \neq 0$ and $b_j \neq 0$ for every $j \in \{1, \dots, n\}$. Assuming that this is the case, we may write

$$\begin{aligned}
\det\left(I_{n\times n} + a \otimes b\right) &= \det \begin{bmatrix} 1+a_1b_1 & a_1b_2 & \cdots & a_1b_n \\ a_2b_1 & 1+a_2b_2 & \cdots & a_2b_n \\ \cdots & \cdots & \cdots & \cdots \\ a_nb_1 & a_nb_2 & \cdots & 1+a_nb_n \end{bmatrix} \\
&= \Big(\prod_{j=1}^{n} a_j\Big) \det \begin{bmatrix} \frac{1}{a_1}+b_1 & b_2 & \cdots & b_n \\ b_1 & \frac{1}{a_2}+b_2 & \cdots & b_n \\ \cdots & \cdots & \cdots & \cdots \\ b_1 & b_2 & \cdots & \frac{1}{a_n}+b_n \end{bmatrix} \\
&= \Big(\prod_{j=1}^{n} a_j\Big)\Big(\prod_{j=1}^{n} b_j\Big) \det \begin{bmatrix} \frac{1}{a_1b_1}+1 & 1 & \cdots & 1 \\ 1 & \frac{1}{a_2b_2}+1 & \cdots & 1 \\ \cdots & \cdots & \cdots & \cdots \\ 1 & 1 & \cdots & \frac{1}{a_nb_n}+1 \end{bmatrix} \\
&= \Big(\prod_{j=1}^{n} a_j\Big)\Big(\prod_{j=1}^{n} b_j\Big)\Big(\prod_{j=1}^{n} \frac{1}{a_jb_j}\Big)\Big(1+\sum_{j=1}^{n} a_jb_j\Big) \\
&= 1 + \sum_{j=1}^{n} a_jb_j = 1 + a \cdot b. \qquad (10.3.5)
\end{aligned}$$

The format of the Lamé system (10.3.1) suggests the following definition and result, shedding light on the conditions imposed on the Lamé moduli in (10.3.2).

Proposition 10.14. *Given any Lamé moduli $\lambda, \mu \in \mathbb{C}$, define the characteristic matrix for the Lamé system (10.3.1) as*

$$L(\xi) := \mu|\xi|^2 I_{n\times n} + (\lambda+\mu)\xi \otimes \xi, \qquad \forall\, \xi \in \mathbb{R}^n. \qquad (10.3.6)$$

Then the following statements are equivalent:

(1) $L(\xi)$ is invertible for every $\xi \in \mathbb{R}^n \setminus \{0\}$;

(2) $L(\xi)$ is invertible for some $\xi \in \mathbb{R}^n \setminus \{0\}$;

(3) $\mu \neq 0$ and $\lambda + 2\mu \neq 0$.

Moreover, if $\mu \neq 0$ and $\lambda + 2\mu \neq 0$, then for each $\xi \in \mathbb{R}^n \setminus \{0\}$ one has

$$\big[L(\xi)\big]^{-1} = \frac{1}{\mu|\xi|^2}\left[I_{n\times n} - \frac{\lambda+\mu}{\lambda+2\mu}\frac{\xi}{|\xi|} \otimes \frac{\xi}{|\xi|}\right]. \qquad (10.3.7)$$

Proof. This is a direct consequence of Exercise 10.13. □

Now we are ready to tackle the issue of finding all fundamental solutions for the Lamé operator (10.3.1) when the Lamé moduli satisfy (10.3.2). In this context, according to Definition 10.1, a fundamental solution for the Lamé operator is a matrix distribution $\mathbb{E} \in \mathcal{M}_{n\times n}(\mathcal{D}'(\mathbb{R}^n))$ with the property that $L\mathbb{E} = \delta I_{n\times n}$ in $\mathcal{M}_{M\times M}\big(\mathcal{D}'(\mathbb{R}^n)\big)$. It follows that the columns $\mathbf{E}_k$, $k = 1, \ldots, n$, of $\mathbb{E}$ satisfy the equations

$$L\mathbf{E}_k = \delta\, \mathbf{e}_k \quad \text{in} \quad \big[\mathcal{D}'(\mathbb{R}^n)\big]^n \quad \text{for} \quad k = 1, \ldots, n, \tag{10.3.8}$$

where $\mathbf{e}_k$ denotes the unit vector in $\mathbb{R}^n$ with 1 on the kth entry, for each $k \in \{1, \ldots, n\}$.

Our goal is to determine, under the standing assumption $n \geq 3$, all the fundamental solutions for the Lamé operator with entries in $\mathcal{S}'(\mathbb{R}^n)$. We do so by relying on the tools developed for scalar operators. To get started, suppose that there exists some matrix $\mathbb{E} = \big(E_{jk}\big)_{1\leq j,k\leq n} \in \mathcal{M}_{n\times n}(\mathcal{S}'(\mathbb{R}^n))$ with columns that satisfy (10.3.8). Since $\delta \in \mathcal{S}'(\mathbb{R}^n)$ and (4.1.25) is true, using (10.3.1), the latter is equivalent to

$$\mu\Delta E_{jk} + (\lambda+\mu)\sum_{\ell=1}^{n} \partial_j\partial_\ell E_{\ell k} = \delta_{jk}\delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \quad j,k \in \{1, \ldots, n\}. \tag{10.3.9}$$

Applying the Fourier transform to each equation in (10.3.9), we further write

$$-\mu|\xi|^2\,\widehat{E_{jk}}(\xi) - (\lambda+\mu)\xi_j \sum_{\ell=1}^{n} \xi_\ell \widehat{E_{\ell k}}(\xi) = \delta_{jk} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{10.3.10}$$

for each $j, k \in \{1, \ldots, n\}$. For k arbitrary, fixed, multiply (10.3.10) with ξ_j and then sum up the resulting identities over $j = 1, \ldots, n$ to obtain that

$$-\mu|\xi|^2 \sum_{j=1}^{n} \xi_j \widehat{E_{jk}}(\xi) - (\lambda+\mu)\bigg(\sum_{j=1}^{n} \xi_j^2\bigg) \sum_{\ell=1}^{n} \xi_\ell \widehat{E_{\ell k}}(\xi) = \xi_k \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{10.3.11}$$

for all $k \in \{1, \ldots, n\}$, or equivalently, that

$$-(\lambda+2\mu)|\xi|^2 \sum_{j=1}^{n} \xi_j \widehat{E_{jk}}(\xi) = \xi_k \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \quad k \in \{1, \ldots, n\}. \tag{10.3.12}$$

If we now multiply (10.3.10) by $(\lambda+2\mu)|\xi|^2$ and make use of (10.3.12), we may conclude that, for each $j, k \in \{1, \ldots, n\}$,

$$-\mu(\lambda+2\mu)|\xi|^4 \widehat{E_{jk}}(\xi) = (\lambda+2\mu)|\xi|^2\delta_{jk} - (\lambda+\mu)\xi_j\xi_k \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{10.3.13}$$

To proceed fix $j, k \in \{1, \ldots, n\}$ and note that since $n \geq 3$, by Exercise 4.4 we have $\frac{1}{|\xi|^2} \in \mathcal{S}'(\mathbb{R}^n)$. Also, $\frac{\xi_j\xi_k}{|\xi|^4} \in L^1_{\rm loc}(\mathbb{R}^n)$ and in view of Example 4.3 one may infer that $\frac{\xi_j\xi_k}{|\xi|^4} \in \mathcal{S}'(\mathbb{R}^n)$. In addition, $|\xi|^4 \in \mathcal{L}(\mathbb{R}^n)$, thus $|\xi|^4 \cdot \frac{\xi_j\xi_k}{|\xi|^4}$,

$|\xi|^4\cdot\frac{1}{|\xi|^2}\in\mathcal{S}'(\mathbb{R}^n)$ (recall *(b)* in Theorem 4.13), and it is not difficult to check that $|\xi|^4\cdot\frac{\xi_j\xi_k}{|\xi|^4}=\xi_j\xi_k$ and $|\xi|^4\cdot\frac{1}{|\xi|^2}=|\xi|^2$ in $\mathcal{S}'(\mathbb{R}^n)$. These conclusions combined with (10.3.13) imply (recall also (10.3.2))

$$\mu(\lambda+2\mu)|\xi|^4\left[\widehat{E_{jk}}(\xi)+\frac{\delta_{jk}}{\mu}\cdot\frac{1}{|\xi|^2}-\frac{(\lambda+\mu)}{\mu(\lambda+2\mu)}\cdot\frac{\xi_j\xi_k}{|\xi|^4}\right]=0\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n). \tag{10.3.14}$$

Thus, by Proposition 5.1 applied with $P(D):=\Delta^2$, it follows that

$$\widehat{E_{jk}}(\xi)+\frac{\delta_{jk}}{\mu}\cdot\frac{1}{|\xi|^2}-\frac{(\lambda+\mu)}{\mu(\lambda+2\mu)}\cdot\frac{\xi_j\xi_k}{|\xi|^4}=\widehat{P_{jk}}(\xi)\quad\text{in }\mathcal{S}'(\mathbb{R}^n), \tag{10.3.15}$$

where P_{jk} is a polynomial in $\mathbb{R}^n$ satisfying $\Delta^2P_{jk}=0$ pointwise in $\mathbb{R}^n$. To continue with the computation of E_{jk} we apply the inverse Fourier transform to (10.3.15) and use Proposition 4.61 with $\lambda=2$ as well as (7.10.7) to write

$$\begin{aligned}
E_{jk}&=-\frac{\delta_{jk}}{\mu}\mathcal{F}^{-1}\left(\frac{1}{|\xi|^2}\right)+\frac{(\lambda+\mu)}{\mu(\lambda+2\mu)}\mathcal{F}^{-1}\left(\frac{\xi_j\xi_k}{|\xi|^4}\right)+P_{jk}\\
&=-\frac{1}{\mu(n-2)\omega_{n-1}}\cdot\frac{\delta_{jk}}{|x|^{n-2}}+P_{jk}(x)\\
&\quad+\frac{(\lambda+\mu)}{\mu(\lambda+2\mu)}\left[\frac{1}{2(n-2)\omega_{n-1}}\cdot\frac{\delta_{jk}}{|x|^{n-2}}-\frac{1}{2\omega_{n-1}}\cdot\frac{x_jx_k}{|x|^n}\right]\\
&=\left[\frac{(\lambda+\mu)}{\mu(\lambda+2\mu)}\cdot\frac{1}{2(n-2)\omega_{n-1}}-\frac{1}{\mu(n-2)\omega_{n-1}}\right]\cdot\frac{\delta_{jk}}{|x|^{n-2}}\\
&\quad-\frac{(\lambda+\mu)}{2\omega_{n-1}\mu(\lambda+2\mu)}\cdot\frac{x_jx_k}{|x|^n}+P_{jk}(x)\\
&=\frac{-1}{2\mu(2\mu+\lambda)\omega_{n-1}}\left[\frac{3\mu+\lambda}{n-2}\frac{\delta_{jk}}{|x|^{n-2}}+\frac{(\mu+\lambda)x_jx_k}{|x|^n}\right]+P_{jk}(x)
\end{aligned}\tag{10.3.16}$$

in $\mathcal{S}'(\mathbb{R}^n)$ (hence, in particular, for all $x\in\mathbb{R}^n\setminus\{0\}$).

Next, we claim that the matrix $\mathbb{F}=\big(F_{jk}\big)_{1\leq j,k\leq n}\in\mathcal{M}_{n\times n}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ with entries defined by

$$F_{jk}:=\frac{-1}{2\mu(2\mu+\lambda)\omega_{n-1}}\left[\frac{3\mu+\lambda}{n-2}\frac{\delta_{jk}}{|x|^{n-2}}+\frac{(\mu+\lambda)x_jx_k}{|x|^n}\right], \tag{10.3.17}$$

for $j,k\in\{1,\dots,n\}$, is a fundamental solution for the Lamé operator. Note that, based on the properties of the Fourier transform, this claim is equivalent with having the entries of $\mathbb{F}$ satisfy

$$-\mu|\xi|^2\,\widehat{F_{jk}}(\xi)-(\lambda+\mu)\xi_j\sum_{\ell=1}^{n}\xi_\ell\widehat{F_{\ell k}}(\xi)=\delta_{jk}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n) \tag{10.3.18}$$

for each $j,k\in\{1,\dots,n\}$. To check (10.3.18) we use (10.3.17), Proposition 4.61 (with $\lambda=n-2$) and (7.10.8) to first write

$$\begin{aligned}\widehat{F_{jk}}(\xi)&=\frac{-1}{2\mu(2\mu+\lambda)\omega_{n-1}}\left[\frac{(3\mu+\lambda)\delta_{jk}}{n-2}\mathcal{F}\left(\frac{1}{|x|^{n-2}}\right)+(\mu+\lambda)\mathcal{F}\left(\frac{x_jx_k}{|x|^n}\right)\right]\\&=\frac{-(3\mu+\lambda)\delta_{jk}}{2\mu(2\mu+\lambda)\omega_{n-1}(n-2)}\cdot\frac{(n-2)\omega_{n-1}}{|\xi|^2}\\&\quad-\frac{\mu+\lambda}{2\mu(2\mu+\lambda)\omega_{n-1}}\left[\omega_{n-1}\frac{\delta_{jk}}{|\xi|^2}-2\omega_{n-1}\frac{\xi_j\xi_k}{|\xi|^4}\right]\\&=-\frac{\delta_{jk}}{\mu}\cdot\frac{1}{|\xi|^2}+\frac{(\lambda+\mu)}{\mu(\lambda+2\mu)}\cdot\frac{\xi_j\xi_k}{|\xi|^4}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n).\end{aligned}\tag{10.3.19}$$

Next, we use (10.3.19) to rewrite the term in the left-hand side of (10.3.18) as

$$\begin{aligned}&-\mu|\xi|^2\widehat{F_{jk}}(\xi)-(\lambda+\mu)\xi_j\sum_{\ell=1}^{n}\xi_\ell\widehat{F_{\ell k}}(\xi)\\&=\mu|\xi|^2\left[\frac{\delta_{jk}}{\mu|\xi|^2}-\frac{(\lambda+\mu)\xi_j\xi_k}{\mu(\lambda+2\mu)|\xi|^4}\right]+(\lambda+\mu)\xi_j\sum_{\ell=1}^{n}\xi_\ell\left[\frac{\delta_{\ell k}}{\mu|\xi|^2}-\frac{(\lambda+\mu)\xi_\ell\xi_k}{\mu(\lambda+2\mu)|\xi|^4}\right]\\&=\delta_{jk}-\frac{(\lambda+\mu)\xi_j\xi_k}{(\lambda+2\mu)|\xi|^2}+(\lambda+\mu)\xi_j\left[\frac{\xi_k}{\mu|\xi|^2}-\frac{(\lambda+\mu)\xi_k}{\mu(\lambda+2\mu)|\xi|^2}\right]\\&=\delta_{jk}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^n),\end{aligned}\tag{10.3.20}$$

proving that the matrix $\mathbb{F}$ is a fundamental solution for the Lamé operator.

The main result emerging from this analysis is summarized next.

Theorem 10.15. *Assume $n\geq 3$ and let L be the Lamé operator from (10.3.1) such that the constants the $\lambda,\mu\in\mathbb{C}$ satisfy (10.3.2). Then the matrix $\mathbb{E}=\big(E_{jk}\big)_{1\leq j,k\leq n}$, which has entries given by the $L^1_{\rm loc}(\mathbb{R}^n)$ functions*

$$E_{jk}(x):=\frac{-1}{2\mu(2\mu+\lambda)\omega_{n-1}}\left[\frac{3\mu+\lambda}{n-2}\frac{\delta_{jk}}{|x|^{n-2}}+\frac{(\mu+\lambda)x_jx_k}{|x|^n}\right]\tag{10.3.21}$$

where $x\in\mathbb{R}^n\setminus\{0\}$, for each $j,k\in\{1,\dots,n\}$, belongs to $\mathcal{M}_{n\times n}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ and is a fundamental solution for the Lamé operator in $\mathbb{R}^n$.

In addition, any fundamental solution $\mathbb{U}\in\mathcal{M}_{n\times n}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ of the Lamé operator in $\mathbb{R}^n$ is of the form $\mathbb{U}=\mathbb{E}+\mathbb{P}$, for some matrix $\mathbb{P}:=\big(P_{jk}\big)_{1\leq j,k\leq n}$ whose entries are polynomials in $\mathbb{R}^n$ and whose columns $\mathbf{P}_k$, $k=1,\dots,n$, satisfy the pointwise equations $L\mathbf{P}_k=(0,\dots,0)\in\mathbb{C}^n$ in $\mathbb{R}^n$ for $k=1,\dots,n$.

Proof. Since the entries of $\mathbb{E}$ as given in (10.3.21) are the same as the expressions from (10.3.17), the earlier analysis shows that $\mathbb{E}$ belongs to $\mathcal{M}_{n\times n}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ and

is a fundamental solution for the Lamé operator in $\mathbb{R}^n$. To justify the claim in the last paragraph of the statement of the theorem we shall invoke Proposition 10.2. Concretely, since condition (10.3.2) and Proposition 10.14 imply $\det(L(\xi)) \neq 0$ for $\xi \in \mathbb{R}^n \setminus \{0\}$, it follows that $\det(L(i\xi)) = -\det(L(\xi)) \neq 0$ for $\xi \in \mathbb{R}^n \setminus \{0\}$. This shows that (10.1.15) is satisfied, hence Proposition 10.2 applies and yields the desired conclusion. □

Note that, as Lemma 10.17 shows, if $\mathbb{P} = \big(P_{jk}\big)_{1\leq j,k\leq n}$ is as in Theorem 10.15, then $\Delta^2 P_{jk} = 0$ pointwise in $\mathbb{R}^n$ for every $j, k \in \{1, \ldots, n\}$.

One may check without using of the Fourier transform that the matrix from Theorem 10.15 is a fundamental solution for the Lamé operator (much as in the spirit of Remark 7.4).

Exercise 10.16. *Follow the outline below to check, without the use of the Fourier transform, that the matrix* $\mathbb{E} = \big(E_{jk}\big)_{1\leq j,k\leq n} \in \mathcal{M}_{n\times n}(\mathcal{S}'(\mathbb{R}^n))$, *with entries of function type defined by the functions from* (10.3.21), *is a fundamental solution for the Lamé operator in* $\mathbb{R}^n$, $n \geq 3$.

Step 1. Show that $\mu\Delta E_{jk} + (\lambda+\mu)\sum_{\ell=1}^{n} \partial_j\partial_\ell E_{\ell k} = 0$ pointwise in $\mathbb{R}^n \setminus \{0\}$ for each $j, k \in \{1, \ldots, n\}$.

Step 2. Show that the desired conclusion (i.e., that the given E is a fundamental solution for the Lamé operator in $\mathbb{R}^n$, $n \geq 3$) is equivalent with the condition thateqnarray

$$\lim_{\varepsilon\to 0^+}\left[\mu\int_{|x|\geq\varepsilon} E_{jk}(x)\Delta\varphi(x)\,dx + (\lambda+\mu)\int_{|x|\geq\varepsilon}\sum_{\ell=1}^{n} E_{\ell k}(x)\partial_j\partial_\ell\varphi(x)\,dx\right] = \varphi(0)\delta_{jk} \tag{10.3.22}$$

for every $j, k \in \{1, \ldots, n\}$ and every $\varphi \in C_0^\infty(\mathbb{R}^n)$.

Step 3. Fix $j, k \in \{1, \ldots, n\}$ and a function $\varphi \in C_0^\infty(\mathbb{R}^n)$. Let $R \in (0, \infty)$ be such that $\operatorname{supp}\varphi \subseteq B(0, R)$, so that one may replace the domain of integration for the integrals in the left-hand side of (10.3.22) with the set $\{x \in \mathbb{R}^n : \varepsilon < |x| < R\}$. Use this domain of integration, (13.7.5), (13.7.4), and the result from Step 1, to prove that (10.3.22) is equivalent to

$$\begin{aligned}\lim_{\varepsilon\to 0^+}\Bigg[&-\mu\int_{\partial B(0,\varepsilon)} E_{jk}(x)\frac{\partial\varphi}{\partial\nu}(x)\,d\sigma(x) + \mu\int_{\partial B(0,\varepsilon)}\varphi(x)\frac{\partial E_{jk}}{\partial\nu}(x)\,d\sigma(x)\\ &-(\lambda+\mu)\int_{\partial B(0,\varepsilon)}\Big[\sum_{\ell=1}^{n} E_{\ell k}(x)\partial_\ell\varphi(x)\Big]\nu_j(x)\,d\sigma(x)\\ &+(\lambda+\mu)\int_{\partial B(0,\varepsilon)}\Big[\sum_{\ell=1}^{n}\nu_\ell(x)\partial_j E_{\ell k}(x)\Big]\varphi(x)\,d\sigma(x)\Bigg] = \varphi(0)\delta_{jk},\end{aligned} \tag{10.3.23}$$

where $\nu(x) = \frac{x}{\varepsilon}$ for each $x \in \partial B(0, \varepsilon)$.

Step 4. Prove that there exists a constant $C \in (0,\infty)$ independent of ε such that each of the quantities:

$$\left|\int_{\partial B(0,\varepsilon)} E_{jk}(x)\frac{\partial\varphi}{\partial\nu}(x)\,d\sigma(x)\right|, \qquad \left|\int_{\partial B(0,\varepsilon)} \frac{\partial E_{jk}}{\partial\nu}(x)[\varphi(x)-\varphi(0)]\,d\sigma(x)\right|,$$

$$\left|\int_{\partial B(0,\varepsilon)} \left[\sum_{\ell=1}^{n} E_{\ell k}(x)\partial_\ell\varphi(x)\right]\nu_j(x)\,d\sigma(x)\right|, \quad \text{and}$$

$$\left|\int_{\partial B(0,\varepsilon)} \left[\sum_{\ell=1}^{n} \nu_\ell(x)\partial_j E_{\ell k}(x)\right][\varphi(x)-\varphi(0)]\,d\sigma(x)\right|, \tag{10.3.24}$$

is bounded by $C\|\nabla\varphi\|_{L^\infty(\mathbb{R}^n)}\,\varepsilon$, thus convergent to zero as $\varepsilon\to 0^+$.

Step 5. Combine Steps 2–4 to reduce matters to proving that

$$\lim_{\varepsilon\to 0^+}\Big[\mu \int_{\partial B(0,\varepsilon)} \sum_{s=1}^{n}\frac{x_s}{\varepsilon}(\partial_s E_{jk})(x)\,d\sigma(x) \tag{10.3.25}$$

$$+(\lambda+\mu)\int_{\partial B(0,\varepsilon)} \sum_{\ell=1}^{n}\frac{x_\ell}{\varepsilon}(\partial_j E_{\ell k})(x)\,d\sigma(x)\Big] = \delta_{jk}.$$

Step 6. Prove that for every $x\in\partial B(0,\varepsilon)$ we have

$$\sum_{s=1}^{n}\frac{x_s}{\varepsilon}(\partial_s E_{jk})(x) = \frac{(3\mu+\lambda)\delta_{jk}}{2\mu(2\mu+\lambda)\,\omega_{n-1}\varepsilon^{n-1}} - \frac{(\mu+\lambda)(2-n)x_j x_k}{2\mu(2\mu+\lambda)\,\omega_{n-1}\varepsilon^{n+1}} \tag{10.3.26}$$

and

$$\sum_{\ell=1}^{n}\frac{x_\ell}{\varepsilon}(\partial_j E_{\ell k})(x) = \frac{(3\mu+\lambda)-(\mu+\lambda)(1-n)}{2\mu(2\mu+\lambda)\,\omega_{n-1}\varepsilon^{n+1}}x_j x_k$$

$$-\frac{(\lambda+\mu)\delta_{jk}}{2\mu(2\mu+\lambda)\,\omega_{n-1}\varepsilon^{n-1}}. \tag{10.3.27}$$

Step 7. Using the fact that (cf. (13.8.44))

$$\int_{\partial B(0,\varepsilon)} x_j x_k\,d\sigma(x) = \frac{\varepsilon^{n+1}\omega_{n-1}}{n}\delta_{jk}, \tag{10.3.28}$$

integrate the expressions in (10.3.26)–(10.3.27) to conclude that

$$\int_{\partial B(0,\varepsilon)} \sum_{s=1}^{n} \frac{x_s}{\varepsilon} (\partial_s E_{jk})(x)\, d\sigma(x) = \left[3\mu + \lambda - \frac{(\mu+\lambda)(2-n)}{n} \right] \frac{\delta_{jk}}{2\mu(2\mu+\lambda)} \tag{10.3.29}$$

and

$$\int_{\partial B(x,\varepsilon)} \sum_{\ell=1}^{n} \frac{x_\ell}{\varepsilon} (\partial_j E_{\ell k})(x)\, d\sigma(x) = \frac{\delta_{jk}}{n(2\mu+\lambda)}, \tag{10.3.30}$$

then finish the proof of (10.3.25).

10.4 Mean Value Formulas and Interior Estimates for the Lamé Operator

The goal here is to prove that solutions of the Lamé system satisfy certain mean value formulas similar in spirit to those for harmonic functions. We start by establishing a few properties of elastic vector fields.

Lemma 10.17. *Let $\lambda, \mu \in \mathbb{C}$ be such that $\mu \neq 0$ and $\lambda + 2\mu \neq 0$. Assume that the vector distribution $\mathbf{u} = (u_1, u_2, \ldots, u_n) \in \big[\mathcal{D}'(\Omega)\big]^n$ satisfies the Lamé system*

$$\mu \Delta \mathbf{u} + (\lambda + \mu) \nabla \operatorname{div} \mathbf{u} = \mathbf{0} \quad \text{in } \big[\mathcal{D}'(\Omega)\big]^n. \tag{10.4.1}$$

Then $\mathbf{u} \in \big[C^\infty(\Omega)\big]^n$ and the following statements are true.

(i) The function $\operatorname{div} \mathbf{u}$ is harmonic in Ω.

(ii) The function u_j is biharmonic in Ω for each $j = 1, \ldots, n$.

(iii) The function $\partial_j(\operatorname{div} \mathbf{u})$ is harmonic in Ω for each $j = 1, \ldots, n$.

Proof. The fact that $\mathbf{u} \in \big[C^\infty(\Omega)\big]^n$ is a consequence of Theorem 10.9, Proposition 10.14, and our assumptions on λ, μ. To show *(i)*, we apply div to (10.4.1). Since $\operatorname{div}\nabla = \Delta$ and $\operatorname{div}\Delta = \Delta \operatorname{div}$ we obtain $(\lambda+2\mu)\Delta \operatorname{div} \mathbf{u} = 0$ in Ω, thus *(i)* follows since $\lambda + 2\mu \neq 0$. Now if we return to (10.4.1) and apply Δ to it, the second term will be zero because of the harmonicity of $\operatorname{div} \mathbf{u}$. Consequently, $\mu \Delta^2 \mathbf{u} = 0$ in Ω which proves *(ii)* since $\mu \neq 0$. Finally, applying Δ to (10.4.1) and using *(ii)* yields $(\lambda + \mu)\Delta(\nabla \operatorname{div} \mathbf{u}) = 0$, and *(iii)* follows from this in the case $\lambda + \mu \neq 0$. If the latter condition fails, then from (10.4.1) we have that u is harmonic, so *(iii)* also holds in this case. □

As an auxiliary step in the direction of Theorem 10.19 that contains the mean value formulas alluded to earlier, we prove the following useful result.

Proposition 10.18. *Let $\lambda, \mu \in \mathbb{C}$ be such that $\mu \neq 0$ and $\lambda + 2\mu \neq 0$. Assume $\mathbf{u} = (u_1, u_2, \ldots, u_n) \in [\mathcal{D}'(\Omega)]^n$ satisfies the Lamé system (10.4.1). Then we have $\mathbf{u} \in [C^\infty(\Omega)]^n$ and, for every $x \in \Omega$, every $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, and every $j \in \{1, \ldots, n\}$, the following formulas hold:*

$$u_j(x) + \frac{(\mu-\lambda)r^2}{2\mu(n+2)}\partial_j(\operatorname{div}\mathbf{u})(x) = \frac{n}{r^2}\fint_{\partial B(x,r)}(y_j - x_j)(y-x)\cdot\mathbf{u}(y)\,d\sigma(y), \tag{10.4.2}$$

and

$$\begin{aligned} u_j(x) - \frac{[(n+3)\mu + (n+1)\lambda]r^2}{2\mu(n+2)(n-1)}\,\partial_j(\operatorname{div}\mathbf{u})(x) \qquad\qquad (10.4.3)\\ = -\frac{n}{(n-1)r^2}\fint_{\partial B(x,r)}\big[(y_j - x_j)(y-x)\cdot\mathbf{u}(y) - r^2 u_j(y)\big]\,d\sigma(y). \end{aligned}$$

Proof. The fact that $\mathbf{u} \in [C^\infty(\Omega)]^n$ follows from Lemma 10.17. To proceed, fix some $x \in \Omega$, $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, and $j \in \{1, \ldots, n\}$. By (iii) in Lemma 10.17, we have that $\partial_j(\operatorname{div}\mathbf{u})(x)$ is harmonic in Ω. Thus, we may apply the mean value formula on solid balls for harmonic function (c.f. the first formula in (7.2.9)), followed by an application of the integration by parts formula (13.7.4), to write

$$\begin{aligned} \partial_j(\operatorname{div}\mathbf{u})(x) &= \frac{n}{\omega_{n-1}r^n}\int_{B(x,r)}\partial_j(\operatorname{div}\mathbf{u})(y)\,dy \\ &= \frac{n}{\omega_{n-1}r^n}\int_{\partial B(x,r)}\frac{y_j - x_j}{r}\operatorname{div}\mathbf{u}(y)\,d\sigma(y) \\ &= \frac{n}{\omega_{n-1}r^{n+1}}\int_{\partial B(x,r)}\operatorname{div}\big[(y_j - x_j)\mathbf{u}(y)\big]\,d\sigma(y) \\ &\quad - \frac{n}{\omega_{n-1}r^{n+1}}\int_{\partial B(x,r)}u_j(y)\,d\sigma(y) =: I + II. \end{aligned} \tag{10.4.4}$$

Since u_j is biharmonic (recall (ii) in Lemma 10.17), we may write (7.4.1) for u_j, then use the latter formula to simplify II, and then replace Δu_j by $-\frac{\lambda+\mu}{\mu}\,\partial_j(\operatorname{div}\mathbf{u})$ (recall that $\mathbf{u}$ satisfies the Lamé system) to obtain

$$II = -\frac{n}{r^2}\,u_j(x) - \frac{1}{2}\,\Delta u_j(x) = -\frac{n}{r^2}\,u_j(x) + \frac{\lambda+\mu}{2\mu}\,\partial_j(\operatorname{div}\mathbf{u})(x). \tag{10.4.5}$$

Combining (10.4.4) and (10.4.5) we see that

$$\begin{aligned} &\frac{\mu-\lambda}{2\mu}\,r^{n+1}\,\partial_j(\operatorname{div}\mathbf{u})(x) + nr^{n-1}u_j(x) \\ &\qquad = \frac{n}{\omega_{n-1}}\int_{\partial B(x,r)}\operatorname{div}\big[(y_j - x_j)\mathbf{u}(y)\big]\,d\sigma(y). \end{aligned} \tag{10.4.6}$$

Fix $R \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$. Integrating (10.4.6) with respect to r for $r \in (0, R)$, applying (13.8.5) and then (13.7.4), we arrive at

$$\begin{aligned} & R^n u_j(x) + \frac{\mu-\lambda}{2\mu(n+2)} R^{n+2}\, \partial_j(\operatorname{div}\mathbf{u})(x) \\ &\quad = \frac{n}{\omega_{n-1}} \int_{B(x,R)} \operatorname{div}\big((y_j - x_j)\mathbf{u}(y)\big)\, dy \\ &\quad = \frac{n}{\omega_{n-1}} \int_{\partial B(x,R)} (y_j - x_j) \frac{y-x}{R} \cdot \mathbf{u}(y)\, d\sigma(y) \end{aligned} \tag{10.4.7}$$

which in turn yields (10.4.2) (with R in place of r) after dividing by R^n.

To prove (10.4.3) we start with the term in the right-hand side of formula (10.4.2) in which we add and subtract $r^2 u_j(y)$ under the integral sign and then split it in two integrals one involving $(y_j - x_j)(y-x)\cdot\mathbf{u}(y) - r^2 u_j(y)$, call this expression I_1, and another one involving $u_j(y)$, call it I_2. For I_2 we recall that u_j is biharmonic and use (7.4.1) after which we replace $\Delta u_j(x)$ by $-\frac{\lambda+\mu}{\mu}\,\partial_j(\operatorname{div}\mathbf{u})(x)$. Hence,

$$\begin{aligned} & \frac{n}{\omega_{n-1} r^{n+1}} \int_{\partial B(x,r)} (y_j - x_j)(y-x)\cdot\mathbf{u}(y)\, d\sigma(y) \\ &\quad = \frac{n}{\omega_{n-1} r^{n+1}} \int_{\partial B(x,r)} \big[(y_j - x_j)(y-x)\cdot\mathbf{u}(y) - r^2 u_j(y)\big]\, d\sigma(y) \\ &\qquad + n u_j(x) - \frac{r^2}{2}\cdot\frac{\lambda+\mu}{\mu}\partial_j(\operatorname{div}\mathbf{u})(x). \end{aligned} \tag{10.4.8}$$

Finally, (10.4.3) follows by adding (10.4.2) and (10.4.8). □

As mentioned before, Proposition 10.18 is an important ingredient in proving the following mean value formulas for the Lamé system.

Theorem 10.19. *Let $\lambda, \mu \in \mathbb{C}$ be such that $\mu \neq 0$, $\lambda + 2\mu \neq 0$, and $(n+1)\mu + \lambda \neq 0$. Assume $\mathbf{u} \in [\mathcal{D}'(\Omega)]^n$ satisfies the Lamé system (10.4.1). Then $\mathbf{u} \in [C^\infty(\Omega)]^n$, and for every $x \in \Omega$ and every $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$ the following formulas hold:*

$$\begin{aligned} \mathbf{u}(x) &= \frac{n(\lambda+\mu)(n+2)}{2[(n+1)\mu+\lambda]r^2} \fint_{\partial B(x,r)} \big[(y-x)\cdot\mathbf{u}(y)\big](y-x)\, d\sigma(y) \\ &\quad + \frac{n(\mu-\lambda)}{2[(n+1)\mu+\lambda]} \fint_{\partial B(x,r)} \mathbf{u}(y)\, d\sigma(y) \end{aligned} \tag{10.4.9}$$

and

$$\begin{aligned} \mathbf{u}(x) &= \frac{n(\lambda+\mu)(n+2)}{2[(n+1)\mu+\lambda]} \fint_{B(x,r)} \left[\frac{y-x}{|y-x|}\cdot\mathbf{u}(y)\right]\frac{y-x}{|y-x|}\, dy \\ &\quad + \frac{n(\mu-\lambda)}{2[(n+1)\mu+\lambda]} \fint_{B(x,r)} \mathbf{u}(y)\, dy. \end{aligned} \tag{10.4.10}$$

Proof. Once again, that $\mathbf{u}\in[C^\infty(\Omega)]^n$ is contained in Lemma 10.17. Formula (10.4.9) follows by taking a suitable linear combination of (10.4.2) and (10.4.3) so that $\partial_j(\operatorname{div}\mathbf{u})$ cancels (here $(n+1)\mu+\lambda\neq 0$ is used). To prove (10.4.10), multiply (10.4.9) by $\omega_{n-1}r^{n-1}$, then move $\frac{1}{r^2}$ from the front of the first integral in the right-hand side of the equality obtained inside of that integral as $\frac{1}{|y-x|^2}$, then integrate with respect to $r\in(0,R)$, where $R\in(0,\operatorname{dist}(x,\partial\Omega))$ and use (13.8.5). Finally divide the very last expression by $\omega_{n-1}R^n/n$ which is precisely the volume of $B(x,R)$. This gives (10.4.10) written with R in place of r. □

We shall now discuss how the mean value formulas from Theorem 10.19 can be used to obtain interior estimates for solutions of the Lamé system. Two such versions are proved in Theorems 10.20 and 10.22 (cf. also Exercise 10.21).

Theorem 10.20 (L^1-Interior estimates for the Lamé operator)**.** *Let $\lambda,\mu\in\mathbb{C}$ be such that $\mu\neq 0$, $\lambda+2\mu\neq 0$, and $(n+1)\mu+\lambda\neq 0$. Assume $\mathbf{u}\in[\mathcal{D}'(\Omega)]^n$ is a vector distribution satisfying the Lamé system (10.4.1). Then $\mathbf{u}\in[C^\infty(\Omega)]^n$ and there exists $C\in(0,\infty)$ depending only on n, λ, and μ, such that for every $x\in\Omega$, every $r\in\big(0,\operatorname{dist}(x,\partial\Omega)\big)$ and each $k\in\{1,\dots,n\}$, one has*

$$|\partial_k\mathbf{u}(x)|\leq\frac{C}{r}\fint_{B(x,r)}|\mathbf{u}(y)|\,dy. \tag{10.4.11}$$

Proof. The fact that $\mathbf{u}\in[C^\infty(\Omega)]^n$ has been established in Lemma 10.17. Fix $k\in\{1,\dots,n\}$ and observe that since $\mathbf{u}=(u_1,\dots,u_n)$ satisfies the Lamé system in Ω, then so does $\partial_k\mathbf{u}$. Fix $x^*\in\Omega$ arbitrary and select $R\in\big(0,\operatorname{dist}(x^*,\partial\Omega)\big)$. Writing formula (10.4.10) for $\mathbf{u}$ replaced by $\partial_k\mathbf{u}$ and r replaced by $R/2$, shows that for each $j\in\{1,\dots,n\}$,

$$\begin{aligned}\partial_k u_j(x^*)&=\frac{c_1 2^n n}{\omega_{n-1}R^n}\int_{B(x^*,\,R/2)}\left(\frac{y-x^*}{|y-x^*|}\cdot\partial_k\mathbf{u}(y)\right)\frac{y_j-x_j^*}{|y-x^*|}\,dy\\&\quad+\frac{c_2\,2^n n}{\omega_{n-1}R^n}\int_{B(x^*,\,R/2)}\partial_k u_j(y)\,dy,\end{aligned} \tag{10.4.12}$$

where

$$c_1:=\frac{n(\lambda+\mu)(n+2)}{2[(n+1)\mu+\lambda]}\quad\text{and}\quad c_2:=\frac{n(\mu-\lambda)}{2[(n+1)\mu+\lambda]}. \tag{10.4.13}$$

Integrating by parts [using (13.7.4)] in both integrals in (10.4.12) yields

$$\partial_k u_j(x^*)=-\frac{c_1 2^n n}{\omega_{n-1}R^n}\int\limits_{B(x^*,\,R/2)}\sum_{\ell=1}^n\partial_{y_k}\left[\frac{(y_j-x_j^*)(y_\ell-x_\ell^*)}{|y-x^*|^2}\right]u_\ell(y)\,dy \tag{10.4.14}$$

$$+\frac{c_1 2^n n}{\omega_{n-1}R^n}\int\limits_{\partial B(x^*,\,R/2)}\frac{y_k-x_k^*}{|y-x^*|}\sum_{\ell=1}^n\frac{(y_j-x_j^*)(y_\ell-x_\ell^*)}{|y-x^*|^2}\,u_\ell(y)\,d\sigma(y)$$

$$+\frac{c_2\,2^n n}{\omega_{n-1}R^n}\int\limits_{\partial B(x^*,\,R/2)}\frac{y_k-x_k^*}{|y-x^*|}\,u_j(y)\,d\sigma(y).$$

Thus,

$$|\partial_k\mathbf{u}(x^*)|\leq CR^{-n}\int\limits_{B(x^*,\,R/2)}\frac{|\mathbf{u}(y)|}{|y-x^*|}\,dy+CR^{-n}\int\limits_{\partial B(x^*,\,R/2)}|\mathbf{u}(y)|\,d\sigma(y),\quad(10.4.15)$$

where C stands for a finite positive constant depending only on n, λ, and μ. Before continuing let us note a useful consequence of (10.4.10). Specifically, for every $x\in\Omega$ and every $r\in\big(0,\operatorname{dist}(x,\partial\Omega)\big)$ one has

$$|\mathbf{u}(x)|\leq C\fint_{B(x,r)}|\mathbf{u}(z)|\,dz.\quad(10.4.16)$$

Taking $y\in\overline{B(x^*,\,R/2)}$ forces $B(y,R/2)\subset B(x^*,\,R)$ which, when used in estimate (10.4.16) written for x replaced by y and r by $R/2$, gives

$$|\mathbf{u}(y)|\leq C\fint_{B(y,R/2)}|\mathbf{u}(z)|\,dz\leq C\fint_{B(x^*,\,R)}|\mathbf{u}(z)|\,dz,\quad\forall\,y\in\overline{B(x^*,\,R/2)}.\quad(10.4.17)$$

This, in turn, allows us to estimate the integrals in (10.4.15) as follows. For the boundary integral, (10.4.17) yields

$$\begin{aligned}\int\limits_{\partial B(x^*,\,R/2)}|\mathbf{u}(y)|\,d\sigma(y)&\leq C\bigg(\fint_{B(x^*,\,R)}|\mathbf{u}(z)|\,dz\bigg)R^{n-1}\\&=\frac{C}{R}\int_{B(x^*,\,R)}|\mathbf{u}(z)|\,dz,\end{aligned}\quad(10.4.18)$$

while for the solid integral

$$\begin{aligned}\int\limits_{B(x^*,\,R/2)}\frac{|\mathbf{u}(y)|}{|y-x^*|}\,dy&\leq C\bigg(\fint_{B(x^*,\,R)}|\mathbf{u}(z)|\,dz\bigg)\bigg(\int_{B(x^*,\,R/2)}\frac{dy}{|y-x^*|}\bigg)\\&\leq\frac{C}{R}\int_{B(x^*,\,R)}|\mathbf{u}(z)|\,dz,\end{aligned}\quad(10.4.19)$$

since $\int_{B(x^*,\,R/2)}\frac{dy}{|y-x^*|}=C\int_0^{\frac{R}{2}}\rho^{n-2}\,d\rho=CR^{n-1}$. Now (10.4.11) (with r replaced by R) follows from (10.4.15), (10.4.18), and (10.4.19). □

Exercise 10.21. *Under the same background assumptions as in Theorem 10.20, prove that for every $\alpha\in\mathbb{N}_0^n$ there exists a finite constant $C_\alpha>0$ that depends only on n,α,λ,μ, with the property that for every $x\in\Omega$ and every $r\in\big(0,\operatorname{dist}(x,\partial\Omega)\big)$*

$$|\partial^\alpha\mathbf{u}(x)|\leq\frac{C_\alpha}{r^{|\alpha|}}\fint_{B(x,r)}|\mathbf{u}(y)|\,dy.\quad(10.4.20)$$

Hint: Use induction on $|\alpha|$, (10.4.11) written for $r/2$, and the fact that $\partial^\alpha \mathbf{u}$ continues to be a solution of the Lamé system (10.4.1)

Theorem 10.22 (L^∞-Interior estimates for the Lamé operator)**.** *Let $\lambda, \mu \in \mathbb{C}$ be such that $\mu \neq 0$, $\lambda + 2\mu \neq 0$, and $(n+1)\mu + \lambda \neq 0$ and suppose $\mathbf{u} \in [\mathcal{D}'(\Omega)]^n$ satisfies the Lamé system (10.4.1). Then $\mathbf{u} \in [C^\infty(\Omega)]^n$ and, with C as in Theorem 10.20, for each $x \in \Omega$, each $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, and each $k \in \mathbb{N}$, we have*

$$|\partial^\alpha \mathbf{u}(x)| \leq \frac{C^k e^{k-1} k!}{r^k} \max_{y \in \overline{B(x,r)}} |\mathbf{u}(y)|, \qquad \forall\, \alpha \in \mathbb{N}_0^n \quad \text{with} \quad |\alpha| = k. \tag{10.4.21}$$

Proof. The fact that $\mathbf{u} \in [C^\infty(\Omega)]^n$ follows from Lemma 10.17. In particular, this implies that $\partial^\alpha \mathbf{u}$ satisfies (10.4.1) for every $\alpha \in \mathbb{N}_0^n$. The case $k = 1$ is an immediate consequence of (10.4.11) since, clearly, $⨍_{B(x,r)} |\mathbf{u}(y)|\, dy \leq \max_{y \in \overline{B(x,r)}} |\mathbf{u}(y)|$. Having established this, the desired conclusion follows by invoking Lemma 6.21 (with $\mathcal{A}$ the class of null-solutions for the Lamé system in Ω). □

Recall Definition 6.22.

Theorem 10.23. *Suppose λ and μ are complex numbers such that $\mu \neq 0$, $\lambda + 2\mu \neq 0$, and $(n+1)\mu + \lambda \neq 0$. Then any null solution of the Lamé system (10.4.1) has components that are real-analytic in Ω.*

Proof. This is an immediate consequence of Theorem 10.22 and Lemma 6.24. □

Theorem 10.24. *Suppose λ and μ are complex numbers such that $\mu \neq 0$, $\lambda + 2\mu \neq 0$, and $(n+1)\mu + \lambda \neq 0$ and suppose $\Omega \subseteq \mathbb{R}^n$ is open and connected. If $\mathbf{u}$ is a null solution of the Lamé system (10.4.1) such that for some $x_0 \in \Omega$ we have $\partial^\alpha \mathbf{u}(x_0) = 0$ for all $\alpha \in \mathbb{N}_0^n$, then $\mathbf{u} = 0$ in Ω.*

Proof. This follows from Theorems 10.23 and 6.25. □

Next we record the analogue of the classical Liouville's theorem for the Laplacian (cf. Theorem 7.15) in the case of the Lamé system.

Theorem 10.25 (Liouville's theorem for the Lamé system)**.** *Let $\lambda, \mu \in \mathbb{C}$ be such that $\mu \neq 0$, $\lambda + 2\mu \neq 0$ and suppose $\mathbf{u} \in [L^\infty(\mathbb{R}^n)]^n$ satisfies the Lamé system*

$$\mu \Delta \mathbf{u} + (\lambda + \mu) \nabla \operatorname{div} \mathbf{u} = \mathbf{0} \quad \text{in} \quad \big[\mathcal{D}'(\mathbb{R}^n)\big]^n. \tag{10.4.22}$$

Then there exists a constant vector $\mathbf{c} \in \mathbb{C}^n$ such that $\mathbf{u}(x) = \mathbf{c}$ for all $x \in \mathbb{R}^n$.

Proof. This is a particular case of Theorem 10.3 since based on the current assumptions on λ and μ, Proposition 10.14 ensures that condition (10.1.15) is satisfied. □

Exercise 10.26. *Assuming that $\lambda, \mu \in \mathbb{C}$ are such that $\mu \neq 0$, $\lambda + 2\mu \neq 0$, and $(n+1)\mu + \lambda \neq 0$, give an alternative proof of Theorem 10.25 by relying on the interior estimates from (10.4.11).*

10.5 The Poisson Equation for the Lamé Operator

Let L be the operator from (10.3.1) with coefficients $\mu, \lambda \in \mathbb{C}$ satisfying (10.3.2). Then the Poisson equation for the Lamé operator in an open subset Ω of $\mathbb{R}^n$ reads

$$L\mathbf{u} = \mathbf{f}, \tag{10.5.1}$$

where the vector $\mathbf{f}$ is given and the vector $\mathbf{u}$ is the unknown. If $\mathbf{u}$ is a priori known to be of class C^2, then the equality in (10.5.1) is considered in the pointwise sense, everywhere in $\mathbb{R}^n$. Such a solution is called classical. Often, one starts with $\mathbf{u}$ simply a vector distribution in which scenario (10.5.1) is interpreted in $\big[\mathcal{D}'(\Omega)\big]^n$. In this case, we shall refer to $\mathbf{u}$ as a distributional solution. In this vein, it is worth pointing out that if the datum $\mathbf{f}$ is of class C^∞ in Ω then any distributional solution of (10.5.1) is also of class C^∞ in Ω, as seen from Theorem 10.9, Proposition 10.14, and (10.3.2).

A key ingredient in solving the Poisson equation (10.5.1) is going to be the fundamental solution for the Lamé system derived in (10.3.21).

Proposition 10.27. *Let L be the Lamé operator from (10.3.1) such that (10.3.2) is satisfied. Assume $n \geq 2$ and let $\mathbb{E} = \big(E_{jk}\big)_{1\leq j,k\leq n}$ be the fundamental solution for L in $\mathbb{R}^n$ with entries as in (10.3.21) for $n \geq 3$ and as in (10.7.1) for $n = 2$. Suppose that Ω is an open set in $\mathbb{R}^n$ and that $\mathbf{f} \in [L^\infty(\Omega)]^n$ vanishes outside a bounded subset of Ω. Then*

$$\mathbf{u}(x) = \int_\Omega \mathbb{E}(x-y)\mathbf{f}(y)\,dy, \qquad \forall\, x \in \Omega, \tag{10.5.2}$$

is a distributional solution of the Poisson equation for the Lamé system in Ω, that is, $L\mathbf{u} = \mathbf{f}$ in $[\mathcal{D}'(\Omega)]^n$. In addition, $\mathbf{u} \in [C^1(\Omega)]^n$.

Proof. This is established by arguing along the lines of the proof of Proposition 7.8. □

The main result in this section is the following well-posedness result for the Poisson problem for the Lamé operator in $\mathbb{R}^n$.

Theorem 10.28. *Assume $n \geq 3$, and let L be the Lamé operator from (10.3.1) with $\lambda, \mu \in \mathbb{C}$ satisfying $\mu \neq 0$ and $\lambda + 2\mu \neq 0$. Also, suppose $\mathbf{f} \in \big[L^\infty_{\rm comp}(\mathbb{R}^n)\big]^n$ and $\mathbf{c} \in \mathbb{C}^n$ are given. Then the Poisson problem for the Lamé operator in $\mathbb{R}^n$,*

$$\begin{cases} \mathbf{u} \in [C^0(\mathbb{R}^n)]^n, \\ L\mathbf{u} = \mathbf{f} \quad \text{in} \quad \big[\mathcal{D}'(\mathbb{R}^n)\big]^n, \\ \lim\limits_{|x|\to\infty} \mathbf{u}(x) = \mathbf{c}, \end{cases} \tag{10.5.3}$$

has a unique solution. Moreover, the solution $\mathbf{u}$ *satisfies the following additional properties.*

(1) The function $\mathbf{u}$ *is of class* C^1 *in* $\mathbb{R}^n$ *and admits the integral representation formula*

$$\mathbf{u}(x)=\mathbf{c}+\int_{\mathbb{R}^n}\mathbb{E}(x-y)\mathbf{f}(y)\,dy,\qquad\forall\,x\in\mathbb{R}^n,\tag{10.5.4}$$

where $\mathbb{E}=\big(E_{jk}\big)_{1\leq j,k\leq n}$ *is the fundamental solution for* L *in* $\mathbb{R}^n$ *with entries as in* (10.3.21). *Moreover, for each* $j\in\{1,\dots,n\}$ *we have*

$$\partial_j\mathbf{u}(x)=\int_{\mathbb{R}^n}(\partial_j\mathbb{E})(x-y)\mathbf{f}(y)\,dy,\qquad\forall\,x\in\mathbb{R}^n.\tag{10.5.5}$$

(2) If in fact $\mathbf{f}\in\big[C_0^\infty(\mathbb{R}^n)\big]^n$ *then* $\mathbf{u}\in\big[C^\infty(\mathbb{R}^n)\big]^n$.

(3) For every $j,k\in\{1,\dots,n\}$ *there exists a matrix* $C_{jk}\in\mathcal{M}_{n\times n}(\mathbb{C})$ *such that*

$$\partial_j\partial_k\mathbf{u}=C_{jk}\,\mathbf{f}+T_{\partial_j\partial_k\mathbb{E}}\,\mathbf{f}\quad\text{in}\quad\big[\mathcal{D}'(\mathbb{R}^n)\big]^n,\tag{10.5.6}$$

where $T_{\partial_j\partial_k\mathbb{E}}$ *is the singular integral operator associated with the matrix-valued function* $\Theta:=\partial_j\partial_k\mathbb{E}$ *(cf. Definition 4.90).*

(4) For every $p\in(1,\infty)$, *the solution* $\mathbf{u}$ *of* (10.5.3) *satisfies* $\partial_j\partial_k\mathbf{u}\in\big[L^p(\mathbb{R}^n)\big]^n$ *for each* $j,k\in\{1,\dots,n\}$, *where the derivatives are taken in* $\big[\mathcal{D}'(\mathbb{R}^n)\big]^n$. *Moreover, there exists a constant* $C=C(p,n)\in(0,\infty)$ *with the property that*

$$\sum_{j,k=1}^{n}\|\partial_j\partial_k\mathbf{u}\|_{[L^p(\mathbb{R}^n)]^n}\leq C\|\mathbf{f}\|_{[L^p(\mathbb{R}^n)]^n}.\tag{10.5.7}$$

Proof. From Proposition 10.27 we have that $\mathbf{u}$ defined as in (10.5.4) is of class C^1 in $\mathbb{R}^n$ and satisfies $L\mathbf{u}=\mathbf{f}$ in $\big[\mathcal{D}'(\mathbb{R}^n)\big]^n$. In addition, by reasoning as in the proof of Proposition 7.8 we also see that formula (10.5.5) holds for each $j\in\{1,\dots,n\}$. Furthermore, the same type of estimate as in (7.2.21) proves that the function $\mathbf{u}$ from (10.5.4) also satisfies the limit condition in (10.5.3). This concludes the treatment of the existence. As for uniqueness, suppose $\mathbf{u}$ is a solution of (10.5.3) for $\mathbf{f}=\mathbf{0}\in\mathbb{C}^n$. Given that $\mathbf{u}$ satisfies (10.5.3), it follows that $\mathbf{u}$ is bounded in $\mathbb{R}^n$. As such, Liouville's theorem (cf. Theorem 10.25) applies and gives that $\mathbf{u}=\mathbf{0}$. This proves that $\mathbf{u}$ defined as in (10.5.4) is the unique solution of (10.5.3). Next, the regularity result in part *(2)* may be seen either directly from (10.5.4), or by relying on Theorem 10.9, Proposition 10.14, and the assumptions on the Lamé moduli λ,μ. This concludes the proof of the claims made in parts *(1)–(2)*.

Consider now the claim made in part *(3)*. Fix $j,k\in\{1,\dots,n\}$ arbitrary. Then, as seen from (10.3.21), the function $\Phi:=\partial_k\mathbb{E}$ is C^∞ and positive homogeneous of degree $1-n$ in $\mathbb{R}^n\setminus\{0\}$. In turn, this implies that the matrix-valued

function $\Theta := \partial_j\partial_k\mathbb{E}$ has entries satisfying the conditions in (4.4.1) (here the discussion in Example 4.68 is relevant). From what we have proved in part *(1)* and Theorem 4.99 we then conclude that, in the sense of $\big[\mathcal{D}'(\mathbb{R}^n)\big]^n$,

$$\partial_j\partial_k\mathbf{u}(x) = \partial_j\Big[\partial_k\mathbf{u}(x)\Big] = \partial_j\bigg[\int_{\mathbb{R}^n}(\partial_k\mathbb{E})(x-y)\mathbf{f}(y)\,\mathrm{d}y\bigg] \tag{10.5.8}$$

$$= \bigg(\int_{S^{n-1}}(\partial_k\mathbb{E})(\omega)\omega_j\,\mathrm{d}\sigma(\omega)\bigg)\mathbf{f}(x) + \lim_{\varepsilon\to 0^+}\int_{|x-y|>\varepsilon}(\partial_j\partial_k\mathbb{E})(x-y)\mathbf{f}(y)\mathrm{d}y.$$

Upon taking $C_{jk} := \int_{S^{n-1}}(\partial_k\mathbb{E})(\omega)\omega_j\,\mathrm{d}\sigma(\omega) \in \mathcal{M}_{n\times n}(\mathbb{C})$, formula (10.5.6) follows, finishing the proof of part *(3)*. Lastly, the claim in *(4)* is a consequence of *(3)* and the boundedness of the singular integral operators $T_{\partial_j\partial_k\mathbb{E}}$ on $\big[L^p(\mathbb{R}^n)\big]^n$ (as seen by applying Theorem 4.97 componentwise). □

10.6 Fundamental Solutions for the Stokes Operator

Let $\mathbf{u} = (u_1,\ldots,u_n) \in \big[\mathcal{D}'(\mathbb{R}^n)\big]^n$ and $p \in \mathcal{D}'(\mathbb{R}^n)$. Then the Stokes operator L_S acting on $(\mathbf{u},p) = (u_1,\ldots,u_n,p)$ is defined by

$$L_S(\mathbf{u},p) := \Big(\Delta u_1 - \partial_1 p,\ldots,\Delta u_n - \partial_n p\,,\sum_{s=1}^n\partial_s u_s\Big) \in \big[\mathcal{D}'(\mathbb{R}^n)\big]^{n+1}. \tag{10.6.1}$$

In practice, $\mathbf{u}$ and p are referred to as the velocity field and the pressure function, respectively.

A fundamental solution for the Stokes operator is given by a pair $(\mathbb{E},\mathbf{p})$, where $\mathbb{E} = \big(E_{jk}\big)_{1\le j,k\le n} \in \mathcal{M}_{n\times n}(\mathcal{D}'(\mathbb{R}^n))$, $\mathbf{p} = (p_1,\ldots,p_n) \in \big[\mathcal{D}'(\mathbb{R}^n)\big]^n$, satisfy the following conditions. If for each $k \in \{1,\ldots,n\}$ we set $\mathbf{\Gamma}_k := \big(E_{1k},\ldots,E_{nk},p_k\big)$ and $\mathbf{e}_k^{n+1}$ denotes the unit vector in $\mathbb{R}^{n+1}$ with 1 on the kth entry, then

$$L_S\mathbf{\Gamma}_k = \delta\,\mathbf{e}_k^{n+1} \ \text{ in } \ \big[\mathcal{D}'(\mathbb{R}^n)\big]^{n+1} \ \text{ for } \ k=1,\ldots,n. \tag{10.6.2}$$

We propose to determine all the fundamental solutions for the Stokes operator with entries in $\mathcal{S}'(\mathbb{R}^n)$ in the case when $n \ge 3$, a condition assumed in this section unless otherwise specified. Suppose $(\mathbb{E},\mathbf{p})$ is a fundamental solution for L_S with the property that $\mathbb{E} = \big(E_{jk}\big)_{1\le j,k\le n} \in \mathcal{M}_{n\times n}(\mathcal{S}'(\mathbb{R}^n))$ and $\mathbf{p} = (p_1,\ldots,p_n) \in \big[\mathcal{S}'(\mathbb{R}^n)\big]^n$. Then (10.6.2) implies

$$\Delta E_{jk} - \partial_j p_k = \delta_{jk}\delta \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n), \quad j,k\in\{1,\ldots,n\}, \tag{10.6.3}$$

$$\sum_{s=1}^n \partial_s E_{sk} = 0 \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n), \quad k\in\{1,\ldots,n\}. \tag{10.6.4}$$

Apply the Fourier transform to each of the equalities in (10.6.3) and (10.6.4) to obtain

$$-|\xi|^2\widehat{E_{jk}} - i\xi_j\widehat{p_k} = \delta_{jk} \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n), \quad j,k\in\{1,\dots,n\}, \tag{10.6.5}$$

$$\sum_{s=1}^{n}\xi_s\widehat{E_{sk}} = 0 \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n), \quad k\in\{1,\dots,n\}. \tag{10.6.6}$$

Fix $k\in\{1,\dots,n\}$ and, for each $j=1,\dots,n$, multiply the identity in (10.6.5) corresponding to this j with ξ_j, then sum up over j and use (10.6.6) to arrive at

$$|\xi|^2\widehat{p_k} = i\xi_k \quad \text{ in } \quad \mathcal{S}'(\mathbb{R}^n). \tag{10.6.7}$$

Reasoning as in the derivation of (10.3.15) from (10.3.13), the last identity implies

$$\widehat{p_k} = i\frac{\xi_k}{|\xi|^2} + \widehat{r_k}(\xi) \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n), \tag{10.6.8}$$

for some harmonic polynomial r_k in $\mathbb{R}^n$. An application of the inverse Fourier transform to (10.6.8) combined with (7.10.5) then yields

$$p_k = i\mathcal{F}^{-1}\left(\frac{\xi_k}{|\xi|^2}\right) + r_k = -\frac{1}{\omega_{n-1}}\cdot\frac{x_k}{|x|^n} + r_k \quad \text{ in } \ \mathcal{S}'(\mathbb{R}^n). \tag{10.6.9}$$

In particular, if we take $p_k = -\frac{1}{\omega_{n-1}}\cdot\frac{x_k}{|x|^n}$ and use it to substitute for $\widehat{p_k}$ back in (10.6.5), we arrive at the condition

$$|\xi|^2\widehat{E_{jk}} = \frac{\xi_j\xi_k}{|\xi|^2} - \delta_{jk} \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n), \quad j\in\{1,\dots,n\}. \tag{10.6.10}$$

Consequently, since $n\geq 3$, by reasoning as in the derivation of (10.3.15) from (10.3.13), identity (10.6.10) implies

$$\widehat{E_{jk}} = \frac{\xi_j\xi_k}{|\xi|^4} - \frac{\delta_{jk}}{|\xi|^2} + \widehat{R_{jk}} \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n), \quad j\in\{1,\dots,n\}, \tag{10.6.11}$$

for some polynomials R_{jk} in $\mathbb{R}^n$ satisfying $\Delta R_{jk}=0$ in $\mathbb{R}^n$, $j=1,\dots,n$. Taking the Fourier transform in (10.6.11), then using (7.10.7) and Proposition 4.61 with $\lambda=n-2$, we obtain

$$\begin{aligned} E_{jk} &= \mathcal{F}^{-1}\left(\frac{\xi_j\xi_k}{|\xi|^4}\right) - \mathcal{F}^{-1}\left(\frac{1}{|\xi|^2}\right)\delta_{jk} + R_{jk} \\ &= -\frac{1}{2(n-2)\omega_{n-1}}\frac{\delta_{jk}}{|x|^{n-2}} - \frac{1}{2\omega_{n-1}}\frac{x_jx_k}{|x|^n} + R_{jk} \ \text{ in } \ \mathcal{S}'(\mathbb{R}^n). \end{aligned} \tag{10.6.12}$$

We are now ready to state our main result regarding fundamental solutions for the Stokes operator that are tempered distributions.

Theorem 10.29. *Let $n \geq 3$ and let L_S be the Stokes operator from (10.6.1). Consider the following functions in $L^1_{\rm loc}(\mathbb{R}^n)$:*

$$E_{jk}(x) := -\frac{1}{2(n-2)\omega_{n-1}}\frac{\delta_{jk}}{|x|^{n-2}} - \frac{1}{2\omega_{n-1}}\frac{x_j x_k}{|x|^n}, \quad j,k \in \{1,\ldots,n\}, \tag{10.6.13}$$

$$p_k(x) := -\frac{1}{\omega_{n-1}}\frac{x_k}{|x|^n}, \quad k \in \{1,\ldots,n\}, \tag{10.6.14}$$

defined for $x \in \mathbb{R}^n\backslash\{0\}$. Then if we set $\mathbb{E} := \big(E_{jk}\big)_{1\leq j,k\leq n}$ and $\mathbf{p} := (p_1,\ldots,p_n)$, we have $\mathbb{E} \in \mathcal{M}_{n\times n}\big(\mathcal{S}'(\mathbb{R}^n)\big)$, $\mathbf{p} \in \big[\mathcal{S}'(\mathbb{R}^n)\big]^n$, and the pair $(\mathbb{E},\mathbf{p})$ is a fundamental solution for the Stokes operator in $\mathbb{R}^n$.

Moreover, any fundamental solution $(\mathbb{U},\mathbf{q})$ for the Stokes operator with the property that $\mathbb{U} \in \mathcal{M}_{n\times n}\big(\mathcal{S}'(\mathbb{R}^n)\big)$, $\mathbf{q} \in \big[\mathcal{S}'(\mathbb{R}^n)\big]^n$ is of the form $\mathbb{U} = \mathbb{E} + \mathbb{P}$, $\mathbf{q} = \mathbf{p} + \mathbf{r}$, for some matrix $\mathbb{P} := \big(P_{jk}\big)_{1\leq j,k\leq n}$ whose entries are polynomials in $\mathbb{R}^n$ satisfying $\Delta^2 P_{jk} = 0$ pointwise in $\mathbb{R}^n$ for $j,k = 1,\ldots,n$, and some vector $\mathbf{r} := (r_1,\ldots,r_n)$, whose entries are polynomials in $\mathbb{R}^n$ satisfying $\Delta r_k = 0$ pointwise in $\mathbb{R}^n$ for $k = 1,\ldots,n$.

Proof. From (10.6.12) and (10.6.9) we have $\mathbb{E} \in \mathcal{M}_{n\times n}\big(\mathcal{S}'(\mathbb{R}^n)\big)$, $\mathbf{p} \in \big[\mathcal{S}'(\mathbb{R}^n)\big]^n$, and

$$\widehat{E_{jk}} = \frac{\xi_j\xi_k}{|\xi|^4} - \frac{\delta_{jk}}{|\xi|^2}, \quad \widehat{p_k} = i\frac{\xi_k}{|\xi|^2}, \quad \text{in } \mathcal{S}'(\mathbb{R}^n), \quad j,k \in \{1,\ldots,n\}. \tag{10.6.15}$$

Based on the properties of the Fourier transform, we have that $(\mathbb{E},\mathbf{p})$ is a fundamental solution for the Stokes operator if and only if its components satisfy (10.6.5) and (10.6.6). By making use of (10.6.15) we may write

$$-|\xi|^2\widehat{E_{jk}} - i\xi_j\widehat{p_k} = -\frac{\xi_j\xi_k}{|\xi|^2} + \delta_{jk} + \frac{\xi_j\xi_k}{|\xi|^2} = \delta_{jk} \text{ in } \mathcal{S}'(\mathbb{R}^n), \quad j,k \in \{1,\ldots,n\} \tag{10.6.16}$$

and

$$\sum_{j=1}^n \xi_j\widehat{E_{jk}} = \sum_{j=1}^n \frac{\xi_j^2\xi_k}{|\xi|^4} - \sum_{j=1}^n \frac{\delta_{jk}\xi_j}{|\xi|^2} = 0 \text{ in } \mathcal{S}'(\mathbb{R}^n), \tag{10.6.17}$$

proving that $(\mathbb{E},\mathbf{p})$ is a fundamental solution for the Stokes operator.

Suppose $(\mathbb{U},\mathbf{q})$ is another fundamental solution for the Stokes operator such that $\mathbb{U} = \big(U_{jk}\big)_{1\leq j,k\leq n} \in \mathcal{M}_{n\times n}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ and $\mathbf{q} = (q_1,\ldots,q_n) \in \big[\mathcal{S}'(\mathbb{R}^n)\big]^n$. Then for each $k \in \{1,\ldots,n\}$ we have $q_k = p_k + r_k$ for some harmonic polynomial r_k in $\mathbb{R}^n$ (recall the computation that lead to (10.6.9)). From this fact and the equations corresponding to (10.6.3) written for the components of $(\mathbb{E},\mathbf{p})$ and $(\mathbb{U},\mathbf{q})$, we further obtain that for each $j,k \in \{1,\ldots,n\}$ there holds $\Delta(U_{jk} - E_{jk}) = \partial_j r_k$ in $\mathcal{S}'(\mathbb{R}^n)$, hence $\Delta^2(U_{jk} - E_{jk}) = 0$ in $\mathcal{S}'(\mathbb{R}^n)$. Consequently, $|\xi|^4(\widehat{U_{jk}} - \widehat{E_{jk}}) = 0$ in $\mathcal{S}'(\mathbb{R}^n)$ for $j,k \in \{1,\ldots,n\}$, which in turn implies $U_{jk} - E_{jk} = R_{jk}$ for some polynomials R_{jk} in $\mathbb{R}^n$ satisfying $\Delta^2 R_{jk} = 0$ pointwise in $\mathbb{R}^n$ for all $j,k \in \{1,\ldots,n\}$. The proof of the theorem is now complete. □

Exercise 10.30. *Follow the outline below to check, without the use of the Fourier transform, that* $(\mathbb{E},\mathbf{p})$ *with entries* $\mathbb{E}=\big(E_{jk}\big)_{1\leq j,k\leq n}$ *and* $\mathbf{p}=(p_1,\ldots,p_n)$ *of function type defined by the functions from* (10.6.13) *and* (10.6.14) *is a fundamental solution for the Stokes operator in* $\mathbb{R}^n$, $n\geq 3$.

Step 1. Show that $\Delta E_{jk}-\partial_j p_k=0$ and $\sum\limits_{\ell=1}^{n}\partial_\ell E_{\ell k}=0$ pointwise in $\mathbb{R}^n\setminus\{0\}$ for every $j,k\in\{1,\ldots,n\}$.

Step 2. Show that the desired conclusion (i.e., that the given $(\mathbb{E},\mathbf{p})$ is a fundamental solution for the Stokes operator in $\mathbb{R}^n$, $n\geq 3$) is equivalent with the conditions that

$$\lim_{\varepsilon\to 0^+}\left[\int_{|x|\geq\varepsilon}E_{jk}(x)\Delta\varphi(x)\,dx+\int_{|x|\geq\varepsilon}p_k(x)\partial_j\varphi(x)\,dx\right]=\varphi(0)\delta_{jk} \tag{10.6.18}$$

and

$$\lim_{\varepsilon\to 0^+}\left[\int_{|x|\geq\varepsilon}\sum_{\ell=1}^{n}E_{\ell k}(x)\partial_\ell\varphi(x)\,dx\right]=0 \tag{10.6.19}$$

for every $j,k\in\{1,\ldots,n\}$ and every $\varphi\in C_0^\infty(\mathbb{R}^n)$.

Step 3. Fix $j,k\in\{1,\ldots,n\}$, $\varphi\in C_0^\infty(\mathbb{R}^n)$, and let $R\in(0,\infty)$ be such that $\operatorname{supp}\varphi\subseteq B(0,R)$, so that one may replace the domain of integration for the integrals in (10.6.18) and (10.6.19) with $\{x\in\mathbb{R}^n:\varepsilon<|x|<R\}$. Use this domain of integration, (13.7.5), (13.7.4), and the result from Step 1, to prove that (10.6.18) and (10.6.19) are equivalent with

$$\lim_{\varepsilon\to 0^+}\left[-\int_{\partial B(0,\varepsilon)}E_{jk}(x)\frac{\partial\varphi}{\partial\nu}(x)\,d\sigma(x)+\int_{\partial B(0,\varepsilon)}\varphi(x)\frac{\partial E_{jk}}{\partial\nu}(x)\,d\sigma(x)\right.$$
$$\left.-\int_{\partial B(0,\varepsilon)}p_k(x)\varphi(x)\nu_j(x)\,d\sigma x\right]=\varphi(0)\delta_{jk} \tag{10.6.20}$$

and

$$\lim_{\varepsilon\to 0^+}\left[\int_{\partial B(0,\varepsilon)}\sum_{\ell=1}^{n}E_{\ell k}(x)\varphi(x)\nu_l(x)\,d\sigma x\right]=0, \tag{10.6.21}$$

where $\nu(x)=\frac{x}{\varepsilon}$ for each $x\in\partial B(0,\varepsilon)$.

Step 4. Prove that there exists a constant $C\in(0,\infty)$ independent of ε such that each of the quantities:

$$\left|\textstyle\int_{\partial B(0,\varepsilon)}E_{jk}(x)\frac{\partial\varphi}{\partial\nu}(x)\,d\sigma(x)\right|,\qquad\left|\textstyle\int_{\partial B(0,\varepsilon)}\frac{\partial E_{jk}}{\partial\nu}(x)[\varphi(x)-\varphi(0)]\,d\sigma(x)\right|,$$
$$\left|\textstyle\int_{\partial B(0,\varepsilon)}p_k(x)[\varphi(x)-\varphi(0)]\nu_j(x)\,d\sigma x\right|, \tag{10.6.22}$$

is bounded by $C\|\nabla\varphi\|_{L^\infty(\mathbb{R}^n)}\,\varepsilon$, thus convergent to zero as $\varepsilon\to 0^+$, and such that

$$\left|\int_{\partial B(0,\varepsilon)}\sum_{\ell=1}^{n}E_{\ell k}(x)\varphi(x)\nu_l(x)\,d\sigma(x)\right|\leq C\|\varphi\|_{L^\infty(\mathbb{R}^n)}\,\varepsilon\xrightarrow[\varepsilon\to 0^+]{}0. \tag{10.6.23}$$

Step 5. Combine Steps 2, 3, and 4, to reduce matters to proving that

$$\lim_{\varepsilon\to 0^+}\left[\int\limits_{\partial B(0,\varepsilon)}\sum_{s=1}^{n}\frac{x_s}{\varepsilon}(\partial_s E_{jk})(x)\,d\sigma(x)-\int\limits_{\partial B(0,\varepsilon)}\frac{x_j}{\varepsilon}p_k(x)\,d\sigma(x)\right]=\delta_{jk}. \tag{10.6.24}$$

Step 6. Prove that for each $x\in\partial B(0,\varepsilon)$ we have

$$\sum_{s=1}^{n}\frac{x_s}{\varepsilon}(\partial_s E_{jk})(x)-\frac{x_j}{\varepsilon}p_k(x)=\frac{\delta_{jk}}{2\,\omega_{n-1}\varepsilon^{n-1}}+\frac{nx_jx_k}{2\,\omega_{n-1}\varepsilon^{n+1}}. \tag{10.6.25}$$

Step 7. Integrate the expression in (10.6.25) and use (10.3.28) to finish the proof of (10.6.24).

Further Notes for Chap. 10. The discussion in Chaps. 5 and 6 about the existence and nature of fundamental solutions has been limited to the case of scalar differential operators, and in Sect. 10.2 these scalar results have been extended to generic constant coefficient systems (of arbitrary order) via an approach that appears to be new. Moreover, this approach offers further options for finding an explicit form for a fundamental solution for a given system, such as the Lamé system. While we shall pursue this idea later, in Sect. 11.1, we felt it is natural and beneficial to first deal with the issue of computing a fundamental solution for the Lamé and Stokes systems via Fourier analysis (as done in Sects. 10.3 and 10.6, respectively).

The inclusion of a section on mean value formulas for the Lamé operator is justified by the fact that such formulas directly yield interior estimates, without resorting to quantitative elliptic regularity, which is typically formulated in the language of Sobolev spaces. In turn, these interior estimates play a pivotal role in establishing uniqueness for the corresponding Poisson problem. A more systematic treatment of the issue of mean value formulas for the Lamé operator may be found in [51].

10.7 Additional Exercises for Chap. 10

Exercise 10.31. *Let L_S be the Stokes operator from (10.6.1). Suppose that $p\in\mathcal{D}'(\Omega)$ and $\mathbf{u}=(u_1,\ldots,u_n)\in\big[\mathcal{D}'(\Omega)\big]^n$ satisfy $L_S(\mathbf{u},p)=\mathbf{0}$ in $\big[\mathcal{D}'(\Omega)\big]^{n+1}$. Under these conditions prove that $p,u_j\in C^\infty(\Omega)$ for $j\in\{1,\ldots,n\}$, and $\Delta p=0$ pointwise in Ω while $\Delta^2 u_j=0$ pointwise in Ω for $j\in\{1,\ldots,n\}$.*

Exercise 10.32. *For each $x \in \mathbb{R}^2 \setminus \{0\}$ consider the functions*

$$E_{jk}(x) := \frac{1}{4\pi\mu(2\mu+\lambda)}\left[(3\mu+\lambda)\delta_{jk}\ln|x| - \frac{(\mu+\lambda)x_jx_k}{|x|^2}\right], \quad j,k\in\{1,2\}. \tag{10.7.1}$$

Prove that $\mathbb{E} := (E_{j,k})_{1\leq j,k\leq 2}$ is a fundamental solution for the Lamé operator in $\mathbb{R}^2$.

Exercise 10.33. *Consider the functions*

$$E_{jk}(x) := \frac{1}{4\pi}\delta_{jk}\ln|x| - \frac{1}{4\pi}\cdot\frac{x_jx_k}{|x|^2}, \qquad p_k(x) := -\frac{1}{2\pi}\cdot\frac{x_k}{|x|^2}, \tag{10.7.2}$$

defined for $x \in \mathbb{R}^2 \setminus \{0\}$ and $j,k \in \{1,2\}$. Prove that if $\mathbb{E} := (E_{j,k})_{1\leq j,k\leq 2}$ and $\mathbf{p} = (p_1,p_2)$, then $(\mathbb{E},\mathbf{p})$ is a fundamental solution for the Stokes operator in $\mathbb{R}^2$.

Exercise 10.34. *Assume $n \geq 3$, let L be the Lamé operator from (10.3.1) such that (10.3.2) is satisfied and consider $\mathbf{f} = (f_1,\ldots,f_n) \in \left[C_0^\infty(\mathbb{R}^n)\right]^n$. Prove that for each $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ and each $b \in \mathbb{R}^n$, there exists a unique solution for the problem*

$$\begin{cases} \mathbf{u} = (u_1,\ldots,u_n) \in [C^\infty(\mathbb{R}^n)]^n, \\ L\mathbf{u} = \mathbf{f} \quad \text{pointwise in} \quad \mathbb{R}^n, \\ \lim\limits_{|x|\to\infty} |\mathbf{u}(x) - Ax - B| = 0. \end{cases} \tag{10.7.3}$$

Exercise 10.35. *Assume $n \geq 3$, let L be the Lamé operator from (10.3.1) such that (10.3.2) is satisfied. Prove that if $m \in \mathbb{N}_0$ and $\mathbf{u}$ is a solution of the problem*

$$\begin{cases} \mathbf{u} \in [C^\infty(\mathbb{R}^n)]^n, \\ L\mathbf{u} = 0 \quad \text{pointwise in} \quad \mathbb{R}^n, \\ |\mathbf{u}(x)| \leq C(|x|^m + 1), \end{cases} \tag{10.7.4}$$

for some $C \in (0,\infty)$, then the components of $\mathbf{u}$ are polynomials of degree $\leq m$.

Chapter 11

More on Fundamental Solutions for Systems

11.1 Computing a Fundamental Solution for the Lamé Operator

In this section we use Theorem 10.8 in order to find a fundamental solution for the Lamé operator of elastostatics in $\mathbb{R}^n$, with $n \geq 2$. This is possible since in the case when $L(\partial)$ is the Lamé operator we have that $\det(L(\partial))$ is a constant factor of Δ^n.

Concretely, fix $n \in \mathbb{N}$, $n \geq 2$, and consider the algebra $\mathfrak{R}$ as in (10.2.15). Fix constants $\lambda, \mu \in \mathbb{C}$ satisfying $\mu \neq 0$, $\lambda + 2\mu \neq 0$, and set

$$L(\partial) := \mu\Delta + (\lambda + \mu)\nabla\mathrm{div} \in \mathcal{M}_{n\times n}(\mathfrak{R}), \tag{11.1.1}$$

[recall that $\nabla = (\partial_1, \ldots, \partial_n)$ and the operator div was defined in Exercise 7.60]. This is precisely the Lamé operator from (10.3.1) and $L(\partial)$ has the form (10.2.12), with $a := \nabla$. Thus, for this choice of a, formula (10.2.13) gives

$$P(\partial) := \det L(\partial) = \mu^{n-1}(\lambda + 2\mu)\Delta^n, \tag{11.1.2}$$

while (10.2.14) implies

$$\mathrm{adj}\,[L(\partial)] = \mu^{n-2}\Big[(\lambda + 2\mu)\Delta I_{n\times n} - (\lambda + \mu)\nabla\mathrm{div}\Big]\Delta^{n-2}. \tag{11.1.3}$$

Going further, for each $m \in \mathbb{N}$, denote by E_{Δ^m} the fundamental solution for the poly-harmonic differential operator Δ^m in $\mathbb{R}^n$ from (7.5.2) and apply Proposition 7.29 to conclude that

$$\Delta^{n-2}\Big[E_{\Delta^n}\Big] = E_{\Delta^2} + c(n) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n) \quad \text{and pointwise in } \mathbb{R}^n \setminus \{0\}, \tag{11.1.4}$$

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where $c(n)=0$ if $n\neq 4$ and $c(4)\in\mathbb{R}$. A direct computation starting with (7.5.2) corresponding to $n=4=m$ and using Lemma 7.20 yields $c(4)=-\frac{5}{24\pi^2}$. Thus, a combination of (11.1.4) and (7.3.8) yields

$$\Delta^{n-2}E_{\Delta^n}(x)=\begin{cases}\dfrac{|x|^{4-n}}{2(n-2)(n-4)\omega_{n-1}} & \text{if } n\geq 3,\ n\neq 4,\\[2ex] -\dfrac{1}{8\pi^2}\cdot\ln|x|-\dfrac{5}{24\pi^2} & \text{if } n=4,\\[2ex] \dfrac{1}{8\pi}\cdot|x|^2\,\ln|x| & \text{if } n=2,\end{cases}\tag{11.1.5}$$

both for each $x\in\mathbb{R}^n\setminus\{0\}$ as well as in $\mathcal{S}'(\mathbb{R}^n)$. Also, by applying one more Δ to the identity in (11.1.4) and recalling (7.1.12), we have

$$\Delta^{n-1}E_{\Delta^n}(x)=E_{\Delta}(x)=\begin{cases}-\dfrac{1}{(n-2)\omega_{n-1}}\cdot|x|^{2-n} & \text{if } n\geq 3,\\[2ex] \dfrac{1}{2\pi}\cdot\ln|x| & \text{if } n=2,\end{cases}\tag{11.1.6}$$

for every $x\in\mathbb{R}^n\setminus\{0\}$ and in $\mathcal{S}'(\mathbb{R}^n)$.

Hence, based on (10.2.23) and (11.1.2), we obtain that a fundamental (matrix) solution $\mathbb{E}^L$ for the operator $L(\partial)$ as in (11.1.1) is given by

$$\mathbb{E}^L(x)=\operatorname{adj}\left[L(\partial)\right]\Big(E_{\mu^{n-1}(\lambda+2\mu)\Delta^n}(x)I_{n\times n}\Big),\quad x\in\mathbb{R}^n\setminus\{0\},\tag{11.1.7}$$

where $E_{\mu^{n-1}(\lambda+2\mu)\Delta^n}$ is a fundamental solution for $P(\partial)=\mu^{n-1}(\lambda+2\mu)\Delta^n$. Using (11.1.3) and (11.1.4) in the rightmost expression in (11.1.7) further gives that, for each $x\in\mathbb{R}^n\setminus\{0\}$,

$$\begin{aligned}\mathbb{E}^L(x)&=\frac{1}{\mu^{n-1}(\lambda+2\mu)}\cdot\mu^{n-2}\Big[(\lambda+2\mu)\Delta I_{n\times n}-(\lambda+\mu)\nabla\operatorname{div}\Big]\Delta^{n-2}\Big(E_{\Delta^n}(x)I_{n\times n}\Big)\\&=\left(\frac{1}{\mu}\delta_{jk}E_{\Delta}(x)-\frac{\lambda+\mu}{\mu(\lambda+2\mu)}(\partial_j\partial_k E_{\Delta^2})(x)\right)_{1\leq j,k\leq n}.\end{aligned}\tag{11.1.8}$$

A quick inspection of (7.3.8) and the last equality in (11.1.6) shows that both for each $x=(x_1,\ldots,x_n)\in\mathbb{R}^n\setminus\{0\}$ as well as in $\mathcal{S}'(\mathbb{R}^n)$, for each $j,k\in\{1,\ldots,n\}$ we have

$$\partial_j\partial_k E_{\Delta^2}(x)=\begin{cases}\dfrac{\delta_{jk}}{2}E_{\Delta}(x)+\dfrac{1}{2\omega_{n-1}}\cdot\dfrac{x_jx_k}{|x|^n} & \text{if } n\geq 3,\\[2ex] \dfrac{\delta_{jk}}{2}E_{\Delta}(x)+\dfrac{1}{2\omega_{n-1}}\cdot\dfrac{x_jx_k}{|x|^n}+\dfrac{\delta_{jk}}{8}, & \text{if } n=2.\end{cases}\tag{11.1.9}$$

Note that, for the purpose of computing E^L, the additive constant $\frac{\delta_{jk}}{8}$ from (11.1.9) may be dropped. Thus, based on (11.1.8) and (11.1.9) (with $\delta_{jk}/8$ dropped), it follows that

$$\mathbb{E}^L(x) = \frac{3\mu+\lambda}{2\mu(2\mu+\lambda)} E_\Delta(x) I_{n\times n} \tag{11.1.10}$$
$$- \frac{\lambda+\mu}{2\mu(2\mu+\lambda)\omega_{n-1}} \cdot \frac{x}{|x|^{n/2}} \otimes \frac{x}{|x|^{n/2}}, \quad \forall\, x \in \mathbb{R}^n \setminus \{0\},$$

is a tempered distribution that is a fundamental solution for the operator Lamé operator (11.1.1) (with $\mu \neq 0$, $\lambda + 2\mu \neq 0$) in $\mathbb{R}^n$. Note that this formula coincides with the one from (10.3.21) if $n \geq 3$ and with the one from (10.7.1) if $n = 2$.

11.2 Computing a Fundamental Solution for the Stokes Operator

In this section, the goal is to compute a fundamental solution for the Stokes operator in $\mathbb{R}^n$ starting from the explicit expression for the fundamental solution for the operator Lamé operator $\mathbb{E}^L$ from (11.1.10). To this end, fix $n \in \mathbb{N}$, $n \geq 2$ and $\lambda, \mu \in \mathbb{C}$ satisfying $\mu \neq 0$, $\lambda + 2\mu \neq 0$, and let $x = (x_1, \ldots, x_n)$ for $x \in \mathbb{R}^n$.

For each $k \in \{1, \ldots, n\}$, denote by $\mathbf{E}_k^L$ the kth column of the fundamental solution $\mathbb{E}^L$ from (11.1.10). Then a straightforward calculation gives that for each $x \in \mathbb{R}^n \setminus \{0\}$ and in $\mathcal{S}'(\mathbb{R}^n)$, there holds

$$\operatorname{div}[\mathbf{E}_k^L(x)] = \frac{1}{(\lambda+2\mu)\omega_{n-1}} \cdot \frac{x_k}{|x|^n}, \quad \forall\, k \in \{1, \ldots, n\}. \tag{11.2.1}$$

Define

$$p_k(x) := -\frac{1}{\omega_{n-1}} \cdot \frac{x_k}{|x|^n}, \quad \forall\, x \in \mathbb{R}^n \setminus \{0\}, \quad \forall\, k \in \{1, \ldots, n\}. \tag{11.2.2}$$

Then, from (11.2.1) it follows that for each $k \in \{1, \ldots, n\}$ we have

$$\lim_{\lambda\to\infty} \operatorname{div}[\mathbf{E}_k^L] = 0 \ \text{ pointwise in } \mathbb{R}^n \setminus \{0\} \text{ and in } \mathcal{S}'(\mathbb{R}^n), \tag{11.2.3}$$

and

$$\lim_{\lambda\to\infty} \Big[-(\lambda+\mu)\operatorname{div}[\mathbf{E}_k^L]\Big] = p_k \quad \text{pointwise in } \mathbb{R}^n \setminus \{0\} \text{ and in } \mathcal{S}'(\mathbb{R}^n). \tag{11.2.4}$$

In addition, from (11.1.10) we obtain

$$\lim_{\lambda\to\infty} \mathbb{E}^L = \frac{1}{\mu}\mathbb{E}^S \ \text{ pointwise in } \mathbb{R}^n \setminus \{0\} \text{ and in } \mathcal{S}'(\mathbb{R}^n), \tag{11.2.5}$$

where, we have set

$$\mathbb{E}^S(x) := \Big(\frac{\delta_{jk}}{2} \cdot E_\Delta(x) - \frac{1}{2\omega_{n-1}} \cdot \frac{x_j x_k}{|x|^n}\Big)_{1\leq j,k\leq n} \quad \text{for } x \in \mathbb{R}^n \setminus \{0\}, \tag{11.2.6}$$

with E_Δ as in (11.1.6). Consequently, a combination of (11.2.4) and (11.2.6) (in view of the definition (11.1.1) of the Lamé operator), implies

$$\Delta E^S_{jk} - \partial_j p_k = \lim_{\lambda\to\infty}\Big[\mu E^L_{jk} + (\lambda+\mu)\sum_{s=1}^{n}\partial_j\partial_s E^L_{sk}\Big]$$
$$= \delta_{jk}\delta \quad \text{in } \mathcal{S}'(\mathbb{R}^n),\ \ \forall\, j,k\in\{1,\dots,n\}. \tag{11.2.7}$$

Also, (11.2.5) and (11.2.3), with the convention that $\mathbf{E}^S_k$ stands for the kth column in the matrix $\mathbb{E}^S$ from (11.2.6), give

$$\operatorname{div}[\mathbf{E}^S_k] = \lim_{\lambda\to\infty}\Big(\operatorname{div}[\mathbf{E}^L_k]\Big) = 0 \quad \text{in } \mathcal{S}'(\mathbb{R}^n),\ \ \forall\, k\in\{1,\dots,n\}. \tag{11.2.8}$$

A quick inspection of (11.2.6) and (11.2.2), shows that the entries in the matrix $\mathbb{E}^S$ and the components of the vector $\mathbf{p} := (p_1,\dots,p_n)$ belong to $L^1_{\rm loc}(\mathbb{R}^n)$, while (11.2.7) guarantees that

$$\begin{cases} \Delta\mathbb{E}^S - \nabla\mathbf{p} = \delta_0 I_{n\times n}, & \text{in } \ \mathcal{M}_{n\times n}\big(\mathcal{S}'(\mathbb{R}^n)\big), \\ \operatorname{div}\mathbb{E}^S = 0, & \text{in } \ [\mathcal{S}'(\mathbb{R}^n)]^n. \end{cases} \tag{11.2.9}$$

Recalling now the definition of the Stokes operator from (10.6.1), we may conclude that

$$(\mathbb{E}^S,\mathbf{p}) \ \text{ is a fundamental solution for the Stokes operator in } \mathbb{R}^n. \tag{11.2.10}$$

Note that the expressions we obtained for $(\mathbb{E}^S,\mathbf{p})$ as given in (11.2.6) and (11.2.2) are the same as the ones from (10.6.13)–(10.6.14) if $n\geq 3$ and as the one from (10.7.2) if $n=2$.

11.3 Fundamental Solutions for Higher-Order Systems

In this section we discuss an approach for computing all fundamental solutions that are tempered distributions for a certain subclass of systems of the form (10.2.16). To specify this class, let as before $n\in\mathbb{N}$ denote the Euclidean dimension, fix $M,m\in\mathbb{N}$, and consider $M\times M$ systems of homogeneous differential operators of order $2m$ with complex constant coefficients that have the form

$$L = L(\partial) := \Big(\sum_{|\alpha|=2m} a^{jk}_\alpha\partial^\alpha\Big)_{1\leq j,k\leq M} =: \big(L_{jk}(\partial)\big)_{1\leq j,k\leq M}, \tag{11.3.1}$$

where $a^{jk}_\alpha\in\mathbb{C}$, $j,k\in\{1,\dots,M\}$. Define the characteristic matrix of L to be

$$L(\xi) := \Big(\sum_{|\alpha|=2m} a^{jk}_\alpha\xi^\alpha\Big)_{1\leq j,k\leq M} \quad \text{for } \ \xi\in\mathbb{R}^n. \tag{11.3.2}$$

The standing assumption under which we will identify all fundamental solutions for L that are tempered distributions is

$$\det[L(\xi)]\neq 0, \quad \forall\,\xi\in\mathbb{R}^n\setminus\{0\}. \tag{11.3.3}$$

Several types of operators discussed earlier fall under the scope of these specifications, including the class of strictly elliptic, homogeneous, second-order,

constant coefficient operators from Sect. 7.8, the poly-harmonic operator from Sect. 7.5, and the Lamé operator from Sect. 10.3. For any operator L as in (11.3.1), its transpose and complex conjugate are, respectively, given by

$$L^{\top} := \Big(\sum_{|\alpha|=2m} a_{\alpha}^{jk} \partial^{\alpha} \Big)_{1\leq k,j\leq M}, \quad \overline{L} = \Big(\sum_{|\alpha|=2m} \overline{a_{\alpha}^{jk}} \partial^{\alpha} \Big)_{1\leq j,k\leq M}, \tag{11.3.4}$$

while the formal adjoint of L is defined as

$$L^{*} := \overline{L}^{\top}. \tag{11.3.5}$$

The main result in this section is Theorem 11.1 below, describing the nature and properties of all fundamental solutions for a higher-order system L as in (11.3.1)–(11.3.3) that are tempered distributions. Before presenting this basic result we first describe a strategy that points to a natural candidate for a fundamental solution for L.

Suppose that Q is a scalar constant coefficient differential operator (of an auxiliary nature) which, from other sources of information, is known to have a fundamental solution in $\mathbb{R}^n$ of the form

$$E_Q(x) = \int_{S^{n-1}} F(x\cdot\xi)\, d\sigma(\xi), \tag{11.3.6}$$

where F is a sufficiently regular scalar-valued function on the real line. Granted this, we then proceed to define

$$E_L(x) := Q\Big[\int_{S^{n-1}} G(x\cdot\xi) \big[L(\xi)\big]^{-1} d\sigma(\xi) \Big], \tag{11.3.7}$$

where G is a scalar-valued function on the real line, chosen in such a way that

$$p(t) := G^{(2m)}(t) - F(t) \ \text{ is a polynomial in } t\in\mathbb{R} \text{ of degree} < \text{order}\, Q. \tag{11.3.8}$$

Then

$$P(x) := \int_{S^{n-1}} p(x\cdot\xi)\, d\sigma(\xi), \quad x\in\mathbb{R}^n, \tag{11.3.9}$$

is a polynomial in $\mathbb{R}^n$ of degree $<$ order Q. Keeping this in mind and observing that

$$LQ = QL \quad \text{and} \quad L\big[G(x\cdot\xi)\big] = G^{(2m)}(x\cdot\xi) L(\xi), \tag{11.3.10}$$

we may compute

$$\begin{aligned} L\big[E_L(x)\big] &= Q\Big[\int_{S^{n-1}} G^{(2m)}(x\cdot\xi) L(\xi) \big[L(\xi)\big]^{-1} d\sigma(\xi) \Big] \\ &= Q\Big[\int_{S^{n-1}} F(x\cdot\xi)\, d\sigma(\xi) + P(x) \Big] I_{M\times M} \\ &= Q[E_Q(x)] I_{M\times M} + (QP)(x) I_{M\times M} = \delta I_{M\times M} + 0\, I_{M\times M} \\ &= \delta I_{M\times M}, \end{aligned} \tag{11.3.11}$$

which shows that E_L is a fundamental solution for the system L in $\mathbb{R}^n$.

To implement the procedure just described, it is natural to take the poly-harmonic operator Δ^N to play the role of the auxiliary scalar differential operator Q. This is not just a matter of convenience, but from Lemma 7.31 we already know that if $N \in \mathbb{N}$ satisfies $N \geq n/2$ then there exists a fundamental solution for this operator that has the form requested in (11.3.6). Specifically, in this case we have (compare with (7.5.24))

$$F(t) := -\frac{1}{(2\pi i)^n (2N-n)!}\, t^{2N-n} \log\big(t/i\big). \tag{11.3.12}$$

To find a function G on the real line that satisfies (11.3.8), we may try

$$G(t) := a\, t^A \log\big(t/i\big), \tag{11.3.13}$$

where $a \in \mathbb{R}$, $A \in \mathbb{N}$ with $A \geq 2m$, and note there exists a constant $C_{A,a,m}$ such that

$$\Big(\frac{\mathrm{d}}{\mathrm{d}t}\Big)^{2m}\big[G(t)\big] = a\frac{A!}{(A-2m)!}t^{A-2m}\log\big(t/i\big) + C_{A,a,m}\, t^{A-2m}, \tag{11.3.14}$$

which means that (11.3.8) is going to hold if we take

$$A := 2N-n+2m \quad \text{and} \quad a := -\frac{1}{(2\pi i)^n (2N-n+2m)!}, \tag{11.3.15}$$

for a polynomial p on $\mathbb{R}$ of degree $2N-n$, which is strictly less than the order of $Q = \Delta^N$.

To bring matters more in line with notation employed in the proof of Theorem 11.1, pick now a number $q \in \mathbb{N}_0$ with the same parity as n, and choose $N := (n+q)/2$ (which satisfies $N \in \mathbb{N}$ and $N \geq n/2$). For this choice of N,

$$\text{the operator } Q \text{ becomes } \Delta^{\frac{n+q}{2}}, \tag{11.3.16}$$

and if A, a are as in (11.3.15) the function G from (11.3.13) takes the form

$$G(t) = -\frac{1}{(2\pi i)^n (2m+q)!}\, t^{2m+q} \log\big(t/i\big). \tag{11.3.17}$$

For these specifications, the function E_L from (11.3.7) then becomes precisely the function

$$\mathbb{E}(x) := \frac{-1}{(2\pi\, i)^n (2m+q)!}\Delta_x^{(n+q)/2}\int_{S^{n-1}} (x\cdot\xi)^{2m+q}\log\Big(\frac{x\cdot\xi}{i}\Big)\big[L(\xi)\big]^{-1}\,\mathrm{d}\sigma(\xi). \tag{11.3.18}$$

Having explained the genesis of this formula, we now proceed to rigorously show that not only is this candidate for a fundamental solution natural but it also does the job.

Theorem 11.1. *Fix $n, m, M \in \mathbb{N}$ with $n \geq 2$, and assume that L is an $M \times M$ system in $\mathbb{R}^n$ of order $2m$ as in (11.3.1)–(11.3.3). Then the $M \times M$ matrix $\mathbb{E}$ defined at each $x \in \mathbb{R}^n \setminus \{0\}$ by*

$$\mathbb{E}(x) := \frac{1}{4(2\pi\, i)^{n-1}(2m-1)!}\Delta_x^{(n-1)/2}\int_{S^{n-1}}(x\cdot\xi)^{2m-1}\operatorname{sgn}(x\cdot\xi)\big[L(\xi)\big]^{-1}\,d\sigma(\xi) \tag{11.3.19}$$

if n is odd, and

$$\mathbb{E}(x) := \frac{-1}{(2\pi\, i)^{n}(2m)!}\Delta_x^{n/2}\int_{S^{n-1}}(x\cdot\xi)^{2m}\ln|x\cdot\xi|\big[L(\xi)\big]^{-1}\,d\sigma(\xi) \tag{11.3.20}$$

if n is even, satisfies the following properties.

1. *One has $\mathbb{E} \in \mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big) \cap \mathcal{M}_{M\times M}\big(C^\infty(\mathbb{R}^n \setminus \{0\})\big)$ and*

$$\mathbb{E}(-x) = \mathbb{E}(x) \quad \text{for all} \quad x \in \mathbb{R}^n \setminus \{0\}. \tag{11.3.21}$$

   Moreover, the entries in $\mathbb{E}$ are real-analytic functions in $\mathbb{R}^n \setminus \{0\}$.

2. *If $I_{M\times M}$ is the $M \times M$ identity matrix, then for each $y \in \mathbb{R}^n$*

$$L^x\big[\mathbb{E}(x-y)\big] = \delta_y(x)\, I_{M\times M} \quad \text{in} \quad \mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big), \tag{11.3.22}$$

   where the superscript x denotes the fact that the operator L in (11.3.22) is applied to each column of $\mathbb{E}$ in the variable x.

3. *Define the $M \times M$ matrix-valued function*

$$\mathbb{P}(x) := \frac{-1}{(2\pi\, i)^{n}(2m-n)!}\int_{S^{n-1}}(x\cdot\xi)^{2m-n}\big[L(\xi)\big]^{-1}\,d\sigma(\xi), \quad \forall\, x \in \mathbb{R}^n. \tag{11.3.23}$$

   Then the entries of $\mathbb{P}$ are identically zero when either n is odd or $n > 2m$, and are homogeneous polynomials of degree $2m-n$ when $n \leq 2m$. Moreover, there exists a function $\Phi \in \mathcal{M}_{M\times M}\big(C^\infty(\mathbb{R}^n \setminus \{0\})\big)$ that is positive homogeneous of degree $2m-n$ such that

$$\mathbb{E}(x) = \Phi(x) + \big(\ln|x|\big)\mathbb{P}(x), \quad \forall\, x \in \mathbb{R}^n \setminus \{0\}. \tag{11.3.24}$$

4. *For each $\beta \in \mathbb{N}_0^n$ with $|\beta| \geq 2m-1$, the restriction to $\mathbb{R}^n \setminus \{0\}$ of the matrix distribution $\partial^\beta\mathbb{E}$ is of class C^∞ and positive homogeneous of degree $2m-n-|\beta|$.*

5. *For each $\beta \in \mathbb{N}_0^n$ there exists $C_\beta \in (0,\infty)$ such that the estimate*

$$|\partial^\beta\mathbb{E}(x)| \leq \begin{cases} \dfrac{C_\beta}{|x|^{n-2m+|\beta|}} & \text{if either } n \text{ is odd, or } n > 2m, \text{ or if } |\beta| > 2m-n, \\[2ex] \dfrac{C_\beta(1+|\ln|x||)}{|x|^{n-2m+|\beta|}} & \text{if } 0 \leq |\beta| \leq 2m-n, \end{cases} \tag{11.3.25}$$

   holds for each $x \in \mathbb{R}^n \setminus \{0\}$.

6. *When restricted to* $\mathbb{R}^n \setminus \{0\}$, *the entries of* $\widehat{\mathbb{E}}$ *(with the "hat" denoting the Fourier transform) are* C^∞ *functions and, moreover,*

$$\widehat{\mathbb{E}}(\xi) = (-1)^m \big[L(\xi)\big]^{-1} \quad \text{for each} \quad \xi \in \mathbb{R}^n \setminus \{0\}. \tag{11.3.26}$$

7. *Writing* $\mathbb{E}_L$ *in place of* $\mathbb{E}$ *to emphasize the dependence on* L, *the fundamental solution* $\mathbb{E}_L$ *with entries as in* (11.3.19)–(11.3.20) *satisfies*

$$\begin{aligned} &\big(\mathbb{E}_L\big)^\top = \mathbb{E}_{L^\top}, \quad \overline{\mathbb{E}_L} = \mathbb{E}_{\overline{L}}, \quad \big(\mathbb{E}_L\big)^* = \mathbb{E}_{L^*}, \\ &\text{and} \quad \mathbb{E}_{\lambda L} = \lambda^{-1}\mathbb{E}_L \quad \forall\, \lambda \in \mathbb{C} \setminus \{0\}. \end{aligned} \tag{11.3.27}$$

8. *Any fundamental solution* $\mathbb{U} \in \mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ *of the system* L *in* $\mathbb{R}^n$ *is of the form* $\mathbb{U} = \mathbb{E} + \mathbb{Q}$ *where* $\mathbb{E}$ *is as in* (11.3.19)–(11.3.20) *and* $\mathbb{Q}$ *is an* $M \times M$ *matrix with entries that are polynomials in* $\mathbb{R}^n$ *and with columns,* $\mathbf{Q}_k$, $k \in \{1, \dots, M\}$, *that satisfy the pointwise equations* $L\mathbf{Q}_k = \mathbf{0} \in \mathbb{C}^M$ *in* $\mathbb{R}^n$ *for each* $k \in \{1, \dots, M\}$.

Proof. To facilitate the subsequent discussion, denote by $\big(P^{jk}(\xi)\big)_{1\leq j,k\leq M}$ the inverse of the characteristic matrix $L(\xi)$, that is,

$$\Big(P^{jk}(\xi)\Big)_{1\leq j,k\leq M} := \big[L(\xi)\big]^{-1}, \quad \forall\, \xi \in \mathbb{R}^n \setminus \{0\}. \tag{11.3.28}$$

Then the entries $\big(E_{jk}\big)_{1\leq j,k\leq M}$ of the matrix $\mathbb{E}$ are given at each point x in $\mathbb{R}^n \setminus \{0\}$ by

$$E_{jk}(x) = \frac{1}{4(2\pi\, i)^{n-1}(2m-1)!}\Delta_x^{(n-1)/2}\int_{S^{n-1}} (x\cdot\xi)^{2m-1}\operatorname{sgn}(x\cdot\xi)P^{jk}(\xi)\,d\sigma(\xi) \tag{11.3.29}$$

if n is odd, and

$$E_{jk}(x) = \frac{-1}{(2\pi\, i)^n(2m)!}\Delta_x^{n/2}\int_{S^{n-1}} (x\cdot\xi)^{2m}\ln|x\cdot\xi|P^{jk}(\xi)\,d\sigma(\xi) \tag{11.3.30}$$

if n is even. The proof of the theorem is completed in eight steps.

Step I. We claim that if we set $q := 0$ *if* n *is even and* $q := 1$ *if* n *is odd, then*

$$E_{jk}(x) = \frac{-1}{(2\pi\, i)^n(2m+q)!}\Delta_x^{(n+q)/2}\int_{S^{n-1}} (x\cdot\xi)^{2m+q}\log\Big(\frac{x\cdot\xi}{i}\Big)P^{jk}(\xi)\,d\sigma(\xi) \tag{11.3.31}$$

for $x \in \mathbb{R}^n \setminus \{0\}$ *and* $j,k \in \{1,\dots,M\}$, *where* log *denotes the principal branch of the complex logarithm defined for points*

$$z \in \mathbb{C} \setminus \{x :\, x \in \mathbb{R},\, x \leq\, 0\}.$$

Suppose first that n is even, thus $q = 0$, and start from (11.3.30), then use the formula $\log\big(\frac{x\cdot\xi}{i}\big) = \ln|x\cdot\xi| - \frac{\pi\, i}{2}\operatorname{sgn}(x\cdot\xi)$ for the term under the integral sign

and the fact that the integral over S^{n-1} of the function $(x\cdot\xi)^{2m}\mathrm{sgn}\,(x\cdot\xi)P^{jk}(\xi)$ (which is odd in ξ) is zero, to obtain that (11.3.31) holds.

Moving on to the case when n is odd, consider the function

$$F(t):=\begin{cases} t^{2m+q}\log(t/i) & \text{if } t\in\mathbb{R}\setminus\{0\},\\ 0 & \text{if } t=0.\end{cases} \tag{11.3.32}$$

It is not difficult to see that

$$F\in C^\infty(\mathbb{R}\setminus\{0\})\cap C^{2m+q-1}(\mathbb{R}),\quad F^{(k)}(0)=0\quad\text{if } k=1,\dots,2m+q-1, \tag{11.3.33}$$

and that for each $k\in\{1,\dots,2m\}$ there exists a constant $C_{m,q,k}$ such that

$$\Big(\frac{\mathrm{d}}{\mathrm{d}t}\Big)^k F(t)=\frac{(2m+q)!}{(2m+q-k)!}t^{2m+q-k}\log\frac{t}{i}+C_{m,q,k}\,t^{2m+q-k}\quad\text{in }\mathbb{R}\setminus\{0\}. \tag{11.3.34}$$

In particular, by the chain rule, if $\beta\in\mathbb{N}_0^n$ is such that $|\beta|\leq 2m+q-1$, then

$$\partial_x^\beta[F(x\cdot\xi)]=\big(F^{(|\beta|)}\big)(x\cdot\xi)\xi^\beta,\quad\forall\,x\in\mathbb{R}^n,\ \forall\,\xi\in S^{n-1}. \tag{11.3.35}$$

Based on (11.3.35) and (11.3.34) for each $x\in\mathbb{R}^n\setminus\{0\}$ we may write

$$\begin{aligned}
&\Delta_x\int_{S^{n-1}}(x\cdot\xi)^{2m+1}\log\Big(\frac{x\cdot\xi}{i}\Big)P^{jk}(\xi)\,\mathrm{d}\sigma(\xi)\\
&\qquad=\Delta_x\int_{S^{n-1}}F(x\cdot\xi)P^{jk}(\xi)\,\mathrm{d}\sigma(\xi)=\int_{S^{n-1}}F''(x\cdot\xi)P^{jk}(\xi)\,\mathrm{d}\sigma(\xi)\\
&\qquad=2m(2m+1)\int_{S^{n-1}}(x\cdot\xi)^{2m-1}\log\Big(\frac{x\cdot\xi}{i}\Big)P^{jk}(\xi)\,\mathrm{d}\sigma(\xi)\\
&\qquad\quad+C_{m,q}\int_{S^{n-1}}(x\cdot\xi)^{2m-1}P^{jk}(\xi)\,\mathrm{d}\sigma(\xi)\\
&\qquad=2m(2m+1)\int_{S^{n-1}}(x\cdot\xi)^{2m-1}\Big[\ln|x\cdot\xi|-\tfrac{\pi i}{2}\mathrm{sgn}(x\cdot\xi)\Big]P^{jk}(\xi)\,\mathrm{d}\sigma(\xi)\\
&\qquad=-\pi im(2m+1)\int_{S^{n-1}}(x\cdot\xi)^{2m-1}\mathrm{sgn}(x\cdot\xi)P^{jk}(\xi)\,\mathrm{d}\sigma(\xi).
\end{aligned}\tag{11.3.36}$$

Hence, if one starts with the expression in the right-hand side of (11.3.31), then one transfers Δ under the integral sign using from (11.3.36), one arrives at the expression of E_{jk} from (11.3.29). This completes the proof of Step I.

Step II. Proof of the fact that the entries of $\mathbb{E}$ *are* C^∞ *and even in* $\mathbb{R}^n\setminus\{0\}$.
That the functions in (11.3.29) and (11.3.30) are even is immediate from their respective expressions. To show that the components of $\mathbb{E}$ belong to $C^\infty(\mathbb{R}^n\setminus\{0\})$, for each $\ell\in\{1,\dots,n\}$ let $\mathbf{e}_\ell$ be the unit vector in $\mathbb{R}^n$ with one on the ℓth component, and consider the open set

$$\mathcal{O}_\ell:=\mathbb{R}^n\setminus\{\lambda\,\mathbf{e}_\ell:\ \lambda\leq 0\}=\big\{x=(x_1,\dots,x_n)\in\mathbb{R}^n:\ x_\ell>0\big\}. \tag{11.3.37}$$

Then for each given $x \in \mathcal{O}_\ell$ define the linear map $R_{\ell,x} : \mathbb{R}^n \to \mathbb{R}^n$ by

$$R_{\ell,x}(\xi) := \xi - \frac{\xi_\ell |x| + \xi \cdot x}{|x|(|x| + x_\ell)} + \frac{(\xi \cdot x)(|x| + 2x_\ell) - \xi_\ell |x|^2}{|x|(|x| + x_\ell)} \, \mathbf{e}_\ell, \quad \xi \in \mathbb{R}^n. \tag{11.3.38}$$

By Exercise 4.125 (with $\zeta = \frac{x}{|x|}$ and $\eta := \mathbf{e}_\ell$) we have that this is a unitary transformation and

$$x \cdot R_{\ell,x}(\xi) = |x| \xi_\ell, \quad \forall\, x \in \mathcal{O}_\ell, \quad \forall\, \xi \in \mathbb{R}^n. \tag{11.3.39}$$

Also,

$$R_{\ell,\lambda\, x} = R_{\ell,x} \quad \text{whenever} \quad x \in \mathcal{O}_\ell \quad \text{and} \quad \lambda > 0 \tag{11.3.40}$$

and the joint application

$$\mathcal{O}_\ell \times \mathbb{R}^n \ni (x, \xi) \mapsto R_{\ell,x}(\xi) \in \mathbb{R}^n \quad \text{is of class } C^\infty. \tag{11.3.41}$$

Fix $j, k \in \{1, \dots, M\}$. Using the invariance under unitary transformations of the operation of integration over S^{n-1}, for each $x \in \mathcal{O}_\ell$ we may then write

$$\begin{aligned}
&\int_{S^{n-1}} \Big[(x \cdot \xi)^{2m+q} \cdot \log\Big(\frac{x \cdot \xi}{i} \Big) \Big] P^{jk}(\xi) \, d\sigma(\xi) \\
&\quad = \int_{S^{n-1}} \Big[(x \cdot R_{\ell,x}(\xi))^{2m+q} \cdot \log\Big(\frac{x \cdot R_{\ell,x}(\xi)}{i} \Big) \Big] P^{jk}(R_{\ell,x}(\xi)) \, d\sigma(\xi) \\
&\quad = \int_{S^{n-1}} \Big[(|x| \xi_\ell)^{2m+q} \cdot \log\Big(\frac{|x| \xi_\ell}{i} \Big) \Big] P^{jk}(R_{\ell,x}(\xi)) \, d\sigma(\xi) \\
&\quad = |x|^{2m+q} \int_{S^{n-1}} \xi_\ell^{2m+q} \left\{ \ln |x| + \log\Big(\frac{\xi_\ell}{i} \Big) \right\} P^{jk}(R_{\ell,x}(\xi)) \, d\sigma(\xi).
\end{aligned} \tag{11.3.42}$$

From this representation, (11.3.41), and (11.3.31), it is clear that we have $E_{jk} \in C^\infty(\mathcal{O}_\ell)$. To complete the proof of Step II, there remains to observe that $\mathbb{R}^n \setminus \{0\} = \bigcup_{\ell=1}^{n} \mathcal{O}_\ell$.

Step III. Proof of part (3) in the statement of the theorem.
To facilitate the discussion, fix $j, k \in \{1, \dots, M\}$, introduce

$$Q_{jk}(x) := \int_{S^{n-1}} (x \cdot \xi)^{2m+q} P^{jk}(\xi) \, d\sigma(\xi), \quad \forall\, x \in \mathbb{R}^n, \tag{11.3.43}$$

and define $\Psi_{jk} : \mathbb{R}^n \setminus \{0\} \to \mathbb{C}$ by setting

$$\Psi_{jk}(x) := \int_{S^{n-1}} \Big[(x \cdot \xi)^{2m+q} \cdot \log\Big(\frac{x \cdot \xi}{i} \Big) \Big] P^{jk}(\xi) \, d\sigma(\xi) - \Big(\ln |x| \Big) Q_{jk}(x) \tag{11.3.44}$$

for each $\mathbb{R}^n \setminus \{0\}$. Observe that Q_{jk} is a polynomial of degree $2m+q$ that vanishes when n is odd [since in that case the integrand in (11.3.43) is odd]. Also, from our earlier analysis in (11.3.42), we know that the integral in the right-hand side of (11.3.44) depends in a C^∞ fashion on the variable $x \in \mathbb{R}^n \setminus \{0\}$. These comments and (11.3.44) imply that Ψ_{jk} is of class C^∞ in $\mathbb{R}^n \setminus \{0\}$.

Our next goal is to prove that

$$\Psi_{jk} \quad \text{is positive homogeneous of degree } 2m+q \text{ in } \mathbb{R}^n \setminus \{0\}. \tag{11.3.45}$$

To this end, first note that for each $\ell \in \{1,\dots,n\}$ we may write

$$Q_{jk}(x) = |x|^{2m+q} \int_{S^{n-1}} \xi_\ell^{2m+q} P^{jk}(R_{\ell,x}(\xi))\, d\sigma(\xi), \quad \forall\, x \in \mathcal{O}_\ell, \tag{11.3.46}$$

by (11.3.43) and (11.3.39). Consequently, from (11.3.46), (11.3.42), and (11.3.44), we deduce that for each $\ell \in \{1,\dots,n\}$,

$$\Psi_{jk}(x) := |x|^{2m+q} \int_{S^{n-1}} \xi_\ell^{2m+q} \log\Big(\frac{\xi_\ell}{i}\Big) P^{jk}(R_{\ell,x}(\xi))\, d\sigma(\xi), \quad \forall\, x \in \mathcal{O}_\ell.$$

In turn, this and (11.3.40) readily show that Ψ_{jk} is positive homogeneous of degree $2m+q$ when restricted to the cone-like region $\mathcal{O}_\ell$. Since $\mathbb{R}^n \setminus \{0\} = \bigcup_{\ell=1}^{n} \mathcal{O}_\ell$, the claim in (11.3.45) follows.

Next, an induction argument shows that for each $\xi \in \mathbb{R}^n \setminus \{0\}$ and $k, N \in \mathbb{N}$ satisfying $N \geq 2k$, the following formulas hold:

$$\Delta_x^k\big[(x\cdot\xi)^N\big] = \frac{N!}{(N-2k)!}(x\cdot\xi)^{N-2k}|\xi|^{2k} \tag{11.3.47}$$

and

$$\Delta_x^k\big[(x\cdot\xi)^N \ln|x|\big] = (\ln|x|)\Delta_x^k\big[(x\cdot\xi)^N\big] + \sum_{r=1}^{k} c(r,k,N,n)|x|^{-2r}(x\cdot\xi)^{N-2k+2r}. \tag{11.3.48}$$

We now observe that if P_{jk} is the (j,k)-entry in the matrix $\mathbb{P}$ from (11.3.23), then

$$P_{jk}(x) = \frac{-1}{(2\pi i)^n (2m+q)!}\Delta_x^{(n+q)/2} \int_{S^{n-1}} (x\cdot\xi)^{2m+q} P^{jk}(\xi)\, d\sigma(\xi), \tag{11.3.49}$$

as seen from identity (11.3.47) with $N := 2m+q$ and $k := (n+q)/2$. It is also immediate from (11.3.49) that P_{jk} is identically zero when either n is odd (due to parity considerations) or $n > 2m$ (due to degree considerations). In addition, formula (11.3.49) shows that P_{jk} is a homogeneous

polynomial of degree $2m-n$ whenever $n\leq 2m$. Finally, we note that combining (11.3.43), (11.3.49), and (11.3.48) used with $N:=2m+q$ and $k:=(n+q)/2$, yields

$$\frac{-1}{(2\pi\,i)^n(2m+q)!}\Delta_x^{(n+q)/2}\big[(\ln|x|)Q_{jk}(x)\big]=(\ln|x|)P_{jk}(x) \tag{11.3.50}$$
$$+\sum_{r=1}^{(n+q)/2}C_r|x|^{-2r}\int_{S^{n-1}}(x\cdot\xi)^{2m-n+2r}P^{jk}(\xi)\,d\sigma(\xi),\quad\forall\,x\in\mathbb{R}^n\setminus\{0\},$$

for some constants C_r depending only on r,n,q, and m. It is easy to see that the sum in the right-hand side of (11.3.50) gives rise to a function that belongs to $C^\infty(\mathbb{R}^n\setminus\{0\})$ and is positive homogeneous of degree $2m-n$. At this point, if we define

$$\Phi_{jk}(x):=\frac{-1}{(2\pi\,i)^n(2m+q)!}\Delta_x^{(n+q)/2}\Psi_{jk}(x) \tag{11.3.51}$$
$$+\sum_{r=1}^{(n+q)/2}C_r|x|^{-2r}\int_{S^{n-1}}(x\cdot\xi)^{2m-n+2r}P^{jk}(\xi)\,d\sigma(\xi),\quad\forall\,x\in\mathbb{R}^n\setminus\{0\},$$

then (11.3.24) follows from (11.3.31), (11.3.44), (11.3.50), and (11.3.51). Moreover, from (11.3.45) and (11.3.51) it is clear that Φ_{jk} is positive homogeneous of degree $2m-n$, while the regularity of Ψ_{jk} established earlier entails $\Phi_{jk}\in C^\infty(\mathbb{R}^n\setminus\{0\})$.

Step IV. Proof of the fact that $\mathbb{E}\in\mathcal{M}_{M\times M}(\mathcal{S}'(\mathbb{R}^n))$.
Fix $j,k\in\{1,\ldots,M\}$ and recall (11.3.24). By what we proved in Step III, Φ_{jk} is positive homogeneous of degree $2m-n$ in $\mathbb{R}^n\setminus\{0\}$, and since $m\geq 1$ we may invoke Exercise 4.51 to conclude that $\Phi_{jk}\in L^1_{\rm loc}(\mathbb{R}^n)$. Now the estimate from Exercise 4.51 and Example 4.3 give $\Phi_{jk}\in\mathcal{S}'(\mathbb{R}^n)$. In addition, by Exercise 4.104, it follows that $(\ln|x|)P_{jk}\in\mathcal{S}'(\mathbb{R}^n)$. In summary, we have $E_{jk}\in\mathcal{S}'(\mathbb{R}^n)$.

Step V. Proof of (11.3.22).
Componentwise, (11.3.22) reads as follows: for each $j,k\in\{1,\ldots,M\}$,

$$\sum_{r=1}^{M}L_{jr}^x\big[E_{rk}(x-y)\big]=\begin{cases}0 & \text{if } j\neq k,\\ \delta_y(x) & \text{if } j=k,\end{cases}\quad\text{in }\ \mathcal{S}'(\mathbb{R}^n), \tag{11.3.52}$$

where the superscript x denotes the fact that the operator L_{jr} (defined as in (11.3.1)) is applied in the variable x. To justify this, fix some numbers $j,k,r\in\{1,\ldots,M\}$. By (11.3.34) and (11.3.35) we have

$$L_{jr}^xF(x\cdot\xi)=\Big[\frac{(2m+q)!}{q!}(x\cdot\xi)^q\log\frac{x\cdot\xi}{i}+C_{m,q}\,(x\cdot\xi)^q\Big]L_{jr}(\xi) \tag{11.3.53}$$

for every $x \in \mathbb{R}^n \setminus \{0\}$ and every $\xi \in S^{n-1}$ such that $x \cdot \xi \neq 0$. To continue, fix $\varphi \in C_0^\infty(\mathbb{R}^n)$ and write

$$\begin{aligned}
&\Big\langle L_{jr}^x \Big[\int_{S^{n-1}} F(x\cdot\xi) P^{rk}(\xi)\, d\sigma(\xi) \Big], \varphi(x) \Big\rangle \\
&\qquad = \Big\langle \int_{S^{n-1}} F(x\cdot\xi) P^{rk}(\xi)\, d\sigma(\xi), L_{jr}\varphi(x) \Big\rangle \\
&\qquad = \int_{\mathbb{R}^n} \int_{S^{n-1}} F(x\cdot\xi) P^{rk}(\xi)\, d\sigma(\xi) L_{jr}\varphi(x)\, dx \\
&\qquad = \int_{S^{n-1}} P^{rk}(\xi) \int_{x\in\mathbb{R}^n,\, x\cdot\xi>0} F(x\cdot\xi) L_{jr}\varphi(x)\, dx\, d\sigma(\xi) \\
&\qquad\quad + \int_{S^{n-1}} P^{rk}(\xi) \int_{x\in\mathbb{R}^n,\, x\cdot\xi<0} F(x\cdot\xi) L_{jr}\varphi(x)\, dx\, d\sigma(\xi).
\end{aligned} \tag{11.3.54}$$

At this point, we integrate by parts repeatedly with respect to x (according to formula (13.7.4)) in the innermost integrals in (11.3.54) until we transfer all the derivatives from φ onto $F(x\cdot\xi)$. This is justified by (11.3.33), the fact that $\partial_x^\alpha \big[F(x\cdot\xi)\big] \in L^1_{\rm loc}(\mathbb{R}^n)$ whenever $\alpha \in \mathbb{N}_0^n$ has $|\alpha| \leq 2m$ (cf. (11.3.35) and Exercise 2.108), and the fact that φ has compact support. Note that in the process, the terms corresponding to boundary integrals (i.e., integrals over the set $\{x \in \mathbb{R}^n : x\cdot\xi = 0\} \cap \operatorname{supp}\varphi$) are zero thanks to the formulas in (11.3.33). Hence, summing up over r the resulting identity in (11.3.54), then using (11.3.53) and the fact that $\sum_{r=1}^M L_{jr}(\xi) P^{rk}(\xi) = \delta_{jk}$ for every $\xi \in S^{n-1}$, we arrive at

$$\begin{aligned}
&\Big\langle \sum_{r=1}^M L_{jr}^x \Big[\int_{S^{n-1}} F(x\cdot\xi) P^{rk}(\xi)\, d\sigma(\xi) \Big], \varphi(x) \Big\rangle \qquad (11.3.55)\\
&= \sum_{r=1}^M \int_{S^{n-1}} P^{rk}(\xi) \int_{x\in\mathbb{R}^n} [L_{jr}^x F(x\cdot\xi)]\varphi(x)\, dx\, d\sigma(\xi) \\
&= \int_{\mathbb{R}^n} \int_{S^{n-1}} \Big[\frac{(2m+q)!}{q!} (x\cdot\xi)^q \log\frac{x\cdot\xi}{i} + C_{m,q}\,(x\cdot\xi)^q \Big] \delta_{jk}\, d\sigma(\xi)\varphi(x)\, dx.
\end{aligned}$$

Consequently, since $\varphi \in C_0^\infty(\mathbb{R}^n)$ is arbitrary, a combination of (11.3.55), (11.3.32), and (11.3.31), yields

$$\sum_{r=1}^M L_{jr} E_{rk} = \Delta^{(n+q)/2} \Big(E_q + P_q \Big) \delta_{jk} \quad \text{in} \quad \mathcal{D}'(\mathbb{R}^n), \tag{11.3.56}$$

where

$$E_q(x) := \frac{-1}{(2\pi\, i)^n q!} \int_{S^{n-1}} (x\cdot\xi)^q \cdot \log\Big(\frac{x\cdot\xi}{i}\Big)\, d\sigma(\xi), \quad \forall\, x\in\mathbb{R}^n\setminus\{0\}, \tag{11.3.57}$$

and

$$P_q(x) := \frac{-C_{m,q}}{(2\pi\, i)^n (2m+q)!} \int_{S^{n-1}} (x\cdot\xi)^q\, d\sigma(\xi), \quad \forall\, x\in\mathbb{R}^n. \tag{11.3.58}$$

Given our choice of q, from Lemma 7.31 we have that E_q is a fundamental solution for $\Delta^{(n+q)/2}$, so $\Delta^{(n+q)/2}E_q = \delta$ in $\mathcal{D}'(\mathbb{R}^n)$. Also, P_q is a homogeneous polynomial of degree q in $\mathbb{R}^n$, thus $\Delta^{(n+q)/2}P_q = 0$ pointwise and in $\mathcal{D}'(\mathbb{R}^n)$. Therefore, (11.3.56) becomes

$$\sum_{r=1}^{M} L_{jr}E_{rk} = \delta_{jk}\delta \quad \text{in} \quad \mathcal{D}'(\mathbb{R}^n), \quad \forall\, j,k\in\{1,\dots,M\}. \tag{11.3.59}$$

The statement in part *(2)* of the theorem follows from (11.3.59), (4.1.25), and the result from Step IV.

Step VI. Proof of claims in parts (4)–(7) in the statement of the theorem. The claim in part *(4)* follows from part *(1)*, (11.3.24), the fact that each P_{jk} is a homogeneous polynomial of degree at most $2m-n$, and the observation that when computing $\partial^\beta E_{jk}$ with $|\beta|\geq 2m-1$ at least one derivative falls on ln. The estimates in (11.3.25) are a direct consequence of (11.3.24). This takes care of parts *(4)* and *(5)* in the statement of the theorem.

Moving on to the proof of part *(6)*, fix $j,k\in\{1,\dots,M\}$ and recall (11.3.24). From the proof in Step IV, we have that Φ_{jk} and $(\ln|x|)P_{jk}$ are tempered distributions in $\mathbb{R}^n$. Since $\Phi_{jk}\in C^\infty(\mathbb{R}^n\setminus\{0\})$ and is positive homogeneous of degree $2m-n$ in $\mathbb{R}^n\setminus\{0\}$, Proposition 4.58 implies that its Fourier transform coincides with a C^∞ function on $\mathbb{R}^n\setminus\{0\}$. To analyze the effect of taking the Fourier transform of $(\ln|x|)P_{jk}$ pick some $\theta\in C^\infty_0(\mathbb{R}^n)$ such that $\operatorname{supp}\theta\subset B(0,2)$ and $\theta\equiv 1$ on $B(0,1)$ and write

$$(\ln|x|)P_{jk} = (1-\theta)(\ln|x|)P_{jk} + \theta(\ln|x|)P_{jk} \quad \text{in} \quad \mathbb{R}^n\setminus\{0\}. \tag{11.3.60}$$

The two terms in the right-hand side of (11.3.60) continue to belong to $\mathcal{S}'(\mathbb{R}^n)$. From Example 2.9 and the fact that θ is compactly supported, we obtain $\theta(\ln|x|)P_{jk}\in L^1(\mathbb{R}^n)$ and has compact support, thus $\mathcal{F}\big(\theta(\ln|x|)P_{jk}\big)\in C^\infty(\mathbb{R}^n)$ (recall Exercise 3.30). Regarding $(1-\theta)(\ln|x|)P_{jk}$, note that this function is of class C^∞ in $\mathbb{R}^n$. Also, for every $\beta\in\mathbb{N}_0^n$, the function $x^\beta\partial^\alpha\big[(1-\theta)(\ln|x|)P_{jk}\big]$ belongs to $L^1(\mathbb{R}^n)$ provided $\alpha\in\mathbb{N}_0^n$ and $|\alpha|$ is sufficiently large. Since the Fourier transform of any L^1 function is continuous, this readily implies that for any $r\in\mathbb{N}$ there exists $\alpha\in\mathbb{N}_0^n$ such that

$$\mathcal{F}\big(\partial^\alpha[(1-\theta)(\ln|x|)P_{jk}]\big) = (i\xi)^\alpha\mathcal{F}\big((1-\theta)(\ln|x|)P_{jk}\big)\in C^r(\mathbb{R}^n). \tag{11.3.61}$$

Thus, we necessarily have $\mathcal{F}\big((1-\theta)(\ln|x|)P_{jk}\big) \in C^\infty(\mathbb{R}^n\setminus\{0\})$.

The reasoning above shows that the Fourier transform of (the matrix-valued tempered distribution) $\mathbb{E}$ when restricted to $\mathbb{R}^n\setminus\{0\}$ is a function of class C^∞. Taking the Fourier transforms of both sides of (11.3.59) gives

$$(-1)^m\sum_{r=1}^{M} L_{jr}(\xi)\widehat{E_{rk}}(\xi) = \delta_{jk} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \quad \forall\, j,k\in\{1,\ldots,M\}. \tag{11.3.62}$$

Restricting (11.3.62) to $\mathbb{R}^n\setminus\{0\}$ then readily implies (11.3.26) and finishes the proof of part *(6)*. Finally, the identities in (11.3.27) can be seen directly from (11.3.29)–(11.3.30).

Step VII. Proof of the claim in part (8) in the statement of the theorem. Let $\mathbb{U}\in\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ be an arbitrary fundamental solution of the system L in $\mathbb{R}^n$ and set $\mathbb{Q}:=\mathbb{U}-\mathbb{E}$. Then $L\mathbb{Q}=0$ in $\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ and, on the Fourier transform side, $\mathbb{Q}$ satisfies $L(\xi)\widehat{\mathbb{Q}}=0$ in $\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)$. In light of (11.3.3), this implies $\operatorname{supp}\widehat{\mathbb{Q}}\subseteq\{0\}$, hence $\mathbb{Q}$ is an $M\times M$ matrix with entries that are polynomials in $\mathbb{R}^n$ by Exercise 4.35 (applied to each entry).

Step VIII. Proof of the fact that the entries of $\mathbb{E}$ are real-analytic in $\mathbb{R}^n\setminus\{0\}$. This is a direct consequence of the fact that (cf. (11.3.22))

$$L\,\mathbb{E} = \delta\, I_{M\times M} \quad \text{in} \quad \mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big), \tag{11.3.63}$$

where the operator L is applied to each column of $\mathbb{E}$, which implies that $L\,\mathbb{E}=0$ in $\mathcal{M}_{M\times M}\big(\mathcal{D}'(\mathbb{R}^n\setminus\{0\})\big)$, and Theorem 11.4, established in the next section.

The proof of Theorem 11.1 is now complete. □

Theorem 11.1 describes all fundamental solutions, which are tempered distributions, for any homogeneous constant coefficient system with an invertible characteristic matrix, and elaborates on the properties of such fundamental solutions. In specific cases, it is possible to use formulas (11.3.19)–(11.3.20) to find an explicit expression for a specific fundamental solution. The case of the poly-harmonic operator is discussed in Exercise 11.21. Here we study in detail the case of the three-dimensional Lamé operator (cf. also Exercises 11.17 and 11.18). We remark that the argument in the proof of Proposition 11.2 is different from the one used to derive the fundamental solution for the Lamé operator in Sect. 11.1.

Proposition 11.2. *Let $\lambda,\,\mu\in\mathbb{C}$ be such that $\mu\neq 0$, $\lambda+2\mu\neq 0$. A fundamental solution for the Lamé operator (10.3.1) in $\mathbb{R}^3$ is*

$$E(x) = -\frac{1}{8\pi}\,\frac{\lambda+3\mu}{\mu(\lambda+2\mu)}\,\frac{1}{|x|}\,I_{3\times 3} - \frac{1}{8\pi}\,\frac{\lambda+\mu}{\mu(\lambda+2\mu)}\,\frac{x\otimes x}{|x|^3}, \quad x\in\mathbb{R}^3\setminus\{0\}. \tag{11.3.64}$$

Proof. We start by recalling formula (11.3.29) that gives an expression for the fundamental solution of a homogeneous differential operator for n odd. In our case, $L = (L_{jk})_{1\leq j,k\leq 3}$ and $L_{jk} = \mu\delta_{jk}\Delta + (\lambda+\mu)\partial_j\partial_k$, thus

$$L(\xi) = \mu I_{3\times 3} + (\lambda+\mu)\xi\otimes\xi \,\text{for}\, \xi\in S^2.$$

Observe that $(x\cdot\xi)\,\text{sgn}\,(x\cdot\xi) = |x\cdot\xi|$. As such, for any $x\in\mathbb{R}^3\setminus\{0\}$ we may compute

$$\begin{aligned}
E(x) &= -\frac{1}{16\pi^2}\Delta_x\int_{S^2}|x\cdot\xi|(L(\xi))^{-1}\,\mathrm{d}\sigma(\xi) \qquad (11.3.65)\\
&= -\frac{1}{16\pi^2}\Delta_x\int_{S^2}|x\cdot\xi|\Big(\frac{1}{\mu}I_{3\times 3}-\frac{\lambda+\mu}{\mu(\lambda+2\mu)}\xi\otimes\xi\Big)\,\mathrm{d}\sigma(\xi)\\
&= -\frac{1}{16\pi^2}\Delta_x\Big[\frac{\omega_1}{\mu}|x|I_{3\times 3}-\frac{\lambda+\mu}{\mu(\lambda+2\mu)}\frac{\omega_1}{4}\Big(|x|I_{3\times 3}+\frac{x\otimes x}{|x|}\Big)\Big]\\
&= -\frac{1}{16\pi}\Delta_x\Big[\frac{4\lambda+8\mu-\lambda-\mu}{2\mu(\lambda+2\mu)}|x|I_{3\times 3}-\frac{\lambda+\mu}{2\mu(\lambda+2\mu)}\frac{x\otimes x}{|x|}\Big]\\
&= -\frac{1}{16\pi}\Big[\frac{3\lambda+7\mu}{2\mu(\lambda+2\mu)}\frac{2}{|x|}I_{3\times 3}-\frac{\lambda+\mu}{2\mu(\lambda+2\mu)}\Big(\frac{2}{|x|}I_{3\times 3}-\frac{4x\otimes x}{|x|^3}\Big)\Big]\\
&= -\frac{1}{8\pi}\frac{\lambda+3\mu}{\mu(\lambda+2\mu)}\frac{1}{|x|}I_{3\times 3}-\frac{1}{8\pi}\frac{\lambda+\mu}{\mu(\lambda+2\mu)}\frac{x\otimes x}{|x|^3}.
\end{aligned}$$

For the second equality in (11.3.65) we have used Proposition 10.14, for the third we have used Proposition 13.47 and Proposition 13.48, while for the fifth we have used (7.3.2) and the readily verified fact that if N is a nonzero integer then

$$\Delta_x\Big[\frac{x\otimes x}{|x|^N}\Big] = \frac{2}{|x|^N}I_{n\times n}+\frac{N(N-n-2)}{|x|^{N+2}}\,x\otimes x, \qquad (11.3.66)$$

for each $x\in\mathbb{R}^n\setminus\{0\}$. □

11.4 Interior Estimates and Real-Analyticity for Null-Solutions of Systems

The aim in this section is to explore the extent to which results such as integral representation formula and interior estimates, proved earlier in Sect. 6.3 in the scalar context, continue to hold for vector-valued functions that are null-solutions of a certain class of systems of differential operators.

Proposition 11.3. *Fix $n,m,M\in\mathbb{N}$ with $n\geq 2$, and assume that L is an $M\times M$ system in $\mathbb{R}^n$ of order $2m$ of the form*

$$L = \sum_{|\alpha|=2m}A_\alpha\partial^\alpha, \quad A_\alpha = \big(a_\alpha^{jk}\big)_{1\leq j,k\leq M}\in\mathcal{M}_{M\times M}(\mathbb{C}), \qquad (11.4.1)$$

with the property that $\det[L(\xi)] \neq 0$ *for each* $\xi \in \mathbb{R}^n \setminus \{0\}$. *In addition, suppose* $\Omega \subseteq \mathbb{R}^n$ *is an open set and* $\mathbf{u} = (u_1, \ldots, u_M) \in \big[\mathcal{D}'(\Omega)\big]^M$ *is such that* $L\mathbf{u} = 0$ *in* $\big[\mathcal{D}'(\Omega)\big]^M$.

Then $\mathbf{u} \in \big[C^\infty(\Omega)\big]^M$ *and for each* $x_0 \in \Omega$, *each* $r \in \big(0, \operatorname{dist}(x_0, \partial\Omega)\big)$, *and each function* $\psi \in C_0^\infty\big(B(x_0, r)\big)$ *such that* $\psi = 1$ *near* $\overline{B(x_0, r/2)}$, *we have*

$$u_\ell(x) = -\sum_{j,k=1}^{M} \sum_{|\alpha|=2m} \sum_{\gamma<\alpha} (-1)^{|\gamma|}\, a_\alpha^{jk} \frac{\alpha!}{\gamma!(\alpha-\gamma)!} \times$$

$$\times \int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}} (\partial^\gamma E_{j\ell})(x-y)\partial^{\alpha-\gamma}\psi(y)\, u_k(y)\, dy \tag{11.4.2}$$

for each $\ell \in \{1, \ldots, M\}$ *and each* $x \in B(x_0, r/2)$, *where*

$$\mathbb{E} = \big(E_{jk}\big)_{1\leq j,k\leq M} \in \mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big) \cap \mathcal{M}_{M\times M}\big(C^\infty(\mathbb{R}^n \setminus \{0\})\big) \tag{11.4.3}$$

is the fundamental matrix for $L^\top$ *(the transpose of* L*) as given by Theorem 11.1.*

In particular, for every $\mu \in \mathbb{N}_0^n$,

$$\partial^\mu u_\ell(x) = -\sum_{j,k=1}^{M} \sum_{|\alpha|=2m} \sum_{\gamma<\alpha} (-1)^{|\gamma|}\, a_\alpha^{jk} \frac{\alpha!}{\gamma!(\alpha-\gamma)!} \times$$

$$\times \int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}} (\partial^{\gamma+\mu} E_{j\ell})(x-y)\partial^{\alpha-\gamma}\psi(y)\, u_k(y)\, dy \tag{11.4.4}$$

for each $\ell \in \{1, \ldots, M\}$ *and each* $x \in B(x_0, r/2)$. *Also, if either* n *is odd, or* $n > 2m$, *or if* $|\mu| > 2m - n$, *we have*

$$|(\partial^\mu \mathbf{u})(x_0)| \leq \frac{C_\mu}{r^{|\mu|}} \fint_{B(x_0,r)} |\mathbf{u}(y)|\, dy, \tag{11.4.5}$$

where $C_\mu \in (0, \infty)$ *is independent of* $\mathbf{u}$, x_0, r, *and* Ω.

Proof. Let $\mathbf{u} = (u_\ell)_{1\leq\ell\leq M}$ and $\mathbb{E} = \big(E_{jk}\big)_{1\leq j,k\leq M}$ be as specified in the statement of the proposition. In particular, by Theorem 11.1,

$$\mathbb{E} \in \mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big) \cap \mathcal{M}_{M\times M}\big(C^\infty(\mathbb{R}^n \setminus \{0\})\big).$$

Also, pick an arbitrary point $x_0 \in \Omega$, a number $r \in \big(0, \operatorname{dist}(x_0, \partial\Omega)\big)$, and fix some function $\psi \in C_0^\infty\big(B(x_0, r)\big)$ such that $\psi = 1$ near $\overline{B(x_0, r/2)}$. Then, from Theorem 10.9 it follows that $\mathbf{u} \in \big[C^\infty(\Omega)\big]^M$, hence also $\psi\mathbf{u} \in \big[C_0^\infty(\Omega)\big]^M$. Granted these, for each $x \in B(x_0, r/2)$ we may then write (keeping in mind that $\mathbb{E}$ is a fundamental solution for $L^\top$, the transposed of L),

$$\mathbf{u}(x) = (\psi\mathbf{u})(x) = \big((\psi u_\ell)(x)\big)_{1\leq\ell\leq M}$$

$$= \Big(\sum_{k=1}^{M} \big\langle \delta_{k\ell}\, \delta\,,\, (\psi u_k)(x-\cdot)\big\rangle\Big)_{1\leq\ell\leq M}$$

$$= \Big(\sum_{k=1}^{M} \big\langle (L^{\top}\mathbb{E})_{k\ell}\,,\, (\psi u_k)(x-\cdot) \big\rangle \Big)_{1\leq \ell \leq M}. \tag{11.4.6}$$

Note that

$$L^{\top} = \sum_{|\alpha|=2m} (-1)^{|\alpha|} A_{\alpha}^{\top} \partial^{\alpha} = \sum_{|\alpha|=2m} A_{\alpha}^{\top} \partial^{\alpha}, \tag{11.4.7}$$

where $A_{\alpha}^{\top}$ is the transposed of the matrix A_{α}, for each α. Thus, (11.4.6) implies

$$\begin{aligned}
\mathbf{u}(x) &= \Big(\sum_{k=1}^{M} \Big\langle \sum_{|\alpha|=2m} \sum_{j=1}^{M} (A_{\alpha}^{\top})_{kj} \partial^{\alpha} E_{j\ell}, (\psi u_k)(x-\cdot) \Big\rangle \Big)_{1\leq \ell \leq M} \qquad (11.4.8)\\
&= \Big(\sum_{k=1}^{M} \Big\langle \sum_{|\alpha|=2m} \sum_{j=1}^{M} a_{\alpha}^{jk} E_{j\ell}, \partial^{\alpha} \big[(\psi u_k)(x-\cdot) \big] \Big\rangle \Big)_{1\leq \ell \leq M}\\
&= \Big(\sum_{j=1}^{M} \Big\langle E_{j\ell}, \sum_{|\alpha|=2m} \sum_{k=1}^{M} a_{\alpha}^{jk} (\partial^{\alpha}(\psi u_k))(x-\cdot) \Big\rangle \Big)_{1\leq \ell \leq M}\\
&= \Big(\sum_{j=1}^{M} \Big\langle E_{j\ell}(x-\cdot), \sum_{|\alpha|=2m} \sum_{k=1}^{M} a_{\alpha}^{jk} \sum_{0<\beta\leq\alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!} \partial^{\beta}\psi \partial^{\alpha-\beta} u_k \Big\rangle \Big)_{1\leq \ell \leq M}\\
&\quad + \Big(\sum_{j=1}^{M} \Big\langle E_{j\ell}(x-\cdot), \psi \sum_{|\alpha|=2m} \sum_{k=1}^{M} a_{\alpha}^{jk} \partial^{\alpha} u_k \Big\rangle \Big)_{1\leq \ell \leq M}\\
&= \Big(\sum_{j,k=1}^{M} \sum_{|\alpha|=2m} \sum_{0<\beta\leq\alpha} a_{\alpha}^{jk} \frac{\alpha!}{\beta!(\alpha-\beta)!} \Big\langle E_{j\ell}(x-\cdot), \partial^{\beta}\psi\, \partial^{\alpha-\beta} u_k \Big\rangle \Big)_{1\leq \ell \leq M}
\end{aligned}$$

since for each $j \in \{1,\ldots,M\}$,

$$\psi \sum_{|\alpha|=2m} \sum_{k=1}^{M} a_{\alpha}^{jk} \partial^{\alpha} u_k = \psi (L\mathbf{u})_j = 0 \ \text{ in } \ \mathbb{R}^n. \tag{11.4.9}$$

Next, for each $\ell, j, k \in \{1,\ldots,M\}$ and each $\alpha, \beta \in \mathbb{N}_0^n$ with $|\alpha| = 2m$ and $0 < \beta \leq \alpha$, we observe that

$$\begin{aligned}
&\big\langle E_{j\ell}(x-\cdot), \partial^{\beta}\psi\, \partial^{\alpha-\beta} u_k \big\rangle\\
&\qquad = \int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}} E_{j\ell}(x-y) \partial^{\beta}\psi(y) \partial^{\alpha-\beta} u_k(y)\, dy\\
&\qquad = (-1)^{|\beta|} \int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}} \partial_y^{\alpha-\beta} \big[E_{j\ell}(x-y) \partial^{\beta}\psi(y) \big] u_k(y)\, dy\\
&\qquad = \sum_{\gamma\leq\alpha-\beta} (-1)^{|\beta|+|\gamma|} \frac{(\alpha-\beta)!}{\gamma!(\alpha-\beta-\gamma)!} \times
\end{aligned}$$

$$\times \int_{B(x_0,r)\setminus\overline{B(x_0,r/2)}} (\partial^\gamma E_{j\ell})(x-y)\partial^{\alpha-\gamma}\psi(y)\,u_k(y)\,dy, \quad (11.4.10)$$

and that, whenever $\gamma < \alpha$,

$$\sum_{0<\beta\leq\alpha-\gamma} (-1)^{|\beta|}\frac{(\alpha-\gamma)!}{\beta!(\alpha-\beta-\gamma)!} = \Big(\sum_{\beta\leq\alpha-\gamma} (-1)^{|\beta|}\frac{(\alpha-\gamma)!}{\beta!(\alpha-\beta-\gamma)!}\Big) - 1$$

$$= 0 - 1 = -1. \quad (11.4.11)$$

At this stage, (11.4.2) follows from (11.4.8), (11.4.10), and (11.4.11). In turn, (11.4.2) readily implies (11.4.4). Finally, (11.4.5) is a consequence of (11.4.4), the assumptions on n and μ, and (11.3.25), by choosing ψ as in (6.3.11)–(6.3.12). □

An L^∞-version of the interior estimate (11.4.5), valid in all space dimensions, is established in the next theorem. In particular, this version allows us to prove the real-analyticity of null-solutions of the systems considered here.

Theorem 11.4. *Let $n, m, M \in \mathbb{N}$ and suppose L is an $M \times M$ constant (complex) coefficient system in $\mathbb{R}^n$, homogeneous of order $2m$, and with the property that $\det[L(\xi)] \neq 0$ for each $\xi \in \mathbb{R}^n \setminus \{0\}$. Assume also that $\Omega \subseteq \mathbb{R}^n$ is an open set and $\mathbf{u} \in \big[\mathcal{D}'(\Omega)\big]^M$ is such that $L\mathbf{u} = 0$ in $\big[\mathcal{D}'(\Omega)\big]^M$.*

Then $\mathbf{u} \in \big[C^\infty(\Omega)\big]^M$ and there exists a constant $C \in (0,\infty)$ such that

$$\max_{y\in\overline{B(x,\lambda r)}} |\partial^\alpha \mathbf{u}(y)| \leq \frac{C^{|\alpha|}(1-\lambda)^{-|\alpha|}|\alpha|!}{r^{|\alpha|}} \max_{y\in\overline{B(x,r)}} |\mathbf{u}(y)|, \quad \forall\,\alpha\in\mathbb{N}_0^n, \quad (11.4.12)$$

for each $x \in \Omega$, each $r \in \big(0, \operatorname{dist}(x,\partial\Omega)\big)$, and each $\lambda \in (0,1)$. In particular, each component of $\mathbf{u}$ is real-analytic in Ω.

Proof. Theorem 10.9 gives that $\mathbf{u} \in \big[C^\infty(\Omega)\big]^M$. As far as (11.4.12) is concerned, consider first the case when either n is odd, or $n > 2m$. In this scenario, (11.4.5) gives that there exists some $C \in (0,\infty)$, independent of $\mathbf{u}$ and Ω, such that

$$|\partial_j \mathbf{u}(x)| \leq \frac{C}{r} \fint_{B(x,r)} |\mathbf{u}(y)|\,dy, \quad \forall\, j \in \{1,\dots,n\}, \quad (11.4.13)$$

whenever $x \in \Omega$ and $r \in \big(0, \operatorname{dist}(x,\partial\Omega)\big)$. A quick inspection reveals that the proof of Lemma 6.21 carries through in the case when the functions involved are vector-valued. When applied with

$$\mathcal{A} := \big\{\mathbf{u} \in \big[C^\infty(\Omega)\big]^M : L\mathbf{u} = 0 \ \text{ in } \ \Omega\big\}, \quad (11.4.14)$$

this lemma gives [thanks to (11.4.13)] that for every $x \in \Omega$, every $r \in \big(0, \operatorname{dist}(x,\partial\Omega)\big)$, every $k \in \mathbb{N}$, and every $\lambda \in (0,1)$, we have (with C as in (11.4.13))

$$\max_{y\in\overline{B(x,\lambda r)}} |\partial^\alpha \mathbf{u}(y)| \leq \frac{C^k(1-\lambda)^{-k}\mathrm{e}^{k-1}k!}{r^k} \max_{y\in\overline{B(x,r)}} |\mathbf{u}(y)|, \quad (11.4.15)$$

for every multi-index $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = k$.

This proves (11.4.12) under the current assumptions on n.

Fix now an arbitrary $n \in \mathbb{N}$ and pick some $k \in \mathbb{N}$. With L as in (11.4.1) consider the constant coefficient, homogeneous, $M \times M$ system in $\mathbb{R}^n$, of order $4m$, given by

$$\mathscr{L}(\partial) := L^* L = \sum_{|\alpha|=|\beta|=2m} A_\alpha^* A_\beta \partial^{\alpha+\beta}, \tag{11.4.16}$$

and note that for each $\xi \in \mathbb{R}^n$ we have

$$\mathscr{L}(\xi) = \sum_{|\alpha|=|\beta|=2m} \xi^{\alpha+\beta} A_\alpha^* A_\beta = \sum_{|\alpha|=|\beta|=2m} \big(\xi^\alpha A_\alpha\big)^* \big(\xi^\beta A_\beta\big) = \big(L(\xi)\big)^* L(\xi). \tag{11.4.17}$$

In particular,

$$\mathscr{L}(\xi)\eta \cdot \overline{\eta} = |L(\xi)\eta|^2, \quad \forall\, \xi \in \mathbb{R}^n, \quad \forall\, \eta \in \mathbb{C}^M. \tag{11.4.18}$$

Finally, define the constant coefficient, homogeneous, $M \times M$ system in $\mathbb{R}^{n+k}$, of order $4m$,

$$\widetilde{\mathscr{L}}(\partial, \partial_{n+1}, \ldots, \partial_{n+k}) := \mathscr{L}(\partial) + \big(\partial_{n+1}^{4m} + \cdots + \partial_{n+k}^{4m}\big) I_{M\times M}. \tag{11.4.19}$$

We claim that

$$\det\big[\widetilde{\mathscr{L}}(\xi, \xi_{n+1}, \ldots, \xi_{n+k})\big] \neq 0, \quad \forall\, (\xi, \xi_{n+1}, \ldots, \xi_{n+k}) \in \mathbb{R}^{n+k} \setminus \{0\}. \tag{11.4.20}$$

To see that this is the case, fix some $(\xi, \xi_{n+1}, \ldots, \xi_{n+k}) \in \mathbb{R}^{n+k} \setminus \{0\}$ and note that it suffices to show that the $M \times M$ complex matrix $\widetilde{\mathscr{L}}(\xi, \xi_{n+1}, \ldots, \xi_{n+k})$ acts injectively on $\mathbb{C}^M$. With this in mind, let $\eta \in \mathbb{C}^M$ be such that $\widetilde{\mathscr{L}}(\xi, \xi_{n+1}, \ldots, \xi_{n+k})\eta = 0 \in \mathbb{C}^M$. Then

$$0 = \widetilde{\mathscr{L}}(\xi, \xi_{n+1}, \ldots, \xi_{n+k})\eta \cdot \overline{\eta} = |L(\xi)\eta|^2 + \big(\xi_{n+1}^{4m} + \cdots + \xi_{n+k}^{4m}\big)|\eta|^2 \tag{11.4.21}$$

which forces

$$L(\xi)\eta = 0 \quad \text{and} \quad \big(\xi_{n+1}^{4m} + \cdots + \xi_{n+k}^{4m}\big)|\eta|^2 = 0. \tag{11.4.22}$$

If $\xi \in \mathbb{R}^n \setminus \{0\}$ then the first condition in (11.4.22) implies $\eta = 0$, given the assumptions on L. If $\xi = 0$ then necessarily there exists $j \in \{1, \ldots, k\}$ such that $\xi_{n+j} \neq 0$, and the second condition in (11.4.22) now implies $\eta = 0$. Thus, $\eta = 0$ in all alternatives which finishes the proof of (11.4.20).

To proceed, assume $\mathbf{u} \in \big[C^\infty(\Omega)\big]^M$ satisfies $L\mathbf{u} = 0$ in Ω, and define

$$\widetilde{\mathbf{u}}(x, x_{n+1}, \ldots, x_{n+k}) := \mathbf{u}(x), \quad \forall\, (x, x_{n+1}, \ldots, x_{n+k}) \in \widetilde{\Omega} := \Omega \times \mathbb{R}^k. \tag{11.4.23}$$

Then $\widetilde{\Omega}$ is an open subset of $\mathbb{R}^{n+k}$ and $\widetilde{\mathbf{u}} \in \big[C^\infty(\widetilde{\Omega})\big]^M$. Moreover, as is apparent from (11.4.16), (11.4.19), and (11.4.23), we have

$$\widetilde{\mathscr{L}}(\partial, \partial_{n+1}, \ldots, \partial_{n+k})\widetilde{\mathbf{u}} = 0 \quad \text{in} \quad \widetilde{\Omega}. \tag{11.4.24}$$

Assume now that $n \in \mathbb{N}$ is an arbitrary given number and pick $k \in \mathbb{N}$ such that $n+k$ is either odd or $n+k > 4m$. Then from the first part in the proof, applied to $\widetilde{\mathscr{L}}$ and $\widetilde{\mathbf{u}}$, we know that there exists a constant $C \in (0,\infty)$ such that

$$\max_{B(\widetilde{x},\lambda r)} |\partial^\alpha \widetilde{\mathbf{u}}| \leq \frac{C^{|\alpha|}(1-\lambda)^{-|\alpha|}|\alpha|!}{r^{|\alpha|}} \max_{B(\widetilde{x},r)} |\widetilde{\mathbf{u}}|, \quad \forall\, \alpha \in \mathbb{N}_0^n, \tag{11.4.25}$$

for each $\widetilde{x} \in \widetilde{\Omega}$, each $r \in \big(0, \operatorname{dist}(\widetilde{x}, \partial\widetilde{\Omega})\big)$, and each $\lambda \in (0,1)$, where the balls appearing in (11.4.25) are considered in $\mathbb{R}^{n+k}$. Now, given $x \in \Omega$ and $r \in \big(0, \operatorname{dist}(x,\partial\Omega)\big)$, specializing (11.4.25) to the case when $\widetilde{x} := (x, 0, \ldots, 0)$ yields (11.4.12) for the current n. Thus, (11.4.12) holds for any n. Finally, the last claim in the statement of the theorem is a consequence of (11.4.12) and Lemma 6.24. □

11.5 Reverse Hölder Estimates for Null-Solutions of Systems

The principal result in this section is the version of interior estimates for null-solutions of certain higher-order systems stated in Theorem 11.9. In particular, this contains the fact that such null-solutions satisfy reverse Hölder estimates (a topic worthy of investigation in its own right). We begin by giving the following definition.

Definition 11.5. *A continuous (complex) vector-valued function* $\mathbf{u}$ *defined in* Ω *is said to be* p-`subaveraging` *for some* $p \in (0,\infty)$ *if there exists a finite constant* $C > 0$ *with the property that*

$$|\mathbf{u}(x)| \leq C \left(\fint_{B(x,r)} |\mathbf{u}(y)|^p \, dy \right)^{\frac{1}{p}} \tag{11.5.1}$$

for every $x \in \Omega$ *and every* $r \in \big(0, \operatorname{dist}(x,\partial\Omega)\big)$.

The class of p-subaveraging functions exhibits a number of self-improvement properties, the first of which is discussed below.

Lemma 11.6. *Assume that* $\mathbf{u}$ *is a continuous (complex) vector-valued function defined in* Ω, *and suppose that* $0 < p < \infty$. *Then* $\mathbf{u}$ *is* p*-subaveraging if and only if there exists a finite constant* $C > 0$ *such that*

$$\sup_{z \in B(x,\lambda r)} |\mathbf{u}(z)| \leq C(1-\lambda)^{-n/p} \left(\fint_{B(x,r)} |\mathbf{u}(y)|^p \, dy \right)^{\frac{1}{p}} \tag{11.5.2}$$

for every $x \in \Omega$, *every* $r \in \big(0, \operatorname{dist}(x,\partial\Omega)\big)$, *and every* $\lambda \in (0,1)$.

Proof. The fact that (11.5.2) implies (11.5.1) is obvious. Conversely, suppose that u is p-subaveraging. Pick $x \in \Omega$, $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, and $z \in B(x, \lambda r)$. Then, if $R := (1-\lambda)r$, it follows that $z \in \Omega$ and $0 < R < \operatorname{dist}(z, \partial\Omega)$. Furthermore, $B(z, R) \subseteq B(x, r)$. Consequently, with C as in (11.5.1),

$$|\mathbf{u}(z)| \leq C \left(\fint_{B(z,R)} |\mathbf{u}(y)|^p \, dy \right)^{\frac{1}{p}} = C \left(\frac{(1-\lambda)^{-n}}{|B(z,r)|} \int_{B(z,R)} |\mathbf{u}(y)|^p \, dy \right)^{\frac{1}{p}}$$

$$\leq C(1-\lambda)^{-n/p} \left(\fint_{B(x,r)} |\mathbf{u}(y)|^p \, dy \right)^{\frac{1}{p}}, \tag{11.5.3}$$

which readily implies (11.5.2) by taking the supremum over $z \in B(x, \lambda r)$. □

The second self-improvement within the class of p-subaveraging functions is the fact that the value of the parameter p is immaterial.

Lemma 11.7. *If there exists $p_0 > 0$ such that $\mathbf{u}$ is p_0-subaveraging function, then $\mathbf{u}$ is p-subaveraging for every $p \in (0, \infty)$.*

In light of Lemma 11.7, it is unequivocal to refer to a function $\mathbf{u}$ as simply being `subaveraging` if it is p-subaveraging for some $p \in (0, \infty)$. The optimal constant which can be used in (11.5.1) is referred to as the p-subaveraging constant of the function $\mathbf{u}$.

Proof of Lemma 11.7. The proof is based on ideas used in the work of Hardy and Littlewood [26] (cf. also [14, Lemma 2, pp. 172–173]). The case $p > p_0$ can be handled directly utilizing Hölder's inequality with $q = \frac{p}{p_0} > 1$. Henceforth we shall focus on the case when $p < p_0$. Replacing $\mathbf{u}$ by a suitable power of $|\mathbf{u}|$, there is no loss of generality in assuming that, in fact, $p_0 = 1$ and $p < 1$. We may also assume (by rescaling and making a translation) that $\overline{B(0,1)} \subseteq \Omega$, $x = 0$, and $\int_{B(0,1)} |\mathbf{u}(y)|^p \, dy = 1$. The goal is then to prove the estimate $|\mathbf{u}(0)| \leq C$ with a finite constant $C > 0$ independent of $\mathbf{u}$. Continuing our series of reductions, we may assume that $|\mathbf{u}(0)| > 1$. Next, for each $r \in (0, 1]$ and $q \in (0, \infty]$, introduce

$$m_q(r) := \begin{cases} \left(\int_{B(0,r)} |\mathbf{u}(y)|^p \, dy \right)^{\frac{1}{p}} & \text{if } \ q < \infty, \\ \\ \sup_{B(0,r)} |\mathbf{u}(y)| & \text{if } \ q = \infty. \end{cases} \tag{11.5.4}$$

Observe that for each $r \in (0, 1)$,

$$m_1(r) \leq \big(m_p(r)\big)^p \big(m_\infty(r)\big)^{1-p} \leq \big(m_\infty(r)\big)^{1-p}, \quad \forall\, p \in (0,1), \tag{11.5.5}$$

where the last inequality holds by virtue of the trivial estimate $m_p(r) \leq m_p(1)$, valid for every $r \in (0, 1)$, and the assumption $m_p(1) = 1$. On the other hand, for every $x \in \Omega$ and every $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$

$$|\mathbf{u}(x)| \leq \frac{C}{r^n} \int_{B(x,r)} |\mathbf{u}(y)| \, dy, \tag{11.5.6}$$

and, consequently,

$$|\mathbf{u}(z)| \leq \frac{C}{(r-\rho)^n} \int_{B(z,r-\rho)} |\mathbf{u}(y)|\,\mathrm{d}y \leq \frac{C}{(r-\rho)^n} \int_{B(0,r)} |\mathbf{u}(y)|\,\mathrm{d}y \quad (11.5.7)$$

whenever $|z| = \rho \in (0, r)$. Then for any $z^* \in B(0, \rho)$ such that $|z^*| = \rho^* < \rho$ we obtain

$$|\mathbf{u}(z^*)| \leq \frac{C}{(r-\rho^*)^n} \int_{B(0,r)} |\mathbf{u}(y)|\,\mathrm{d}y \leq \frac{C}{(r-\rho)^n} \int_{B(0,r)} |\mathbf{u}(y)|\,\mathrm{d}y, \quad (11.5.8)$$

which, in concert with (11.5.5), yields the estimate

$$m_\infty(\rho) \leq \frac{C}{(r-\rho)^n}\, m_\infty(r)^{1-p}. \quad (11.5.9)$$

To continue, set $\rho := r^a$ for some $a \in (1, \infty)$ to be specified momentarily. Then (11.5.9) entails

$$\begin{aligned} \int_{1/2}^1 \ln\, m_\infty(r^a)\, \frac{\mathrm{d}r}{r} &\leq C + n \int_{1/2}^1 \ln\, \frac{1}{(r-r^a)}\, \frac{\mathrm{d}r}{r} \\ &\quad + (1-p) \int_{1/2}^1 \ln\, m_\infty(r)\, \frac{\mathrm{d}r}{r}, \end{aligned} \quad (11.5.10)$$

and for the first integral above the change of variables $t := r^a$ gives

$$\int_{1/2}^1 \ln\, m_\infty(r^a)\, \frac{\mathrm{d}r}{r} = \frac{1}{a} \int_{(1/2)^a}^1 \ln\, m_\infty(t)\, \frac{\mathrm{d}t}{t}. \quad (11.5.11)$$

Since our assumption $|\mathbf{u}(0)| > 1$ implies $m_\infty(t) \geq 1$, the right-hand side of (11.5.11) is bounded from below by

$$\frac{1}{a} \int_{1/2}^1 \ln\, m_\infty(r)\, \frac{\mathrm{d}r}{r}. \quad (11.5.12)$$

Therefore, (11.5.10)–(11.5.12) imply

$$\Big(\frac{1}{a} - 1 + p\Big) \int_{1/2}^1 \ln\, m_\infty(r)\, \frac{\mathrm{d}r}{r} \leq C + C \int_{1/2}^1 \ln\, \frac{1}{(r-r^a)}\, \frac{\mathrm{d}r}{r} \leq C < \infty. \quad (11.5.13)$$

Choose now $a > 1$ such that $\frac{1}{a} - 1 + p > 0$. Then (11.5.13) forces

$$\int_{1/2}^1 \ln\, m_\infty(r)\, \mathrm{d}r \leq C, \quad (11.5.14)$$

and hence, $\ln\, m_\infty(1/2) \leq C$ for some finite constant $C > 0$ independent of initial function $\mathbf{u}$. In concert with the inequality $|\mathbf{u}(0)| \leq m_\infty(1/2)$, this finishes the proof of the lemma. □

There are certain connections between the subaveraging property and reverse Hölder estimates, brought to light by our next two results.

Lemma 11.8. *Let* $\mathbf{u}$ *be a subaveraging function. Then for every* $p, q \in (0, \infty)$ *and* $\lambda \in (0, 1)$ *the following reverse Hölder estimate holds*

$$\Big(\fint_{B(x,\lambda r)} |\mathbf{u}(y)|^q \, dy\Big)^{\frac{1}{q}} \leq C\Big(\fint_{B(x,r)} |\mathbf{u}(y)|^p \, \mathrm{d}y\Big)^{\frac{1}{p}}, \tag{11.5.15}$$

for $x \in \Omega$ *and* $0 < r < \operatorname{dist}(x, \partial\Omega)$, *where* $C > 0$ *is a finite constant depending only on* p, q, λ, n *and the* p*-subaveraging constant of* $\mathbf{u}$.

Proof. Given $x \in \Omega$ and $0 < r < \operatorname{dist}(x, \partial\Omega)$, we write

$$\Big(\fint_{B(x,\lambda r)} |\mathbf{u}(y)|^q \, dy\Big)^{\frac{1}{q}} \leq \sup_{z \in B(x,\lambda r)} |\mathbf{u}(z)| \leq C\Big(\fint_{B(x,r)} |\mathbf{u}(y)|^p \, \mathrm{d}y\Big)^{\frac{1}{p}}, \tag{11.5.16}$$

by Lemma 11.6. □

The main result in this section is the combination of interior estimates and reverse Hölder estimates contained in the following theorem.

Theorem 11.9. *Let* $n, m, M \in \mathbb{N}$ *and suppose* L *is an* $M \times M$ *constant (complex) coefficient system in* $\mathbb{R}^n$, *homogeneous of order* $2m$, *and with the property that* $\det[L(\xi)] \neq 0$ *for each* $\xi \in \mathbb{R}^n \setminus \{0\}$. *Assume also that* $\Omega \subseteq \mathbb{R}^n$ *is an open set and* $\mathbf{u} \in \big[\mathcal{D}'(\Omega)\big]^M$ *is such that* $L\mathbf{u} = 0$ *in* $\big[\mathcal{D}'(\Omega)\big]^M$.

Then $\mathbf{u} \in \big[C^\infty(\Omega)\big]^M$ *and* $\mathbf{u}$ *is subaveraging. Moreover, there exists a constant* $C = C(L, n) \in (0, \infty)$ *such that given* $p \in (0, \infty)$ *there exists* $c = c(L, n, p) \in (0, \infty)$ *with the property that*

$$\max_{y \in \overline{B(x,\lambda r)}} |\partial^\alpha \mathbf{u}(y)| \leq c\,(1-\lambda)^{-|\alpha|-n/p} \frac{C^{|\alpha|}|\alpha|!}{r^{|\alpha|}} \Big(\fint_{B(x,r)} |\mathbf{u}(y)|^p \, \mathrm{d}y\Big)^{1/p}, \tag{11.5.17}$$

whenever $x \in \Omega$, $0 < r < \operatorname{dist}(x, \partial\Omega)$, $\lambda \in (0, 1)$, *and* $\alpha \in \mathbb{N}_0^n$.

Proof. As in the past, Theorem 10.9 gives that $\mathbf{u} \in \big[C^\infty(\Omega)\big]^M$. By working with the system $\widetilde{\mathscr{L}}$ defined as in (11.4.19) and the function $\widetilde{\mathbf{u}}$ defined as in (11.4.23), the same type of reasoning as in the proof of Theorem 11.4 shows that estimate (11.4.5) actually holds without any restrictions on the dimension n. Keeping this in mind, the version of this estimate corresponding to $\mu = (0, \ldots, 0)$ may be interpreted as saying that $\mathbf{u}$ is 1-subaveraging. Hence, by Lemma 11.7 and the ensuing remark, it follows that $\mathbf{u}$ is subaveraging. Based on this and (11.4.12), for each $x \in \Omega$, each $r \in \big(0, \operatorname{dist}(x, \partial\Omega)\big)$, each $\theta, \eta \in (0, 1)$ and each $\alpha \in \mathbb{N}_0^n$ we may write

$$\max_{y \in \overline{B(x,\theta\eta r)}} |\partial^\alpha \mathbf{u}(y)| \leq \frac{C^{|\alpha|}(1-\theta)^{-|\alpha|}|\alpha|!}{(\eta r)^{|\alpha|}} \max_{y \in \overline{B(x,\eta r)}} |\mathbf{u}(y)| \tag{11.5.18}$$

$$\leq c(1-\eta)^{-n/p}\frac{C^{|\alpha|}(1-\theta)^{-|\alpha|}|\alpha|!}{(\eta r)^{|\alpha|}}\Big(\fint_{B(x,r)}|\mathbf{u}(y)|^p\,dy\Big)^{1/p},$$

where $c=c(L,n,p)\in(0,\infty)$. Next, given any $\lambda\in(0,1)$, specialize (11.5.18) to the case when $\theta:=2\lambda/(\lambda+1)$ and $\eta:=(\lambda+1)/2$. Note that $\theta,\eta\in(0,1)$, $\theta\eta=\lambda$, $1-\eta=(1-\lambda)/2$, and $1-\theta=(1-\lambda)/(1+\lambda)$. Based on these, we may transform (11.5.18) into

$$\max_{y\in\overline{B(x,\lambda r)}}|\partial^\alpha\mathbf{u}(y)|\leq c\,2^{n/p}(1-\lambda)^{-|\alpha|-n/p}\frac{(2C)^{|\alpha|}|\alpha|!}{r^{|\alpha|}}\Big(\fint_{B(x,r)}|\mathbf{u}(y)|^p\,dy\Big)^{1/p},\tag{11.5.19}$$

from which (11.5.17) follows after adjusting notation. □

11.6 Layer Potentials and Jump Relations for Systems

The goal of this section is to introduce and study certain types of integral operators, typically called layer potentials, that are particularly useful in the treatment of boundary value problems for systems. This is also an excellent opportunity to illustrate how a good understanding of the nature of fundamental solutions coupled with a versatile command of distribution theory yield powerful tools in the study of partial differential equations.

To set the stage, we introduce some notation and make some background assumptions. Throughout this section we let

$$L=\sum_{j,k=1}^{n}A_{jk}\partial_j\partial_k,\quad A_{jk}=\big(a_{jk}^{\alpha\beta}\big)_{1\leq\alpha,\beta\leq M}\in\mathcal{M}_{M\times M}(\mathbb{C}),\tag{11.6.1}$$

be a second-order, homogeneous, complex constant coefficient, $M\times M$ system, where $M\in\mathbb{N}$, with the property that its characteristic matrix, $L(\xi)$, defined for each vector $\xi=(\xi_1,\ldots,\xi_n)\in\mathbb{R}^n$ by the formula

$$L(\xi):=\sum_{j,k=1}^{n}\xi_j\xi_kA_{jk}=\Big(\sum_{j,k=1}^{n}a_{jk}^{\alpha\beta}\xi_j\xi_k\Big)_{1\leq\alpha,\beta\leq M}\in\mathcal{M}_{M\times M}(\mathbb{C}),\tag{11.6.2}$$

satisfies the ellipticity condition

$$\det\big[L(\xi)\big]\neq 0,\quad\forall\,\xi\in\mathbb{R}^n\setminus\{0\}.\tag{11.6.3}$$

In particular, since $L(\mathbf{e}_j)=A_{jj}$ for each $j\in\{1,\ldots,n\}$, the ellipticity condition (11.6.3) entails

$$A_{jj}\in\mathcal{M}_{M\times M}(\mathbb{C})\ \text{ is an invertible matrix for each }\ j\in\{1,\ldots,n\}.\tag{11.6.4}$$

Thanks to (11.6.3), Theorem 11.1 is applicable (with $m=1$) and we denote by

$$\mathbb{E}=\big(E_{\alpha\beta}\big)_{1\leq\alpha,\beta\leq M}\in\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)\tag{11.6.5}$$

the matrix-valued fundamental solution of L in $\mathbb{R}^n$ where the entries are constructed according to the recipe devised there. Thus, among other things, for each $\alpha, \beta \in \{1, \ldots, M\}$ we have

$$E_{\alpha\beta}\big|_{\mathbb{R}^n\setminus\{0\}} \text{ is even and belongs to } C^\infty(\mathbb{R}^n \setminus \{0\}), \tag{11.6.6}$$

$$\nabla E_{\alpha\beta} \text{ is positive homogeneous of degree } 1-n. \tag{11.6.7}$$

First, by specializing Theorem 4.76 to the case when Φ is a first-order partial derivative of $\mathbb{E}$, we obtain the following basic result.

Theorem 11.10. *Assume that the system L is as in (11.6.1)–(11.6.3) and suppose that $\mathbb{E}$ is the fundamental solution for L in $\mathbb{R}^n$ constructed in Theorem 11.1. Then for each $j \in \{1, \ldots, n\}$ one has*

$$\lim_{\varepsilon\to 0^\pm} (\partial_j \mathbb{E})(x', \varepsilon) = \pm\tfrac{1}{2}\,\delta_{jn}\delta(x')\big(A_{nn}\big)^{-1} + \text{P.V.}\,(\partial_j\mathbb{E})(x', 0) \tag{11.6.8}$$

in $\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^{n-1})\big)$.

Proof. Fix $j \in \{1, \ldots, n\}$. Then $\Phi := \partial_j\mathbb{E}$ is C^∞, odd, and positive homogeneous of degree $1-n$ in $\mathbb{R}^n \setminus \{0\}$ (cf. (11.6.6)–(11.6.7)). Moreover, by virtue of (11.3.26) used with $m = 1$, we have

$$\widehat{\Phi}(\xi) = \widehat{\partial_j\mathbb{E}}(\xi) = i\,\xi_j\widehat{\mathbb{E}}(\xi) = -i\,\xi_j\big[L(\xi)\big]^{-1} \quad \text{in} \quad \mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^{n-1})\big). \tag{11.6.9}$$

This further implies that

$$\widehat{\Phi}(0', 1) = \widehat{\Phi}(\mathbf{e}_n) = -i\,\delta_{jn}\big[L(\mathbf{e}_n)\big]^{-1} = -i\,\delta_{jn}\big(A_{nn}\big)^{-1}. \tag{11.6.10}$$

With this in hand, (11.6.8) follows with the help of (4.7.1). □

In the next result, we introduce the so-called single-layer potential operator associated with the system L and study the boundary behavior of its first-order derivatives. Here and elsewhere, the integral of a vector-valued function is applied to each individual component.

Theorem 11.11. *Suppose that the system L is as in (11.6.1)–(11.6.3) and assume that $\mathbb{E}$ is the fundamental solution for L in $\mathbb{R}^n$ constructed in Theorem 11.1. Given any $\mathbb{C}^M$-valued function $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$, define the $\mathbb{C}^M$-valued function*

$$(\mathscr{S}\varphi)(x) := \int_{\mathbb{R}^{n-1}} \mathbb{E}(x' - y', t)\varphi(y')\,dy', \quad \forall\, x = (x', t) \in \mathbb{R}^n \ \text{ with } \ t \neq 0. \tag{11.6.11}$$

Then for any $\mathbb{C}^M$-valued function $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$,

$$\mathscr{S}\varphi \in C^\infty(\mathbb{R}^n_\pm), \quad L(\mathscr{S}\varphi) = 0 \ \text{ pointwise in } \ \mathbb{R}^n_\pm, \tag{11.6.12}$$

and for each $j \in \{1,\dots,n\}$ *and each* $x' \in \mathbb{R}^{n-1}$ *one has*

$$\lim_{t\to 0^{\pm}} \partial_j(\mathscr{S}\varphi)(x',t) = \pm\tfrac{1}{2}\,\delta_{jn}\big(A_{nn}\big)^{-1}\varphi(x') + \lim_{\varepsilon\to 0^+} \int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\ |x'-y'|>\varepsilon}} (\partial_j\mathbb{E})(x'-y',0)\varphi(y')\,dy'. \tag{11.6.13}$$

Proof. Let $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$ be an arbitrary $\mathbb{C}^M$-valued function. That $\mathscr{S}\varphi$ is C^∞ in $\mathbb{R}^n \setminus \{x_n = 0\}$ is clear from (11.6.11), (11.6.6), and estimates for the derivatives of $\mathbb{E}$ (see (11.3.25)). In addition, for each $x = (x', x_n) \in \mathbb{R}^n$ with $x_n \neq 0$,

$$L(\mathscr{S}\varphi)(x) = \int_{\mathbb{R}^{n-1}} (L\mathbb{E})(x'-y',x_n)\varphi(y')\,dy' = 0 \tag{11.6.14}$$

since $L\mathbb{E} = \delta\, I_{M\times M}$ in $\mathcal{D}'(\mathbb{R}^n)$ which (upon recalling that $\mathbb{E} \in C^\infty(\mathbb{R}^n \setminus \{0\})$) forces

$$L\mathbb{E} = 0 \quad \text{pointwise in} \quad \mathbb{R}^n_{\pm}. \tag{11.6.15}$$

Finally, (11.6.13) follows from Corollary 4.78 applied to the same function Φ used in the proof of Theorem 11.10. □

Remark 11.12. *The integral operator* $\mathscr{S}$ *from* (11.6.11) *is called the single-layer potential (or single layer operator) associated to* L *in the upper and lower half-spaces in* $\mathbb{R}^n$*, while* (11.6.13) *may be naturally regarded as the jump formula for the gradient of the single-layer potential in this setting.*

Our next result brings to the forefront another basic integral operator, typically referred to as the double layer potential.

Theorem 11.13. *Let the system* L *be as in* (11.6.1)–(11.6.3) *and assume that* $\mathbb{E}$ *is the fundamental solution for* L *in* $\mathbb{R}^n$ *constructed in Theorem 11.1. Given any* $\mathbb{C}^M$*-valued function* $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$*, define for each* $x = (x',t) \in \mathbb{R}^n$ *with* $t \neq 0$

$$(\mathscr{D}\varphi)(x) := \int_{\mathbb{R}^{n-1}} \sum_{k=1}^{n} (\partial_k\mathbb{E})(x'-y',t)A_{kn}\varphi(y')\,dy'. \tag{11.6.16}$$

Then, for any $\mathbb{C}^M$*-valued function* $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$*,*

$$\mathscr{D}\varphi \in C^\infty(\mathbb{R}^n_{\pm}), \quad L(\mathscr{D}\varphi) = 0 \ \text{ pointwise in } \ \mathbb{R}^n_{\pm}, \tag{11.6.17}$$

and for every $x' \in \mathbb{R}^{n-1}$ *one has*

$$\lim_{t\to 0^{\pm}} (\mathscr{D}\varphi)(x',t) = \pm\tfrac{1}{2}\,\varphi(x') + \lim_{\varepsilon\to 0^+} \int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\ |x'-y'|>\varepsilon}} \sum_{k=1}^{n} (\partial_k\mathbb{E})(x'-y',0)A_{kn}\varphi(y')\,dy'. \tag{11.6.18}$$

Proof. Fix some $\mathbb{C}^M$-valued function $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$. Then the fact that $\mathscr{D}\varphi$ is C^∞ in $\mathbb{R}^n \setminus \{x_n = 0\}$ is seen from (11.6.16), (11.6.6), and estimates for the derivatives of $\mathbb{E}$ (cf. (11.3.25)). Moreover, for each $x = (x', x_n) \in \mathbb{R}^n$ with $x_n \neq 0$,

$$L(\mathscr{D}\varphi)(x) = \int_{\mathbb{R}^{n-1}} \sum_{k=1}^{n} \partial_{x_k} \big[(L\mathbb{E})(x' - y', x_n)\big] A_{kn}\varphi(y')\,\mathrm{d}y' = 0 \tag{11.6.19}$$

where the last equality uses (11.6.15).

At this stage, there remains to prove (11.6.18). In this regard, if $x' \in \mathbb{R}^{n-1}$ has been fixed, making use of Corollary 11.11 we may write

$$\begin{aligned}
\lim_{t\to 0^\pm} (\mathscr{D}\varphi)(x', t) &= \sum_{k=1}^{n} \lim_{t\to 0^\pm} \int_{\mathbb{R}^{n-1}} (\partial_k \mathbb{E})(x' - y', t) A_{kn}\varphi(y')\,\mathrm{d}y' \\
&= \sum_{k=1}^{n} \lim_{t\to 0^\pm} \partial_k (\mathscr{S} A_{kn}\varphi)(x', t) \\
&= \pm\frac{1}{2} \sum_{k=1}^{n} \delta_{kn} \big(A_{nn}\big)^{-1} A_{kn}\varphi(x') \qquad (11.6.20) \\
&\quad + \lim_{\varepsilon\to 0^+} \int\limits_{\substack{y'\in\mathbb{R}^{n-1} \\ |x'-y'|>\varepsilon}} \sum_{k=1}^{n} (\partial_k \mathbb{E})(x' - y', 0) A_{kn}\varphi(y')\,\mathrm{d}y' \\
&= \pm\tfrac{1}{2}\,\varphi(x') + \lim_{\varepsilon\to 0^+} \int\limits_{\substack{y'\in\mathbb{R}^{n-1} \\ |x'-y'|>\varepsilon}} \sum_{k=1}^{n} (\partial_k \mathbb{E})(x' - y', 0) A_{kn}\varphi(y')\,\mathrm{d}y',
\end{aligned}$$

as wanted. □

Remark 11.14. *The mapping $\mathscr{D}$ from (11.6.16) is called the double-layer potential (or double layer operator) associated to the system L in $\mathbb{R}^n_\pm$ and (11.6.18) may be naturally regarded as the jump formula for the double-layer potential in this setting.*

When dealing with the so-called Neumann boundary value problem for the system L [discussed later in (11.6.34)], as boundary condition one prescribes a certain combination of first-order derivatives of the unknown function, namely $-\sum_{k=1}^{n} A_{nk}\partial_k$, amounting to what is called the conormal derivative associated with the system L. Our next result elaborates on the nature of boundary behavior of this conormal derivative of the single-layer potential $\mathscr{S}$ introduced earlier in (11.6.11).

Corollary 11.15. *Assume that the system L is as in (11.6.1)–(11.6.3) and let $\mathbb{E}$ be the fundamental solution for L in $\mathbb{R}^n$ constructed in Theorem 11.1. Then for every $\mathbb{C}^M$-valued function $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$,*

$$\lim_{t\to 0^{\pm}} \sum_{k=1}^{n} A_{nk}\partial_k(\mathscr{S}\varphi)(x',t) = \pm\tfrac{1}{2}\,\varphi(x') + \lim_{\varepsilon\to 0^+} \int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\ |x'-y'|>\varepsilon}} \sum_{k=1}^{n} A_{nk}(\partial_k\mathbb{E})(x'-y',0)\varphi(y')\,dy'. \tag{11.6.21}$$

Proof. For each $\mathbb{C}^M$-valued function $\varphi \in \mathcal{S}(\mathbb{R}^{n-1})$, formula (11.6.21) follows using Corollary 11.11 by writing

$$\begin{aligned}
\lim_{t\to 0^{\pm}} \sum_{k=1}^{n} A_{nk}\,\partial_k(\mathscr{S}\varphi)(x',t) &= \sum_{k=1}^{n} A_{nk} \lim_{t\to 0^{\pm}} \partial_k(\mathscr{S}\varphi)(x',t) \\
&= \pm\frac{1}{2}\sum_{k=1}^{n} A_{nk}\delta_{kn}\big(A_{nn}\big)^{-1}\varphi(x') \\
&\quad + \lim_{\varepsilon\to 0^+} \int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\ |x'-y'|>\varepsilon}} \sum_{k=1}^{n} A_{nk}(\partial_k\mathbb{E})(x'-y',0)\varphi(y')\,dy' \\
&= \pm\tfrac{1}{2}\,\varphi(x') + \lim_{\varepsilon\to 0^+} \int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\ |x'-y'|>\varepsilon}} \sum_{k=1}^{n} A_{nk}(\partial_k\mathbb{E})(x'-y',0)\varphi(y')\,dy',
\end{aligned} \tag{11.6.22}$$

at every point $x' \in \mathbb{R}^{n-1}$. □

Exercise 11.16. *State and prove the analogue of Theorem 4.96 in the scenario when $\Theta : \mathbb{R}^n \setminus \{0\} \to \mathcal{M}_{M\times M}(\mathbb{C})$, for some $M \in \mathbb{N}$, is a matrix-valued function with scalar entries that satisfy the conditions in (4.4.1). In particular, pay attention to the fact that the format of (4.9.42) now becomes*

$$\big(T_\Theta\big)^* = T_{\overline{\Theta^\top}^\vee} \quad \text{in} \quad \big[L^2(\mathbb{R}^n)\big]^M \tag{11.6.23}$$

where, as before, $\Theta^\top$ denotes the transposed of the matrix Θ.

Hint: Prove and use the fact that $\big[m_\Theta\big]^\top = m_{\Theta^\top}$.

The fact that for any $\varphi \in \big[\mathcal{S}(\mathbb{R}^{n-1})\big]^M$ we have $L(\mathscr{D}\varphi) = 0$ in $\mathbb{R}^n_\pm$ (cf. (11.6.17)) means that we may regard the double-layer potential operator $\mathscr{D}$ defined in (11.6.16) as a mechanism for generating an abundance of solutions of the partial differential equation $L\mathbf{u} = 0$ in $\mathbb{R}^n_\pm$. In addition, the jump formula (11.6.18) fully clarifies the boundary behavior of the function

$$\mathbf{u} := \mathscr{D}\varphi. \tag{11.6.24}$$

This is particularly relevant in the context of the Dirichlet problem for the $M\times M$ system L (assumed to be as in (11.6.1)–(11.6.3)), the formulation of which in, say, the upper half-space reads (compare with (4.8.17) in the case $L:=\Delta$)

$$\begin{cases} \mathbf{u}\in\big[C^{\infty}(\mathbb{R}^n_+)\big]^M, & \\ L\mathbf{u}=0 & \text{in } \mathbb{R}^n_+, \\ \mathbf{u}\Big|^{ver}_{\partial\mathbb{R}^n_+}=\psi & \text{on } \mathbb{R}^{n-1}\equiv\partial\mathbb{R}^n_+. \end{cases} \tag{11.6.25}$$

Above, $\psi\in\big[\mathcal{S}(\mathbb{R}^{n-1})\big]^M$ is a given function (called the boundary datum) and, much as in the case of (4.8.17), the symbol $\mathbf{u}\Big|^{ver}_{\partial\mathbb{R}^n_+}$ stands for the "vertical limit" of $\mathbf{u}$ to the boundary of the upper half-space, understood at each point $x'\in\mathbb{R}^{n-1}$ as $\lim_{x_n\to 0^+}\mathbf{u}(x',x_n)$.

Indeed, focusing the search for a solution of (11.6.25) in the class of functions $\mathbf{u}$ defined as in (11.6.24) (with $\varphi\in\big[\mathcal{S}(\mathbb{R}^{n-1})\big]^M$ yet to be determined) has the distinct advantage that the first two conditions in (11.6.25) are automatically satisfied (thanks to (11.6.17)), irrespective of the choice of φ. Keeping in mind (11.6.18), matters have been therefore reduced to solving the boundary integral equation

$$\tfrac{1}{2}\,\varphi(x')+\lim_{\varepsilon\to 0^+}\int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\ |x'-y'|>\varepsilon}}\sum_{k=1}^{n}(\partial_k\mathbb{E})(x'-y',0)A_{kn}\varphi(y')\,dy'=\psi(x'),\quad x'\in\mathbb{R}^{n-1}, \tag{11.6.26}$$

so named since $\mathbb{R}^{n-1}$ may be regarded as the boundary of $\mathbb{R}^n_+$. To streamline notation, introduce the singular integral operator

$$(K\varphi)(x'):=\lim_{\varepsilon\to 0^+}\int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\ |x'-y'|>\varepsilon}}\sum_{k=1}^{n}(\partial_k\mathbb{E})(x'-y',0)A_{kn}\varphi(y')\,dy',\quad x'\in\mathbb{R}^{n-1}, \tag{11.6.27}$$

typically referred to as the principal value (or boundary version) of the double layer associated with the system L in the upper half-space. Employing (11.6.27), we may write in place of (11.6.26)

$$\big(\tfrac{1}{2}I_{M\times M}+K\big)\varphi=\psi\quad\text{in}\quad\mathbb{R}^{n-1}. \tag{11.6.28}$$

Reducing the entire boundary value problem (11.6.25) to solving the boundary integral equation (11.6.28) is perhaps the most distinguished feature of the technology described thus far for dealing with (11.6.25), called the method of boundary layer potentials (in standard partial differential equations parlance).

Regarding the boundary integral equation (11.6.28), involving the singular integral operator K introduced in (11.6.27), we wish to note that by taking the

Fourier transform (in $\mathbb{R}^{n-1}$), then invoking Theorem 4.71 (the applicability of which in the present setting is justified by virtue of the properties of $\mathbb{E}$ from (11.6.6)–(11.6.7)), we arrive at

$$\big(\tfrac{1}{2}I_{M\times M}+a_L\big)\widehat{\varphi}=\widehat{\psi}\quad\text{in}\quad\mathcal{S}'(\mathbb{R}^{n-1}),\tag{11.6.29}$$

where the $M\times M$ matrix-valued function a_L is defined by the formula

$$a_L(\xi'):=-\sum_{k=1}^{n}\Big(\int_{S^{n-2}}(\partial_k\mathbb{E})(\omega',0)\log(i(\xi'\cdot\omega'))\,d\sigma(\omega')\Big)A_{kn}\tag{11.6.30}$$

for each $\xi'\in\mathbb{R}^{n-1}\setminus\{0'\}$. Thus, in the case when the matrix $\frac{1}{2}I_{M\times M}+a_L$ is invertible, at least at the formal level we may express the solution of (11.6.29) in the form

$$\varphi=\mathcal{F}^{-1}\Big(\big(\tfrac{1}{2}I_{M\times M}+a_L\big)^{-1}\widehat{\psi}\Big),\tag{11.6.31}$$

then ultimately conclude from (11.6.24) and (11.6.31) that

$$\mathbf{u}=\mathscr{D}\Big[\mathcal{F}^{-1}\Big(\big(\tfrac{1}{2}I_{M\times M}+a_L\big)^{-1}\widehat{\psi}\Big)\Big]\tag{11.6.32}$$

solves the Dirichlet problem (11.6.25) for the system L in $\mathbb{R}^n_+$ with ψ as boundary datum.

In certain concrete cases of practical importance, the "symbol" function a_L from (11.6.30) is simple enough in order for us to make sense of the expression appearing in the right-hand side of (11.6.31). For example, it is visible from (11.6.30) that $a_L=0$ in $\mathbb{R}^{n-1}\setminus\{0'\}$ whenever

$$\sum_{k=1}^{n}(\partial_k\mathbb{E})(x',0)A_{kn}=0\quad\text{for all}\quad x'\in\mathbb{R}^{n-1}\setminus\{0'\}.\tag{11.6.33}$$

This is indeed the case when $L=\Delta$, the Laplacian in $\mathbb{R}^n$. To see that this happens, note that in this scalar ($M=1$) case, $A_{jk}=\delta_{jk}$ for each $j,k\in\{1,\dots,n\}$, and if E_Δ denotes the fundamental solution for the Laplacian from (7.1.12) then for each $x=(x',x_n)\in\mathbb{R}^n\setminus\{0\}$ we have $(\partial_nE_\Delta)(x',x_n)=\frac{1}{\omega_{n-1}}\frac{x_n}{|x|^n}$, hence $(\partial_nE_\Delta)(x',0)=0$ for each $x'\in\mathbb{R}^{n-1}\setminus\{0'\}$. The bottom line is that the symbol function a_Δ vanishes identically. In light of (11.6.31), this shows that for the Laplacian we must chose the (originally unspecified) function φ to be equal to the (given) boundary datum ψ. For this choice, the general formula (11.6.32) then reduces precisely to the classical expression (4.8.19), as the reader may verify without difficulty.

However, in general, the assignment $\psi\mapsto\mathcal{F}^{-1}\Big(\big(\frac{1}{2}I_{M\times M}+a_L\big)^{-1}\widehat{\psi}\Big)$ can be of an intricate nature, leading to operators that are well beyond the class of singular integral operators introduced in Definition 4.90. This leads to the consideration of more exotic classes of operators, such as pseudodifferential operators, Fourier integral operators, etc., which are outside of the scope of the present monograph.

This being said, the material developed here serves both as preparation and motivation for the reader interested in further pursuing such matters.

Similar considerations apply in the case of the Neumann problem for a system L as in (11.6.1)–(11.6.3), where the formulation reads as

$$\begin{cases} \mathbf{u} \in \big[C^\infty(\mathbb{R}^n_+)\big]^M, \\ L\mathbf{u} = 0 & \text{in } \mathbb{R}^n_+, \\ -\sum\limits_{k=1}^{n} A_{nk}\big[\partial_k \mathbf{u}\big]\Big|^{ver}_{\partial\mathbb{R}^n_+} = \psi & \text{on } \mathbb{R}^{n-1} \equiv \partial\mathbb{R}^n_+, \end{cases} \tag{11.6.34}$$

where $\psi \in \big[\mathcal{S}(\mathbb{R}^{n-1})\big]^M$ is the boundary datum. Given the results proved in Theorem 11.11 and Corollary 11.15, this time it is natural to seek a solution in the form (compare with (11.6.24))

$$\mathbf{u} := \mathscr{S}\varphi \quad \text{in} \quad \mathbb{R}^n_+, \tag{11.6.35}$$

where the function $\varphi \in \big[\mathcal{S}(\mathbb{R}^{n-1})\big]^M$ is subject to the boundary integral equation (implied by (11.6.21))

$$-\tfrac{1}{2}\varphi(x') - \lim_{\varepsilon\to 0^+} \int\limits_{\substack{y'\in\mathbb{R}^{n-1}\\ |x'-y'|>\varepsilon}} \sum_{k=1}^{n} A_{nk}(\partial_k \mathbb{E})(x'-y',0)\varphi(y')\,dy' = \psi(x') \tag{11.6.36}$$

for each $x' \in \mathbb{R}^{n-1}$. Once again, Theorem 4.71 may be used in order to rewrite (11.6.36) on the Fourier transform side as

$$\big(-\tfrac{1}{2} I_{M\times M} + \widetilde{a}_L\big)\widehat{\varphi} = \widehat{\psi} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^{n-1}), \tag{11.6.37}$$

where the $M \times M$ matrix-valued function $\widetilde{a}_L$ is now defined by the formula

$$\widetilde{a}_L(\xi') := \sum_{k=1}^{n} A_{nk}\Big(\int_{S^{n-2}} (\partial_k \mathbb{E})(\omega',0)\log(i(\xi'\cdot\omega'))\,d\sigma(\omega')\Big) \tag{11.6.38}$$

for each $\xi' \in \mathbb{R}^{n-1}\setminus\{0'\}$. As regards the symbol function $\widetilde{a}_L$ for the Neumann problem, one can see from (11.6.38), (11.6.30), and (11.3.27) that

$$\widetilde{a}_L = -\big[a_{L^\top}\big]^\top \tag{11.6.39}$$

which shows that $\widetilde{a}_L$ has, up to transpositions, the same nature as the symbol function for the Dirichlet problem. Hence, the same type of considerations as in the latter case apply. For example, corresponding to the case when $L = \Delta$ we have $\widetilde{a}_\Delta = 0$ which means that a solution to the Neumann problem for the Laplacian in the upper half-space, that is,

$$\begin{cases} u \in C^\infty(\mathbb{R}^n_+), \\ \Delta u = 0 & \text{in } \mathbb{R}^n_+, \\ -\big[\partial_n u\big]\Big|^{ver}_{\partial\mathbb{R}^n_+} = \psi & \text{on } \mathbb{R}^{n-1} \equiv \partial\mathbb{R}^n_+, \end{cases} \tag{11.6.40}$$

where $\psi \in \mathcal{S}(\mathbb{R}^{n-1})$ is the boundary datum, is given by

$$u(x) := \int_{\mathbb{R}^{n-1}} E_{\Delta}(x'-y',x_n)\psi(y')\,dy' \quad \text{for} \quad x=(x',x_n)\in\mathbb{R}^n_+, \tag{11.6.41}$$

with E_{Δ} denoting the standard fundamental solution for the Laplacian (cf. (7.1.12)).

Further Notes for Chap. 11. As already mentioned, the approach developed in Sect. 10.2 contains a constructive procedure for reducing the task of finding a fundamental solution for the Lamé operator to the scalar case. This scheme has been implemented in Sect. 11.1 based on the knowledge of a fundamental solution for the poly-harmonic operator from Sect. 7.5. Subsequently, in Sect. 11.2 a fundamental solution for the Stokes operator is computed indirectly by rigorously making sense of the informal observation that this operator is a limiting case of the Lamé operator (taking $\mu=1$ and sending $\lambda\to\infty$).

From the considerations in Sect. 10.2 it was also already known that the higher-order homogeneous elliptic constant coefficient linear systems considered in Sect. 11.3 possess fundamental solutions that are tempered distributions with singular support at the origin. The new issue addressed in Theorem 11.1 is to find an explicit formula and to study other properties of such fundamental solutions. This theorem refines and further builds on the results proved in [31, 35, 50, 54, 61], in various degrees of generality. Fundamental solutions for variable coefficient elliptic operators on manifolds have been studied in [52].

11.7 Additional Exercises for Chap. 11

Exercise 11.17. *Use Theorem 11.1 in a similar manner as in the proof of Proposition 11.2 in order to derive a formula for a fundamental solution for the Lamé operator in $\mathbb{R}^n$ when $n\geq 3$ is arbitrary and odd.*

Exercise 11.18. *Similarly to Exercise 11.17, derive a formula for a fundamental solution for the Lamé operator in $\mathbb{R}^n$ when n is even.*

Exercise 11.19. *Let L be a homogeneous differential operator in $\mathbb{R}^n$, $n\geq 2$, of order $2m$, $m\in\mathbb{N}$, with complex constant coefficients as in (11.3.1) that satisfies (11.3.3), and recall (11.3.28). Also, let $q\in\mathbb{N}_0$ be such that $n+q$ is even. Consider the matrix $\mathbb{E}:=\big(E_{jk}\big)_{1\leq j,k\leq M}$ with entries given by*

$$E_{jk}(x)=\frac{-1}{(2\pi\,i)^n(2m+q)!}\int_{S^{n-1}}(x\cdot\xi)^{2m+q}\log\Big(\frac{x\cdot\xi}{i}\Big)P^{jk}(\xi)\,d\sigma(\xi) \tag{11.7.1}$$

for $x\in\mathbb{R}^n\setminus\{0\}$ and $j,k\in\{1,\dots,M\}$, where $\log$ denotes the principal branch of the complex logarithm defined for points $z\in\mathbb{C}\setminus\{x:\,x\in\mathbb{R},\,x\leq 0\}$. Prove that $\mathbb{E}\in\mathcal{M}_{M\times M}\big(\mathcal{S}'(\mathbb{R}^n)\big)$ is a fundamental solution for the system $\Delta_x^{(n+q)/2}L$ in $\mathbb{R}^n$.

Exercise 11.20. *Let L_1, L_2 be two homogeneous $M \times M$ systems of differential operators in $\mathbb{R}^n$, $n \geq 2$, of orders $2m_1$ and $2m_2$, respectively, where $m_1, m_2 \in \mathbb{N}$, with complex constant coefficients, satisfying $\det L_1(\xi) \neq 0$ and $\det L_2(\xi) \neq 0$ for every $\xi \in \mathbb{R}^n \setminus \{0\}$. Prove that, with the notational convention employed in part (7) of Theorem 11.1, one has*

$$L_2 \mathbb{E}_{L_1 L_2} = \mathbb{E}_{L_1} + P \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{11.7.2}$$

where P is zero if either n is odd or $2m_1 < n$, and P is an $M \times M$ matrix whose entries are homogeneous polynomials of degree $2m_1 - n$ if $2m_1 \geq n$.

Exercise 11.21. *Recall the notational convention employed in part (7) of Theorem 11.1 as well as $F_{m,n}$ from (7.5.2). Prove that for every $m \in \mathbb{N}$ one has*

$$\mathbb{E}_{\Delta^m} = F_{m,n} + P \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n), \tag{11.7.3}$$

where P is zero if either n is odd or $2m < n$, and P is a homogeneous polynomial of degree $2m - n$ if $2m \geq n$.

Chapter 12

Solutions to Selected Exercises

12.1 Solutions to Exercises from Sect. 1.4

Exercise 1.24 $a = 0$; see Examples 1.4 and 1.5.

Exercise 1.25 No; see Examples 1.4 and 1.5.

Exercise 1.26 Let $\varphi \in C_0^\infty(\mathbb{R}^2)$. Based on integration by parts and the fact that φ is compactly supported we have

$$\begin{aligned}\langle \partial_1\partial_2 f, \varphi\rangle &= \int_{\mathbb{R}^2} f(x,y)(\partial_1\partial_2\varphi)(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_0^\infty \int_{\mathbb{R}} \partial_2(\partial_1\varphi)(x,y)\,\mathrm{d}y\,\mathrm{d}x + \int_0^\infty \int_{\mathbb{R}} \partial_1(\partial_2\varphi)(x,y)\,\mathrm{d}x\,\mathrm{d}y = 0.\end{aligned} \tag{12.1.1}$$

This shows that $\partial_1\partial_2 f = 0$ in the weak sense. Similarly, the other weak derivatives are $\partial_2(\partial_1\partial_2 f) = 0$ and $\partial_1^2\partial_2 f = 0$. Suppose now that $\partial_1 f$ exists in the weak sense, i.e., there exists $g \in L^1_{\mathrm{loc}}(\mathbb{R}^2)$ such that

$$\int_{\mathbb{R}^2} f(x,y)(\partial_1\varphi)(x,y)\,\mathrm{d}x\,\mathrm{d}y = -\int_{\mathbb{R}^2} g(x,y)\varphi(x,y)\,\mathrm{d}x\,\mathrm{d}y, \quad \forall\,\varphi \in C_0^\infty(\mathbb{R}^2). \tag{12.1.2}$$

Since $\int_{\mathbb{R}^2} f(x,y)(\partial_1\varphi)(x,y)\,\mathrm{d}x\,\mathrm{d}y = -\int_{\mathbb{R}} \varphi(0,y)\,\mathrm{d}y$ (as seen using the definition of f and integration by parts), (12.1.2) becomes

$$\int_{\mathbb{R}} \varphi(0,y)\,\mathrm{d}y = \int_{\mathbb{R}^2} g(x,y)\varphi(x,y)\,\mathrm{d}x\,\mathrm{d}y, \qquad \forall\,\varphi \in C_0^\infty(\mathbb{R}^2). \tag{12.1.3}$$

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Taking now φ in (12.1.3) to be supported in $\{(x,y)\in\mathbb{R}^2 : x\neq 0\}$ and recalling Theorem 1.3, it follows that $g=0$ almost everywhere on $\{(x,y)\in\mathbb{R}^2 : x\neq 0\}$. Thus, $g=0$ almost everywhere in $\mathbb{R}^2$. The latter, when used in (12.1.3), implies $\int_{\mathbb{R}}\varphi(0,y)\,\mathrm{d}y=0$ for every $\varphi\in C_0^\infty(\mathbb{R}^2)$. However this leads to a contradiction by taking $\varphi(x,y)=\varphi_1(x)\varphi_2(y)$ for $(x,y)\in\mathbb{R}^2$, where φ_1, $\varphi_2\in C_0^\infty(\mathbb{R})$ are such that $\varphi_1(0)=1$ and $\int_{\mathbb{R}}\varphi_2(y)\,\mathrm{d}y=1$.

Exercise 1.27 Let $\varphi\in C_0^\infty\big((-1,1)\big)$ be arbitrary. Then Lebesgue's dominated convergence theorem and integration by parts give

$$\begin{aligned}
-\int_{-1}^{1} f(x)\varphi'(x)\,\mathrm{d}x &= \lim_{\varepsilon\to 0^+}\int_{-1}^{-\varepsilon}\sqrt{-x}\,\varphi'(x)\,\mathrm{d}x-\lim_{\varepsilon\to 0^+}\int_{\varepsilon}^{1}\sqrt{x}\,\varphi'(x)\,\mathrm{d}x\\
&= \lim_{\varepsilon\to 0^+}\left[\sqrt{-x}\,\varphi(x)|_{-1}^{\varepsilon}+\int_{-1}^{-\varepsilon}\frac{1}{2\sqrt{-x}}\,\varphi(x)\,\mathrm{d}x\right]\\
&\quad-\lim_{\varepsilon\to 0^+}\left[\sqrt{x}\varphi(x)|_{-1}^{\varepsilon}-\int_{\varepsilon}^{1}\frac{1}{2\sqrt{x}}\,\varphi(x)\,\mathrm{d}x\right]\\
&= \int_{-1}^{1}\frac{1}{2\sqrt{|x|}}\,\varphi(x)\,\mathrm{d}x, \qquad (12.1.4)
\end{aligned}$$

where we have used the fact that $\frac{1}{2\sqrt{|x|}}\in L^1\big((-1,1)\big)$. Hence the weak derivative of order one of f is the function given by $g(x):=\frac{1}{2\sqrt{|x|}}$ for $x\in(-1,1)\setminus\{0\}$.

Exercise 1.28 Consider a function $\varphi\in C_0^\infty(\mathbb{R}^2)$ and pick a number $R>0$ such that $\operatorname{supp}\varphi\subset(-R,R)\times(-R,R)$. Then

$$-\int_{\mathbb{R}^2} x|y|(\partial_1^2\partial_2\varphi)(x,y)\,\mathrm{d}x\,\mathrm{d}y=-\int_{\mathbb{R}}|y|\int_{\mathbb{R}} x(\partial_1^2\partial_2\varphi)(x,y)\,\mathrm{d}x\,\mathrm{d}y=0, \quad (12.1.5)$$

where for the last equality is obtained integrating by parts twice with respect to x, keeping in mind that φ has compact support. This proves that $\partial_1^2\partial_2 f=0$ in the weak sense. On the other hand,

$$\begin{aligned}
-\int_{\mathbb{R}^2} x|y|(\partial_1\partial_2^2\varphi)(x,y)\,\mathrm{d}x\,\mathrm{d}y &= \int_{\mathbb{R}^2}|y|(\partial_2^2\varphi)(x,y)\,\mathrm{d}x\,\mathrm{d}y\\
&= \int_{\mathbb{R}}\left[-\int_{-R}^{0} y(\partial_2^2\varphi)(x,y)\,\mathrm{d}y+\int_{0}^{R} y(\partial_2^2\varphi)(x,y)\,\mathrm{d}y\right]\mathrm{d}x\\
&= -\int_{\mathbb{R}^2}\operatorname{sgn}(y)(\partial_2\varphi)(x,y)\,\mathrm{d}y\,\mathrm{d}x=2\int_{\mathbb{R}}\varphi(x,0)\,\mathrm{d}x \qquad (12.1.6)
\end{aligned}$$

after integrating by parts, first with respect to x, then twice with respect to y. Now reasoning as in the proof of Exercise 1.26 one can show that if $g\in L^1_{\mathrm{loc}}(\mathbb{R}^2)$ is such that

$$2\int_{\mathbb{R}} \varphi(x,0)\,\mathrm{d}x = \int_{\mathbb{R}^2} g(x,y)\varphi(x,y)\,\mathrm{d}x\,\mathrm{d}y, \qquad \forall\,\varphi \in C_0^\infty(\mathbb{R}^2),$$

leads to a contradiction. Hence, $\partial_1\partial_2^2 f$ does not exist in the weak sense.

Exercise 1.29 Reasoning as in the proofs of Exercise 1.26 and Exercise 1.28, one can show that $\partial_1^k f$ and $\partial_2^k f$ do not exist in the weak sense for any $k \in \mathbb{N}$, and that $\partial^\alpha f = 0$ in the weak sense whenever $\alpha = (\alpha_1, \alpha_2) \in \mathbb{N}_0^2$ with $\alpha_1 \neq 0$ and $\alpha_2 \neq 0$.

Exercise 1.30 Let $\varphi \in C^\infty(\mathbb{R})$ with $\operatorname{supp}\varphi \subset (-R, R)$ for some $R \in (0,\infty)$. Then,

$$\begin{aligned} -\int_{\mathbb{R}} f(x)\varphi'(x)\,\mathrm{d}x &= -\int_{-R}^{0} \sin(-x)\varphi'(x)\,dx - \int_0^R \sin(x)\varphi'(x)\,\mathrm{d}x \\ &= -\int_{-R}^{0} \cos(x)\varphi(x)\,dx + \int_0^R \cos(x)\varphi(x)\,\mathrm{d}x \\ &= -\int_{\mathbb{R}} \operatorname{sgn}(x)\cos(x)\varphi(x)\,\mathrm{d}x. \end{aligned} \tag{12.1.7}$$

Since $\operatorname{sgn}(x)\cos(x) \in L^1_{\mathrm{loc}}(\mathbb{R})$, we deduce from (12.1.7) that f' exists in the weak sense and that $f'(x) = \operatorname{sgn}(x)\cos(x)$ for $x \in \mathbb{R}$. Also,

$$\begin{aligned} \int_{\mathbb{R}} f(x)\varphi''(x)\,\mathrm{d}x &= \int_{-R}^{0} \cos(x)\varphi'(x)\,\mathrm{d}x - \int_0^R \cos(x)\varphi'(x)\,\mathrm{d}x \\ &= 2\varphi(0) - \int_{\mathbb{R}} \operatorname{sgn}(x)\sin(x)\varphi(x)\,\mathrm{d}x. \end{aligned} \tag{12.1.8}$$

If the weak derivative f'' were to exist we could find $g \in L^1_{\mathrm{loc}}(\mathbb{R})$ such that

$$\int_{\mathbb{R}} g(x)\varphi(x)\,\mathrm{d}x = 2\varphi(0) - \int_{\mathbb{R}} \operatorname{sgn}(x)\sin(x)\varphi(x)\,\mathrm{d}x, \quad \forall\,\varphi \in C_0^\infty(\mathbb{R}). \tag{12.1.9}$$

In particular, (12.1.9) would also hold for all $\varphi \in C_0^\infty(\mathbb{R})$ supported in $(0,\infty)$, which would give $g\big|_{(0,\infty)} = -\sin(x)$. Similarly, we would conclude that $g\big|_{(-\infty,0)} = \sin(x)$ and, hence, returning to (12.1.9), we would obtain $\varphi(0) = 0$ for all $\varphi \in C_0^\infty(\mathbb{R})$. The latter is false, thus the weak derivative f'' does not exist.

Exercise 1.32 Note that $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ for every $n \in \mathbb{N}$. Suppose $n \geq 2$, and fix some index $j \in \{1,\ldots,n\}$ along with a function $\varphi \in C_0^\infty(\mathbb{R}^n)$. If $R > 0$ is such that $\operatorname{supp}\varphi \subset B(0,R)$, then using Lebesgue's dominated convergence theorem and (13.7.4) we obtain

$$-\int_{\mathbb{R}^n} f(x)\partial_j\varphi(x)\,\mathrm{d}x = -\lim_{r\to 0^+} \int_{r<|x|<R} f(x)\partial_j\varphi(x)\,\mathrm{d}x$$

$$= \lim_{r\to 0^+}\left[\int_{\partial B(0,r)} \frac{x_j}{r^{\varepsilon+1}}\varphi(x)\,d\sigma(x) - \varepsilon\int_{r<|x|<R}\frac{x_j}{|x|^{\varepsilon+2}}\varphi(x)\,dx\right]$$

$$= -\varepsilon\int_{\mathbb{R}^n}\frac{x_j}{|x|^{\varepsilon+2}}\varphi(x)\,dx, \tag{12.1.10}$$

since, given that under the current assumptions $n-1-\varepsilon>0$, we have

$$\left|\int_{\partial B(0,r)}\frac{x_j}{r^{\varepsilon+1}}\varphi(x)\,d\sigma(x)\right| \leq \omega_{n-1}\|\varphi\|_{L^\infty(\mathbb{R}^n)}r^{n-1-\varepsilon}\to 0 \quad \text{as} \quad r\to 0^+. \tag{12.1.11}$$

This proves that if $n\geq 2$ then the weak derivative $\partial_j f$ exists and is equal to the function $-\varepsilon\frac{x_j}{|x|^{\varepsilon+2}}\in L^1_{\rm loc}(\mathbb{R}^n)$.

Assume next that $n=1$ and suppose that there exists some $g\in L^1_{\rm loc}(\mathbb{R})$ with the property that $-\int_{\mathbb{R}} f\varphi'\,dx=\int_{\mathbb{R}} g\varphi\,dx$ for every $\varphi\in C^\infty_0(\mathbb{R})$. Then, with $R>0$ such that $\operatorname{supp}\varphi\subset(-R,R)$, using Lebesgue's dominated convergence theorem we may write

$$\int_{\mathbb{R}} f\varphi'\,dx = \lim_{r\to 0^+}\left[\int_{-R}^{-r}\frac{\varphi'(x)}{(-x)^\varepsilon}\,dx+\int_r^R\frac{\varphi'(x)}{x^\varepsilon}\,dx\right]$$

$$= \lim_{r\to 0^+}\left[\frac{\varphi(-r)}{r^\varepsilon}-\varepsilon\int_{-R}^{-r}\frac{\varphi(x)}{(-x)^{\varepsilon+1}}\,dx-\frac{\varphi(r)}{r^\varepsilon}+\varepsilon\int_r^R\frac{\varphi(x)}{x^{\varepsilon+1}}\,dx\right]. \tag{12.1.12}$$

In particular, identity (12.1.12) holds if $\operatorname{supp}\varphi\subset(0,\infty)$, in which case (12.1.12) implies $g(x)=\varepsilon x^{-\varepsilon-1}$ for $x>0$, and if $\operatorname{supp}\varphi\subset(-\infty,0)$, when we obtain $g(x)=-\varepsilon(-x)^{-\varepsilon-1}$ for $x<0$ (recall Theorem 1.3). However, the function g thus obtained does not belong to $L^1_{\rm loc}(\mathbb{R})$. Consequently, f does not have a weak derivative of order one in the case when $n=1$.

Exercise 1.33

(a) Let $f\in L^1_{\rm loc}\big((a,b)\big)$ be such that the weak derivative f' exists and equals 0. That is,

$$\int_a^b f(x)\varphi'(x)\,dx=0, \quad \forall\,\varphi\in C^\infty_0\big((a,b)\big). \tag{12.1.13}$$

Select $\varphi_0\in C^\infty_0\big((a,b)\big)$ such that $\int_a^b\varphi_0(s)\,ds=1$. Then, each $\varphi\in C^\infty_0\big((a,b)\big)$ may be written as

$$\varphi(t)=\varphi_0(t)\int_a^b\varphi(s)\,ds+\psi'(t), \quad \forall\,t\in(a,b), \tag{12.1.14}$$

where

$$\psi(x):=\int_a^x\Big[\varphi(t)-\varphi_0(t)\int_a^b\varphi(s)\,ds\Big]dt, \quad \forall\,x\in(a,b). \tag{12.1.15}$$

An inspection of (12.1.15) reveals that $\psi \in C^\infty\big((a,b)\big)$, and if $[a_0,b_0] \subseteq (a,b)$ is an interval such that $\operatorname{supp}\varphi$, $\operatorname{supp}\varphi_0 \subseteq [a_0,b_0]$, then we also have $\operatorname{supp}\psi \subseteq [a_0,b_0]$ (as seen by analyzing the cases $x \in (a,a_0)$ and $x \in (b_0,b)$). Now define the constant $c := \int_a^b f(x)\varphi_0(x)\,dx$ and use (12.1.13) to obtain

$$\int_a^b f(x)\varphi(x)\,dx = \int_a^b c\,\varphi(s)\,ds, \quad \forall\,\varphi \in C_0^\infty\big((a,b)\big). \tag{12.1.16}$$

Hence, by invoking Theorem 1.3 we obtain $f = c$ almost everywhere in (a,b).

(b) Let $g \in L^1_{\rm loc}\big((a,b)\big)$, $x_0 \in (a,b)$, and set $f(x) := \int_{x_0}^x g(t)\,dt$ for every $x \in (a,b)$. By Lebesgue's dominated convergence theorem, it follows that f is continuous on (a,b), hence $f \in L^1_{\rm loc}\big((a,b)\big)$. There remains to prove that g is the weak derivative of order one of f.

Parenthetically, we note that under the current assumptions, one may not expect f to necessarily be pointwise differentiable in (a,b). As an example, take the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) := x$ for $x \geq 0$ and $f(x) := 0$ for $x < 0$. Then f is not differentiable at zero but its weak derivative f' exists and is equal to $\chi_{(0,\infty)} \in L^1_{\rm loc}(\mathbb{R})$. Also, observe that the fundamental theorem of calculus does not apply in this case since while g is in $L^1_{\rm loc}(\mathbb{R})$ it is not continuous.

Let $\varphi \in C_0^\infty\big((a,b)\big)$ and $c \in (a,b)$ such that $\operatorname{supp}\varphi \subset (c,b)$. Consider two cases.

Case 1: Assume $c > x_0$. Then $\operatorname{supp}\varphi \subset (x_0,b)$ and using the expression for f and Fubini's theorem we may write

$$\begin{aligned}\int_a^b f(x)\varphi'(x)\,dx &= \int_{x_0}^b \left(\int_{x_0}^x g(t)\,dt\right)\varphi'(x)\,dx = \int_{x_0}^b \left(\int_t^b \varphi'(x)\,dx\right) g(t)\,dt \\ &= -\int_{x_0}^b \varphi(t)g(t)\,dt = -\int_a^b g(t)\varphi(t)\,dt.\end{aligned} \tag{12.1.17}$$

Case 2: Assume $c < x_0$. Again, using the expression for f and Fubini's theorem it follows that

$$\begin{aligned}\int_a^b f(x)\varphi'(x)\,dx &= \int_a^{x_0} f(x)\varphi'(x)\,dx + \int_{x_0}^b f(x)\varphi'(x)\,dx \\ &= \int_a^{x_0} \left(\int_{x_0}^x g(t)\,dt\right)\varphi'(x)\,dx + \int_{x_0}^b \left(\int_{x_0}^x g(t)\,dt\right)\varphi'(x)\,dx \\ &= -\int_a^{x_0} g(t)\int_a^t \varphi'(x)\,dx\,dt + \int_{x_0}^b g(t)\int_t^b \varphi'(x)\,dx\,dt \\ &= -\int_a^b g(t)\varphi(t)\,dt.\end{aligned} \tag{12.1.18}$$

From (12.1.17) and (12.1.18) the desired conclusion follows.

(c) Denote by $h \in L^1_{\mathrm{loc}}\big((a,b)\big)$ the weak derivative $f^{(k)}$ and fix an arbitrary point $x_0 \in (a,b)$. If one sets $g(x) := \int_{x_0}^x h(s)\,\mathrm{d}s$ for $x \in (a,b)$, then by Lebesgue's dominated convergence theorem, it follows that g is continuous. Thus, in particular, $g \in L^1_{\mathrm{loc}}\big((a,b)\big)$. Fix $\varphi_0 \in C_0^\infty\big((a,b)\big)$ with $\int_a^b \varphi_0(t)\,\mathrm{d}t = 1$. Using induction and what we have proved in part (b), the desired conclusion will follow if we show that

$$f^{(k-1)} = g - \int_a^b g(t)\varphi_0(t)\,\mathrm{d}t + (-1)^{k-1}\int_a^b f(t)\varphi_0^{(k-1)}(t)\,\mathrm{d}t \quad \text{a.e. on } (a,b). \tag{12.1.19}$$

Let $\varphi \in C_0^\infty\big((a,b)\big)$ and write it as in (12.1.14)–(12.1.15). Then

$$\begin{aligned}
\int_a^b g(x)\varphi(x)\,\mathrm{d}x &= \Big(\int_a^b g(x)\varphi_0(x)\,\mathrm{d}x\Big)\Big(\int_a^b \varphi(t)\,\mathrm{d}t\Big) + \int_a^b g(x)\psi'(x)\,\mathrm{d}x \\
&= \Big(\int_a^b g(x)\varphi_0(x)\,\mathrm{d}x\Big)\Big(\int_a^b \varphi(t)\,\mathrm{d}t\Big) - \int_a^b h(x)\psi(x)\,\mathrm{d}x \\
&= \Big(\int_a^b g(x)\varphi_0(x)\,\mathrm{d}x\Big)\Big(\int_a^b \varphi(t)\,\mathrm{d}t\Big) + (-1)^{k-1}\int_a^b f(x)\psi^{(k)}(x)\,\mathrm{d}x \\
&= \Big(\int_a^b g(x)\varphi_0(x)\,\mathrm{d}x\Big)\Big(\int_a^b \varphi(t)\,\mathrm{d}t\Big) + (-1)^{k-1}\int_a^b f(x)\varphi^{(k-1)}(x)\,\mathrm{d}x \\
&\quad + (-1)^k\Big(\int_a^b f(x)\varphi_0^{(k-1)}(x)\,\mathrm{d}x\Big)\Big(\int_a^b \varphi(t)\,\mathrm{d}t\Big).
\end{aligned} \tag{12.1.20}$$

Now (12.1.19) follows from (12.1.20) and Theorem 1.3.

(d) Using what we have proved in part (c), we obtain that the weak derivative $f^{(j)}$ exists for each $j \in \mathbb{N}$ satisfying $j \le k$. From what we proved in part (a) we know that $f^{(k-1)} = a_{k-1} \in \mathbb{C}$, while from part (c) we know that $f^{(k-2)} = a_{k-1}x - a_{k-1}x_0 - c$. The proof may be now completed by induction.

(e) No; see Exercise 1.26.

Exercise 1.34 Let $R > 0$ be such that $\operatorname{supp}\theta \subseteq \overline{B(0,R)}$. Then $\varphi_j \in C_0^\infty(\mathbb{R}^n)$ and $\operatorname{supp}\varphi_j \subseteq \overline{B(0,R)}$ for each $j \in \mathbb{N}$. Also, given any $\alpha \in \mathbb{N}_0^n$ we have

$$\partial^\alpha \varphi_j = \mathrm{e}^{-j} j^{m+|\alpha|}(\partial^\alpha\theta)(j\cdot) \qquad \text{for every } j \in \mathbb{N}. \tag{12.1.21}$$

Hence,

$$\sup_{x\in\overline{B(0,R)}} |\partial^\alpha\varphi_j(x)| \le \mathrm{e}^{-j} j^{m+|\alpha|}\|\theta\|_{L^\infty(\mathbb{R}^n)} \xrightarrow[j\to\infty]{} 0.$$

This shows that $\varphi_j \xrightarrow[j\to\infty]{\mathcal{D}(\mathbb{R}^n)} 0$.

Exercise 1.35 Clearly, for every $j \in \mathbb{N}$, $\varphi_j \in C_0^\infty(\mathbb{R}^n)$ and $\operatorname{supp}\varphi_j = \operatorname{supp}\theta - jh$. Hence, for every $R > 0$ there exists $j_0 \in \mathbb{N}$ such that $(\operatorname{supp}\theta - jh) \cap B(0,R) = \emptyset$ for $j \geq j_0$. This shows that there is no compact set $K \subset \mathbb{R}^n$ such that $\operatorname{supp}\varphi_j \subseteq K$ for all $j \in \mathbb{N}$, which implies that $\{\varphi_j\}_{j\in\mathbb{N}}$ does not converge in $\mathcal{D}(\mathbb{R}^n)$. In addition, if $\alpha \in \mathbb{N}_0^n$, K is a compact set in $\mathbb{R}^n$, and $R > 0$ is such that $K \subseteq B(0,R)$, then $\sup_{x\in K} |\partial^\alpha \varphi_j(x)| = \sup_{x\in K} |(\partial^\alpha\theta)(x + jh)| = 0$ for all $j \geq j_0$, where $j_0 \in \mathbb{N}$ is such that $(\operatorname{supp}\theta - jh) \cap B(0,R) = \emptyset$. This proves that $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} 0$.

Exercise 1.36 Suppose there exists $\varphi \in C_0^\infty(\mathbb{R}^n)$ such that $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} \varphi$. Then necessarily $\varphi_j \xrightarrow[j\to\infty]{} \varphi$ pointwise (explain why). Since $\lim_{j\to\infty} \varphi_j(x) = 0$ for every $x \in \mathbb{R}^n$, this forces $\varphi = 0$. However, if $\alpha \in \mathbb{N}_0$ is such that $|\alpha| = 1$, then $\partial^\alpha \varphi_j = (\partial^\alpha\theta)(j\cdot)$ for every $j \in \mathbb{N}$, hence for each compact K we have $\sup_{x\in K} |\partial^\alpha \varphi_j| = \|\partial^\alpha\theta\|_{L^\infty(K)}$ which does not converge to zero as $j \to \infty$, leading to a contradiction.

Exercise 1.37 One implication is immediate. If $\theta \neq 0$, then $\operatorname{supp}\varphi_j = jh + \frac{1}{j}\operatorname{supp}\theta$ for each $j \in \mathbb{N}$, and since $h \neq 0$, there is no compact $K \subset \mathbb{R}^n$ such that $\operatorname{supp}\varphi_j \subseteq K$ for all $j \in \mathbb{N}$. This shows that $\{\varphi_j\}_{j\in\mathbb{N}}$ does not converge in $\mathcal{D}(\mathbb{R}^n)$.

Exercise 1.38 $\{\varphi_j\}_{j\in\mathbb{N}}$ does not converges in $\mathcal{D}(\mathbb{R}^n)$ since $\operatorname{supp}\varphi_j = j\operatorname{supp}\theta$ for every $j \in \mathbb{N}$, thus there is no compact $K \subset \mathbb{R}^n$ such that $\operatorname{supp}\varphi_j \subseteq K$ for all $j \in \mathbb{N}$. On the other hand, if $\alpha \in \mathbb{N}_0^n$ then $|\partial^\alpha \varphi_j(x)| \leq j^{-1-|\alpha|}\|\partial^\alpha\theta\|_{L^\infty(\mathbb{R}^n)}$ for every $x \in \mathbb{R}^n$. Thus, $\partial^\alpha\varphi_j$ converges to zero uniformly on $\mathbb{R}^n$, proving that $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} 0$.

Exercise 1.39 Let $x_0 \in \Omega$ and $R > 0$ be such that $B(x_0,R) \subset \Omega$. Consider a nonzero function $\varphi \in C_0^\infty(\mathbb{R})$ with the property that $\operatorname{supp}\varphi \subset B(0,R)$ and define the sequence $\varphi_j(x) := \frac{1}{j}\varphi\big(j(x - x_0)\big)$, $x \in \Omega$, $j \in \mathbb{N}$. Show that the sequence $\{\varphi_j\}_{j\in\mathbb{N}}$ satisfies the hypotheses in the problem but it does not converge in $\mathcal{D}(\mathbb{R}^n)$.

12.2 Solutions to Exercises from Sect. 2.10

Exercise 2.105 Let K be a compact set in $\mathbb{R}$ and fix an arbitrary function $\varphi \in C_0^\infty(\mathbb{R})$ with $\operatorname{supp}\varphi \subseteq K$. By the mean value theorem we have

$$|\varphi\big(\frac{1}{j^2}\big) - \varphi(0)| \leq \frac{1}{j^2}\|\varphi'\|_{L^\infty(K)}.$$

This shows that the series in (2.10.1) is absolutely convergent, hence u is well-defined. Clearly, u is linear and since $|u(\varphi)| \leq \Big(\sum_{j=1}^{\infty} j^{-2}\Big) \sup_{x\in K} |\varphi'(x)|$ it follows

that $u \in \mathcal{D}'(\mathbb{R})$, it has finite order, and its order is at most 1. To see that u is not of order zero, consider the sequence of functions $\{\varphi_k\}_{k\in\mathbb{N}}$ satisfying, for each $k \in \mathbb{N}$, the conditions

$$\begin{aligned} &\varphi_k \in C_0^\infty\big((0,2)\big),\ \ 0 \le \varphi_k \le \tfrac{1}{k}, \\ &\varphi_k(x) = 0 \text{ if } x \le \tfrac{1}{(k+1)^2},\ \ \varphi_k(x) = \tfrac{1}{k} \text{ if } x \in \Big[\tfrac{1}{k^2}, 1\Big]. \end{aligned} \tag{12.2.1}$$

Then $\langle u, \varphi_k\rangle = \sum\limits_{j=1}^{k} \varphi_k\big(\frac{1}{j^2}\big) = 1$. If we assume that u has order zero, then there exists a finite positive number C such that $|\langle u, \varphi\rangle| \le C \sup_{x\in[0,2]} |\varphi(x)|$ for every $\varphi \in C_0^\infty(\mathbb{R})$ with $\operatorname{supp}\varphi \subseteq [0,2]$. This implies $1 = |\langle u, \varphi_k\rangle| \le \frac{C}{k}$ for every $k \in \mathbb{N}$, which leads to a contradiction. Hence the order of u is 1.

Exercise 2.106 Take u to be the distribution given by the constant function 1. Then this u does not satisfy (2.10.2). Indeed, for any compact $K \subset \Omega$ we can choose $\varphi \in C_0^\infty(\Omega)$ with $\operatorname{supp}\varphi \subseteq \Omega \setminus K$ and $\int_\Omega \varphi\,\mathrm{d}x \neq 0$ which would lead to a contradiction if (2.10.2) were true for $u = 1$.

Exercise 2.107 Use a reasoning similar in spirit to the one from Example 2.9.

Exercise 2.108 Estimate $\int_{B(0,R)} |f(x)|\,\mathrm{d}x$ by working in polar coordinates [based on formula (13.8.7)] and using Proposition 13.46, as well as (2.1.9).

Exercise 2.109 Let $\varphi \in C_0^\infty(\mathbb{R})$ with $\operatorname{supp}\varphi \subset (-R, R)$. Then using integration by parts we have

$$\begin{aligned} \big\langle (\ln|x|)', \varphi\big\rangle &= -\big\langle \ln|x|, \varphi'\big\rangle \\ &= -\int_{\mathbb{R}} \varphi'(x)\ln|x|\,\mathrm{d}x = -\lim_{\varepsilon\to 0^+} \int_{\varepsilon<|x|<R} \varphi'(x)\ln|x|\,\mathrm{d}x \\ &= \lim_{\varepsilon\to 0^+} \Big[\int_{\varepsilon<|x|<R} \frac{\varphi(x)}{x}\,\mathrm{d}x - \varphi(x)\ln|x|\Big|_{-R}^{-\varepsilon} - \varphi(x)\ln|x|\Big|_{\varepsilon}^{R} \Big] \\ &= \Big\langle \mathrm{P.V.}\,\frac{1}{x}, \varphi\Big\rangle + \lim_{\varepsilon\to 0^+} \big[\big(\varphi(\varepsilon) - \varphi(-\varepsilon)\big)\ln\varepsilon\big] \\ &= \Big\langle \mathrm{P.V.}\,\frac{1}{x}, \varphi\Big\rangle. \end{aligned} \tag{12.2.2}$$

For the last equality in (12.2.2) note that

$$\Big|\big(\varphi(\varepsilon) - \varphi(-\varepsilon)\big)\ln\varepsilon\Big| \le 2\|\varphi'\|_{L^\infty(\mathbb{R})}\,\varepsilon|\ln\varepsilon| \xrightarrow[\varepsilon\to 0^+]{} 0. \tag{12.2.3}$$

Exercise 2.110 It is immediate that f is continuous on $\mathbb{R}\setminus\{0\}$, while its continuity at $x = 0$ follows from the fact that

$$\lim_{x\to 0} x\ln|x| = 0. \tag{12.2.4}$$

Fix $\varphi \in C_0^\infty(\mathbb{R})$. Starting with the definition of distributional derivative, then applying Lebesgue's dominated convergence theorem (in concert with the properties of φ), then integrating by parts and using (12.2.4), we obtain

$$\begin{aligned}\big\langle f', \varphi\big\rangle &= -\big\langle f, \varphi'\big\rangle \\ &= -\lim_{\varepsilon\to 0^+}\Big[\int_{-\infty}^{-\varepsilon}(x\ln(-x)-x)\varphi'(x)\,dx + \int_{\varepsilon}^{\infty}(x\ln x - x)\varphi'(x)\,dx\Big] \\ &= \lim_{\varepsilon\to 0^+}\Big[\int_{-\infty}^{-\varepsilon}\ln(-x)\varphi(x)\,dx + \int_{\varepsilon}^{\infty}(\ln x)\varphi(x)\,dx\Big] \\ &= \big\langle \ln|x|, \varphi\big\rangle. \end{aligned} \tag{12.2.5}$$

The last equality in (12.2.5) uses the fact that $\ln|x| \in L^1_{\rm loc}(\mathbb{R})$.

Exercise 2.111 Fix $j \in \{1,\dots,n\}$ and $\varphi \in C_0^\infty(\mathbb{R}^n)$. Then

$$\begin{aligned}\big\langle \partial_j(\ln|x|), \varphi\big\rangle &= \big\langle \ln|x|, \partial_j\varphi\big\rangle = -\int_{\mathbb{R}^n}(\ln|x|)\partial_j\varphi(x)\,dx \\ &= -\lim_{\varepsilon\to 0^+}\int_{|x|\geq\varepsilon}(\ln|x|)\partial_j\varphi(x)\,dx \\ &= \lim_{\varepsilon\to 0^+}\int_{|x|\geq\varepsilon}\frac{x_j}{|x|^2}\varphi(x)\,dx + \lim_{\varepsilon\to 0^+}\int_{|x|=\varepsilon}(\ln\varepsilon)\frac{x_j}{\varepsilon}\varphi(x)\,d\sigma(x) \\ &= \int_{\mathbb{R}^n}\frac{x_j}{|x|^2}\varphi(x)\,dx + \lim_{\varepsilon\to 0^+}\varepsilon^{n-1}(\ln\varepsilon)\int_{S^{n-1}}\omega_j\varphi(\varepsilon\omega)\,d\sigma(\omega) \end{aligned} \tag{12.2.6}$$

For the third and last equality in (12.2.6) we used Lebesgue's dominated convergence theorem [note that $\frac{x_j}{|x|^2}$, $\ln|x| \in L^1_{\rm loc}(\mathbb{R}^n)$ and φ is compactly supported]. The fourth equality uses the integration by parts formula (13.7.4). Also, for the last equality, in the integral on $\partial B(0,\varepsilon)$ we made the change of variables $x = \varepsilon\omega$ with $\omega \in S^{n-1}$. One more application of Lebesgue's dominated convergence theorem yields

$$\lim_{\varepsilon\to 0^+}\int_{S^{n-1}}\omega_j\varphi(\varepsilon\omega)\,d\sigma(\omega) = \varphi(0)\int_{S^{n-1}}\omega_j\,d\sigma(\omega) = 0, \tag{12.2.7}$$

where the last equality is due to the fact that the integral of an odd function over the unit sphere is zero. Moreover, (2.1.9) implies $\lim_{\varepsilon\to 0^+}\varepsilon^{n-1}(\ln\varepsilon) = 0$ (recall that we are assuming $n \geq 2$). Returning with all these comments to (12.2.6) we arrive at the conclusion that

$$\big\langle \partial_j(\ln|x|), \varphi\big\rangle = \Big\langle \frac{x_j}{|x|^2}, \varphi\Big\rangle \quad \text{for every} \quad \varphi \in C_0^\infty(\mathbb{R}^n).$$

Hence, if $n \geq 2$ then $\partial_j(\ln|x|) = \frac{x_j}{|x|^2}$ in $\mathcal{D}'(\mathbb{R}^n)$.

Exercise 2.112 Fix $j \in \mathbb{N}$. Using the change of variables $y = jx - jt$ the expression for ψ_j becomes

$$\psi_j(x) = \int_{jx-j^2}^{jx-1} \theta(y)\,\mathrm{d}y \quad \text{for} \quad x \in \mathbb{R}.$$

Hence $\psi_j(x) = 0$ if $x \leq 0$, while for $x > 0$ we have $\psi_j(x) = \int_{-1}^{1} \theta(y)\,\mathrm{d}y = 1$ if $j \geq j_0$, where $j_0 \in \mathbb{N}$ is such that $j_0 x - j_0^2 \leq -1$ and $j_0 x - 1 \geq 1$. This shows that ψ_j converges pointwise to H as $j \to \infty$. In addition, $|\psi_j(x)| \leq \int_{-1}^{1} |\theta(y)|\,\mathrm{d}y < \infty$ for every $j \in \mathbb{N}$. Hence, by applying Lebesgue's dominated convergence theorem,

$$\lim_{j\to\infty} \int_{\mathbb{R}} \psi_j(x)\varphi(x)\,\mathrm{d}x = \int_{\mathbb{R}} H(x)\varphi(x)\,\mathrm{d}x \quad \text{for each} \quad \varphi \in C_0^\infty(\mathbb{R}).$$

This proves that $\psi_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} H$ as wanted.

Exercise 2.113 If $\varphi \in C_0^\infty(\mathbb{R})$ then the support condition for φ guarantees that only finitely many terms in the sum $\sum_{j=1}^{\infty} \varphi^{(j)}(j)$ are nonzero, hence u is well-defined. Clearly u is linear. If K is a compact set in $\mathbb{R}$ and $k \in \mathbb{N}$ is such that $K \subseteq [-k, k]$, then

$$|u(\varphi)| \leq \sum_{j=1}^{k} |\varphi^{(j)}(j)| \leq k \sup_{x\in K,\, j\leq k} |\varphi^{(j)}(x)| \quad \text{for} \quad \varphi \in C_0^\infty(\mathbb{R}) \quad \text{with } \operatorname{supp}\varphi \subseteq K.$$

This proves that $u \in \mathcal{D}'(\mathbb{R})$.

Suppose the distribution u has finite order. Then there exists a nonnegative integer k_0 with the property that for each compact set $K \subset \mathbb{R}$ there is a finite constant $C_K \geq 0$ such that $|\langle u, \varphi\rangle| \leq C_K \sup_{x\in K,\, j\leq k_0} |\varphi^{(j)}(x)|$ for every $\varphi \in C_0^\infty(\mathbb{R})$ with $\operatorname{supp}\varphi \subseteq K$. In particular, from this and the definition of u it follows that there exists $C \in [0, \infty)$ such that whenever $\varphi \in C_0^\infty(\mathbb{R})$ satisfies the condition $\operatorname{supp}\varphi \subseteq \left[k_0 + \frac{1}{2}, k_0 + \frac{3}{2}\right]$ we have

$$|\varphi^{(k_0+1)}(k_0+1)| = |\langle u, \varphi\rangle| \leq C \sup_{x\in\left[k_0+\frac{1}{2}, k_0+\frac{1}{3}\right],\, \ell\leq k_0} |\varphi^{(\ell)}(x)|. \tag{12.2.8}$$

Now consider $\theta \in C_0^\infty\big((-1/2, 1/2)\big)$ satisfying $\theta(0) = 1$ and construct the sequence of smooth functions $\varphi_j(x) := \theta(jx - j(k_0+1))$ for $x \in \mathbb{R}$ and $j \in \mathbb{N}$. Then, for each $j \in \mathbb{N}$, we have

$$\begin{aligned}
&\operatorname{supp}\varphi_j \subseteq \left[k_0 + \frac{1}{2}, k_0 + \frac{3}{2}\right],\\
&\varphi_j^{(\ell)} = j^\ell \theta^{(\ell)}(j\cdot - j(k_0+1)), \quad \forall\, \ell \in \mathbb{N}_0,\\
&\text{and} \quad \varphi_j^{(k_0+1)}(k_0+1) = j^{k_0+1}.
\end{aligned}$$

Combining all these facts with (12.2.8) it follows that for each $j\in\mathbb{N}$,

$$\begin{aligned} j^{k_0+1} &\leq C \sup_{x\in\left[-\frac{1}{2},\frac{1}{2}\right],\,\ell\leq k_0} j^{\ell}|\theta^{(\ell)}(x)| \\ &\leq C\max\left\{\|\theta^{(\ell)}\|_{L^\infty([-1/2,1/2])}:\ \ell\leq k_0\right\}j^{k_0}. \end{aligned} \tag{12.2.9}$$

Choosing now j sufficiently large in (12.2.9) leads to a contradiction. Hence u does not have finite order.

Exercise 2.114 Note that for each $j\in\mathbb{N}$ we have $f_j\in L^1_{\rm loc}(\mathbb{R}^n)$. Take some $\varphi\in C^\infty_0(\mathbb{R}^n)$ and suppose $R>0$ is such that $\operatorname{supp}\varphi\subset B(0,R)$. Then

$$\langle f_j,\varphi\rangle=\frac{1}{j}\int_{B(0,R)}\frac{\varphi(x)-\varphi(0)}{|x|^{n-\frac{1}{j}}}\,{\rm d}x+\frac{\varphi(0)}{j}\int_{B(0,R)}\frac{1}{|x|^{n-\frac{1}{j}}}\,{\rm d}x. \tag{12.2.10}$$

Using the mean value theorem and then (13.8.6) we may further write

$$\begin{aligned} \left|\frac{1}{j}\int_{B(0,R)}\frac{\varphi(x)-\varphi(0)}{|x|^{n-\frac{1}{j}}}\,{\rm d}x\right| &\leq \frac{\|\nabla\varphi\|_{L^\infty(\mathbb{R}^n)}}{j}\int_{B(0,R)}\frac{1}{|x|^{n-1-\frac{1}{j}}}\,{\rm d}x \\ &=\frac{\omega_{n-1}\|\nabla\varphi\|_{L^\infty(\mathbb{R}^n)}}{j}\int_0^R\rho^{\frac{1}{j}}\,{\rm d}\rho=\frac{\omega_{n-1}\|\nabla\varphi\|_{L^\infty(\mathbb{R}^n)}R^{\frac{1}{j}+1}}{j+1}\xrightarrow[j\to\infty]{}0. \end{aligned} \tag{12.2.11}$$

One more use of (13.8.6) implies

$$\begin{aligned} \frac{1}{j}\int_{B(0,R)}\frac{1}{|x|^{n-\frac{1}{j}}}\,{\rm d}x&=\frac{\omega_{n-1}}{j}\int_0^R\rho^{\frac{1}{j}-1}\,{\rm d}\rho \\ &=\omega_{n-1}R^{\frac{1}{j}}\xrightarrow[j\to\infty]{}\omega_{n-1}. \end{aligned} \tag{12.2.12}$$

By combining (12.2.10)–(12.2.12) we obtain $\lim_{j\to\infty}\langle f_j,\varphi\rangle=\omega_{n-1}\varphi(0)=\langle\omega_{n-1}\delta,\varphi\rangle$. The desired conclusion now follows.

Exercise 2.115 Note that $f_\varepsilon\in L^1_{\rm loc}(\mathbb{R})$ for every $\varepsilon>0$, hence $f_\varepsilon\in\mathcal{D}'(\mathbb{R})$. Also, $\lim_{\varepsilon\to0^+}f_\varepsilon(x)=0$ for every $x\in\mathbb{R}\setminus\{0\}$. For a given function $\varphi\in C^\infty_0(\mathbb{R})$ let $R>0$ be such that $\operatorname{supp}\varphi\subset(-R,R)$ and for $\varepsilon>0$ write

$$\langle f_\varepsilon,\varphi\rangle=\frac{1}{\pi}\int_{-R}^R\frac{x\varepsilon}{x^2+\varepsilon^2}\cdot\frac{\varphi(x)-\varphi(0)}{x}\,{\rm d}x+\frac{\varphi(0)\varepsilon}{\pi}\int_{-R}^R\frac{{\rm d}x}{x^2+\varepsilon^2}=:I+II. \tag{12.2.13}$$

Then $\left|\frac{x\varepsilon}{x^2+\varepsilon^2}\right|\leq 1/2$ for each $x\in\mathbb{R}$ if $\varepsilon>0$, so we may apply Lebesgue's dominated convergence theorem to conclude that $\lim_{\varepsilon\to0^+}I=0$. Also, integrating the

second term and then taking the limit yields $\lim_{\varepsilon\to 0^+} II = \frac{2\varphi(0)}{\pi} \lim_{\varepsilon\to 0^+} \arctan(R/\varepsilon) = \varphi(0)$. In conclusion, $\lim_{\varepsilon\to 0^+} \langle f_\varepsilon, \varphi\rangle = \varphi(0) = \langle \delta, \varphi\rangle$ as desired.

Exercise 2.116 Let $\varphi \in C_0^\infty(\mathbb{R}^n)$ and $\varepsilon > 0$. Using first the change of variables $x = 2\sqrt{\varepsilon}y$ and then Lebesgue's dominated convergence theorem, we have

$$\begin{aligned}\langle f_\varepsilon, \varphi\rangle &= (4\pi\varepsilon)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \mathrm{e}^{-\frac{|x|^2}{4\varepsilon}} \varphi(x)\,\mathrm{d}x \qquad (12.2.14)\\ &= \pi^{-\frac{n}{2}} \int_{\mathbb{R}^n} \mathrm{e}^{-|y|^2} \varphi(2\sqrt{\varepsilon}y)\,\mathrm{d}y \xrightarrow[\varepsilon\to 0^+]{} \pi^{-\frac{n}{2}} \int_{\mathbb{R}^n} \mathrm{e}^{-|y|^2} \varphi(0)\,\mathrm{d}y = \varphi(0).\end{aligned}$$

Exercise 2.117 Note that for every $\varepsilon > 0$ we have $|f_\varepsilon^\pm| \le \frac{1}{\varepsilon}$, hence these are functions in $L^1_{\mathrm{loc}}(\mathbb{R}) \subset \mathcal{D}'(\mathbb{R})$. Let $\varphi \in C_0^\infty(\mathbb{R})$ and let $R > 0$ be such that $\operatorname{supp}\varphi \subset (-R, R)$. Then

$$\langle f_\varepsilon^\pm(x), \varphi\rangle = \varphi(0) \int_{-R}^{R} \frac{x \mp i\varepsilon}{x^2+\varepsilon^2}\,\mathrm{d}x + \int_{-R}^{R} \frac{x \mp i\varepsilon}{x^2+\varepsilon^2}[\varphi(x) - \varphi(0)]\,\mathrm{d}x =: I + II. \qquad (12.2.15)$$

Since $\frac{x}{x^2+\varepsilon^2}$ is odd, we further obtain

$$I = \mp 2i\varphi(0) \arctan(R/\varepsilon) \xrightarrow[\varepsilon\to 0^+]{} \mp i\pi\varphi(0) = \langle \mp i\pi\delta, \varphi\rangle. \qquad (12.2.16)$$

As for II, since $\lim_{\varepsilon\to 0^+} \frac{x\mp i\varepsilon}{x^2+\varepsilon^2} = \frac{1}{x}$ for every $x \neq 0$ and

$$\left|\frac{x \mp i\varepsilon}{x^2+\varepsilon^2}\big[\varphi(x) - \varphi(0)\big]\right| \le \frac{|\varphi(x) - \varphi(0)|}{|x|} \in L^1\big((-R, R)\big),$$

we may apply Lebesgue's dominated convergence theorem to obtain

$$\lim_{\varepsilon\to 0^+} II = \int_{-R}^{R} \frac{\varphi(x) - \varphi(0)}{x}\,\mathrm{d}x = \Big\langle \mathrm{P.V.}\,\frac{1}{x}, \varphi\Big\rangle. \qquad (12.2.17)$$

Exercise 2.118 You may use the following outline:

(a) Show that for $f \in L^1(\mathbb{R})$ one has $\lim_{j\to\infty} \int_{\mathbb{R}} f(x)\sin(jx)\,\mathrm{d}x = 0$ by reducing to the case $f \in C_0^\infty(\mathbb{R})$ based on density arguments.

(b) For $\varphi \in C_0^\infty(\mathbb{R})$ and $R > 0$, write the expression $\big\langle \frac{1}{\pi}\frac{\sin jx}{x}, \varphi(x)\big\rangle$ as the sum of two integrals, one over the region $\{x \in \mathbb{R} : |x| \le R\}$, the other over $\{x \in \mathbb{R} : |x| > R\}$, and use (a) to obtain the desired conclusion. Recall that $\int_{\mathbb{R}} \frac{\sin x}{x}\,\mathrm{d}x = \pi$.

Exercise 2.119 Let $\varphi \in C_0^\infty(\mathbb{R})$.

(a) In the expression for $\langle f_j, \varphi\rangle$ use the change of variables $x = y/\sqrt{j}$ and then Lebesgue's dominated convergence theorem to conclude $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} \delta$.

(b) Integrate by parts $m+1$ times to conclude that

$$|\langle f_j, \varphi\rangle| = j^m \left| \int_{\mathbb{R}} \cos(jx)\varphi(x)\,\mathrm{d}x \right| \le j^{-1} \int_{\mathbb{R}} |\varphi^{(m+1)}(x)|\,\mathrm{d}x \xrightarrow[j\to\infty]{} 0,$$

hence $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} 0$.

(c) Use a reasoning similar to the one in the proof of (a) to conclude that $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} \delta$, this time via the change of variables $x = y/j$.

(d) Not convergent since if the function $\varphi \in C_0^\infty(\mathbb{R})$ is such that $\varphi(0) \neq 0$, then $\langle u_j, \varphi\rangle = (-1)^j \varphi(1/j)$ and the sequence $\{(-1)^j \varphi(1/j)\}_{j\in\mathbb{N}}$ is not convergent.

(e) Use the mean value theorem to obtain that $\langle u_j, \varphi\rangle \xrightarrow[j\to\infty]{} \varphi'(0) = \langle -\delta\,', \varphi\rangle$.

(f) $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} \text{P.V.}\,\frac{1}{x}$.

(g) Use a reasoning similar to the one in the proof of (a) to conclude that $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} \delta$, this time via the change of variables $x = y/j$ and recalling that $\int_{\mathbb{R}} \frac{(\sin y)^2}{y^2}\,\mathrm{d}y = \pi$.

(h) Integrate by parts $m+1$ times and then use Lebesgue's dominated convergence theorem to conclude that $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} 0$.

(j) Integrate by parts twice to obtain

$$\begin{aligned}\langle u_j, \varphi\rangle &= i\varphi(0) + i\int_0^\infty \mathrm{e}^{ijx}\varphi'(x)\,\mathrm{d}x \\ &= i\varphi(0) - \frac{1}{j}\varphi'(0) - \frac{1}{j}\int_0^\infty \mathrm{e}^{ijx}\varphi''(x)\,\mathrm{d}x \xrightarrow[j\to\infty]{} i\varphi(0) = \langle i\delta, \varphi\rangle.\end{aligned} \tag{12.2.18}$$

Exercise 2.120 $(H(\cdot - a))' = \delta_a$ in $\mathcal{D}'(\mathbb{R})$.

Exercise 2.121 $(u_f)' = a\delta_a + H(\cdot - a)$ in $\mathcal{D}'(\mathbb{R})$.

Exercise 2.122 Let $\varphi \in C_0^\infty(\mathbb{R})$. Then integration by parts yields

$$-\int_{\mathbb{R}} f(x)\varphi'(x)\,\mathrm{d}x = \int_{-\infty}^{0} \sin(x)\varphi'(x)\,\mathrm{d}x - \int_{0}^{\infty} \sin(x)\varphi'(x)\,\mathrm{d}x$$
$$= -\int_{-\infty}^{0} \cos(x)\varphi(x)\,\mathrm{d}x + \int_{0}^{\infty} \cos(x)\varphi(x)\,\mathrm{d}x, \qquad (12.2.19)$$

hence $(u_f)' = \cos(x)H(x) - \cos(x)H(-x)$ in $\mathcal{D}'(\mathbb{R})$. Similarly,

$$\int_{\mathbb{R}} f(x)\varphi''(x)\,\mathrm{d}x = 2\varphi(0) + \int_{-\infty}^{0} \sin(x)\varphi(x)\,\mathrm{d}x - \int_{0}^{\infty} \sin(x)\varphi(x)\,\mathrm{d}x, \quad (12.2.20)$$

hence $(u_f)'' = 2\delta - \sin(x)H(x) + \sin(x)H(-x)$ in $\mathcal{D}'(\mathbb{R})$.

Exercise 2.123 Let $a, b \in I$ be such that $a < x_0 < b$. Since for each $x \in [a, x_0)$ we have $f(x) = f(a) + \int_a^x f'(t)\,\mathrm{d}t$, Lebesgue's dominated convergence theorem gives that $\lim_{x \to x_0^-} f(x)$ exists and equals $f(a) + \int_a^{x_0} f'(t)\,\mathrm{d}t$. A similar argument proves that $\lim_{x \to x_0^+} f(x) = f(b) - \int_{x_0}^b f'(t)\,\mathrm{d}t$. Note that $f \in L^1\big([a,b]\big)$. Suppose now that $\varphi \in C_0^\infty(I)$ satisfies $\operatorname{supp}\varphi \subset [c,d]$ for some $c < x_0 < d$. Then for $\varepsilon > 0$ small enough we have

$$\langle (u_f)', \varphi\rangle = -\langle u_f, \varphi'\rangle = -\int_c^d f(t)\varphi'(t)\,\mathrm{d}t$$
$$= -f(x_0-\varepsilon)\varphi(x_0-\varepsilon) + \int_c^{x_0-\varepsilon} f'(t)\varphi(t)\,\mathrm{d}t - \int_{x_0-\varepsilon}^{x_0+\varepsilon} f(t)\varphi'(t)\,\mathrm{d}t$$
$$+ f(x_0+\varepsilon)\varphi(x_0+\varepsilon) + \int_{x_0+\varepsilon}^d f'(t)\varphi(t)\,\mathrm{d}t. \qquad (12.2.21)$$

Send $\varepsilon \to 0^+$ in (12.2.21) and observe that $\lim_{\varepsilon \to 0^+} \int_{x_0-\varepsilon}^{x_0+\varepsilon} f(t)\varphi'(t)\,\mathrm{d}t = 0$ by Lebesgue's dominated convergence theorem. The case when x_0 is not in the interior of the support of φ is simpler, and can be handled via a direct integration by parts.

Exercise 2.125 Use Exercise 2.123 and induction.

Exercise 2.126 Use Exercise 2.123 and the fact that since $\{x_k\}_{k\in\mathbb{N}}$ has no accumulation point in I, for each $R > 0$ only finitely many terms of the sequence $\{x_k\}_{k\in\mathbb{N}}$ are contained in $(-R, R)$.

Exercise 2.127 Use Exercise 2.126.

Exercise 2.128 Clearly δ_Σ is well-defined and linear. Also, for each compact set $K \subset \mathbb{R}^n$ and each $\varphi \in C_0^\infty(\mathbb{R}^n)$ satisfying $\operatorname{supp}\varphi \subseteq K$ we have

$$|\delta_\Sigma(\varphi)| \leq \sigma(\Sigma \cap K) \sup_{x\in K} |\varphi(x)|. \tag{12.2.22}$$

This shows that $\delta_\Sigma \in \mathcal{D}'(\mathbb{R}^n)$ and has order zero. Also $\langle \delta_\Sigma, \varphi\rangle = 0$ if $\operatorname{supp}\varphi \cap \Sigma = \emptyset$, thus $\operatorname{supp}\delta_\Sigma \subseteq \Sigma$.

To prove that $\Sigma \subseteq \operatorname{supp}\delta_\Sigma$, note that if $x^* \in \Sigma$, then there exists a neighborhood $U(x^*)$ of x^* and a local parametrization P of class C^1 near x^* as in (13.6.2)–(13.6.3). In particular, if $u_0 \in \mathcal{O}$ is such that $P(u_0) = x^*$, then the vectors $\partial_1 P(u_0), \ldots, \partial_{n-1}P(u_0)$, are linearly independent. Upon recalling the cross product from (13.6.4), this ensures that

$$c_0 := \|\partial_1 P(u_0) \times \cdots \times \partial_{n-1}P(u_0)\| \neq 0. \tag{12.2.23}$$

Since P is of class C^1, it follows that $\|\partial_1 P(u) \times \cdots \times \partial_{n-1}P(u)\| \geq c_0/2$ for each u belonging to some small open neighborhood $\widetilde{\mathcal{O}} \subseteq \mathcal{O}$ of u_0 in $\mathbb{R}^{n-1}$. Using the fact that $P : \mathcal{O} \to P(\mathcal{O})$ is a homeomorphism (see Proposition 13.37), it follows that there exists some open set $V(x^*)$ in $\mathbb{R}^n$ contained in $U(x^*)$ and containing x^* with the property that $P(\widetilde{\mathcal{O}}) = V(x^*) \cap \Sigma$. By further shrinking $\widetilde{\mathcal{O}}$ if necessary, there is no loss of generality in assuming that $V(x^*)$ is bounded.

Now consider $0 < r_1 < r_2 < \infty$ such that $V(x^*) \subseteq \overline{B(x^*, r_1)} \subseteq B(x^*, r_2)$. Pick a function $\varphi \in C_0^\infty\big(B(x^*, r_2)\big)$ with $\varphi \geq 0$, $\varphi = 1$ in a neighborhood of $\overline{B(x^*, r_1)}$ (see Proposition 13.26). Then

$$\langle \delta_\Sigma, \varphi\rangle = \int_\Sigma \varphi(x)\, d\sigma(x) \geq \int_{\Sigma\cap V(x^*)} \varphi(x)\, d\sigma(x) \tag{12.2.24}$$

$$= \int_{\widetilde{\mathcal{O}}} \|\partial_1 P(u) \times \cdots \times \partial_{n-1}P(u)\|\, du \geq c_0|\widetilde{\mathcal{O}}| > 0. \tag{12.2.25}$$

In a similar manner, for each function g satisfying $g \in L^\infty(K \cap \Sigma)$ for any compact set $K \subseteq \mathbb{R}^n$, we have $g\delta_\Sigma \in \mathcal{D}'(\mathbb{R}^n)$ and for each compact K one has

$$|(g\delta_\Sigma)(\varphi)| \leq \|g\|_{L^\infty(K)}\sigma(\Sigma \cap K) \sup_{x\in K} |\varphi(x)|, \quad \forall\, \varphi \in C_0^\infty(\mathbb{R}^n), \ \operatorname{supp}\varphi \subseteq K.$$

Exercise 2.129 Use integration by parts (see Theorem 13.41) and Exercise 2.128.

Exercise 2.130 Use the definition of distributional derivative, integration by parts (cf. Theorem 13.41), and Exercise 2.128.

Exercise 2.131 Let $\varphi \in C_0^\infty(\mathbb{R}^n)$ be such that $\operatorname{supp}\varphi \cap \partial B(0, R) = \emptyset$. In this scenario, we have $\frac{\varphi}{|x|^2-R^2} \in C_0^\infty(\mathbb{R}^n)$, hence we may write

$$\langle u, \varphi\rangle = \Big\langle (|x|^2 - R^2)u, \frac{\varphi}{|x|^2 - R^2}\Big\rangle = 0.$$

This proves that $\operatorname{supp} u \subseteq \partial B(0,R)$, thus u is compactly supported. Examples of distributions satisfying the given equation include $\delta_{\partial B(0,R)}$ and δ_{x_0}, for any $x_0 \in \mathbb{R}^n$ with $|x_0| = R$.

Exercise 2.132 Use Example 2.50.

Exercise 2.133 The derivative of order m is equal to:

(a) $\operatorname{sgn}(x)$ if $m = 1$ and $2\delta^{(m-2)}$ if $m \geq 2$;

(b) $2\delta^{(m-1)}$;

(c) $\sum_{j=0}^{n} \delta^{(2j)} - \sin x\, H$ if $m = 2n+1$, $n \in \mathbb{N}$, and $\sum_{j=1}^{n} \delta^{(2j-1)} + \cos x\, H$ if $m = 2n$, $n \in \mathbb{N}$;

(d) $(\sin x\, H)' = \cos x\, H$ and use (c);

(e) $\delta_1 - \delta_{-1} - 2x\chi_{[-1,1]}$ if $m = 1$, $(\delta_1)' - (\delta_{-1})' + 2\delta_1 + 2\delta_{-1} + 2\chi_{[-1,1]}$ if $m = 2$, and $\delta_1^{(m-1)} - \delta_{-1}^{(m-1)} + 2\delta_1^{(m-2)} + 2\delta_{-1}^{(m-2)} - 2\delta_1^{(m-3)} + 2\delta_{-1}^{(m-3)}$ if $m \geq 3$.

Exercise 2.134 For $\varphi \in C_0^\infty(\mathbb{R}^2)$ use the change of variables $u = x+y$, $v = x-y$ and the reasoning in (1.1.6)–(1.1.8) with $\psi(u,v) := \varphi\big(\frac{u+v}{2}, \frac{u-v}{2}\big)$ for $u \in [2,4]$, $v \in [0,2]$, to write

$$\begin{aligned}
\big\langle (\partial_1^2 - \partial_2^2)\chi_A, \varphi \big\rangle &= \int\int_A [(\partial_1^2\varphi)(x,y) - (\partial_2^2\varphi)(x,y)]\,\mathrm{d}x\,\mathrm{d}y \\
&= 2\int_2^4 \int_0^2 \partial_u\partial_v\psi(u,v)\,\mathrm{d}v\,\mathrm{d}u \\
&= 2[\psi(4,2) - \psi(2,2) - \psi(4,0) + \psi(2,0)] \\
&= 2[\varphi(3,1) - \varphi(2,0) - \varphi(2,2) + \varphi(1,1)] \\
&= \big\langle 2[\delta_{(3,1)} - \delta_{(2,0)} - \delta_{(2,2)} + \delta_{(1,1)}], \varphi \big\rangle. \qquad (12.2.26)
\end{aligned}$$

Exercise 2.135 Fix $\varphi \in C_0^\infty(\mathbb{R}^2)$. Then, using the change of variables $x = t+y$ we obtain

$$\begin{aligned}
\big\langle \partial_1(u_f), \varphi \big\rangle &= -\int\int_{\mathbb{R}^2} \chi_{[0,1]}(x-y)\partial_1\varphi(x,y)\,\mathrm{d}x\,\mathrm{d}y \\
&= -\int\int_{\mathbb{R}^2} \chi_{[0,1]}(t)(\partial_1\varphi)(t+y,y)\,\mathrm{d}t\,\mathrm{d}y \\
&= \int_{\mathbb{R}} [\varphi(y,y) - \varphi(1+y,y)]\,\mathrm{d}y. \qquad (12.2.27)
\end{aligned}$$

Similarly,

$$\big\langle \partial_2(u_f), \varphi \big\rangle = \int_{\mathbb{R}} [\varphi(x, x-1) - \varphi(x,x)]\, dx. \tag{12.2.28}$$

Combining (12.2.27)–(12.2.28) it follows that $\partial_1(u_f) - \partial_2(u_f) = 0$ in $\mathcal{D}'(\mathbb{R}^2)$ and, in turn, that $\partial_1^2(u_f) - \partial_2^2(u_f) = (\partial_1 + \partial_2)\big(\partial_1(u_f) - \partial_2(u_f)\big) = 0$ in $\mathcal{D}'(\mathbb{R}^2)$.

Exercise 2.138 The uniqueness statement is a consequence of Proposition 2.45. Let K be a compact set in $\mathbb{R}^n$ such that $K \subset \Omega$. Refine $\{\Omega_j\}_{j\in I}$ to a finite subcover $\{\Omega_{\ell_k}\}_{k=1}^N$, with $\ell_k \in I$ for $k = 1, \ldots, N$, of K. Consider a partition of unity $\{\varphi_k\}_{1\leq k\leq N}$ subordinate to the cover $\{\Omega_{\ell_k}\}_{k=1}^N$ of K, as given by Theorem 13.29. Hence, for each $k \in \mathbb{N}$ we have $\varphi_k \in C_0^\infty(\Omega)$ satisfies $\operatorname{supp}\varphi_k \subset \Omega_{\ell_k}$ and $\sum_{k=1}^N \varphi_k = 1$ in a neighborhood of K. Next, for each function $\varphi \in C_0^\infty(\Omega)$ with $\operatorname{supp}\varphi \subseteq K$ define $u_K(\varphi) := \sum_{k=1}^N \langle u_{\ell_k}, \varphi_k \varphi\rangle$. Show that $\sum_{k=1}^N \langle u_{\ell_k}, \varphi_k\varphi\rangle$ is independent of the cover of K chosen and of the partition of unity, thus $u_K : \mathcal{D}_K(\Omega) \to \mathbb{C}$ is well-defined. The map u_K is clearly linear. Now set $u : \mathcal{D}(\Omega) \to \mathbb{C}$ to be the map given by $u(\varphi) := u_K(\varphi)$ for each $\varphi \in C_0^\infty(\Omega)$ such that $\operatorname{supp}\varphi \subseteq K$. Show that this map is well-defined, satisfies $u \in \mathcal{D}'(\Omega)$, and $u\big|_{\Omega_j} = u_j$ in $\mathcal{D}'(\Omega_j)$ for every $j \in I$.

Exercise 2.139 Fix $\varphi_0 \in C_0^\infty(\mathbb{R}^n)$ with the property that $\int_{\mathbb{R}^n} \varphi_0(x)\, dx = 1$. Next, let $\varphi \in C_0^\infty(\mathbb{R}^n)$ be arbitrary and set $\lambda := \int_{\mathbb{R}^n} \varphi(x)\, dx$. Hence, if we set $\psi := \varphi - \lambda\varphi_0$, then $\psi \in C_0^\infty(\mathbb{R}^n)$ and $\int_{\mathbb{R}^n} \psi(x)\, dx = 0$, so

$$0 = \langle u, \psi\rangle = \langle u, \varphi\rangle - \lambda\langle u, \varphi_0\rangle. \tag{12.2.29}$$

Consequently, $\langle u, \varphi\rangle = \lambda\langle u, \varphi_0\rangle = \langle c, \varphi\rangle$, where $c := \langle u, \varphi_0\rangle \in \mathbb{C}$.

Exercise 2.140 You may proceed by completing the following steps. Fix some arbitrary $\varphi \in C_0^\infty(\mathbb{R}^n)$.

Step I. Prove that $\int_{\mathbb{R}^n} \varphi(x)\, dx = 0$ if and only if there exist $\varphi_1, \ldots, \varphi_n \in C_0^\infty(\mathbb{R}^n)$ such that $\varphi = \sum_{j=1}^n \partial_j\varphi_j$. Do so by induction over n. Corresponding to $n = 1$ show that if $a, b \in \mathbb{R}$ satisfy $a < b$ and for $\varphi \in C_0^\infty\big((a,b)\big)$ one defines $\phi_1(x) := \int_a^x \varphi(t)\, dt$, $x \in (a,b)$, then $\phi_1 \in C_0^\infty\big((a,b)\big)$ if and only if $\int_a^b \varphi(x)\, dx = 0$.

Step II. Show that the statement from Step I continues to hold if $\mathbb{R}^n$ is replaced by $(a_1, b_1) \times \cdots \times (a_n, b_n)$, where $a_j, b_j \in \mathbb{R}$, $a_j < b_j$ for each $j = 1, \ldots, n$.

Step III. Fix $a_j, b_j \in \mathbb{R}$, $a_j < b_j$ for $j = 1, \ldots, n$ and consider the n-dimensional rectangle $Q := (a_1, b_1) \times \cdots \times (a_n, b_n)$. Let $\varphi_0 \in C_0^\infty(Q)$ be such that

$\int_Q \varphi_0\,dx = 1$. Then, if $\varphi \in C_0^\infty(Q)$ is arbitrary, the function defined by $\psi := \varphi - \big[\int_Q \varphi\,dx\big]\varphi_0$ belongs to $C_0^\infty(Q)$ and satisfies $\int_Q \psi\,dx = 0$. As such, Step II applies and shows that there exist $\varphi_1,\dots,\varphi_n \in C_0^\infty(Q)$ such that $\varphi = \big[\int_Q \varphi\,dx\big]\varphi_0 + \sum_{j=1}^n \partial_j\varphi_j$. In turn, this permits us to write

$$\langle u,\varphi\rangle = \langle u,\varphi_0\rangle \int_Q \varphi\,dx - \sum_{j=1}^n \langle \partial_j u,\varphi_j\rangle = \big\langle \langle u,\varphi_0\rangle,\varphi\big\rangle, \tag{12.2.30}$$

which shows if $c_Q := \langle u,\varphi_0\rangle \in \mathbb{C}$, then $u\big|_Q = c_Q$ in $\mathcal{D}'(Q)$.

Step IV. Since Ω is connected and open, it is path connected. Now combine this with the fact that u is locally constant (as proved in Step III) to finish the proof.

Exercise 2.141 By Proposition 2.73, it suffices to prove that there exists some $v \in \mathcal{D}'(\mathbb{R}^{n-1})$ such that $\langle u,\varphi\otimes\psi\rangle = \langle v,\varphi\rangle\langle\delta,\psi\rangle$ for every $\varphi \in C_0^\infty(\mathbb{R}^{n-1})$ and every $\psi \in C_0^\infty(\mathbb{R})$. Fix $\varphi \in C_0^\infty(\mathbb{R}^{n-1})$, $\psi \in C_0^\infty(\mathbb{R})$, and consider $\psi_0 \in C_0^\infty(\mathbb{R})$ with the property that $\psi_0(0) = 1$. Then there exists some $h \in C_0^\infty(\mathbb{R})$ satisfying $\psi(x_n) - \psi(0)\psi_0(x_n) = x_n h(x_n)$ for every $x_n \in \mathbb{R}$. This and the fact that $x_n u = 0$ allows us to write

$$\begin{aligned}\langle u,\varphi\psi\rangle &= \langle u,\varphi(\psi-\psi(0)\psi_0)\rangle + \langle u,\varphi\psi_0\rangle\psi(0)\\ &= \langle x_n u,\varphi h\rangle + \langle u,\varphi\psi_0\rangle\psi(0) = \langle u,\varphi\psi_0\rangle\langle\delta,\psi\rangle.\end{aligned} \tag{12.2.31}$$

Now define $v : \mathcal{D}(\mathbb{R}^{n-1}) \to \mathbb{C}$ by $v(\varphi) := \langle u,\varphi\psi_0\rangle$ for $\varphi \in C_0^\infty(\mathbb{R}^{n-1})$, and show that $v \in \mathcal{D}(\mathbb{R}^{n-1})$. By (12.2.31) this v does the job.

Exercise 2.142 Fix $\psi \in C_0^\infty(\mathbb{R})$ with the property that $\psi(0) = 1$. Use Exercise 2.141 and induction to show that $u = c\,\delta(x_1)\otimes\cdots\otimes\delta(x_n)$, where $c := \langle u,\psi\otimes\cdots\otimes\psi\rangle \in \mathbb{C}$.

Exercise 2.143 Fix $\psi \in C_0^\infty(\mathbb{R})$ with the property that $\int_{\mathbb{R}} \psi(s)\,ds = 1$. Given any function $\varphi \in C_0^\infty(\mathbb{R}^n)$, at each point $x = (x',x_n) \in \mathbb{R}^{n-1}\times\mathbb{R}$ we may write

$$\begin{aligned}\varphi(x) &= \varphi(x) - \psi(x_n)\int_{\mathbb{R}} \varphi(x',s)\,ds + \psi(x_n)\int_{\mathbb{R}} \varphi(x',s)\,ds\\ &= \partial_{x_n}\Big[\int_{-\infty}^{x_n}\Big[\varphi(x',t) - \psi(t)\int_{\mathbb{R}}\varphi(x',s)\,ds\Big]\,dt\Big] + \psi(x_n)\int_{\mathbb{R}}\varphi(x',s)\,ds.\end{aligned}$$

Since $\partial_n u = 0$ in $\mathcal{D}'(\mathbb{R}^n)$, this yields

$$\langle u,\varphi\rangle = \Big\langle u\,,\,\psi(x_n)\int_{\mathbb{R}}\varphi(x',s)\,ds\Big\rangle. \tag{12.2.32}$$

In particular, if $\varphi=\varphi_1\otimes\varphi_2$ for some $\varphi_1\in C_0^\infty(\mathbb{R}^{n-1})$ and some $\varphi_2\in C_0^\infty(\mathbb{R})$, then

$$\langle u,\varphi\rangle=\Big\langle u\,,\,\varphi_1\otimes\psi\Big(\int_{\mathbb{R}}\varphi_2(s)\,ds\Big)\Big\rangle=\langle u,\varphi_1\otimes\psi\rangle\langle 1,\varphi_2\rangle. \tag{12.2.33}$$

Define $v:C_0^\infty(\mathbb{R}^{n-1})\to\mathbb{C}$ by $v(\theta):=\langle u,\theta\otimes\psi\rangle$ for every $\theta\in C_0^\infty(\mathbb{R}^{n-1})$. Then $v\in\mathcal{D}'(\mathbb{R}^{n-1})$ and $u(x',x_n)=v(x')\otimes 1$ when restricted to $C_0^\infty(\mathbb{R}^{n-1})\otimes C_0^\infty(\mathbb{R})$. The desired conclusion follows by recalling Proposition 2.73.

Exercise 2.146 Fix $j\in\mathbb{N}$ and note that for each $x=(x_1,\dots,x_n)\in\mathbb{R}^n$ we may write

$$f_j(x)=\frac{1}{2\pi}\int_{-j}^{j}e^{ix_1\xi_1}d\xi_1\otimes\cdots\otimes\frac{1}{2\pi}\int_{-j}^{j}e^{ix_n\xi_n}d\xi_n. \tag{12.2.34}$$

Also, for each $j\in\mathbb{N}$ and each $k\in\{1,\dots,n\}$, the fundamental theorem of calculus gives

$$\int_{-j}^{j}e^{ix_k\xi_k}d\xi_k=\int_{-j}^{j}\cos(x_k\xi_k)\,d\xi_k=2\frac{\sin(jx_k)}{x_k},\quad\text{assuming}\quad x_k\neq 0. \tag{12.2.35}$$

Now use (12.2.34), (12.2.35), Exercise 2.118, and part *(d)* in Theorem 2.80 to finish the proof.

Exercise 2.147 Note that $u=-\delta$ is a solution for the equation in (1). Hence, if u is any other solution of the equation $(x-1)u=\delta$, then setting $v:=u+\delta$ it follows that $(x-1)v=0$ in $\mathcal{D}'(\mathbb{R})$. Next use this and the reasoning from Example 2.68 to conclude that the general solution for the equation in (1) is $-\delta+c\,\delta_1$, with $c\in\mathbb{C}$.

Fix $\psi\in C_0^\infty(\mathbb{R})$ with the property that $\psi(0)=1$. Show that any solution u of the equation in (2) satisfies

$$\langle u,\varphi\rangle=\int_{\mathbb{R}}a(x)\frac{\varphi(x)-\varphi(0)\psi(x)}{x}\,dx+\big\langle\langle u,\psi\rangle\delta,\varphi\big\rangle,\qquad\forall\,\varphi\in C_0^\infty(\mathbb{R}) \tag{12.2.36}$$

and use this to obtain that the general solution of the equation in (2) is $v_a+c\,\delta$, $c\in\mathbb{C}$, where v_a is the distribution given by

$$\langle v_a,\varphi\rangle:=\int_{\mathbb{R}}a(x)\frac{\varphi(x)-\varphi(0)\psi(x)}{x}\,dx,\qquad\forall\,\varphi\in C_0^\infty(\mathbb{R}).$$

Similarly, any solution u of the equation in (3) satisfies

$$\langle u,\varphi\rangle=\Big\langle v,\frac{\varphi-\varphi(0)\psi}{x}\Big\rangle+\big\langle\langle u,\psi\rangle\delta,\varphi\big\rangle,\qquad\forall\,\varphi\in C_0^\infty(\mathbb{R}), \tag{12.2.37}$$

so the general solution for (3) is $w+c\,\delta$, where $c\in\mathbb{C}$ and w is the distribution given by

$$\langle w,\varphi\rangle:=\Big\langle v,\frac{\varphi-\varphi(0)\psi}{x}\Big\rangle,\qquad\forall\,\varphi\in C_0^\infty(\mathbb{R}).$$

Exercise 2.148
(a) Since $H \in L^1_{\rm loc}(\mathbb{R})$ and

$$M_R := \big\{(x,y) \in [0,\infty) \times [0,\infty) : \, |x+y| \leq R\big\}$$

is a compact set in $\mathbb{R}^2$ for each $R > 0$, by Remark 2.84 and Theorem 2.85, it follows that $H * H$ is well-defined and belongs to $\mathcal{D}'(\mathbb{R})$. Fix a compact set K in $\mathbb{R}$ and let $R > 0$ be such that $K \subset (-R, R)$. Pick now $\varphi \in C^\infty_0(\mathbb{R})$ with $\operatorname{supp}\varphi \subseteq K$, and suppose that $\psi \in C^\infty_0(\mathbb{R}^2)$ satisfies $\psi = 1$ on

$$M_K := \big\{(x,y) \in [0,\infty) \times [0,\infty) : \, x+y \in K\big\}.$$

Then

$$\begin{aligned} \langle H * H, \varphi\rangle &= \int_{\mathbb{R}} \int_{\mathbb{R}} H(x)H(y)\varphi(x+y)\psi(x,y)\,{\rm d}y\,{\rm d}x \\ &= \int_0^\infty \int_0^\infty \varphi(x+y)\psi(x,y)\,{\rm d}y\,{\rm d}x \end{aligned} \tag{12.2.38}$$

Note that

$$\big\{(x,y) \in [0,\infty) \times [0,\infty) : \, |x+y| \leq R\big\} = \big\{(x,y) : \, 0 \leq x \leq R, \, 0 \leq y \leq R-x\big\},$$

hence

$$\begin{aligned} \langle H * H, \varphi\rangle &= \int_0^R \int_0^{R-x} \varphi(x+y)\psi(x,y)\,{\rm d}y\,{\rm d}x = \int_0^R \int_0^{R-x} \varphi(x+y)\,{\rm d}y\,{\rm d}x \\ &= \int_0^R \int_x^R \varphi(z)\,{\rm d}z\,{\rm d}x = \int_0^R \int_0^z \varphi(z)\,{\rm d}x\,{\rm d}z = \int_0^R z\varphi(z)\,{\rm d}z \\ &= \langle xH, \varphi\rangle. \end{aligned} \tag{12.2.39}$$

Alternatively, one may use Remark 2.86, to observe that $H * H$ in the distributional sense is the distribution given by the function obtained by taking the convolution, in the sense of (2.8.2), of the function H with itself. Hence,

$$(H * H)(x) = \int_0^\infty \chi_{[0,\infty)}(x-y)\,{\rm d}y = xH(x) \quad \text{ for every } \ x \in \mathbb{R}. \tag{12.2.40}$$

(b) $-xH(-x)$
(c) $(x^2 - 2 + 2\cos x)H(x)$
(d) $\frac{x^2}{2}H(x) - \frac{(x-1)^2}{2}H(x-1)$
(e) Exercise 2.128 ensures that $\delta_{\partial B(0,r)}$ is compactly supported, so the given convolution is well-defined. Also, by Exercise 2.94, $|x|^2 * \delta_{\partial B(0,r)} \in C^\infty(\mathbb{R}^n)$ and equals

$$\begin{aligned}\langle \delta_{\partial B(0,r)}, |x-y|^2\rangle &= \int_{\partial B(0,r)} |x-y|^2\,\mathrm{d}\sigma(y) = r^{n-1}\int_{S^{n-1}} |x-r\omega|^2\,\mathrm{d}\sigma(\omega)\\ &= r^{n-1}\int_{S^{n-1}} [r^2+|x|^2-2rx\cdot\omega]\,\mathrm{d}\sigma(\omega)\\ &= (r^{n+1}+|x|^2 r^{n-1})\omega_{n-1}, \qquad \forall\, x\in\mathbb{R}^n. \end{aligned} \tag{12.2.41}$$

For the second equality in (12.2.41) we used the change of variables $y = r\omega$, $\omega \in S^{n-1}$, while for the last equality we used the fact that since $x\cdot\omega$ as a function in ω is odd, its integral over S^{n-1} is zero.

Exercise 2.149 $u_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} 0$, $v_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} 0$. Given that $u_j, v_j \in \mathcal{E}'(\mathbb{R}^n)$, we deduce that $u_j * v_j \in \mathcal{E}'(\mathbb{R}^n)$ for each $j\in\mathbb{N}$. Also, $u_j * v_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} \delta$.

Exercise 2.150 The limits in (a) and (b) do not exist. Since for each $j\in\mathbb{N}$ we have $f_j \in \mathcal{E}'(\mathbb{R})$ and $g_j \in C^\infty(\mathbb{R})$, Exercise 2.94 may be used to conclude that $f_j * g_j \in C^\infty(\mathbb{R})$ and that $f_j * g_j = 1$ for every j. Thus, $f_j * g_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} 1$.

Exercise 2.151 Note that Λ is well-defined based on Proposition 2.93. You may want to use Theorem 13.5 to prove the continuity of Λ.

Exercise 2.152 For $f:\mathbb{R}^n\to\mathbb{C}$ set $f^\vee(x) := f(-x)$, $x\in\mathbb{R}^n$. If $u\in\mathcal{D}'(\mathbb{R}^n)$ is such that $u*\varphi = 0$ for every $\varphi\in C_0^\infty(\mathbb{R}^n)$, then $0 = (u*\varphi^\vee)(0) = \langle u,\varphi\rangle$ for every $\varphi\in C_0^\infty(\mathbb{R}^n)$, thus $u=0$. This proves the uniqueness part in the statement. As for existence, given Λ as specified, define $u_0 : \mathcal{D}(\mathbb{R}^n)\to\mathbb{C}$ by $u_0(\varphi) := \langle \delta, \Lambda(\varphi^\vee)\rangle$ for $\varphi\in C_0^\infty(\mathbb{R}^n)$. Being a composition of linear and continuous maps, u_0 is linear and continuous, thus $u_0\in\mathcal{D}'(\mathbb{R}^n)$. Also, if $\varphi\in C_0^\infty(\mathbb{R}^n)$ is fixed, we have

$$\begin{aligned}(u_0*\varphi)(x) &= \langle u_0, \varphi(x-\cdot)\rangle = \big\langle u_0, (\varphi^\vee)(\cdot - x)\big\rangle = \big\langle \delta, \Lambda(\varphi(\cdot+x))\big\rangle\\ &= \big\langle \delta, (\Lambda\varphi)(\cdot+x)\big\rangle = \Lambda(\varphi)(x), \quad \forall\, x\in\mathbb{R}^n. \end{aligned} \tag{12.2.42}$$

For the first equality in (12.2.42) we used Proposition 2.93, the third equality is based on the definition of u_0, while the forth equality uses the fact that Λ is commutes with translations.

Exercise 2.153 From hypotheses we obtain $\langle u, P\rangle = 0$ for every polynomial P in $\mathbb{R}^n$. Now use Lemma 2.75 to conclude that $\langle u,\varphi\rangle = 0$ for every $\varphi\in C_0^\infty(\mathbb{R}^n)$. Let $\psi\in C_0^\infty(\mathbb{R}^n)$ be such that $\psi = 1$ in a neighborhood of $\operatorname{supp} u$. Then for every function $\varphi\in C^\infty(\mathbb{R}^n)$ we have $0 = \langle u, \psi\varphi\rangle = \langle u,\varphi\rangle$ since the support condition on u implies $\langle u, (1-\psi)\varphi\rangle = 0$.

Exercise 2.154 Fix $\varphi \in C^\infty(\mathbb{R})$ and write

$$\sum_{j=1}^{k} \varphi\big(\tfrac{1}{j}\big) - k\varphi(0) - \varphi'(0)\ln k$$

$$= \sum_{j=1}^{k} \Big[\varphi\big(\tfrac{1}{j}\big) - \varphi(0) - \tfrac{1}{j}\varphi'(0)\Big] + \varphi'(0)\left[\sum_{j=1}^{k} \tfrac{1}{j} - \ln k\right]. \tag{12.2.43}$$

Since

$$\Big|\varphi\big(\tfrac{1}{j}\big) - \varphi(0) - \tfrac{1}{j}\varphi'(0)\Big| \leq \tfrac{1}{j^2}\|\varphi''\|_{L^\infty([0,1])} \qquad \forall\, j \in \mathbb{N},$$

taking the limit as $k \to \infty$ in (12.2.43) [also recall Euler's constant γ from (4.6.23)] we obtain

$$u(\varphi) = \sum_{j=1}^{\infty} \Big[\varphi\big(\tfrac{1}{j}\big) - \varphi(0) - \tfrac{1}{j}\varphi'(0)\Big] + \gamma\,\varphi'(0).$$

Now apply Fact 2.55 with $K := [0,1]$, $m := 2$ and $C := \sum_{j=1}^{\infty} \frac{1}{j^2} + \gamma$ to conclude $u \in \mathcal{E}'(\mathbb{R})$. Also, show that $\operatorname{supp} u = \{0\} \cup \Big\{\frac{1}{j} : j \in \mathbb{N}\Big\}$.

Exercise 2.155 Note that $f_j \in L^1_{\rm comp}(\mathbb{R})$ for each $j \in \mathbb{N}$, hence $\{f_j\}_{j\in\mathbb{N}} \subset \mathcal{E}'(\mathbb{R})$. If $\varphi \in C^\infty(\mathbb{R})$ then we may write

$$\big|\langle f_j, \varphi\rangle - \langle \delta, \varphi\rangle\big| = \left|\int_{-1/j}^{1/j} \frac{j}{2}\big(\varphi(x) - \varphi(0)\big)\,dx\right| \leq \frac{j}{2}\int_{-1/j}^{1/j} |\varphi(x) - \varphi(0)|\,dx$$

$$\leq \frac{1}{j}\|\varphi'\|_{L^\infty([-1,1])} \to 0 \quad \text{as} \quad j \to \infty. \tag{12.2.44}$$

Exercise 2.156 Since $f_j \in L^1_{\rm comp}(\mathbb{R})$ for each $j \in \mathbb{N}$, we have $\{f_j\}_{j\in\mathbb{N}} \subset \mathcal{E}'(\mathbb{R})$. Let $\varphi \in C^\infty_0(\mathbb{R})$ and suppose $R \in (0,\infty)$ is such that $\operatorname{supp}\varphi \subset (-R, R)$. Then for $j \in \mathbb{N}$ with $j \geq R$ we have

$$|\langle f_j, \varphi\rangle| = \left|\int_{-j}^{j} \frac{\varphi(x)}{j}\,dx\right| = \left|\int_{-R}^{R} \frac{\varphi(x)}{j}\,dx\right| \leq \frac{2R\|\varphi\|_{L^\infty(\mathbb{R})}}{j} \tag{12.2.45}$$

which proves that $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R}^n)} 0$. Suppose next that there exists a distribution $u \in \mathcal{E}'(\mathbb{R}^n)$ such that $f_j \xrightarrow[j\to\infty]{\mathcal{E}'(\mathbb{R}^n)} u$. In particular, we have $\lim_{j\to\infty}\langle f_j, \varphi\rangle = \langle u, \varphi\rangle$ for every $\varphi \in C^\infty_0(\mathbb{R})$. Together with what we have proved before, this forces

$u = 0$. However, $\langle f_j, 1\rangle = 2$ for every $j \in \mathbb{N}$, which contradicts the fact that $u = 0$. Thus, $\{f_j\}_{j\in\mathbb{N}}$ does not converge in $\mathcal{E}'(\mathbb{R}^n)$.

Exercise 2.157 Since $f_j \in C_0^\infty(\mathbb{R})$ for each $j \in \mathbb{N}$, we have $\{f_j\}_{j\in\mathbb{N}} \subset \mathcal{E}'(\mathbb{R})$. Let $\varphi \in C^\infty(\mathbb{R}^n)$. Then using the change of variables $jx = y$ we may write

$$\begin{aligned} |\langle f_j, \varphi\rangle - \langle \delta, \varphi\rangle| &= \left| \int_{\mathbb{R}^n} j^n \psi(jx)\varphi(x)\,\mathrm{d}x - \varphi(0) \int_{\mathbb{R}^n} \psi(x)\,\mathrm{d}x \right| \\ &= \left| \int_{\mathbb{R}^n} \psi(y)\varphi(y/j)\,\mathrm{d}y - \varphi(0) \int_{\mathbb{R}^n} \psi(x)\,\mathrm{d}x \right| \\ &\le \int_{\operatorname{supp}\psi} |\varphi(y/j) - \varphi(0)||\psi(y)|\,\mathrm{d}y \to 0 \ \text{ as } \ j \to \infty, \quad (12.2.46) \end{aligned}$$

where the convergence is based on Lebesgue's dominated convergence theorem.

Exercise 2.158 For each $k \in \{1, \ldots, n\}$ consider the sequence $\{\langle \delta_{x_j}, \varphi_k\rangle\}_{j\in\mathbb{N}}$ where the function φ_k is defined by $\varphi_k(x) := x_k$ for each $x = (x_1 \ldots, x_n) \in \mathbb{R}^n$.

12.3 Solutions to Exercises from Sect. 3.3

Exercise 3.32 Let $R > 0$ be such that $\operatorname{supp}\varphi \subset B(0, R)$. Then $\varphi_j \in C_0^\infty(\mathbb{R}^n)$ and $\operatorname{supp}\varphi_j \subset B(0, jR)$ for each $j \in \mathbb{N}$. Since there is no compact $K \subset \mathbb{R}^n$ such that $\operatorname{supp}\varphi_j \subset K$ for all $j \in \mathbb{N}$, the sequence $\{\varphi_j\}_{j\in\mathbb{N}}$ does not converge in $\mathcal{D}(\mathbb{R}^n)$. Also, for each $\alpha \in \mathbb{N}_0^n$ we have $\partial^\alpha \varphi_j(x) = \frac{1}{j^{|\alpha|}} \mathrm{e}^{-j}(\partial^\alpha\varphi)(x/j)$. Hence if we also take $\beta \in \mathbb{N}_0^n$ arbitrary, then

$$\begin{aligned} \sup_{x\in\mathbb{R}^n} |x^\beta \partial^\alpha \varphi_j(x)| &\le \mathrm{e}^{-j} j^{-|\alpha|} \sup_{x\in B(0,jR)} |x^\beta (\partial^\alpha\varphi)(x/j)| \\ &\le \mathrm{e}^{-j} j^{-|\alpha|} \sup_{x\in B(0,jR)} |x^\beta| \|\partial^\alpha\varphi\|_{L^\infty(B(0,R))} \\ &\le \mathrm{e}^{-j} j^{|\beta|-|\alpha|} R^{|\beta|} \|\partial^\alpha\varphi\|_{L^\infty(B(0,R))} \xrightarrow[j\to\infty]{} 0, \qquad (12.3.1) \end{aligned}$$

which implies that $\{\varphi_j\}_{j\in\mathbb{N}}$ converges to zero in $\mathcal{S}(\mathbb{R}^n)$.

Exercise 3.33 For each $\alpha \in \mathbb{N}_0^n$ we have $\partial^\alpha \varphi_j(x) = \frac{1}{j^{|\alpha|+1}} (\partial^\alpha\varphi)(x/j)$, for each $x \in \mathbb{R}^n$. Consequently, for every compact subset K of $\mathbb{R}^n$ we may write

$$\begin{aligned} \sup_{x\in K} |\partial^\alpha \varphi_j(x)| &\le j^{-|\alpha|-1} \sup_{x\in K} |(\partial^\alpha\varphi)(x/j)| \\ &\le j^{-|\alpha|-1} \|\partial^\alpha\varphi\|_{L^\infty(\mathbb{R}^n)} \xrightarrow[j\to\infty]{} 0, \qquad (12.3.2) \end{aligned}$$

which proves that $\varphi_j \xrightarrow[j\to\infty]{\mathcal{E}(\mathbb{R}^n)} 0$. Moreover, if

$$\mathcal{O} := \big\{ x = (x_1, \ldots, x_n) \in \mathbb{R}^n : \, x_j \neq 0 \ \text{ for } \ j = 1, \ldots, n \big\}$$

we claim that there exists $x_* \in \mathcal{O}$ with the property that $\varphi(x_*) \neq 0$. Indeed, if this were not to be the case, we would have $\varphi = 0$ in $\mathcal{O}$, hence $\varphi = 0$ in $\mathbb{R}^n$ since φ is continuous and $\mathcal{O}$ is dense in $\mathbb{R}^n$. Having fixed such a point x_* we then proceed to estimate

$$\begin{aligned}\sup_{x\in\mathbb{R}^n}|x^\beta\varphi_j(x)| &= j^{-1}\sup_{x\in\mathbb{R}^n}|x^\beta\varphi(x/j)| = j^{|\beta|-1}\sup_{x\in\mathbb{R}^n}|x^\beta\varphi(x)| \\ &\geq j^{|\beta|-1}|x_*^\beta|\,|\varphi(x_*)| \xrightarrow[j\to\infty]{} \infty, \end{aligned} \tag{12.3.3}$$

for $\beta \in \mathbb{N}_0^n$ satisfying $|\beta| > 1$. Thus, $\{\varphi_j\}_{j\in\mathbb{N}}$ does not converge in $\mathcal{S}(\mathbb{R}^n)$.

Exercise 3.34 Clearly $\{\varphi_j\}_{j\in\mathbb{N}} \subset C^\infty(\mathbb{R})$. If $m \in \mathbb{N}$ then for each $j \in \mathbb{N}_0$ we have

$$\varphi_j^{(m)}(x) = \frac{1}{j}\sum_{k=0}^{m}\frac{m!}{k!(m-k)!}\psi^{(k)}(x)\theta^{(m-k)}(x/j)\frac{1}{j^{m-k}}, \qquad \forall\, x\in\mathbb{R}. \tag{12.3.4}$$

Hence, if $\ell \in \mathbb{N}_0$, then using the properties of θ and ψ we may write

$$\begin{aligned}\sup_{x\in\mathbb{R}}|x^\ell\varphi_j^{(m)}(x)| &= \frac{1}{j}\sup_{x\in\mathbb{R}}\Big|\sum_{k=0}^{m}\frac{m!}{k!(m-k)!}\psi^{(k)}(x)\theta^{(m-k)}(x/j)\frac{x^\ell}{j^{m-k}}\Big| \\ &\leq \frac{1}{j}\sup_{|x|\leq 1}|\psi^{(m)}(x)| + \frac{1}{j}\sup_{x>1}\Big|\sum_{k=0}^{m}\frac{m!}{k!(m-k)!}\mathrm{e}^{-x}\theta^{(m-k)}(x/j)\frac{x^\ell}{j^{m-k}}\Big| \\ &\leq \frac{C}{j} + \frac{C}{j}\sup_{x>1}|\mathrm{e}^{-x}x^\ell| \leq \frac{C}{j}\xrightarrow[j\to\infty]{} 0. \end{aligned} \tag{12.3.5}$$

In conclusion, $\varphi_j \xrightarrow[j\to\infty]{\mathcal{S}(\mathbb{R}^n)} 0$.

Exercise 3.35

(a) Not in $\mathcal{S}(\mathbb{R}^n)$ since it is not bounded given that $\lim_{x_1\to-\infty}\mathrm{e}^{-x_1} = \infty$.

(b) Since $\mathrm{e}^{-|x|^2} \in \mathcal{S}(\mathbb{R}^n)$ and $|x|^{2n!} \in \mathcal{L}(\mathbb{R}^n)$, by (a) in Theorem 3.13, their product is in $\mathcal{S}(\mathbb{R}^n)$.

(c) $(1+|x|^2)^{2^n}$ is a polynomial function and if $(1+|x|^2)^{-2^n} \in \mathcal{S}(\mathbb{R}^n)$ then their product, which is equal to 1, would belong to $\mathcal{S}(\mathbb{R}^n)$, which is not true.

(d) Show first that $\sin(\mathrm{e}^{-|x|^2}) \in \mathcal{S}(\mathbb{R}^n)$. Then, given that $\frac{1}{1+|x|^2} \in \mathcal{L}(\mathbb{R}^n)$, it follows that $\frac{\sin(\mathrm{e}^{-|x|^2})}{1+|x|^2} \in \mathcal{S}(\mathbb{R}^n)$.

(e) Not in $\mathcal{S}(\mathbb{R}^n)$ since $\lim_{|x|\to\infty}(1+|x|^2)^{n+1}\frac{\cos(\mathrm{e}^{-|x|^2})}{(1+|x|^2)^n}=\infty$.

(f) Set $\varphi(x):=\mathrm{e}^{-|x|^2}\sin(\mathrm{e}^{x_1^2})$, $x\in\mathbb{R}^n$. If φ were to belong to $\mathcal{S}(\mathbb{R}^n)$ then $\partial_1\varphi$ and $x_1\partial\varphi$ would be bounded. However, for every $x=(x_1,\ldots,x_n)\in\mathbb{R}^n$,

$$\partial_1\varphi(x)=2\mathrm{e}^{-x_2^2-x_3^2-\cdots-x_n^2}x_1[\cos(\mathrm{e}^{x_1^2})-\mathrm{e}^{-x_1^2}\sin(\mathrm{e}^{x_1^2})]. \tag{12.3.6}$$

In particular,

$$(\partial_1\varphi)(x_1,0,\ldots,0)+2x_1\varphi(x_1,0,\ldots,0)=2x_1\cos(\mathrm{e}^{x_1^2})$$

which is not bounded.

(g) Since A is positive definite, there exists a real, symmetric, positive definite $n\times n$ matrix B such that $B^2=A$. Hence, $\varphi(x)=\mathrm{e}^{-|Bx|^2}$ for every $x\in\mathbb{R}^n$, which means that φ is the composition between the function $\mathrm{e}^{-|x|^2}\in\mathcal{S}(\mathbb{R}^n)$ (recall Exercise 3.3) and the linear transformation B that maps $\mathcal{S}(\mathbb{R}^n)$ into itself. Recalling Exercise 3.16 we may conclude that $\varphi\in\mathcal{S}(\mathbb{R}^n)$.

Exercise 3.36 From part *(g)* in Exercise 3.35 we have $f\in\mathcal{S}(\mathbb{R}^n)$. Let $B\in\mathcal{M}_{n\times n}(\mathbb{R})$ be such that $B^2=A$, so that $f(x)=\mathrm{e}^{-|Bx|^2}$ for every $x\in\mathbb{R}^n$. Now use Exercise 3.23 and Example 3.21.

Exercise 3.37 Let $A:=\begin{pmatrix}1&1/2\\1/2&1\end{pmatrix}$. Then $f(x)=\mathrm{e}^{-(Ax)\cdot x}$ for every $x\in\mathbb{R}^2$ and Exercise 3.36 applies and yields

$$\widehat{f}(\xi)=\frac{2\pi}{\sqrt{3}}\,\mathrm{e}^{-(\xi_1^2-\xi_1\xi_2+\xi_2^2)/3}\qquad\text{for }\xi=(\xi_1,\xi_2)\in\mathbb{R}^2. \tag{12.3.7}$$

Exercise 3.38 Use (b) in Theorem 3.20 and Example 3.21.

Exercise 3.39 Since $\sin(x\cdot x_0)=\frac{1}{2i}\big[\mathrm{e}^{ix\cdot x_0}-\mathrm{e}^{-ix\cdot x_0}\big]$ matters reduced to the case $n=1$ after which we may apply Exercise 3.22. Hence, if $x=(x_1,\ldots,x_n)$, $x_0=(x_{01},\ldots,x_{0n})$, and $\xi=(\xi_1,\ldots,\xi_n)$, we may write

$$\begin{aligned}\mathcal{F}\big(\mathrm{e}^{-a|x|^2}\sin(x\cdot x_0)\big)(\xi)&=\tfrac{1}{2i}\Big[\prod_{j=1}^n\mathcal{F}\big(\mathrm{e}^{-ax_j^2+ix_jx_{0j}}\big)-\prod_{j=1}^n\mathcal{F}\big(\mathrm{e}^{-ax_j^2-ix_jx_{0j}}\big)\Big]\\&=\tfrac{1}{2i}\big(\tfrac{\pi}{a}\big)^{\frac{n}{2}}\Big[\prod_{j=1}^n\mathrm{e}^{-\frac{(\xi_j-x_{0j})^2}{4a}}-\prod_{j=1}^n\mathrm{e}^{-\frac{(\xi_j+x_{0j})^2}{4a}}\Big]\\&=\tfrac{1}{2i}\big(\tfrac{\pi}{a}\big)^{\frac{n}{2}}\Big[\mathrm{e}^{-\frac{|\xi-x_0|^2}{4a}}-\mathrm{e}^{-\frac{|\xi+x_0|^2}{4a}}\Big].\end{aligned} \tag{12.3.8}$$

Exercise 3.40 Fix $\varphi \in \mathcal{S}(\mathbb{R})$. Suppose $\psi \in \mathcal{S}(\mathbb{R})$ is such that $\psi' = \varphi$. Then

$$\int_{\mathbb{R}} \varphi(x)\,dx = \lim_{R_1,R_2\to\infty} \int_{R_1}^{R_2} \psi'(x)\,dx = \lim_{R_1,R_2\to\infty} [\psi(R_2) - \psi(R_1)] = 0.$$

Conversely, if $\int_{\mathbb{R}} \varphi(x)\,dx = 0$, then the function $\psi(x) := \int_0^x \varphi(t)\,dt$, for $x \in \mathbb{R}$, belongs to $\mathcal{S}(\mathbb{R})$ and $\psi' = \varphi$.

Exercise 3.41 Use Exercise 3.40.

12.4 Solutions to Exercises from Sect. 4.11

Exercise 4.103 Using the change of variables $y = -x$ on the region corresponding to $x < 0$, we have

$$\int_{|x|\geq\varepsilon} \frac{\varphi(x)}{x}\,dx = \int_{\varepsilon}^{\infty} \frac{\varphi(x) - \varphi(-x)}{x}\,dx, \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{12.4.1}$$

Moreover, for each $\varepsilon > 0$ and each $\varphi \in \mathcal{S}(\mathbb{R})$ we may write

$$\begin{aligned}\left|\int_{|x|\geq\varepsilon} \frac{\varphi(x)}{x}\,dx\right| &\leq \int_{\varepsilon}^{1} \frac{|\varphi(x) - \varphi(-x)|}{x}\,dx + \int_{1}^{\infty} \frac{|\varphi(x) - \varphi(-x)|x}{x^2}\,dx \\ &\leq \sup_{x\in\mathbb{R}} |\varphi'(x)| + 2\Big(\int_1^{\infty} \frac{dx}{x^2}\Big) \sup_{x\in\mathbb{R}} |x\varphi(x)| \end{aligned} \tag{12.4.2}$$

so that P.V. $\frac{1}{x}$ is well-defined and

$$\left|\left\langle \text{P.V.}\,\frac{1}{x}, \varphi\right\rangle\right| \leq \Big(1 + \int_{|x|>1} \frac{dx}{|x|^2}\Big) \sup_{x\in\mathbb{R},\, k=0,1,\, j=0,1} |x^k \varphi^{(j)}(x)|, \quad \forall\, \varphi \in \mathcal{S}(\mathbb{R}), \tag{12.4.3}$$

hence (4.1.2) holds for $m = k = 1$. Since P.V. $\frac{1}{x}$ is also linear, we necessarily have P.V. $\frac{1}{x} \in \mathcal{S}'(\mathbb{R}^n)$.

Exercise 4.104 Use Exercise 2.107, (2.1.9), and Exercise 4.5.

Exercise 4.105 Use Exercise 2.109 and (4.1.25).

Exercise 4.106 Use a reasoning similar to the proof of the fact that the function in (4.1.27) is not a tempered distribution to conclude that $\mathrm{e}^{ax}H(x)$ and $\mathrm{e}^{-ax}H(-x)$ do not belong to $\mathcal{S}'(\mathbb{R})$ while $\mathrm{e}^{-ax}H(x)$, $\mathrm{e}^{ax}H(-x) \in \mathcal{S}'(\mathbb{R})$.

Exercise 4.107 Let $\varphi \in \mathcal{S}(\mathbb{R})$ and $j \in \mathbb{N}$.

(a) We may write

$$\left\langle \frac{x}{x^2+j^{-2}}, \varphi \right\rangle = \int_{|x|>1} \frac{x}{x^2+j^{-2}}\,\varphi(x)\,\mathrm{d}x + \int_{|x|\le 1} \frac{x^2}{x^2+j^{-2}} \cdot \frac{\varphi(x)-\varphi(0)}{x}\,\mathrm{d}x + \varphi(0)\int_{|x|\le 1} \frac{x}{x^2+j^{-2}}\,\mathrm{d}x. \tag{12.4.4}$$

The last integral in (12.4.4) is equal to zero (the function integrated is odd) while for the other two integrals in (12.4.4) apply Lebesgue's dominated convergence theorem to obtain that the first converges to $\int_{|x|>1} \frac{\varphi(x)}{x}\,\mathrm{d}x$ and the second converges to $\int_{|x|\le 1} \frac{\varphi(x)-\varphi(0)}{x}\,\mathrm{d}x$. In conclusion, $\frac{x}{x^2+j^{-2}} \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R})} \text{P.V.}\,\frac{1}{x}$.

(b) Making the change of variables $x = y/j$ and then applying Lebesgue's dominated convergence theorem write

$$\left\langle \frac{1}{j(x^2+j^{-2})}, \varphi \right\rangle = \int_{\mathbb{R}} \frac{\varphi(x)}{j(x^2+j^{-2})}\,\mathrm{d}x = \int_{\mathbb{R}} \frac{\varphi(y/j)}{y^2+1}\,\mathrm{d}y \xrightarrow[j\to\infty]{} \pi\varphi(0), \tag{12.4.5}$$

hence $\frac{1}{j(x^2+j^{-2})} \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R})} \pi\delta$.

(c) See Exercise 2.118. $\frac{\sin(jx)}{\pi x} \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R})} \delta$.

(d) Prove first that $\mathrm{e}^{j^2}\delta_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} 0$. If $\{\mathrm{e}^{j^2}\delta_j\}_{j\in\mathbb{N}}$ would converge in $\mathcal{S}'(\mathbb{R})$ it would have to converge to 0. However, for the test function $\mathrm{e}^{-\frac{x^2}{2}} \in \mathcal{S}(\mathbb{R})$ one has

$$\left\langle \mathrm{e}^{j^2}\delta_j, \mathrm{e}^{-\frac{x^2}{2}} \right\rangle = \mathrm{e}^{\frac{j^2}{2}} \xrightarrow[j\to\infty]{} \infty.$$

Exercise 4.108 Use the change of variables $jx = y$ and Lebesgue's dominated convergence theorem to show that $j^n\theta(jx) \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R})} \delta$.

Exercise 4.109 Based on Lebesgue's dominated convergence theorem, it follows that $f_j \xrightarrow[j\to\infty]{\mathcal{D}'(\mathbb{R})} \mathrm{e}^{x^2}$. On the other hand, the sequence $\{f_j\}_{j\in\mathbb{N}}$ does not convergence in $\mathcal{S}'(\mathbb{R})$ since

$$\left\langle f_j, \mathrm{e}^{-x^2/2} \right\rangle = \int_{-j}^{j} \mathrm{e}^{x^2/2}\,\mathrm{d}x \xrightarrow[j\to\infty]{} \int_{\mathbb{R}} \mathrm{e}^{x^2/2}\,\mathrm{d}x = \infty.$$

Exercise 4.110 If there were some $f \in L^p(\mathbb{R})$ such that $\lim_{j\to\infty} \|f_j - f\|_{L^p(\mathbb{R})} = 0$, then $\lim_{j\to\infty} f_j(x) = f(x)$ for almost every $x \in \mathbb{R}$. Since $\lim_{j\to\infty} f_j(x) = 0$ for every $x \in \mathbb{R}$, we have that $f = 0$ almost everywhere on $\mathbb{R}$. This leads to a contradiction given that we would have $0 = \lim_{j\to\infty} \|f_j\|^p_{L^p(\mathbb{R})} = \int_{j-1}^{j} 1\,\mathrm{d}x = 1$. Hence, the sequence $\{f_j\}_{j\in\mathbb{N}}$ does not converge in $L^p(\mathbb{R})$. If $\varphi \in \mathcal{S}(\mathbb{R})$ then

$$|\langle f_j, \varphi\rangle| \leq \sup_{x\in\mathbb{R}} |x^2\varphi(x)| \int_{j-1}^{j} \frac{\mathrm{d}x}{x^2} \leq \frac{C}{j(j-1)} \xrightarrow[j\to\infty]{} 0, \tag{12.4.6}$$

which shows that $f_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R})} 0$.

Exercise 4.111 Using Lebesgue's dominated convergence theorem it is not difficult to check that $\langle u_j, \varphi\rangle \xrightarrow[j\to\infty]{} \langle \mathrm{e}^x \sin(\mathrm{e}^x), \varphi\rangle$ for every $\varphi \in C_0^\infty(\mathbb{R})$.

If $\varphi \in \mathcal{S}(\mathbb{R})$, integration by parts yields

$$\int_{-j}^{j} \mathrm{e}^x \sin(\mathrm{e}^x)\varphi(x)\,\mathrm{d}x = -\cos(\mathrm{e}^x)\varphi(x)\big|_{-j}^{j} + \int_{-j}^{j} \cos(\mathrm{e}^x)\varphi'(x)\,\mathrm{d}x. \tag{12.4.7}$$

Since $\varphi \in \mathcal{S}(\mathbb{R})$ one has $\cos(\mathrm{e}^x)\varphi' \in L^1(\mathbb{R})$ and

$$\Big|\cos(\mathrm{e}^x)\varphi(x)\big|_{-j}^{j}\Big| = \Big|\frac{\cos(\mathrm{e}^x)}{x}\, x\varphi(x)\big|_{-j}^{j}\Big| \leq \frac{C}{j} \xrightarrow[j\to\infty]{} 0.$$

Thus, $\lim_{j\to\infty} \langle u_j, \varphi\rangle = \int_{\mathbb{R}} \cos(\mathrm{e}^x)\varphi'(x)\,\mathrm{d}x$ and if $u : \mathcal{S}(\mathbb{R}) \to \mathbb{C}$ is such that

$$\langle u, \varphi\rangle := \int_{\mathbb{R}} \cos(\mathrm{e}^x)\varphi'(x)\,\mathrm{d}x \qquad \text{for every} \quad \varphi \in \mathcal{S}(\mathbb{R}), \tag{12.4.8}$$

then u is a well-defined, linear mapping and

$$|\langle u, \varphi\rangle| \leq \Big(\int_{\mathbb{R}} \frac{\mathrm{d}x}{1+x^2}\Big) \sup_{x\in\mathbb{R}} |(1+x^2)\varphi'(x)| \qquad \text{for every} \quad \varphi \in \mathcal{S}(\mathbb{R}). \tag{12.4.9}$$

Hence $u \in \mathcal{S}'(\mathbb{R})$ and $u_j \xrightarrow[j\to\infty]{\mathcal{S}'(\mathbb{R})} u$.

Exercise 4.113 Fix $f \in \mathcal{S}'(\mathbb{R})$ and let $\psi \in \mathcal{S}(\mathbb{R})$ be such that $\psi(0) = 1$. Then, if u is a solution of the equation $xu = f$ in $\mathcal{S}'(\mathbb{R})$, for each $\varphi \in \mathcal{S}(\mathbb{R})$ we have

$$\langle u, \varphi\rangle = \Big\langle u, x\,\frac{\varphi - \varphi(0)\psi}{x}\Big\rangle + \langle u, \varphi(0)\psi\rangle = \Big\langle f, \frac{\varphi - \varphi(0)\psi}{x}\Big\rangle + \langle u, \psi\rangle\langle \delta, \varphi\rangle. \tag{12.4.10}$$

Set

$$g(x) := \begin{cases} \frac{\varphi(x) - \varphi(0)\psi(x)}{x} & \text{if } x \in \mathbb{R}\setminus\{0\}, \\ \varphi'(0) - \varphi(0)\psi'(0) & \text{if } x = 0. \end{cases} \tag{12.4.11}$$

Note that $g \in \mathcal{S}(\mathbb{R})$, since $g(x) = \displaystyle\int_0^1 \big[\varphi'(tx) - \varphi(0)\psi'(tx)\big]\,\mathrm{d}t$ for $x \in \mathbb{R}$. Thus, if we define

$$\widetilde{\varphi}(x) := \begin{cases} \frac{\varphi(x)-\varphi(0)\psi(x)}{x} & \text{if } |x| > 1, \\ g(x) & \text{if } |x| \le 1, \end{cases} \tag{12.4.12}$$

then $\widetilde{\varphi} \in \mathcal{S}(\mathbb{R})$ and $\widetilde{x\widetilde{\varphi}} = \varphi$. There remains to observe that the mapping

$$w_f : \mathcal{S}(\mathbb{R}) \to \mathbb{C}, \qquad \langle w_f, \varphi\rangle := \langle f, \widetilde{\varphi}\,\rangle \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}), \tag{12.4.13}$$

where $\widetilde{\varphi}$ is as in (12.4.12) satisfies $w_f \in \mathcal{S}'(\mathbb{R})$, in order to conclude that $u = w_f + c\delta$, some $c \in \mathbb{R}$.

Exercise 4.114 Suppose $u \in \mathcal{S}'(\mathbb{R}^n)$ is a solution of the equation $\mathrm{e}^{-|x|^2}u = 1$ in $\mathcal{S}'(\mathbb{R}^n)$. Then for $\varphi \in C_0^\infty(\mathbb{R}^n)$ we have

$$\langle u, \varphi\rangle = \big\langle u, e^{-|x|^2}\mathrm{e}^{|x|^2}\varphi\big\rangle = \big\langle 1, \mathrm{e}^{|x|^2}\varphi\big\rangle = \big\langle e^{|x|^2}, \varphi\big\rangle, \tag{12.4.14}$$

which shows that $u\big|_{C_0^\infty(\mathbb{R}^n)} = \mathrm{e}^{|x|^2}$. Since $C_0^\infty(\mathbb{R}^n)$ is sequentially dense in $\mathcal{S}(\mathbb{R}^n)$, this would imply that $u = \mathrm{e}^{|x|^2}$. However, as proved following Remark 4.15, $\mathrm{e}^{|x|^2} \notin \mathcal{S}'(\mathbb{R}^n)$.

Exercise 4.115 Since H is Lebesgue measurable and $(1+x^2)^{-1}H \in L^1(\mathbb{R})$, by Exercise 4.5 it follows that H defines a tempered distribution. We have seen that $H \in \mathcal{D}'(\mathbb{R})$ and $H' = \delta$ in $\mathcal{D}'(\mathbb{R})$ (recall Example 2.37). Since $\delta \in \mathcal{S}'(\mathbb{R})$, by (4.1.25) it follows that $H' = \delta$ in $\mathcal{S}'(\mathbb{R})$. Taking the Fourier transform of the last equation we arrive at $i\xi\widehat{H} = 1$ in $\mathcal{S}'(\mathbb{R})$. On the other hand, $\xi\left(\mathrm{P.V.}\,\frac{1}{\xi}\right) = 1$ in $\mathcal{S}'(\mathbb{R})$ by Exercise 4.103 and (2.3.6). Consequently, $\xi\big(i\widehat{H} - \mathrm{P.V.}\,\frac{1}{\xi}\big) = 0$ in $\mathcal{S}'(\mathbb{R})$. Now Example 2.68 may be used to conclude that $i\widehat{H} - \mathrm{P.V.}\,\frac{1}{\xi} = c\delta$, for some $c \in \mathbb{C}$. Hence

$$\widehat{H} = -i\mathrm{P.V.}\,\frac{1}{\xi} - c\delta \qquad \text{in } \mathcal{S}'(\mathbb{R}). \tag{12.4.15}$$

To determine c apply $\widehat{H}$ to the function $\varphi(x) := \mathrm{e}^{-x^2} \in \mathcal{S}(\mathbb{R})$. First, by Example 3.21 write

$$\big\langle \widehat{H}, \mathrm{e}^{-x^2}\big\rangle = \big\langle H, \widehat{\mathrm{e}^{-x^2}}\,\big\rangle = \big\langle H, \sqrt{\pi}\mathrm{e}^{-x^2/4}\big\rangle = \sqrt{\pi}\int_0^\infty \mathrm{e}^{-x^2/4}\,\mathrm{d}x$$

$$= \tfrac{\sqrt{\pi}}{2}\int_{-\infty}^{\infty} e^{-x^2/4}\,\mathrm{d}x = \widehat{e^{-x^2/4}}(0) = \pi. \tag{12.4.16}$$

Second, from (12.4.15) and the fact that $\frac{\mathrm{e}^{-x^2}}{x}$ is an odd function, we obtain

$$\Big\langle -i\,\mathrm{P.V.}\,\frac{1}{\xi} - c\delta, \mathrm{e}^{-x^2}\Big\rangle = -i\lim_{\varepsilon\to 0^+}\int_{|x|>\varepsilon}\frac{\mathrm{e}^{-x^2}}{x}\,\mathrm{d}x - c = -c. \tag{12.4.17}$$

Combining (12.4.15)–(12.4.17), we arrive at $\widehat{H} = -i\,\mathrm{P.V.}\,\frac{1}{\xi} + \pi\delta$.

Exercise 4.116 You may use Exercise 4.115 to show that

$$\mathcal{F}\Big(\text{P.V.}\,\frac{1}{x}\Big)(\xi) = 2\pi i H(\xi) - i\pi = i\pi\,\text{sgn}\,(\xi) \quad \text{in} \quad \mathcal{S}'(\mathbb{R}).$$

Exercise 4.117

(a) $\text{sgn} = H - H^{\vee}$, hence using Exercise 4.115 one may write

$$\widehat{\text{sgn}} = \widehat{H} - \widehat{H^{\vee}} = \pi\delta - i\,\text{P.V.}\,\frac{1}{\xi} - \pi\delta - i\,\text{P.V.}\,\frac{1}{\xi} = -2i\,\text{P.V.}\,\frac{1}{\xi} \quad \text{in } \mathcal{S}'(\mathbb{R}). \tag{12.4.18}$$

Alternatively, you may use Exercise 4.116 and take another Fourier transform.

(b) If k is even, then $|x|^k = x^k$, so that $\widehat{|x|^k} = \widehat{x^k\delta} = (-D)^k\widehat{\delta} = 2\pi i^k\delta^{(k)}$ in $\mathcal{S}'(\mathbb{R})$. If k is odd, then $|x|^k = x^k\text{sgn}\,x$, thus

$$\widehat{|x|^k} = \widehat{x^k\text{sgn}\,x} = (-D)^k\widehat{\text{sgn}\,x} = -2i^{k+1}\Big(\text{P.V.}\,\frac{1}{\xi}\Big)^{(k)} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}). \tag{12.4.19}$$

(c) You may use Example 4.34. Alternatively, take the Fourier transform of the identity $x\cdot\frac{\sin(ax)}{x} = \sin(ax)$ in $\mathcal{S}'(\mathbb{R})$, then use *(c)* in Theorem 4.25 and (4.2.42) to conclude that

$$\Big[\mathcal{F}\Big(\frac{\sin(ax)}{x}\Big)\Big]' = \pi\delta_{-a} - \pi\delta_a \qquad \text{in } \mathcal{S}'(\mathbb{R}).$$

Now invoke Exercise 2.120 and Proposition 2.40 to arrive at

$$\mathcal{F}\Big(\frac{\sin(ax)}{x}\Big) = \pi\,\text{sgn}(a)\chi_{[-|a|,|a|]} + c \quad \text{in } \mathcal{S}'(\mathbb{R}),$$

and then show that $c = 0$ by applying the Fourier transform to the last identity and recalling (4.2.39).

(d) Take the Fourier transform of the identity

$$x\cdot\frac{\sin(ax)\sin(bx)}{x} = \sin(ax)\sin(bx) \qquad \text{in } \mathcal{S}'(\mathbb{R}),$$

then use *(c)* in Theorem 4.25 and (4.2.45) to conclude that

$$\Big[\mathcal{F}\Big(\frac{\sin(ax)\sin(bx)}{x}\Big)\Big]' = \frac{i\pi}{2}\left[\delta_{a+b} - \delta_{a-b} - \delta_{b-a} + \delta_{-a-b}\right] \qquad \text{in } \mathcal{S}'(\mathbb{R}).$$

Now use Exercise 2.120 and Proposition 2.40 to obtain

$$\begin{aligned}
&\mathcal{F}\Big(\frac{\sin(ax)\sin(bx)}{x}\Big)\\
&\quad=\frac{i\pi}{2}\left[H(x-a-b)-H(x-a+b)-H(x-b+a)+H(x+a+b)\right]+c_0\\
&\quad=\frac{i\pi}{2}\operatorname{sgn}(b)\left[\chi_{[-a-|b|,-a+|b|]}-\chi_{[a-|b|,a+|b|]}\right]+c_0 \qquad (12.4.20)
\end{aligned}$$

in $\mathcal{S}'(\mathbb{R})$, and then show that $c_0=0$ by applying the Fourier transform to the last identity.

(e) Using the identity $\sin(x^2)=\frac{1}{2i}\left[\mathrm{e}^{ix^2}-\mathrm{e}^{-ix^2}\right]$, (4.2.21) and (4.2.50), we may write

$$\begin{aligned}
\widehat{\sin(x^2)}&=\frac{1}{2i}\left[\widehat{\mathrm{e}^{ix^2}}-\widehat{\mathrm{e}^{-ix^2}}\right]\\
&=\frac{\sqrt{\pi}}{2i}\left[\mathrm{e}^{i\frac{\pi}{4}}\mathrm{e}^{-i\frac{\xi^2}{4}}-\mathrm{e}^{-i\frac{\pi}{4}}\mathrm{e}^{i\frac{\xi^2}{4}}\right]\\
&=\frac{\sqrt{\pi}}{2i}\left[\mathrm{e}^{i\frac{\pi-\xi^2}{4}}-\mathrm{e}^{-i\frac{\pi-\xi^2}{4}}\right]. \qquad (12.4.21)
\end{aligned}$$

(f) Use (4.6.25), (4.2.3), and Proposition 4.30 to show

$$\widehat{\ln|x|}=-2\pi\gamma\delta-\pi w_{\chi_{(-1,1)}} \qquad \text{in } \mathcal{S}'(\mathbb{R}).$$

Exercise 4.118 From Exercise 2.128 we know that $\delta_{\partial B(0,R)}\in\mathcal{E}'(\mathbb{R}^3)$, thus by *(b)* in Theorem 4.33 it follows that

$$\widehat{\delta_{\partial B(0,R)}}(\xi)=\left\langle\delta_{\partial B(0,R)},\mathrm{e}^{-ix\cdot\xi}\right\rangle=\int_{\partial B(0,R)}\mathrm{e}^{-ix\cdot\xi}\,\mathrm{d}\sigma(x),\qquad\forall\,\xi\in\mathbb{R}^3. \tag{12.4.22}$$

Check that $\delta_{\partial B(0,R)}$ is invariant under orthogonal transformations, and conclude that $\widehat{\delta_{\partial B(0,R)}}$ is invariant under orthogonal transformations. Fix $\xi\in\mathbb{R}^3\setminus\{0\}$ and show that there exists an orthogonal transformation $A\in\mathcal{M}_{3\times3}(\mathbb{R})$ such that $A\xi=(|\xi|,0,0)$, and furthermore that $\widehat{\delta_{\partial B(0,R)}}(\xi)=\int_{\partial B(0,R)}\mathrm{e}^{-ix_1|\xi|}\,\mathrm{d}\sigma(x)$. Now use spherical coordinates to compute the latter integral and conclude that $\widehat{\delta_{\partial B(0,R)}}(\xi)=4\pi R\,\frac{\sin(R|\xi|)}{|\xi|}$. Treat separately the case $\xi=(0,0,0)$.

Exercise 4.119 Use Lemma 4.27.

Exercise 4.120 Since $\chi_{[-1,1]}\in L^1(\mathbb{R})$, from (4.1.8) it follows that $\chi_{[-1,1]}\in\mathcal{S}'(\mathbb{R})$. Also, by Exercise 4.26, we have that $\widehat{\chi_{[-1,1]}}$ in $\mathcal{S}'(\mathbb{R})$ is the tempered distribution given by the function

$$\int_{-1}^{1} \mathrm{e}^{-ix\xi}\,\mathrm{d}x = \frac{2\sin\xi}{\xi}, \quad \xi\in\mathbb{R}\setminus\{0\}.$$

In particular, since $\frac{\sin\xi}{\xi}\notin L^1(\mathbb{R})$ we have $\widehat{\chi_{[-1,1]}}\notin L^1(\mathbb{R})$.

Exercise 4.122 See Example 4.36 and deduce that $\widehat{g}=\pi\delta_a+\pi\delta_{-a}$ in $\mathcal{S}'(\mathbb{R})$.

Exercise 4.123 Use Exercise 4.119.

Exercise 4.124 If $P(x)=\sum_{|\alpha|\leq m}a_\alpha x^\alpha$, then the condition $P(tx)=t^{-k}P(x)$ in $\mathbb{R}^n$ for each $t>0$ implies $\sum_{|\alpha|\leq m}t^{|\alpha|}a_\alpha x^\alpha = t^{-k}\sum_{|\alpha|\leq m}a_\alpha x^\alpha$ in $\mathbb{R}^n$ for each $t>0$. Hence, for each $\alpha\in\mathbb{N}_0^n$ we have $t^{|\alpha|}a_\alpha=t^{-k}a_\alpha$ for every $t>0$, or equivalently that $t^{|\alpha|+k}a_\alpha=a_\alpha$ for every $t>0$. Now take $t\to 0^+$ in the last equality to obtain $a_\alpha=0$.

Exercise 4.125 Check, via a direct calculation, that

$$R^\top w := w-\Big[\frac{w\cdot[\zeta-(1+2\eta\cdot\zeta)\eta]}{1+\eta\cdot\zeta}\Big]\zeta-\Big[\frac{w\cdot(\zeta+\eta)}{1+\eta\cdot\zeta}\Big]\eta, \quad \forall\, w\in\mathbb{R}^n, \tag{12.4.23}$$

and then use (12.4.23) to prove that $R^\top R=RR^\top=I_{n\times n}$. Now (4.11.4) is easy to verify.

Exercise 4.126 Fix $c\in\mathbb{R}\setminus\{0\}$ and use the fact that cosine is an odd function, the fundamental theorem of calculus, and Fubini's theorem to write

$$\begin{aligned}\int_\varepsilon^R \frac{\cos(c\rho)-\cos\rho}{\rho}\,\mathrm{d}\rho &= \int_\varepsilon^R \frac{\cos(|c|\rho)-\cos\rho}{\rho}\,\mathrm{d}\rho\\ &= -\int_\varepsilon^R\int_1^{|c|}\sin(r\rho)\,\mathrm{d}r\,\mathrm{d}\rho\\ &= -\int_1^{|c|}\int_\varepsilon^R \sin(r\rho)\,\mathrm{d}\rho\,\mathrm{d}r\\ &= \int_1^{|c|}\Big[\frac{\cos(Rr)}{r}-\frac{\cos(\varepsilon r)}{r}\Big]\,\mathrm{d}r. \end{aligned}\tag{12.4.24}$$

In particular, estimate (4.11.7) readily follows from this formula. Going further, we note that an integration by parts gives

$$\int_1^{|c|}\frac{\cos(Rr)}{r}\,\mathrm{d}r = \Big[\frac{\sin(Rr)}{Rr}\Big]\Big|_{r=1}^{r=|c|}+\frac{1}{R}\int_1^{|c|}\frac{\sin(Rr)}{r^2}\,\mathrm{d}r. \tag{12.4.25}$$

By combining (12.4.24) with (12.4.25) we arrive at

$$\int_{\varepsilon}^{R} \frac{\cos(c\rho)-\cos\rho}{\rho}\,{\rm d}\rho = \frac{\sin(|c|R)}{|c|R} - \frac{\sin R}{R} + \frac{1}{R}\int_{1}^{|c|}\frac{\sin(Rr)}{r^2}\,{\rm d}r$$
$$- \int_{1}^{|c|}\frac{\cos(\varepsilon r)}{r}\,{\rm d}r. \tag{12.4.26}$$

Passing to limit $R\to\infty$ and $\varepsilon\to 0^+$ in (12.4.26) yields (4.11.5) after observing (e.g., by Lebesgue's dominated convergence theorem) that

$$\lim_{\varepsilon\to 0^+}\int_{1}^{|c|}\frac{\cos(\varepsilon r)}{r}\,{\rm d}r = \int_{1}^{|c|}\frac{{\rm d}r}{r} = \ln|c|. \tag{12.4.27}$$

Next,

$$\lim_{\substack{\varepsilon\to 0^+\\ R\to\infty}}\int_{\varepsilon}^{R}\frac{\sin(c\rho)}{\rho}\,{\rm d}\rho = \lim_{\substack{\varepsilon\to 0^+\\ R\to\infty}}\int_{\varepsilon c}^{Rc}\frac{\sin t}{t}\,{\rm d}t$$
$$= \big(\operatorname{sgn} c\big)\lim_{\substack{\varepsilon\to 0^+\\ R\to\infty}}\int_{\varepsilon}^{R}\frac{\sin t}{t}\,{\rm d}t = \frac{\pi}{2}\operatorname{sgn} c, \tag{12.4.28}$$

by a suitable change of variables and the well-known fact (based on a residue calculation) that $\lim_{\substack{\varepsilon\to 0^+\\ R\to\infty}}\int_{\varepsilon}^{R}\frac{\sin t}{t}\,{\rm d}t = \frac{\pi}{2}$. This proves (4.11.6). Finally, to justify (4.11.8) use the fact that whenever $0<a<b<\infty$ an integration by parts gives

$$\int_{a}^{b}\frac{\sin t}{t}\,{\rm d}t = -\Big[\frac{\cos t}{t}\Big]\Big|_{t=a}^{t=b} - \int_{a}^{b}\frac{\cos t}{t^2}\,{\rm d}t. \tag{12.4.29}$$

12.5 Solutions to Exercises from Sect. 5.4

Exercise 5.15 The ordinary differential equation $\big(\frac{\rm d}{{\rm d}x}\big)^m v = 0$, with initial conditions $v(0)=v'(0)=\cdots=v^{(m-2)}(0)=0$, and $v^{(m-1)}(0)=1$, has the unique solution $v(x)=\frac{1}{(m-1)!}x^{m-1}$ for $x\in\mathbb{R}$. Hence, by Example 5.11 the function $u:=\frac{1}{(m-1)!}x^{m-1}H$ is a fundamental solution for $\big(\frac{\rm d}{{\rm d}x}\big)^m$ in $\mathbb{R}$. In addition, by Exercise 4.115 we have $H\in\mathcal{S}'(\mathbb{R})$ which, when combined with the fact that $\frac{1}{(m-1)!}x^{m-1}\in\mathcal{L}(\mathbb{R})$, implies (as a consequence of *(b)* in Theorem 4.13) that $\frac{1}{(m-1)!}x^{m-1}H\in\mathcal{S}'(\mathbb{R})$. If $v\in\mathcal{S}'(\mathbb{R})$ is an arbitrary fundamental solution for $\big(\frac{\rm d}{{\rm d}x}\big)^m$, then by Proposition 5.7 we have that $v-u=P$ for some polynomial in $\mathbb{R}$ satisfying $\big(\frac{\rm d}{{\rm d}x}\big)^m P=0$ in $\mathbb{R}$. This forces P to be a polynomial of degree less than or equal to $m-1$.

Exercise 5.16 First prove that

$$\langle P_1(D_x)\otimes P_2(D_y),\varphi_1(x)\otimes\varphi_2(y)\rangle = \langle\delta_x\otimes\delta_y,\varphi_1(x)\otimes\varphi_2(y)\rangle$$

for every $\varphi_1\in C_0^\infty(\mathbb{R}^n)$ and every $\varphi_2\in C_0^\infty(\mathbb{R}^m)$, and then use Proposition 2.73.

12.6 Solutions to Exercises from Sect. 6.4

Exercise 6.27

(a) The general solution for the ordinary differential equation $v'' - a^2 v = 0$ in $\mathbb{R}$ is $v(x) := c_1 \mathrm{e}^{ax} + c_2 \mathrm{e}^{-ax}$ for $x \in \mathbb{R}$, where $c_1, c_2 \in \mathbb{C}$. Hence, if we further impose the conditions $v(0) = 0$ and $v'(0) = 1$ then $c_1 = \frac{1}{2a} = -c_2$, and by Example 5.11, we have that $u := \frac{1}{2a}\mathrm{e}^{ax} H - \frac{1}{2a}\mathrm{e}^{-ax} H$ satisfies $u'' - a^2 u = \delta$ in $\mathcal{D}'(\mathbb{R})$.

(b) By Theorem 5.13 there exists $E \in \mathcal{S}'(\mathbb{R})$ such that $\left(\frac{\mathrm{d}^2}{\mathrm{d}x^2} - a^2\right) E = \delta$ in $\mathcal{S}'(\mathbb{R})$. Fix such an E and apply the Fourier transform to the last equation. Then $-(\xi^2 + a^2)\widehat{E} = 1$ in $\mathcal{S}'(\mathbb{R})$. Since $\frac{1}{\xi^2 + a^2} \in \mathcal{L}(\mathbb{R})$ we may multiply the last equality by $\frac{1}{\xi^2 + a^2}$ to conclude that $-\widehat{E} = \frac{1}{\xi^2 + a^2} = f$, and, furthermore, that $\widehat{f} = -2\pi E^\vee$. Therefore we can determine $\widehat{f}$ as soon as we find a fundamental solution $E \in \mathcal{S}'(\mathbb{R})$ for the operator $\frac{\mathrm{d}^2}{\mathrm{d}x^2} - a^2$.

By Theorem 6.15, $\frac{\mathrm{d}^2}{\mathrm{d}x^2} - a^2$ is a hypoelliptic operator in $\mathbb{R}$. As a consequence of Remark 6.7 we have [with u as in (a)] that $u - E \in C^\infty(\mathbb{R})$ and is a classical solution of the ordinary differential equation $v'' - a^2 v = 0$ in $\mathbb{R}$. Thus,

$$\begin{aligned} E &= u + c_1 \mathrm{e}^{ax} + c_2 \mathrm{e}^{-ax} \\ &= \frac{1}{2a}\mathrm{e}^{ax} H - \frac{1}{2a}\mathrm{e}^{-ax} H + c_1 \mathrm{e}^{ax} H + c_1 \mathrm{e}^{ax} H^\vee + c_2 \mathrm{e}^{-ax} H + c_2 \mathrm{e}^{-ax} H^\vee \\ &= \left[\frac{1}{2a} + c_1\right] \mathrm{e}^{ax} H + \left[-\frac{1}{2a} + c_2\right] \mathrm{e}^{-ax} H + c_1 \mathrm{e}^{ax} H^\vee + c_2 \mathrm{e}^{-ax} H^\vee. \end{aligned} \tag{12.6.1}$$

The condition $E \in \mathcal{S}'(\mathbb{R})$ implies (in view of Exercise 4.106) that $\frac{1}{2a} + c_1 = 0$ and $c_2 = 0$, which when used in (12.6.1) give $E = -\frac{1}{2a}\mathrm{e}^{-a|x|}$. Consequently, $\widehat{f}(x) = \frac{\pi}{a}\mathrm{e}^{-a|x|}$ for $x \in \mathbb{R}$.

Exercise 6.29 Recall Exercise 6.4. Consider $u := 1 + \delta$ in $\mathcal{D}'(\mathbb{R}^n)$. It follows that $\operatorname{sing\,supp} u = \{0\}$ and $\operatorname{supp} u = \mathbb{R}^n$, thus $\operatorname{sing\,supp} u \subset \operatorname{supp} u$.

Exercise 6.30 Since $\partial^\alpha \delta_{x_0}\big|_{\mathbb{R}^n \setminus \{x_0\}} = 0$ we have $\operatorname{sing\,supp} \partial^\alpha \delta_{x_0} \subseteq \{x_0\}$. To see that the latter inclusion is in fact equality, use Example 2.13 if $\alpha = (0, \ldots, 0)$ and if $|\alpha| > 0$ use the fact that the order of the distribution $\partial^\alpha \delta_{x_0}$ equals $|\alpha|$ while any distribution of function type is of order zero.

Exercise 6.31 If $x \in \Omega$ is such that there exists an open set $\omega \subseteq \Omega$ with $x \in \omega$ and $u\big|_\omega \in C^\infty(\omega)$, then $(au)\big|_\omega \in C^\infty(\omega)$, which gives $\Omega \setminus \operatorname{sing\,supp} u \subseteq \Omega \setminus \operatorname{sing\,supp}(au)$.

Exercise 6.32 Since $\left(\text{P.V.}\,\frac{1}{x}\right)\Big|_{\mathbb{R}^n\setminus\{0\}} = \frac{1}{x}$ and $\frac{1}{x}$ belongs to $C^\infty(\mathbb{R}^n\setminus\{0\})$, it follows that $\operatorname{sing\,supp}\left(\text{P.V.}\,\frac{1}{x}\right)\subseteq\{0\}$. Now using Example 2.11 we have that the distribution P.V. $\frac{1}{x}$ is not of function type.

Exercise 6.33 First prove that $P(x)\cdot\frac{1}{P(x)} = 1$ in $\mathcal{S}'(\mathbb{R}^n)$. Then take the Fourier transform to arrive at

$$P(-D)\mathcal{F}\left(\frac{1}{P}\right)(\xi) = (2\pi)^n\delta \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{12.6.2}$$

Use (12.6.2), (6.1.3), and Example 6.4, to show that it is not possible to have

$$\operatorname{sing\,supp}\left(\mathcal{F}\left(\frac{1}{P}\right)\right) = \emptyset.$$

Exercise 6.34 $\operatorname{sing\,supp} u = \mathbb{R}\times\{0\}$. To prove this you may want to use the fact that

$$\langle u,\varphi\rangle = \left\langle \text{P.V.}\,\frac{1}{x}, \varphi(x,0)\right\rangle = \int_{|x|\geq 1}\frac{\varphi(x,0)}{x}\,dx + \int_{|x|\leq 1}\frac{\varphi(x,0)-\varphi(0,0)}{x}\,dx \tag{12.6.3}$$

for every $\varphi\in C_0^\infty(\mathbb{R}^2)$.

12.7 Solutions to Exercises from Sect. 7.10

Exercise 7.58 If the equation has two solutions $E_1, E_2\in\mathcal{S}'(\mathbb{R}^n)$, set $E := E_1 - E_2$. Then $\Delta E - E = 0$ in $\mathcal{S}'(\mathbb{R}^n)$ implies (after an application of the Fourier transform) $-(|\xi|^2+1)\widehat{E} = 0$ in $\mathcal{S}'(\mathbb{R}^n)$. Since $\frac{1}{1+|\xi|^2}\in\mathcal{L}(\mathbb{R}^n)$ (recall Exercise 3.12), by part *(a)* in Theorem 3.13 we have $\frac{1}{1+|\xi|^2}\,\varphi\in\mathcal{S}(\mathbb{R}^n)$ for every $\varphi\in\mathcal{S}(\mathbb{R}^n)$. Thus,

$$\langle\widehat{E},\varphi\rangle = \left\langle(1+|\xi|^2)\widehat{E}, \frac{1}{1+|\xi|^2}\varphi\right\rangle = 0, \qquad \forall\,\varphi\in\mathcal{S}(\mathbb{R}^n).$$

This proves that $\widehat{E} = 0$ in $\mathcal{S}'(\mathbb{R}^n)$; thus, after applying the Fourier transform we obtain $E = 0$ in $\mathcal{S}'(\mathbb{R}^n)$. In turn, the latter implies $E_1 = E_2$ in $\mathcal{S}'(\mathbb{R}^n)$ and the proof of the uniqueness statement is complete. Regarding the existence statement, suppose $E\in\mathcal{S}'(\mathbb{R}^n)$ satisfies $\Delta E - E = \delta$ in $\mathcal{S}'(\mathbb{R}^n)$. This equation, via the Fourier transform, is equivalent to $-(|\xi|^2+1)\widehat{E} = 1$ in $\mathcal{S}'(\mathbb{R}^n)$. Since $\frac{1}{1+|\xi|^2}\in\mathcal{S}'(\mathbb{R}^n)$ (use Exercise 4.6) and $(|\xi|^2+1)\,\frac{1}{1+|\xi|^2} = 1$ in $\mathcal{S}'(\mathbb{R}^n)$, it follows that $E := -\mathcal{F}^{-1}\left(\frac{1}{1+|\xi|^2}\right)$ is a solution of $\Delta E - E = \delta$ in $\mathcal{S}'(\mathbb{R}^n)$, which must be unique based on the earlier reasoning.

Exercise 7.59 Suppose there exists $E \in L^1(\mathbb{R}^n)$ with $\Delta E = \delta$. Then $-|\xi|^2 \widehat{E} = 1$. We know that $\widehat{E} \in \widehat{L^1} \subset C^0(\mathbb{R}^n)$ [recall (3.1.3)], hence $[|\xi|^2 \widehat{E}]\big|_{\xi=(0,\dots,0)} = 0$. This leads to a contradiction.

Exercise 7.62 Since $E_\Delta \in \mathcal{S}'(\mathbb{R}^n)$ is a fundamental solution for Δ, we have $\Delta E_\Delta = \delta$ in $\mathcal{S}'(\mathbb{R}^n)$. Fix $j \in \{1, \dots, n\}$. Then, $\Delta(\partial_j E_\Delta) = \partial_j \delta$ in $\mathcal{S}'(\mathbb{R}^n)$. Take the Fourier transform of the last equation to arrive at

$$|\xi|^2 \mathcal{F}(\partial_j E_\Delta) = -i\xi_j \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{12.7.1}$$

Since $n \geq 2$, by Example 4.3 we have $\frac{\xi_j}{|\xi|^2} \in \mathcal{S}'(\mathbb{R}^n)$. Also, $|\xi|^2 \in \mathcal{L}(\mathbb{R}^n)$, thus $|\xi|^2 \cdot \frac{\xi_j}{|\xi|^2} \in \mathcal{S}'(\mathbb{R}^n)$ (recall *(b)* in Theorem 4.13), and it is not difficult to check that $|\xi|^2 \cdot \frac{\xi_j}{|\xi|^2} = \xi_j$ in $\mathcal{S}'(\mathbb{R}^n)$. These facts combined with (12.7.1) imply

$$|\xi|^2 \left(\mathcal{F}(\partial_j E_\Delta) + i \frac{\xi_j}{|\xi|^2} \right) = 0 \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{12.7.2}$$

Thus, $\operatorname{supp} \left(\mathcal{F}(\partial_j E_\Delta) + i \frac{\xi_j}{|\xi|^2} \right) \subseteq \{0\}$ and by Exercise 4.35, it follows that

$$\mathcal{F}(\partial_j E_\Delta) = -i \frac{\xi_j}{|\xi|^2} + \widehat{P_j}(\xi) \quad \text{in } \mathcal{S}'(\mathbb{R}^n), \tag{12.7.3}$$

where P_j is a polynomial in $\mathbb{R}^n$. Now a direct computation gives that if $n \geq 2$, then

$$\partial_j E_\Delta = \frac{1}{\omega_{n-1}} \cdot \frac{x_j}{|x|^n} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{12.7.4}$$

Hence, $\partial_j E_\Delta$ is positive homogeneous of degree $1 - n$, which in turn when combined with Proposition 4.57 implies that $\mathcal{F}(\partial_j E_\Delta)$ is positive homogeneous of degree -1. Thus, the term in the right-hand side of (12.7.3) is positive homogeneous of degree -1. Since $\frac{\xi_j}{|\xi|^2}$ is positive homogeneous of degree -1, we conclude that $\widehat{P_j}(\xi)$ is positive homogeneous of degree -1, and furthermore, by Proposition 4.57 and Exercise 4.55, that the polynomial P_j is positive homogeneous of degree $1 - n \leq -1$. Now invoking Exercise 4.124 we obtain $P_j \equiv 0$. The latter when used in (12.7.3) proves (7.10.3). Identities (7.10.4) and (7.10.5) follow from (7.10.3) and (12.7.4).

Exercise 7.63 Since $E_{\Delta^2} \in \mathcal{S}'(\mathbb{R}^n)$ is a fundamental solution for Δ^2, we have $\Delta^2 E_{\Delta^2} = \delta$ in $\mathcal{S}'(\mathbb{R}^n)$. Fix $j, k \in \{1, \dots, n\}$. Then, $\Delta^2(\partial_j \partial_k E_{\Delta^2}) = \partial_j \partial_k \delta$ in $\mathcal{S}'(\mathbb{R}^n)$. Take the Fourier transform of the last equation to arrive at

$$|\xi|^4 \mathcal{F}(\partial_j \partial_k E_{\Delta^2}) = -\xi_j \xi_k \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{12.7.5}$$

Under the current assumption on n, by Exercise 4.4 we have $\frac{1}{|\xi|^2} \in \mathcal{S}'(\mathbb{R}^n)$. Also, $\frac{\xi_j \xi_k}{|\xi|^4} \in L^1_{\text{loc}}(\mathbb{R}^n)$ and in view of Example 4.3 one may infer that $\frac{\xi_j \xi_k}{|\xi|^4} \in \mathcal{S}'(\mathbb{R}^n)$.

In addition, $|\xi|^4 \in \mathcal{L}(\mathbb{R}^n)$, thus $|\xi|^4 \cdot \frac{\xi_j\xi_k}{|\xi|^4}$, $|\xi|^4 \cdot \frac{1}{|\xi|^2} \in \mathcal{S}'(\mathbb{R}^n)$ (recall *(b)* in Theorem 4.13), and $|\xi|^4 \cdot \frac{\xi_j\xi_k}{|\xi|^4} = \xi_j\xi_k$, $|\xi|^4 \cdot \frac{1}{|\xi|^2} = |\xi|^2$ in $\mathcal{S}'(\mathbb{R}^n)$. These facts combined with (12.7.5) imply

$$|\xi|^4 \left(\mathcal{F}(\partial_j\partial_k E_{\Delta^2}) + \frac{\xi_j\xi_k}{|\xi|^4} \right) = 0 \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{12.7.6}$$

Thus, $\operatorname{supp}\left(\mathcal{F}(\partial_j\partial_k E_{\Delta^2}) + \frac{\xi_j\xi_k}{|\xi|^4} \right) \subseteq \{0\}$ and by Exercise 4.35, it follows that

$$\mathcal{F}(\partial_j\partial_k E_{\Delta^2}) = -\frac{\xi_j\xi_k}{|\xi|^4} + \widehat{R_{jk}}(\xi) \quad \text{in } \mathcal{S}'(\mathbb{R}^n), \tag{12.7.7}$$

where R_{jk} is a polynomial in $\mathbb{R}^n$.

It is easy to check that

$$\partial_j\partial_k E_{\Delta^2} = -\frac{1}{2(n-2)\omega_{n-1}} \cdot \frac{\delta_{jk}}{|x|^{n-2}} + \frac{1}{2\omega_{n-1}} \cdot \frac{x_j x_k}{|x|^n} \quad \text{in} \quad \mathcal{S}'(\mathbb{R}^n). \tag{12.7.8}$$

Hence, $\partial_j\partial_k E_{\Delta^2}$ is positive homogeneous of degree $2-n$ which, in turn, when combined with Proposition 4.57, implies that $\mathcal{F}(\partial_j\partial_k E_{\Delta^2})$ is positive homogeneous of degree -2. Thus, the term in the right-hand side of (12.7.7) is positive homogeneous of degree -2. Since $\frac{\xi_j\xi_k}{|\xi|^4}$ is positive homogeneous of degree -2, we may conclude that $\widehat{R_{jk}}(\xi)$ is positive homogeneous of degree -2 and furthermore (by Proposition 4.57 and Exercise 4.55) that the polynomial R_{jk} is positive homogeneous of degree $2-n \leq -1$. Now invoking Exercise 4.124 we obtain $R_{jk} \equiv 0$. The latter when used in (12.7.7) proves (7.10.6). Identity (7.10.7) follows from (7.10.6) and (12.7.8). As for identity (7.10.8), apply the Fourier transform to (7.10.7) and then use Proposition 4.61 with $\lambda = n-2$.

Exercise 7.64 If $P(D)$ is elliptic, then first show that there exists $C \in (0,\infty)$ such that $P_m(\xi) \geq C$ for $\xi \in S^{n-1}$; thus, conclude $P_m(\xi) \geq C|\xi|^m$ for every $\xi \in \mathbb{R}^n \setminus \{0\}$. Use the latter and the fact that

$$|P(\xi)| \geq |P_m(\xi)| - \Big(\sum_{|\alpha|\leq m-1} a_\alpha \xi^\alpha \Big)$$

to obtain the desired conclusion. Conversely, suppose that there exist $C, R \in (0,\infty)$ such that $|P(\xi)| \geq C|\xi|^m$ for every $\xi \in \mathbb{R}^n \setminus B(0,R)$ and that $P_m(\xi_*) = 0$ for some $\xi_* \in \mathbb{R}^n \setminus \{0\}$. Then for every $\lambda > R/|\xi_*|$ we have

$$0 < C\lambda^m|\xi_*|^m \leq |P(\lambda\xi_*)| = \Big| \sum_{|\alpha|<m} a_\alpha \lambda^{|\alpha|}\xi_*^\alpha \Big| \leq \sum_{j=1}^{m-1} c_j\lambda^j$$

and obtain a contradiction by dividing with λ^m and then letting $\lambda \to \infty$.

Exercise 7.65 Fix $\varphi\in C_0^\infty(\mathbb{R}^n)$. Using the fact that φ is compactly supported, Lebesgue's dominated convergence theorem, formula (13.7.4) twice, and the fact that A is symmetric, we may write

$$\begin{aligned}\int_{\mathbb{R}^n} E_A(x)L_A\varphi(x)\,dx &= \lim_{\varepsilon\to 0^+}\int_{\mathbb{R}^n\setminus\overline{B(0,\varepsilon)}} E_A(x)L_A\varphi(x)\,dx\\ &=\lim_{\varepsilon\to 0^+}\Bigg[\int_{\mathbb{R}^n\setminus\overline{B(0,\varepsilon)}}(L_AE_A)(x)\varphi(x)\,dx-\frac{1}{\varepsilon}\int_{\partial B(0,\varepsilon)}E_A(x)(A\nabla\varphi(x))\cdot x\,d\sigma(x)\\ &\qquad+\frac{1}{\varepsilon}\int_{\partial B(0,\varepsilon)}[\varphi(x)-\varphi(0)]\big(A\nabla E_A(x)\big)\cdot x\,d\sigma(x)\\ &\qquad+\frac{\varphi(0)}{\varepsilon}\int_{\partial B(0,\varepsilon)}\big(A\nabla E_A(x)\big)\cdot x\,d\sigma(x)\Bigg].\end{aligned}\tag{12.7.9}$$

For the second equality in (12.7.9) we also used the fact that the outward unit normal to $\partial B(0,\varepsilon)$ is $\nu(x)=\frac{1}{\varepsilon}x$, for $x\in\partial B(0,\varepsilon)$.

Analyze each of the integrals in the rightmost side of (12.7.9). First, a direct computation shows that $LE_A=0$ pointwise in $\mathbb{R}^n\setminus\{0\}$. Second, (7.8.48), (7.8.25), and the mean value theorem for φ, imply

$$\left|\frac{1}{\varepsilon}\int_{\partial B(0,\varepsilon)}[\varphi(x)-\varphi(0)]\big(A\nabla E_A(x)\big)\cdot x\,d\sigma(x)\right|\leq C\varepsilon\xrightarrow[\varepsilon\to 0^+]{}0,\tag{12.7.10}$$

where the constant C in (12.7.10) depends on A, $\|\nabla\varphi\|_{L^\infty(\mathbb{R}^n)}$ and n but not on ε. Third, thanks to (7.8.25) we have

$$\begin{aligned}&\left|\frac{1}{\varepsilon}\int_{\partial B(0,\varepsilon)}E_A(x)(A\nabla\varphi(x))\cdot x\,d\sigma(x)\right|\\ &\qquad\leq\begin{cases}C\varepsilon & \text{if } n\geq 3\\ C(|\ln\varepsilon|+1)\varepsilon & \text{if } n=2\end{cases}\xrightarrow[\varepsilon\to 0^+]{}0.\end{aligned}\tag{12.7.11}$$

Fourth, (7.8.48) and a natural change of variables allow us to write

$$\begin{aligned}\frac{1}{\varepsilon}\int_{\partial B(0,\varepsilon)}\big(A\nabla E_A(x)\big)\cdot x\,d\sigma(x)&=\frac{\varepsilon}{\omega_{n-1}\sqrt{\det A}}\int_{\partial B(0,\varepsilon)}\frac{d\sigma(x)}{[(A^{-1}x)\cdot x]^{\frac{n}{2}}}\\ &=\frac{1}{\omega_{n-1}\sqrt{\det A}}\int_{S^{n-1}}\frac{d\sigma(x)}{[(A^{-1}x)\cdot x]^{\frac{n}{2}}}.\end{aligned}\tag{12.7.12}$$

Combining (12.7.9)–(12.7.12) and (7.8.47), we conclude that, under the current assumptions on A, for every $\varphi\in C_0^\infty(\mathbb{R}^n)$ we have

$$\int_{\mathbb{R}^n}E_A(x)L_A\varphi(x)\,dx=\frac{\varphi(0)}{\omega_{n-1}\sqrt{\det A}}\int_{S^{n-1}}\frac{d\sigma(x)}{[(A^{-1}x)\cdot x]^{\frac{n}{2}}}.\tag{12.7.13}$$

In particular, (12.7.13) holds if A has real entries, is symmetric, and satisfies (7.8.8). In this latter situation, we already proved that E_A is a fundamental solution for L_A, thus the left-hand side in (12.7.13) is equal to $\langle E_A, L_A\varphi\rangle = \langle L_A E_A, \varphi\rangle = \langle \delta, \varphi\rangle = \varphi(0)$, which means that (7.8.50) holds in the case when A has real entries, is symmetric and satisfies (7.8.8). As in the proof of Theorem 7.54, the identity (7.8.50) may be extended to the class of complex symmetric matrices satisfying condition (7.8.8).

Based on what we have just proved and (12.7.13) we deduce that

$$\int_{\mathbb{R}^n} E_A(x) L_A\varphi(x)\,dx = \varphi(0) \tag{12.7.14}$$

holds for every $\varphi \in C_0^\infty(\mathbb{R}^n)$ and every matrix $A \in \mathcal{M}_{n\times n}(\mathbb{C}^n)$ satisfying $A = A^\top$ as well as condition (7.8.8). The latter, combined with the fact that $E_A \in L^1_{\rm loc}(\mathbb{R}^n)$ further implies that E_A is a fundamental solution for L_A whenever $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is symmetric and satisfies (7.8.8).

12.8 Solutions to Exercises from Sect. 9.3

Exercise 9.12 Let E be the fundamental solution for the heat operator $\partial_t - \Delta_x$ in $\mathbb{R}^{n+1}$ as in (8.1.7). Then $E \in L^1_{\rm loc}(\mathbb{R}^{n+1})$ and we claim that for each $x \in \mathbb{R}^n \setminus \{0\}$, the function $E(x,t)$ is absolutely integrable with respect to $t \in \mathbb{R}$. Indeed, using the change of variables $\tau = \frac{|x|^2}{4t}$, (13.5.1), the fact that $\Gamma\left(\frac{n}{2}\right) = \left(\frac{n}{2}-1\right)\Gamma\left(\frac{n}{2}-1\right)$, and (13.5.6), we obtain

$$\begin{aligned}\int_{-\infty}^{\infty} E(x,t)\,dt &= -\frac{1}{4\pi^{n/2}|x|^{n-2}}\int_0^\infty \mathrm{e}^{-\tau}\tau^{\frac{n}{2}-2}\,d\tau = -\frac{\Gamma(n/2-1)}{4\pi^{n/2}|x|^{n-2}}\\ &= -\frac{1}{(n-2)\omega_{n-1}}\cdot\frac{1}{|x|^{n-2}} \qquad \text{if } \ x\in\mathbb{R}^n\setminus\{0\}.\end{aligned} \tag{12.8.1}$$

Hence, we may apply Proposition 9.6 and Proposition 9.7, to conclude that the distribution

$$-\int_{-\infty}^{\infty} E(x,t)\,dt = \frac{1}{(n-2)\omega_{n-1}}\cdot\frac{1}{|x|^{n-2}}, \quad x\in\mathbb{R}^n\setminus\{0\},$$

is a fundamental solution for Δ in $\mathbb{R}^n$, when $n \geq 3$.

12.9 Solutions to Exercises from Sect. 10.7

Exercise 10.31 We have $\Delta\mathbf{u} + \nabla p = 0$ in $\left[\mathcal{D}'(\Omega)\right]^n$ and $\operatorname{div} u = 0$ in $\mathcal{D}'(\Omega)$. Apply div to the first equation and conclude that $\Delta p = 0$ in $\mathcal{D}'(\Omega)$. Next apply Δ to the first equation. Finally, use Theorem 7.10 and Theorem 7.23.

Exercise 10.32 Follow the outline from Exercise 10.16 with the adjustment that the integrals in (10.3.24) this time will be bounded by $C\|\nabla\varphi\|_{L^\infty(\mathbb{R}^n)}\,\varepsilon(1+|\ln\varepsilon|)$.

Exercise 10.33 Follow the outline from Exercise 10.30 with the adjustment that the integrals in (10.6.22) this time will be bounded by $C\|\nabla\varphi\|_{L^\infty(\mathbb{R}^n)}\,\varepsilon(1+|\ln\varepsilon|)$.

Exercise 10.34 Note that $\mathbf{u}$ is a solution for (10.7.3) if and only if $\mathbf{u}-Ax-b$ is a solution for (10.5.3).

Exercise 10.35 Show that Theorem 10.3 applies.

12.10 Solutions to Exercises from Sect. 11.7

Exercise 11.19 Follow the outline of the proof of Theorem 11.1.

Exercise 11.20 Compute $L_2\mathbb{E}_{L_1L_2}$ using the analogue of (11.3.31) written for the operator $L:=L_1L_2$, and make use of a suitable version of (11.3.53). Deduce that

$$P(x)=c_{m_1,m_2,n,q}\Delta^{(n+q)/2}\int_{S^{n-1}}(x\cdot\xi)^{2m_1+q}\big[L_2(\xi)\big]^{-1}\,\mathrm{d}\sigma(\xi) \tag{12.10.1}$$

for every $x\in\mathbb{R}^n$, from which all the desired properties of P follow.

Exercise 11.21 Start with (11.3.31) written for $L=\Delta^m$ and use (7.5.24), (7.5.25), and (7.5.13), in order to deduce that (11.7.3) holds with

$$P(x)=\mathcal{P}_{m+\frac{n+q}{2},\frac{n+q}{2}}(x)+C(n,q)\Delta^{(n+q)/2}\big[|x|^q\big],\qquad x\in\mathbb{R}^n, \tag{12.10.2}$$

where $C(n,q)$ is as in (7.5.26).

Chapter 13

Appendix

13.1 Summary of Topological and Functional Analytic

A `topology` on a set X is a family τ of subsets of X satisfying the following properties:

1. $X, \emptyset \in \tau$;
2. If $\{A_\alpha\}_{\alpha\in I}$ is a family of sets contained in τ, then $\bigcup_{\alpha\in I} A_\alpha \in \tau$;
3. If $A_1, A_2 \in \tau$, then $A_1 \cap A_2 \in \tau$.

If X and τ are as above, the pair (X, τ) is called a `topological space` and the elements of τ are called `open` sets. If x is an element of X (sometimes referred to as point), then any open set containing x is called an `open neighborhood` of x. Any set that contains an open neighborhood of x is called a neighborhood of x. A sequence $\{x_j\}_{j\in\mathbb{N}}$ of elements of a topological space (X, τ) are said to converge to $x \in X$, and we write $\lim_{j\to\infty} x_j = x$, if every neighborhood of x contains all but a finite number of the elements of the sequence.

A family $\mathcal{B}$ of subsets of X is a `base for a topology` τ on X if $\mathcal{B}$ is a subfamily of τ and for each $x \in X$, and each neighborhood U of x, there is $V \in \mathcal{B}$ such that $x \in V \subset U$. We say that the base $\mathcal{B}$ generates the topology τ. An equivalent characterization of bases that is useful in applications is as follows. A base for a topological space (X, τ) is a collection $\mathcal{B}$ of open sets in τ such that every open set in τ can be written as a union of elements of $\mathcal{B}$.

It is important to note that not any family of subsets of a given set is a base for some topology on the set. Given a set X and a family $\mathcal{B}$ of subsets of X, this family $\mathcal{B}$ is a base for some topology on X if and only if

$$\begin{cases} \bigcup\limits_{B\in\mathcal{B}} B = X, \\ \forall\, B_1,\, B_2 \in \mathcal{B} \text{ and } \forall\, x \in B_1 \cap B_2,\ \exists\, B \in \mathcal{B} \text{ such that } x \in B \subseteq B_1 \cap B_2. \end{cases} \tag{13.1.1}$$

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If τ_1 and τ_2 are two topologies on a set X, such that every member of τ_2 is also a member of τ_1, then we say that τ_1 is `finer` (or, larger) than τ_2, and that τ_2 is `coarser` (or, smaller) than τ_1. If (X, τ) is a topological space and $E \subseteq X$, then the `topology induced` by τ on E is the topology for which all open sets are intersections of open sets in τ with E.

Let (X, τ) and (Y, τ') be two topological spaces. The topology on $X \times Y$ with base the collection of products of open sets in X and open sets in Y [which satisfies (13.1.1)] is called the `product topology`. A function $f : X \to Y$ is called `continuous at` $x \in X$ if the inverse image under f of every open neighborhood of $f(x)$ is an open neighborhood of x. It is called continuous on X if it is continuous at every $x \in X$. In particular, it is easy to see that if $f : X \to Y$ is continuous then $\lim_{j\to\infty} f(x_j) = f(x)$ for every convergent sequence $\{x_j\}_{j\in\mathbb{N}}$ of X converging to $x \in X$, that is, f is `sequentially continuous`. While in general the converse is false, in the case when X is in fact a metric space (see below), if $f : X \to Y$ is sequential continuous, then f is continuous.

A topological space (X, τ) is said to be `separated`, or `Hausdorff`, if for any two distinct elements x and y of X, there exist U, a neighborhood of x and V, a neighborhood of y, such that $U \cap V = \emptyset$. In a Hausdorff space the limit of a convergent sequence is unique.

A `metric space` is a set X equipped with a `distance` function (also called metric). This is a function $d : X \times X \to [0, \infty)$ satisfying the following properties:

1. If $x, y \in X$, then $d(x, y) = 0$ if and only if $x = y$ (nondegeneracy);
2. $d(x, y) = d(y, x)$ for every $x, y \in X$ (symmetry);
3. $d(x, y) \leq d(x, z) + d(z, y)$ for every $x, y, z \in X$ (triangle inequality).

Let d be a metric on X. The open balls $B(x, r) := \{y \in X : d(x, y) < r\}$, $x \in X$, $r \in (0, \infty)$, are then the base of a Hausdorff topology on X (since it satisfies (13.1.1)), denoted τ_d. For a sequence $\{x_j\}_{j\in\mathbb{N}}$ of points in X and $x \in X$ one has $\lim_{j\to\infty} x_j = x$ in the topology τ_d, if and only if the sequence of numbers $\{d(x_j, x)\}_{j\in\mathbb{N}}$ converges to 0 as $j \to \infty$. A sequence $\{x_j\}_{j\in\mathbb{N}}$ of points in (X, τ_d) is called `Cauchy` if $d(x_j, x_k) \to 0$ as $j, k \to \infty$. A metric space is called `complete` if every Cauchy sequence is convergent.

A topological space X is called `metrizable` if there exists a distance function for which the open balls form a base for the topology (i.e., the topology on X coincides with τ_d).

A vector space X over $\mathbb{C}$ becomes a `topological vector space` if equipped with a topology that is compatible with its vector structure, that is, the operation of vector addition, and also the operation of multiplication by a complex number, are continuous maps $X \times X \to X$ and $\mathbb{C} \times X \to X$, respectively (where $X \times X$ and $\mathbb{C} \times X$ are endowed each with the corresponding product topology). Observe that in order to specify a topology on a vector space, it suffices to give the system of neighborhoods of the zero element $0 \in X$, since the system of

neighborhoods of any other element of X is obtained from this via translation. In fact, it suffices to give a base of neighborhoods of 0. This is a family $\mathcal{B}$ of neighborhoods of 0 such that every neighborhood of 0 contains a member of $\mathcal{B}$. A set $E \subseteq X$ is then open if and only if, for every $x \in E$, there exists $U \in \mathcal{B}$ such that $x + U \subseteq E$, where $x + U := \{x + y : y \in U\}$.

A topological vector space is called `locally convex` if it has a base of neighborhoods of 0 consisting of sets that are convex, balanced, and absorbing. A set U is called `convex` provided $tx + (1-t)y \in U$ whenever $x, y \in U$ and $t \in [0,1]$. A set U is called `balanced` if $cx \in U$ whenever $x \in U$, $c \in \mathbb{C}$ and $|c| \leq 1$. A set U is called `absorbing` provided for each $x \in X$ there exists $t > 0$ such that $x \in tU := \{ty : y \in U\}$.

A `seminorm` on a vector space X is a function $p : X \to \mathbb{R}$ satisfying the properties

1. $p(cx) = |c|p(x)$, for every $c \in \mathbb{C}$ and every $x \in X$ (positively homogeneous);

2. $p(x + y) \leq p(x) + p(y)$, for every $x, y \in X$ (sub-additive).

In particular, a seminorm p on X satisfies $p(0) = 0$ and $p(x) \geq 0$ for every $x \in X$.

A family $\mathcal{P}$ of seminorms on X is called `separating` if, for each $x \in X$, $x \neq 0$, there exists $p \in \mathcal{P}$ such that $p(x) \neq 0$. Given a separating family of seminorms $\mathcal{P}$ on X, let B be the collection of sets of the form

$$\big\{x : p(x) < \varepsilon \ \ \forall\, p \in \mathcal{P}_0\big\}, \quad \mathcal{P}_0 \subseteq \mathcal{P},\ \mathcal{P}_0 \ \text{ finite},\ \ \varepsilon > 0. \tag{13.1.2}$$

Then B is a base of neighborhoods of 0 of a locally convex vector space topology $\tau_{\mathcal{P}}$ on X called the `topology generated by the family of seminorms` $\mathcal{P}$. In this context, it may be readily verified that if Y is a linear subspace of X and $\mathcal{P}|_Y := \{p|_Y : p \in \mathcal{P}\}$, then

$$\text{the topology induced on } Y \text{ by } \tau_{\mathcal{P}} \text{ coincides with } \tau_{\mathcal{P}|_Y}. \tag{13.1.3}$$

Conversely, if X is a locally convex topological vector space, for each U convex, balanced, and absorbing neighborhood of 0, the mapping $p_U : X \to \mathbb{R}$ defined by $p_U(x) := \inf\{t > 0 : t^{-1}x \in U\}$, $x \in X$ (called the Minkowski functional associated with U) is a seminorm. It is then not hard to see that the topology on X is generated by this family of seminorms (in the manner described above).

Let $\mathcal{P} = \{p_j\}_{j\in\mathbb{N}}$ be a countable family of seminorms that is also separating (thus, if $x \in X$ and $p_j(x) = 0$ for all $j \in \mathbb{N}$, then $x = 0$). The topology generated by this family is metrizable. Indeed, the function $d : X \times X \to \mathbb{R}$ defined by

$$d(x,y) := \sum_{j=1}^{\infty} 2^{-j} \frac{p_j(x-y)}{1 + p_j(x-y)}, \quad \text{for each } \ x, y \in X, \tag{13.1.4}$$

is a distance on X and the topology τ_d induced by the metric d coincides with the topology generated by $\mathcal{P}$. In the converse direction, it can be shown that the

topology of a locally convex space that is metrizable, endowed with a translation invariant metric, can be generated by a countable family of seminorms.

A locally convex topological vector space that is metrizable and complete is called a `Fréchet` space. Thus, if a family $\mathcal{P} = \{p_j\}_{j\in\mathbb{N}}$ of seminorms generates the topology of a Fréchet space X, then whenever $p_j(x_k - x_l) \to 0$ as $k, l \to \infty$ for every $j \in \mathbb{N}$, there exists $x \in X$ such that $p_j(x_k - x) \to 0$ as $k \to \infty$ for every $j \in \mathbb{N}$.

Let X be a vector space, $\{X_j\}_{j\in J}$ be a family of vector subspaces of X such that $X = \bigcup_{j\in J} X_j$ and if $j_1, j_2 \in J$ then there exists $j_3 \in J$ with $X_{j_1} \subset X_{j_3}$ and $X_{j_2} \subset X_{j_3}$. Also assume that there exist topologies τ_j on X_j such that if $X_{j_1} \subset X_{j_2}$; then the topology τ_{j_1} is finer than the topology $\tau_{j_2 j_1}$ induced on X_{j_1} by X_{j_2}. Let

$$\mathcal{W} := \big\{W \subset X : W \text{ balanced, convex and such that } W \cap X_j \text{ is a neighborhood of 0 in } X_j,\ \forall\, j \in \mathbb{N}\big\}. \tag{13.1.5}$$

Then $\mathcal{W}$ is a base of neighborhoods of 0 in a locally convex topology τ on X. Call this topology the `inductive limit topology` on X.

If in addition whenever $X_{j_1} \subset X_{j_2}$ we also have $\tau_{j_2 j_1} = \tau_{j_1}$, we call the inductive limit strict. If the topology τ on X is the strict inductive limit of the topologies of an increasing sequence $\{X_n\}_{n\in\mathbb{N}}$, then the topology induced on X_n by the topology τ on X coincides with the initial topology τ_n on X_n, for every $n \in \mathbb{N}$.

In general, if X is a topological vector space, its `dual space` is the collection of all linear mappings $f : X \to \mathbb{C}$ (also referred to as functionals) that are continuous with respect to the topology on X. In the case when X is a locally convex topological vector space and $\mathcal{P}$ is a family of seminorms generating the topology of X, then a seminorm q on X is continuous if and only if there exist $N \in \mathbb{N}$, seminorms $p_1, \ldots, p_N \in \mathcal{P}$, and a constant $C \in (0, \infty)$, such that

$$|q(x)| \leq C \max\{p_1(x), \ldots, p_n(x)\}, \quad \forall\, x \in X. \tag{13.1.6}$$

This fact then gives a criterion for continuity for functionals on X, since if $f : X \to \mathbb{C}$ is a linear mapping, then $q(x) := |f(x)|$, for $x \in X$, is a seminorm on X. In addition, if the family of seminorms $\mathcal{P}$ has the property that for any p_1, $p_2 \in \mathcal{P}$ there exists $p_3 \in \mathcal{P}$ with the property that $\max\{p_1(x), p_2(x)\} \leq p_3(x)$ for every $x \in X$, then a linear functional $f : X \to \mathbb{C}$ is continuous if and only if there exist a seminorm $p \in \mathcal{P}$ and a constant $C \in (0, \infty)$, such that

$$|f(x)| \leq Cp(x), \quad \forall\, x \in X. \tag{13.1.7}$$

Given a nonempty set X and a family $\mathcal{F}$ of mappings $f : X \to \mathbb{C}$, denote by $\tau_{\mathcal{F}}$ the collection of all unions of finite intersections of sets $f^{-1}(V)$, for $f \in \mathcal{F}$ and V open set in $\mathbb{C}$. Then $\tau_{\mathcal{F}}$ is a topology on X, and is the weakest topology on X that makes every $f \in \mathcal{F}$ continuous. We will refer to it as the $\mathcal{F}$-`topology on` X.

Let X be a vector space, $\mathcal{F}$ be a separating vector space of linear functionals on X and $\tau_{\mathcal{F}}$ be the $\mathcal{F}$-topology on X. Then $(X,\tau_{\mathcal{F}})$ is a locally convex topological space and its dual is $\mathcal{F}$. In particular, a sequence $\{x_j\}_j$ in X satisfies $x_j\to 0$ in $\tau_{\mathcal{F}}$ as $j\to\infty$ if and only if $f(x_j)\to 0$ as $j\to\infty$ for every $f\in\mathcal{F}$.

If X is a topological vector space and X' denotes its dual space, then every $x\in X$ induces a linear functional F_x on X' defined by $F_x(\Lambda):=\Lambda(x)$ for every $\Lambda\in X'$ and if we set $\mathcal{F}:=\{F_x:\,x\in X\}$, then $\mathcal{F}$ separates points in X'. In particular, the $\mathcal{F}$-topology on X', called the `weak*-topology` on X', is locally convex and every linear functional on X' that is continuous with respect to the weak$*$-topology is precisely of the form F_x, for some $x\in X$. Moreover, a sequence $\{\Lambda_j\}_j\subset X'$ converges to some $\Lambda\in X'$, weak$*$-topology on X', if and only if $\Lambda_j(x)\to\Lambda(x)$ as $j\to\infty$ for every $x\in X$.

An inspection of the definition of the weak$*$-topology yields a description of the open sets in this topology. More precisely, if X is a topological vector space and for $A\subseteq X$ and $\varepsilon\in(0,\infty)$ we set

$$\mathcal{O}_{A,\varepsilon}:=\big\{f\in X':\,|f(x)|<\varepsilon,\ \forall\,x\in A\big\},\tag{13.1.8}$$

then the following equivalence holds:

$$\begin{aligned}&\mathcal{O}\subseteq X'\ \text{ is a weak}*\text{-open neighborhood of } 0\in X'\iff\ \text{there exist a set } I,\\ &A_j\subseteq X\ \text{finite and } \varepsilon_j>0\ \text{for each } j\in I,\ \text{such that }\ \mathcal{O}=\bigcup_{j\in I}\mathcal{O}_{A_j,\varepsilon_j}.\end{aligned}\tag{13.1.9}$$

The transpose of any linear and continuous operator between two topological vector spaces is always continuous at the level of dual spaces equipped with weak$*$-topologies.

Proposition 13.1. *Assume X and Y are two given topological vector spaces, and denote by X', Y' their duals, each endowed with the corresponding weak$*$-topology. Also, suppose $T:X\to Y$ is a linear and continuous operator, and define its transpose T^t as the mapping*

$$T^t:Y'\to X',\quad T^t(y'):=y'\circ T\quad\text{for each}\quad y'\in Y'.\tag{13.1.10}$$

Then T^t is well-defined, linear, and continuous.

Proof. For each $y'\in Y'$ it follows that $y'\circ T$ is a composition of two linear and continuous mappings. Hence, $y'\circ T\in X'$ which proves that $T^t:Y'\to X'$ is well-defined. It is also clear from (13.1.10) that T^t is linear. There remains to prove that T^t is continuous. By linearity it suffices to check that T^t is continuous at 0. With this goal in mind, fix an arbitrary finite subset A of X along with an arbitrary number $\varepsilon\in(0,\infty)$, and define

$$\mathcal{O}_{A,\varepsilon}:=\big\{x'\in X':\,|x'(x)|<\varepsilon,\ \forall\,x\in A\big\}.\tag{13.1.11}$$

Furthermore, introduce $\widetilde{A}:=\{Tx:\,x\in A\}$ and set

$$\widetilde{\mathcal{O}}_{\widetilde{A},\varepsilon}:=\big\{y'\in Y':\,|y'(y)|<\varepsilon,\ \forall\,y\in\widetilde{A}\,\big\}.\tag{13.1.12}$$

Then $T^t(\widetilde{\mathcal{O}}_{\widetilde{A},\varepsilon}) \subseteq \mathcal{O}_{A,\varepsilon}$. Invoking the description from (13.1.9) we may conclude that T^t is continuous at 0, and this finishes the proof of the proposition. □

Proposition 13.2. *Suppose that X, Y, Z are topological vector spaces, and denote by X', Y', Z' their duals, each endowed with the corresponding weak∗-topology. In addition, assume that $T : X \to Y$ and $R : Y \to Z$ are two linear and continuous operators. Then*

$$(R \circ T)^t = T^t \circ R^t. \tag{13.1.13}$$

In particular, if $T : X \to Y$ is a linear, continuous, bijective map, with continuous inverse $T^{-1} : Y \to X$, then $T^t : Y' \to X'$ is also bijective and has a continuous inverse $(T^t)^{-1} : X' \to Y'$ that satisfies $(T^t)^{-1} = (T^{-1})^t$.

Proof. Formula (13.1.13) is immediate from definitions, while the claims in the last part of the statement are direct consequences of (13.1.13) and the fact that the transpose of the identity is also the identity. □

We also state and prove an embedding result at the level of dual spaces endowed with the weak∗-topology.

Proposition 13.3. *Suppose X and Y are topological vector spaces such that $X \subseteq Y$ densely and the inclusion map $\iota : X \to Y$, $\iota(x) := x$ for each $x \in X$, is continuous. Then Y' endowed with the weak∗-topology embeds continuously in the space X' endowed with the weak∗-topology, in the sense that the mapping*

$$\iota^t : Y' \longrightarrow X' \tag{13.1.14}$$

is well-defined, linear, injective and continuous. Under the identification of Y' with $\iota^t(Y') \subseteq X'$, we therefore have

$$Y' \hookrightarrow X'. \tag{13.1.15}$$

Proof. The fact that ι^t is well-defined, linear, and continuous follows directly from Proposition 13.1 and assumptions. Assume now that $y' \in Y'$ is such that $\iota^t(y') = 0$. Then from the fact that $y' \circ \iota : X \to \mathbb{C}$ is zero we deduce that $y'\big|_X = 0$. Since $y' : Y \to \mathbb{C}$ is continuous and X is dense in Y, we necessarily have that y' vanishes on Y, forcing $y' = 0$ in Y'. Keeping in mind that ι^t is linear, this implies that ι^t is injective. □

Theorem 13.4 (Hahn–Banach Theorem)**.** *Let X be a vector space (over complex numbers) and suppose $p : X \to [0, \infty)$ is a seminorm on X. Also assume that Y is a linear subspace of X and that $\varphi : Y \to \mathbb{C}$ is a linear functional dominated by p on Y, that is,*

$$|\phi(y)| \leq p(y), \quad \forall\, y \in Y. \tag{13.1.16}$$

Then there exists a linear functional $\Phi : X \to \mathbb{C}$ satisfying

$$\Phi(y) = \phi(y) \quad \forall\, y \in Y, \quad |\Phi(x)| \leq p(x) \quad \forall\, x \in X. \tag{13.1.17}$$

In the next installment we shall specialize the above considerations to various specific settings used in the book. The reader is reminded that $\Omega \subseteq \mathbb{R}^n$ denotes a fixed, nonempty, arbitrary open set.

The Topological Vector Space $\mathcal{E}(\Omega)$
By τ we denote the topology on $C^\infty(\Omega)$ generated by the following family of seminorms:

$$\begin{cases} p_{K,m} : C^\infty(\Omega) \to \mathbb{R}, \\ p_{K,m}(\varphi) := \sup\limits_{x\in K,\, \alpha\in\mathbb{N}_0^n,\, |\alpha|\leq m} |\partial^\alpha \varphi(x)|, \quad \forall\, \varphi \in C^\infty(\Omega), \end{cases} \tag{13.1.18}$$

where $K \subset \Omega$ is a compact set and $m \in \mathbb{N}_0$. Consequently, a sequence $\varphi_j \in C^\infty(\Omega)$, $j \in \mathbb{N}$, converges in τ to a function $\varphi \in C^\infty(\Omega)$ as $j \to \infty$, if and only if for any compact set $K \subset \Omega$ and any $m \in \mathbb{N}_0$ one has

$$\lim_{j\to\infty} \sup_{\alpha\in\mathbb{N}_0^n,\, |\alpha|\leq m} \sup_{x\in K} |\partial^\alpha(\varphi_j - \varphi)(x)| = 0. \tag{13.1.19}$$

We will use the notation

$$\mathcal{E}(\Omega) = \big(C^\infty(\Omega), \tau\big). \tag{13.1.20}$$

The space $\mathcal{E}(\Omega)$ is locally convex and metrizable since its topology is defined by the family of countable seminorms $\{p_{K_m,m}\}_{m\in\mathbb{N}_0}$ where

$$K_m \subset \mathring{K}_{m+1} \subset \Omega, \quad K_m \text{ is compact for each } m \in \mathbb{N}_0 \text{ and } \Omega = \bigcup_{m=0}^{\infty} K_m. \tag{13.1.21}$$

In addition, τ is independent of the family $\{K_m\}_{m\in\mathbb{N}_0}$ with the above properties and $\mathcal{E}(\Omega)$ is complete. Thus, $\mathcal{E}(\Omega)$ is a Frechét space.

The Topological Vector Space $\mathcal{E}'(\Omega)$
Based on the discussion on dual spaces for locally convex topological vector spaces (see (13.1.7) and the remarks preceding it), it follows that the dual space of $\mathcal{E}(\Omega)$ is the collection of all linear functionals $u : \mathcal{E}(\Omega) \to \mathbb{C}$ for which there exist $m \in \mathbb{N}$, a compact set $K \subset \mathbb{R}^n$ with $K \subset \Omega$, and a constant $C > 0$ such that

$$|u(\varphi)| \leq C \sup_{\alpha\in\mathbb{N}_0^n,\, |\alpha|\leq m} \sup_{x\in K} \big|\partial^\alpha(\varphi)(x)\big|, \quad \forall\, \varphi \in C^\infty(\Omega). \tag{13.1.22}$$

This dual space will be endowed with the weak$*$-topology induced by $\mathcal{E}(\Omega)$ and we denote this topological space by $\mathcal{E}'(\Omega)$. Hence, if for each $\varphi \in C^\infty(\Omega)$ we consider the evaluation mapping F_φ taking any functional u from the dual of $\mathcal{E}(\Omega)$ into the number $F_\varphi(u) := u(\varphi) \in \mathbb{C}$, then the family $\mathcal{F} := \{F_\varphi : \varphi \in \mathcal{E}(\Omega)\}$ separates points in the dual of $\mathcal{E}(\Omega)$, and the weak$*$-topology on this dual is the $\mathcal{F}$-topology on it. In particular, if $\{u_j\}_{j\in\mathbb{N}}$ is a sequence in $\mathcal{E}'(\Omega)$ and $u \in \mathcal{E}'(\Omega)$, then

$$\begin{aligned} &u_j \to u \text{ in } \mathcal{E}'(\Omega) \text{ as } j \to \infty \iff \\ &u_j(\varphi) \to u(\varphi) \text{ as } j \to \infty, \ \forall\, \varphi \in C^\infty(\Omega). \end{aligned} \tag{13.1.23}$$

Moreover, a sequence $\{u_j\}_{j\in\mathbb{N}}$ in $\mathcal{E}'(\Omega)$ is Cauchy provided $\lim_{j,k\to\infty}(u_j-u_k)(\varphi)=0$ for every $\varphi\in C^\infty(\Omega)$ and the weak$*$-topology on the dual of $\mathcal{E}(\Omega)$ is locally convex and complete.

The Topological Vector Space $\mathcal{D}_K(\Omega)$
Let $K\subseteq\Omega$ be a compact set in $\mathbb{R}^n$. Denote by $\mathcal{D}_K(\Omega)$ the topological vector space of functions $\{f\in C^\infty(\Omega):\operatorname{supp} f\subseteq K\}$ with the topology induced by τ, the topology in $\mathcal{E}(\Omega)$. Then the topology on $\mathcal{D}_K(\Omega)$ is generated by the family of seminorms

$$\begin{cases} p_m:\mathcal{D}_K(\Omega)\to\mathbb{R}, \\ p_m(\varphi):=\sup\limits_{x\in K,\,\alpha\in\mathbb{N}_0^n,\,|\alpha|\le m}|\partial^\alpha\varphi(x)|, \quad \forall\,\varphi\in C^\infty(\Omega),\quad \operatorname{supp}\varphi\subseteq K, \end{cases} \tag{13.1.24}$$

where $m\in\mathbb{N}_0$. Hence, $\mathcal{D}_K(\Omega)$ is a Fréchet space. In addition, a linear mapping $u:\mathcal{D}_K(\Omega)\to\mathbb{C}$ is continuous if and only if there exist $m\in\mathbb{N}$ and a constant $C>0$, both depending on K, such that

$$|u(\varphi)|\le C\sup_{x\in K,\,\alpha\in\mathbb{N}_0^n,\,|\alpha|\le m}|\partial^\alpha(\varphi)(x)|, \quad \forall\,\varphi\in C^\infty(\Omega),\quad \operatorname{supp}\varphi\subseteq K. \tag{13.1.25}$$

The Topological Vector Space $\mathcal{D}(\Omega)$
The topological vector space on $C_0^\infty(\Omega)$ endowed with the inductive limit topology of the Frechét spaces $\mathcal{D}_K(\Omega)$ will be denoted by $\mathcal{D}(\Omega)$. In this setting, we have

$$\begin{aligned} &\varphi_j\to\varphi \quad \text{in}\ \ \mathcal{D}(\Omega)\ \ \text{as}\ \ j\to\infty \\ &\iff \begin{cases} \exists\, K\subseteq\Omega \text{ compact set such that } \varphi_j\in\mathcal{D}_K(\Omega),\ \ \forall\, j,\ \ \text{and} \\ \varphi_j\to\varphi \text{ in } \mathcal{D}_K(\Omega) \text{ as } j\to\infty. \end{cases} \end{aligned} \tag{13.1.26}$$

The topology $\mathcal{D}(\Omega)$ is locally convex and complete but not metrizable (thus not Fréchet) and is the strict inductive limit of the topologies $\{\mathcal{D}_{K_j}(\Omega)\}_{j\in\mathbb{N}}$, where the family $\{K_j\}_{j\in\mathbb{N}}$ is as in (13.1.21).

We also record an important result that is proved in [59, Theorem 6.6, p. 155].

Theorem 13.5. *Let X be a locally convex topological vector space and suppose the map $\Lambda:\mathcal{D}(\Omega)\to X$ is linear. Then Λ is continuous if and only if for every sequence $\{\varphi_j\}_{j\in\mathbb{N}}$ in $C_0^\infty(\Omega)$ satisfying $\varphi_j\xrightarrow[j\to\infty]{\mathcal{D}(\Omega)}0$ we have $\lim_{j\to\infty}\Lambda(\varphi_j)=0$ in X.*

The Topological Vector Space $\mathcal{D}'(\Omega)$
The dual space of $\mathcal{D}(\Omega)$ endowed with the weak$*$-topology is denoted by $\mathcal{D}'(\Omega)$. Hence, if $\{u_j\}_{j\in\mathbb{N}}$ is a sequence in $\mathcal{D}'(\Omega)$ and $u\in\mathcal{D}'(\Omega)$, then

$$\begin{aligned} &u_j\to u \text{ in } \ \mathcal{D}'(\Omega)\ \ \text{as}\ \ j\to\infty \iff \\ &u_j(\varphi)\to u(\varphi)\ \ \text{as}\ \ j\to\infty,\ \forall\,\varphi\in C_0^\infty(\Omega). \end{aligned} \tag{13.1.27}$$

In addition, a sequence $\{u_j\}_{j\in\mathbb{N}}$ in $\mathcal{D}'(\Omega)$ is called Cauchy provided

$$\lim_{j,k\to\infty}(u_j-u_k)(\varphi)=0$$

for every $\varphi\in C_0^\infty(\Omega)$. The weak$*$-topology on the dual of $\mathcal{D}(\Omega)$ is locally convex and complete and an inspection of this topology reveals that it coincides with the topology defined by the family of seminorms

$$\mathcal{D}'(\Omega)\ni u\mapsto \max_{1\leq j\leq m}|u(\varphi_j)|,\quad m\in\mathbb{N},\ \varphi_1,\ldots,\varphi_m\in\mathcal{D}(\Omega). \tag{13.1.28}$$

The Topological Vector Space $\mathcal{S}(\mathbb{R}^n)$
The Schwartz class of rapidly decreasing functions is the vector space

$$\mathcal{S}(\mathbb{R}^n):=\Big\{\varphi\in C^\infty(\mathbb{R}^n):\forall\,\alpha,\beta\in\mathbb{N}_0^n,\ \sup_{x\in\mathbb{R}^n}|x^\beta\partial^\alpha\varphi(x)|<\infty\Big\}, \tag{13.1.29}$$

endowed with the topology generated by the family of seminorms $\{p_{k,m}\}_{k,m\in\mathbb{N}_0}$ defined by

$$\begin{cases} p_{k,m}:\mathcal{S}(\mathbb{R}^n)\to\mathbb{R}, \\ p_{k,m}(\varphi):=\sup\limits_{x\in\mathbb{R}^n,\,|\alpha|\leq m,\,|\beta|\leq k}|x^\beta\partial^\alpha\varphi(x)|,\quad\forall\,\varphi\in\mathcal{S}(\mathbb{R}^n). \end{cases} \tag{13.1.30}$$

Hence, the topology generated by the family of seminorms $\{p_{k,m}\}_{k,m\in\mathbb{N}_0}$ on $\mathcal{S}(\mathbb{R}^n)$ is locally convex, metrizable, and since it is also complete, the space $\mathcal{S}(\mathbb{R}^n)$ is Frechét. Moreover, a sequence $\varphi_j\in\mathcal{S}(\mathbb{R}^n)$, $j\in\mathbb{N}$, converges in $\mathcal{S}(\mathbb{R}^n)$ to a function $\varphi\in\mathcal{S}(\mathbb{R}^n)$ as $j\to\infty$, if and only if for every $m,k\in\mathbb{N}_0$ one has

$$\lim_{j\to\infty}\sup_{x\in\mathbb{R}^n,\,|\alpha|\leq m,\,|\beta|\leq k}\left|x^\beta\partial^\alpha(\varphi_j-\varphi)(x)\right|=0. \tag{13.1.31}$$

The Topological Vector Space $\mathcal{S}'(\mathbb{R}^n)$
By the discussion about dual spaces for locally convex topological vector spaces, it follows that the dual space of $\mathcal{S}(\mathbb{R}^n)$ is the collection of all linear functions $u:\mathcal{S}(\mathbb{R}^n)\to\mathbb{C}$ for which there exist $m,k\in\mathbb{N}_0$, and a finite constant $C>0$, such that

$$|u(\varphi)|\leq C\sup_{\alpha,\beta\in\mathbb{N}_0^n,\,|\alpha|\leq m,\,|\beta|\leq k}\ \sup_{x\in\mathbb{R}^n}\left|x^\beta\partial^\alpha\varphi(x)\right|,\quad\forall\,\varphi\in\mathcal{S}(\mathbb{R}^n). \tag{13.1.32}$$

We endow the dual of $\mathcal{S}(\mathbb{R}^n)$ with the weak$*$-topology and denote the resulting locally convex topological vector space by $\mathcal{S}'(\mathbb{R}^n)$. Hence, if $\{u_j\}_{j\in\mathbb{N}}$ is a sequence in $\mathcal{S}'(\mathbb{R}^n)$ and $u\in\mathcal{S}'(\mathbb{R}^n)$, then

$$\begin{aligned} &u_j\to u\text{ in }\ \mathcal{S}'(\mathbb{R}^n)\ \text{ as }\ j\to\infty\ \iff \\ &u_j(\varphi)\to u(\varphi)\ \text{ as }\ j\to\infty,\ \forall\,\varphi\in\mathcal{S}(\mathbb{R}^n). \end{aligned} \tag{13.1.33}$$

Also, a sequence $\{u_j\}_{j\in\mathbb{N}}$ in $\mathcal{S}'(\mathbb{R}^n)$ is called Cauchy provided

$$\lim_{j,k\to\infty}(u_j-u_k)(\varphi)=0\text{ for every }\varphi\in\mathcal{S}(\mathbb{R}^n).$$

13.2 Summary of Basic Results from Calculus, Measure Theory, and Topology

Proposition 13.6 (Multinomial theorem). *If $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ and $N \in \mathbb{N}$ are arbitrary, then*

$$\Big(\sum_{j=1}^{n} x_j\Big)^N = \sum_{|\alpha|=N} \frac{N!}{\alpha!}\, x^\alpha. \tag{13.2.1}$$

Theorem 13.7 (Binomial theorem). *For any $x, y \in \mathbb{C}^n$ and any $\gamma \in \mathbb{N}_0^n$ we have*

$$(x+y)^\gamma = \sum_{\alpha+\beta=\gamma} \frac{\gamma!}{\alpha!\beta!}\, x^\alpha y^\beta, \tag{13.2.2}$$

(with the convention that $z^0 := 1$ for each $z \in \mathbb{C}$).

In the particular case when $x = (1, \ldots, 1) \in \mathbb{C}^n$ and $y = (-1, \ldots, -1) \in \mathbb{C}^n$, formula (13.2.2) yields

$$0 = \sum_{\alpha+\beta=\gamma} \frac{\gamma!}{\alpha!\beta!}(-1)^{|\beta|}, \quad \forall\, \gamma \in \mathbb{N}_0^n \ \text{ with } \ |\gamma| > 0. \tag{13.2.3}$$

Proposition 13.8 (Leibniz's formula). *Suppose that $U \subseteq \mathbb{R}^n$ is an open set, $N \in \mathbb{N}$, and $f, g : U \to \mathbb{C}$ are two functions of class C^N in U. Then*

$$\partial^\alpha(fg) = \sum_{\beta \le \alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!}\, (\partial^\beta f)(\partial^{\alpha-\beta} g) \quad \text{ in } \ U, \tag{13.2.4}$$

for every multi-index $\alpha \in \mathbb{N}_0^n$ of length $\le N$.

It is useful to note that for each $\alpha, \beta \in \mathbb{N}_0^n$ and $x \in \mathbb{R}^n$,

$$\partial^\beta(x^\alpha) = \begin{cases} \dfrac{\alpha!}{(\alpha-\beta)!}\, x^{\alpha-\beta} & \text{if } \ \beta \le \alpha, \\ 0 & \text{otherwise.} \end{cases} \tag{13.2.5}$$

Theorem 13.9 (Taylor's formula). *Assume $U \subseteq \mathbb{R}^n$ is an open convex set, and that $N \in \mathbb{N}$. Also, suppose that $f : U \to \mathbb{C}$ is a function of class C^{N+1} on U. Then for every $x, y \in U$ one has*

$$\begin{aligned} f(x) = {} & \sum_{|\alpha| \le N} \frac{1}{\alpha!}(x-y)^\alpha (\partial^\alpha f)(y) \\ & + \sum_{|\alpha|=N+1} \frac{N+1}{\alpha!} \int_0^1 (1-t)^N (x-y)^\alpha (\partial^\alpha f)(tx + (1-t)y)\, dt. \end{aligned} \tag{13.2.6}$$

In particular, for each $x, y \in U$ there exists $\theta \in (0,1)$ with the property that

$$f(x) = \sum_{|\alpha|\leq N} \frac{1}{\alpha!}(x-y)^{\alpha}(\partial^{\alpha} f)(y) + \sum_{|\alpha|=N+1} \frac{1}{\alpha!}(x-y)^{\alpha}(\partial^{\alpha} f)(\theta x + (1-\theta)y). \tag{13.2.7}$$

Theorem 13.10 (Rademacher's theorem)**.** *If $f : \mathbb{R}^n \to \mathbb{R}$ is a Lipschitz function with Lipschitz constant less than or equal to M, for some $M \in (0,\infty)$, then f is differentiable almost everywhere and $\|\partial_k f\|_{L^\infty(\mathbb{R}^n)} \leq M$ for each $k = 1, \ldots, n$.*

Theorem 13.11 (Lebesgue's differentiation theorem)**.** *If $f \in L^1_{\rm loc}(\mathbb{R}^n)$, then*

$$\lim_{\varepsilon\to 0^+} \frac{1}{|B(x,\varepsilon)|} \int_{B(x,\varepsilon)} |f(y) - f(x)|\, {\rm d}y = 0 \quad \text{for almost every } x \in \mathbb{R}^n. \tag{13.2.8}$$

In particular,

$$\lim_{\varepsilon\to 0^+} \frac{1}{|B(x,\varepsilon)|} \int_{B(x,\varepsilon)} f(y)\, {\rm d}y = f(x) \quad \text{for almost every } x \in \mathbb{R}^n. \tag{13.2.9}$$

Theorem 13.12 (Lebesgue's dominated convergence theorem)**.** *Let (X,μ) be a positive measure space and assume that $g \in L^1(X,\mu)$ is a nonnegative function. If $\{f_j\}_{j\in\mathbb{N}}$ is a sequence of μ-measurable, complex valued functions on X, such that $|f_j(x)| \leq g(x)$ for μ-almost every $x \in X$ and $f(x) := \lim_{j\to\infty} f_j(x)$ exists (in $\mathbb{C}$) for μ-almost every $x \in X$, then $f \in L^1(X,\mu)$ and $\lim_{j\to\infty} \int_X |f_j - f|\, {\rm d}\mu = 0$. In particular, $\lim_{j\to\infty} \int_X f_j\, {\rm d}\mu = \int_X f\, {\rm d}\mu$.*

Theorem 13.13 (Young's inequality)**.** *Assume that $1 \leq p, q, r \leq \infty$ are such that $\frac{1}{p} + \frac{1}{q} = \frac{1}{r} + 1$. Then for every $f \in L^p(\mathbb{R}^n)$, $g \in L^q(\mathbb{R}^n)$ it follows that $f * g$ is well-defined almost everywhere in $\mathbb{R}^n$, $f * g \in L^r(\mathbb{R}^n)$, and $\|f * g\|_{L^r(\mathbb{R}^n)} \leq \|f\|_{L^p(\mathbb{R}^n)} \|g\|_{L^q(\mathbb{R}^n)}$.*

Theorem 13.14. *Let X and Y be two Hausdorff topological spaces.*

(a) If $\Lambda : X \to Y$ is continuous, then Λ is sequentially continuous.

(b) If $\Lambda : X \to Y$ is sequentially continuous and X is metrizable, then Λ is continuous.

For a proof of Theorem 13.14 see [59, Theorem, p. 395].

Definition 13.15. *A* `rigid transformation`*, or* `isometry`*, of the Euclidean space $\mathbb{R}^n$ is any distance preserving mapping of $\mathbb{R}^n$, that is, any function $T : \mathbb{R}^n \to \mathbb{R}^n$ satisfying*

$$|T(x) - T(y)| = |x - y|, \quad \forall\, x, y \in \mathbb{R}^n. \tag{13.2.10}$$

A rigid transformation of $\mathbb{R}^n$ is any distance preserving mapping of $\mathbb{R}^n$, i.e., any function $T : \mathbb{R}^n \to \mathbb{R}^n$ satisfying $|Tx - Ty| = |x - y|$ for every $x, y \in \mathbb{R}^n$. The rigid transformations of the Euclidean space $\mathbb{R}^n$ are precisely those obtained by composing a translation with a mapping in $\mathbb{R}^n$ given by an orthogonal matrix. In other words, a mapping $T : \mathbb{R}^n \to \mathbb{R}^n$ is a rigid transformation of $\mathbb{R}^n$ if and only if there exist $x_0 \in \mathbb{R}^n$ and an orthogonal matrix $A : \mathbb{R}^n \to \mathbb{R}^n$ with the property that

$$T(x) = x_0 + Ax, \quad \forall\, x \in \mathbb{R}^n. \tag{13.2.11}$$

For a proof of the following version of the Arzelà–Ascoli theorem see [57, Corollary 34, p. 179].

Theorem 13.16 (Arzelà–Ascoli theorem)**.** *Let $\mathcal{F}$ be an equicontinuous family of real-valued functions on a separable space X. Then each sequence $\{f_j\}_{j\in\mathbb{N}}$ in $\mathcal{F}$ which is bounded at each point has a subsequence $\{f_{j_k}\}_{k\in\mathbb{N}}$ that converges pointwise to a continuous function, the converges being uniform on each compact subset of X.*

Theorem 13.17 (Riesz's representation theorem for positive functionals)**.** *Let X be a locally compact Hausdorff topological space and Λ a positive linear functional on the space of continuous, compactly supported functions on X (denoted by $C_0^0(X)$). Then there exists a unique σ-algebra $\mathfrak{M}$ on X, which contains all Borel sets on X, and a unique measure $\mu : \mathfrak{M} \to [0, \infty]$ that represents Λ, that is, the following hold:*

(i) $\Lambda f = \int_X f \, \mathrm{d}\mu$ *for every continuous, compactly supported function f on X;*

(ii) $\mu(K) < \infty$ *for every compact $K \subset X$;*

(iii) For every $E \in \mathfrak{M}$ we have $\mu(E) = \inf\{\mu(V) : E \subset V, V \text{ open}\}$;

(iv) $\mu(E) = \sup\{\mu(K) : K \subset E, K \text{ compact}\}$ *for every open set E and every $E \in \mathfrak{M}$ with $\mu(E) < \infty$;*

(v) If $E \in \mathfrak{M}$, $A \subset E$, and $\mu(E) = 0$, then $\mu(A) = 0$.

Theorem 13.18 (Riesz's representation theorem for complex functionals)**.** *Let X be a locally compact Hausdorff topological space and consider the space of continuous functions on X vanishing at infinity, that is,*

$$C_{oo}(X) := \{f \in C^0(X) : \forall\, \varepsilon > 0, \exists \text{ compact } K \subset X \text{ such that } |f(x)| < \varepsilon \text{ for } x \in X \setminus K\}. \tag{13.2.12}$$

Then $C_{oo}(X)$ is the closure in the uniform norm of $C_0^0(X)$ and for every bounded linear functional $\Lambda : C_{oo}(X) \to \mathbb{C}$ there exists a unique regular complex Borel measure μ on X such that $\Lambda f = \int_X f \, \mathrm{d}\mu$ for every $f \in C_{oo}(X)$ and $\|\Lambda\| = |\mu|(X)$.

Theorem 13.19 (Riesz's representation theorem for locally bounded functionals)**.** *Let X be a locally compact Hausdorff topological space and assume that $\Lambda : C_0^0(X) \to \mathbb{R}$ is a linear functional that is locally bounded, in the sense that for each compact set $K \subset X$ there exists a constant $C_K \in (0, \infty)$ such that*

$$|\Lambda f| \leq C_K \sup_{x \in K} |f(x)|, \quad \forall\, f \in C_0^0(X) \quad \text{with} \quad \operatorname{supp} f \subseteq K. \tag{13.2.13}$$

Then there exist two measures μ_1, μ_2, taking Borel sets from X into $[0, \infty]$, and satisfying properties (ii)–(iv) in Theorem 13.17, such that

$$\Lambda f = \int_X f \, \mathrm{d}\mu_1 - \int_X f \, \mathrm{d}\mu_2 \quad \text{for every} \quad f \in C_0^0(X). \tag{13.2.14}$$

The reader is warned that since both μ_1 and μ_2 are allowed to take the value ∞, their difference $\mu_1 - \mu_2$ is not well-defined in general. This being said, $\mu_1 - \mu_2$ is a well-defined finite signed measure on each compact subset of X.

Proposition 13.20 (Urysohn's lemma)**.** *If X is a locally compact Hausdorff space and $K \subset U \subset X$ are such that K is compact and U is open, then there exists a function $f \in C_0^0(U)$ that satisfies $f = 1$ on K and $0 \leq f \leq 1$.*

Theorem 13.21 (Vitali's convergence theorem)**.** *Let (X, μ) be a positive measure space with $\mu(X) < \infty$. Suppose $\{f_k\}_{k \in \mathbb{N}}$ is a sequence of functions in $L^1(X, \mu)$ and that f is a function on X (all complex-valued) satisfying:*

(i) $f_k(x) \to f(x)$ *for μ-almost every $x \in X$ as $k \to \infty$;*

(ii) $|f| < \infty$ *μ-almost everywhere in X;*

(iii) $\{f_k\}_{k \in \mathbb{N}}$ *is uniformly integrable, in the sense that for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every $k \in \mathbb{N}$ we have $\left| \int_E f_k \, \mathrm{d}\mu \right| < \varepsilon$ whenever $E \subseteq X$ is a μ-measurable set with $\mu(E) < \delta$.*

Then $f \in L^1(X, \mu)$ and

$$\lim_{k \to \infty} \int_X \left| f_k - f \right| \mathrm{d}\mu = 0. \tag{13.2.15}$$

In particular, $\lim_{k \to \infty} \int_X f_k \, \mathrm{d}\mu = \int_X f \, \mathrm{d}\mu$.

See, for example, [58, p. 133].

Proposition 13.22. *Let (X, μ) be a positive measure space and suppose that $f \in L^1(X, \mu)$. Then for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every μ-measurable set $A \subseteq X$ satisfying $\mu(A) < \delta$ we have $\int_A |f| \, \mathrm{d}\mu < \varepsilon$.*

Proof. Consider the measure $\lambda := |f|\mu$ on X. Then λ is absolutely continuous with respect to μ and the (ε, δ) characterization of absolute continuity of measures (see, e.g., [58, Theorem 6.11, p. 124]) yields the desired conclusion. □

13.3 Custom-Designing Smooth Cut-Off Functions

Lemma 13.23. *Let $f:\mathbb{R}\to\mathbb{R}$ be the function defined by*

$$f(x):=\begin{cases}\mathrm{e}^{-1/x}, & \text{if} \quad x>0,\\ 0, & \text{if} \quad x\leq 0,\end{cases}\qquad \forall\, x\in\mathbb{R}. \tag{13.3.1}$$

Then f is of class C^∞ on $\mathbb{R}$.

Proof. Denote by $\mathcal{C}$ the collection of functions $g:\mathbb{R}\to\mathbb{R}$ for which there exists a polynomial P such that

$$g(x):=\begin{cases}\mathrm{e}^{-1/x}\,P(1/x), & \text{if} \quad x>0,\\ 0, & \text{if} \quad x\leq 0,\end{cases}\qquad \forall\, x\in\mathbb{R}. \tag{13.3.2}$$

Recall that if $h:\mathbb{R}\to\mathbb{R}$ is a continuous function that is differentiable on $\mathbb{R}\setminus\{0\}$ and for which there exists $L\in\mathbb{R}$ such that $\lim_{x\to 0^-} h'(x)=L=\lim_{x\to 0^+} h'(x)$, then h is also differentiable at the origin and $h'(0)=L$. An immediate consequence of this fact is that any $g\in\mathcal{C}$ is differentiable and $g'\in\mathcal{C}$. In turn, this readily gives that any $g\in\mathcal{C}$ is of class C^∞ on $\mathbb{R}$. Since f defined in (13.3.1) clearly belongs to $\mathcal{C}$, it follows that f is of class C^∞ on $\mathbb{R}$. □

Lemma 13.24. *The function $\phi:\mathbb{R}^n\to\mathbb{R}$ defined by*

$$\phi(x):=\begin{cases}C\mathrm{e}^{\frac{1}{|x|^2-1}} & \text{if} \quad x\in B(0,1),\\ 0 & \text{if} \quad x\in\mathbb{R}^n\setminus\overline{B(0,1)},\end{cases} \tag{13.3.3}$$

where $C:=\left(\omega_{n-1}\int_0^\infty \mathrm{e}^{1/(\rho^2-1)}\rho^{n-1}\,\mathrm{d}\rho\right)^{-1}\in(0,\infty)$, satisfies the following properties:

$$\phi\in C^\infty(\mathbb{R}^n),\quad \phi\geq 0,\quad \operatorname{supp}\phi\subseteq\overline{B(0,1)},\quad \text{and}\quad \int_{\mathbb{R}^n}\phi(x)\,\mathrm{d}x=1. \tag{13.3.4}$$

Proof. That $\phi\geq 0$ and $\operatorname{supp}\phi\subseteq\overline{B(0,1)}$ is immediate from its definition. Also, since $\phi(x)=f(1-|x|^2)$ for $x\in\mathbb{R}^n$ where f is as in (13.3.1), invoking Lemma 13.23 it follows that $\phi\in C^\infty(\mathbb{R}^n)$. Finally, the condition $\int_{\mathbb{R}^n}\phi(x)\,\mathrm{d}x=1$ follows upon observing that based on (13.8.9) we have $\int_{\mathbb{R}^n}\mathrm{e}^{\frac{1}{|x|^2-1}}\,\mathrm{d}x=1/C$. □

Proposition 13.25. *Let $F_0,F_1\subset\mathbb{R}^n$ be two nonempty sets with the property that $\operatorname{dist}(F_0,F_1)>0$. Then there exists a function $\psi:\mathbb{R}^n\to\mathbb{R}$ with the following properties:*

$$\psi\in C^\infty(\mathbb{R}^n),\quad 0\leq\psi\leq 1,\quad \psi=0 \text{ on } F_0,\quad \psi=1 \text{ on } F_1,\quad \text{and}$$

$$\forall\,\alpha\in\mathbb{N}_0^n\ \ \exists\, C_\alpha\in(0,\infty)\quad \text{such that}\quad |\partial^\alpha\psi(x)|\leq\frac{C_\alpha}{\operatorname{dist}(F_0,F_1)^{|\alpha|}}\quad \forall\, x\in\mathbb{R}^n. \tag{13.3.5}$$

Proof. Let $r := \operatorname{dist}(F_0, F_1) > 0$ and set $\widetilde{F_1} := \big\{x \in \mathbb{R}^n : \operatorname{dist}(x, F_1) \leq r/4\big\}$. Also, with ϕ as in Lemma 13.24, define the function $\theta(x) := \big(\frac{4}{r}\big)^n \phi(4x/r)$ for $x \in \mathbb{R}^n$. Then

$$\theta \in C^\infty(\mathbb{R}^n), \quad \theta \geq 0, \quad \operatorname{supp}\theta \subseteq \overline{B(0, r/4)}, \quad \text{and} \quad \int_{\mathbb{R}^n} \theta(x)\,dx = 1. \tag{13.3.6}$$

We claim that the function $\psi : \mathbb{R}^n \to \mathbb{R}$ defined by $\psi := \chi_{\widetilde{F_1}} * \theta$ has the desired properties. To see why this is true, note that since

$$\psi(x) = \int_{\widetilde{F_1}} \theta(x - y)\,dy \quad \forall\, x \in \mathbb{R}^n, \tag{13.3.7}$$

from the properties of θ it is immediate that $\psi \in C^\infty(\mathbb{R}^n)$ and $0 \leq \psi \leq 1$. Furthermore, for $\alpha \in \mathbb{N}_0^n$ and $x \in \mathbb{R}^n$ we may write

$$\begin{aligned} |\partial^\alpha \psi(x)| &\leq \left(\frac{4}{r}\right)^{|\alpha|} \int_{\widetilde{F_1}} \left(\frac{4}{r}\right)^n \big|(\partial^\alpha \phi)\big(4(x-y)/r\big)\big|\,dy \\ &\leq \big(\tfrac{4}{r}\big)^{|\alpha|} \int_{\mathbb{R}^n} \big|\partial^\alpha \phi(y)\big|\,dy = C_\alpha r^{-|\alpha|} = \frac{C_\alpha}{\operatorname{dist}(F_0, F_1)^{|\alpha|}}, \end{aligned} \tag{13.3.8}$$

where $C_\alpha := 4^{|\alpha|} \int_{\mathbb{R}^n} \big|\partial^\alpha \phi(y)\big|\,dy$ is a positive, finite number independent of r.

We are left with checking the fact that $\psi = j$ on F_j, $j = 0, 1$. First, if $x \in F_0$, then $|x - y| \geq 3r/4$ for every $y \in \widetilde{F_1}$, hence $\theta(x - y) = 0$ for every $y \in \widetilde{F_1}$, which when combined with (13.3.7) implies $\psi(x) = 0$. Second, if $x \in F_1$, then $\overline{B(x, r/4)} \subseteq \widetilde{F_1}$. Since the support of $\theta(x - \cdot) \subset \overline{B(x, r/4)}$ the latter implies $\psi(x) = \int_{\mathbb{R}^n} \theta(x - y)\,dy = 1$. The proof of the proposition is complete. □

A trivial yet useful consequence of the above result is as follows.

Proposition 13.26. *If $U \subseteq \mathbb{R}^n$ is open and $K \subset U$ is compact, then there exists a function $\psi : \mathbb{R}^n \to \mathbb{R}$ that is of class C^∞, satisfies $0 \leq \psi(x) \leq 1$ for every $x \in \mathbb{R}^n$ and $\psi(x) = 1$ for every $x \in K$, and which has compact support, contained in U.*

Proof. Since K is a compact contained in U we have $r := \operatorname{dist}(K, \mathbb{R}^n \setminus U)$ is contained in $(0, \infty)$. Proposition 13.25 applied with $F_1 := K$ and with set $F_0 := \{x \in \mathbb{R}^n : \operatorname{dist}(x, K) > r/2\}$ then yields the desired function ψ. □

13.4 Partition of Unity

Lemma 13.27. *If $C \subset \mathbb{R}^n$ is compact, and $U \subseteq \mathbb{R}^n$ is an open set such that $C \subset U$, then there exists a compact set $D \subseteq \mathbb{R}^n$ such that $C \subset \mathring{D} \subset D \subset U$.*

Proof. Let $V = U^c \cap \overline{B(0, R)}$, where $R > 0$ is large enough so that $\overline{U} \subset B(0, R)$. Then V is compact and disjoint from C so $r := \operatorname{dist}(V, C) = \inf\limits_{\substack{x \in C \\ y \in V}} \|x - y\| > 0$.

Hence, if we set

$$D := \bigcup_{x\in C} \overline{B(x, r/4)}, \tag{13.4.1}$$

then D is compact and $C \subset \mathring{D} \subset D \subset U$. □

Lemma 13.28. *Suppose that $K \subseteq \mathbb{R}^n$ is a compact set and that $\{O_j\}_{1\leq j\leq k}$ is a finite open cover of K. Then there are compact sets $D_j \subset O_j$, $1 \leq j \leq k$, with the property that*

$$K \subset \bigcup_{j=1}^{k} \mathring{D}_j. \tag{13.4.2}$$

Proof. Set $C_1 := K \setminus \bigcup_{j=2}^{k} O_j \subseteq O_1$. Since C_1 is compact, Lemma 13.27 shows that there exists a compact set D_1 with the property that $C_1 \subset \mathring{D}_1 \subset D_1 \subset O_1$. Next, proceeding inductively, suppose that $m \in \mathbb{N}$ is such that $1 \leq m < k$ and that m compact sets $D_1, \ldots, D_m$ have been constructed with the property that $K \subset \big(\bigcup_{j=1}^{m} \mathring{D}_j\big) \cup \big(\bigcup_{j=m+1}^{k} O_j\big)$. Introduce

$$C_{m+1} := K \setminus \left(\bigcup_{j=1}^{m} \mathring{D}_j \cup \bigcup_{j=m+2}^{k} O_j \right). \tag{13.4.3}$$

Clearly, C_{m+1} is a compact subset of O_{m+1}. By once again invoking Lemma 13.27, there exists a compact set $D_{m+1} \subset O_{m+1}$ such that $C_{m+1} \subset \mathring{D}_{m+1}$. After k iterations, this procedure yields a family of sets $C_1, \ldots, C_k$ which have all the desired properties. □

Theorem 13.29 (Partition of unity for compact sets)**.** *Let $K \subset \mathbb{R}^n$ be a compact set, and let $\{O_j\}_{1\leq j\leq N}$ be a finite open cover of K. Then there exists a finite collection of C^∞ functions $\varphi_j : \mathbb{R}^n \to \mathbb{R}$, $1 \leq j \leq N$, satisfying the following properties:*

(i) For every $1 \leq j \leq N$, the set $\operatorname{supp}(\varphi_j)$ is compact and contained in O_j;

(ii) For every $1 \leq j \leq N$, one has $0 \leq \varphi_j \leq 1$;

(iii) $\sum_{j=1}^{N} \varphi_j(x) = 1$ for every $x \in K$.

The family $\{\varphi_j : 1 \leq j \leq N\}$ is called a `partition of unity subordinate to the cover` *$\{O_j\}_{1\leq j\leq N}$ of K.*

Proof. Let $\{O_j\}_{1\leq j\leq N}$ be any finite open cover for K. From Lemma 13.28 we know that there exist compact sets $D_j \subseteq O_j$, $1 \leq j \leq N$, such that $K \subset \bigcup_{j=1}^{N} \mathring{D}_j$.

By Proposition 13.26, for each $1 \leq j \leq N$, choose a C^∞ function $\eta_j : \mathbb{R}^n \to [0,\infty)$ that is positive on D_j and has compact support in O_j. It follows that $\sum_{j=1}^{N} \eta_j(x) > 0$ for all $x \in \bigcup_{j=1}^{N} \mathring{D}_j$, so we can define for each $j \in \{1,\ldots,N\}$ the function

$$\psi_j : \bigcup_{i=1}^{N} \mathring{D}_j \to \mathbb{R}, \quad \psi_j(x) := \frac{\eta_j(x)}{\sum_{k=1}^{N} \eta_k(x)}, \quad \forall\, x \in \bigcup_{i=1}^{N} \mathring{D}_j. \tag{13.4.4}$$

By Lemma 13.28, there exists a compact set $U \subseteq \mathbb{R}^n$ with $K \subseteq \mathring{U} \subseteq U \subseteq \bigcup_{j=1}^{N} \mathring{D}_j$. We apply Proposition 13.26 to obtain a C^∞ function $f : \mathbb{R}^n \to [0,1]$ that satisfies $f(x) = 1$ for $x \in K$ and having compact support contained in $\mathring{U}$. Then for each $j \in \{1,\ldots,N\}$ we define the function $\varphi_j := f\psi_j$ acting from $\mathbb{R}^n$ into $\mathbb{R}$. It is not hard to see that each φ_j is C^∞ in $\mathbb{R}^n$, has compact support, contained in O_j, $0 \leq \varphi_j \leq 1$, and $\sum_{1\leq j\leq N} \varphi_j(x) = 1$ for all $x \in K$. □

Definition 13.30.

(i) A family $\{F_j\}_{j\in I}$ of subsets of $\mathbb{R}^n$ is said to be `locally finite in` *$E \subseteq \mathbb{R}^n$ provided every $x \in E$ has a neighborhood $\mathcal{O} \subseteq \mathbb{R}^n$ with the property that the set $\{j \in I : F_j \cap \mathcal{O} \neq \varnothing\}$ is finite.*

(ii) Given a collection of functions $f_j : \Omega \to \mathbb{R}$, $j \in I$, defined in some fixed subset Ω of $\mathbb{R}^n$, the sum $\sum_{j\in I} f_j$ is called `locally finite in` *$E \subseteq \mathbb{R}^n$ provided the family of sets $\{x \in \Omega : f_j(x) \neq 0\}$, indexed by $j \in I$, is locally finite in E.*

Exercise 13.31. *Show that a family $\{F_j\}_{j\in I}$ of subsets of $\mathbb{R}^n$ is locally finite in the open set $E \subseteq \mathbb{R}^n$ if and only if for every compact $K \subseteq E$ the collection $\{j \in I : F_j \cap K \neq \varnothing\}$ is finite.*

Exercise 13.32. *Show that if the family $\{F_j\}_{j\in I}$ of subsets of $\mathbb{R}^n$ is locally finite in $E \subseteq \mathbb{R}^n$, then $\{\overline{F_j}\}_{j\in I}$ is also locally finite in E.*

Exercise 13.33. *Show that if the family $\{F_j\}_{j\in I}$ of closed subsets of $\mathbb{R}^n$ is locally finite in $E \subseteq \mathbb{R}^n$, then $\bigcup_{j\in I}(E \cap F_j)$ is a relatively closed subset of E.*

Proof. Let $x^* \in E$ be such that $x^* \notin F_j$ for every $j \in I$. Then there exists $r > 0$ with the property that $B(x^*, r) \cap F_j = \varnothing$ for every $j \in I \setminus I_*$, where I_* is a finite subset of I. Hence, by eventually further decreasing r, it can be assumed that $B(x^*, r) \cap F_j = \varnothing$ for every I_* as well. Thus, $B(x^*, r) \cap E$ is a relative neighborhood of x^* in E that is disjoint from $\bigcup_{j\in I} F_j$. This proves that $E \setminus \left(\bigcup_{j\in I} F_j\right)$ is relatively open in E hence, $\bigcup_{j\in I}(E \cap F_j)$ is a relatively closed subset of E. □

Theorem 13.34 (Partition of unity for arbitrary open covers)**.** *Let $\{O_k\}_{k\in I}$ be an arbitrary family of open sets in $\mathbb{R}^n$ and set $\Omega := \bigcup_{k\in I} O_k$. Then there exists an at most countable collection $\{\varphi_j\}_{j\in J}$ of C^∞ functions $\varphi_j : \Omega \to \mathbb{R}$ satisfying the following properties:*

(i) For every $j \in J$ there exists $k \in I$ such that φ_j is compactly supported in O_k;

(ii) For every $j \in J$, one has $0 \leq \varphi_j \leq 1$ in Ω;

(iii) The family of sets $\{x \in \Omega : \varphi_j(x) \neq 0\}$, indexed by $j \in J$, is locally finite in Ω;

(iv) $\sum_{j\in J} \varphi_j(x) = 1$ for every $x \in \Omega$.

The family $\{\varphi_j\}_{j\in J}$ is called a `partition of unity subordinate to the family` *$\{O_k\}_{k\in I}$.*

Proof. Start by defining

$$\Omega_j := \Big\{x \in \Omega : |x| \leq j \text{ and } \operatorname{dist}(x, \partial\Omega) \geq \tfrac{1}{j}\Big\}, \quad j \in \mathbb{N}, \tag{13.4.5}$$

Then $\Omega = \bigcup_{j=1}^{\infty} \Omega_j$ and

$$\Omega_j \subseteq \mathbb{R}^n \text{ is compact}, \quad \Omega_j \subset \mathring{\Omega}_{j+1} \quad \text{for every } j \in \mathbb{N}. \tag{13.4.6}$$

Proceed now to define the compact sets

$$K_2 := \Omega_2, \quad K_j := \Omega_j \setminus \mathring{\Omega}_{j-1} \quad \text{for every } j \geq 3. \tag{13.4.7}$$

As such, from (13.4.7) and (13.4.6) we have

$$K_j = \Omega_j \cap \big(\mathring{\Omega}_{j-1}\big)^c \subseteq \mathring{\Omega}_{j+1} \cap (\Omega_{j-2})^c \quad \text{for every } j \geq 3. \tag{13.4.8}$$

Finally, we define the following families of open sets

$$\mathcal{O}_2 := \big\{O_k \cap \mathring{\Omega}_3 : k \in I\big\}, \quad \mathcal{O}_j := \big\{O_k \cap \big(\Omega_{j+1} \setminus \mathring{\Omega}_{j-2}\big) : k \in I\big\} \quad \forall\, j \geq 3. \tag{13.4.9}$$

Making use of (13.4.6), for every $j \geq 3$ we have that

$$\begin{aligned}\big(\Omega_{j+1} \setminus \Omega_{j-2}\big)^\circ &= \big(\Omega_{j+1} \cap (\Omega_{j-2})^c\big)^\circ = \mathring{\Omega}_{j+1} \cap \big((\Omega_{j-2})^c\big)^\circ \\ &= \mathring{\Omega}_{j+1} \cap \big(\overline{\Omega_{j-2}}\big)^c = \mathring{\Omega}_{j+1} \cap (\Omega_{j-2})^c.\end{aligned} \tag{13.4.10}$$

Hence, (13.4.7), (13.4.9), and (13.4.10), imply that $\mathcal{O}_j$ is an open cover for K_j for every $j \geq 3$. Also, from the definitions of K_2 and $\mathcal{O}_2$ and (13.4.6), we obtain that $\mathcal{O}_2$ is an open cover for K_2. Since the K_j's are compact, these open covers can be refined to finite subcovers in each case. As such, for each

$j = 2, 3, \ldots$, we can apply Theorem 13.29 to obtain a finite partition of unity $\{\varphi : \varphi \in \Phi_j\}$ for K_j subordinate to $\mathcal{O}_j$. Also note that due to (13.4.10) and (13.4.9), we necessarily have that, for each $j \in \{2, 3, \ldots\}$, $\Omega_j \cap O = \varnothing$ for every $O \in \mathcal{O}_k$, for every $k \in \mathbb{N}$ satisfying $k \geq j + 2$. This ensures that the family $\{\operatorname{supp} \varphi : \varphi \in \Phi_j,\, j \geq 2\}$ is locally finite in Ω, so we can define

$$s(x) := \sum_{j \geq 2} \sum_{\varphi \in \Phi_j} \varphi(x), \quad \text{for every } x \in \Omega. \tag{13.4.11}$$

Given that differentiability is a local property, it follows that s is of class C^∞ in Ω. Moreover, note that $s(x) > 0$ for every $x \in \Omega$, since $0 \leq \varphi \leq 1$ for all $\varphi \in \Phi_j$, $j = 2, 3, \ldots$, and if $x \in \Omega_j$, some $j \in \{2, 3, \ldots\}$, then $\sum_{\varphi \in \Phi_j} \varphi(x) = 1$. Consequently, $1/s$ is also a C^∞ function in Ω. It is then clear that the collection

$$\Phi := \big\{\varphi/s : \varphi \in \Phi_j,\, j = 2, 3, \ldots\big\}$$

is a partition of unity subordinate to the family of open sets $\{O_k\}_{k \in I}$ (in the sense described in the statement of the theorem). □

Theorem 13.35 (Partition of unity with preservation of indexes)**.** *Let $\{O_k\}_{k \in I}$ be an arbitrary family of open sets in $\mathbb{R}^n$ and set $\Omega := \bigcup_{k \in I} O_k$. Then there exists a collection $\{\psi_k\}_{k \in I}$ of C^∞ functions $\psi_k : \Omega \to \mathbb{R}$ satisfying the following properties:*

(i) For every $k \in I$ the function ψ_k vanishes outside of a relatively closed subset of O_k;

(ii) For every $k \in I$, one has $0 \leq \psi_k \leq 1$ in Ω;

(iii) The family of sets $\{x \in \Omega : \psi_k(x) \neq 0\}$, indexed by $k \in I$, is locally finite in Ω;

(iv) $\sum_{k \in I} \psi_k(x) = 1$ for every $x \in \Omega$.

Proof. Let $\{\varphi_j\}_{j \in J}$ be a partition of unity subordinate to the family $\{O_k\}_{k \in I}$, and denote by $f : J \to I$ a function with the property that, for every $j \in J$, the function φ_j is compactly supported in $O_{f(j)}$. That this exists is guaranteed by Theorem 13.34. For every $k \in I$ then define

$$\psi_k(x) := \sum_{j \in f^{-1}(\{k\})} \varphi_j(x), \quad \forall\, x \in \Omega. \tag{13.4.12}$$

Note that the sum is locally finite in Ω, hence $\psi_k : \Omega \to \mathbb{R}$ is a well-defined nonnegative function, of class C^∞ in Ω, for every $k \in I$. In addition, the result from Exercise 13.33 shows that, for every $k \in I$, φ_k vanishes outside a relatively

closed subset of O_k. Furthermore, since $\{f^{-1}(\{k\})\}_{k\in I}$ is a partition of J into mutually disjoint subsets, we may compute

$$\sum_{k\in I}\psi_k(x)=\sum_{k\in I}\sum_{j\in f^{-1}(\{k\})}\varphi_j(x)=\sum_{j\in J}\varphi_j(x)=1\quad\forall\,x\in\Omega. \tag{13.4.13}$$

Incidentally, this also shows that, necessarily, $0\leq\psi_k(x)\leq 1$ for every $k\in I$ and $x\in\Omega$. Finally, the fact that the family of sets $\{x\in\Omega:\,\psi_k(x)\neq 0\}$, $k\in I$, is locally finite in Ω is inherited from the corresponding property of the φ_j's. □

13.5 The Gamma and Beta Functions

The `gamma function` is defined as

$$\Gamma(z):=\int_0^\infty t^{z-1}\mathrm{e}^{-t}\,\mathrm{d}t,\quad\text{for } z\in\mathbb{C},\ \operatorname{Re}z>0. \tag{13.5.1}$$

It is easy to check that $\Gamma(1)=1$, $\Gamma(1/2)=\sqrt{\pi}$ and via integration by parts that $\Gamma(z+1)=z\Gamma(z)$ for $z\in\mathbb{C}$, $\operatorname{Re}z>0$. By analytic continuation, the function $\Gamma(z)$ is extended to a meromorphic function defined for all complex numbers z except for $z=-n$, $n\in\mathbb{N}_0$, where the extended function has simple poles with residue $(-1)^n/n!$ and this extension satisfies

$$\Gamma(z+1)=z\,\Gamma(z)\quad\text{for}\quad z\in\mathbb{C}\setminus\{-n:\,n\in\mathbb{N}_0\}. \tag{13.5.2}$$

By induction it follows that for every $n\in\mathbb{N}$ we have

$$\Gamma(n)=(n-1)!, \tag{13.5.3}$$

$$\Gamma\Big(\frac{1}{2}+n\Big)=\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2^n}\sqrt{\pi}=\frac{(2n)!}{2^{2n}\,n!}\sqrt{\pi}, \tag{13.5.4}$$

$$\Gamma\Big(\frac{1}{2}-n\Big)=\frac{(-1)^n 2^n}{1\cdot 3\cdot 5\cdots(2n-1)}\sqrt{\pi}=\frac{(-1)^n 2^{2n}\,n!}{(2n)!}\sqrt{\pi}. \tag{13.5.5}$$

The volume of the unit ball $B(0,1)$ in $\mathbb{R}^n$, which we denote by $|B(0,1)|$, and the surface area of the unit sphere in $\mathbb{R}^n$, denoted here by ω_{n-1}, have the following formulas:

$$|B(0,1)|=\frac{\pi^{\frac{n}{2}}}{\Gamma\big(\frac{n}{2}+1\big)},\quad\omega_{n-1}=n|B(0,1)|=\frac{2\pi^{\frac{n}{2}}}{\Gamma\big(\frac{n}{2}\big)}. \tag{13.5.6}$$

Next, consider the so-called `beta function`

$$B(z,w):=\int_0^1 t^{z-1}(1-t)^{w-1}\,\mathrm{d}t,\quad\operatorname{Re}z,\operatorname{Re}w>0. \tag{13.5.7}$$

Clearly $B(z,w) = B(w,z)$. Making the change of variables $t = u/(u+1)$, $u \in (0,\infty)$, it follows that

$$B(z,w) = \int_0^\infty u^{w-1}\Big(\frac{1}{u+1}\Big)^{z+w}\,\mathrm{d}u \tag{13.5.8}$$

whenever $\operatorname{Re} z$, $\operatorname{Re} w > 0$.

The basic identity relating the gamma and beta functions reads

$$B(z,w) = \frac{\Gamma(z)\Gamma(w)}{\Gamma(z+w)}, \quad \operatorname{Re} z, \operatorname{Re} w > 0. \tag{13.5.9}$$

This is easily proved starting with (13.5.8), writing $\Gamma(z+w) = \int_0^\infty t^{z+w-1}\mathrm{e}^{-t}\,\mathrm{d}t$ and expressing $B(z,w)\Gamma(z+w)$ as a double integral, then making the change of variables $s := t/(u+1)$. A useful consequence of identity (13.5.9) is the following formula

$$\int_0^{\pi/2}(\sin\theta)^a(\cos\theta)^b\,\mathrm{d}\theta = \frac{1}{2}B\left(\frac{a+1}{2},\frac{b+1}{2}\right) = \frac{1}{2}\frac{\Gamma(\frac{a+1}{2})\Gamma(\frac{b+1}{2})}{\Gamma(\frac{a+b+2}{2})} \quad \text{if } a,b>-1. \tag{13.5.10}$$

Indeed, making the change of variables $u := (\sin\theta)^2$, the integral in the leftmost side of (13.5.10) becomes $\frac{1}{2}\int_0^1 u^{(a-1)/2}(1-u)^{(b-1)/2}\,\mathrm{d}u$.

For further reference, let us also note here that

$$\begin{aligned}\int_0^{\pi}(\sin\theta)^a(\cos\theta)^b\,\mathrm{d}\theta &= \frac{1+(-1)^b}{2}\cdot B\left(\frac{a+1}{2},\frac{b+1}{2}\right)\\ &= \frac{1+(-1)^b}{2}\cdot\frac{\Gamma(\frac{a+1}{2})\Gamma(\frac{b+1}{2})}{\Gamma(\frac{a+b+2}{2})},\end{aligned} \tag{13.5.11}$$

whenever $a,b>-1$. This is proved by splitting the domain of integration into $(0,\pi/2)\cup(\pi/2,\pi)$, making a change of variables $\theta\mapsto\theta-\pi/2$ in the second integral, and invoking (13.5.10).

13.6 Surfaces in $\mathbb{R}^n$ and Surface Integrals

Definition 13.36. *Given $n\geq 2$ and $k\in\mathbb{N}\cup\{\infty\}$, a C^k* `surface` *(or, surface of class C^k) in $\mathbb{R}^n$ is a subset Σ of $\mathbb{R}^n$ with the property that for every $x^*\in\Sigma$ there exists an open neighborhood $U(x^*)$ such that*

$$\Sigma\cap U(x^*) = P(\mathcal{O}) \tag{13.6.1}$$

where $\mathcal{O}$ is an open subset of $\mathbb{R}^{n-1}$ and

$$\begin{aligned}&P:\mathcal{O}\longrightarrow\mathbb{R}^n \quad \text{is an injective function of class } C^k \text{ satisfying}\\ &\quad \operatorname{rank}[DP(u)] = n-1, \text{ for all } u=(u_1,\dots,u_{n-1})\in\mathcal{O},\end{aligned} \tag{13.6.2}$$

where DP is the Jacobian matrix of P, that is,

$$DP(u) = \left(\frac{D(P_1,\dots,P_n)}{D(u_1,\dots,u_{n-1})}\right)(u), \quad u \in \mathcal{O}. \tag{13.6.3}$$

The function P in (13.6.2) is called a `local parametrization of class` C^k `near` x^* *and $\Sigma \cap U(x^*)$ a* `parametrizable patch`.

In the case when (13.6.1) holds when we formally take $r = +\infty$, that is, in the case when $\Sigma = P(\mathcal{O})$, we call P a `global parametrization` *of the surface Σ.*

Proposition 13.37. *If $\mathcal{O}$ is an open subset of $\mathbb{R}^{n-1}$ and $P : \mathcal{O} \to \mathbb{R}^n$ satisfies (13.6.2)–(13.6.3) for $k = 1$, then $P : \mathcal{O} \to P(\mathcal{O})$ is a homeomorphism.*

Definition 13.38. *Assume $n \geq 3$. If $v_1 = (v_{11},\dots,v_{1n}),\dots,v_{n-1} = (v_{n-1\,1},\dots,v_{n-1\,n})$ are $n-1$ vectors in $\mathbb{R}^n$, their* `cross product` *is defined as*

$$v_1 \times v_2 \times \cdots \times v_{n-1} := \det \begin{pmatrix} v_{11} & v_{12} & \dots & v_{1n} \\ v_{21} & v_{22} & \dots & v_{2n} \\ \vdots & \vdots & \dots & \vdots \\ v_{n-1\,1} & v_{n-1\,2} & \dots & v_{n-1\,n} \\ \mathbf{e}_1 & \mathbf{e}_2 & \dots & \mathbf{e}_n \end{pmatrix}, \tag{13.6.4}$$

where the determinant is understood as computed by formally expanding it with respect to the last row, the result being a vector in $\mathbb{R}^n$. More precisely,

$$v_1 \times \cdots \times v_{n-1} \tag{13.6.5}$$
$$:= \sum_{j=1}^{n} (-1)^{j+1} \det \begin{pmatrix} v_{11} & \dots & v_{j-1} & v_{j+1} & \dots & v_{1n} \\ \vdots & \dots & \vdots & \dots & \vdots & \\ v_{n-1\,1} & \dots & v_{n-1\,j-1} & v_{n-1\,j+1} & \dots & v_{n-1\,n} \end{pmatrix} \mathbf{e}_j.$$

Definition 13.39. *Let $\Sigma \subset \mathbb{R}^n$, $n \geq 2$, be a C^1 surface and assume that $P : \mathcal{O} \to \mathbb{R}^n$, with $\mathcal{O}$ open subset of $\mathbb{R}^{n-1}$, is a local parametrization of Σ of class C^1 near some point $X^* \in \Sigma$. Also, suppose that $f : \Sigma \to \mathbb{R}$ is a continuous function on Σ that vanishes outside of a compact subset of $P(\mathcal{O})$. We then define*

$$\int_\Sigma f(x)\,\mathrm{d}\sigma(x) := \int_{\mathcal{O}} (f \circ P)(u)\ |\partial_1 P(u) \times \cdots \times \partial_{n-1} P(u)|\ \mathrm{d}u_1 \dots \mathrm{d}u_{n-1} \tag{13.6.6}$$

if $n \geq 3$, and

$$\int_\Sigma f(x)\,\mathrm{d}\sigma(x) := \int_{\mathcal{O}} (f \circ P)(u) |P'(u)|\,\mathrm{d}u \quad \text{if} \quad n = 2. \tag{13.6.7}$$

In (13.6.6), $d\sigma$ stands for the `surface measure` *(or, surface area element), whereas in (13.6.7), $d\sigma$ stands for the* `arc-length measure`.

13.7 Integration by Parts and Green's Formula

Definition 13.40. *We say that a nonempty open set $\Omega \subseteq \mathbb{R}^n$, where $n \geq 2$, is a C^k domain (or, a domain of class C^k), for some $k \in \mathbb{N}_0 \cup \{\infty\}$, provided the following holds. For every point $x^* \in \partial\Omega$ there exist $R > 0$, an open interval $I \subset \mathbb{R}$ with $0 \in I$, a rigid transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ with $T(x^*) = 0$, along with a function ϕ of class C^k that maps $B(0,R) \subseteq \mathbb{R}^{n-1}$ into I with the property that $\phi(0) = 0$ and, if $\mathcal{C}$ denotes the (open) cylinder $B(0,R) \times I \subseteq \mathbb{R}^{n-1} \times \mathbb{R} = \mathbb{R}^n$, then*

$$\mathcal{C} \cap T(\Omega) = \{x = (x', x_n) \in \mathcal{C} : x_n > \phi(x')\}, \tag{13.7.1}$$

$$\mathcal{C} \cap \partial T(\Omega) = \{x = (x', x_n) \in \mathcal{C} : x_n = \phi(x')\}, \tag{13.7.2}$$

$$\mathcal{C} \cap (\overline{T(\Omega)})^c = \{x = (x', x_n) \in \mathcal{C} : x_n < \phi(x')\}. \tag{13.7.3}$$

If $\Omega \subseteq \mathbb{R}^n$ is a C^k domain for some $k \in \mathbb{N}_0 \cup \{\infty\}$, then it may be easily verified that $\partial\Omega$ is a C^k surface.

Theorem 13.41 (Integration by parts formula)**.** *Suppose $\Omega \subset \mathbb{R}^n$ is a domain of class C^1 and $\nu = (\nu_1, \ldots, \nu_n)$ denotes its outward unit normal. Let $k \in \{1, \ldots, n\}$ and assume $f, g \in C^1(\Omega) \cap C^0(\overline{\Omega})$ are such that $\partial_k f$, $\partial_k g \in L^1(\Omega)$ and there exists a compact set $K \subset \mathbb{R}^n$ with the property that $f = 0$ on $\Omega \setminus K$. Then,*

$$\int_\Omega (\partial_k f) g \, dx = -\int_\Omega f(\partial_k g) \, dx + \int_{\partial\Omega} fg\nu_k \, d\sigma, \tag{13.7.4}$$

where σ is the surface measure on $\partial\Omega$.

For the sense in which the last integral in (13.7.4) should be understood see Definition 13.39.

An immediate corollary of Theorem 13.41 is Green's formula that is stated next (recall (7.1.14)).

Theorem 13.42 (Green's formula)**.** *Suppose $\Omega \subset \mathbb{R}^n$ is a bounded domain of class C^1 and ν denotes its outward unit normal. If $f, g \in C^2(\overline{\Omega})$, then*

$$\int_\Omega f \Delta g \, dx = \int_\Omega g \Delta f \, dx + \int_{\partial\Omega} \left(f \frac{\partial g}{\partial \nu} - g \frac{\partial f}{\partial \nu} \right) d\sigma. \tag{13.7.5}$$

13.8 Polar Coordinates and Integrals on Spheres

Assume that $n \geq 3$ and $R > 0$ are fixed. For $\rho \in (0, R)$, $\theta_j \in (0, \pi)$, $j \in \{1, \ldots, n-2\}$, and $\theta_{n-1} \in (0, 2\pi)$, set

$$\begin{aligned} x_1 &:= \rho \cos\theta_1, \\ x_2 &:= \rho \sin\theta_1 \cos\theta_2, \\ x_3 &:= \rho \sin\theta_1 \sin\theta_2 \cos\theta_3, \end{aligned} \tag{13.8.1}$$

$$\vdots$$

$$x_{n-1} := \rho \sin\theta_1 \sin\theta_2 \dots \sin\theta_{n-2} \cos\theta_{n-1},$$
$$x_n := \rho \sin\theta_1 \sin\theta_2 \dots \sin\theta_{n-2} \sin\theta_{n-1}.$$

The variables $\theta_1, \dots, \theta_{n-1}, \rho$ are called `polar coordinates`.

Definition 13.43. *Assume that $x_* \in \mathbb{R}^n$, $n \geq 3$, is a fixed point, and $R > 0$ is arbitrary. The* `standard parametrization of the ball` $B(x_*, R)$ *is defined as*

$$\begin{aligned} &\mathcal{P} : (0,\pi)^{n-2} \times (0,2\pi) \times (0,R) \longrightarrow \mathbb{R}^n, \\ &\mathcal{P}(\theta_1, \theta_2, \dots, \theta_{n-1}, \rho) := x_* + (x_1, x_2, \dots, x_n), \end{aligned} \tag{13.8.2}$$

where $x_1, \dots, x_n$ are as in (13.8.1).

The function $\mathcal{P}$ in (13.8.2) is injective, of class C^∞, takes values in $B(x_*, R)$, its image differs from $B(x_*, R)$ by a subset of measure zero and

$$\det(D\mathcal{P})(\theta_1, \theta_2, \cdots, \theta_{n-1}, \rho) = \rho^{n-1} (\sin\theta_1)^{n-2} (\sin\theta_2)^{n-3} \cdots (\sin\theta_{n-2}), \tag{13.8.3}$$

at every point in its domain, where $D\mathcal{P}$ denotes the Jacobian of $\mathcal{P}$. Using this standard parametrization for the unit sphere in $\mathbb{R}^n$, we see that

$$\omega_{n-1} = \int_0^\pi \int_0^\pi \dots \int_0^\pi \int_0^{2\pi} (\sin\varphi_1)^{n-2} (\sin\varphi_2)^{n-3} \dots (\sin\varphi_{n-2}) \, \mathrm{d}\varphi_{n-1} \, \mathrm{d}\varphi_{n-2} \dots \mathrm{d}\varphi_1. \tag{13.8.4}$$

This parametrization of the sphere $B(x_*, R)$ may also be used to prove the following theorem.

Theorem 13.44 (Spherical Fubini and polar coordinates)**.** *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$, $n \geq 2$. Then for each $x_* \in \mathbb{R}^n$ and each $R > 0$ the following formulas hold:*

$$\int_{B(x_*,R)} f \, \mathrm{d}x = \int_0^R \Big(\int_{\partial B(x_*,\rho)} f \, \mathrm{d}\sigma \Big) \mathrm{d}\rho, \tag{13.8.5}$$

$$\int_{B(x_*,R)} f \, \mathrm{d}x = \int_0^R \int_{S^{n-1}} f(x_* + \rho\omega) \rho^{n-1} \, \mathrm{d}\sigma(\omega) \, \mathrm{d}\rho \tag{13.8.6}$$

$$= \int_0^R \int_{\partial B(x_*,1)} f(\rho\omega) \rho^{n-1} \, \mathrm{d}\sigma(\omega) \, \mathrm{d}\rho. \tag{13.8.7}$$

Moreover, if $f \in L^1(\mathbb{R}^n)$, then

$$\int_{\mathbb{R}^n} f \, \mathrm{d}x = \int_0^\infty \Big(\int_{\partial B(0,\rho)} f \, \mathrm{d}\sigma \Big) \mathrm{d}\rho, \tag{13.8.8}$$

$$\int_{\mathbb{R}^n} f \, \mathrm{d}x = \int_0^\infty \int_{S^{n-1}} f(\rho\omega) \rho^{n-1} \, \mathrm{d}\sigma(\omega) \, \mathrm{d}\rho. \tag{13.8.9}$$

Note that if $P : \mathcal{O} \to S^{n-1}$ is a parametrization of the unit sphere in $\mathbb{R}^n$ and if $\mathcal{R}$ is a unitary transformation in $\mathbb{R}^n$, then $\mathcal{R} \circ P : \mathcal{O} \to S^{n-1}$ is also a parametrization of the unit sphere in $\mathbb{R}^n$. Indeed, this function is injective, has C^1 components, its image is S^{n-1} (up to a negligible set) and

$$\begin{aligned} \operatorname{Rank}\Big[D(\mathcal{R}\circ P)\Big] &= \dim\Big(\operatorname{Im}\big[\mathcal{R}(DP)\big]\Big) \\ &= \dim\Big(\operatorname{Im}(DP)\Big) = \operatorname{Rank}(DP) = n-1. \end{aligned} \tag{13.8.10}$$

Hence,

$$\begin{aligned} \int_{S^{n-1}} f\circ\mathcal{R}\,\mathrm{d}\sigma &= \int_{\mathcal{O}} f(\mathcal{R}\circ P)|\partial_{u_1}P\times\cdots\times\partial_{u_{n-1}}P|\,\mathrm{d}u_1\ldots\mathrm{d}u_{n-1} \\ &= \int_{\mathcal{O}} f(\mathcal{R}\circ P)|\partial_{u_1}(\mathcal{R}\circ P)\times\cdots\times\partial_{u_{n-1}}(\mathcal{R}\circ P)|\,\mathrm{d}u_1\cdots\mathrm{d}u_{n-1} \\ &= \int_{S^{n-1}} f\,\mathrm{d}\sigma. \end{aligned} \tag{13.8.11}$$

The same type of reasoning also yields the following result.

Proposition 13.45. *For each $x = (x_1,\ldots,x_n) \in \mathbb{R}^n$, define*

$$\begin{aligned} &R_j(x) := (x_1,\ldots,x_{j-1},-x_j,x_{j+1},\ldots,x_n), \quad 1\le j\le n, \\ &R_{jk}(x) := (x_1,\ldots,x_{j-1},x_k,x_{j+1},\ldots,x_{k-1},x_j,x_{k+1},\ldots,x_n), \quad 1\le j\le k\le n. \end{aligned} \tag{13.8.12}$$

Then

$$\int_{S^{n-1}} f\circ R_j\,\mathrm{d}\sigma = \int_{S^{n-1}} f\,\mathrm{d}\sigma, \quad j=1,2,\ldots,n, \tag{13.8.13}$$

$$\int_{S^{n-1}} f\circ R_{jk}\,\mathrm{d}\sigma = \int_{S^{n-1}} f\,\mathrm{d}\sigma, \quad 1\le j\le k\le n. \tag{13.8.14}$$

Proposition 13.46. *Let $v\in\mathbb{R}^n\setminus\{0\}$, $n\ge 2$, be fixed. Then for any real-valued function f defined on the real line, there holds*

$$\int_{S^{n-1}} f(v\cdot\theta)\,\mathrm{d}\sigma(\theta) = \omega_{n-2}\int_{-1}^{1} f(s|v|)(\sqrt{1-s^2})^{n-3}\,\mathrm{d}s. \tag{13.8.15}$$

Proof. Since integrals over the unit sphere are invariant under orthogonal transformations, we may assume that $v/|v| = e_1$ and, hence, using polar coordinates and (13.8.3), we have

$$\begin{aligned} \int_{S^{n-1}} f(v\cdot\theta)\,\mathrm{d}\sigma(\theta) &= \int_{S^{n-1}} f(|v|\theta_1)\,\mathrm{d}\sigma(\theta) \\ &= \int_0^{2\pi}\Big(\int_0^{\pi}\cdots\int_0^{\pi} f(|v|\cos\varphi_1)\prod_{j=1}^{n-2}(\sin\varphi_j)^{n-1-j}\mathrm{d}\varphi_1\cdots\mathrm{d}\varphi_{n-2}\Big)\mathrm{d}\varphi_{n-1} \\ &= \omega_{n-2}\int_0^{\pi} f(|v|\cos\varphi_1)(\sin\varphi_1)^{n-2}\,\mathrm{d}\varphi_1. \end{aligned} \tag{13.8.16}$$

Making the change of variables $s := \cos\varphi_1$ in the last integral above shows that this matches the right-hand side of (13.8.15). □

Proposition 13.47. *Let $f \in C^0(\mathbb{R})$ be positive homogeneous of degree $m \in \mathbb{R}$ and fix $\eta = (\eta_1, \dots, \eta_n) \in \mathbb{R}^n$. Then if $n \in \mathbb{N}$ with $n \geq 2$, for $j, k \in \{1, \dots, n\}$ one has*

$$\int_{S^{n-1}} f(\eta \cdot \xi)\xi_j\xi_k \, d\sigma(\xi) = \alpha|\eta|^m\delta_{jk} + \beta|\eta|^{m-2}\eta_j\eta_k \tag{13.8.17}$$

where

$$\begin{aligned} \alpha &= \frac{1}{n-1}\int_{S^{n-1}} f(\xi_1)(1-\xi_1^2)\, d\sigma(\xi), \\ \beta &= \frac{1}{n-1}\int_{S^{n-1}} f(\xi_1)(n\xi_1^2 - 1)\, d\sigma(\xi). \end{aligned} \tag{13.8.18}$$

Proof. For $j, k \in \{1, \dots, n\}$ set

$$q_{jk}(\eta) := \int_{S^{n-1}} f(\eta \cdot \xi)\xi_j\xi_k \, d\sigma(\xi), \quad \eta \in \mathbb{R}^n, \tag{13.8.19}$$

and define the quadratic form

$$Q(\zeta, \eta) := \sum_{j,k=1}^{n} q_{jk}(\eta)\zeta_j\zeta_k, \quad \forall\, \zeta,\, \eta \in \mathbb{R}^n. \tag{13.8.20}$$

Observe that we can write $Q(\zeta, \eta) = \int_{S^{n-1}} f(\eta\cdot\xi)(\zeta\cdot\xi)^2 \, d\sigma(\xi)$. By the invariance under rotations of integrals over S^{n-1} (see (13.8.11)), we have that for any rotation R in $\mathbb{R}^n$

$$Q(\zeta, \eta) = \int_{S^{n-1}} f(\eta\cdot R^\top\xi)(\zeta\cdot R^\top\xi)^2 \, d\sigma(\xi) = Q(R\zeta, R\eta), \quad \zeta,\, \eta \in \mathbb{R}^n. \tag{13.8.21}$$

A direct computation also gives that

$$Q(\lambda_1\zeta, \lambda_2\eta) = \lambda_1^2\lambda_2^m Q(\zeta, \eta) \quad \text{for all } \lambda_1,\, \lambda_2 > 0,\, \zeta,\, \eta \in \mathbb{R}^n. \tag{13.8.22}$$

Next we claim that

$$Q(\zeta, \eta) = \alpha + \beta(\eta \cdot \zeta)^2, \quad \forall\, \zeta,\, \eta \in S^{n-1}. \tag{13.8.23}$$

To show that (13.8.23) holds, we first observe that it suffices to prove (13.8.23) when $\eta = e_1$. Indeed, if we assume that (13.8.23) is true for $\eta = e_1$, then for an arbitrary η let R be the rotation such that $R\eta = e_1$. Then if we also take into account (13.8.20), we have

$$\begin{aligned} Q(\zeta, \eta) = Q(R\zeta, R\eta) = Q(R\zeta, e_1) &= \alpha + \beta(R\zeta \cdot e_1)^2 \\ &= \alpha + \beta(\zeta \cdot R^\top e_1)^2 = \alpha + \beta(\eta \cdot \zeta)^2. \end{aligned} \tag{13.8.24}$$

Hence, (13.8.23) will follow if we prove that

$$\int_{S^{n-1}} f(\xi_1)(\zeta\cdot\xi)^2\,d\sigma(\xi)=\alpha+\beta\,\xi_1^2,\quad\forall\,\zeta\in S^{n-1}. \tag{13.8.25}$$

To see the later we let $A_{jk}:=\int_{S^{n-1}} f(\xi_1)\xi_j\xi_k\,d\sigma(\xi)$ for $j,k\in\{1,\dots,n\}$. Assume $j\neq k$. Then either $j\neq 1$ or $k\neq 1$. If, say, $j\neq 1$ we use (13.8.13) to conclude that $A_{jk}=0$ in this case. A similar reasoning applies if $k\neq 1$. Clearly, $A_{11}=\int_{S^{n-1}} f(\xi_1)\xi_1^2\,d\sigma(\xi)$. As for the case $j=k\neq 1$, we first observe that $A_{22}=\cdots=A_{nn}$ since A_{jj} is independent of $j\in\{2,\dots,n\}$ due to (13.8.14). Moreover, $\sum\limits_{j=1}^{n} A_{jj}=\int_{S^{n-1}} f(\xi_1)\,d\sigma(\xi)$, which in turn implies that $A_{jj}=\frac{1}{n-1}\Big(\int_{S^{n-1}} f(\xi_1)\,d\sigma(\xi)-A_{11}\Big)=\alpha$ for each $j=2,\dots,n$. Combining all these we have that for $\zeta\in S^{n-1}$,

$$\begin{aligned}\int_{S^{n-1}} f(\xi_1)(\zeta\cdot\xi)^2\,d\sigma(\xi)&=\sum_{j,k=1}^{n}\zeta_j\zeta_k A_{jk}\\&=\zeta_1^2\int_{S^{n-1}} f(\xi_1)\xi_1^2\,d\sigma(\xi)+\alpha\sum_{j=2}^{n}\zeta_j^2\\&=\alpha+\beta\zeta_1^2.\end{aligned} \tag{13.8.26}$$

This concludes the proof of (13.8.25) which, in turn, implies (13.8.23). Now if $\zeta,\,\eta\in\mathbb{R}^n\setminus\{0\}$, we make use of (13.8.22) and (13.8.23) to write

$$\begin{aligned}Q(\zeta,\eta)&=|\zeta|^2|\eta|^m Q\Big(\frac{\zeta}{|\zeta|},\frac{\eta}{|\eta|}\Big)=\alpha|\zeta|^2|\eta|^m+\beta|\eta|^{m-2}(\zeta\cdot\eta)^2\\&=\sum_{j,k=1}^{n}\big(\alpha|\eta|^m\delta_{jk}+\beta|\eta|^{m-2}\eta_j\eta_k\big)\zeta_j\zeta_k\end{aligned} \tag{13.8.27}$$

which proves (13.8.17). □

Proposition 13.48. *Consider $f(t):=|t|$ for $t\in\mathbb{R}$, and let α,β be as in (13.8.17) for this choice of f. Then*

$$\alpha=\beta=\frac{2\omega_{n-2}}{n^2-1}, \tag{13.8.28}$$

where ω_{n-2} denotes the surface measure of the unit ball in $\mathbb{R}^{n-1}$.

Proof. Using the standard parametrization of S^{n-1} (see (13.8.1) with $R=1$) we have

$$\alpha = \frac{1}{n-1}\int_0^{\pi}\int_0^{\pi}\cdots\int_0^{\pi}\int_0^{2\pi} |\cos\varphi_1|(1-(\cos\varphi_1)^2)(\sin\varphi_1)^{n-2}(\sin\varphi_2)^{n-3}\times$$
$$\cdots\times(\sin\varphi_{n-2})\,d\varphi_{n-1}\,d\varphi_{n-2}\cdots d\varphi_1$$
$$= \frac{\omega_{n-2}}{n-1}\int_0^{\pi}|\cos\varphi_1|(\sin\varphi_1)^n\,d\varphi_1. \tag{13.8.29}$$

The change of variables $\theta=\pi-y$ yields $-\int_{\frac{\pi}{2}}^{\pi}\cos\theta(\sin\theta)^n\,d\theta=\int_0^{\frac{\pi}{2}}\cos y(\sin y)^n\,dy$. Using this back in (13.8.29) then gives

$$\alpha = \frac{2\,\omega_{n-2}}{n-1}\int_0^{\frac{\pi}{2}}\cos\theta\,(\sin\theta)^n\,d\theta = \frac{2\,\omega_{n-2}}{n-1}\,\frac{(\sin\theta)^{n+1}}{n+1}\Big|_0^{\frac{\pi}{2}} = \frac{2\,\omega_{n-2}}{n^2-1}. \tag{13.8.30}$$

Similar arguments in computing β give that

$$\beta = \frac{\omega_{n-2}}{n-1}\int_0^{\pi}|\cos\theta|[n(\cos\theta)^2-1](\sin\theta)^{n-2}\,d\theta \tag{13.8.31}$$
$$= \omega_{n-2}\int_0^{\pi}|\cos\theta|(\sin\theta)^{n-2}\,d\theta - \frac{n\,\omega_{n-2}}{n-1}\int_0^{\pi}|\cos\theta|(\sin\theta)^n\,d\theta$$
$$= \frac{2\,\omega_{n-2}}{n-1}\Big[(n-1)\int_0^{\frac{\pi}{2}}\cos\theta\,(\sin\theta)^{n-2}\,d\theta - n\int_0^{\frac{\pi}{2}}\cos\theta\,(\sin\theta)^n\,d\theta\Big]$$
$$= \frac{2\,\omega_{n-2}}{n-1}\Big[(\sin\theta)^{n-1}\Big|_0^{\frac{\pi}{2}} - n\,\frac{(\sin\theta)^{n+1}}{n+1}\Big|_0^{\frac{\pi}{2}}\Big] = \frac{2\,\omega_{n-2}}{n^2-1}.$$

This finishes the proof. □

Recall from 0.0.7 that$z^\alpha = z_1^{\alpha_1}z_2^{\alpha_2}\cdots z_n^{\alpha_n}$ whenever $z=(z_1,\ldots,z_n)\in\mathbb{R}^n$ and $\alpha=(\alpha_1,\alpha_2,\ldots,\alpha_n)\in\mathbb{N}_0^n$. Let us also introduce

$$2\mathbb{N}_0^n := \big\{(2\alpha_1,2\alpha_2,\ldots,2\alpha_n):\,\alpha=(\alpha_1,\alpha_2,\ldots,\alpha_n)\in\mathbb{N}_0^n\big\}. \tag{13.8.32}$$

The next proposition deals with the issue of integrating arbitrary monomials on the unit sphere centered at the origin.

Proposition 13.49. *For each multi-index $\alpha=(\alpha_1,\ldots,\alpha_n)\in\mathbb{N}_0^n$,*

$$\int_{S^{n-1}} z^\alpha\,d\sigma(z) = \begin{cases} 0 & \text{if } \alpha\notin 2\mathbb{N}_0^n,\\[2ex] \dfrac{\left(\frac{|\alpha|}{2}\right)!}{|\alpha|!}\cdot\dfrac{\alpha!}{\left(\frac{\alpha}{2}\right)!}\,\dfrac{2\pi^{\frac{n-1}{2}}\Gamma\left(\frac{1+|\alpha|}{2}\right)}{\Gamma\left(\frac{|\alpha|+n}{2}\right)} & \text{if } \alpha\in 2\mathbb{N}_0^n, \end{cases} \tag{13.8.33}$$

where Γ is the gamma function introduced in (13.5.1).

Proof. Fix an arbitrary $k \in \mathbb{N}$ and set

$$q_\alpha := \int_{S^{n-1}} z^\alpha \, d\sigma(z), \quad \forall\, \alpha \in \mathbb{N}_0^n \text{ with } |\alpha| = k. \tag{13.8.34}$$

Also, with a "dot" standing for the standard inner product in $\mathbb{R}^n$, introduce

$$Q_k(x) := \sum_{\substack{\alpha\in\mathbb{N}_0^n \\ |\alpha|=k}} \frac{k!}{\alpha!}\, q_\alpha x^\alpha, \quad x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n. \tag{13.8.35}$$

Then (13.2.1) implies that

$$Q_k(x) = \int_{S^{n-1}} (z \cdot x)^k \, d\sigma(z), \quad \forall\, x \in \mathbb{R}^n. \tag{13.8.36}$$

Let us also observe here that, if $x \in S^{n-1}$ is arbitrary but fixed and if $\mathcal{R}$ is a rotation about the origin in $\mathbb{R}^n$ such that $\mathcal{R}^{-1}x = e_n := (0, \ldots, 0, 1) \in \mathbb{R}^n$, then by (13.8.35) and the rotation invariance of integrals on S^{n-1} (cf. (13.8.11)), we have

$$Q_k(x) = \int_{S^{n-1}} (\mathcal{R}z \cdot x)^k \, d\sigma(z) = Q_k(e_n). \tag{13.8.37}$$

By the homogeneity of Q_k, (13.8.37) implies that $Q_k(x) = |x|^k Q_k(e_n)$ for all $x \in \mathbb{R}^n$ and, hence,

$$\sum_{|\alpha|=k} \frac{k!}{\alpha!}\, q_\alpha x^\alpha = |x|^k Q_k(e_n) \quad \text{for all } x \in \mathbb{R}^n. \tag{13.8.38}$$

We now consider two cases:

Case I: k is odd. In this scenario, the mapping $S^{n-1} \ni z \mapsto z_n^k \in \mathbb{R}$ is odd. In particular, $Q_k(x) = \int_{S^{n-1}} z_n^k \, d\sigma(z) = 0$. This, in turn, along with (13.8.38) then force $\sum_{|\alpha|=k} \frac{k!}{\alpha!}\, q_\alpha x^\alpha = 0$ for every $x \in \mathbb{R}^n$. From (13.2.5) it is easy to deduce that for each $\beta \in \mathbb{N}_0^n$ we have

$$\partial_x^\gamma [x^\alpha]\Big|_{x=0} = \begin{cases} 0 & \text{if } \alpha \neq \gamma, \\ \alpha! & \text{if } \alpha = \gamma. \end{cases} \tag{13.8.39}$$

We may therefore conclude that $q_\gamma = \partial_x^\gamma \left[\sum_{|\alpha|=k} \frac{1}{\alpha!}\, q_\alpha x^\alpha \right] = 0$ for each $\gamma \in \mathbb{N}_0^n$ with $|\gamma| = k$, in agreement with (13.8.33).

Case II: k is even. Suppose $k = 2m$ for some $m \in \mathbb{N}$. Then, $|x|^k = \sum_{|\beta|=m} \frac{m!}{\beta!} x^{2\beta}$ and (13.8.38) becomes

$$\sum_{|\beta|=m} \frac{m!}{\beta!} x^{2\beta} Q_k(e_n) = \sum_{|\alpha|=k} \frac{k!}{\alpha!} q_\alpha x^\alpha \quad \text{for all } x \in \mathbb{R}^n. \tag{13.8.40}$$

Fix $\gamma \in \mathbb{N}_0^n$ such that $|\gamma| = k$ and observe that

$$\partial_x^\gamma \left[\frac{m!}{\beta!} x^{2\beta} Q_k(e_n)\right]\Big|_{x=0} = \begin{cases} \frac{m!}{\beta!}(2\beta)! Q_k(e_n) & \text{if } \gamma = 2\beta, \text{ some } \beta \in \mathbb{N}_0^n, \ |\beta| = m, \\ 0 & \text{otherwise.} \end{cases} \tag{13.8.41}$$

This, (13.8.40), and (13.8.39) then imply that

$$q_\gamma = \begin{cases} 0 & \text{if } \gamma \notin 2\mathbb{N}_0^n, \\ \frac{\left(\frac{|\gamma|}{2}\right)!}{\left(\frac{\gamma}{2}\right)!} \cdot \frac{\gamma!}{|\gamma|!} Q_{|\gamma|}(e_n) & \text{if } \gamma \in 2\mathbb{N}_0^n. \end{cases} \tag{13.8.42}$$

We are now left with computing $Q_{2m}(e_n)$ when $m \in \mathbb{N}$. Using spherical coordinates, a direct computation gives that

$$\begin{aligned} Q_{2m}(e_n) &= \int_{S^{n-1}} (z \cdot e_n)^{2m} \, d\sigma(z) \\ &= \int_0^{2\pi} \Big(\int_0^\pi \dots \int_0^\pi (\cos\theta_1)^{2m} \prod_{j=1}^{n-2} (\sin\theta_j)^{n-1-j} \, d\theta_1 \cdots d\theta_{n-2} \Big) \, d\theta_{n-1} \\ &= \omega_{n-2} \int_0^\pi (\cos\theta)^{2m} (\sin\theta)^{n-2} \, d\theta = \frac{2\pi^{\frac{n-1}{2}} \Gamma(\frac{1}{2} + m)}{\Gamma(m + \frac{n}{2})}, \end{aligned} \tag{13.8.43}$$

by (13.8.4) and (13.5.6) (considered with $n-1$ in place of n), and (13.5.11). This once again agrees with (13.8.33), and the proof of Proposition 13.49 is finished. □

A simple useful consequence of the general formula (13.8.33) is the fact that

$$\int_{S^{n-1}} z_j z_k \, d\sigma(z) = \frac{\omega_{n-1}}{n} \, \delta_{jk} \quad \text{whenever } 1 \leq j, k \leq n. \tag{13.8.44}$$

13.9 Tables of Fourier Transforms

Fourier Transforms of Schwartz Functions

f	$\mathcal{F}(f)$	Location
$\mathrm{e}^{-\lambda\lvert x\rvert^2}$, $\lambda \in \mathbb{C}$, $\mathrm{Re}(\lambda) > 0$	$\left(\frac{\pi}{\lambda}\right)^{\frac{n}{2}} \mathrm{e}^{-\frac{\lvert\xi\rvert^2}{4\lambda}}$	Example 3.21
e^{-ax^2+ibx}, $a > 0$ and $b \in \mathbb{R}$ fixed	$\left(\frac{\pi}{a}\right)^{\frac{1}{2}} \mathrm{e}^{-\frac{(\xi-b)^2}{4a}}$	Exercise 3.22
$\mathrm{e}^{-(Ax)\cdot x}$, $A \in \mathcal{M}_{n\times n}(\mathbb{R})$, $A = A^\top$, $(Ax)\cdot x > 0$ for $x \in \mathbb{R}^n \setminus \{0\}$	$\frac{\pi^{\frac{n}{2}}}{\sqrt{\det A}} \mathrm{e}^{-\frac{(A^{-1}\xi)\cdot\xi}{4}}$	Exercise 3.36
$\mathrm{e}^{-(x_1^2+x_1x_2+x_2^2)}$ for $(x_1, x_2) \in \mathbb{R}^2$	$\frac{2\pi}{\sqrt{3}} \mathrm{e}^{-(\xi_1^2-\xi_1\xi_2+\xi_2^2)/3}$	Exercise 3.37
$\mathrm{e}^{-a\lvert x\rvert^2} \sin(x\cdot x_0)$, for $x \in \mathbb{R}^n$, if $a > 0$ and $x_0 \in \mathbb{R}^n$ are fixed	$\frac{1}{2i}\left(\frac{\pi}{a}\right)^{\frac{n}{2}} \left[\mathrm{e}^{-\frac{\lvert\xi-x_0\rvert^2}{4a}} - \mathrm{e}^{-\frac{\lvert\xi+x_0\rvert^2}{4a}}\right]$	Exercise 3.39

Fourier Transforms of Tempered Distributions in $\mathbb{R}$
Below $x, \xi \in \mathbb{R}$.

u	$\mathcal{F}(u)$	Location
$\frac{1}{x^2+a^2}$, $a \in (0,\infty)$ given	$\frac{\pi}{a}\mathrm{e}^{-a\lvert\xi\rvert}$	Example 4.23
$\frac{x}{x^2+a^2}$, $a \in (0,\infty)$ given	$-\pi i(\operatorname{sgn}\xi)\mathrm{e}^{-a\lvert\xi\rvert}$	Example 4.29
$\chi_{[-a,a]}$, $a \in (0,\infty)$ given	$\begin{cases} 2\frac{\sin(a\xi)}{\xi} & \text{for } \xi \in \mathbb{R}\setminus\{0\} \\ 2a & \text{for } \xi = 0 \end{cases}$	Example 4.34
$\sin(b\,x)$, $b \in \mathbb{R}$ fixed	$-i\pi\delta_b + i\pi\delta_{-b}$	Example 4.36
$\cos(b\,x)$, $b \in \mathbb{R}$	$\pi\delta_b + \pi\delta_{-b}$	Exercise 4.122
$\sin(b\,x)\sin(c\,x)$, $b, c \in \mathbb{R}$ fixed	$-\frac{\pi}{2}\left(\delta_{b+c} - \delta_{b-c} - \delta_{c-b} + \delta_{-b-c}\right)$	Example 4.37
H	$-i\text{P.V.}\,\frac{1}{\xi} - c\,\delta$	Exercise 4.115
P.V. $\frac{1}{x}$	$i\pi\,\operatorname{sgn}(\xi)$	Exercise 4.116
$\operatorname{sgn} x$	$-2i\,\text{P.V.}\,\frac{1}{\xi}$	Exercise 4.117
$\lvert x\rvert^k$, $k \in \mathbb{N}$ even	$2\pi i^k\delta^{(k)}$	Exercise 4.117
$\lvert x\rvert^k$, $k \in \mathbb{N}$ odd	$-2i^{k+1}\left(\text{P.V.}\,\frac{1}{\xi}\right)^{(k)}$	Exercise 4.117
$\frac{\sin(b\,x)}{x}$, $b \in \mathbb{R}$ fixed	$\pi\,\operatorname{sgn}(b)\chi_{[-\lvert b\rvert,\lvert b\rvert]}$	Exercise 4.117
$\frac{\sin(b\,x)\sin(c\,x)}{x}$, $b, c \in \mathbb{R}$ fixed	$\frac{i\pi}{2}\operatorname{sgn}(c)\left[\chi_{[-b-\lvert c\rvert,-b+\lvert c\rvert]} - \chi_{[b-\lvert c\rvert,b+\lvert c\rvert]}\right]$	Exercise 4.117
$\sin(x^2)$	$\frac{\sqrt{\pi}}{2i}\left[\mathrm{e}^{i\frac{\pi-\xi^2}{4}} - \mathrm{e}^{-i\frac{\pi-\xi^2}{4}}\right]$	Exercise 4.117
$\ln\lvert x\rvert$	$-2\pi\gamma\delta - \pi w_{\chi_{(-1,1)}}$ $w_{\chi_{(-1,1)}}$ from (4.6.21), γ from (4.6.23)	Exercise 4.117

Fourier Transforms of Tempered Distributions
Below $x, \xi \in \mathbb{R}^n$ and $x', \xi' \in \mathbb{R}^{n-1}$.

u	$\mathcal{F}(u)$	Location
δ	1	Example 4.21
1	$(2\pi)^n \delta$	Exercise 4.31
$\mathrm{e}^{-ia\lvert x\rvert^2}$, $a \in (0,\infty)$ given	$\left(\frac{\pi}{a}\right)^{\frac{n}{2}} \mathrm{e}^{-i\frac{n\pi}{4}} \mathrm{e}^{i\frac{\lvert\xi\rvert^2}{4a}}$	Example 4.24
$\mathrm{e}^{ia\lvert x\rvert^2}$, $a \in (0,\infty)$ given	$\left(\frac{\pi}{a}\right)^{\frac{n}{2}} \mathrm{e}^{i\frac{n\pi}{4}} \mathrm{e}^{-i\frac{\lvert\xi\rvert^2}{4a}}$	Example 4.38
$\lvert x\rvert^{-\lambda}$, $\lambda \in (0,n)$ fixed	$2^{n-\lambda}\pi^{\frac{n}{2}} \frac{\Gamma\left(\frac{n-\lambda}{2}\right)}{\Gamma\left(\frac{\lambda}{2}\right)} \lvert\xi\rvert^{\lambda-n}$	Proposition 4.61
$\frac{x_j}{\lvert x\rvert^n}$, $n \geq 2$	$-i\,\omega_{n-1} \frac{\xi_j}{\lvert\xi\rvert^2}$	Corollary 4.62
$\frac{x_j}{\lvert x\rvert^{\lambda+2}}$, $n \geq 2$, $\lambda \in [0, n-1)$, $j \in \{1,\dots,n\}$	$-i\,2^{n-\lambda-1}\pi^{\frac{n}{2}} \frac{\Gamma\left(\frac{n-\lambda}{2}\right)}{\Gamma\left(\frac{\lambda}{2}+1\right)} \frac{\xi_j}{\lvert\xi\rvert^{n-\lambda}}$	Corollary 4.62
$\frac{x_j x_k}{\lvert x\rvert^n}$, $n \geq 3$, $j,k \in \{1,\dots,n\}$	$\omega_{n-1} \frac{\delta_{jk}}{\lvert\xi\rvert^2} - 2\omega_{n-1} \frac{\xi_j\xi_k}{\lvert\xi\rvert^4}$	Exercise 7.63
P.V. $\left(\partial_j\left[\frac{x_k}{\lvert x\rvert^n}\right]\right)$, $j,k \in \{1,\dots,n\}$	$\omega_{n-1} \frac{\xi_j\xi_k}{\lvert\xi\rvert^2} - \frac{\omega_{n-1}}{n}\delta_{jk}$	Proposition 4.70
P.V. Θ, with Θ as in (4.4.1)	$-\int_{S^{n-1}} \Theta(\omega) \log(i(\xi\cdot\omega))\, \mathrm{d}\sigma(\omega)$	Theorem 4.71
P.V. $\left(\frac{x_j}{\lvert x\rvert^{n+1}}\right)$, $j \in \{1,\dots,n\}$	$-\frac{i\omega_n}{2} \frac{\xi_j}{\lvert\xi\rvert}$	Proposition 4.81
$\frac{2}{\omega_{n-1}} \frac{t}{(t^2+\lvert x'\rvert^2)^{\frac{n}{2}}}$ with $t > 0$ fixed	$\mathrm{e}^{-t\lvert\xi'\rvert}$	Proposition 4.87
$\frac{2}{\omega_{n-1}} \frac{x_j}{(t^2+\lvert x'\rvert^2)^{\frac{n}{2}}}$ with $t > 0$ fixed, $j \in \{1,\dots,n-1\}$	$-i\frac{\xi_j}{\lvert\xi'\rvert}\mathrm{e}^{-t\lvert\xi'\rvert}$	Proposition 4.89

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Subject Index

absorbing set, 417
Arzelà-Ascoli's Theorem, 426
balanced set, 417
base for a topology, 415
Beta function, 434
bi-harmonic, 237
Binomial Theorem, 424
Cauchy problem
  heat operator, 280
  vibrating infinite string, 1
  wave operator, 307
Cauchy–Clifford operator, 257
Cauchy operator, 252
Cauchy sequence, 416
Clifford algebra, 253
conjugate Poisson kernels, 165
conormal derivative, 271
continuous, 416
convergence in
  $\mathcal{D}(\Omega)$, 11
  $\mathcal{E}(\Omega)$, 9
  $\mathcal{S}(\mathbb{R}^n)$, 93
convex set, 417
convolution of distributions, 61
convolution of functions, 59
convolution, 115, 116
cross product in $\mathbb{R}^n$, 436
cut-off functions, 428
differential operators
  bi-Laplacian, 207, 233
  Cauchy–Riemann, 248
  Dirac, 253, 255
  heat, 277
  higher order systems, 344
  Lamé, 319
  Laplacian, 207, 217, 220
  poly-harmonic, 207, 240
  scalar second order, 260
  Schrödinger, 287
  second order parabolic, 282
  Stokes, 334
  wave, 289
dilations, 130
Dirac distribution δ, 21, 31, 36, 120
Dirichlet problem, 165, 370
distance, 416
distribution
  compactly supported, 40, 41
  definition of, 17
  of finite order, 19
  of function type, 19
  tempered, 109
distributional derivative, 29
elliptic operators, 206, 207
Euler's constant, 150
Fourier transform, 89, 99, 105, 120, 123, 125, 142, 160
  partial, 128
Fréchet space, 418
$\mathcal{F}$-topology, 418
function spaces
  $\mathcal{D}(\Omega)$, 11, 422
  $\mathcal{D}_K(\Omega)$, 422
  $\mathcal{E}(\Omega)$, 9, 421
  $\mathcal{S}(\mathbb{R}^n)$, 91, 423
fundamental solutions
  definition of, 191
  for higher order systems, 348
  for scalar second order operators, 261
  for second order parabolic operators, 286
  for strongly elliptic operators, 270
  for the bi-Laplacian, 235
  for the Cauchy operator, 249
  for the Dirac operator, 256
  for the heat operator, 279
  for the Lamé operator, 323, 341
  for the Laplacian, 220
  for the poly-harmonic operator, 241
  for the Schrödinger operator, 288
  for the Stokes operator, 336, 343
  for the wave operator, 305
gamma function, 434
generalized volume potential, 179
Green's formula, 437
Hahn-Banach Theorem, 420
harmonic function, 217, 225
Heaviside function
  definition of, 6
  derivative, 31

D. Mitrea, *Distributions, Partial Differential Equations, and Harmonic Analysis*, Universitext, DOI 10.1007/978-1-4614-8208-6,

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Hilbert transform, 167, 171, 253
hypoelliptic operators, 201, 203, 207, 279
inductive limit topology, 418
integral representation formula, 210, 357
integration by parts formula, 437
integration on surfaces, 436
interior estimates
for elliptic homogeneous systems, 359
for hypoelliptic operators, 211
for the bi-Laplacian, 238
for the Lamé operator, 329, 331
for the Laplacian, 226
for the poly-harmonic operator, 245
invariant under orthogonal transformations, 131
inverse Fourier transform, 102
isometry, 425
layer potential operators
double layer, 232, 367
harmonic double-layer, 162
single layer, 232, 366
layer potential representation formula, 229, 272
Lebesgue's Differentiation Theorem, 425
Lebesgue's Dominated Convergence Theorem, 425
Leibniz's Formula, 424
Liouville's theorem
for bi-Laplacian, 238
for general operators, 191
for Lamé, 331
for Laplacian, 227
for systems, 312
for the heat operator, 280
Lipschitz function, 30, 77
locally finite, 431
Malgrange-Ehrenpreis theorem, 195
mean value formulas
for bi-Laplacian, 237
for Lamé, 328
for Laplacian, 225
metric space
complete, 416
definition of, 416
Multinomial Theorem, 424
multiplier, 168
Neumann problem, 372
Newtonian potential, 175, 177, 178, 232
open neighborhood, 415
open set, 415
order of a distribution, 19
orthogonal transformation, 131
parametrix, 205
parametrization, 436
Parseval's identity, 104
partition of unity
for arbitrary open covers, 432
for compact sets, 430
with preservation of indexes, 433
Planchérel's identity, 106
Plemelj formula, 252
Poisson equation, 192, 223, 332
Poisson kernel, 162
Poisson problem
for Lamé, 332
for the bi-Laplacian, 239
for the Laplacian, 227
polar coordinates, 438
positive definite matrix, 264
positive homogeneous
distribution, 132
function, 131
principal symbol, 189
principal value distribution, 20, 136
projection, 253, 259
***p*-subaveraging**, 361
Rademacher's Theorem, 425
real-analytic function, 214, 216, 227, 238, 245, 331, 347, 359
restriction of a distribution, 34
reverse Hölder estimate, 364
Riesz's Representation Theorem
for complex functionals, 426
for locally bounded functionals, 427
for positive functionals, 426
Riesz transforms, 167, 168, 221, 222, 228, 258
rigid transformation, 425
seminorm, 417
separating seminorms, 417
sequentially continuous, 416

sign function, 14
singular integral operator, 167, 172
singular support, 202
slowly increasing functions, 93
strongly elliptic matrix, 263
strongly elliptic operators, 263
p**-subaveraging**, 362, 364
support of an arbitrary function, 37
support of a distribution, 36
surface, 435
Taylor's Formula, 424
tensor product
of distributions, 55, 112
of functions, 48
the method of descent, 299
topological space
definition of, 415
Hausdorff, 416
metrizable, 416
separated, 416
topological vector space
definition, 416
dual, 418
locally convex, 417
topology
coarser, 416
definition of, 415
finer, 416
induced, 416
product, 416
transpose of an operator, 7
unique continuation property, 215
Urysohn's Lemma, 427
Vitali's Convergencen Theorem, 427
weak
derivative, 4
solution, 2
weak$*$-topology, 419
Young's Inequality, 425

Symbol Index

α ! defined on page, xix
$|\alpha|$ defined on page, xix

$\mathscr{C}$
  Cauchy operator in $\mathbb{C}$, 252
  Cauchy-Clifford operator, 257
χ_E characteristic function of the set E, xx
$f^\vee$, 104
$u^\vee$, 124
$C^k(\Omega)$, xx
$C^k(\overline{\Omega})$, xx
$C_0^k(\Omega)$, xx
$C_0^\infty(\Omega)$, xx
$\mathcal{C}\ell_n$ Clifford algebra with n generators, 253
$\odot$ Clifford algebra multiplication, 253
$*$ convolution of
  $\mathcal{S}(\mathbb{R}^n)$ with $\mathcal{S}'(\mathbb{R}^n)$, 116
  distributions, 61
  functions, 59
$v_1 \times v_2 \times \cdots \times v_{n-1}$ cross product in $\mathbb{R}^n$, 436

D Dirac operator, 255
$D = \left(\frac{1}{i}\partial_1, \ldots \frac{1}{i}\partial_n\right)$, 91
δ_{jk} Kronecker symbol, xix
δ Dirac distribution, 36
$\mathcal{D}(\Omega)$ test functions, 11, 422
$\mathcal{D}_K(\Omega)$ test functions supported in K, 11, 422
$\mathcal{D}'(\Omega)$ general distributions, 26, 422

$\{\mathbf{e}_j\}_{1\leq j\leq n}$ orthonormal basis in $\mathbb{R}^n$, xix
$\mathcal{E}(\Omega)$, 9, 421
$\mathcal{E}'(\Omega)$ compactly supported distributions, 41, 421
E_A fundamental solution for L_A
  elliptic, 270
  parabolic, 286
γ Euler's constant, 150

! factorial, xix
$\mathcal{F}$ Fourier transform, 89, 123
$\widehat{\cdot}$ Fourier transform, 89, 120
$\mathcal{F}^{-1}$ inverse Fourier transform, 102
$\mathcal{F}_x$ partial Fourier transform, 128
$\mathcal{F}_x^{-1}$ inverse partial Fourier transform, 129
$F_{m,n}$ fundamental solution for Δ^m in $\mathbb{R}^n$, 241

Γ gamma function, 434

H Heaviside function, 6
H Hilbert transform, 167

i complex imaginary unit, xix
$\fint_A f\, d\mu$ integral average, xx

Δ Laplace operator, 165
Δ^2 bi-Laplace operator, 207
Δ^m poly-harmonic operator, 207
$\partial_t - \Delta$ heat operator, 277
$\partial_t^2 - \Delta$ wave operator, 289
$L^p_{comp}(\Omega)$, xx
$L^p_{loc}(\Omega)$, xx
L_A
  elliptic, 262
  parabolic, 282

$\mathcal{M}_{n\times m}(\mathfrak{R})$ matrices with entries from the ring $\mathfrak{R}$, xxi, 314
$\mathcal{M}_{n\times m}\big(\mathcal{D}'(\Omega)\big)$, 309
$\mathcal{M}_{n\times m}\big(\mathcal{E}'(\Omega)\big)$, 311
m_Θ the Fourier transform of P.V. Θ, 142

N the Newtonian potential, 175
$\frac{\partial f}{\partial \nu}$ normal derivative of f, 220
∂_ν^A conormal derivative, 271

ω_{n-1} surface measure of unit ball in $\mathbb{R}^n$, xx, 434

∂^α partial derivative of order α, xix
φ^Δ defined on page, 60
Π_Φ generalized volume potential, 179
P
  harmonic Poisson kernel, 162
  projection, 253, 259
p_t defined on page, 163
$\mathscr{P}$ harmonic double layer, 162

D. Mitrea, *Distributions, Partial Differential Equations, and Harmonic Analysis*, Universitext, DOI 10.1007/978-1-4614-8208-6,

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$P(D)$ linear constant coefficient partial differential operator, 101, 189
$P(x, \partial)$ linear partial differential operator, 7
$P(\xi)$, 101, 189
$P_m(\xi)$ principal symbol of $P(D)$, 189
P.V. Θ principal value distribution associated with Θ, 136
P.V. $\frac{1}{x}$ principal value distribution associated with $\frac{1}{x}$, 20

Q_j defined on page, 165
$(q_j)_t$ defined on page, 166

Re real part of a complex number, xix
Im imaginary part of a complex number, xix
$u|_\omega$ restriction of the distribution u to the open set ω , 34
R_j Riesz transform, 167

$\mathcal{S}(\mathbb{R}^n)$ Schwartz functions, 91, 423
$\mathcal{S}'(\mathbb{R}^n)$ tempered distributions, 111, 423

$\operatorname{sgn} x$ sign function, 14
$\operatorname{sing\,supp} u$ singular support of the distribution u, 202
$\mathcal{L}(\mathbb{R}^n)$ slowly increasing functions, 93
$\operatorname{supp} f$ support of an arbitrary function f, 37
$\operatorname{supp} u$ support of the distribution u, 36

τ_t dilation, 130
$\otimes$ tensor product
of distributions, 55
of functions, 48
of tempered distributions, 112
t_{x_0} translation by x_0 map
of distributions, 66
of functions, 12
$P^\top$ transpose of the operator P, 7
T_Θ defined on page, 167

u_f distribution associated with f, 19, 110
$u|^{ver}_{\partial\mathbb{R}^n_+}$ vertical limit of u to $\partial\mathbb{R}^n_+$, 165